Lie Groups

***EYNTKA* Series**

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Introduction

A Lie group is a group that is at the same time a smooth manifold, with the two structures compatible: multiplication and inversion are smooth maps. The subject built on this definition runs on a single device, the passage between the group and its Lie algebra. Differentiating at the identity replaces a curved, nonlinear, global object by a finite-dimensional vector space with a bracket; the exponential map carries linear-algebraic conclusions back to the group. Everything this book does is an application of that dictionary: establishing that it is faithful (the correspondence theorems), computing what it translates structure into (root systems and the classification), and applying it to the representation theory of compact groups and the structure of noncompact ones.

The reader is assumed to know smooth manifolds, vector fields and flows, and the Frobenius theorem at the level of *Differential Geometry* [13]; groups and linear algebra at the level of *Core Algebra* [2]; the representation theory of finite groups from *Core Representation Theory* [3], whose orthogonality relations Part III generalizes; covering spaces and the fundamental group from *Topology* [9]; and, for the analytic material, Hilbert spaces and L^2 from *Functional Analysis* [14]. The existence of Haar measure is cited to *Real Analysis* [15, Existence of Haar Measure] and then specialized to compact groups, where the averaging arguments are carried out in full; the Frobenius theorem is cited, but the closed-subgroup theorem and the integration of subalgebras that rest on it are proved; the spectral theorem for compact self-adjoint operators is cited, but the Peter-Weyl theorem it powers is proved end to end.

The book has four parts. Part I builds the dictionary. It starts with matrix groups, where the exponential and the bracket are computable in coordinates, rebuilds both intrinsically on an abstract Lie group, and establishes the correspondence theorems: the closed-subgroup theorem, the integration of subalgebras, and the equivalence between simply connected groups and their Lie algebras. Homogeneous spaces and covering groups close the part, turning the correspondence into a machine for recognizing manifolds. Part II is Lie-algebra structure theory: Engel and Lie on the solvable side, the Cartan criteria and Weyl's complete reducibility on the semisimple one, then the root space decomposition and the classification of root systems by Dynkin diagram. Part III returns to groups and adds analysis: invariant integration forces complete reducibility, Peter-Weyl decomposes $L^2(G)$, maximal tori reduce class functions to the torus, and the Weyl character formula computes every irreducible character in closed form. Part IV removes compactness: real forms and

the Cartan decomposition split a semisimple group into a maximal compact subgroup and a Euclidean complement, the Iwasawa decomposition organizes the noncompact directions, and chapter 15 reads the same data geometrically, as a Riemannian symmetric space, taking advantage of *Riemannian Geometry* [7].

One example runs through the whole book: the group SU(2). It is met in chapter 1 as a sphere, covers SO(3) twice, has a visible maximal torus and Weyl group, carries the irreducible representations that the character formula first computes, and acquires a noncompact sibling, $\mathrm{SL}(2,\mathbb{R})$, on which every theorem of Part IV is tested. When an abstract statement lands, its meaning for SU(2) is worked out nearby; the scaling-up family $\mathrm{SU}(n)$ and $\mathrm{SO}(n)$ carries the general case.

The algebraic development of the same subject (algebraic groups over an arbitrary field, the same root systems reached by polynomial rather than analytic methods) is the companion volume *Algebraic Groups and Representations* [10]; care will be given in both books to show their overlap.

Notation and Conventions

The manifold-theoretic notation follows *Differential Geometry* [13]. The conventions particular to this book are collected here.

- Lie groups are G, H, K; their Lie algebras are the corresponding fraktur letters, $\mathfrak{g} = \mathrm{Lie}(G) = T_eG$, with the bracket $[X,Y]$. For a matrix group the bracket is the commutator $[X,Y] = XY - YX$, and exp denotes both the matrix exponential and the intrinsic exponential map; the two are identified in chapter 2.
- Classical groups are typeset as operators with the field in parentheses: $\mathrm{GL}(n,\mathbb{R})$, $\mathrm{SL}(n,\mathbb{C})$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, $\mathrm{Sp}(2n,\mathbb{F})$, where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.
- In Part II the ground field is $\mathbb{C}$ (algebraically closed, characteristic 0) unless stated otherwise; the places where characteristic 0 is key are flagged. Real Lie algebras return in Part IV, where $\mathfrak{g}_0$ denotes a real form with complexification $\mathfrak{g}$. a hardcoded numeral. The single hyphen is the only dash in the text.

Part I

Lie Groups and Their Lie Algebras

The object of this part is the dictionary between a Lie group and its Lie algebra. Chapter 1 builds it for matrix groups, where the exponential map and the bracket are computed with bare hands; chapter 2 rebuilds both intrinsically on an abstract Lie group and identifies the two pictures. Chapter 3 proves the theorems that make the dictionary trustworthy: closed subgroups are Lie subgroups, subalgebras integrate to subgroups, and a simply connected group is recovered from its Lie algebra. Chapter 4 puts the dictionary to work, manufacturing manifolds as quotients G/H and organizing the groups sharing a Lie algebra into a tower of coverings.

1

Matrix Lie Groups and the Classical Groups

In the middle of the nineteenth century Galois had shown that the solvability of a polynomial equation is controlled by a finite group of symmetries: the permutations of the roots preserving every algebraic relation among them. Sophus Lie set out to build the analogue for differential equations, where the natural symmetries form not a finite set but a continuous family, the rotations of the plane, the translations along a flow, the scalings carrying one solution of an equation to another. The object governing such a family is a group that is at the same time a smooth manifold, so that one may differentiate along it, and the interplay between the group law and the differentiable structure is the source of the theory's power. Lie's own goal, a theory of integration by continuous symmetry parallel to Galois's theory of solvability, proved harder to carry through; but the objects he introduced organize a large part of modern mathematics, from the harmonic analysis of functions on a symmetric space, through the gauge fields of physics and the automorphic forms at the center of number theory, to the classification of the algebraic groups.

This book approaches Lie theory from its analytic and differential-geometric side: a Lie group is a manifold carrying a compatible group law, and the aim is to understand such groups, their infinitesimal counterparts (the Lie algebras), and their representations. The parallel algebraic development, in which the same groups are studied as varieties over an arbitrary field, is the subject of the companion volume [10]; the two meet at the theory of root systems in Part II.

The present chapter takes the most concrete route in. Before a Lie group is defined in the generality of an abstract manifold, the task of the next chapter, the classical groups are met as what they are at the surface: sets of matrices cut out by explicit polynomial or analytic constraints. The general linear group, and inside it the orthogonal, unitary, and symplectic groups, are exhibited directly as smooth manifolds, and the two objects around which the whole theory turns, the matrix exponential and the Lie algebra, are produced by differentiating matrix-valued functions rather than by invoking

any abstract apparatus. The reward for this concreteness is that when the general theory comes in the next chapter it finds a fully worked family of examples already in place. The chapter closes with the group SU(2) and its two-to-one relation with the rotation group of three-space, the example carried through the rest of the book.

1.1 The General Linear Group as a Manifold

The starting object of the whole subject is the set of all $n \times n$ matrices over a field, and the invertible ones among them. Throughout this chapter $\mathbb{F}$ denotes either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$, and $M_n(\mathbb{F})$ denotes the set of $n \times n$ matrices with entries in $\mathbb{F}$. Addition and scalar multiplication make $M_n(\mathbb{F})$ an $\mathbb{F}$-vector space of dimension n^2, and recording a matrix by its n^2 entries identifies it with $\mathbb{F}^{n^2}$. Matrix multiplication makes it moreover an associative algebra with unit the identity matrix I. The identification with $\mathbb{F}^{n^2}$ equips $M_n(\mathbb{F})$ with a topology and a smooth structure, those of a finite-dimensional vector space, and every construction below is made against this fixed background.

What singles out the invertible matrices is that they form a group.

Definition 1.1.1: General Linear Group

Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. The *general linear group* of rank n over $\mathbb{F}$ is the group of invertible matrices

$$\mathrm{GL}_n(\mathbb{F}) = \{A \in M_n(\mathbb{F}) \mid \det A \neq 0\}$$

under matrix multiplication, with identity element the identity matrix I and the inverse of A its matrix inverse A^{-1}.

That the condition $\det A \neq 0$ characterizes invertibility is the content of the cofactor formula: a matrix has a two-sided inverse if and only if its determinant is a unit of $\mathbb{F}$, and every nonzero element of a field is a unit. The determinant is a polynomial in the n^2 entries of A, so it is a continuous, indeed smooth, function $\det\colon M_n(\mathbb{F}) \to \mathbb{F}$. The set $\mathbb{F} \setminus \{0\}$ is open in $\mathbb{F}$, and $\mathrm{GL}_n(\mathbb{F})$ is its preimage under this continuous map. An open subset of a smooth manifold is itself a smooth manifold of the same dimension, an *open submanifold*, with charts obtained by restricting the ambient charts (see [13, Smooth Manifolds]).

Proposition 1.1.2: The General Linear Group is an Open Submanifold

The group $\mathrm{GL}_n(\mathbb{R})$ is an open subset of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, hence a smooth manifold of dimension n^2. The group $\mathrm{GL}_n(\mathbb{C})$ is an open subset of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$, hence a smooth manifold of real dimension $2n^2$. In each case the group operations

$$m\colon \mathrm{GL}_n(\mathbb{F}) \times \mathrm{GL}_n(\mathbb{F}) \to \mathrm{GL}_n(\mathbb{F}), \quad (A, B) \mapsto AB, \qquad \iota\colon \mathrm{GL}_n(\mathbb{F}) \to \mathrm{GL}_n(\mathbb{F}), \quad A \mapsto A^{-1}$$

are smooth.

***Proof*:**

Openness and the resulting manifold structure were established above: $\mathrm{GL}_n(\mathbb{F}) = \det^{-1}(\mathbb{F} \setminus \{0\})$ is the preimage of an open set under the continuous map det, hence open, and an open subset of

the vector space $M_n(\mathbb{F})$ is a smooth manifold of the same dimension. Over $\mathbb{R}$ this dimension is n^2; over $\mathbb{C}$, identifying each complex entry with its real and imaginary parts identifies $M_n(\mathbb{C})$ with $\mathbb{R}^{2n^2}$, so the real dimension is $2n^2$.

For the multiplication, the (i,j) entry of AB is $\sum_{k=1}^{n} A_{ik}B_{kj}$, a polynomial in the entries of A and of B. A map into $\mathbb{F}^{n^2}$ whose every component is a polynomial in the coordinates is smooth, so m is smooth. For the inversion, Cramer's rule gives

$$A^{-1} = \frac{1}{\det A}\,\operatorname{adj}(A)$$

where the *adjugate* $\operatorname{adj}(A)$ has as its entries the $(n-1)\times(n-1)$ signed minors of A, themselves polynomials in the entries of A. On $\mathrm{GL}_n(\mathbb{F})$ the denominator $\det A$ never vanishes, so each entry of A^{-1} is a ratio of a polynomial by a nowhere-zero polynomial, hence a smooth function of the entries of A. Over $\mathbb{C}$ the same formulas exhibit the two maps as smooth in the real and imaginary parts of the entries. Thus m and ι are smooth, completing the proof.

A manifold that is also a group whose multiplication and inversion are smooth is precisely what the next chapter will call a *Lie group*, and proposition 1.1.2 is the statement that $\mathrm{GL}_n(\mathbb{F})$ is one. Every group met in this chapter is a subgroup of some $\mathrm{GL}_n(\mathbb{F})$, and the general linear group is the ambient object inside which the classical groups will be cut out. For that reason its own geometry is worth settling first. The most basic geometric invariant, the number of connected pieces, already distinguishes the real from the complex case.

Proposition 1.1.3: Connected Components of the General Linear Group

The group $\mathrm{GL}_n(\mathbb{C})$ is path-connected. The group $\mathrm{GL}_n(\mathbb{R})$ has exactly two connected components, namely

$$\mathrm{GL}_n^{+}(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid \det A > 0\}, \qquad \mathrm{GL}_n^{-}(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid \det A < 0\},$$

each of which is open and path-connected. The component $\mathrm{GL}_n^{+}(\mathbb{R})$ is a normal subgroup of index two.

***Proof*:**
The argument rests on one fact from linear algebra: every matrix of determinant 1 is a product of *transvections* (one of the elementary matrix forms), the matrices $I + \lambda E_{ij}$ with $i \neq j$ and $\lambda \in \mathbb{F}$, where E_{ij} is the matrix with a single 1 in position (i,j) and zeros elsewhere. This is the statement that Gaussian elimination reduces a determinant-one matrix to the identity using only the row operation that adds a multiple of one row to another. Given any $A \in \mathrm{GL}_n(\mathbb{F})$, set $D = \operatorname{diag}(\det A, 1, \ldots, 1)$; then $D^{-1}A$ has determinant 1, so

$$A = D \cdot S, \qquad D = \operatorname{diag}(\det A, 1, \ldots, 1), \quad S \in \mathrm{SL}_n(\mathbb{F}) \tag{1.1}$$

with $S = D^{-1}A$ a product of transvections.

Step 1: each factor is joined to the identity by a path. A single transvection $I + \lambda E_{ij}$ is joined to I by $t \mapsto I + t\lambda E_{ij}$ for $t \in [0,1]$, a path lying in $\mathrm{GL}_n(\mathbb{F})$ because each of these matrices has determinant 1. A product of transvections is therefore joined to I by the pointwise product of the corresponding paths, which again lies in $\mathrm{GL}_n(\mathbb{F})$; so every $S \in \mathrm{SL}_n(\mathbb{F})$ is path-connected to I within $\mathrm{SL}_n(\mathbb{F})$. The diagonal factor $D = \operatorname{diag}(d, 1, \ldots, 1)$ with $d = \det A$ is handled by moving

d: for any path $d\colon [0,1] \to \mathbb{F} \setminus \{0\}$ from $d(0) = d$ to $d(1) = 1$, the matrices $\operatorname{diag}(d(t), 1, \dots, 1)$ form a path in $\mathrm{GL}_n(\mathbb{F})$ from D to I.

Step 2: the complex case. When $\mathbb{F} = \mathbb{C}$ the punctured plane $\mathbb{C} \setminus \{0\}$ is path-connected, so a path from $d = \det A$ to 1 as required in Step 1 always exists. Concatenating the path from $A = DS$ to S (moving D to I) with the path from S to I shows that every $A \in \mathrm{GL}_n(\mathbb{C})$ is path-connected to I. Hence $\mathrm{GL}_n(\mathbb{C})$ is path-connected.

Step 3: the real case splits according to the sign of the determinant. When $\mathbb{F} = \mathbb{R}$ the set $\mathbb{R} \setminus \{0\}$ is the disjoint union of the two open rays $\mathbb{R}_{>0}$ and $\mathbb{R}_{<0}$, and $\det\colon \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R} \setminus \{0\}$ is continuous, so the two preimages $\mathrm{GL}_n^+(\mathbb{R})$ and $\mathrm{GL}_n^-(\mathbb{R})$ are disjoint, nonempty, open, and cover $\mathrm{GL}_n(\mathbb{R})$. No connected set meets both, since det would then take a value of each sign and, being continuous, the value 0 in between, which is impossible on $\mathrm{GL}_n(\mathbb{R})$. It remains to see that each piece is path-connected. If $\det A > 0$, then in (1.1) the scalar $d = \det A$ lies in $\mathbb{R}_{>0}$, which is path-connected, so the path of Step 1 from D to I may be taken with $d(t) > 0$ throughout; the concatenated path from A to I then satisfies $\det(\operatorname{diag}(d(t), 1, \dots, 1)S(t)) = d(t) > 0$ at every time, so it stays in $\mathrm{GL}_n^+(\mathbb{R})$. Thus $\mathrm{GL}_n^+(\mathbb{R})$ is path-connected. For $\mathrm{GL}_n^-(\mathbb{R})$, fix the reflection $J = \operatorname{diag}(-1, 1, \dots, 1)$, which has determinant -1. Given A with $\det A < 0$, the matrix $J^{-1}A$ has determinant > 0, hence is joined to I by a path γ in $\mathrm{GL}_n^+(\mathbb{R})$; then $t \mapsto J\gamma(t)$ joins J to A and has determinant $-\det \gamma(t) < 0$ throughout, so it lies in $\mathrm{GL}_n^-(\mathbb{R})$. Hence $\mathrm{GL}_n^-(\mathbb{R})$ is path-connected as well.

Finally, $\mathrm{GL}_n^+(\mathbb{R})$ is the kernel of the composite homomorphism $\mathrm{GL}_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R} \setminus \{0\} \xrightarrow{\mathrm{sgn}} \{\pm 1\}$, hence a normal subgroup; its two cosets are $\mathrm{GL}_n^+(\mathbb{R})$ and $\mathrm{GL}_n^-(\mathbb{R}) = J \cdot \mathrm{GL}_n^+(\mathbb{R})$, so it has index two. This exhibits the two path-connected pieces as the two components, completing the proof.

The disconnectedness of $\mathrm{GL}_n(\mathbb{R})$ is the matrix expression of orientation: the sign of the determinant records whether a linear automorphism of $\mathbb{R}^n$ preserves or reverses orientation, and no continuous deformation through invertible maps can turn one into the other without passing through a singular matrix. Over $\mathbb{C}$ this obstruction dissolves, because a single complex scalar can be routed around the origin; the punctured line $\mathbb{C} \setminus \{0\}$ is connected where $\mathbb{R} \setminus \{0\}$ is not. The same split will reappear for the orthogonal group in the next section, where $\mathrm{SO}(n)$ plays the role of the identity component.

Example 1.1.4: The Groups GL_1 and $\mathrm{GL}_2(\mathbb{R})$

For $n = 1$ a matrix is a scalar and invertibility means nonvanishing, so $\mathrm{GL}_1(\mathbb{F}) = \mathbb{F}^\times$, the multiplicative group of the field. Then $\mathrm{GL}_1(\mathbb{R}) = \mathbb{R}^\times$ is the disjoint union of the two half-lines $\mathbb{R}_{>0}$ and $\mathbb{R}_{<0}$, while $\mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$ is the connected punctured plane.

For $n = 2$ the group $\mathrm{GL}_2(\mathbb{R})$ is a four-dimensional manifold, the open subset of $M_2(\mathbb{R}) \cong \mathbb{R}^4$ where $ad - bc \neq 0$ for the matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Its identity component $\mathrm{GL}_2^+(\mathbb{R})$ contains every rotation $\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, of determinant 1, and every positive scaling. A reflection such as $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, of determinant -1, lies in the other component $\mathrm{GL}_2^-(\mathbb{R})$ and cannot be connected to the identity without its determinant passing through zero.

Two features of $\mathrm{GL}_n(\mathbb{F})$ separate it from the groups studied in the rest of the chapter and are worth recording now. It is *noncompact*: it contains the matrices $\operatorname{diag}(t, 1, \dots, 1)$ for all $t \neq 0$, an unbounded family, so $\mathrm{GL}_n(\mathbb{F})$ is an unbounded subset of $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$ and cannot be compact. It is also *dense* in $M_n(\mathbb{F})$: any matrix can be perturbed to an invertible one, since the determinant of $A - tI$ is a nonzero polynomial in t and so vanishes for only finitely many t. The classical groups introduced next trade this openness for the opposite geometry: cut out by equations rather than inequalities,

several of them are compact, and it is compactness that will make their representation theory in Part III so tractable.

Exercise 1.1.1

1. Show that $\mathrm{GL}_n(\mathbb{F})$ is dense in $M_n(\mathbb{F})$: given any $A \in M_n(\mathbb{F})$, prove that $A - tI$ is invertible for all but finitely many $t \in \mathbb{F}$, and deduce that every matrix is a limit of invertible ones.
2. Prove that $\mathrm{GL}_n(\mathbb{F})$ is not compact, in two ways: by exhibiting an unbounded sequence in it, and by showing it is a nonempty proper open subset of the connected space $M_n(\mathbb{F})$ and hence not closed.
3. Determine the center of $\mathrm{GL}_n(\mathbb{F})$. Show that a matrix commuting with every element of $\mathrm{GL}_n(\mathbb{F})$ (equivalently, with every transvection $I + E_{ij}$) is a scalar multiple λI of the identity with $\lambda \in \mathbb{F}^\times$.
4. Prove that $\det\colon \mathrm{GL}_n(\mathbb{F}) \to \mathbb{F}^\times$ is a surjective group homomorphism with kernel $\mathrm{SL}_n(\mathbb{F})$, and conclude that $\mathrm{SL}_n(\mathbb{F})$ is a normal subgroup with $\mathrm{GL}_n(\mathbb{F})/\mathrm{SL}_n(\mathbb{F}) \cong \mathbb{F}^\times$.
5. Using proposition 1.1.3, show that $A \mapsto \operatorname{sgn}(\det A)$ induces an isomorphism $\mathrm{GL}_n(\mathbb{R})/\mathrm{GL}_n^+(\mathbb{R}) \cong \{\pm 1\}$, and that $\mathrm{GL}_n^+(\mathbb{R})$ is exactly the set of products of transvections and positive scalings.

1.2 The Classical Groups

The general linear group is cut out of the space of all matrices by an *inequality*, $\det A \neq 0$, which is why it is open. The groups of present interest are cut out instead by *equations*: they are the subgroups of GL_n preserving some piece of structure, a determinant normalization, a symmetric or Hermitian inner product, or a skew form. Preserving structure is a "closed" condition, and several of these groups are furthermore compact. Each is presented here as the solution set of a system of polynomial (or, in the complex case, real-analytic) equations, and the uniform tool that converts such a solution set into a smooth manifold is the regular value theorem.

Recall the statement in the form used throughout: if $F\colon M \to N$ is a smooth map of manifolds and $c \in N$ is a *regular value*, meaning that the differential DF_p is surjective at every $p \in F^{-1}(c)$, then $F^{-1}(c)$ is an embedded submanifold of M of dimension $\dim M - \dim N$, with tangent space $T_pF^{-1}(c) = \ker DF_p$ at each of its points (see [13, Regular Level Set Theorem, Tangent Space via a Defining Map]). For each classical group the recipe is the same: write the group as $F^{-1}(c)$ for an explicit F into a vector space of "constraint values", check surjectivity of DF along the group by a single substitution, and read off the dimension. The closed-subgroup theorem of chapter 1 will later subsume all of these computations under one statement, that every closed subgroup of a Lie group is a Lie subgroup; the point of doing them by hand now is to make the manifold structure, and above all the dimension, of each principal example explicit and concrete.

Definition 1.2.1: Special Linear Group

The *special linear group* is the kernel of the determinant,

$$\mathrm{SL}_n(\mathbb{F}) = \{A \in \mathrm{GL}_n(\mathbb{F}) \mid \det A = 1\}$$

the group of matrices of determinant 1.

The determinant is the simplest structure a matrix can preserve, namely volume with orientation, and the special linear group is the first application of the regular value recipe.

Proposition 1.2.2: The Special Linear Group is a Submanifold

$\mathrm{SL}_n(\mathbb{R})$ is a closed embedded submanifold of $\mathrm{GL}_n(\mathbb{R})$ of dimension n^2-1, and $\mathrm{SL}_n(\mathbb{C})$ is a closed embedded submanifold of $\mathrm{GL}_n(\mathbb{C})$ of real dimension $2(n^2-1)$ (complex dimension n^2-1).

***Proof*:**
The determinant restricts to a smooth map $\det\colon \mathrm{GL}_n(\mathbb{F}) \to \mathbb{F}^\times$, and $\mathrm{SL}_n(\mathbb{F}) = \det^{-1}(1)$. The derivative of the determinant at A is computed from the expansion

$$\det(A+tX) = \det(A)\det(I+tA^{-1}X) = \det(A)\big(1+t\mathrm{tr}(A^{-1}X)+O(t^2)\big)$$

valid for A invertible, where the middle equality factors out A and the last uses $\det(I+tB) = 1+t\mathrm{tr}(B)+O(t^2)$. Differentiating at $t=0$ gives Jacobi's formula

$$D\det_A(X) = \det(A)\,\mathrm{tr}(A^{-1}X)$$

As X ranges over $M_n(\mathbb{F})$ the quantity $\mathrm{tr}(A^{-1}X)$ ranges over all of $\mathbb{F}$, for instance $X = AE_{11}$ gives $\mathrm{tr}(E_{11}) = 1$; hence $D\det_A$ is surjective onto $\mathbb{F}$ at every $A \in \mathrm{GL}_n(\mathbb{F})$, so every value, in particular 1, is regular. Therefore $\mathrm{SL}_n(\mathbb{F}) = \det^{-1}(1)$ is an embedded submanifold, closed because $\{1\}$ is closed, of dimension $\dim \mathrm{GL}_n(\mathbb{F}) - \dim \mathbb{F}$. Over $\mathbb{R}$ this is n^2-1; over $\mathbb{C}$, where det is a map of real manifolds onto $\mathbb{C} \cong \mathbb{R}^2$, the real codimension is 2, giving real dimension $2n^2-2$, completing the proof.

The next three groups preserve an inner product. Over $\mathbb{R}$ the relevant structure is a symmetric bilinear form; over $\mathbb{C}$, a Hermitian form; and the groups preserving them are the orthogonal and unitary groups. These are the compact classical groups, and the mechanism producing their compactness, that an isometry sends an orthonormal basis to an orthonormal one and so has bounded entries, is visible already in the proofs.

Definition 1.2.3: Orthogonal and Special Orthogonal Groups

The *orthogonal group* is the group of real matrices preserving the standard inner product on $\mathbb{R}^n$,

$$\mathrm{O}(n) = \left\{A \in \mathrm{GL}_n(\mathbb{R}) \;\middle|\; A^{\mathsf{T}}A = I\right\}$$

equivalently the matrices whose columns form an orthonormal basis. The *special orthogonal group* is the subgroup of determinant 1,

$$\mathrm{SO}(n) = \mathrm{O}(n) \cap \mathrm{SL}_n(\mathbb{R}) = \{A \in \mathrm{O}(n) \mid \det A = 1\}$$

the group of rotations of $\mathbb{R}^n$.

The relation $A^{\mathsf{T}}A = I$ forces $(\det A)^2 = \det(A^{\mathsf{T}})\det(A) = 1$, so an orthogonal matrix has determinant ± 1 and the condition $\det A = 1$ picks out one half. The orthogonal group is the prototype for the regular value computation, carried out here in full detail.

Proposition 1.2.4: The Orthogonal Group is a Compact Submanifold of Dimension $\binom{n}{2}$

$\mathrm{O}(n)$ is a compact embedded submanifold of $M_n(\mathbb{R})$ of dimension $\binom{n}{2} = \frac{n(n-1)}{2}$. It is disconnected: the determinant takes the values ± 1 on it, and the two clopen pieces $\det^{-1}(1) = \mathrm{SO}(n)$ and $\det^{-1}(-1)$ are each a compact submanifold of the same dimension.

***Proof*:**

Let $\mathrm{Sym}_n(\mathbb{R}) = \left\{S \in M_n(\mathbb{R}) \;\middle|\; S^{\mathsf{T}} = S\right\}$ be the space of symmetric matrices, a real vector space of dimension $\frac{n(n+1)}{2}$, and define

$$F\colon M_n(\mathbb{R}) \to \mathrm{Sym}_n(\mathbb{R}), \qquad F(A) = A^{\mathsf{T}}A$$

which lands in $\mathrm{Sym}_n(\mathbb{R})$ because $(A^{\mathsf{T}}A)^{\mathsf{T}} = A^{\mathsf{T}}A$, and is smooth because its entries are polynomials. By construction $\mathrm{O}(n) = F^{-1}(I)$.

Step 1: I is a regular value. Fix $A \in \mathrm{O}(n)$. The differential of F at A is

$$DF_A(X) = X^{\mathsf{T}}A + A^{\mathsf{T}}X$$

as one sees by expanding $F(A + tX) = A^{\mathsf{T}}A + t(X^{\mathsf{T}}A + A^{\mathsf{T}}X) + t^2X^{\mathsf{T}}X$ and differentiating at $t = 0$. To see that DF_A is onto $\mathrm{Sym}_n(\mathbb{R})$, substitute $X = AZ$ and use $A^{\mathsf{T}}A = I$:

$$DF_A(AZ) = (AZ)^{\mathsf{T}}A + A^{\mathsf{T}}(AZ) = Z^{\mathsf{T}}A^{\mathsf{T}}A + A^{\mathsf{T}}AZ = Z^{\mathsf{T}} + Z$$

Given any symmetric S, taking $Z = \frac{1}{2}S$ yields $Z^{\mathsf{T}} + Z = S$; since $X \mapsto Z = A^{-1}X$ is a bijection of $M_n(\mathbb{R})$, the map DF_A is surjective. Hence I is a regular value.

Step 2: the manifold structure and dimension. By the regular value theorem $\mathrm{O}(n) = F^{-1}(I)$ is an embedded submanifold of $M_n(\mathbb{R})$ of dimension

$$\dim M_n(\mathbb{R}) - \dim \mathrm{Sym}_n(\mathbb{R}) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2} = \binom{n}{2}$$

Step 3: compactness. The set $\mathrm{O}(n)$ is closed in $M_n(\mathbb{R})$, being $F^{-1}(I)$ for continuous F. It is bounded: if $A^{\mathsf{T}}A = I$ then the diagonal entries of $A^{\mathsf{T}}A$ give $\sum_i a_{ij}^2 = 1$ for each column j, so the Frobenius norm satisfies $\|A\|^2 = \sum_{i,j} a_{ij}^2 = n$. A closed bounded subset of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$ is compact by the Heine-Borel theorem.

Step 4: the two pieces. The determinant is continuous on $\mathrm{O}(n)$ and takes only the values ± 1, both attained (I has determinant 1 and $\mathrm{diag}(-1, 1, \dots, 1)$ has determinant -1), so $\det^{-1}(1) = \mathrm{SO}(n)$ and $\det^{-1}(-1)$ are disjoint, nonempty, and both open and closed in $\mathrm{O}(n)$. Each is a submanifold of dimension $\binom{n}{2}$, being open in $\mathrm{O}(n)$, and each is compact, being closed in the compact space $\mathrm{O}(n)$. That $\mathrm{SO}(n)$ is connected, so that these two pieces are exactly the connected components, is proposition 1.2.10 below, completing the proof.

The complex analogue replaces the transpose by the conjugate transpose and the symmetric form by a Hermitian one. Write $A^* = \overline{A}^{\mathsf{T}}$ for the conjugate transpose.

Definition 1.2.5: Unitary and Special Unitary Groups

The *unitary group* is the group of complex matrices preserving the standard Hermitian inner product on $\mathbb{C}^n$,

$$\mathrm{U}(n) = \{A \in \mathrm{GL}_n(\mathbb{C}) \mid A^*A = I\}$$

the matrices whose columns form a Hermitian-orthonormal basis. The *special unitary group* is the subgroup of determinant 1,

$$\mathrm{SU}(n) = \mathrm{U}(n) \cap \mathrm{SL}_n(\mathbb{C}) = \{A \in \mathrm{U}(n) \mid \det A = 1\}$$

The relation $A^*A = I$ gives $|\det A|^2 = \overline{\det A} \det A = \det(A^*)\det(A) = 1$, so a unitary matrix has determinant on the unit circle, and $\mathrm{SU}(n)$ is the fibre of the determinant over $1 \in S^1$.

Proposition 1.2.6: The Unitary and Special Unitary Groups are Compact Submanifolds

$\mathrm{U}(n)$ is a compact embedded submanifold of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$ of real dimension n^2. Its subgroup $\mathrm{SU}(n)$ is a compact embedded submanifold of real dimension $n^2 - 1$.

***Proof*:**
Let $\mathrm{Herm}_n = \{H \in M_n(\mathbb{C}) \mid H^* = H\}$ be the space of Hermitian matrices. This is a *real* vector space (it is not closed under multiplication by i) of real dimension n^2: a Hermitian matrix has n real diagonal entries and $\binom{n}{2}$ free complex entries above the diagonal, contributing $n + 2\binom{n}{2} = n + n(n-1) = n^2$ real parameters. Define

$$F\colon M_n(\mathbb{C}) \to \mathrm{Herm}_n, \qquad F(A) = A^*A$$

a smooth map (its entries are real-polynomial in the real and imaginary parts of the entries of A) with $\mathrm{U}(n) = F^{-1}(I)$.

Step 1: I is a regular value. For $A \in \mathrm{U}(n)$ the differential is $DF_A(X) = X^*A + A^*X$, and the substitution $X = AZ$ with $A^*A = I$ gives

$$DF_A(AZ) = (AZ)^*A + A^*(AZ) = Z^*A^*A + A^*AZ = Z^* + Z$$

Given Hermitian H, taking $Z = \frac{1}{2}H$ gives $Z^* + Z = H$, so DF_A is surjective onto Herm_n; hence I is a regular value and $\mathrm{U}(n)$ is an embedded submanifold of real dimension $2n^2 - n^2 = n^2$.

Step 2: compactness. As with the orthogonal group, $\mathrm{U}(n) = F^{-1}(I)$ is closed, and the columns of a unitary matrix are unit vectors, so $\|A\|^2 = n$ and $\mathrm{U}(n)$ is bounded; it is therefore compact by Heine-Borel.

Step 3: the special unitary group. By the regular value theorem the tangent space at $A \in \mathrm{U}(n)$ is $T_A\,\mathrm{U}(n) = \ker DF_A = \{AZ \mid Z^* + Z = 0\}$, that is $A \cdot \mathfrak{u}(n)$ where $\mathfrak{u}(n)$ is the space of skew-Hermitian matrices. Consider the smooth map $\det\colon \mathrm{U}(n) \to S^1$. Its derivative at A, along a tangent vector $X = AZ$ with Z skew-Hermitian, is by Jacobi's formula

$$D\det_A(AZ) = \det(A)\,\mathrm{tr}(A^{-1}AZ) = \det(A)\,\mathrm{tr}(Z)$$

A skew-Hermitian matrix has purely imaginary diagonal, so $\mathrm{tr}(Z) \in i\mathbb{R}$, and $\mathrm{tr}(Z)$ takes every value in $i\mathbb{R}$ (for example $Z = \mathrm{diag}(it, 0, \dots, 0)$). Thus $D\det_A$ maps $T_A\,\mathrm{U}(n)$ onto $\det(A) \cdot i\mathbb{R}$, which is exactly the tangent line to S^1 at the point $\det(A)$. Hence $\det\colon \mathrm{U}(n) \to S^1$ is a submersion, 1 is a regular value, and $\mathrm{SU}(n) = \det^{-1}(1)$ is an embedded submanifold of $\mathrm{U}(n)$ of dimension $n^2 - 1$. It is closed in the compact group $\mathrm{U}(n)$, hence compact, as we sought to show.

The last family preserves a skew-symmetric form. Its notation is a persistent source of confusion, so the two groups sharing the name "symplectic" are fixed side by side, together with a convention.

Definition 1.2.7: Symplectic Groups

Let $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right) \in M_{2n}(\mathbb{F})$, the matrix of the standard skew form $\omega(x, y) = x^{\mathsf{T}} J y$ on $\mathbb{F}^{2n}$. The *symplectic group* is the group preserving ω,

$$\mathrm{Sp}(2n, \mathbb{F}) = \left\{ A \in \mathrm{GL}_{2n}(\mathbb{F}) \;\middle|\; A^{\mathsf{T}} J A = J \right\}$$

The *compact symplectic group* is the group of quaternionic-unitary matrices,

$$\mathrm{Sp}(n) = \{A \in M_n(\mathbb{H}) \mid A^* A = I\}$$

where $\mathbb{H}$ is the quaternions and $A^* = \overline{A}^{\mathsf{T}}$ is the quaternionic conjugate transpose.

1.2.8 Convention: Two Meanings of "Symplectic Group" The symbol $\mathrm{Sp}(2n, \mathbb{F})$ denotes the (noncompact) group of $2n \times 2n$ matrices over $\mathbb{F}$ preserving the standard skew form; the symbol $\mathrm{Sp}(n)$ denotes the (compact) group of $n \times n$ quaternionic matrices preserving the standard quaternionic-Hermitian form. Both have dimension $n(2n+1)$, but they are different groups: $\mathrm{Sp}(2n, \mathbb{R})$ is noncompact and $\mathrm{Sp}(n)$ is compact. The compact group is a real form of the complex group $\mathrm{Sp}(2n, \mathbb{C})$; concretely, under the standard identification $\mathbb{H}^n \cong \mathbb{C}^{2n}$ one has $\mathrm{Sp}(n) = \mathrm{U}(2n) \cap \mathrm{Sp}(2n, \mathbb{C})$, the verification of which is a routine matter of unwinding the quaternionic structure, taken up where the compact group is needed in Part III. Throughout this book $\mathrm{Sp}(2n, \mathbb{F})$ always carries an explicit field and even size $2n$; the bare symbol $\mathrm{Sp}(n)$ always means the compact group.

The noncompact group is another instance of the regular value recipe, with the skew-symmetric matrices as the space of constraint values.

Proposition 1.2.9: The Symplectic Group is a Submanifold of Dimension $n(2n+1)$

$\mathrm{Sp}(2n,\mathbb{F})$ is a closed embedded submanifold of $\mathrm{GL}_{2n}(\mathbb{F})$ of dimension $n(2n+1)$ over $\mathbb{R}$ (and complex dimension $n(2n+1)$ over $\mathbb{C}$). The compact group $\mathrm{Sp}(n)$ is a compact subgroup of $\mathrm{U}(2n)$, hence compact, of dimension $n(2n+1)$.

***Proof*:**
Let $\mathrm{Skew}_{2n}(\mathbb{F}) = \left\{S \in M_{2n}(\mathbb{F}) \;\middle|\; S^{\mathsf{T}} = -S\right\}$, of dimension $\binom{2n}{2} = n(2n-1)$, and define $F\colon M_{2n}(\mathbb{F}) \to \mathrm{Skew}_{2n}(\mathbb{F})$ by $F(A) = A^{\mathsf{T}}JA$; this is skew-symmetric because $J^{\mathsf{T}} = -J$, and $\mathrm{Sp}(2n,\mathbb{F}) = F^{-1}(J)$. The differential at A is $DF_A(X) = X^{\mathsf{T}}JA + A^{\mathsf{T}}JX$, and for $A \in \mathrm{Sp}(2n,\mathbb{F})$ the substitution $X = AZ$ with $A^{\mathsf{T}}JA = J$ gives

$$DF_A(AZ) = Z^{\mathsf{T}}A^{\mathsf{T}}JA + A^{\mathsf{T}}JAZ = Z^{\mathsf{T}}J + JZ$$

Given skew-symmetric S, set $Z = -\frac{1}{2}JS$. Using $J^{\mathsf{T}} = -J$, $J^2 = -I$, and $S^{\mathsf{T}} = -S$, the transpose is $Z^{\mathsf{T}} = -\frac{1}{2}S^{\mathsf{T}}J^{\mathsf{T}} = -\frac{1}{2}(-S)(-J) = -\frac{1}{2}SJ$, so

$$Z^{\mathsf{T}}J + JZ = -\tfrac{1}{2}SJ^2 - \tfrac{1}{2}J^2S = \tfrac{1}{2}S + \tfrac{1}{2}S = S$$

Hence DF_A is surjective onto $\mathrm{Skew}_{2n}(\mathbb{F})$, the value J is regular, and $\mathrm{Sp}(2n,\mathbb{F}) = F^{-1}(J)$ is a closed embedded submanifold of dimension

$$(2n)^2 - n(2n-1) = 4n^2 - 2n^2 + n = 2n^2 + n = n(2n+1)$$

For the compact group, the identification $\mathbb{H}^n \cong \mathbb{C}^{2n}$ of box 1.2.8 realizes $\mathrm{Sp}(n)$ as $\mathrm{U}(2n) \cap \mathrm{Sp}(2n,\mathbb{C})$, a closed subset of the compact group $\mathrm{U}(2n)$, hence compact; its dimension $n(2n+1)$ is computed from its Lie algebra in chapter 1. This completes the proof.

It remains to settle the connectivity left open for $\mathrm{SO}(n)$, $\mathrm{U}(n)$, and $\mathrm{SU}(n)$. Each is proved by the same idea: diagonalize, or bring to real normal form, and then contract the eigenvalues to 1 along a path that stays inside the group.

Proposition 1.2.10: Connectivity of the Compact Classical Groups

The groups $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and $\mathrm{SO}(n)$ are path-connected. Consequently $\mathrm{O}(n)$ has exactly two connected components, $\mathrm{SO}(n)$ and $\det^{-1}(-1)$.

***Proof*:**
Step 1: the unitary group. A unitary matrix is normal, so by the spectral theorem it is unitarily diagonalizable: $A = PDP^{-1}$ with $P \in \mathrm{U}(n)$ and $D = \mathrm{diag}(e^{i\theta_1},\dots,e^{i\theta_n})$ for real $\theta_1,\dots,\theta_n$, the eigenvalues lying on the unit circle because A is unitary. The path $D(s) = \mathrm{diag}(e^{is\theta_1},\dots,e^{is\theta_n})$ for $s \in [0,1]$ runs inside $\mathrm{U}(n)$ from $D(1) = D$ to $D(0) = I$, and conjugating, $s \mapsto PD(s)P^{-1}$ is a path in $\mathrm{U}(n)$ from A to I. Hence $\mathrm{U}(n)$ is path-connected.

Step 2: the special unitary group. For $A \in \mathrm{SU}(n)$ diagonalize as above; now $\det A = \prod_j e^{i\theta_j} = 1$, so $\sum_j \theta_j \in 2\pi\mathbb{Z}$, and subtracting a suitable integer multiple of 2π from one of the angles (which does not change the corresponding eigenvalue) arranges $\sum_j \theta_j = 0$ exactly. Then the same path

$D(s)$ satisfies $\det D(s) = e^{is\sum_j \theta_j} = 1$ for all s, so it lies in $\mathrm{SU}(n)$, and $s \mapsto PD(s)P^{-1}$ joins A to I within $\mathrm{SU}(n)$. Hence $\mathrm{SU}(n)$ is path-connected.

Step 3: the special orthogonal group. A matrix $A \in \mathrm{SO}(n)$ is real orthogonal, hence normal, and the real normal form for orthogonal matrices [4, The Real Normal Form for an Orthogonal Matrix] provides an orthogonal Q with $Q^{-1}AQ$ block-diagonal, the blocks being 2×2 rotations $R(\theta_k) = \left(\begin{smallmatrix} \cos\theta_k & -\sin\theta_k \\ \sin\theta_k & \cos\theta_k \end{smallmatrix}\right)$ together with diagonal entries ± 1; the real eigenvalues ± 1 occur because they are the only real numbers on the unit circle, and the number of -1 entries is even because $\det A = 1$, so those entries pair up into rotation blocks $R(\pi) = \mathrm{diag}(-1,-1)$. Thus $Q^{-1}AQ$ is a direct sum of rotation blocks $R(\theta_1), \dots, R(\theta_m)$ and a $+1$ identity block. Replacing each angle θ_k by $s\theta_k$ gives a path of block matrices in $\mathrm{SO}(n)$ from $Q^{-1}AQ$ (at $s = 1$) to I (at $s = 0$), and conjugating by Q produces a path in $\mathrm{SO}(n)$ from A to I. Hence $\mathrm{SO}(n)$ is path-connected.

Since $\mathrm{SO}(n) = \det^{-1}(1)$ is path-connected and $\det^{-1}(-1) = \mathrm{diag}(-1,1,\dots,1) \cdot \mathrm{SO}(n)$ is its homeomorphic image under left multiplication, both clopen pieces of $\mathrm{O}(n)$ from proposition 1.2.4 are connected, so they are precisely the two components of $\mathrm{O}(n)$, completing the proof.

The dimensions and topological types of the classical groups are collected in fig. 1.1 for reference; the entries for $\mathrm{O}(n)$ and $\mathrm{U}(n)$ recur constantly in Part III, where the representation theory of the compact groups is developed. Every entry has been established above, with one exception: the connectedness of the two symplectic groups is recorded as a standard fact, to be proved from the polar decomposition of Part IV, where $\mathrm{Sp}(2n,\mathbb{R})$ and $\mathrm{Sp}(n)$ are analyzed alongside the other real forms.

Group	Defining relation	Field	$\dim_{\mathbb{R}}$	Topology
$\mathrm{GL}_n(\mathbb{R})$	$\det \neq 0$	$\mathbb{R}$	n^2	open, 2 components
$\mathrm{GL}_n(\mathbb{C})$	$\det \neq 0$	$\mathbb{C}$	$2n^2$	open, connected
$\mathrm{SL}_n(\mathbb{R})$	$\det = 1$	$\mathbb{R}$	$n^2 - 1$	closed, connected
$\mathrm{SL}_n(\mathbb{C})$	$\det = 1$	$\mathbb{C}$	$2(n^2 - 1)$	closed, connected
$\mathrm{O}(n)$	$A^{\mathsf{T}}A = I$	$\mathbb{R}$	$\binom{n}{2}$	compact, 2 components
$\mathrm{SO}(n)$	$A^{\mathsf{T}}A = I,\ \det = 1$	$\mathbb{R}$	$\binom{n}{2}$	compact, connected
$\mathrm{U}(n)$	$A^*A = I$	$\mathbb{C}$	n^2	compact, connected
$\mathrm{SU}(n)$	$A^*A = I,\ \det = 1$	$\mathbb{C}$	$n^2 - 1$	compact, connected
$\mathrm{Sp}(2n,\mathbb{R})$	$A^{\mathsf{T}}JA = J$	$\mathbb{R}$	$n(2n+1)$	closed, connected
$\mathrm{Sp}(n)$	$A^*A = I$ (over $\mathbb{H}$)	$\mathbb{H}$	$n(2n+1)$	compact, connected

Figure 1.1: The classical groups, their defining relations, real dimensions, and topological types

The smallest orthogonal and unitary groups already exhibit the two shapes that recur throughout: a circle and its higher analogues.

Example 1.2.11: The Circle Groups $\mathrm{SO}(2)$ and $\mathrm{U}(1)$

The group $\mathrm{SO}(2)$ consists of the plane rotations

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in \mathbb{R}$$

and $\theta \mapsto R(\theta)$ is a surjective homomorphism $\mathbb{R} \to \mathrm{SO}(2)$ with kernel $2\pi\mathbb{Z}$, so $\mathrm{SO}(2) \cong \mathbb{R}/2\pi\mathbb{Z} \cong S^1$, a circle, of dimension $\binom{2}{2} = 1$ as predicted. Likewise $\mathrm{U}(1) = \{z \in \mathbb{C} \mid |z| = 1\} = S^1$ is the unit

circle in $\mathbb{C}$, of real dimension 1, and $z \mapsto \left(\begin{smallmatrix} \operatorname{Re} z & -\operatorname{Im} z \\ \operatorname{Im} z & \operatorname{Re} z \end{smallmatrix}\right)$ is an isomorphism $\mathrm{U}(1) \xrightarrow{\sim} \mathrm{SO}(2)$ realizing the standard fact that multiplication by a unit complex number is a rotation of the plane. The next dimension up, the group $\mathrm{SU}(2)$, turns out to be a 3-sphere and is the subject of section 1.5.

Exercise 1.2.1

1. Verify directly that $\mathrm{SO}(3)$ has dimension 3 and $\mathrm{SU}(2)$ has dimension 3, and that $\mathrm{O}(1) = \{\pm 1\}$ is zero-dimensional with two components, consistent with $\binom{1}{2} = 0$.
2. Show that $\det\colon \mathrm{U}(n) \to S^1$ is a surjective group homomorphism with kernel $\mathrm{SU}(n)$, and deduce that $\mathrm{U}(n)/\mathrm{SU}(n) \cong S^1$.
3. Prove that the center of $\mathrm{SU}(n)$ is the group of scalar matrices ζI with $\zeta^n = 1$, so that $Z(\mathrm{SU}(n)) \cong \mu_n$, the group of nth roots of unity.
4. Show that $A \in \mathrm{Sp}(2n, \mathbb{F})$ if and only if A preserves the skew form $\omega(x, y) = x^{\mathsf{T}} J y$, that is $\omega(Ax, Ay) = \omega(x, y)$ for all $x, y \in \mathbb{F}^{2n}$, and conclude that $\det A = 1$ for every symplectic matrix.
5. Prove that
$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \;\middle|\; \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$$
and deduce that $\mathrm{SU}(2)$ is diffeomorphic to the unit sphere $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$.

1.3 The Matrix Exponential and Logarithm: One-Parameter Subgroups

The classical groups of the previous section were described statically, as solution sets of matrix equations. The subject acquires its dynamics from a single construction that turns a matrix into a motion: the exponential of a matrix X is a point of $\mathrm{GL}_n(\mathbb{F})$, and letting X vary linearly, through tX for $t \in \mathbb{R}$, sweeps out a curve through the identity whose velocity at time zero is X itself. This is the bridge between the linear world of $M_n(\mathbb{F})$, where one adds and takes brackets, and the curved world of the group, where one multiplies. Everything in the group theory of the following chapters is read off this bridge: the tangent space at the identity becomes a Lie algebra, and the exponential map carries its linear structure onto the group.

The construction copies the scalar power series $e^x = \sum_{k \geq 0} x^k / k!$ verbatim, replacing the scalar x by a matrix X and scalar powers by matrix powers. The first task is to make sense of the resulting infinite sum of matrices, and the tool for that is the operator norm, whose submultiplicativity tames every convergence question in this section.

Definition 1.3.1: Operator Norm on $M_n(\mathbb{F})$

The *operator norm* of $X \in M_n(\mathbb{F})$, defined and related to the singular values in [4, chapter 11.4], is

$$\|X\| = \sup_{v \neq 0} \frac{\|Xv\|}{\|v\|} = \sup_{\|v\|=1} \|Xv\|$$

where $\|v\|$ is the Euclidean norm on $\mathbb{F}^n$. It makes $M_n(\mathbb{F})$ a normed vector space, and the supremum is finite and attained because the unit sphere of $\mathbb{F}^n$ is compact and $v \mapsto \|Xv\|$ is continuous.

Two properties of this norm carry the entire section. First, it dominates each matrix entry, so convergence in operator norm is the same as entrywise convergence and hence as convergence in $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$ with its usual topology; a sequence of matrices converges in operator norm if and only if each of its n^2 entry-sequences converges. Second, and this is the property with no scalar analogue, the norm is *submultiplicative*:

$$\|XY\| \leq \|X\| \, \|Y\| \qquad \text{for all } X, Y \in M_n(\mathbb{F})$$

This holds because $\|XYv\| \leq \|X\| \, \|Yv\| \leq \|X\| \, \|Y\| \, \|v\|$ for every v; iterating gives $\|X^k\| \leq \|X\|^k$, which is what forces the exponential series to converge as fast as the scalar series $\sum_k \|X\|^k / k!$.

Definition 1.3.2: Matrix Exponential

For $X \in M_n(\mathbb{F})$ the *matrix exponential* is the sum of the series

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots$$

where $X^0 = I$ by convention. It defines a map $\exp\colon M_n(\mathbb{F}) \to M_n(\mathbb{F})$.

For the definition to be legitimate the series must converge, and it must converge to a matrix depending smoothly on X; both follow from the norm estimates just recorded.

Proposition 1.3.3: Convergence and Smoothness of the Exponential

For every $X \in M_n(\mathbb{F})$ the series $\sum_{k \geq 0} X^k / k!$ converges absolutely in $M_n(\mathbb{F})$, so $\exp(X)$ is a well-defined matrix, and it satisfies the bound $\|\exp(X)\| \leq e^{\|X\|}$. Moreover the map $\exp\colon M_n(\mathbb{F}) \to M_n(\mathbb{F})$ is smooth, and its value at 0 is $\exp(0) = I$.

***Proof*:**

The space $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$ is a finite-dimensional normed vector space, hence complete, so a series converges as soon as it converges absolutely; it therefore suffices to bound the norms of the terms. Submultiplicativity gives $\|X^k\| \leq \|X\|^k$, so

$$\sum_{k=0}^{\infty} \left\| \frac{X^k}{k!} \right\| = \sum_{k=0}^{\infty} \frac{\|X^k\|}{k!} \leq \sum_{k=0}^{\infty} \frac{\|X\|^k}{k!} = e^{\|X\|}$$

The right-hand series is the convergent scalar exponential of the real number $\|X\|$, so the series

of norms converges; the matrix series converges absolutely, and its sum $\exp(X)$ has norm at most $e^{\|X\|}$ by the triangle inequality applied to the partial sums.

For smoothness, each partial sum $S_N(X) = \sum_{k=0}^{N} X^k/k!$ is a polynomial map $M_n(\mathbb{F}) \to M_n(\mathbb{F})$, hence smooth. On any ball $\|X\| \le r$ the tail is bounded uniformly by $\sum_{k>N} r^k/k!$, which tends to 0 as $N \to \infty$ independently of X; the same estimate applied to the series of any fixed partial derivative (itself an exponential-type series in the entries, dominated in the same way) shows that all partial derivatives converge uniformly on compact sets. A limit of smooth maps whose derivatives to every order converge uniformly on compact sets is smooth, so exp is smooth. Setting $X = 0$ leaves only the $k = 0$ term, $\exp(0) = I$, completing the proof.

The bound $\|\exp(X)\| \le e^{\|X\|}$ is the running estimate of the section: it lets one differentiate the series term by term, rearrange it, and multiply two such series together, all justified by absolute convergence. The algebraic properties collected next are exactly those of the scalar exponential that persist for matrices, and the one that does not, the additivity $e^{x+y} = e^x e^y$, is isolated afterward because its failure is the source of the bracket.

Proposition 1.3.4: Algebraic Properties of the Exponential

For all $X \in M_n(\mathbb{F})$ and all $s, t \in \mathbb{R}$ (or $\mathbb{C}$) the matrix exponential satisfies:

1. $\exp(0) = I$, and $t \mapsto \exp(tX)$ is a smooth curve in $M_n(\mathbb{F})$ with $\dfrac{d}{dt}\exp(tX) = X\exp(tX) = \exp(tX)X$.
2. $\exp((s+t)X) = \exp(sX)\exp(tX)$; in particular $\exp(X)$ is invertible with inverse $\exp(X)^{-1} = \exp(-X)$, so $\exp(X) \in \mathrm{GL}_n(\mathbb{F})$.
3. For every $g \in \mathrm{GL}_n(\mathbb{F})$, conjugation is intertwined: $\exp(gXg^{-1}) = g\,\exp(X)\,g^{-1}$.
4. The determinant of the exponential is the exponential of the trace: $\det\exp(X) = e^{\mathrm{tr}X}$.

***Proof*:**
We prove the four statements in turn; each rests on the absolute convergence of proposition 1.3.3, which permits term-by-term differentiation and the rearrangement of the double series obtained from a product.

1. That $\exp(0) = I$ is proposition 1.3.3. Differentiating the series $\exp(tX) = \sum_k t^k X^k/k!$ term by term, legitimate because the differentiated series $\sum_k t^{k-1}X^k/(k-1)!$ converges uniformly on compact t-intervals by the same domination as before, gives
$$\frac{d}{dt}\exp(tX) = \sum_{k=1}^{\infty} \frac{t^{k-1}X^k}{(k-1)!} = X\sum_{j=0}^{\infty}\frac{t^j X^j}{j!} = X\exp(tX)$$
where the third equality reindexes $j = k-1$ and factors one X to the left; factoring it to the right instead gives $\exp(tX)X$, and the two agree because X commutes with every power of itself.
2. Fix X and consider $f(t) = \exp((s+t)X)$ and $h(t) = \exp(sX)\exp(tX)$ as functions of t. By part (1) both satisfy the same linear differential equation $y'(t) = Xy(t)$ with the same

initial value $y(0) = \exp(sX)$: for f this is the chain rule applied to part (1), and for h it is part (1) applied to the right-hand factor with $\exp(sX)$ constant. The initial value problem $y' = Xy$, $y(0) = \exp(sX)$ has a unique solution (its difference from any other solution has derivative X times itself and vanishes at 0, hence vanishes identically, a point proved in full in theorem 1.3.10), so $f = h$; that is $\exp((s+t)X) = \exp(sX)\exp(tX)$. Setting $t = -s$ and using $\exp(0) = I$ gives $\exp(sX)\exp(-sX) = I$, so $\exp(sX)$ is invertible with the stated inverse; taking $s = 1$ puts $\exp(X)$ in $\mathrm{GL}_n(\mathbb{F})$.

3. Since $(gXg^{-1})^k = gX^kg^{-1}$ for every k, and conjugation $A \mapsto gAg^{-1}$ is a continuous linear map, it commutes with the convergent sum:

$$\exp(gXg^{-1}) = \sum_{k=0}^{\infty} \frac{(gXg^{-1})^k}{k!} = \sum_{k=0}^{\infty} \frac{gX^kg^{-1}}{k!} = g\left(\sum_{k=0}^{\infty} \frac{X^k}{k!}\right) g^{-1} = g\,\exp(X)\,g^{-1}$$

4. Consider $\varphi(t) = \det\exp(tX)$, a smooth real (or complex) function of t. By Jacobi's formula for the derivative of the determinant (established in proposition 1.2.2), and part (1),

$$\begin{aligned}\varphi'(t) &= D_{\exp(tX)}\det\left(\frac{d}{dt}\exp(tX)\right)\\ &= \det\big(\exp(tX)\big)\,\mathrm{tr}\big(\exp(tX)^{-1}X\exp(tX)\big)\\ &= \varphi(t)\,\mathrm{tr}(X)\end{aligned}$$

where the last equality uses that trace is invariant under conjugation, $\mathrm{tr}(\exp(tX)^{-1}X\exp(tX)) = \mathrm{tr}(X)$. Thus φ solves the scalar equation $\varphi' = (\mathrm{tr}X)\varphi$ with $\varphi(0) = \det I = 1$, whose unique solution is $\varphi(t) = e^{t\mathrm{tr}X}$. Setting $t = 1$ gives $\det\exp(X) = e^{\mathrm{tr}X}$, completing the proof.

Part (2) is the statement that exp lands in the group and, restricted to a single line $\mathbb{R}X$, is a homomorphism from the additive group $\mathbb{R}$ to $\mathrm{GL}_n(\mathbb{F})$; this is the germ of the one-parameter subgroups taken up at the end of the section. Part (4) already pays a concrete dividend: it identifies the trace-zero matrices as exactly those whose exponentials have determinant 1, so the exponential of the space of trace-zero matrices lands in $\mathrm{SL}_n(\mathbb{F})$, the first instance of the pattern by which the defining equation of a classical group is the linearization of the group condition. What part (2) does *not* say is that $\exp(X+Y) = \exp(X)\exp(Y)$ for two different matrices X, Y; that identity requires the two to commute, and its failure is the mechanism behind the entire theory.

Proposition 1.3.5: Exponential of a Sum of Commuting Matrices

Let $X, Y \in M_n(\mathbb{F})$. If $XY = YX$ then

$$\exp(X+Y) = \exp(X)\exp(Y) = \exp(Y)\exp(X)$$

The commutativity hypothesis cannot be dropped: there exist X, Y with $\exp(X+Y) \neq \exp(X)\exp(Y)$.

***Proof*:**
Step 1: the commuting case. Suppose $XY = YX$. Both series converge absolutely by proposition 1.3.3, so their product may be expanded as a double series and summed in any order, in particular grouped by the total degree $m = j + k$:

$$\exp(X)\exp(Y) = \left(\sum_{j=0}^{\infty} \frac{X^j}{j!}\right)\left(\sum_{k=0}^{\infty} \frac{Y^k}{k!}\right) = \sum_{m=0}^{\infty} \sum_{j+k=m} \frac{X^j Y^k}{j!\,k!} = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{j=0}^{m} \binom{m}{j} X^j Y^{m-j}$$

The inner sum is the binomial expansion of $(X+Y)^m$, which is valid *precisely because* X *and* Y *commute*: only then do the factors in $(X+Y)^m$ reorder into powers of X times powers of Y with binomial coefficients. Hence the last sum is $\sum_m (X+Y)^m/m! = \exp(X+Y)$. The same computation with the roles of X and Y interchanged gives $\exp(Y)\exp(X) = \exp(Y+X) = \exp(X+Y)$, so all three agree.

Step 2: a non-commuting counterexample. Take the nilpotent matrices

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

which do not commute, since $XY = \operatorname{diag}(1,0)$ while $YX = \operatorname{diag}(0,1)$. As $X^2 = Y^2 = 0$, the series truncate: $\exp(X) = I + X = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\exp(Y) = I + Y = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, so

$$\exp(X)\exp(Y) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

On the other hand $X + Y = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ satisfies $(X+Y)^2 = I$, so its exponential sums the even and odd parts to hyperbolic functions,

$$\exp(X+Y) = \cosh(1)\, I + \sinh(1)\,(X+Y) = \begin{pmatrix} \cosh 1 & \sinh 1 \\ \sinh 1 & \cosh 1 \end{pmatrix}$$

with $\cosh 1 \approx 1.543$ and $\sinh 1 \approx 1.175$. These two matrices are unequal, so $\exp(X+Y) \neq \exp(X)\exp(Y)$, as we sought to show.

The gap between $\exp(X+Y)$ and $\exp(X)\exp(Y)$ measures the failure of X and Y to commute, and unwinding it to second order will produce the commutator $[X,Y] = XY - YX$ in the next section; the Baker-Campbell-Hausdorff formula of Chapter 3 is the exact accounting of this gap to all orders. For now the counterexample serves as a warning label on part (2) of proposition 1.3.4: along a single line the exponential is a homomorphism, but the map on all of $M_n(\mathbb{F})$ is not.

Having built exp, one wants to invert it, both to recover X from $\exp(X)$ and to introduce coordinates on the group near the identity. The inverse is the matrix logarithm, defined by the series for $\log(1+x)$ with the same substitution of a matrix for the scalar. Because that scalar series has radius of convergence 1, the matrix version converges only near the identity, which is exactly the regime needed.

Proposition 1.3.6: The Matrix Logarithm

For $A \in M_n(\mathbb{F})$ with $\|A\| < 1$ the series

$$\log(I + A) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} A^k = A - \frac{A^2}{2} + \frac{A^3}{3} - \cdots$$

converges absolutely and defines a smooth map on the open unit ball $\{A \in M_n(\mathbb{F}) \mid \|A\| < 1\}$. It inverts the exponential where both are defined: $\exp\big(\log(I + A)\big) = I + A$ for $\|A\| < 1$, and $\log\big(\exp(X)\big) = X$ whenever $\|\exp(X) - I\| < 1$, in particular whenever $\|X\| < \log 2$.

***Proof*:**

Step 1: convergence and smoothness. For $\|A\| < 1$, submultiplicativity gives $\|A^k\| \le \|A\|^k$, so

$$\sum_{k=1}^{\infty} \left\| \frac{(-1)^{k+1}}{k} A^k \right\| \le \sum_{k=1}^{\infty} \frac{\|A\|^k}{k} = -\log(1 - \|A\|) < \infty$$

so the series converges absolutely, and the same domination on the ball $\|A\| \le r < 1$ makes the convergence, together with that of every differentiated series, uniform on compact subsets; the limit is therefore smooth by the argument of proposition 1.3.3.

Step 2: the two composites are formal inverses. The scalar identities

$$\exp\big(\log(1 + a)\big) = 1 + a, \qquad \log\big(\exp(x)\big) = x$$

hold as identities of formal power series in a single indeterminate, because exp and log are inverse as formal power series over $\mathbb{Q}$. Both composites $\exp \circ \log$ and $\log \circ \exp$, applied to a *single* matrix A (respectively X), only ever combine powers of that one matrix, which commute with one another; so every rearrangement used to verify the formal identity is a rearrangement of an absolutely convergent series of commuting matrices, valid by proposition 1.3.5 and Step 1. Substituting the matrix for the indeterminate therefore transports the two formal identities to genuine matrix identities on their domains of convergence: $\exp(\log(I + A)) = I + A$ for $\|A\| < 1$, and $\log(\exp(X)) = X$ whenever $\|\exp(X) - I\| < 1$.

Step 3: the explicit domain. If $\|X\| < \log 2$ then, using $\exp(X) - I = \sum_{k \ge 1} X^k/k!$ and $\|X^k\| \le \|X\|^k$,

$$\|\exp(X) - I\| \le \sum_{k=1}^{\infty} \frac{\|X\|^k}{k!} = e^{\|X\|} - 1 < e^{\log 2} - 1 = 1$$

so the hypothesis $\|\exp(X) - I\| < 1$ of Step 2 is met and $\log(\exp(X)) = X$, completing the proof.

The logarithm inverts the exponential on overlapping neighborhoods, and the two maps are smooth; what remains is to package this into the statement that exp is a diffeomorphism near 0, which gives the analytic charts on the group used throughout the book. The engine is the value of the differential of exp at the origin, and it is the simplest possible value.

Theorem 1.3.7: The Exponential is a Local Diffeomorphism at the Origin

The differential of $\exp\colon M_n(\mathbb{F}) \to M_n(\mathbb{F})$ at 0 is the identity, $D\exp_0 = \mathrm{id}_{M_n(\mathbb{F})}$. Consequently there are open neighborhoods U of 0 in $M_n(\mathbb{F})$ and V of I in $\mathrm{GL}_n(\mathbb{F})$ such that

$$\exp\colon U \to V \qquad \text{is a diffeomorphism, with inverse } \log\colon V \to U$$

***Proof*:**
The differential $D\exp_0$ acts on a tangent vector $X \in M_n(\mathbb{F})$ by differentiating along the line $t \mapsto tX$ through the origin:

$$D\exp_0(X) = \frac{d}{dt}\Big|_{t=0} \exp(tX) = \frac{d}{dt}\Big|_{t=0} \left(I + tX + \tfrac{t^2}{2}X^2 + \cdots\right) = X$$

the higher terms all carrying a factor t that vanishes at $t = 0$. Thus $D\exp_0 = \mathrm{id}$, which is invertible. By the inverse function theorem ([13, Inverse Function Theorem for Manifolds]), exp restricts to a diffeomorphism from some open neighborhood U of 0 onto some open neighborhood $V = \exp(U)$ of $\exp(0) = I$. Shrinking U if necessary so that $\|X\| < \log 2$ on U, proposition 1.3.6 identifies the inverse diffeomorphism with log, since $\log(\exp(X)) = X$ there; and shrinking V so that $\|A - I\| < 1$ on V makes log defined on all of V with $\exp(\log(A)) = A$. This exhibits $\exp\colon U \to V$ and $\log\colon V \to U$ as mutually inverse diffeomorphisms, completing the proof.

This is the analytic foundation for everything that follows. Near the identity, log is a coordinate chart on $\mathrm{GL}_n(\mathbb{F})$ valued in the vector space $M_n(\mathbb{F})$, and exp reads those coordinates back onto the group; the group multiplication, transported through log into a neighborhood of 0, becomes the operation $X * Y = \log(\exp X \exp Y)$ that Baker-Campbell-Hausdorff will expand in brackets. That $D\exp_0 = \mathrm{id}$ is what makes the linearization exact to first order: to first order the group law near I is just addition in $M_n(\mathbb{F})$, and the correction terms are governed by the noncommutativity measured in proposition 1.3.5. The chart is local, valid only near I; the question of how far exp reaches globally, in particular whether it is onto, depends on the group and is settled group by group in later chapters (for the compact groups it is onto, established in Part III).

Before turning to the theorem that organizes all of this, it is worth computing the exponential in the cases where it can be summed by hand, both to show the definition and to see the algebraic properties confirmed on examples.

Example 1.3.8: Rotations, Nilpotents, and $\det = e^{\mathrm{tr}}$

1. *A plane-rotation generator.* Let $X = \left(\begin{smallmatrix} 0 & -\theta \\ \theta & 0 \end{smallmatrix}\right)$ for $\theta \in \mathbb{R}$. Then $X^2 = -\theta^2 I$, so the powers cycle through $I, X, -\theta^2 I, -\theta^2 X, \theta^4 I, \ldots$, and splitting the exponential series into even and odd powers gives

   $$\exp(X) = \Big(\sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m}}{(2m)!}\Big) I + \Big(\sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!}\Big) \frac{X}{\theta} = \cos\theta\, I + \sin\theta\, \frac{X}{\theta} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

   so $\exp(X) = R(\theta) \in \mathrm{SO}(2)$ is the rotation by angle θ. The generator X is skew-symmetric with trace 0, and indeed $\det\exp(X) = \cos^2\theta + \sin^2\theta = 1 = e^0 = e^{\mathrm{tr}X}$, as proposition 1.3.4(4) requires. This recovers, from the single generator $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the entire circle group $\mathrm{SO}(2)$ met in example 1.2.11.

2. *A nilpotent matrix.* Let $N = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, the 3×3 nilpotent Jordan block. Then $N^2 = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ and $N^3 = 0$, so the series terminates after three terms:

$$\exp(tN) = I + tN + \frac{t^2}{2}N^2 = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

a polynomial in t, not a transcendental function. The trace of N is 0 and $\det \exp(tN) = 1$, again matching part (4); the curve $t \mapsto \exp(tN)$ is a one-parameter subgroup consisting of unipotent upper-triangular matrices.

3. *A diagonal check of* $\det = e^{\mathrm{tr}}$. For $D = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ the powers are $D^k = \mathrm{diag}(\lambda_1^k, \dots, \lambda_n^k)$, so summing entrywise gives $\exp(D) = \mathrm{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})$. Its determinant is $\prod_i e^{\lambda_i} = e^{\sum_i \lambda_i} = e^{\mathrm{tr}D}$, confirming proposition 1.3.4(4) directly. Since every matrix over $\mathbb{C}$ is conjugate to an upper-triangular one, with the eigenvalues on the diagonal, the conjugation equivariance of part (3) and the conjugation-invariance of both det and tr propagate this identity from the diagonal case to all of $M_n(\mathbb{C})$, which is the second proof of $\det \exp X = e^{\mathrm{tr}X}$ promised in outline.

The three computations exhibit the two extreme behaviors of the exponential. When the generator has imaginary spectrum, as in (1), the exponential is periodic and traces a circle; when the generator is nilpotent, as in (2), the exponential is polynomial and the orbit runs off to infinity. Every matrix group interpolates between these, and the interpolation is organized by the observation that each of these curves $t \mapsto \exp(tX)$ is a homomorphism $\mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$. The final result of the section is that these are the *only* such homomorphisms: the exponential does not just produce one-parameter subgroups, it produces all of them.

Definition 1.3.9: One-Parameter Subgroup

A *one-parameter subgroup* of $\mathrm{GL}_n(\mathbb{F})$ is a smooth group homomorphism $\gamma\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$; that is, a smooth curve with

$$\gamma(s+t) = \gamma(s)\gamma(t) \quad \text{for all } s, t \in \mathbb{R}, \qquad \gamma(0) = I$$

By proposition 1.3.4(2), for each fixed X the curve $\gamma_X(t) = \exp(tX)$ is a one-parameter subgroup, with velocity $\gamma_X'(0) = X$ at the identity by theorem 1.3.7. The content of the next theorem is the converse: prescribing the initial velocity X determines the whole homomorphism, so the assignment $X \mapsto \exp(tX)$ is a bijection between $M_n(\mathbb{F})$ and the set of one-parameter subgroups.

Theorem 1.3.10: Classification of One-Parameter Subgroups

Every one-parameter subgroup $\gamma\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$ has the form

$$\gamma(t) = \exp(tX), \qquad X = \gamma'(0) \in M_n(\mathbb{F})$$

Conversely every $X \in M_n(\mathbb{F})$ arises this way from a unique one-parameter subgroup, namely $t \mapsto \exp(tX)$. Thus $X \mapsto (t \mapsto \exp(tX))$ is a bijection from $M_n(\mathbb{F})$ onto the set of one-parameter subgroups of $\mathrm{GL}_n(\mathbb{F})$, with inverse $\gamma \mapsto \gamma'(0)$.

***Proof*:**
Step 1: a one-parameter subgroup solves a linear differential equation. Let γ be a one-parameter subgroup and set $X = \gamma'(0)$. Differentiating the homomorphism relation $\gamma(s+t) = \gamma(s)\gamma(t)$ with respect to s and then setting $s = 0$ gives, since $\gamma(0) = I$,

$$\gamma'(t) = \frac{d}{ds}\Big|_{s=0} \gamma(s+t) = \frac{d}{ds}\Big|_{s=0} \big(\gamma(s)\gamma(t)\big) = \gamma'(0)\gamma(t) = X\gamma(t)$$

So γ solves the linear matrix initial value problem

$$\gamma'(t) = X\gamma(t), \qquad \gamma(0) = I \tag{1.2}$$

Step 2: the exponential solves the same problem. By proposition 1.3.4(1) the curve $\eta(t) = \exp(tX)$ satisfies $\eta'(t) = X\exp(tX) = X\eta(t)$ and $\eta(0) = I$, so η is also a solution of (1.2).

Step 3: uniqueness of the solution. It remains to show (1.2) has only one solution, so that $\gamma = \eta$. Suppose γ_1, γ_2 both solve it, and set $\delta(t) = \exp(-tX)\gamma_1(t) - \exp(-tX)\gamma_2(t) = \exp(-tX)\big(\gamma_1(t) - \gamma_2(t)\big)$. Using $\frac{d}{dt}\exp(-tX) = -X\exp(-tX)$ from proposition 1.3.4(1) and the product rule, the derivative of $u(t) = \exp(-tX)\gamma_i(t)$ is

$$u'(t) = -X\exp(-tX)\gamma_i(t) + \exp(-tX)\,X\gamma_i(t) = \exp(-tX)\big(-X + X\big)\gamma_i(t) = 0$$

where the middle step uses $\exp(-tX)X = X\exp(-tX)$, again from part (1). So each $\exp(-tX)\gamma_i(t)$ is constant, equal to its value $\exp(0)\gamma_i(0) = I$ at $t = 0$; hence $\exp(-tX)\gamma_i(t) = I$ for all t, giving $\gamma_i(t) = \exp(tX)$ for both i. In particular the solution is unique, so $\gamma(t) = \exp(tX)$.

Step 4: the bijection. Step 1 through Step 3 show every one-parameter subgroup is $t \mapsto \exp(tX)$ with $X = \gamma'(0)$, so the map $X \mapsto (t \mapsto \exp(tX))$ is surjective onto one-parameter subgroups. It is injective because its inverse is the well-defined operation $\gamma \mapsto \gamma'(0)$: from $\gamma(t) = \exp(tX)$ one recovers $\gamma'(0) = X$ by theorem 1.3.7. Hence the assignment is a bijection with the stated inverse, completing the proof.

The theorem is the first crossing of the bridge announced at the start: it identifies a purely group-theoretic object, the smooth homomorphisms $\mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$, with a purely linear one, the matrices $X = \gamma'(0)$. The matrix X is the *infinitesimal generator* of the flow γ, and the differential equation (1.2) is the statement that the group element moves, at each instant, by left translation of its current velocity X. This is the seed of the Lie algebra: the next section defines the Lie algebra of a matrix group G as the set of generators X whose one-parameter subgroups $\exp(tX)$ stay inside G, and the classification theorem is what guarantees this set captures all the infinitesimal motions of G. The uniqueness argument of Step 3, that $\exp(-tX)\gamma(t)$ is constant, recurs verbatim when the flow of a left-invariant vector field is constructed intrinsically in chapter 2; here it is available by bare hands because everything lives in the linear space $M_n(\mathbb{F})$.

Exercise 1.3.1

1. Show that $\exp(X^{\mathsf{T}}) = \exp(X)^{\mathsf{T}}$ and $\exp(X^*) = \exp(X)^*$ for all $X \in M_n(\mathbb{F})$. Deduce that if X is skew-symmetric ($X^{\mathsf{T}} = -X$) then $\exp(X) \in \mathrm{SO}(n)$, and if X is skew-Hermitian ($X^* = -X$)

then $\exp(X) \in \mathrm{U}(n)$.

2. Show that $\exp\colon M_2(\mathbb{R}) \to \mathrm{GL}_2(\mathbb{R})$ is not surjective by proving that no real X satisfies $\exp(X) = \left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$. (Hint: $\det \exp X = e^{\operatorname{tr} X} > 0$ forces the image into $\mathrm{GL}_2^+(\mathbb{R})$; then analyze the real Jordan form of a putative X with $\exp(X)$ having the single eigenvalue -1.)

3. Let $X = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$ be block-diagonal. Prove that $\exp(X) = \left(\begin{smallmatrix} \exp(A) & 0 \\ 0 & \exp(B) \end{smallmatrix}\right)$. Use this together with example 1.3.8(1) to write down $\exp(tX)$ for $X = \operatorname{diag}\left(\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\right) \in \mathfrak{so}(4)$, and identify its image as a one-parameter subgroup of $\mathrm{SO}(4)$.

4. For $X, Y \in M_n(\mathbb{F})$ define $\beta(t) = \exp(tX)\, Y \exp(-tX)$. Compute $\beta(0)$ and $\beta'(0)$, showing $\beta'(0) = XY - YX$. (This is the computation that will define the bracket in the next section; do it here by differentiating the product.)

5. Let $\gamma\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$ be a one-parameter subgroup with generator $X = \gamma'(0)$, and let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a closed subgroup. Prove that if $\gamma(t) \in G$ for all t in some open interval around 0, then $\gamma(t) \in G$ for all $t \in \mathbb{R}$.

6. Using the logarithm series of proposition 1.3.6, compute $\log(R(\theta))$ for the rotation $R(\theta) = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ with $|\theta|$ small, and verify that it equals the generator $\left(\begin{smallmatrix} 0 & -\theta \\ \theta & 0 \end{smallmatrix}\right)$ of example 1.3.8(1), consistent with exp and log being mutually inverse near I.

1.4 The Lie Algebra of a Matrix Group

The exponential map of section 1.3 turns a single matrix X into a one-parameter family $t \mapsto \exp(tX)$ of group elements, a curve through the identity whose velocity at time zero is X itself. This section reverses the emphasis: rather than fix X and watch the curve, fix a matrix group G and ask which matrices X generate a one-parameter family that stays inside G for all time. The answer is a vector space, the *Lie algebra* of G, and it is the first appearance of the object around which the entire book is built. For each classical group of section 1.2 the space will be a concrete list of matrix conditions, and its dimension will match the manifold dimension already computed, so that the algebra recovers by pure linear algebra what the regular value theorem found by calculus.

The vector-space structure is only half the story. The set of matrices generating one-parameter subgroups of G carries a second operation, the commutator $[X, Y] = XY - YX$, and this bracket is what promotes the tangent space to a Lie algebra in the abstract sense. The bracket measures the failure of $\exp(X)$ and $\exp(Y)$ to commute, and its closure inside the tangent space follows from the closure of G under multiplication. By the end of the section the reader will have both the concrete matrix Lie algebras and the abstract definition they instantiate, together with the running example $\mathfrak{su}(2)$, whose bracket is the cross product on $\mathbb{R}^3$.

Definition 1.4.1: Lie Algebra of a Matrix Group

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group. The *Lie algebra* of G is the set

$$\mathfrak{g} = \{X \in M_n(\mathbb{F}) \mid \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}$$

of matrices whose associated one-parameter subgroup lies entirely in G.

The definition is engineered to be checkable: to decide whether X belongs to $\mathfrak{g}$ one differentiates the defining equations of G along the curve $\exp(tX)$ at $t = 0$. Before extracting those conditions for the classical groups, the first order of business is to see that $\mathfrak{g}$ is a real vector space at all, which is not obvious from the definition: it asks about the exponential of tX, and the exponential is neither additive nor linear.

Proposition 1.4.2: The Lie Algebra is a Real Vector Space

For any matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$, the Lie algebra $\mathfrak{g}$ of definition 1.4.1 is a real vector subspace of $M_n(\mathbb{F})$. It contains 0, and if $X, Y \in \mathfrak{g}$ and $\lambda \in \mathbb{R}$ then $\lambda X \in \mathfrak{g}$ and $X + Y \in \mathfrak{g}$.

***Proof*:**
The zero matrix lies in $\mathfrak{g}$ because $\exp(t \cdot 0) = I \in G$ for every t. For closure under real scalars, fix $X \in \mathfrak{g}$ and $\lambda \in \mathbb{R}$; then for every $t \in \mathbb{R}$ the product $t\lambda$ is again a real number, so

$$\exp\big(t(\lambda X)\big) = \exp\big((t\lambda)X\big) \in G$$

because $X \in \mathfrak{g}$. Hence $\lambda X \in \mathfrak{g}$. This uses only that the scalar is real; it is the reason $\mathfrak{g}$ is a real vector space even when $\mathbb{F} = \mathbb{C}$.

Closure under addition is where the exponential's nonlinearity must be confronted. The tool is the *Lie product formula*

$$\exp(X + Y) = \lim_{m\to\infty} \Big(\exp\Big(\tfrac{X}{m}\Big) \exp\Big(\tfrac{Y}{m}\Big)\Big)^m \tag{1.3}$$

valid for all $X, Y \in M_n(\mathbb{F})$, which expresses the exponential of a sum as a limit of products of exponentials, precisely the kind of expression a group is closed under.

Step 1: proof of the Lie product formula. Fix X, Y and set $A_m = \exp(X/m)\exp(Y/m)$. Expanding each factor to second order in the small parameter $1/m$ gives

$$\exp\Big(\tfrac{X}{m}\Big) = I + \tfrac{X}{m} + O\Big(\tfrac{1}{m^2}\Big), \qquad \exp\Big(\tfrac{Y}{m}\Big) = I + \tfrac{Y}{m} + O\Big(\tfrac{1}{m^2}\Big)$$

where the $O(1/m^2)$ terms are matrices of operator norm bounded by a constant (depending on $\|X\|$, $\|Y\|$) times $1/m^2$. Multiplying,

$$A_m = I + \frac{X + Y}{m} + O\Big(\tfrac{1}{m^2}\Big)$$

Since $\|A_m - I\| \to 0$, for m large the matrix A_m lies in the domain where the logarithm series of proposition 1.3.6 converges, and there $\log(I + B) = B + O(\|B\|^2)$; applying this with $B = A_m - I = \frac{X+Y}{m} + O(1/m^2)$ gives

$$\log A_m = \frac{X + Y}{m} + O\Big(\tfrac{1}{m^2}\Big)$$

Because log and exp are mutually inverse near I and 0 respectively (theorem 1.3.7), one has $A_m = \exp(\log A_m)$, and therefore

$$A_m^m = \exp(m \log A_m) = \exp\Big(X + Y + O\Big(\tfrac{1}{m}\Big)\Big)$$

As $m \to \infty$ the argument $X + Y + O(1/m)$ converges to $X + Y$ in $M_n(\mathbb{F})$, and exp is continuous, so $A_m^m \to \exp(X + Y)$. This is (1.3).

Step 2: additive closure. Suppose $X, Y \in \mathfrak{g}$ and fix $t \in \mathbb{R}$. Applying (1.3) to the matrices tX and tY,

$$\exp\big(t(X+Y)\big) = \exp(tX + tY) = \lim_{m\to\infty} \Big(\exp\Big(\tfrac{tX}{m}\Big)\exp\Big(\tfrac{tY}{m}\Big)\Big)^m$$

For each fixed m, the matrices $\exp(tX/m)$ and $\exp(tY/m)$ lie in G, because $X, Y \in \mathfrak{g}$ and t/m is a real number; their product raised to the mth power is a product of elements of G, hence lies in G, a subgroup. The one-parameter subgroup $\exp(t(X + Y))$ is thus a limit of elements of G. A matrix group G is defined by polynomial equations in the entries (the classical groups of section 1.2 are each $F^{-1}(c)$ for a continuous F), so G is closed in $\mathrm{GL}_n(\mathbb{F})$, and a limit of elements of G that remains invertible lies in G. Since $\exp(t(X + Y))$ is invertible with inverse $\exp(-t(X + Y))$, it lies in G. As t was arbitrary, $X + Y \in \mathfrak{g}$, completing the proof.

The role of the Lie product formula is worth dwelling on. Addition of matrices is a linear operation with no group-theoretic meaning, yet (1.3) realizes $\exp(X + Y)$ entirely through multiplication and limits, the two operations a closed subgroup respects. This is the mechanism that transfers the linear structure of $M_n(\mathbb{F})$ onto the tangent space $\mathfrak{g}$: the sum $X + Y$ is legitimate inside $\mathfrak{g}$ precisely because its exponential can be manufactured from the exponentials of X and Y without ever leaving G. The same interplay between the additive world of $\mathfrak{g}$ and the multiplicative world of G drives the bracket computation below.

For the classical groups the definition of $\mathfrak{g}$ unwinds into a linear condition on X, one per group, obtained by differentiating the defining relation. These conditions are the working description of the classical Lie algebras, and the point of the theorem is that they reproduce the dimensions of section 1.2 exactly.

Theorem 1.4.3: The Classical Lie Algebras

The Lie algebras of the classical groups are the following matrix spaces, with the indicated real dimensions.

1. $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$, the full matrix algebra; $\dim_{\mathbb{R}} \mathfrak{gl}_n(\mathbb{R}) = n^2$ and $\dim_{\mathbb{R}} \mathfrak{gl}_n(\mathbb{C}) = 2n^2$.
2. $\mathfrak{sl}_n(\mathbb{F}) = \{X \in M_n(\mathbb{F}) \mid \mathrm{tr} X = 0\}$, the traceless matrices; $\dim_{\mathbb{R}} \mathfrak{sl}_n(\mathbb{R}) = n^2 - 1$ and $\dim_{\mathbb{R}} \mathfrak{sl}_n(\mathbb{C}) = 2(n^2 - 1)$.
3. $\mathfrak{so}(n) = \big\{X \in M_n(\mathbb{R}) \;\big|\; X^{\mathsf{T}} = -X\big\}$, the real skew-symmetric matrices; $\dim_{\mathbb{R}} \mathfrak{so}(n) = \binom{n}{2}$.
4. $\mathfrak{u}(n) = \{X \in M_n(\mathbb{C}) \mid X^* = -X\}$, the skew-Hermitian matrices; $\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2$.
5. $\mathfrak{su}(n) = \{X \in M_n(\mathbb{C}) \mid X^* = -X,\ \mathrm{tr} X = 0\}$, the traceless skew-Hermitian matrices; $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$.
6. $\mathfrak{sp}(2n, \mathbb{F}) = \big\{X \in M_{2n}(\mathbb{F}) \;\big|\; X^{\mathsf{T}} J + JX = 0\big\}$, where $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$; $\dim_{\mathbb{R}} \mathfrak{sp}(2n, \mathbb{R}) = n(2n + 1)$.

In each case the dimension of $\mathfrak{g}$ equals the manifold dimension of the corresponding group computed in section 1.2.

***Proof*:**
The argument is the same for each group: differentiate the defining relation of G along $t \mapsto \exp(tX)$ at $t = 0$, using $\frac{d}{dt}\big|_0 \exp(tX) = X$ and the derivative rules for the exponential from proposition 1.3.4.

1. *The general linear group.* Every $X \in M_n(\mathbb{F})$ has $\exp(tX) \in \mathrm{GL}_n(\mathbb{F})$ for all t, since $\exp(tX)$ is invertible with inverse $\exp(-tX)$ by proposition 1.3.4. Hence $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$, of real dimension n^2 over $\mathbb{R}$ and $2n^2$ over $\mathbb{C}$, matching $\dim \mathrm{GL}_n(\mathbb{F})$ from proposition 1.1.2.

2. *The special linear group.* The relation defining $\mathrm{SL}_n(\mathbb{F})$ is $\det = 1$. By the identity $\det \exp(tX) = e^{t\mathrm{tr}X}$ of proposition 1.3.4, the curve $\exp(tX)$ lies in $\mathrm{SL}_n(\mathbb{F})$ for all t if and only if $e^{t\mathrm{tr}X} = 1$ for all $t \in \mathbb{R}$, which holds if and only if $\mathrm{tr}X = 0$. Thus $\mathfrak{sl}_n(\mathbb{F})$ is the space of traceless matrices. The trace is one linear equation on $M_n(\mathbb{F})$, and it is surjective onto $\mathbb{F}$, so its kernel has real codimension 1 over $\mathbb{R}$ and real codimension 2 over $\mathbb{C}$, giving $n^2 - 1$ and $2(n^2 - 1)$, matching proposition 1.2.2.

3. *The orthogonal and special orthogonal groups.* The relation is $A^\mathsf{T}A = I$. Substituting $A = \exp(tX)$ and using $\exp(tX)^\mathsf{T} = \exp(tX^\mathsf{T})$, the curve satisfies $\exp(tX^\mathsf{T})\exp(tX) = I$ for all t. Differentiating at $t = 0$, the product rule gives $X^\mathsf{T} + X = 0$, so X is skew-symmetric. Conversely, if $X^\mathsf{T} = -X$ then tX^T and tX commute (they are negatives), so $\exp(tX^\mathsf{T})\exp(tX) = \exp\big(t(X^\mathsf{T} + X)\big) = \exp(0) = I$ by proposition 1.3.5, and $\exp(tX) \in \mathrm{O}(n)$ for all t. A skew-symmetric matrix has zero diagonal and its below- diagonal entries are the negatives of those above, so its free parameters are the $\binom{n}{2}$ strictly-upper entries; hence $\dim \mathfrak{so}(n) = \binom{n}{2}$, matching proposition 1.2.4. The determinant-one constraint cutting $\mathrm{SO}(n)$ out of $\mathrm{O}(n)$ imposes nothing further on the algebra: every skew-symmetric X already has $\mathrm{tr}X = 0$, so $\det \exp(tX) = e^{t\mathrm{tr}X} = 1$ automatically, and $\mathfrak{so}(n)$ is at once the Lie algebra of $\mathrm{O}(n)$ and of $\mathrm{SO}(n)$, reflecting that the two groups share the identity component.

4. *The unitary group.* The relation is $A^*A = I$. With $A = \exp(tX)$ and $\exp(tX)^* = \exp(tX^*)$ (the conjugate transpose of $\exp(tX)$ is exp of the conjugate transpose, as the power series has real coefficients), the same differentiation gives $X^* + X = 0$, so X is skew-Hermitian; the converse is the commuting exponential identity as in (3). A skew-Hermitian matrix has purely imaginary diagonal entries (n real parameters) and $\binom{n}{2}$ free complex entries above the diagonal ($n(n-1)$ real parameters), the below-diagonal entries being determined by $X^* = -X$; the total is $n + n(n-1) = n^2$, matching proposition 1.2.6.

5. *The special unitary group.* Here X must be skew-Hermitian and additionally satisfy the special condition. By (2) the determinant constraint $\det \exp(tX) = 1$ forces $\mathrm{tr}X = 0$, so $\mathfrak{su}(n)$ is the space of traceless skew-Hermitian matrices. Tracelessness removes one real parameter from $\mathfrak{u}(n)$: a skew-Hermitian matrix has purely imaginary diagonal $\mathrm{diag}(i\theta_1, \ldots, i\theta_n)$, and $\mathrm{tr}X = i(\theta_1 + \cdots + \theta_n) = 0$ is the single real equation $\sum_j \theta_j = 0$. Hence $\dim \mathfrak{su}(n) = n^2 - 1$, matching proposition 1.2.6.

6. *The symplectic group.* The relation is $A^\mathsf{T}JA = J$. With $A = \exp(tX)$, the condition $\exp(tX)^\mathsf{T} J \exp(tX) = J$ for all t, differentiated at $t = 0$, yields $X^\mathsf{T}J + JX = 0$; the converse again follows because $X^\mathsf{T}J + JX = 0$ makes $t \mapsto \exp(tX)^\mathsf{T} J \exp(tX)$ have vanishing derivative and value J at $t = 0$, hence constantly J. The condition $X^\mathsf{T}J = -JX$ says exactly that JX is symmetric: $(JX)^\mathsf{T} = X^\mathsf{T}J^\mathsf{T} = -X^\mathsf{T}J = JX$, using $J^\mathsf{T} = -J$.

The map $X \mapsto JX$ is a linear bijection of $M_{2n}(\mathbb{F})$ (with inverse $Y \mapsto J^{-1}Y = -JY$, since $J^2 = -I$), so $\mathfrak{sp}(2n, \mathbb{F})$ is linearly isomorphic to the space of symmetric $2n \times 2n$ matrices, of dimension $\binom{2n+1}{2} = \frac{2n(2n+1)}{2} = n(2n+1)$, matching proposition 1.2.9.

In every case the linearized condition reproduces the dimension found by the regular value theorem in section 1.2, completing the proof.

The coincidence of dimensions follows from a general principle. The Lie algebra $\mathfrak{g}$ is the tangent space to G at the identity: differentiating a defining equation of G along a curve through I is exactly the operation the regular value theorem performs when it identifies the tangent space $T_I G$ with $\ker DF_I$, and the two computations agree because they are the same computation. The exponential description of definition 1.4.1 and the tangent-space description of section 1.2 therefore define the same object; the next chapter will make the identification $\mathfrak{g} = T_I G$ the definition of the Lie algebra of an abstract Lie group. One reads off from theorem 1.4.3 that the skew-symmetric matrices form the infinitesimal rotations and the skew-Hermitian matrices the infinitesimal unitaries: exponentiating a small skew matrix produces a rotation close to the identity, the linear-algebra content of "a rotation is generated by an angular velocity".

So far $\mathfrak{g}$ is only a vector space. The operation that makes it a Lie algebra is the commutator, and its presence is forced by the same closure under multiplication that gave additive closure. If $X, Y \in \mathfrak{g}$, then conjugating the one-parameter subgroup of Y by an element of the one-parameter subgroup of X stays in G, and differentiating this conjugation produces a new element of $\mathfrak{g}$, which turns out to be the commutator.

Proposition 1.4.4: The Lie Algebra is Closed under the Commutator

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g}$. For all $X, Y \in \mathfrak{g}$, the commutator

$$[X, Y] = XY - YX$$

again lies in $\mathfrak{g}$. The bracket $[-,-]$ is $\mathbb{R}$-bilinear, alternating (so that $[X, X] = 0$ and $[Y, X] = -[X, Y]$), and satisfies the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all $X, Y, Z \in \mathfrak{g}$.

***Proof*:**

Step 1: conjugation stays in the group. Fix $X, Y \in \mathfrak{g}$ and $s \in \mathbb{R}$, and consider the curve

$$c(s) = \exp(sX)\, Y \exp(-sX) \in M_n(\mathbb{F})$$

obtained by conjugating Y. For each real s the elements $\exp(sX)$ and $\exp(-sX) = \exp(sX)^{-1}$ lie in G, so conjugation by $\exp(sX)$ is an automorphism of G carrying the one-parameter subgroup $t \mapsto \exp(tY)$ to $t \mapsto \exp(sX)\exp(tY)\exp(-sX)$. This last curve lies in G, and by the conjugation equivariance of the exponential (proposition 1.3.4),

$$\exp(sX)\exp(tY)\exp(-sX) = \exp\big(t\, c(s)\big) \tag{1.4}$$

where $c(s) = \exp(sX)\, Y \exp(-sX)$. The left side lies in G for all t, so (1.4) shows $\exp(t\, c(s)) \in G$ for all t; that is, $c(s) \in \mathfrak{g}$ for every $s \in \mathbb{R}$.

Step 2: differentiate the conjugation. The set $\mathfrak{g}$ is a vector subspace of the finite-dimensional space $M_n(\mathbb{F})$ (proposition 1.4.2), hence closed, and $s \mapsto c(s)$ is a smooth curve lying in $\mathfrak{g}$. A closed subspace contains the derivative of any differentiable curve lying in it, so $c'(0) \in \mathfrak{g}$. Computing the derivative by the product rule, with $\frac{d}{ds}\exp(sX) = X\exp(sX)$ and $\frac{d}{ds}\exp(-sX) = -\exp(-sX)X$,

$$c'(s) = X\exp(sX)\,Y\exp(-sX) - \exp(sX)\,Y\exp(-sX)\,X$$

Evaluating at $s = 0$, where $\exp(0) = I$, gives

$$c'(0) = XY - YX = [X, Y]$$

Therefore $[X, Y] \in \mathfrak{g}$, which is the closure claim.

Step 3: the algebraic identities. Bilinearity of $(X, Y) \mapsto XY - YX$ is inherited from the bilinearity of matrix multiplication: $[aX + bX', Y] = a[X, Y] + b[X', Y]$ and likewise in the second slot, for $a, b \in \mathbb{R}$. The bracket is alternating because $[X, X] = XX - XX = 0$; expanding $[X+Y, X+Y] = 0$ and using bilinearity gives $[X, Y] + [Y, X] = 0$, so $[Y, X] = -[X, Y]$. For the Jacobi identity, expand each of the three double brackets. Writing out $[X, [Y, Z]] = X(YZ - ZY) - (YZ - ZY)X$ and the two cyclic analogues, the sum

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]]$$

consists of the twelve triple products $XYZ, XZY, \ldots$, each appearing once with a plus sign and once with a minus sign; they cancel in pairs, leaving 0. This establishes all three identities, completing the proof.

The conjugation curve $c(s)$ in the proof is the first appearance of the adjoint action, which the next chapter will develop into the representations Ad and ad that encode the bracket intrinsically. What matters now is the structural conclusion: the tangent space $\mathfrak{g}$ is not just a vector space but is closed under an antisymmetric bilinear product satisfying the Jacobi identity. This package of data has a name, and it is the central algebraic object of the theory.

Definition 1.4.5: Abstract Lie Algebra

A *Lie algebra* over a field k is a k-vector space $\mathfrak{g}$ equipped with a bilinear map

$$[-,-]\colon \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g}, \qquad (X,Y)\mapsto [X,Y]$$

called the *bracket*, satisfying two axioms:

1. the bracket is *alternating*: $[X,X]=0$ for all $X\in\mathfrak{g}$ (equivalently, over any field of characteristic $\neq 2$, antisymmetric: $[Y,X]=-[X,Y]$);
2. the bracket satisfies the *Jacobi identity*:

$$[X,[Y,Z]]+[Y,[Z,X]]+[Z,[X,Y]]=0$$

for all $X,Y,Z\in\mathfrak{g}$.

A *homomorphism* of Lie algebras is a k-linear map $\varphi\colon\mathfrak{g}\to\mathfrak{h}$ preserving the bracket, $\varphi([X,Y])=[\varphi(X),\varphi(Y)]$.

proposition 1.4.4 says precisely that the Lie algebra $\mathfrak{g}$ of a matrix group, with the commutator bracket $[X,Y]=XY-YX$, is an abstract real Lie algebra in the sense of definition 1.4.5. The classical algebras $\mathfrak{gl}_n,\mathfrak{sl}_n,\mathfrak{so}(n),\mathfrak{u}(n),\mathfrak{su}(n)$, and $\mathfrak{sp}(2n,\mathbb{F})$ are all real Lie algebras under the commutator, and one checks in each case that the bracket preserves the defining condition: for instance if X,Y are skew-symmetric then $[X,Y]^{\mathsf{T}}=(XY-YX)^{\mathsf{T}}=Y^{\mathsf{T}}X^{\mathsf{T}}-X^{\mathsf{T}}Y^{\mathsf{T}}=YX-XY=-[X,Y]$, so $[X,Y]\in\mathfrak{so}(n)$, consistent with proposition 1.4.4. The full matrix algebra $\mathfrak{gl}_n(\mathbb{F})=M_n(\mathbb{F})$ under the commutator is the ambient Lie algebra of which all the others are subalgebras, the exact analogue of $\mathrm{GL}_n(\mathbb{F})$ being the ambient group.

The abstract definition frees the bracket from its origin as a commutator of matrices. In chapter 2 the Lie algebra of a general Lie group is constructed intrinsically, as the space of left-invariant vector fields under the Lie bracket of vector fields, with no matrices in sight, and definition 1.4.5 is the target of that construction: the intrinsic bracket will satisfy exactly these two axioms. The matrix Lie algebras of this chapter are then recovered as the special case where the group sits inside some $\mathrm{GL}_n(\mathbb{F})$, and the intrinsic bracket restricts to the commutator. The reader who wants a preview should keep in mind that the abstract axioms are all that survive of the group structure at the infinitesimal level, and that the entire structure theory of Part II and beyond is carried out with the axioms alone.

The bridge between the group and its algebra is the exponential map. The defining property of $\mathfrak{g}$ is that exp sends it into G, and the local diffeomorphism of section 1.3 pins down the dimension of the correspondence.

Corollary 1.4.6: The Exponential Bridge and the Dimension Match

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g}$. Then exp maps $\mathfrak{g}$ into G, and the tangent space to G at the identity contains $\mathfrak{g}$:

$$\mathfrak{g} \subseteq T_I G$$

Suppose in addition that G is presented as a regular-value preimage $G = F^{-1}(c)$ for a smooth map F, as each classical group of section 1.2 is. Then $\mathfrak{g} = T_I G$, the map exp restricts to a diffeomorphism from a neighborhood of 0 in $\mathfrak{g}$ onto a neighborhood of I in G, and in particular

$$\dim_{\mathbb{R}} \mathfrak{g} = \dim G$$

***Proof*:**

That $\exp(\mathfrak{g}) \subseteq G$ is immediate from definition 1.4.1: for $X \in \mathfrak{g}$, taking $t = 1$ gives $\exp X = \exp(1 \cdot X) \in G$. For the inclusion into the tangent space, let $X \in \mathfrak{g}$. The curve $t \mapsto \exp(tX)$ lies in G by definition 1.4.1, passes through I at $t = 0$, and has velocity X there by proposition 1.3.4; a velocity vector of a curve in G through I is by definition an element of $T_I G$, so $X \in T_I G$. Hence $\mathfrak{g} \subseteq T_I G$.

Suppose now $G = F^{-1}(c)$ with c a regular value of a smooth F, so that $T_I G = \ker DF_I$. The reverse inclusion $T_I G \subseteq \mathfrak{g}$ is the content of the discussion following theorem 1.4.3: differentiating the defining relation $F = c$ along a curve in G through I shows that the linearized condition cutting out $\mathfrak{g}$ is exactly $\ker DF_I$, so $\mathfrak{g} = \ker DF_I = T_I G$. With this equality in hand, by theorem 1.3.7 the map exp is a diffeomorphism from a neighborhood U of 0 in $M_n(\mathbb{F})$ onto a neighborhood of I in $\mathrm{GL}_n(\mathbb{F})$, with $D\exp_0 = \mathrm{id}$; restricting to $\mathfrak{g}$, the differential $D\exp_0 = \mathrm{id}$ carries $\mathfrak{g} = T_0\mathfrak{g}$ isomorphically onto $\mathfrak{g} = T_I G$, so the inverse function theorem applied within G makes the restriction $\exp\colon U \cap \mathfrak{g} \to G$ a diffeomorphism onto a neighborhood of I in G. Two manifolds admitting a diffeomorphism between open sets have equal dimension, so $\dim_{\mathbb{R}} \mathfrak{g} = \dim G$, completing the proof.

The corollary is the quantitative form of the slogan that the Lie algebra is an infinitesimal copy of the group: near the identity, G looks exactly like its tangent space $\mathfrak{g}$, with exp providing the chart. The inclusion $\mathfrak{g} \subseteq T_I G$ holds for every matrix group and is the easy half, read off by differentiating $\exp(tX)$; the reverse inclusion, which upgrades it to the equality $\mathfrak{g} = T_I G$ and thereby forces the dimensions to match, is what requires the group to be presented as a regular-value preimage, and it is established above exactly for the classical groups of section 1.2. For a matrix group given only as an abstract embedded submanifold, with no defining map F at hand, the equality $\mathfrak{g} = T_I G$ is still true but rests on the closed-subgroup theorem, proved in chapter 3, which produces the exponential chart from the closedness of G alone. The dimension equality has already been verified group by group in theorem 1.4.3, but the corollary explains why it had to hold: exp is a local chart from $\mathfrak{g}$ to G, so the two spaces cannot have different dimensions. The global behavior of exp is more delicate, and whether it is surjective onto G depends on the group; section 1.5 will show exp is onto for the compact group $\mathrm{SU}(2)$, while for $\mathrm{GL}_n(\mathbb{R})$ it is not, as the negative-determinant matrices are never exponentials.

Everything comes together in the smallest nonabelian compact example, $\mathfrak{su}(2)$, the Lie algebra of the running-example group $\mathrm{SU}(2)$. Its bracket is the cross product, and this identification is the first instance of the dictionary the book exists to develop.

Example 1.4.7: The Lie Algebra $\mathfrak{su}(2)$ and the Cross Product

By theorem 1.4.3, $\mathfrak{su}(2)$ is the space of traceless skew-Hermitian 2×2 complex matrices. A traceless skew-Hermitian matrix has purely imaginary diagonal summing to zero and off-diagonal entries $b, -\bar{b}$, so it has the form

$$\begin{pmatrix} ia & b \\ -\bar{b} & -ia \end{pmatrix}, \qquad a \in \mathbb{R},\ b \in \mathbb{C}$$

a real three-dimensional space, consistent with $\dim \mathfrak{su}(2) = 2^2 - 1 = 3$. A convenient basis is built from the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

which are Hermitian and traceless; multiplying by $-\frac{i}{2}$ produces traceless skew-Hermitian matrices. Set

$$X_k = -\tfrac{i}{2}\,\sigma_k, \qquad k = 1, 2, 3$$

explicitly

$$X_1 = \tfrac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \qquad X_2 = \tfrac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad X_3 = \tfrac{1}{2}\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

Each X_k is traceless and satisfies $X_k^* = -X_k$, so $X_1, X_2, X_3 \in \mathfrak{su}(2)$, and they are linearly independent over $\mathbb{R}$, hence a basis.

The bracket relations are computed directly from the Pauli identity $\sigma_j\sigma_k = \delta_{jk} I + i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell$, where $\varepsilon_{jk\ell}$ is the sign of the permutation (j,k,ℓ). The commutator of Pauli matrices is therefore $[\sigma_j, \sigma_k] = 2i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell$, and so

$$[X_j, X_k] = \left(-\tfrac{i}{2}\right)^2 [\sigma_j, \sigma_k] = -\tfrac{1}{4}\cdot 2i \sum_\ell \varepsilon_{jk\ell}\sigma_\ell = \sum_\ell \varepsilon_{jk\ell}\left(-\tfrac{i}{2}\sigma_\ell\right) = \sum_\ell \varepsilon_{jk\ell} X_\ell$$

In cyclic form,

$$[X_1, X_2] = X_3, \qquad [X_2, X_3] = X_1, \qquad [X_3, X_1] = X_2 \tag{1.5}$$

These are exactly the multiplication rules of the standard basis e_1, e_2, e_3 of $\mathbb{R}^3$ under the cross product: $e_1 \times e_2 = e_3$, $e_2 \times e_3 = e_1$, $e_3 \times e_1 = e_2$.

Define the linear map $\Phi\colon \mathfrak{su}(2) \to \mathbb{R}^3$ by $\Phi(X_k) = e_k$ on the basis and extending by linearity. It is a linear isomorphism of three-dimensional real vector spaces, and it preserves the bracket: on basis elements (1.5) matches the cross product term by term, and both operations are bilinear, so $\Phi([X,Y]) = \Phi(X) \times \Phi(Y)$ for all X, Y. Therefore

$$\Phi\colon \mathfrak{su}(2) \xrightarrow{\sim} (\mathbb{R}^3, \times)$$

is an isomorphism of Lie algebras, where $\mathbb{R}^3$ carries the cross product as its bracket. The cross product does satisfy the Jacobi identity, which is the familiar vector identity $a \times (b \times c) + b \times (c \times a) + c \times (a \times b) = 0$; the isomorphism Φ makes this identity and the Jacobi identity for $\mathfrak{su}(2)$ the same statement. This identification is the infinitesimal form of the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ taken up in section 1.5: the algebra $(\mathbb{R}^3, \times)$ is also $\mathfrak{so}(3)$ (the skew-symmetric 3×3 matrices act on $\mathbb{R}^3$ as $v \mapsto \omega \times v$), so $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ even though $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ are not isomorphic as groups.

The isomorphism $\mathfrak{su}(2) \cong \mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ is a compressed statement of the entire local theory of

these groups: two nonisomorphic groups share a Lie algebra, so they must agree near the identity and differ only in their global topology. That the infinitesimal object cannot distinguish SU(2) from SO(3), while the groups themselves are distinguished by a factor of $\mathbb{Z}/2$, is the phenomenon section 1.5 analyzes in full, and it is the reason the passage from group to algebra loses information that only the topology of the group can give.

Exercise 1.4.1

1. Write down the standard basis of the skew-symmetric matrices $\mathfrak{so}(3)$, namely the three matrices L_1, L_2, L_3 with $(L_k)_{ij} = -\varepsilon_{kij}$, and verify the bracket relations $[L_1, L_2] = L_3$, $[L_2, L_3] = L_1$, $[L_3, L_1] = L_2$. Conclude that $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$, and hence $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ as real Lie algebras.

2. Show that for the commutator bracket the trace gives a Lie algebra homomorphism $\operatorname{tr}\colon \mathfrak{gl}_n(\mathbb{F}) \to \mathbb{F}$ (with $\mathbb{F}$ abelian, so $[\lambda, \mu] = 0$), and deduce that $\mathfrak{sl}_n(\mathbb{F}) = \ker \operatorname{tr}$ is an ideal. Prove the direct sum decomposition $\mathfrak{gl}_n(\mathbb{F}) = \mathfrak{sl}_n(\mathbb{F}) \oplus \mathbb{F}I$ of vector spaces, and determine the bracket of a scalar matrix λI with an arbitrary $X \in \mathfrak{gl}_n(\mathbb{F})$.

3. Using $\det \exp X = e^{\operatorname{tr} X}$, show directly from definition 1.4.1 that a matrix $X \in M_n(\mathbb{R})$ lies in $\mathfrak{sl}_n(\mathbb{R})$ if and only if $\operatorname{tr} X = 0$, without invoking theorem 1.4.3. Give an explicit $X \in \mathfrak{gl}_2(\mathbb{R})$ with $\operatorname{tr} X \neq 0$ and verify that $\exp(tX) \notin \mathrm{SL}_2(\mathbb{R})$ for some t.

4. For $X \in \mathfrak{su}(2)$ with $\Phi(X) = v \in \mathbb{R}^3$ under the isomorphism of example 1.4.7, show that $X^2 = -\frac{1}{4}|v|^2 I$, where $|v|$ is the Euclidean norm. Deduce the closed form

$$\exp(X) = \cos\!\left(\tfrac{|v|}{2}\right) I + \frac{\sin\!\left(\frac{|v|}{2}\right)}{|v|/2}\, X$$

   for $v \neq 0$, and confirm that $\exp(X) \in \mathrm{SU}(2)$.

5. Verify that $\mathfrak{sp}(2n, \mathbb{F})$ is closed under the commutator: if $X^{\mathsf{T}}J + JX = 0$ and $Y^{\mathsf{T}}J + JY = 0$, show $[X, Y]^{\mathsf{T}}J + J[X, Y] = 0$. This confirms $\mathfrak{sp}(2n, \mathbb{F})$ is a Lie subalgebra of $\mathfrak{gl}_{2n}(\mathbb{F})$, as proposition 1.4.4 predicts.

6. Show that $\mathfrak{u}(n)$ is a *real* but not a complex vector subspace of $M_n(\mathbb{C})$: exhibit a skew-Hermitian matrix X for which iX is not skew-Hermitian. Explain how this is consistent with proposition 1.4.2, which asserts only that the Lie algebra of a matrix group is a real vector space.

1.5 $SU(2)$, $SO(3)$, and the Double Cover $SU(2) \to SO(3)$

Every structural feature that the later chapters develop in generality is already present, in fully computable form, in the smallest noncommutative compact group, SU(2). This section studies it directly: as a manifold it is a 3-sphere, and its Lie algebra $\mathfrak{su}(2)$ is three-dimensional, small enough that the exponential map, the adjoint action, and the passage to SO(3) can all be written out with explicit matrices. The centerpiece is the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, a two-to-one surjective homomorphism that realizes SO(3) as the quotient $\mathrm{SU}(2)/\{\pm I\}$ and identifies SU(2) as the universal cover of the rotation group. This one map explains why a rotation of ordinary space cannot be tracked continuously by a single square root, why $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2$, and where the spin of quantum mechanics comes from. It is the first place in the book where the correspondence between a group

and its Lie algebra does concrete work, and SU(2) becomes the running example that every part returns to: its maximal torus and Weyl group in the compact theory, its irreducible representations and their characters at the analytic summit, and its noncompact sibling $\mathrm{SL}_2(\mathbb{R})$ in the theory of real forms.

The starting point is the explicit description of SU(2) obtained at the end of the previous section, which realizes it as a sphere.

Proposition 1.5.1: SU(2) **is the 3-Sphere**

The special unitary group in dimension two is

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \;\middle|\; \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$$

and the map $(\alpha, \beta) \mapsto \left(\begin{smallmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{smallmatrix}\right)$ is a diffeomorphism from the unit sphere $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$ onto SU(2). In particular SU(2) is compact, connected, and simply connected.

***Proof*:**

A matrix $A \in \mathrm{U}(2)$ has orthonormal columns; writing the first column as $(\alpha, \beta)^{\mathsf{T}}$ with $|\alpha|^2 + |\beta|^2 = 1$, the second column is a unit vector orthogonal to it, hence a unit scalar multiple $\lambda(-\overline{\beta}, \overline{\alpha})^{\mathsf{T}}$ with $|\lambda| = 1$. The determinant of the resulting matrix is $\lambda(|\alpha|^2 + |\beta|^2) = \lambda$, so the constraint $\det A = 1$ defining SU(2) forces $\lambda = 1$, giving the displayed form. Conversely every matrix of that form is special unitary, as a direct check of $A^* A = I$ and $\det A = 1$ shows. The parametrization $(\alpha, \beta) \mapsto A$ is thus a bijection from $\left\{(\alpha, \beta) \in \mathbb{C}^2 \;\middle|\; |\alpha|^2 + |\beta|^2 = 1\right\} = S^3$ onto SU(2); it is the restriction of a real-linear injection $\mathbb{C}^2 \to M_2(\mathbb{C})$, hence a smooth embedding, and its inverse (read off α and β from the first column) is smooth, so it is a diffeomorphism.
The sphere S^3 is compact, being closed and bounded in $\mathbb{R}^4$, and it is connected and simply connected. Connectedness holds because the two stereographic chart domains are each diffeomorphic to $\mathbb{R}^3$, hence connected, and they overlap. For simple connectivity, subdivide a loop into finitely many arcs, each contained in one of the two chart domains, and homotope each arc rel endpoints to the straight segment between its endpoints in the chart's coordinates. The straightened loop is a finite union of smooth arcs, so its image has measure zero in S^3: each arc extends to a smooth map of a neighborhood of $[0,1] \times \{0\} \subseteq \mathbb{R}^3$ into S^3, constant in the two extra coordinates, and $[0,1] \times \{0\}$ is a null set of $\mathbb{R}^3$, so the arc's image is null by [13, Measure Zero under Smooth Maps]. The loop therefore omits some point p. A loop omitting p lies in the chart $S^3 \setminus \{p\} \cong \mathbb{R}^3$ and contracts there. The systematic sphere computations $\pi_1(S^n) = 0$ for $n \geq 2$ are deferred to [11]. These facts transfer to SU(2) through the diffeomorphism, completing the proof.

That SU(2) is a sphere is the geometric fact underlying everything that follows. Compactness is what will later force its representation theory to be completely reducible and its irreducible representations to be finite-dimensional; connectedness means it is generated by any neighborhood of the identity, so that its structure is determined by its Lie algebra; and simple connectedness is precisely the property that makes SU(2), rather than SO(3), the group corresponding to the shared Lie algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ under the equivalence of chapter 3. The three-sphere is also the unit group of the quaternions: writing a quaternion as $q = \alpha + \beta j$ with $\alpha, \beta \in \mathbb{C}$ identifies $\{q \in \mathbb{H} \mid |q| = 1\}$ with the matrices above, so SU(2), S^3, and the unit quaternions are three names for one object.

The Lie algebra of SU(2) was identified in section 1.4 as the space of traceless skew-Hermitian 2×2 matrices. Its concrete shape is worth having in view before exponentiating.

Example 1.5.2: The Lie Algebra $\mathfrak{su}(2)$ in Coordinates

By theorem 1.4.3 the Lie algebra of SU(2) is

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) \mid X^* = -X,\ \operatorname{tr} X = 0\}$$

the traceless skew-Hermitian matrices. Skew-Hermiticity forces the diagonal entries to be purely imaginary and the off-diagonal entries to be conjugate-negatives, and tracelessness makes the two diagonal entries opposite, so a general element is

$$X = \begin{pmatrix} ic & -\bar{b} \\ b & -ic \end{pmatrix}, \qquad c \in \mathbb{R},\ b \in \mathbb{C}$$

depending on the three real parameters c and $\operatorname{Re} b, \operatorname{Im} b$; thus $\dim_{\mathbb{R}} \mathfrak{su}(2) = 3 = 2^2 - 1$, matching the dimension of SU(2) from proposition 1.2.6. A convenient basis is provided by the matrices

$$E_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad E_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad E_3 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$$

which are $\pm\frac{1}{2i}$ times the Pauli matrices, and whose brackets are $[E_1, E_2] = E_3$, $[E_2, E_3] = E_1$, $[E_3, E_1] = E_2$, the cross-product relations of $\mathbb{R}^3$; this is the identification $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$ recorded in example 1.4.7.

The determinant provides a natural quadratic form on this space, and its shape is what makes SU(2) act by rotations below. For $X = \left(\begin{smallmatrix} ic & -\bar{b} \\ b & -ic \end{smallmatrix}\right) \in \mathfrak{su}(2)$,

$$\det X = (ic)(-ic) - (-\bar{b})(b) = c^2 + |b|^2 \tag{1.6}$$

a positive-definite quadratic form. Evaluating on the basis gives $\det E_j = \frac{1}{4}$ for each j, and since $\det(sX) = s^2 \det X$ for a scalar s, the form det is $\frac{1}{4}$ times the standard Euclidean squared norm in the coordinates dual to E_1, E_2, E_3; equivalently $\det X = -\frac{1}{2}\operatorname{tr}(X^2)$, because for traceless X the Cayley-Hamilton relation gives $X^2 = -(\det X)I$ and hence $\operatorname{tr}(X^2) = -2 \det X$. In either form the quadratic form is a scalar multiple of the trace form and is invariant under conjugation, the property that drives the double cover.

The exponential map sends $\mathfrak{su}(2)$ into SU(2), and because $\mathfrak{su}(2)$ is three-dimensional the exponential admits a closed formula in the spirit of Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$. The mechanism is that every $X \in \mathfrak{su}(2)$ satisfies a quadratic relation, so its powers cycle and the exponential series resums into cosine and sine.

Proposition 1.5.3: The Exponential is Onto SU(2)

For $X \in \mathfrak{su}(2)$ set $|X| = \sqrt{\det X}$, so that $X^2 = -|X|^2 I$. Then

$$\exp X = \cos(|X|)\, I + \frac{\sin(|X|)}{|X|}\, X \tag{1.7}$$

interpreted as $\exp 0 = I$ when $X = 0$. The map $\exp\colon \mathfrak{su}(2) \to \mathrm{SU}(2)$ is surjective.

***Proof*:**
Step 1: the quadratic relation. By the Cayley-Hamilton theorem a traceless 2×2 matrix X satisfies $X^2 - (\mathrm{tr}X)X + (\det X)I = 0$, which with $\mathrm{tr}X = 0$ reads $X^2 = -(\det X)I$. For $X \in \mathfrak{su}(2)$ the determinant is $|X|^2 \geq 0$ by (1.6), so writing $r = |X|$ gives $X^2 = -r^2 I$.

Step 2: resumming the series. Assume $r > 0$. The relation $X^2 = -r^2 I$ gives $X^{2k} = (-1)^k r^{2k} I$ and $X^{2k+1} = (-1)^k r^{2k} X$ for all $k \geq 0$. Splitting the exponential series (absolutely convergent by definition 1.3.2) into even and odd powers,

$$\exp X = \sum_{k\geq 0} \frac{X^{2k}}{(2k)!} + \sum_{k\geq 0} \frac{X^{2k+1}}{(2k+1)!} = \left(\sum_{k\geq 0} \frac{(-1)^k r^{2k}}{(2k)!}\right) I + \left(\sum_{k\geq 0} \frac{(-1)^k r^{2k}}{(2k+1)!}\right) X$$

The first bracketed sum is $\cos r$; the second is $\frac{1}{r}\sum_k \frac{(-1)^k r^{2k+1}}{(2k+1)!} = \frac{\sin r}{r}$. This is (1.7). For $X = 0$ both sides equal I, and the right-hand side has the removable singularity $\frac{\sin r}{r} \to 1$ as $r \to 0$, so the formula holds uniformly with the stated convention.

Step 3: surjectivity. Let $A \in \mathrm{SU}(2)$; by proposition 1.5.1 write $A = \left(\begin{smallmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{smallmatrix}\right)$. Then $\mathrm{tr}A = \alpha + \overline{\alpha} = 2\,\mathrm{Re}\,\alpha \in [-2, 2]$, so there is $r \in [0, \pi]$ with $\cos r = \mathrm{Re}\,\alpha$. If $A = I$ then $A = \exp 0$; if $A = -I$ then $\mathrm{Re}\,\alpha = -1$, $r = \pi$, and $A = \exp(\pi Y)$ for any $Y \in \mathfrak{su}(2)$ with $|Y| = 1$, since (1.7) gives $\exp(\pi Y) = \cos\pi\, I + \sin\pi\, Y = -I$. Otherwise $\mathrm{Re}\,\alpha \in (-1, 1)$, so $\sin r > 0$, and the matrix

$$X = \frac{r}{\sin r}\left(A - \mathrm{Re}(\alpha)\, I\right)$$

lies in $\mathfrak{su}(2)$: the matrix $A - \mathrm{Re}(\alpha)I$ is traceless because $\mathrm{tr}A = 2\,\mathrm{Re}\,\alpha$, and skew-Hermitian because its diagonal entries are $\pm i\,\mathrm{Im}\,\alpha$ (purely imaginary) while its off-diagonal entries $-\overline{\beta}, \beta$ are conjugate-negatives; scaling by the real factor $r/\sin r$ preserves both properties. This X has $|X| = r$, since $\det X = (r/\sin r)^2 \det(A - \mathrm{Re}(\alpha)I) = (r/\sin r)^2(\mathrm{Im}\,\alpha)^2 + (r/\sin r)^2|\beta|^2$ and $\sin^2 r = 1 - \mathrm{Re}(\alpha)^2 = (\mathrm{Im}\,\alpha)^2 + |\beta|^2$ (using $|\alpha|^2 + |\beta|^2 = 1$), giving $\det X = r^2$. Substituting into (1.7),

$$\exp X = \cos(r)\, I + \frac{\sin r}{r} \cdot \frac{r}{\sin r}\big(A - \mathrm{Re}(\alpha) I\big) = \mathrm{Re}(\alpha)\, I + A - \mathrm{Re}(\alpha)\, I = A$$

so $A = \exp X$. Every element of SU(2) is therefore an exponential, as we sought to show.

Formula (1.7) is the noncommutative avatar of Euler's formula: as t runs over $\mathbb{R}$, the curve $t \mapsto \exp(tX)$

traces a great circle on the sphere SU(2), periodic with period $2\pi/|X|$, and its image is the one-parameter subgroup generated by X from theorem 1.3.10. Surjectivity of the exponential is special to compact connected groups and is proved in general in Part III; for SU(2) it is visible by hand because the group is a sphere and every point lies on a great circle through the identity. The formula also makes the covering map computable: a rotation angle on the SU(2) side is $|X|$, and it corresponds to a rotation angle of $2|X|$ on the SO(3) side, the source of the factor of two in the double cover.

Example 1.5.4: One-Parameter Subgroups Through the Poles

Take $X = E_1 = \frac{1}{2}\operatorname{diag}(i, -i)$, so $|E_1| = \sqrt{\det E_1} = \frac{1}{2}$ and $tX = \frac{t}{2}\operatorname{diag}(i, -i)$ has $|tX| = |t|/2$. Formula (1.7) gives

$$\exp(tE_1) = \cos\left(\frac{t}{2}\right) I + 2\sin\left(\frac{t}{2}\right) E_1 = \begin{pmatrix} e^{it/2} & 0 \\ 0 & e^{-it/2} \end{pmatrix}$$

directly recovering the diagonal exponential $\exp \operatorname{diag}(a, -a) = \operatorname{diag}(e^a, e^{-a})$ with $a = it/2$. This one-parameter subgroup is the standard maximal torus of SU(2), the diagonal circle, which the compact theory of Part III returns to. It has period 4π in t, not 2π: the parameter must run to 4π before $\exp(tE_1)$ returns to I, whereas the corresponding rotation of $\mathbb{R}^3$ has period 2π. The doubling is again the signature of the cover.

The group SO(3) is 3-dimensional, connected, and compact, with Lie algebra of the same dimension. Its Lie algebra $\mathfrak{so}(3)$, the real skew-symmetric 3×3 matrices, is spanned by the infinitesimal rotations about the three coordinate axes,

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

whose brackets are $[L_1, L_2] = L_3$, $[L_2, L_3] = L_1$, $[L_3, L_1] = L_2$, the identical cross-product relations found for E_1, E_2, E_3 in example 1.5.2. The linear map $L_i \mapsto E_i$ is therefore an isomorphism of Lie algebras, so

$$\mathfrak{su}(2) \;\cong\; \mathfrak{so}(3) \;\cong\; (\mathbb{R}^3, \times) \tag{1.8}$$

all three being the three-dimensional real Lie algebra with the cross-product bracket. Concretely $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ via $v \mapsto (w \mapsto v \times w)$: the matrix L_v of "cross with v" acts on w as $v \times w$, and $L_{v \times v'} = [L_v, L_{v'}]$ recovers the Jacobi identity for the cross product. The isomorphism (1.8) predicts, by the correspondence of chapter 3, a covering homomorphism between the associated connected groups. The rest of the section constructs it explicitly.

The construction is the adjoint action: SU(2) acts on its own Lie algebra by conjugation, and because $\mathfrak{su}(2)$ carries the invariant quadratic form (1.6), this action lands in the orthogonal group of that form, which is a copy of SO(3).

Theorem 1.5.5: The Double Cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$

For $g \in \mathrm{SU}(2)$ let $\varphi(g)$ be the conjugation map

$$\varphi(g)\colon \mathfrak{su}(2) \to \mathfrak{su}(2), \qquad \varphi(g)(X) = gXg^{-1}$$

Identifying $\mathfrak{su}(2)$ with $\mathbb{R}^3$ via the basis E_1, E_2, E_3, with the Euclidean structure given by the form det of (1.6), the assignment $g \mapsto \varphi(g)$ is a homomorphism

$$\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$$

It is surjective, and its kernel is exactly the center $\{\pm I\}$. Hence φ induces an isomorphism

$$\mathrm{SU}(2)/\{\pm I\} \xrightarrow{\sim} \mathrm{SO}(3)$$

and is a two-to-one covering map.

***Proof*:**

Step 1: $\varphi(g)$ is a well-defined orthogonal transformation. For $g \in \mathrm{SU}(2)$ and $X \in \mathfrak{su}(2)$ the matrix gXg^{-1} is again traceless (trace is conjugation-invariant) and skew-Hermitian, since $g^{-1} = g^*$ gives $(gXg^*)^* = gX^*g^* = -gXg^*$; so $\varphi(g)$ maps $\mathfrak{su}(2)$ to itself, linearly, with inverse $\varphi(g^{-1})$. It preserves the form (1.6) because the determinant is conjugation-invariant, $\det(\varphi(g)X) = \det(gXg^{-1}) = \det X$. Thus $\varphi(g)$ is a linear isometry of the Euclidean space $(\mathfrak{su}(2), \det) \cong \mathbb{R}^3$, that is $\varphi(g) \in \mathrm{O}(3)$. That φ is a homomorphism is immediate: $\varphi(gh)(X) = (gh)X(gh)^{-1} = g(hXh^{-1})g^{-1} = \varphi(g)\varphi(h)(X)$, and $\varphi(I) = \mathrm{id}$.

Step 2: the image lies in $\mathrm{SO}(3)$. The group $\mathrm{SU}(2)$ is connected by proposition 1.5.1, and $\varphi\colon \mathrm{SU}(2) \to \mathrm{O}(3)$ is continuous (its matrix entries are quadratic polynomials in the entries of g), so the image $\varphi(\mathrm{SU}(2))$ is a connected subset of $\mathrm{O}(3)$. The determinant $\det \circ \varphi\colon \mathrm{SU}(2) \to \{\pm 1\}$ is continuous on a connected space, hence constant, and it takes the value $\det \varphi(I) = \det(\mathrm{id}) = 1$; therefore $\det \varphi(g) = 1$ for all g, so $\varphi(\mathrm{SU}(2)) \subseteq \mathrm{SO}(3)$ and $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$.

Step 3: the kernel is $\{\pm I\}$. Suppose $\varphi(g) = \mathrm{id}$, that is $gXg^{-1} = X$ for all $X \in \mathfrak{su}(2)$, so g commutes with every traceless skew-Hermitian matrix. The matrices E_1, E_2, E_3 span $\mathfrak{su}(2)$ over $\mathbb{R}$, and together with I they span $M_2(\mathbb{C})$ over $\mathbb{C}$ (they are, up to scalars, the Pauli matrices, which with I form a basis of $M_2(\mathbb{C})$); so g commutes with all of $M_2(\mathbb{C})$. An element commuting with every matrix is a scalar, $g = \lambda I$, and $g \in \mathrm{SU}(2)$ forces $\lambda^2 = \det g = 1$, hence $\lambda = \pm 1$. Thus $\ker \varphi = \{\pm I\}$, and since $\pm I$ are the only scalars in $\mathrm{SU}(2)$, this is exactly the center.

Step 4: surjectivity. Fix a unit vector $u \in \mathfrak{su}(2)$, so $u^2 = -I$ by Step 1 of proposition 1.5.3. The one-parameter group $s \mapsto \varphi(\exp(su))$ in $\mathrm{SO}(3)$ has derivative at $s = 0$ equal to the operator $\mathrm{ad}(u)(X) = uX - Xu = [u, X]$ on $\mathfrak{su}(2)$, since $\frac{d}{ds}\big|_0 \exp(su)X\exp(-su) = uX - Xu$ by proposition 1.3.4; hence $\varphi(\exp(su)) = \exp(s\,\mathrm{ad}(u))$ as maps of $\mathfrak{su}(2)$. Under the identification (1.8) the operator $\mathrm{ad}(u)$ is rotation generator about the axis u; because the structure constants of E_1, E_2, E_3 in the orthonormal frame of the form (1.6) are twice those of the geometric generators L_i (each E_j has $\det E_j = \frac{1}{4}$, i.e. Euclidean length $\frac{1}{2}$, while the L_i have length 1), the flow $\exp(s\,\mathrm{ad}(u))$ is the rotation of $\mathbb{R}^3$ by angle $2s$ about the axis u. Consequently, given $R \in \mathrm{SO}(3)$,

Euler's theorem [4, Euler's Rotation Theorem] writes R as rotation by some angle θ about some unit axis $u \in S^2 \subset \mathfrak{su}(2)$, and then $g = \exp(\frac{\theta}{2}u)$ satisfies $\varphi(g) = \exp(\frac{\theta}{2}\operatorname{ad}(u)) =$ rotation by θ about $u = R$. As R was arbitrary, φ is surjective.

Step 5: the covering. By Steps 2 through 4, $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ is a surjective homomorphism with kernel $\{\pm I\}$, so the first isomorphism theorem for groups ([2, First Isomorphism Theorem]) gives a group isomorphism $\mathrm{SU}(2)/\{\pm I\} \xrightarrow{\sim} \mathrm{SO}(3)$. Both sides are compact three-dimensional manifolds, and the induced map is a continuous bijection between compact Hausdorff spaces, hence a homeomorphism; its differential at the identity is the Lie algebra isomorphism (1.8) (Step 4 computed $d\varphi_I = \operatorname{ad}$ under the frames, a linear isomorphism), so it is a diffeomorphism. Finally φ is two-to-one with fibers $\{g, -g\} = g \cdot \ker\varphi$, and since $d\varphi_I$ is an isomorphism φ is a local diffeomorphism at I, spread to every point by left translation; a two-to-one local diffeomorphism between compact manifolds with discrete fibers is a two-sheeted covering map, completing the proof.

The factor of two is the arithmetic heart of the cover, and it is worth isolating. For a unit vector $u \in \mathfrak{su}(2)$ the element $\exp(\frac{\theta}{2}u)$ on the SU(2) side maps under φ to rotation by the full angle θ on the SO(3) side: the group parameter is half the spatial rotation angle. This is exactly the discrepancy seen in example 1.5.4, where $\exp(tE_1)$ had period 4π while its image rotation of $\mathbb{R}^3$ had period 2π. Passing to $-I$ at the halfway point $t = 2\pi$ is the mechanism: the two elements $\pm g$ of a fiber are the two lifts of a single rotation, and no continuous choice of lift exists over all of SO(3).

The theorem says that SO(3) is SU(2) with antipodal points of the sphere S^3 glued together, which is the standard model of three-dimensional real projective space. So $\mathrm{SO}(3) \cong \mathbb{R}P^3$ as a manifold: the rotation group of ordinary space is real projective three-space. The two preimages $\pm g$ of a rotation $R = \varphi(g)$ are the two ways to represent that rotation by a unit quaternion, differing by a sign, and the impossibility of choosing the sign continuously over all of SO(3) is the failure of SO(3) to be simply connected, computed next.

Corollary 1.5.6: The Fundamental Group of SO(3)

The map $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ is the universal cover of SO(3), and

$$\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2$$

The generator is any loop in SO(3) that lifts to a path in SU(2) from I to $-I$, such as the image under φ of $t \mapsto \exp(tE_1)$, $t \in [0, 2\pi]$; this loop is not contractible, but traversing it twice is.

***Proof*:**
By proposition 1.5.1 the total space $\mathrm{SU}(2) \cong S^3$ is simply connected, and by theorem 1.5.5 the map φ is a covering with fiber the two-element group $\{\pm I\}$. A covering map with simply-connected total space is the universal cover of its base, and for a universal cover arising from a central subgroup (here $\{\pm I\}$) the fundamental group of the base is the deck group, which is the kernel: $\pi_1(\mathrm{SO}(3)) \cong \ker\varphi = \{\pm I\} \cong \mathbb{Z}/2$. These covering-space facts are proved in [9, The Deck Group of the Universal Cover].
For the explicit generator, the path $\gamma(t) = \exp(tE_1)$ for $t \in [0, 2\pi]$ runs in SU(2) from $\gamma(0) = I$ to $\gamma(2\pi) = \operatorname{diag}(e^{i\pi}, e^{-i\pi}) = -I$ by example 1.5.4, so $\varphi \circ \gamma$ is a loop in SO(3) (it returns to

$\varphi(-I) = \varphi(I) = \mathrm{id}$) whose lift γ has distinct endpoints $I \neq -I$. A loop whose lift to the universal cover is not closed is noncontractible, so $\varphi \circ \gamma$ generates $\pi_1(\mathrm{SO}(3))$. Doubling it, $t \in [0, 4\pi]$, produces a lift $\gamma(t)$ from I back to $\gamma(4\pi) = I$, a closed lift, so the doubled loop is contractible; the generator therefore has order two, completing the proof.

The order-two element of $\pi_1(\mathrm{SO}(3))$ is the mathematics behind the physical fact that a continuous 2π rotation of a rigid body is not homotopic to no rotation, while a 4π rotation is: the "belt trick", in which a 360° twist in a belt cannot be undone without moving the ends but a 720° twist can, is a demonstration of a noncontractible loop in SO(3) together with its contractible square. In quantum mechanics the same two-to-one relationship is why the state of a spin-$\frac{1}{2}$ particle acquires a sign under a 2π rotation and returns to itself only after 4π: the physical symmetry group is SO(3), but the group acting on wavefunctions is its double cover SU(2), and the sign is the nontrivial element of the kernel. The representations that see this sign, the ones that do not descend to SO(3), are the half-integer-spin representations of SU(2) classified in Part III.

This section launches the running example that threads the rest of the book. The group SU(2) reappears in each part, always as the smallest place to see a new phenomenon in full: its diagonal circle is the maximal torus whose conjugates cover the group, and the reflection swapping $\mathrm{diag}(z, \overline{z})$ to $\mathrm{diag}(\overline{z}, z)$ is the Weyl group $\mathbb{Z}/2$, both taken up in the compact theory of maximal tori. Its irreducible representations are the symmetric powers of the standard two-dimensional representation, one in each dimension, and their characters are the functions $\chi_n(\theta) = \sin((n+1)\theta)/\sin\theta$ that the Weyl character formula produces as its first nontrivial instance. Its noncompact sibling $\mathrm{SL}_2(\mathbb{R})$, sharing the complexified Lie algebra $\mathfrak{sl}_2(\mathbb{C})$ but replacing the sphere by a group with a hyperbolic plane inside it, is the running example of the theory of real forms and the Iwasawa decomposition in Part IV. Every one of these returns to the concrete matrix computations of this section.

Exercise 1.5.1

1. Using (1.7), compute $\exp(\theta E_3)$ explicitly as a 2×2 complex matrix, and verify that $\exp(2\pi E_3) = -I$ while $\exp(4\pi E_3) = I$. Interpret the period 4π in terms of the double cover.

2. Show that the map sending the quaternion $q = a+bi+cj+dk$ (with $a, b, c, d \in \mathbb{R}$) to the matrix $\left(\begin{smallmatrix} a+bi & c+di \\ -c+di & a-bi \end{smallmatrix}\right)$ restricts to an isomorphism from the group of unit quaternions $\{q \mid |q| = 1\}$ onto SU(2), and that under this identification $\mathfrak{su}(2)$ is the space of purely imaginary quaternions $bi + cj + dk$.

3. Fix $g = \exp(\theta E_3)$. Compute $\varphi(g)$ as a 3×3 matrix in the basis E_1, E_2, E_3 by evaluating gE_1g^{-1}, gE_2g^{-1}, and gE_3g^{-1}, and confirm that $\varphi(g)$ is the rotation of $\mathbb{R}^3$ by angle θ about the third axis.

4. Prove that the center of SU(2) is exactly $\{\pm I\}$, and deduce from theorem 1.5.5 that SO(3) has trivial center.

5. Show that there is no continuous group homomorphism $\psi\colon \mathrm{SO}(3) \to \mathrm{SU}(2)$ with $\varphi\circ\psi = \mathrm{id}_{\mathrm{SO}(3)}$; that is, the double cover does not split. Deduce that SU(2) and $\mathrm{SO}(3)\times\{\pm I\}$ are not isomorphic as Lie groups.

6. Explain why SU(2) and SO(3) are not isomorphic as Lie groups despite having isomorphic Lie algebras, and reconcile this with the principle that a Lie algebra determines its simply-connected group uniquely.

2

Lie Groups and Their Lie Algebras

The previous chapter met the classical groups at the surface, as sets of matrices cut out by explicit equations, and produced the two objects on which the theory turns, the exponential map and the Lie algebra, by differentiating matrix-valued functions. Nothing in those constructions truly required the ambient matrices. The tangent space at the identity, the one-parameter subgroups, the bracket: each was a manifestation of the group law and the smooth structure, and each survives when the matrices are removed and only a manifold carrying a compatible group law remains. That manifold is a Lie group in the sense of this chapter, and the task now is to rebuild the machinery of chapter 1 intrinsically, on any Lie group, with no embedding into GL_n presupposed.

The mechanism that replaces the matrix computations is left-invariance. A single tangent vector at the identity spreads, by left translation, to a vector field defined over the whole group. The vector fields arising this way are closed under the Lie bracket of vector fields, and this bracket makes the tangent space at the identity a Lie algebra $\mathfrak{g}$, the intrinsic counterpart of the matrix Lie algebra of chapter 1. The flows of these left-invariant fields are the one-parameter subgroups, and assembling them at time one defines the exponential map $\exp\colon \mathfrak{g} \to G$. Differentiating the action of the group on itself by conjugation produces the adjoint representations Ad and ad, and recovers the bracket in the form $[X, Y] = \mathrm{ad}(X)Y$. For a matrix group every one of these constructions returns the answer already computed by hand: the intrinsic exponential is the matrix exponential, $\mathrm{Ad}(g)$ is conjugation $X \mapsto gXg^{-1}$, and $\mathrm{ad}(X)$ is the commutator $Y \mapsto XY - YX$. The abstract theory is thus anchored at every step to the concrete picture of the first chapter, and the reader who keeps SU(2) in view loses nothing by the ascent to generality.

2.1 Lie Groups: Definition, Examples, and the Category of Lie Groups

Chapter 1 studied groups of matrices: closed subgroups of $\mathrm{GL}_n(\mathbb{F})$, cut out by polynomial equations, whose smooth structure came for free from the ambient vector space $M_n(\mathbb{F})$. Every construction there was anchored in coordinates. The payoff of dropping them is generality: the theorems apply to groups that are not presented as matrix groups, and the constructions become functorial. The cost is that the smoothness which was automatic for matrix groups must now be built into the definition.

Multiplication is a smooth map, so it can be differentiated, and left translation by a group element is at once a group-theoretic bijection and a diffeomorphism. Left translation transports the neighborhood of the identity to the neighborhood of any point, forcing the manifold to be homogeneous. It trivializes the tangent bundle, forcing parallelizability. It preserves orientation, forcing orientability. Each of these is a geometric fact about G read off from a purely algebraic operation.

Definition 2.1.1: Lie Group

A *Lie group* is a smooth manifold G that is also a group, such that the multiplication and inversion maps

$$m\colon G\times G\to G,\quad (g,h)\mapsto gh,\qquad \iota\colon G\to G,\quad g\mapsto g^{-1}$$

are smooth, where $G\times G$ carries the product smooth structure. The dimension of G as a Lie group is its dimension as a manifold.

Smoothness of m and ι is equivalent to smoothness of the single map

$$\mu\colon G\times G\to G,\qquad (g,h)\mapsto gh^{-1}$$

Indeed, if m and ι are smooth then $\mu = m\circ(\mathrm{id}_G\times\iota)$ is a composite of smooth maps, hence smooth. Conversely, given μ smooth, inversion is recovered as $\iota(g)=\mu(e,g)$, the restriction of μ to the smooth submanifold $\{e\}\times G$, so ι is smooth. Then $m(g,h)=\mu(g,\iota(h))=\mu(g,h^{-1})$ is again a composite of smooth maps. The one-map formulation is the more economical to verify in examples, and it is the version used to check that a subset of a Lie group inheriting the two operations is itself a Lie group.

The smoothness axiom makes the group elements act on G by diffeomorphisms. This is the intrinsic replacement for the observation, made throughout chapter 1, that multiplication by a fixed matrix is a smooth invertible self-map of $\mathrm{GL}_n(\mathbb{F})$ with smooth inverse, visible directly from proposition 1.1.2.

Proposition 2.1.2: Translations and Conjugations are Diffeomorphisms

Let G be a Lie group and $g \in G$. The *left translation*, *right translation*, and *conjugation* maps

$$L_g \colon G \to G, \quad h \mapsto gh, \qquad R_g \colon G \to G, \quad h \mapsto hg, \qquad C_g \colon G \to G, \quad h \mapsto ghg^{-1}$$

are diffeomorphisms of G, with inverses $L_g^{-1} = L_{g^{-1}}$, $R_g^{-1} = R_{g^{-1}}$, and $C_g^{-1} = C_{g^{-1}}$. Consequently:

1. G is *homogeneous*: for any two points $g, h \in G$ the diffeomorphism $L_{hg^{-1}}$ carries g to h, so G looks the same near every point.
2. G is *parallelizable*: its tangent bundle TG is trivial, $TG \cong G \times T_eG$.
3. G is *orientable*.

***Proof*:**

Step 1: the translations are diffeomorphisms. Fix $g \in G$. Left translation L_g is the composite

$$G \xrightarrow{h \mapsto (g,h)} G \times G \xrightarrow{m} G$$

of the smooth map $h \mapsto (g, h)$ (smooth because its two components, the constant g and the identity, are smooth) with the smooth multiplication m of definition 2.1.1, so L_g is smooth. The group axioms give $L_g \circ L_{g^{-1}} = L_{gg^{-1}} = L_e = \mathrm{id}_G$ and likewise $L_{g^{-1}} \circ L_g = \mathrm{id}_G$, so L_g is a bijection with inverse $L_{g^{-1}}$, which is smooth by the same argument. Hence L_g is a diffeomorphism. The identical argument, using $h \mapsto (h, g)$ and the relation $R_g \circ R_{g^{-1}} = R_{g^{-1}g} = \mathrm{id}_G$, shows R_g is a diffeomorphism with inverse $R_{g^{-1}}$. Note that $C_g = L_g \circ R_{g^{-1}}$ is a composite of two diffeomorphisms, hence a diffeomorphism, with inverse $C_{g^{-1}} = L_{g^{-1}} \circ R_g$.

Step 2: homogeneity. Given $g, h \in G$, set $k = hg^{-1}$. Then L_k is a diffeomorphism by Step 1, and $L_k(g) = (hg^{-1})g = h$. A diffeomorphism carrying g to h identifies a neighborhood of g with a neighborhood of h compatibly with the smooth structure, so every point of G has a neighborhood diffeomorphic to a neighborhood of any other point. This is the assertion that G is homogeneous.

Step 3: parallelizability. For each $g \in G$ the differential of left translation at the identity,

$$(dL_g)_e \colon T_eG \to T_gG$$

is a linear isomorphism, since L_g is a diffeomorphism carrying e to g. Define

$$\Psi \colon G \times T_eG \to TG, \qquad (g, v) \mapsto (dL_g)_e(v)$$

The map Ψ is a bijection: for each g it restricts to the linear isomorphism $(dL_g)_e \colon \{g\} \times T_eG \to T_gG$ onto the fiber over g. That Ψ and its inverse are smooth follows from the smoothness of m: the assignment $(g, h) \mapsto (dL_g)_h$ is smooth because L_g is built from m, so Ψ is a smooth bundle isomorphism covering the identity of G. Its existence is exactly the statement that TG is trivial, $TG \cong G \times T_eG$, so G is parallelizable.

Step 4: orientability. A parallelizable manifold is orientable. Fix any basis $v_1, \dots, v_n$ of T_eG, where $n = \dim G$. By Step 3 the vector fields $g \mapsto (dL_g)_e(v_i)$ are smooth and, at each point g, their values $(dL_g)_e(v_1), \dots, (dL_g)_e(v_n)$ form a basis of T_gG, since $(dL_g)_e$ is a linear isomorphism.

This global smooth frame determines an orientation of G: declare a basis of T_gG positively oriented when it has positive determinant relative to the frame. The frame is smooth, so the orientation is consistent across overlaps, and G is orientable, completing the proof.

Homogeneity says G has no distinguished point apart from the identity, and even the identity is distinguished only by the group structure, not by the geometry: near e and near any other g the manifold is indistinguishable. This is why so much of Lie theory is local analysis at the identity, the tangent space T_eG being the germ from which the whole group is reconstructed. The trivialization $TG \cong G \times T_eG$ says that a tangent vector at e can be translated to a well-defined tangent vector at every point, producing a global vector field. These translated fields are the left-invariant vector fields of section 2.2, and their bracket is what makes T_eG a Lie algebra. Orientability then follows for free, which already rules out some manifolds as Lie groups: no even-dimensional real projective space $\mathbb{RP}^{2k}$ with $k \geq 1$ can be a Lie group, since it is not orientable.

The matrix groups of chapter 1 are the first source of examples, and every group named there is a Lie group in the sense of definition 2.1.1.

Example 2.1.3: First Examples of Lie Groups
The following are Lie groups.

1. *The general linear group* $\mathrm{GL}_n(\mathbb{F})$, for $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. proposition 1.1.2 establishes exactly the content of definition 2.1.1: $\mathrm{GL}_n(\mathbb{F})$ is an open submanifold of $M_n(\mathbb{F})$, and multiplication and inversion are smooth. Its dimension is n^2 over $\mathbb{R}$ and $2n^2$ over $\mathbb{C}$. This is the ambient Lie group inside which the classical groups sit.

2. *The circle* $S^1 = \mathrm{U}(1)$. Regard $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ as the unit circle in $\mathbb{C}$, a one-dimensional smooth manifold, with group operation the multiplication of complex numbers. In the angle coordinate $z = e^{i\theta}$ the product is $e^{i\theta}e^{i\varphi} = e^{i(\theta+\varphi)}$ and the inverse is $e^{-i\theta}$, both smooth in θ, so S^1 is a one-dimensional abelian Lie group. As a group of 1×1 unitary matrices it is exactly $\mathrm{U}(1)$ of definition 1.2.5, and example 1.2.11 already identified $\mathrm{SO}(2) \cong \mathrm{U}(1) \cong S^1$.

3. *The tori* $\mathbb{T}^n = (S^1)^n$. The n-fold product of circles is a compact connected abelian Lie group of dimension n, with coordinatewise multiplication. Equivalently $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$, the quotient of the additive group $\mathbb{R}^n$ by the lattice $\mathbb{Z}^n$, with the quotient smooth structure. These are the maximal tori that control the structure theory of compact groups in Part III.

4. *Direct products.* If G and H are Lie groups, the product manifold $G \times H$ with componentwise multiplication $(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2)$ is a Lie group of dimension $\dim G + \dim H$. Multiplication and inversion are smooth because they are so in each factor. The tori are the special case $G = H = S^1$ iterated.

5. *Discrete groups.* Any countable group Γ equipped with the discrete topology is a zero-dimensional Lie group: a 0-manifold is a countable set of points, and every map out of a discrete space is smooth, so m and ι are automatically smooth. Finite groups and the additive group $\mathbb{Z}$ are Lie groups in this way, of dimension 0.

6. *The classical groups.* The groups $\mathrm{SL}_n(\mathbb{F})$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and $\mathrm{Sp}(2n, \mathbb{F})$ of section 1.2 are each a smooth embedded submanifold of some $\mathrm{GL}_N(\mathbb{F})$, cut out as a regular-value preimage, on which the ambient multiplication and inversion restrict to smooth maps. Each is therefore a Lie group. They are moreover *Lie subgroups* of $\mathrm{GL}_N(\mathbb{F})$: embedded submanifolds that are also subgroups. That every closed subgroup of a Lie group is

automatically an embedded Lie subgroup, with no separate verification of the manifold structure required, is Cartan's closed subgroup theorem, proved in chapter 3. Here the manifold structure was already given by the regular value theorem in section 1.2.

7. *The running example* $\mathrm{SU}(2)$. By proposition 1.5.1, $\mathrm{SU}(2)$ is diffeomorphic to the 3-sphere S^3, and it is a compact, connected, simply connected Lie group of dimension 3. It is the smallest simply connected nonabelian compact Lie group and the thread this book follows from here.

A Lie group need not come with a preferred embedding into $\mathrm{GL}_n(\mathbb{F})$. The description $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ presents the torus as a quotient, with no matrices at all, and the two descriptions of S^1, as unit complex numbers and as $\mathbb{R}/\mathbb{Z}$, are related by the smooth surjective homomorphism $\mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, whose kernel is $\mathbb{Z}$. This homomorphism is the exponential map of the circle in disguise, and the fact that S^1 is a quotient of its own tangent line by a discrete subgroup is the one-dimensional case of the covering-group theory of chapter 4. That the same abstract Lie group admits several concrete presentations is precisely the freedom the intrinsic definition gives.

Requiring smoothness of a group homomorphism is a strong condition, and chapter 3 will show it is even automatic for continuous homomorphisms, so the class of morphisms is more robust than it first appears.

Definition 2.1.4: Homomorphism of Lie Groups

Let G and H be Lie groups. A *homomorphism of Lie groups* is a map $\varphi\colon G \to H$ that is both a group homomorphism, $\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2)$ for all $g_1, g_2 \in G$, and a smooth map of manifolds. It is an *isomorphism of Lie groups* if it is in addition a bijection whose inverse $\varphi^{-1}\colon H \to G$ is also a homomorphism of Lie groups, that is, a group isomorphism that is a diffeomorphism.

A homomorphism of Lie groups automatically commutes with the translation structure of proposition 2.1.2: from $\varphi(gh) = \varphi(g)\varphi(h)$ one reads $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$, so φ intertwines left translation on G with left translation on H. This compatibility is what will force the differential $d\varphi_e$ at the identity to determine φ on all of a connected G, the theme of section 2.4. The determinant furnishes the running example of a homomorphism: $\det\colon \mathrm{GL}_n(\mathbb{F}) \to \mathbb{F}^\times$ is a group homomorphism, and it is smooth because the determinant is a polynomial in the matrix entries, so it is a homomorphism of Lie groups whose kernel is the Lie subgroup $\mathrm{SL}_n(\mathbb{F})$ of definition 1.2.1. The exponential $\mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, and the inclusion $\mathrm{SU}(2) \hookrightarrow \mathrm{GL}_2(\mathbb{C})$ are two further homomorphisms already in hand.

Proposition 2.1.5: The Category of Lie Groups

Lie groups and their homomorphisms form a category **LieGrp**: the composite of two Lie group homomorphisms is a Lie group homomorphism, the identity map of a Lie group is a homomorphism, and composition is associative. An isomorphism in this category, in the categorical sense of a morphism with a two-sided inverse morphism, is exactly an isomorphism of Lie groups in the sense of definition 2.1.4.

***Proof*:**
If $\varphi\colon G \to H$ and $\psi\colon H \to K$ are Lie group homomorphisms, then $\psi \circ \varphi$ is a group homomorphism (the composite of group homomorphisms) and a smooth map (the composite of smooth maps), hence a Lie group homomorphism. For each Lie group G the identity map id_G is a group

homomorphism and is smooth, so it is a Lie group homomorphism, and it is a two-sided identity for composition. Associativity of composition holds because composition of set maps is associative. Thus **LieGrp** is a category.

For the identification of isomorphisms: a morphism $\varphi\colon G \to H$ with a two-sided inverse morphism $\varphi^{-1}\colon H \to G$ in **LieGrp** is a bijection whose inverse is a Lie group homomorphism, hence a group isomorphism that is a diffeomorphism, which is a Lie group isomorphism by definition 2.1.4. Conversely a Lie group isomorphism is by definition a homomorphism admitting an inverse homomorphism, which is a two-sided inverse morphism in **LieGrp**, completing the proof.

One structural feature of a Lie group has no analogue for a bare topological group and organizes much of what follows: the piece of G containing the identity is an open subgroup. Let G^0 denote the connected component of G containing the identity e, the *identity component*. Because G is a manifold it is locally connected, so its connected components are open. In particular G^0 is an open subset of G. It is a subgroup: the map $(g,h) \mapsto gh^{-1}$ is continuous and sends $G^0 \times G^0$, a connected set containing (e,e), into a connected subset of G containing $ee^{-1} = e$. A connected set containing e lies in the component G^0, so $gh^{-1} \in G^0$ for all $g, h \in G^0$, and G^0 is a subgroup. It is normal: for any $g \in G$ the conjugation C_g of proposition 2.1.2 is a homeomorphism fixing e, so it carries the component G^0 onto the component of $C_g(e) = e$, which is again G^0. Thus $gG^0g^{-1} = G^0$. Being open and a subgroup, G^0 has open cosets, so its cosets partition G into the connected components, and the quotient G/G^0 is a discrete group, the *component group*. For $\mathrm{GL}_n(\mathbb{R})$ this recovers the two components of proposition 1.1.3, with $G^0 = \mathrm{GL}_n^+(\mathbb{R})$ and component group $\mathbb{Z}/2$. For the connected classical groups of proposition 1.2.10 one has $G^0 = G$. The subgroup theory that turns G^0 into the first tool for dissecting a Lie group is developed in full in chapter 3, and the quotient theory that does the same for G/G^0 in chapter 4.

Definition 2.1.6: Identity Component

The *identity component* G^0 of a Lie group G is the connected component of G containing the identity element e. It is an open normal subgroup of G, and the quotient G/G^0 is the *component group*, a discrete group whose elements are the connected components of G.

The identity component isolates the part of G accessible to the infinitesimal methods of this book. Everything the Lie algebra sees lies in G^0: the exponential image, the one-parameter subgroups, and the neighborhood of the identity all sit inside the connected piece, and section 2.3 will show that a connected Lie group is generated by any neighborhood of e, so G^0 is generated by exponentials. The passage from G^0 to G is the passage from the differential-geometric to the discrete: it is governed by the finite or countable component group, not by the Lie algebra, and this is why the theorems of chapters 11 and 12 are stated for connected groups. Two connected Lie groups with the same Lie algebra can differ only by the discrete central subgroup of their common simply connected cover by which each is realized as a quotient, the datum the tangent space cannot detect, exactly as $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ share the Lie algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ of example 1.4.7 while differing by the factor $\mathbb{Z}/2$ of theorem 1.5.5.

Exercise 2.1.1

1. Let G and H be Lie groups. Prove in full that the product manifold $G \times H$, with the componentwise group operation, is a Lie group of dimension $\dim G + \dim H$, by exhibiting the map $\big((g_1,h_1),(g_2,h_2)\big) \mapsto (g_1g_2^{-1}, h_1h_2^{-1})$ as smooth and invoking the one-map criterion

following definition 2.1.1. Identify the tangent space $T_{(e,e)}(G \times H)$ with $T_eG \oplus T_eH$.

2. Let G be a Lie group with identity e and inversion $\iota\colon G \to G$. Using that $m(g, \iota(g)) = e$ is constant, differentiate at e to prove that the differential of inversion at the identity is minus the identity,
$$d\iota_e = -\mathrm{id}_{T_eG}$$
(Hint: the differential of m at (e, e) is $dm_{(e,e)}(v, w) = v + w$ under the identification $T_{(e,e)}(G \times G) \cong T_eG \oplus T_eG$. Establish this first from $m(g, e) = g$ and $m(e, h) = h$.)

3. Let $\varphi\colon G \to H$ be a homomorphism of Lie groups. Prove that φ has constant rank, that is, the rank of the differential $d\varphi_g\colon T_gG \to T_{\varphi(g)}H$ is the same for every $g \in G$. (Hint: use the intertwining relation $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$ and that translations are diffeomorphisms, proposition 2.1.2, to relate $d\varphi_g$ to $d\varphi_e$ by isomorphisms.)

4. Prove that $\mathbb{T}^n = (S^1)^n$ is a compact, connected, abelian Lie group of dimension n, and that the map $\mathbb{R}^n \to \mathbb{T}^n$, $(t_1, \dots, t_n) \mapsto (e^{2\pi i t_1}, \dots, e^{2\pi i t_n})$, is a surjective homomorphism of Lie groups with kernel $\mathbb{Z}^n$. Conclude that $\mathbb{T}^n \cong \mathbb{R}^n/\mathbb{Z}^n$ as groups.

5. Let G be a Lie group. Prove that the identity component G^0 is open in G, and that it is closed. Deduce that the connected components of G are exactly the cosets gG^0, that each is open and closed, and that G^0 is the smallest open subgroup of G. (Hint: an open subgroup is closed because its complement is a union of its nontrivial cosets, each of which is open.)

6. Using proposition 2.1.2, explain why the even-dimensional real projective spaces $\mathbb{RP}^{2k}$ with $k \geq 1$ and the Klein bottle carry no Lie group structure, while the odd-dimensional spheres S^1 and S^3 do. Which single property from proposition 2.1.2 does the obstruction rest on, and why is that property alone not enough to guarantee a manifold is a Lie group?

2.2 Left-Invariant Vector Fields and the Lie Algebra

Chapter 1 built the Lie algebra of a matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$ by hand, as the space of matrices X whose one-parameter subgroups $t \mapsto \exp(tX)$ stay inside G, and identified it with the tangent space T_IG. That construction leaned on the ambient matrix ring at every step: the exponential series, the commutator $XY - YX$, the trace. A general Lie group carries no ambient ring, so the same object must be built from the manifold alone. The raw material is the tangent space T_eG at the identity, a vector space of the right dimension but with no bracket in sight.

The mechanism is left translation. A single tangent vector $v \in T_eG$ can be transported to every point g of the group by pushing it forward along the diffeomorphism $L_g\colon h \mapsto gh$, producing a vector field on all of G that is, by construction, insensitive to the group's own left multiplication. Such *left-invariant* vector fields form a vector space canonically isomorphic to T_eG, and, because the Lie bracket of two of them is again one, they form a Lie subalgebra of all vector fields. Transporting that bracket back to T_eG turns the identity tangent space into a Lie algebra in the abstract sense of definition 1.4.5, with no choice of coordinates, no exponential, and no matrices.

Definition 2.2.1: Left-Invariant Vector Field

Let G be a Lie group with identity e, and for $g \in G$ let $L_g \colon G \to G$, $L_g(h) = gh$, denote left translation, a diffeomorphism by proposition 2.1.2. A vector field X on G is *left-invariant* if it is L_g-related to itself for every $g \in G$, that is,

$$(L_g)_* X = X \qquad \text{for all } g \in G$$

Written out at a point, this is the condition

$$d(L_g)_h\big(X_h\big) = X_{gh} \qquad \text{for all } g, h \in G$$

Setting $h = e$ shows that a left-invariant field is determined by its value at the identity through

$$X_g = d(L_g)_e\big(X_e\big) \tag{2.1}$$

The set of left-invariant vector fields on G is written $\mathfrak{g}$.

The defining equation says that the flow of X commutes with left multiplication: transporting a tangent vector from h to gh by the differential of L_g lands on the value the field already assigns at gh. A left-invariant field is thus rigid data, pinned down everywhere by its single value X_e at the identity through (2.1).

Example 2.2.2: Left-Invariant Fields on the Circle

Take $G = S^1$, realized as the unit complex numbers under multiplication, with identity $e = 1$. The tangent space $T_1 S^1$ is one-dimensional, spanned by the velocity $v = i$ of the curve $\theta \mapsto e^{i\theta}$ at $\theta = 0$. Left translation by $g = e^{i\alpha}$ is $L_g(e^{i\theta}) = e^{i(\alpha+\theta)}$, whose differential rotates tangent vectors by α. Pushing $v = i$ forward from 1 to g gives the velocity ig of the curve $\theta \mapsto e^{i(\alpha+\theta)}$ at its start. In the angular coordinate θ, this is the constant coordinate field

$$X = \frac{\partial}{\partial \theta}$$

the unit tangent that circulates the circle at constant angular speed. It is nowhere zero, so a single left-invariant field trivializes the tangent bundle of S^1. Every left-invariant field on S^1 is a constant real multiple of $\partial/\partial\theta$, matching $\dim S^1 = 1$.

Theorem 2.2.3: Left-Invariant Fields and the Bracket on T_eG

Let G be a Lie group.

1. The evaluation map
$$\varepsilon\colon \mathfrak{g} \to T_eG, \qquad X \mapsto X_e$$
is a linear isomorphism from the space of left-invariant vector fields to the tangent space at the identity. Its inverse sends $v \in T_eG$ to the field $\widetilde{v}$ defined by $\widetilde{v}_g = d(L_g)_e v$, which is smooth.

2. If X and Y are left-invariant, so is their Lie bracket $[X, Y]$. Hence $\mathfrak{g}$ is a Lie subalgebra of the Lie algebra of all vector fields on G.

3. Transporting the bracket through ε makes T_eG a Lie algebra: for $u, v \in T_eG$, set
$$[u, v] := \big[\widetilde{u}, \widetilde{v}\big]_e$$
This bracket is bilinear, alternating, and satisfies the Jacobi identity, so T_eG is a Lie algebra in the sense of definition 1.4.5.

***Proof*:**
Step 1: evaluation is a linear isomorphism. The map ε is linear because the value of a sum or scalar multiple of vector fields at e is the corresponding sum or scalar multiple of values. It is injective: if $X_e = 0$ then (2.1) gives $X_g = d(L_g)_e(0) = 0$ for every g, so X is the zero field. For surjectivity, fix $v \in T_eG$ and define $\widetilde{v}_g = d(L_g)_e v$. This assignment satisfies the left-invariance identity: for $g, h \in G$, the chain rule applied to $L_{gh} = L_g \circ L_h$ gives $d(L_{gh})_e = d(L_g)_h \circ d(L_h)_e$, whence
$$d(L_g)_h\big(\widetilde{v}_h\big) = d(L_g)_h\big(d(L_h)_e v\big) = d(L_{gh})_e v = \widetilde{v}_{gh}$$
so $\widetilde{v}$ is left-invariant provided it is a smooth vector field. Smoothness is the one point that uses more than the chain rule, and it is where the smoothness of multiplication enters. Let $m\colon G \times G \to G$ be the smooth multiplication. The value $\widetilde{v}_g = d(L_g)_e v$ is the image of v under the partial differential of m in its second argument at (g, e). A smooth map has smooth partial differentials, so $g \mapsto \widetilde{v}_g$ is a smooth section of TG. Concretely, choose a smooth curve $c\colon (-\epsilon, \epsilon) \to G$ with $c(0) = e$ and $c'(0) = v$. Then
$$\widetilde{v}_g = \frac{d}{dt}\Big|_0 m\big(g, c(t)\big) = \frac{d}{dt}\Big|_0 g\, c(t)$$
and the right side depends smoothly on g because m is smooth and the t-derivative of the smooth family $g \mapsto m(g, c(\cdot))$ is smooth in g (see [13, The Tangent Functor] for smooth dependence of derivatives on parameters). Thus $\widetilde{v} \in \mathfrak{g}$ and $\varepsilon(\widetilde{v}) = \widetilde{v}_e = d(L_e)_e v = v$, so ε is surjective with inverse $v \mapsto \widetilde{v}$, hence a linear isomorphism.

Step 2: the bracket of left-invariant fields is left-invariant. The input is the naturality of the Lie bracket under smooth maps, from [13, Naturality of the Bracket]: if $\varphi\colon M \to N$ is smooth, X_1 on M is φ-related to Y_1 on N, and X_2 is φ-related to Y_2, then $[X_1, X_2]$ is φ-related to $[Y_1, Y_2]$. Apply this with $\varphi = L_g$ and $M = N = G$: a left-invariant field X is by definition L_g-related to itself, and likewise Y, so $[X, Y]$ is L_g-related to $[X, Y]$. As this holds for every $g \in G$, the bracket $[X, Y]$ is left-invariant. The bracket of vector fields is already bilinear, alternating, and satisfies

the Jacobi identity on all vector fields ([13, The Lie Algebra of Vector Fields]). Restricting to the subspace $\mathfrak{g}$, closed under the bracket by what was just shown, exhibits $\mathfrak{g}$ as a Lie subalgebra.

Step 3: transport to T_eG. By Step 1 the map ε is a linear isomorphism, so $[u, v] := [\widetilde{u}, \widetilde{v}]_e$ is a well-defined bilinear operation on T_eG, the image of the bracket on $\mathfrak{g}$ under ε. Transporting a Lie bracket through a linear isomorphism preserves each of its axioms: bilinearity, the alternating property $[u, u] = [\widetilde{u}, \widetilde{u}]_e = 0$, and the Jacobi identity, since each is a multilinear identity that ε^{-1} carries from $\mathfrak{g}$ to T_eG and ε carries back. Therefore $(T_eG, [-, -])$ satisfies the two axioms of definition 1.4.5, completing the proof.

The theorem is the intrinsic replacement for the entire vector-space and bracket apparatus of chapter 1. Evaluation at e collapses the space of left-invariant fields, a subspace of the infinite-dimensional space of all vector fields on G, onto the finite-dimensional T_eG with no loss, and the bracket of fields, which a priori knows about the whole manifold, is completely recorded by its value at a single point. That value is the intrinsic bracket. The construction is canonical: it required choosing neither coordinates nor a curve representing each tangent vector (the curve c in Step 1 was an auxiliary device whose t-derivative is coordinate-free), so the Lie-algebra structure on T_eG belongs to the group and to nothing else.

Definition 2.2.4: The Lie Algebra of a Lie Group

Let G be a Lie group. Its *Lie algebra* is the tangent space at the identity,

$$\mathfrak{g} = \mathrm{Lie}(G) := T_eG$$

equipped with the bracket $[u, v] = [\widetilde{u}, \widetilde{v}]_e$ transported from left-invariant vector fields by theorem 2.2.3. Via the isomorphism ε of that theorem, $\mathrm{Lie}(G)$ is identified with the Lie algebra of left-invariant vector fields, and the two descriptions are used interchangeably.

Because ε is a linear isomorphism onto T_eG, the Lie algebra has the dimension of the group:

$$\dim \mathfrak{g} = \dim T_eG = \dim G \tag{2.2}$$

As a tangent space, $\mathfrak{g} = T_eG$ is where velocities of curves through e live and where the exponential map of section 2.3 will land its derivative. As a space of left-invariant fields, $\mathfrak{g}$ is where the bracket is visibly a Lie bracket of vector fields, with its Jacobi identity inherited rather than recomputed. The identification ε lets one pass freely between the pointwise picture and the global-field picture, and an element of $\mathfrak{g}$ should be read as both at once: a tangent vector X_e and the entire field X it generates.

If $v_1, \dots, v_n$ is a basis of T_eG, then the left-invariant fields $\widetilde{v}_1, \dots, \widetilde{v}_n$ are, at every point g, the images $d(L_g)_e v_1, \dots, d(L_g)_e v_n$ of a basis under the isomorphism $d(L_g)_e$, hence a basis of T_gG. They therefore form a *global frame*

$$T_gG = \mathrm{span}_{\mathbb{R}}\{(\widetilde{v}_1)_g, \dots, (\widetilde{v}_n)_g\} \quad \text{for every } g \in G \tag{2.3}$$

which trivializes the tangent bundle, $TG \cong G \times \mathbb{R}^n$. Every Lie group is thus *parallelizable*, the property already asserted in proposition 2.1.2, and (2.3) is its proof: left translation spreads any frame at e to a frame everywhere. This is a strong restriction on which manifolds can be groups. The circle S^1, the torus $\mathbb{T}^n$, and every Lie group are parallelizable, whereas the even-dimensional sphere S^2 is not: the hairy-ball theorem states that every continuous vector field on S^2 vanishes at some point, so S^2 admits no nowhere-zero field, let alone a global frame, and therefore carries no Lie group structure.

A one-parameter subgroup will be the integral curve of a left-invariant field through the identity, and for it to be defined for every real time, the field must be complete. On a general manifold an integral curve can run off to infinity in finite time. Left-invariance forbids this, because the flow near the identity can be translated to reproduce the flow near any point, extending every integral curve indefinitely.

Proposition 2.2.5: Left-Invariant Fields Are Complete

Every left-invariant vector field X on a Lie group G is complete: for each $g \in G$ the maximal integral curve of X starting at g is defined on all of $\mathbb{R}$. Consequently X generates a global flow $\theta\colon \mathbb{R} \times G \to G$, and $\dim \mathfrak{g} = \dim G$.

***Proof*:**

Step 1: integral curves are translates of one another. Let $\gamma\colon (a, b) \to G$ be an integral curve of X, so $\gamma'(t) = X_{\gamma(t)}$, and fix $g \in G$. The left-translated curve $\sigma(t) = g\,\gamma(t) = L_g(\gamma(t))$ is again an integral curve of X. Indeed, differentiating and using the chain rule,

$$\sigma'(t) = d(L_g)_{\gamma(t)}\big(\gamma'(t)\big) = d(L_g)_{\gamma(t)}\big(X_{\gamma(t)}\big) = X_{g\gamma(t)} = X_{\sigma(t)}$$

where the third equality is the left-invariance of X from definition 2.2.1. Thus L_g carries integral curves of X to integral curves of X, defined on the same time interval.

Step 2: extend the flow near e to all time. By the existence and uniqueness theorem for integral curves [13, Fundamental Theorem of Flows], there is an $\epsilon > 0$ and an integral curve $\gamma\colon (-\epsilon, \epsilon) \to G$ with $\gamma(0) = e$. Suppose, for contradiction, that the maximal integral curve γ_g of X with $\gamma_g(0) = g$ had a finite upper endpoint T for its domain. Choose a time t_0 with $T - \epsilon < t_0 < T$, and set $h = \gamma_g(t_0)$. By Step 1 applied to the base curve γ, the translate $t \mapsto h\,\gamma(t) = L_h(\gamma(t))$ is an integral curve of X defined on $(-\epsilon, \epsilon)$ with value h at $t = 0$. By uniqueness of integral curves it agrees with $\gamma_g(t_0 + t)$ wherever both are defined, and it extends γ_g to times up to $t_0 + \epsilon > T$, contradicting the maximality of T. The same argument rules out a finite lower endpoint. Hence γ_g is defined on all of $\mathbb{R}$, so X is complete.

Step 3: the global flow and the dimension. A complete vector field generates a global flow $\theta\colon \mathbb{R} \times G \to G$, $\theta(t, g) = \gamma_g(t)$, smooth by the smooth dependence of solutions on initial conditions [13, Fundamental Theorem of Flows]. Finally $\dim \mathfrak{g} = \dim T_eG = \dim G$ because ε of theorem 2.2.3 is a linear isomorphism, restating (2.2), which completes the proof.

The completeness of every left-invariant field is what will let the exponential map be defined on all of $\mathfrak{g}$, not just on a neighborhood of the origin: for each $X \in \mathfrak{g}$ the integral curve through the identity runs for all time, and its value at $t = 1$ is $\exp X$ in section 2.3. On a general manifold no such global object exists, because a vector field's integral curves may escape in finite time. The group structure removes this obstruction entirely. The flow of a left-invariant field is determined near the identity and then copied to every other point by left translation, so there is no room for finite-time blowup. This homogeneity of the flow is the analytic counterpart of the algebraic fact that a group looks the same at every point.

Proposition 2.2.6: The Intrinsic Lie Algebra of $\mathrm{GL}_n(\mathbb{F})$

Let $G = \mathrm{GL}_n(\mathbb{F})$, an open subset of $M_n(\mathbb{F})$ by proposition 1.1.2, so that its tangent space at the identity is canonically

$$T_I\mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F}) = \mathfrak{gl}_n(\mathbb{F})$$

the space of theorem 1.4.3. For $X \in M_n(\mathbb{F})$ the associated left-invariant vector field $\widetilde{X}$ is given at $A \in \mathrm{GL}_n(\mathbb{F})$ by

$$\widetilde{X}_A = AX \tag{2.4}$$

that is, $\widetilde{X}$ is the linear vector field $A \mapsto AX$ on $\mathrm{GL}_n(\mathbb{F})$.

***Proof*:**
Because $\mathrm{GL}_n(\mathbb{F})$ is open in the vector space $M_n(\mathbb{F})$, its tangent space at any point A is canonically $M_n(\mathbb{F})$ itself, and a tangent vector at A is the velocity of a curve of matrices through A. Left translation $L_A(B) = AB$ is the restriction to $\mathrm{GL}_n(\mathbb{F})$ of the linear map $B \mapsto AB$ on $M_n(\mathbb{F})$. The differential of a linear map is the map itself under this canonical identification, so

$$d(L_A)_I \colon M_n(\mathbb{F}) \to M_n(\mathbb{F}), \qquad Z \mapsto AZ$$

Applying (2.1) with $g = A$ and $X_e = X \in T_I\mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F})$ gives $\widetilde{X}_A = d(L_A)_I(X) = AX$, which is (2.4). This is smooth in A, confirming directly that $\widetilde{X}$ is a genuine vector field, as theorem 2.2.3 guarantees in general, completing the proof.

The left-invariant field of X on $\mathrm{GL}_n(\mathbb{F})$ is the matrix version of the constant field on the circle in example 2.2.2: it assigns to each group element A the tangent vector AX, spreading the single matrix $X = \widetilde{X}_I$ over the whole group by left multiplication. With (2.4) in hand, the intrinsic bracket is the commutator. The Lie bracket of two vector fields on an open set of $M_n(\mathbb{F})$ is computed from their directional derivatives: writing the fields as $M_n(\mathbb{F})$-valued functions $A \mapsto AX$ and $A \mapsto AY$, the bracket is

$$[\widetilde{X}, \widetilde{Y}]_A = D(\widetilde{Y})_A\big(\widetilde{X}_A\big) - D(\widetilde{X})_A\big(\widetilde{Y}_A\big)$$

where $D(\widetilde{Y})_A$ is the derivative of $A \mapsto AY$, namely the linear map $Z \mapsto ZY$. Substituting $\widetilde{X}_A = AX$ and $\widetilde{Y}_A = AY$,

$$[\widetilde{X}, \widetilde{Y}]_A = (AX)Y - (AY)X = A(XY - YX)$$

Evaluating at $A = I$ gives $[\widetilde{X}, \widetilde{Y}]_I = XY - YX$. Through the identification ε of theorem 2.2.3, the intrinsic bracket on $T_I\mathrm{GL}_n(\mathbb{F}) = \mathfrak{gl}_n(\mathbb{F})$ is therefore the commutator

$$[X, Y] = XY - YX \tag{2.5}$$

recovering exactly the matrix bracket of proposition 1.4.4. The value $[\widetilde{X}, \widetilde{Y}]_A = A(XY - YX)$ at every A, not just at I, is itself the left-invariant field of the commutator, a computation that reconfirms Step 2 of theorem 2.2.3 in this case. The full, coordinate-free explanation of why the intrinsic bracket agrees with the commutator, expressed through the adjoint representation, is deferred to section 2.5, where it appears as theorem 2.5.4. The direct computation above is enough for now to see that the intrinsic Lie algebra of $\mathrm{GL}_n(\mathbb{F})$ is $\mathfrak{gl}_n(\mathbb{F})$ with its commutator, and a matrix subgroup $G \subseteq \mathrm{GL}_n(\mathbb{F})$ inherits the bracket of its ambient $\mathfrak{gl}_n(\mathbb{F})$.

The intrinsic Lie algebra of $\mathrm{SU}(2)$ is $T_I\mathrm{SU}(2)$, and this tangent space is precisely the space of traceless skew-Hermitian matrices of example 1.4.7: a tangent vector at I is the velocity $c'(0)$ of a curve c in $\mathrm{SU}(2)$ through I, and differentiating the relations $c(t)^*c(t) = I$ and $\det c(t) = 1$ at $t = 0$ returns $X^* + X = 0$ and $\mathrm{tr}X = 0$. By (2.5) the intrinsic bracket on $T_I\mathrm{SU}(2)$ is the commutator, so the intrinsic Lie algebra of $\mathrm{SU}(2)$ is $\mathfrak{su}(2)$ with the bracket of example 1.4.7, and in particular it is the cross product on $\mathbb{R}^3$. The abstract machine of this section, applied to $\mathrm{SU}(2)$, returns the concrete algebra chapter 1 computed by hand.

Exercise 2.2.1

1. On the n-torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$, with group operation addition of coordinates modulo 1 and identity $e = 0$, show that the constant coordinate fields $\partial/\partial\theta_1, \dots, \partial/\partial\theta_n$ are left-invariant, and that they form a basis of the Lie algebra $\mathrm{Lie}(\mathbb{T}^n)$. Compute all brackets $[\partial/\partial\theta_i, \partial/\partial\theta_j]$ and conclude that $\mathrm{Lie}(\mathbb{T}^n)$ is abelian, that is, $[u, v] = 0$ for all u, v.

2. Let G be a Lie group of dimension n and let $v_1, \dots, v_n$ be a basis of T_eG. Prove in detail that the left-invariant fields $\widetilde{v}_1, \dots, \widetilde{v}_n$ form a global frame, using that $d(L_g)_e$ is a linear isomorphism for every g. Deduce that TG is trivial, and explain why this shows the even-dimensional spheres S^{2k} with $k \geq 1$ carry no Lie group structure.

3. For $G = \mathrm{GL}_n(\mathbb{F})$ and $X \in M_n(\mathbb{F})$, verify by a direct differential computation that the field $\widetilde{X}_A = AX$ of proposition 2.2.6 is left-invariant: show that $d(L_B)_A(\widetilde{X}_A) = \widetilde{X}_{BA}$ for all $A, B \in \mathrm{GL}_n(\mathbb{F})$. Then take $X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ in $\mathfrak{gl}_2(\mathbb{R})$ and write out $\widetilde{X}_A$ for a general $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$.

4. A vector field X on G is *right-invariant* if $(R_g)_*X = X$ for all g, where $R_g(h) = hg$. State and prove the right-invariant analogue of part (1) of theorem 2.2.3: evaluation at e is a linear isomorphism from right-invariant fields to T_eG, with inverse $v \mapsto \widehat{v}$ where $\widehat{v}_g = d(R_g)_e v$. For $G = \mathrm{GL}_n(\mathbb{F})$, compute $\widehat{X}_A$ and contrast it with the left-invariant field of proposition 2.2.6.

5. Show that a connected Lie group G is abelian if and only if every left-invariant vector field is also right-invariant. (Left and right translations always commute by associativity. For the harder direction the exponential map of section 2.3 is used to conclude that the elements moved by these flows are central and generate G, so a clear argument reducing the claim to the commuting of left and right translations is what is asked for.)

6. Using only the left-invariant fields $\widetilde{X}_A = AX$ on $\mathrm{GL}_2(\mathbb{R})$ and the coordinate expression for the Lie bracket of vector fields, recompute $[\widetilde{X}, \widetilde{Y}]_I$ for $X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, and confirm it equals the commutator $XY - YX$ of proposition 1.4.4.

2.3 One-Parameter Subgroups and the Exponential Map

chapter 1 produced the exponential of a matrix group by summing a power series and produced the one-parameter subgroups of $\mathrm{GL}_n(\mathbb{F})$ as the curves $t \mapsto \exp(tX)$. The classification theorem theorem 1.3.10 showed these are all of them. Neither the series nor the classification made any use of the ambient matrices beyond the linear differential equation $\gamma' = X\gamma$ that a homomorphism $\mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$ must satisfy. On a general Lie group there is no power series, but there is still a differential equation: a smooth homomorphism $\gamma\colon \mathbb{R} \to G$ is determined by its velocity $X = \gamma'(0)$, because X generates a left-invariant vector field whose flow line through the identity is γ.

By theorem 2.2.3 and proposition 2.2.5, a tangent vector $X \in \mathfrak{g} = T_eG$ determines a left-invariant vector field X^L whose flow exists for all time. Because the field is left-invariant, its flow interacts with the group law in the simplest possible way: translating a flow line by a group element produces another flow line.

Theorem 2.3.1: One-Parameter Subgroups and the Lie Algebra

Let G be a Lie group with Lie algebra $\mathfrak{g} = T_eG$. A *one-parameter subgroup* of G is a smooth homomorphism $\gamma\colon \mathbb{R} \to G$, that is, a smooth curve with $\gamma(s+t) = \gamma(s)\gamma(t)$ for all $s, t \in \mathbb{R}$ (so in particular $\gamma(0) = e$). For each $X \in \mathfrak{g}$ let X^L be the left-invariant vector field with $X^L_e = X$ (definition 2.2.1) and let γ_X be the integral curve of X^L through e. Then:

1. γ_X is a one-parameter subgroup of G, defined for all $t \in \mathbb{R}$, with $\gamma_X'(0) = X$.
2. Every one-parameter subgroup γ of G equals γ_X for $X = \gamma'(0)$.

Consequently $X \mapsto \gamma_X$ is a bijection from $\mathfrak{g}$ onto the set of one-parameter subgroups of G, with inverse $\gamma \mapsto \gamma'(0)$.

***Proof*:**

1. *The integral curve is a homomorphism.* By proposition 2.2.5 the field X^L is complete, so its integral curve $\gamma_X\colon \mathbb{R} \to G$ through $\gamma_X(0) = e$ is defined for all $t \in \mathbb{R}$. It satisfies

$$\gamma_X'(t) = X^L_{\gamma_X(t)}, \qquad \gamma_X(0) = e \tag{2.6}$$

and $\gamma_X'(0) = X^L_e = X$ is the velocity at the identity.

Fix $s \in \mathbb{R}$ and consider the two curves

$$\alpha(t) = \gamma_X(s+t), \qquad \beta(t) = \gamma_X(s)\gamma_X(t) = L_{\gamma_X(s)}\big(\gamma_X(t)\big)$$

The claim is $\alpha = \beta$. Since s is arbitrary, this claim is exactly the homomorphism property. Both curves start at the same point: $\alpha(0) = \gamma_X(s)$ and $\beta(0) = \gamma_X(s)\gamma_X(0) = \gamma_X(s)e = \gamma_X(s)$. Both satisfy (2.6). For α this is the chain rule applied to the time-translation $t \mapsto s+t$: $\alpha'(t) = \gamma_X'(s+t) = X^L_{\gamma_X(s+t)} = X^L_{\alpha(t)}$. For β this is where left-invariance enters. Writing $g = \gamma_X(s)$ and using that L_g is a diffeomorphism (proposition 2.1.2),

$$\beta'(t) = \frac{d}{dt}\, L_g\big(\gamma_X(t)\big) = (dL_g)_{\gamma_X(t)}\big(\gamma_X'(t)\big) = (dL_g)_{\gamma_X(t)}\big(X^L_{\gamma_X(t)}\big)$$

The defining property of a left-invariant field is that it is L_g-related to itself, $(dL_g)_h(X^L_h) = X^L_{gh}$ for all g, h (definition 2.2.1). Applying this with $h = \gamma_X(t)$ turns the last expression into $X^L_{g\gamma_X(t)} = X^L_{\beta(t)}$. Thus β also solves (2.6), now with initial point $\gamma_X(s)$.

Both α and β are integral curves of X^L through the common initial point $\gamma_X(s)$ at $t = 0$. By the uniqueness of integral curves of a smooth vector field ([13, Fundamental Theorem of Flows]), $\alpha = \beta$ for all t. That is, $\gamma_X(s+t) = \gamma_X(s)\gamma_X(t)$ for all $s, t \in \mathbb{R}$. Hence γ_X is a smooth homomorphism $\mathbb{R} \to G$, a one-parameter subgroup, with $\gamma_X'(0) = X$.

2. *Every one-parameter subgroup is an integral curve.* Let $\gamma\colon \mathbb{R} \to G$ be a one-parameter subgroup and set $X = \gamma'(0) \in T_eG = \mathfrak{g}$. Differentiating $\gamma(s+t) = \gamma(s)\gamma(t) = L_{\gamma(s)}(\gamma(t))$ with respect to t and evaluating at $t = 0$, where $\gamma(0) = e$ and $\gamma'(0) = X$,

$$\gamma'(s) = \frac{d}{dt}\bigg|_{t=0} L_{\gamma(s)}\big(\gamma(t)\big) = (dL_{\gamma(s)})_e\big(\gamma'(0)\big) = (dL_{\gamma(s)})_e(X) = X^L_{\gamma(s)}$$

the last equality being the definition of the left-invariant field X^L generated by X. So γ solves (2.6) with $\gamma(0) = e$, the same problem γ_X solves. Uniqueness of integral curves gives $\gamma = \gamma_X$.

Parts (1) and (2) show $X \mapsto \gamma_X$ is a surjection onto the one-parameter subgroups with a two-sided inverse $\gamma \mapsto \gamma'(0)$: part (1) gives $\gamma'_X(0) = X$, and part (2) gives $\gamma_{\gamma'(0)} = \gamma$. Hence the assignment is a bijection with the stated inverse, completing the proof.

The theorem is the intrinsic form of the classification theorem 1.3.10, and the parallel is exact. There, a one-parameter subgroup of $\mathrm{GL}_n(\mathbb{F})$ solved the linear equation $\gamma' = X\gamma$ and was pinned down by its initial velocity. Here it solves the equation $\gamma' = X^L_\gamma$ of an integral curve and is pinned down the same way. The matrix version is the special case in which the left-invariant field X^L is the explicit vector field $A \mapsto AX$ and the flow can be summed as a power series. What has been removed is any need for that series: the group law forces the integral curve to be a homomorphism, and completeness forces it to run for all time, so the correspondence between generators and one-parameter subgroups holds on any Lie group.

Definition 2.3.2: The Exponential Map

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The *exponential map* $\exp\colon \mathfrak{g} \to G$ sends $X \in \mathfrak{g}$ to the value at time 1 of its one-parameter subgroup:

$$\exp(X) = \gamma_X(1)$$

where γ_X is the one-parameter subgroup of theorem 2.3.1 with $\gamma'_X(0) = X$.

The definition compresses the entire family $\{\gamma_X\}_{X\in\mathfrak{g}}$ into a single map on $\mathfrak{g}$, and the family is recovered from it by rescaling. For $X \in \mathfrak{g}$ and $t \in \mathbb{R}$, the curve $s \mapsto \gamma_X(ts)$ is a one-parameter subgroup with velocity $t\gamma'_X(0) = tX$ at $s = 0$. By the uniqueness in theorem 2.3.1 it is γ_{tX}, so $\gamma_{tX}(s) = \gamma_X(ts)$. Taking $s = 1$ gives

$$\exp(tX) = \gamma_{tX}(1) = \gamma_X(t) \tag{2.7}$$

so that $t \mapsto \exp(tX)$ is exactly the one-parameter subgroup γ_X, and exp traces out the whole flow as t varies.

Proposition 2.3.3: Properties of the Exponential Map

Let G be a Lie group with Lie algebra $\mathfrak{g}$ and exponential map $\exp\colon \mathfrak{g} \to G$. For all $X \in \mathfrak{g}$ and all $s, t \in \mathbb{R}$:

1. $\exp(0) = e$.
2. $\exp\big((s+t)X\big) = \exp(sX)\exp(tX)$, and consequently $\exp(-X) = \exp(X)^{-1}$. The curve $t \mapsto \exp(tX)$ is the one-parameter subgroup γ_X with velocity $\frac{d}{dt}\big|_0 \exp(tX) = X$.
3. $\exp\colon \mathfrak{g} \to G$ is smooth.

***Proof*:**

Part (1). The one-parameter subgroup generated by $X = 0$ is the constant curve $\gamma_0 \equiv e$, since the zero vector field has the constant integral curve through e. Hence $\exp(0) = \gamma_0(1) = e$.

Part (2). By (2.7), $t \mapsto \exp(tX) = \gamma_X(t)$ is the one-parameter subgroup γ_X, so the homomorphism property $\gamma_X(s+t) = \gamma_X(s)\gamma_X(t)$ of theorem 2.3.1 reads

$$\exp\big((s+t)X\big) = \exp(sX)\exp(tX)$$

Setting $t = -s$ and using part (1) gives $\exp(sX)\exp(-sX) = \exp(0) = e$, so $\exp(-sX) = \exp(sX)^{-1}$. Taking $s = 1$ gives $\exp(-X) = \exp(X)^{-1}$. The velocity statement is $\frac{d}{dt}\big|_0 \exp(tX) = \gamma_X'(0) = X$.

Part (3). Smoothness of exp is the smooth dependence of a flow on its initial data, applied to a single well-chosen field. Consider the Lie group G together with its Lie algebra $\mathfrak{g}$, and form on the product manifold $\mathfrak{g} \times G$ the vector field V whose value at (X, g) is $(0, X^L_g)$: at each point it is the zero vector in the $\mathfrak{g}$-direction and the left-invariant field X^L evaluated at g in the G-direction. The assignment $(X, g) \mapsto X^L_g$ is smooth in both variables, because $X^L_g = (dL_g)_e(X)$ depends smoothly on g through the smooth multiplication of G and linearly, hence smoothly, on X. So V is a smooth vector field on $\mathfrak{g} \times G$. Its integral curve through (X, e) is $t \mapsto (X, \gamma_X(t))$, since the first coordinate is constant and the second solves (2.6). This field is complete, as each X^L is complete by proposition 2.2.5. The flow of a complete smooth vector field is smooth jointly in time and initial condition ([13, Fundamental Theorem of Flows]), so the time-1 flow $(X, e) \mapsto (X, \gamma_X(1)) = (X, \exp X)$ is smooth. Projecting to G shows $X \mapsto \exp(X)$ is smooth, completing the proof.

Part (2) is the reason the map deserves the name "exponential": restricted to a single line $\mathbb{R}X$ in $\mathfrak{g}$ it converts addition of real multiples of X into multiplication in G, exactly as the scalar exponential converts addition into multiplication. The restriction to a line is essential. The full map exp is not a homomorphism from $(\mathfrak{g}, +)$ to G unless $\mathfrak{g}$ is abelian, because $\exp(X)\exp(Y)$ and $\exp(X+Y)$ disagree already at second order when X and Y fail to commute, precisely the failure of $\exp(A+B) = \exp(A)\exp(B)$ recorded for matrices in proposition 1.3.5. What survives universally is the one-parameter law of part (2), and it is enough to make exp a chart near the identity.

Theorem 2.3.4: The Exponential is a Local Diffeomorphism

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Identifying $T_0\mathfrak{g}$ with $\mathfrak{g}$ and T_eG with $\mathfrak{g}$, the differential of exp at the origin is the identity,

$$d(\exp)_0 = \mathrm{id}_{\mathfrak{g}}$$

Consequently there are open neighborhoods U of 0 in $\mathfrak{g}$ and V of e in G such that $\exp\colon U \to V$ is a diffeomorphism.

***Proof*:**
The differential $d(\exp)_0$ acts on a tangent vector $X \in T_0\mathfrak{g} = \mathfrak{g}$ by differentiating exp along any curve through 0 with velocity X. The line $t \mapsto tX$ is such a curve, so by (2.7) and proposition 2.3.3(2),

$$d(\exp)_0(X) = \left.\frac{d}{dt}\right|_{t=0} \exp(tX) = \gamma'_X(0) = X$$

under the identification $T_eG = \mathfrak{g}$. Thus $d(\exp)_0 = \mathrm{id}_{\mathfrak{g}}$, which is invertible. Since exp is smooth (proposition 2.3.3) with invertible differential at 0, the inverse function theorem ([13, Inverse Function Theorem for Manifolds]) provides open neighborhoods U of 0 and $V = \exp(U)$ of $\exp(0) = e$ on which exp is a diffeomorphism, completing the proof.

The theorem installs exp as a chart on G centered at the identity, with $\mathfrak{g}$ as the model vector space, so that near e the group is described by the linear data of its Lie algebra. This is the intrinsic version of the exponential chart of theorem 1.3.7, and it is what makes the Lie algebra a faithful record of the group near the identity. A connected group hides nothing beyond what this chart already shows: the whole group is assembled from arbitrarily small pieces near e.

Corollary 2.3.5: Connected Groups Are Generated Near the Identity

Let G be a connected Lie group. Then G is generated by the image $\exp(\mathfrak{g})$. Equivalently, G is generated by any neighborhood of the identity. Explicitly, every $g \in G$ is a finite product $g = \exp(X_1)\exp(X_2)\cdots\exp(X_k)$ with $X_1, \ldots, X_k \in \mathfrak{g}$.

***Proof*:**
Let V be the neighborhood of e on which exp is a diffeomorphism from a neighborhood U of 0, as in theorem 2.3.4. Then $V \subseteq \exp(\mathfrak{g})$, so it suffices to show that any neighborhood W of e generates G. Let $H = \bigcup_{k\geq 1} W_{\mathrm{sym}}^k$ be the subgroup generated by $W_{\mathrm{sym}} = W \cap W^{-1}$, which is a symmetric neighborhood of e (the inversion map ι is smooth by definition 2.1.1 and satisfies $\iota\circ\iota = \mathrm{id}_G$, hence is a diffeomorphism, so $W^{-1} = \iota(W)$ is open and W_{sym} is an open neighborhood of e) and W_{sym}^k denotes the set of k-fold products. Then H is a subgroup of G.

The subgroup H is open: for $h \in H$, the translate $hW_{\mathrm{sym}} = L_h(W_{\mathrm{sym}})$ is an open neighborhood of h contained in H, since L_h is a diffeomorphism (proposition 2.1.2) and $hW_{\mathrm{sym}} \subseteq H$ as a product of elements of H. An open subgroup is also closed: its complement $G \setminus H = \bigsqcup_{gH\neq H} gH$ is a union of cosets, each open (being a translate of the open set H), hence $G \setminus H$ is open and H is closed. Thus H is a nonempty subset of the connected space G that is both open and closed, so $H = G$. Every $g \in G$ therefore lies in some $W_{\mathrm{sym}}^k \subseteq \exp(\mathfrak{g})^k$, giving the product representation with each factor in $\exp(\mathfrak{g})$, as we sought to show.

The corollary explains why the Lie algebra is so powerful a tool for connected groups: knowing $\mathfrak{g}$ determines the local structure of G near e through the exponential chart, and connectedness then propagates that local structure to all of G by taking products. A homomorphism of connected groups is thereby determined by its behavior near the identity, and ultimately by its differential at e, a theme the next section develops in full. The corollary does not assert that exp is surjective. The elements of G are products of exponentials, not necessarily single exponentials, and the gap between the two is real, as the exercises show.

Proposition 2.3.6: The Intrinsic Exponential is the Matrix Exponential

Let $G = \mathrm{GL}_n(\mathbb{F})$, whose Lie algebra is $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ identified with $T_I\mathrm{GL}_n(\mathbb{F})$. For every $X \in \mathfrak{gl}_n(\mathbb{F})$ the intrinsic exponential of definition 2.3.2 agrees with the matrix exponential of definition 1.3.2:

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

The same holds on every matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$: the intrinsic exponential of G is the restriction of the matrix exponential to $\mathfrak{g}$.

***Proof*:**

Step 1: the left-invariant field on $\mathrm{GL}_n(\mathbb{F})$. The group $\mathrm{GL}_n(\mathbb{F})$ is an open subset of the vector space $M_n(\mathbb{F})$ (proposition 1.1.2), so its tangent space at any point A is canonically $M_n(\mathbb{F})$, and left translation $L_A(B) = AB$ is the restriction of a *linear* map $M_n(\mathbb{F}) \to M_n(\mathbb{F})$. The differential of a linear map is the map itself, so $(dL_A)_I(X) = AX$ for every $X \in M_n(\mathbb{F})$. Hence the left-invariant field X^L generated by $X \in T_I\mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F})$ is

$$X^L_A = (dL_A)_I(X) = AX \tag{2.8}$$

the vector field $A \mapsto AX$ on $\mathrm{GL}_n(\mathbb{F})$.

Step 2: the integral curve is the matrix exponential. By (2.8) the integral curve γ_X of X^L through I solves the initial value problem

$$\gamma_X'(t) = X^L_{\gamma_X(t)} = \gamma_X(t)\,X, \qquad \gamma_X(0) = I$$

which is exactly the linear matrix differential equation (1.2) governing one-parameter subgroups of $\mathrm{GL}_n(\mathbb{F})$, written on the right rather than the left. Its unique solution is the matrix exponential $t \mapsto \sum_{k\geq 0} t^k X^k/k!$: by proposition 1.3.4(1) the matrix curve $\eta(t) = \sum_k t^k X^k/k!$ satisfies $\eta'(t) = \eta(t)X$ and $\eta(0) = I$. Uniqueness for this right-multiplication equation follows by the identical argument that theorem 1.3.10 runs for the left form $\gamma' = X\gamma$ (if η_1, η_2 both solve it, then $(\eta_1\eta_2^{-1})' = 0$, so $\eta_1\eta_2^{-1}$ is constant, equal to I at $t = 0$). Therefore $\gamma_X(t) = \sum_k t^k X^k/k!$, and evaluating at $t = 1$,

$$\exp(X) = \gamma_X(1) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

the matrix exponential of definition 1.3.2.

Step 3: passage to a subgroup. Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{F})$, and let $X \in \mathfrak{g}$. The one-parameter subgroup of G generated by X is a one-parameter subgroup of the ambient $\mathrm{GL}_n(\mathbb{F})$ generated by the same X, because the inclusion $G \hookrightarrow \mathrm{GL}_n(\mathbb{F})$ is a

homomorphism of Lie groups carrying the left-invariant field of X on G to that on $\mathrm{GL}_n(\mathbb{F})$: concretely, an integral curve of X^L in G through I is also an integral curve of the ambient field $A \mapsto AX$ through I, so by uniqueness the two one-parameter subgroups coincide as curves in $\mathrm{GL}_n(\mathbb{F})$. Evaluating at $t = 1$, the intrinsic exponential of G at X equals the intrinsic exponential of $\mathrm{GL}_n(\mathbb{F})$ at X, which by Steps 1 and 2 is the matrix exponential. Hence the intrinsic exponential of G is the restriction of the matrix exponential to $\mathfrak{g}$, completing the proof.

The proposition answers the question left open in chapter 1: the exponential built there by summing a series and the exponential built here by flowing a left-invariant field are one and the same. The two constructions look unrelated, one a computation in a normed algebra and the other the integration of a vector field on a manifold, but both solve the identical differential equation, and the uniqueness of its solution forces them to agree. From now on the notation exp carries no ambiguity: on a matrix group it is the matrix exponential, and the abstract properties proved above specialize to the concrete properties of chapter 1.

Example 2.3.7: The Exponential on $\mathfrak{su}(2)$

For $G = \mathrm{SU}(2)$ the intrinsic Lie algebra is $\mathfrak{su}(2) = T_I\mathrm{SU}(2)$, the traceless skew-Hermitian 2×2 matrices (example 1.4.7). By proposition 2.3.6 the intrinsic exponential is the matrix exponential restricted to $\mathfrak{su}(2)$, so the closed form already computed in chapter 1 is the intrinsic answer. Writing $X \in \mathfrak{su}(2)$ with $\Phi(X) = v \in \mathbb{R}^3$ under the isomorphism $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$ of example 1.4.7, the computation $X^2 = -\frac{1}{4}|v|^2 I$ splits the exponential series into its even and odd parts exactly as cos and sin split, giving, for $v \neq 0$,

$$\exp(X) = \cos\left(\tfrac{|v|}{2}\right) I + \frac{\sin\left(\frac{|v|}{2}\right)}{|v|/2}\, X \tag{2.9}$$

This is the map of proposition 1.5.3, which established that exp is *onto* $\mathrm{SU}(2)$: every element of $\mathrm{SU}(2)$ has the form (2.9) for a suitable X, so $\exp\colon \mathfrak{su}(2) \to \mathrm{SU}(2)$ is surjective. The surjectivity is a feature of $\mathrm{SU}(2)$ as a compact connected group, not a general fact. It sharpens the product representation of corollary 2.3.5 from a finite product of exponentials down to a single one. The intrinsic theorem 2.3.1 recovers, as its content on $\mathrm{SU}(2)$, the concrete one-parameter subgroups computed in example 1.5.4: the curve $t \mapsto \exp(tX)$ with $\Phi(X) = v$ is a great circle on the sphere $\mathrm{SU}(2) \cong S^3$, closing up with period $4\pi/|v|$ in t.

The closed form (2.9) shows the abstract theory returning the concrete chapter 1 formula with nothing lost. The period $4\pi/|v|$, twice the $\mathrm{SO}(3)$-side period $2\pi/|v|$, is the same factor of 2 that separates $\mathrm{SU}(2)$ from $\mathrm{SO}(3)$ in the double cover theorem 1.5.5: a one-parameter subgroup of $\mathrm{SU}(2)$ must wind twice before it closes, because a rotation by 2π lifts to $-I$ rather than I upstairs. The exponential map is where this doubling first becomes visible in coordinates, and it foreshadows the role exp will play in Chapter 3 in translating between a group and its covers through their common Lie algebra.

Exercise 2.3.1

1. Show that $\exp\colon \mathfrak{g} \to G$ is a group homomorphism from $(\mathfrak{g}, +)$ to G if and only if $\exp(X)\exp(Y) = \exp(Y)\exp(X)$ for all $X, Y \in \mathfrak{g}$. Using the matrices $X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ in $\mathfrak{gl}_2(\mathbb{R})$, verify that $\exp(X)\exp(Y) \neq \exp(X + Y)$, so that exp is not a homomorphism on $\mathrm{GL}_2(\mathbb{R})$.

2. Prove that if G is a connected abelian Lie group then $\exp\colon \mathfrak{g} \to G$ is a surjective group homomorphism. (You may use corollary 2.3.5 for surjectivity after establishing the homomorphism property. For the homomorphism property, show first that for commuting X, Y the curve $t \mapsto \exp(tX)\exp(tY)$ is a one-parameter subgroup with velocity $X + Y$.)

3. Exhibit a connected Lie group G on which exp is *not* surjective, by showing that no $X \in \mathfrak{sl}_2(\mathbb{R})$ satisfies $\exp(X) = \begin{pmatrix} -2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$, even though this matrix lies in the connected group $\mathrm{SL}_2(\mathbb{R})$. (Hint: analyze the eigenvalues of $\exp(X)$ for a traceless real X according to the sign of $\det X$.)

4. Show that $\exp\colon \mathfrak{g} \to G$ need not be injective by computing its fibers for $G = S^1 = \{z \in \mathbb{C} \,|\, |z| = 1\}$ with $\mathfrak{g} = i\mathbb{R}$. Determine the kernel of exp as a homomorphism $(\mathfrak{g}, +) \to S^1$ and identify it with a lattice in $\mathfrak{g}$.

5. Prove the reparametrization identity (2.7), that $\exp(tX) = \gamma_X(t)$, directly from the uniqueness in theorem 2.3.1, by checking that $s \mapsto \gamma_X(ts)$ is the one-parameter subgroup generated by tX.

6. Combining example 2.3.7 with corollary 2.3.5, explain why every element of $\mathrm{SU}(2)$ is a *single* exponential $\exp(X)$, whereas corollary 2.3.5 guarantees only a finite product of exponentials in general. State the property of $\mathrm{SU}(2)$ responsible, and contrast with $\mathrm{SL}_2(\mathbb{R})$ from exercise 2.3.1 (3).

2.4 The Differential of a Homomorphism and the Naturality of the Exponential

The previous three sections built, for a single Lie group G, the dictionary that assigns to G its Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ and the exponential map $\exp\colon \mathfrak{g} \to G$. A homomorphism $\varphi\colon G \to H$ of Lie groups is a map of the group structure, and being smooth it also carries tangent data. Its differential at the identity is a linear map $d\varphi_e\colon \mathfrak{g} \to \mathfrak{h}$. This linear map respects the bracket, so it is a homomorphism of Lie algebras, and it commutes with the exponential, so the diagram relating $\exp_G$ and $\exp_H$ closes. These two compatibilities together say that passing from a group to its Lie algebra is a functor, and that exp is a natural transformation.

Because exp generates a connected group from a neighborhood of the identity, a homomorphism out of a connected group is pinned down completely by the single linear map $d\varphi_e$: the infinitesimal data determines the global map. This is the first half of the correspondence the book exists to develop, and it reduces questions about maps of groups to questions about maps of vector spaces equipped with a bracket.

Definition 2.4.1: The Differential of a Homomorphism

Let $\varphi\colon G \to H$ be a homomorphism of Lie groups, with Lie algebras $\mathfrak{g} = T_eG$ and $\mathfrak{h} = T_eH$. The *differential* of φ is the linear map

$$d\varphi_e\colon \mathfrak{g} \longrightarrow \mathfrak{h}$$

given by the tangent map of φ at the identity, sending a tangent vector $X \in T_eG$, represented by a curve $t \mapsto c(t)$ with $c(0) = e$ and $c'(0) = X$, to the velocity $d\varphi_e(X) = \frac{d}{dt}\big|_0 \varphi(c(t)) \in T_eH$ of its image.

The differential is well defined precisely because φ preserves the identity: $\varphi(e_G) = e_H$, so φ maps a curve through e_G to a curve through e_H, and the tangent map at that point lands in $\mathfrak{h} = T_{e_H}H$ as claimed. It is a linear map by the chain rule of [13, Properties of the Tangent Map], being the restriction of the smooth map φ's tangent map $d\varphi\colon TG \to TH$ to the fibre over the identity. For a homomorphism of matrix groups the differential is computed by differentiating φ along $t \mapsto \exp(tX)$, and for the classical inclusion homomorphisms it is nothing but the corresponding inclusion of matrix Lie algebras from theorem 1.4.3.

The mechanism that makes $d\varphi_e$ respect the bracket is the notion of φ-relatedness of vector fields. Recall from [13, f-Related Vector Fields, Naturality of the Bracket] that two vector fields ξ on G and η on H are *φ-related* when $d\varphi_g(\xi_g) = \eta_{\varphi(g)}$ for every $g \in G$, and that the bracket of vector fields is natural under this relation: if ξ_1 is φ-related to η_1 and ξ_2 to η_2, then $[\xi_1, \xi_2]$ is φ-related to $[\eta_1, \eta_2]$. The Lie algebra $\mathfrak{g}$ was built in section 2.2 as the left-invariant vector fields, and the left-invariant field with value X at e is φ-related to the left-invariant field with value $d\varphi_e X$.

Proposition 2.4.2: The Differential is a Lie Algebra Homomorphism

Let $\varphi\colon G \to H$ be a homomorphism of Lie groups. Then the differential $d\varphi_e\colon \mathfrak{g} \to \mathfrak{h}$ is a homomorphism of Lie algebras:

$$d\varphi_e[X, Y] = [\, d\varphi_e X,\ d\varphi_e Y\,]$$

for all $X, Y \in \mathfrak{g}$, where the bracket on the left is that of $\mathfrak{g}$ and the bracket on the right is that of $\mathfrak{h}$.

***Proof*:**

Write L_g for left translation by g, and for a vector $X \in \mathfrak{g} = T_eG$ let $\widetilde{X}$ denote the left-invariant vector field on G with $\widetilde{X}_e = X$, characterized by $\widetilde{X}_g = d(L_g)_e X$ as in definition 2.2.1. By theorem 2.2.3, the bracket on $\mathfrak{g}$ is defined by transporting the vector-field bracket, so $\widetilde{[X, Y]} = [\widetilde{X}, \widetilde{Y}]$. The same holds on H with fields denoted $\widehat{(\cdot)}$.

Step 1: the invariant fields are φ-related. Fix $X \in \mathfrak{g}$ and set $X' = d\varphi_e X \in \mathfrak{h}$. The claim is that $\widetilde{X}$ on G is φ-related to $\widehat{X'}$ on H, that is

$$d\varphi_g\big(\widetilde{X}_g\big) = \widehat{X'}_{\varphi(g)} \qquad \text{for all } g \in G \tag{2.10}$$

Because φ is a homomorphism, $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$: both sides send $h \mapsto \varphi(gh) = \varphi(g)\varphi(h)$.

Taking differentials at e and applying the chain rule of [13, Properties of the Tangent Map],

$$d\varphi_g \circ d(L_g)_e = d(L_{\varphi(g)})_e \circ d\varphi_e$$

as maps $T_eG \to T_{\varphi(g)}H$. Evaluating both sides on X and using $\widetilde{X}_g = d(L_g)_e X$ on the left and $\widehat{X'}_{\varphi(g)} = d(L_{\varphi(g)})_e X'$ with $X' = d\varphi_e X$ on the right gives exactly (2.10). Thus $\widetilde{X}$ is φ-related to $\widehat{d\varphi_e X}$ for every X.

Step 2: apply naturality of the bracket. Let $X, Y \in \mathfrak{g}$. By Step 1, $\widetilde{X}$ is φ-related to $\widehat{d\varphi_e X}$ and $\widetilde{Y}$ is φ-related to $\widehat{d\varphi_e Y}$. The naturality of the vector-field bracket under φ-relatedness ([13, Naturality of the Bracket]) then gives that $[\widetilde{X}, \widetilde{Y}]$ is φ-related to $[\widehat{d\varphi_e X}, \widehat{d\varphi_e Y}]$. Evaluating the φ-relatedness at $g = e$, where $\varphi(e) = e$, yields

$$d\varphi_e\big([\widetilde{X}, \widetilde{Y}]_e\big) = [\widehat{d\varphi_e X}, \widehat{d\varphi_e Y}]_e$$

The left side is $d\varphi_e[X, Y]$, since $[\widetilde{X}, \widetilde{Y}]_e = \widetilde{[X, Y]}_e = [X, Y]$ by the definition of the bracket on $\mathfrak{g}$. The right side is $[\, d\varphi_e X,\, d\varphi_e Y\,]$ by the same definition on $\mathfrak{h}$, since $\widehat{Z}_e = Z$ for any $Z \in \mathfrak{h}$. Therefore $d\varphi_e[X, Y] = [\, d\varphi_e X,\, d\varphi_e Y\,]$, as we sought to show.

The relation (2.10) holds because φ intertwines left translations, which is a direct consequence of φ being a homomorphism. Everything after that is the general naturality of the bracket for smooth maps, imported from [13, Naturality of the Bracket]. This is why the differential of a homomorphism respects the bracket while the differential of an arbitrary smooth map need not: only a homomorphism carries left-invariant fields to left-invariant fields. The result says that the assignment $\varphi \mapsto d\varphi_e$ converts the analytic datum of a smooth group homomorphism into the purely algebraic datum of a Lie algebra homomorphism, and it is the reason the structure theory of Parts II and III can be conducted at the level of Lie algebras and then transported back to groups.

The exponential of G is built from one-parameter subgroups, and φ carries one-parameter subgroups of G to one-parameter subgroups of H. Tracking the velocity through this transport is exactly what the differential records.

Theorem 2.4.3: Naturality of the Exponential Map

Let $\varphi\colon G \to H$ be a homomorphism of Lie groups with differential $d\varphi_e\colon \mathfrak{g} \to \mathfrak{h}$. Then for every $X \in \mathfrak{g}$,

$$\varphi\big(\exp_G X\big) = \exp_H\big(d\varphi_e X\big)$$

equivalently, the square

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\;d\varphi_e\;} & \mathfrak{h} \\ {\scriptstyle \exp_G}\downarrow & & \downarrow{\scriptstyle \exp_H} \\ G & \xrightarrow[\;\varphi\;]{} & H \end{array}$$

commutes.

***Proof*:**
Fix $X \in \mathfrak{g}$ and consider the curve

$$\sigma\colon \mathbb{R} \to H, \qquad \sigma(t) = \varphi\big(\exp_G(tX)\big)$$

Step 1: σ is a one-parameter subgroup of H. By proposition 2.3.3 the map $t \mapsto \exp_G(tX)$ is the one-parameter subgroup of G with velocity X, so $\exp_G((s+t)X) = \exp_G(sX)\exp_G(tX)$ and $\exp_G(0) = e$. Since φ is a homomorphism,

$$\sigma(s+t) = \varphi\big(\exp_G(sX)\exp_G(tX)\big) = \varphi\big(\exp_G(sX)\big)\varphi\big(\exp_G(tX)\big) = \sigma(s)\sigma(t)$$

and $\sigma(0) = \varphi(e_G) = e_H$. Being a composite of smooth maps, σ is smooth. Hence σ is a one-parameter subgroup of H.

Step 2: identify its velocity. Differentiating σ at $t = 0$ by the chain rule of [13, Properties of the Tangent Map],

$$\sigma'(0) = \frac{d}{dt}\Big|_0 \varphi\big(\exp_G(tX)\big) = d\varphi_e\left(\frac{d}{dt}\Big|_0 \exp_G(tX)\right) = d\varphi_e(X)$$

where the last equality uses that $t \mapsto \exp_G(tX)$ has velocity X at $t = 0$, again by proposition 2.3.3.

Step 3: conclude by uniqueness. By theorem 2.3.1, a one-parameter subgroup of H is determined by its velocity at $t = 0$, and the one with velocity $d\varphi_e X$ is precisely $t \mapsto \exp_H(t\, d\varphi_e X)$ by proposition 2.3.3. Steps 1 and 2 show σ is a one-parameter subgroup with this same velocity, so

$$\varphi\big(\exp_G(tX)\big) = \sigma(t) = \exp_H\big(t\, d\varphi_e X\big) \qquad \text{for all } t \in \mathbb{R}$$

Setting $t = 1$ gives $\varphi(\exp_G X) = \exp_H(d\varphi_e X)$, completing the proof.

The theorem is the computational engine of the correspondence. It converts the exponential of H into the exponential of G across any homomorphism, so that to evaluate φ on an element of the form $\exp_G X$, which is to say on any element near the identity, one need only know the single linear map $d\varphi_e$ and the exponential of the target. For matrix groups, where by proposition 2.3.6 the intrinsic exponential is the matrix exponential, naturality reads: $\varphi(\exp X) = \exp(d\varphi_e X)$ with both exponentials the matrix power series, and the classical identities of chapter 1 fall out as special cases. The commuting square also explains the terminology of exp as a natural transformation: it is a family of maps $\exp_G\colon \mathfrak{g} \to G$, one per group, that commutes with every morphism, which is exactly the condition for a natural transformation. To be precise, both endpoints are functors into smooth manifolds: the source is Lie followed by the forgetful functor $\mathbf{LieAlg}_{\mathbb{R}} \to \mathbf{Man}$ sending $\mathfrak{g}$ to its underlying vector space, written informally $G \mapsto \mathfrak{g}$, and the target is the forgetful functor $\mathbf{LieGrp} \to \mathbf{Man}$, written $G \mapsto G$. The square above is the naturality condition relating them.

Corollary 2.4.4: A Homomorphism is Determined by its Differential

Let G be a *connected* Lie group and H any Lie group. If $\varphi, \psi\colon G \to H$ are homomorphisms of Lie groups with $d\varphi_e = d\psi_e$, then $\varphi = \psi$. Consequently the map $\varphi \mapsto d\varphi_e$ from homomorphisms $G \to H$ to Lie algebra homomorphisms $\mathfrak{g} \to \mathfrak{h}$ is injective when G is connected.

***Proof*:**
Suppose $d\varphi_e = d\psi_e$. For every $X \in \mathfrak{g}$, naturality (theorem 2.4.3) applied to each homomorphism gives

$$\varphi(\exp_G X) = \exp_H(d\varphi_e X) = \exp_H(d\psi_e X) = \psi(\exp_G X)$$

so φ and ψ agree on the image $\exp_G(\mathfrak{g})$. By theorem 2.3.4, $\exp_G$ is a local diffeomorphism at 0, so its image contains an open neighborhood U of e in G, which we may take symmetric ($U = U^{-1}$) by replacing it with $U \cap U^{-1}$. By corollary 2.3.5, a connected group G is generated by any neighborhood of e, so $G = \langle U \rangle = \bigcup_{k \geq 1} U^k$ where $U^k = \{u_1 \cdots u_k : u_i \in U\}$.

It remains to propagate agreement on U to all of G. Let $W = \{g \in G \mid \varphi(g) = \psi(g)\}$ be the set on which the two homomorphisms agree. Since φ, ψ are homomorphisms, W is a subgroup of G: if $g, g' \in W$ then $\varphi(gg'^{-1}) = \varphi(g)\varphi(g')^{-1} = \psi(g)\psi(g')^{-1} = \psi(gg'^{-1})$. The subgroup W contains U, hence contains every product $u_1 \cdots u_k$ of elements of U, hence contains $\bigcup_{k \geq 1} U^k = G$. Therefore $W = G$, that is $\varphi = \psi$. The final claim restates this: two homomorphisms with equal image under $\varphi \mapsto d\varphi_e$, that is with $d\varphi_e = d\psi_e$, are themselves equal, so the map is injective, completing the proof.

The corollary is the reason the Lie algebra deserves to be called the infinitesimal group. For a connected group there is at most one homomorphism with a prescribed differential, so the passage $\varphi \mapsto d\varphi_e$ loses no information about φ. The only question left open is existence, whether a given Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{h}$ arises from some group homomorphism. Existence requires more, and it fails in general, because global topology intervenes. The exercises exhibit the witness on the circle: by exercise 2.4.1 (4) the only homomorphisms $S^1 \to S^1$ are the power maps $z \mapsto z^n$ with $n \in \mathbb{Z}$, so the differentials that arise are exactly multiplication by an integer on $\mathrm{Lie}(S^1) = \mathbb{R}$, while every real scalar λ is a Lie algebra homomorphism $\mathbb{R} \to \mathbb{R}$. For $\lambda \notin \mathbb{Z}$ that homomorphism is $d\varphi_e$ for no φ. The sharp existence statement, that when G is simply connected every Lie algebra homomorphism does integrate, is Lie's second theorem, proved in chapter 3. Injectivity here plus that existence theorem there is the full equivalence.

Proposition 2.4.5: The Lie Functor

Let **LieGrp** be the category of Lie groups and Lie group homomorphisms, and $\mathbf{LieAlg}_{\mathbb{R}}$ the category of finite-dimensional real Lie algebras and Lie algebra homomorphisms. The assignment

$$\mathrm{Lie}\colon \mathbf{LieGrp} \to \mathbf{LieAlg}_{\mathbb{R}}, \qquad G \mapsto \mathfrak{g} = T_eG, \qquad (\varphi\colon G \to H) \mapsto (d\varphi_e\colon \mathfrak{g} \to \mathfrak{h})$$

is a functor. That is, $d(\mathrm{id}_G)_e = \mathrm{id}_{\mathfrak{g}}$, and for composable homomorphisms $\varphi\colon G \to H$ and $\psi\colon H \to K$,

$$d(\psi \circ \varphi)_e = d\psi_e \circ d\varphi_e$$

***Proof*:**
Each object $\mathfrak{g} = T_eG$ is a finite-dimensional real Lie algebra by theorem 2.2.3 and proposition 2.2.5, and each morphism $d\varphi_e$ is a Lie algebra homomorphism by proposition 2.4.2, so Lie lands in $\mathbf{LieAlg}_{\mathbb{R}}$ on both objects and morphisms. For the identity, the differential of the identity map $\mathrm{id}_G\colon G \to G$ at e is the identity linear map on T_eG, since id_G sends every curve to itself and hence every velocity to itself. Thus $d(\mathrm{id}_G)_e = \mathrm{id}_{\mathfrak{g}}$. For composition, the chain rule of [13, Properties of the Tangent Map] applied to the composite $\psi \circ \varphi$ at the identity, using $\varphi(e_G) = e_H$,

gives

$$d(\psi \circ \varphi)_e = d\psi_{\varphi(e_G)} \circ d\varphi_e = d\psi_{e_H} \circ d\varphi_e$$

which is $d\psi_e \circ d\varphi_e$. Both functor axioms hold, completing the proof.

The functor Lie is the formal container for the entire correspondence. corollary 2.4.4 says Lie is faithful on the full subcategory of connected Lie groups, since it is injective on each morphism set out of a connected group. The reverse direction, reconstructing a group homomorphism, and even a group, from Lie algebra data, is the substance of Lie's theorems. When G is simply connected, every Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{h}$ is $d\varphi_e$ for a unique φ, so Lie becomes fully faithful there. Every finite-dimensional real Lie algebra is $\mathrm{Lie}(G)$ for some simply connected G, so Lie is essentially surjective onto $\mathbf{LieAlg}_{\mathbb{R}}$. These two facts upgrade Lie to an equivalence between the category of simply connected Lie groups and the category of finite-dimensional real Lie algebras, which is Lie's third theorem. The equivalence, together with the integration of Lie algebra homomorphisms, is proved in chapter 3.

Example 2.4.6: The Determinant and the Trace

Take $G = \mathrm{GL}_n(\mathbb{F})$ and $H = \mathbb{F}^\times$, the multiplicative group of nonzero scalars, a Lie group of dimension 1 for $\mathbb{F} = \mathbb{R}$ and 2 for $\mathbb{F} = \mathbb{C}$, with Lie algebra $\mathfrak{h} = T_1\mathbb{F}^\times = \mathbb{F}$ and exponential map the ordinary scalar exponential $\exp_H = \exp\colon \mathbb{F} \to \mathbb{F}^\times$, $z \mapsto e^z$. The determinant

$$\det\colon \mathrm{GL}_n(\mathbb{F}) \longrightarrow \mathbb{F}^\times$$

is a smooth homomorphism, since $\det(AB) = \det A \det B$ and the determinant is polynomial in the entries.

The differential $d(\det)_I\colon \mathfrak{gl}_n(\mathbb{F}) \to \mathbb{F}$ is the trace. To see this, differentiate det along the curve $t \mapsto \exp(tX) = I + tX + O(t^2)$ through I. Expand the determinant of $I + tX$ as the sum over permutations σ of $\pm\prod_i (I+tX)_{i\sigma(i)}$. Any $\sigma \neq \mathrm{id}$ moves at least two indices, so its term has at least two off-diagonal factors $tX_{i\sigma(i)}$, each carrying a factor t, and hence contributes only at order t^2. Only the identity permutation contributes a linear term, namely $\prod_i (1+tX_{ii}) = 1+t\sum_i X_{ii}+O(t^2)$, so

$$\det\big(I + tX + O(t^2)\big) = 1 + t\,\mathrm{tr}X + O(t^2)$$

whence

$$d(\det)_I(X) = \frac{d}{dt}\Big|_0 \det\exp(tX) = \mathrm{tr}X$$

Thus $d(\det)_I = \mathrm{tr}\colon \mathfrak{gl}_n(\mathbb{F}) \to \mathbb{F}$, which is indeed a Lie algebra homomorphism to the abelian $\mathbb{F}$: $\mathrm{tr}[X,Y] = \mathrm{tr}(XY) - \mathrm{tr}(YX) = 0 = [\mathrm{tr}X, \mathrm{tr}Y]$, consistent with proposition 2.4.2.

Now apply naturality (theorem 2.4.3) to $\varphi = \det$. The square reads $\det(\exp_G X) = \exp_H(d(\det)_I X)$, and with $\exp_G$ the matrix exponential (proposition 2.3.6), $\exp_H$ the scalar exponential, and $d(\det)_I = \mathrm{tr}$, this is

$$\det\exp X = e^{\mathrm{tr}X}$$

the identity of proposition 1.3.4. The chapter 1 computation, which proved $\det\exp X = e^{\mathrm{tr}X}$ by triangularizing X and multiplying eigenvalue exponentials, is here revealed as a single instance of naturality: the determinant is a homomorphism, so it must intertwine the two exponentials, and its differential can only be the trace.

The example is the template for reading any classical homomorphism through its differential. The

inclusion $\mathrm{SU}(2) \hookrightarrow \mathrm{SU}(3)$ has differential the inclusion $\mathfrak{su}(2) \hookrightarrow \mathfrak{su}(3)$. The projection of a product $G \times H \to G$ has differential the projection $\mathfrak{g} \oplus \mathfrak{h} \to \mathfrak{g}$. As the exercises develop, conjugation $C_g \colon x \mapsto gxg^{-1}$ has a differential that is a homomorphism of $\mathfrak{g}$ to itself, the value at g of the adjoint representation constructed in section 2.5. In every case the differential is easier to compute than the map, and naturality recovers the map on exponentials from it.

Exercise 2.4.1

1. For a fixed g in a Lie group G, let $C_g \colon G \to G$ be conjugation, $C_g(x) = gxg^{-1}$. Verify that C_g is a Lie group homomorphism (indeed an automorphism), and show that its differential $d(C_g)_e \colon \mathfrak{g} \to \mathfrak{g}$ is a Lie algebra automorphism. For a matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$, compute $d(C_g)_e(X)$ explicitly in terms of matrix multiplication, and use theorem 2.4.3 to write $g \exp(X) g^{-1}$ as a single exponential. (This differential, as a function of g, is the adjoint representation $\mathrm{Ad}(g)$ constructed in section 2.5.)

2. The double cover $p \colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5 is a homomorphism of Lie groups. Explain why its differential $dp_e \colon \mathfrak{su}(2) \to \mathfrak{so}(3)$ is an isomorphism of Lie algebras, even though p is not injective as a map of groups. Reconcile this with corollary 2.4.4, and state what additional global datum distinguishes $\mathrm{SU}(2)$ from $\mathrm{SO}(3)$.

3. Classify the connected Lie groups of dimension 1 up to isomorphism: show there are exactly two, namely $\mathbb{R}$ (under addition) and the circle S^1. Use that a one-dimensional Lie algebra is abelian, that exp is a homomorphism onto the identity component for an abelian group, and corollary 2.4.4.

4. Determine all Lie group homomorphisms $S^1 \to S^1$, where $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$. Identify $\mathrm{Lie}(S^1)$ with $\mathbb{R}$ so that $\exp(t) = e^{2\pi i t}$, compute the differential of the candidate map $z \mapsto z^n$, and use corollary 2.4.4 to show these are all of them.

5. Let $\det \colon \mathrm{U}(n) \to S^1$ be the determinant, a homomorphism onto the circle. Compute its differential $d(\det)_I \colon \mathfrak{u}(n) \to \mathrm{Lie}(S^1)$, and verify theorem 2.4.3 directly on a skew-Hermitian $X \in \mathfrak{u}(n)$ by comparing $\det \exp X$ with the scalar exponential of $d(\det)_I(X)$.

6. Let $\varphi \colon G \to H$ be a homomorphism of Lie groups with G connected. Show that if $d\varphi_e = 0$ then φ is the constant map $g \mapsto e_H$. Deduce that if $d\varphi_e$ is injective then φ is a local diffeomorphism onto its image near e, and give an example (from exercise 2.4.1 (2)) where $d\varphi_e$ is injective but φ is not globally injective.

2.5 The Adjoint Representations Ad and ad. The Bracket Intrinsically

The intrinsic Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ of definition 2.2.4 was built by transporting the bracket of vector fields from left-invariant fields to the tangent space T_eG, an operation with no visible connection to the commutator $XY - YX$ that governed the matrix Lie algebras of chapter 1. The group acts on itself by conjugation, and differentiating that action twice produces two representations, Ad on the group level and ad on the algebra level, whose interplay expresses the bracket.

For each $g \in G$ the map $C_g \colon G \to G$, $C_g(h) = ghg^{-1}$, is a Lie group automorphism fixing the identity, so its differential at e is a linear automorphism of $T_eG = \mathfrak{g}$. Collecting these differentials as g varies

gives a representation of G on its own Lie algebra.

Definition 2.5.1: The Adjoint Representation Ad

Let G be a Lie group with Lie algebra $\mathfrak{g} = T_eG$. For $g \in G$ let $C_g \colon G \to G$ be the conjugation automorphism $C_g(h) = ghg^{-1}$, a diffeomorphism fixing e (proposition 2.1.2). Its differential at e is a linear isomorphism of $\mathfrak{g}$, denoted

$$\mathrm{Ad}(g) = d(C_g)_e \colon \mathfrak{g} \to \mathfrak{g}$$

The resulting map $\mathrm{Ad} \colon G \to \mathrm{GL}(\mathfrak{g})$, into the group of linear automorphisms of the finite-dimensional vector space $\mathfrak{g}$, is the *adjoint representation* of G.

Proposition 2.5.2: Ad **is a Lie Group Homomorphism**

The map $\mathrm{Ad} \colon G \to \mathrm{GL}(\mathfrak{g})$ of definition 2.5.1 is a homomorphism of Lie groups: it is smooth, and

$$\mathrm{Ad}(gh) = \mathrm{Ad}(g)\ \mathrm{Ad}(h), \qquad \mathrm{Ad}(e) = \mathrm{id}_{\mathfrak{g}}$$

for all $g, h \in G$.

***Proof*:**
The conjugation maps compose by $C_{gh} = C_g \circ C_h$, since $C_{gh}(x) = (gh)x(gh)^{-1} = g(hxh^{-1})g^{-1} = C_g(C_h(x))$, and $C_e = \mathrm{id}_G$. Both C_g and C_h fix e, so the chain rule for differentials applies at e:

$$\mathrm{Ad}(gh) = d(C_{gh})_e = d(C_g \circ C_h)_e = d(C_g)_e \circ d(C_h)_e = \mathrm{Ad}(g)\ \mathrm{Ad}(h)$$

and $\mathrm{Ad}(e) = d(\mathrm{id}_G)_e = \mathrm{id}_{\mathfrak{g}}$. In particular $\mathrm{Ad}(g)\,\mathrm{Ad}(g^{-1}) = \mathrm{Ad}(e) = \mathrm{id}$, so each $\mathrm{Ad}(g)$ is invertible and Ad lands in $\mathrm{GL}(\mathfrak{g})$. Smoothness is a consequence of the smoothness of the group operations: the conjugation map $(g, h) \mapsto ghg^{-1}$ is smooth on $G \times G$ as a composite of multiplication and inversion, and $\mathrm{Ad}(g)$ is the partial differential in the h-variable at $h = e$, which depends smoothly on g because differentiation of a smooth family of maps produces a smooth family of differentials ([13, The Tangent Functor]). Thus Ad is smooth, completing the proof.

The homomorphism Ad packages the conjugation action of G on itself into its infinitesimal effect on the tangent space at e. It sees only the germ of conjugation at the identity, which is exactly the data a Lie algebra is built to remember. Because $\mathrm{GL}(\mathfrak{g})$ is itself a Lie group, with Lie algebra the endomorphism algebra $\mathfrak{gl}(\mathfrak{g}) = \mathrm{End}(\mathfrak{g})$, the machinery of section 2.4 applies to Ad: it has a differential at e, and that differential is the second of the two representations.

Definition 2.5.3: The Adjoint Representation ad

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The *adjoint representation* of $\mathfrak{g}$ is the differential of Ad at the identity,

$$\mathrm{ad} = d(\mathrm{Ad})_e \colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}) = \mathrm{End}(\mathfrak{g})$$

Explicitly, for $X \in \mathfrak{g}$ the endomorphism $\mathrm{ad}(X)$ of $\mathfrak{g}$ is

$$\mathrm{ad}(X) = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp tX)$$

the velocity at $t = 0$ of the curve $t \mapsto \mathrm{Ad}(\exp tX)$ in $\mathrm{GL}(\mathfrak{g})$.
For a Lie algebra $\mathfrak{g}$ over an arbitrary field, presented by its bracket alone with no group attached, the *adjoint representation* of $\mathfrak{g}$ is the map

$$\mathrm{ad} \colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}) = \mathrm{End}(\mathfrak{g}), \qquad \mathrm{ad}(X)Y = [X, Y]$$

which is a homomorphism of Lie algebras, the Jacobi identity of definition 1.4.5 rewritten. When $\mathfrak{g} = T_eG$ is the Lie algebra of a Lie group, the two descriptions of ad agree (theorem 2.5.4).

The explicit formula is the naturality of exp from theorem 2.4.3 read for the homomorphism Ad: the one-parameter subgroup $t \mapsto \exp(tX)$ of G maps under Ad to the one-parameter subgroup $t \mapsto \mathrm{Ad}(\exp tX)$ of $\mathrm{GL}(\mathfrak{g})$, whose generator is $d(\mathrm{Ad})_e(X) = \mathrm{ad}(X)$. Since $\mathrm{GL}(\mathfrak{g})$ is a matrix group, its exponential is the matrix exponential (proposition 2.3.6), so the naturality square specializes to

$$\mathrm{Ad}(\exp tX) = \exp\big(t\,\mathrm{ad}(X)\big) \tag{2.11}$$

an identity of endomorphisms of $\mathfrak{g}$ that will reappear in the exercises as the statement $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}X}$. Differentiating (2.11) at $t = 0$ recovers the definition of $\mathrm{ad}(X)$, and integrating it recovers Ad from ad on the identity component.

By proposition 2.4.2 the differential of a Lie group homomorphism is a Lie algebra homomorphism, so ad preserves brackets: $\mathrm{ad}([X, Y]) = [\mathrm{ad}(X), \mathrm{ad}(Y)]$, where the bracket on the right is the commutator of endomorphisms in $\mathrm{End}(\mathfrak{g})$. This is the Jacobi identity in disguise, a point developed in the exercises.

The bracket on $\mathfrak{g}$ of definition 2.2.4 was defined by carrying the bracket of vector fields across the isomorphism $X \mapsto \tilde{X}$ of theorem 2.2.3 between T_eG and the left-invariant fields. The bracket induced by ad is a second candidate $\mathrm{ad}(X)Y$.

Theorem 2.5.4: The Bracket is the Adjoint Action

Let G be a Lie group with Lie algebra $\mathfrak{g} = T_eG$, and let $[-,-]$ denote the bracket of definition 2.2.4, transported from the Lie bracket of left-invariant vector fields via theorem 2.2.3. Then for all $X, Y \in \mathfrak{g}$,

$$[X,Y] = \mathrm{ad}(X)Y = \left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(\exp tX)\, Y$$

The abstract bracket satisfying the axioms of definition 1.4.5 is therefore the infinitesimal conjugation action ad, and the two brackets on $\mathfrak{g}$ coincide.

***Proof*:**

Write $\tilde{X}$ for the left-invariant vector field with $\tilde{X}_e = X$, so that by definition of the bracket on $\mathfrak{g}$,

$$[X,Y] = [\tilde{X},\tilde{Y}]_e \tag{2.12}$$

the right side being the value at e of the Lie bracket of the two left-invariant fields. It suffices to show $[\tilde{X},\tilde{Y}]_e = \mathrm{ad}(X)Y$.

Step 1: the flow of a left-invariant field is right translation. Let $\gamma_X(t) = \exp(tX)$ be the integral curve of $\tilde{X}$ through e, which is the one-parameter subgroup with velocity X (proposition 2.3.3). Because $\tilde{X}$ is left-invariant, its integral curve through an arbitrary point g is $t \mapsto g\exp(tX) = L_g(\exp tX)$: applying the diffeomorphism L_g carries the integral curve through e to one through g, and L_g sends the field $\tilde{X}$ to itself. Hence the flow of $\tilde{X}$ at time t is right translation by $\exp(tX)$,

$$\Phi_t^{\tilde{X}} = R_{\exp(tX)}, \qquad g \mapsto g\exp(tX) \tag{2.13}$$

These flows exist for all $t \in \mathbb{R}$ because left-invariant fields are complete (proposition 2.2.5).

Step 2: the bracket as a Lie derivative. The Lie bracket of two vector fields is the Lie derivative of one along the flow of the other: for any smooth vector fields $\tilde{X}, \tilde{Y}$,

$$[\tilde{X},\tilde{Y}]_g = \left.\frac{d}{dt}\right|_{t=0} \left(d\Phi_{-t}^{\tilde{X}}\right)_{\Phi_t^{\tilde{X}}(g)} \left(\tilde{Y}_{\Phi_t^{\tilde{X}}(g)}\right)$$

the standard formula $[\tilde{X},\tilde{Y}] = \mathcal{L}_{\tilde{X}}\tilde{Y}$ for the bracket in terms of the pushforward of $\tilde{Y}$ by the backward flow of $\tilde{X}$ ([13, The Lie Derivative is the Bracket]). Evaluate at $g = e$ and substitute the flow (2.13), so $\Phi_t^{\tilde{X}}(e) = \exp(tX)$ and $\Phi_{-t}^{\tilde{X}} = R_{\exp(-tX)}$:

$$[\tilde{X},\tilde{Y}]_e = \left.\frac{d}{dt}\right|_{t=0} \left(dR_{\exp(-tX)}\right)_{\exp(tX)} \left(\tilde{Y}_{\exp(tX)}\right) \tag{2.14}$$

Step 3: rewrite the inner expression through conjugation. The field $\tilde{Y}$ is left-invariant, so its value at $\exp(tX)$ is the left translate of Y,

$$\tilde{Y}_{\exp(tX)} = \left(dL_{\exp(tX)}\right)_e Y$$

Insert this into (2.14). The differential $\left(dR_{\exp(-tX)}\right)_{\exp(tX)}$ acts on the result, and since the maps $L_{\exp(tX)}$ and $R_{\exp(-tX)}$ commute as diffeomorphisms of G (left and right translations

always commute), their composite is the conjugation map:

$$R_{\exp(-tX)} \circ L_{\exp(tX)} = \big(h \mapsto \exp(tX)\, h\, \exp(-tX)\big) = C_{\exp(tX)}$$

Both $L_{\exp(tX)}$ and $R_{\exp(-tX)}$ are diffeomorphisms, so the chain rule gives, for the composite evaluated on $Y \in T_eG$,

$$\big(dR_{\exp(-tX)}\big)_{\exp(tX)}\big(dL_{\exp(tX)}\big)_e\, Y = d\big(C_{\exp(tX)}\big)_e\, Y = \mathrm{Ad}(\exp tX)\, Y$$

the last equality by the definition of Ad in definition 2.5.1. Substituting into (2.14),

$$[\tilde{X}, \tilde{Y}]_e = \left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(\exp tX)\, Y \tag{2.15}$$

Step 4: identify with ad. By definition 2.5.3 the right side of (2.15) is precisely $\mathrm{ad}(X)Y$, the velocity at $t = 0$ of the curve $t \mapsto \mathrm{Ad}(\exp tX)Y$ in $\mathfrak{g}$. Combining with (2.12) and (2.15),

$$[X, Y] = [\tilde{X}, \tilde{Y}]_e = \left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(\exp tX)\, Y = \mathrm{ad}(X)Y$$

as we sought to show.

The bracket on $\mathfrak{g}$ was first defined abstractly, as whatever operation the vector-field bracket induces on T_eG, with the alternating and Jacobi axioms of definition 1.4.5 guaranteed but its meaning opaque. By theorem 2.5.4, $[X, Y]$ is the rate at which conjugation by $\exp(tX)$ moves Y, measured at $t = 0$. The bracket is thus the infinitesimal record of the failure of the two one-parameter subgroups $\exp(tX)$ and $\exp(sY)$ to commute, exactly the interpretation the matrix commutator carried in proposition 1.4.4, now established with no matrices in sight. Every structural fact about the bracket, the Jacobi identity included, is now a fact about the conjugation action of the group on itself.

The operator $\mathrm{ad}(X)$ acts on $\mathfrak{g}$ by $Y \mapsto [X, Y]$, so ad is the map sending X to left-bracketing by X. That ad is a Lie algebra homomorphism (proposition 2.4.2) then reads $\mathrm{ad}([X, Y]) = [\mathrm{ad}(X), \mathrm{ad}(Y)]$, which unwinds to $[[X, Y], Z] = [X, [Y, Z]] - [Y, [X, Z]]$ for all Z, a rearrangement of the Jacobi identity. The single homomorphism ad therefore encodes the entire bracket and its one nontrivial axiom at once.

Proposition 2.5.5: Ad and ad for a Matrix Group

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g} \subseteq M_n(\mathbb{F})$, identified with T_IG as in corollary 1.4.6. Then for all $g \in G$ and $X, Y \in \mathfrak{g}$,

$$\mathrm{Ad}(g)X = gXg^{-1}, \qquad \mathrm{ad}(X)Y = XY - YX = [X, Y]$$

The intrinsic adjoint action is conjugation of matrices, and the intrinsic bracket $\mathrm{ad}(X)Y$ is the commutator of proposition 1.4.4. The intrinsic bracket on $\mathfrak{gl}_n(\mathbb{F})$ is therefore the matrix commutator.

***Proof*:**

Step 1: the formula for Ad. Fix $g \in G$. The conjugation map $C_g(h) = ghg^{-1}$ is the restriction to G of the ambient linear map $M \mapsto gMg^{-1}$ on $M_n(\mathbb{F})$, which is its own differential at every point since it is linear. For $X \in \mathfrak{g} = T_I G$, represent X as the velocity of the curve $t \mapsto \exp(tX)$ through I (proposition 2.3.6 identifies the intrinsic exp with the matrix exponential). The conjugation-equivariance of the exponential (proposition 1.3.4) gives

$$C_g(\exp tX) = g\exp(tX)g^{-1} = \exp\big(t\,gXg^{-1}\big) \tag{2.16}$$

the last equality because $g\exp(tX)g^{-1} = \exp(g(tX)g^{-1})$ by the series computation $g\big(\sum_k \frac{(tX)^k}{k!}\big)g^{-1} = \sum_k \frac{(g\,tX\,g^{-1})^k}{k!}$. Differentiating (2.16) at $t = 0$,

$$\mathrm{Ad}(g)X = d(C_g)_I\,X = \left.\frac{d}{dt}\right|_{t=0} \exp\big(t\,gXg^{-1}\big) = gXg^{-1}$$

using $\left.\frac{d}{dt}\right|_0 \exp(tZ) = Z$ with $Z = gXg^{-1}$. The result gXg^{-1} lies in $\mathfrak{g}$, consistent with $\mathrm{Ad}(g)$ being an endomorphism of $\mathfrak{g}$.

Step 2: the formula for ad. Apply Step 1 with $g = \exp(sX)$, so $\mathrm{Ad}(\exp sX)Y = \exp(sX)\,Y\exp(-sX)$ for $Y \in \mathfrak{g}$. Differentiating in s at $s = 0$, with $\left.\frac{d}{ds}\right|_0 \exp(sX) = X$ and $\left.\frac{d}{ds}\right|_0 \exp(-sX) = -X$, the product rule gives

$$\mathrm{ad}(X)Y = \left.\frac{d}{ds}\right|_{s=0} \exp(sX)\,Y\exp(-sX) = XY - YX = [X,Y]$$

which is the commutator. By theorem 2.5.4 the left side is the intrinsic bracket, so the intrinsic bracket equals the matrix commutator, recovering proposition 1.4.4. For $G = \mathrm{GL}_n(\mathbb{F})$ this says the intrinsic Lie algebra $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ carries the commutator bracket, completing the proof.

The bracket that section 2.2 manufactured from left-invariant vector fields, with no reference to matrices, becomes the ordinary commutator $XY - YX$ the moment G is realized inside some $\mathrm{GL}_n(\mathbb{F})$. Two constructions that looked unrelated, the vector-field bracket of theorem 2.2.3 and the commutator of proposition 1.4.4, are the same operation seen from two sides. The intrinsic definition is the more general one, applying to any Lie group, and proposition 2.5.5 certifies that it specializes correctly on the matrix groups where an independent computation exists.

Example 2.5.6: Ad and ad for SU(2) and the Rotation Action

Take $G = \mathrm{SU}(2)$, with Lie algebra $\mathfrak{su}(2)$ the traceless skew-Hermitian 2×2 matrices of example 1.4.7, and its basis X_1, X_2, X_3 with $[X_j, X_k] = \sum_\ell \varepsilon_{jk\ell} X_\ell$. By proposition 2.5.5 the adjoint representation is conjugation,

$$\mathrm{Ad}(g)X = gXg^{-1}, \qquad g \in \mathrm{SU}(2),\ X \in \mathfrak{su}(2)$$

which is precisely the map $\varphi(g)\colon X \mapsto gXg^{-1}$ of theorem 1.5.5. That theorem showed $\varphi(g)$ preserves the Euclidean structure on $\mathfrak{su}(2) \cong \mathbb{R}^3$ given by the determinant form and lands in $\mathrm{SO}(3)$. In the present language this says

$$\mathrm{Ad}\colon \mathrm{SU}(2) \to \mathrm{SO}(3) \subset \mathrm{GL}(\mathfrak{su}(2))$$

is the adjoint representation, and the double cover $\varphi = \mathrm{Ad}$ is nothing but Ad regarded as a homomorphism onto the rotation group of the three-dimensional space $\mathfrak{su}(2)$. The kernel of Ad is

the center $\{\pm I\}$, the two-to-one fibers of the cover.

On the algebra level, proposition 2.5.5 gives $\mathrm{ad}(X)Y = [X, Y]$, so on the basis

$$\mathrm{ad}(X_1)X_2 = [X_1, X_2] = X_3, \qquad \mathrm{ad}(X_1)X_3 = [X_1, X_3] = -X_2, \qquad \mathrm{ad}(X_1)X_1 = 0$$

In the ordered basis (X_1, X_2, X_3) the operator $\mathrm{ad}(X_1)$ has matrix

$$\mathrm{ad}(X_1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

the generator of rotation about the first axis, and $\mathrm{ad}(X_2)$, $\mathrm{ad}(X_3)$ are the analogous generators about the other two axes. Under the isomorphism $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$ of example 1.4.7 the operator $\mathrm{ad}(X) = [X, -]$ becomes $v \mapsto v \times (-)$, cross product with the vector $v = \Phi(X)$, which is exactly the infinitesimal rotation about the axis v. The identity $\mathrm{Ad}(\exp tX) = \exp(t\,\mathrm{ad}(X))$ of (2.11) then states that conjugating by $\exp(tX)$ is the finite rotation whose generator is cross product with v, recovering the computation in Step 4 of theorem 1.5.5 that for u with $u^2 = -I$, so that $|\Phi(u)| = 2$, the element $\exp(\frac{\theta}{2}u)$ maps under Ad to the rotation by θ about u. The factor of two between the group parameter and the rotation angle, the arithmetic heart of the double cover, is precisely that normalization: $\mathrm{ad}(u)$ is cross product with a vector of Euclidean length 2, so it generates rotation at twice unit speed.

The example returns to where the book began. In chapter 1 the conjugation action of SU(2) on $\mathfrak{su}(2)$ was constructed by hand to build the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, before the words "adjoint representation" were available. That action is Ad, and the double cover is Ad itself, the homomorphism onto the orthogonal group of the invariant form on $\mathfrak{su}(2)$. What chapter 1 saw as a geometric accident, that a unit-quaternion group double-covers the rotation group, is the adjoint representation of SU(2) read through the isomorphism $\mathfrak{su}(2) \cong \mathbb{R}^3$.

The bracket now stands identified in three forms. It is the matrix commutator $XY - YX$ of proposition 1.4.4, the operation chapter 1 extracted by differentiating a conjugation curve. It is the left-invariant-field bracket $[\tilde{X}, \tilde{Y}]_e$ of theorem 2.2.3, the intrinsic definition that requires no embedding. And it is the adjoint action $\mathrm{ad}(X)Y = \frac{d}{dt}\big|_0 \mathrm{Ad}(\exp tX)Y$ of theorem 2.5.4, the infinitesimal conjugation of the group on its own algebra. These are one operation, and the equivalence of the three descriptions is what lets a computation with brackets in $\mathfrak{g}$ stand in for conjugation in G. The bilinear form $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y)$ built from ad, invariant under the adjoint action of G, is the *Killing form* (defined and developed in exercise 2.5.1 (4) below). It measures the semisimplicity of $\mathfrak{g}$ and is the tool that drives the structure theory of Part II and the classification of chapter 6 onward.

Exercise 2.5.1

1. Prove the identity $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}X}$ for all $X \in \mathfrak{g}$, where the right side is the exponential in $\mathrm{GL}(\mathfrak{g})$ of the endomorphism $\mathrm{ad}X \in \mathrm{End}(\mathfrak{g})$. Deduce that for a matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$, conjugation $e^X Y e^{-X}$ is given by the convergent series $Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \cdots$.

2. Show that the kernel of $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ contains the center $Z(G)$, and that for a *connected* Lie group G the kernel is exactly $Z(G)$. Conclude that a connected G acts trivially by conjugation on its Lie algebra if and only if G is abelian.

3. Show that for each $X \in \mathfrak{g}$ the operator $\mathrm{ad}(X)\colon \mathfrak{g} \to \mathfrak{g}$ is a *derivation* of the bracket, that is

$$\mathrm{ad}(X)[Y,Z] = [\mathrm{ad}(X)Y, Z] + [Y, \mathrm{ad}(X)Z]$$

   for all $Y, Z \in \mathfrak{g}$, and prove that this identity is equivalent to the Jacobi identity of definition 1.4.5.

4. Define the *Killing form* $\kappa(X,Y) = \mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y)$ on $\mathfrak{g}$. Show that κ is a symmetric bilinear form, and that it is Ad-*invariant*: $\kappa(\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y) = \kappa(X,Y)$ for all $g \in G$. Deduce the infinitesimal invariance $\kappa([X,Y],Z) + \kappa(Y,[X,Z]) = 0$, the associativity (or invariance) property of the Killing form.

5. For $\mathfrak{su}(2)$ with basis X_1, X_2, X_3 as in example 1.4.7, write out the three matrices $\mathrm{ad}(X_1), \mathrm{ad}(X_2), \mathrm{ad}(X_3)$ in the ordered basis (X_1, X_2, X_3), and compute the Killing form κ explicitly. Verify that κ is negative definite, and identify it as a negative multiple of the standard inner product on $\mathbb{R}^3$.

6. Give an example of a Lie group G for which $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ is not injective, and describe its kernel. Contrast with the case $G = \mathrm{SU}(2)$, where ad is injective. What feature of the Lie algebra distinguishes the two cases?

3

The Lie Group-Lie Algebra Correspondence

Chapter 2 attached to every Lie group its Lie algebra and to every homomorphism its differential, assembling the assignment into a functor from Lie groups to Lie algebras. The question this chapter settles is how much of the curved object G survives in the linear object $\mathfrak{g}$. The answer, the content of Lie's fundamental theorems, is that almost everything does: subgroups correspond to subalgebras, homomorphisms to their differentials, and, up to a single global subtlety carried by the fundamental group, a connected Lie group can be reconstructed in full from its Lie algebra. The correspondence is what makes Lie theory effective, for it converts questions about a nonlinear manifold with a group law into questions of linear algebra over $\mathfrak{g}$.

Three theorems carry the chapter. Cartan's closed-subgroup theorem removes smoothness from the hypotheses entirely: a subgroup closed as a subset is automatically an embedded submanifold, hence a Lie subgroup, so that topology alone certifies the geometry. Lie's integration theorems run the functor backwards: every subalgebra of $\mathfrak{g}$ integrates to a connected subgroup of G, by way of the Frobenius theorem cited to [13, Global Frobenius Theorem], and every homomorphism of Lie algebras out of a simply-connected group integrates to a homomorphism of groups. Together they yield Lie's third theorem, the equivalence between the category of simply-connected Lie groups and the category of finite-dimensional real Lie algebras. The chapter ends with the Baker-Campbell-Hausdorff formula, which makes the reconstruction explicit by expressing $\exp X \exp Y$ as the exponential of a universal series in X, Y, and their iterated brackets, so that the group law near the identity is literally written in the bracket. The one datum the Lie algebra cannot recover, the fundamental group, is what motivates the covering-group theory of chapter 4.

3.1 Lie Subgroups and Lie Subalgebras

Chapter 2 completed the passage from a Lie group G to its Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$, and showed the passage is functorial: a homomorphism $\varphi\colon G \to H$ has a differential $d\varphi_e\colon \mathfrak{g} \to \mathfrak{h}$ that respects the bracket (proposition 2.4.5). This chapter turns the dictionary into a tool for recognizing and building groups. The guiding question is how the subobjects match up: which subgroups of G carry a Lie algebra sitting inside $\mathfrak{g}$, and conversely which subalgebras of $\mathfrak{g}$ arise from a subgroup. The present section sets up the two notions of subobject and establishes the easy direction of their correspondence. A subgroup of G that is also a submanifold has a Lie algebra, a subalgebra of $\mathfrak{g}$; the hard converse, that every subalgebra integrates to a subgroup, is the business of section 3.3.

The subtlety that organizes the section is topological. A submanifold of a manifold comes in two strengths: an *embedded* submanifold, carrying the subspace topology, and an *immersed* one, carrying its own possibly finer topology under an injective immersion. For subgroups both strengths occur, and the distinction is not a technicality. The immersed subgroups can be topologically wild, winding densely through the ambient group, and it is exactly this wildness that the closed-subgroup theorem of section 3.2 tames. The canonical example separating the two, an irrational line on the torus, is worked out at the end of the section and threads through the rest of the chapter.

Definition 3.1.1: Lie Subgroup

Let G be a Lie group. A *Lie subgroup* of G is a subgroup $H \subseteq G$ equipped with a smooth manifold structure making H a Lie group, such that the inclusion

$$\iota\colon H \hookrightarrow G$$

is a smooth homomorphism that is an *immersion*, that is, $d\iota_h\colon T_hH \to T_{\iota(h)}G$ is injective at every $h \in H$. The Lie subgroup is *embedded* if in addition ι is a topological embedding, meaning the manifold topology on H agrees with the subspace topology it inherits from G; otherwise it is (properly) *immersed*.

The definition carries two independent pieces of data that must not be conflated. The set H is a subgroup of G in the algebraic sense, and it is simultaneously the underlying set of a Lie group whose smooth structure need not be induced from G. What ties the two together is that the inclusion is required to be a smooth immersion, so the abstract Lie group H maps injectively into G with injective differential, and the Lie algebra $\mathfrak{h} = T_eH$ maps injectively into $\mathfrak{g} = T_eG$. In the embedded case the topology on H is forced to be the subspace topology and nothing is lost by regarding H as a subset with its inherited geometry. In the immersed but non-embedded case the topology on H is strictly finer than the subspace topology, and H as a subset of G can fail to be locally closed: its points are visited by the immersion, but a subset open in H need not be the trace of an open subset of G; the irrational line of example 3.1.5 exhibits exactly this failure. Every subgroup that is a regular level set of a submersion, and in particular every classical matrix group inside $\mathrm{GL}_n(\mathbb{F})$, is embedded, so the immersed-only phenomenon is invisible until one meets a subgroup whose image folds back on itself.

The dual notion on the algebra side is purely linear-algebraic and requires no topology at all. A Lie algebra is a vector space with a bracket, and its subobjects are the subspaces that the bracket preserves.

Definition 3.1.2: Lie Subalgebra

Let $\mathfrak{g}$ be a Lie algebra over a field k. A *Lie subalgebra* of $\mathfrak{g}$ is a k-linear subspace $\mathfrak{h} \subseteq \mathfrak{g}$ that is closed under the bracket: $[X, Y] \in \mathfrak{h}$ for all $X, Y \in \mathfrak{h}$. With the restricted bracket $\mathfrak{h}$ is a Lie algebra in its own right, and the inclusion $\mathfrak{h} \hookrightarrow \mathfrak{g}$ is a homomorphism of Lie algebras.

A subalgebra is thus the linearization of the subgroup notion: where a Lie subgroup is a subgroup closed under the smooth group operations, a Lie subalgebra is a subspace closed under the one operation the algebra carries. The bracket-closure condition is the only requirement, and it is easy to verify in practice by computing brackets on a spanning set. Two matrix examples show the definition before the general correspondence is proved.

Example 3.1.3: Subalgebras of $\mathfrak{gl}_n$

Inside $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ with the commutator bracket $[X, Y] = XY - YX$, the following are Lie subalgebras.

1. The traceless matrices $\mathfrak{sl}_n(\mathbb{F}) = \{X \in M_n(\mathbb{F}) \mid \operatorname{tr} X = 0\}$. This is a subspace of codimension one, and it is bracket-closed because $\operatorname{tr}(XY - YX) = 0$ for all X, Y, so every commutator is traceless. This is the Lie algebra of $\mathrm{SL}(n, \mathbb{F})$ recorded in theorem 1.4.3.
2. The skew-symmetric matrices $\mathfrak{so}(n) = \left\{X \in M_n(\mathbb{R}) \mid X^{\mathsf{T}} = -X\right\}$. If $X^{\mathsf{T}} = -X$ and $Y^{\mathsf{T}} = -Y$ then
$$[X, Y]^{\mathsf{T}} = (XY - YX)^{\mathsf{T}} = Y^{\mathsf{T}}X^{\mathsf{T}} - X^{\mathsf{T}}Y^{\mathsf{T}} = YX - XY = -[X, Y]$$
so $[X, Y]$ is again skew-symmetric. This is the Lie algebra of $\mathrm{SO}(n)$ and of $\mathrm{O}(n)$ from theorem 1.4.3.
3. The upper-triangular matrices, and separately the strictly upper-triangular ones. If X, Y have zero entries below the diagonal, so does XY, hence so does $[X, Y]$; the strictly upper-triangular case is identical with zeros on and below the diagonal. These two subalgebras will reappear in Part II as the models of solvable and nilpotent Lie algebras.

By contrast, the symmetric matrices are a subspace but not a subalgebra: the commutator of two symmetric matrices is skew-symmetric, not symmetric, so bracket-closure fails.

The examples exhibit the two directions the section connects. Each of $\mathfrak{sl}_n$ and $\mathfrak{so}(n)$ is not only a subalgebra of $\mathfrak{gl}_n$ but the Lie algebra of an actual subgroup, $\mathrm{SL}(n, \mathbb{F})$ and $\mathrm{SO}(n)$ respectively, and these subgroups are embedded, being regular level sets inside $\mathrm{GL}_n(\mathbb{F})$. That the subalgebra inclusion $\mathfrak{so}(n) \hookrightarrow \mathfrak{gl}_n$ is exactly the differential of the group inclusion $\mathrm{SO}(n) \hookrightarrow \mathrm{GL}_n(\mathbb{R})$ is the content of the next proposition, in general form.

The proposition is the easy half of the Lie group–Lie algebra correspondence for subobjects. A Lie subgroup, being a Lie group with a homomorphic inclusion into G, feeds into the functor Lie of proposition 2.4.5, and the functor turns the inclusion of groups into an inclusion of algebras. The only points to check are that the differential of the inclusion is injective, so that $\mathfrak{h}$ sits inside $\mathfrak{g}$ as a true copy, and that the exponential maps are compatible.

Proposition 3.1.4: The Lie Algebra of a Lie Subgroup

Let H be a Lie subgroup of a Lie group G, with inclusion $\iota\colon H \hookrightarrow G$. Then:

1. The differential $d\iota_e\colon \mathfrak{h} \to \mathfrak{g}$ is an injective homomorphism of Lie algebras, so its image $d\iota_e(\mathrm{Lie}H)$ is a Lie subalgebra of $\mathfrak{g}$. Identifying $\mathrm{Lie}(H)$ with this image, one writes $\mathfrak{h} = \mathrm{Lie}(H) \subseteq \mathfrak{g}$.

2. The exponential of G restricts to that of H: for all $X \in \mathfrak{h}$,

$$\exp_G(X) = \iota\big(\exp_H(X)\big)$$

so the one-parameter subgroups of G tangent to $\mathfrak{h}$ are exactly those of H, and $\exp_G(\mathfrak{h}) \subseteq H$.

***Proof*:**

1. *The differential is an injective algebra homomorphism.* By definition 3.1.1 the inclusion ι is a smooth homomorphism of Lie groups and an immersion, so its differential $d\iota_e\colon T_eH \to T_eG$ (definition 2.4.1) is injective by the immersion hypothesis. It is a homomorphism of Lie algebras by proposition 2.4.2 applied to the homomorphism ι: for $X, Y \in \mathfrak{h}$,

$$d\iota_e[X,Y]_{\mathfrak{h}} = [\, d\iota_e X,\ d\iota_e Y \,]_{\mathfrak{g}}$$

where the left bracket is that of $\mathrm{Lie}(H)$ and the right that of $\mathfrak{g}$. An injective linear map is an isomorphism onto its image, so the image $d\iota_e(\mathfrak{h})$ is a linear subspace of $\mathfrak{g}$; and the displayed identity, read on the image, says that this subspace is closed under the bracket of $\mathfrak{g}$, since $[d\iota_e X, d\iota_e Y] = d\iota_e[X,Y]$ lies in the image. Thus $d\iota_e(\mathfrak{h})$ is a Lie subalgebra of $\mathfrak{g}$ in the sense of definition 3.1.2, and $d\iota_e$ is a Lie algebra isomorphism of $\mathrm{Lie}(H)$ onto it. Identifying $\mathrm{Lie}(H)$ with its image is then harmless, which is the first assertion.

2. *Compatibility of the exponentials.* The exponential map is natural under homomorphisms: for a homomorphism $\varphi\colon G_1 \to G_2$ one has $\varphi \circ \exp_{G_1} = \exp_{G_2} \circ d\varphi_e$ (theorem 2.4.3). Apply this to $\varphi = \iota\colon H \to G$, whose differential is the inclusion $d\iota_e$ of part (1). For $X \in \mathfrak{h}$,

$$\iota\big(\exp_H(X)\big) = \exp_G\big(d\iota_e X\big) = \exp_G(X)$$

the last equality under the identification of $X \in \mathfrak{h}$ with $d\iota_e X \in \mathfrak{g}$. Hence $\exp_G$ restricted to $\mathfrak{h}$ takes values in $\iota(H) = H$ and coincides with $\exp_H$ followed by the inclusion. In particular the one-parameter subgroup $t \mapsto \exp_G(tX)$ of G, for $X \in \mathfrak{h}$, is the image under ι of the one-parameter subgroup $t \mapsto \exp_H(tX)$ of H, as we sought to show.

The proposition records that the functor Lie sends a Lie subgroup to a Lie subalgebra, with the subgroup's own exponential nothing but the restriction of the ambient one. The immersion hypothesis in definition 3.1.1 is doing real work here: it is precisely what forces $d\iota_e$ to be injective, so that $\mathfrak{h}$ is a genuine copy of $\mathrm{Lie}(H)$ inside $\mathfrak{g}$ rather than a quotient. A reader who has met the matrix case will recognize the content: for a closed matrix group the intrinsic subalgebra is the space of matrices X with $\exp(tX)$ in the group for all t, which is exactly the tangent-space description of definition 1.4.1, and part (2) is the statement that this description is coordinate-free. This identification is made

precise in section 3.2, where the closed-subgroup theorem gives the missing embedded structure.

The converse question is the one this chapter exists to answer, and it is not easy. Given a Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, is there a Lie subgroup $H \leq G$ with $\mathrm{Lie}(H) = \mathfrak{h}$? The answer is yes, and it is the integration theorem of Lie proved in section 3.3: every Lie subalgebra integrates to a unique connected Lie subgroup, constructed as the maximal integral manifold through the identity of the left-invariant distribution spanned by $\mathfrak{h}$. The subgroup so produced is in general only immersed, not embedded, and the example below shows this outcome is unavoidable: there are subalgebras whose integral subgroup is a dense wind.

The distinction between embedded and immersed is not idle. The prototype that forces the distinction, and the running counterexample for the rest of the chapter, is a line of irrational slope on the two-dimensional torus. It is a one-parameter subgroup, hence a connected immersed Lie subgroup, but its image is dense, so it is as far from embedded as a subgroup can be.

Example 3.1.5: The Irrational Line on the Torus

Let $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ be the two-dimensional torus, a compact abelian Lie group of dimension two, with quotient map $\pi\colon \mathbb{R}^2 \to \mathbb{T}^2$. Its Lie algebra is $\mathbb{R}^2$ with the zero bracket, and the exponential map is the quotient projection $\exp = \pi$, since $\mathbb{T}^2$ is abelian and π is a homomorphism whose differential at 0 is the identity: by theorem 2.3.1 a one-parameter subgroup of $\mathbb{T}^2$ is $t \mapsto \pi(tv)$ for a velocity vector $v \in \mathbb{R}^2$. Fix a slope $\alpha \in \mathbb{R}$ and consider the one-dimensional subalgebra $\mathfrak{h}_\alpha = \mathbb{R} \cdot (1, \alpha) \subseteq \mathbb{R}^2$ spanned by the vector $v = (1, \alpha)$. It is trivially a subalgebra, the bracket being zero. The corresponding one-parameter subgroup is

$$\gamma_\alpha\colon \mathbb{R} \to \mathbb{T}^2, \qquad \gamma_\alpha(t) = \pi(t, t\alpha) \tag{3.1}$$

the image of the line of slope α through the origin of $\mathbb{R}^2$. Whether the resulting Lie subgroup $H_\alpha = \gamma_\alpha(\mathbb{R})$ is embedded or only immersed depends entirely on the arithmetic of α.

Rational slope: an embedded circle. Suppose $\alpha = p/q$ in lowest terms with $q > 0$. Then $\gamma_\alpha(q) = \pi(q, p) = \pi(0, 0) = e$, since $(q, p) \in \mathbb{Z}^2$, and q is the least positive such time. Thus γ_α is periodic with period q, and it factors through an injective immersion of the circle $\mathbb{R}/q\mathbb{Z}$ onto H_α. The image is a compact embedded circle in $\mathbb{T}^2$, a closed one-dimensional submanifold winding p times around one generator while going once around the other. Being compact, its manifold topology and its subspace topology agree, so H_α is an embedded Lie subgroup, isomorphic to the circle S^1.

Irrational slope: a dense immersed line. Suppose instead that α is irrational. Then γ_α is injective: if $\gamma_\alpha(s) = \gamma_\alpha(t)$ then $(s - t, (s - t)\alpha) \in \mathbb{Z}^2$, so $s - t = m$ and $(s - t)\alpha = n$ for integers m, n, forcing $m\alpha = n$; with α irrational this gives $m = n = 0$, hence $s = t$. So γ_α is an injective immersion of the line $\mathbb{R}$ into $\mathbb{T}^2$, and $H_\alpha = \gamma_\alpha(\mathbb{R})$ is a connected immersed Lie subgroup carried diffeomorphically by γ_α from $\mathbb{R}$. The image is *dense*. To see this, examine where the wind meets the vertical circle $\{\pi(0, y) \mid y \in \mathbb{R}\}$: the point $\gamma_\alpha(t) = \pi(t, t\alpha)$ lies on it precisely when $t \in \mathbb{Z}$, that is at the return times $t = n$ for $n \in \mathbb{Z}$, and the second coordinates of these return points are $\{n\alpha \bmod 1 : n \in \mathbb{Z}\}$. This set is dense in $\mathbb{R}/\mathbb{Z}$ because α is irrational: the rotation $y \mapsto y + \alpha$ of the circle has no finite orbit, so by pigeonhole two of its iterates lie within any prescribed distance, producing an arbitrarily small nonzero return, whose integer multiples then fall within that distance of every point. Density of the return points on one circle, together with the flow filling the transverse direction, makes H_α dense in all of $\mathbb{T}^2$.

Consequently H_α for irrational α is an immersed but *not* embedded Lie subgroup. Its manifold topology is that of $\mathbb{R}$ under γ_α, in which a bounded open interval is open, whereas in the subspace

topology from $\mathbb{T}^2$ every neighborhood of a point of H_α meets infinitely many far-apart pieces of the wind, since the image is dense. The two topologies disagree, and H_α is not locally closed in $\mathbb{T}^2$: its closure is all of $\mathbb{T}^2$, so H_α is a proper dense subgroup and cannot be a submanifold in the subspace topology.

The example is the fault line of the whole subject. The same one-dimensional subalgebra $\mathfrak{h}_\alpha = \mathbb{R}\cdot(1,\alpha)$ integrates to a circle for rational α and to a dense line for irrational α, and no purely infinitesimal datum can tell the two apart: the subalgebra is the same one-dimensional space in both cases, and the integrating subgroup exists in both cases by the same construction. What differs is a global, arithmetic fact about the slope, invisible to the Lie algebra. This is why the integration theorem of section 3.3 can promise only an immersed subgroup, and why the immersed-versus-embedded distinction of definition 3.1.1 had to be built in from the start.

The example also motivates the theorem that dominates the next section. Immersed subgroups can be topologically wild, dense in the ambient group, neither closed nor locally closed. The remedy is to impose a topological hypothesis and watch it force geometric regularity. The closed-subgroup theorem of section 3.2 does exactly this: a subgroup that is *closed* as a subset of G, with no smoothness assumed at all, is automatically an embedded Lie subgroup. The dense line H_α is not closed, so it is not a counterexample; and the theorem explains why the classical groups, defined as closed subsets of $\mathrm{GL}_n(\mathbb{F})$ cut out by equations, were embedded submanifolds without any winding pathology. The usefulness of the closed-subgroup theorem is measured precisely against how badly immersed subgroups can behave, and H_α is the measure.

Exercise 3.1.1

1. Let $\varphi\colon G \to H$ be a homomorphism of Lie groups. Show that the image $\varphi(G)$ is a subgroup of H, and that it carries the structure of an immersed Lie subgroup of H with Lie algebra $d\varphi_e(\mathfrak{g}) \subseteq \mathfrak{h}$. (Realize $\varphi(G)$ as the image of the induced injective homomorphism $G/\ker\varphi \to H$; you may cite the closed-subgroup theorem of section 3.2 to know that $\ker\varphi$ is an embedded Lie subgroup and $G/\ker\varphi$ a Lie group.) When is the image embedded?

2. Show that a subgroup $H \subseteq G$ that is also an embedded submanifold of G is a Lie subgroup, that is, that the subspace smooth structure makes H a Lie group and the inclusion a smooth immersion. (The one thing to verify is that multiplication and inversion on H are smooth for the subspace structure; use that a smooth map into G whose image lies in an embedded submanifold is smooth as a map into that submanifold.)

3. Verify directly that $\mathfrak{sl}_n(\mathbb{F})$ and $\mathfrak{so}(n)$ are Lie subalgebras of $\mathfrak{gl}_n(\mathbb{F})$, recovering the computations of example 3.1.3, and confirm that these are the Lie algebras of the embedded Lie subgroups $\mathrm{SL}(n,\mathbb{F})$ and $\mathrm{SO}(n)$ named in theorem 1.4.3. Then show over $\mathbb{R}$ that $\mathfrak{so}(n) \subseteq \mathfrak{sl}_n(\mathbb{R})$, and interpret this inclusion of subalgebras as the inclusion of subgroups $\mathrm{SO}(n) \subseteq \mathrm{SL}(n,\mathbb{R})$.

4. For the irrational line H_α of example 3.1.5, show that H_α is not a closed subset of $\mathbb{T}^2$ by exhibiting a point of $\mathbb{T}^2$ in the closure of H_α but not in H_α. Conclude that the closure $\overline{H_\alpha}$ is a larger subgroup than H_α, and identify it.

5. Inside $G = \mathbb{T}^2 \times \mathbb{T}^2$, describe the subalgebra $\mathfrak{h} = \{(v,v) \mid v \in \mathbb{R}^2\}$ of the diagonal, and the subgroup it integrates to. Is the diagonal subgroup embedded? Contrast with the subalgebra $\{(v,Av) \mid v \in \mathbb{R}^2\}$, where $A = \mathrm{diag}(1,\lambda)$ scales the second coordinate by an irrational factor λ, and explain which feature produces a dense immersed subgroup.

3.2 The Closed Subgroup Theorem

Section 3.1 drew the distinction between an immersed Lie subgroup and an embedded one, and example 3.1.5 showed the gap is real: the irrational-slope winding of a line on the torus is an immersed subgroup that is not embedded, its image dense rather than a submanifold. The example carries a warning, that a subgroup of a Lie group need be no manifold at all in any natural way, and it isolates the culprit, which is that the image of the winding is not closed. Cartan's theorem, the subject of this section, says that closedness is the whole story. A subgroup of a Lie group that happens to be a closed subset is automatically an embedded submanifold, hence a Lie group in its own right, with no further hypothesis. This is a statement about topology forcing smooth structure, and it has no counterpart for manifolds in general.

The force of the theorem is best measured against how much it gives for how little. To exhibit a subgroup as an embedded Lie subgroup by hand, one would produce submanifold charts, verify the group operations are smooth in those charts, and match the smooth structure to the subspace topology. Cartan's theorem replaces all of that with a single point-set condition. Every classical group of section 1.2 is closed in $\mathrm{GL}_n(\mathbb{F})$ by inspection, being the zero set of continuous matrix equations, so the theorem subsumes the entire manifold apparatus of chapter 1 under one hypothesis, and its companion proposition 3.2.2 recovers the concrete definition of the matrix Lie algebra intrinsically. The proof organizes itself around a single object, the set of infinitesimal directions along which H can be approached from the identity.

The candidate for the Lie algebra of H is forced by the requirement that exp carry it into H. Mirroring the matrix definition definition 1.4.1, one collects exactly those tangent vectors whose one-parameter subgroup never leaves H.

Theorem 3.2.1: The Closed Subgroup Theorem (Cartan)

Let G be a Lie group and let $H \subseteq G$ be a subgroup that is closed as a subset of G. Then H is an embedded Lie subgroup of G: it is an embedded smooth submanifold, and with the induced smooth structure it is a Lie group whose inclusion into G is a smooth embedding. Its tangent space at the identity is the linear subspace

$$\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

of $\mathfrak{g}$, and exp carries a neighborhood of 0 in $\mathfrak{h}$ diffeomorphically onto a neighborhood of e in H.

Before the proof, the shape of the argument. The set $\mathfrak{h}$ is defined by a condition on all of $\mathbb{R}$, so its linearity is not visible from the definition; the first task is to show $\mathfrak{h}$ is a linear subspace of $\mathfrak{g}$, and the mechanism is the same Lie product formula that carried the linear structure onto the matrix Lie algebra in proposition 1.4.2, now with the closedness of H giving the step that closedness of the matrix group gave there. The heart of the matter is a lemma that identifies $\mathfrak{h}$ with the directions along which H accumulates at e: any sequence in H approaching the identity approaches it along a direction lying in $\mathfrak{h}$. The second task is to promote this infinitesimal information into a chart. Choosing a vector-space complement $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$, the map $(X, Y) \mapsto \exp X \exp Y$ is a local diffeomorphism at the origin, and the lemma forces its H-part to be exactly the $\mathfrak{h}$-slice near e, which is the submanifold chart. Left translation then spreads the chart over all of H.

***Proof*:**
Fix an inner product on $\mathfrak{g}$ and write $\|\cdot\|$ for the associated norm; because $\mathfrak{g}$ is finite-dimensional the choice is immaterial, and continuity or convergence statements below are independent of it. By theorem 2.3.4 there are open neighborhoods U_0 of 0 in $\mathfrak{g}$ and V_0 of e in G with $\exp\colon U_0 \to V_0$ a diffeomorphism; write $\log\colon V_0 \to U_0$ for its inverse. Shrink U_0 to an open ball about 0 so that exp restricted to it is a diffeomorphism onto its image.

Step 1: the key lemma on accumulation directions. The claim is the following. Suppose $(X_n)_{n\geq 1}$ is a sequence in $\mathfrak{g}$ with $X_n \to 0$, $X_n \neq 0$, and $\exp(X_n) \in H$ for every n, and suppose the normalized directions converge,

$$\frac{X_n}{\|X_n\|} \longrightarrow X \in \mathfrak{g}$$

Then $X \in \mathfrak{h}$, that is $\exp(tX) \in H$ for every $t \in \mathbb{R}$.

Fix $t \in \mathbb{R}$. For each n set $k_n = \lfloor t/\|X_n\| \rfloor$, the greatest integer not exceeding $t/\|X_n\|$, so that $k_n\|X_n\|$ differs from t by less than $\|X_n\|$. Then

$$k_n X_n = \big(k_n \|X_n\|\big)\, \frac{X_n}{\|X_n\|}$$

Since $k_n\|X_n\| \to t$ and $X_n/\|X_n\| \to X$, the product converges, $k_n X_n \to tX$. The exponential is continuous, so $\exp(k_n X_n) \to \exp(tX)$. But $\exp(k_n X_n) = \exp(X_n)^{k_n}$ by the one-parameter subgroup law (proposition 2.3.3), and $\exp(X_n) \in H$, so $\exp(X_n)^{k_n} \in H$ because H is a subgroup. Thus $\exp(tX)$ is a limit of a sequence in H, and H is closed, so $\exp(tX) \in H$. As t was arbitrary, $X \in \mathfrak{h}$, proving the lemma.

The lemma is where closedness is used: it converts a sequence of discrete group elements shrinking toward e into a genuine one-parameter subgroup lying in H, by riding the integer multiples $k_n X_n$ up to the target tX and closing the limit inside H. Every subsequent step uses it.

Step 2: $\mathfrak{h}$ is a linear subspace of $\mathfrak{g}$. That $0 \in \mathfrak{h}$ is immediate, since $\exp(t \cdot 0) = e \in H$ for all t. For closure under real scalars, let $X \in \mathfrak{h}$ and $\lambda \in \mathbb{R}$; then $\exp(t\lambda X) = \exp((t\lambda)X) \in H$ for every t because $t\lambda \in \mathbb{R}$ and $X \in \mathfrak{h}$, so $\lambda X \in \mathfrak{h}$. Closure under addition is the point where the nonlinearity of exp must be handled, and the tool is the Lie product formula, the intrinsic analogue of the matrix formula that carried the linear structure onto the matrix Lie algebra in proposition 1.4.2. It holds in the form

$$\exp(X+Y) = \lim_{m\to\infty} \Big(\exp\Big(\tfrac{X}{m}\Big)\exp\Big(\tfrac{Y}{m}\Big)\Big)^m \tag{3.2}$$

for all $X, Y \in \mathfrak{g}$, and is proved in any Lie group directly from the exponential chart of theorem 2.3.4. Fix $X, Y \in \mathfrak{g}$ and set $a_m = \exp(X/m)\exp(Y/m) \in G$. Since $X/m, Y/m \to 0$ and exp is continuous, $a_m \to e$, so $a_m \in V_0$ for m large and $Z_m := \log(a_m) \in U_0$ is defined with $Z_m \to \log(e) = 0$. The map $m \mapsto \log\big(\exp(X/m)\exp(Y/m)\big)$ extends to a smooth function of the real parameter $s = 1/m$ near $s = 0$, namely $s \mapsto \log\big(\exp(sX)\exp(sY)\big)$, whose value at $s = 0$ is 0; its derivative there is computed by the chain rule from $d(\exp)_0 = \mathrm{id}$ (theorem 2.3.4), the product rule at e, and $d(\log)_e = (d(\exp)_0)^{-1} = \mathrm{id}$, giving

$$\frac{d}{ds}\bigg|_{s=0} \log\big(\exp(sX)\exp(sY)\big) = X + Y$$

Hence $Z_m = \log(a_m) = \frac{1}{m}(X+Y) + o(1/m)$ as $m \to \infty$, so $mZ_m \to X+Y$. Now $a_m = \exp(Z_m)$, and by the one-parameter subgroup law (proposition 2.3.3) $a_m^m = \exp(Z_m)^m = \exp(mZ_m)$; continuity of exp then gives $a_m^m = \exp(mZ_m) \to \exp(X+Y)$, which is (3.2).

With the formula in hand, closure under addition follows exactly as in the matrix case. Let $X, Y \in \mathfrak{h}$ and fix $t \in \mathbb{R}$. Applying (3.2) to tX and tY,

$$\exp\big(t(X+Y)\big) = \lim_{m\to\infty} \Big(\exp\Big(\tfrac{tX}{m}\Big)\exp\Big(\tfrac{tY}{m}\Big)\Big)^m$$

For each fixed m the elements $\exp(tX/m)$ and $\exp(tY/m)$ lie in H, since $tX, tY \in \mathfrak{h}$ and $t/m \in \mathbb{R}$; their product raised to the mth power is a product of elements of the subgroup H, hence lies in H. The left side is therefore a limit of elements of H, and since H is closed, $\exp(t(X+Y)) \in H$. As t was arbitrary, $X + Y \in \mathfrak{h}$. Thus $\mathfrak{h}$ is a linear subspace of $\mathfrak{g}$; in particular it has a finite dimension, say $k = \dim \mathfrak{h}$, and admits a vector-space complement $\mathfrak{m}$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$.

Step 3: $\mathfrak{h}$ captures every accumulation direction of H at e. The claim, which the chart construction will use in contrapositive form, is that

$$\text{if } g_n \in H,\ g_n \to e,\ g_n \neq e,\ \text{then } \log(g_n) \text{ has all its accumulation directions in } \mathfrak{h} \tag{3.3}$$

Precisely: for $g_n \in H$ with $g_n \to e$ and $g_n \neq e$, write $Z_n = \log(g_n) \in \mathfrak{g}$ for n large (so that $g_n \in V_0$); then $Z_n \to 0$, $Z_n \neq 0$, and any subsequential limit of the unit vectors $Z_n/\|Z_n\|$ lies in $\mathfrak{h}$. Indeed $\exp(Z_n) = g_n \in H$, and $Z_n = \log(g_n) \to \log(e) = 0$ by continuity of log on V_0; moreover $Z_n \neq 0$ because exp is injective on U_0 and $g_n \neq e = \exp(0)$. The unit vectors $Z_n/\|Z_n\|$ lie on the compact unit sphere of $\mathfrak{g}$, so any subsequence has a further subsequence converging to some unit vector Z; along that subsequence the hypotheses of Step 1 hold, whence $Z \in \mathfrak{h}$. This proves (3.3).

Step 4: the exponential of $\mathfrak{m}$ meets H only at e near the origin. The claim is that there is an open neighborhood W of 0 in $\mathfrak{m}$ such that

$$\exp(Y) \in H \text{ and } Y \in W \implies Y = 0 \tag{3.4}$$

Suppose not. Then there is a sequence $Y_n \in \mathfrak{m}$ with $Y_n \neq 0$, $Y_n \to 0$, and $\exp(Y_n) \in H$. Applying Step 1 to the sequence $X_n = Y_n$ (its exponentials lie in H, it tends to 0, and it is nonzero), pass to a subsequence along which $Y_n/\|Y_n\|$ converges to a unit vector $Y \in \mathfrak{g}$; Step 1 gives $Y \in \mathfrak{h}$. But each $Y_n \in \mathfrak{m}$, so $Y_n/\|Y_n\| \in \mathfrak{m}$, and $\mathfrak{m}$ is closed as a linear subspace, so $Y \in \mathfrak{m}$. Then $Y \in \mathfrak{h} \cap \mathfrak{m} = \{0\}$, contradicting $\|Y\| = 1$. Hence no such sequence exists, which is (3.4).

Step 5: the submanifold chart at e. Consider the map

$$\Psi\colon \mathfrak{h} \oplus \mathfrak{m} \to G, \qquad \Psi(X, Y) = \exp(X)\exp(Y)$$

It is smooth, and its differential at $(0,0)$ sends (X, Y) to $X+Y$: differentiating $\exp(sX)\exp(0) = \exp(sX)$ in s gives X in the first slot, differentiating $\exp(0)\exp(sY) = \exp(sY)$ gives Y in the second, and the product rule at the identity adds them, using $d(\exp)_0 = \mathrm{id}$ from theorem 2.3.4. Under the identification $\mathfrak{h} \oplus \mathfrak{m} = \mathfrak{g}$ this differential is the identity, so by the inverse function theorem ([13, Inverse Function Theorem for Manifolds]) there are open balls $B_{\mathfrak{h}} \subseteq \mathfrak{h}$ and $B_{\mathfrak{m}} \subseteq \mathfrak{m}$ about the origins, with $B_{\mathfrak{m}} \subseteq W$ (the set of Step 4), such that Ψ restricts to a diffeomorphism of $B_{\mathfrak{h}} \times B_{\mathfrak{m}}$ onto an open neighborhood V of e in G. Shrinking further if needed, arrange also that $\exp(B_{\mathfrak{h}}) \subseteq H$, which is possible because $B_{\mathfrak{h}} \subseteq \mathfrak{h}$ and hence $\exp(X) \in H$ for every $X \in B_{\mathfrak{h}}$ by definition of $\mathfrak{h}$.

The chart is $\Psi^{-1}\colon V \to B_{\mathfrak{h}} \times B_{\mathfrak{m}}$, and the claim is that it carries $H \cap V$ onto the slice $B_{\mathfrak{h}} \times \{0\}$. One inclusion is immediate: if $X \in B_{\mathfrak{h}}$ then $\Psi(X, 0) = \exp(X) \in H$, so $\Psi(B_{\mathfrak{h}} \times \{0\}) \subseteq H \cap V$.

For the reverse, let $h \in H \cap V$ and write $h = \Psi(X, Y) = \exp(X)\exp(Y)$ with $(X, Y) \in B_{\mathfrak{h}} \times B_{\mathfrak{m}}$. Then $\exp(X) \in H$ (as $X \in B_{\mathfrak{h}} \subseteq \mathfrak{h}$), and $h \in H$, so

$$\exp(Y) = \exp(X)^{-1} h \in H$$

because H is a subgroup. Now $Y \in B_{\mathfrak{m}} \subseteq W$ and $\exp(Y) \in H$, so (3.4) forces $Y = 0$. Hence $h = \exp(X) = \Psi(X, 0)$ lies in the slice. Therefore

$$\Psi^{-1}(H \cap V) = B_{\mathfrak{h}} \times \{0\} \tag{3.5}$$

which is exactly a submanifold chart for H at e: in the coordinates Ψ^{-1}, the subset H is cut out by the vanishing of the $\mathfrak{m}$-coordinate. This shows H is an embedded k-dimensional submanifold near e, with tangent space $\mathfrak{h}$ at e, and that exp maps $B_{\mathfrak{h}}$ diffeomorphically onto the neighborhood $\exp(B_{\mathfrak{h}}) = H \cap V$ of e in H.

Step 6: spreading the chart by left translation, and the Lie group structure. The construction of Step 5 produced a submanifold chart only at e, but the homogeneity of a group upgrades it to a chart at every point. Let $h_0 \in H$ be arbitrary. Left translation $L_{h_0} \colon G \to G$ is a diffeomorphism (proposition 2.1.2) carrying e to h_0, and it carries H to H because H is a subgroup, so $L_{h_0}(H \cap V) = (h_0 H) \cap (h_0 V) = H \cap (h_0 V)$. Thus $L_{h_0} \circ \Psi^{-1}$, restricted appropriately, is a submanifold chart for H on the neighborhood $h_0 V$ of h_0, with the same slice description (3.5) transported by the diffeomorphism L_{h_0}. Every point of H therefore has a submanifold chart, so H is an embedded smooth submanifold of G of dimension $k = \dim \mathfrak{h}$, and the subspace topology on H agrees with its manifold topology.

It remains to see that H with this smooth structure is a Lie group. The multiplication $m \colon G \times G \to G$ and inversion $i \colon G \to G$ are smooth, and they restrict to maps $H \times H \to H$ and $H \to H$ because H is a subgroup. Since H is an embedded submanifold, a smooth map from a manifold into G whose image lies in H is smooth as a map into H ([13, Corestriction to a Regular Submanifold]); applying this to $m|_{H \times H}$ and $i|_H$, which are smooth into G with image in H, shows the group operations of H are smooth for its own manifold structure. Hence H is a Lie group, and the inclusion $H \hookrightarrow G$ is a smooth embedding by construction, completing the proof.

The theorem is the mechanism that makes Lie theory a theory of groups rather than of specially prepared manifolds. It says the category of Lie groups is closed under the most basic algebraic operation, passing to a closed subgroup, with no analytic bookkeeping required of the user. The contrast with general manifold theory is sharp: a closed subset of a manifold that happens to be an image of some group need not be a submanifold, and it is exactly the compatibility of the algebraic structure (subgroup) with the point-set structure (closed) that forces smoothness. Nothing like this holds for an arbitrary closed subset. The engine is the accumulation lemma of Step 1, which turns the failure of H to be open, its elements crowding toward e, into positive information about the tangent directions H occupies.

The subalgebra $\mathfrak{h}$ that emerged in the proof is more than a tangent space; it is a Lie subalgebra of $\mathfrak{g}$, and it deserves to be recorded as the Lie algebra of the closed subgroup H. Its defining formula is identical in shape to the matrix definition of chapter 1, and identifying the two is what ties the intrinsic theory back to the concrete computations that opened the book.

Proposition 3.2.2: The Lie Algebra of a Closed Subgroup

Let H be a closed subgroup of a Lie group G, an embedded Lie subgroup by theorem 3.2.1. Its Lie algebra, the tangent space T_eH with the bracket inherited from $\mathfrak{g}$, is

$$\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

and this is a Lie subalgebra of $\mathfrak{g}$. When $G = \mathrm{GL}_n(\mathbb{F})$ and $H \subseteq \mathrm{GL}_n(\mathbb{F})$ is a closed matrix group, $\mathfrak{h}$ coincides with the matrix Lie algebra of definition 1.4.1.

***Proof*:**

The equality $T_eH = \mathfrak{h}$ was established in Step 5 of theorem 3.2.1: the chart Ψ^{-1} identifies T_eH with the slice direction $\mathfrak{h}$. That $\mathfrak{h}$ is a Lie subalgebra, that is closed under the bracket of $\mathfrak{g}$, is the general fact (proposition 3.1.4) that the Lie algebra of an embedded Lie subgroup is a subalgebra: the inclusion $H \hookrightarrow G$ is a Lie group homomorphism, and its differential at e is the injective Lie algebra homomorphism $\mathfrak{h} \hookrightarrow \mathfrak{g}$, whose image is therefore closed under the bracket.

For the last claim, let $G = \mathrm{GL}_n(\mathbb{F})$, so that $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ and the intrinsic exponential of G is the matrix exponential (proposition 2.3.6). For a closed matrix group $H \subseteq \mathrm{GL}_n(\mathbb{F})$, the set $\mathfrak{h}$ above reads

$$\mathfrak{h} = \{X \in M_n(\mathbb{F}) \mid \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

which is verbatim the definition of the matrix Lie algebra in definition 1.4.1. The two constructions therefore agree, as we sought to show.

The proposition closes the loop opened in chapter 1. There the Lie algebra of a matrix group was defined by fiat as the set of matrices whose one-parameter subgroup stays in G, a definition justified only by its usefulness. Cartan's theorem now shows that this ad hoc formula is the intrinsic Lie algebra T_eH of H as a manifold, computed with no reference to matrices, and that the matrix formula is its specialization to $G = \mathrm{GL}_n(\mathbb{F})$. The classical groups of section 1.2 are all closed in $\mathrm{GL}_n(\mathbb{F})$, so each of them is covered twice over, once by the explicit regular-value construction of Chapter 1 and once by the single hypothesis of theorem 3.2.1, and the two computations of the Lie algebra necessarily agree.

A homomorphism's kernel is the closed subgroup met most often in practice, and its Lie algebra has a name-worthy computation.

Corollary 3.2.3: The Lie Algebra of a Kernel

Let $\varphi\colon G \to H$ be a homomorphism of Lie groups. Then $\ker\varphi$ is a closed normal subgroup of G, an embedded Lie subgroup, and its Lie algebra is

$$T_e(\ker\varphi) = \ker d\varphi_e$$

***Proof*:**

The kernel is the preimage of the closed set $\{e_H\}$ under the continuous φ, hence a closed normal subgroup, and an embedded Lie subgroup by theorem 3.2.1. By proposition 3.2.2, a vector $X \in \mathfrak{g}$ lies in the Lie algebra of $\ker\varphi$ if and only if $\exp_G(tX) \in \ker\varphi$ for all t, that is, $\varphi(\exp_G(tX)) = e_H$ for all t. By naturality of the exponential (theorem 2.4.3) this reads $\exp_H(t\,d\varphi_e X) = e_H$ for

all t. If $d\varphi_e X = 0$ the condition holds trivially. Conversely, $\exp_H$ is a diffeomorphism near 0 (theorem 2.3.4), so $\exp_H(t\, d\varphi_e X) = e_H$ for all small t forces $t\, d\varphi_e X = 0$ and hence $d\varphi_e X = 0$, completing the proof.

The group SU(2) was built in Chapter 1 as a matrix group and shown to be diffeomorphic to S^3; it is a closed subgroup of $\mathrm{GL}_2(\mathbb{C})$, so Cartan's theorem applies and returns its Lie algebra intrinsically.

Example 3.2.4: SU(2) and SO(3) as Closed Subgroups

The unitary group U(2) is closed in $\mathrm{GL}_2(\mathbb{C})$, being the set of matrices g with $g^* g = I$, a system of continuous equations in the entries; and $\mathrm{SU}(2) = \{g \in \mathrm{U}(2) : \det g = 1\}$ is closed in U(2), hence in $\mathrm{GL}_2(\mathbb{C})$, as the further zero set of the continuous function $\det - 1$. By theorem 3.2.1, SU(2) is an embedded Lie subgroup of $\mathrm{GL}_2(\mathbb{C})$, and by proposition 3.2.2 its Lie algebra is

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) \mid \exp(tX) \in \mathrm{SU}(2) \text{ for all } t\}$$

Differentiating the two defining conditions $\exp(tX)^* \exp(tX) = I$ and $\det \exp(tX) = 1$ at $t = 0$ returns $X^* + X = 0$ and $\mathrm{tr} X = 0$, the traceless skew-Hermitian matrices already found in theorem 1.4.3, a three-dimensional real space matching $\dim \mathrm{SU}(2) = \dim S^3 = 3$. The same argument applies to SO(3), closed in $\mathrm{GL}_3(\mathbb{R})$ as the set of g with $g^{\mathsf{T}} g = I$ and $\det g = 1$, whose Lie algebra $\mathfrak{so}(3)$ is the skew-symmetric 3×3 matrices. The double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5 is then a homomorphism between two closed subgroups, and it induces the isomorphism $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ of their Lie algebras, both being the cross-product algebra on $\mathbb{R}^3$; the groups differ only by the central kernel $\{\pm I\}$, invisible to the Lie algebra.

The example exhibits the theorem doing exactly the work chapter 1 did by hand, now for free. What required a regular-value computation there, verifying that SU(2) is a submanifold and extracting its tangent space, is here the single observation that SU(2) is a closed subset of a Lie group. The Lie algebra comes out identical because proposition 3.2.2 guarantees it.

A second consequence of Cartan's theorem is about maps rather than subgroups, and it is of the same character: a purely topological hypothesis forces smoothness. A homomorphism between Lie groups that is continuous, with no assumption of differentiability, is automatically smooth. The proof uses the same device as the theorem itself, realizing the map as a closed subgroup.

Corollary 3.2.5: Continuous Homomorphisms Are Smooth

Let G and H be Lie groups and let $\varphi\colon G \to H$ be a continuous group homomorphism. Then φ is smooth, and hence a homomorphism of Lie groups in the sense of definition 2.1.4.

***Proof*:**

Consider the graph of φ,

$$\Gamma = \{(g, \varphi(g)) \mid g \in G\} \subseteq G \times H$$

inside the product Lie group $G \times H$. It is a subgroup: since φ is a homomorphism, $(g, \varphi(g))(g', \varphi(g')) = (gg', \varphi(g)\varphi(g')) = (gg', \varphi(gg')) \in \Gamma$, and $(e, e) \in \Gamma$, and $(g, \varphi(g))^{-1} = (g^{-1}, \varphi(g)^{-1}) = (g^{-1}, \varphi(g^{-1})) \in \Gamma$. It is closed: Γ is the preimage of the diagonal $\Delta_H = \{(h, h) : h \in H\} \subseteq H \times H$ under the continuous map $G \times H \to H \times H$, $(g, h) \mapsto (\varphi(g), h)$, and Δ_H is closed because H is Hausdorff. Being a closed subgroup of $G \times H$, the graph Γ is an embedded Lie subgroup by theorem 3.2.1.

Now consider the two coordinate projections restricted to Γ,

$$\pi_G \colon \Gamma \to G, \quad (g, \varphi(g)) \mapsto g, \qquad \pi_H \colon \Gamma \to H, \quad (g, \varphi(g)) \mapsto \varphi(g)$$

Both are smooth, as restrictions of the smooth projections of the product to the embedded submanifold Γ. The first, π_G, is a smooth bijective homomorphism of Lie groups: it is injective because the first coordinate determines the pair $(g, \varphi(g))$, and surjective because every g has the image $\varphi(g)$. Its set-theoretic inverse is $g \mapsto (g, \varphi(g))$.

First, $\dim \Gamma = \dim G$. Indeed π_G is a continuous bijection whose set-theoretic inverse $g \mapsto (g, \varphi(g))$ is continuous, by continuity of φ; so π_G is a homeomorphism, and an embedded submanifold Γ of $G \times H$ homeomorphic to G has G's dimension by invariance of dimension.

Second, $d(\pi_G)_{(e,e)}$ is injective. This is the Lie-group fact that an injective homomorphism of Lie groups has injective differential at e: the kernel of $d(\pi_G)_{(e,e)}$ is the Lie algebra of the closed subgroup $\ker \pi_G$ (corollary 3.2.3), and $\ker \pi_G = \{(e, e)\}$ because π_G is injective, so this Lie algebra is $\{0\}$ and $d(\pi_G)_{(e,e)}$ has trivial kernel. An injective linear map between spaces of equal dimension is an isomorphism, so by $\dim \Gamma = \dim G$ the differential $d(\pi_G)_{(e,e)}$ is a linear isomorphism, and left-translating shows the differential is an isomorphism at every point, so π_G is a diffeomorphism by the inverse function theorem ([13, Global Rank Theorem]). Therefore $\pi_G^{-1} \colon G \to \Gamma$, $g \mapsto (g, \varphi(g))$, is smooth. Composing,

$$\varphi = \pi_H \circ \pi_G^{-1}$$

is a composite of smooth maps, hence smooth, completing the proof.

The corollary removes smoothness from the definition of a homomorphism of Lie groups. Whatever category one starts in, continuous or smooth homomorphisms of Lie groups are the same maps, so the passage from topology to differential geometry is automatic at the level of morphisms as well as objects. The device is the same one that powers Cartan's theorem: an object one wants to control, here a bare continuous map, is repackaged as a closed subgroup, whose smooth structure is then handed over by theorem 3.2.1. The graph device will recur in the next section, where the graph of a Lie algebra homomorphism inside $\mathfrak{g} \oplus \mathfrak{h}$ integrates to the graph of the group homomorphism it induces.

Taken together, the closed subgroup theorem and its corollary express a rigidity peculiar to Lie groups. Neither subgroups nor homomorphisms carry a smoothness hypothesis: closedness of a subset and continuity of a map, both purely topological, suffice to produce the full differential-geometric structure. There is no analogue for manifolds in general, where a closed subset or a continuous map is under no obligation to be smooth. The reason is that a Lie group's smooth structure is rigidly determined near the identity by its algebraic structure, through the exponential chart, and this rigidity propagates: the algebra of the group leaves the topology no freedom.

Exercise 3.2.1

1. Show that $\mathrm{O}(n)$, $\mathrm{U}(n)$, and $\mathrm{Sp}(2n, \mathbb{R})$ are closed subsets of the appropriate $\mathrm{GL}_N(\mathbb{F})$, and conclude from theorem 3.2.1 that each is an embedded Lie subgroup. Recover in each case the Lie algebra of theorem 1.4.3 from proposition 3.2.2 by differentiating the defining equations along $\exp(tX)$ at $t = 0$.
2. Let S be a subgroup of a Lie group G (not assumed closed). Prove that its topological closure

$\overline{S}$ is again a subgroup of G, and hence, by theorem 3.2.1, an embedded Lie subgroup. Deduce that the closure of the irrational-slope one-parameter subgroup of the torus $\mathbb{T}^2$ (example 3.1.5) is all of $\mathbb{T}^2$.

3. Let H be a closed subgroup of a connected Lie group G, and suppose H has countable index in G (that is, G/H is countable). Prove that $H = G$. (Hint: G is a countable union of the cosets of H; use theorem 3.2.1 to compare dimensions, and a Baire-category argument to rule out $\dim H < \dim G$.)

4. Use corollary 3.2.5 to show that the only continuous automorphisms of the additive group $(\mathbb{R}, +)$ are the maps $x \mapsto cx$ for $c \in \mathbb{R}^\times$. Contrast with the existence of discontinuous additive maps $\mathbb{R} \to \mathbb{R}$, and explain why the continuity hypothesis cannot be dropped even though the conclusion of smoothness is automatic once continuity is present.

5. Prove that the center $Z(G)$ of a Lie group G is a closed subgroup, hence an embedded Lie subgroup, and identify its Lie algebra as $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \text{ for all } Y \in \mathfrak{g}\}$ when G is connected. (Use the characterization $\ker \mathrm{Ad} = Z(G)$ for connected G from exercise 2.5.1 (2).)

6. Explain precisely which hypothesis of theorem 3.2.1 the irrational-slope one-parameter subgroup of $\mathbb{T}^2$ (example 3.1.5) violates, and identify the step of the proof that consequently fails. Why do the accumulation directions of this subgroup at the identity fill out all of $\mathfrak{g}$ rather than a one-dimensional line?

3.3 Homomorphisms and Lie's Integration Theorems

The closed subgroup theorem of section 3.2 answered a question about *closed* subsets: a closed subgroup of a Lie group is automatically a submanifold, hence carries a Lie algebra $\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H \ \forall t\}$ (proposition 3.2.2). This section runs the correspondence in the reverse direction and drops the closedness. Starting from a Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, an algebraic object with no a priori geometric content, the goal is to manufacture a subgroup $H \leq G$ whose Lie algebra is exactly $\mathfrak{h}$. The subgroup produced need not be closed, as the irrational line of example 3.1.5 already warned, so the natural category is that of immersed Lie subgroups (definition 3.1.1). The mechanism is the Frobenius theorem: a subalgebra $\mathfrak{h}$ determines a left-invariant distribution on G, and the bracket-closure of $\mathfrak{h}$ is precisely the involutivity that Frobenius needs to integrate that distribution to a foliation, one of whose leaves is the sought subgroup.

Once subalgebras integrate to subgroups, the same device integrates *homomorphisms*. A Lie algebra homomorphism $\psi\colon \mathfrak{g} \to \mathfrak{h}$ has a graph, a subalgebra of $\mathfrak{g} \oplus \mathfrak{h}$; the subgroup integrating that graph is the graph of the group homomorphism sought, provided the graph subgroup projects isomorphically onto G. That projection is a covering map, and simple connectivity of G is exactly what forces a covering of G to be trivial. This is Lie's second theorem, and it upgrades the faithfulness of the Lie functor (corollary 2.4.4) to a bijection $\mathrm{Hom}(G, H) \xrightarrow{\sim} \mathrm{Hom}(\mathfrak{g}, \mathfrak{h})$ for G simply connected. The object-level companion, that every Lie algebra is $\mathrm{Lie}(G)$ for some G, is Lie's third theorem and is deferred to section 3.4, where the same integration is applied one last time to the adjoint representation.

The starting observation is that a subspace of $\mathfrak{g}$ spreads out over all of G by left translation, producing a field of tangent subspaces, that is a distribution. Everything below is organized around it.

Definition 3.3.1: The Left-Invariant Distribution of a Subspace

Let G be a Lie group with Lie algebra $\mathfrak{g} = T_eG$, and let $\mathfrak{h} \subseteq \mathfrak{g}$ be a linear subspace of dimension k. The *left-invariant distribution* $D^{\mathfrak{h}}$ determined by $\mathfrak{h}$ assigns to each $g \in G$ the k-dimensional subspace of T_gG

$$D^{\mathfrak{h}}_g = (dL_g)_e\, \mathfrak{h} \subseteq T_gG$$

the image of $\mathfrak{h}$ under the differential of left translation $L_g \colon x \mapsto gx$. Equivalently, if $X_1, \dots, X_k$ is a basis of $\mathfrak{h}$ with left-invariant extensions $X_1^L, \dots, X_k^L$ (definition 2.2.1), then $D^{\mathfrak{h}}_g = \operatorname{span}\{(X_1^L)_g, \dots, (X_k^L)_g\}$.

The distribution $D^{\mathfrak{h}}$ is smooth because it is globally framed by the left-invariant fields $X_1^L, \dots, X_k^L$, which are smooth and, being the translates of a basis of $\mathfrak{h}$, linearly independent at every point (left translation is a diffeomorphism, so $(dL_g)_e$ is a linear isomorphism). A smooth k-plane field is exactly the data the Frobenius theorem operates on. The one hypothesis Frobenius requires is involutivity: the bracket of two vector fields tangent to the distribution must again be tangent to it. For a left-invariant distribution this hypothesis translates directly into an algebraic condition on $\mathfrak{h}$.

Proposition 3.3.2: Involutivity Equals Bracket-Closure

Let $\mathfrak{h} \subseteq \mathfrak{g}$ be a linear subspace and $D^{\mathfrak{h}}$ its left-invariant distribution (definition 3.3.1). Then $D^{\mathfrak{h}}$ is involutive, that is $[V, W]$ is a section of $D^{\mathfrak{h}}$ whenever V, W are, if and only if $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$ (definition 3.1.2).

***Proof*:**

Fix a basis $X_1, \dots, X_k$ of $\mathfrak{h}$ and let X_i^L be the left-invariant extensions, which frame $D^{\mathfrak{h}}$ by definition 3.3.1.

Step 1: bracket-closure of $\mathfrak{h}$ implies involutivity. Suppose $\mathfrak{h}$ is closed under the bracket, so $[X_i, X_j] \in \mathfrak{h}$ for all i, j. Under the identification of $\mathfrak{g}$ with left-invariant fields (theorem 2.2.3) the bracket on $\mathfrak{g}$ is computed by the bracket of the fields, $[X_i, X_j]^L = [X_i^L, X_j^L]$, and $[X_i, X_j] \in \mathfrak{h}$ means this field is a constant-coefficient combination of the frame, $[X_i^L, X_j^L] = \sum_\ell c_{ij}^\ell X_\ell^L$ with $c_{ij}^\ell \in \mathbb{R}$ the structure constants. Any two sections of $D^{\mathfrak{h}}$ have the form $V = \sum_i f_i X_i^L$ and $W = \sum_j g_j X_j^L$ with f_i, g_j smooth functions on G. Expanding the bracket by the Leibniz rule,

$$[V, W] = \sum_{i,j} f_i g_j [X_i^L, X_j^L] + \sum_{i,j} f_i (X_i^L g_j) X_j^L - \sum_{i,j} g_j (X_j^L f_i) X_i^L$$

The second and third sums are already sections of $D^{\mathfrak{h}}$, being combinations of the frame X_ℓ^L with smooth coefficients; and the first sum equals $\sum_{i,j,\ell} f_i g_j c_{ij}^\ell X_\ell^L$, again a section of $D^{\mathfrak{h}}$. Hence $[V, W]$ is a section of $D^{\mathfrak{h}}$, so $D^{\mathfrak{h}}$ is involutive.

Step 2: involutivity implies bracket-closure. Conversely suppose $D^{\mathfrak{h}}$ is involutive. The fields X_i^L are themselves sections of $D^{\mathfrak{h}}$, so $[X_i^L, X_j^L]$ is a section, that is $[X_i^L, X_j^L]_g \in D^{\mathfrak{h}}_g = (dL_g)_e \mathfrak{h}$ for every g. Evaluating at $g = e$ gives $[X_i^L, X_j^L]_e \in \mathfrak{h}$. But $[X_i^L, X_j^L]_e = [X_i, X_j]$ is the value at e of the bracket field, which is the bracket on $\mathfrak{g}$ by theorem 2.2.3. Thus $[X_i, X_j] \in \mathfrak{h}$ for all basis vectors, and by bilinearity $[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$, so $\mathfrak{h}$ is a subalgebra, completing the proof.

The proposition is the pivot on which the entire integration turns. It says that the algebraic condition defining a subalgebra, closure under the bracket, is not a formal accident but the exact geometric condition that lets the associated plane field be integrated into a foliation. The bracket, which Chapter 2 built as the infinitesimal record of noncommutativity, reappears here as the obstruction to fitting the subspaces $D^{\mathfrak{h}}_g$ together into submanifolds. When the obstruction vanishes, which is precisely when $\mathfrak{h}$ absorbs its own brackets, the subspaces knit into leaves, and the leaf through e will be the subgroup. The Frobenius theorem gives the knitting; the burden of proof here is entirely the group theory that follows, since the analytic integration is imported.

3.3.3 Note: The Frobenius theorem as a cited input The Frobenius theorem states that a smooth involutive distribution D of rank k on a manifold M is integrable: through each point $p \in M$ there is a unique maximal connected *integral manifold*, an immersed k-dimensional submanifold L everywhere tangent to D ($T_qL = D_q$ for all $q \in L$), and these leaves partition M into a foliation. Its proof, via the simultaneous straightening of commuting flows, belongs to the theory of smooth manifolds and is taken as given here; see [13, Frobenius Theorem, Global Frobenius Theorem]. Everything below uses Frobenius as a black box and proves the group-theoretic content in full.

With involutivity secured by proposition 3.3.2, the Frobenius theorem hands back the leaf through the identity. The work that remains is to show this leaf is a subgroup and that its intrinsic Lie algebra is the $\mathfrak{h}$ that was started with. Left-invariance of the distribution is what makes the leaf a subgroup: translating a leaf carries it to a leaf, and the translate through e is again the leaf through e.

Theorem 3.3.4: Integration of Subalgebras

Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $\mathfrak{h} \subseteq \mathfrak{g}$ be a Lie subalgebra (definition 3.1.2). Then there is a unique connected immersed Lie subgroup $H \leq G$ (definition 3.1.1) whose Lie algebra is $\mathfrak{h}$, that is $\mathrm{Lie}(H) = \mathfrak{h} \subseteq \mathfrak{g}$ under the inclusion of proposition 3.1.4. Concretely, H is the maximal connected integral manifold through e of the left-invariant distribution $D^{\mathfrak{h}}$ of definition 3.3.1.

***Proof*:**

By proposition 3.3.2 the distribution $D^{\mathfrak{h}}$ is smooth and involutive, so the Frobenius theorem (box 3.3.3, [13, Global Frobenius Theorem]) applies. Let H be the maximal connected integral manifold of $D^{\mathfrak{h}}$ through e; it is an immersed connected k-dimensional submanifold with $T_gH = D^{\mathfrak{h}}_g$ for all $g \in H$. The proof shows H is a subgroup, computes its Lie algebra, and establishes uniqueness.

Step 1: left translation permutes the leaves. The distribution $D^{\mathfrak{h}}$ is left-invariant: for any $a \in G$ the diffeomorphism L_a carries $D^{\mathfrak{h}}$ to itself, because

$$(dL_a)_g\, D^{\mathfrak{h}}_g = (dL_a)_g (dL_g)_e\, \mathfrak{h} = (dL_{ag})_e\, \mathfrak{h} = D^{\mathfrak{h}}_{ag}$$

using $L_a \circ L_g = L_{ag}$ and the chain rule. A diffeomorphism carrying a distribution to itself carries integral manifolds to integral manifolds and maximal connected ones to maximal connected ones. Hence $L_a(H)$ is the maximal connected integral manifold through $L_a(e) = a$; that is, the leaf through a is aH.

Step 2: H is a subgroup. Let $g \in H$. By Step 1 the leaf through g is gH, and it also equals H itself, because $g \in H$ and the leaves partition G into disjoint sets, so the unique leaf containing g is H; thus $gH = H$. Consequently for any $h \in H$ the product $gh \in gH = H$, so H is closed under multiplication. For inverses, $gH = H$ gives $e = g^{-1}g \in g^{-1}H$, so the leaf $g^{-1}H$ through g^{-1} contains e; the leaf through e is H, and distinct leaves are disjoint, so $g^{-1}H = H$, whence $g^{-1} = g^{-1}e \in H$. Thus H is a subgroup of G. The multiplication and inversion of G restrict to smooth maps on the immersed submanifold H, because $H \hookrightarrow G$ is an immersion whose image is a union of leaves and the group operations preserve the foliation by Step 1; this is the standard fact that a subgroup which is an integral manifold of a left-invariant distribution is a Lie subgroup ([13, Leaves Are Weakly Embedded]). Hence H is an immersed Lie subgroup in the sense of definition 3.1.1.

Step 3: the Lie algebra of H is $\mathfrak{h}$. By construction H is an integral manifold of $D^{\mathfrak{h}}$, so its tangent space at the identity is

$$T_eH = D^{\mathfrak{h}}_e = (dL_e)_e\,\mathfrak{h} = \mathfrak{h}$$

since $L_e = \mathrm{id}_G$. Under the inclusion of proposition 3.1.4, $\mathrm{Lie}(H) = T_eH$ is identified with a subalgebra of $\mathfrak{g}$, and the identification is compatible with the bracket because the inclusion $\iota\colon H \hookrightarrow G$ is a Lie group homomorphism whose differential $d\iota_e$ is the inclusion $T_eH = \mathfrak{h} \hookrightarrow \mathfrak{g}$; by proposition 2.4.2 this differential preserves brackets. Therefore $\mathrm{Lie}(H) = \mathfrak{h}$ as a subalgebra of $\mathfrak{g}$.

Step 4: uniqueness. Suppose $H' \leq G$ is another connected immersed Lie subgroup with $\mathrm{Lie}(H') = \mathfrak{h}$. Then $T_eH' = \mathfrak{h}$, and by left-invariance of the tangent bundle of a subgroup, for each $g \in H'$ the tangent space $T_gH' = (dL_g)_eT_eH' = (dL_g)_e\mathfrak{h} = D^{\mathfrak{h}}_g$. So H' is a connected integral manifold of $D^{\mathfrak{h}}$ containing e, hence contained in the maximal one, $H' \subseteq H$. Both are connected immersed subgroups of the same dimension k with the same tangent space at e; the inclusion $H' \hookrightarrow H$ is therefore an immersion between manifolds of equal dimension, an open map, so H' is an open subgroup of the connected group H. An open subgroup of a connected group is the whole group, since its cosets are open and partition H, forcing a single coset. Hence $H' = H$, establishing uniqueness and completing the proof.

The theorem completes the local dictionary between Lie groups and Lie algebras at the level of subobjects: subalgebras of $\mathfrak{g}$ correspond exactly to connected immersed subgroups of G. The correspondence is not perfect in the category of embedded subgroups, precisely because integration can produce a subgroup that fails to be closed, but in the category of immersed subgroups it is a clean bijection. The subgroup H is intrinsically determined by $\mathfrak{h}$ and is often called the *integral subgroup* or *analytic subgroup* generated by $\mathfrak{h}$; it may be described concretely as the subgroup generated by $\exp(\mathfrak{h})$, the smallest subgroup containing all one-parameter subgroups whose generators lie in $\mathfrak{h}$.

The irrational line shows the necessity of the immersed category. Its Lie algebra is a one-dimensional subalgebra of the abelian $\mathfrak{g} = \mathbb{R}^2$ of the torus, and integration returns exactly that dense, non-embedded winding line.

Example 3.3.5: The Irrational Line as an Integral Subgroup

Take $G = \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, whose Lie algebra $\mathfrak{g} = \mathbb{R}^2$ is abelian, so every linear subspace is a subalgebra. Fix an irrational α and let $\mathfrak{h} = \mathbb{R} \cdot (1, \alpha) \subseteq \mathbb{R}^2$, a one-dimensional subalgebra. The left-invariant distribution $D^{\mathfrak{h}}$ is translation-invariant, since the group is abelian: $D^{\mathfrak{h}}_g$ is the $(1, \alpha)$-direction at every g, the constant line field of slope α. Its integral manifold through 0 is the image of

$$\mathbb{R} \to \mathbb{T}^2, \qquad t \mapsto (t, \alpha t) \bmod \mathbb{Z}^2$$

the one-parameter subgroup H with generator $(1, \alpha)$. This is exactly the immersed, non-embedded subgroup of example 3.1.5: it is a subgroup, an injectively immersed copy of $\mathbb{R}$, and dense in $\mathbb{T}^2$ because α is irrational, so its subspace topology from $\mathbb{T}^2$ is strictly coarser than its manifold topology as $\mathbb{R}$. The integration theorem produces H with $\mathrm{Lie}(H) = \mathfrak{h}$, and the failure of H to be embedded is not a defect of the theorem but a feature of the subalgebra: no embedded subgroup of $\mathbb{T}^2$ has Lie algebra the irrational-slope line, because the only embedded connected subgroups are $\{0\}$, the rational circles, and $\mathbb{T}^2$ itself. The immersed category is forced.

For a matrix group the integral subgroup of a matrix subalgebra is generated by exponentials, matching the concrete picture of chapter 1. If $G \subseteq \mathrm{GL}_n(\mathbb{F})$ and $\mathfrak{h} \subseteq \mathfrak{g} \subseteq M_n(\mathbb{F})$ is a subalgebra of matrices closed under the commutator, the integral subgroup H is the subgroup generated by $\{\exp X : X \in \mathfrak{h}\}$, since these one-parameter subgroups are tangent to $\mathfrak{h}$ at e and generate a connected subgroup whose Lie algebra is $\mathfrak{h}$ by the same argument. When $\mathfrak{h} = \mathfrak{g}$ this recovers the standard fact that the identity component of a matrix group is generated by exponentials.

With subalgebras integrating to subgroups, the integration of homomorphisms follows by a device that recurs throughout Lie theory: encode a map as its graph, and integrate the graph. A Lie algebra homomorphism $\psi\colon \mathfrak{g} \to \mathfrak{h}$ is not itself a subalgebra, but its graph is a subalgebra of the direct sum, and the subgroup integrating that graph is, up to the projection isomorphism, the graph of the homomorphism sought. This is Lie's second theorem. The role of simple connectivity is isolated in a single step, where a covering map with connected simply connected base is shown to be a diffeomorphism.

Before the theorem, one preparatory fact about the direct product. The Lie algebra of $G \times H$ is $\mathfrak{g} \oplus \mathfrak{h}$ with the componentwise bracket, and a homomorphism $\psi\colon \mathfrak{g} \to \mathfrak{h}$ produces a subalgebra of this direct sum.

Lemma 3.3.6: The Graph of a Lie Algebra Homomorphism

Let $\mathfrak{g}, \mathfrak{h}$ be Lie algebras and $\psi\colon \mathfrak{g} \to \mathfrak{h}$ a Lie algebra homomorphism. The graph

$$\mathfrak{k} = \{(X, \psi X) \mid X \in \mathfrak{g}\} \subseteq \mathfrak{g} \oplus \mathfrak{h}$$

is a Lie subalgebra of $\mathfrak{g} \oplus \mathfrak{h} = \mathrm{Lie}(G \times H)$, isomorphic to $\mathfrak{g}$ via the projection to the first factor.

***Proof*:**
The set $\mathfrak{k}$ is a linear subspace, the graph of the linear map ψ. To see it is closed under the componentwise bracket of $\mathfrak{g} \oplus \mathfrak{h}$, compute for $X, X' \in \mathfrak{g}$

$$[(X, \psi X), (X', \psi X')] = \big([X, X'], [\psi X, \psi X']\big) = \big([X, X'], \psi[X, X']\big)$$

the second equality because ψ is a Lie algebra homomorphism, $[\psi X, \psi X'] = \psi[X, X']$. The result has the form $(Z, \psi Z)$ with $Z = [X, X']$, so it lies in $\mathfrak{k}$; hence $\mathfrak{k}$ is a subalgebra. The projection $\mathrm{pr}_{\mathfrak{g}}\colon \mathfrak{k} \to \mathfrak{g}$, $(X, \psi X) \mapsto X$, is a linear bijection with inverse $X \mapsto (X, \psi X)$, and it preserves brackets by the computation just made, so it is a Lie algebra isomorphism $\mathfrak{k} \cong \mathfrak{g}$, completing the proof.

The lemma is the algebraic half of the graph trick; the theorem is the geometric half. The identification $\mathrm{Lie}(G \times H) = \mathfrak{g} \oplus \mathfrak{h}$ used above is the componentwise statement that the exponential and bracket of

a product are computed factorwise, a direct consequence of the naturality of exp (theorem 2.4.3) applied to the two projections $G \times H \to G$ and $G \times H \to H$. Now the graph subalgebra integrates to a subgroup of $G \times H$, and reading off its two projections produces the homomorphism.

Theorem 3.3.7: Lie's Second Theorem: Integration of Homomorphisms

Let G be a *simply connected* Lie group, H any Lie group, and $\psi\colon \mathfrak{g} \to \mathfrak{h}$ a homomorphism of their Lie algebras. Then there is a unique Lie group homomorphism $\varphi\colon G \to H$ with $d\varphi_e = \psi$.

***Proof*:**
Uniqueness is immediate from the connectivity of G: if $\varphi_1, \varphi_2\colon G \to H$ both have differential ψ at e, then $d(\varphi_1)_e = d(\varphi_2)_e$, and by corollary 2.4.4 a homomorphism out of a connected group is determined by its differential, so $\varphi_1 = \varphi_2$. It remains to construct φ.

Step 1: integrate the graph. Form the product Lie group $G \times H$, with Lie algebra $\mathfrak{g} \oplus \mathfrak{h}$. By lemma 3.3.6 the graph $\mathfrak{k} = \{(X, \psi X) \mid X \in \mathfrak{g}\}$ is a Lie subalgebra of $\mathfrak{g} \oplus \mathfrak{h}$. By the integration theorem (theorem 3.3.4) there is a unique connected immersed Lie subgroup $K \leq G \times H$ with $\mathrm{Lie}(K) = \mathfrak{k}$.

Step 2: the first projection restricts to a homomorphism with invertible differential. Let $p = \mathrm{pr}_G\colon G \times H \to G$ be the projection, a Lie group homomorphism with differential the projection $\mathfrak{g} \oplus \mathfrak{h} \to \mathfrak{g}$. Restrict p to K and call it $q = p|_K\colon K \to G$, a homomorphism of Lie groups. Its differential at e is the composite

$$\mathfrak{k} \hookrightarrow \mathfrak{g} \oplus \mathfrak{h} \xrightarrow{\mathrm{pr}_{\mathfrak{g}}} \mathfrak{g}, \qquad (X, \psi X) \mapsto X$$

which is the isomorphism $\mathfrak{k} \xrightarrow{\sim} \mathfrak{g}$ of lemma 3.3.6. Thus dq_e is a linear isomorphism.

Step 3: q is a covering map. Since $dq_e\colon \mathfrak{k} \to \mathfrak{g}$ is an isomorphism and q is a homomorphism, the differential at an arbitrary $g \in K$ factors as $dq_g = (dL_{q(g)})_e \circ dq_e \circ (dL_{g^{-1}})_g$ (translation-equivariance of a homomorphism), a composite of isomorphisms, so dq_g is an isomorphism at every point. Hence q is a local diffeomorphism, and its image is an open subgroup of G; as G is connected, an open subgroup is all of G, so q is surjective. The kernel $\Gamma = \ker q$ is discrete: q is a local diffeomorphism near e, so e is isolated in $\Gamma = q^{-1}(e)$, and left translation by elements of Γ spreads this to isolation everywhere. A surjective homomorphism of Lie groups that is a local diffeomorphism with discrete kernel realizes G as the quotient K/Γ, with Γ acting freely and properly discontinuously by the deck transformations $k \mapsto k\gamma$; the covering-space criterion of [9, Covering Quotient Theorem] identifies $q\colon K \to K/\Gamma = G$ as a covering map.

Step 4: simple connectivity trivializes the cover. The base G is simply connected. A connected covering of a connected, simply connected, locally path connected space is a homeomorphism, by the lifting theory of [9, Lifting Correspondence and Path-Connected Spaces]: the number of sheets of q equals the index of the image of $\pi_1(K)$ in $\pi_1(G) = 1$, hence equals one, so the discrete fiber Γ is trivial and q is bijective. A bijective local diffeomorphism is a diffeomorphism, so $q\colon K \to G$ is an isomorphism of Lie groups, with inverse $q^{-1}\colon G \to K$ a Lie group homomorphism.

Step 5: read off φ. Define

$$\varphi = \mathrm{pr}_H \circ q^{-1}\colon G \to H$$

the composite of the isomorphism $q^{-1}\colon G \xrightarrow{\sim} K$ with the projection $\mathrm{pr}_H\colon G \times H \to H$ (restricted to K). As a composite of Lie group homomorphisms, φ is a Lie group homomorphism. Its

differential at e is

$$d\varphi_e = d(\mathrm{pr}_H)_e \circ (dq_e)^{-1}$$

Now $(dq_e)^{-1}\colon \mathfrak{g} \to \mathfrak{k}$ is $X \mapsto (X, \psi X)$ by Step 2, and $d(\mathrm{pr}_H)_e\colon \mathfrak{g} \oplus \mathfrak{h} \to \mathfrak{h}$ is the projection $(A, B) \mapsto B$; composing,

$$d\varphi_e(X) = d(\mathrm{pr}_H)_e(X, \psi X) = \psi X$$

so $d\varphi_e = \psi$. This is the homomorphism sought, completing the proof.

The theorem is the existence half of the morphism-level correspondence, and its hypothesis pinpoints the exact obstruction. The differential $d\varphi_e = \psi$ can always be reconstructed on a neighborhood of e, where exp is a diffeomorphism and one sets $\varphi(\exp X) = \exp(\psi X)$; the difficulty is globalizing this local homomorphism to all of G, and the obstruction is exactly $\pi_1(G)$. Running the local homomorphism around a noncontractible loop in G can return an inconsistent value, and the covering $K \to G$ of the proof is precisely the space on which the local homomorphism unwinds consistently. Simple connectivity says every loop is contractible, so no inconsistency arises and the local homomorphism extends. When G is not simply connected the extension can fail, and the failure is measured by which homomorphisms $\pi_1(G) \to H$ the deck group forces.

Packaging both halves, faithfulness from corollary 2.4.4 and now existence, gives the morphism-level equivalence for simply connected groups.

Corollary 3.3.8: The Correspondence on Homomorphisms

Let G be a simply connected Lie group and H any Lie group. The differential map

$$\mathrm{Hom}(G, H) \longrightarrow \mathrm{Hom}(\mathfrak{g}, \mathfrak{h}), \qquad \varphi \mapsto d\varphi_e$$

from Lie group homomorphisms to Lie algebra homomorphisms is a bijection.

***Proof*:**
The map is well defined, sending a Lie group homomorphism to a Lie algebra homomorphism by proposition 2.4.2. It is injective by corollary 2.4.4, since G is connected (being simply connected). It is surjective by theorem 3.3.7: every Lie algebra homomorphism $\psi\colon \mathfrak{g} \to \mathfrak{h}$ equals $d\varphi_e$ for some $\varphi \in \mathrm{Hom}(G, H)$. An injective and surjective map is a bijection, completing the proof.

The corollary is the sharp form of the statement that the Lie functor is fully faithful on simply connected groups. For a simply connected G, the group homomorphisms out of G are in bijection with the algebra homomorphisms out of $\mathfrak{g}$: nothing is lost and nothing is gained by passing to differentials. This is an equivalence at the level of *morphisms only*. The complementary object-level statement, that every finite-dimensional real Lie algebra is $\mathfrak{g} = \mathrm{Lie}(G)$ for some simply connected G, is Lie's third theorem, which requires one further integration, of the adjoint representation, together with Ado's embedding theorem as an input; that argument is carried out in section 3.4. Together the morphism bijection here and the object statement there assemble into an equivalence of categories between simply connected Lie groups and finite-dimensional real Lie algebras.

The running example is the cleanest illustration of both the power and the limit of the corollary. The isomorphism of Lie algebras $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ integrates in one direction but not the other, and the asymmetry is exactly simple connectivity.

Example 3.3.9: Integrating and Failing to Integrate for SU(2) and SO(3)

The Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are isomorphic (example 1.4.7), both being $(\mathbb{R}^3, \times)$; fix the isomorphism $\psi\colon \mathfrak{su}(2) \xrightarrow{\sim} \mathfrak{so}(3)$. Consider its two directions as candidate inputs to theorem 3.3.7.

1. Take $G = \mathrm{SU}(2)$, which is diffeomorphic to S^3 (proposition 1.5.1) and hence simply connected. Then theorem 3.3.7 applies: the isomorphism $\psi\colon \mathfrak{su}(2) \to \mathfrak{so}(3)$ integrates to a unique Lie group homomorphism $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ with $d\varphi_e = \psi$. This φ is exactly the double cover of theorem 1.5.5, recovered here from purely algebraic data. The integration theorem produces it without any of the explicit quaternion computation of chapter 1; the algebra isomorphism alone forces the covering homomorphism into existence.

2. Take instead $G = \mathrm{SO}(3)$, which is *not* simply connected: $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2\mathbb{Z}$ (corollary 1.5.6). Now theorem 3.3.7 does not apply, and indeed the inverse isomorphism $\psi^{-1}\colon \mathfrak{so}(3) \to \mathfrak{su}(2)$ does *not* integrate to a group homomorphism $\mathrm{SO}(3) \to \mathrm{SU}(2)$. If it did, that homomorphism ρ would satisfy $d\rho_e = \psi^{-1} = (d\varphi_e)^{-1}$, so $d(\varphi \circ \rho)_e = \mathrm{id}$; by corollary 2.4.4 on the connected $\mathrm{SO}(3)$, $\varphi \circ \rho = \mathrm{id}_{\mathrm{SO}(3)}$, making ρ a continuous section of the double cover $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$. A connected double cover with nontrivial π_1 downstairs admits no continuous section, since a section would exhibit $\mathrm{SO}(3)$ as a retract of $\mathrm{SU}(2)$ and force $\pi_1(\mathrm{SO}(3))$ to inject into $\pi_1(\mathrm{SU}(2)) = 1$, contradicting $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2\mathbb{Z}$. The obstruction is precisely this nontrivial fundamental group that the double cover measures.

The asymmetry is the entire content of the simple connectivity hypothesis: the algebra isomorphism is symmetric, but integration is possible only from the simply connected side. $\mathrm{SU}(2)$ is the universal cover, and morphisms out of it realize every algebra morphism; $\mathrm{SO}(3)$ realizes only those algebra morphisms compatible with its fundamental group.

The example shows that corollary 3.3.8 is not a restatement of the algebra data. For $\mathrm{SU}(2)$ the set $\mathrm{Hom}(\mathrm{SU}(2), \mathrm{SO}(3))$ contains the double cover, and its bijection with $\mathrm{Hom}(\mathfrak{su}(2), \mathfrak{so}(3))$ accounts for it; for $\mathrm{SO}(3)$ the map $\mathrm{Hom}(\mathrm{SO}(3), \mathrm{SU}(2)) \to \mathrm{Hom}(\mathfrak{so}(3), \mathfrak{su}(2))$ is not surjective, missing the algebra isomorphism ψ^{-1}. Simple connectivity is what removes the topological obstruction and turns the differential map into a bijection.

Exercise 3.3.1

1. Show that for any Lie group G, the Lie group homomorphisms $\mathbb{R} \to G$ are exactly the one-parameter subgroups of G, and deduce from corollary 3.3.8 that $\mathrm{Hom}(\mathbb{R}, G) \to \mathrm{Hom}(\mathbb{R}, \mathfrak{g}) = \mathfrak{g}$, $\varphi \mapsto d\varphi_e(1)$, is a bijection. Verify that this bijection is the one of theorem 2.3.1, and check that the hypothesis of corollary 3.3.8 holds for the source group $\mathbb{R}$.

2. Let $\mathfrak{n} \subseteq \mathfrak{gl}_n(\mathbb{R})$ be the subalgebra of strictly upper-triangular matrices, closed under the commutator. Show that its integral subgroup (theorem 3.3.4) inside $\mathrm{GL}_n(\mathbb{R})$ is the group N of upper-triangular matrices with 1's on the diagonal (the unipotent group), that $\exp\colon \mathfrak{n} \to N$ is a diffeomorphism, and that N is embedded and closed. Contrast this with the irrational line of example 3.3.5: what makes the unipotent case embedded?

3. Let G be a connected Lie group with center $Z(G)$, and let $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \ \forall Y\}$ be the center of $\mathfrak{g}$. Show that $\mathfrak{z}(\mathfrak{g})$ is a subalgebra and that its integral subgroup is the identity component of $Z(G)$. (Use that $X \in \mathfrak{z}(\mathfrak{g})$ iff $\mathrm{ad}X = 0$ iff $\mathrm{Ad}(\exp tX) = \mathrm{id}$ for all t, via (2.11) of section 2.5.)

4. Let G be simply connected with Lie algebra a direct sum of ideals $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$. Using theorem 3.3.7 and theorem 3.3.4, show that $G \cong G_1 \times G_2$ for simply connected G_1, G_2 with $\mathrm{Lie}(G_i) = \mathfrak{g}_i$. (Integrate the two projections $\mathfrak{g} \to \mathfrak{g}_i$.)

5. Let $H = S^1 = \mathbb{R}/\mathbb{Z}$ and consider a one-parameter subgroup $\gamma_c \colon \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i c t}$, generated by $c \in \mathbb{R} = \mathfrak{h}$. Determine for which c the map γ_c descends to a Lie group homomorphism $S^1 = \mathbb{R}/\mathbb{Z} \to S^1$, and explain how the answer is consistent with corollary 3.3.8 failing when the source S^1 is not simply connected. Identify the resulting image of the differential map $\mathrm{Hom}(S^1, S^1) \to \mathrm{Hom}(\mathbb{R}, \mathbb{R}) = \mathbb{R}$.

6. Prove that the integral subgroup H of theorem 3.3.4 is embedded (a closed submanifold) if and only if H is closed in G, and that when H is closed the integration theorem and the closed subgroup theorem (theorem 3.2.1) produce the same subgroup. Use $\mathbb{T}^2$ to exhibit both an embedded and a non-embedded integral subgroup.

3.4 Simply-Connected Groups and Lie's Third Theorem

The correspondence built so far runs in one direction with a caveat. Passing from a Lie group to its Lie algebra is a functor Lie (proposition 2.4.5), and theorem 3.3.7 showed that on a *simply connected* domain this functor is faithful and full at the level of homomorphisms: a Lie-algebra map integrates to a unique group map. Two questions remain, and they are the two halves of Lie's third theorem. First, does every finite-dimensional real Lie algebra arise as $\mathrm{Lie}(G)$ for some Lie group G, so that the functor is essentially surjective? Second, what is the relationship among the possibly many groups sharing a fixed Lie algebra, once simple-connectivity is dropped? This section answers both. The mechanism is the universal cover. Every connected Lie group G has a universal covering space $\tilde{G}$, and the group structure lifts to make $\tilde{G}$ a simply connected Lie group with the same Lie algebra; the group G is recovered from $\tilde{G}$ by quotienting by a discrete central subgroup isomorphic to $\pi_1(G)$. The upshot is an equivalence of categories between simply connected Lie groups and finite-dimensional real Lie algebras, and a complete classification of all connected groups with a given algebra as the quotients of one universal object. The Lie algebra determines the group up to exactly one piece of data the linear theory cannot see: the discrete central subgroup that is the fundamental group.

The starting point is a topological fact upgraded to a group-theoretic one. A connected, locally path-connected, semilocally simply connected space has a universal covering space ([9, Existence of a Universal Cover]); a Lie group, being a connected manifold, satisfies these hypotheses, so $\tilde{G}$ exists as a space. The content of the next proposition is that the group structure of G transports to $\tilde{G}$ through the covering, and does so uniquely once a basepoint over the identity is fixed.

Proposition 3.4.1: The Universal Covering Group

Let G be a connected Lie group and let $\pi\colon \tilde{G} \to G$ be the universal covering space, with a chosen point $\tilde{e} \in \tilde{G}$ over the identity $e \in G$. Then $\tilde{G}$ carries a unique Lie group structure with identity $\tilde{e}$ for which π is a homomorphism of Lie groups. Moreover:

1. the kernel $\ker \pi$ is a discrete subgroup of the center $Z(\tilde{G})$, and $\ker \pi \cong \pi_1(G)$ as groups;
2. the differential $d\pi_{\tilde{e}}\colon \mathrm{Lie}(\tilde{G}) \to \mathrm{Lie}(G)$ is an isomorphism of Lie algebras.

The Lie group $\tilde{G}$ is called the *universal covering group* of G, and it is simply connected.

***Proof*:**
The strategy is to lift the multiplication and inversion maps of G through the covering, using the lifting criterion for maps into a covering space. Recall ([9, The Lifting Criterion]) that if Y is connected and locally path connected and $f\colon Y \to G$ is a continuous map, then f lifts to a map $\tilde{f}\colon Y \to \tilde{G}$ with $\pi \circ \tilde{f} = f$ and a prescribed value at one point precisely when $f_*\pi_1(Y) \subseteq \pi_*\pi_1(\tilde{G})$; since $\tilde{G}$ is simply connected the right side is trivial, so the criterion is met exactly when $f_*\pi_1(Y) = 0$. When Y is itself simply connected this holds automatically, and the lift is unique once its value at one point is fixed. Every lift produced below is unique for this reason.

Step 1: lifting the multiplication. Consider the composite

$$\mu \circ (\pi \times \pi)\colon \tilde{G} \times \tilde{G} \to G, \qquad (\tilde{a}, \tilde{b}) \mapsto \pi(\tilde{a})\,\pi(\tilde{b})$$

where $\mu\colon G \times G \to G$ is the multiplication of G. The domain $\tilde{G} \times \tilde{G}$ is simply connected: the universal cover of a connected manifold is simply connected, and a product of simply connected spaces is simply connected ([9, Fundamental Group of a Product]). By the lifting criterion there is a unique smooth map

$$\tilde{\mu}\colon \tilde{G} \times \tilde{G} \to \tilde{G}$$

with $\pi \circ \tilde{\mu} = \mu \circ (\pi \times \pi)$ and $\tilde{\mu}(\tilde{e}, \tilde{e}) = \tilde{e}$. That $\tilde{\mu}$ is smooth, not only continuous, is automatic: a continuous lift of a smooth map through a covering by a smooth manifold is smooth, since π is a local diffeomorphism and the lift is locally $\pi^{-1} \circ f$ in charts. Define inversion similarly by lifting $\iota \circ \pi\colon \tilde{G} \to G$, $\tilde{a} \mapsto \pi(\tilde{a})^{-1}$, to the unique smooth $\tilde{\iota}\colon \tilde{G} \to \tilde{G}$ with $\tilde{\iota}(\tilde{e}) = \tilde{e}$.

Step 2: the group axioms. The axioms hold because their two sides are lifts of the same map and agree at $\tilde{e}$, so they coincide by uniqueness of lifts. For associativity, both $(\tilde{a}, \tilde{b}, \tilde{c}) \mapsto \tilde{\mu}(\tilde{\mu}(\tilde{a}, \tilde{b}), \tilde{c})$ and $(\tilde{a}, \tilde{b}, \tilde{c}) \mapsto \tilde{\mu}(\tilde{a}, \tilde{\mu}(\tilde{b}, \tilde{c}))$ are maps $\tilde{G}^3 \to \tilde{G}$ that project under π to the single product $(\tilde{a}, \tilde{b}, \tilde{c}) \mapsto \pi(\tilde{a})\pi(\tilde{b})\pi(\tilde{c})$ on G, associative in G, and both send $(\tilde{e}, \tilde{e}, \tilde{e})$ to $\tilde{e}$; since $\tilde{G}^3$ is simply connected, the two lifts agree. The identity and inverse axioms follow the same way: $\tilde{a} \mapsto \tilde{\mu}(\tilde{e}, \tilde{a})$ and the identity map $\tilde{a} \mapsto \tilde{a}$ both lift $\tilde{a} \mapsto \pi(\tilde{a})$ and agree at $\tilde{e}$, hence are equal, so $\tilde{e}$ is a left identity; and $\tilde{a} \mapsto \tilde{\mu}(\tilde{\iota}\tilde{a}, \tilde{a})$ and the constant map at $\tilde{e}$ both lift the constant map e and agree at $\tilde{e}$, so $\tilde{\iota}\tilde{a}$ is a left inverse. Right identity and right inverse are identical. Thus $\tilde{G}$ is a Lie group with $\tilde{e}$ its identity, and π is a homomorphism by construction: $\pi(\tilde{\mu}(\tilde{a}, \tilde{b})) = \pi(\tilde{a})\pi(\tilde{b})$. Uniqueness of the group structure is uniqueness of the lifted multiplication.

Step 3: the kernel is discrete, central, and $\cong \pi_1(G)$. As the kernel of a homomorphism, $\ker \pi$ is a subgroup of $\tilde{G}$; as the fiber $\pi^{-1}(e)$ of a covering map, it is a discrete subset. So $\ker \pi$ is a discrete normal subgroup of the connected group $\tilde{G}$. A discrete normal subgroup N of a connected

topological group is central: for fixed $n \in N$ the map $\tilde{G} \to N$, $\tilde{a} \mapsto \tilde{a}\, n\, \tilde{a}^{-1}$, is continuous with connected domain $\tilde{G}$ and image inside the discrete set N (normality gives $\tilde{a}\, n\, \tilde{a}^{-1} \in N$), hence constant; its value at $\tilde{a} = \tilde{e}$ is n, so $\tilde{a}\, n\, \tilde{a}^{-1} = n$ for all $\tilde{a}$, that is $n \in Z(\tilde{G})$. Thus $\ker \pi \subseteq Z(\tilde{G})$. Finally, π is the universal cover of the connected group G; the deck transformation group of the universal cover acts simply transitively on the fiber $\pi^{-1}(e) = \ker \pi$ and is isomorphic to $\pi_1(G, e)$, and for a covering group these deck transformations are exactly left translations by the elements of $\ker \pi$, so the deck group is $\ker \pi$ itself and $\ker \pi \cong \pi_1(G)$ as groups ([9, The Deck Group of the Universal Cover]).

Step 4: the differential is an isomorphism. A covering map is a local diffeomorphism, so $d\pi_{\tilde{e}} \colon T_{\tilde{e}}\tilde{G} \to T_e G$ is a linear isomorphism of tangent spaces. It is a Lie-algebra homomorphism because π is a group homomorphism, so its differential preserves the bracket (proposition 2.4.2, via the differential of definition 2.4.1). A bijective Lie-algebra homomorphism is an isomorphism of Lie algebras, so $d\pi_{\tilde{e}}$ identifies $\mathrm{Lie}(\tilde{G})$ with $\mathrm{Lie}(G)$, completing the proof.

The proposition is the technical heart of the section, and its consequences organize everything that follows. It says that among the connected Lie groups with a given Lie algebra there is a distinguished one, the simply connected cover, from which all the others descend by a discrete central quotient. The covering map is not just a topological device: it is a homomorphism, so $\pi_1(G)$ enters the algebra of the situation as a central subgroup of $\tilde{G}$. The differential being an isomorphism records the invisibility of the covering to the linear theory. The groups G and $\tilde{G}$ have literally the same Lie algebra, and no infinitesimal computation can tell them apart; the distinction lives entirely in the discrete data of the kernel.

The centrality argument in Step 3 deserves a second look, because it is the recurring reason discrete kernels behave so rigidly. A discrete normal subgroup of a connected group cannot be twisted by conjugation: conjugation is a continuous motion, the subgroup is a scatter of isolated points, and a continuous path cannot slide one isolated point to another without leaving the subgroup. So conjugation fixes each element, which is centrality. This is why the fundamental group of a Lie group is always abelian, a fact that follows the moment $\pi_1(G) \cong \ker \pi \subseteq Z(\tilde{G})$.

The concrete instance to keep in mind is the circle and its cover, which exhibits every feature of the proposition in the smallest possible setting.

Example 3.4.2: The Real Line Covering the Circle

Let $G = S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ with multiplication of complex numbers, a connected abelian Lie group of dimension one with Lie algebra $\mathrm{Lie}(S^1) = \mathbb{R}$, the tangent line at 1 with trivial bracket. The universal cover is

$$\pi \colon \mathbb{R} \to S^1, \qquad \pi(t) = e^{2\pi i t}$$

with $\tilde{e} = 0$. The lifted multiplication on $\mathbb{R}$ is ordinary addition: the map $(s, t) \mapsto s + t$ satisfies $\pi(s + t) = e^{2\pi i(s+t)} = e^{2\pi i s} e^{2\pi i t} = \pi(s)\pi(t)$ and sends $(0, 0)$ to 0, so by uniqueness of lifts it is $\tilde{\mu}$. Thus $\tilde{G} = (\mathbb{R}, +)$, the simply connected group with Lie algebra $\mathbb{R}$. The kernel is $\ker \pi = \mathbb{Z} \subset \mathbb{R}$, a discrete central subgroup (central because $\mathbb{R}$ is abelian), and $\pi_1(S^1) \cong \mathbb{Z}$ matches $\ker \pi = \mathbb{Z}$ exactly. The differential $d\pi_0 \colon \mathbb{R} = T_0\mathbb{R} \to T_1 S^1 = i\mathbb{R}$ sends the generator 1 to the velocity $\frac{d}{dt}\big|_{t=0} e^{2\pi i t} = 2\pi i$, so it is multiplication by $2\pi i$ into the tangent line $i\mathbb{R}$, a linear isomorphism, as the proposition requires. The circle is recovered as the quotient $S^1 = \mathbb{R}/\mathbb{Z}$, the archetype of a connected group presented as $\tilde{G}/\Gamma$ for a discrete central Γ.

With the universal cover in hand, the essential-surjectivity half of Lie's third theorem is within reach.

The remaining input is purely algebraic: every finite-dimensional real Lie algebra can be realized inside a matrix algebra. This is Ado's theorem, whose proof is a substantial piece of Lie-algebra structure theory. The tools it requires are built across Part II, and section 8.9 proves the theorem in full once they are in place. Here it is stated and used as a black box.

Theorem 3.4.3: Ado's Theorem

Every finite-dimensional real Lie algebra $\mathfrak{g}$ admits a faithful finite-dimensional representation: there is an injective Lie-algebra homomorphism $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(\mathbb{R})$ for some n. In particular every finite-dimensional real Lie algebra is isomorphic to a Lie subalgebra of $\mathfrak{gl}_n(\mathbb{R}) = \mathrm{Lie}(\mathrm{GL}_n(\mathbb{R}))$. The same holds over $\mathbb{C}$, with $\mathfrak{gl}_n(\mathbb{C})$ as the target.

***Proof Key ideas*:**
The proof uses the structure theory of Lie algebras, in particular the Levi decomposition into a solvable radical and a semisimple part together with a delicate extension of representations across that decomposition. The tools it requires are developed in Part II of this book, downstream of the present chapter, and the full argument is carried out in section 8.9, the closing section of that part. The argument there is purely algebraic and draws on none of the group theory between here and there, so no circularity arises. The result is stated here because it is the one external input to the existence half of Lie's third theorem, entering only as the assertion that a faithful finite-dimensional representation exists. That single assertion, a faithful finite-dimensional representation for every $\mathfrak{g}$, is all the integration arguments below will draw on.

The interplay of Ado's theorem with the integration theorem of the previous section is the crux of the existence argument. Ado places $\mathfrak{g}$ inside $\mathfrak{gl}_n(\mathbb{R})$, the Lie algebra of the concrete group $\mathrm{GL}_n(\mathbb{R})$; the integration theorem (theorem 3.3.4) turns that subalgebra into an actual subgroup; and the universal cover (proposition 3.4.1) upgrades that subgroup to a simply connected group with the same algebra. Chaining these produces a group for *every* algebra, which is essential surjectivity. Combined with the full faithfulness already established, this is the equivalence of categories.

Theorem 3.4.4: Lie's Third Theorem

The functor Lie sending a simply connected Lie group G to its Lie algebra $\mathrm{Lie}(G)$, and a homomorphism $\varphi\colon G \to H$ to its differential $d\varphi_e\colon \mathrm{Lie}(G) \to \mathrm{Lie}(H)$, is an equivalence of categories

$$\mathrm{Lie}\colon \{\text{simply connected Lie groups}\} \xrightarrow{\sim} \{\text{finite-dimensional real Lie algebras}\}$$

Concretely: every finite-dimensional real Lie algebra $\mathfrak{g}$ is $\mathrm{Lie}(G)$ for a simply connected Lie group G, unique up to isomorphism, and for simply connected G, H the map $\varphi \mapsto d\varphi_e$ is a bijection $\mathrm{Hom}(G,H) \xrightarrow{\sim} \mathrm{Hom}(\mathrm{Lie}(G), \mathrm{Lie}(H))$.

***Proof*:**
A functor is an equivalence of categories precisely when it is fully faithful and essentially surjective. Both properties are established by assembling results already in hand.

Step 1: full faithfulness. For simply connected G and H, the map $\varphi \mapsto d\varphi_e$ from $\mathrm{Hom}(G,H)$ to $\mathrm{Hom}(\mathrm{Lie}(G), \mathrm{Lie}(H))$ is a bijection: this is exactly corollary 3.3.8, the corollary of Lie's second

theorem. Its injectivity is that a homomorphism from a connected group is determined by its differential (corollary 2.4.4), and its surjectivity is that a Lie-algebra map integrates to a group map when the domain is simply connected (theorem 3.3.7). Full faithfulness is precisely this bijection on each hom-set.

Step 2: essential surjectivity. Let $\mathfrak{g}$ be a finite-dimensional real Lie algebra. By Ado's theorem (theorem 3.4.3) there is an injective Lie-algebra homomorphism $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(\mathbb{R}) = \mathrm{Lie}(\mathrm{GL}_n(\mathbb{R}))$ for some n, identifying $\mathfrak{g}$ with a Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{gl}_n(\mathbb{R})$. By the integration theorem (theorem 3.3.4) there is a connected immersed Lie subgroup $H \leq \mathrm{GL}_n(\mathbb{R})$ with $\mathrm{Lie}(H) = \mathfrak{h} \cong \mathfrak{g}$. The group H is connected but need not be simply connected; pass to its universal covering group $\tilde{H}$ (proposition 3.4.1). Then $\tilde{H}$ is simply connected, and the covering map $\pi\colon \tilde{H} \to H$ has differential an isomorphism, so

$$\mathrm{Lie}(\tilde{H}) \cong \mathrm{Lie}(H) = \mathfrak{h} \cong \mathfrak{g}$$

Thus $\mathfrak{g} \cong \mathrm{Lie}(\tilde{H})$ with $\tilde{H}$ simply connected, and Lie hits every algebra up to isomorphism.

Step 3: uniqueness of the realizing group. If G_1 and G_2 are simply connected with $\mathrm{Lie}(G_1) \cong \mathrm{Lie}(G_2)$, then the isomorphism $\psi\colon \mathrm{Lie}(G_1) \to \mathrm{Lie}(G_2)$ and its inverse integrate by Step 1 to homomorphisms $\varphi\colon G_1 \to G_2$ and $\varphi'\colon G_2 \to G_1$ with $d\varphi_e = \psi$ and $d\varphi'_e = \psi^{-1}$. Then $\varphi' \circ \varphi$ and $\varphi \circ \varphi'$ have differentials $\psi^{-1}\psi = \mathrm{id}$ and $\psi\psi^{-1} = \mathrm{id}$, so by corollary 2.4.4 they equal the identity maps of G_1 and G_2; hence φ is an isomorphism and $G_1 \cong G_2$. A fully faithful, essentially surjective functor is an equivalence, completing the proof.

Lie's third theorem is the summit of the correspondence and closes Part I's central question. The passage from a group to its Lie algebra, which began in Chapter 1 as the extraction of a tangent space of matrices, is now known to lose no information at all on the simply connected side: the categories of simply connected Lie groups and of finite-dimensional real Lie algebras are the same category wearing two descriptions. A theorem about groups can be proved by computing with brackets, and a finite-dimensional Lie algebra, an object of finite linear data, is guaranteed to be the Lie algebra of an actual manifold-with-multiplication. The one hypothesis that cannot be dropped is simple connectivity; without it the functor forgets the fundamental group, and it is precisely that forgotten data the next corollary restores.

Dropping simple connectivity, the theorem still controls the situation completely, because every connected group is a quotient of its simply connected cover. The classification of connected groups with a fixed Lie algebra therefore reduces to the classification of discrete central subgroups of a single object.

Corollary 3.4.5: Connected Groups with a Given Lie Algebra

Let $\mathfrak{g}$ be a finite-dimensional real Lie algebra and let $\tilde{G}$ be the simply connected Lie group with $\mathrm{Lie}(\tilde{G}) \cong \mathfrak{g}$ of theorem 3.4.4. Then the connected Lie groups G with $\mathrm{Lie}(G) \cong \mathfrak{g}$ are, up to isomorphism, exactly the quotients

$$G \cong \tilde{G}/\Gamma, \qquad \Gamma \subseteq Z(\tilde{G}) \text{ a discrete subgroup}$$

and $\pi_1(G) \cong \Gamma$.

***Proof*:**
Let G be a connected Lie group with $\mathrm{Lie}(G) \cong \mathfrak{g}$. By proposition 3.4.1 its universal covering group $\tilde{G}'$ is simply connected with $\mathrm{Lie}(\tilde{G}') \cong \mathrm{Lie}(G) \cong \mathfrak{g}$, so by the uniqueness in theorem 3.4.4 there is an isomorphism $\tilde{G}' \cong \tilde{G}$; identify the two. The covering map $\pi\colon \tilde{G} \to G$ is then a surjective homomorphism with discrete central kernel $\Gamma = \ker\pi \subseteq Z(\tilde{G})$, and $\Gamma \cong \pi_1(G)$ by proposition 3.4.1. The first isomorphism theorem ([2, First Isomorphism Theorem] for the algebraic content) gives $G \cong \tilde{G}/\Gamma$, and the quotient is a Lie group with π its covering map because Γ is discrete.

Conversely, for any discrete subgroup $\Gamma \subseteq Z(\tilde{G})$ the quotient $\tilde{G}/\Gamma$ is a Lie group, and the quotient map $\tilde{G} \to \tilde{G}/\Gamma$ is a covering: a discrete central subgroup acts on $\tilde{G}$ by translations, and this action is free and properly discontinuous, so the orbit map is a covering ([9, Covering Quotient Theorem]). A covering map is a local diffeomorphism, so $\mathrm{Lie}(\tilde{G}/\Gamma) \cong \mathrm{Lie}(\tilde{G}) \cong \mathfrak{g}$, and $\tilde{G}$ being simply connected makes the quotient map the universal cover, whence $\pi_1(\tilde{G}/\Gamma) \cong \Gamma$ by proposition 3.4.1. Thus the connected groups with algebra $\mathfrak{g}$ are exactly the $\tilde{G}/\Gamma$, and their fundamental groups are the corresponding Γ, as we sought to show.

The corollary reduces a differential-geometric classification to a discrete computation inside one group. To list every connected Lie group with a given Lie algebra, one finds the simply connected model $\tilde{G}$, computes its center $Z(\tilde{G})$, and enumerates the discrete subgroups of that center. Each such subgroup Γ produces a group $\tilde{G}/\Gamma$ with fundamental group Γ; the choice $\Gamma = \{e\}$ recovers $\tilde{G}$ itself, and the largest possible $\Gamma = Z(\tilde{G})$ produces the smallest group in the family, the adjoint group $\tilde{G}/Z(\tilde{G})$ when $Z(\tilde{G})$ is finite, whose fundamental group $\pi_1 \cong Z(\tilde{G})$ is correspondingly the largest. Everything hinges on the center of the simply connected group, a finite-dimensional and often finite piece of arithmetic.

The running example makes the enumeration explicit and recovers the double cover of chapter 1 as an instance of the corollary.

Example 3.4.6: The Two Connected Groups with Algebra $\mathfrak{su}(2)$
Take $\mathfrak{g} = \mathfrak{su}(2)$, the Lie algebra of example 1.4.7, isomorphic to $(\mathbb{R}^3, \times)$. Its simply connected model is $\tilde{G} = \mathrm{SU}(2)$: the group $\mathrm{SU}(2)$ is diffeomorphic to the 3-sphere S^3 and hence simply connected (proposition 1.5.1), and its Lie algebra is $\mathfrak{su}(2)$, so by the uniqueness in theorem 3.4.4 it is *the* simply connected group with this algebra. To classify all connected groups with algebra $\mathfrak{su}(2)$, compute the center. The center of $\mathrm{SU}(2)$ is

$$Z(\mathrm{SU}(2)) = \{I, -I\} \cong \mathbb{Z}/2$$

as a direct computation shows: a matrix $g \in \mathrm{SU}(2)$ commuting with the diagonal $\mathrm{diag}(i, -i) \in \mathrm{SU}(2)$ must itself be diagonal, say $g = \mathrm{diag}(a, d)$, since $\mathrm{diag}(i, -i)$ has distinct eigenvalues; and commuting in addition with $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SU}(2)$ forces $a = d$, so $g = aI$ with $a^2 = 1$ and $|a| = 1$, whence $g = \pm I$. The discrete subgroups of $Z(\mathrm{SU}(2)) = \mathbb{Z}/2$ are only two, the trivial group $\{I\}$ and the whole center $\{\pm I\}$. By corollary 3.4.5 there are therefore exactly two connected Lie groups with Lie algebra $\mathfrak{su}(2)$:

1. $\Gamma = \{I\}$ gives $\mathrm{SU}(2)/\{I\} = \mathrm{SU}(2)$ itself, simply connected, $\pi_1 = 0$;
2. $\Gamma = \{\pm I\}$ gives $\mathrm{SU}(2)/\{\pm I\} \cong \mathrm{SO}(3)$, with $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2$.

The second quotient is the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5: that map is a surjective homomorphism with kernel exactly $\{\pm I\}$, so $\mathrm{SO}(3) \cong \mathrm{SU}(2)/\{\pm I\}$, and its fundamental group

> $\mathbb{Z}/2$ (corollary 1.5.6) is the kernel Γ, as the corollary predicts. There is no third connected group with this algebra, because $\mathbb{Z}/2$ has no other subgroups. The pair $\mathrm{SU}(2), \mathrm{SO}(3)$ is the complete list, and the double cover is the only nontrivial central quotient the algebra $\mathfrak{su}(2)$ permits.

The example is the sharpest illustration of the section's thesis. The two groups $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ are indistinguishable by their Lie algebra: both have $\mathrm{Lie} = \mathfrak{su}(2)$, the same exponential map on a neighborhood of the identity, the same brackets, the same local structure. What separates them is a single bit of global topology, $\pi_1 = 0$ against $\pi_1 = \mathbb{Z}/2$, encoded as the central subgroup $\{\pm I\}$ one quotients by. Chapter 1 constructed the double cover by an explicit computation with quaternions and rotations; the present corollary explains why that computation had to produce exactly a $\mathbb{Z}/2$ and no more, and why $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ exhaust the connected groups with this algebra. The linear theory sees $\mathfrak{su}(2)$; the topology sees the choice of Γ.

This is the exact boundary of the Lie group-Lie algebra dictionary. The Lie algebra determines a connected group completely except for the discrete central data of the fundamental group, the one invariant the infinitesimal, linear theory cannot detect. Two connected groups with isomorphic Lie algebras are locally isomorphic and differ only in which discrete central subgroup of the common simply connected cover has been divided out. Reading the correspondence in this light turns the fundamental group from a topological afterthought into the governing invariant of the classification, and it is the covering-group theory of the next chapter that develops the tools to compute it: group actions, quotient manifolds, and the systematic study of covering groups and their centers. The center of the simply connected group, whose subgroups index the connected forms, becomes an object of study in its own right, and the fundamental group $\pi_1(G) \cong \Gamma$ becomes a piece of the group's structure rather than a fact about its underlying space.

Exercise 3.4.1

1. Verify in detail that $\pi\colon \mathbb{R} \to S^1$, $\pi(t) = e^{2\pi i t}$, is the universal covering group of the circle: show it is a Lie group homomorphism, that its kernel is $\mathbb{Z}$, and that $\pi_1(S^1) \cong \mathbb{Z}$. Identify explicitly the discrete central subgroup $\Gamma \subseteq \mathbb{R}$ with $S^1 \cong \mathbb{R}/\Gamma$.

2. Prove that every connected abelian Lie group is isomorphic to $\mathbb{R}^k \times \mathbb{T}^m$ for some $k, m \geq 0$, where $\mathbb{T}^m = (S^1)^m$. Use that the simply connected abelian group with Lie algebra $\mathbb{R}^n$ is $(\mathbb{R}^n, +)$, obtained by integrating the trivial bracket via theorem 3.4.4, then apply corollary 3.4.5 to a discrete central subgroup.

3. Deduce from the previous exercise that a *simply connected* connected abelian Lie group of dimension n is isomorphic to $(\mathbb{R}^n, +)$. Which value of m in the description $\mathbb{R}^k \times \mathbb{T}^m$ corresponds to simple connectivity, and why?

4. State the fact that $\mathrm{SU}(n)$ is simply connected for every $n \geq 1$ (a topological fact deferred to [11]). Using corollary 3.4.5 and the center $Z(\mathrm{SU}(n)) \cong \mathbb{Z}/n$, describe how many connected Lie groups have Lie algebra $\mathfrak{su}(n)$, and identify $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ as the two groups in the case $n = 2$.

5. Let $\mathfrak{g}$ be a finite-dimensional real Lie algebra whose simply connected group $\tilde{G}$ has *trivial* center $Z(\tilde{G}) = \{e\}$. Prove that $\tilde{G}$ is the only connected Lie group with Lie algebra $\mathfrak{g}$. Give an example of such a $\mathfrak{g}$ together with its unique connected group.

6. With $G = \mathrm{SO}(3)$, $H = \mathrm{SU}(2)$, and $\psi = \mathrm{id}$ the identity on $\mathfrak{su}(2) = \mathrm{Lie}(\mathrm{SO}(3)) = \mathrm{Lie}(\mathrm{SU}(2))$, show that ψ does *not* arise as $d\varphi_e$ for any Lie group homomorphism $\varphi\colon \mathrm{SO}(3) \to \mathrm{SU}(2)$.

Explain how this is consistent with corollary 3.3.8.

3.5 The Baker-Campbell-Hausdorff Formula

Lie's third theorem (theorem 3.4.4) asserts that a simply-connected Lie group is recovered from its Lie algebra, an equivalence of categories that trades an analytic object for an algebraic one. The proof runs through Ado's theorem and the integration of subalgebras, and it is in a sense non-constructive: it produces the group but does not exhibit its multiplication. This section gives what is missing near the identity. The exponential map is a local diffeomorphism (theorem 2.3.4), so the product of two group elements $\exp X$ and $\exp Y$ close to e is encoded by a single element Z of the Lie algebra with $\exp Z = \exp X \exp Y$; the content of this section is a closed formula for $Z = \log(\exp X \exp Y)$ purely in terms of X, Y, and their iterated brackets. The formula is universal, the same combination of brackets for every Lie group, and its existence makes precise the claim that the bracket determines the group law near the identity.

The formula was found in incomplete form by Campbell, Poincare, Baker, and Hausdorff around the turn of the twentieth century, with the explicit bracket-series expression due to Dynkin in 1947. Its structural message is sharper than any single coefficient: the product of two exponentials is again an exponential, and the exponent lies in the Lie subalgebra generated by X and Y under the bracket alone. Everything the group multiplication does infinitesimally is bracketing.

The starting point is a direct expansion in the matrix setting, where exp and log are the convergent series of definition 1.3.2 and proposition 1.3.6. Multiplying the two exponential series and feeding the product into the logarithm series produces, order by order in X and Y, a polynomial in X and Y; the content of the theorem is that after collecting terms this polynomial is expressible entirely through commutators. Before stating the general formula the low-order terms are worth isolating, since they carry the whole conceptual load and are what one uses in practice.

Theorem 3.5.1: The Baker-Campbell-Hausdorff Formula

Let G be a Lie group with Lie algebra $\mathfrak{g}$. There is a neighborhood V of 0 in $\mathfrak{g}$ such that for all $X, Y \in V$ the product $\exp X \exp Y$ lies in the image of the exponential chart, and

$$\log(\exp X \exp Y) = X + Y + \tfrac{1}{2}[X,Y] + \tfrac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big) - \tfrac{1}{24}[Y,[X,[X,Y]]] + \cdots$$

where every term after $X + Y$ is a rational multiple of an iterated bracket of X and Y, and the omitted terms have bracket-degree ≥ 5. The right-hand side is a single *universal Lie series*: the same rational coefficients occur for every Lie group, and the series depends on G only through its bracket. In Dynkin's closed form,

$$\log(\exp X \exp Y) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sum_{\substack{p_i+q_i \geq 1 \\ 1 \leq i \leq n}} \frac{[X^{p_1} Y^{q_1} \cdots X^{p_n} Y^{q_n}]}{\left(\sum_{i=1}^{n}(p_i+q_i)\right) \prod_{i=1}^{n} p_i!\, q_i!} \tag{3.6}$$

where $[X^{p_1} Y^{q_1} \cdots X^{p_n} Y^{q_n}]$ denotes the right-nested iterated bracket $[Z_1, [Z_2, [\ldots, [Z_{N-1}, Z_N] \cdots]]]$ whose arguments $Z_1, \ldots, Z_N$ are, read left to right, the letters X occurring p_i times and Y occurring q_i times in the ith block; a bracket whose innermost two arguments coincide is understood to be zero.

The three lowest-order terms are the ones that appear in nearly every application, and they already display the structure. To first order the group is abelian: $\log(\exp X \exp Y) = X + Y + O(2)$, so infinitesimally exp turns addition in $\mathfrak{g}$ into multiplication in G. The first correction is $\frac{1}{2}[X, Y]$, the leading measure of noncommutativity, and the third-order terms $\frac{1}{12}([X, [X, Y]] + [Y, [Y, X]])$ are its refinement. That no term ever involves anything but iterated brackets is the substance of the theorem: a product of exponentials returns to the Lie subalgebra generated by X and Y, never leaving it for the ambient associative algebra. This is why the Lie algebra, a purely bracket-theoretic object, suffices to reconstruct the local group.

The Dynkin expression (3.6) is explicit but combinatorially dense; in practice one computes the first few terms directly from the exp/log series and invokes the closed form only to certify that all higher terms are brackets. The convergence of the series is a separate matter, taken up next, and the recursion that generates the coefficients is developed in the proof.

Before the general proof, the convergence claim deserves its own statement, since it is what makes the formula an identity between elements of $\mathfrak{g}$ rather than a formal manipulation of symbols.

Proposition 3.5.2: Convergence of the BCH Series

Let G be a Lie group with Lie algebra $\mathfrak{g}$, and fix any norm $\|\cdot\|$ on the finite-dimensional space $\mathfrak{g}$. There is a constant $\rho > 0$, depending only on the bracket of $\mathfrak{g}$ and the chosen norm, such that the Dynkin series (3.6) converges absolutely whenever $\|X\| + \|Y\| < \rho$, and its sum is $\log(\exp X \exp Y)$. In particular the series defines a real-analytic $\mathfrak{g}$-valued function of (X, Y) on the ball $\|X\| + \|Y\| < \rho$.

***Proof*:**

Step 1: reduction to the matrix case. By theorem 3.4.3 (proved in section 8.9), the Lie algebra $\mathfrak{g}$ admits a faithful finite-dimensional representation $\iota\colon \mathfrak{g} \hookrightarrow \mathfrak{gl}_m(\mathbb{R}) = M_m(\mathbb{R})$ for some m, a Lie-algebra embedding into a matrix algebra. Both sides of (3.6) are built from X, Y, and the bracket, and ι preserves the bracket, so the series for $\mathfrak{g}$ converges if and only if the corresponding series for the matrices $\iota(X), \iota(Y) \in M_m(\mathbb{R})$ converges, and to the image under ι of the same element. It therefore suffices to prove absolute convergence and the identity in $M_m(\mathbb{R})$, where exp and log are the concrete series of chapter 1. Fix the operator norm $\|\cdot\|$ on $M_m(\mathbb{R})$, which is submultiplicative, so that $\|[A, B]\| \le 2\|A\|\,\|B\|$.

Step 2: absolute convergence of the term-by-term series. Set $W = \exp X \exp Y - I$. The bound $\|\exp X\| \le e^{\|X\|}$ of proposition 1.3.3 gives

$$\|W\| = \|\exp X \exp Y - I\| \le e^{\|X\|} e^{\|Y\|} - 1 = e^{\|X\| + \|Y\|} - 1$$

so if $\|X\| + \|Y\| < \log 2$ then $\|W\| < 1$ and the logarithm series $\log(I + W) = \sum_{k \ge 1} (-1)^{k+1} W^k / k$ of proposition 1.3.6 converges absolutely to a well-defined matrix $Z = \log(\exp X \exp Y)$. Expanding $W = \sum_{p+q \ge 1} X^p Y^q / (p!\, q!)$ from the two exponential series and substituting into the logarithm series produces, after grouping monomials by total degree, exactly the summands of (3.6) before the passage from associative monomials to brackets; the associated series of absolute values is dominated term by term by the scalar series obtained by replacing X, Y with $\|X\|$, $\|Y\|$, namely $-\log(2 - e^{s+t})$ with $s = \|X\|$, $t = \|Y\|$, which is finite precisely when $e^{s+t} < 2$. Hence the associative form of the series converges absolutely for $\|X\| + \|Y\| < \log 2$.

Step 3: the bracket form and its radius. The passage from the associative monomials to the iterated brackets of (3.6) is a finite regrouping within each fixed total degree n: the Dynkin

coefficient of a bracket of degree n is a fixed rational number, and there are finitely many brackets of each degree, so the bracket series is the absolutely convergent associative series collected into finite blocks. Each iterated bracket of degree n is bounded by $\|[X^{p_1}\cdots Y^{q_n}]\| \leq 2^{n-1}\|X\|^{\sum p_i}\|Y\|^{\sum q_i}$, by iterating $\|[A,B]\| \leq 2\|A\|\|B\|$. Summing these bounds against the Dynkin coefficients gives a scalar majorant convergent on a ball $\|X\| + \|Y\| < \rho$ for some $\rho \in (0, \log 2]$ determined by the coefficient growth and the factor 2 from the bracket estimate. On that ball the bracket series converges absolutely to the same sum $Z = \log(\exp X \exp Y)$ as the associative series, since the two are absolute rearrangements of one another. The transferred constant ρ for the original $\mathfrak{g}$ absorbs the distortion of the norm under the fixed linear isomorphism ι onto its image. A power series converging absolutely to a matrix-valued (hence $\mathfrak{g}$-valued) function is real-analytic there, completing the proof.

The proposition upgrades the formula from a formal identity to an equation between elements of $\mathfrak{g}$, valid on an explicit neighborhood. The radius ρ is not sharp and depends on the algebra: for a nilpotent Lie algebra the series terminates and $\rho = \infty$, while for $\mathfrak{sl}_2(\mathbb{R})$ the exponential fails to be globally surjective and no global formula can hold (exercise 3.5.1 (6)). What matters for the structure theory is only that some positive radius exists, so that the group law is analytic in the canonical coordinates exp provides.

The proof of the theorem now proceeds in two parts: the explicit low-order computation, done from the series, and the recursion that produces the higher coefficients and certifies they are brackets. The low-order part is the one the reader should be able to reproduce; the recursion is where Dynkin's closed form comes from, and it is stated and its key step verified, with the closed evaluation of the coefficients cited.

***Proof Proof of theorem 3.5.1*:**
By proposition 3.5.2 it suffices to work in $M_m(\mathbb{R})$ after the faithful embedding of its Step 1, so throughout $X, Y \in M_m(\mathbb{R})$ and exp, log are the series of definition 1.3.2 and proposition 1.3.6.

Step 1: the low-order terms by direct expansion. Write the two exponential series to third order,

$$\exp X = I + X + \tfrac{1}{2}X^2 + \tfrac{1}{6}X^3 + \cdots, \qquad \exp Y = I + Y + \tfrac{1}{2}Y^2 + \tfrac{1}{6}Y^3 + \cdots$$

and multiply, collecting by total degree,

$$\exp X \exp Y = I + (X+Y) + \tfrac{1}{2}\big(X^2 + 2XY + Y^2\big) + \tfrac{1}{6}\big(X^3 + 3X^2Y + 3XY^2 + Y^3\big) + \cdots$$

where in each degree the cross terms carry X's to the left of Y's, since $\exp X$ is the left factor. Denote by $W = \exp X \exp Y - I$ the sum of the terms of degree ≥ 1 displayed above, and apply the logarithm series $\log(I+W) = W - \frac{1}{2}W^2 + \frac{1}{3}W^3 - \cdots$, retaining only terms of total degree ≤ 3.

The degree-one part of $\log(I+W)$ is the degree-one part of W, namely $X+Y$. For degree two, W contributes $\frac{1}{2}(X^2+2XY+Y^2)$ and $-\frac{1}{2}W^2$ contributes $-\frac{1}{2}(X+Y)^2 = -\frac{1}{2}(X^2+XY+YX+Y^2)$; adding,

$$\tfrac{1}{2}(X^2 + 2XY + Y^2) - \tfrac{1}{2}(X^2 + XY + YX + Y^2) = \tfrac{1}{2}(XY - YX) = \tfrac{1}{2}[X,Y]$$

The squares X^2 and Y^2 cancel exactly and only the commutator survives, the second-order term of the theorem.

Step 2: the third-order term. Collect the degree-three contributions from W, $-\frac{1}{2}W^2$, and $\frac{1}{3}W^3$. Write $W = W_1 + W_2 + W_3 + \cdots$ by degree, with $W_1 = X + Y$ and $W_2 = \frac{1}{2}(X^2 + 2XY + Y^2)$.

The degree-three part of W is $W_3 = \frac{1}{6}(X^3 + 3X^2Y + 3XY^2 + Y^3)$; the degree-three part of $-\frac{1}{2}W^2$ is $-\frac{1}{2}(W_1W_2 + W_2W_1)$; the degree-three part of $\frac{1}{3}W^3$ is $\frac{1}{3}W_1^3 = \frac{1}{3}(X+Y)^3$. Summing these and tallying every associative monomial of degree three, the pure powers X^3, Y^3 cancel and the mixed monomials assemble into

$$\big[\log(\exp X \exp Y)\big]_{\deg 3} = \tfrac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big)$$

with the stated coefficient $\frac{1}{12}$. The same bookkeeping at degree four yields the term $-\frac{1}{24}[Y,[X,[X,Y]]]$ recorded in the theorem.

Step 3: the recursion and universality. That the phenomenon of Steps 1 and 2 persists to all orders, so that every term is a bracket, follows from a differential recursion. Fix X and set $Z(t) = \log(\exp X \exp tY)$, a curve in $M_m(\mathbb{R})$ with $Z(0) = X$. Differentiating $\exp Z(t) = \exp X \exp tY$ in t and using the standard formula for the differential of the exponential,

$$\frac{d}{dt}\exp Z(t) = \left(\int_0^1 \exp(s\,\mathrm{ad}Z(t))\,ds\right)\big(\dot Z(t)\big)\,\exp Z(t)$$

together with $\frac{d}{dt}(\exp X \exp tY) = \exp X \exp tY \cdot Y = \exp Z(t)\,Y$, one uses the conjugation identity $\exp Z(t)\,Y = \big(e^{\mathrm{ad}Z(t)}Y\big)\exp Z(t)$ to move the right factor across, and then, cancelling $\exp Z(t)$ on the right, obtains the identity

$$\left(\int_0^1 \exp(s\,\mathrm{ad}Z(t))\,ds\right)\big(\dot Z(t)\big) = e^{\mathrm{ad}Z(t)}\,Y$$

The operator $\int_0^1 \exp(s\,\mathrm{ad}Z)\,ds$ equals $g(\mathrm{ad}Z)$ for the analytic function $g(u) = (e^u - 1)/u$, which is invertible near $u = 0$; composing its inverse $u/(e^u - 1)$ with the factor $e^{\mathrm{ad}Z}$ on the right produces the analytic function $u/(1 - e^{-u})$, the generating function of the Bernoulli numbers in the convention $B_1 = +\frac{1}{2}$,

$$\frac{u}{1-e^{-u}} = \sum_{k=0}^{\infty}\frac{B_k}{k!}\,u^k = 1 + \tfrac{1}{2}u + \tfrac{1}{12}u^2 - \tfrac{1}{720}u^4 + \cdots \tag{3.7}$$

Inverting gives the differential equation, whose operator ratio is read off from its power series on the right, so that the kernel of $\mathrm{ad}Z$, forced by $[Z,Z] = 0$, poses no problem of dividing by a noninvertible $I - e^{-\mathrm{ad}Z}$,

$$\dot Z(t) = \frac{\mathrm{ad}Z(t)}{I - \exp(-\mathrm{ad}Z(t))}\,(Y) = \sum_{k=0}^{\infty}\frac{B_k}{k!}\,(\mathrm{ad}Z(t))^k\,(Y) \tag{3.8}$$

with initial condition $Z(0) = X$. Every term on the right of (3.8) is an iterated bracket of $Z(t)$ and Y, and $Z(0) = X$ is a single letter; solving by Picard iteration therefore produces $Z(1) = \log(\exp X \exp Y)$ as a series each of whose terms is an iterated bracket of X and Y, with rational coefficients built from the B_k. Because (3.8) refers only to the bracket and the universal constants B_k, the resulting series is the same for every Lie algebra, which is the asserted universality. Integrating (3.8) order by order reproduces $X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}([X,[X,Y]] + [Y,[Y,X]]) + \cdots$ and, carried out in closed form, yields precisely Dynkin's expression (3.6); the combinatorial evaluation of the coefficients is a bookkeeping computation whose details are in [10]. This establishes the formula and that all its terms are brackets, completing the proof.

The proof exhibits two faces of the formula. The direct expansion of Steps 1 and 2 is elementary and self-checking: multiply the series, take the logarithm, watch the pure powers cancel. The recursion of Step 3 is the engine, encoding the entire series in a single ordinary differential equation (3.8) whose only ingredients are the bracket and the Bernoulli numbers. The appearance of the Bernoulli generating function (3.7) is not accidental: it is the same series that governs the differential of the exponential map, and it is why the coefficients $\frac{1}{2}, \frac{1}{12}, 0, -\frac{1}{720}, \dots$ of BCH are exactly the Bernoulli numbers divided by factorials. That $B_3 = 0$, and more generally $B_k = 0$ for odd $k \geq 3$, is the reason the series thins out at odd orders.

The formula answers the question left open by Lie's third theorem: not only does a Lie algebra determine a simply-connected group, it determines the group multiplication explicitly near the identity. This is the local reconstruction, recorded as the following corollary.

Corollary 3.5.3: The Local Group Law from the Bracket

Let G and G' be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{g}'$, and suppose $\varphi\colon \mathfrak{g} \to \mathfrak{g}'$ is an isomorphism of Lie algebras. Then there are neighborhoods U of e in G and U' of e' in G' and a diffeomorphism $\Phi\colon U \to U'$ that is a *local isomorphism*: $\Phi(e) = e'$, and $\Phi(gh) = \Phi(g)\Phi(h)$ whenever g, h, and gh all lie in U. Consequently the multiplication of G on a neighborhood of e is determined by the bracket of $\mathfrak{g}$ alone, and two Lie groups with isomorphic Lie algebras are locally isomorphic.

***Proof*:**
Choose the exponential charts $\exp\colon V \to U$ and $\exp'\colon V' \to U'$ of theorem 2.3.4, shrinking V, V' so that the BCH series of proposition 3.5.2 converges on V and on $\varphi(V)$ and so that $\varphi(V) \subseteq V'$. Define $\Phi = \exp' \circ \varphi \circ \exp^{-1}\colon U \to U'$, a diffeomorphism carrying $e = \exp 0$ to $e' = \exp' 0$. For $g = \exp X$ and $h = \exp Y$ with $X, Y \in V$ small enough that $\log(gh) \in V$, the formula (3.6) gives $\log(gh) = B(X, Y)$, where $B(X, Y) = X + Y + \frac{1}{2}[X, Y] + \cdots$ is the universal bracket series. The map φ is a Lie-algebra isomorphism and so commutes with every bracket, and B is a universal series in the bracket with the same coefficients on both sides, so $\varphi(B(X, Y)) = B(\varphi X, \varphi Y)$. Therefore

$$\Phi(g)\Phi(h) = \exp'(\varphi X)\exp'(\varphi Y) = \exp'\big(B(\varphi X, \varphi Y)\big) = \exp'\big(\varphi B(X, Y)\big) = \exp'\big(\varphi(\log(gh))\big) = \Phi(gh)$$

using theorem 3.5.1 in G' for the second equality and universality for the third. Thus Φ is a local homomorphism, hence a local isomorphism, since φ^{-1} produces the inverse the same way. The multiplication on U is then $gh = \exp\big(B(\log g, \log h)\big)$, a formula in the bracket alone, as we sought to show.

This is the local half of Lie's third theorem made explicit. Lie's third theorem (theorem 3.4.4) says an isomorphism of Lie algebras integrates to an isomorphism of the simply-connected groups; the corollary extracts the part that needs no global topology, namely that the algebras being isomorphic already forces the groups to look identical near the identity. Global obstructions, differences in the fundamental group or the center, are invisible to the bracket and appear only when the local isomorphism is continued around noncontractible loops. The pair SU(2) and SO(3) is the standing example: their Lie algebras are isomorphic (both isomorphic to $(\mathbb{R}^3, \times)$, with $\mathfrak{so}(3) \cong \mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$), so BCH gives the same local multiplication, and the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5 is a local isomorphism that fails to be global precisely because $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2$ is nontrivial

(corollary 1.5.6). The classification of the connected groups sharing one algebra as quotients $\tilde{G}/\Gamma$ (corollary 3.4.5) is the global completion of this local picture.

The formula also isolates exactly when the group is abelian near the identity. If $[X,Y]=0$ then every bracket in the BCH series vanishes and the formula collapses to its first term.

Corollary 3.5.4: Commuting Exponentials

Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $X,Y\in\mathfrak{g}$ with $[X,Y]=0$. Then, for X,Y in the domain of convergence of proposition 3.5.2,

$$\exp X\exp Y=\exp(X+Y)=\exp Y\exp X$$

Conversely, if $\exp(sX)$ and $\exp(tY)$ commute for all small s,t, then $[X,Y]=0$.

***Proof*:**
If $[X,Y]=0$ then every iterated bracket in (3.6) of degree ≥ 2 contains a factor $[X,Y]$ or $[Y,X]$ as an innermost commutator, hence vanishes; the series reduces to $X+Y$, so $\log(\exp X\exp Y)=X+Y$ and $\exp X\exp Y=\exp(X+Y)$. Exchanging the roles of X and Y and using $\exp(X+Y)=\exp(Y+X)$ gives $\exp X\exp Y=\exp Y\exp X$. For the converse, the group commutator $\exp(sX)\exp(tY)\exp(-sX)\exp(-tY)$ has BCH expansion $st[X,Y]+O(3)$ in s,t; if it is the identity for all small s,t then its second-order term $st[X,Y]$ vanishes identically, forcing $[X,Y]=0$, completing the proof.

In the matrix case this recovers the additivity of the exponential on commuting matrices proved directly in proposition 1.3.5: there the identity $\exp X\exp Y=\exp(X+Y)$ for $XY=YX$ was obtained by rearranging the product of two absolutely convergent series, and BCH now explains it structurally, as the statement that all correction terms are brackets and therefore disappear the instant the bracket does. The bracket is the entire obstruction to the naive exponential law, measured to leading order by $\frac{1}{2}[X,Y]$ and to all orders by the BCH series.

Example 3.5.5: BCH on the Heisenberg Algebra

Let $\mathfrak{h}_3$ be the three-dimensional *Heisenberg algebra*: the Lie algebra with basis P,Q,C and single nontrivial bracket $[P,Q]=C$, all other brackets among basis elements zero. It is realized concretely as the strictly upper-triangular 3×3 real matrices, with

$$P=\begin{pmatrix}0&1&0\\0&0&0\\0&0&0\end{pmatrix},\quad Q=\begin{pmatrix}0&0&0\\0&0&1\\0&0&0\end{pmatrix},\quad C=\begin{pmatrix}0&0&1\\0&0&0\\0&0&0\end{pmatrix}$$

so that $[P,Q]=PQ-QP=C$ and C is central. Every triple bracket vanishes: $[Z,[Z',Z'']]$ for basis elements is a bracket involving C, which brackets to zero with everything. Hence in (3.6) every term of bracket-degree ≥ 3 is zero, and the series terminates after the $\frac{1}{2}[X,Y]$ term:

$$\log(\exp X\exp Y)=X+Y+\tfrac{1}{2}[X,Y]\qquad\text{for all } X,Y\in\mathfrak{h}_3$$

an exact identity with no domain restriction, since a terminating series converges everywhere. Writing $X=aP+bQ+cC$ and $X'=a'P+b'Q+c'C$, the bracket is $[X,X']=(ab'-a'b)C$, so

$$\exp X\exp X'=\exp\Big((a+a')P+(b+b')Q+\big(c+c'+\tfrac{1}{2}(ab'-a'b)\big)C\Big)$$

The corresponding Heisenberg group H_3 of unipotent upper-triangular matrices therefore has the multiplication law $(a,b,c)\cdot(a',b',c') = (a+a', b+b', c+c'+\tfrac{1}{2}(ab'-a'b))$ in exponential coordinates, the standard cocycle presentation. This is the smallest nonabelian example in which BCH is visible in full: the group is nonabelian, yet the failure of commutativity is captured entirely by the single quadratic term $\tfrac{1}{2}(ab'-a'b)$.

The Heisenberg example is the cleanest illustration of the formula because the series stops after two terms. More generally, on any *nilpotent* Lie algebra the iterated brackets eventually vanish, so the BCH series is a polynomial and converges everywhere; the exponential map is then a global diffeomorphism onto a simply-connected nilpotent group, and BCH presents its multiplication as a polynomial map, a point developed with the nilpotent Lie algebras of Part II. At the opposite extreme, on an abelian Lie algebra every bracket is zero, BCH reduces to $X+Y$, and exp is a homomorphism from $(\mathfrak{g},+)$ to G, recovering that a connected abelian Lie group is a quotient of its own Lie algebra. Between these poles the BCH series measures exactly how far a group departs from being the additive group of its Lie algebra.

Exercise 3.5.1

1. Verify the third-order term of the BCH formula directly. Using $\log(\exp X\exp Y) = W - \tfrac{1}{2}W^2 + \tfrac{1}{3}W^3 - \cdots$ with $W = \exp X\exp Y - I$, collect all degree-three associative monomials in X and Y and confirm they sum to $\tfrac{1}{12}([X,[X,Y]]+[Y,[Y,X]])$.

2. Show from $\exp X\exp Y = \exp Z$ that BCH satisfies the symmetry $\log(\exp X\exp Y) = -\log(\exp(-Y)\exp(-X))$. Deduce that the degree-n term $Z_n(X,Y)$ obeys $Z_n(X,Y) = (-1)^{n+1}Z_n(Y,X)$, and check this against the terms $X+Y$, $\tfrac{1}{2}[X,Y]$, and $\tfrac{1}{12}([X,[X,Y]]+[Y,[Y,X]])$.

3. Prove the conjugation identity $\exp X\,\exp Y\,\exp(-X) = \exp\big(\mathrm{Ad}(\exp X)\,Y\big) = \exp\big(e^{\mathrm{ad}X}Y\big)$ for a matrix group, and explain why, unlike BCH, it holds globally and needs no logarithm. Use proposition 2.5.5 and the identity $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}X}$.

4. On the Heisenberg algebra $\mathfrak{h}_3$ of example 3.5.5, compute $\exp(P)$, $\exp(Q)$, and $\exp(P)\exp(Q)$ explicitly as 3×3 matrices, and confirm by direct multiplication that $\exp(P)\exp(Q) = \exp(P+Q+\tfrac{1}{2}C)$. Then verify $\exp(P)\exp(Q)\exp(-P)\exp(-Q) = \exp(C)$, the group commutator, and interpret it in light of corollary 3.5.4.

5. Let $\mathfrak{g}$ be an abelian Lie algebra and G a connected Lie group with $\mathrm{Lie}(G) = \mathfrak{g}$. Show $\exp\colon(\mathfrak{g},+)\to G$ is a surjective Lie group homomorphism with discrete kernel, and conclude $G\cong\mathfrak{g}/\Gamma$ for a discrete subgroup $\Gamma\subseteq\mathfrak{g}$. Identify Γ for $G=\mathbb{T}=\mathbb{R}/\mathbb{Z}$ and for $G=\mathbb{R}_{>0}$.

6. Explain why the BCH series cannot converge on all of $\mathfrak{sl}_2(\mathbb{R})\times\mathfrak{sl}_2(\mathbb{R})$. Show that the matrix $g = \left(\begin{smallmatrix}-1 & 1\\ 0 & -1\end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$ is not $\exp Z$ for any $Z\in\mathfrak{sl}_2(\mathbb{R})$, then exhibit $X,Y\in\mathfrak{sl}_2(\mathbb{R})$ with $\exp X\exp Y = g$, so that no $Z=\log(\exp X\exp Y)$ exists for these X,Y. Relate this to the failure of exp to be surjective on $\mathrm{SL}(2,\mathbb{R})$.

4

Homogeneous Spaces, Actions, and Covering Groups

The correspondence of chapter 3 turned the Lie algebra into a tool for building and recognizing groups. This chapter turns the same machinery outward, onto the spaces a Lie group acts on. The organizing observation is that a great many of the manifolds of geometry, the spheres, the projective spaces, the Grassmannians and flag manifolds, do more than carry a Lie group action: they are *homogeneous*, meaning the group acts transitively, so that every point looks like every other, and the manifold is recovered as a quotient G/H by the stabilizer of a single point. The quotient construction, and the theorem that G/H is again a smooth manifold whenever H is closed, converts group theory into a factory for manifolds with a large symmetry group already built in.

Two threads run through the chapter. The first is the quotient itself: for a closed subgroup $H \leq G$ the coset space G/H carries a unique smooth structure making $G \to G/H$ a submersion, indeed a principal H-bundle, and the orbit-stabilizer theorem identifies every transitive action with such a quotient. The isotropy representation of H on the tangent space $\mathfrak{g}/\mathfrak{h}$ then linearizes the geometry at a point, the first step toward the curvature computations of the symmetric-space theory in Part IV. The second thread returns to the loose end of chapter 3: the fundamental group. Covering maps of a connected Lie group are themselves homomorphisms, with central discrete kernel, so the connected groups sharing a fixed Lie algebra are organized by the subgroups of π_1, and the universal cover sits atop the tower. The running example gathers both threads at once, for the Hopf fibration $\mathrm{SU}(2) \to \mathrm{SU}(2)/\mathrm{U}(1) = S^2$ and the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ exhibit the same group as a bundle over a sphere and as a two-sheeted cover of a rotation group.

4.1 Group Actions and the Quotient Manifold G/H

Chapter 3 turned the algebraic structure of a Lie group into differential-geometric structure: a closed subgroup is automatically an embedded submanifold (theorem 3.2.1), and a continuous homomorphism is automatically smooth (corollary 3.2.5). This chapter puts that machinery to work building manifolds. The construction that does the building is the coset space G/H of a closed subgroup, and the principle behind it is that a Lie group is a producer of symmetric spaces: whenever a group acts smoothly and transitively on a set, the set is a coset space of the group, and if the stabilizer is closed, that coset space is a manifold. Spheres, projective spaces, Grassmannians, and flag manifolds all arise this way, each as G/H for a suitable classical group G and a suitable closed subgroup H. This section builds the factory; the next two run the examples through it.

The starting point is the notion of a group acting on a manifold. Actions formalize symmetry, and the two invariants attached to a point, its orbit and its stabilizer, are the raw material from which coset spaces are assembled. The first task is to set down the vocabulary of smooth actions carefully, because the quotient construction depends on knowing that stabilizers are closed and that orbits are the fibers of a smooth map.

Definition 4.1.1: Smooth Action, Orbit, and Stabilizer

Let G be a Lie group and M a smooth manifold. A *smooth (left) action* of G on M is a smooth map

$$\theta\colon G \times M \to M, \qquad \theta(g,x) = g \cdot x$$

satisfying $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in M$. For each fixed g the map $\theta_g\colon M \to M$, $x \mapsto g \cdot x$, is a diffeomorphism with inverse $\theta_{g^{-1}}$. Given $x \in M$:

- the *orbit* of x is $G \cdot x = \{g \cdot x \mid g \in G\} \subseteq M$;
- the *stabilizer* (or isotropy subgroup) of x is $G_x = \{g \in G \mid g \cdot x = x\} \subseteq G$.

If G also acts smoothly on a manifold N, a smooth map $F\colon M \to N$ is *G-equivariant* if $F(g \cdot x) = g \cdot F(x)$ for all $g \in G$ and $x \in M$. The action is

- *effective* (or faithful) if $g \cdot x = x$ for all x forces $g = e$, that is, only the identity acts trivially;
- *free* if $g \cdot x = x$ for some x forces $g = e$, that is, every stabilizer is trivial;
- *transitive* if $G \cdot x = M$ for some (equivalently every) $x \in M$, that is, there is a single orbit.

The two invariants pull in opposite directions. The orbit $G \cdot x$ measures how far G can move x; a transitive action has a single orbit and so identifies M with the shape of G as seen from x. The stabilizer G_x measures the redundancy in this identification, the group elements that fix x and are therefore invisible to it. The relation between them is the orbit-stabilizer correspondence: the elements of the orbit are in bijection with the cosets G/G_x, since $g \cdot x = g' \cdot x$ exactly when $g^{-1}g' \in G_x$. This is the set-theoretic skeleton of every homogeneous space (section 4.2 upgrades it to a diffeomorphism). A free action is one with no redundancy, so the orbit is a faithful copy of G; an effective action is the weaker condition that no single group element is invisible at every point at once. The conjugation action below is effective whenever the center is trivial but rarely free, and this

contrast is exactly why the center of a group is worth naming.

The definition is abstract, so it is worth pinning it to the actions this chapter uses. Four are basic, and each will reappear.

Example 4.1.2: The Four Basic Actions

1. $\mathrm{GL}_n(\mathbb{R})$ *on* $\mathbb{R}^n$. The map $(A, v) \mapsto Av$ is smooth, bilinear in fact, and satisfies $Iv = v$ and $A(Bv) = (AB)v$; it is the defining representation. The orbit of 0 is $\{0\}$, and the orbit of any $v \neq 0$ is all of $\mathbb{R}^n \setminus \{0\}$, since an invertible matrix carries v to any other nonzero vector (extend v and the target each to a basis and take the change-of-basis matrix). The action is effective: if $Av = v$ for every v then $A = I$. It is not transitive on $\mathbb{R}^n$ (the origin is its own orbit) and not free (the stabilizer of e_1 contains every A fixing e_1, a copy of the affine group of $\mathbb{R}^{n-1}$).

2. $\mathrm{O}(n)$ *on* S^{n-1}. An orthogonal matrix preserves the Euclidean norm, so it maps the unit sphere $S^{n-1} = \{v \in \mathbb{R}^n \mid |v| = 1\}$ to itself; the restriction $(A, v) \mapsto Av$ is a smooth action on S^{n-1}. It is transitive: given unit vectors v, w, extend each to an orthonormal basis and let A be the change-of-basis matrix, which is orthogonal and carries v to w. The stabilizer of $e_n \in S^{n-1}$ is the copy of $\mathrm{O}(n-1)$ acting on the orthogonal complement $e_n^{\perp}$, so this action realizes S^{n-1} as the coset space $\mathrm{O}(n)/\mathrm{O}(n-1)$, the theme of section 4.3.

3. *G on itself by left translation.* The map $(g, x) \mapsto gx$ is smooth because multiplication is smooth, and $\theta_g = L_g$ is the left translation, a diffeomorphism by proposition 2.1.2. This action is transitive (any x is carried to any y by $g = yx^{-1}$) and free (if $gx = x$ then $g = e$). Homogeneity of a Lie group, the fact that every point looks like every other, is precisely the transitivity of this action, and it is the reason a chart at e can be spread over the whole group.

4. *G on itself by conjugation.* The map $(g, x) \mapsto gxg^{-1}$ is smooth, and $\theta_g = C_g$ is the inner automorphism, a diffeomorphism again by proposition 2.1.2. The orbits are the conjugacy classes of G, and the stabilizer of x is its centralizer $\{g \mid gx = xg\}$. The action is effective on a group with trivial center but never free unless $G = \{e\}$, since every element fixes e and, more generally, every element of the center acts trivially. Its differential at e is the adjoint representation Ad of definition 2.5.1.

The left-translation action is transitive and free, so G is a single orbit with no redundancy, which is why the quotient by a subgroup below is so well behaved: left translation by G descends to a transitive action on G/H, and it is this residual symmetry that spreads a chart near the identity coset over the whole quotient. The conjugation action, by contrast, is the source of the adjoint representation and, further downstream, of the entire structure theory of Part II; its orbits are the conjugacy classes, and its infinitesimal version Ad already appeared in chapter 2.

With actions in hand, the central construction of the section is the quotient. Fix a closed subgroup H of G and let

$$G/H = \{gH \mid g \in G\}$$

be the set of left cosets, with the quotient topology induced by the projection $\pi\colon G \to G/H$, $g \mapsto gH$. The group G acts on G/H on the left by $a \cdot (gH) = (ag)H$, and this action is transitive by construction. The question is whether G/H carries a smooth structure making this a smooth action and π a smooth map. It does, and the closedness of H is exactly what makes it work: H is an embedded

submanifold by theorem 3.2.1, and this is what allows the cosets to be separated by a smooth slice.

Theorem 4.1.3: The Quotient Manifold Theorem

Let G be a Lie group and $H \subseteq G$ a closed subgroup, with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$. Then the coset space G/H carries a unique smooth manifold structure for which the projection $\pi\colon G \to G/H$ is a smooth submersion. With this structure:

1. $\dim(G/H) = \dim G - \dim H$;
2. the left action $G \times G/H \to G/H$, $(a, gH) \mapsto (ag)H$, is smooth and transitive;
3. the quotient topology on G/H is Hausdorff and second countable, so G/H is a smooth manifold in the usual sense.

The idea of the proof is to build a chart on G/H near the identity coset eH by choosing a slice through e transverse to H. Since H is an embedded submanifold with tangent space $\mathfrak{h}$ at e, a complementary subspace $\mathfrak{m}$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ gives, via the exponential, a submanifold $S = \exp(\text{ball in } \mathfrak{m})$ that crosses H transversally at e. The point is that S meets each coset near eH exactly once, so S is a local model for G/H and the map $\pi|_S$ is the chart. Left translation by G then spreads this chart over the whole quotient, and the smoothness of the transition maps is where the group structure and the exponential chart do their work. The submersion and constant-rank facts about smooth maps are cited to [13]; the coset charts are built here.

***Proof*:**
Fix an inner product on $\mathfrak{g}$, giving norms and open balls, and choose a vector-space complement $\mathfrak{m}$ to $\mathfrak{h}$, so that

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$$

The strategy is to use $\mathfrak{m}$ as the coordinate directions transverse to H and π as the chart map. There are five steps: build the slice, show it meets nearby cosets once, define the identity chart, spread it by translation, and verify the transitions and the submersion property.

Step 1: the slice through e. Consider the map

$$\Phi\colon \mathfrak{m} \oplus \mathfrak{h} \to G, \qquad \Phi(Y, X) = \exp(Y)\exp(X)$$

Its differential at $(0,0)$ sends (Y, X) to $Y + X$, by the same product-rule computation as in theorem 3.2.1: differentiating $\exp(sY)\exp(0)$ in s gives Y, differentiating $\exp(0)\exp(sX)$ gives X, and the product rule at e adds them, using $d(\exp)_0 = \mathrm{id}$ from theorem 2.3.4. Under $\mathfrak{m} \oplus \mathfrak{h} = \mathfrak{g}$ this differential is the identity, so the inverse function theorem ([13, Inverse Function Theorem for Manifolds]) furnishes open balls $B_{\mathfrak{m}} \subseteq \mathfrak{m}$ and $B_{\mathfrak{h}} \subseteq \mathfrak{h}$ about the origins on which Φ restricts to a diffeomorphism onto an open neighborhood Ω of e in G. Set

$$S = \exp(B_{\mathfrak{m}}) \subseteq G$$

an embedded submanifold of G through e with tangent space $\mathfrak{m}$ at e; it is the *slice*. Because Φ is a diffeomorphism on $B_{\mathfrak{m}} \times B_{\mathfrak{h}}$, every element $g \in \Omega$ factors *uniquely* as $g = \exp(Y)\exp(X)$ with $Y \in B_{\mathfrak{m}}$ and $X \in B_{\mathfrak{h}}$; since $\exp(X) \in H$ for $X \in B_{\mathfrak{h}} \subseteq \mathfrak{h}$, this says g lies in the coset $\exp(Y)H$ with $Y \in B_{\mathfrak{m}}$ determined by g.

Step 2: the slice meets nearby cosets exactly once. The claim is that, after shrinking $B_{\mathfrak{m}}$ if

necessary, the exponentials of distinct points of $B_{\mathfrak{m}}$ lie in distinct cosets:

$$Y_1, Y_2 \in B_{\mathfrak{m}}, \quad \exp(Y_1)H = \exp(Y_2)H \implies Y_1 = Y_2 \tag{4.1}$$

The equality of cosets means $h := \exp(Y_2)^{-1}\exp(Y_1) \in H$. Suppose no shrinking made (4.1) hold. Then there are sequences $Y_1^{(n)}, Y_2^{(n)} \in \mathfrak{m}$ both tending to 0, with $Y_1^{(n)} \neq Y_2^{(n)}$, such that $h_n := \exp(Y_2^{(n)})^{-1}\exp(Y_1^{(n)}) \in H$ for all n. Now $h_n \to e$, so for large n the element h_n lies in Ω and factors as $h_n = \exp(Y_n)\exp(X_n)$ with $Y_n \in B_{\mathfrak{m}}$, $X_n \in B_{\mathfrak{h}}$, both tending to 0. But $h_n \in H$ forces $\exp(Y_n) = h_n \exp(X_n)^{-1} \in H$ with $Y_n \in \mathfrak{m}$; the transversality established in theorem 3.2.1 (its Step 4, that exp of a small nonzero element of $\mathfrak{m}$ never lands in H) then gives $Y_n = 0$ once n is large. Hence $h_n = \exp(X_n) \in \exp(B_{\mathfrak{h}})$ for large n. Feeding this back, $\exp(Y_1^{(n)}) = \exp(Y_2^{(n)})\exp(X_n)$ expresses one element of Ω in two ways as $\Phi(\cdot,\cdot)$; the uniqueness of the factorization in Step 1 forces $Y_1^{(n)} = Y_2^{(n)}$ and $X_n = 0$, contradicting $Y_1^{(n)} \neq Y_2^{(n)}$. Therefore (4.1) holds on a small enough ball, which we fix and continue to call $B_{\mathfrak{m}}$.

Combining Steps 1 and 2, the restriction $\pi|_S \colon S \to G/H$ is injective, and its image $\pi(S)$ is the set of cosets $\exp(Y)H$ with $Y \in B_{\mathfrak{m}}$. Moreover $\pi(S)$ is open in G/H. By definition of the quotient topology it suffices that the saturation $\pi^{-1}(\pi(S)) = S \cdot H = \bigcup_{h\in H} Sh$ be open in G, and it is: each $Sh = R_h(S)$ is the image of the open-in-its-slice set under a diffeomorphism, and the union $S \cdot H \supseteq \Omega$ contains the open neighborhood Ω of e translated across H, so $S \cdot H = \bigcup_{h\in H} \Omega h$ is a union of open sets.

Step 3: the identity chart. Define

$$\varphi_e \colon \pi(S) \to B_{\mathfrak{m}}, \qquad \varphi_e(\exp(Y)H) = Y$$

This is well defined and injective by (4.1), and surjective onto $B_{\mathfrak{m}}$ by construction, hence a bijection from the open set $\pi(S)$ onto the open ball $B_{\mathfrak{m}} \subseteq \mathfrak{m} \cong \mathbb{R}^{\dim G - \dim H}$. It is a homeomorphism. Its inverse $Y \mapsto \exp(Y)H = \pi(\exp Y)$ is continuous, being a composite of the continuous exp and π. For continuity of φ_e itself, note that $\varphi_e = \log_{\mathfrak{m}} \circ \sigma$, where $\sigma \colon \pi(S) \to S$ is the section $\exp(Y)H \mapsto \exp(Y)$ and $\log_{\mathfrak{m}} = (\exp|_{B_{\mathfrak{m}}})^{-1}$ is a diffeomorphism $S \to B_{\mathfrak{m}}$; σ is continuous because $\pi|_S$ is a continuous open bijection onto $\pi(S)$, hence a homeomorphism, and σ is its inverse followed by the inclusion $S \hookrightarrow G$. Thus φ_e is a chart on G/H around the coset eH, valued in $\mathbb{R}^{\dim G - \dim H}$.

Step 4: spreading the chart by left translation. For each $a \in G$ let $\tau_a \colon G/H \to G/H$ be the map $gH \mapsto (ag)H$; it is a homeomorphism, a bijection with continuous inverse $\tau_{a^{-1}}$, both induced by the continuous open π and left translation L_a on G via $\pi \circ L_a = \tau_a \circ \pi$. Define charts around an arbitrary coset aH by transporting the identity chart:

$$\varphi_a \colon \tau_a(\pi(S)) \to B_{\mathfrak{m}}, \qquad \varphi_a = \varphi_e \circ \tau_a^{-1}$$

The domains $\tau_a(\pi(S))$ are open and cover G/H: given a coset bH, take $a = b$, and then $bH = \tau_b(eH) \in \tau_b(\pi(S))$ since $eH \in \pi(S)$. So $\{(\tau_a(\pi(S)), \varphi_a)\}_{a\in G}$ is an atlas into $\mathbb{R}^{\dim G - \dim H}$.

Step 5: smooth transitions and the submersion property. Two charts φ_a, φ_b with overlapping domains have transition map

$$\varphi_b \circ \varphi_a^{-1} = \varphi_e \circ (\tau_b^{-1} \circ \tau_a) \circ \varphi_e^{-1} = \varphi_e \circ \tau_c \circ \varphi_e^{-1}, \qquad c = b^{-1}a$$

on its domain of definition. To see this is smooth, unwind it through G. On its domain, $\varphi_e^{-1}(Y) = \exp(Y)H$; applying τ_c gives $(c\exp(Y))H$; and applying φ_e requires factoring $c\exp(Y)$

into slice-times-H form. Where the overlap is nonempty, $c\exp(Y)$ lies in the region Ω where Φ is a diffeomorphism, so its slice component is $P_{\mathfrak{m}}\big(\Phi^{-1}(c\exp(Y))\big)$, the $\mathfrak{m}$-part of Φ^{-1}, where $P_{\mathfrak{m}}\colon \mathfrak{m}\oplus\mathfrak{h}\to\mathfrak{m}$ is the linear projection. This is a composite of the smooth maps $Y\mapsto c\exp(Y)$ (smooth, as multiplication and exp are smooth), the smooth Φ^{-1}, and the linear $P_{\mathfrak{m}}$. Hence $\varphi_b\circ\varphi_a^{-1}$ is smooth, and being invertible with a symmetrically smooth inverse, it is a diffeomorphism between open subsets of $\mathbb{R}^{\dim G-\dim H}$. The atlas is therefore smooth, and it defines a smooth manifold structure on G/H of dimension $\dim G-\dim H$, giving part (1).

In these charts the projection is a submersion. Read π near a point of G in the source chart Φ (coordinates $(Y,X)\in B_{\mathfrak{m}}\times B_{\mathfrak{h}}$, after left-translating so the point is e) and the target chart φ_e. The composite $\varphi_e\circ\pi\circ\Phi$ sends $(Y,X)\mapsto\exp(Y)\exp(X)H\mapsto Y$, that is, it is the linear projection $(Y,X)\mapsto Y$. This is a smooth submersion, its differential the surjection $\mathfrak{m}\oplus\mathfrak{h}\to\mathfrak{m}$; left translation carries the same local form to every point of G, so π is a smooth submersion everywhere, and its differential at e is the quotient map $\mathfrak{g}\to\mathfrak{g}/\mathfrak{h}\cong\mathfrak{m}$ with kernel $\mathfrak{h}$.

The left action $G\times G/H\to G/H$ is smooth: in the charts above, $(a,gH)\mapsto(ag)H$ reads, after the same local factoring, as a composite of the smooth multiplication on G and the smooth chart maps, exactly as the transition maps were shown smooth. It is transitive by construction, since $(gx^{-1})\cdot(xH)=gH$ for any x,g. This is part (2).

For part (3), the quotient is Hausdorff and second countable. The projection π is an open continuous surjection: for $U\subseteq G$ open, $\pi^{-1}(\pi(U))=UH=\bigcup_{h\in H}Uh$ is a union of open sets, hence open, so $\pi(U)$ is open. Second countability then passes from G to its continuous open image G/H. For Hausdorffness the closedness of H is used. The set

$$R=\big\{(g,g')\in G\times G\;\big|\;g^{-1}g'\in H\big\}$$

is the preimage of the closed set H under the continuous map $(g,g')\mapsto g^{-1}g'$, hence closed in $G\times G$; R is the graph of the coset equivalence relation, and the quotient of a topological group by a closed subgroup, with its open projection, is Hausdorff precisely when this relation is closed ([9, Separation of the Quotient]). Concretely, if $gH\neq g'H$ then $g^{-1}g'\notin H$; as H is closed there is a symmetric neighborhood W of e with $Wg^{-1}g'W\cap H=\varnothing$, and then $\pi(gW)$ and $\pi(g'W)$ are disjoint open neighborhoods separating gH and $g'H$. Thus G/H is a Hausdorff, second-countable smooth manifold.

Uniqueness. Suppose G/H carries a second smooth structure for which π is a smooth submersion. A submersion admits local smooth sections ([13, Submersions and Local Sections]); using such a section s for the first structure, the identity map from the first structure to the second equals (π-descent of)$\pi\circ s$ locally and is smooth, and symmetrically its inverse is smooth, so the identity is a diffeomorphism between the two structures, which therefore coincide. (This uniqueness is restated cleanly through the universal property in proposition 4.1.5.) This completes the proof.

The theorem is the manifold factory promised at the start. It converts a purely algebraic datum, a closed subgroup $H\leq G$, into a manifold G/H that comes with a transitive G-action already installed. The dimension count $\dim G-\dim H$ is the tangible payoff: to build a manifold of a prescribed dimension, one finds a group G and a closed subgroup H whose dimensions differ by the target, arranges transitivity, and reads off the space. The slice $S=\exp(B_{\mathfrak{m}})$ is the geometric heart of the construction, a submanifold transverse to H that meets each nearby coset once, so that the $\mathfrak{m}$-directions become coordinates on the quotient while the $\mathfrak{h}$-directions are quotiented away. Closedness of H enters in two places, both indispensable: it makes H an embedded submanifold, which is what gives the transversality of Step 2, and it makes the relation R closed, which is what

makes the quotient Hausdorff. Drop closedness, as in the irrational winding line on the torus (example 3.1.5), and the quotient is non-Hausdorff with no manifold structure.

A concrete instance is worth carrying through immediately, before the systematic examples of the next two sections. The sphere is the first coset manifold, and it exhibits every ingredient of the theorem.

Example 4.1.4: The Sphere $S^{n-1} = \mathrm{O}(n)/\mathrm{O}(n-1)$

Take $G = \mathrm{O}(n)$ acting on $S^{n-1} \subseteq \mathbb{R}^n$ by $(A, v) \mapsto Av$, as in example 4.1.2. The action is transitive, and the stabilizer of the north pole $e_n = (0, \dots, 0, 1)$ is

$$\mathrm{O}(n)_{e_n} = \{A \in \mathrm{O}(n) \mid Ae_n = e_n\} = \left\{ \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix} \;\middle|\; B \in \mathrm{O}(n-1) \right\} \cong \mathrm{O}(n-1)$$

since an orthogonal matrix fixing e_n must preserve its orthogonal complement $e_n^{\perp} = \mathbb{R}^{n-1}$, on which it acts by an arbitrary element of $\mathrm{O}(n-1)$. This stabilizer is closed in $\mathrm{O}(n)$, being the intersection of $\mathrm{O}(n)$ with the closed condition $Ae_n = e_n$, so theorem 4.1.3 makes $\mathrm{O}(n)/\mathrm{O}(n-1)$ a smooth manifold with

$$\dim\big(\mathrm{O}(n)/\mathrm{O}(n-1)\big) = \binom{n}{2} - \binom{n-1}{2} = n - 1$$

using $\dim \mathrm{O}(k) = \binom{k}{2}$ from section 1.2. The dimension matches $\dim S^{n-1} = n-1$, exactly as it must: the orbit map $A\,\mathrm{O}(n-1) \mapsto Ae_n$ is the G-equivariant diffeomorphism $\mathrm{O}(n)/\mathrm{O}(n-1) \xrightarrow{\sim} S^{n-1}$ that section 4.2 will establish in general. The projection $\pi\colon \mathrm{O}(n) \to S^{n-1}$ is $A \mapsto Ae_n$, the map reading off the last column, and it is a submersion: its fibers are the cosets $A\,\mathrm{O}(n-1)$, and its slice at e_n is a small patch of directions transverse to the stabilizer.

The example is the pattern every homogeneous space will follow: a transitive action, a closed stabilizer, and a dimension count that confirms the coset space is the space one started with. Here S^{n-1} was already a manifold, so the coset presentation is a new description rather than a new object, but the same machine, applied to abstract cosets, manufactures Grassmannians and flag manifolds that have no prior definition.

What makes the smooth structure on G/H canonical, rather than one choice among many, is a universal property. The projection π is a submersion, and submersions are characterized by how smooth maps factor through them: a map out of G/H is smooth exactly when its pullback along π is smooth. This is the working criterion for smoothness on a quotient, and it is used constantly, since a map defined on cosets is almost always specified by a formula in g that happens to be constant on H-cosets.

Proposition 4.1.5: Universal Property of the Quotient

Let $H \subseteq G$ be a closed subgroup and $\pi\colon G \to G/H$ the quotient projection of theorem 4.1.3. Then:

1. for any smooth manifold N, a map $f\colon G/H \to N$ is smooth if and only if $f \circ \pi\colon G \to N$ is smooth;
2. consequently the smooth structure of theorem 4.1.3 is the unique one making π a submersion, and G/H is determined up to a unique G-equivariant diffeomorphism by this property;
3. π admits local smooth sections: every point of G/H has an open neighborhood V and a smooth map $s\colon V \to G$ with $\pi \circ s = \mathrm{id}_V$.

***Proof*:**

1. If f is smooth then $f \circ \pi$ is smooth, a composite of smooth maps. Conversely suppose $f \circ \pi$ is smooth. Smoothness of f is local, so fix a coset gH. By part (3), proved independently below, there is an open neighborhood V of gH and a smooth section $s\colon V \to G$ with $\pi \circ s = \mathrm{id}_V$. Then on V,

$$f = f \circ \mathrm{id}_V = f \circ (\pi \circ s) = (f \circ \pi) \circ s$$

   is a composite of the smooth map $f \circ \pi$ with the smooth section s, hence smooth on V. As gH was arbitrary, f is smooth.

2. Any smooth structure on G/H making π a submersion satisfies the factoring property of part (1), by the same argument, which used only that π is a submersion (to obtain local sections). Given two such structures, apply part (1) to $f = \mathrm{id}$ viewed from the first structure to the second: $\mathrm{id} \circ \pi = \pi$ is smooth into the target structure, so id is smooth from the first structure to the second, and symmetrically its inverse is smooth, so id is a diffeomorphism and the structures coincide. The diffeomorphism intertwining any two realizations of G/H is G-equivariant because π is G-equivariant, and it is unique because a G-equivariant map is determined by its value at the base point eH, whose orbit is all of G/H. Hence G/H is canonical up to a unique G-equivariant diffeomorphism.

3. The section is read off the slice of theorem 4.1.3. At the identity coset, the chart $\varphi_e\colon \pi(S) \to B_{\mathfrak{m}}$ has continuous inverse $\varphi_e^{-1}(Y) = \exp(Y)H$, and the map

$$s_e\colon \pi(S) \to G, \qquad s_e(\exp(Y)H) = \exp(Y)$$

   is smooth (it equals $\exp \circ \varphi_e$, a composite of smooth maps) and satisfies $\pi \circ s_e = \mathrm{id}_{\pi(S)}$, since $\pi(\exp(Y)) = \exp(Y)H$. This is a smooth section over the neighborhood $\pi(S)$ of eH. For an arbitrary coset aH, transport by left translation: $s_a = L_a \circ s_e \circ \tau_{a^{-1}}$ is a smooth section over the neighborhood $\tau_a(\pi(S))$ of aH, using that $L_a\colon G \to G$ is a diffeomorphism (proposition 2.1.2) and $\pi \circ L_a = \tau_a \circ \pi$, so $\pi \circ s_a = \tau_a \circ \pi \circ s_e \circ \tau_{a^{-1}} = \tau_a \circ \tau_{a^{-1}} = \mathrm{id}$. Every point of G/H thus has a local smooth section, as we sought to show.

The universal property is the practical tool. It says G/H is the best quotient of G by the equivalence relation of H-cosets: smoothness downward is free, and smoothness upward is the criterion one checks. In practice a map on a coset space is defined by an H-invariant formula on G, a function of g that does not change when g is replaced by gh, and the proposition certifies that such a formula defines a smooth map on G/H exactly when it is smooth on G. Uniqueness up to canonical diffeomorphism means the notation G/H is unambiguous: any two constructions of a manifold with a transitive G-action and stabilizer H are the same, so one may speak of the homogeneous space G/H. Local sections are the recurring technical device, the smooth choice of coset representative that trivializes π over a small open set. They are precisely what can fail to exist globally, and that failure is the twisting that makes $G \to G/H$ a nontrivial bundle in section 4.5.

The construction has converted a closed subgroup into a manifold carrying a built-in transitive G-action, and this is the mechanism behind every example of section 4.3. Two structural facts round out the picture, both consequences of the theorem worth isolating. When H is normal, the quotient is not just a manifold but a Lie group, and π is a homomorphism: the group operations on G/H are smooth because they pull back to the smooth operations on G, and the universal property upgrades this to smoothness on the quotient. And the tangent space of G/H at the identity coset is the quotient vector space $\mathfrak{g}/\mathfrak{h}$, since $d\pi_e$ is the surjection $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ with kernel $\mathfrak{h}$; this identification is the seed of the isotropy representation in section 4.5, where H acts on $\mathfrak{g}/\mathfrak{h}$ and the geometry of G/H near eH is read from that action. Both facts appear as exercises below.

Exercise 4.1.1

1. Let G be a Lie group and G^0 its identity component (definition 2.1.6), a closed normal subgroup. Show that G/G^0 is a 0-dimensional manifold, that is, a discrete group, and identify it with the component group of G. More generally, prove that if a closed subgroup $H \leq G$ is open, then G/H is discrete.

2. Let H be a *closed normal* subgroup of a Lie group G. Prove that the quotient group G/H, with the smooth structure of theorem 4.1.3, is a Lie group and that $\pi\colon G \to G/H$ is a homomorphism of Lie groups. (Use proposition 4.1.5 to check that multiplication and inversion on G/H are smooth, and corollary 3.2.5 to reduce smoothness of π to its continuity.)

3. With G, H, $\mathfrak{g}$, $\mathfrak{h}$ as in theorem 4.1.3, prove that the differential $d\pi_e\colon \mathfrak{g} \to T_{eH}(G/H)$ is surjective with kernel exactly $\mathfrak{h}$, and conclude that there is a canonical linear isomorphism $T_{eH}(G/H) \cong \mathfrak{g}/\mathfrak{h}$. Verify the dimension count of theorem 4.1.3 against this.

4. Show that $\mathrm{GL}_n(\mathbb{R})$ acts smoothly and transitively on the set $\mathcal{B}$ of ordered bases of $\mathbb{R}^n$ by $A \cdot (v_1, \dots, v_n) = (Av_1, \dots, Av_n)$, and compute the stabilizer of the standard basis. Conclude that $\mathcal{B}$, given the coset-space structure, is diffeomorphic to $\mathrm{GL}_n(\mathbb{R})$ itself, and explain why the stabilizer being trivial forces this.

5. Let $H = \{(e^{2\pi i t}, e^{2\pi i \alpha t}) \mid t \in \mathbb{R}\} \subseteq \mathbb{T}^2$ be the image of the irrational-slope one-parameter subgroup with α irrational (example 3.1.5). Explain why the quotient $\mathbb{T}^2/H$ is not a Hausdorff manifold, identifying the step of theorem 4.1.3 that fails and the hypothesis responsible.

6. Give an example of a coset space G/H for which $\pi\colon G \to G/H$ admits no global smooth section $s\colon G/H \to G$ with $\pi \circ s = \mathrm{id}$, and explain how this is consistent with proposition 4.1.5, which guarantees only *local* sections. (Consider $\mathrm{SO}(3) \to S^2$ and what a global section would say about the tangent bundle of S^2.)

4.2 Transitive Actions and Homogeneous Spaces

The quotient construction of section 4.1 started from a subgroup and produced a manifold G/H. This section reverses the arrow. Many of the manifolds one meets in geometry, the spheres and projective spaces and Grassmannians of the next section, come already carrying an action of a Lie group, and the action has a feature that makes the manifold rigid: it is transitive. Every point can be moved to every other by the group, so there is no intrinsic way to tell two points apart, and the manifold looks the same seen from anywhere. A space with this degree of symmetry is called homogeneous, and the central result of the section is that a homogeneous space is nothing other than a coset space G/H in disguise. The disguise is stripped by a single computation: fix a point, take its stabilizer H, and the orbit map $gH \mapsto g \cdot x$ is a diffeomorphism $G/H \to M$.

This is the observation that turns the abstract quotient of the previous section into a practical tool. To recognize a given manifold M as a coset space, one does not construct charts or verify a submersion by hand. It suffices to exhibit a Lie group acting transitively on M and to compute the stabilizer of one point. The orbit-stabilizer diffeomorphism then gives the identification $M \cong G/H$ for free, together with the smooth structure and the group action. The section develops this identification, first for transitive actions, where it is a clean diffeomorphism, and then for general actions, where each individual orbit is a homogeneous space for G and sits inside M as an immersed submanifold.

Definition 4.2.1: Homogeneous Space

A smooth action of a Lie group G on a smooth manifold M (definition 4.1.1) is *transitive* if for every pair of points $x, y \in M$ there is an element $g \in G$ with $g \cdot x = y$; equivalently, M consists of a single orbit. A smooth manifold M equipped with a transitive smooth action of G is a *homogeneous G-space*, or simply a *homogeneous space* when the group is understood.

Transitivity is a strong hypothesis with an immediate consequence for the shape of M: because G moves any point to any other by a diffeomorphism (the action of a fixed g is a diffeomorphism of M with inverse the action of g^{-1}), every local or infinitesimal feature at one point is carried to every other. A homogeneous space has no distinguished point, no special locus, no place where it is more curved or more singular than elsewhere. This uniformity is what makes such spaces the natural setting for the geometry of symmetry, and it is the source of the eventual identification with a coset space, where the distinguished coset eH is a choice, not an intrinsic feature.

The prototypical homogeneous space is the coset space itself. Before the general theorem, it is worth seeing that G/H is homogeneous under the action that defines it, since this is the target of the identification to come.

Example 4.2.2: The Coset Space G/H

Let H be a closed subgroup of a Lie group G, so that G/H is a smooth manifold by theorem 4.1.3. The group G acts on G/H by left translation,

$$g \cdot (g'H) = (gg')H$$

which is well defined because left multiplication by g carries the coset $g'H$ to $(gg')H$ regardless of the representative g', and which is smooth because it is induced from the smooth left multiplication $G \times G \to G$ through the submersion $\pi \colon G \to G/H$ (proposition 4.1.5). The action is transitive: given two cosets $g'H$ and $g''H$, the element $g = g''(g')^{-1}$ satisfies $g \cdot (g'H) = g''H$. Thus G/H is a homogeneous G-space. The stabilizer of the base coset eH is exactly H, since $g \cdot eH = eH$

means $gH = H$, that is $g \in H$. This example is the model to which the next theorem reduces every homogeneous space.

The theorem is the converse of the example. Any homogeneous space, however it arises, is diffeomorphic to a coset space of the acting group, and the diffeomorphism respects the action. The mechanism is the orbit map through a chosen point: it factors through the coset space of the stabilizer and lands in the manifold as an equivariant bijection, which the manifold structure then forces to be a diffeomorphism.

Theorem 4.2.3: Smooth Orbit-Stabilizer

Let G be a Lie group acting smoothly and transitively on a smooth manifold M, and fix a point $x \in M$. Let $H = G_x = \{g \in G \mid g \cdot x = x\}$ be the stabilizer of x. Then H is a closed subgroup of G, so that G/H is a smooth manifold, and the orbit map

$$\Phi \colon G/H \longrightarrow M, \qquad \Phi(gH) = g \cdot x$$

is a well-defined G-equivariant diffeomorphism. In particular $\dim M = \dim G - \dim H$.

***Proof*:**

Step 1: the stabilizer is closed, so G/H is a manifold. The orbit map of the point x, namely $\theta_x \colon G \to M$ with $\theta_x(g) = g \cdot x$, is smooth, being the composite of $g \mapsto (g, x)$ with the smooth action $G \times M \to M$ (definition 4.1.1); in particular it is continuous. The stabilizer $H = \theta_x^{-1}(\{x\})$ is the preimage of the one-point set $\{x\}$, which is closed because M is Hausdorff, so H is a closed subset of G. It is a subgroup: $e \cdot x = x$, and if $g \cdot x = x$ and $g' \cdot x = x$ then $(gg') \cdot x = g \cdot (g' \cdot x) = g \cdot x = x$ and $g^{-1} \cdot x = g^{-1} \cdot (g \cdot x) = (g^{-1}g) \cdot x = x$. Being a closed subgroup, H is an embedded Lie subgroup by theorem 3.2.1, and theorem 4.1.3 then endows G/H with its smooth manifold structure, with $\pi \colon G \to G/H$ a smooth submersion.

Step 2: the orbit map descends and is a smooth equivariant bijection. The map $\theta_x \colon G \to M$ is constant on the cosets of H: if $g' = gh$ with $h \in H$, then $\theta_x(gh) = (gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = \theta_x(g)$. Hence θ_x factors through the quotient, inducing a well-defined map $\Phi \colon G/H \to M$ with $\Phi(gH) = g \cdot x$ and $\Phi \circ \pi = \theta_x$. Because θ_x is smooth and π is a surjective submersion, Φ is smooth by the universal property of the quotient (proposition 4.1.5). It is G-equivariant: for $g_0 \in G$,

$$\Phi\big(g_0 \cdot (gH)\big) = \Phi\big((g_0 g)H\big) = (g_0 g) \cdot x = g_0 \cdot (g \cdot x) = g_0 \cdot \Phi(gH)$$

It is surjective because the action is transitive: every $y \in M$ equals $g \cdot x = \Phi(gH)$ for some g. It is injective because $\Phi(gH) = \Phi(g'H)$ means $g \cdot x = g' \cdot x$, whence $(g^{-1}g') \cdot x = x$, so $g^{-1}g' \in H$ and $gH = g'H$. Thus Φ is a smooth equivariant bijection.

Step 3: a smooth equivariant bijection of homogeneous spaces has constant rank. It remains to promote the bijection Φ to a diffeomorphism, and the obstruction is that a smooth bijection need not have smooth inverse (the map $t \mapsto t^3$ on $\mathbb{R}$ is the standard warning). Equivariance removes the obstruction by making the rank of Φ the same at every point. Write $\ell_g \colon G/H \to G/H$ for the diffeomorphism $g'H \mapsto (gg')H$ given by the action of g on G/H, and $L_g \colon M \to M$ for the diffeomorphism $y \mapsto g \cdot y$ given by the action of g on M. Equivariance is the identity $\Phi \circ \ell_g = L_g \circ \Phi$. Differentiating at an arbitrary point $g'H$ and using the chain rule,

$$d\Phi_{(gg')H} \circ (d\ell_g)_{g'H} = (dL_g)_{\Phi(g'H)} \circ d\Phi_{g'H}$$

Both $(d\ell_g)_{g'H}$ and $(dL_g)_{\Phi(g'H)}$ are linear isomorphisms, being differentials of diffeomorphisms, so $d\Phi_{(gg')H}$ and $d\Phi_{g'H}$ have the same rank. Since G acts transitively on G/H (every coset is $\ell_g(eH)$ for a suitable g), the rank of $d\Phi$ is the same at every point of G/H. Thus Φ is a smooth map of constant rank.

Step 4: constant rank plus bijectivity forces a diffeomorphism. A smooth map of constant rank r that is injective is an immersion, and one that is surjective is a submersion; this is the global rank theorem ([13, Global Rank Theorem]). The bijection Φ is both injective and surjective, so it is simultaneously an immersion and a submersion, which forces

$$r = \dim(G/H) = \dim M$$

so that $d\Phi$ is an isomorphism at every point. A bijective local diffeomorphism is a diffeomorphism, so Φ is a diffeomorphism. The dimension count follows from $\dim(G/H) = \dim G - \dim H$ (theorem 4.1.3), giving $\dim M = \dim G - \dim H$, completing the proof.

The theorem is the master tool for the examples of the next section, and its practical content is worth isolating. To recognize a manifold M as a coset space G/H, one need only produce a Lie group G acting smoothly and transitively on M and compute the stabilizer H of a single point. The theorem then delivers the diffeomorphism $M \cong G/H$ with no further work, and in particular the dimension identity $\dim M = \dim G - \dim H$ becomes a consistency check: if the arithmetic fails, the proposed action cannot be transitive. The choice of base point x is immaterial up to isomorphism, since different base points give conjugate stabilizers, a point the exercises make precise. What the theorem does not require is any preexisting knowledge of the smooth structure of M beyond the manifold axioms; the structure is matched automatically to that of G/H.

The archetype is the action of the rotation group on the sphere, which the next section develops in full. A first instance is the running example, where the identification of $\mathrm{SU}(2)$ with the three-sphere lets one read off a homogeneous-space presentation of the two-sphere.

Example 4.2.4: $\mathrm{SU}(2)$ Acting on S^2

The group $\mathrm{SU}(2)$ acts on the two-sphere $S^2 \subseteq \mathbb{R}^3$ through the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5: an element $g \in \mathrm{SU}(2)$ maps to a rotation $R(g) \in \mathrm{SO}(3)$, which rotates S^2. The action is transitive because $\mathrm{SO}(3)$ acts transitively on S^2 (any unit vector can be rotated to any other), and $\mathrm{SU}(2) \to \mathrm{SO}(3)$ is surjective. Fix the north pole $x = (0, 0, 1)$. Its stabilizer in $\mathrm{SO}(3)$ is the group of rotations about the vertical axis, a circle $\mathrm{SO}(2)$; its stabilizer in $\mathrm{SU}(2)$ is the preimage of this circle, which is the diagonal torus

$$\mathrm{U}(1) = \left\{ \begin{pmatrix} e^{i\vartheta} & 0 \\ 0 & e^{-i\vartheta} \end{pmatrix} \;\middle|\; \vartheta \in \mathbb{R} \right\}$$

a closed subgroup isomorphic to the circle. By theorem 4.2.3,

$$S^2 \cong \mathrm{SU}(2)/\mathrm{U}(1)$$

as homogeneous spaces, and the dimension arithmetic checks: $\dim \mathrm{SU}(2) - \dim \mathrm{U}(1) = 3 - 1 = 2 = \dim S^2$. Since $\mathrm{SU}(2)$ is diffeomorphic to S^3 (proposition 1.5.1), the quotient map $\mathrm{SU}(2) \to \mathrm{SU}(2)/\mathrm{U}(1)$ is a smooth map $S^3 \to S^2$ with circle fibers, the Hopf fibration; the next section returns to it as example 4.3.3.

The example is a template. The manifold S^2 was presented as a quotient without ever writing down a chart on it, purely by exhibiting the transitive action of SU(2) and identifying one stabilizer. Every homogeneous-space identification in the next section follows this pattern, and the dimension count $\dim G - \dim H = \dim M$ is the routine sanity check that the action is transitive.

The orbit-stabilizer theorem assumed transitivity. Dropping that hypothesis, a smooth action need not sweep out all of M from a single point; instead M decomposes into orbits, and each orbit is a homogeneous space in its own right. The question is how an individual orbit sits inside M. The theorem's argument applies verbatim to the action restricted to one orbit, identifying the orbit with a coset space, but the orbit carries the subspace topology from M, which may or may not agree with its intrinsic manifold topology. This is exactly the immersed-versus-embedded distinction of section 3.1, and the outcome is that orbits are always immersed submanifolds and are embedded under a mild closedness condition.

Proposition 4.2.5: Orbits as Homogeneous Submanifolds

Let G be a Lie group acting smoothly on a smooth manifold M, and let $x \in M$. Then the stabilizer G_x is a closed subgroup of G, and the orbit map

$$\Phi_x \colon G/G_x \longrightarrow M, \qquad g\,G_x \mapsto g \cdot x$$

is an injective smooth immersion whose image is the orbit $G \cdot x$. Thus $G \cdot x$ is an injectively immersed submanifold of M, diffeomorphic as a homogeneous G-space to G/G_x, and its dimension is $\dim G - \dim G_x$. The orbit is an embedded submanifold whenever it is locally closed in M; in particular, every orbit of a compact Lie group is embedded.

***Proof*:**

Step 1: the orbit is a homogeneous space and Φ_x is an injective immersion. That G_x is a closed subgroup is the argument of Step 1 of theorem 4.2.3, which used only smoothness of the action, not transitivity: $G_x = \theta_x^{-1}(\{x\})$ is closed, and it is a subgroup, so it is an embedded Lie subgroup by theorem 3.2.1 and G/G_x is a manifold. The map $\Phi_x \colon G/G_x \to M$ is smooth, well defined, injective, and G-equivariant by Steps 1 and 2 of theorem 4.2.3, whose verifications used only smoothness of the action, never transitivity of G on all of M; the same descent through the submersion $\pi \colon G \to G/G_x$ and the same injectivity computation apply. By the constant-rank argument of Step 3 there, equivariance makes $d\Phi_x$ of constant rank, and injectivity of Φ_x forces that constant rank to equal $\dim(G/G_x)$, so Φ_x is an injective immersion. Its image is the set of all $g \cdot x$, which is precisely the orbit $G \cdot x$. Thus $G \cdot x$ is an injectively immersed submanifold, carrying via Φ_x the manifold structure of G/G_x, and $\dim(G \cdot x) = \dim(G/G_x) = \dim G - \dim G_x$.

Step 2: local closedness upgrades the immersion to an embedding. An injective immersion is an embedding precisely when it is a homeomorphism onto its image with the subspace topology, and for an orbit this holds exactly when the orbit is locally closed, that is open in its closure. Suppose $G \cdot x$ is locally closed. The homogeneity of the orbit makes the embedding condition need checking at only one point: the transitive G-action carries a neighborhood of x in $G \cdot x$ to a neighborhood of any other point, so if the immersion is a homeomorphism onto its image near x it is so everywhere. Local closedness at x provides an open set $U \subseteq M$ with $G \cdot x \cap U$ closed in U; a standard Baire-category argument on the second-countable group G ([13, Orbits with Locally Closed Image Are Embedded]) then shows the injective immersion Φ_x is a homeomorphism onto $G \cdot x$ with the subspace topology, hence an embedding.

Step 3: orbits of a compact group are embedded. Suppose G is compact. The orbit map $\theta_x\colon G \to M$ is continuous and G is compact, so the orbit $G\cdot x = \theta_x(G)$ is a compact subset of the Hausdorff space M, hence closed; a closed set is in particular locally closed, so by Step 2 the orbit is embedded. The conclusion is also visible directly: $\Phi_x\colon G/G_x \to M$ is a continuous injection from the compact space G/G_x (a continuous image of the compact G) into the Hausdorff space M, and a continuous injection from a compact space to a Hausdorff space is a homeomorphism onto its image; combined with the immersion property of Step 1, Φ_x is an embedding. Therefore every orbit of a compact Lie group is an embedded submanifold, completing the proof.

The proposition records the general shape of an action: the manifold M is partitioned into orbits, each a homogeneous space G/G_x immersed inside M, with the stabilizer varying from orbit to orbit. The immersion is always injective, so each orbit is an image of a coset space, but the way it sits in M can be complicated, exactly as an immersed Lie subgroup can wind densely rather than embed. The irrational winding of example 3.1.5 is the paradigm: the additive group $\mathbb{R}$ acting on the torus $\mathbb{T}^2$ by an irrational-slope flow has a single orbit that is dense, an injectively immersed but non-embedded copy of $\mathbb{R}$, and the orbit fails to be locally closed. The compact case is clean because a continuous image of a compact group is compact, hence closed, hence locally closed, and the pathology cannot occur.

The two results together say that transitivity is what turns an immersed orbit into a diffeomorphism onto the whole manifold. When G acts transitively there is one orbit, it is all of M, and the immersion of proposition 4.2.5 becomes the diffeomorphism of theorem 4.2.3. The orbit-stabilizer diffeomorphism is thus the transitive special case where the immersed orbit fills the manifold, and it is this special case that the next section exploits: each of the spheres, projective spaces, Grassmannians, and flag manifolds is exhibited as a single orbit of a classical group acting transitively, with the manifold recovered as G/H by computing one stabilizer.

Exercise 4.2.1

1. Verify directly, without invoking theorem 4.2.3, that the left-translation action of G on the coset space G/H (example 4.2.2) is transitive and that the stabilizer of the base coset eH is exactly H. Conclude that under the identification of theorem 4.2.3 a homogeneous space is recovered as the coset space of the stabilizer of any chosen point.

2. Let G act transitively and smoothly on M, and let $x, y \in M$. Prove that the stabilizers G_x and G_y are conjugate subgroups of G: if $y = g\cdot x$ then $G_y = g\,G_x\,g^{-1}$. Deduce that the diffeomorphism type of the presentation $M \cong G/G_x$ does not depend on the base point x, and conversely that if $H, H' \le G$ are closed subgroups then G/H and G/H' are G-equivariantly diffeomorphic if and only if H and H' are conjugate in G.

3. Show that the standard action of $\mathrm{O}(n)$ on $\mathbb{R}^n$ restricts to a transitive action on the unit sphere S^{n-1}, compute the stabilizer of the point $e_n = (0,\dots,0,1)$, and conclude from theorem 4.2.3 that $S^{n-1} \cong \mathrm{O}(n)/\mathrm{O}(n-1)$. Check the dimension count.

4. Let the additive group $\mathbb{R}$ act on the torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ by $t\cdot(\bar u, \bar v) = (\overline{u+t}, \overline{v+\alpha t})$, with α irrational. Show that every orbit is an injectively immersed copy of $\mathbb{R}$ that is dense in $\mathbb{T}^2$, hence not embedded, and identify precisely which hypothesis of the embeddedness conclusion of proposition 4.2.5 fails. Explain why replacing $\mathbb{R}$ by the compact group S^1 (available when α is rational) makes every orbit embedded.

5. Let G be a connected Lie group acting transitively on a manifold M. Prove that M is

connected. Then, using $\dim M = \dim G - \dim G_x$, show that if a compact connected Lie group G acts transitively on M with $\dim M = \dim G$ and trivial stabilizer $G_x = \{e\}$, then the action is free and $G \to M$ is a diffeomorphism, so $M \cong G$ as a homogeneous space.

6. Suppose a Lie group G acts smoothly and transitively on M. Prove that the orbit map $\theta_x \colon G \to M$ is a surjective submersion, and use this to give a second proof that $\dim M \leq \dim G$, with equality if and only if the stabilizer G_x is discrete.

4.3 Examples: Spheres, Projective Spaces, Grassmannians, and Flag Manifolds

The two preceding sections built the machine. Theorem 4.1.3 manufactures a smooth manifold G/H from a Lie group G and a closed subgroup H, and theorem 4.2.3 identifies any manifold M carrying a transitive G-action with such a quotient, through the orbit map $gH \mapsto g \cdot x$. This section runs the machine on the manifolds a geometer meets first. Each of the classical spaces, the spheres, the projective spaces, the Grassmannians of k-planes, and the flag manifolds, turns out to be a single orbit of a classical group, and reading off the stabilizer of a base point presents it as a coset space G/H with the symmetry group displayed in the notation itself.

The gain is not just uniform notation. A presentation $M = G/H$ makes the full group G act on M by isometries of whatever G-invariant geometry M carries, and it reduces questions about M, its dimension, its tangent space at a point, its cohomology, its invariant metrics, to questions about the pair $(\mathfrak{g}, \mathfrak{h})$ of Lie algebras. Every computation below follows the same three steps demanded by theorem 4.2.3: exhibit a transitive action, compute the stabilizer of a base point, and confirm the dimension arithmetic $\dim M = \dim G - \dim H$ against the manifold dimensions of chapter 1. The running example $\mathrm{SU}(2)$ reappears at the end as the smallest nontrivial instance, where the coset presentation is exactly the Hopf fibration.

The most basic homogeneous space is the sphere. The orthogonal group acts on $\mathbb{R}^n$ preserving lengths, so it permutes the vectors of any fixed length; the transitivity of this action on the unit sphere is the statement that any unit vector can be rotated to any other, which is the content of the Gram-Schmidt process. Fixing the north pole and reading off which orthogonal transformations fix it presents the sphere as a coset space of $\mathrm{O}(n)$, and its determinant-one subgroup gives a second presentation.

Example 4.3.1: Spheres as Quotients of Orthogonal and Unitary Groups

The unit sphere $S^{n-1} = \{x \in \mathbb{R}^n \mid \|x\| = 1\}$ is a homogeneous space for both the orthogonal group and the rotation group,

$$S^{n-1} \;\cong\; \mathrm{O}(n)/\mathrm{O}(n-1) \;\cong\; \mathrm{SO}(n)/\mathrm{SO}(n-1)$$

and the unit sphere $S^{2n-1} = \{z \in \mathbb{C}^n \mid \|z\| = 1\}$ in $\mathbb{C}^n \cong \mathbb{R}^{2n}$ is a homogeneous space for both the unitary and the special unitary group,

$$S^{2n-1} \;\cong\; \mathrm{U}(n)/\mathrm{U}(n-1) \;\cong\; \mathrm{SU}(n)/\mathrm{SU}(n-1)$$

all four presentations being G-equivariant diffeomorphisms in the sense of theorem 4.2.3.

The verification of the real case runs through the three required checks. *Transitivity.* Let $x, y \in S^{n-1}$ be unit vectors. Extend x to an orthonormal basis $x = u_1, u_2, \dots, u_n$ and y to an orthonormal basis $y = v_1, v_2, \dots, v_n$ of $\mathbb{R}^n$, possible by the Gram-Schmidt process. The matrix g carrying $u_i \mapsto v_i$ is orthogonal, since it sends one orthonormal basis to another, and it satisfies $gx = y$. Thus $\mathrm{O}(n)$ acts transitively on S^{n-1}; the action is smooth, being the restriction of the linear action $\mathrm{O}(n) \times \mathbb{R}^n \to \mathbb{R}^n$.

Stabilizer. Take the base point $e_n = (0, \dots, 0, 1)$, the north pole. A matrix $g \in \mathrm{O}(n)$ fixes e_n if and only if its last column is e_n; orthogonality then forces the last row to be e_n^{T} as well, since the columns are orthonormal and the nth column is e_n, so the top-left $(n-1) \times (n-1)$ block is itself orthogonal and the off-diagonal blocks vanish. Hence the stabilizer is

$$\mathrm{Stab}_{\mathrm{O}(n)}(e_n) = \left\{ \begin{pmatrix} h & 0 \\ 0 & 1 \end{pmatrix} \;\middle|\; h \in \mathrm{O}(n-1) \right\} \;\cong\; \mathrm{O}(n-1)$$

embedded as the upper-left block. This is a closed subgroup, as proposition 4.2.5 guarantees and as one checks directly, and theorem 4.2.3 delivers the diffeomorphism $\mathrm{O}(n)/\mathrm{O}(n-1) \cong S^{n-1}$.

Dimension. By proposition 1.2.4, $\dim \mathrm{O}(n) = \binom{n}{2}$, and the dimension count reads

$$\dim \mathrm{O}(n) - \dim \mathrm{O}(n-1) = \binom{n}{2} - \binom{n-1}{2} = \frac{n(n-1)}{2} - \frac{(n-1)(n-2)}{2} = n - 1$$

which is the dimension of S^{n-1}. The same base point e_n works for the rotation group $\mathrm{SO}(n)$: the action is still transitive, because for $n \geq 2$ the matrix g above can be corrected to have determinant 1 by negating one of the auxiliary basis vectors $v_2, \dots, v_n$, none of which is the target $y = v_1$; and the stabilizer of e_n in $\mathrm{SO}(n)$ consists of the block matrices above whose block h has determinant 1, that is $\mathrm{SO}(n-1)$. The dimension count is identical, $\binom{n}{2} - \binom{n-1}{2} = n - 1$, since $\mathrm{SO}(m)$ and $\mathrm{O}(m)$ have the same dimension by proposition 1.2.4.

The complex case is the exact analogue with the Hermitian inner product in place of the Euclidean one. The unitary group $\mathrm{U}(n)$ preserves $\|z\|^2 = \sum |z_i|^2$ on $\mathbb{C}^n$, hence acts on the unit sphere $S^{2n-1} \subset \mathbb{C}^n$, and the action is transitive by the Hermitian Gram-Schmidt process: any unit vector extends to a unitary basis, so any unit vector is carried to any other by a unitary matrix. Fixing the base point $e_n \in \mathbb{C}^n$, a unitary matrix fixes it if and only if it has the block form $\left(\begin{smallmatrix} h & 0 \\ 0 & 1 \end{smallmatrix}\right)$ with $h \in \mathrm{U}(n-1)$, by the same column-orthonormality argument, so the stabilizer is $\mathrm{U}(n-1)$. The dimension check uses $\dim \mathrm{U}(n) = n^2$ from proposition 1.2.6:

$$\dim \mathrm{U}(n) - \dim \mathrm{U}(n-1) = n^2 - (n-1)^2 = 2n - 1$$

the dimension of S^{2n-1}. For $\mathrm{SU}(n)$ the action on S^{2n-1} is again transitive for $n \geq 2$: build the unitary g carrying z to w so that w is its first column, exactly as above, and then correct its determinant to 1 by rescaling one of the auxiliary columns $2, \dots, n$ (none of which is the target column) by a unit scalar, which adjusts $\det g$ freely while leaving the first column, hence $gz = w$, untouched. Its stabilizer of e_n is the block group with $h \in \mathrm{U}(n-1)$ and $\det h = 1$, that is $\mathrm{SU}(n-1)$. By proposition 1.2.6, $\dim \mathrm{SU}(n) = n^2 - 1$, and

$$\dim \mathrm{SU}(n) - \dim \mathrm{SU}(n-1) = (n^2 - 1) - \big((n-1)^2 - 1\big) = 2n - 1$$

again the dimension of S^{2n-1}.

The four presentations of the sphere are not idle repetition; they record which symmetries survive as one shrinks the group. That $S^{n-1} = \mathrm{SO}(n)/\mathrm{SO}(n-1)$ already exhibits the round metric as

$\mathrm{SO}(n)$-invariant, so the isometry group of the sphere contains $\mathrm{SO}(n)$, and the coset picture will let Part IV read off the sphere as a symmetric space. That the same sphere is $\mathrm{O}(n)/\mathrm{O}(n-1)$ records the extra reflection symmetry. The complex presentations $S^{2n-1} = \mathrm{U}(n)/\mathrm{U}(n-1)$ and $\mathrm{SU}(n)/\mathrm{SU}(n-1)$ see the odd sphere as a unitary orbit, and this is the presentation that will produce the Hopf fibration below, once the scalar circle inside $\mathrm{U}(n)$ is quotiented to pass from S^{2n-1} to complex projective space.

Passing to that quotient is the subject of the next family of examples. Projective space is the space of lines through the origin, and a Grassmannian is the space of k-dimensional subspaces. A classical group acts transitively on each because any subspace of a given dimension can be moved to any other by an orthogonal (or unitary) transformation, and the stabilizer of a fixed subspace splits into the symmetries of the subspace and the symmetries of its orthogonal complement. This block structure is the source of the product subgroups appearing below.

Example 4.3.2: Projective Spaces, Grassmannians, and Flag Manifolds
The real projective space, complex projective space, real Grassmannian of k-planes, and full complex flag manifold are the homogeneous spaces

$$\mathbb{R}\mathrm{P}^n \cong \mathrm{O}(n+1)/\big(\mathrm{O}(1)\times\mathrm{O}(n)\big) \tag{4.2}$$
$$\mathbb{C}\mathrm{P}^n \cong \mathrm{SU}(n+1)/\mathrm{S}\big(\mathrm{U}(1)\times\mathrm{U}(n)\big) \tag{4.3}$$
$$\mathrm{Gr}_k(\mathbb{R}^n) \cong \mathrm{O}(n)/\big(\mathrm{O}(k)\times\mathrm{O}(n-k)\big) \tag{4.4}$$
$$\mathrm{Fl}(n) \cong \mathrm{U}(n)/\mathbb{T}^n \tag{4.5}$$

of real dimensions n, $2n$, $k(n-k)$, and $n(n-1)$ respectively, where $\mathrm{Gr}_k(\mathbb{R}^n)$ is the set of k-dimensional linear subspaces of $\mathbb{R}^n$, $\mathrm{Fl}(n)$ is the set of complete flags $0 \subset L_1 \subset \cdots \subset L_n = \mathbb{C}^n$ with $\dim_{\mathbb{C}} L_j = j$, and $\mathbb{T}^n \subset \mathrm{U}(n)$ is the diagonal maximal torus of unitary diagonal matrices.

Real projective space $\mathbb{R}\mathrm{P}^n$ is the set of lines through the origin in $\mathbb{R}^{n+1}$, equivalently the Grassmannian $\mathrm{Gr}_1(\mathbb{R}^{n+1})$, so it is the $k=1$ case of (4.4), and it is worth treating first to fix the pattern. The orthogonal group $\mathrm{O}(n+1)$ acts on lines by $g\cdot\mathbb{R}x = \mathbb{R}(gx)$. *Transitivity*: any line $\mathbb{R}x$ with $\|x\|=1$ is carried to any other line $\mathbb{R}y$ by an orthogonal g with $gx=y$, which exists by the sphere case above. *Stabilizer*: fix the line $\ell = \mathbb{R}e_1$. A matrix $g \in \mathrm{O}(n+1)$ maps ℓ to itself if and only if $ge_1 = \pm e_1$; orthogonality then forces g to preserve the orthogonal complement $e_1^{\perp} = \mathbb{R}^n$ and to act orthogonally on it, so

$$\mathrm{Stab}_{\mathrm{O}(n+1)}(\ell) = \left\{ \begin{pmatrix} \varepsilon & 0 \\ 0 & h \end{pmatrix} \;\middle|\; \varepsilon \in \{\pm 1\},\ h \in \mathrm{O}(n) \right\} \;\cong\; \mathrm{O}(1)\times\mathrm{O}(n)$$

recording the sign $\varepsilon = \pm 1$ on the line ($\mathrm{O}(1) = \{\pm 1\}$) and the orthogonal symmetry h of the complementary hyperplane. *Dimension*: with $\dim \mathrm{O}(1) = 0$ and $\dim \mathrm{O}(n) = \binom{n}{2}$,

$$\dim \mathrm{O}(n+1) - \dim\big(\mathrm{O}(1)\times\mathrm{O}(n)\big) = \binom{n+1}{2} - \binom{n}{2} = n$$

the dimension of $\mathbb{R}\mathrm{P}^n$, and theorem 4.2.3 gives (4.2).

The general real Grassmannian $\mathrm{Gr}_k(\mathbb{R}^n)$ repeats this argument with a k-plane in place of a line. The group $\mathrm{O}(n)$ acts on k-dimensional subspaces by $g\cdot W = gW$. Transitivity is again Gram-Schmidt: given two k-planes W, W', extend orthonormal bases of W and W' to orthonormal bases of $\mathbb{R}^n$; the orthogonal matrix carrying the first basis to the second carries W to W'. Fix the base point $W_0 = \mathbb{R}^k = \mathrm{span}(e_1,\dots,e_k)$. A matrix $g \in \mathrm{O}(n)$ preserves W_0 if and only if it also preserves

its orthogonal complement $W_0^\perp = \operatorname{span}(e_{k+1}, \dots, e_n)$, because g orthogonal and $gW_0 = W_0$ forces $gW_0^\perp = (gW_0)^\perp = W_0^\perp$; on each summand g acts orthogonally, so

$$\operatorname{Stab}_{\mathrm{O}(n)}(W_0) = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \;\middle|\; a \in \mathrm{O}(k),\ d \in \mathrm{O}(n-k) \right\} \;\cong\; \mathrm{O}(k) \times \mathrm{O}(n-k)$$

This is a closed subgroup, being the intersection of $\mathrm{O}(n)$ with the closed subgroup of block-diagonal matrices, so theorem 4.2.3 yields (4.4). The dimension count is

$$\dim \mathrm{O}(n) - \dim\big(\mathrm{O}(k) \times \mathrm{O}(n-k)\big) = \binom{n}{2} - \binom{k}{2} - \binom{n-k}{2} = k(n-k)$$

where the last equality is the identity $\binom{n}{2} = \binom{k}{2} + \binom{n-k}{2} + k(n-k)$, which counts the pairs $\{i, j\}$ from $\{1, \dots, n\}$ according to whether both indices lie in the first block, both in the second, or one in each. Setting $k = 1$ recovers $\dim \mathbb{R}\mathrm{P}^n = 1 \cdot n = n$.

Complex projective space $\mathbb{C}\mathrm{P}^n$ is the set of complex lines through the origin in $\mathbb{C}^{n+1}$, on which $\mathrm{U}(n+1)$ acts transitively by the Hermitian version of the same argument. The presentation (4.3) uses the special unitary group, so the stabilizer requires slightly more care. Fix the base line $\ell_0 = \mathbb{C}e_1$. A matrix $g \in \mathrm{SU}(n+1)$ preserves ℓ_0 if and only if $ge_1 = \lambda e_1$ for some unit scalar $\lambda \in \mathrm{U}(1)$; unitarity forces g to preserve $e_1^\perp = \mathbb{C}^n$ and act unitarily there, so g has the block form $\left(\begin{smallmatrix} \lambda & 0 \\ 0 & B \end{smallmatrix}\right)$ with $\lambda \in \mathrm{U}(1)$ and $B \in \mathrm{U}(n)$, and the determinant constraint $\det g = \lambda \det B = 1$ ties the two blocks together. The stabilizer is therefore

$$\operatorname{Stab}_{\mathrm{SU}(n+1)}(\ell_0) = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & B \end{pmatrix} \;\middle|\; \lambda \in \mathrm{U}(1),\ B \in \mathrm{U}(n),\ \lambda \det B = 1 \right\} =: \mathrm{S}\big(\mathrm{U}(1) \times \mathrm{U}(n)\big)$$

the group $\mathrm{S}(\mathrm{U}(1) \times \mathrm{U}(n))$ of block-diagonal unitary matrices of total determinant 1. Its dimension is $\dim(\mathrm{U}(1) \times \mathrm{U}(n)) - 1 = (n^2 + 1) - 1 = n^2$, the determinant condition cutting one real dimension. The dimension count is

$$\dim \mathrm{SU}(n+1) - \dim \mathrm{S}\big(\mathrm{U}(1) \times \mathrm{U}(n)\big) = \big((n+1)^2 - 1\big) - n^2 = 2n$$

the real dimension of $\mathbb{C}\mathrm{P}^n$, which as a complex n-manifold has real dimension $2n$. This gives (4.3).

The full flag manifold $\mathrm{Fl}(n)$ carries the pattern to its extreme, refining a single subspace into a complete tower. A complete flag in $\mathbb{C}^n$ is a nested chain $0 \subset L_1 \subset L_2 \subset \cdots \subset L_n = \mathbb{C}^n$ of subspaces with $\dim_{\mathbb{C}} L_j = j$. The unitary group acts by $g \cdot (L_j) = (gL_j)$, and transitivity follows because any orthonormal basis $v_1, \dots, v_n$ adapted to a flag (with $L_j = \operatorname{span}(v_1, \dots, v_j)$) can be carried to any other adapted orthonormal basis by a unitary matrix. Fix the standard flag $L_j^0 = \operatorname{span}(e_1, \dots, e_j)$. A unitary g fixes this flag if and only if it maps each L_j^0 into itself, which forces g to be upper triangular; a unitary upper-triangular matrix is diagonal, and its diagonal entries are unit scalars, so

$$\operatorname{Stab}_{\mathrm{U}(n)}\big((L_j^0)\big) = \{\operatorname{diag}(t_1, \dots, t_n) \mid t_j \in \mathrm{U}(1)\} = \mathbb{T}^n$$

the diagonal maximal torus. The dimension count, with $\dim \mathrm{U}(n) = n^2$ and $\dim \mathbb{T}^n = n$, is

$$\dim \mathrm{U}(n) - \dim \mathbb{T}^n = n^2 - n = n(n-1)$$

so (4.5) holds and $\mathrm{Fl}(n)$ has real dimension $n(n-1)$, an even number, as befits a complex manifold (of complex dimension $\binom{n}{2}$).

The four presentations of example 4.3.2 share one grammar: a base point is a distinguished piece of linear-algebraic data, a line, a k-plane, or a flag, and its stabilizer is the group of symmetries preserving that data, which decomposes along the pieces the data cuts out. The invariant that varies is how finely the data slices $\mathbb{R}^n$ or $\mathbb{C}^n$: a Grassmannian cuts it into two blocks and gets a product stabilizer $\mathrm{O}(k) \times \mathrm{O}(n-k)$, while the full flag cuts it into n ordered lines and gets the finest stabilizer available, a maximal torus. This last presentation, $\mathrm{Fl}(n) = \mathrm{U}(n)/\mathbb{T}^n$, is the geometric object at the center of the highest-weight theory of Part III: the coset space of a compact group by its maximal torus is where the Weyl character formula and the Borel-Weil construction of irreducible representations live.

The running example SU(2) is the smallest case where the sphere presentation and the projective presentation collide, and their collision is the Hopf fibration. The special unitary group SU(2) acts on the complex projective line $\mathbb{C}\mathrm{P}^1$, which is a 2-sphere, and the stabilizer of a point is a circle; the resulting coset presentation, read as a map of total spaces, is the fibration $S^3 \to S^2$ with circle fibers. This is the same group SU(2) whose double cover of SO(3) organized chapter 1, now seen from the side of its action on the sphere rather than its action on its own Lie algebra.

Example 4.3.3: The Hopf Fibration $S^3 \to S^2$

Complex projective space $\mathbb{C}\mathrm{P}^1$ is diffeomorphic to the 2-sphere S^2, and the action of SU(2) on $\mathbb{C}\mathrm{P}^1$ is transitive with stabilizer the diagonal maximal torus $\mathrm{U}(1) \subset \mathrm{SU}(2)$, giving

$$S^2 \;\cong\; \mathbb{C}\mathrm{P}^1 \;\cong\; \mathrm{SU}(2)/\,\mathrm{U}(1)$$

Combined with the diffeomorphism $\mathrm{SU}(2) \cong S^3$ of proposition 1.5.1, the quotient map $\pi\colon \mathrm{SU}(2) \to \mathrm{SU}(2)/\,\mathrm{U}(1)$ is the *Hopf fibration*

$$S^1 \hookrightarrow S^3 \xrightarrow{\;\pi\;} S^2$$

a smooth surjection with every fiber a circle.

The presentation is the $n = 1$ case of the complex projective example, specialized to the special unitary group. Complex projective space $\mathbb{C}\mathrm{P}^1$ is the set of complex lines in $\mathbb{C}^2$; specializing (4.3) to $n = 1$ gives $\mathbb{C}\mathrm{P}^1 \cong \mathrm{SU}(2)/\mathrm{S}(\mathrm{U}(1) \times \mathrm{U}(1))$, and the stabilizer $\mathrm{S}(\mathrm{U}(1) \times \mathrm{U}(1)) = \{\mathrm{diag}(\lambda, \mu) \mid \lambda, \mu \in \mathrm{U}(1),\ \lambda\mu = 1\}$ is exactly the diagonal subgroup $\left\{\mathrm{diag}(\lambda, \lambda^{-1}) \;\middle|\; \lambda \in \mathrm{U}(1)\right\}$, a circle, which is the diagonal maximal torus of SU(2). The dimension count $\dim \mathrm{SU}(2) - \dim \mathrm{U}(1) = 3 - 1 = 2$ confirms that the quotient is a surface, and it is the 2-sphere: $\mathbb{C}\mathrm{P}^1$ is diffeomorphic to S^2 by stereographic coordinates, sending the line through $(z_0, z_1) \neq 0$ to the point $z_1/z_0 \in \mathbb{C} \cup \{\infty\}$, the Riemann sphere, a standard identification from the theory of Riemann surfaces (*Complex Analysis* [12]).

Reading the coset map through the identifications $\mathrm{SU}(2) \cong S^3$ (proposition 1.5.1) and $\mathrm{SU}(2)/\,\mathrm{U}(1) \cong S^2$ produces a smooth surjection $\pi\colon S^3 \to S^2$ whose fibers are the cosets of U(1), each a circle S^1. In coordinates: writing a point of $S^3 \subset \mathbb{C}^2$ as (α, β) with $|\alpha|^2 + |\beta|^2 = 1$, the Hopf map sends (α, β) to the line $\mathbb{C}(\alpha, \beta) \in \mathbb{C}\mathrm{P}^1$, equivalently to the ratio $[\alpha : \beta] \in \mathbb{C}\mathrm{P}^1 \cong S^2$; two points (α, β) and (α', β') have the same image precisely when $(\alpha', \beta') = \lambda(\alpha, \beta)$ for a unit scalar $\lambda \in \mathrm{U}(1)$, so the fiber over each point is a circle, the U(1)-orbit. The principal-bundle structure of this map, that π is a principal U(1)-bundle, is the general fact recorded for every coset projection and proved in the next section.

The Hopf fibration completes the portrait of SU(2) begun in chapter 1. There the two-to-one map $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5 realized SU(2) as the double cover of the rotation group, a covering of a 3-manifold by a 3-manifold. Here the map $\mathrm{SU}(2) \to S^2$ realizes the same group as the total space of a circle bundle over a 2-manifold, a fibration that drops dimension. The two maps out of

$\mathrm{SU}(2) \cong S^3$ are complementary: one is the running example seen as a covering group, the object of the next section, and the other is the running example seen as a fiber bundle, the object of the section after. That a single 3-sphere fibers nontrivially over the 2-sphere with connected circle fibers is the origin of the fact that $\pi_3(S^2) \neq 0$, generated by the Hopf class, a computation from homotopy theory (*Algebraic Topology* [11]); the Hopf fibration is the geometric demonstration of the nontriviality of that homotopy group.

The examples of this section make one point about the whole subject. A single classical group presents an entire family of the manifolds a geometer cares about, each as a quotient G/H, and the presentation is not a bookkeeping convenience but a statement of symmetry: the group G acts on G/H, and every G-invariant structure on the quotient, a metric, a volume form, a complex structure, is thereby forced to be homogeneous, looking the same at every point. The isotropy subgroup H controls the local geometry through its action on the tangent space at the base coset, and this is the mechanism that the next two sections and the symmetric-space theory of Part IV will exploit: an invariant Riemannian metric on G/H is exactly an $\mathrm{Ad}(H)$-invariant inner product on the complementary subspace $\mathfrak{m}$ of $\mathfrak{h}$ in $\mathfrak{g}$, so the differential geometry of the quotient is encoded in the finite-dimensional representation theory of H. The sphere $S^2 = \mathrm{SU}(2)/\mathrm{U}(1)$ carries its round metric $\mathrm{SO}(3)$-invariantly for exactly this reason, and the Grassmannians and flag manifolds inherit their canonical metrics by the same principle.

Exercise 4.3.1

1. Verify the identity $\binom{n}{2} = \binom{k}{2} + \binom{n-k}{2} + k(n-k)$ used in example 4.3.2, and conclude that $\dim \mathrm{Gr}_k(\mathbb{R}^n) = k(n-k)$ from the presentation (4.4). Check directly that this dimension is symmetric in the exchange $k \leftrightarrow n-k$, and explain geometrically why $\mathrm{Gr}_k(\mathbb{R}^n)$ and $\mathrm{Gr}_{n-k}(\mathbb{R}^n)$ are diffeomorphic.

2. Show that the complex flag manifold $\mathrm{SU}(n)/\mathbb{T}$, where $\mathbb{T} = \mathbb{T}^{n-1}$ is the diagonal maximal torus of $\mathrm{SU}(n)$, is even-dimensional, and compute its dimension. (Note $\mathrm{SU}(n)$ has dimension n^2-1 and its maximal torus has dimension $n-1$.) Reconcile your answer with the presentation $\mathrm{Fl}(n) = \mathrm{U}(n)/\mathbb{T}^n$ of (4.5) by checking that the two quotients have the same dimension.

3. Using the presentation $S^2 = \mathrm{SU}(2)/\mathrm{U}(1)$ of example 4.3.3, explain why the round metric on S^2 is invariant under the action of $\mathrm{SO}(3)$. (Hint: $\mathrm{SU}(2)$ acts on $\mathbb{CP}^1 = S^2$ through its quotient $\mathrm{SU}(2)/\{\pm I\} = \mathrm{SO}(3)$ of theorem 1.5.5, since the center $\{\pm I\} \subset \mathrm{U}(1)$ acts trivially on lines.)

4. The *Stiefel manifold* $V_k(\mathbb{R}^n)$ is the set of orthonormal k-frames in $\mathbb{R}^n$, that is, ordered k-tuples $(v_1, \dots, v_k)$ of orthonormal vectors. Show that $\mathrm{O}(n)$ acts transitively on $V_k(\mathbb{R}^n)$ with stabilizer $\mathrm{O}(n-k)$, giving $V_k(\mathbb{R}^n) \cong \mathrm{O}(n)/\mathrm{O}(n-k)$, and compute its dimension. Identify the special cases $k=1$ and $k=n$.

5. With notation as in exercise 4.3.1 (4), exhibit a natural surjection $V_k(\mathbb{R}^n) \to \mathrm{Gr}_k(\mathbb{R}^n)$ and identify its fibers with a group. Verify that the resulting dimension arithmetic is consistent: $\dim V_k(\mathbb{R}^n) = \dim \mathrm{Gr}_k(\mathbb{R}^n) + \dim \mathrm{O}(k)$.

6. Confirm the dimension count for (4.3) by computing $\dim \mathrm{S}(\mathrm{U}(1) \times \mathrm{U}(n))$ directly, and check that $\dim \mathbb{CP}^n = 2n$ agrees with the count of complex homogeneous coordinates $[z_0 : \cdots : z_n]$ modulo scaling.

4.4 Covering Groups, the Fundamental Group, and Discrete Central Kernels

The correspondence of theorem 3.4.4 identifies a simply connected Lie group with its Lie algebra, and corollary 3.4.5 extended this to all connected groups by dividing the simply connected model $\tilde{G}$ by a discrete central subgroup. That corollary answered the algebraic question, which connected groups carry a given Lie algebra, by reducing it to a computation inside $Z(\tilde{G})$. This section reads the same data from the topological side. The subgroup one quotients by is exactly the fundamental group $\pi_1(G)$, sitting as the kernel of the covering homomorphism $\tilde{G} \to G$, and the covers of a fixed group G are organized not by an arbitrary list but by the lattice of subgroups of $\pi_1(G)$. The result is a tower of Lie groups, all sharing the single Lie algebra $\mathrm{Lie}(G)$, indexed by the subgroup structure of one finite or finitely generated abelian group.

Two facts drive everything. The first is that the kernel of a covering homomorphism is not just discrete but central, so that $\pi_1(G)$ lives inside the center of $\tilde{G}$ and, being a fiber of a covering of a topological group, is abelian. The second is the covering-space correspondence itself, imported from *Topology* [9, Classification of Coverings], which matches connected covers of G with subgroups of $\pi_1(G)$; the content of this section is that the correspondence respects Lie group structure, so every connected cover of a Lie group is again a Lie group covering homomorphically, with the same Lie algebra. The fundamental group, the one datum Lie's third theorem could not detect, thereby becomes the deck group governing a tower of covers.

The centrality fact appeared once already, embedded in Step 3 of the construction of the universal covering group (proposition 3.4.1). It deserves to be recorded on its own, because it holds for any discrete normal subgroup of any connected group and is the reason discrete kernels behave so rigidly. The argument is a single application of the principle that a continuous map from a connected space into a discrete space is constant.

Proposition 4.4.1: Discrete Central Kernels and the Abelian Fundamental Group

Let G be a connected Lie group.

1. Every discrete normal subgroup $N \trianglelefteq G$ is contained in the center $Z(G)$.
2. If $\pi\colon \tilde{G} \to G$ is a covering homomorphism of connected Lie groups, its kernel $\ker \pi$ is a discrete subgroup of $Z(\tilde{G})$, and $\ker \pi \cong \pi_1(G)/\pi_*\pi_1(\tilde{G})$.
3. The fundamental group $\pi_1(G)$ is abelian.

***Proof*:**
The three parts are proved in turn; each rests on the same principle, that a continuous map from a connected space to a discrete space is constant.

1. Fix $n \in N$ and consider the conjugation map
$$c_n\colon G \to G, \qquad c_n(g) = g\,n\,g^{-1}$$
It is continuous, being a composition of multiplications and inversion. Because N is normal, $g\,n\,g^{-1} \in N$ for every $g \in G$, so c_n restricts to a continuous map $G \to N$ into the discrete subspace N. The domain G is connected, and a continuous map from a connected space

into a discrete space is constant, so c_n is constant on G; its value at $g = e$ is $c_n(e) = n$. Hence $g\,n\,g^{-1} = n$ for all $g \in G$, that is $n \in Z(G)$. As $n \in N$ was arbitrary, $N \subseteq Z(G)$.

2. As the kernel of a homomorphism, $K = \ker \pi$ is a normal subgroup of $\tilde{G}$. As the fiber $\pi^{-1}(e)$ of a covering map, it is a discrete subset of $\tilde{G}$: a covering map is a local homeomorphism, so each point of the fiber over e has a neighborhood meeting the fiber only in that point. Thus K is a discrete normal subgroup of the connected group $\tilde{G}$, and part (1) applied to $\tilde{G}$ places $K \subseteq Z(\tilde{G})$. The isomorphism $K \cong \pi_1(G)/\pi_*\pi_1(\tilde{G})$ is the covering-space computation of the fiber: for a covering $\pi\colon \tilde{G} \to G$ of connected, locally path-connected spaces the fiber $\pi^{-1}(e)$ is in bijection with the coset space $\pi_1(G, e)/\pi_*\pi_1(\tilde{G}, \tilde{e})$, and for a covering *homomorphism* this bijection is a group isomorphism because the deck transformations are left translations by the elements of K ([9, The Fibre as a Coset Space]). When $\tilde{G}$ is the universal cover, $\pi_*\pi_1(\tilde{G})$ is trivial and this recovers $K \cong \pi_1(G)$ of proposition 3.4.1.

3. Apply part (2) to the universal covering homomorphism $\pi\colon \tilde{G} \to G$ of proposition 3.4.1, whose existence is guaranteed there. Its kernel $\ker \pi$ is contained in $Z(\tilde{G})$ and is therefore abelian, being a subgroup of an abelian group. By proposition 3.4.1 the kernel is isomorphic to $\pi_1(G)$, so $\pi_1(G)$ is abelian, as we sought to show.

The proposition explains why the fundamental group of a Lie group is so much better behaved than the fundamental group of a general space. For an arbitrary path-connected space π_1 can be any group at all, and is typically nonabelian, as the figure-eight already shows with its free group on two generators. A Lie group is a topological group, and the group structure forces its fundamental group to be abelian. The mechanism is entirely contained in part (3): $\pi_1(G)$ is realized as a subgroup of the center of $\tilde{G}$, and central subgroups are abelian. This is the same phenomenon that makes the fundamental group of any topological group abelian, seen here through the covering homomorphism rather than through the Eckmann-Hilton argument on loop concatenation.

The centrality of the kernel is what makes the covering theory of Lie groups a theory of central extensions. A connected group G and its universal cover fit into a short exact sequence

$$1 \longrightarrow \pi_1(G) \longrightarrow \tilde{G} \xrightarrow{\pi} G \longrightarrow 1$$

in which the kernel $\pi_1(G)$ is central. The infinitesimal theory sees only the middle and right terms as the same object, since $d\pi_{\tilde{e}}$ is an isomorphism of Lie algebras (proposition 3.4.1); the entire content of the extension is the discrete central subgroup on the left. Chapter 3 read this sequence algebraically, as $G \cong \tilde{G}/\Gamma$ with Γ a discrete subgroup of $Z(\tilde{G})$; the covering-space language reads the same Γ as $\pi_1(G)$, the deck group of the cover.

A first instance is already at hand. The circle S^1 is a connected Lie group, and its universal cover is the additive group $\mathbb{R}$ via $t \mapsto e^{2\pi i t}$, with kernel the discrete central subgroup $\mathbb{Z} \subseteq \mathbb{R}$. By proposition 4.4.1 this kernel is central, which is automatic since $\mathbb{R}$ is abelian, and $\pi_1(S^1) \cong \mathbb{Z}$, abelian as the proposition requires. The winding number of a loop, the classical invariant $\pi_1(S^1) = \mathbb{Z}$, is literally the element of the kernel through which the loop lifts.

With the kernel understood, the covers of a fixed group G can be classified. The covering-space correspondence of *Topology* [9, Classification of Coverings] matches connected covers of a reasonable space X with subgroups of $\pi_1(X)$; the theorem below is that for a Lie group this correspondence lands inside the category of Lie groups, so every connected cover is again a Lie group covering G by a homomorphism. This is the global, discrete refinement of the Lie group-Lie algebra dictionary, complementing corollary 3.4.5: where the corollary fixed the Lie algebra and varied the group, the theorem fixes the group G and organizes the groups lying above it.

Theorem 4.4.2: Classification of Connected Covering Groups

Let G be a connected Lie group with universal covering group $\tilde{G}$ and fundamental group $\pi_1(G) \cong \ker(\tilde{G} \to G) \subseteq Z(\tilde{G})$ (propositions 3.4.1 and 4.4.1). Assigning to a subgroup $\Lambda \subseteq \pi_1(G)$ the quotient $\tilde{G}/\Lambda$ gives an order-reversing bijection

$$\{\text{subgroups } \Lambda \subseteq \pi_1(G)\} \longleftrightarrow \{\text{connected covering groups of } G\}/\cong$$

under which:

1. each $\tilde{G}/\Lambda$ is a connected Lie group and the induced map $p_\Lambda \colon \tilde{G}/\Lambda \to G$ is a covering homomorphism with kernel $\pi_1(G)/\Lambda$, so $\pi_1(\tilde{G}/\Lambda) \cong \Lambda$;
2. $\Lambda = \{e\}$ gives the universal cover $\tilde{G}$ (on top) and $\Lambda = \pi_1(G)$ gives G itself (on the bottom);
3. every group in the tower has the same Lie algebra: $\mathrm{Lie}(\tilde{G}/\Lambda) \cong \mathrm{Lie}(G)$, and every covering homomorphism in the tower is a local diffeomorphism inducing this identity on Lie algebras.

***Proof*:**

Write $\Pi = \pi_1(G)$, identified with the discrete central subgroup $\ker(\pi \colon \tilde{G} \to G) \subseteq Z(\tilde{G})$ via proposition 3.4.1. Since Π is central, every subgroup $\Lambda \subseteq \Pi$ is normal in $\tilde{G}$, so the quotients below are groups.

Step 1: each quotient $\tilde{G}/\Lambda$ is a Lie group and p_Λ is a covering homomorphism. Let $\Lambda \subseteq \Pi$ be a subgroup. As a subgroup of the discrete group Π, it is itself discrete, and it is central in $\tilde{G}$. A discrete central subgroup acts on $\tilde{G}$ by left translations, freely (translation by a nonidentity element has no fixed point) and properly discontinuously (discreteness of Λ separates the orbit of any point), so the quotient map $q_\Lambda \colon \tilde{G} \to \tilde{G}/\Lambda$ is a covering ([9, Covering Quotient Theorem]) and $\tilde{G}/\Lambda$ inherits a smooth manifold structure for which q_Λ is a local diffeomorphism (the smooth refinement, with the quotient's Hausdorffness as its extra hypothesis, is [13, Covering Quotient Theorem]). The group operations descend because Λ is normal, and they are smooth because they lift through the local diffeomorphism q_Λ to the smooth operations of $\tilde{G}$; hence $\tilde{G}/\Lambda$ is a Lie group. Because $\Lambda \subseteq \Pi = \ker \pi$, the universal covering homomorphism π factors through q_Λ: there is a unique homomorphism $p_\Lambda \colon \tilde{G}/\Lambda \to G$ with $p_\Lambda \circ q_\Lambda = \pi$, given on cosets by $p_\Lambda(\tilde{g}\Lambda) = \pi(\tilde{g})$. It is well defined since $\Lambda \subseteq \ker \pi$, surjective since π is, and its kernel is $\ker \pi/\Lambda = \Pi/\Lambda$. That p_Λ is a covering follows from $\pi = p_\Lambda \circ q_\Lambda$ being a covering with q_Λ a covering: a local homeomorphism p_Λ whose composite with the surjective covering q_Λ is a covering is itself a covering, with fiber the discrete group Π/Λ ([9]). Since $\tilde{G}$ is simply connected, q_Λ is the universal cover of $\tilde{G}/\Lambda$, so $\pi_1(\tilde{G}/\Lambda) \cong \Lambda$ by proposition 3.4.1.

Step 2: surjectivity of the assignment. Let $p \colon H \to G$ be any connected covering homomorphism. Its universal cover is a simply connected Lie group covering G, so by uniqueness of the universal covering group (proposition 3.4.1) it is isomorphic to $\tilde{G}$, and the covering $\tilde{G} \to H \xrightarrow{p} G$ realizes $\tilde{G}$ as the universal cover of H compatibly with π. The kernel $\Lambda := \ker(\tilde{G} \to H)$ is then a discrete central subgroup of $\tilde{G}$ contained in $\ker \pi = \Pi$ (since the composite $\tilde{G} \to H \to G$ is π), and $H \cong \tilde{G}/\Lambda$ as covering groups of G by the first isomorphism theorem, with the isomorphism commuting with the maps to G. Thus every connected covering group of G arises as some $\tilde{G}/\Lambda$.

Step 3: injectivity and order-reversal. Suppose $\tilde{G}/\Lambda \cong \tilde{G}/\Lambda'$ as covering groups of G, that is, there is an isomorphism ψ with $p_{\Lambda'} \circ \psi = p_\Lambda$. Lifting to the common universal cover $\tilde{G}$, the isomorphism ψ lifts to an automorphism of $\tilde{G}$ commuting with π; the only such automorphism fixing the basepoint is the identity, since a lift of the identity map of G through the universal cover that fixes $\tilde{e}$ is the identity ([9, Uniqueness of Lifts]). Hence $q_{\Lambda'} = \psi \circ q_\Lambda$ forces $\Lambda = \Lambda'$ as the respective kernels. Order-reversal is immediate from the construction: if $\Lambda \subseteq \Lambda'$ then $\tilde{G}/\Lambda'$ is a further quotient of $\tilde{G}/\Lambda$, so $\tilde{G}/\Lambda$ sits above $\tilde{G}/\Lambda'$ in the tower; larger subgroups give smaller covers. This proves the bijection is order-reversing and establishes claims (1) and (2), with $\Lambda = \{e\}$ giving $\tilde{G}$ on top and $\Lambda = \Pi$ giving $\tilde{G}/\Pi \cong G$ on the bottom.

Step 4: the common Lie algebra. Each covering homomorphism in the tower is a covering map, hence a local diffeomorphism, so its differential at the identity is a linear isomorphism of tangent spaces and a Lie-algebra homomorphism because the map is a group homomorphism. A bijective Lie-algebra homomorphism is an isomorphism, so $\mathrm{Lie}(\tilde{G}/\Lambda) \cong \mathrm{Lie}(\tilde{G}) \cong \mathrm{Lie}(G)$ for every Λ, with the identifications compatible across the tower. This is claim (3), completing the proof.

The theorem completes the picture begun in chapter 3. There the varying object was the Lie algebra's group, listed as central quotients of $\tilde{G}$; here the fixed object is G, and the theorem organizes the groups above it as a lattice indexed by the subgroups of $\pi_1(G)$. The two descriptions are the same short exact sequence read from opposite ends. Corollary 3.4.5 started from $\mathfrak{g}$ and produced $\tilde{G}/\Gamma$ for each discrete $\Gamma \subseteq Z(\tilde{G})$; the present theorem starts from $G = \tilde{G}/\pi_1(G)$ and produces the intermediate covers $\tilde{G}/\Lambda$ for $\Lambda \subseteq \pi_1(G)$. Since $\pi_1(G)$ is one particular such Γ, its subgroups are exactly the discrete central subgroups sitting between $\{e\}$ and $\pi_1(G)$, and the tower of covers of G is the interval $[\,\{e\},\, \pi_1(G)\,]$ in the subgroup lattice of $Z(\tilde{G})$.

The order-reversal is the content worth remembering. A larger subgroup Λ of $\pi_1(G)$ means a larger quotient $\tilde{G}/\Lambda$, hence a smaller cover, hence fewer sheets: passing up the subgroup lattice of $\pi_1(G)$ passes down the tower of covers toward G. The universal cover, with $\Lambda = \{e\}$, is maximally unwound; G itself, with $\Lambda = \pi_1(G)$, is maximally wound. Everything in between is a cover of G and simultaneously a quotient of $\tilde{G}$, and none of these distinctions is visible to the Lie algebra, which is the same $\mathrm{Lie}(G)$ at every stage.

The worked examples show the tower, from the one-line case of the circle to the double cover of the rotation group and the cases that reach beyond the matrix groups entirely.

Example 4.4.3: Covering Groups: Circle, Rotations, Spin, and $\mathrm{SL}_2(\mathbb{R})$

The four examples run from the simplest infinite tower to a case that leaves the matrix groups behind.

1. *The circle.* The map $\pi\colon \mathbb{R} \to S^1$, $\pi(t) = e^{2\pi i t}$, is a homomorphism of Lie groups with kernel the discrete central subgroup $\mathbb{Z}$, so it is the universal covering group of S^1 and $\pi_1(S^1) \cong \mathbb{Z}$. Since $\mathbb{R}$ is abelian its center is all of $\mathbb{R}$, and the subgroups of $\pi_1(S^1) = \mathbb{Z}$ are exactly $n\mathbb{Z}$ for $n \geq 0$. By theorem 4.4.2 the connected covering groups of S^1 are the quotients $\mathbb{R}/n\mathbb{Z}$: for $n = 0$ this is $\mathbb{R}$ itself (the universal cover), and for $n \geq 1$ the quotient $\mathbb{R}/n\mathbb{Z} \cong S^1$ is the circle covering itself by the n-fold map $z \mapsto z^n$, with $\pi_1 \cong n\mathbb{Z} \cong \mathbb{Z}$ realized as the index-n subgroup. The tower is the lattice of subgroups of $\mathbb{Z}$, an infinite chain of circles above S^1 topped by the line $\mathbb{R}$.

2. *The rotation group.* The double cover $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5 has kernel exactly the center $\{\pm I\}$ of $\mathrm{SU}(2)$, a discrete central subgroup isomorphic to $\mathbb{Z}/2$. Since $\mathrm{SU}(2) \cong S^3$ is simply connected (proposition 1.5.1), φ is the universal cover of $\mathrm{SO}(3)$, and $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2$

(corollary 1.5.6). The subgroups of $\mathbb{Z}/2$ are only $\{e\}$ and $\mathbb{Z}/2$, so by theorem 4.4.2 the tower of connected covers of $\mathrm{SO}(3)$ has exactly two floors: $\mathrm{SU}(2)$ on top ($\Lambda = \{e\}$) and $\mathrm{SO}(3)$ itself on the bottom ($\Lambda = \mathbb{Z}/2$). There is no intermediate cover, because $\mathbb{Z}/2$ has no proper nontrivial subgroup. The nontrivial loop in $\mathrm{SO}(3)$, a full 2π rotation about a fixed axis, lifts to a path in $\mathrm{SU}(2)$ from I to $-I$, and the element $-I \in \ker\varphi$ is the generator of $\pi_1(\mathrm{SO}(3))$.

3. *The spin groups.* The pattern of $\mathrm{SU}(2) \to \mathrm{SO}(3)$ persists in every dimension. For $n \geq 3$ the special orthogonal group $\mathrm{SO}(n)$ has fundamental group $\pi_1(\mathrm{SO}(n)) \cong \mathbb{Z}/2$, and its universal cover is the spin group $\mathrm{Spin}(n)$, fitting into a double cover

$$1 \longrightarrow \mathbb{Z}/2 \longrightarrow \mathrm{Spin}(n) \longrightarrow \mathrm{SO}(n) \longrightarrow 1$$

with central kernel $\mathbb{Z}/2$. The case $n = 3$ is $\mathrm{Spin}(3) = \mathrm{SU}(2)$, the double cover of this example, so the running example is the first spin group. As with $\mathrm{SO}(3)$, the tower over $\mathrm{SO}(n)$ has just two floors for $n \geq 3$, since $\pi_1 = \mathbb{Z}/2$; the case $n = 2$ is different, with $\mathrm{SO}(2) = S^1$ and the infinite tower of item (1). The spin groups are the arena of the spinor representations, which live on $\mathrm{Spin}(n)$ but not on $\mathrm{SO}(n)$ precisely because they fail to descend across the central $\mathbb{Z}/2$; the construction of $\mathrm{Spin}(n)$ inside the Clifford algebra is developed in the Riemannian and index-theoretic sequels to this book.

4. *A cover that is not a matrix group.* The examples above all have matrix groups at every floor of the tower, but this is not forced. The group $\mathrm{SL}_2(\mathbb{R})$ is connected with fundamental group $\pi_1(\mathrm{SL}_2(\mathbb{R})) \cong \mathbb{Z}$, so by theorem 4.4.2 its tower of covers is the infinite chain indexed by the subgroups $n\mathbb{Z} \subseteq \mathbb{Z}$, topped by the universal cover $\widetilde{\mathrm{SL}_2(\mathbb{R})}$. This universal cover admits no faithful finite-dimensional representation, and in particular is not isomorphic to any matrix group: any continuous homomorphism $\widetilde{\mathrm{SL}_2(\mathbb{R})} \to \mathrm{GL}_N(\mathbb{C})$ must kill an infinite subgroup of the central $\mathbb{Z}$, because the center of every connected matrix group with this Lie algebra is finite. Thus $\widetilde{\mathrm{SL}_2(\mathbb{R})}$ is a Lie group that cannot be realized as a closed subgroup of any GL_N, the first object in this book outside the matrix world of chapter 1. Its representation theory belongs to the analysis of noncompact real groups in Part IV, where $\mathrm{SL}_2(\mathbb{R})$ and its covers organize the noncompact side of the theory.

The examples span the range of behavior the classification allows. The circle and $\mathrm{SL}_2(\mathbb{R})$ have infinite fundamental group $\mathbb{Z}$ and so infinite towers; the rotation and spin groups have $\pi_1 = \mathbb{Z}/2$ and so two-floor towers. In every case the Lie algebra is blind to the height in the tower, and the topology, encoded in π_1, is what distinguishes the floors. The last example is the one that breaks the frame of chapter 1: $\widetilde{\mathrm{SL}_2(\mathbb{R})}$ is a bona fide Lie group produced by the abstract correspondence, yet it is not a matrix group, which is why the correspondence had to be developed intrinsically rather than through matrices, and why Part IV must handle noncompact groups by tools beyond the classical-group computations.

The section closes the account of Part I's dictionary. Lie's third theorem (theorem 3.4.4) established that the Lie algebra determines the simply connected group, and named the single invisible invariant: the fundamental group, which the infinitesimal theory cannot see. That invariant is now organized. It is the discrete central kernel of the universal cover (proposition 4.4.1), it is abelian, and its subgroups index a tower of connected covers (theorem 4.4.2), all sharing the one Lie algebra $\mathrm{Lie}(G)$. A Lie group is thus a pair of data: a Lie algebra, carrying all the local structure, and a discrete central subgroup of the associated simply connected group, carrying the global topology. The rest of the book computes with the first; this section named and organized the second.

Exercise 4.4.1

1. Show that the n-torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ is a connected Lie group with universal cover $\mathbb{R}^n$, and that $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$. Using theorem 4.4.2, describe the connected covering groups of $\mathbb{T}^n$ in terms of the subgroups of $\mathbb{Z}^n$, and verify directly that the kernel of the covering $\mathbb{R}^n \to \mathbb{T}^n$ is central.

2. Let G be a connected Lie group with finite center $Z(G)$. Prove that G has only finitely many connected covering groups (up to isomorphism) if and only if $\pi_1(G)$ is finite. (Use theorem 4.4.2 to reduce to counting subgroups of $\pi_1(G)$, and note that $\pi_1(G) \subseteq Z(\tilde{G})$; the finiteness of $Z(G)$ and of $\pi_1(G)$ together are equivalent to finiteness of $Z(\tilde{G})$.)

3. Prove that the unitary group $\mathrm{U}(n)$ is isomorphic to the quotient $(\mathrm{SU}(n) \times \mathrm{U}(1))/\mu_n$, where $\mu_n = \{(\zeta^{-1}I, \zeta) \mid \zeta^n = 1\}$ is the subgroup of the center generated by $(\omega^{-1}I, \omega)$ for $\omega = e^{2\pi i/n}$. Identify the map $\mathrm{SU}(n) \times \mathrm{U}(1) \to \mathrm{U}(n)$ realizing this quotient and compute its kernel.

4. Prove that the center of $\mathrm{SU}(n)$ is the cyclic group of order n consisting of the scalar matrices ζI with $\zeta^n = 1$, that is $Z(\mathrm{SU}(n)) = \{\zeta I \mid \zeta^n = 1\} \cong \mathbb{Z}/n$. Deduce that the adjoint group $\mathrm{SU}(n)/Z(\mathrm{SU}(n)) = \mathrm{PSU}(n)$ has fundamental group $\pi_1(\mathrm{PSU}(n)) \cong \mathbb{Z}/n$, and that $\mathrm{SU}(n)$ is its universal cover.

5. Using $\pi_1(\mathrm{SO}(4)) \cong \mathbb{Z}/2$ and the exceptional isomorphism $\mathrm{Spin}(4) \cong \mathrm{SU}(2) \times \mathrm{SU}(2)$, describe the tower of connected covers of $\mathrm{SO}(4)$. In particular, identify the subgroup of $\pi_1(\mathrm{SO}(4))$ corresponding to the intermediate group $\mathrm{SO}(4)$'s relation to $\mathrm{SU}(2) \times \mathrm{SU}(2)$, and explain why $\mathrm{SO}(4)$ has exactly two connected covers despite $\mathrm{Spin}(4)$ splitting as a product.

6. Prove that for connected Lie groups G and H the fundamental group of the product satisfies $\pi_1(G \times H) \cong \pi_1(G) \times \pi_1(H)$, and that the universal cover of $G \times H$ is $\tilde{G} \times \tilde{H}$. Use this and theorem 4.4.2 to compute the tower of covers of the torus $\mathbb{T}^2 = S^1 \times S^1$ from the towers of its factors, and reconcile the answer with exercise 4.4.1 (1).

4.5 Fiber Bundles from Subgroups and the Isotropy Representation

The quotient G/H of theorem 4.1.3 was built as a smooth manifold, and the orbit-stabilizer theorem theorem 4.2.3 identified it with any homogeneous G-space carrying a point of stabilizer H. This section reads the projection $\pi\colon G \to G/H$ not as a bare submersion but as a bundle: the group H acts on G by right translation, its orbits are exactly the fibers of π, and near each point of the base this action looks like the trivial action of H on a product. That is the content of being a principal H-bundle, and it is the structure that organizes the geometry attached to a homogeneous space. Every H-space F then produces an associated bundle $G \times_H F$ over G/H by gluing copies of F along the transition data of π, and the associated bundle attached to the most geometrically loaded F is the tangent bundle of G/H itself.

The payoff of this reading is the isotropy representation. A homogeneous space is the same at every point, so its local geometry is encoded at the single base point eH, in the tangent space $T_{eH}(G/H)$ together with the linear action of the stabilizer H on it. This section identifies that tangent space with the quotient $\mathfrak{g}/\mathfrak{h}$ of Lie algebras and the action with $\mathrm{Ad}(H)$ descended to the quotient. When a complement to $\mathfrak{h}$ can be chosen $\mathrm{Ad}(H)$-invariantly, which happens whenever H is compact, the tangent space is realized inside $\mathfrak{g}$ itself as a subspace $\mathfrak{m}$ carrying an H-representation,

and a G-invariant metric on G/H becomes the same datum as an $\mathrm{Ad}(H)$-invariant inner product on $\mathfrak{m}$. This is the input to the theory of Riemannian symmetric spaces developed in Part IV.

The right action of H on G is the starting point. For $h \in H$ the map $R_h \colon g \mapsto gh$ is a diffeomorphism of G, it preserves each left coset gH, and $\pi(gh) = gh\,H = gH = \pi(g)$, so R_h acts within the fibers of π. The action is free, since $gh = g$ forces $h = e$, and its orbits are precisely the fibers $\pi^{-1}(gH) = gH$. A principal bundle is the assertion that this fiberwise free action is, in suitable local coordinates, the translation action of H on a product $U \times H$.

Theorem 4.5.1: $G \to G/H$ Is a Principal H-Bundle

Let G be a Lie group and $H \leq G$ a closed subgroup, and equip G/H with the smooth structure of theorem 4.1.3. Then the projection $\pi \colon G \to G/H$, together with the right translation action of H on G, is a smooth principal H-bundle. That is:

1. H acts smoothly and freely on G on the right by $g \cdot h = gh$, preserving each fiber of π, and acts simply transitively on each fiber.
2. Every point of G/H has an open neighborhood U and a diffeomorphism $\Phi \colon \pi^{-1}(U) \xrightarrow{\sim} U \times H$ that is H-equivariant and covers the identity, meaning $\Phi(g) = (\pi(g), \varphi(g))$ with $\varphi(gh) = \varphi(g)h$ for all $h \in H$.

Consequently, for every smooth left H-space F the associated bundle $G \times_H F = (G \times F)/H$, with H acting by $h \cdot (g, f) = (gh^{-1}, h \cdot f)$, is a smooth fiber bundle over G/H with fiber F and projection $[g, f] \mapsto \pi(g)$.

***Proof*:**

Step 1: the right action and freeness. The right action $G \times H \to G$, $(g, h) \mapsto gh$, is the restriction of the smooth multiplication of G, hence smooth. If $gh = g$ then $h = e$ by cancellation, so the action is free. For $g, g' \in G$ the equality $\pi(g) = \pi(g')$ means $g^{-1}g' \in H$, so $g' = g(g^{-1}g') = g \cdot (g^{-1}g')$ lies in the H-orbit of g; and any gh lies in the same fiber as g since $\pi(gh) = gH = \pi(g)$. Thus the orbit of g is exactly $\pi^{-1}(\pi(g))$, and freeness makes the action simply transitive on that fiber.

Step 2: a local section through the base point. The construction of theorem 4.1.3 produced, from a vector-space complement $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ and the exponential chart of theorem 2.3.4, an open ball $B \subseteq \mathfrak{m}$ about 0 for which $Y \mapsto \pi(\exp Y)$ is a diffeomorphism of B onto an open neighborhood $U_0 := \pi(\exp B)$ of the base point eH, the coset chart of theorem 4.1.3. Its inverse followed by exp is a smooth map

$$s_0 \colon U_0 \to G, \qquad \pi(\exp Y) \mapsto \exp Y \quad (Y \in B)$$

with $\pi \circ s_0 = \mathrm{id}_{U_0}$: the slice $\exp(B)$ is transverse to H near e and meets each nearby fiber in exactly one point. A map $s \colon U \to G$ on an open set $U \subseteq G/H$ with $\pi \circ s = \mathrm{id}_U$ is a *local section* of π; the map s_0 is such a section on U_0.

Step 3: the trivialization from a local section. Given a smooth local section $s \colon U \to G$, define

$$\Psi \colon U \times H \to \pi^{-1}(U), \qquad \Psi(x, h) = s(x)\,h$$

This is smooth, being built from s and the multiplication of G. It is a bijection onto $\pi^{-1}(U)$: given $g \in \pi^{-1}(U)$, set $x = \pi(g) \in U$; then $\pi(s(x)) = x = \pi(g)$, so $s(x)^{-1}g \in H$, and

$g = s(x)\,(s(x)^{-1}g) = \Psi(x, s(x)^{-1}g)$, giving surjectivity, while if $s(x)h = s(x')h'$ then applying π gives $x = x'$ and then $h = h'$ by cancellation, giving injectivity. Its inverse is

$$\Phi\colon \pi^{-1}(U) \to U \times H, \qquad \Phi(g) = \big(\pi(g),\ s(\pi(g))^{-1}g\big)$$

which is smooth because π, s, inversion, and multiplication all are; here $s(\pi(g))^{-1}g \in H$ lies in the embedded Lie subgroup H, and the corestriction to H is smooth by the embedded-submanifold property ([13, Corestriction to a Regular Submanifold]). Thus Φ is a diffeomorphism. Writing $\varphi(g) = s(\pi(g))^{-1}g$ for the H-component, equivariance is the computation

$$\varphi(gh) = s(\pi(gh))^{-1}(gh) = s(\pi(g))^{-1}g\,h = \varphi(g)\,h$$

using $\pi(gh) = \pi(g)$. This establishes the trivialization over U_0 from the section s_0 of Step 2.

Step 4: trivializing over every point by translation. Let $gH \in G/H$ be arbitrary. Left translation $L_g\colon G \to G$ is a diffeomorphism, and it descends to a diffeomorphism $\bar{L}_g\colon G/H \to G/H$, $\bar{L}_g(g'H) = gg'H$, because L_g carries fibers to fibers and $\pi \circ L_g = \bar{L}_g \circ \pi$ by proposition 4.1.5. Then $s := L_g \circ s_0 \circ \bar{L}_g^{-1}$ is a smooth local section over $U := \bar{L}_g(U_0)$, an open neighborhood of gH: indeed $\pi \circ s = \pi \circ L_g \circ s_0 \circ \bar{L}_g^{-1} = \bar{L}_g \circ \pi \circ s_0 \circ \bar{L}_g^{-1} = \bar{L}_g \circ \bar{L}_g^{-1} = \mathrm{id}_U$, using $\pi \circ s_0 = \mathrm{id}$. Step 3 turns this section into an H-equivariant trivialization over U. Since gH was arbitrary, π is locally trivial, and with the free fiberwise action of Step 1 this is precisely a principal H-bundle.

Step 5: the associated bundle. Let F be a smooth left H-space. The group H acts on $G \times F$ by $h \cdot (g, f) = (gh^{-1}, h \cdot f)$, freely, since freeness in the first factor already forces $h = e$. The quotient $G \times_H F := (G \times F)/H$ carries the quotient topology, and the map $q\colon G \times F \to G/H$, $(g, f) \mapsto \pi(g)$, is constant on H-orbits, hence descends to $p\colon G \times_H F \to G/H$, $[g, f] \mapsto \pi(g)$. Over a trivializing set U with section s, the map

$$U \times F \to p^{-1}(U), \qquad (x, f) \mapsto [s(x), f]$$

is a bijection with inverse $[g, f] \mapsto (\pi(g), \varphi(g) \cdot f)$, where $\varphi(g) = s(\pi(g))^{-1}g$ as in Step 3: this is well defined because $[gh^{-1}, h \cdot f] \mapsto (\pi(g), \varphi(gh^{-1}) \cdot (h \cdot f)) = (\pi(g), \varphi(g)h^{-1}h \cdot f) = (\pi(g), \varphi(g) \cdot f)$, using the equivariance $\varphi(gh^{-1}) = \varphi(g)h^{-1}$. These charts are smoothly compatible because on overlaps the two local sections differ by a smooth H-valued function, which acts smoothly on F; transporting the smooth structure of $U \times F$ through them makes $G \times_H F$ a smooth manifold and p a smooth fiber bundle with fiber F, completing the proof.

The theorem promotes the submersion $\pi\colon G \to G/H$ to a bundle whose structure group is H acting on itself by translation, and it does so with no new analytic input: the single local section s_0 built from the coset chart of theorem 4.1.3, spread over the base by the homogeneity of G, is the entire construction. The mechanism is the one that recurs throughout the chapter, that a group acts transitively on its own quotient and so a chart at one point suffices everywhere. The associated-bundle construction $F \rightsquigarrow G \times_H F$ then manufactures a vector bundle, a sphere bundle, or a bundle of frames over G/H from any H-space F, all sharing the transition functions of π; the geometry of a homogeneous space is thereby reduced to the representation theory of the single group H.

The archetype is the Hopf fibration, where $G = \mathrm{SU}(2)$, $H = \mathrm{U}(1)$, and the base is the two-sphere.

Example 4.5.2: The Hopf Fibration as a Principal U(1)-Bundle

Realize $\mathrm{U}(1) = \{\mathrm{diag}(e^{i\theta}, e^{-i\theta}) \,|\, \theta \in \mathbb{R}\}$ as the diagonal maximal torus of SU(2), a closed subgroup. By proposition 1.5.1 the group SU(2) is diffeomorphic to S^3, and by example 4.3.3 the coset space $\mathrm{SU}(2)/\mathrm{U}(1)$ is diffeomorphic to S^2, with the projection realizing the Hopf map $S^3 \to S^2$. Theorem 4.5.1 identifies this projection as a principal U(1)-bundle

$$S^1 \hookrightarrow S^3 \longrightarrow S^2$$

the circle $\mathrm{U}(1) \cong S^1$ acting freely on S^3 on the right by $g \mapsto g\,\mathrm{diag}(e^{i\theta}, e^{-i\theta})$. Freeness is immediate from Step 1, and the fibers are the circles $g\,\mathrm{U}(1)$. Local triviality is not global: were the bundle trivial, S^3 would be diffeomorphic to $S^2 \times S^1$, but these are distinguished by their fundamental groups, $\pi_1(S^3) = 0$ while $\pi_1(S^2 \times S^1) = \mathbb{Z}$ ([9, Fundamental Groups Of The Circle, Fundamental Group of a Product], with the sphere computations $\pi_1(S^2) = \pi_1(S^3) = 0$ from [11]). The Hopf bundle is therefore a nontrivial principal circle bundle, the smallest example in which the local product structure of theorem 4.5.1 fails to be global.

The example shows the principal-bundle language capturing a nontriviality that the plain diffeomorphism $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$ does not see: the total space S^3 is assembled from circles over S^2 in a way that cannot be undone globally. The obstruction is measured by the topology of the base and structure group, and it is the kind of invariant that the associated-bundle construction converts into characteristic classes in a later theory. Here the point is more elementary, that $\pi\colon G \to G/H$ is a genuine bundle and its local sections need not patch to a global one.

The associated bundle attached to the geometrically decisive H-space produces the tangent bundle of G/H itself. To see which H-space that is, fix attention at the base point $o := eH \in G/H$. Its stabilizer under the G-action of theorem 4.2.3 is exactly H, so H acts on the tangent space $T_o(G/H)$ by differentials of the maps $\bar{L}_h\colon G/H \to G/H$ fixing o. This linear action is the datum that linearizes the homogeneous space at o.

Definition 4.5.3: The Linear Isotropy Representation

Let $H \leq G$ be a closed subgroup, $o = eH$ the base point of G/H, and for $h \in H$ let $\bar{L}_h\colon G/H \to G/H$, $gH \mapsto hgH$, be the diffeomorphism induced by left translation by h. Each $\bar{L}_h$ fixes o, since $hH = H$, so its differential at o is a linear automorphism of $T_o(G/H)$. The *linear isotropy representation* of H is the homomorphism

$$\chi\colon H \to \mathrm{GL}\big(T_o(G/H)\big), \qquad \chi(h) = d(\bar{L}_h)_o$$

Under the canonical isomorphism $T_o(G/H) \cong \mathfrak{g}/\mathfrak{h}$ induced by the submersion $d\pi_e\colon \mathfrak{g} \to T_o(G/H)$ (whose kernel is $\mathfrak{h} = T_eH$), the isotropy representation is $\mathrm{Ad}(h)$ descended to the quotient: writing $\overline{\mathrm{Ad}}(h)\colon \mathfrak{g}/\mathfrak{h} \to \mathfrak{g}/\mathfrak{h}$ for the map induced by $\mathrm{Ad}(h)$, well defined because $\mathrm{Ad}(H)$ preserves $\mathfrak{h}$ (definition 2.5.1), the relation

$$\chi(h) = d\pi_e \circ \mathrm{Ad}(h) \circ (d\pi_e)^{-1} = \overline{\mathrm{Ad}}(h)$$

holds for every $h \in H$.

The identification requires two verifications, and both are short enough to record at once. First, $d\pi_e\colon \mathfrak{g} \to T_o(G/H)$ is surjective because π is a submersion (theorem 4.1.3), and its kernel is $\mathfrak{h}$: the

fiber $\pi^{-1}(o) = H$ has tangent space $\mathfrak{h}$ at e, and a tangent vector at e lies in $\ker d\pi_e$ iff it is tangent to the fiber, so $\ker d\pi_e = T_eH = \mathfrak{h}$. Hence $d\pi_e$ descends to the isomorphism $\mathfrak{g}/\mathfrak{h} \xrightarrow{\sim} T_o(G/H)$. Second, that $\mathrm{Ad}(H)$ preserves $\mathfrak{h}$: for $h \in H$ the conjugation C_h maps H to H, so its differential $\mathrm{Ad}(h) = d(C_h)_e$ maps $T_eH = \mathfrak{h}$ into $\mathfrak{h}$, and therefore $\mathrm{Ad}(h)$ descends to $\overline{\mathrm{Ad}}(h)$ on $\mathfrak{g}/\mathfrak{h}$. The naturality square $\pi \circ C_h = \bar{L}_h \circ \pi$, valid because $C_h(g)H = hgh^{-1}H = hgH = \bar{L}_h(gH)$ (the middle step using $h^{-1} \in H$), differentiates at e to $d\pi_e \circ \mathrm{Ad}(h) = \chi(h) \circ d\pi_e$, which is exactly the claimed relation $\chi(h) = \overline{\mathrm{Ad}}(h)$ under the identification.

The isotropy representation is the linearization of a homogeneous space at a point. Because G moves any point to any other, $T_o(G/H)$ together with the H-action χ carries the pointwise structure of G/H: a G-invariant tensor field on G/H, evaluated at o, is an H-invariant tensor on $T_o(G/H)$ for the isotropy action, and conversely each such H-invariant tensor at o spreads by G to a unique G-invariant field. The formula $\chi = \overline{\mathrm{Ad}}$ makes this computable: the isotropy representation is nothing but the adjoint representation of G, restricted to H and pushed to $\mathfrak{g}/\mathfrak{h}$. The linear algebra of G/H is thereby read off from the pair $(\mathfrak{g}, \mathfrak{h})$ and the adjoint action.

The running example illustrates the identification in both of its extremes, a discrete stabilizer where the quotient does not collapse and a one-dimensional stabilizer where it does.

Example 4.5.4: Isotropy for $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ and for S^2

Take first $H = \{\pm I\}$, the center of $G = \mathrm{SU}(2)$, a closed discrete subgroup with $\mathfrak{h} = 0$. Then $G/H = \mathrm{SO}(3)$ by theorem 1.5.5, and $\mathfrak{g}/\mathfrak{h} = \mathfrak{su}(2) \cong \mathbb{R}^3$ with no collapse. The isotropy representation $\chi\colon \{\pm I\} \to \mathrm{GL}(\mathfrak{su}(2))$ is $\overline{\mathrm{Ad}}$; since $\mathrm{Ad}(\pm I)X = (\pm I)X(\pm I)^{-1} = X$, the representation is trivial, as it must be because a discrete stabilizer acts trivially on the tangent space of a covering quotient. This recovers that $\mathrm{SO}(3)$ has the same infinitesimal structure at every point as $\mathrm{SU}(2)$, the two differing only by the central kernel.

Take next $H = \mathrm{U}(1)$, the diagonal torus, so $G/H = S^2$ by example 4.3.3. Use the basis X_1, X_2, X_3 of $\mathfrak{su}(2)$ from example 1.4.7, with $X_3 = \frac{1}{2}\,\mathrm{diag}(-i, i)$ and $[X_1, X_2] = X_3$, $[X_2, X_3] = X_1$, $[X_3, X_1] = X_2$. Since $\exp(tX_3) = \mathrm{diag}(e^{-it/2}, e^{it/2})$, the diagonal torus is $\mathrm{U}(1) = \{\exp(-2\theta X_3) : \theta \in \mathbb{R}\} = \{\mathrm{diag}(e^{i\theta}, e^{-i\theta})\}$, so $\mathfrak{h} = \mathfrak{u}(1) = \mathbb{R}X_3$ is one-dimensional and $\mathfrak{g}/\mathfrak{h}$ is two-dimensional, spanned by the images of X_1, X_2, matching $\dim S^2 = 2$. For $u_\theta = \exp(-2\theta X_3) \in \mathrm{U}(1)$ the adjoint action is $\mathrm{Ad}(u_\theta) = \exp(-2\theta\,\mathrm{ad}X_3)$; on the plane $\mathrm{span}(X_1, X_2)$ the operator $\mathrm{ad}X_3$ sends $X_1 \mapsto X_2$ and $X_2 \mapsto -X_1$, a rotation generator, so

$$\mathrm{Ad}(u_\theta)X_1 = \cos(2\theta)X_1 - \sin(2\theta)X_2, \qquad \mathrm{Ad}(u_\theta)X_2 = \sin(2\theta)X_1 + \cos(2\theta)X_2$$

a rotation through angle 2θ of the (X_1, X_2)-plane, while $\mathrm{Ad}(u_\theta)X_3 = X_3$ fixes $\mathfrak{h}$. Descending to $\mathfrak{g}/\mathfrak{h}$, the isotropy representation $\chi(u_\theta)$ is exactly this rotation through 2θ of the tangent plane T_oS^2. The circle $\mathrm{U}(1)$ thus acts on the tangent plane to the sphere at the base point by rotations, the isotropy action under which the round metric on S^2 is invariant.

The example separates the two extremes: a discrete stabilizer gives a trivial isotropy representation, so a covering quotient is infinitesimally indistinguishable from its total space, whereas a positive-dimensional stabilizer acts by a nontrivial rotation, encoding the rotational symmetry of the round sphere about each point. The doubling of the angle, rotation by 2θ under $\mathrm{Ad}(u_\theta)$, is the same factor of two that made $\mathrm{SU}(2) \to \mathrm{SO}(3)$ a double cover, now seen infinitesimally in how the torus turns the tangent plane. In both cases the tangent space and its symmetry are computed entirely inside $\mathfrak{g}$, with no reference to charts on G/H.

The quotient $\mathfrak{g}/\mathfrak{h}$ is intrinsic, but working with a quotient of vector spaces is less convenient than

working with a subspace. A *reductive* homogeneous space is one where the quotient can be split off as an $\mathrm{Ad}(H)$-invariant complement, realizing $T_o(G/H)$ as a concrete subspace of $\mathfrak{g}$ on which H acts by Ad with no descent. The splitting exists whenever H is compact, by an averaging argument that will recur when invariant integration is developed for compact groups in Part III.

Proposition 4.5.5: Reductive Decompositions and the Isotropy Representation

Let $H \leq G$ be a closed subgroup with Lie algebras $\mathfrak{h} \subseteq \mathfrak{g}$. Suppose there is a vector-space decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m} \qquad \text{with} \qquad \mathrm{Ad}(h)\mathfrak{m} \subseteq \mathfrak{m} \ \text{ for all } h \in H \tag{4.6}$$

Then $\mathfrak{m}$ is called a *reductive complement* and G/H a *reductive homogeneous space*. In this case the composite

$$\mathfrak{m} \hookrightarrow \mathfrak{g} \xrightarrow{d\pi_e} T_o(G/H)$$

is an isomorphism of vector spaces, and it intertwines the restriction $\mathrm{Ad}\,|_H$ of the adjoint representation to $\mathfrak{m}$ with the linear isotropy representation χ of definition 4.5.3. That is, $T_o(G/H) \cong \mathfrak{m}$ as H-representations, with H acting on $\mathfrak{m}$ by $h \mapsto \mathrm{Ad}(h)|_{\mathfrak{m}}$. A reductive complement exists whenever H is compact.

***Proof*:**

Step 1: the complement realizes the tangent space. By the identification in definition 4.5.3, $d\pi_e \colon \mathfrak{g} \to T_o(G/H)$ is surjective with kernel $\mathfrak{h}$. Restricting to $\mathfrak{m}$, the map $d\pi_e|_{\mathfrak{m}} \colon \mathfrak{m} \to T_o(G/H)$ has kernel $\mathfrak{m} \cap \mathfrak{h} = 0$ by the directness of the sum (4.6), so it is injective, and $\dim \mathfrak{m} = \dim \mathfrak{g} - \dim \mathfrak{h} = \dim T_o(G/H)$ by theorem 4.1.3, so it is an isomorphism.

Step 2: it intertwines the two actions. For $h \in H$ the invariance $\mathrm{Ad}(h)\mathfrak{m} \subseteq \mathfrak{m}$ of (4.6) makes $\mathrm{Ad}(h)|_{\mathfrak{m}}$ a linear automorphism of $\mathfrak{m}$. The naturality relation $d\pi_e \circ \mathrm{Ad}(h) = \chi(h) \circ d\pi_e$ established in definition 4.5.3, restricted to $\mathfrak{m}$, reads

$$d\pi_e\big(\mathrm{Ad}(h)|_{\mathfrak{m}}\, Y\big) = \chi(h)\, d\pi_e(Y) \qquad (Y \in \mathfrak{m})$$

which says precisely that the isomorphism $d\pi_e|_{\mathfrak{m}}$ carries the action $\mathrm{Ad}\,|_H$ on $\mathfrak{m}$ to the isotropy action χ on $T_o(G/H)$. Hence $\mathfrak{m} \cong T_o(G/H)$ as H-representations.

Step 3: existence for compact H. Suppose H is compact. Choose any inner product $\langle -, - \rangle_0$ on $\mathfrak{g}$ and average it over H against the bi-invariant (Haar) probability measure dh on H:

$$\langle X, Y \rangle = \int_H \langle \mathrm{Ad}(h)X, \mathrm{Ad}(h)Y \rangle_0 \, dh$$

A compact group carries such a measure, its Haar measure, whose construction is developed for compact groups in Part III and whose existence for locally compact groups is cited to [15, Existence of Haar Measure]; the integrand is continuous in h, and the integral of a positive-definite form against a probability measure is again positive definite, so $\langle -, - \rangle$ is an inner product on $\mathfrak{g}$. It is $\mathrm{Ad}(H)$-invariant: for $h_0 \in H$ the substitution $h \mapsto hh_0$ preserves the measure and gives $\langle \mathrm{Ad}(h_0)X, \mathrm{Ad}(h_0)Y \rangle = \int_H \langle \mathrm{Ad}(hh_0)X, \mathrm{Ad}(hh_0)Y \rangle_0 \, dh = \langle X, Y \rangle$. Now $\mathfrak{h}$ is $\mathrm{Ad}(H)$-invariant (definition 4.5.3), so its orthogonal complement

$$\mathfrak{m} = \mathfrak{h}^{\perp} = \{ Y \in \mathfrak{g} \mid \langle Y, X \rangle = 0 \text{ for all } X \in \mathfrak{h} \}$$

is $\mathrm{Ad}(H)$-invariant as well, since $\mathrm{Ad}(h)$ preserves the inner product and maps $\mathfrak{h}$ to itself, hence maps $\mathfrak{h}^\perp$ to $\mathfrak{h}^\perp$. The orthogonal decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ then satisfies (4.6), completing the proof.

The proposition trades the quotient $\mathfrak{g}/\mathfrak{h}$ for a subspace $\mathfrak{m}$ carrying the same H-representation, and this is the form in which the isotropy data is used in practice. The isotropy representation of a reductive homogeneous space is simply $\mathrm{Ad}(H)$ acting on $\mathfrak{m}$, a representation of H on a piece of the Lie algebra of G, and every G-invariant geometric structure on G/H is now an H-invariant algebraic structure on $\mathfrak{m}$. The compactness of H is a sufficient condition by averaging, not a necessary one; the split (4.6) can exist for noncompact H, and the symmetric spaces of Part IV give a canonical reductive complement, the negative eigenspace of the Cartan involution, with no compactness assumption on H.

The construction identifies exactly the data from which a G-invariant metric is assembled. A G-invariant Riemannian metric on a reductive G/H is the same thing as an $\mathrm{Ad}(H)$-invariant inner product on $\mathfrak{m}$: the value of the metric at the base point o is an inner product on $T_o(G/H) \cong \mathfrak{m}$, it must be invariant under the isotropy action $\mathrm{Ad}(H)|_{\mathfrak{m}}$ for the metric to be G-invariant, and any such inner product spreads by the transitive G-action to a unique G-invariant metric on all of G/H. When H is compact the averaging of Step 3 produces such an inner product, so every reductive homogeneous space with compact isotropy carries a G-invariant metric. This is the entry point to the theory of Riemannian symmetric spaces: there the reductive complement $\mathfrak{m}$ is distinguished by a further bracket condition, $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{h}$, which turns the geodesic symmetry at o into an isometry and pins down the curvature of the invariant metric in terms of the Lie bracket alone. Part IV takes up that theory, with the Cartan decomposition giving $\mathfrak{m}$ and the isotropy representation governing the geometry.

Exercise 4.5.1

1. Verify directly that the Hopf fibration is a principal $\mathrm{U}(1)$-bundle in the sense of theorem 4.5.1: realize $\mathrm{U}(1)$ as the diagonal torus of $\mathrm{SU}(2)$, describe the right action on $S^3 \cong \mathrm{SU}(2)$ (proposition 1.5.1) explicitly, and exhibit the fibers as circles. Explain why the bundle cannot be globally trivial by comparing the fundamental groups of S^3 and $S^2 \times S^1$.

2. Show that the tangent bundle of G/H is the associated bundle $T(G/H) \cong G \times_H (\mathfrak{g}/\mathfrak{h})$, where H acts on $\mathfrak{g}/\mathfrak{h}$ by the isotropy representation $\overline{\mathrm{Ad}}$ of definition 4.5.3. When G/H is reductive with complement $\mathfrak{m}$, rewrite this as $T(G/H) \cong G \times_H \mathfrak{m}$ and identify the fiber over gH with $g \cdot \mathfrak{m}$.

3. Let G/H be reductive with H compact, and let $\langle -, - \rangle$ be an $\mathrm{Ad}(H)$-invariant inner product on the reductive complement $\mathfrak{m}$ (which exists by proposition 4.5.5). Construct from it a G-invariant Riemannian metric on G/H, and prove it is well defined, that is, independent of the coset representative g used to identify $T_{gH}(G/H)$ with $\mathfrak{m}$.

4. Let M be a smooth n-manifold and let $\mathrm{Fr}(M)$ be its frame bundle, whose fiber over $p \in M$ is the set of ordered bases (frames) of T_pM. Show that the right action of $\mathrm{GL}_n(\mathbb{R})$ on frames makes $\mathrm{Fr}(M) \to M$ a principal $\mathrm{GL}_n(\mathbb{R})$-bundle, and identify the tangent bundle TM as the associated bundle $\mathrm{Fr}(M) \times_{\mathrm{GL}_n(\mathbb{R})} \mathbb{R}^n$ for the standard representation of $\mathrm{GL}_n(\mathbb{R})$ on $\mathbb{R}^n$.

5. For the sphere $S^{n-1} = \mathrm{SO}(n)/\mathrm{SO}(n-1)$ (example 4.3.1), compute the isotropy representation of $H = \mathrm{SO}(n-1)$ on $\mathfrak{g}/\mathfrak{h} = \mathfrak{so}(n)/\mathfrak{so}(n-1)$. Identify the reductive complement $\mathfrak{m}$ explicitly and show the isotropy representation is the standard representation of $\mathrm{SO}(n-1)$ on $\mathbb{R}^{n-1}$.

Conclude that S^{n-1} carries an $\mathrm{SO}(n)$-invariant metric, the round metric.

6. Give an example of a closed subgroup $H \leq G$ for which no reductive complement exists, so that G/H is not reductive, and explain which step of the compact-case argument in proposition 4.5.5 fails. (Hint: arrange the adjoint action of H on $\mathfrak{g}$ so that $\mathfrak{h}$ is an invariant subspace with no invariant complement, which is possible only when H is noncompact.)

Part II

The Structure of Lie Algebras

The four chapters of this part carry out the structure theory of finite-dimensional Lie algebras, from the two descending series to the Dynkin diagrams. Chapter 5 develops solvability and nilpotency and the triangularization theorems of Engel and Lie; chapter 6 introduces the Killing form, characterizes semisimplicity through the Cartan criteria, and proves Weyl's complete reducibility and the Levi decomposition; chapter 7 splits a semisimple algebra against a Cartan subalgebra into root spaces; chapter 8 axiomatizes the resulting root systems and classifies them. The data assembled along the way (Cartan subalgebras, roots, the Weyl group) is exactly what Part III re-derives analytically inside a compact group.

5

Nilpotent and Solvable Lie Algebras

Part I built the dictionary between a Lie group and its Lie algebra. From here through the end of Part II the group recedes and the Lie algebra is studied on its own terms, as a finite-dimensional vector space equipped with a bracket. The destination of the part is the classification of the semisimple Lie algebras by their Dynkin diagrams, the combinatorial skeleton that will drive the representation theory of Part III. The route to it runs through the opposite of semisimplicity: to isolate the semisimple algebras one must first understand the solvable ones, the algebras built by successive abelian extensions, and the two classical theorems, Engel's and Lie's, that bring a solvable or nilpotent algebra of operators into triangular form.

This chapter develops solvability and nilpotency from the ground up. Two descending chains of ideals, the derived series and the lower central series, measure how far a Lie algebra stands from being abelian; the algebra is called solvable or nilpotent precisely when the respective chain reaches zero. Engel's theorem characterizes nilpotency infinitesimally, through the nilpotence of the operators $\operatorname{ad}X$, while Lie's theorem shows that over an algebraically closed field of characteristic zero every solvable algebra of matrices can be put simultaneously into upper-triangular form. The chapter closes by extracting from an arbitrary Lie algebra its radical, the largest solvable ideal, whose vanishing is the definition of the semisimple algebras taken up in chapter 6. Unless stated otherwise the ground field throughout Part II is $\mathbb{C}$, or any algebraically closed field of characteristic zero where only those hypotheses are used; the descent from $\mathfrak{g} \otimes \mathbb{C}$ back to a real Lie algebra is the business of Part IV.

5.1 Ideals, Quotients, and the Derived Series

Part I built the dictionary between a Lie group and its Lie algebra and turned it into a tool. Part II now spends the whole of its effort inside the algebra. The reason is one of leverage: the structure theory of Lie algebras is finite-dimensional linear algebra with a bracket, and once it is complete it transfers back to groups through the correspondence theorems of chapter 3 without further analysis.

The goal of the next four chapters is a classification of the simplest interesting Lie algebras, the semisimple ones, and the route runs through their opposite. A semisimple algebra is one with no solvable part at all; to isolate it one must first understand what solvable means, which is the business of this chapter.

The organizing idea is that a Lie algebra can be dismantled the way a group is, by quotienting out a substructure and studying the pieces. For groups the substructures that permit quotients are the normal subgroups; for Lie algebras they are the ideals. This section installs ideals, the quotient algebra, and the homomorphism theorems that make the quotient behave, then uses them to build the first structural invariant, the derived series, whose termination is the definition of solvability. Solvable algebras are the ones assembled from abelian layers by repeated extension, and they are exactly the algebras that Lie's theorem of section 5.4 will triangularize.

The subalgebra of definition 3.1.2 was the linear analogue of a subgroup, a subspace closed under the one operation the algebra carries. It is the wrong notion for forming quotients. To divide $\mathfrak{g}$ by a subspace $\mathfrak{a}$ and inherit a bracket on the cosets, the value $[X+\mathfrak{a}, Y+\mathfrak{a}]$ must not depend on the coset representatives, and this forces a stronger closure condition than bracket closure within $\mathfrak{a}$: the bracket of anything in $\mathfrak{g}$ with anything in $\mathfrak{a}$ must land back in $\mathfrak{a}$. That condition is what an ideal records, and it is the exact linearization of normality.

Definition 5.1.1: Ideal and Quotient Lie Algebra

Let $\mathfrak{g}$ be a Lie algebra over a field k. An *ideal* of $\mathfrak{g}$ is a k-linear subspace $\mathfrak{a} \subseteq \mathfrak{g}$ such that

$$[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}, \qquad \text{that is} \qquad [X, Y] \in \mathfrak{a} \ \text{ for all } X \in \mathfrak{g},\ Y \in \mathfrak{a}$$

written $\mathfrak{a} \trianglelefteq \mathfrak{g}$. Because the bracket is antisymmetric, the one-sided condition $[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}$ already gives $[\mathfrak{a}, \mathfrak{g}] \subseteq \mathfrak{a}$, so there is no distinction between left and right ideals. Every ideal is in particular a subalgebra.
Given an ideal $\mathfrak{a} \trianglelefteq \mathfrak{g}$, the *quotient Lie algebra* $\mathfrak{g}/\mathfrak{a}$ is the quotient vector space with the bracket

$$[X+\mathfrak{a},\ Y+\mathfrak{a}] := [X, Y] + \mathfrak{a}$$

This is well defined, and it makes $\mathfrak{g}/\mathfrak{a}$ a Lie algebra for which the projection $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$, $X \mapsto X + \mathfrak{a}$, is a homomorphism.

The one point requiring proof in the definition is that the bracket on cosets is well defined; everything else is inherited from $\mathfrak{g}$. Suppose $X+\mathfrak{a} = X'+\mathfrak{a}$ and $Y+\mathfrak{a} = Y'+\mathfrak{a}$, so $X' = X + A$ and $Y' = Y + B$ with $A, B \in \mathfrak{a}$. Bilinearity expands the bracket of the primed representatives,

$$[X', Y'] = [X + A,\ Y + B] = [X, Y] + [X, B] + [A, Y] + [A, B]$$

and each of the last three terms lies in $\mathfrak{a}$: the terms $[X, B]$ and $[A, B]$ because $B \in \mathfrak{a}$ and $\mathfrak{a}$ is an ideal, and $[A, Y]$ because $A \in \mathfrak{a}$. Hence $[X', Y'] - [X, Y] \in \mathfrak{a}$, so $[X', Y'] + \mathfrak{a} = [X, Y] + \mathfrak{a}$, and the coset bracket does not depend on the representatives. The bracket axioms of definition 1.4.5, bilinearity, the alternating law, and the Jacobi identity, then pass to the quotient because each is an identity that holds representative by representative. This is the same mechanism that produces the quotient group of a group by a normal subgroup, transported to the additive setting where the subspace $\mathfrak{a}$ plays the role of the normal subgroup and the ideal condition plays the role of normality.

The parallel with groups is more than formal; it is the source of the homomorphism theorems, which hold for Lie algebras in exactly the form the reader knows from [2, First Isomorphism Theorem].

These are the tools that let one recognize an ideal as a kernel and a quotient as an image, and they are used at every turn in the structure theory to follow.

Proposition 5.1.2: Kernel, Image, and the First Isomorphism Theorem

Let $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ be a homomorphism of Lie algebras.

1. The kernel $\ker\varphi = \{X \in \mathfrak{g} \mid \varphi(X) = 0\}$ is an ideal of $\mathfrak{g}$.
2. The image $\varphi(\mathfrak{g})$ is a subalgebra of $\mathfrak{h}$.
3. The induced map $\overline{\varphi}\colon \mathfrak{g}/\ker\varphi \to \varphi(\mathfrak{g})$, $X + \ker\varphi \mapsto \varphi(X)$, is an isomorphism of Lie algebras.

***Proof*:**
The three statements are proved in order, each a short verification once the ideal condition is written out.

1. *Kernel is an ideal.* The kernel is a linear subspace, being the null space of the linear map φ. For the ideal condition, let $X \in \mathfrak{g}$ and $Y \in \ker\varphi$. Then
$$\varphi([X,Y]) = [\varphi(X),\varphi(Y)] = [\varphi(X),0] = 0$$
using that φ preserves brackets and $\varphi(Y) = 0$, so $[X,Y] \in \ker\varphi$. Hence $[\mathfrak{g},\ker\varphi] \subseteq \ker\varphi$, and $\ker\varphi$ is an ideal.
2. *Image is a subalgebra.* The image is a linear subspace, being the range of a linear map. If $\varphi(X),\varphi(Y)$ are two elements of $\varphi(\mathfrak{g})$, then their bracket $[\varphi(X),\varphi(Y)] = \varphi([X,Y])$ is again in the image, so $\varphi(\mathfrak{g})$ is closed under the bracket and is a subalgebra. It need not be an ideal of $\mathfrak{h}$: the inclusion of a subalgebra that is not an ideal, such as the diagonal matrices inside $\mathfrak{gl}_2$, is a homomorphism whose image is not an ideal.
3. *First isomorphism theorem.* By part (1) the quotient $\mathfrak{g}/\ker\varphi$ is a Lie algebra. The linear map $\overline{\varphi}(X + \ker\varphi) = \varphi(X)$ is well defined, since $X - X' \in \ker\varphi$ gives $\varphi(X) = \varphi(X')$, and it is a linear isomorphism onto $\varphi(\mathfrak{g})$ by the linear-algebra first isomorphism theorem of [2, Isomorphism Theorems For Modules]. It preserves the bracket:
$$\overline{\varphi}\big([X+\ker\varphi,\ Y+\ker\varphi]\big) = \overline{\varphi}\big([X,Y]+\ker\varphi\big) = \varphi([X,Y]) = [\varphi(X),\varphi(Y)]$$
which equals $[\overline{\varphi}(X+\ker\varphi),\ \overline{\varphi}(Y+\ker\varphi)]$. A bracket-preserving linear isomorphism is an isomorphism of Lie algebras, as we sought to show.

The proposition says that ideals and kernels are the same objects seen from two sides: every kernel is an ideal by part (1), and conversely every ideal $\mathfrak{a}$ is the kernel of its own projection $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$. So the ideals of $\mathfrak{g}$ are precisely the substructures one can quotient by, and each quotient realizes $\mathfrak{g}$ as an extension of $\mathfrak{g}/\mathfrak{a}$ by $\mathfrak{a}$. The contrast with the subalgebra of definition 3.1.2 is the whole point of introducing the two notions separately: a subalgebra sits inside $\mathfrak{g}$ and is a Lie algebra in its own right, but one cannot in general form $\mathfrak{g}$ modulo a subalgebra, because the coset bracket is ill defined; an ideal is a subalgebra strong enough to divide by. The diagonal matrices $\mathfrak{d} \subseteq \mathfrak{gl}_2$ mark the gap concretely: $\mathfrak{d}$ is a subalgebra of $\mathfrak{gl}_2$, being abelian and closed under the bracket, but it is not an ideal,

since $[E_{11}, E_{12}] = E_{12} \notin \mathfrak{d}$ places the bracket of a diagonal matrix with an off-diagonal one outside $\mathfrak{d}$. This same $\mathfrak{d}$ reappears inside $\mathfrak{b}_2$ in the computation of example 5.1.7 below as a subalgebra that is not an ideal of the solvable algebra $\mathfrak{b}_2$.

Ideals come readily from the bracket itself. The most important is built by bracketing $\mathfrak{g}$ with itself and then iterating, and it measures how far $\mathfrak{g}$ is from being abelian.

Definition 5.1.3: The Derived Algebra and the Derived Series

Let $\mathfrak{g}$ be a Lie algebra. The *derived algebra* of $\mathfrak{g}$ is the linear span

$$[\mathfrak{g}, \mathfrak{g}] = \operatorname{span}_k \{[X, Y] \mid X, Y \in \mathfrak{g}\}$$

The *derived series* of $\mathfrak{g}$ is the descending chain of subalgebras defined by

$$\mathfrak{g}^{(0)} = \mathfrak{g}, \qquad \mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$$

so that $\mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}]$ is the derived algebra, $\mathfrak{g}^{(2)} = [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]]$, and so on. The chain

$$\mathfrak{g} = \mathfrak{g}^{(0)} \supseteq \mathfrak{g}^{(1)} \supseteq \mathfrak{g}^{(2)} \supseteq \cdots$$

is decreasing, and each $\mathfrak{g}^{(k)}$ is an ideal of $\mathfrak{g}$.

Two claims in the definition need verification: that each $\mathfrak{g}^{(k)}$ is an ideal of $\mathfrak{g}$ (not only of $\mathfrak{g}^{(k-1)}$), and that the chain descends. Both follow from a single fact about how the bracket construction interacts with symmetries of $\mathfrak{g}$.

Lemma 5.1.4: The Derived Series Consists of Characteristic Ideals

For every Lie algebra $\mathfrak{g}$ and every $k \geq 0$, the subspace $\mathfrak{g}^{(k)}$ is an ideal of $\mathfrak{g}$, and it is moreover *characteristic*: every derivation $D \colon \mathfrak{g} \to \mathfrak{g}$ satisfies $D(\mathfrak{g}^{(k)}) \subseteq \mathfrak{g}^{(k)}$. In particular $\mathfrak{g}^{(k)}$ is invariant under $\operatorname{ad}X$ for every $X \in \mathfrak{g}$, and the chain descends, $\mathfrak{g}^{(k+1)} \subseteq \mathfrak{g}^{(k)}$.

***Proof*:**

Recall that a *derivation* of $\mathfrak{g}$ is a linear map $D \colon \mathfrak{g} \to \mathfrak{g}$ satisfying the Leibniz rule $D[X, Y] = [DX, Y] + [X, DY]$; for each $X \in \mathfrak{g}$ the operator $\operatorname{ad}X$ of definition 2.5.3 is a derivation, which is precisely the Jacobi identity of definition 1.4.5 read as $\operatorname{ad}X[Y, Z] = [\operatorname{ad}X\, Y, Z] + [Y, \operatorname{ad}X\, Z]$. The proof is by induction on k, the characteristic property being the strengthening that makes the induction close.

Base case. For $k = 0$, $\mathfrak{g}^{(0)} = \mathfrak{g}$ is carried into itself by any linear map $\mathfrak{g} \to \mathfrak{g}$, so it is characteristic.

Inductive step. Suppose $\mathfrak{g}^{(k)}$ is characteristic, and let D be any derivation. A general element of $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$ is a sum of brackets $[U, V]$ with $U, V \in \mathfrak{g}^{(k)}$, so by linearity of D it suffices to treat one such bracket. The Leibniz rule gives

$$D[U, V] = [DU, V] + [U, DV]$$

By the inductive hypothesis $DU, DV \in \mathfrak{g}^{(k)}$, so both $[DU, V]$ and $[U, DV]$ lie in $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] = \mathfrak{g}^{(k+1)}$. Hence $D(\mathfrak{g}^{(k+1)}) \subseteq \mathfrak{g}^{(k+1)}$, and $\mathfrak{g}^{(k+1)}$ is characteristic.

Specializing $D = \operatorname{ad}X$ shows $[X, W] \in \mathfrak{g}^{(k)}$ for every $X \in \mathfrak{g}$ and $W \in \mathfrak{g}^{(k)}$, which is the statement

$[\mathfrak{g}, \mathfrak{g}^{(k)}] \subseteq \mathfrak{g}^{(k)}$, so each $\mathfrak{g}^{(k)}$ is an ideal of $\mathfrak{g}$. The chain descends because $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] \subseteq \mathfrak{g}^{(k)}$, the bracket of the subalgebra $\mathfrak{g}^{(k)}$ with itself landing inside $\mathfrak{g}^{(k)}$, completing the proof.

The lemma isolates the reason the derived series is a good invariant: it is built canonically from the bracket, so it is preserved not only by inner automorphisms but by every derivation, and in particular by every automorphism of $\mathfrak{g}$. An invariant this rigid cannot be moved by a change of basis or by any symmetry of the algebra, which is what allows the next definition to read off a genuine property of $\mathfrak{g}$ from where the series stops.

The derived algebra $[\mathfrak{g}, \mathfrak{g}]$ vanishes exactly when $\mathfrak{g}$ is abelian, so the first step of the derived series measures the failure of commutativity. Iterating asks whether that failure itself commutes, and so on down. If the process reaches zero, the algebra has been resolved into abelian layers.

Definition 5.1.5: Solvable Lie Algebra

A Lie algebra $\mathfrak{g}$ is *solvable* if its derived series reaches zero: there is an integer $n \geq 0$ with

$$\mathfrak{g}^{(n)} = 0$$

The least such n is the *derived length* of $\mathfrak{g}$.

An abelian Lie algebra has $\mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}] = 0$ and so is solvable of derived length at most one; conversely a solvable algebra of derived length at most one is abelian. Solvability is thus the property of being built from abelian pieces by iterated extension. Reading the derived series from the bottom, $\mathfrak{g}^{(n)} = 0$ says $\mathfrak{g}^{(n-1)}$ is abelian; each quotient $\mathfrak{g}^{(k)}/\mathfrak{g}^{(k+1)}$ is abelian by construction, since $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] = \mathfrak{g}^{(k+1)}$ is exactly what is divided out; and $\mathfrak{g}$ is assembled from the abelian quotients $\mathfrak{g}^{(0)}/\mathfrak{g}^{(1)}$, $\mathfrak{g}^{(1)}/\mathfrak{g}^{(2)}, \ldots, \mathfrak{g}^{(n-1)}/\mathfrak{g}^{(n)}$ by a tower of extensions. This is the sense in which solvable algebras are the ones with no rigid, non-abelian core, and it is the exact linear counterpart of a solvable group, whose derived series of iterated commutator subgroups terminates. The name is inherited from Galois theory through that group-theoretic parallel, where solvability of a group governs solvability of a polynomial by radicals.

The closure properties of the solvable class are what make it usable. They say solvability is preserved by the two operations that build and dismantle algebras, passing to subalgebras and quotients, and that it is stable under extension and under summing ideals. The last of these is the fact that section 5.5 will use to produce a unique maximal solvable ideal, the radical, whose vanishing defines semisimplicity.

Proposition 5.1.6: Closure Properties of Solvable Lie Algebras

Let $\mathfrak{g}$ be a Lie algebra.

1. Every subalgebra and every quotient of a solvable Lie algebra is solvable.
2. If $\mathfrak{a} \trianglelefteq \mathfrak{g}$ is an ideal such that both $\mathfrak{a}$ and $\mathfrak{g}/\mathfrak{a}$ are solvable, then $\mathfrak{g}$ is solvable.
3. If $\mathfrak{a}, \mathfrak{b} \trianglelefteq \mathfrak{g}$ are solvable ideals, then their sum $\mathfrak{a} + \mathfrak{b}$ is a solvable ideal.

***Proof*:**
The proofs turn on one observation about how the derived series behaves under homomorphisms. If $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ is a homomorphism, then

$$\varphi(\mathfrak{g}^{(k)}) = \varphi(\mathfrak{g})^{(k)} \qquad \text{for every } k \geq 0 \tag{5.1}$$

because φ preserves brackets: $\varphi([\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]) = [\varphi(\mathfrak{g}^{(k)}), \varphi(\mathfrak{g}^{(k)})]$, and induction on k propagates the equality up the series from the base case $\varphi(\mathfrak{g}^{(0)}) = \varphi(\mathfrak{g})$. Two consequences are used repeatedly. For a subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, applying the bracket inside $\mathfrak{h}$ gives $\mathfrak{h}^{(k)} \subseteq \mathfrak{g}^{(k)}$. For the quotient projection $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$, equation (5.1) reads $(\mathfrak{g}/\mathfrak{a})^{(k)} = \pi(\mathfrak{g}^{(k)})$.

1. *Subalgebras and quotients.* Let $\mathfrak{g}$ be solvable with $\mathfrak{g}^{(n)} = 0$. If $\mathfrak{h} \subseteq \mathfrak{g}$ is a subalgebra, then $\mathfrak{h}^{(n)} \subseteq \mathfrak{g}^{(n)} = 0$, so $\mathfrak{h}$ is solvable. If $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$ is the quotient by an ideal, then $(\mathfrak{g}/\mathfrak{a})^{(n)} = \pi(\mathfrak{g}^{(n)}) = \pi(0) = 0$, so the quotient is solvable.

2. *Extensions.* Suppose $\mathfrak{a}$ and $\mathfrak{g}/\mathfrak{a}$ are solvable, say $(\mathfrak{g}/\mathfrak{a})^{(m)} = 0$ and $\mathfrak{a}^{(p)} = 0$. Let $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$ be the projection. From $(\mathfrak{g}/\mathfrak{a})^{(m)} = \pi(\mathfrak{g}^{(m)}) = 0$ one gets $\mathfrak{g}^{(m)} \subseteq \ker \pi = \mathfrak{a}$. The derived series composes with itself, in the sense that $(\mathfrak{g}^{(m)})^{(p)} = \mathfrak{g}^{(m+p)}$, since applying the derived operation p times to $\mathfrak{g}^{(m)}$ continues the same recursion. Because $\mathfrak{g}^{(m)} \subseteq \mathfrak{a}$ is a subalgebra, the subalgebra case of the opening observation gives $(\mathfrak{g}^{(m)})^{(p)} \subseteq \mathfrak{a}^{(p)}$, and therefore

$$\mathfrak{g}^{(m+p)} = (\mathfrak{g}^{(m)})^{(p)} \subseteq \mathfrak{a}^{(p)} = 0$$

   Hence $\mathfrak{g}^{(m+p)} = 0$ and $\mathfrak{g}$ is solvable.

3. *Sum of solvable ideals.* That $\mathfrak{a} + \mathfrak{b}$ is an ideal is immediate: it is a subspace, and $[\mathfrak{g}, \mathfrak{a} + \mathfrak{b}] \subseteq [\mathfrak{g}, \mathfrak{a}] + [\mathfrak{g}, \mathfrak{b}] \subseteq \mathfrak{a} + \mathfrak{b}$. For solvability, apply the second isomorphism theorem, valid for Lie algebras by the same coset argument as for modules ([2, Isomorphism Theorems For Modules]): since $\mathfrak{a}$ and $\mathfrak{b}$ are ideals, $\mathfrak{b} \trianglelefteq \mathfrak{a} + \mathfrak{b}$ and

$$(\mathfrak{a} + \mathfrak{b})/\mathfrak{b} \;\cong\; \mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$$

   as Lie algebras. The right-hand side is a quotient of the solvable algebra $\mathfrak{a}$, hence solvable by part (1). Thus $(\mathfrak{a} + \mathfrak{b})/\mathfrak{b}$ is solvable, and $\mathfrak{b}$ is solvable by hypothesis, so $\mathfrak{a} + \mathfrak{b}$ is solvable by part (2), completing the proof.

The three properties fit together into a single structural picture. Parts (1) and (2) say the solvable algebras form a class closed under sub-objects, quotients, and extensions, which is precisely the closure a class must have to support a decomposition theory. Part (3) is the one that does not follow formally from the others, and it carries the most weight downstream: among all ideals of $\mathfrak{g}$, the solvable ones are closed under addition, so their sum is again solvable. Since $\mathfrak{g}$ is finite-dimensional this sum stabilizes, and section 5.5 takes the sum of *all* solvable ideals to define the radical $\mathrm{rad}\mathfrak{g}$, the unique maximal solvable ideal, well defined precisely because of part (3). Semisimplicity is then the vanishing of this radical, so the entire dichotomy that organizes the classification, solvable versus semisimple, rests on the closure statement just proved.

The definitions have so far been abstract; the class of examples that shows them, and that recurs through the whole chapter, is the upper-triangular matrices. This is where solvability is not just a formal property but a visible one, the derived series descending step by step through the strictly-upper-triangular layers until it reaches zero.

Example 5.1.7: The Upper Triangular Algebra $\mathfrak{b}_n$ is Solvable

Let $\mathfrak{b}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$ be the subalgebra of upper-triangular matrices, those with $X_{ij} = 0$ for $i > j$, and let $\mathfrak{n}_n \subseteq \mathfrak{b}_n$ be the *strictly* upper-triangular matrices, with $X_{ij} = 0$ for $i \geq j$. Both are subalgebras of $\mathfrak{gl}_n$: the product of two upper-triangular matrices is upper-triangular, so the commutator is too, and likewise for strictly upper-triangular matrices. Write E_{ij} for the matrix unit with a single 1 in position (i, j); the bracket of matrix units is

$$[E_{ij}, E_{k\ell}] = \delta_{jk} E_{i\ell} - \delta_{\ell i} E_{kj} \tag{5.2}$$

The claim is that the derived algebra of $\mathfrak{b}_n$ is exactly $\mathfrak{n}_n$,

$$[\mathfrak{b}_n, \mathfrak{b}_n] = \mathfrak{n}_n$$

One inclusion is a diagonal count. Write $X = D_X + N_X$ with D_X diagonal and $N_X \in \mathfrak{n}_n$; the diagonal entries of a product of upper-triangular matrices are the products of the diagonal entries, so the diagonal part of XY is $D_X D_Y$, and diagonal matrices commute. Hence the diagonal part of $[X, Y]$ is $D_X D_Y - D_Y D_X = 0$, so every bracket $[X, Y]$ has zero diagonal, giving $[\mathfrak{b}_n, \mathfrak{b}_n] \subseteq \mathfrak{n}_n$. For the reverse inclusion, produce each generator E_{ij} with $i < j$ as a bracket. Using (5.2) with the diagonal unit $E_{ii} \in \mathfrak{b}_n$,

$$[E_{ii}, E_{ij}] = \delta_{ii} E_{ij} - \delta_{ji} E_{ii} = E_{ij} \qquad (i < j)$$

since $\delta_{ii} = 1$ and $\delta_{ji} = 0$ for $i \neq j$, and $E_{ij} \in \mathfrak{b}_n$. Thus every strictly-upper-triangular generator is a bracket of two elements of $\mathfrak{b}_n$, so $\mathfrak{n}_n \subseteq [\mathfrak{b}_n, \mathfrak{b}_n]$, and the two inclusions give $[\mathfrak{b}_n, \mathfrak{b}_n] = \mathfrak{n}_n$.

The derived series now descends through the strictly-upper-triangular layers. Bracketing raises the superdiagonal on which the nonzero entries sit: write $\mathfrak{n}_n^{[p]}$ for the matrices supported on the pth superdiagonal and beyond (entries X_{ij} with $j - i \geq p$), so $\mathfrak{n}_n = \mathfrak{n}_n^{[1]}$. If $X \in \mathfrak{n}_n^{[p]}$ and $Y \in \mathfrak{n}_n^{[q]}$, then (5.2) applied entrywise shows every nonzero entry of $[X, Y]$ has row-to-column gap at least $p + q$, so $[\mathfrak{n}_n^{[p]}, \mathfrak{n}_n^{[q]}] \subseteq \mathfrak{n}_n^{[p+q]}$. Consequently the derived series of $\mathfrak{b}_n$ satisfies

$$\mathfrak{b}_n^{(0)} = \mathfrak{b}_n, \quad \mathfrak{b}_n^{(1)} = \mathfrak{n}_n = \mathfrak{n}_n^{[1]}, \quad \mathfrak{b}_n^{(2)} \subseteq \mathfrak{n}_n^{[2]}, \quad \mathfrak{b}_n^{(k)} \subseteq \mathfrak{n}_n^{[2^{k-1}]} \quad (k \geq 1)$$

the last inclusion by induction, since $\mathfrak{b}_n^{(k+1)} = [\mathfrak{b}_n^{(k)}, \mathfrak{b}_n^{(k)}] \subseteq [\mathfrak{n}_n^{[2^{k-1}]}, \mathfrak{n}_n^{[2^{k-1}]}] \subseteq \mathfrak{n}_n^{[2^k]}$. Once $2^{k-1} \geq n$ the superdiagonal index exceeds the matrix size and $\mathfrak{n}_n^{[2^{k-1}]} = 0$, so $\mathfrak{b}_n^{(k)} = 0$ for k large. Therefore $\mathfrak{b}_n$ is solvable. For $n = 2$ the computation is fully explicit: $\mathfrak{b}_2$ is spanned by E_{11}, E_{22}, E_{12}, its derived algebra is $\mathfrak{n}_2 = \mathbb{C}E_{12}$, which is one-dimensional and abelian, so $\mathfrak{b}_2^{(2)} = 0$ and $\mathfrak{b}_2$ has derived length 2.

The example is the concrete shape of solvability. The upper-triangular algebra $\mathfrak{b}_n$ is not abelian, yet its derived series collapses to zero, and the descent is legible: each application of the bracket pushes the nonzero entries a full block up toward the corner, and the finite size of the matrix forces termination. Two features of this example are worth recording. The subalgebra $\mathfrak{b}_n$ is the standard model of a *Borel subalgebra*, the maximal solvable subalgebra of $\mathfrak{gl}_n$; and its diagonal subalgebra $\mathfrak{d}$, the diagonal matrices, is a subalgebra of $\mathfrak{b}_n$ that is not an ideal of $\mathfrak{b}_n$, since $[E_{11}, E_{12}] = E_{12}$ lands outside $\mathfrak{d}$. This is the concrete instance of the subalgebra-not-ideal gap promised after proposition 5.1.2.

The other feature is a warning that separates this chapter's two central notions. Although $\mathfrak{b}_n$ is solvable, it is *not* nilpotent in the sense of section 5.2: its lower central series does not reach zero,

because bracketing the off-diagonal entries against the diagonal part of $\mathfrak{b}_n$ reproduces those same off-diagonal entries indefinitely rather than shrinking them, as the computation $[E_{ii}, E_{ij}] = E_{ij}$ above shows. The strictly upper-triangular algebra $\mathfrak{n}_n$, by contrast, is nilpotent, as example 5.2.5 will verify. Solvability and nilpotency are therefore distinct properties, and the pair $\mathfrak{n}_n \subseteq \mathfrak{b}_n$ is the minimal example exhibiting the difference: nilpotency will imply solvability in section 5.2, but the diagonal directions of $\mathfrak{b}_n$ obstruct the reverse implication.

Solvability is the first of the two structural properties Part II uses to carve out the semisimple algebras. In the language that section 5.5 will make precise, a Lie algebra is *semisimple* when it has no nonzero solvable ideal, so semisimplicity is the exact negation of harboring a solvable part. In this sense the upper-triangular algebra $\mathfrak{b}_n$ sits at the opposite pole from the semisimple algebras: it is solvable, hence equal to its own radical, hence as far from semisimple as an algebra can be. The whole of Part II studies how a general Lie algebra is glued from a solvable piece and a semisimple piece, a splitting (the Levi decomposition) that the Cartan criteria of chapter 6 make available; the equivalence of the radical-zero definition of semisimplicity with the nondegeneracy of the Killing form is proved there as well. Everything downstream begins with the solvable end of the spectrum, which is why it comes first.

Exercise 5.1.1

1. Prove that a Lie algebra $\mathfrak{g}$ is solvable if and only if there is a chain of subalgebras

$$\mathfrak{g} = \mathfrak{g}_0 \supseteq \mathfrak{g}_1 \supseteq \cdots \supseteq \mathfrak{g}_m = 0$$

   in which each $\mathfrak{g}_{i+1}$ is an ideal of $\mathfrak{g}_i$ and each quotient $\mathfrak{g}_i/\mathfrak{g}_{i+1}$ is abelian. (This characterizes solvability by an abelian tower; note that the $\mathfrak{g}_i$ need only be ideals in the preceding term, not in $\mathfrak{g}$.)

2. Let $\mathfrak{g}$ be the two-dimensional Lie algebra over $\mathbb{C}$ with basis X, Y and bracket $[X, Y] = X$, the remaining brackets fixed by antisymmetry. Verify that this is a Lie algebra, compute its derived series, and conclude that it is solvable but not abelian. Realize $\mathfrak{g}$ as a subalgebra of $\mathfrak{b}_2$.

3. Show that the quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian, and that it is the *maximal* abelian quotient of $\mathfrak{g}$: for any ideal $\mathfrak{a} \trianglelefteq \mathfrak{g}$ with $\mathfrak{g}/\mathfrak{a}$ abelian, one has $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{a}$, so $\mathfrak{g}/\mathfrak{a}$ is a quotient of $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$. (This quotient is the *abelianization* of $\mathfrak{g}$.)

4. Let $\mathfrak{a} \trianglelefteq \mathfrak{g}$ be an ideal. Prove that $\mathfrak{g}$ is solvable if and only if both $\mathfrak{a}$ and $\mathfrak{g}/\mathfrak{a}$ are solvable. (One direction is proposition 5.1.6; give the other and assemble the equivalence.)

5. Give an example of two solvable subalgebras $\mathfrak{h}_1, \mathfrak{h}_2$ of a Lie algebra $\mathfrak{g}$ whose sum $\mathfrak{h}_1 + \mathfrak{h}_2$ is not solvable, and explain why this does not contradict part (3) of proposition 5.1.6. (Hint: inside $\mathfrak{sl}_2(\mathbb{C})$, take the upper-triangular and lower-triangular traceless matrices.)

6. Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. Using proposition 5.1.6(3) and finite-dimensionality, prove that $\mathfrak{g}$ has a unique maximal solvable ideal, that is, a solvable ideal containing every solvable ideal of $\mathfrak{g}$. (This is the radical of section 5.5; the point here is that finite-dimensionality plus closure under sums forces existence.)

5.2 The Lower Central Series and Nilpotency

The derived series of section 5.1 measures how far a Lie algebra stands from being abelian by iterating the commutator on the whole of the previous term: at each stage one brackets $\mathfrak{g}^{(k)}$ with itself. There is a second, finer descending chain that brackets against all of $\mathfrak{g}$ at every stage, refusing to let the terms shrink so quickly. This is the lower central series, and the algebras it drives to zero are the *nilpotent* ones. Nilpotency is a strictly stronger condition than solvability, and the gap between the two chains is exactly the gap between the two structure theorems that organize this chapter: solvability is the hypothesis of Lie's theorem, nilpotency the hypothesis of Engel's. The nilpotent algebras are the closest a nonabelian Lie algebra comes to being abelian, built by iterated *central* extensions rather than abelian ones, and they are the algebras on which the exponential map behaves best, as the Heisenberg computation of example 3.5.5 already hinted.

The definition below brackets against $\mathfrak{g}$ at every stage. Where the derived series replaces $\mathfrak{g}^{(k)}$ by $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$, keeping both slots inside the shrinking term, the lower central series keeps one slot pinned to the ambient algebra. This single change is what separates the two notions.

Definition 5.2.1: Lower Central Series

Let $\mathfrak{g}$ be a Lie algebra. Its *lower central series* is the descending chain of subspaces $\mathfrak{g}^1 \supseteq \mathfrak{g}^2 \supseteq \cdots$ defined recursively by

$$\mathfrak{g}^1 = \mathfrak{g}, \qquad \mathfrak{g}^{k+1} = [\mathfrak{g}, \mathfrak{g}^k]$$

where $[\mathfrak{g}, \mathfrak{g}^k]$ denotes the subspace spanned by all brackets $[X, Y]$ with $X \in \mathfrak{g}$ and $Y \in \mathfrak{g}^k$. The second term $\mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}]$ is the derived algebra of definition 5.1.3.

Each bracket in the recursion pins its first slot to the whole of $\mathfrak{g}$, so the lower central series never brackets a small term against itself; it always drags all of $\mathfrak{g}$ across. Two facts about the terms need recording before the definition of nilpotency can be stated cleanly: every $\mathfrak{g}^k$ is an ideal, and the series descends at least as slowly as the derived series.

Proposition 5.2.2: The Terms of the Lower Central Series

Let $\mathfrak{g}$ be a Lie algebra with lower central series $\mathfrak{g}^k$ and derived series $\mathfrak{g}^{(k)}$.

1. Each $\mathfrak{g}^k$ is an ideal of $\mathfrak{g}$, and $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ for all $i, j \geq 1$.
2. $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$ for every $k \geq 0$; in particular the derived series descends at least as fast as the lower central series.

***Proof*:**

1. *Each term is an ideal, and the bracket inclusion.* That $\mathfrak{g}^1 = \mathfrak{g}$ is an ideal is immediate. Proceeding by induction, suppose $\mathfrak{g}^k$ is an ideal. Then $\mathfrak{g}^{k+1} = [\mathfrak{g}, \mathfrak{g}^k]$ is a subspace, and for $X \in \mathfrak{g}$ and a spanning bracket $[Y, Z] \in \mathfrak{g}^{k+1}$ with $Y \in \mathfrak{g}$, $Z \in \mathfrak{g}^k$, the Jacobi identity gives

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]]$$

The first term lies in $[\mathfrak{g}, \mathfrak{g}^k] = \mathfrak{g}^{k+1}$ because $[X, Y] \in \mathfrak{g}$ and $Z \in \mathfrak{g}^k$; the second lies in $[\mathfrak{g}, \mathfrak{g}^k] = \mathfrak{g}^{k+1}$ because $Y \in \mathfrak{g}$ and $[X, Z] \in \mathfrak{g}^k$, the latter since $\mathfrak{g}^k$ is an ideal. Thus

$[X,[Y,Z]] \in \mathfrak{g}^{k+1}$; as such brackets span $\mathfrak{g}^{k+1}$ and X was arbitrary, $[\mathfrak{g},\mathfrak{g}^{k+1}] \subseteq \mathfrak{g}^{k+1}$, so $\mathfrak{g}^{k+1}$ is an ideal.

For the inclusion $[\mathfrak{g}^i,\mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$, argue by induction on j, the statement being taken to hold for all $i \geq 1$ simultaneously at each fixed j. The base case $j = 1$ reads $[\mathfrak{g}^i,\mathfrak{g}] \subseteq \mathfrak{g}^{i+1}$ for all i, which holds because $\mathfrak{g}^i$ is an ideal, so $[\mathfrak{g}^i,\mathfrak{g}] = [\mathfrak{g},\mathfrak{g}^i] = \mathfrak{g}^{i+1}$. Assume the statement for j, again for all i. A spanning generator of $\mathfrak{g}^{j+1} = [\mathfrak{g},\mathfrak{g}^j]$ has the form $[U,V]$ with $U \in \mathfrak{g}$, $V \in \mathfrak{g}^j$, and for $W \in \mathfrak{g}^i$ the Jacobi identity gives

$$[W,[U,V]] = [[W,U],V] + [U,[W,V]]$$

In the first term, $[W,U] \in [\mathfrak{g}^i,\mathfrak{g}] \subseteq \mathfrak{g}^{i+1}$ by the base case, and $V \in \mathfrak{g}^j$, so the inductive hypothesis at index pair $(i+1,j)$ gives $[[W,U],V] \in [\mathfrak{g}^{i+1},\mathfrak{g}^j] \subseteq \mathfrak{g}^{i+1+j}$. In the second term, $[W,V] \in [\mathfrak{g}^i,\mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ by the inductive hypothesis at (i,j), so $[U,[W,V]] \in [\mathfrak{g},\mathfrak{g}^{i+j}] = \mathfrak{g}^{i+j+1}$. Both terms lie in $\mathfrak{g}^{i+j+1}$, and since such generators span $[\mathfrak{g}^i,\mathfrak{g}^{j+1}]$, one concludes $[\mathfrak{g}^i,\mathfrak{g}^{j+1}] \subseteq \mathfrak{g}^{i+j+1}$ for all i, completing the induction.

2. *The derived series descends faster.* Argue by induction on k. For $k = 0$, $\mathfrak{g}^{(0)} = \mathfrak{g} = \mathfrak{g}^1$, so $\mathfrak{g}^{(0)} \subseteq \mathfrak{g}^1 = \mathfrak{g}^{0+1}$. Suppose $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$. Then

$$\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)},\mathfrak{g}^{(k)}] \subseteq [\mathfrak{g}^{k+1},\mathfrak{g}^{k+1}] \subseteq [\mathfrak{g},\mathfrak{g}^{k+1}] = \mathfrak{g}^{k+2}$$

the first inclusion from the inductive hypothesis (bracket is monotone in each slot), the second because $\mathfrak{g}^{k+1} \subseteq \mathfrak{g}$. Thus $\mathfrak{g}^{(k+1)} \subseteq \mathfrak{g}^{(k+1)+1}$, as we sought to show.

The bracket inclusion $[\mathfrak{g}^i,\mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ is the arithmetic fact the whole theory rests on: it says the lower central filtration is compatible with the bracket in the way a grading would be, the index adding under multiplication. It is exactly this additivity that will force iterated brackets to vanish once the filtration reaches zero, and it is why a nilpotent Lie algebra has all sufficiently long brackets equal to zero regardless of how the factors are associated. The containment $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$ records the coarser fact that a chain bracketing against all of $\mathfrak{g}$ cannot outrun the chain bracketing against the smaller term: reaching zero along the slower chain is the harder condition.

Definition 5.2.3: Nilpotent Lie Algebra

A Lie algebra $\mathfrak{g}$ is *nilpotent* if $\mathfrak{g}^k = 0$ for some $k \geq 1$; the least such $k-1$ is the *nilpotency class* of $\mathfrak{g}$. Equivalently, $\mathfrak{g}$ is nilpotent of class $\leq c$ if every iterated bracket of $c+1$ elements vanishes,

$$[X_1,[X_2,[\cdots,[X_c,X_{c+1}]\cdots]]] = 0 \qquad \text{for all } X_1,\ldots,X_{c+1} \in \mathfrak{g}$$

A nonzero abelian Lie algebra is nilpotent of class 1: there $\mathfrak{g}^1 = \mathfrak{g} \neq 0$ but $\mathfrak{g}^2 = [\mathfrak{g},\mathfrak{g}] = 0$, so the least k with $\mathfrak{g}^k = 0$ is $k = 2$ and the class is $k-1 = 1$ (class 0 is reserved for the zero algebra). A nilpotent algebra of class 2 has $\mathfrak{g}^2 \neq 0$ but $\mathfrak{g}^3 = [\mathfrak{g},\mathfrak{g}^2] = 0$, meaning $[\mathfrak{g},\mathfrak{g}]$ is central; the Heisenberg algebra of example 3.5.5 is the smallest such, and its lower central series is computed below. The equivalence between $\mathfrak{g}^{c+1} = 0$ and the vanishing of every length-$(c+1)$ iterated bracket is a direct unwinding of the recursion $\mathfrak{g}^{k+1} = [\mathfrak{g},\mathfrak{g}^k]$: the term $\mathfrak{g}^k$ is spanned by left-normed brackets $[X_1,[X_2,[\cdots,X_k]\cdots]]$ of length k, so $\mathfrak{g}^{c+1} = 0$ says exactly that all such brackets of length $c+1$ vanish, and the inclusion $[\mathfrak{g}^i,\mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ of proposition 5.2.2 promotes this to arbitrarily-associated brackets.

Comparing the two chains term by term settles the relationship between the two structural notions.

Since $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$, once the lower central series reaches zero so does the derived series, one step later at worst. Nilpotency therefore implies solvability. The converse fails, and the failure is not exotic: the upper-triangular matrices, solvable by example 5.1.7, are not nilpotent, a point the next proposition proves and the running matrix example shows.

Proposition 5.2.4: Basic Properties of Nilpotent Lie Algebras

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra.

1. If $\mathfrak{g}$ is nilpotent then it is solvable. The converse fails: $\mathfrak{b}_n$ (example 5.1.7) is solvable but not nilpotent for $n \geq 2$.
2. Every subalgebra and every quotient of a nilpotent Lie algebra is nilpotent.
3. If $\mathfrak{g} \neq 0$ is nilpotent, then its center $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \text{ for all } Y \in \mathfrak{g}\}$ is nonzero.

***Proof*:**

1. *Nilpotent implies solvable.* If $\mathfrak{g}^k = 0$ then by proposition 5.2.2(2), $\mathfrak{g}^{(k-1)} \subseteq \mathfrak{g}^k = 0$, so $\mathfrak{g}^{(k-1)} = 0$ and $\mathfrak{g}$ is solvable in the sense of definition 5.1.5. That the converse fails is exhibited by $\mathfrak{b}_n$; the computation is example 5.2.5 below, whose lower central series stabilizes at the nonzero ideal $\mathfrak{n}_n$ and never reaches zero.
2. *Subalgebras and quotients.* Let $\mathfrak{h} \subseteq \mathfrak{g}$ be a subalgebra. A straightforward induction shows $\mathfrak{h}^k \subseteq \mathfrak{g}^k$: the base case is $\mathfrak{h}^1 = \mathfrak{h} \subseteq \mathfrak{g} = \mathfrak{g}^1$, and if $\mathfrak{h}^k \subseteq \mathfrak{g}^k$ then $\mathfrak{h}^{k+1} = [\mathfrak{h}, \mathfrak{h}^k] \subseteq [\mathfrak{g}, \mathfrak{g}^k] = \mathfrak{g}^{k+1}$. Hence if $\mathfrak{g}^k = 0$ then $\mathfrak{h}^k = 0$, and $\mathfrak{h}$ is nilpotent.

   For a quotient, let $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$ be the projection onto the quotient by an ideal $\mathfrak{a}$ (definition 5.1.1). Since π is a surjective homomorphism, $\pi([\mathfrak{g}, \mathfrak{g}^k]) = [\pi(\mathfrak{g}), \pi(\mathfrak{g}^k)]$, and an induction gives $(\mathfrak{g}/\mathfrak{a})^k = \pi(\mathfrak{g}^k)$: the base case is $(\mathfrak{g}/\mathfrak{a})^1 = \mathfrak{g}/\mathfrak{a} = \pi(\mathfrak{g})$, and the inductive step reads

$$(\mathfrak{g}/\mathfrak{a})^{k+1} = [\mathfrak{g}/\mathfrak{a}, (\mathfrak{g}/\mathfrak{a})^k] = [\pi(\mathfrak{g}), \pi(\mathfrak{g}^k)] = \pi([\mathfrak{g}, \mathfrak{g}^k]) = \pi(\mathfrak{g}^{k+1})$$

   Hence if $\mathfrak{g}^k = 0$ then $(\mathfrak{g}/\mathfrak{a})^k = \pi(0) = 0$, so the quotient is nilpotent.
3. *Nonzero center.* Let $\mathfrak{g} \neq 0$ be nilpotent, and let $k \geq 2$ be the least index with $\mathfrak{g}^k = 0$; such k exists because $\mathfrak{g}^1 = \mathfrak{g} \neq 0$, and $k \geq 2$ since $\mathfrak{g}^1 \neq 0$. Then $\mathfrak{g}^{k-1} \neq 0$ by minimality, while

$$[\mathfrak{g}, \mathfrak{g}^{k-1}] = \mathfrak{g}^k = 0$$

   which says precisely that every element of $\mathfrak{g}^{k-1}$ brackets to zero against all of $\mathfrak{g}$, that is $\mathfrak{g}^{k-1} \subseteq \mathfrak{z}(\mathfrak{g})$. Since $\mathfrak{g}^{k-1} \neq 0$, the center is nonzero, completing the proof.

The third statement is the structural heart of nilpotency. The last nonzero term of the lower central series is central, so a nilpotent Lie algebra always has a nontrivial center to quotient by; the quotient is again nilpotent by part (2), of one lower class, and one may recurse. This is the precise sense in which a nilpotent Lie algebra is built by *iterated central extensions*: repeatedly quotienting by a central ideal reduces $\mathfrak{g}$ to 0. One such chain of central ideals is the lower central series itself,

$$\mathfrak{g} = \mathfrak{g}_0 \supseteq \mathfrak{g}_1 \supseteq \cdots \supseteq \mathfrak{g}_r = 0, \qquad \mathfrak{g}_i = \mathfrak{g}^{i+1}$$

whose terms are the lower-central-series ideals rather than the successively quotiented centers, yet which is still a central chain: each successive quotient $\mathfrak{g}_i/\mathfrak{g}_{i+1}$ is central in $\mathfrak{g}/\mathfrak{g}_{i+1}$, since $[\mathfrak{g}, \mathfrak{g}^{i+1}] \subseteq \mathfrak{g}^{i+2}$. Solvability, by contrast, only guarantees the successive quotients are *abelian*, not central. The distinction, central versus abelian, is exactly the extra strength nilpotency carries over solvability, and it is why Engel's theorem in section 5.3 can force a full flag of ideals where Lie's theorem in section 5.4 produces only a flag of subspaces. The nonvanishing of the center also rules out $\mathfrak{sl}_2$ from ever being nilpotent, since $\mathfrak{sl}_2$ has trivial center; nilpotency is a condition satisfied only by algebras far from the semisimple ones.

The canonical example of a nilpotent Lie algebra, and the one that fixes the intuition for the whole subject, is the algebra of strictly upper-triangular matrices. Its lower central series can be computed by hand, and each bracket literally pushes the nonzero entries one step further above the diagonal until they run off the top of the matrix. Set against the upper-triangular algebra $\mathfrak{b}_n$ of example 5.1.7, the computation displays the difference between solvable and nilpotent in coordinates.

Example 5.2.5: The Strictly Upper-Triangular Algebra $\mathfrak{n}_n$

Let $\mathfrak{n}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$ be the subalgebra of strictly upper-triangular matrices, those with zeros on and below the main diagonal. It is a subalgebra: a product of two strictly upper-triangular matrices is again strictly upper-triangular, so the commutator is too, and $\dim \mathfrak{n}_n = \binom{n}{2}$. For $0 \le d \le n-1$ write

$$\mathfrak{n}_n(d) = \operatorname{span}\{E_{ij} \mid j - i \ge d\}$$

for the span of the matrix units E_{ij} lying at least d steps above the diagonal, so that $\mathfrak{n}_n(1) = \mathfrak{n}_n$, the chain $\mathfrak{n}_n(1) \supseteq \mathfrak{n}_n(2) \supseteq \cdots \supseteq \mathfrak{n}_n(n-1) \supseteq \mathfrak{n}_n(n) = 0$ descends by one super-diagonal at each step, and $\mathfrak{n}_n(n) = 0$ because no entry sits n or more steps above the diagonal in an $n \times n$ matrix.

The claim is that the lower central series is exactly this chain:

$$\mathfrak{n}_n^k = \mathfrak{n}_n(k) \qquad \text{for } 1 \le k \le n \tag{5.3}$$

The computation rests on the multiplication rule for matrix units, $E_{ij}E_{kl} = \delta_{jk}E_{il}$, which gives the bracket

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}$$

Bracketing a unit E_{ab} with $a < b$ (a generator of $\mathfrak{n}_n$) against a unit E_{cd} with $d - c \ge k$ (a generator of $\mathfrak{n}_n(k)$) produces $\delta_{bc}E_{ad} - \delta_{da}E_{cb}$. In the first term $b = c$ forces $d - a = (d-c) + (b-a) \ge k+1$, so $E_{ad} \in \mathfrak{n}_n(k+1)$; in the second $d = a$ forces $b - c = (b-a) + (d-c) \ge k+1$, so $E_{cb} \in \mathfrak{n}_n(k+1)$. Hence $[\mathfrak{n}_n, \mathfrak{n}_n(k)] \subseteq \mathfrak{n}_n(k+1)$. Conversely every generator E_{il} of $\mathfrak{n}_n(k+1)$, with $l - i \ge k+1$, is produced: choosing any j with $i < j$ and $l - j \ge k$ (possible since $l - i \ge k+1$), one has $[E_{ij}, E_{jl}] = E_{il}$ with $E_{ij} \in \mathfrak{n}_n$ and $E_{jl} \in \mathfrak{n}_n(k)$. Therefore $[\mathfrak{n}_n, \mathfrak{n}_n(k)] = \mathfrak{n}_n(k+1)$, and starting from $\mathfrak{n}_n^1 = \mathfrak{n}_n = \mathfrak{n}_n(1)$ the recursion $\mathfrak{n}_n^{k+1} = [\mathfrak{n}_n, \mathfrak{n}_n^k]$ yields (5.3) by induction.

In particular $\mathfrak{n}_n^n = \mathfrak{n}_n(n) = 0$, so $\mathfrak{n}_n$ is nilpotent of class $n-1$. For $n = 3$ the series reads

$$\mathfrak{n}_3^1 = \mathfrak{n}_3 \ (\dim 3), \qquad \mathfrak{n}_3^2 = \operatorname{span}\{E_{13}\} \ (\dim 1), \qquad \mathfrak{n}_3^3 = 0$$

each bracket raising the super-diagonal by one until the top corner E_{13} is all that survives, and then vanishing.

This is the same algebra as the Heisenberg algebra $\mathfrak{h}_3$ of example 3.5.5, under the identification $P = E_{12}$, $Q = E_{23}$, $C = E_{13}$: its only nontrivial bracket is $[P, Q] = C$, its center is $\operatorname{span}\{C\} = \mathfrak{n}_3^2$, and it is the smallest nonabelian nilpotent Lie algebra, of dimension 3 and class 2. The nonzero center predicted by proposition 5.2.4(3) is precisely the last nonzero term $\mathfrak{n}_3^2 = \operatorname{span}\{C\}$.

The contrast with $\mathfrak{b}_n$ is the point of the example. For the full upper-triangular algebra $\mathfrak{b}_n$ the derived series terminates (example 5.1.7), so $\mathfrak{b}_n$ is solvable, but its lower central series does not reach zero. Concretely $\mathfrak{b}_n^2 = [\mathfrak{b}_n, \mathfrak{b}_n] = \mathfrak{n}_n$, and thereafter the diagonal part of $\mathfrak{b}_n$ keeps the series from shrinking: writing $H = \operatorname{diag}(h_1, \dots, h_n)$ for a diagonal matrix, one computes $[H, E_{ij}] = (h_i - h_j)E_{ij}$, so for $i < j$ and a choice of H with $h_i \neq h_j$ the bracket recovers E_{ij}. Thus $[\mathfrak{b}_n, \mathfrak{n}_n] = \mathfrak{n}_n$, and the lower central series stabilizes,

$$\mathfrak{b}_n^1 = \mathfrak{b}_n, \qquad \mathfrak{b}_n^k = \mathfrak{n}_n \quad \text{for all } k \geq 2$$

never reaching zero when $n \geq 2$, since $\mathfrak{n}_n \neq 0$. For $n = 3$ the dimensions are $6, 3, 3, 3, \dots$, a chain that drops once and then freezes at $\mathfrak{n}_3$, whereas the derived series $6, 3, 1, 0$ runs to zero. The obstruction is exactly the diagonal torus: it is abelian, so the derived series kills it in one step, but bracketing against it never diminishes $\mathfrak{n}_n$, so the lower central series cannot pass below $\mathfrak{n}_n$. Solvability tolerates such a nonzero stable floor; nilpotency does not.

Nilpotency admits a reading that will organize the next section. The condition $\mathfrak{g}^{c+1} = 0$ says every left-normed bracket $[X_1, [X_2, [\cdots, X_{c+1}] \cdots]]$ of length $c+1$ vanishes, which is the statement that the operators $\operatorname{ad}X_1, \dots, \operatorname{ad}X_c$, applied in succession to any X_{c+1}, eventually annihilate it. Nilpotency of the algebra is thus a statement about the operators $\operatorname{ad}X$ collectively driving everything to zero, an iterated centrality: each application of an $\operatorname{ad}X$ moves deeper into the filtration, and after c steps one has fallen out the bottom. The infinitesimal form of this observation, that $\mathfrak{g}$ is nilpotent precisely when every single operator $\operatorname{ad}X$ is a nilpotent endomorphism, is Engel's theorem, proved in section 5.3. One direction is immediate and recorded in the exercises: if $\mathfrak{g}$ is nilpotent of class c then $(\operatorname{ad}X)^c = 0$ for every X, since $(\operatorname{ad}X)^c Y$ is a length-$(c+1)$ iterated bracket. The content of Engel's theorem is the converse, that the nilpotence of each $\operatorname{ad}X$ separately forces the collective nilpotence of the algebra.

Exercise 5.2.1

1. Prove that a finite-dimensional Lie algebra $\mathfrak{g}$ is nilpotent if and only if the lower central series reaches 0, if and only if there is a finite chain of ideals

   $$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \cdots \supseteq \mathfrak{a}_m = 0$$

   of $\mathfrak{g}$ with $[\mathfrak{g}, \mathfrak{a}_i] \subseteq \mathfrak{a}_{i+1}$ for every i. (Such a chain is a *central chain*: each step drops into the next term under bracketing with all of $\mathfrak{g}$.)

2. Show that if $\mathfrak{g}$ is nilpotent of class c, then $\operatorname{ad}X$ is a nilpotent operator for every $X \in \mathfrak{g}$, with $(\operatorname{ad}X)^c = 0$. (This is the easy direction of Engel's theorem, section 5.3.) Verify it explicitly on the Heisenberg algebra $\mathfrak{n}_3$ of example 5.2.5 by writing out $\operatorname{ad}P$, $\operatorname{ad}Q$, $\operatorname{ad}C$ as 3×3 matrices in the basis P, Q, C and checking each squares to zero.

3. Let $\mathfrak{g}_1$ and $\mathfrak{g}_2$ be nilpotent Lie algebras. Prove that the direct sum $\mathfrak{g}_1 \oplus \mathfrak{g}_2$, with bracket $[(X_1, X_2), (Y_1, Y_2)] = ([X_1, Y_1], [X_2, Y_2])$, is nilpotent, and that its class is the maximum of the two classes.

4. Let $\mathfrak{z} = \mathfrak{z}(\mathfrak{g})$ be the center of $\mathfrak{g}$. Prove that $\mathfrak{g}$ is nilpotent if and only if $\mathfrak{g}/\mathfrak{z}$ is nilpotent. (This is the recursive "peel off the center" structure of nilpotent algebras; contrast with solvability, where the analogous statement uses the derived algebra rather than the center.)

5. Give an example of a solvable Lie algebra $\mathfrak{g}$ that is not nilpotent, other than $\mathfrak{b}_n$, and identify the term at which its lower central series stabilizes above zero. (Hint: the two-dimensional nonabelian Lie algebra with basis X, Y and $[X, Y] = Y$ is the smallest example.)

6. Using the inclusion $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ of proposition 5.2.2, prove that if $\mathfrak{g}$ is nilpotent then every iterated bracket of length $c+1$ vanishes regardless of how the brackets are associated, where c is the nilpotency class. Deduce that the nilpotency class defined via the lower central series agrees with the longest nonvanishing iterated bracket.

5.3 Engel's Theorem

The lower central series of section 5.2 measures nilpotency from the outside, by iterating the bracket against all of $\mathfrak{g}$ until the chain reaches zero. Engel's theorem replaces this global condition with a pointwise one, phrased entirely in terms of a single operator attached to each element. The bridge is the adjoint representation ad of definition 2.5.3: to each $x \in \mathfrak{g}$ it assigns the endomorphism $\mathrm{ad}x\colon y \mapsto [x, y]$ of $\mathfrak{g}$, and nilpotency of the algebra turns out to be detectable from the nilpotency of these operators one at a time. One direction is already in hand. The lower central series of definition 5.2.1 makes iterated brackets vanish once $\mathfrak{g}$ is nilpotent, so $(\mathrm{ad}x)^k = 0$ for k large and every $\mathrm{ad}x$ is a nilpotent operator; this is the easy implication recorded in exercise 5.2.1 (2). Engel's theorem is the converse, and it is the harder half: from the nilpotence of each $\mathrm{ad}x$ separately one must reconstruct the vanishing of the whole lower central series.

The converse cannot be a formal manipulation of the series, because the hypothesis says nothing about how the operators $\mathrm{ad}x$ interact for different x. What it does say is that $\mathrm{ad}(\mathfrak{g})$ is a Lie subalgebra of $\mathfrak{gl}(\mathfrak{g})$ consisting of nilpotent operators, and the crux of Engel's theorem is a statement of pure linear algebra about any such algebra of operators: a Lie algebra of nilpotent endomorphisms of a nonzero space always kills a common vector. That linear-algebra fact, proved first, is then applied to $\mathrm{ad}(\mathfrak{g})$ acting on $\mathfrak{g}$ itself to run the induction that collapses the lower central series. The pattern will recur in section 5.4: there a solvable algebra of operators is shown to have a common eigenvector, and Lie's theorem follows by the same passage from a pointwise spectral fact to a global structural one.

Definition 5.3.1: ad-nilpotent Element

Let $\mathfrak{g}$ be a Lie algebra over a field k. An element $x \in \mathfrak{g}$ is *ad-nilpotent* if the operator $\mathrm{ad}x\colon \mathfrak{g} \to \mathfrak{g}$, $y \mapsto [x, y]$, is nilpotent, that is, $(\mathrm{ad}x)^m = 0$ for some $m \geq 1$.

The definition isolates exactly the property that exercise 5.2.1 (2) produced as a consequence of nilpotency. It is a condition on one element at a time, referring only to the operator $\mathrm{ad}x$ and not to the algebra as a whole, and this locality is what makes it a usable hypothesis. The strictly upper-triangular matrices give the guiding picture.

Example 5.3.2: ad-nilpotence in $\mathfrak{n}_n$

Let $\mathfrak{n}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$ be the strictly upper-triangular matrices of example 5.2.5, and take the elementary matrix $E_{12} \in \mathfrak{n}_3$ (a single 1 in the $(1,2)$ slot). The operator $\mathrm{ad}E_{12}$ acts on the basis of $\mathfrak{n}_3$ given by E_{12}, E_{13}, E_{23} by

$$[E_{12}, E_{12}] = 0, \qquad [E_{12}, E_{13}] = 0, \qquad [E_{12}, E_{23}] = E_{13}$$

using $[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}$. Thus $\mathrm{ad}E_{12}$ sends $E_{23} \mapsto E_{13} \mapsto 0$ and annihilates the rest, so $(\mathrm{ad}E_{12})^2 = 0$ and E_{12} is ad-nilpotent. The same holds for every element of $\mathfrak{n}_n$: nilpotency of the matrix X forces nilpotency of the operator $\mathrm{ad}X = L_X - R_X$ (left minus right multiplication), since L_X and R_X commute and are each nilpotent when X is, so their difference is nilpotent by the binomial expansion. Every element of $\mathfrak{n}_n$ is therefore ad-nilpotent, in agreement with the

nilpotency of $\mathfrak{n}_n$ established in example 5.2.5.

The example already exhibits the two faces of the coming theorem. On the one hand $\mathfrak{n}_n$ is nilpotent as an algebra, so all its elements are ad-nilpotent; on the other, its elements are literally nilpotent matrices, and they share the common null vector e_1 of $\mathbb{C}^n$, the first standard basis vector, which every strictly upper-triangular matrix kills. The theorem will show these two features are two aspects of one phenomenon. The technical heart is the second: a common null vector for a whole algebra of nilpotent operators.

A single nilpotent operator N on a nonzero space V has a nonzero kernel, since $N^m = 0$ with $N^{m-1} \neq 0$ produces a nonzero vector $N^{m-1}v$ killed by N. The content of the next lemma is that this survives passage from one operator to a whole Lie algebra of them: the kernels can be arranged to overlap in a common nonzero vector. Neither commutativity nor any relation among the operators is assumed beyond closure under the bracket, and that closure is exactly what makes the induction go.

Lemma 5.3.3: Common Null Vector for Nilpotent Operators

Let V be a nonzero finite-dimensional vector space over a field k, and let $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra all of whose elements are nilpotent endomorphisms of V. Then there is a nonzero vector $v \in V$ with $Xv = 0$ for every $X \in \mathfrak{g}$.

Before the proof, the mechanism deserves a word, because the induction is on $\dim \mathfrak{g}$ and every ingredient is arranged to make the algebra one dimension smaller act on a strictly smaller space. The engine has three moving parts. First, a nilpotent operator on $\mathfrak{gl}(V)$ arises from the adjoint action of a nilpotent operator on V, so $\mathfrak{g}$ acts on itself by nilpotent operators, and a maximal proper subalgebra $\mathfrak{h}$ acquires a nonzero eigenvector in $\mathfrak{g}/\mathfrak{h}$ for that action, forcing $\mathfrak{h}$ to be an ideal of codimension one. Second, the inductive hypothesis applied to $\mathfrak{h}$ on V produces a nonzero common null space W of $\mathfrak{h}$. Third, choosing $z \in \mathfrak{g}$ outside $\mathfrak{h}$, the ideal property makes W invariant under z, and the nilpotent operator $z|_W$ has a null vector, which is the common null vector sought.

***Proof*:**

The argument is by induction on $\dim \mathfrak{g}$. If $\dim \mathfrak{g} = 0$ there is nothing to prove, and if $\dim \mathfrak{g} = 1$, say $\mathfrak{g} = kX$ with X nilpotent, then X has a nonzero kernel as noted above, and any nonzero $v \in \ker X$ works. Assume the statement for all Lie algebras of nilpotent operators of dimension less than $n := \dim \mathfrak{g}$, with $n \geq 2$.

Step 1: $\mathfrak{g}$ acts on itself by nilpotent operators. For $X \in \mathfrak{g} \subseteq \mathfrak{gl}(V)$, the endomorphism $\mathrm{ad}X$ of $\mathfrak{g}$ is the restriction to $\mathfrak{g}$ of $\mathrm{ad}X$ on all of $\mathfrak{gl}(V)$, and on $\mathfrak{gl}(V)$ one has $\mathrm{ad}X = L_X - R_X$, where $L_X(A) = XA$ and $R_X(A) = AX$ are left and right multiplication. These two operators commute, and each is nilpotent because X is: if $X^m = 0$ then $L_X^m = R_X^m = 0$. A difference of two commuting nilpotent operators is nilpotent, by the binomial theorem $(L_X - R_X)^{2m} = \sum_i \binom{2m}{i}(-1)^i L_X^{2m-i} R_X^i$, in which every term has $L_X^{2m-i} = 0$ or $R_X^i = 0$. Hence $\mathrm{ad}X$ is a nilpotent operator on $\mathfrak{gl}(V)$, and its restriction to the invariant subspace $\mathfrak{g}$ is nilpotent as well. Thus $\mathrm{ad}(\mathfrak{g})$ is a Lie algebra of nilpotent operators on the vector space $\mathfrak{g}$.

Step 2: a maximal proper subalgebra is an ideal of codimension one. Since $n \geq 2$, $\mathfrak{g}$ has proper subalgebras (for instance 0), so choose $\mathfrak{h} \subsetneq \mathfrak{g}$ a proper subalgebra of maximal dimension. The subalgebra $\mathfrak{h}$ acts on $\mathfrak{g}$ by ad, and it preserves both $\mathfrak{h}$ (as $\mathfrak{h}$ is a subalgebra) and hence the quotient space $\mathfrak{g}/\mathfrak{h}$. This gives a representation

$$\overline{\mathrm{ad}}\colon \mathfrak{h} \to \mathfrak{gl}(\mathfrak{g}/\mathfrak{h}), \qquad \overline{\mathrm{ad}}(H)(y + \mathfrak{h}) = [H, y] + \mathfrak{h}$$

By Step 1 each $\mathrm{ad}H$ is nilpotent on $\mathfrak{g}$, so each $\overline{\mathrm{ad}}(H)$ is nilpotent on the quotient $\mathfrak{g}/\mathfrak{h}$. Now $\dim \mathfrak{h} < n$, so the inductive hypothesis applies to the Lie algebra $\overline{\mathrm{ad}}(\mathfrak{h}) \subseteq \mathfrak{gl}(\mathfrak{g}/\mathfrak{h})$ acting on the nonzero space $\mathfrak{g}/\mathfrak{h}$ (nonzero because $\mathfrak{h}$ is proper): there is a nonzero coset $\overline{y} = y + \mathfrak{h}$ with $\overline{\mathrm{ad}}(H)\overline{y} = 0$ for all $H \in \mathfrak{h}$. Unwrapping, $y \in \mathfrak{g} \setminus \mathfrak{h}$ satisfies

$$[H, y] \in \mathfrak{h} \qquad \text{for every } H \in \mathfrak{h} \tag{5.4}$$

Consider the subspace $\mathfrak{h}' = \mathfrak{h} + ky \subseteq \mathfrak{g}$. It is a subalgebra: brackets within $\mathfrak{h}$ stay in $\mathfrak{h}$, while $[H, y] \in \mathfrak{h} \subseteq \mathfrak{h}'$ by (5.4), and $[y, y] = 0$. Since $y \notin \mathfrak{h}$, the subalgebra $\mathfrak{h}'$ strictly contains $\mathfrak{h}$, and maximality of $\mathfrak{h}$ forces $\mathfrak{h}' = \mathfrak{g}$. Therefore $\mathfrak{h}$ has codimension one in $\mathfrak{g}$, and (5.4) says precisely that $[\mathfrak{h}, y] \subseteq \mathfrak{h}$; combined with $[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$ and $\mathfrak{g} = \mathfrak{h} + ky$, this gives $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$, so $\mathfrak{h}$ is an ideal of $\mathfrak{g}$.

Step 3: the common null space of $\mathfrak{h}$ is nonzero and $\mathfrak{g}$-invariant. As a Lie algebra of nilpotent operators on V of dimension $\dim \mathfrak{h} = n - 1 < n$, the ideal $\mathfrak{h}$ falls under the inductive hypothesis on the original space V: there is a nonzero vector killed by every element of $\mathfrak{h}$. Collect all of them into

$$W = \{w \in V \mid Hw = 0 \text{ for every } H \in \mathfrak{h}\}$$

which is a linear subspace of V, nonzero by the inductive hypothesis. The claim is that W is invariant under all of $\mathfrak{g}$, extending the invariance under $\mathfrak{h}$ that its definition already provides. Let $X \in \mathfrak{g}$ and $w \in W$; the task is to show $Xw \in W$, that is, $H(Xw) = 0$ for every $H \in \mathfrak{h}$. Write

$$H(Xw) = X(Hw) + [H, X]w$$

from $HX = XH + [H, X]$ in the associative algebra $\mathfrak{gl}(V)$. Now $Hw = 0$ because $w \in W$, so the first term vanishes. For the second, $[H, X] \in \mathfrak{h}$ because $\mathfrak{h}$ is an ideal of $\mathfrak{g}$ (Step 2), and therefore $[H, X]w = 0$ since $w \in W$. Hence $H(Xw) = 0$ for every $H \in \mathfrak{h}$, so $Xw \in W$. Thus $XW \subseteq W$ for every $X \in \mathfrak{g}$, and W is $\mathfrak{g}$-invariant.

Step 4: an eigenvector of $z|_W$ completes the descent. Because $\mathfrak{h}$ has codimension one, choose $z \in \mathfrak{g}$ with $z \notin \mathfrak{h}$, so that $\mathfrak{g} = \mathfrak{h} \oplus kz$ as vector spaces. By Step 3 the operator z preserves W, so it restricts to an endomorphism $z|_W \colon W \to W$. This restriction is nilpotent, being the restriction of the nilpotent operator z on V to an invariant subspace. A nilpotent operator on the nonzero space W has a nonzero kernel, so there is a nonzero $v \in W$ with $zv = 0$. This v is the common null vector: every $X \in \mathfrak{g}$ is $X = H + \lambda z$ with $H \in \mathfrak{h}$ and $\lambda \in k$, and then

$$Xv = Hv + \lambda\, zv = 0 + 0 = 0$$

since $v \in W$ kills $\mathfrak{h}$ and $zv = 0$. Thus $Xv = 0$ for all $X \in \mathfrak{g}$, and $v \neq 0$, completing the proof.

The lemma is the linear-algebra core of the whole section, and the induction is worth reading twice, because it runs on two dimensions at once: the algebra $\mathfrak{g}$ shrinks to the codimension-one ideal $\mathfrak{h}$, while the space V shrinks to the common null space W. The hypothesis that $\mathfrak{g}$ be a Lie subalgebra, closed under the bracket rather than an unstructured set of nilpotent operators, is what powers Step 2, where the self-action by ad manufactures the codimension-one ideal; without closure under the bracket there would be no such ideal, and the descent would have no first step. The conclusion is geometric: the nonzero vector v spans a line on which every operator in $\mathfrak{g}$ acts as zero, a common fixed direction for the entire algebra. Iterating this observation produces the flag that Engel's theorem is usually stated with. That iteration is now the content of the main theorem, first in its intrinsic form for a Lie algebra and then in its operator form.

Theorem 5.3.4: Engel's Theorem

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over an arbitrary field k. Then $\mathfrak{g}$ is nilpotent if and only if every element $x \in \mathfrak{g}$ is ad-nilpotent. Equivalently, in operator form: if $V \neq 0$ is a finite-dimensional vector space and $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ is a Lie subalgebra all of whose elements are nilpotent endomorphisms, then there is a basis of V in which every element of $\mathfrak{g}$ is strictly upper-triangular; that is, a full flag of subspaces

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_m = V, \qquad \dim V_i = i$$

with $\mathfrak{g}\, V_i \subseteq V_{i-1}$ for every i.

***Proof*:**

The intrinsic equivalence, forward direction. Suppose $\mathfrak{g}$ is nilpotent, so $\mathfrak{g}^{k+1} = 0$ for some k in the lower central series of definition 5.2.1. By definition $\mathfrak{g}^{k+1} = [\mathfrak{g}, [\mathfrak{g}, [\cdots, [\mathfrak{g}, \mathfrak{g}] \cdots]]]$ with $k+1$ copies of $\mathfrak{g}$, so for any $x, y \in \mathfrak{g}$ the iterated bracket $(\mathrm{ad}x)^k y = [x, [x, \cdots, [x, y] \cdots]]$ lies in $\mathfrak{g}^{k+1} = 0$. Hence $(\mathrm{ad}x)^k = 0$ and every x is ad-nilpotent. This is exactly the implication recorded in exercise 5.2.1 (2).

The intrinsic equivalence, converse direction. Suppose every $x \in \mathfrak{g}$ is ad-nilpotent, so $\mathrm{ad}(\mathfrak{g}) \subseteq \mathfrak{gl}(\mathfrak{g})$ is a Lie subalgebra consisting of nilpotent operators (it is the image of $\mathfrak{g}$ under the Lie algebra homomorphism ad of definition 2.5.3, hence a subalgebra). Apply the operator form, proved next, to $\mathrm{ad}(\mathfrak{g})$ acting on $V = \mathfrak{g}$. It yields a full flag of subspaces

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_m = \mathfrak{g}, \qquad \dim V_i = i$$

with $\mathrm{ad}(\mathfrak{g})\, V_i \subseteq V_{i-1}$, that is $[\mathfrak{g}, V_i] \subseteq V_{i-1}$ for every i. In particular each V_i is an ideal of $\mathfrak{g}$, and the descending chain of the flag forces the lower central series down: $\mathfrak{g}^1 = \mathfrak{g} = V_m$, and inductively if $\mathfrak{g}^j \subseteq V_{m-j+1}$ then

$$\mathfrak{g}^{j+1} = [\mathfrak{g}, \mathfrak{g}^j] \subseteq [\mathfrak{g}, V_{m-j+1}] \subseteq V_{m-j}$$

so after m steps $\mathfrak{g}^{m+1} \subseteq V_0 = 0$. Therefore $\mathfrak{g}$ is nilpotent, which establishes the converse and with it the intrinsic equivalence.

The operator form, existence of the flag. Let $V \neq 0$ and $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra of nilpotent operators. The flag is built one step at a time, from the bottom, by repeatedly invoking lemma 5.3.3. By that lemma there is a nonzero $v_1 \in V$ with $Xv_1 = 0$ for all $X \in \mathfrak{g}$; set $V_1 = kv_1$, a one-dimensional subspace with $\mathfrak{g}\, V_1 = 0 = V_0$. Suppose inductively that a chain $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_i$ has been constructed with $\dim V_j = j$ and $\mathfrak{g}\, V_j \subseteq V_{j-1}$ for $j \leq i$; each V_j is then a $\mathfrak{g}$-invariant subspace. If $V_i = V$ the flag is complete. Otherwise form the nonzero quotient space V/V_i. Every $X \in \mathfrak{g}$ preserves V_i, so it descends to an operator $\overline{X}$ on V/V_i, and $\overline{X}$ is nilpotent because X is. The map $X \mapsto \overline{X}$ is a Lie algebra homomorphism, so its image $\overline{\mathfrak{g}} \subseteq \mathfrak{gl}(V/V_i)$ is a Lie algebra of nilpotent operators on the nonzero space V/V_i. Lemma 5.3.3 gives a nonzero coset $\overline{v} = v + V_i$ with $\overline{X}\,\overline{v} = 0$ for all X, that is $Xv \in V_i$ for all $X \in \mathfrak{g}$. Set $V_{i+1} = V_i + kv$. Then $\dim V_{i+1} = i+1$ since $v \notin V_i$, and $\mathfrak{g}\, V_{i+1} \subseteq V_i$: on V_i this holds by the inductive hypothesis ($\mathfrak{g}\, V_i \subseteq V_{i-1} \subseteq V_i$), and $\mathfrak{g}\, v \subseteq V_i$ by construction. The chain has grown by one dimension. Continuing until the top is reached produces the full flag $0 = V_0 \subsetneq \cdots \subsetneq V_m = V$ with $\dim V_i = i$ and $\mathfrak{g}\, V_i \subseteq V_{i-1}$.

Choosing a basis $v_1, \dots, v_m$ of V adapted to the flag, so that $v_1, \dots, v_i$ span V_i, the condition $\mathfrak{g}\, V_i \subseteq V_{i-1}$ says that every $X \in \mathfrak{g}$ sends v_i into the span of $v_1, \dots, v_{i-1}$, which is exactly the statement that the matrix of X in this basis is strictly upper-triangular. This completes the operator form, and hence the proof.

Engel's theorem completes the reduction of nilpotency to a pointwise condition. The lower central series is a global object, a descending chain of ideals whose terms are computed from all of $\mathfrak{g}$ at once, yet its termination is now certified by inspecting each operator $\mathrm{ad}x$ in isolation and asking only whether it is nilpotent. The theorem thereby converts a structural property of the algebra into a spectral property of individual operators, and the conversion runs both ways: nilpotency of $\mathfrak{g}$ produces nilpotent $\mathrm{ad}x$, and nilpotent $\mathrm{ad}x$ reconstructs nilpotency of $\mathfrak{g}$. The A Lie algebra of nilpotent operators does more than collect individually triangularizable matrices; the theorem asserts they can be triangularized *simultaneously*, in a single basis, with zeros forced onto the diagonal. This is the precise sense in which $\mathfrak{n}_n$ of example 5.2.5 is the universal model.

The place where the hypotheses were used is worth marking, because it is narrower than for the theorems to come. The entire argument, from lemma 5.3.3 through the flag, used only that k is a field: nilpotent operators have nonzero kernels over any field, the induction on dimension needs no more, and at no point was an eigenvalue extracted or a polynomial factored. Engel's theorem therefore holds over an arbitrary field, with no assumption of algebraic closure and no restriction on the characteristic. This is the sharp contrast with Lie's theorem of section 5.4, where the passage from nilpotent to a general solvable algebra forces one to find a common *eigenvector* rather than a common null vector, and eigenvalues exist only once the field is algebraically closed; the characteristic-zero hypothesis enters there as well, for a reason that has no analogue here. The strict upper-triangular form of Engel and the plain upper-triangular form of Lie stand on opposite sides of exactly this divide.

The flag makes the reconstruction of the model explicit. For the running strictly-upper-triangular algebra, the theorem recovers the standard flag directly.

Example 5.3.5: The Standard Flag for $\mathfrak{n}_n$

Take $V = \mathbb{C}^n$ with standard basis $e_1, \dots, e_n$, and let $\mathfrak{n}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$ be the strictly upper-triangular matrices of example 5.2.5, a Lie algebra of nilpotent operators. The subspaces

$$V_i = \mathrm{span}(e_1, \dots, e_i), \qquad i = 0, 1, \dots, n$$

form a full flag, and a strictly upper-triangular X satisfies $Xe_i \in V_{i-1}$ for each i: the ith column of X has nonzero entries only in rows $1, \dots, i-1$. Hence $\mathfrak{n}_n\, V_i \subseteq V_{i-1}$, and this is the flag produced by theorem 5.3.4. The bottom step recovers the common null vector of lemma 5.3.3: $\mathfrak{n}_n\, V_1 = 0$ says every element of $\mathfrak{n}_n$ kills e_1, which is the common null vector observed after example 5.3.2. The example shows Engel's theorem is exact rather than qualitative: it does not just assert that some flag exists, it returns, for the model algebra, the flag one would write down by hand.

A first structural payoff of the theorem is the existence of a center, obtained without leaving the algebra $\mathfrak{g}$ at all. When every $\mathrm{ad}x$ is nilpotent, the operator form applied to $\mathrm{ad}(\mathfrak{g})$ on $\mathfrak{g}$ gives a nonzero common null vector, and a common null vector for $\mathrm{ad}(\mathfrak{g})$ is by definition a nonzero central element. This reproves the nonzero-center statement of proposition 5.2.4 from the operator side, and it is the form in which the fact will be used repeatedly in the structure theory.

Corollary 5.3.6: Nonzero Center of a Nilpotent Lie Algebra

Let $\mathfrak{g} \neq 0$ be a finite-dimensional Lie algebra all of whose elements are ad-nilpotent (in particular, any nonzero nilpotent Lie algebra). Then the center $\mathfrak{z}(\mathfrak{g}) = \{z \in \mathfrak{g} \mid [z, y] = 0 \text{ for all } y \in \mathfrak{g}\}$ is nonzero. Moreover any nonzero ideal $\mathfrak{a} \trianglelefteq \mathfrak{g}$ meets the center nontrivially: $\mathfrak{a} \cap \mathfrak{z}(\mathfrak{g}) \neq 0$.

***Proof*:**
Apply the operator form of theorem 5.3.4 to the Lie algebra $\mathrm{ad}(\mathfrak{g}) \subseteq \mathfrak{gl}(\mathfrak{g})$, whose elements are nilpotent by hypothesis, acting on the nonzero space $V = \mathfrak{g}$. It provides a nonzero $z \in \mathfrak{g}$ with $(\mathrm{ad}x)(z) = 0$ for all $x \in \mathfrak{g}$, that is $[x, z] = 0$ for all x, so $z \in \mathfrak{z}(\mathfrak{g})$ and the center is nonzero.

For the second statement, let $\mathfrak{a}$ be a nonzero ideal. Since $\mathfrak{a}$ is an ideal, $[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}$, so each $\mathrm{ad}x$ ($x \in \mathfrak{g}$) restricts to an operator $\mathrm{ad}x|_{\mathfrak{a}}$ on $\mathfrak{a}$, and this restriction is nilpotent because $\mathrm{ad}x$ is. The image $\{\mathrm{ad}x|_{\mathfrak{a}} \mid x \in \mathfrak{g}\} \subseteq \mathfrak{gl}(\mathfrak{a})$ is a Lie algebra of nilpotent operators on the nonzero space $\mathfrak{a}$, so lemma 5.3.3 yields a nonzero $z \in \mathfrak{a}$ with $(\mathrm{ad}x)(z) = [x, z] = 0$ for all $x \in \mathfrak{g}$. Then $z \in \mathfrak{a} \cap \mathfrak{z}(\mathfrak{g})$ and $z \neq 0$, completing the proof.

The corollary is the induction lever for nilpotent algebras. Passing to $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ drops the dimension while staying within the class, since quotients of nilpotent algebras are nilpotent by proposition 5.2.4, so a nonzero center guarantees the quotient is strictly smaller and arguments by induction on dimension have a base to descend on. The refinement that every nonzero ideal meets the center will be used when splitting off central subalgebras: any minimal ideal of a nilpotent algebra is in fact central, hence one-dimensional. These consequences flow from the same common-null-vector lemma, now read on the adjoint action rather than on an external representation, which is the recurring move of the subject.

Exercise 5.3.1

1. Show that the conclusion of lemma 5.3.3 can fail if $\mathfrak{g}$ is not assumed to be a Lie subalgebra. Concretely, exhibit a single non-nilpotent operator on $\mathbb{C}^2$ with no nonzero kernel, and explain why this does not contradict the lemma. Then exhibit two nilpotent operators on $\mathbb{C}^2$ that span a subspace (not closed under the bracket) with no common null vector, showing that the closure hypothesis, not just nilpotence of the individual operators, is doing work.

2. Prove directly, without invoking the full flag, that if every $\mathrm{ad}x$ is nilpotent then $\mathfrak{g}$ has a nonzero center, by applying lemma 5.3.3 to $\mathrm{ad}(\mathfrak{g})$. Then deduce Engel's theorem (the converse direction) by induction on $\dim \mathfrak{g}$, using the quotient $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$.

3. Let $\mathfrak{h}_3 \subseteq \mathfrak{n}_3$ be the Heisenberg algebra, spanned by E_{12}, E_{23}, E_{13} with the single nonzero bracket $[E_{12}, E_{23}] = E_{13}$. Verify that every element is ad-nilpotent, compute the lower central series to confirm nilpotency, and exhibit explicitly the flag of ideals in $\mathfrak{g} = \mathfrak{h}_3$ produced by theorem 5.3.4 for the adjoint action.

4. Let $\mathfrak{b}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$ be the upper-triangular matrices of example 5.1.7. Show that a diagonal matrix $D = \mathrm{diag}(d_1, \ldots, d_n) \in \mathfrak{b}_n$ with distinct entries is *not* ad-nilpotent, and conclude that $\mathfrak{b}_n$ is not nilpotent, in agreement with example 5.1.7. Why does this not contradict the operator form of Engel's theorem applied to $\mathfrak{b}_n$ acting on $\mathbb{C}^n$?

5. Using corollary 5.3.6, prove that a minimal nonzero ideal $\mathfrak{a}$ of a nilpotent Lie algebra $\mathfrak{g}$ is contained in the center and is one-dimensional.

5.4 Lie's Theorem

Engel's theorem (theorem 5.3.4) governs a Lie algebra of nilpotent operators: such an algebra has a common null vector, and so a full flag of invariant subspaces on which every operator is strictly upper triangular. Lie's theorem is the solvable analogue. A solvable algebra of operators need not consist of nilpotent endomorphisms, so "common null vector" has no chance of surviving, but its natural weakening does: a solvable algebra has a common *eigenvector*, and hence a full flag on which every operator is *upper* triangular, not necessarily strictly. Where Engel triangularizes with zeros on the diagonal, Lie triangularizes and lets the diagonal record the eigenvalues. This is the second of the two triangularization theorems that open the structure theory, and between them they pin down exactly what solvability and nilpotency give at the level of concrete matrices.

The price of passing from null vectors to eigenvectors is two hypotheses that Engel did not need, both of which are necessary. First, the field must be algebraically closed, because an eigenvector requires an eigenvalue to exist, and over $\mathbb{R}$ a rotation has none. Second, the field must have characteristic zero, for a subtler reason that the proof will isolate precisely: a trace-counting step divides by the dimension of a subspace, and in characteristic p that dimension can be a multiple of p, collapsing the argument. Both failures are exhibited by explicit examples at the end of the section. Throughout, the field is k, algebraically closed of characteristic zero, and the reader may take $k = \mathbb{C}$ without losing anything; V is a nonzero finite-dimensional vector space over k.

The heart of the matter is an invariance property of weight spaces. Given an ideal $\mathfrak{a}$ of a Lie algebra of operators and a linear functional $\lambda\colon \mathfrak{a} \to k$, the vectors on which every $A \in \mathfrak{a}$ acts as the scalar $\lambda(A)$ form the *weight space* attached to λ. The following lemma says that once such a weight space is nonzero, the whole algebra preserves it. This is the mechanism that will let the induction in Lie's theorem restrict from V to a smaller invariant subspace without losing the eigenvector it is building.

Definition 5.4.1: Weight and Weight Space

Let $\mathfrak{a} \subseteq \mathfrak{gl}(V)$ be a Lie algebra of operators and $\lambda\colon \mathfrak{a} \to k$ a linear functional (a *weight* of $\mathfrak{a}$). The associated *weight space* is

$$V_\lambda = \{v \in V \mid Av = \lambda(A)\, v \text{ for all } A \in \mathfrak{a}\}$$

the set of simultaneous eigenvectors of $\mathfrak{a}$ whose eigenvalue for each A is $\lambda(A)$, together with 0.

A weight space is the simultaneous-eigenspace notion for a whole algebra of operators. For a single operator A with eigenvalue c, the weight space of the one-dimensional algebra kA under $\lambda(A) = c$ is the ordinary eigenspace $\ker(A - c)$. The linearity of λ is forced: if $v \neq 0$ lies in V_λ, then $A \mapsto \lambda(A)$ is the composite of $A \mapsto Av$ with the scalar-extraction $cv \mapsto c$, both linear, so a nonzero weight space determines its weight uniquely as a linear functional on $\mathfrak{a}$. A common eigenvector for $\mathfrak{a}$ is precisely a nonzero element of some V_λ, so producing a common eigenvector is the same as producing a weight whose weight space is nonzero.

Consider the upper-triangular algebra $\mathfrak{b}_2 \subseteq \mathfrak{gl}_2(k)$ acting on $V = k^2$. The first basis vector e_1 is a common eigenvector: a matrix $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$ sends e_1 to $a\,e_1$, so $e_1 \in V_\lambda$ for the weight $\lambda\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) = a$ reading off the top-left entry. This weight is a linear functional on $\mathfrak{b}_2$, and the line ke_1 is its weight space. That e_1 is a common eigenvector, and that the associated diagonal entry is a Lie-algebra homomorphism to k, is Lie's theorem in miniature for $\mathfrak{b}_2$; the theorem asserts this pattern for every solvable algebra once k is algebraically closed of characteristic zero.

5.4.2 Note: the weight vanishes on brackets inside $\mathfrak{a}$ If $V_\lambda \neq 0$ for an ideal $\mathfrak{a}$, then λ annihilates $[\mathfrak{a},\mathfrak{a}]$: for $A, B \in \mathfrak{a}$ and $0 \neq v \in V_\lambda$, the operator $[A,B]$ acts on v as $AB - BA$, giving $\lambda(A)\lambda(B)v - \lambda(B)\lambda(A)v = 0$, so $\lambda([A,B])v = 0$ forces $\lambda([A,B]) = 0$. The content of the invariance lemma below is the stronger statement that λ vanishes on $[\mathfrak{g},\mathfrak{a}]$, brackets of the *ambient* algebra with $\mathfrak{a}$, which is what makes V_λ invariant under all of $\mathfrak{g}$.

The lemma that carries the theorem is the invariance of V_λ under the ambient algebra. The subtlety is that $x \in \mathfrak{g}$ need not commute with $\mathfrak{a}$, so it is not obvious that x maps a simultaneous eigenvector of $\mathfrak{a}$ to another one. The proof builds an x-cyclic subspace, shows $\mathfrak{a}$ acts on it by upper-triangular matrices with λ on the diagonal, and extracts the invariance from a trace computation. This is the one place where characteristic zero is used, and the proof flags it.

Lemma 5.4.3: The Invariance Lemma

Let $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ be a solvable Lie algebra over k, let $\mathfrak{a} \trianglelefteq \mathfrak{g}$ be an ideal, and let $\lambda\colon \mathfrak{a} \to k$ be a weight with nonzero weight space V_λ. Then V_λ is invariant under $\mathfrak{g}$; that is, $x\,V_\lambda \subseteq V_\lambda$ for every $x \in \mathfrak{g}$. Equivalently, $\lambda\big([x,A]\big) = 0$ for all $x \in \mathfrak{g}$ and $A \in \mathfrak{a}$.

***Proof*:**
Fix $x \in \mathfrak{g}$ and a nonzero $v \in V_\lambda$. The argument produces an x-stable subspace containing v, on which every $A \in \mathfrak{a}$ is upper triangular with constant diagonal $\lambda(A)$, and reads the equation $\lambda([x,A]) = 0$ off the trace of $[x,A]$ on that subspace. The invariance $x\,V_\lambda \subseteq V_\lambda$ then follows.

Step 1: the x-cyclic flag. Consider the subspaces

$$W_0 = 0, \qquad W_i = \operatorname{span}_k\{v,\, xv,\, x^2v,\, \ldots,\, x^{i-1}v\} \quad (i \geq 1)$$

so $W_1 = kv \subseteq W_2 \subseteq \cdots$ is an increasing chain, each W_i spanned by the first i vectors of the orbit of v under powers of x. Since V is finite-dimensional the chain stabilizes; let n be the least index with $W_n = W_{n+1}$, so that $v, xv, \ldots, x^{n-1}v$ are linearly independent and $x^n v \in W_n$. Then $\dim W_n = n$, the space W_n is x-invariant by construction (each $xW_i \subseteq W_{i+1}$, and $xW_n \subseteq W_n$ because $x^n v \in W_n$), and $v \in W_n$.

Step 2: every $A \in \mathfrak{a}$ is upper triangular on W_n with diagonal $\lambda(A)$. The claim is that, in the ordered basis $v, xv, \ldots, x^{n-1}v$,

$$A\,x^i v \equiv \lambda(A)\,x^i v \pmod{W_i} \qquad (A \in \mathfrak{a},\ 0 \leq i \leq n-1) \tag{5.5}$$

that is, A carries $x^i v$ to $\lambda(A)x^i v$ plus a combination of the earlier basis vectors $v, \ldots, x^{i-1}v$. In particular each W_i is A-invariant, and $A|_{W_n}$ is upper triangular with every diagonal entry equal to $\lambda(A)$.

The proof of (5.5) is by induction on i. For $i = 0$: $Av = \lambda(A)v$ since $v \in V_\lambda$, and $W_0 = 0$, so the congruence holds. Assume it holds for all indices up to $i-1$ and all $A \in \mathfrak{a}$. Then

$$A\,x^i v = Ax(x^{i-1}v) = xA(x^{i-1}v) + [A,x]\,(x^{i-1}v)$$

Treat the two terms separately. By the inductive hypothesis applied to A at index $i-1$, $A(x^{i-1}v) = \lambda(A)x^{i-1}v + w$ with $w \in W_{i-1}$; applying x gives $xA(x^{i-1}v) = \lambda(A)x^i v + xw$, and $xw \in xW_{i-1} \subseteq W_i$, so this term is $\lambda(A)x^i v$ modulo W_i. For the second term, $[A,x] = -[x,A] \in \mathfrak{a}$

because $\mathfrak{a}$ is an ideal of $\mathfrak{g}$ and $x \in \mathfrak{g}$; the inductive hypothesis applied to the element $[A, x] \in \mathfrak{a}$ at index $i-1$ gives $[A, x](x^{i-1}v) \equiv \lambda([A, x])x^{i-1}v \pmod{W_{i-1}}$, so in particular $[A, x](x^{i-1}v) \in W_i$. Adding, $A\,x^i v \equiv \lambda(A)x^i v \pmod{W_i}$, which is (5.5) at index i. The induction is complete, and it shows simultaneously that each W_i is A-invariant for every $A \in \mathfrak{a}$.

Step 3: the trace computation and the characteristic-zero step. Consider the commutator $[x, A]$ for a fixed $A \in \mathfrak{a}$. Since $\mathfrak{a}$ is an ideal, $[x, A] \in \mathfrak{a}$, so (5.5) applies to it: on W_n, the operator $[x, A]$ is upper triangular with every diagonal entry equal to $\lambda([x, A])$. Its trace on W_n is therefore the sum of n copies of that entry,

$$\operatorname{tr}\big([x, A]\big|_{W_n}\big) = n\,\lambda([x, A]) \tag{5.6}$$

On the other hand, W_n is invariant under x (Step 1) and under A (Step 2), so both $x|_{W_n}$ and $A|_{W_n}$ are endomorphisms of W_n, and the commutator of their restrictions is the restriction of their commutator: $[x, A]|_{W_n} = [\,x|_{W_n},\, A|_{W_n}\,]$. The trace of any commutator of endomorphisms of a finite-dimensional space vanishes, since $\operatorname{tr}(PQ) = \operatorname{tr}(QP)$, so

$$\operatorname{tr}\big([x, A]\big|_{W_n}\big) = \operatorname{tr}\big(x|_{W_n} A|_{W_n} - A|_{W_n} x|_{W_n}\big) = 0 \tag{5.7}$$

Comparing (5.6) and (5.7) gives $n\,\lambda([x, A]) = 0$. Here is the one point where the field matters: $n = \dim W_n$ is a positive integer, and in characteristic zero it is invertible in k, so dividing by n yields

$$\lambda([x, A]) = 0$$

In characteristic p the integer n could be a multiple of p, hence zero in k, and the conclusion would fail; this is exactly the crack that the characteristic-p counterexample at the end of the section exploits.

Step 4: invariance of V_λ. It remains to deduce $x\,V_\lambda \subseteq V_\lambda$. Let $A \in \mathfrak{a}$ and $w \in V_\lambda$. Then

$$A(xw) = x(Aw) + [A, x]\,w = x\,\lambda(A)\,w + [A, x]\,w = \lambda(A)\,(xw) + \lambda([A, x])\,w$$

using $Aw = \lambda(A)w$ and, in the last step, that $[A, x] \in \mathfrak{a}$ acts on $w \in V_\lambda$ as the scalar $\lambda([A, x])$. Step 3 gives $\lambda([A, x]) = -\lambda([x, A]) = 0$, so $A(xw) = \lambda(A)\,(xw)$. As $A \in \mathfrak{a}$ was arbitrary, $xw \in V_\lambda$. Hence $x\,V_\lambda \subseteq V_\lambda$, and V_λ is $\mathfrak{g}$-invariant, completing the proof.

Lie's theorem rests on this lemma, and it is worth isolating what it accomplishes. A weight space V_λ of an ideal $\mathfrak{a}$ is invariant under $\mathfrak{a}$ itself, by definition. The lemma promotes this to invariance under the entire ambient algebra $\mathfrak{g}$, by way of the trace computation. Once V_λ is $\mathfrak{g}$-invariant, the whole algebra $\mathfrak{g}$ acts on it, and one may look for a common eigenvector inside V_λ using the remaining directions of $\mathfrak{g}$ beyond $\mathfrak{a}$. This is the reduction step that makes the induction on $\dim \mathfrak{g}$ run. Solvability enters only through the existence of the codimension-one ideal that the induction requires; the lemma itself needs $\mathfrak{a}$ to be an ideal, but not $\mathfrak{g}$ to be solvable, and its proof used solvability nowhere.

With the invariance lemma in place, the theorem is a clean induction. A solvable Lie algebra has a codimension-one ideal, because $[\mathfrak{g}, \mathfrak{g}]$ is a proper subspace for a nonzero solvable algebra, and any subspace of codimension one containing $[\mathfrak{g}, \mathfrak{g}]$ is automatically an ideal. Peeling off a one-dimensional complement to such an ideal splits the algebra as $\mathfrak{h} \oplus kz$, and the eigenvector for $\mathfrak{g}$ is assembled from an eigenvector for $\mathfrak{h}$ and then a single eigenvalue of z on the invariant weight space.

Theorem 5.4.4: Lie's Theorem

Let k be algebraically closed of characteristic zero, $V \neq 0$ a finite-dimensional vector space over k, and $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ a solvable Lie algebra. Then $\mathfrak{g}$ has a common eigenvector: there exist $0 \neq v \in V$ and a weight $\lambda\colon \mathfrak{g} \to k$ with $xv = \lambda(x)\, v$ for all $x \in \mathfrak{g}$. Consequently there is a full flag

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V, \qquad \dim V_i = i$$

of $\mathfrak{g}$-invariant subspaces; equivalently, in a suitable basis every element of $\mathfrak{g}$ is upper triangular, so $\mathfrak{g}$ is simultaneously triangularizable.

***Proof*:**

The common eigenvector is produced by induction on $\dim \mathfrak{g}$; the flag is then a short deduction.

Step 1: the base case. If $\dim \mathfrak{g} = 0$, then $\mathfrak{g} = 0$ and every nonzero $v \in V$ is a common eigenvector with $\lambda = 0$. If $\dim \mathfrak{g} = 1$, write $\mathfrak{g} = kx$. Since k is algebraically closed, the single operator x has an eigenvector $v \neq 0$, say $xv = cv$; then v is a common eigenvector for $\mathfrak{g}$ with weight $\lambda(x) = c$. Algebraic closure is used here and nowhere else in the induction.

Step 2: a codimension-one ideal. Assume $\dim \mathfrak{g} \geq 1$ and the result for all solvable subalgebras of smaller dimension acting on any nonzero space. Since $\mathfrak{g}$ is solvable and nonzero, $[\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$: if $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, the derived series (definition 5.1.3) would be constant and never reach 0, contradicting solvability (definition 5.1.5). The quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is therefore a nonzero abelian Lie algebra, in which *every* subspace is an ideal. Choose a subspace of codimension one in $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ and let $\mathfrak{h}$ be its preimage in $\mathfrak{g}$. Then $\mathfrak{h}$ is a subspace of codimension one containing $[\mathfrak{g}, \mathfrak{g}]$, and any such subspace is an ideal: for $x \in \mathfrak{g}$ and $y \in \mathfrak{h}$, $[x, y] \in [\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{h}$. Being a subalgebra of the solvable algebra $\mathfrak{g}$, the ideal $\mathfrak{h}$ is itself solvable (proposition 5.1.6), and $\dim \mathfrak{h} = \dim \mathfrak{g} - 1$. Fix any $z \in \mathfrak{g} \setminus \mathfrak{h}$, so that

$$\mathfrak{g} = \mathfrak{h} \oplus kz$$

as vector spaces.

Step 3: a common eigenvector for $\mathfrak{h}$. By the inductive hypothesis applied to $\mathfrak{h} \subseteq \mathfrak{gl}(V)$, there is a common eigenvector for $\mathfrak{h}$, hence a weight $\lambda\colon \mathfrak{h} \to k$ with $V_\lambda \neq 0$, where

$$V_\lambda = \{w \in V \mid Aw = \lambda(A)\, w \text{ for all } A \in \mathfrak{h}\}$$

is the weight space of definition 5.4.1.

Step 4: restrict to V_λ and diagonalize z there. The algebra $\mathfrak{h}$ is an ideal of $\mathfrak{g}$ (Step 2), and $\mathfrak{g}$ is solvable, so the invariance lemma (lemma 5.4.3) applies: V_λ is invariant under all of $\mathfrak{g}$. In particular $z\, V_\lambda \subseteq V_\lambda$, so z restricts to an operator on the nonzero space V_λ. Since k is algebraically closed, $z|_{V_\lambda}$ has an eigenvector: there is $0 \neq v \in V_\lambda$ and $c \in k$ with $zv = cv$. This v is the sought common eigenvector for $\mathfrak{g}$. Indeed $v \in V_\lambda$ gives $Av = \lambda(A)v$ for all $A \in \mathfrak{h}$, and $zv = cv$; since every element of $\mathfrak{g} = \mathfrak{h} \oplus kz$ is $A + tz$ with $A \in \mathfrak{h}$ and $t \in k$, one has

$$(A + tz)v = \lambda(A)v + tc\, v = \big(\lambda(A) + tc\big)v$$

so v is a common eigenvector, with weight $\widetilde{\lambda}(A + tz) = \lambda(A) + tc$ extending λ from $\mathfrak{h}$ to $\mathfrak{g}$. This weight is linear, being λ on the first summand and $t \mapsto tc$ on the second. The induction is complete.

Step 5: the invariant flag. The flag follows by induction on $\dim V = m$, using the common eigenvector at each stage. For $m = 1$ take $0 \subsetneq V$. For $m > 1$, the common eigenvector v spans a one-dimensional $\mathfrak{g}$-invariant subspace $V_1 = kv$. The algebra $\mathfrak{g}$ acts on the quotient $\overline{V} = V/V_1$, which has dimension $m-1$, and the image of $\mathfrak{g}$ in $\mathfrak{gl}(\overline{V})$ is a homomorphic image of a solvable algebra, hence solvable (proposition 5.1.6). By the inductive hypothesis on dimension of the space, $\overline{V}$ carries a full flag $0 = \overline{V}_0 \subsetneq \overline{V}_1 \subsetneq \cdots \subsetneq \overline{V}_{m-1} = \overline{V}$ of subspaces invariant under the image of $\mathfrak{g}$. Pulling back along the quotient map $V \to \overline{V}$, and prepending $V_0 = 0$, produces a full flag $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_m = V$ of $\mathfrak{g}$-invariant subspaces with $\dim V_i = i$. Choosing a basis $v_1, \ldots, v_m$ adapted to this flag, so that $v_1, \ldots, v_i$ span V_i, every $x \in \mathfrak{g}$ maps V_i into itself and so is upper triangular in this basis. Thus $\mathfrak{g}$ is simultaneously triangularizable, completing the proof.

Lie's theorem is the exact solvable counterpart of Engel's theorem, and the comparison is the quickest way to hold both in mind. Engel takes an algebra of nilpotent operators and produces a common null vector, a full flag on which every element is *strictly* upper triangular, and it works over any field. Lie takes a solvable algebra and produces a common eigenvector, a full flag on which every element is upper triangular but not necessarily strictly, and it requires the field to be algebraically closed of characteristic zero. The diagonal entries that Engel forces to zero are, in Lie's theorem, the values of the weight λ, a Lie-algebra homomorphism $\mathfrak{g} \to k$ reading each element's eigenvalue on the flag. That the diagonal is a homomorphism, and not just a function, is the residue of lemma 5.4.3: a bracket $[x, y]$ acts by $\lambda([x, y])$ on the top line of the flag, and that value is 0, so the strictly-upper-triangular part carries all the bracketing. This last observation is precisely the input to the corollary below.

The abstract content of the theorem, stripped of the choice of representation, is that a solvable Lie algebra over $\mathbb{C}$ admits a filtration by ideals with one-dimensional successive quotients. Applying Lie's theorem to the adjoint representation shows this and yields the structural corollary that separates solvable from semisimple: the derived algebra of a solvable Lie algebra is nilpotent.

Corollary 5.4.5: Corollaries of Lie's Theorem

Let $\mathfrak{g}$ be a finite-dimensional solvable Lie algebra over k, algebraically closed of characteristic zero.

1. There is a chain of ideals $0 = \mathfrak{g}_0 \subsetneq \mathfrak{g}_1 \subsetneq \cdots \subsetneq \mathfrak{g}_n = \mathfrak{g}$ with $\dim \mathfrak{g}_i = i$ and each $\mathfrak{g}_i$ an ideal of $\mathfrak{g}$.
2. The derived algebra $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent. Equivalently, $\mathrm{ad}x$ is a nilpotent operator for every $x \in [\mathfrak{g}, \mathfrak{g}]$.

***Proof*:**
Apply Lie's theorem to the adjoint representation $\mathrm{ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ of definition 2.5.3, whose image $\mathrm{ad}\mathfrak{g} \subseteq \mathfrak{gl}(\mathfrak{g})$ is a solvable Lie algebra, being a homomorphic image of the solvable algebra $\mathfrak{g}$ (proposition 5.1.6). Here $V = \mathfrak{g}$ is the representation space.

1. By theorem 5.4.4 applied to $\mathrm{ad}\mathfrak{g}$ acting on $\mathfrak{g}$, there is a full flag $0 = \mathfrak{g}_0 \subsetneq \mathfrak{g}_1 \subsetneq \cdots \subsetneq \mathfrak{g}_n = \mathfrak{g}$ of subspaces invariant under $\mathrm{ad}\mathfrak{g}$, with $\dim \mathfrak{g}_i = i$. Invariance under $\mathrm{ad}\mathfrak{g}$ means $[\mathfrak{g}, \mathfrak{g}_i] = (\mathrm{ad}\mathfrak{g})(\mathfrak{g}_i) \subseteq \mathfrak{g}_i$, which is exactly the statement that each $\mathfrak{g}_i$ is an ideal of $\mathfrak{g}$. This proves (1).
2. Fix a basis of $\mathfrak{g}$ adapted to the flag of (1). In this basis every $\mathrm{ad}x$ is upper triangular, by theorem 5.4.4. For $x, y \in \mathfrak{g}$ the operator $\mathrm{ad}[x, y] = [\mathrm{ad}x, \mathrm{ad}y]$ is the commutator of two upper-triangular matrices, hence *strictly* upper triangular: the diagonal of a commutator

of upper-triangular matrices vanishes, since the (i,i) entry of a product UW of upper triangular matrices is $\sum_j u_{ij}w_{ji}$, and $u_{ij}w_{ji}$ can be nonzero only when $i \le j$ (from U upper triangular) and $j \le i$ (from W), forcing $j = i$, so the (i,i) entry of $[U,W]$ is $u_{ii}w_{ii} - w_{ii}u_{ii} = 0$ when U, W are upper triangular. Therefore $\mathrm{ad}z$ is strictly upper triangular for every $z \in [\mathfrak{g},\mathfrak{g}]$, being a linear combination of the $\mathrm{ad}[x,y]$. A strictly-upper-triangular matrix is nilpotent, so $\mathrm{ad}z$ is nilpotent for every $z \in [\mathfrak{g},\mathfrak{g}]$.

It remains to pass from "$\mathrm{ad}z$ nilpotent for $z \in [\mathfrak{g},\mathfrak{g}]$" to "$[\mathfrak{g},\mathfrak{g}]$ is nilpotent as a Lie algebra". Set $\mathfrak{k} = [\mathfrak{g},\mathfrak{g}]$. For $z \in \mathfrak{k}$, the operator $\mathrm{ad}_{\mathfrak{g}}z$ is nilpotent on $\mathfrak{g}$, so its restriction $\mathrm{ad}_{\mathfrak{k}}z = (\mathrm{ad}_{\mathfrak{g}}z)|_{\mathfrak{k}}$, which is defined because $\mathfrak{k}$ is an ideal and so $\mathrm{ad}_{\mathfrak{g}}z$ preserves $\mathfrak{k}$, is also nilpotent. Thus every element of $\mathfrak{k}$ is ad-nilpotent in $\mathfrak{k}$ (definition 5.3.1), and Engel's theorem (theorem 5.3.4) gives that $\mathfrak{k} = [\mathfrak{g},\mathfrak{g}]$ is nilpotent, completing the proof.

The corollary is the structural payoff, and it draws the line the whole chapter has been approaching. Solvability is a weaker condition than nilpotency, as the upper-triangular algebra $\mathfrak{b}_n$ shows: $\mathfrak{b}_n$ is solvable but not nilpotent (example 5.1.7). Lie's theorem measures exactly how far solvability falls short. It says that solvability fails to be nilpotency only in the diagonal directions, because the entire bracket $[\mathfrak{g},\mathfrak{g}]$, where all the nonabelian structure lives, is nilpotent. A solvable algebra is thus a nilpotent algebra $[\mathfrak{g},\mathfrak{g}]$ with an abelian quotient $\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ stacked on top; the abelian part carries the eigenvalues, the nilpotent part carries the brackets. Over $\mathbb{C}$ this reads off the concrete matrix picture directly: a solvable Lie algebra of operators is one that can be put into upper-triangular form, and the class of such algebras is exactly the triangularizable ones. This is the structural fact that isolates the opposite extreme. Semisimple algebras, the subject of chapter 6, are those with no nonzero solvable ideal at all, hence no invariant flag of this kind, and the theory splits cleanly along this divide: every finite-dimensional Lie algebra is an extension of a semisimple algebra by its maximal solvable ideal, the radical, and Lie's theorem has just described the internal structure of the solvable part in full.

Both hypotheses on the field are needed, and pinning down where each fails is the surest way to remember why they appear. Over $\mathbb{R}$, a common eigenvector may simply not exist: the rotation algebra $\mathbb{R} \cdot \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is abelian, hence solvable, yet the rotation has no real eigenvector. In characteristic p, the abstract statement fails for a different reason, traced precisely to the division by n in lemma 5.4.3: there is a p-dimensional solvable, even nilpotent, representation of a two-dimensional solvable algebra with no common eigenvector. The exercises construct both examples in detail and extract the positive consequences that survive: a one-dimensional invariant subspace in every representation, and the classification of the smallest solvable algebras.

Exercise 5.4.1

1. Over $\mathbb{R}$, let $\mathfrak{g} = \mathbb{R} \cdot R \subseteq \mathfrak{gl}_2(\mathbb{R})$ where $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Show that $\mathfrak{g}$ is solvable but has no common eigenvector in $\mathbb{R}^2$, so the conclusion of Lie's theorem fails over $\mathbb{R}$. Which hypothesis of theorem 5.4.4 is violated, and at which step of the proof does the argument break? Verify that after extending scalars to $\mathbb{C}$ the theorem is restored, and exhibit the common eigenvector in $\mathbb{C}^2$.

2. Fix a prime p and work over a field k of characteristic p. On $V = k^p$ with basis $e_0, \dots, e_{p-1}$, let X be the cyclic shift $Xe_i = e_{i+1}$ (indices mod p, so $Xe_{p-1} = e_0$) and let H be the diagonal operator $He_i = i\,e_i$. Compute $[H,X]$ and show it equals X, so that $\mathfrak{g} = kX + kH$ is a two-dimensional solvable Lie algebra. Prove that $\mathfrak{g}$ has no common eigenvector in V. Identify the exact step of the proof of theorem 5.4.4 (via lemma 5.4.3) that fails, and pinpoint where

characteristic p enters.

3. Deduce from theorem 5.4.4 that every finite-dimensional representation $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ of a solvable Lie algebra $\mathfrak{g}$ over k (algebraically closed, characteristic zero), with $V \neq 0$, has a one-dimensional $\mathfrak{g}$-invariant subspace. Conversely, show that if every finite-dimensional nonzero representation of $\mathfrak{g}$ has a one-dimensional invariant subspace, must $\mathfrak{g}$ be solvable? Decide the question, using the standard representation of $\mathfrak{sl}_2(\mathbb{C})$ on $\mathbb{C}^2$ to inform your answer.

4. Prove that the only simple Lie algebra that is also solvable is impossible: no Lie algebra is both simple and solvable. Deduce that a solvable Lie algebra with no proper nonzero ideals is one-dimensional (necessarily abelian), and conversely that every one-dimensional Lie algebra is abelian, solvable, and has no proper nonzero ideals.

5. Let $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ be solvable over k (algebraically closed, characteristic zero) and let $\lambda\colon \mathfrak{g} \to k$ be the weight of the common eigenvector produced by theorem 5.4.4. Show directly that λ is a Lie-algebra homomorphism, that is $\lambda([\mathfrak{g},\mathfrak{g}]) = 0$, and use this to give a second proof that every $x \in [\mathfrak{g},\mathfrak{g}]$ acts on V as a nilpotent operator (equivalently, vanishes on the associated graded of the flag of theorem 5.4.4).

5.5 The Radical and the Nilradical

The three preceding sections built the machinery of solvability from the inside: the derived series measures how a Lie algebra decays toward zero under repeated bracketing, Engel's theorem reads nilpotency off the operators $\mathrm{ad}x$, and Lie's theorem triangularizes a solvable algebra of matrices. All of this describes what it means for an algebra to be solvable. The present section turns the question outward. Given an arbitrary finite-dimensional Lie algebra $\mathfrak{g}$, which need be neither solvable nor anything close to it, how much solvability does it contain? The answer is a single canonical ideal, the largest solvable ideal, and its extraction is the pivot on which the entire structure theory of Part II turns. Isolating this ideal splits every Lie algebra into two pieces of opposite character: a solvable part that the tools of this chapter already control, and a quotient with no solvable part at all, the semisimple algebras that chapter 6 takes up.

The existence of a largest solvable ideal is not obvious. A Lie algebra can carry many solvable ideals at once, and to speak of a single largest one requires that they combine rather than compete. The mechanism is the last clause of proposition 5.1.6: the sum of two solvable ideals is again a solvable ideal. Iterating this over all solvable ideals produces one ideal containing every other, and that ideal is what the theory names.

Proposition 5.5.1: Existence of a Maximal Solvable Ideal

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. The sum of all solvable ideals of $\mathfrak{g}$ is itself a solvable ideal. Consequently $\mathfrak{g}$ possesses a unique maximal solvable ideal, one that contains every solvable ideal of $\mathfrak{g}$.

***Proof*:**
Let $\mathfrak{r}$ denote the sum of all solvable ideals of $\mathfrak{g}$, that is the set of finite sums $a_1 + \cdots + a_k$ with each a_i lying in some solvable ideal. This is an ideal: a sum of ideals is an ideal, since it is a subspace and $[\mathfrak{g}, \mathfrak{a}+\mathfrak{b}] \subseteq [\mathfrak{g},\mathfrak{a}] + [\mathfrak{g},\mathfrak{b}] \subseteq \mathfrak{a}+\mathfrak{b}$ whenever $\mathfrak{a},\mathfrak{b}$ are ideals. The task is to show $\mathfrak{r}$ is

solvable.

Because $\mathfrak{g}$ is finite-dimensional, the ascending chain of partial sums must stabilize: there are finitely many solvable ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_m$ with

$$\mathfrak{r} = \mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_m$$

Indeed, choosing solvable ideals one at a time so that each strictly enlarges the running sum, dimension forces the process to halt after at most $\dim \mathfrak{g}$ steps, and the resulting sum already contains every solvable ideal (any further one would either enlarge the sum, contradicting maximality of the running dimension, or be absorbed). It therefore suffices to prove that a sum of finitely many solvable ideals is solvable, which follows by induction on m from the two-ideal case. For $m = 1$ there is nothing to prove. For the inductive step, write $\mathfrak{r} = (\mathfrak{a}_1 + \cdots + \mathfrak{a}_{m-1}) + \mathfrak{a}_m$; the first summand is solvable by the inductive hypothesis and is an ideal, $\mathfrak{a}_m$ is a solvable ideal, and the last clause of proposition 5.1.6 states that the sum of two solvable ideals is solvable. Hence $\mathfrak{r}$ is solvable.

Uniqueness of a maximal solvable ideal is now immediate. The ideal $\mathfrak{r}$ is solvable and contains every solvable ideal of $\mathfrak{g}$ by construction, so it is the largest one; any other maximal solvable ideal would be contained in $\mathfrak{r}$ and, being maximal, would equal it, completing the proof.

The argument is entirely a bookkeeping consequence of one algebraic fact, that solvable ideals add to solvable ideals, together with the finite dimension that keeps the sum from running away. What makes the conclusion carry weight is that it singles out a piece of $\mathfrak{g}$ with no arbitrary choice involved: the maximal solvable ideal is determined by $\mathfrak{g}$ alone, not by any basis, generating set, or auxiliary decomposition. This canonicity is what allows the ideal to be given a name and treated as an invariant of the algebra.

Definition 5.5.2: The Radical

The *radical* of a finite-dimensional Lie algebra $\mathfrak{g}$, written $\operatorname{rad}\mathfrak{g}$, is its unique maximal solvable ideal, the ideal produced by proposition 5.5.1. It contains every solvable ideal of $\mathfrak{g}$.

The radical collects all of the solvability in $\mathfrak{g}$ into one ideal. Parallel to it, and contained in it, sits a second canonical ideal that collects all of the nilpotency. The largest nilpotent ideal exists on grounds analogous to those for the radical, though the argument that a sum of two nilpotent ideals is again nilpotent differs from its solvable counterpart and is given after the definition.

Definition 5.5.3: The Nilradical

The *nilradical* of a finite-dimensional Lie algebra $\mathfrak{g}$, written $\operatorname{nil}\mathfrak{g}$, is its unique maximal nilpotent ideal: the sum of all nilpotent ideals of $\mathfrak{g}$, which is again a nilpotent ideal and contains every nilpotent ideal.

That the nilradical exists is proved as proposition 5.5.1 was, once one knows that the sum of two nilpotent ideals is a nilpotent ideal. Here the argument is not the extension argument used for solvability, because an extension of a nilpotent algebra by a nilpotent ideal need not be nilpotent, as the two-dimensional non-abelian algebra $\langle x, y | [x, y] = x \rangle$ shows. What replaces it uses that both summands are ideals of the same ambient algebra. Let $\mathfrak{a}$ and $\mathfrak{b}$ be nilpotent ideals of $\mathfrak{g}$, with lower central series terminating at steps p and q respectively. Because both are ideals, $[\mathfrak{a}, \mathfrak{b}] \subseteq \mathfrak{a} \cap \mathfrak{b}$, so in an iterated bracket of $p + q$ elements drawn from $\mathfrak{a} \cup \mathfrak{b}$ the factors may be regrouped so that either

at least p come from $\mathfrak{a}$ or at least q come from $\mathfrak{b}$; the first case lands the bracket in the p-th term of the lower central series of $\mathfrak{a}$, which is zero, and the second lands it in the q-th term of $\mathfrak{b}$'s, also zero. Hence the $(p+q)$-th term of the lower central series of $\mathfrak{a}+\mathfrak{b}$ vanishes, so $\mathfrak{a}+\mathfrak{b}$ is nilpotent, and it is an ideal as a sum of ideals. The summation argument of proposition 5.5.1 then produces a unique maximal nilpotent ideal. Since every nilpotent ideal is solvable by proposition 5.2.4, the nilradical is a solvable ideal, and the maximality of the radical among solvable ideals gives the containment

$$\operatorname{nil}\mathfrak{g} \subseteq \operatorname{rad}\mathfrak{g} \tag{5.8}$$

The containment (5.8) can be strict. The two ideals measure different failures of semisimplicity, one solvable and one nilpotent, and the gap between them appears already for the two-dimensional non-abelian algebra $\mathfrak{g} = \langle x, y \big| [x,y] = x \rangle$: it is solvable, so $\operatorname{rad}\mathfrak{g} = \mathfrak{g}$, while its nilradical is the abelian ideal $\mathbb{C}x$, strictly smaller (the computation is carried out in exercise 5.5.1 (5)). That $\mathfrak{g}$ is solvable without being nilpotent is the same distinction seen in the upper-triangular algebra $\mathfrak{b}_n$ of example 5.1.7 against its strictly-upper-triangular ideal $\mathfrak{n}_n$ of example 5.2.5.

A worked case grounds both definitions at once. The full matrix algebra $\mathfrak{gl}_n$ is the standard test object of Part II, and its radical and nilradical can be computed by hand.

Example 5.5.4: The Radical of $\mathfrak{gl}_n$

Let $\mathfrak{gl}_n = \mathfrak{gl}_n(\mathbb{C})$ and let $\mathfrak{z} = \mathbb{C}I$ be its center, the scalar matrices, which is the center of $\mathfrak{gl}_n$ because a matrix commuting with every matrix is scalar. The center is an abelian ideal, hence solvable, so $\mathfrak{z} \subseteq \operatorname{rad}\mathfrak{gl}_n$. The claim is that this containment is an equality,

$$\operatorname{rad}\mathfrak{gl}_n = \mathbb{C}I$$

To see there is nothing more, use the decomposition $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$ into traceless matrices and scalars, an identity of ideals recorded in theorem 1.4.3. The intersection $\operatorname{rad}\mathfrak{gl}_n \cap \mathfrak{sl}_n$ is a solvable ideal of $\mathfrak{sl}_n$. For $n \geq 2$ the algebra $\mathfrak{sl}_n$ is simple in the sense of definition 5.5.5 below, a standard fact proved for general n through the root-space decomposition of chapter 6 and verified directly for $n = 2$ in example 5.5.6; its only ideals are then 0 and $\mathfrak{sl}_n$. A solvable ideal cannot be all of $\mathfrak{sl}_n$, since a simple algebra is not solvable (its derived algebra equals itself), so $\operatorname{rad}\mathfrak{gl}_n \cap \mathfrak{sl}_n = 0$. Any $X \in \operatorname{rad}\mathfrak{gl}_n$ therefore has its $\mathfrak{sl}_n$-component in $\operatorname{rad}\mathfrak{gl}_n \cap \mathfrak{sl}_n = 0$, leaving only its scalar part, so $\operatorname{rad}\mathfrak{gl}_n \subseteq \mathbb{C}I$. Combined with the reverse containment, $\operatorname{rad}\mathfrak{gl}_n = \mathbb{C}I$. The nilradical here coincides with the radical. The center $\mathbb{C}I$ is an abelian ideal, hence nilpotent as a Lie algebra, so it is a nilpotent ideal and $\mathbb{C}I \subseteq \operatorname{nil}\mathfrak{gl}_n$; the containment (5.8) then closes the circle, giving

$$\operatorname{nil}\mathfrak{gl}_n = \operatorname{rad}\mathfrak{gl}_n = \mathbb{C}I$$

so that for $\mathfrak{gl}_n$ the two ideals agree.

The example shows the radical detecting exactly the scalar directions of $\mathfrak{gl}_n$, the part of the algebra that commutes with everything; here the radical and nilradical coincide because that solvable part is abelian, and an abelian ideal is nilpotent. The gap between the two ideals in (5.8) therefore does not open for $\mathfrak{gl}_n$; it opens when the radical carries a semisimple direction that fails to be nilpotent, as in the non-abelian solvable algebras of the exercises. The computation also isolates the object the whole theory is built to study. The traceless algebra $\mathfrak{sl}_n$ had radical zero, and an algebra whose radical is zero is precisely one with no solvable part to strip away. This is the definition that organizes the rest of the book.

Definition 5.5.5: Semisimple and Simple Lie Algebras

A finite-dimensional Lie algebra $\mathfrak{g}$ is *semisimple* if its radical is zero, $\mathrm{rad}\,\mathfrak{g} = 0$; equivalently, if $\mathfrak{g}$ has no nonzero solvable ideal. It is *simple* if it is non-abelian and its only ideals are 0 and $\mathfrak{g}$ itself.

Semisimplicity is defined here by a negation: $\mathfrak{g}$ is semisimple when it contains no solvable ideal other than the zero ideal, so that the radical, the receptacle for all solvability, is empty. Simplicity is the stronger condition that $\mathfrak{g}$ have no proper nonzero ideal at all, together with the non-abelian requirement that excludes the one-dimensional algebra (whose only ideals are trivially 0 and itself, but which carries no interesting structure). A simple Lie algebra is semisimple: its only ideals are 0 and $\mathfrak{g}$, and $\mathfrak{g}$ is not solvable because $[\mathfrak{g}, \mathfrak{g}]$ is a nonzero ideal, hence all of $\mathfrak{g}$, so the derived series never descends; the radical, being a solvable ideal, cannot be $\mathfrak{g}$ and so is 0. The simple algebras are thus the atoms among the semisimple ones, and chapter 6 will show that every semisimple algebra is a direct sum of simple ideals.

The first example is the one the reader has carried since Part I. The algebra $\mathfrak{sl}_2$ is the smallest semisimple Lie algebra, and it is in fact simple.

Example 5.5.6: $\mathfrak{sl}_2(\mathbb{C})$ Is Semisimple

The algebra $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$ has the standard basis

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

with the bracket relations $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$, computed directly from the matrix commutator. To see $\mathfrak{sl}_2$ is semisimple it suffices to show $\mathrm{rad}\mathfrak{sl}_2 = 0$, and the cleanest route is to show $\mathfrak{sl}_2$ has no proper nonzero ideal, so that it is simple, whence semisimple. Let $\mathfrak{a} \neq 0$ be an ideal and pick $0 \neq x = \alpha h + \beta e + \gamma f \in \mathfrak{a}$. Applying $\mathrm{ad}e$ twice,

$$[e, x] = -2\alpha e + \gamma h, \qquad [e, [e, x]] = [e, -2\alpha e + \gamma h] = \gamma[e, h] = -2\gamma e$$

and applying $\mathrm{ad}f$ twice symmetrically gives $[f, [f, x]] = -2\beta f$. If $\gamma \neq 0$ then $[e, [e, x]] = -2\gamma e \in \mathfrak{a}$ shows $e \in \mathfrak{a}$; then $[f, e] = -h \in \mathfrak{a}$ and $[e, f] = h$, so also $f = -[h, f]/2 \in \mathfrak{a}$ (from $[h, f] = -2f$), and $\mathfrak{a} = \mathfrak{sl}_2$. If instead $\gamma = 0$ but $\beta \neq 0$, the symmetric computation puts $f \in \mathfrak{a}$ and the same closure gives $\mathfrak{a} = \mathfrak{sl}_2$. If $\beta = \gamma = 0$ but $\alpha \neq 0$, then $h \in \mathfrak{a}$, and $[h, e] = 2e$, $[h, f] = -2f$ put $e, f \in \mathfrak{a}$, again $\mathfrak{a} = \mathfrak{sl}_2$. In every case a nonzero ideal is all of $\mathfrak{sl}_2$, so $\mathfrak{sl}_2$ is simple, hence semisimple.

The verification that $\mathrm{rad}\mathfrak{sl}_2 = 0$ was reduced to the sharper fact that $\mathfrak{sl}_2$ has no proper nonzero ideal, and the computation is a model for the general phenomenon: bracketing against e and f propagates any nonzero element out to a spanning set. This is why $\mathfrak{sl}_2$ is the running example of the semisimple theory, the smallest place where root spaces, the Cartan subalgebra $\mathbb{C}h$, and the two root directions $\mathbb{C}e$, $\mathbb{C}f$ are all present. By contrast the ambient algebra $\mathfrak{gl}_2$ is not semisimple: it splits as $\mathfrak{sl}_2 \oplus \mathbb{C}I$, and the scalar summand $\mathbb{C}I$ is a nonzero abelian, hence solvable, ideal, so $\mathrm{rad}\mathfrak{gl}_2 = \mathbb{C}I \neq 0$. More generally $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$ is never semisimple, its radical being the center $\mathbb{C}I$ computed in example 5.5.4; the presence of the central direction is exactly the obstruction.

The radical was constructed to be the largest solvable ideal, and its defining role is to make the quotient by it semisimple. Passing to $\mathfrak{g}/\mathrm{rad}\mathfrak{g}$ removes all of the solvability from $\mathfrak{g}$ at one stroke, and the resulting algebra has no solvability left. This is the structural statement toward which the section

has been building.

Proposition 5.5.7: The Quotient by the Radical Is Semisimple

For any finite-dimensional Lie algebra $\mathfrak{g}$, the quotient $\mathfrak{g}/\mathrm{rad}\mathfrak{g}$ is semisimple.

***Proof*:**
Write $\mathfrak{r} = \mathrm{rad}\mathfrak{g}$ and let $\pi\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{r}$ be the quotient map, a Lie algebra homomorphism by definition 5.1.1. The claim is that $\mathrm{rad}(\mathfrak{g}/\mathfrak{r}) = 0$. Suppose instead that $\mathfrak{g}/\mathfrak{r}$ has a nonzero solvable ideal $\bar{\mathfrak{s}}$. Its preimage $\mathfrak{s} = \pi^{-1}(\bar{\mathfrak{s}})$ is an ideal of $\mathfrak{g}$: for $x \in \mathfrak{g}$ and $s \in \mathfrak{s}$ one has $\pi([x,s]) = [\pi(x), \pi(s)] \in \bar{\mathfrak{s}}$ because $\bar{\mathfrak{s}}$ is an ideal and $\pi(s) \in \bar{\mathfrak{s}}$, so $[x,s] \in \pi^{-1}(\bar{\mathfrak{s}}) = \mathfrak{s}$.

This ideal $\mathfrak{s}$ is solvable. It sits in the short exact sequence

$$0 \longrightarrow \mathfrak{r} \longrightarrow \mathfrak{s} \xrightarrow{\pi} \bar{\mathfrak{s}} \longrightarrow 0$$

since $\ker(\pi|_{\mathfrak{s}}) = \mathfrak{s} \cap \mathfrak{r} = \mathfrak{r}$ (because $\mathfrak{r} = \pi^{-1}(0) \subseteq \pi^{-1}(\bar{\mathfrak{s}}) = \mathfrak{s}$) and $\pi(\mathfrak{s}) = \bar{\mathfrak{s}}$. Thus $\mathfrak{s}$ is an extension of the solvable algebra $\bar{\mathfrak{s}}$ by the solvable ideal $\mathfrak{r}$, and an extension of a solvable algebra by a solvable ideal is solvable by proposition 5.1.6. So $\mathfrak{s}$ is a solvable ideal of $\mathfrak{g}$.

By maximality of the radical, $\mathfrak{s} \subseteq \mathfrak{r}$. But $\mathfrak{r} \subseteq \mathfrak{s}$ already, so $\mathfrak{s} = \mathfrak{r}$, whence $\bar{\mathfrak{s}} = \pi(\mathfrak{s}) = \pi(\mathfrak{r}) = 0$, contradicting $\bar{\mathfrak{s}} \neq 0$. Therefore $\mathfrak{g}/\mathfrak{r}$ has no nonzero solvable ideal, that is $\mathrm{rad}(\mathfrak{g}/\mathfrak{r}) = 0$, and $\mathfrak{g}/\mathfrak{r}$ is semisimple, as we sought to show.

The mechanism of the proof is the correspondence between ideals of a quotient and ideals of $\mathfrak{g}$ lying above the radical: a nonzero solvable ideal of $\mathfrak{g}/\mathrm{rad}\mathfrak{g}$ would pull back to a solvable ideal of $\mathfrak{g}$ strictly larger than $\mathrm{rad}\mathfrak{g}$, which the maximality of the radical forbids. The radical is thus exactly the obstruction to semisimplicity, and dividing it out removes the obstruction and nothing more.

This is the first structural dichotomy of the subject. Every finite-dimensional Lie algebra $\mathfrak{g}$ fits into a short exact sequence

$$0 \longrightarrow \mathrm{rad}\mathfrak{g} \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\mathrm{rad}\mathfrak{g} \longrightarrow 0$$

whose left term is solvable and whose right term is semisimple. Every Lie algebra is, in this sense, an extension of a semisimple algebra by a solvable one, and the two ends are governed by disjoint theories: the solvable end by Engel and Lie, the machinery of this chapter, and the semisimple end by the root-space theory to come. The sequence does more than exist. The Levi decomposition of chapter 6 will show it splits, exhibiting $\mathfrak{g}$ as a semidirect product $\mathfrak{g} = \mathfrak{s} \ltimes \mathrm{rad}\mathfrak{g}$ of the radical with a semisimple subalgebra $\mathfrak{s}$ isomorphic to $\mathfrak{g}/\mathrm{rad}\mathfrak{g}$, so that the classification of all Lie algebras reduces to the classification of the semisimple ones together with their solvable extensions.

Semisimplicity has been defined here negatively, as the absence of solvable ideals, and at this stage that is all the reader can be told. The definition is usable, as the computations for $\mathfrak{sl}_2$ and $\mathfrak{gl}_n$ show, but it gives no positive handle on what a semisimple algebra looks like from the inside. The recharacterization comes in chapter 6, where the Killing form, an invariant symmetric bilinear form built from the traces of the operators $\mathrm{ad}x\,\mathrm{ad}y$, converts the negative condition $\mathrm{rad}\mathfrak{g} = 0$ into the positive and computable condition that the Killing form be nondegenerate. That equivalence, Cartan's criterion for semisimplicity, is what makes the semisimple algebras tractable: nondegeneracy of a bilinear form is checked by a single determinant, whereas the absence of solvable ideals is a statement about every ideal at once. With the nondegenerate form in hand, chapter 6 proves that

a semisimple algebra decomposes as a direct sum of simple ideals, closing the gap between the semisimple algebras and their atoms.

Exercise 5.5.1

1. Verify in full that $\mathrm{rad}\mathfrak{gl}_n(\mathbb{C}) = \mathbb{C}I$, the scalar matrices, and identify the semisimple quotient $\mathfrak{gl}_n/\mathrm{rad}\mathfrak{gl}_n$. Using $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$ from theorem 1.4.3, apply the second isomorphism theorem to identify $\mathfrak{gl}_n/\mathbb{C}I$ with $\mathfrak{sl}_n/(\mathfrak{sl}_n \cap \mathbb{C}I)$, describe precisely what $\mathfrak{sl}_n \cap \mathbb{C}I$ is over $\mathbb{C}$, and conclude that $\mathfrak{gl}_n/\mathrm{rad}\mathfrak{gl}_n \cong \mathfrak{sl}_n$. Explain why the characteristic-zero hypothesis is what makes $\mathfrak{sl}_n \cap \mathbb{C}I = 0$, so that the projective special linear algebra $\mathfrak{psl}_n = \mathfrak{sl}_n/\mathfrak{z}(\mathfrak{sl}_n)$ coincides with $\mathfrak{sl}_n$ here.

2. Give a complete proof that $\mathfrak{sl}_2(\mathbb{C})$ is simple by showing directly that it has no ideal other than 0 and itself, following the strategy of example 5.5.6 but writing out every case ($\gamma \neq 0$, $\gamma = 0 \neq \beta$, $\beta = \gamma = 0 \neq \alpha$). Conclude that $\mathfrak{sl}_2(\mathbb{C})$ is semisimple and that its radical and nilradical are both zero.

3. Prove that a Lie algebra of dimension 1 or 2 is never semisimple. Treat the one-dimensional case (necessarily abelian) and both two-dimensional cases (the abelian algebra, and the unique non-abelian one with basis x, y and $[x, y] = x$), in each case exhibiting a nonzero solvable ideal that is the whole radical.

4. Prove that the radical is preserved by every automorphism: if $\varphi\colon \mathfrak{g} \to \mathfrak{g}$ is a Lie algebra automorphism, then $\varphi(\mathrm{rad}\mathfrak{g}) = \mathrm{rad}\mathfrak{g}$. Deduce the same for the nilradical. (Hint: an automorphism carries solvable ideals to solvable ideals bijectively, hence carries the maximal one to the maximal one.)

5. Exhibit a Lie algebra for which the containment $\mathrm{nil}\,\mathfrak{g} \subsetneq \mathrm{rad}\mathfrak{g}$ is strict, and one for which it is an equality. For the strict case, use the two-dimensional non-abelian algebra of exercise 5.5.1 (3); for the equality case, use the strictly-upper-triangular algebra $\mathfrak{n}_n$ of example 5.2.5. Compute both ideals in each case.

6. Show that although $\mathfrak{g}/\mathrm{rad}\mathfrak{g}$ is always semisimple, the analogous quotient by the nilradical, $\mathfrak{g}/\,\mathrm{nil}\,\mathfrak{g}$, need not be semisimple. Use $\mathfrak{g} = \mathfrak{b}_2$, the 2×2 upper-triangular algebra of example 5.1.7, computing $\mathrm{nil}\,\mathfrak{b}_2$, the quotient, and its radical.

6

Semisimple Lie Algebras and the Cartan Criteria

Chapter 5 defined a semisimple Lie algebra negatively, by the vanishing of its radical, a condition that is hard to check directly because it quantifies over all solvable ideals. This chapter replaces that definition with a positive and computable one. Attached to any Lie algebra is a symmetric bilinear form built purely from the bracket, the Killing form $\kappa(X, Y) = \operatorname{tr}(\operatorname{ad}X \operatorname{ad}Y)$, and Cartan's criterion says that the algebra is semisimple exactly when this form is nondegenerate. Nondegeneracy is a condition one can test by computing a determinant, and it is the single property from which the entire theory of semisimple algebras unfolds.

That leverage is the theme of the chapter. Cartan's criterion for solvability, proved through the Jordan decomposition of a linear operator, detects solvability by the vanishing of certain traces; its contrapositive yields the nondegeneracy criterion for semisimplicity, and nondegeneracy in turn forces a semisimple algebra to break into an orthogonal direct sum of simple ideals. The same invariant form produces a distinguished central element in every representation, the Casimir operator, and the Casimir drives Weyl's theorem that every finite-dimensional representation of a semisimple algebra is completely reducible, the exact algebraic counterpart of the unitarian trick of Part III. The chapter closes with the Levi decomposition, which splits an arbitrary Lie algebra as a semidirect product of its solvable radical and a semisimple subalgebra, so that the classification of all Lie algebras rests on the two extremes treated in this part. The internal anatomy of a single semisimple algebra, its decomposition under a maximal abelian subalgebra, is the subject of chapter 7.

6.1 The Killing Form and Invariant Bilinear Forms

Chapter 5 isolated the two extremes of Lie algebra structure, the solvable algebras at one end and, at the other, the algebras whose radical vanishes, which definition 5.5.5 named semisimple. That definition is negative: it declares an algebra semisimple by the absence of a solvable ideal, and it offers no way to test the condition without first computing the radical. This chapter gives the missing instrument. Attached to every finite-dimensional Lie algebra is a symmetric bilinear form, the Killing form, built entirely from the bracket, and the Cartan criteria of the next two sections will read solvability and semisimplicity directly off its degeneracy. The present section constructs the form, records the invariance that makes it a Lie-algebraic object rather than a mere pairing of vectors, and computes it on the running examples so that the criteria to come have something concrete to certify.

The form does not appear from nowhere. On a compact Lie group there is an Ad-invariant inner product on the Lie algebra, obtained by averaging any inner product over the group with respect to Haar measure, and its infinitesimal signature is a bilinear form B on $\mathfrak{g}$ satisfying an invariance identity under the bracket. The Killing form is the canonical such form, the one manufactured from the adjoint representation with no choices at all. Before constructing it the invariance property it exemplifies deserves a name of its own, because several forms share it and the arguments below use only the invariance, never the particular construction.

Definition 6.1.1: Invariant Bilinear Form

Let $\mathfrak{g}$ be a Lie algebra over a field k. A bilinear form $B\colon \mathfrak{g}\times\mathfrak{g}\to k$ is *invariant* (or *associative*) if

$$B([X,Y],Z) = B(X,[Y,Z]) \qquad \text{for all } X,Y,Z\in\mathfrak{g} \tag{6.1}$$

It is *symmetric* if $B(X,Y)=B(Y,X)$ for all X,Y.

The name "associative" is the reason for the label: writing the form as a product, (6.1) reads $[X,Y]\cdot Z = X\cdot[Y,Z]$, the associative law with one factor a bracket. The equivalent name "invariant" points at what the identity secures. Rewriting (6.1) with W in the middle slot, $B([W,X],Y) = -B(X,[W,Y])$ is, after moving both brackets to one side, the statement

$$B(\operatorname{ad}(W)X,Y) + B(X,\operatorname{ad}(W)Y) = 0 \qquad \text{for all } W,X,Y\in\mathfrak{g} \tag{6.2}$$

using $\operatorname{ad}(W)X=[W,X]$ from definition 2.5.3 and the antisymmetry $[X,W]=-[W,X]$. Thus each operator $\operatorname{ad}(W)$ is skew with respect to B: it behaves like an infinitesimal isometry, which is what differentiating the invariance of B under $\operatorname{Ad}(g)$ gives. An invariant form is one for which the adjoint action is by infinitesimal isometries, and (6.1) and (6.2) are two encodings of this single fact. The equivalence is immediate: unwinding ad, (6.2) reads $B([W,X],Y) = -B(X,[W,Y]) = B(X,[Y,W])$, the last step by the antisymmetry $[W,Y]=-[Y,W]$. The right side $B(X,[Y,W])$ is (6.1) in the form $B([X,Y],W) = B(X,[Y,W])$, and the two identities pass into each other on applying the antisymmetry $[W,X]=-[X,W]$ of the outer bracket, so (6.1) and (6.2) determine one another.

Before constructing the canonical example, a small invariant form on the smallest interesting algebra fixes the picture.

Example 6.1.2: The Trace Form on $\mathfrak{sl}_2$

On $\mathfrak{gl}_n(\mathbb{C})$ the pairing $B(X,Y)=\operatorname{tr}(XY)$ is symmetric, since $\operatorname{tr}(XY)=\operatorname{tr}(YX)$, and invariant: using the trace identity $\operatorname{tr}([A,B]C)=\operatorname{tr}(A[B,C])$, which holds because both sides equal $\operatorname{tr}(ABC)-$

$\operatorname{tr}(BAC) = \operatorname{tr}(ABC) - \operatorname{tr}(ACB)$, one has

$$B([X,Y],Z) = \operatorname{tr}([X,Y]Z) = \operatorname{tr}(X[Y,Z]) = B(X,[Y,Z])$$

Restricting to $\mathfrak{sl}_2(\mathbb{C})$ with the standard basis

$$e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \qquad h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right), \qquad f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$$

direct multiplication gives $\operatorname{tr}(h^2) = 2$, $\operatorname{tr}(ef) = \operatorname{tr}(fe) = 1$, and $\operatorname{tr}(e^2) = \operatorname{tr}(f^2) = \operatorname{tr}(eh) = \operatorname{tr}(fh) = 0$. So $B(h,h) = 2$, $B(e,f) = 1$, and all other basis pairings vanish. In the ordered basis (e,h,f) the Gram matrix is

$$\left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$$

of determinant $-2 \neq 0$, so this trace form is nondegenerate on $\mathfrak{sl}_2$.

The trace form depends on the embedding $\mathfrak{g} \subseteq \mathfrak{gl}_n$ that produced it, and a Lie algebra given abstractly carries no such embedding. What is needed is an invariant form built from the bracket alone, intrinsic to $\mathfrak{g}$. Every Lie algebra does carry one distinguished representation, the adjoint representation of definition 2.5.3, and feeding it into the trace form of $\mathfrak{gl}(\mathfrak{g})$ produces exactly the intrinsic object.

Definition 6.1.3: Killing Form

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k. Its *Killing form* is the bilinear form $\kappa\colon \mathfrak{g} \times \mathfrak{g} \to k$ defined by

$$\kappa(X,Y) = \operatorname{tr}\big(\operatorname{ad}X \circ \operatorname{ad}Y\big) \tag{6.3}$$

the trace of the composite operator $\operatorname{ad}X \circ \operatorname{ad}Y$ on the vector space $\mathfrak{g}$.

The Killing form is manufactured from the bracket and nothing else: $\operatorname{ad}X$ and $\operatorname{ad}Y$ are the operators $Z \mapsto [X,Z]$ and $Z \mapsto [Y,Z]$, so $\kappa(X,Y)$ records how the two adjoint actions compose on $\mathfrak{g}$. It is intrinsic, changing only by the automorphism that any change of basis of $\mathfrak{g}$ induces on the operators, and the trace is unaffected by that. The two properties that make κ a genuine invariant form, symmetry and the invariance of definition 6.1.1, are both consequences of the trace identity used in example 6.1.2.

Proposition 6.1.4: Symmetry and Invariance of the Killing Form

The Killing form κ of a finite-dimensional Lie algebra $\mathfrak{g}$ is symmetric and invariant: $\kappa(X,Y) = \kappa(Y,X)$ and $\kappa([X,Y],Z) = \kappa(X,[Y,Z])$ for all $X,Y,Z \in \mathfrak{g}$.

***Proof*:**
Symmetry is the commutation of trace under composition: $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad}X\operatorname{ad}Y) = \operatorname{tr}(\operatorname{ad}Y\operatorname{ad}X) = \kappa(Y,X)$, since $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for any operators A, B on $\mathfrak{g}$. For invariance, the key is that ad is a Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$, so $\operatorname{ad}[X,Y] = [\operatorname{ad}X, \operatorname{ad}Y] = \operatorname{ad}X\,\operatorname{ad}Y - \operatorname{ad}Y\,\operatorname{ad}X$ by the definition of the bracket on operators. Write $A = \operatorname{ad}X$, $B = \operatorname{ad}Y$, $C = \operatorname{ad}Z$. Then

$$\kappa([X,Y],Z) = \operatorname{tr}\big(\operatorname{ad}[X,Y]\operatorname{ad}Z\big) = \operatorname{tr}\big([A,B]C\big)$$

and the trace identity $\operatorname{tr}([A,B]C) = \operatorname{tr}(A[B,C])$, valid for any three operators because both

sides equal $\operatorname{tr}(ABC) - \operatorname{tr}(BAC)$, gives

$$\operatorname{tr}\big([A,B]C\big) = \operatorname{tr}\big(A[B,C]\big) = \operatorname{tr}\big(\operatorname{ad}X \operatorname{ad}[Y,Z]\big) = \kappa(X,[Y,Z])$$

using $[B,C] = [\operatorname{ad}Y, \operatorname{ad}Z] = \operatorname{ad}[Y,Z]$ once more. Thus $\kappa([X,Y],Z) = \kappa(X,[Y,Z])$, as we sought to show.

The proof isolates why the adjoint representation is the right ingredient: it is the homomorphism property $\operatorname{ad}[X,Y] = [\operatorname{ad}X, \operatorname{ad}Y]$ that converts the outer bracket $[X,Y]$ into an operator commutator, at which point the trace identity does the rest. Any representation φ of $\mathfrak{g}$ would produce an invariant symmetric form $(X,Y) \mapsto \operatorname{tr}(\varphi(X)\varphi(Y))$ by the same argument, a trace form of φ; the Killing form is the trace form of the adjoint representation, singled out because ad is available on every Lie algebra without any auxiliary choice. This canonicity is what the invariance promotes into a symmetry of the whole algebra, recorded next.

The invariance of κ is an infinitesimal statement, expressed through ad. Its integrated form is a genuine invariance under symmetries of $\mathfrak{g}$: the Killing form cannot distinguish an element from its image under any automorphism. On the group side this is the Ad-invariance that motivated the whole construction, and the automorphism statement contains it, since each $\operatorname{Ad}(g)$ is an automorphism of $\mathfrak{g}$. The invariance also forces the degenerate directions of κ to assemble into an ideal, the first structural payoff of the form and the object the Cartan criteria will measure.

Proposition 6.1.5: Automorphism Invariance and the Radical

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k, with Killing form κ.

1. For every automorphism $\varphi \in \operatorname{Aut}(\mathfrak{g})$ of the Lie algebra,

   $$\kappa(\varphi X, \varphi Y) = \kappa(X,Y) \qquad \text{for all } X, Y \in \mathfrak{g}$$

   In particular κ is Ad-invariant: $\kappa(\operatorname{Ad}(g)X, \operatorname{Ad}(g)Y) = \kappa(X,Y)$ whenever $\mathfrak{g}$ is the Lie algebra of a Lie group G and $g \in G$.

2. The *radical* of κ,

   $$\operatorname{rad}\kappa = \{X \in \mathfrak{g} \mid \kappa(X,Y) = 0 \text{ for all } Y \in \mathfrak{g}\}$$

   is an ideal of $\mathfrak{g}$.

***Proof*:**

1. *Automorphism invariance.* The defining relation of an automorphism, $\varphi[X,Z] = [\varphi X, \varphi Z]$, says exactly that $\operatorname{ad}(\varphi X) = \varphi \circ \operatorname{ad}X \circ \varphi^{-1}$ as operators on $\mathfrak{g}$: applying the left side to φZ gives $[\varphi X, \varphi Z] = \varphi[X,Z] = \varphi(\operatorname{ad}X)(Z)$, and since φ is a bijection this identifies the two operators. Consequently

   $$\operatorname{ad}(\varphi X)\operatorname{ad}(\varphi Y) = \varphi\,(\operatorname{ad}X)\,\varphi^{-1}\,\varphi\,(\operatorname{ad}Y)\,\varphi^{-1} = \varphi\,(\operatorname{ad}X \operatorname{ad}Y)\,\varphi^{-1}$$

   a conjugate of $\operatorname{ad}X \operatorname{ad}Y$, and the trace is a conjugation invariant, so

   $$\kappa(\varphi X, \varphi Y) = \operatorname{tr}\big(\varphi\,(\operatorname{ad}X \operatorname{ad}Y)\,\varphi^{-1}\big) = \operatorname{tr}(\operatorname{ad}X \operatorname{ad}Y) = \kappa(X,Y)$$

For the Ad-invariance, each $\mathrm{Ad}(g)\colon \mathfrak{g} \to \mathfrak{g}$ is an automorphism of the Lie algebra (it is the differential of the inner automorphism $x \mapsto gxg^{-1}$, hence respects the bracket), so the general statement applies with $\varphi = \mathrm{Ad}(g)$.

2. *The radical is an ideal.* That $\operatorname{rad}\kappa$ is a linear subspace is immediate from the bilinearity of κ. To see it is an ideal, take $X \in \operatorname{rad}\kappa$ and $W \in \mathfrak{g}$; the claim is $[W, X] \in \operatorname{rad}\kappa$, that is $\kappa([W, X], Y) = 0$ for every $Y \in \mathfrak{g}$. Writing $[W, X] = \mathrm{ad}(W)X$, the skew-adjointness (6.2) of $\mathrm{ad}(W)$ with respect to κ gives

$$\kappa([W, X], Y) = \kappa(\mathrm{ad}(W)X, Y) = -\kappa(X, \mathrm{ad}(W)Y) = -\kappa(X, [W, Y])$$

the middle equality being (6.2), itself a form of the invariance of proposition 6.1.4. Since $X \in \operatorname{rad}\kappa$, the pairing $\kappa(X, \cdot)$ vanishes identically, so $\kappa(X, [W, Y]) = 0$, whence $\kappa([W, X], Y) = 0$. As Y was arbitrary, $[W, X] \in \operatorname{rad}\kappa$, and $\operatorname{rad}\kappa$ is an ideal, completing the proof.

The two statements pull in the same direction. Automorphism invariance says the Killing form is a canonical invariant of the isomorphism class of $\mathfrak{g}$, not an artifact of any presentation: any two isomorphic Lie algebras have Gram matrices related by a change of basis, so quantities like the rank and, over the reals, the signature of κ are genuine invariants of the algebra. The radical is the first such invariant put to structural use. It is the obstruction to nondegeneracy of κ, and being an ideal, it is a candidate for the kind of object semisimplicity forbids. Theorem 6.3.1 will make this precise, showing that $\operatorname{rad}\kappa = 0$ is equivalent to semisimplicity; the radical of the Killing form and the radical of the algebra from definition 5.5.2, distinct objects in general, will there be shown to determine each other.

Structure theory proceeds by cutting a Lie algebra along its ideals, and the Killing form must be compatible with this dissection for it to be a useful tool. The compatibility is as clean as possible: the Killing form of an ideal, formed intrinsically inside the ideal, agrees with the ambient Killing form restricted to it. The point is a trace computation, hinging on the fact that the adjoint operators of an ideal preserve the ideal, so the trace over $\mathfrak{g}$ of a product that lands in the ideal already equals the trace over the ideal.

Proposition 6.1.6: The Killing Form on Ideals and Orthogonal Complements

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k and let $\mathfrak{a} \trianglelefteq \mathfrak{g}$ be an ideal.

1. The Killing form of $\mathfrak{a}$, computed intrinsically in the Lie algebra $\mathfrak{a}$, equals the restriction of the Killing form of $\mathfrak{g}$:

$$\kappa_{\mathfrak{a}}(X, Y) = \kappa_{\mathfrak{g}}(X, Y) \qquad \text{for all } X, Y \in \mathfrak{a}$$

2. The Killing-orthogonal complement

$$\mathfrak{a}^{\perp} = \{X \in \mathfrak{g} \mid \kappa_{\mathfrak{g}}(X, Y) = 0 \text{ for all } Y \in \mathfrak{a}\}$$

is an ideal of $\mathfrak{g}$.

***Proof*:**

1. *Restriction of the Killing form.* Fix $X, Y \in \mathfrak{a}$. On the ambient algebra, the operator $\mathrm{ad}_{\mathfrak{g}}X \,\mathrm{ad}_{\mathfrak{g}}Y$ maps $\mathfrak{g}$ into $\mathfrak{a}$: for any $Z \in \mathfrak{g}$ one has $\mathrm{ad}_{\mathfrak{g}}Y(Z) = [Y, Z] \in \mathfrak{a}$ because $\mathfrak{a}$ is an ideal and $Y \in \mathfrak{a}$, and then $\mathrm{ad}_{\mathfrak{g}}X$ maps $\mathfrak{a}$ into $\mathfrak{a}$ for the same reason. Thus $\mathrm{ad}_{\mathfrak{g}}X \,\mathrm{ad}_{\mathfrak{g}}Y$ is an endomorphism of $\mathfrak{g}$ whose image lies in $\mathfrak{a}$, and whose restriction to $\mathfrak{a}$ is precisely $\mathrm{ad}_{\mathfrak{a}}X \,\mathrm{ad}_{\mathfrak{a}}Y$ (the composite of the intrinsic adjoint operators of $\mathfrak{a}$, which are the restrictions of $\mathrm{ad}_{\mathfrak{g}}X$ and $\mathrm{ad}_{\mathfrak{g}}Y$ to the invariant subspace $\mathfrak{a}$). Choose a basis of $\mathfrak{g}$ extending a basis of $\mathfrak{a}$. In this basis the matrix of $\mathrm{ad}_{\mathfrak{g}}X \,\mathrm{ad}_{\mathfrak{g}}Y$ is block upper-triangular of the form

$$\begin{pmatrix} \mathrm{ad}_{\mathfrak{a}}X \,\mathrm{ad}_{\mathfrak{a}}Y & * \\ 0 & 0 \end{pmatrix}$$

   the lower blocks vanishing because the image lies in $\mathfrak{a}$. Hence

$$\kappa_{\mathfrak{g}}(X, Y) = \mathrm{tr}\big(\mathrm{ad}_{\mathfrak{g}}X \,\mathrm{ad}_{\mathfrak{g}}Y\big) = \mathrm{tr}\big(\mathrm{ad}_{\mathfrak{a}}X \,\mathrm{ad}_{\mathfrak{a}}Y\big) = \kappa_{\mathfrak{a}}(X, Y)$$

   the trace of a block-triangular matrix being the sum of the traces of its diagonal blocks.

2. *The orthogonal complement is an ideal.* Let $X \in \mathfrak{a}^{\perp}$ and $W \in \mathfrak{g}$; the goal is $[W, X] \in \mathfrak{a}^{\perp}$, that is $\kappa_{\mathfrak{g}}([W, X], Y) = 0$ for every $Y \in \mathfrak{a}$. By the skew-adjointness (6.2) of $\mathrm{ad}(W)$ (proposition 6.1.4),

$$\kappa_{\mathfrak{g}}([W, X], Y) = -\kappa_{\mathfrak{g}}(X, [W, Y])$$

   Now $[W, Y] \in \mathfrak{a}$ because $\mathfrak{a}$ is an ideal and $Y \in \mathfrak{a}$, so $[W, Y]$ is one of the elements against which X is orthogonal; hence $\kappa_{\mathfrak{g}}(X, [W, Y]) = 0$. Therefore $\kappa_{\mathfrak{g}}([W, X], Y) = 0$ for all $Y \in \mathfrak{a}$, so $[W, X] \in \mathfrak{a}^{\perp}$, and $\mathfrak{a}^{\perp}$ is an ideal, as we sought to show.

Both statements are used constantly in the sequel. The restriction property (1) means the Killing form is a hereditary invariant: to compute it on an ideal one need not re-embed the ideal in its own adjoint representation, the ambient form already sees the right answer. This is what lets an inductive argument on ideals carry the Killing form along without renegotiating its definition at each step. The orthogonal-complement property (2) is what produces the semisimple decomposition. When κ is nondegenerate, an ideal $\mathfrak{a}$ and its orthogonal complement $\mathfrak{a}^{\perp}$ are complementary ideals summing to $\mathfrak{g}$, and theorem 6.3.3 will iterate exactly this splitting to break a semisimple algebra into simple pieces. The proof of (2) uses only invariance, so the same statement holds for any invariant form, but it is the Killing form's nondegeneracy in the semisimple case that turns the orthogonal complement into a true complement.

The abstract picture is complete once it is grounded in a computation. The running examples $\mathfrak{sl}_2$ and $\mathfrak{gl}_n$ show the Killing form explicitly and, in the case of $\mathfrak{sl}_2$, exhibit the nondegeneracy that the semisimplicity criterion will require. The first computation confirms that the intrinsic Killing form of $\mathfrak{sl}_2$ agrees, up to a scalar, with the concrete trace form of example 6.1.2, an instance of a general proportionality on simple algebras.

Example 6.1.7: The Killing Form of $\mathfrak{sl}_2(\mathbb{C})$

Take $\mathfrak{sl}_2(\mathbb{C})$ with the standard basis (e, h, f) of example 6.1.2, whose brackets are

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h$$

The Killing form is computed from the matrices of $\mathrm{ad}e, \mathrm{ad}h, \mathrm{ad}f$ in this basis. Reading off $\mathrm{ad}h$

from $[h,e]=2e$, $[h,h]=0$, $[h,f]=-2f$, its matrix in the ordered basis (e,h,f) is $\operatorname{diag}(2,0,-2)$, so

$$\kappa(h,h)=\operatorname{tr}\big((\operatorname{ad}h)^2\big)=2^2+0^2+(-2)^2=8$$

For $\operatorname{ad}e$, the brackets $[e,e]=0$, $[e,h]=-2e$, $[e,f]=h$ give the matrix

$$\operatorname{ad}e=\begin{pmatrix}0&-2&0\\0&0&1\\0&0&0\end{pmatrix}$$

and symmetrically $[f,e]=-h$, $[f,h]=2f$, $[f,f]=0$ give

$$\operatorname{ad}f=\begin{pmatrix}0&0&0\\-1&0&0\\0&2&0\end{pmatrix}$$

Then $\operatorname{ad}e\operatorname{ad}f$ has diagonal entries obtained by pairing row i of $\operatorname{ad}e$ with column i of $\operatorname{ad}f$: the entries are $(-2)(-1)=2$ at $(1,1)$, $1\cdot 2=2$ at $(2,2)$, and 0 at $(3,3)$, so the diagonal is $(2,2,0)$; carrying out the multiplication,

$$\operatorname{ad}e\operatorname{ad}f=\begin{pmatrix}2&0&0\\0&2&0\\0&0&0\end{pmatrix},\qquad \kappa(e,f)=\operatorname{tr}(\operatorname{ad}e\operatorname{ad}f)=4$$

Both $\operatorname{ad}e$ and $\operatorname{ad}f$ are strictly triangular nilpotent operators, so $(\operatorname{ad}e)^2$ and $(\operatorname{ad}f)^2$ are traceless, giving $\kappa(e,e)=\kappa(f,f)=0$; and $\operatorname{ad}e\operatorname{ad}h$, $\operatorname{ad}f\operatorname{ad}h$ are strictly upper- and lower-triangular respectively, hence traceless, so $\kappa(e,h)=\kappa(f,h)=0$. The Gram matrix in the basis (e,h,f) is therefore

$$[\kappa]=\begin{pmatrix}0&0&4\\0&8&0\\4&0&0\end{pmatrix}$$

with determinant $-128\neq 0$. The Killing form of $\mathfrak{sl}_2$ is nondegenerate. Comparing with example 6.1.2, where the trace form had Gram matrix $\left(\begin{smallmatrix}0&0&1\\0&2&0\\1&0&0\end{smallmatrix}\right)$, the two forms are proportional, $\kappa=4\operatorname{tr}(XY)$, the instance $n=2$ of the identity computed next.

The nondegeneracy just witnessed is the entire content of the Cartan criterion in this smallest case: $\mathfrak{sl}_2$ is semisimple, and its Killing form is nondegenerate, exactly as theorem 6.3.1 will require in general. The proportionality $\kappa=4\operatorname{tr}(XY)$ does not depend on the choice of basis; it reflects the general behavior of trace forms on $\mathfrak{gl}_n$, which the next computation records in closed form.

Example 6.1.8: The Killing Form of $\mathfrak{gl}_n$ and $\mathfrak{sl}_n$

For $\mathfrak{gl}_n(\mathbb{C})$ the Killing form has the closed expression

$$\kappa(X,Y)=2n\operatorname{tr}(XY)-2\operatorname{tr}(X)\operatorname{tr}(Y) \tag{6.4}$$

To derive it, write $\operatorname{ad}X=L_X-R_X$ on $\mathfrak{gl}_n$, where $L_X(A)=XA$ and $R_X(A)=AX$ are left and right multiplication, which commute. Then

$$\operatorname{ad}X\operatorname{ad}Y=(L_X-R_X)(L_Y-R_Y)=L_XL_Y-L_XR_Y-R_XL_Y+R_XR_Y$$

and the four traces are computed on the space $\mathfrak{gl}_n\cong\mathbb{C}^n\otimes(\mathbb{C}^n)^*$. Left multiplications compose as

$L_X L_Y = L_{XY}$, and L_{XY} acts as XY on each of the n columns, so $\operatorname{tr}(L_X L_Y) = n \operatorname{tr}(XY)$; likewise $R_X R_Y = R_{YX}$ acts as YX on each of the n rows, giving $\operatorname{tr}(R_X R_Y) = n \operatorname{tr}(YX) = n \operatorname{tr}(XY)$. For the mixed terms, $L_X R_Y$ sends the matrix unit E_{ij} to $X E_{ij} Y$, whose E_{ij} coefficient is $X_{ii} Y_{jj}$, so $\operatorname{tr}(L_X R_Y) = \sum_{i,j} X_{ii} Y_{jj} = \operatorname{tr}(X) \operatorname{tr}(Y)$, and by the same count $\operatorname{tr}(R_X L_Y) = \operatorname{tr}(Y) \operatorname{tr}(X)$. Assembling the four,

$$\kappa(X, Y) = n \operatorname{tr}(XY) - \operatorname{tr}(X)\operatorname{tr}(Y) - \operatorname{tr}(Y)\operatorname{tr}(X) + n \operatorname{tr}(XY) = 2n \operatorname{tr}(XY) - 2 \operatorname{tr}(X)\operatorname{tr}(Y)$$

which is (6.4). Restricting to $\mathfrak{sl}_n$, where $\operatorname{tr}(X) = \operatorname{tr}(Y) = 0$, the second term drops and

$$\kappa_{\mathfrak{sl}_n}(X, Y) = 2n \operatorname{tr}(XY) \tag{6.5}$$

so the Killing form of $\mathfrak{sl}_n$ is $2n$ times the trace form. Although (6.5) was obtained by dropping a term from the $\mathfrak{gl}_n$ formula, it is the genuine intrinsic Killing form of $\mathfrak{sl}_n$: since $\mathfrak{sl}_n$ is an ideal, proposition 6.1.6 guarantees the restriction of $\kappa_{\mathfrak{gl}_n}$ agrees with the intrinsic $\kappa_{\mathfrak{sl}_n}$, and the direct $\operatorname{ad}_{\mathfrak{sl}_n}$ computation inside $\mathfrak{sl}_n$ reproduces (6.5). For $n = 2$ this gives $\kappa = 4 \operatorname{tr}(XY)$, confirming the proportionality found in example 6.1.7. On $\mathfrak{gl}_n$ itself the form is degenerate: scalar matrices λI satisfy $\operatorname{ad}(\lambda I) = 0$, so $\kappa(\lambda I, Y) = 0$ for all Y, and the center $\mathbb{C}I$ lies in $\operatorname{rad} \kappa$; this is the imprint of the abelian direction that keeps $\mathfrak{gl}_n$ from being semisimple.

The two computations bracket the phenomenon the chapter is organized around. On $\mathfrak{sl}_n$, a simple algebra, the Killing form is nondegenerate. On $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$, the abelian summand $\mathbb{C}I$ sits inside the radical of κ and destroys nondegeneracy, and this abelian direction is precisely the radical of the algebra in the sense of definition 5.5.2. The alignment of the two radicals in these examples is the instance of the general theorem: nondegeneracy of the Killing form detects the absence of a solvable ideal. That equivalence, Cartan's criterion for semisimplicity, is the destination of the next two sections, and the Killing form built here is the instrument that makes it computable.

Exercise 6.1.1

1. Show that the Killing form of an abelian Lie algebra is identically zero. More generally, if $\mathfrak{g}$ is nilpotent, show that $\kappa \equiv 0$. (Use theorem 5.3.4: every $\operatorname{ad}x$ is nilpotent, so the operator form of Engel's theorem makes them simultaneously strictly upper-triangular, and then each product $\operatorname{ad}x \operatorname{ad}y$ is strictly upper-triangular, hence traceless.)

2. Let $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$ be a direct sum of Lie algebras, meaning $\mathfrak{a}, \mathfrak{b}$ are ideals with $[\mathfrak{a}, \mathfrak{b}] = 0$. Prove that the Killing form is block-diagonal: $\kappa_{\mathfrak{g}}(A + B, A' + B') = \kappa_{\mathfrak{a}}(A, A') + \kappa_{\mathfrak{b}}(B, B')$ for $A, A' \in \mathfrak{a}$ and $B, B' \in \mathfrak{b}$, and in particular $\kappa_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b}) = 0$. Deduce that $\kappa_{\mathfrak{g}}$ is nondegenerate if and only if both $\kappa_{\mathfrak{a}}$ and $\kappa_{\mathfrak{b}}$ are.

3. Compute the Killing form of the two-dimensional nonabelian Lie algebra $\mathfrak{g} = \operatorname{span}(X, H)$ with the single nonzero bracket $[H, X] = X$. Determine its radical and verify that $\mathfrak{g}$ is solvable but κ is nonzero, so the vanishing of κ is not equivalent to solvability. (This is why the criterion of section 6.2 tests traces on the derived algebra, not the Killing form itself.)

4. Let $\mathfrak{g}$ be a real Lie algebra and $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ its complexification. Show that the Killing form of $\mathfrak{g}_{\mathbb{C}}$ is the $\mathbb{C}$-bilinear extension of the Killing form of $\mathfrak{g}$; that is, in a real basis of $\mathfrak{g}$ (also a complex basis of $\mathfrak{g}_{\mathbb{C}}$) the two Gram matrices coincide. Conclude that $\kappa_{\mathfrak{g}}$ is nondegenerate if and only if $\kappa_{\mathfrak{g}_{\mathbb{C}}}$ is.

5. Compute the Killing form of $\mathfrak{so}_3(\mathbb{R})$ in the basis L_1, L_2, L_3 with brackets $[L_1, L_2] = L_3$,

$[L_2, L_3] = L_1$, $[L_3, L_1] = L_2$. Show that κ is negative definite, and relate this to the compactness of the group $SO(3)$.

6. Suppose κ is nondegenerate on a Lie algebra $\mathfrak{g}$ with basis $X_1, \dots, X_n$. Let $X^1, \dots, X^n$ be the dual basis with respect to κ, so $\kappa(X_i, X^j) = \delta_i^j$. Show that for every $W \in \mathfrak{g}$ the two families $([W, X_i])_i$ and $([W, X^j])_j$ are related by $\kappa([W, X_i], X^j) = -\kappa(X_i, [W, X^j])$, and use this to prove that the element $\sum_i X_i \otimes X^i$ is independent of the choice of basis. (This element is the seed of the Casimir operator of section 6.4.)

6.2 Cartan's Criterion for Solvability

Lie's theorem of section 5.4 certified solvability of a linear Lie algebra by producing a simultaneous upper-triangular form: over $\mathbb{C}$, a solvable $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ can be conjugated into the upper-triangular matrices $\mathfrak{b}_n$. That form is a structural conclusion, and it is not directly a test one can run on a given algebra, because verifying it requires finding the flag. Cartan's criterion turns solvability into a computation. It replaces the search for a flag with the vanishing of finitely many numbers, the traces $\operatorname{tr}(XY)$ as X ranges over the derived algebra $[\mathfrak{g}, \mathfrak{g}]$ and Y over $\mathfrak{g}$. A single bilinear form, evaluated on a basis, decides the question. This is the first place where the trace pairing of section 6.1 does structural work, and it is the template for the sharper criterion for semisimplicity that follows in section 6.3.

Throughout this section the ground field is $\mathbb{C}$, algebraically closed and of characteristic zero. Both hypotheses are used, and at different points. Algebraic closure is needed for the Jordan decomposition to split every operator into a diagonalizable and a nilpotent part with eigenvalues in the field, and for Lie's theorem in the forward direction. Characteristic zero enters through a division by an integer that is not invertible in positive characteristic, and it is the reason the criterion fails over $\overline{\mathbb{F}}_p$: the trace of a sum of p equal terms vanishes not because the terms are zero but because $p = 0$ in the field. The point at which characteristic zero is used is flagged explicitly in the proof of the lemma below.

The whole criterion rests on one fact of linear algebra, isolated so that its proof, the only subtle argument in the section, can be given once and cleanly. The setting is a pair of nested subspaces of $\mathfrak{gl}(V)$ and the operators that carry the larger into the smaller under the bracket. The claim is that an operator in this configuration whose self-traces against the configuration all vanish must be nilpotent. The reader should think of B as $\mathfrak{g}$, of A as $[\mathfrak{g}, \mathfrak{g}]$, and of the operators x carrying B into A as containing the derived algebra $[\mathfrak{g}, \mathfrak{g}]$; the lemma will then say that every element of $[\mathfrak{g}, \mathfrak{g}]$ acts nilpotently, which is exactly the leverage needed to prove the theorem by Engel.

Lemma 6.2.1: Trace Criterion for Nilpotence

Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $A \subseteq B \subseteq \mathfrak{gl}(V)$ be subspaces. Set

$$M = \{y \in \mathfrak{gl}(V) \mid [y, B] \subseteq A\}$$

the space of operators carrying B into A under the bracket. If $x \in M$ satisfies $\operatorname{tr}(xy) = 0$ for every $y \in M$, then x is a nilpotent endomorphism of V.

The mechanism deserves a preview, because the proof manufactures an auxiliary operator out of nothing but the eigenvalues of x. Let $x = s+n$ be the Jordan decomposition of x, with s diagonalizable and n nilpotent (the decomposition is cited to [2, Jordan-Chevalley Decomposition]). The task is

to show $s = 0$, for then $x = n$ is nilpotent. Fix a basis $v_1, \dots, v_m$ of V in which s is diagonal with eigenvalues $\lambda_1, \dots, \lambda_m$. To force the λ_i to vanish one pairs s against the operator y that is diagonal in the same basis with the *conjugate* eigenvalues $\overline{\lambda_i}$; then $\operatorname{tr}(sy) = \sum|\lambda_i|^2$, a sum of nonnegative reals which is zero only if all λ_i vanish. The one thing that must be checked is that this conjugate operator y lies in M, so that the hypothesis $\operatorname{tr}(xy) = 0$ applies to it; that verification, achieved by writing $\operatorname{ad}y$ as a polynomial in $\operatorname{ad}s$ with no constant term, is the substance of the proof.

***Proof*:**

Let $x = s + n$ be the Jordan decomposition of $x \in \mathfrak{gl}(V)$, with s diagonalizable, n nilpotent, and $sn = ns$; existence and uniqueness of this decomposition over the algebraically closed field $\mathbb{C}$, together with the fact that s and n are polynomials in x without constant term, are proved in [2, Jordan-Chevalley Decomposition]. Fix a basis $v_1, \dots, v_m$ of V consisting of eigenvectors of s, say $sv_i = \lambda_i v_i$. The goal is $s = 0$, that is $\lambda_i = 0$ for all i, which gives $x = n$ nilpotent.

Step 1: the conjugate-diagonal operator and its adjoint. Define $y \in \mathfrak{gl}(V)$ by $yv_i = \overline{\lambda_i} v_i$, where the bar is complex conjugation; y is diagonal in the basis $v_1, \dots, v_m$ with eigenvalues $\overline{\lambda_1}, \dots, \overline{\lambda_m}$. Let e_{ij} denote the endomorphism sending $v_j \mapsto v_i$ and the other basis vectors to 0, so that $\{e_{ij} \mid 1 \le i, j \le m\}$ is a basis of $\mathfrak{gl}(V)$. A direct computation gives the action of the adjoint of a diagonal operator on this basis: for the diagonal operator s,

$$(\operatorname{ad}s)(e_{ij}) = se_{ij} - e_{ij}s = (\lambda_i - \lambda_j)e_{ij}$$

so $\operatorname{ad}s$ is itself diagonal on $\mathfrak{gl}(V)$ with eigenvalue $\lambda_i - \lambda_j$ on e_{ij}. The same computation for y gives

$$(\operatorname{ad}y)(e_{ij}) = (\overline{\lambda_i} - \overline{\lambda_j})e_{ij} = \overline{\lambda_i - \lambda_j}\, e_{ij} \tag{6.6}$$

so $\operatorname{ad}y$ is diagonal on the same basis with eigenvalue $\overline{\lambda_i - \lambda_j}$ on e_{ij}.

Step 2: $\operatorname{ad}y$ is a polynomial in $\operatorname{ad}s$ with no constant term. Let $\mu_1, \dots, \mu_r$ be the distinct values among the finitely many differences $\lambda_i - \lambda_j$. By Lagrange interpolation there is a polynomial $r(T) \in \mathbb{C}[T]$ with

$$r(\mu_k) = \overline{\mu_k} \qquad \text{for each } k = 1, \dots, r$$

and, since 0 occurs among the μ_k (take $i = j$) and $\overline{0} = 0$, the interpolation can be arranged with $r(0) = 0$, so r has no constant term. Applying r to the diagonal operator $\operatorname{ad}s$ acts eigenvalue by eigenvalue: $r(\operatorname{ad}s)$ sends $e_{ij} \mapsto r(\lambda_i - \lambda_j)e_{ij} = \overline{\lambda_i - \lambda_j}\, e_{ij}$, which by (6.6) is exactly $(\operatorname{ad}y)(e_{ij})$. Since the two operators agree on a basis,

$$\operatorname{ad}y = r(\operatorname{ad}s), \qquad r(0) = 0 \tag{6.7}$$

Step 3: y lies in M. By hypothesis $x \in M$, so $[x, B] \subseteq A$, that is $(\operatorname{ad}x)(B) \subseteq A$. The key point is that $\operatorname{ad}s$ is a constant-term-free polynomial in $\operatorname{ad}x$. This does *not* follow from applying ad to the polynomial expressing s in terms of x: the map $z \mapsto \operatorname{ad}z$ preserves brackets, not products, so for a nonlinear polynomial p one has $\operatorname{ad}(p(x)) \neq p(\operatorname{ad}x)$ in general. The correct route is through the Jordan decomposition of $\operatorname{ad}x$ itself. Because $x = s + n$ with s semisimple, n nilpotent, and $sn = ns$, the operators $\operatorname{ad}s$ and $\operatorname{ad}n$ commute; moreover $\operatorname{ad}s$ is semisimple, since s is (its eigenvalues are the differences $\lambda_i - \lambda_j$ computed in Step 1), and $\operatorname{ad}n$ is nilpotent, since n is. Thus $\operatorname{ad}x = \operatorname{ad}s + \operatorname{ad}n$ is the Jordan decomposition of $\operatorname{ad}x$, and by uniqueness its semisimple part $\operatorname{ad}s$ is a polynomial in $\operatorname{ad}x$ with no constant term (the semisimple part of any operator is a constant-term-free polynomial in it, cited to [2, Jordan-Chevalley Decomposition], the constant term vanishing because a nilpotent operator has semisimple part 0). Hence $\operatorname{ad}s$ carries B into A:

each power $(\mathrm{ad}x)^k$ with $k \geq 1$ sends B into A, since $(\mathrm{ad}x)(B) \subseteq A \subseteq B$ forces $(\mathrm{ad}x)^k(B) \subseteq A$, and $\mathrm{ad}s$ is a constant-term-free combination of these. Now (6.7) writes $\mathrm{ad}y = r(\mathrm{ad}s)$ with $r(0) = 0$, so $\mathrm{ad}y$ is a constant-term-free polynomial in $\mathrm{ad}s$, and therefore $\mathrm{ad}y$ also carries B into A:

$$[y, B] = (\mathrm{ad}y)(B) \subseteq A$$

Thus $y \in M$.

Step 4: the trace pairing forces the eigenvalues to vanish. Since $x \in M$ and $y \in M$, the hypothesis gives $\mathrm{tr}(xy) = 0$. Compute this trace in the eigenbasis. The nilpotent part n of x commutes with s and is nilpotent, and y is diagonal; a nilpotent operator has trace zero, and so does the product ny: writing n as strictly upper-triangular in a basis compatible with the eigenspaces of s, in which y is diagonal, the product ny is strictly upper-triangular, so $\mathrm{tr}(ny) = 0$. Hence the trace splits as

$$0 = \mathrm{tr}(xy) = \mathrm{tr}(sy) + \mathrm{tr}(ny) = \mathrm{tr}(sy) = \sum_{i=1}^{m} \lambda_i \overline{\lambda_i} = \sum_{i=1}^{m} |\lambda_i|^2$$

using $sy\, v_i = \lambda_i \overline{\lambda_i} v_i$. The right-hand side is a sum of nonnegative real numbers, and such a sum vanishes only when every summand does. This is where the reality and ordering of $\mathbb{R}$ inside $\mathbb{C}$ enter, a positivity with no analogue in characteristic p. Therefore $|\lambda_i|^2 = 0$, so $\lambda_i = 0$ for every i, giving $s = 0$ and $x = n$. Hence x is nilpotent, as we sought to show.

The lemma is a statement about a single operator and a linear configuration around it, with no Lie algebra in sight, and its proof is the one place in the section where the field must be $\mathbb{C}$ rather than an abstract algebraically closed field of characteristic zero: complex conjugation and the resulting sum of squares of absolute values are the device that converts the abstract vanishing $\mathrm{tr}(xy) = 0$ into the concrete vanishing of each λ_i. An argument avoiding conjugation runs over any algebraically closed field of characteristic zero by pairing against a $\mathbb{Q}$-linear functional on the $\mathbb{Q}$-span of the eigenvalues instead; the substance is identical, and the conjugation version is chosen here because $\mathbb{C}$ is the standing ground field. What the lemma delivers is nilpotence extracted purely from trace data, and that is precisely the shape of Engel's hypothesis, which is why the criterion can close the loop through Engel.

Example 6.2.2: The Trace Lemma on a Diagonal Operator

Take $V = \mathbb{C}^2$ and $B = A = \mathfrak{gl}(V)$, so that $M = \{y \mid [y, \mathfrak{gl}(V)] \subseteq \mathfrak{gl}(V)\} = \mathfrak{gl}(V)$ is all of $\mathfrak{gl}(V)$. The hypothesis of lemma 6.2.1 then asks for an x with $\mathrm{tr}(xy) = 0$ for *all* y, which since the trace form is nondegenerate on $\mathfrak{gl}(V)$ forces $x = 0$, trivially nilpotent. The content of the lemma is visible only when M is a proper subspace. Consider instead $x = \mathrm{diag}(1, -1)$ and ask whether the hypothesis can hold: pairing against $y = \mathrm{diag}(1, -1) = \overline{x}$ (here x is real) gives $\mathrm{tr}(xy) = 1 + 1 = 2 \neq 0$, so the hypothesis fails and the lemma correctly declines to conclude, since x is diagonalizable with nonzero eigenvalues and is not nilpotent. The conjugate-diagonal pairing $y = \overline{x}$ of the proof is exactly the witness that detects the nonzero eigenvalues: it is the operator whose existence in M obstructs the vanishing hypothesis unless x is nilpotent.

With the linear-algebra core in hand, the criterion itself is a short passage between two theorems already proved. The forward direction is Lie's theorem: a solvable linear algebra is upper-triangularizable, and the derived algebra of an upper-triangular algebra is strictly upper-triangular, on which every trace against an upper-triangular matrix vanishes. The converse is where the lemma

earns its place: the trace hypothesis feeds lemma 6.2.1 to show every element of $[\mathfrak{g},\mathfrak{g}]$ acts nilpotently, whence $[\mathfrak{g},\mathfrak{g}]$ is a nilpotent algebra by Engel, and an algebra with nilpotent derived algebra is solvable.

Theorem 6.2.3: Cartan's Criterion for Solvability

Let V be a finite-dimensional vector space over $\mathbb{C}$ and let $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra. Then $\mathfrak{g}$ is solvable if and only if

$$\operatorname{tr}(XY) = 0 \qquad \text{for all } X \in [\mathfrak{g},\mathfrak{g}] \text{ and all } Y \in \mathfrak{g}$$

***Proof*:**

Forward direction: solvability implies vanishing traces. Suppose $\mathfrak{g}$ is solvable. By Lie's theorem (theorem 5.4.4), applied over the algebraically closed field $\mathbb{C}$ of characteristic zero, there is a basis of V in which every element of $\mathfrak{g}$ is upper-triangular; that is, after conjugation $\mathfrak{g} \subseteq \mathfrak{b}_n$, the upper-triangular matrices, where $n = \dim V$. The commutator of two upper-triangular matrices is strictly upper-triangular, since the diagonal of a product AB of upper-triangular matrices is the entrywise product of the diagonals of A and B, which is symmetric in A and B, so the diagonal of $[A,B] = AB - BA$ vanishes. Therefore $[\mathfrak{g},\mathfrak{g}] \subseteq \mathfrak{n}_n$, the strictly upper-triangular matrices. Now let $X \in [\mathfrak{g},\mathfrak{g}]$ and $Y \in \mathfrak{g}$. Then X is strictly upper-triangular and Y is upper-triangular, so the product XY is strictly upper-triangular, with zero diagonal, and $\operatorname{tr}(XY) = 0$. This proves the vanishing of all such traces.

Converse direction: vanishing traces imply solvability. Suppose $\operatorname{tr}(XY) = 0$ for all $X \in [\mathfrak{g},\mathfrak{g}]$ and all $Y \in \mathfrak{g}$. The strategy is to prove that every element of $[\mathfrak{g},\mathfrak{g}]$ is a nilpotent operator on V; then $[\mathfrak{g},\mathfrak{g}]$ is a Lie algebra of nilpotent operators, nilpotent as an algebra by Engel's theorem, and $\mathfrak{g}$ is solvable because its derived algebra is nilpotent.

Apply the trace lemma with the nested subspaces

$$A = [\mathfrak{g},\mathfrak{g}] \subseteq B = \mathfrak{g} \subseteq \mathfrak{gl}(V)$$

and the associated space

$$M = \{y \in \mathfrak{gl}(V) \mid [y,\mathfrak{g}] \subseteq [\mathfrak{g},\mathfrak{g}]\}$$

Observe that $\mathfrak{g} \subseteq M$: for $y \in \mathfrak{g}$ one has $[y,\mathfrak{g}] \subseteq [\mathfrak{g},\mathfrak{g}] = A$ directly. The plan is to verify the hypothesis of lemma 6.2.1 for an arbitrary $X \in [\mathfrak{g},\mathfrak{g}]$, which will yield nilpotence of every such X at once.

The lemma requires, for such an X, that $\operatorname{tr}(Xy) = 0$ for every $y \in M$. This is stronger than the given hypothesis, which gives vanishing only against $y \in \mathfrak{g}$, so the trace against elements of the larger space M must be produced. The identity that does this is the associativity of the trace form under the bracket, namely

$$\operatorname{tr}([a,b]\,y) = \operatorname{tr}(a\,[b,y]) \tag{6.8}$$

which follows from $\operatorname{tr}((ab-ba)y) = \operatorname{tr}(aby) - \operatorname{tr}(bay) = \operatorname{tr}(a(by-yb)) = \operatorname{tr}(a[b,y])$ using the cyclic invariance $\operatorname{tr}(bay) = \operatorname{tr}(ayb)$. Now let $y \in M$ and let $X = \sum_k [a_k,b_k]$ be an arbitrary element of $[\mathfrak{g},\mathfrak{g}]$ with $a_k, b_k \in \mathfrak{g}$. By (6.8) applied termwise,

$$\operatorname{tr}(Xy) = \sum_k \operatorname{tr}([a_k,b_k]\,y) = \sum_k \operatorname{tr}(a_k\,[b_k,y])$$

For $y \in M$ the bracket $[b_k, y] = -[y, b_k]$ lies in $[y, \mathfrak{g}] \subseteq [\mathfrak{g}, \mathfrak{g}]$, since $b_k \in \mathfrak{g}$ and $[y, \mathfrak{g}] \subseteq [\mathfrak{g}, \mathfrak{g}]$ by the defining condition of M. Thus each $[b_k, y] \in [\mathfrak{g}, \mathfrak{g}]$ and each $a_k \in \mathfrak{g}$, so the given hypothesis applies to the pair $([b_k, y], a_k)$ and gives $\operatorname{tr}(a_k[b_k, y]) = \operatorname{tr}([b_k, y]\, a_k) = 0$ (using cyclicity of the trace to match the hypothesis, which pairs an element of $[\mathfrak{g}, \mathfrak{g}]$ with an element of $\mathfrak{g}$). Summing, $\operatorname{tr}(Xy) = 0$ for every $y \in M$.

The hypothesis of lemma 6.2.1 is therefore met by every $X \in [\mathfrak{g}, \mathfrak{g}]$, so every such X is a nilpotent endomorphism of V. Consequently $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{gl}(V)$ is a Lie subalgebra all of whose elements are nilpotent operators, and by the operator form of Engel's theorem (theorem 5.3.4) the algebra $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent. A Lie algebra whose derived algebra is nilpotent is solvable: nilpotency of $[\mathfrak{g}, \mathfrak{g}]$ gives a chain terminating in 0 under repeated bracketing, and the derived series of $\mathfrak{g}$ from $\mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}]$ onward is dominated by the lower central series of $[\mathfrak{g}, \mathfrak{g}]$, hence also reaches 0. Therefore $\mathfrak{g}$ is solvable, completing the proof.

The criterion is the exact converse in spirit to Lie's theorem, and the two should be read together. Lie's theorem takes solvability and produces a normal form; Cartan's criterion reads traces off that normal form and recovers solvability. The equivalence is not a restatement, because the hypothesis in the converse direction is weaker than upper-triangularizability: one is handed only that certain traces vanish, and the machinery of the trace lemma and Engel is what reconstitutes the structural conclusion. The hypothesis on $[\mathfrak{g}, \mathfrak{g}]$ cannot be weakened to $\mathfrak{g}$. It is not true that a solvable algebra has $\operatorname{tr}(XY) = 0$ for all $X, Y \in \mathfrak{g}$; the upper-triangular algebra $\mathfrak{b}_n$ is solvable, yet $\operatorname{tr}(D^2) = \sum d_i^2 \neq 0$ for a diagonal D with $\sum d_i^2 \neq 0$. It is only against the *derived* algebra, which lands in the strictly upper-triangular part, that the traces are forced to vanish. This asymmetry, traces of derived-against-whole versus whole-against-whole, is the germ of the semisimplicity criterion of the next section, where the Killing form measures precisely the whole-against-whole pairing through the adjoint representation.

Example 6.2.4: Solvability of $\mathfrak{b}_2$ by the Criterion

Let $\mathfrak{b}_2 \subseteq \mathfrak{gl}_2(\mathbb{C})$ be the upper-triangular 2×2 matrices, with basis $H_1 = e_{11}$, $H_2 = e_{22}$, $E = e_{12}$. The brackets are $[H_1, E] = E$, $[H_2, E] = -E$, and $[H_1, H_2] = 0$, so the derived algebra is $[\mathfrak{b}_2, \mathfrak{b}_2] = \mathbb{C}E$, the strictly upper-triangular line. To test solvability by theorem 6.2.3, compute $\operatorname{tr}(XY)$ for $X = E \in [\mathfrak{b}_2, \mathfrak{b}_2]$ and $Y = \alpha H_1 + \beta H_2 + \gamma E$ a general element of $\mathfrak{b}_2$. Since $E = e_{12}$ is strictly upper-triangular and Y is upper-triangular, the product EY is strictly upper-triangular:

$$EY = e_{12}(\alpha e_{11} + \beta e_{22} + \gamma e_{12}) = \beta e_{12}$$

which has zero diagonal, so $\operatorname{tr}(EY) = 0$ for every $Y \in \mathfrak{b}_2$. The criterion is satisfied and confirms $\mathfrak{b}_2$ is solvable, in agreement with its derived series $\mathfrak{b}_2 \supseteq \mathbb{C}E \supseteq 0$. The computation makes the mechanism transparent: the derived algebra sits in the strictly upper-triangular matrices, and every trace against it vanishes exactly because a strictly-upper-triangular times upper-triangular product has no diagonal.

The criterion as proved is a statement about a *linear* Lie algebra, one already presented as operators on a space V. Its most consequential use is the passage to an abstract Lie algebra, one given only by its bracket with no representation attached. The bridge is the adjoint representation of definition 2.5.3: every Lie algebra $\mathfrak{g}$ acts on itself by ad, and the image $\operatorname{ad}(\mathfrak{g}) \subseteq \mathfrak{gl}(\mathfrak{g})$ is a linear Lie algebra to which the criterion applies. The trace pairing on $\operatorname{ad}(\mathfrak{g})$ is by definition the Killing form κ of definition 6.1.3, so the linear criterion becomes a criterion phrased through κ. The one point to handle is the kernel of ad, which is the center $\mathfrak{z}(\mathfrak{g})$; because the center is abelian, hence solvable, solvability of $\operatorname{ad}(\mathfrak{g})$ transfers back to $\mathfrak{g}$ across this abelian kernel.

Corollary 6.2.5: Cartan's Criterion, Abstract Form

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{C}$ with Killing form κ. Then $\mathfrak{g}$ is solvable if and only if

$$\kappa(X, Y) = 0 \qquad \text{for all } X \in [\mathfrak{g}, \mathfrak{g}] \text{ and all } Y \in \mathfrak{g}$$

***Proof*:**

Consider the adjoint representation $\mathrm{ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ of definition 2.5.3, a Lie algebra homomorphism whose image $\mathrm{ad}(\mathfrak{g})$ is a Lie subalgebra of $\mathfrak{gl}(\mathfrak{g})$ and whose kernel is the center $\mathfrak{z}(\mathfrak{g}) = \{Z \mid [Z, W] = 0 \text{ for all } W\}$. By definition of the Killing form (definition 6.1.3), for $X, Y \in \mathfrak{g}$,

$$\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}X\, \mathrm{ad}Y) \tag{6.9}$$

and ad carries brackets to brackets, so $\mathrm{ad}([\mathfrak{g}, \mathfrak{g}]) = [\mathrm{ad}(\mathfrak{g}), \mathrm{ad}(\mathfrak{g})]$.

Forward direction. Suppose $\mathfrak{g}$ is solvable. Then its homomorphic image $\mathrm{ad}(\mathfrak{g})$ is solvable, since solvability passes to homomorphic images (proposition 5.1.6). By theorem 6.2.3 applied to the linear algebra $\mathrm{ad}(\mathfrak{g}) \subseteq \mathfrak{gl}(\mathfrak{g})$, the traces $\mathrm{tr}(PQ)$ vanish for all $P \in [\mathrm{ad}(\mathfrak{g}), \mathrm{ad}(\mathfrak{g})]$ and $Q \in \mathrm{ad}(\mathfrak{g})$. Taking $P = \mathrm{ad}X$ with $X \in [\mathfrak{g}, \mathfrak{g}]$ and $Q = \mathrm{ad}Y$ with $Y \in \mathfrak{g}$, and using (6.9), gives $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}X\, \mathrm{ad}Y) = 0$ for all such X, Y.

Converse direction. Suppose $\kappa(X, Y) = 0$ for all $X \in [\mathfrak{g}, \mathfrak{g}]$ and $Y \in \mathfrak{g}$. Translating through (6.9), this says $\mathrm{tr}(\mathrm{ad}X\, \mathrm{ad}Y) = 0$ for all $X \in [\mathfrak{g}, \mathfrak{g}]$, $Y \in \mathfrak{g}$, that is $\mathrm{tr}(PQ) = 0$ for all $P \in [\mathrm{ad}(\mathfrak{g}), \mathrm{ad}(\mathfrak{g})]$ and $Q \in \mathrm{ad}(\mathfrak{g})$. By theorem 6.2.3 the linear Lie algebra $\mathrm{ad}(\mathfrak{g})$ is solvable. Now $\mathrm{ad}(\mathfrak{g}) \cong \mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ by the first isomorphism theorem, so $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ is solvable. The center $\mathfrak{z}(\mathfrak{g})$ is abelian, hence solvable. An extension of a solvable algebra by a solvable ideal is solvable (proposition 5.1.6): from the short exact sequence $0 \to \mathfrak{z}(\mathfrak{g}) \to \mathfrak{g} \to \mathfrak{g}/\mathfrak{z}(\mathfrak{g}) \to 0$ with both ends solvable, the middle $\mathfrak{g}$ is solvable. This completes the proof.

The abstract corollary is the form in which Cartan's criterion is used throughout the structure theory, because it refers to nothing but the intrinsic bracket of $\mathfrak{g}$ through the Killing form. It converts solvability, a property defined by an iterated construction (the derived series of definition 5.1.3), into a single linear-algebra condition on one bilinear form, testable by computing κ on a basis and checking that the restriction to $[\mathfrak{g}, \mathfrak{g}] \times \mathfrak{g}$ vanishes. The center is the only obstruction to reading the criterion off $\mathrm{ad}(\mathfrak{g})$ directly, and it is a harmless one precisely because abelian algebras are solvable, so no information is lost in the passage through ad. The next section inverts the emphasis: rather than detecting when κ collapses (solvability), it asks when κ is as nondegenerate as possible, and that condition will characterize semisimplicity, the polar opposite of solvability.

Exercise 6.2.1

1. Show that the derived algebra cannot be dropped from theorem 6.2.3: exhibit a solvable Lie subalgebra $\mathfrak{g} \subseteq \mathfrak{gl}_2(\mathbb{C})$ and elements $X, Y \in \mathfrak{g}$ (not both in $[\mathfrak{g}, \mathfrak{g}]$) with $\mathrm{tr}(XY) \neq 0$. Explain why this does not contradict the criterion.

2. Let $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ consist entirely of nilpotent operators and be a Lie subalgebra. Using theorem 6.2.3, give a one-line proof that $\mathfrak{g}$ is solvable, and identify where the trace hypothesis is verified.

3. Compute $\mathrm{tr}(XY)$ for $X = [e, f] = h$ and $Y = h$ in the standard basis $\{e, h, f\}$ of $\mathfrak{sl}_2(\mathbb{C})$ acting

on $\mathbb{C}^2$ (the defining representation), where $e = e_{12}$, $f = e_{21}$, $h = e_{11} - e_{22}$. Conclude from theorem 6.2.3 that $\mathfrak{sl}_2(\mathbb{C})$ is *not* solvable.

4. Explain, in one paragraph, where the proof of lemma 6.2.1 uses characteristic zero, and sketch why the criterion can fail over an algebraically closed field of characteristic p. (Consider a diagonal operator whose p eigenvalues sum to zero in the field but are not all zero.)

5. In the proof of corollary 6.2.5, the converse passes from solvability of $\mathrm{ad}(\mathfrak{g})$ to solvability of $\mathfrak{g}$ across the center. Show by an explicit example that the center can be nonzero and nontrivial, so this step is not vacuous: compute $\mathfrak{z}(\mathfrak{g})$ and $\mathrm{ad}(\mathfrak{g})$ for the three-dimensional Heisenberg algebra $\mathfrak{h}_3$ with basis X, Y, Z and single nonzero bracket $[X, Y] = Z$.

6.3 Cartan's Criterion for Semisimplicity

The solvability criterion of section 6.2 tested a linear Lie algebra for solvability by a trace condition, phrased entirely through the Killing form of the standard representation. The target of the present section is the complementary structural notion: semisimplicity, defined in definition 5.5.5 as the vanishing of the radical. That definition is intrinsic but hard to check, since it asks one to compute the largest solvable ideal and confirm it is zero. Cartan's criterion for semisimplicity replaces this with a single verifiable condition on the Killing form κ of definition 6.1.3: an algebra is semisimple exactly when κ is nondegenerate. Nondegeneracy of a bilinear form is a determinant computation, so semisimplicity becomes something one can decide by writing down a matrix and checking that it is invertible.

The criterion does more than provide a test. Once κ is known to be nondegenerate it becomes a working tool: it identifies $\mathfrak{g}$ with its dual as a $\mathfrak{g}$-module, it splits off ideals by taking orthogonal complements, and it forces every derivation to be inner. These consequences, assembled here, reduce the entire theory of semisimple Lie algebras to the theory of *simple* ones, the algebras with no proper nonzero ideals, which Part II's remaining chapters classify by their root systems. This section is where semisimplicity turns from a definition into a structure theory.

Theorem 6.3.1: Cartan's Criterion for Semisimplicity

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k of characteristic zero. Then $\mathfrak{g}$ is semisimple, in the sense of definition 5.5.5 that its radical is zero, if and only if its Killing form κ is nondegenerate.

Before the proof, it is worth naming the two ideals in play, because the argument is a pincer between them. On one side sits the radical $\operatorname{rad}\mathfrak{g}$ of definition 5.5.2, the largest solvable ideal, whose vanishing is semisimplicity by fiat. On the other sits the radical of the Killing form,

$$\operatorname{rad}\kappa = \{X \in \mathfrak{g} \mid \kappa(X, Y) = 0 \text{ for all } Y \in \mathfrak{g}\}$$

the degenerate subspace of κ, whose vanishing is nondegeneracy. By proposition 6.1.5 the subspace $\operatorname{rad}\kappa$ is an ideal of $\mathfrak{g}$. The theorem asserts these two ideals vanish together, and each direction is a containment: the forward direction shows $\operatorname{rad}\kappa \subseteq \operatorname{rad}\mathfrak{g}$, the converse shows $\operatorname{rad}\mathfrak{g} \subseteq \operatorname{rad}\kappa$ once $\operatorname{rad}\mathfrak{g} \neq 0$ is assumed. Characteristic zero enters through Cartan's solvability criterion in the forward direction, which is the only place a trace argument is invoked.

***Proof*:**
Write $\mathfrak{r} = \operatorname{rad}\mathfrak{g}$ for the radical and $\mathfrak{s} = \operatorname{rad}\kappa$ for the radical of the Killing form.

Step 1: nondegeneracy implies semisimplicity. Assume κ is nondegenerate, so $\mathfrak{s} = 0$. The claim is $\mathfrak{r} = 0$. Suppose not, so $\mathfrak{r}$ is a nonzero solvable ideal. A nonzero solvable ideal contains a nonzero *abelian* ideal of $\mathfrak{g}$: the derived series of definition 5.1.3 of $\mathfrak{r}$ terminates at 0, so there is a last nonzero term $\mathfrak{a} = \mathfrak{r}^{(m)}$ with $\mathfrak{r}^{(m+1)} = [\mathfrak{a},\mathfrak{a}] = 0$, and $\mathfrak{a}$ is abelian. Each derived term $\mathfrak{r}^{(i)}$ is a characteristic ideal of $\mathfrak{r}$ hence an ideal of $\mathfrak{g}$ (it is preserved by every derivation of $\mathfrak{r}$, in particular by $\operatorname{ad}X|_{\mathfrak{r}}$ for $X \in \mathfrak{g}$, since $\mathfrak{r}$ is an ideal), so $\mathfrak{a}$ is a nonzero abelian ideal of $\mathfrak{g}$. It suffices to show $\mathfrak{a} \subseteq \mathfrak{s} = 0$, a contradiction.

Let $X \in \mathfrak{a}$ and $Y \in \mathfrak{g}$. The task is to show $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad}X \operatorname{ad}Y) = 0$. Because $\mathfrak{a}$ is an ideal, $\operatorname{ad}Y$ carries $\mathfrak{g}$ into $\mathfrak{g}$ and $\operatorname{ad}X$ carries $\mathfrak{g}$ into $\mathfrak{a}$; hence the composite $\operatorname{ad}X \operatorname{ad}Y$ maps $\mathfrak{g}$ into $\mathfrak{a}$. Applying it again keeps the image inside $\mathfrak{a}$, and the next chain of inclusions tracks that image down into $[\mathfrak{a},\mathfrak{a}]$:

$$(\operatorname{ad}X \operatorname{ad}Y)^2(\mathfrak{g}) = \operatorname{ad}X \operatorname{ad}Y\big((\operatorname{ad}X \operatorname{ad}Y)(\mathfrak{g})\big) \subseteq \operatorname{ad}X \operatorname{ad}Y(\mathfrak{a}) \subseteq [X,[Y,\mathfrak{a}]] \subseteq [\mathfrak{a},\mathfrak{a}] = 0$$

the last step using that $[Y,\mathfrak{a}] \subseteq \mathfrak{a}$ (again $\mathfrak{a}$ is an ideal) and that $\mathfrak{a}$ is abelian. Thus $\operatorname{ad}X \operatorname{ad}Y$ is a nilpotent operator on $\mathfrak{g}$, its square being zero. A nilpotent operator has trace zero, so $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad}X \operatorname{ad}Y) = 0$. As Y was arbitrary, $X \in \operatorname{rad}\kappa = \mathfrak{s} = 0$, so $X = 0$. Hence $\mathfrak{a} = 0$, contradicting $\mathfrak{a} \neq 0$. Therefore $\mathfrak{r} = 0$ and $\mathfrak{g}$ is semisimple.

Step 2: semisimplicity implies nondegeneracy. Assume $\mathfrak{g}$ is semisimple, so $\mathfrak{r} = 0$. The claim is $\mathfrak{s} = 0$. The subspace $\mathfrak{s} = \operatorname{rad}\kappa$ is an ideal of $\mathfrak{g}$ by proposition 6.1.5, and by construction κ vanishes identically on $\mathfrak{s} \times \mathfrak{s}$; in particular $\kappa(X,Y) = 0$ for all $X, Y \in \mathfrak{s}$, so certainly for all $X \in [\mathfrak{s},\mathfrak{s}]$ and $Y \in \mathfrak{s}$.

Consider the image $\operatorname{ad}(\mathfrak{s}) \subseteq \mathfrak{gl}(\mathfrak{g})$ under the adjoint representation of definition 2.5.3. By proposition 6.1.6 the Killing form of the ideal $\mathfrak{s}$, computed inside $\mathfrak{s}$, agrees with the restriction of κ to $\mathfrak{s} \times \mathfrak{s}$, which vanishes. What the solvability criterion needs, however, is the trace form of $\operatorname{ad}(\mathfrak{s})$ acting on $V = \mathfrak{g}$, namely $\operatorname{tr}(\operatorname{ad}X \operatorname{ad}Y)$ for $X, Y \in \mathfrak{s}$, and this is exactly $\kappa(X,Y)$ computed in $\mathfrak{g}$, which is zero. In particular $\operatorname{tr}(\operatorname{ad}X \operatorname{ad}Y) = 0$ for all $X \in [\mathfrak{s},\mathfrak{s}]$ and $Y \in \mathfrak{s}$. By Cartan's criterion for solvability (theorem 6.2.3, applied to the linear Lie algebra $\operatorname{ad}(\mathfrak{s}) \subseteq \mathfrak{gl}(\mathfrak{g})$, which uses characteristic zero), the algebra $\operatorname{ad}(\mathfrak{s})$ is solvable. The kernel of $\operatorname{ad}|_{\mathfrak{s}}$ is $\mathfrak{s} \cap \mathfrak{z}(\mathfrak{g})$, an abelian hence solvable ideal, so $\mathfrak{s}$ is an extension of the solvable algebra $\operatorname{ad}(\mathfrak{s})$ by a solvable kernel, and therefore solvable by proposition 5.1.6.

Thus $\mathfrak{s}$ is a solvable ideal of $\mathfrak{g}$. By maximality of the radical, $\mathfrak{s} \subseteq \operatorname{rad}\mathfrak{g} = \mathfrak{r} = 0$, so $\mathfrak{s} = 0$ and κ is nondegenerate, as we sought to show.

The proof exposes the mechanism behind the criterion, and each direction uses a different half of the Cartan machinery. The forward direction (nondegeneracy ⇒ semisimple) is elementary linear algebra: a nonzero abelian ideal $\mathfrak{a}$ produces operators $\operatorname{ad}X \operatorname{ad}Y$ that are nilpotent because they push everything into $\mathfrak{a}$ and then die on the second application, and nilpotent operators are invisible to the trace, so $\mathfrak{a}$ lands in the Killing radical. The converse (semisimple ⇒ nondegenerate) is where the solvability criterion earns its place: the Killing radical is an ideal on which κ vanishes, and theorem 6.2.3 converts that vanishing directly into solvability, which a semisimple algebra cannot tolerate. This is the payoff of having proved the solvability criterion first.

The running example shows the criterion.

Example 6.3.2: Semisimplicity of $\mathfrak{sl}_2$ via the Killing Form
Take $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$ with the standard basis

$$e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \qquad h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right), \qquad f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$$

and brackets $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. The Killing form was computed in section 6.1 to have

$$\kappa(h, h) = 8, \qquad \kappa(e, f) = \kappa(f, e) = 4$$

with all other pairings among $\{e, h, f\}$ equal to zero. In the ordered basis (e, h, f) the Gram matrix of κ is therefore

$$[\kappa] = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 8 & 0 \\ 4 & 0 & 0 \end{pmatrix}$$

whose determinant, by cofactor expansion along the middle row, is

$$\det[\kappa] = 8 \cdot \det\left(\begin{smallmatrix} 0 & 4 \\ 4 & 0 \end{smallmatrix}\right) = 8 \cdot (-16) = -128 \neq 0$$

The form is nondegenerate, so theorem 6.3.1 certifies that $\mathfrak{sl}_2$ is semisimple. This matches the direct check: $\mathfrak{sl}_2$ has no nonzero proper ideal at all (any ideal is $\mathrm{ad}h$-invariant, hence a sum of the eigenlines $\mathbb{C}e$, $\mathbb{C}h$, $\mathbb{C}f$ for the eigenvalues $2, 0, -2$, and the brackets $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$ force any nonzero ideal to be all of $\mathfrak{sl}_2$), so its radical is zero and it is in fact simple. The determinant computation reaches the same conclusion without enumerating ideals.

The example illustrates the practical content of the criterion: deciding semisimplicity has become a determinant, and for $\mathfrak{sl}_2$ that determinant is -128. The same computation runs for the classical algebras of theorem 1.4.3. For $\mathfrak{sl}_n$, $\mathfrak{so}_n$, and $\mathfrak{sp}_{2n}$ the Killing form is a nonzero scalar multiple of the trace form $(X, Y) \mapsto \mathrm{tr}(XY)$ of the defining representation, and that trace form is nondegenerate on each of them, so all are semisimple; the coefficients appear in exercise 6.3.1 (1). With semisimplicity now checkable, the next result harvests its structural content: a semisimple algebra breaks into simple pieces, and the Killing form organizes the decomposition.

The decomposition into simple ideals is the structural heart of the theory, and its building blocks are the *simple* Lie algebras of definition 5.5.5: a Lie algebra $\mathfrak{g}$ is simple if it is not abelian and its only ideals are 0 and $\mathfrak{g}$ itself.

The nonabelian clause excludes the one-dimensional algebra, which has no proper ideals but is solvable, hence not a building block of the semisimple world. The theorem below says the simple algebras are precisely those blocks, and that a semisimple algebra is assembled from them with no overlap and no interaction.

Theorem 6.3.3: Decomposition into Simple Ideals

Let $\mathfrak{g}$ be a semisimple Lie algebra over a field of characteristic zero. Then there are ideals $\mathfrak{g}_1, \dots, \mathfrak{g}_r$ of $\mathfrak{g}$, each simple as a Lie algebra, such that

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$$

as Lie algebras, with $[\mathfrak{g}_i, \mathfrak{g}_j] = 0$ for $i \neq j$ and the summands pairwise orthogonal under the Killing form κ. The decomposition is unique: the $\mathfrak{g}_i$ are exactly the minimal nonzero ideals of $\mathfrak{g}$, and every ideal of $\mathfrak{g}$ is the direct sum of the $\mathfrak{g}_i$ it contains.

The proof is an induction that peels one simple ideal off at a time, and the Killing form gives the peeling tool. If $\mathfrak{a}$ is any ideal, its Killing-orthogonal complement $\mathfrak{a}^\perp$ is again an ideal by proposition 6.1.6, and the two intersect in an ideal on which κ vanishes. Nondegeneracy forces that intersection to be zero, so $\mathfrak{g}$ splits as $\mathfrak{a} \oplus \mathfrak{a}^\perp$. Applying this to a *minimal* nonzero ideal makes the first factor simple, and the second factor is a semisimple algebra of smaller dimension on which the induction proceeds.

***Proof*:**
The Killing form κ of $\mathfrak{g}$ is nondegenerate by theorem 6.3.1. The argument has three stages: the orthogonal splitting off an ideal, the induction producing the decomposition, and the uniqueness statement.

Step 1: an ideal and its Killing-orthogonal complement split $\mathfrak{g}$. Let $\mathfrak{a} \trianglelefteq \mathfrak{g}$ be an ideal and set

$$\mathfrak{a}^\perp = \{X \in \mathfrak{g} \mid \kappa(X, Y) = 0 \text{ for all } Y \in \mathfrak{a}\}$$

By proposition 6.1.6 the subspace $\mathfrak{a}^\perp$ is an ideal of $\mathfrak{g}$. The intersection $\mathfrak{a} \cap \mathfrak{a}^\perp$ is an ideal on which κ vanishes identically, so it is contained in $\operatorname{rad}\kappa$, which is 0 by nondegeneracy; hence $\mathfrak{a} \cap \mathfrak{a}^\perp = 0$. A dimension count settles the sum: nondegeneracy of κ gives $\dim \mathfrak{a}^\perp = \dim \mathfrak{g} - \dim \mathfrak{a}$ (the map $\mathfrak{g} \to \mathfrak{a}^*$, $X \mapsto \kappa(X, -)|_{\mathfrak{a}}$ is surjective with kernel $\mathfrak{a}^\perp$, again because κ is nondegenerate on $\mathfrak{g}$), so $\dim(\mathfrak{a} \oplus \mathfrak{a}^\perp) = \dim \mathfrak{g}$ and $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$. Because both are ideals with zero intersection, $[\mathfrak{a}, \mathfrak{a}^\perp] \subseteq \mathfrak{a} \cap \mathfrak{a}^\perp = 0$, so the sum is a direct sum of ideals as Lie algebras: $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$ with the two factors commuting.

Step 2: induction on dimension produces the simple summands. Proceed by induction on $\dim \mathfrak{g}$. If $\mathfrak{g} = 0$ the empty decomposition serves. Otherwise, since $\mathfrak{g} \neq 0$ is finite dimensional it has a minimal nonzero ideal $\mathfrak{a}$ (any nonzero ideal of least dimension). First, $\mathfrak{a}$ is not abelian: an abelian ideal lies in $\operatorname{rad}\mathfrak{g} = 0$, as one sees from Step 1 of the proof of theorem 6.3.1 (a nonzero abelian ideal lands in $\operatorname{rad}\kappa = 0$), so a nonzero minimal ideal cannot be abelian. Next, $\mathfrak{a}$ is simple: any ideal $\mathfrak{b}$ of $\mathfrak{a}$ that is also an ideal of $\mathfrak{g}$ is either 0 or $\mathfrak{a}$ by minimality, and in fact every ideal of $\mathfrak{a}$ is automatically an ideal of $\mathfrak{g}$. To see the latter, apply Step 1 with this $\mathfrak{a}$: $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$ with $[\mathfrak{a}, \mathfrak{a}^\perp] = 0$, so for an ideal $\mathfrak{b} \trianglelefteq \mathfrak{a}$ one has $[\mathfrak{g}, \mathfrak{b}] = [\mathfrak{a}, \mathfrak{b}] + [\mathfrak{a}^\perp, \mathfrak{b}] \subseteq \mathfrak{b} + 0 = \mathfrak{b}$, using $[\mathfrak{a}^\perp, \mathfrak{b}] \subseteq [\mathfrak{a}^\perp, \mathfrak{a}] = 0$. Thus every ideal of $\mathfrak{a}$ is an ideal of $\mathfrak{g}$, and minimality of $\mathfrak{a}$ leaves only 0 and $\mathfrak{a}$; being nonabelian, $\mathfrak{a}$ is simple.

Now $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$ with $\mathfrak{a}$ simple. The complement $\mathfrak{a}^\perp$ is an ideal, and its Killing form is $\kappa|_{\mathfrak{a}^\perp \times \mathfrak{a}^\perp}$ by proposition 6.1.6; this restriction is nondegenerate, for if $X \in \mathfrak{a}^\perp$ were κ-orthogonal to all of $\mathfrak{a}^\perp$ it would also be orthogonal to $\mathfrak{a}$ (by definition of $\mathfrak{a}^\perp$), hence to all of $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$, forcing $X = 0$. By theorem 6.3.1, $\mathfrak{a}^\perp$ is semisimple, of strictly smaller dimension. By the inductive hypothesis $\mathfrak{a}^\perp = \mathfrak{g}_2 \oplus \cdots \oplus \mathfrak{g}_r$ with each $\mathfrak{g}_j$ a simple ideal of $\mathfrak{a}^\perp$; each such $\mathfrak{g}_j$ is an ideal of $\mathfrak{g}$

as well, since $[\mathfrak{g}, \mathfrak{g}_j] = [\mathfrak{a}, \mathfrak{g}_j] + [\mathfrak{a}^\perp, \mathfrak{g}_j] \subseteq 0 + \mathfrak{g}_j$. Setting $\mathfrak{g}_1 = \mathfrak{a}$ gives $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$ with all summands simple ideals, and the distinct summands commute and are κ-orthogonal because each $\mathfrak{g}_j \subseteq \mathfrak{a}^\perp$ for $j \geq 2$ is orthogonal to $\mathfrak{g}_1 = \mathfrak{a}$, and orthogonality among the $\mathfrak{g}_j$ for $j \geq 2$ holds inside $\mathfrak{a}^\perp$ by induction.

Step 3: uniqueness and the ideal-classification. Let $\mathfrak{b}$ be any ideal of $\mathfrak{g}$. For each index i, the bracket $[\mathfrak{b}, \mathfrak{g}_i]$ lies in $\mathfrak{b} \cap \mathfrak{g}_i$, since $\mathfrak{b}$ and $\mathfrak{g}_i$ are both ideals. Now $\mathfrak{b} \cap \mathfrak{g}_i$ is an ideal of the simple algebra $\mathfrak{g}_i$, hence equals 0 or $\mathfrak{g}_i$. Consider the index set $S = \{i \mid \mathfrak{g}_i \subseteq \mathfrak{b}\}$. The claim is $\mathfrak{b} = \bigoplus_{i \in S} \mathfrak{g}_i$. The containment $\bigoplus_{i \in S} \mathfrak{g}_i \subseteq \mathfrak{b}$ is immediate. For the reverse, take $X \in \mathfrak{b}$ and write $X = \sum_i X_i$ with $X_i \in \mathfrak{g}_i$. Since $\mathfrak{g}_i$ is nonabelian and simple, $[\mathfrak{g}_i, \mathfrak{g}_i] = \mathfrak{g}_i$ (it is an ideal of the simple algebra $\mathfrak{g}_i$, nonzero because $\mathfrak{g}_i$ is nonabelian); so each X_i is a sum of brackets $[U, V]$ with $U, V \in \mathfrak{g}_i$. Choosing $U \in \mathfrak{g}_i$, the bracket $[U, X] = [U, X_i] \in [\mathfrak{g}_i, \mathfrak{b}] \subseteq \mathfrak{b} \cap \mathfrak{g}_i$, because brackets across distinct summands vanish. If $\mathfrak{b} \cap \mathfrak{g}_i = \mathfrak{g}_i$, then $i \in S$; if $\mathfrak{b} \cap \mathfrak{g}_i = 0$, then $[U, X_i] = 0$ for all $U \in \mathfrak{g}_i$, so X_i is central in $\mathfrak{g}_i$, and a simple algebra has trivial center, whence $X_i = 0$. Thus $X = \sum_{i \in S} X_i \in \bigoplus_{i \in S} \mathfrak{g}_i$, giving $\mathfrak{b} = \bigoplus_{i \in S} \mathfrak{g}_i$.

Applying this to a minimal nonzero ideal $\mathfrak{b}$ forces S to be a singleton, so $\mathfrak{b} = \mathfrak{g}_i$ for one index: the minimal nonzero ideals are exactly the $\mathfrak{g}_i$. This pins the collection $\{\mathfrak{g}_1, \ldots, \mathfrak{g}_r\}$ down intrinsically, independent of the choices made in Step 2, which is the uniqueness of the decomposition. The description of every ideal as a sum of the $\mathfrak{g}_i$ is the displayed identity $\mathfrak{b} = \bigoplus_{i \in S} \mathfrak{g}_i$, completing the proof.

The decomposition is the organizing theorem of the subject. It says a semisimple Lie algebra carries no information beyond the list of its simple summands: there is no gluing data, no extension, no interaction between the pieces, because the brackets across distinct summands all vanish. The classification of semisimple Lie algebras is therefore the classification of simple ones together with the choice of a multiset, and Part II's remaining chapters carry out the former by attaching to each simple algebra a root system and reading off a Dynkin diagram. Uniqueness of the decomposition matters for that program: because the simple summands are exactly the minimal ideals, they are canonical invariants, so two semisimple algebras are isomorphic if and only if their multisets of simple summands match.

Example 6.3.4: The Decomposition of $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$

Let $\mathfrak{g} = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, the direct sum of two copies of $\mathfrak{sl}_2$, with the bracket taken componentwise and $[(X, 0), (0, Y)] = 0$. Each copy $\mathfrak{g}_1 = \mathfrak{sl}_2 \oplus 0$ and $\mathfrak{g}_2 = 0 \oplus \mathfrak{sl}_2$ is a simple ideal, by example 6.3.2 applied to each factor, and $[\mathfrak{g}_1, \mathfrak{g}_2] = 0$. The Killing form of $\mathfrak{g}$ is the orthogonal sum $\kappa_{\mathfrak{g}}((X_1, X_2), (Y_1, Y_2)) = \kappa_{\mathfrak{g}_1}(X_1, Y_1) + \kappa_{\mathfrak{g}_2}(X_2, Y_2)$, because $\operatorname{ad}(X_1, X_2)$ is block-diagonal and the trace of the product splits across blocks; its Gram matrix is the block-diagonal 6×6 matrix with two copies of the $\mathfrak{sl}_2$ block of example 6.3.2, of determinant $(-128)^2 \neq 0$, so $\mathfrak{g}$ is semisimple. The minimal nonzero ideals are precisely $\mathfrak{g}_1$ and $\mathfrak{g}_2$: any nonzero ideal $\mathfrak{b}$ satisfies $\mathfrak{b} = \bigoplus_{i \in S} \mathfrak{g}_i$ for $S \subseteq \{1, 2\}$ by theorem 6.3.3, so the only ideals are 0, $\mathfrak{g}_1$, $\mathfrak{g}_2$, and $\mathfrak{g}$. There is no ideal "diagonally" embedded across the two factors, because such a subspace would fail to be an ideal: bracketing a diagonal element (X, X) against $(Y, 0)$ gives $([X, Y], 0)$, which is not diagonal unless $[X, Y] = 0$.

The example warns against a natural error. The diagonal copy of $\mathfrak{sl}_2$ inside $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ is a subalgebra isomorphic to $\mathfrak{sl}_2$, but it is not an ideal, so it does not appear in the decomposition; the ideals are rigidly the coordinate factors. This rigidity is exactly what makes the simple summands canonical. The remaining consequences of nondegeneracy are collected next, and they are the properties that make semisimple algebras pleasant to compute with.

Corollary 6.3.5: Structural Consequences of Semisimplicity

Let $\mathfrak{g}$ be a semisimple Lie algebra over a field of characteristic zero. Then:

1. $\mathfrak{g}$ is perfect: $[\mathfrak{g},\mathfrak{g}]=\mathfrak{g}$.
2. Every ideal and every quotient of $\mathfrak{g}$ is semisimple.
3. Every derivation of $\mathfrak{g}$ is inner: the map $\operatorname{ad}\colon \mathfrak{g}\to \operatorname{Der}(\mathfrak{g})$ is an isomorphism of Lie algebras.

Each statement reads off the decomposition or the nondegeneracy of κ. Perfectness is because a simple algebra equals its own derived subalgebra, and $\mathfrak{g}$ is a sum of simple ideals. The stability of semisimplicity under passing to ideals and quotients is because both operations select or collapse whole simple summands. The inner-derivation statement is the one that genuinely uses nondegeneracy, and it is the property that lets one identify the Lie algebra of the automorphism group with $\mathfrak{g}$ itself.

***Proof*:**
Let $\mathfrak{g}=\mathfrak{g}_1\oplus\cdots\oplus\mathfrak{g}_r$ be the decomposition into simple ideals of theorem 6.3.3.

1. *Perfectness.* Each simple summand satisfies $[\mathfrak{g}_i,\mathfrak{g}_i]=\mathfrak{g}_i$: the subspace $[\mathfrak{g}_i,\mathfrak{g}_i]$ is an ideal of the simple algebra $\mathfrak{g}_i$, hence 0 or $\mathfrak{g}_i$, and it is not 0 because $\mathfrak{g}_i$ is nonabelian. Since the summands commute, $[\mathfrak{g},\mathfrak{g}]=\bigoplus_i[\mathfrak{g}_i,\mathfrak{g}_i]=\bigoplus_i\mathfrak{g}_i=\mathfrak{g}$, so $\mathfrak{g}$ is perfect.

2. *Ideals and quotients.* Let $\mathfrak{b}$ be an ideal of $\mathfrak{g}$. By theorem 6.3.3, $\mathfrak{b}=\bigoplus_{i\in S}\mathfrak{g}_i$ for some subset S, a direct sum of simple ideals, hence semisimple by theorem 6.3.1 applied to $\mathfrak{b}$ (its Killing form is the orthogonal sum of the nondegenerate forms $\kappa_{\mathfrak{g}_i}$, and a direct sum of algebras with nondegenerate Killing form has nondegenerate Killing form). For the quotient $\mathfrak{g}/\mathfrak{b}$, write $\mathfrak{b}=\bigoplus_{i\in S}\mathfrak{g}_i$ and let T be the complementary index set; then $\mathfrak{g}/\mathfrak{b}\cong\bigoplus_{i\in T}\mathfrak{g}_i$ as Lie algebras, since $\mathfrak{g}=\mathfrak{b}\oplus\bigoplus_{i\in T}\mathfrak{g}_i$ and the projection restricts to an isomorphism on the complementary summand. A direct sum of simple ideals is semisimple by the same reasoning, so every quotient of $\mathfrak{g}$ is semisimple.

3. *Derivations are inner.* The map $\operatorname{ad}\colon\mathfrak{g}\to\operatorname{Der}(\mathfrak{g})$ is a Lie algebra homomorphism into the derivation algebra, with image the inner derivations $\operatorname{ad}(\mathfrak{g})=\operatorname{Inn}(\mathfrak{g})$. Its kernel is the center $\mathfrak{z}(\mathfrak{g})$, which is a solvable (indeed abelian) ideal, hence contained in $\operatorname{rad}\mathfrak{g}=0$; so ad is injective and $\mathfrak{g}\cong\operatorname{Inn}(\mathfrak{g})$.

   It remains to show $\operatorname{Inn}(\mathfrak{g})=\operatorname{Der}(\mathfrak{g})$. Write $\mathfrak{d}=\operatorname{Der}(\mathfrak{g})$ and $\mathfrak{i}=\operatorname{Inn}(\mathfrak{g})=\operatorname{ad}(\mathfrak{g})$. The inner derivations form an ideal of $\mathfrak{d}$: for a derivation D and an element $X\in\mathfrak{g}$ one has the identity $[D,\operatorname{ad}X]=\operatorname{ad}(DX)$ in $\mathfrak{gl}(\mathfrak{g})$, since for $Y\in\mathfrak{g}$,

   $$[D,\operatorname{ad}X](Y)=D[X,Y]-[X,DY]=[DX,Y]+[X,DY]-[X,DY]=[DX,Y]=\operatorname{ad}(DX)(Y)$$

   using the derivation property $D[X,Y]=[DX,Y]+[X,DY]$. Thus $[\mathfrak{d},\mathfrak{i}]\subseteq\mathfrak{i}$, so $\mathfrak{i}$ is an ideal of $\mathfrak{d}$.

   Now form the Killing-orthogonal complement of $\mathfrak{i}$ inside $\mathfrak{d}$, taken with respect to the Killing form $\kappa_{\mathfrak{d}}$ of $\mathfrak{d}$; call it $\mathfrak{i}^{\perp}$. It is an ideal of $\mathfrak{d}$ by proposition 6.1.6. The Killing form of $\mathfrak{g}$ is nondegenerate and, by proposition 6.1.6 applied to the ideal $\mathfrak{i}\trianglelefteq\mathfrak{d}$, the restriction $\kappa_{\mathfrak{d}}|_{\mathfrak{i}\times\mathfrak{i}}$ equals the Killing form $\kappa_{\mathfrak{i}}$ of $\mathfrak{i}$; under the isomorphism $\operatorname{ad}\colon\mathfrak{g}\xrightarrow{\sim}\mathfrak{i}$ this is $\kappa_{\mathfrak{g}}$, which is

nondegenerate. Hence $\mathfrak{i} \cap \mathfrak{i}^{\perp} = 0$, because it is an ideal on which $\kappa_{\mathfrak{d}}|_{\mathfrak{i}\times\mathfrak{i}}$ vanishes and that form is nondegenerate on $\mathfrak{i}$.

Take any $D \in \mathfrak{i}^{\perp}$. For every $X \in \mathfrak{g}$, the bracket $[D, \mathrm{ad}X] = \mathrm{ad}(DX)$ lies in $\mathfrak{i} \cap \mathfrak{i}^{\perp} = 0$: it is in $\mathfrak{i}$ as the inner derivation $\mathrm{ad}(DX)$, and it is in $\mathfrak{i}^{\perp}$ because $D \in \mathfrak{i}^{\perp}$, $\mathrm{ad}X \in \mathfrak{i} \subseteq \mathfrak{d}$, and $\mathfrak{i}^{\perp}$ is an ideal of $\mathfrak{d}$. Therefore $\mathrm{ad}(DX) = 0$ for all X, and injectivity of ad gives $DX = 0$ for all X, so $D = 0$. Hence $\mathfrak{i}^{\perp} = 0$.

The proof concludes with a dimension count that does not require $\kappa_{\mathfrak{d}}$ itself to be nondegenerate. The pairing

$$\mathfrak{d} \to \mathfrak{i}^{*}, \qquad D \mapsto \kappa_{\mathfrak{d}}(D, -)|_{\mathfrak{i}}$$

has kernel exactly $\mathfrak{i}^{\perp} = 0$, so it is injective and $\dim \mathfrak{d} \leq \dim \mathfrak{i}$. The reverse inequality holds because $\mathfrak{i} \subseteq \mathfrak{d}$, so $\mathfrak{d} = \mathfrak{i}$. Hence every derivation of $\mathfrak{g}$ is inner, and $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{Der}(\mathfrak{g})$ is an isomorphism, completing the proof.

The inner-derivation statement is the one with the longest reach. For a Lie group G with semisimple Lie algebra $\mathfrak{g}$, the derivation algebra $\mathrm{Der}(\mathfrak{g})$ is the Lie algebra of the automorphism group $\mathrm{Aut}(\mathfrak{g})$, and the corollary identifies it with $\mathfrak{g} = \mathrm{ad}(\mathfrak{g})$ acting through the adjoint representation. The automorphism group of a semisimple Lie algebra is therefore, up to components, the adjoint group itself, with no extra continuous symmetries beyond those already present inside G. This closes the loop opened in definition 2.5.3: the adjoint action, which began as one representation among many, turns out to exhaust the infinitesimal symmetries of a semisimple algebra.

With the criterion, the decomposition, and these consequences in place, semisimplicity has been transformed from the intrinsic condition $\mathrm{rad}\,\mathfrak{g} = 0$ into a checkable nondegeneracy condition on a single explicit bilinear form, and the class of semisimple algebras has been resolved into the class of simple ones with no residual gluing data. Everything that follows in Part II, the Cartan subalgebra, the root space decomposition, and the Dynkin classification, operates on a single simple algebra at a time, knowing that the general semisimple case is recovered by taking direct sums. The simple algebras are the atoms, and the next chapters weigh them one by one.

Exercise 6.3.1

1. Using theorem 6.3.1, show that the classical algebras $\mathfrak{sl}_n(\mathbb{C})$ $(n \geq 2)$, $\mathfrak{so}_n(\mathbb{C})$ $(n \geq 3)$, and $\mathfrak{sp}_{2n}(\mathbb{C})$ $(n \geq 1)$ are semisimple. You may use the fact that for each of these $\kappa(X, Y) = c\,\mathrm{tr}(XY)$ for an explicit nonzero constant c (namely $c = 2n$ for $\mathfrak{sl}_n$, $c = n - 2$ for $\mathfrak{so}_n$, and $c = 2n + 2$ for $\mathfrak{sp}_{2n}$), where $\mathrm{tr}(XY)$ is the trace form of the defining representation. The $\mathfrak{sl}_n$ proportionality is (6.5) of section 6.1; the constants for $\mathfrak{so}_n$ and $\mathfrak{sp}_{2n}$ are standard values computed by the same adjoint-trace method (for $\mathfrak{so}_3$ this recovers the numerical Killing form of exercise 6.1.1 (5)) and may be taken as given. Reduce semisimplicity to the nondegeneracy of that trace form on each algebra.

2. Prove that the only semisimple abelian Lie algebra is 0. Deduce that a nonzero abelian Lie algebra has nonzero radical and is never semisimple, and reconcile this with the fact that its Killing form is identically zero.

3. Let k be a field of characteristic zero, let $\mathfrak{g}$ be a Lie algebra over k, and let $K \supseteq k$ be a field extension. Show that the Killing form of the scalar extension $\mathfrak{g}_K = \mathfrak{g} \otimes_k K$ is the K-bilinear extension of $\kappa_{\mathfrak{g}}$, and conclude that $\mathfrak{g}$ is semisimple if and only if $\mathfrak{g}_K$ is semisimple. Why does this show semisimplicity is unaffected by extending the scalar field?

4. Prove that a semisimple Lie algebra has trivial center, $\mathfrak{z}(\mathfrak{g}) = 0$, directly from the definition of semisimplicity, and again from corollary 6.3.5(3). Give an example of a perfect Lie algebra that is not semisimple, showing that perfectness alone does not imply semisimplicity (so the converse of corollary 6.3.5(1) fails).

5. Let $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$ be the decomposition of a semisimple $\mathfrak{g}$ into simple ideals. Prove that a surjective Lie algebra homomorphism $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ onto a nonzero Lie algebra $\mathfrak{h}$ is, up to isomorphism, the projection onto a subsum $\bigoplus_{i \in T} \mathfrak{g}_i$ for some nonempty $T \subseteq \{1, \ldots, r\}$, and hence $\mathfrak{h}$ is semisimple with simple summands a subset of those of $\mathfrak{g}$.

6.4 Complete Reducibility: the Casimir Element and Weyl's Theorem

The Cartan criteria of the previous three sections isolate the semisimple algebras and dissect them into simple pieces, but they say nothing about how such an algebra *acts*. A Lie algebra is only as useful as its representations, and the central structural fact about the representations of a semisimple algebra is that they never tangle. Every finite-dimensional representation of a semisimple Lie algebra over a field of characteristic zero splits as a direct sum of irreducible pieces, with no residual gluing data left over. This is Weyl's theorem on complete reducibility, and it is the algebraic counterpart of a phenomenon the reader has met before: over a finite group in characteristic zero, Maschke's theorem ([3]) forces every representation to decompose, and averaging over the group produces an invariant inner product that makes the decomposition orthogonal. A semisimple Lie algebra has no group to average over, but it carries an element that plays the role the group average played there, the Casimir element, built from a trace form generalizing the Killing form of definition 6.1.3. The Casimir is a single operator that commutes with the entire action and detects nontrivial pieces by acting on them as a nonzero scalar; this scalar is exactly the leverage needed to peel off a summand.

This section builds the proof in three moves. First it fixes the language of representations and complete reducibility. Then it constructs the Casimir element and establishes its two governing properties: it is central in the action, and it acts by a nonzero scalar on any nontrivial irreducible. Finally it deploys the Casimir to prove Weyl's theorem, reducing the splitting of an arbitrary submodule to the splitting off of a codimension-one piece with trivial quotient, where the Casimir does the work directly. The theorem is what makes the highest-weight theory of Part III possible: because everything decomposes, classifying representations reduces to classifying the irreducible ones, and that is a problem the root space decomposition will make combinatorial.

Definition 6.4.1: Representation of a Lie Algebra

Let $\mathfrak{g}$ be a Lie algebra over a field k. A *representation* of $\mathfrak{g}$ on a vector space V is a Lie algebra homomorphism $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$, that is, a linear map satisfying $\varphi([X,Y]) = \varphi(X)\varphi(Y) - \varphi(Y)\varphi(X)$ for all $X, Y \in \mathfrak{g}$. Equivalently, V is a $\mathfrak{g}$-*module*: an action $\mathfrak{g} \times V \to V$, written $(X, v) \mapsto X \cdot v$, that is bilinear and satisfies $[X,Y] \cdot v = X \cdot (Y \cdot v) - Y \cdot (X \cdot v)$. A representation is *faithful* if φ is injective. A subspace $W \subseteq V$ is a *subrepresentation* (or *submodule*) if $\varphi(X)W \subseteq W$ for every $X \in \mathfrak{g}$. The representation is *irreducible* if $V \neq 0$ and its only submodules are 0 and V, and *completely reducible* if V is a direct sum $V = W_1 \oplus \cdots \oplus W_r$ of irreducible submodules.

A representation records how $\mathfrak{g}$ acts by linear operators, exactly as a module over a ring records how the ring acts, and the two viewpoints are interchangeable: the homomorphism φ and the action $X \cdot v = \varphi(X)v$ carry the same information. The Jacobi identity for $\mathfrak{g}$ is precisely what makes the operators $\varphi(X)$ close under commutator into a copy of $\mathfrak{g}$ inside $\mathfrak{gl}(V)$. The two extreme representations bracket the whole theory. The *trivial* representation acts by $\varphi(X) = 0$ for all X on any vector space, so every subspace is a submodule and the one-dimensional trivial module is irreducible; it is the representation that carries no information about $\mathfrak{g}$ at all. At the other extreme stands the *adjoint* representation $\mathrm{ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ of definition 2.5.3, in which $\mathfrak{g}$ acts on itself by $\mathrm{ad}(X)(Y) = [X, Y]$; its submodules are exactly the ideals of definition 5.1.1, so the adjoint representation of a semisimple algebra is completely reducible precisely because such an algebra is a direct sum of simple ideals (theorem 6.3.3).

Complete reducibility is the property Weyl's theorem will establish for all finite-dimensional representations of a semisimple algebra. It is the algebraic form of the unitarian trick: for a compact group one averages an arbitrary inner product over the group to make it invariant, and orthogonal complements of submodules are then themselves submodules, forcing the decomposition. Part III will run this analytic argument in full for compact Lie groups. Here the same conclusion is reached purely algebraically, without any group or any integration, using only the Casimir element and characteristic zero. The standard representation of $\mathfrak{sl}_2(\mathbb{C})$ shows the language.

Example 6.4.2: The Standard and Adjoint Representations of $\mathfrak{sl}_2(\mathbb{C})$

Let $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ with the standard basis

$$e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \qquad h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right), \qquad f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$$

obeying $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. The inclusion $\mathfrak{sl}_2(\mathbb{C}) \hookrightarrow \mathfrak{gl}_2(\mathbb{C})$ is the *standard* representation on $V = \mathbb{C}^2$: here φ is the identity, so $\varphi(e), \varphi(h), \varphi(f)$ are the matrices above. It is faithful and irreducible. Indeed a nonzero submodule W contains some vector $w = (a, b)^{\mathsf{T}} \neq 0$; if $b \neq 0$ then $e \cdot w = (b, 0)^{\mathsf{T}}$ is a nonzero multiple of the first basis vector, and if $b = 0$ then w already is, so W contains $(1, 0)^{\mathsf{T}}$, whence $f \cdot (1, 0)^{\mathsf{T}} = (0, 1)^{\mathsf{T}}$ lies in W too, giving $W = \mathbb{C}^2$. The *adjoint* representation on $\mathfrak{g}$ itself is three-dimensional; in the basis e, h, f one computes $\mathrm{ad}h = \mathrm{diag}(2, 0, -2)$ (from $[h, e] = 2e$, $[h, h] = 0$, $[h, f] = -2f$), while $\mathrm{ad}e$ and $\mathrm{ad}f$ are nilpotent. It too is irreducible: a submodule is an ideal of the simple algebra $\mathfrak{sl}_2(\mathbb{C})$, and the only ideals are 0 and $\mathfrak{g}$. Both representations are already irreducible, hence trivially completely reducible; the force of Weyl's theorem is that their tensor products, sums, and every other finite-dimensional representation split the same way.

The construction of the Casimir element rests on a nondegenerate bilinear form attached to a faithful representation. For the adjoint representation this form is the Killing form of definition 6.1.3, and theorem 6.3.1 guarantees its nondegeneracy for semisimple $\mathfrak{g}$. The same trace construction applied to any faithful representation yields a form that is again nondegenerate, by the very argument that proved the Killing form nondegenerate, and this is the input the Casimir requires.

Proposition 6.4.3: Nondegeneracy of the Trace Form

Let $\mathfrak{g}$ be a semisimple Lie algebra over a field k of characteristic zero and let $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a faithful finite-dimensional representation. The *trace form*

$$\beta_\varphi(X, Y) = \operatorname{tr}\big(\varphi(X)\varphi(Y)\big)$$

is a symmetric, invariant bilinear form on $\mathfrak{g}$, and it is nondegenerate.

***Proof*:**

Symmetry is immediate from $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Invariance in the sense of definition 6.1.1, namely $\beta_\varphi([X, Y], Z) = \beta_\varphi(X, [Y, Z])$, follows from the identity $\operatorname{tr}([A, B]C) = \operatorname{tr}(A[B, C])$ for operators, applied with $A = \varphi(X)$, $B = \varphi(Y)$, $C = \varphi(Z)$ after using that φ is a homomorphism, so $\varphi([X, Y]) = [\varphi(X), \varphi(Y)]$.

For nondegeneracy, let $\mathfrak{s} = \{X \in \mathfrak{g} \mid \beta_\varphi(X, Y) = 0 \text{ for all } Y \in \mathfrak{g}\}$ be the radical of β_φ. Invariance makes $\mathfrak{s}$ an ideal: if $X \in \mathfrak{s}$ and $W \in \mathfrak{g}$ then for all Y, $\beta_\varphi([W, X], Y) = -\beta_\varphi(X, [W, Y]) = 0$, so $[W, X] \in \mathfrak{s}$. Consider the image $\varphi(\mathfrak{s}) \subseteq \mathfrak{gl}(V)$. For $X \in \mathfrak{s}$ and $Y \in \mathfrak{s}$ one has $\operatorname{tr}(\varphi(X)\varphi(Y)) = \beta_\varphi(X, Y) = 0$; in particular this holds for all $X \in [\mathfrak{s}, \mathfrak{s}]$ and $Y \in \mathfrak{s}$. By Cartan's criterion for solvability (theorem 6.2.3), applied to the linear Lie algebra $\varphi(\mathfrak{s}) \subseteq \mathfrak{gl}(V)$, the algebra $\varphi(\mathfrak{s})$ is solvable, and since φ is faithful its restriction to $\mathfrak{s}$ is an isomorphism onto $\varphi(\mathfrak{s})$, so $\mathfrak{s}$ is a solvable ideal of $\mathfrak{g}$. A semisimple algebra has no nonzero solvable ideal (definition 5.5.5), so $\mathfrak{s} = 0$ and β_φ is nondegenerate, as we sought to show.

The proposition is the same theorem as Cartan's semisimplicity criterion, one notch more general: the Killing form is the trace form of the adjoint representation, which is faithful exactly when $\mathfrak{g}$ has trivial center, and a semisimple algebra has trivial center. What the proposition adds is that *any* faithful representation gives a usable nondegenerate form, not only the adjoint one, and the freedom to change representation is what will let the Casimir act correctly on the piece being split off. With a nondegenerate form in hand, dual bases exist, and the Casimir element is their contraction.

Definition 6.4.4: Casimir Element

Let $\mathfrak{g}$ be a semisimple Lie algebra over a field of characteristic zero and $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ a faithful finite-dimensional representation, with nondegenerate trace form β_φ of proposition 6.4.3. Fix a basis $X_1, \dots, X_n$ of $\mathfrak{g}$ and let $Y_1, \dots, Y_n$ be the *dual basis* with respect to β_φ, characterized by $\beta_\varphi(X_i, Y_j) = \delta_{ij}$. The *Casimir element* of φ is the operator

$$c_\varphi = \sum_{i=1}^{n} \varphi(X_i)\,\varphi(Y_i) \in \mathfrak{gl}(V)$$

The dual basis is well defined precisely because β_φ is nondegenerate: nondegeneracy makes the map $Y \mapsto \beta_\varphi(-, Y)$ an isomorphism $\mathfrak{g} \to \mathfrak{g}^*$, so the linear functionals $\beta_\varphi(X_i, -)$ have a unique preimage basis Y_j dualizing them. The definition looks like it depends on the choice of basis, but c_φ does not: it is the image in $\mathfrak{gl}(V)$ of the canonical element of $\mathfrak{g} \otimes \mathfrak{g}$ associated to the isomorphism $\mathfrak{g} \cong \mathfrak{g}^*$, which is basis-independent. Two properties make the Casimir the operator the proof needs, and both are recorded next: it commutes with the whole action, so its eigenspaces are submodules, and its trace pins down its scalar value on an irreducible.

Proposition 6.4.5: Properties of the Casimir Element

Let $\mathfrak{g}$, φ, and c_φ be as in definition 6.4.4, with $n = \dim \mathfrak{g}$. Then:

1. c_φ commutes with $\varphi(X)$ for every $X \in \mathfrak{g}$; that is, c_φ is a $\mathfrak{g}$-module endomorphism of V.
2. $\operatorname{tr}(c_\varphi) = \dim \mathfrak{g}$.
3. If V is irreducible and the representation is nontrivial (that is, $\varphi \neq 0$), then c_φ acts on V as a nonzero scalar, equal to $\dfrac{\dim \mathfrak{g}}{\dim V}$.

***Proof*:**
Throughout, write $[A, B] = AB - BA$ for the commutator in $\mathfrak{gl}(V)$, and abbreviate $\varphi(X)$ by juxtaposition where no confusion arises.

1. *Commutation with the action.* The key is a compatibility between the two dual bases under the bracket. For $X \in \mathfrak{g}$ write the structure constants of $\operatorname{ad}X$ in the chosen bases:

$$[X, X_i] = \sum_j a_{ij}\, X_j, \qquad [X, Y_i] = \sum_j b_{ij}\, Y_j$$

Invariance of β_φ gives a relation between the coefficient matrices $a = (a_{ij})$ and $b = (b_{ij})$. Indeed

$$a_{ik} = \beta_\varphi\Big(\sum_j a_{ij} X_j,\, Y_k\Big) = \beta_\varphi([X, X_i], Y_k) = -\beta_\varphi(X_i, [X, Y_k]) = -\beta_\varphi\Big(X_i, \sum_j b_{kj} Y_j\Big) = -b_{ki}$$

using $\beta_\varphi([X, X_i], Y_k) = -\beta_\varphi([X_i, X], Y_k) = -\beta_\varphi(X_i, [X, Y_k])$ from invariance. Hence $a_{ik} = -b_{ki}$ for all i, k. Now compute the commutator of $\varphi(X)$ with c_φ using the Leibniz rule $[\varphi(X), AB] = [\varphi(X), A]B + A[\varphi(X), B]$ and $[\varphi(X), \varphi(Y_i)] = \varphi([X, Y_i])$:

$$[\varphi(X), c_\varphi] = \sum_i \big[\varphi(X), \varphi(X_i)\varphi(Y_i)\big] = \sum_i \varphi([X, X_i])\varphi(Y_i) + \sum_i \varphi(X_i)\varphi([X, Y_i])$$

Substituting the structure constants, the first sum is $\sum_{i,j} a_{ij}\, \varphi(X_j)\varphi(Y_i)$ and the second is $\sum_{i,j} b_{ij}\, \varphi(X_i)\varphi(Y_j)$. Reindexing the second sum by swapping the names $i \leftrightarrow j$ turns it into $\sum_{i,j} b_{ji}\, \varphi(X_j)\varphi(Y_i)$, and since $b_{ji} = -a_{ij}$ the two sums cancel term by term:

$$[\varphi(X), c_\varphi] = \sum_{i,j} (a_{ij} + b_{ji})\, \varphi(X_j)\varphi(Y_i) = 0$$

Thus c_φ commutes with every $\varphi(X)$, so it is an endomorphism of V as a $\mathfrak{g}$-module.

2. *The trace.* Taking traces in the definition and using $\operatorname{tr}(\varphi(X_i)\varphi(Y_i)) = \beta_\varphi(X_i, Y_i) = 1$ from the dual-basis relation,

$$\operatorname{tr}(c_\varphi) = \sum_{i=1}^{n} \operatorname{tr}\big(\varphi(X_i)\varphi(Y_i)\big) = \sum_{i=1}^{n} \beta_\varphi(X_i, Y_i) = \sum_{i=1}^{n} 1 = n = \dim \mathfrak{g}$$

3. *The scalar on an irreducible.* By part (1) the operator c_φ is a $\mathfrak{g}$-module endomorphism of the irreducible module V. Since the ground field is algebraically closed (or, in general, after extending scalars, which does not change traces), Schur's lemma forces $c_\varphi = \lambda \,\mathrm{id}_V$ for a scalar $\lambda \in k$. Taking traces, $\mathrm{tr}(c_\varphi) = \lambda \dim V$, and by part (2) this equals $\dim \mathfrak{g}$, so $\lambda = \dim \mathfrak{g} / \dim V$. It remains to see $\lambda \neq 0$. Since φ is a homomorphism from the semisimple algebra $\mathfrak{g}$ and V is nontrivial, $\varphi(\mathfrak{g}) \neq 0$; but $\mathfrak{g}$ is perfect, $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$ by corollary 6.3.5, so $\varphi(\mathfrak{g}) = \varphi([\mathfrak{g},\mathfrak{g}]) = [\varphi(\mathfrak{g}), \varphi(\mathfrak{g})]$ consists of commutators, each of trace zero. If λ were zero then $c_\varphi = 0$, whence $\mathrm{tr}(c_\varphi) = 0$; but $\mathrm{tr}(c_\varphi) = \dim \mathfrak{g} > 0$, a contradiction. Concretely $\lambda = \dim \mathfrak{g} / \dim V \neq 0$, completing the proof.

The three properties are exactly calibrated to the splitting argument. Part (1) says the Casimir respects the module structure, so its kernel and image are submodules and its eigenspaces are submodules; this is what lets it separate V into invariant pieces. Part (3) says it separates the nontrivial from the trivial: on a nontrivial irreducible it is invertible, being a nonzero scalar, whereas on any trivial module $\varphi = 0$ makes $c_\varphi = 0$ act as zero. A single operator that is invertible on the interesting part and zero on the trivial part is precisely a projector waiting to be read off, and reading it off is the content of Weyl's theorem. The computation for $\mathfrak{sl}_2(\mathbb{C})$ shows the scalar in a case one can check by hand.

Example 6.4.6: The Casimir of the Standard Representation of $\mathfrak{sl}_2(\mathbb{C})$

Take $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ with basis e, h, f as in example 6.4.2, and let φ be the standard representation on $V = \mathbb{C}^2$, so $\varphi(e), \varphi(h), \varphi(f)$ are the matrices there. The trace form is $\beta_\varphi(X,Y) = \mathrm{tr}(XY)$ on 2×2 matrices; a direct computation gives

$$\beta_\varphi(h,h) = \mathrm{tr}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = 2, \qquad \beta_\varphi(e,f) = \mathrm{tr}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) = 1, \qquad \beta_\varphi(e,e) = \beta_\varphi(f,f) = \beta_\varphi(h,e) = \beta_\varphi(h,f) = 0$$

The dual basis to (e,h,f) under β_φ is $(f, \frac{1}{2}h, e)$: one checks $\beta_\varphi(e,f) = 1$, $\beta_\varphi(h, \frac{1}{2}h) = 1$, $\beta_\varphi(f,e) = 1$, with all cross terms zero. Hence

$$c_\varphi = \varphi(e)\varphi(f) + \varphi(h)\varphi(\tfrac{1}{2}h) + \varphi(f)\varphi(e) = \varphi(e)\varphi(f) + \tfrac{1}{2}\,\varphi(h)^2 + \varphi(f)\varphi(e)$$

Computing the products,

$$\varphi(e)\varphi(f) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \qquad \varphi(f)\varphi(e) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right), \qquad \tfrac{1}{2}\,\varphi(h)^2 = \tfrac{1}{2}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$$

so $c_\varphi = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right) + \frac{1}{2}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = \frac{3}{2}\,\mathrm{id}_{\mathbb{C}^2}$. The Casimir acts as the scalar $\frac{3}{2}$, in exact agreement with proposition 6.4.5(3): here $\dim \mathfrak{g} = 3$ and $\dim V = 2$, and $\dim \mathfrak{g} / \dim V = \frac{3}{2}$. The scalar is nonzero, as it must be on a nontrivial irreducible, and $\mathrm{tr}(c_\varphi) = \frac{3}{2} \cdot 2 = 3 = \dim \mathfrak{sl}_2(\mathbb{C})$.

Everything is now in place for Weyl's theorem. The strategy is to show that every submodule of a finite-dimensional representation admits a complementary submodule, which by an easy induction gives the full decomposition. The splitting of a general submodule is reduced in two stages to a single sharp case, a codimension-one submodule whose quotient is trivial, and there the Casimir splits it directly. The first stage is a formal device that converts the search for a complement into the search for a submodule complement of a codimension-one piece inside a space of homomorphisms; the second stage handles that piece with the Casimir. The reduction to codimension one is recorded first as a lemma, since it is the technical crux and its proof is where the Casimir enters.

Lemma 6.4.7: Splitting a Codimension-One Trivial Quotient

Let $\mathfrak{g}$ be a semisimple Lie algebra over a field of characteristic zero, and let $\psi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a finite-dimensional representation with a submodule $W \subseteq V$ of codimension one on which V/W is the trivial module. Then W has a one-dimensional $\mathfrak{g}$-invariant complement: there is a one-dimensional submodule $L \subseteq V$ with $V = W \oplus L$.

***Proof*:**

Because V/W is trivial and one-dimensional, $\psi(\mathfrak{g})V \subseteq W$: every X sends V into W, since it acts as zero on the quotient. The proof runs by induction on $\dim V$, reducing at each step to a smaller instance of the same configuration and finishing with the Casimir when W is irreducible.

Step 1: reduction to W irreducible. Suppose W contains a proper nonzero submodule $W' \subsetneq W$. Then W/W' is a submodule of V/W' of codimension one, and the quotient $(V/W')/(W/W') \cong V/W$ is trivial. By the inductive hypothesis applied to V/W' (of dimension $\dim V - \dim W' < \dim V$), the submodule W/W' has a one-dimensional invariant complement, so there is a submodule $\widetilde{W} \subseteq V$ with $W' \subseteq \widetilde{W}$, $\dim \widetilde{W} = \dim W' + 1$, and $V/W' = (W/W') \oplus (\widetilde{W}/W')$. In particular $\widetilde{W} \cap W = W'$ and $\widetilde{W} + W = V$, so $\widetilde{W}$ meets W in the codimension-one submodule W' of $\widetilde{W}$, and $\widetilde{W}/W'$ is the trivial one-dimensional module. Applying the inductive hypothesis a second time, now to $\widetilde{W}$ with its codimension-one submodule W' (and $\dim \widetilde{W} < \dim V$ because $W' \neq 0$), yields a one-dimensional submodule $L \subseteq \widetilde{W}$ with $\widetilde{W} = W' \oplus L$. Then $L \cap W = L \cap \widetilde{W} \cap W = L \cap W' = 0$ and $\dim L = 1 = \dim V - \dim W$, so $V = W \oplus L$, as required. This reduces the lemma to the case where W has no proper nonzero submodule, that is, W is irreducible (or zero, in which case V is trivial one-dimensional and any $L = V$ works).

Step 2: the irreducible case via the Casimir. Assume now W is a nonzero irreducible submodule of codimension one with V/W trivial. If $\mathfrak{g}$ acts trivially on W as well, then $\psi(\mathfrak{g})V \subseteq W$ and $\psi(\mathfrak{g})W = 0$ give $\psi(\mathfrak{g})^2 V = 0$, so $\psi(\mathfrak{g})$ is a commutative algebra of operators; but $\psi(X) = \psi([\dots])$ is a sum of commutators since $\mathfrak{g}$ is perfect (corollary 6.3.5), forcing $\mathrm{tr}\psi(X) = 0$, and more strongly $\psi(\mathfrak{g}) = \psi([\mathfrak{g},\mathfrak{g}]) = [\psi(\mathfrak{g}), \psi(\mathfrak{g})] = 0$ because $\psi(\mathfrak{g})$ commutes with itself. Thus $\psi = 0$, V is a trivial two-dimensional module, and any line L complementary to W is a submodule. So assume W is nontrivial. Then $\ker\psi$ is an ideal of $\mathfrak{g}$ not containing all of $\mathfrak{g}$, and $\mathfrak{g}/\ker\psi$ acts faithfully; since a quotient of a semisimple algebra is semisimple (corollary 6.3.5), the representation factors through a faithful representation of a semisimple algebra. Replacing $\mathfrak{g}$ by $\mathfrak{g}/\ker\psi$ if necessary, assume ψ is faithful, so the Casimir element $c = c_\psi$ of definition 6.4.4 is defined. By proposition 6.4.5(1), c is a $\mathfrak{g}$-module endomorphism of V. Since $\psi(\mathfrak{g})V \subseteq W$ and c is a sum of products $\psi(X_i)\psi(Y_i)$, the image of c lies in W; and c acts as 0 on the trivial quotient V/W, since each $\psi(X_i)$ does. Because c commutes with the action by proposition 6.4.5(1) and W is irreducible, Schur's lemma forces $c|_W = \lambda\,\mathrm{id}_W$ for some scalar $\lambda \in k$. The nonvanishing of λ is where semisimplicity and characteristic zero enter, and it comes from the trace of c over all of V, not from any statement about c restricted to the proper submodule W: since c maps V into W and acts as zero on V/W, its trace over V equals its trace over W, so $\mathrm{tr}_V(c) = \lambda \dim W$. By proposition 6.4.5(2), $\mathrm{tr}_V(c) = \dim\mathfrak{g}$, which is nonzero because ψ is faithful and $\mathfrak{g} \neq 0$. Hence $\lambda = \dim\mathfrak{g}/\dim W \neq 0$, so c is invertible on W. Therefore $\ker c$ is a submodule of V with $\ker c \cap W = 0$ and, since c acts as zero on the one-dimensional V/W, the operator c has rank equal to $\dim W$, so $\dim\ker c = \dim V - \dim W = 1$. Hence $L := \ker c$ is a one-dimensional submodule with $V = W \oplus L$, completing the proof.

The lemma is where semisimplicity and characteristic zero do their work, and it is worth marking exactly where. Semisimplicity enters twice: through $[\mathfrak{g},\mathfrak{g}]=\mathfrak{g}$, which kills the trivial-on-W degenerate case and forces $\operatorname{tr}\psi(X)=0$, and through the Casimir, whose existence rests on the nondegenerate trace form of a faithful representation of a semisimple algebra. Characteristic zero enters through the nonvanishing of $\lambda=\dim\mathfrak{g}/\dim W$; over a field of characteristic p dividing $\dim W$, the scalar could be zero and the argument would collapse, which is exactly why Weyl's theorem is false in positive characteristic. With the codimension-one case settled, the general theorem follows by the Hom-space device that converts an arbitrary complement problem into this one.

Theorem 6.4.8: Weyl's Theorem on Complete Reducibility

Let $\mathfrak{g}$ be a semisimple Lie algebra over a field k of characteristic zero. Then every finite-dimensional representation of $\mathfrak{g}$ is completely reducible: it decomposes as a direct sum of irreducible submodules. Equivalently, every submodule W of a finite-dimensional $\mathfrak{g}$-module V has a $\mathfrak{g}$-invariant complement.

The two formulations are equivalent by a standard induction. If every submodule has an invariant complement, then given $V\neq 0$ pick a nonzero submodule of smallest dimension, which is irreducible, split it off, and repeat on the complement; the dimension drops each time, so after finitely many steps V is exhausted as a direct sum of irreducibles. Conversely a completely reducible module has the complementation property, since a submodule of $\bigoplus W_i$ is complemented by a sub-sum of the W_i. So it suffices to prove the complementation statement, and that is what the Casimir delivers through the codimension-one lemma.

***Proof*:**
Let $W\subseteq V$ be a submodule of a finite-dimensional $\mathfrak{g}$-module; the task is to produce an invariant complement. If $W=0$ or $W=V$ there is nothing to prove, so assume $0\subsetneq W\subsetneq V$.

Step 1: the Hom-space model of a complement. Consider the vector space $\operatorname{Hom}_k(V,W)$ of all k-linear maps $V\to W$, made into a $\mathfrak{g}$-module by the standard action on a Hom-space,

$$(X\cdot T)(v)=X\cdot\big(T(v)\big)-T\big(X\cdot v\big),\qquad X\in\mathfrak{g},\ T\in\operatorname{Hom}_k(V,W),\ v\in V$$

which satisfies the module axiom because $\mathfrak{g}$ acts on both V and W. A linear map $T\colon V\to W$ is a $\mathfrak{g}$-module homomorphism precisely when $X\cdot T=0$ for all X, that is, when T lies in the submodule of invariants. Inside $\operatorname{Hom}_k(V,W)$ single out two subspaces:

$$\mathcal{V}=\{T\in\operatorname{Hom}_k(V,W)\mid T|_W=\mu\,\mathrm{id}_W\text{ for some }\mu\in k\},\qquad \mathcal{W}=\{T\in\mathcal{V}\mid T|_W=0\}$$

Thus $\mathcal{V}$ consists of those linear maps whose restriction to W is a scalar, and $\mathcal{W}$ of those killing W. The map $T\mapsto\mu$ (the scalar by which T acts on W) is linear from $\mathcal{V}$ onto k with kernel $\mathcal{W}$, so $\mathcal{W}$ has codimension one in $\mathcal{V}$.

Step 2: $\mathcal{V}$ and $\mathcal{W}$ are submodules and $\mathcal{V}/\mathcal{W}$ is trivial. Both $\mathcal{V}$ and $\mathcal{W}$ are $\mathfrak{g}$-submodules of $\operatorname{Hom}_k(V,W)$. To see this, let $T\in\mathcal{V}$ with $T|_W=\mu\,\mathrm{id}_W$ and $X\in\mathfrak{g}$; for $w\in W$,

$$(X\cdot T)(w)=X\cdot(Tw)-T(X\cdot w)=X\cdot(\mu w)-\mu(X\cdot w)=0$$

using $Tw=\mu w$ and $X\cdot w\in W$ so $T(X\cdot w)=\mu(X\cdot w)$. Hence $(X\cdot T)|_W=0$, which shows both that $X\cdot T\in\mathcal{V}$ (its restriction to W is the scalar 0) and that $X\cdot T\in\mathcal{W}$. Therefore $\mathcal{V}$ and $\mathcal{W}$ are submodules, $X\cdot\mathcal{V}\subseteq\mathcal{W}$, and the quotient $\mathcal{V}/\mathcal{W}$ is one-dimensional with trivial $\mathfrak{g}$-action.

Step 3: split off the trivial line and read the complement. The pair $\mathcal{W} \subseteq \mathcal{V}$ satisfies the hypotheses of lemma 6.4.7: $\mathcal{W}$ is a codimension-one submodule with trivial quotient $\mathcal{V}/\mathcal{W}$. The lemma provides a one-dimensional invariant complement $\mathcal{L} \subseteq \mathcal{V}$ with $\mathcal{V} = \mathcal{W} \oplus \mathcal{L}$, and $\mathfrak{g}$ acts trivially on $\mathcal{L}$ (any one-dimensional complement to $\mathcal{W}$ in $\mathcal{V}$ maps isomorphically to the trivial $\mathcal{V}/\mathcal{W}$, and $X \cdot \mathcal{L} \subseteq \mathcal{W} \cap \mathcal{L} = 0$). Pick a nonzero $T \in \mathcal{L}$. Scaling, arrange $T|_W = \mathrm{id}_W$: this is possible because $T \notin \mathcal{W}$ means the scalar μ for T is nonzero, so $\mu^{-1}T$ restricts to id_W. Since $\mathfrak{g}$ acts trivially on $\mathcal{L}$, $X \cdot T = 0$ for all X, that is, T is a $\mathfrak{g}$-module homomorphism $V \to W$ restricting to the identity on W. Then T is a $\mathfrak{g}$-equivariant projection of V onto W, and its kernel

$$L = \ker T = \{v \in V \mid Tv = 0\}$$

is a submodule (kernels of module homomorphisms are submodules) satisfying $L \cap W = 0$ (as $T|_W = \mathrm{id}_W$) and $\dim L = \dim V - \dim W$ (as T is surjective onto W). Hence $V = W \oplus L$ is a decomposition into submodules, so W has an invariant complement.

Step 4: from complementation to complete reducibility. The complementation property just proved yields the full decomposition by induction on $\dim V$, as explained before the proof: choose an irreducible submodule W_1 (a nonzero submodule of least dimension is irreducible), split $V = W_1 \oplus V'$ with V' a submodule, and apply the inductive hypothesis to V'. Assembling the pieces expresses V as a direct sum of irreducible submodules, completing the proof.

The proof deserves a second reading for the shape of its reduction, because the Hom-space device is the move that recurs whenever one wants to promote a numerical splitting to an equivariant one. A complement to W inside V is the same data as a $\mathfrak{g}$-equivariant projection $V \to W$ that is the identity on W; the linear projections onto W form an affine subspace of $\mathrm{Hom}_k(V, W)$, and the question is whether an equivariant one exists among them. Recasting this as splitting the codimension-one submodule $\mathcal{W} \subseteq \mathcal{V}$ turns an abstract equivariance question into the one concrete case the Casimir can settle, and the Casimir settles it by acting invertibly on the nontrivial part and by zero on the trivial line. The whole theorem is thus a single scalar computation, the nonvanishing of the Casimir's eigenvalue $\lambda = \dim \mathfrak{g}/\dim W$ on the irreducible piece, which rests on the trace identity $\mathrm{tr}(c) = \dim \mathfrak{g}$ of proposition 6.4.5(2), propagated through two formal reductions.

The consequence is that the representation theory of a semisimple Lie algebra looks like the representation theory of a finite group in characteristic zero: everything decomposes, and to know all finite-dimensional representations it suffices to know the irreducible ones and how they combine. This is the structural fact that makes the highest-weight theory of Part III possible. Once Chapter 7 breaks a semisimple algebra into a Cartan subalgebra and root spaces, each finite-dimensional module will decompose into weight spaces under the Cartan subalgebra, and because Weyl's theorem guarantees the module is a sum of irreducibles, classifying the irreducibles by their highest weight classifies all of them. Without complete reducibility one would be forced to track extension data between irreducibles, an infinite regress that the theorem eliminates in one stroke. The same conclusion will return analytically in Part III, where the compactness of a group replaces semisimplicity of an algebra and Haar-measure averaging replaces the Casimir, but the algebraic route here is the one that continues to work in noncompact and purely algebraic settings.

The theorem also identifies the precise obstruction in the non-semisimple case. A Lie algebra that is not semisimple has a nonzero solvable radical, and its representations need not decompose; the smallest example already occurs in dimension two, where a non-abelian solvable algebra carries a representation with a submodule that no invariant complement completes. The final exercise exhibits this failure explicitly, isolating semisimplicity as the exact hypothesis Weyl's theorem cannot do

without.

Exercise 6.4.1

1. For $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ acting by the standard representation on $V = \mathbb{C}^2$ (example 6.4.2), recompute the Casimir element c_φ from its definition using the basis (e, h, f) and its β_φ-dual basis, and verify that $c_\varphi = \frac{3}{2}\,\mathrm{id}$. Then compute the Casimir scalar of the *adjoint* representation of $\mathfrak{sl}_2(\mathbb{C})$, using that its trace form is proportional to the Killing form; confirm that the scalar equals $\dim \mathfrak{g}/\dim V = 3/3 = 1$ and reconcile this with the Killing-form values $\kappa(h,h) = 8$, $\kappa(e,f) = 4$ computed in section 6.1.

2. In proposition 6.4.5(3) Schur's lemma was invoked to conclude c_φ acts as a scalar on an irreducible module. State Schur's lemma for $\mathfrak{g}$-modules over an algebraically closed field and prove it in the form used: a $\mathfrak{g}$-module endomorphism of a finite-dimensional irreducible module is a scalar. Where is algebraic closure needed?

3. Let $\mathfrak{g}$ be the two-dimensional non-abelian Lie algebra with basis $\{x, y\}$ and bracket $[x, y] = y$ (solvable, not semisimple). Define a representation $\psi\colon \mathfrak{g} \to \mathfrak{gl}_2(\mathbb{C})$ by

$$\psi(x) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \qquad \psi(y) = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$$

   Verify that ψ is a representation. Show that $W = \operatorname{span}\big((1,0)^{\mathsf{T}}\big)$ is a submodule, and prove that W has *no* $\mathfrak{g}$-invariant complement in $\mathbb{C}^2$. Explain why this does not contradict Weyl's theorem.

4. Prove that Weyl's theorem is equivalent to the statement that every short exact sequence of finite-dimensional $\mathfrak{g}$-modules $0 \to W \to V \to U \to 0$ splits, that is, admits a $\mathfrak{g}$-module section $s\colon U \to V$ with $V = W \oplus s(U)$.

5. Explain, without a full construction, where the proof of lemma 6.4.7 uses characteristic zero, and why the scalar $\lambda = \dim \mathfrak{g}/\dim V$ can vanish in characteristic p. Illustrate with the observation that for $\mathfrak{sl}_2$ over a field of characteristic 2, the Casimir scalar $\dim \mathfrak{g}/\dim V = 3/2$ is not defined, so the argument as written breaks down.

6. Show that the Casimir element c_φ of a faithful representation is independent of the choice of basis $X_1, \dots, X_n$: if $X_i' = \sum_j g_{ji} X_j$ is a second basis with β_φ-dual basis Y_i', then $\sum_i \varphi(X_i')\varphi(Y_i') = \sum_i \varphi(X_i)\varphi(Y_i)$. Conclude that c_φ depends only on φ, not on the chosen basis.

6.5 The Levi Decomposition

The two preceding parts of the chapter have pushed the theory of Lie algebras toward two opposite poles. Chapter 5 treated the solvable algebras, where the bracket is so nearly trivial that every irreducible representation is one-dimensional and the algebra can be triangularized; the present chapter has treated the semisimple algebras, where the bracket is so nondegenerate that the Killing form is invertible and every representation splits. Between these poles sits a general finite-dimensional Lie algebra, which is neither. It carries a largest solvable ideal, its radical $\operatorname{rad}\mathfrak{g}$ of definition 5.5.2, and the quotient $\mathfrak{g}/\operatorname{rad}\mathfrak{g}$ is semisimple by proposition 5.5.7. The natural question is whether these two pieces assemble the whole: does $\mathfrak{g}$ contain a copy of the semisimple quotient sitting inside it, complementary to the radical, so that the general algebra is literally a semisimple algebra acting on a

solvable one? The Levi decomposition answers this in the affirmative over any field of characteristic zero, and its proof is the last place in this part where Weyl's complete reducibility theorem does the decisive work.

The structural statement is that the short exact sequence of Lie algebras

$$0 \longrightarrow \mathrm{rad}\mathfrak{g} \longrightarrow \mathfrak{g} \xrightarrow{\pi} \mathfrak{g}/\mathrm{rad}\mathfrak{g} \longrightarrow 0 \tag{6.10}$$

splits: there is a Lie subalgebra $\mathfrak{s} \subseteq \mathfrak{g}$ mapped isomorphically onto the semisimple quotient by π. Splitting a short exact sequence of vector spaces is automatic, since one can always choose a linear complement to a subspace, but splitting it as Lie algebras is a genuine constraint: the complement must be closed under the bracket. The mechanism that gives a bracket-closed complement is exactly the vanishing of the low-degree cohomology of the semisimple quotient with coefficients in the radical, and that vanishing is a restatement of Weyl's theorem. This is the algebraic content that lets one build the splitting map by the same reasoning that split off invariant complements in section 6.4.

Before the theorem, the semidirect-product language must be fixed, so that the conclusion can be stated as an internal decomposition of $\mathfrak{g}$ rather than as an abstract splitting of (6.10).

Definition 6.5.1: Semidirect Product of Lie Algebras

Let $\mathfrak{r}$ and $\mathfrak{s}$ be Lie algebras and let $\theta\colon \mathfrak{s} \to \mathrm{Der}(\mathfrak{r})$ be a Lie algebra homomorphism into the derivations of $\mathfrak{r}$. The *semidirect product* $\mathfrak{r} \rtimes_\theta \mathfrak{s}$ is the vector space $\mathfrak{r} \oplus \mathfrak{s}$ equipped with the bracket

$$[(X_1, Y_1), (X_2, Y_2)] = \big([X_1, X_2]_{\mathfrak{r}} + \theta(Y_1)X_2 - \theta(Y_2)X_1,\ [Y_1, Y_2]_{\mathfrak{s}}\big)$$

This is a Lie algebra in which $\mathfrak{r}$ (as $\mathfrak{r} \oplus 0$) is an ideal, $\mathfrak{s}$ (as $0 \oplus \mathfrak{s}$) is a subalgebra, and the adjoint action of $\mathfrak{s}$ on $\mathfrak{r}$ is θ. Conversely, a Lie algebra $\mathfrak{g}$ is said to be the *internal semidirect product* $\mathfrak{g} = \mathfrak{r} \rtimes \mathfrak{s}$ of an ideal $\mathfrak{r} \trianglelefteq \mathfrak{g}$ and a subalgebra $\mathfrak{s} \subseteq \mathfrak{g}$ if $\mathfrak{g} = \mathfrak{r} \oplus \mathfrak{s}$ as vector spaces; then $\theta(Y) = \mathrm{ad}Y|_{\mathfrak{r}}$ recovers the external description.

The Jacobi identity for the displayed bracket is exactly the statement that θ lands in $\mathrm{Der}(\mathfrak{r})$ and is a homomorphism of Lie algebras, so the definition is consistent; the semidirect product for groups in [2, Semi-Direct Product] is what this exponentiates to. An internal semidirect product is precisely a splitting of the projection $\mathfrak{g} \to \mathfrak{g}/\mathfrak{r}$: the subalgebra $\mathfrak{s}$ maps isomorphically onto the quotient because $\mathfrak{g} = \mathfrak{r} \oplus \mathfrak{s}$ and $\mathfrak{r}$ is the kernel. The concrete model to keep in view is the algebra of the affine group, where a one-dimensional piece acts on an abelian ideal by scaling. It illustrates the semidirect-product structure without yet being a Levi decomposition, since the acting factor there is not semisimple.

Example 6.5.2: The Affine Lie Algebra $\mathfrak{gl}_1 \ltimes \mathbb{C}^n$

Let $\mathfrak{r} = \mathbb{C}^n$ be the abelian Lie algebra (all brackets zero) and let $\mathfrak{s} = \mathfrak{gl}_1 = \mathbb{C}$ act on $\mathfrak{r}$ by scalar multiplication, $\theta(t) = t \cdot \mathrm{id}_{\mathbb{C}^n}$, which is a derivation of the abelian algebra $\mathbb{C}^n$ because every linear map is a derivation when the target bracket vanishes. The semidirect product $\mathfrak{aff}_n = \mathbb{C}^n \rtimes_\theta \mathbb{C}$ has bracket

$$[(v_1, t_1), (v_2, t_2)] = (t_1 v_2 - t_2 v_1,\ 0)$$

realized concretely as the Lie algebra of block matrices $\left(\begin{smallmatrix} tI_n & v \\ 0 & 0 \end{smallmatrix}\right)$ acting on $\mathbb{C}^n \oplus \mathbb{C}$, the Lie algebra of the affine group $x \mapsto e^t x + v$. Here $\mathfrak{r} = \mathbb{C}^n$ is an abelian ideal, but it is *not* the radical: the acting line $\mathfrak{s} = \mathbb{C}$ is itself solvable, not semisimple, so the semidirect product $\mathfrak{aff}_n$ is solvable rather than a genuine Levi decomposition. Indeed the derived algebra is $[\mathfrak{aff}_n, \mathfrak{aff}_n] = \mathbb{C}^n$ (from $[(0,t),(v,0)] = (tv, 0)$), and the next derived algebra $[\mathbb{C}^n, \mathbb{C}^n] = 0$ vanishes, so $\mathfrak{aff}_n$ is solvable of derived length 2. A solvable algebra is its own radical, so $\mathrm{rad}\mathfrak{aff}_n = \mathfrak{r} \oplus \mathfrak{s} = \mathbb{C}^n \oplus \mathbb{C}$ for every

$n \geq 0$, the quotient $\mathfrak{aff}_n/\mathbb{C}^n \cong \mathbb{C}$ being one-dimensional abelian and hence solvable, not semisimple. Consequently the Levi subalgebra of $\mathfrak{aff}_n$ is 0: the example exhibits the semidirect-product structure itself, not a splitting off of a semisimple factor.

The example shows that a semidirect product need not have a semisimple acting factor. Levi's theorem is the statement that one can always *choose* the acting factor to be semisimple when the ideal is taken as large as possible, that is, equal to the radical. The proof rests on translating the existence and uniqueness of the splitting of (6.10) into the vanishing of two cohomology groups of the semisimple quotient, and that vanishing is where semisimplicity and characteristic zero enter, through Weyl's theorem theorem 6.4.8. The next lemma treats the first of these groups, $H^1(\mathfrak{s}, M)$, which controls the conjugacy of a would-be splitting; the second group $H^2(\mathfrak{s}, M)$, which controls its existence, is handled in the proof of the main theorem. The lemma isolates the cohomological input in the exact form the argument needs.

Lemma 6.5.3: Whitehead's First Lemma

Let $\mathfrak{s}$ be a semisimple Lie algebra over a field of characteristic zero, and let M be a finite-dimensional $\mathfrak{s}$-module (a representation in the sense of definition 6.4.1). Let $f\colon \mathfrak{s} \to M$ be a linear map satisfying the cocycle identity

$$f([X,Y]) = X \cdot f(Y) - Y \cdot f(X) \qquad \text{for all } X, Y \in \mathfrak{s} \tag{6.11}$$

Then there is a vector $m \in M$ with $f(X) = X \cdot m$ for all $X \in \mathfrak{s}$; that is, every cocycle is a coboundary, and $H^1(\mathfrak{s}, M) = 0$.

The cocycle identity is the infinitesimal condition that makes the graph of f a Lie subalgebra of $M \rtimes \mathfrak{s}$, and a vector m with $f = (X \mapsto X \cdot m)$ conjugates that graph onto the standard copy of $\mathfrak{s}$. Whitehead's lemma is therefore precisely the algebraic fact needed to straighten a would-be splitting into an actual one. Its proof is Weyl's theorem in disguise, applied to a cleverly chosen module.

***Proof*:**
The strategy is to reduce, by Weyl's complete reducibility, to the case of an irreducible module M, and there to produce the vector m either by perfectness (trivial module) or by the Casimir element (nontrivial module). The cocycle condition is used in the form (6.11) throughout; no identification of cocycles with invariants is made or needed.

Step 1: reduce to an irreducible module. By Weyl's theorem theorem 6.4.8 the module M decomposes as a direct sum $M = \bigoplus_i M_i$ of irreducible $\mathfrak{s}$-submodules. Write $f = \sum_i f_i$ where $f_i = p_i \circ f\colon \mathfrak{s} \to M_i$ is the composite of f with the projection $p_i\colon M \to M_i$. Since each p_i is an $\mathfrak{s}$-module map, applying p_i to the cocycle identity (6.11) shows that each f_i is again a cocycle valued in M_i; and if each $f_i(X) = X \cdot m_i$ for some $m_i \in M_i$, then $f(X) = X \cdot m$ with $m = \sum_i m_i$. It therefore suffices to prove the lemma when M is irreducible.

Step 2: the trivial module. Suppose M is the trivial one-dimensional module, so $X \cdot v = 0$ for all $X \in \mathfrak{s}$ and $v \in M$. The cocycle identity (6.11) reads $f([X,Y]) = X \cdot f(Y) - Y \cdot f(X) = 0$ for all X, Y. Because $\mathfrak{s}$ is semisimple it is perfect, $[\mathfrak{s}, \mathfrak{s}] = \mathfrak{s}$ by corollary 6.3.5, so every element of $\mathfrak{s}$ is a sum of brackets; hence f vanishes identically, and f is the coboundary $X \mapsto X \cdot 0 = 0$ of the vector $m = 0$. More generally the same argument shows $f = 0$ on any summand where $\mathfrak{s}$ acts trivially.

Step 3: the nontrivial irreducible module. Suppose M is irreducible and nontrivial, so the

action $\varphi\colon \mathfrak{s} \to \mathfrak{gl}(M)$ is nonzero; since $\mathfrak{s}$ is semisimple and $\ker\varphi$ is an ideal, φ restricted to the sum of the simple ideals on which it is nonzero is faithful, and the Casimir element c of definition 6.4.4 for this action is defined. By Schur's lemma c acts on the irreducible M as a scalar λ, and $\lambda \neq 0$ because $\lambda \dim M = \operatorname{tr}_M(c) = \dim \mathfrak{s}' \neq 0$, where $\mathfrak{s}'$ is the sum of the simple ideals acting nontrivially, by the trace identity from the proof of Weyl's theorem theorem 6.4.8. Write $c = \sum_a X_a Y_a$ in a basis $\{X_a\}$ of $\mathfrak{s}'$ with κ_φ-dual basis $\{Y_a\}$, and set

$$m = \frac{1}{\lambda} \sum_a X_a \cdot f(Y_a) \in M \tag{6.12}$$

Then $f(X) = X \cdot m$ for every $X \in \mathfrak{s}$: this is a direct computation with the Casimir, using the cocycle identity (6.11) to rewrite $X \cdot f(Y_a) = f([X, Y_a]) + Y_a \cdot f(X)$ and the centrality of c in $U(\mathfrak{s})$ (which makes $\sum_a [X, X_a] \otimes Y_a + \sum_a X_a \otimes [X, Y_a]$ act as zero) to collapse the sum to $\lambda f(X)$. The computation is carried out in full as exercise 6.5.1 (5). Assembling the summands by Step 1 gives $f(X) = X \cdot m$ on all of M, so every cocycle is a coboundary and $H^1(\mathfrak{s}, M) = 0$, completing the proof.

With the cohomological vanishing in hand, the splitting of the extension (6.10) follows from a two-part reduction that is the real content of Levi's theorem. The plan is to first peel off the derived algebra of the radical, reducing by induction on $\dim \operatorname{rad}\mathfrak{g}$ to the case where the radical is abelian, and then to split the abelian case by killing a single obstruction class. When the radical $\mathfrak{r}$ is abelian it is a module over the semisimple quotient $\overline{\mathfrak{g}}$, and a chosen linear section of π fails to be a homomorphism by a 2-cocycle with values in $\mathfrak{r}$; the section can be corrected to a homomorphism, hence to a Levi subalgebra, exactly when that cocycle is a coboundary, that is, when its class in $H^2(\overline{\mathfrak{g}}, \mathfrak{r})$ vanishes. Whitehead's lemmas make both relevant cohomology groups vanish for a semisimple $\overline{\mathfrak{g}}$ in characteristic zero, H^1 governing uniqueness of the splitting up to conjugacy and H^2 governing its existence, and each vanishing is Weyl's theorem theorem 6.4.8 applied to a cochain module. The first lemma was proved above; the parallel H^2 computation is where the abelian case is settled.

Theorem 6.5.4: Levi Decomposition

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k of characteristic zero. Then there is a semisimple subalgebra $\mathfrak{s} \subseteq \mathfrak{g}$, called a *Levi subalgebra*, such that

$$\mathfrak{g} = \operatorname{rad}\mathfrak{g} \rtimes \mathfrak{s}$$

is an internal semidirect product: $\mathfrak{g} = \operatorname{rad}\mathfrak{g} \oplus \mathfrak{s}$ as vector spaces, $\operatorname{rad}\mathfrak{g}$ is an ideal, and $\mathfrak{s}$ is a subalgebra mapped isomorphically onto $\mathfrak{g}/\operatorname{rad}\mathfrak{g}$ by the quotient map. In particular the extension (6.10) splits and $\mathfrak{s} \cong \mathfrak{g}/\operatorname{rad}\mathfrak{g}$.

***Proof*:**
Write $\mathfrak{r} = \operatorname{rad}\mathfrak{g}$ and $\overline{\mathfrak{g}} = \mathfrak{g}/\mathfrak{r}$, which is semisimple by proposition 5.5.7. The task is to produce a subalgebra $\mathfrak{s} \subseteq \mathfrak{g}$ with $\pi|_{\mathfrak{s}}\colon \mathfrak{s} \to \overline{\mathfrak{g}}$ an isomorphism, where $\pi\colon \mathfrak{g} \to \overline{\mathfrak{g}}$ is the projection. The argument is by induction on $\dim \mathfrak{r}$; the base case $\mathfrak{r} = 0$ is immediate, as then $\mathfrak{g}$ is itself semisimple and $\mathfrak{s} = \mathfrak{g}$ works.

Step 1: reduce to an abelian radical. Suppose $\mathfrak{r} \neq 0$. The derived algebra $[\mathfrak{r}, \mathfrak{r}]$ is a proper ideal of $\mathfrak{r}$ (proper because $\mathfrak{r}$ is solvable, so proposition 5.1.6 forbids $[\mathfrak{r}, \mathfrak{r}] = \mathfrak{r}$ for $\mathfrak{r} \neq 0$), and it is an ideal of $\mathfrak{g}$: since $\mathfrak{r}$ is an ideal of $\mathfrak{g}$, the Jacobi identity gives $[\mathfrak{g}, [\mathfrak{r}, \mathfrak{r}]] \subseteq [[\mathfrak{g}, \mathfrak{r}], \mathfrak{r}] \subseteq [\mathfrak{r}, \mathfrak{r}]$. Suppose

first that $[\mathfrak{r},\mathfrak{r}] \neq 0$. Pass to $\mathfrak{g}' = \mathfrak{g}/[\mathfrak{r},\mathfrak{r}]$. Its radical is $\mathfrak{r}/[\mathfrak{r},\mathfrak{r}]$, which is nonzero but of strictly smaller dimension than $\mathfrak{r}$, so by induction $\mathfrak{g}'$ has a Levi subalgebra: there is a subalgebra $\mathfrak{s}' \subseteq \mathfrak{g}'$ complementary to $\mathfrak{r}/[\mathfrak{r},\mathfrak{r}]$. Let $\mathfrak{h} \subseteq \mathfrak{g}$ be the preimage of $\mathfrak{s}'$ under $\mathfrak{g} \to \mathfrak{g}'$. Then $\mathfrak{h}$ is a subalgebra containing $[\mathfrak{r},\mathfrak{r}]$, and $\mathfrak{h}/[\mathfrak{r},\mathfrak{r}] = \mathfrak{s}' \cong \overline{\mathfrak{g}}$ is semisimple; the radical of $\mathfrak{h}$ is $[\mathfrak{r},\mathfrak{r}]$, which has strictly smaller dimension than $\mathfrak{r}$. A second application of the inductive hypothesis to $\mathfrak{h}$ gives a Levi subalgebra $\mathfrak{s} \subseteq \mathfrak{h}$ complementary to $[\mathfrak{r},\mathfrak{r}]$, and then $\mathfrak{s}$ maps isomorphically onto $\mathfrak{h}/[\mathfrak{r},\mathfrak{r}] \cong \overline{\mathfrak{g}}$, so $\mathfrak{s}$ is a Levi subalgebra of $\mathfrak{g}$. Thus the induction reduces the whole theorem to the case $[\mathfrak{r},\mathfrak{r}] = 0$, that is, $\mathfrak{r}$ abelian.

Step 2: make the radical a module and encode the splitting as a cocycle. Assume now $\mathfrak{r}$ is abelian and nonzero. Since $\mathfrak{r}$ is abelian and an ideal, the adjoint action of $\mathfrak{g}$ on $\mathfrak{r}$ descends to an action of $\overline{\mathfrak{g}} = \mathfrak{g}/\mathfrak{r}$ on $\mathfrak{r}$: for $\overline{X} \in \overline{\mathfrak{g}}$ and $v \in \mathfrak{r}$, the formula $\overline{X} \cdot v = [X, v]$ is well defined because $[\mathfrak{r},\mathfrak{r}] = 0$ makes the choice of lift X irrelevant. This turns $\mathfrak{r}$ into a finite-dimensional $\overline{\mathfrak{g}}$-module. Choose any vector-space section $\sigma\colon \overline{\mathfrak{g}} \to \mathfrak{g}$ of π (a linear right inverse, obtained by choosing a linear complement to $\mathfrak{r}$). The section σ fails to be a Lie algebra homomorphism by an amount measured by

$$f(\overline{X},\overline{Y}) = [\sigma(\overline{X}),\sigma(\overline{Y})] - \sigma([\overline{X},\overline{Y}]) \tag{6.13}$$

which lands in $\mathfrak{r} = \ker\pi$ because $\pi \circ \sigma = \mathrm{id}$ makes π of the right side vanish. Replacing σ by $\sigma + \tau$ for a linear map $\tau\colon \overline{\mathfrak{g}} \to \mathfrak{r}$ produces the new subspace $\sigma(\overline{\mathfrak{g}}) + \tau(\overline{\mathfrak{g}})$, and $\sigma + \tau$ is a Lie algebra homomorphism exactly when the associated f vanishes; its image is then a subalgebra mapped isomorphically onto $\overline{\mathfrak{g}}$, that is, a Levi subalgebra. The task is to find τ making $f \equiv 0$.

Step 3: the obstruction lives in H^2, which vanishes. The alternating map f of (6.13) satisfies the 2-cocycle identity for the $\overline{\mathfrak{g}}$-module $\mathfrak{r}$: expanding the Jacobi identity $[[\sigma\overline{X},\sigma\overline{Y}],\sigma\overline{Z}] + \text{cyclic} = 0$ and projecting into $\mathfrak{r}$ yields

$$\overline{X}\cdot f(\overline{Y},\overline{Z}) - \overline{Y}\cdot f(\overline{X},\overline{Z}) + \overline{Z}\cdot f(\overline{X},\overline{Y}) - f([\overline{X},\overline{Y}],\overline{Z}) + f([\overline{X},\overline{Z}],\overline{Y}) - f([\overline{Y},\overline{Z}],\overline{X}) = 0$$

the defining relation for a Lie algebra 2-cocycle. As noted in Step 2, a Levi subalgebra exists exactly when the class $[f] \in H^2(\overline{\mathfrak{g}},\mathfrak{r})$ is zero. That class vanishes because $\overline{\mathfrak{g}}$ is semisimple in characteristic zero: this is *Whitehead's second lemma*, $H^2(\mathfrak{s},M) = 0$ for every finite-dimensional module M over a semisimple $\mathfrak{s}$ in characteristic zero. The proof of the second lemma runs parallel to that of the first (lemma 6.5.3): one packages the 2-cochains as an $\mathfrak{s}$-module, decomposes the coefficient module by Weyl's complete reducibility theorem 6.4.8 into trivial and nontrivial irreducible summands, and handles each summand as before, perfectness killing the cocycle on the trivial part and the Casimir of definition 6.4.4, acting as an invertible scalar, correcting it to a coboundary on the nontrivial part. The full 2-cochain bookkeeping is longer than the first-lemma computation but conceptually identical and adds nothing new; it is deferred to [2], where the Lie algebra cohomology $H^n(\mathfrak{s},M)$ is developed. What is used here is only the vanishing $H^2(\overline{\mathfrak{g}},\mathfrak{r}) = 0$.

Step 4: extract the Levi subalgebra. Since $[f] = 0$, there is a linear map $\tau\colon \overline{\mathfrak{g}} \to \mathfrak{r}$ whose coboundary equals f; that is, replacing σ by $\sigma' = \sigma - \tau$ makes the corresponding obstruction $f'(\overline{X},\overline{Y}) = [\sigma'\overline{X},\sigma'\overline{Y}] - \sigma'([\overline{X},\overline{Y}])$ vanish identically. Concretely, unwinding the coboundary condition $f(\overline{X},\overline{Y}) = \overline{X}\cdot\tau(\overline{Y}) - \overline{Y}\cdot\tau(\overline{X}) - \tau([\overline{X},\overline{Y}])$ gives $[\sigma'\overline{X},\sigma'\overline{Y}] = \sigma'([\overline{X},\overline{Y}])$ for all $\overline{X},\overline{Y}$. Thus $\sigma'\colon \overline{\mathfrak{g}} \to \mathfrak{g}$ is a Lie algebra homomorphism, and it is a section of π because $\pi\circ\sigma' = \pi\circ\sigma - \pi\circ\tau = \mathrm{id} - 0 = \mathrm{id}$ (as τ lands in $\mathfrak{r} = \ker\pi$). Set $\mathfrak{s} = \sigma'(\overline{\mathfrak{g}})$. It is a subalgebra, being the image of a homomorphism, and $\pi|_{\mathfrak{s}}\colon \mathfrak{s} \to \overline{\mathfrak{g}}$ is an isomorphism with inverse σ'. Hence $\mathfrak{s} \cong \overline{\mathfrak{g}}$ is semisimple, $\mathfrak{g} = \mathfrak{r} \oplus \mathfrak{s}$ as vector spaces, and $\mathfrak{s}$ is a Levi subalgebra: $\mathfrak{g} = \mathfrak{r} \rtimes \mathfrak{s}$, completing the proof.

The Levi decomposition is the structural payoff of the entire part. It says that a general finite-dimensional Lie algebra in characteristic zero is built from exactly two kinds of pieces already understood in isolation: a solvable radical, whose theory occupied chapter 5, and a semisimple Levi factor, which the rest of Part II dissects. The two are not simply stacked; the semisimple factor *acts* on the solvable radical by derivations, and the whole algebra is that action packaged as a semidirect product. Nothing else can occur. The proof also exposes why characteristic zero is indispensable: it was used through Weyl's complete reducibility theorem theorem 6.4.8, which fails in positive characteristic, and the Levi decomposition itself can fail there. Over a field of characteristic zero the theorem holds without any assumption of algebraic closure, since the radical and the notion of semisimplicity are stable under field extension, as noted for the Killing form in section 6.3.

The immediate consequence is a clean reduction of the classification problem for Lie algebras.

Corollary 6.5.5: Reduction of the Classification Problem

Over a field of characteristic zero, the classification of finite-dimensional Lie algebras reduces to two problems already isolated: the classification of solvable Lie algebras (chapter 5) and the classification of semisimple Lie algebras, together with the data of a homomorphism $\theta\colon \mathfrak{s} \to \mathrm{Der}(\mathfrak{r})$ describing how the semisimple factor acts on the radical. Moreover the Levi subalgebra is unique up to conjugacy: if $\mathfrak{s}$ and $\mathfrak{s}'$ are two Levi subalgebras of $\mathfrak{g}$, there is an inner automorphism $\exp(\mathrm{ad}z)$ with z in the nilradical $\mathrm{nil}\,\mathfrak{g}$ (definition 5.5.3) carrying $\mathfrak{s}$ onto $\mathfrak{s}'$. This is the Malcev-Harish-Chandra theorem.

***Proof*:**
The reduction is a restatement of theorem 6.5.4: every $\mathfrak{g}$ is $\mathrm{rad}\mathfrak{g} \rtimes_\theta \mathfrak{s}$ with $\mathfrak{s}$ semisimple, and conversely any such data $(\mathfrak{r}, \mathfrak{s}, \theta)$ with $\mathfrak{r}$ solvable and $\mathfrak{s}$ semisimple builds a Lie algebra whose radical is $\mathfrak{r}$ (the semidirect factor $\mathfrak{s}$ being semisimple has no nonzero solvable ideal, so $\mathfrak{r}$ absorbs the whole radical). The uniqueness statement is the Malcev-Harish-Chandra theorem; its proof is a conjugacy argument by induction on $\dim \mathrm{rad}\mathfrak{g}$, running the operator $\exp(\mathrm{ad}z)$ with z ranging over the nilradical to move one Levi subalgebra onto another, and is a standard part of the structure theory, established for Lie algebras over characteristic-zero fields in [2]. The existence half is theorem 6.5.4; the uniqueness half is cited, completing the proof.

The corollary closes the first arc of Part II. The Cartan criteria separated the solvable from the semisimple by the nondegeneracy of a single bilinear form; Weyl's theorem made semisimple algebras completely reducible; and Levi's theorem now certifies that these two classes, joined by a semidirect action, exhaust all Lie algebras. The solvable side, with its abelian and nilpotent substructure, is by comparison wilder and less rigid, and no clean classification of it exists in general. The semisimple side, by contrast, is about to reveal itself as governed by finite combinatorial data. This is the justification for the exclusive attention the following chapters give to semisimple algebras: the general theory has been reduced to them, and they alone admit a complete classification. The internal anatomy of a semisimple algebra, its decomposition into root spaces under a maximal abelian subalgebra, is where that combinatorial rigidity first appears, and it is the subject of the next chapter.

The uniqueness of the Levi subalgebra up to conjugacy deserves a concrete illustration, because it shows that although the splitting (6.10) is not canonical, the ambiguity is exactly an inner automorphism and nothing more. The reductive algebras furnish the cleanest examples, where the radical is the center and the Levi subalgebra is the derived algebra.

Example 6.5.6: The Levi Decomposition of $\mathfrak{gl}_n$

Take $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{C})$, all $n \times n$ complex matrices. Its center is the line $\mathbb{C}I_n$ of scalar matrices, and the trace splits off this center: every matrix X decomposes uniquely as

$$X = \left(X - \tfrac{\operatorname{tr}X}{n} I_n\right) + \tfrac{\operatorname{tr}X}{n} I_n$$

with the first summand traceless and the second scalar. Thus $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I_n$ as vector spaces. The line $\mathbb{C}I_n$ is an abelian ideal (indeed central), so it is solvable, and it is the full radical of $\mathfrak{gl}_n$: the quotient $\mathfrak{gl}_n/\mathbb{C}I_n \cong \mathfrak{sl}_n$ is semisimple by theorem 6.3.1 once one checks the Killing form of $\mathfrak{sl}_n$ is nondegenerate, so no larger solvable ideal exists. Hence $\operatorname{rad}\mathfrak{gl}_n = \mathbb{C}I_n$. The subalgebra $\mathfrak{sl}_n$ of traceless matrices is semisimple and complementary to the radical, and the scalars commute with everything, so the semidirect product degenerates to a direct product of Lie algebras:

$$\mathfrak{gl}_n = \mathbb{C}I_n \rtimes \mathfrak{sl}_n = \mathbb{C}I_n \oplus \mathfrak{sl}_n$$

with $\mathfrak{sl}_n$ the Levi subalgebra. Here the acting homomorphism $\theta\colon \mathfrak{sl}_n \to \operatorname{Der}(\mathbb{C}I_n)$ is zero, since $[\mathfrak{sl}_n, \mathbb{C}I_n] = 0$, so the Levi decomposition is a genuine direct sum. The algebra $\mathfrak{gl}_n$ is *reductive*, its radical equalling its center, not semisimple, precisely because the center $\mathbb{C}I_n$ is nonzero. The nilradical of $\mathfrak{gl}_n$ is $\operatorname{nil}\mathfrak{gl}_n = \mathbb{C}I_n$, coinciding with the radical because $\mathbb{C}I_n$ is abelian, hence a nilpotent ideal, as computed in section 5.5. Even so, the Malcev-Harish-Chandra conjugations $\exp(\operatorname{ad}z)$ with $z \in \mathbb{C}I_n$ are all trivial, since $\operatorname{ad}z = 0$ for a scalar z, so they act as the identity; the Levi subalgebra is therefore *unique*, not just unique up to conjugacy. Indeed $\mathfrak{sl}_n$ is the only semisimple complement to $\mathbb{C}I_n$, as one checks directly: any semisimple complement is perfect, hence lies in the derived algebra $[\mathfrak{gl}_n, \mathfrak{gl}_n] = \mathfrak{sl}_n$, and has the same dimension $n^2 - 1$, so equals $\mathfrak{sl}_n$.

The two extreme cases of Levi's theorem are worth recording as a boundary check on the definition. The Levi subalgebra is trivial exactly when the algebra is solvable, and it is the whole algebra exactly when the algebra is semisimple, so the size of the Levi factor interpolates between these poles and measures how far $\mathfrak{g}$ is from being solvable.

Exercise 6.5.1

1. Prove the two boundary cases of the Levi decomposition directly from the definitions. Show that the Levi subalgebra $\mathfrak{s}$ of a Lie algebra $\mathfrak{g}$ is zero if and only if $\mathfrak{g}$ is solvable, and that $\mathfrak{s} = \mathfrak{g}$ (equivalently $\operatorname{rad}\mathfrak{g} = 0$) if and only if $\mathfrak{g}$ is semisimple. Conclude that $\dim\mathfrak{s} = \dim\mathfrak{g} - \dim\operatorname{rad}\mathfrak{g}$ measures the semisimple part of $\mathfrak{g}$.

2. For $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{C})$, verify in detail that $\mathfrak{sl}_n$ is the unique semisimple subalgebra complementary to the center $\mathbb{C}I_n$. Show that any Lie subalgebra $\mathfrak{s} \subseteq \mathfrak{gl}_n$ with $\mathfrak{gl}_n = \mathbb{C}I_n \oplus \mathfrak{s}$ as vector spaces and $\mathfrak{s}$ semisimple must consist of traceless matrices, hence equal $\mathfrak{sl}_n$, using that a semisimple Lie algebra is perfect (corollary 6.3.5).

3. Compute the Levi decomposition of the affine Lie algebra $\mathfrak{aff}_1 = \mathbb{C} \rtimes \mathbb{C}$ of example 6.5.2 with $n = 1$, spanned by $X = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ with $[X, Y] = Y$. Determine $\operatorname{rad}\mathfrak{aff}_1$ and the Levi subalgebra, and explain why $\mathfrak{aff}_1$ has no nonzero semisimple subalgebra.

4. Give an example of a Lie algebra whose Levi decomposition is a nontrivial semidirect product (the acting homomorphism θ is nonzero), contrasting with $\mathfrak{gl}_n$ where $\theta = 0$. Take $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C}) \ltimes \mathbb{C}^2$, where $\mathfrak{sl}_2$ acts on $\mathbb{C}^2$ by the standard (defining) representation. Identify the

radical, verify that $\mathfrak{sl}_2$ is a Levi subalgebra, and confirm that θ is nonzero.

5. Complete the deferred Casimir computation of Whitehead's first lemma lemma 6.5.3, Step 3. Let $\mathfrak{s}$ be semisimple in characteristic zero and M a nontrivial irreducible $\mathfrak{s}$-module, with Casimir element $c = \sum_a X_a Y_a$ of definition 6.4.4 acting as the nonzero scalar λ. Suppose $f\colon \mathfrak{s} \to M$ satisfies the cocycle identity (6.11), and let $m = \frac{1}{\lambda}\sum_a X_a \cdot f(Y_a)$ be the vector (6.12). Show directly, using (6.11) to rewrite $X \cdot f(Y_a)$ and the centrality of c, that $f(X) = X \cdot m$ for all $X \in \mathfrak{s}$, so f is a coboundary. Then, for the trivial one-dimensional module, show separately that any cocycle vanishes, using perfectness (corollary 6.3.5).

7

Cartan Subalgebras and the Root Space Decomposition

Chapter 6 made the semisimple Lie algebras tractable but studied them from the outside, through the nondegeneracy of a single bilinear form. This chapter opens one up and exhibits its internal anatomy. The instrument is a Cartan subalgebra $\mathfrak{h}$, a maximal abelian subalgebra whose elements act on $\mathfrak{g}$ by commuting diagonalizable operators. Because the operators $\mathrm{ad}H$, $H \in \mathfrak{h}$, commute and are each diagonalizable, they are simultaneously diagonalizable, and $\mathfrak{g}$ breaks into their simultaneous eigenspaces. The eigenvalues are linear functionals on $\mathfrak{h}$, and the nonzero ones among them, the roots, index the decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

that turns a semisimple Lie algebra into a finite configuration of vectors in a Euclidean space.

The chapter constructs this decomposition and reads off its combinatorial shape. Cartan subalgebras exist, and for a semisimple algebra they are exactly the maximal toral subalgebras, abelian and diagonalizable under the adjoint action. Each root space is one-dimensional; the Killing form pairs $\mathfrak{g}_\alpha$ with $\mathfrak{g}_{-\alpha}$ and restricts to a nondegenerate form on $\mathfrak{h}$, which transports the roots into an inner product space. Every root α then generates a copy of $\mathfrak{sl}_2$ inside $\mathfrak{g}$, and the representation theory of that $\mathfrak{sl}_2$ forces the Cartan integers $\langle \beta, \alpha^\vee \rangle = 2(\beta, \alpha)/(\alpha, \alpha)$ to be integers and the root strings to be unbroken. These are precisely the axioms of an abstract root system; the roots span $\mathfrak{h}^*$ and are permuted by the Weyl group generated by the reflections s_α. The configuration so obtained is the object classified, entirely combinatorially, in chapter 8.

7.1 Toral and Cartan Subalgebras: Existence

Chapter 6 reduced the study of semisimple Lie algebras to that of simple ones and equipped every semisimple $\mathfrak{g}$ with a nondegenerate Killing form and inner derivations. What it did not yet give is a coordinate system. A single simple algebra is still an opaque bracket; to see its internal structure one needs a distinguished abelian subalgebra whose adjoint action diagonalizes $\mathfrak{g}$, splitting it into simultaneous eigenspaces indexed by linear functionals. Those functionals are the roots, and the eigenspace decomposition is the root space decomposition of the next section. This section builds the diagonalizing subalgebra: a Cartan subalgebra $\mathfrak{h}$, the "maximal diagonalizable piece" of $\mathfrak{g}$, and proves that it exists.

The construction proceeds in three moves. First, the abstract Jordan decomposition lets one speak of an element of $\mathfrak{g}$ being semisimple (diagonalizable under ad) or nilpotent, intrinsically, without reference to any embedding into matrices. Second, a toral subalgebra is one built entirely from semisimple elements, so that ad acts on $\mathfrak{g}$ by commuting diagonalizable operators; such a subalgebra turns out to be abelian, and a maximal one is the object sought. Third, the key technical result identifies a maximal toral subalgebra with its own centralizer, which is what makes its eigenspace decomposition of $\mathfrak{g}$ leave nothing over in degree zero beyond $\mathfrak{h}$ itself. Throughout, $\mathfrak{g}$ denotes a semisimple Lie algebra over $\mathbb{C}$, and the diagonal traceless matrices in $\mathfrak{sl}_n$ serve as the concrete model of every abstraction.

The Jordan decomposition of a single operator writes it as a sum of a diagonalizable part and a commuting nilpotent part. For an abstract Lie algebra there is no ambient space of matrices to decompose in, but the adjoint representation gives one: each $x \in \mathfrak{g}$ acts as $\mathrm{ad}x$ on the vector space $\mathfrak{g}$, and $\mathrm{ad}x$ has an operator Jordan decomposition. The content of the following definition is that when $\mathfrak{g}$ is semisimple, the two operator parts of $\mathrm{ad}x$ are again of the form $\mathrm{ad}(\,\cdot\,)$ for elements of $\mathfrak{g}$, so the decomposition descends from operators to elements. This rests on the fact, recorded in corollary 6.3.5, that $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{Der}(\mathfrak{g})$ is injective (its kernel is the center, which is zero) and surjects onto all derivations.

Definition 7.1.1: Semisimple and Nilpotent Elements

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$. For $x \in \mathfrak{g}$, let $(\mathrm{ad}x)_s$ and $(\mathrm{ad}x)_n$ be the semisimple (diagonalizable) and nilpotent parts of the operator $\mathrm{ad}x \in \mathfrak{gl}(\mathfrak{g})$ in its Jordan decomposition, so that

$$\mathrm{ad}x = (\mathrm{ad}x)_s + (\mathrm{ad}x)_n, \qquad [(\mathrm{ad}x)_s, (\mathrm{ad}x)_n] = 0$$

Then there are unique elements $x_s, x_n \in \mathfrak{g}$ with

$$\mathrm{ad}x_s = (\mathrm{ad}x)_s, \qquad \mathrm{ad}x_n = (\mathrm{ad}x)_n$$

and one writes $x = x_s + x_n$, the *abstract Jordan decomposition* of x, with $[x_s, x_n] = 0$. The element x is called *semisimple* if $x = x_s$ (equivalently $\mathrm{ad}x$ is diagonalizable) and *nilpotent* if $x = x_n$ (equivalently $\mathrm{ad}x$ is nilpotent).

The definition is well-posed only because of the two facts it invokes, and they deserve to be spelled out. The operator parts $(\mathrm{ad}x)_s$ and $(\mathrm{ad}x)_n$ are polynomials in $\mathrm{ad}x$ with zero constant term, by the standard construction of the Jordan decomposition of an operator (see [2, Jordan-Chevalley Decomposition]); in particular each is again a derivation of $\mathfrak{g}$, because the semisimple and nilpotent parts of a derivation are themselves derivations, a fact proved by decomposing $\mathfrak{g}$ into generalized

eigenspaces for $\mathrm{ad}x$ and checking the derivation identity on each pair of eigenspaces. Since $\mathfrak{g}$ is semisimple, corollary 6.3.5 says every derivation is inner, so $(\mathrm{ad}x)_s = \mathrm{ad}x_s$ and $(\mathrm{ad}x)_n = \mathrm{ad}x_n$ for unique $x_s, x_n \in \mathfrak{g}$; the uniqueness is injectivity of ad. The word "intrinsic" is the payoff: the elements x_s and x_n depend on nothing but the Lie algebra structure of $\mathfrak{g}$. In particular, if $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ is a faithful representation, the abstract Jordan parts agree with the matrix Jordan parts of x acting on V, a compatibility theorem of chapter 6's circle of ideas that the reader may take as given for the classical algebras and whose general form is deferred to [10].

The concrete meaning of these notions is transparent for $\mathfrak{sl}_n$.

Example 7.1.2: Jordan Parts in $\mathfrak{sl}_n$

Let $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$, the traceless $n \times n$ matrices, with bracket $[X, Y] = XY - YX$. Take a diagonal traceless matrix

$$H = \mathrm{diag}(a_1, \ldots, a_n), \qquad \sum_i a_i = 0$$

Its adjoint operator acts on the elementary matrix E_{ij} by $\mathrm{ad}H\,(E_{ij}) = (a_i - a_j)E_{ij}$, a direct commutator computation, so $\mathrm{ad}H$ is diagonal in the basis $\{E_{ij}\}_{i \neq j} \cup \{\text{diagonal traceless matrices}\}$ with eigenvalues $a_i - a_j$. Thus $\mathrm{ad}H$ is diagonalizable, and H is a semisimple element of $\mathfrak{sl}_n$. By contrast, a strictly upper triangular matrix such as $N = E_{12}$ has $\mathrm{ad}N$ nilpotent (conjugation-type operators built from a nilpotent matrix are nilpotent), so N is a nilpotent element.

For a mixed matrix $X = H + N$ with $H = \mathrm{diag}(1, -1)$ and $N = E_{12}$ inside $\mathfrak{sl}_2$, that is $X = \left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$, the matrix Jordan decomposition of X is $X = X_s + X_n$ with $X_s = \mathrm{diag}(1, -1)$ and $X_n = E_{12}$, since X_s is diagonalizable, X_n is nilpotent, and $[X_s, X_n] = X_s X_n - X_n X_s = E_{12} - (-E_{12}) = 2E_{12} \neq 0$. The two matrices do not commute, so this is *not* the Jordan decomposition. The correct one uses that X itself is diagonalizable: its eigenvalues are ± 1, distinct, so X is semisimple as a matrix, whence $X_s = X$ and $X_n = 0$. This is the abstract Jordan decomposition of X as an element of $\mathfrak{sl}_2$ as well, by the compatibility noted above. The lesson is that "diagonal plus strictly upper triangular" is not the Jordan decomposition unless the two pieces commute.

The example isolates the recurring pitfall and its resolution: a matrix's visible splitting into a diagonal and an off-diagonal part is a Jordan decomposition only when the parts commute, and otherwise one must carry out the diagonalization directly. With semisimple and nilpotent elements in hand, the subalgebras built from them can be defined.

A toral subalgebra is the abstract counterpart of a family of commuting diagonalizable matrices. The name comes from the compact-group picture of later chapters, where such a subalgebra exponentiates to a torus; here it is purely algebraic. The requirement is that every element act diagonalizably under ad, so that the whole subalgebra can be simultaneously diagonalized on $\mathfrak{g}$. A Cartan subalgebra is then, for a general Lie algebra, defined by two abstract properties, nilpotency and self-normalization, that will be shown to single out the maximal toral subalgebras in the semisimple case.

Definition 7.1.3: Toral Subalgebra

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$. A *toral subalgebra* of $\mathfrak{g}$ is a subalgebra $\mathfrak{t} \subseteq \mathfrak{g}$ all of whose elements are semisimple, in the sense of definition 7.1.1. Equivalently, $\mathfrak{t}$ is a subalgebra such that $\mathrm{ad}H$ acts diagonalizably on $\mathfrak{g}$ for every $H \in \mathfrak{t}$. A toral subalgebra is *maximal* if it is not properly contained in a larger toral subalgebra.

The equivalence of the two formulations is the definition of a semisimple element unwound. The

word "toral" is justified because the operators $\mathrm{ad}H$ for $H \in \mathfrak{t}$ form a commuting family, as the next result shows, and a commuting family of diagonalizable operators over $\mathbb{C}$ is simultaneously diagonalizable ([2, Simultaneous Diagonalization]); their common eigenspace decomposition of $\mathfrak{g}$ is what produces the roots. A maximal toral subalgebra exists by dimension: any nonzero semisimple element generates a one-dimensional toral subalgebra, and an ascending chain of toral subalgebras must stabilize since $\mathfrak{g}$ is finite dimensional. That a nonzero semisimple element exists at all in a nonzero semisimple $\mathfrak{g}$ is the content of the existence theorem below.

Definition 7.1.4: Cartan Subalgebra

Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$. A *Cartan subalgebra* of $\mathfrak{g}$ is a subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ that is

1. nilpotent as a Lie algebra, in the sense of definition 5.2.3; and
2. self-normalizing: its normalizer

$$N_{\mathfrak{g}}(\mathfrak{h}) = \{X \in \mathfrak{g} \mid [X, \mathfrak{h}] \subseteq \mathfrak{h}\}$$

equals $\mathfrak{h}$ itself.

The definition applies to any Lie algebra, not only a semisimple one, and its two clauses encode a maximality principle. Nilpotency asks that $\mathfrak{h}$ be as close to abelian as a subalgebra can be while still being detected by ad; self-normalization asks that $\mathfrak{h}$ cannot be enlarged by adjoining an element that normalizes it without already lying inside it. The tension between "as abelian as possible" and "as large as possible" is exactly the tension a maximal toral subalgebra resolves: for semisimple $\mathfrak{g}$ the two conditions will collapse into "maximal toral," and the nilpotency will strengthen all the way to abelianness. The self-normalizing condition is the one that does structural work later, because it is what forces the eigenspace $\mathfrak{g}_0$ of $\mathrm{ad}\mathfrak{h}$ for the eigenvalue 0 to be $\mathfrak{h}$ and nothing larger.

Two facts must be secured before a maximal toral subalgebra can be recognized as a Cartan subalgebra: that toral subalgebras are abelian, and that a semisimple algebra contains a nonzero semisimple element in the first place.

Lemma 7.1.5: Toral Subalgebras Are Abelian

Let $\mathfrak{t}$ be a toral subalgebra of a semisimple Lie algebra $\mathfrak{g}$ over $\mathbb{C}$. Then $\mathfrak{t}$ is abelian: $[\mathfrak{t}, \mathfrak{t}] = 0$.

The argument is the standard one and turns on a clean observation: a diagonalizable operator with an eigenvector of nonzero eigenvalue cannot itself have all its brackets land back inside a space of semisimple elements. The idea is that if $\mathrm{ad}H$ had a nonzero eigenvalue on $\mathfrak{t}$, the eigenvector would be a semisimple element whose ad nonetheless acts nilpotently on part of $\mathfrak{t}$, a contradiction.

***Proof*:**
It suffices to show $\mathrm{ad}_{\mathfrak{t}}H = 0$ for every $H \in \mathfrak{t}$, where $\mathrm{ad}_{\mathfrak{t}}H$ denotes the restriction of $\mathrm{ad}H$ to $\mathfrak{t}$ (which maps $\mathfrak{t}$ into itself because $\mathfrak{t}$ is a subalgebra); this says $[H, \mathfrak{t}] = 0$ for all H, hence $[\mathfrak{t}, \mathfrak{t}] = 0$.

Step 1: reduction to a fixed H with $\mathrm{ad}_{\mathfrak{t}}H \neq 0$. Suppose for contradiction that some $H \in \mathfrak{t}$ has $\mathrm{ad}_{\mathfrak{t}}H \neq 0$. Since H is a semisimple element, $\mathrm{ad}H$ is diagonalizable on $\mathfrak{g}$, hence its restriction $\mathrm{ad}_{\mathfrak{t}}H$ is diagonalizable on the invariant subspace $\mathfrak{t}$. Being nonzero, it has a nonzero eigenvalue:

there is $Y \in \mathfrak{t}$, $Y \neq 0$, and a scalar $a \neq 0$ with

$$[H, Y] = a\,Y$$

Step 2: $\mathrm{ad}Y$ *acts nilpotently on the two-dimensional span, forcing a contradiction.* The element $Y \in \mathfrak{t}$ is semisimple, so $\mathrm{ad}Y$ is diagonalizable on $\mathfrak{g}$. On the other hand, consider how $\mathrm{ad}Y$ acts inside the subalgebra $\mathfrak{t}$. From $[H, Y] = aY$ one has $[Y, H] = -aY$, so $\mathrm{ad}Y$ sends H to $-aY$ and sends Y to $[Y, Y] = 0$. Thus in the subspace $U = \mathbb{C}H + \mathbb{C}Y \subseteq \mathfrak{t}$, the operator $\mathrm{ad}_{\mathfrak{t}}Y$ maps $H \mapsto -aY \mapsto 0$ and $Y \mapsto 0$, so $(\mathrm{ad}_{\mathfrak{t}}Y)^2 = 0$ on U and $\mathrm{ad}_{\mathfrak{t}}Y$ is nilpotent on U but not the zero operator there (it sends H to $-aY \neq 0$). Now Y is semisimple, so $\mathrm{ad}Y$ is diagonalizable on all of $\mathfrak{g}$; its restriction to the $\mathrm{ad}Y$-invariant subspace U (note $\mathrm{ad}_{\mathfrak{t}}Y$ preserves U, since $[Y, H] = -aY \in U$ and $[Y, Y] = 0 \in U$) is therefore also diagonalizable. A diagonalizable nilpotent operator is zero. This contradicts $\mathrm{ad}_{\mathfrak{t}}Y \neq 0$ on U. Hence no such H exists, every $\mathrm{ad}_{\mathfrak{t}}H = 0$, and $\mathfrak{t}$ is abelian, as we sought to show.

The lemma is what makes "toral" a workable notion: a toral subalgebra is more than a set of commuting semisimple operators; it is an abelian subalgebra, so the family $\{\mathrm{ad}H : H \in \mathfrak{t}\}$ consists of commuting diagonalizable operators and can be simultaneously diagonalized on $\mathfrak{g}$. This is the mechanism the root space decomposition of section 7.2 runs on. It remains to guarantee that a nonzero maximal toral subalgebra exists, which is where semisimplicity enters through the abstract Jordan decomposition.

Theorem 7.1.6: Existence of Cartan Subalgebras

Every nonzero semisimple Lie algebra $\mathfrak{g}$ over $\mathbb{C}$ contains a nonzero semisimple element, and hence a nonzero maximal toral subalgebra. Any maximal toral subalgebra of $\mathfrak{g}$ is a Cartan subalgebra; in particular Cartan subalgebras of $\mathfrak{g}$ exist.

The heart of the theorem is the first clause: a semisimple algebra is not built from nilpotent elements alone. If every element of $\mathfrak{g}$ were nilpotent, then ad would map $\mathfrak{g}$ into the nilpotent operators, and Engel's theorem would force $\mathfrak{g}$ to be nilpotent as a Lie algebra, hence solvable, hence to have nonzero radical, contradicting semisimplicity. The second clause, that a maximal toral subalgebra is a Cartan subalgebra, is the substantive proposition proved just below, so the theorem's proof cites it.

***Proof*:**

Step 1: a nonzero semisimple element exists. Suppose not, so that $x_s = 0$ for every $x \in \mathfrak{g}$, that is, every element of $\mathfrak{g}$ is nilpotent (definition 7.1.1). Then $\mathrm{ad}x$ is a nilpotent operator on $\mathfrak{g}$ for every $x \in \mathfrak{g}$, so the linear Lie algebra $\mathrm{ad}(\mathfrak{g}) \subseteq \mathfrak{gl}(\mathfrak{g})$ consists of nilpotent operators. By Engel's theorem, $\mathrm{ad}(\mathfrak{g})$ is a nilpotent Lie algebra; and $\mathfrak{g} \cong \mathrm{ad}(\mathfrak{g})$ (the kernel of ad is the center, zero by corollary 6.3.5), so $\mathfrak{g}$ is itself a nilpotent Lie algebra. A nilpotent Lie algebra is solvable, so $\mathrm{rad}\,\mathfrak{g} = \mathfrak{g} \neq 0$, contradicting the semisimplicity of $\mathfrak{g}$ (definition 5.5.5). Hence some $x \in \mathfrak{g}$ has $x_s \neq 0$, and x_s is a nonzero semisimple element.

Step 2: a maximal toral subalgebra exists. The line $\mathbb{C}x_s$ is a toral subalgebra: it is abelian, hence a subalgebra, and every scalar multiple cx_s is semisimple because $\mathrm{ad}(cx_s) = c\,\mathrm{ad}x_s$ is a scalar multiple of a diagonalizable operator, again diagonalizable. Thus the set of toral subalgebras of $\mathfrak{g}$ is nonempty. Since $\mathfrak{g}$ is finite dimensional, a toral subalgebra of maximal dimension exists; it is a maximal toral subalgebra $\mathfrak{h}$, and it is nonzero because it contains $\mathbb{C}x_s$.

Step 3: a maximal toral subalgebra is a Cartan subalgebra. By proposition 7.1.7, proved next, a

maximal toral subalgebra $\mathfrak{h}$ is abelian and self-centralizing, and is a Cartan subalgebra. This establishes the existence of Cartan subalgebras, completing the proof.

The theorem guarantees the raw material; the following proposition gives the structural facts that make a maximal toral subalgebra usable and confirm it is a Cartan subalgebra in the sense of definition 7.1.4. Its central assertion, self-centralization $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, is the technical result on which the entire root space decomposition rests: it is the statement that the zero eigenspace of $\mathrm{ad}\mathfrak{h}$ on $\mathfrak{g}$ is exactly $\mathfrak{h}$, so that no root space hides inside the centralizer.

Proposition 7.1.7: Maximal Toral Subalgebras Are Cartan Subalgebras

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ and let $\mathfrak{h}$ be a maximal toral subalgebra. Then:

1. $\mathfrak{h}$ is abelian.
2. $\mathfrak{h}$ is self-centralizing: $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, where $C_{\mathfrak{g}}(\mathfrak{h}) = \{X \in \mathfrak{g} \mid [X, H] = 0 \text{ for all } H \in \mathfrak{h}\}$.
3. $\mathfrak{h}$ is self-normalizing, $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, and is a Cartan subalgebra.

Conversely, every Cartan subalgebra of $\mathfrak{g}$ is a maximal toral subalgebra. Thus the Cartan subalgebras of a semisimple Lie algebra are exactly its maximal toral subalgebras.

The abelianness in (1) is lemma 7.1.5. The self-centralization in (2) is the crux, and its proof is a careful application of the abstract Jordan decomposition together with an Engel-type argument, run inside the centralizer. The self-normalization in (3) follows from (2) once one notes that for an abelian $\mathfrak{h}$ the normalizer and centralizer coincide up to the 0-eigenspace bookkeeping that (2) controls. The converse pins down the notion: having proved that maximal toral subalgebras are Cartan subalgebras, one checks that a Cartan subalgebra can be nothing else.

***Proof*:**
Statement (1) is lemma 7.1.5. The work is in (2); (3) and the converse follow.

Step 1: the centralizer $\mathfrak{c} = C_{\mathfrak{g}}(\mathfrak{h})$ is stable under Jordan parts and contains $\mathfrak{h}$. Since $\mathfrak{h}$ is abelian by (1), $[\mathfrak{h}, \mathfrak{h}] = 0$, so $\mathfrak{h} \subseteq \mathfrak{c}$. The centralizer $\mathfrak{c}$ is a subalgebra: if X, Y centralize $\mathfrak{h}$ then so does $[X, Y]$, by the Jacobi identity $[[X, Y], H] = [X, [Y, H]] - [Y, [X, H]] = 0$. The key stability property is this: if $x \in \mathfrak{c}$, then both Jordan parts x_s, x_n lie in $\mathfrak{c}$. Indeed, $\mathrm{ad}x_s$ and $\mathrm{ad}x_n$ are polynomials in $\mathrm{ad}x$ with zero constant term (the construction of the Jordan decomposition of the operator $\mathrm{ad}x$), so for $H \in \mathfrak{h}$, $[x, H] = 0$ gives $\mathrm{ad}x\,(H) = 0$, hence any polynomial in $\mathrm{ad}x$ without constant term also kills H: $[x_s, H] = (\mathrm{ad}x_s)(H) = 0$ and likewise $[x_n, H] = 0$. Thus $x_s, x_n \in \mathfrak{c}$.

Step 2: $\mathfrak{h}$ contains every semisimple element of $\mathfrak{c}$. Let $s \in \mathfrak{c}$ be a semisimple element. Then $\mathfrak{h} + \mathbb{C}s$ is a subalgebra: $\mathfrak{h}$ is abelian and s commutes with $\mathfrak{h}$ (as $s \in \mathfrak{c}$), so all brackets among $\mathfrak{h} \cup \{s\}$ vanish, making $\mathfrak{h} + \mathbb{C}s$ an abelian subalgebra. Every element of it is a sum $H + cs$ with $[H, s] = 0$, and both $\mathrm{ad}H$ and $\mathrm{ad}s$ are diagonalizable and commute (their bracket is $\mathrm{ad}[H, s] = 0$), so $\mathrm{ad}(H + cs) = \mathrm{ad}H + c\,\mathrm{ad}s$ is a sum of commuting diagonalizable operators, hence diagonalizable. Therefore $\mathfrak{h} + \mathbb{C}s$ is a toral subalgebra containing $\mathfrak{h}$. By maximality of $\mathfrak{h}$, $\mathfrak{h} + \mathbb{C}s = \mathfrak{h}$, so $s \in \mathfrak{h}$. Every semisimple element of $\mathfrak{c}$ lies in $\mathfrak{h}$.

Step 3: $\mathfrak{c}$ is nilpotent. Consider $\mathrm{ad}_{\mathfrak{c}}x$, the restriction to $\mathfrak{c}$ of $\mathrm{ad}x$, for $x \in \mathfrak{c}$; this is well-defined since $\mathfrak{c}$ is a subalgebra. For $x \in \mathfrak{c}$ write $x = x_s + x_n$ with $x_s, x_n \in \mathfrak{c}$ (Step 1). By Step 2, $x_s \in \mathfrak{h}$, and $\mathfrak{h}$ centralizes $\mathfrak{c}$: indeed for $H \in \mathfrak{h}$ and $y \in \mathfrak{c}$ one has $[H, y] = 0$ by the very definition of $\mathfrak{c} = C_{\mathfrak{g}}(\mathfrak{h})$. Hence $\mathrm{ad}_{\mathfrak{c}}x_s = 0$, and so $\mathrm{ad}_{\mathfrak{c}}x = \mathrm{ad}_{\mathfrak{c}}x_n$, which is the restriction of the nilpotent

operator $\mathrm{ad}x_n$ and is therefore nilpotent on $\mathfrak{c}$. Thus every $\mathrm{ad}_{\mathfrak{c}}x$ is a nilpotent operator on $\mathfrak{c}$. By Engel's theorem, $\mathfrak{c}$ is a nilpotent Lie algebra.

Step 4: $\mathfrak{c} = \mathfrak{h}$. Suppose $\mathfrak{c} \neq \mathfrak{h}$. Because $\mathfrak{c}$ is nilpotent and $\mathfrak{h} \subsetneq \mathfrak{c}$ is a proper subalgebra, the normalizer of $\mathfrak{h}$ in $\mathfrak{c}$ strictly contains $\mathfrak{h}$: a proper subalgebra of a nilpotent Lie algebra is properly contained in its normalizer, since the lower central series descends into $\mathfrak{h}$ and the last stage of the descent that still leaves $\mathfrak{h}$ produces a normalizing element outside it. Choose $z \in N_{\mathfrak{c}}(\mathfrak{h}) \setminus \mathfrak{h}$. Then $[z, \mathfrak{h}] \subseteq \mathfrak{h}$, and since $z \in \mathfrak{c}$ centralizes $\mathfrak{h}$, in fact $[z, \mathfrak{h}] = 0$, consistent with $z \in \mathfrak{c}$. Decompose $z = z_s + z_n$ with $z_s, z_n \in \mathfrak{c}$ (Step 1). By Step 2, $z_s \in \mathfrak{h}$. Therefore $z_n = z - z_s \in \mathfrak{c}$, and $z_n \notin \mathfrak{h}$ because $z \notin \mathfrak{h}$ while $z_s \in \mathfrak{h}$. So $\mathfrak{c}$ contains a nonzero nilpotent element z_n outside $\mathfrak{h}$.

Now z_n is a nilpotent element commuting with $\mathfrak{h}$, so $\mathrm{ad}z_n$ is nilpotent and $[z_n, H] = 0$ for all $H \in \mathfrak{h}$. The contradiction comes from the Killing form κ of $\mathfrak{g}$, which is nondegenerate by theorem 6.3.1. First let $y \in \mathfrak{c}$. The map ad is injective (corollary 6.3.5), so $\mathrm{ad}(\mathfrak{c}) \subseteq \mathfrak{gl}(\mathfrak{g})$ is isomorphic to the nilpotent algebra $\mathfrak{c}$ (Step 3), hence solvable (proposition 5.2.4(1)). By Lie's theorem (theorem 5.4.4) there is one basis of $\mathfrak{g}$ in which every $\mathrm{ad}y$ with $y \in \mathfrak{c}$ is upper triangular. In that basis the nilpotent operator $\mathrm{ad}z_n$ is upper triangular with its eigenvalues, all 0, on the diagonal, so it is strictly upper triangular, and the product $\mathrm{ad}z_n \,\mathrm{ad}y$ of a strictly upper triangular and an upper triangular matrix is strictly upper triangular. Hence $\kappa(z_n, y) = \mathrm{tr}(\mathrm{ad}z_n \,\mathrm{ad}y) = 0$ for all $y \in \mathfrak{c}$.

It remains to see $\kappa(z_n, w) = 0$ for w outside $\mathfrak{c}$ as well. Decompose $\mathfrak{g}$ under the commuting family $\{\mathrm{ad}H : H \in \mathfrak{h}\}$ (simultaneously diagonalizable by lemma 7.1.5 and the semisimplicity of each H): $\mathfrak{g} = \mathfrak{c} \oplus \left(\bigoplus \mathfrak{g}^{\lambda}\right)$, where $\mathfrak{c} = \mathfrak{g}^0$ is the common 0-eigenspace and each $\mathfrak{g}^{\lambda}$ is a common eigenspace for a nonzero eigenvalue-functional λ. For $w \in \mathfrak{g}^{\lambda}$ with $\lambda \neq 0$, the operator $\mathrm{ad}z_n \,\mathrm{ad}w$ shifts the weight by λ: it maps $\mathfrak{g}^{\mu}$ into $\mathfrak{g}^{\mu+\lambda}$, because z_n has weight 0 (it commutes with $\mathfrak{h}$) and w has weight λ. An operator that strictly raises the weight has no diagonal blocks, hence trace zero: $\kappa(z_n, w) = \mathrm{tr}(\mathrm{ad}z_n \,\mathrm{ad}w) = 0$. Combining, $\kappa(z_n, u) = 0$ for all $u \in \mathfrak{g} = \mathfrak{c} \oplus \bigoplus_{\lambda \neq 0} \mathfrak{g}^{\lambda}$. Thus z_n lies in the radical of κ, which is 0 by nondegeneracy, forcing $z_n = 0$. This contradicts $z_n \neq 0$. Therefore $\mathfrak{c} = \mathfrak{h}$.

Step 5: self-normalization and the Cartan conclusion. Let $X \in N_{\mathfrak{g}}(\mathfrak{h})$, so $[X, \mathfrak{h}] \subseteq \mathfrak{h}$. The map $\mathrm{ad}_{\mathfrak{h}/} X \colon \mathfrak{h} \to \mathfrak{h}$, $H \mapsto [X, H]$, is a linear map; consider the induced action of $\mathfrak{h}$ on the quotient recording how X sits over $\mathfrak{h}$. Concretely, for each $H \in \mathfrak{h}$ one has $[H, X] \in \mathfrak{h}$, so the coset $\bar{X} \in \mathfrak{g}/\mathfrak{h}$ satisfies $\mathrm{ad}H\,(\bar{X}) = 0$. Since $\mathfrak{h}$ acts on $\mathfrak{g}$ by commuting semisimple (diagonalizable) operators, it acts semisimply on the quotient $\mathfrak{g}/\mathfrak{h}$ as well, and $\bar{X}$ is annihilated by every $\mathrm{ad}H$. But the 0-weight space of $\mathfrak{h}$ acting on $\mathfrak{g}/\mathfrak{h}$ is $C_{\mathfrak{g}}(\mathfrak{h})/\mathfrak{h} = \mathfrak{c}/\mathfrak{h} = 0$ by Step 4. Hence $\bar{X} = 0$, that is $X \in \mathfrak{h}$. Therefore $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, so $\mathfrak{h}$ is self-normalizing. Being abelian, $\mathfrak{h}$ is nilpotent, and being self-normalizing it satisfies both clauses of definition 7.1.4; so $\mathfrak{h}$ is a Cartan subalgebra.

Step 6: the converse. Let $\mathfrak{k}$ be a Cartan subalgebra of $\mathfrak{g}$, so $\mathfrak{k}$ is nilpotent and $N_{\mathfrak{g}}(\mathfrak{k}) = \mathfrak{k}$. Take $x \in \mathfrak{k}$. The operators $\mathrm{ad}x_s = (\mathrm{ad}x)_s$ and $\mathrm{ad}x_n = (\mathrm{ad}x)_n$ (definition 7.1.1) are polynomials in $\mathrm{ad}x$ without constant term.[a] Since $\mathrm{ad}x$ maps $\mathfrak{k}$ into $\mathfrak{k}$, so do they, and therefore $x_s, x_n \in N_{\mathfrak{g}}(\mathfrak{k}) = \mathfrak{k}$. The restriction of $\mathrm{ad}x$ to the nilpotent algebra $\mathfrak{k}$ is nilpotent (theorem 5.3.4), so the restriction of $\mathrm{ad}x_s$, a polynomial without constant term in it, is nilpotent too. That restriction is also diagonalizable, as the restriction of the diagonalizable $\mathrm{ad}x_s$ to an invariant subspace, and an operator that is both nilpotent and diagonalizable is 0. So $[x_s, \mathfrak{k}] = 0$: the semisimple part of every element of $\mathfrak{k}$ is central in $\mathfrak{k}$.

Let $\mathfrak{t}$ be the set of semisimple elements of $\mathfrak{k}$. Each $s \in \mathfrak{t}$ equals its own semisimple part, so it is central in $\mathfrak{k}$ by the previous paragraph, and in particular the elements of $\mathfrak{t}$ commute pairwise. For $s, s' \in \mathfrak{t}$ the operators $\mathrm{ad}s$ and $\mathrm{ad}s'$ are diagonalizable and commute, since

$[\mathrm{ad}s, \mathrm{ad}s'] = \mathrm{ad}[s, s'] = 0$, so $\mathrm{ad}(s + s')$ is diagonalizable as in Step 2, and $s + s' \in \mathfrak{t}$. Likewise $cs \in \mathfrak{t}$ for $c \in \mathbb{C}$. Thus $\mathfrak{t}$ is a subspace closed under the (zero) bracket, a toral subalgebra. It contains x_s for every $x \in \mathfrak{k}$, since $x_s \in \mathfrak{k}$ and $\mathrm{ad}x_s = (\mathrm{ad}x)_s$ is diagonalizable, and $\mathfrak{k} \subseteq \mathfrak{c}' = C_{\mathfrak{g}}(\mathfrak{t})$ because $\mathfrak{t}$ is central in $\mathfrak{k}$. The centralizer $\mathfrak{c}'$ is a subalgebra containing $\mathfrak{k}$, so each $\mathrm{ad}x$ with $x \in \mathfrak{k}$ preserves $\mathfrak{c}'$, and on $\mathfrak{c}'$ it agrees with $\mathrm{ad}x_n$ because $x_s \in \mathfrak{t}$ kills $\mathfrak{c}'$. Hence $\mathfrak{k}$ acts on $\mathfrak{c}'$ by nilpotent operators preserving $\mathfrak{k}$, and so on the quotient $\mathfrak{c}'/\mathfrak{k}$ by nilpotent operators, which form a Lie subalgebra of $\mathfrak{gl}(\mathfrak{c}'/\mathfrak{k})$ as the image of the induced representation. If $\mathfrak{c}' \neq \mathfrak{k}$, the operator form of theorem 5.3.4 gives a nonzero vector of $\mathfrak{c}'/\mathfrak{k}$ killed by all of $\mathfrak{k}$, that is some $y \in \mathfrak{c}' \setminus \mathfrak{k}$ with $[\mathfrak{k}, y] \subseteq \mathfrak{k}$. Such a y lies in $N_{\mathfrak{g}}(\mathfrak{k}) \setminus \mathfrak{k}$, which is empty. So $\mathfrak{k} = C_{\mathfrak{g}}(\mathfrak{t})$.

Finally, let $\mathfrak{t}'$ be a maximal toral subalgebra containing $\mathfrak{t}$, which exists by dimension. It is abelian (lemma 7.1.5), so it centralizes $\mathfrak{t}$, and $\mathfrak{t}' \subseteq C_{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{k}$. Its elements are semisimple elements of $\mathfrak{k}$, so $\mathfrak{t}' \subseteq \mathfrak{t}$, and $\mathfrak{t} = \mathfrak{t}'$ is maximal toral. By (2) applied to $\mathfrak{t}$, $\mathfrak{k} = C_{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{t}$. Hence $\mathfrak{k}$ is a maximal toral subalgebra, and the two notions coincide, completing the proof.

[a] This is part (2) of [2, Jordan-Chevalley Decomposition], applied to the operator $\mathrm{ad}x$ on $\mathfrak{g}$.

The proposition is the technical foundation of the whole chapter, and the piece worth dwelling on is the self-centralization $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$ of statement (2). Every subsequent result rests on it. When $\mathfrak{g}$ is decomposed into simultaneous eigenspaces for $\mathrm{ad}\mathfrak{h}$ in the next section, the eigenspace for the eigenvalue 0 is precisely the centralizer $C_{\mathfrak{g}}(\mathfrak{h})$, and the proposition says that this 0-eigenspace is exactly $\mathfrak{h}$. That is what guarantees the root space decomposition has the clean form $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_\alpha \mathfrak{g}_\alpha$, with $\mathfrak{h}$ alone in the center and every other summand carrying a nonzero root. Without self-centralization there could be extra pieces of $\mathfrak{g}$ commuting with $\mathfrak{h}$ but not lying in it, and the roots would fail to see the whole algebra. The proof's mechanism, that the nilpotent elements of the centralizer are forced into the radical of the Killing form by a weight argument, is the point where the nondegeneracy earned in chapter 6 does its most consequential work.

For the remainder of the chapter, *fix* a maximal toral subalgebra $\mathfrak{h}$ of the semisimple Lie algebra $\mathfrak{g}$, which by proposition 7.1.7 is a Cartan subalgebra: abelian, self-centralizing, and self-normalizing. Its dimension is called the *rank* of $\mathfrak{g}$. The rank does not depend on the choice of $\mathfrak{h}$, because any two maximal toral subalgebras are carried onto each other by an inner automorphism of $\mathfrak{g}$ (theorem 13.2.7). That conjugacy is proved in Part IV, from the compact real form and the maximal torus theorem, and its proof does not use the independence of the rank. The concrete computations below fix $\mathfrak{h}$ once and for all in each case.

The whole apparatus is transparent for $\mathfrak{sl}_n$, the running model.

Example 7.1.8: The Diagonal Cartan Subalgebra of $\mathfrak{sl}_n$

Let $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$ and let

$$\mathfrak{h} = \left\{ \mathrm{diag}(a_1, \ldots, a_n) \;\middle|\; a_i \in \mathbb{C},\ \sum_i a_i = 0 \right\}$$

be the traceless diagonal matrices. This $\mathfrak{h}$ is a maximal toral subalgebra, hence a Cartan subalgebra, of dimension $n - 1$.

First, $\mathfrak{h}$ is toral. It is abelian, since diagonal matrices commute, and every $H = \mathrm{diag}(a_1, \ldots, a_n) \in \mathfrak{h}$ is a semisimple element: from example 7.1.2, $\mathrm{ad}H\,(E_{ij}) = (a_i - a_j)E_{ij}$ and $\mathrm{ad}H$ kills every diagonal matrix, so $\mathrm{ad}H$ is diagonalized by the basis $\{E_{ij}\}_{i \neq j}$ together with a basis of $\mathfrak{h}$. Thus each H is semisimple and $\mathfrak{h}$ is toral, of dimension $n - 1$ (the traceless condition removes one dimension from the n diagonal entries).

Next, $\mathfrak{h}$ is maximal toral, which is fastest to see through self-centralization. Compute $C_{\mathfrak{g}}(\mathfrak{h})$: a matrix $X = (x_{ij}) \in \mathfrak{sl}_n$ centralizes $\mathfrak{h}$ if and only if $[H, X] = 0$ for all $H \in \mathfrak{h}$, that is $(a_i - a_j)x_{ij} = 0$ for all i, j and all admissible $(a_1, \ldots, a_n)$ with $\sum a_i = 0$. For $i \neq j$ one can choose $H \in \mathfrak{h}$ with $a_i \neq a_j$ (the traceless diagonal matrices separate any two distinct indices), forcing $x_{ij} = 0$. Hence X is diagonal and traceless, so $X \in \mathfrak{h}$: the centralizer is $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. Any toral subalgebra containing $\mathfrak{h}$ is abelian and centralizes $\mathfrak{h}$, hence lies in $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, so $\mathfrak{h}$ is maximal toral. By proposition 7.1.7 it is a Cartan subalgebra, of rank $n - 1$. For $n = 2$ this is the one-dimensional $\mathbb{C}h$ with $h = \operatorname{diag}(1, -1)$, the Cartan subalgebra of $\mathfrak{sl}_2$ from example 7.1.2; the eigenvalues $a_i - a_j$ that appeared here as $\operatorname{ad}H$-eigenvalues become the roots $\varepsilon_i - \varepsilon_j$ of the next section.

The computation shows the abstract self-centralization property in its most concrete form: for $\mathfrak{sl}_n$ the centralizer of the diagonal matrices is just the diagonal matrices, precisely because the eigenvalue differences $a_i - a_j$ do not all vanish for $i \neq j$. This is the finite-dimensional linear algebra behind proposition 7.1.7(2), and it makes visible why the diagonal subalgebra is the natural coordinate system on $\mathfrak{sl}_n$. The $\operatorname{ad}H$-eigenvalues $a_i - a_j$ read off here are the roots; the next section formalizes them and decomposes $\mathfrak{g}$ into their eigenspaces.

Exercise 7.1.1

1. Let $X = \left(\begin{smallmatrix} 2 & 3 \\ 0 & 2 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 3 & 1 \\ 0 & -3 \end{smallmatrix}\right)$, both regarded inside $\mathfrak{gl}_2(\mathbb{C})$ (and $Y \in \mathfrak{sl}_2$). For each, determine the Jordan decomposition into semisimple and nilpotent parts, deciding in particular whether the "diagonal plus strictly upper triangular" splitting is or is not the Jordan decomposition, and explain the difference between the two cases in terms of the eigenvalues.

2. Exhibit a maximal toral (Cartan) subalgebra of $\mathfrak{so}_{2n}(\mathbb{C})$ and of $\mathfrak{sp}_{2n}(\mathbb{C})$ and compute its dimension (the rank), where $\mathfrak{so}_{2n}$ is taken with respect to the split symmetric form $\left(\begin{smallmatrix} 0 & I_n \\ I_n & 0 \end{smallmatrix}\right)$ and $\mathfrak{sp}_{2n}$ with respect to $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$. In each case verify self-centralization on the diagonal subalgebra.

3. The Lie algebra $\mathfrak{gl}_n(\mathbb{C})$ is reductive but not semisimple (it has a nonzero center). Show that its full diagonal subalgebra, all diagonal matrices, is self-centralizing, and explain why this is the correct Cartan subalgebra of $\mathfrak{gl}_n$ even though proposition 7.1.7 is stated for the semisimple case. Contrast with $\mathfrak{sl}_n$: why is the $\mathfrak{gl}_n$ Cartan one dimension larger?

4. Give an example of a nonzero abelian subalgebra of $\mathfrak{sl}_2(\mathbb{C})$ that is nilpotent and self-normalizing among subalgebras of a proper solvable subalgebra, yet is *not* a Cartan subalgebra of $\mathfrak{sl}_2$, and identify which clause fails. Use it to explain why the definition of a Cartan subalgebra is relative to the ambient algebra.

5. Prove directly, without invoking proposition 7.1.7, that any two maximal toral subalgebras of $\mathfrak{sl}_2(\mathbb{C})$ have the same dimension. (Conjugacy of Cartan subalgebras, theorem 13.2.7, gives the general statement. Here a hands-on argument suffices.)

7.2 The Root Space Decomposition

The previous section produced, for a semisimple Lie algebra $\mathfrak{g}$ over $\mathbb{C}$, a Cartan subalgebra $\mathfrak{h}$: a maximal toral subalgebra, which by proposition 7.1.7 is abelian and self-centralizing. The word "toral" means every element of $\mathfrak{h}$ acts diagonalizably on $\mathfrak{g}$ through the adjoint representation, and

"abelian" means these operators commute with one another. Commuting diagonalizable operators can be simultaneously diagonalized, and this section cashes that linear-algebra fact into the central structural statement of the whole theory. The operators $\mathrm{ad}H$ for $H \in \mathfrak{h}$ break $\mathfrak{g}$ into simultaneous eigenspaces, and each eigenvalue is not a single scalar but a linear functional on $\mathfrak{h}$, recording the eigenvalue $\mathrm{ad}H$ produces for each H at once. These functionals are the *roots*, and the eigenspace decomposition they organize is the *root space decomposition*.

The decomposition converts a semisimple Lie algebra into combinatorial data. The whole nonabelian complexity of $\mathfrak{g}$ is compressed into a finite set Φ of vectors in the dual space $\mathfrak{h}^*$, together with the rule for how the eigenspaces bracket against one another. The remaining chapters of Part II show that this finite set of vectors, the root system, determines $\mathfrak{g}$ up to isomorphism and is classified completely by a Dynkin diagram. Everything downstream, the Cartan integers, the Weyl group, and the Cartan-Killing classification, is read off the decomposition established here.

The starting point is to name the simultaneous eigenspaces. Because the operators $\mathrm{ad}H$ commute and are diagonalizable, a simultaneous eigenvector X satisfies $[H, X] = \mathrm{ad}H(X) = \lambda(H)X$ for each H, where the eigenvalue $\lambda(H)$ depends linearly on H; that dependence packages into a single element of $\mathfrak{h}^*$.

Definition 7.2.1: Root Space

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$. For a linear functional $\alpha \in \mathfrak{h}^*$, the associated *root space* is

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} \mid [H, X] = \alpha(H)\, X \text{ for all } H \in \mathfrak{h}\}$$

the simultaneous eigenspace on which every $\mathrm{ad}H$ acts by the scalar $\alpha(H)$.

The functional $\alpha = 0$ is allowed: $\mathfrak{g}_0$ is the subspace killed by every $\mathrm{ad}H$, which is exactly the centralizer $C_{\mathfrak{g}}(\mathfrak{h}) = \{X \mid [H, X] = 0 \text{ for all } H \in \mathfrak{h}\}$. For a nonzero α, the space $\mathfrak{g}_\alpha$ may or may not be zero; only finitely many α can give a nonzero space, since the $\mathfrak{g}_\alpha$ are independent subspaces of the finite-dimensional $\mathfrak{g}$. The nonzero ones carry the interesting information, and they are named next.

Definition 7.2.2: Roots

The *set of roots* of $\mathfrak{g}$ relative to $\mathfrak{h}$ is

$$\Phi = \{\alpha \in \mathfrak{h}^* \mid \alpha \neq 0 \text{ and } \mathfrak{g}_\alpha \neq 0\}$$

the set of nonzero functionals whose root space is nonzero. An element of Φ is called a *root*.

A root is thus a nonzero simultaneous eigenvalue of the family $\{\mathrm{ad}H : H \in \mathfrak{h}\}$, recorded as a functional on $\mathfrak{h}$. The condition $\alpha \neq 0$ deliberately excludes the zero eigenvalue, which behaves differently from the rest: the theorem below shows $\mathfrak{g}_0$ is exactly $\mathfrak{h}$ itself, so it is the diagonal part of $\mathfrak{g}$ rather than an "off-diagonal" eigenspace. Separating $\mathfrak{h}$ from Φ is what makes the decomposition useful. Before proving the general theorem, the running example shows every object.

Example 7.2.3: Roots of $\mathfrak{sl}_n$

Let $\mathfrak{g} = \mathfrak{sl}_n = \mathfrak{sl}_n(\mathbb{C})$, the trace-zero $n \times n$ complex matrices, and let $\mathfrak{h}$ be the subalgebra of diagonal traceless matrices, which is the Cartan subalgebra identified in exercise 7.1.1. Write

a diagonal element as $H = \operatorname{diag}(h_1, \dots, h_n)$ with $\sum_i h_i = 0$, and let $\varepsilon_i \in \mathfrak{h}^*$ be the functional reading off the i-th diagonal entry, $\varepsilon_i(H) = h_i$. These satisfy the single relation $\varepsilon_1 + \cdots + \varepsilon_n = 0$ on $\mathfrak{h}$ (the trace condition), so $\mathfrak{h}^*$ has dimension $n - 1$.

For $i \neq j$ let E_{ij} be the elementary matrix with a single 1 in position (i, j). A direct computation with the commutator gives, for diagonal $H = \operatorname{diag}(h_1, \dots, h_n)$,

$$[H, E_{ij}] = HE_{ij} - E_{ij}H = h_i E_{ij} - h_j E_{ij} = (h_i - h_j)\, E_{ij}$$

since HE_{ij} scales row i by h_i and $E_{ij}H$ scales column j by h_j. Thus E_{ij} is a simultaneous eigenvector with eigenvalue $(\varepsilon_i - \varepsilon_j)(H) = h_i - h_j$. Setting $\alpha_{ij} = \varepsilon_i - \varepsilon_j$, this says $E_{ij} \in \mathfrak{g}_{\alpha_{ij}}$, and $\alpha_{ij} \neq 0$ because for $i \neq j$ one can choose $H \in \mathfrak{h}$ with $h_i \neq h_j$. Each $\mathfrak{g}_{\alpha_{ij}}$ is one-dimensional, spanned by E_{ij}: the matrices $\{E_{ij} : i \neq j\}$ together with a basis of $\mathfrak{h}$ span $\mathfrak{sl}_n$, and distinct α_{ij} pick out distinct eigenlines. Hence

$$\Phi = \{\varepsilon_i - \varepsilon_j \mid 1 \leq i, j \leq n,\ i \neq j\}, \qquad \mathfrak{g}_{\varepsilon_i - \varepsilon_j} = \mathbb{C}E_{ij}$$

There are $n(n-1)$ ordered pairs (i, j) with $i \neq j$, and the functionals $\varepsilon_i - \varepsilon_j$ are pairwise distinct, so $\mathfrak{sl}_n$ has exactly $n(n-1)$ roots. The diagonal part $\mathfrak{h}$ is the zero eigenspace $\mathfrak{g}_0$: a matrix commutes with every diagonal H (with distinct entries available) exactly when it is itself diagonal, and the trace-zero condition then places it in $\mathfrak{h}$. Assembling the pieces,

$$\mathfrak{sl}_n = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathbb{C}E_{ij}$$

which is the familiar decomposition of a traceless matrix into its diagonal part and its off-diagonal entries, now read as an eigenspace decomposition for $\operatorname{ad}\mathfrak{h}$.

The example already exhibits the shape of the general theorem: $\mathfrak{sl}_n$ splits as $\mathfrak{h}$ plus a sum of one-dimensional root spaces, one for each off-diagonal position. The content of the general theorem is that this always happens, for any semisimple $\mathfrak{g}$ and any Cartan subalgebra $\mathfrak{h}$, and that the zero eigenspace is always exactly $\mathfrak{h}$. The one-dimensionality of the root spaces is a finer fact, deferred to section 7.4; what the decomposition itself needs is only that the eigenspaces exhaust $\mathfrak{g}$ and that $\mathfrak{g}_0 = \mathfrak{h}$.

The mechanism behind the theorem is a standard result of linear algebra: a commuting family of diagonalizable operators on a finite-dimensional vector space is simultaneously diagonalizable. This is the point at which $\mathfrak{h}$ being abelian (so the operators commute) and toral (so each is diagonalizable) both enter. The proof of the theorem includes the argument in full, since it is the engine of the whole decomposition.

Theorem 7.2.4: The Root Space Decomposition

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, and let $\Phi \subseteq \mathfrak{h}^*$ be its set of roots. Then

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha$$

a direct sum over $\mathfrak{h}^*$ in which all but finitely many summands vanish. The zero root space is the centralizer of $\mathfrak{h}$ and equals $\mathfrak{h}$ itself:

$$\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$$

Consequently

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

the *root space decomposition* of $\mathfrak{g}$ relative to $\mathfrak{h}$.

***Proof*:**

The proof has two stages. First, simultaneous diagonalization of the family $\{\mathrm{ad}H : H \in \mathfrak{h}\}$ yields the direct sum over all of $\mathfrak{h}^*$. Second, the self-centralizing property of $\mathfrak{h}$ identifies the zero eigenspace with $\mathfrak{h}$.

Step 1: the family $\{\mathrm{ad}H : H \in \mathfrak{h}\}$ *is simultaneously diagonalizable.* Each $H \in \mathfrak{h}$ lies in the toral subalgebra $\mathfrak{h}$, so by definition 7.1.3 the operator $\mathrm{ad}H$ is diagonalizable on $\mathfrak{g}$. The subalgebra $\mathfrak{h}$ is abelian by proposition 7.1.7, so for $H, H' \in \mathfrak{h}$ the identity

$$[\mathrm{ad}H, \mathrm{ad}H'] = \mathrm{ad}[H, H'] = \mathrm{ad}0 = 0$$

shows the operators $\mathrm{ad}H$ and $\mathrm{ad}H'$ commute in $\mathfrak{gl}(\mathfrak{g})$; here the first equality is the fact that ad is a Lie algebra homomorphism (definition 2.5.3). Thus $\{\mathrm{ad}H : H \in \mathfrak{h}\}$ is a commuting family of diagonalizable operators on the finite-dimensional space $\mathfrak{g}$.

Recall the linear-algebra fact that such a family is simultaneously diagonalizable ([2, Simultaneous Diagonalization]). The argument is an induction on the number of operators, resting on the observation that a diagonalizable operator restricts to a diagonalizable operator on any invariant subspace, and that commuting operators preserve one another's eigenspaces. In detail, fix one operator $\mathrm{ad}H_1$ and let $\mathfrak{g} = \bigoplus_\lambda V_\lambda$ be its eigenspace decomposition, where $V_\lambda = \ker(\mathrm{ad}H_1 - \lambda)$. For a second operator $\mathrm{ad}H_2$ commuting with $\mathrm{ad}H_1$, each V_λ is $\mathrm{ad}H_2$-invariant: if $\mathrm{ad}H_1(v) = \lambda v$ then $\mathrm{ad}H_1(\mathrm{ad}H_2(v)) = \mathrm{ad}H_2(\mathrm{ad}H_1(v)) = \lambda\,\mathrm{ad}H_2(v)$, so $\mathrm{ad}H_2(v) \in V_\lambda$. Because $\mathrm{ad}H_2$ is diagonalizable on $\mathfrak{g}$, its minimal polynomial has distinct roots; the minimal polynomial of $\mathrm{ad}H_2|_{V_\lambda}$ divides it, so it too has distinct roots, and $\mathrm{ad}H_2|_{V_\lambda}$ is diagonalizable. Diagonalizing $\mathrm{ad}H_2$ within each V_λ refines the decomposition into joint eigenspaces of $\mathrm{ad}H_1$ and $\mathrm{ad}H_2$. Iterating over a finite spanning set $H_1, \ldots, H_m$ of $\mathfrak{h}$ (a basis suffices, since the eigenvalue of $\mathrm{ad}H$ depends linearly on H) produces a decomposition of $\mathfrak{g}$ into simultaneous eigenspaces of all of $\mathfrak{h}$.

A vector X in a simultaneous eigenspace satisfies $[H, X] = \mathrm{ad}H(X) = \chi(H)\,X$ for each basis vector $H = H_k$, with eigenvalue $\chi(H_k)$; extending χ linearly gives a functional $\alpha \in \mathfrak{h}^*$ with $[H, X] = \alpha(H)X$ for all $H \in \mathfrak{h}$, that is, $X \in \mathfrak{g}_\alpha$. Conversely every $\mathfrak{g}_\alpha$ is a simultaneous eigenspace.

Therefore $\mathfrak{g}$ is the direct sum of the $\mathfrak{g}_\alpha$ over the finitely many $\alpha \in \mathfrak{h}^*$ that occur:

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha$$

The sum is direct because eigenvectors for distinct simultaneous eigenvalues are linearly independent: if $\alpha \neq \beta$ then $\alpha(H) \neq \beta(H)$ for some H, and the $\mathfrak{g}_\alpha, \mathfrak{g}_\beta$ lie in distinct eigenspaces of that single $\mathrm{ad}H$.

Step 2: $\mathfrak{g}_0 = C_\mathfrak{g}(\mathfrak{h}) = \mathfrak{h}$. By definition $\mathfrak{g}_0 = \{X \mid [H, X] = 0 \text{ for all } H \in \mathfrak{h}\}$, which is precisely the centralizer $C_\mathfrak{g}(\mathfrak{h})$. The Cartan subalgebra $\mathfrak{h}$ is self-centralizing by proposition 7.1.7, meaning $C_\mathfrak{g}(\mathfrak{h}) = \mathfrak{h}$. Hence $\mathfrak{g}_0 = \mathfrak{h}$. This is the step that uses semisimplicity in an essential way, through proposition 7.1.7; without it the zero eigenspace would only be known to contain $\mathfrak{h}$, and the decomposition would not separate the diagonal part cleanly.

Combining the two steps, split off the $\alpha = 0$ term:

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

where the second sum ranges only over $\alpha \in \Phi$ because the vanishing root spaces contribute nothing, completing the proof.

The theorem is the pivot of the entire structure theory. It presents a semisimple Lie algebra as its Cartan subalgebra together with a finite collection of root spaces indexed by Φ, and this presentation is what all subsequent analysis manipulates. The abelian, self-centralizing $\mathfrak{h}$ is the "maximal torus" of the Lie algebra, playing the role that a maximal torus plays for a compact Lie group in Part III; the root spaces are the directions transverse to it, each labeled by the functional α describing how $\mathfrak{h}$ acts there. That a single abelian subalgebra can diagonalize all of $\mathfrak{g}$ at once is the payoff of the two properties of $\mathfrak{h}$: toral makes each $\mathrm{ad}H$ diagonalizable, abelian makes them commute, and self-centralizing forces the leftover diagonal block to be $\mathfrak{h}$ and nothing more.

The decomposition also equips $\mathfrak{g}$ with a grading. The bracket of two root spaces lands in a third root space, indexed by the sum of the two functionals, so Φ behaves like an additive bookkeeping device for the Lie bracket. This is the second structural fact of the section, and it is what turns the finite set Φ into a combinatorial gadget rather than a mere list of eigenvalues. The proof is a single application of the Jacobi identity.

Proposition 7.2.5: The Bracket Grades the Root Spaces

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$. For all $\alpha, \beta \in \mathfrak{h}^*$,

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$$

In particular $[\mathfrak{h}, \mathfrak{g}_\alpha] = \mathfrak{g}_\alpha$ for every $\alpha \in \Phi$, and $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathfrak{h}$.

***Proof*:**
Let $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$, and $H \in \mathfrak{h}$. The Jacobi identity, in the form $[H, [X, Y]] = [[H, X], Y] + [X, [H, Y]]$, gives

$$[H, [X, Y]] = [\alpha(H)X, Y] + [X, \beta(H)Y] = \alpha(H)[X, Y] + \beta(H)[X, Y] = (\alpha + \beta)(H)\,[X, Y]$$

using $[H, X] = \alpha(H)X$ and $[H, Y] = \beta(H)Y$ and bilinearity of the bracket. As $H \in \mathfrak{h}$ was arbitrary, $[X, Y]$ is a simultaneous eigenvector with eigenvalue $\alpha + \beta$, that is, $[X, Y] \in \mathfrak{g}_{\alpha+\beta}$. Since X, Y were arbitrary, $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$.

For the two special cases, recall $\mathfrak{h} = \mathfrak{g}_0$ by theorem 7.2.4. Taking $\beta = 0$ gives $[\mathfrak{h}, \mathfrak{g}_\alpha] = [\mathfrak{g}_0, \mathfrak{g}_\alpha] \subseteq \mathfrak{g}_{0+\alpha} = \mathfrak{g}_\alpha$. For a root $\alpha \in \Phi$ the reverse containment $\mathfrak{g}_\alpha \subseteq [\mathfrak{h}, \mathfrak{g}_\alpha]$ holds as well: choose $H \in \mathfrak{h}$ with $\alpha(H) \neq 0$ (possible because $\alpha \neq 0$), and then for any $X \in \mathfrak{g}_\alpha$ one has $X = \alpha(H)^{-1}[H, X] \in [\mathfrak{h}, \mathfrak{g}_\alpha]$; hence $[\mathfrak{h}, \mathfrak{g}_\alpha] = \mathfrak{g}_\alpha$. Finally, taking $\beta = -\alpha$ gives $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathfrak{g}_{\alpha+(-\alpha)} = \mathfrak{g}_0 = \mathfrak{h}$, completing the proof.

The grading is the source of nearly every subsequent computation. Read as a rule, it says: to bracket two eigenvectors, add their labels. The two special cases are the ones used constantly. That $[\mathfrak{h}, \mathfrak{g}_\alpha] = \mathfrak{g}_\alpha$ restates that $\mathfrak{h}$ acts on the root space $\mathfrak{g}_\alpha$ by the nonzero scalar functional α, so $\mathfrak{h}$ moves $\mathfrak{g}_\alpha$ nontrivially and never annihilates it. That $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathfrak{h}$ is the crucial bridge back into the Cartan subalgebra: brackets of oppositely-labeled root vectors return to $\mathfrak{h}$, and these brackets are precisely the coroots that section 7.4 extracts to build $\mathfrak{sl}_2$-triples. The containment $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$ has a silent subtlety worth flagging: when $\alpha + \beta \notin \Phi \cup \{0\}$ the target root space is zero, so the bracket must vanish identically, even though α and β are roots. The bracket of two root vectors can therefore be zero for purely combinatorial reasons, a phenomenon the exercises make explicit.

The grading closes the loop with the running example. In $\mathfrak{sl}_n$ the rule $[\mathfrak{g}_{\varepsilon_i-\varepsilon_j}, \mathfrak{g}_{\varepsilon_k-\varepsilon_l}] \subseteq \mathfrak{g}_{(\varepsilon_i-\varepsilon_j)+(\varepsilon_k-\varepsilon_l)}$ is the elementary-matrix identity $[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}$: bracketing E_{ij} with E_{kl} produces a matrix in position (i, l) when $j = k$, whose label is $\varepsilon_i - \varepsilon_l = (\varepsilon_i - \varepsilon_j) + (\varepsilon_j - \varepsilon_l)$, exactly the sum of the two root labels. When the indices do not chain up ($j \neq k$ and $l \neq i$), the bracket is zero, matching the combinatorial vanishing noted above: the sum $(\varepsilon_i - \varepsilon_j) + (\varepsilon_k - \varepsilon_l)$ is then not of the form $\varepsilon_a - \varepsilon_b$ and so is not a root.

Exercise 7.2.1

1. Carry out the root space decomposition of $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$ explicitly. Take $\mathfrak{h} = \mathbb{C}h$ with $h = \operatorname{diag}(1, -1)$, and let $\alpha \in \mathfrak{h}^*$ be the functional with $\alpha(h) = 2$. Verify that $[h, e] = 2e$ and $[h, f] = -2f$ for the standard $e = E_{12}$, $f = E_{21}$, identify the root spaces $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$, and confirm that $\Phi = \{\alpha, -\alpha\}$ and $\mathfrak{sl}_2 = \mathfrak{g}_{-\alpha} \oplus \mathfrak{h} \oplus \mathfrak{g}_\alpha$. How many roots does $\mathfrak{sl}_2$ have?

2. For $\mathfrak{sl}_3 = \mathfrak{sl}_3(\mathbb{C})$ with $\mathfrak{h}$ the diagonal traceless matrices, list all the roots $\varepsilon_i - \varepsilon_j$ ($1 \leq i, j \leq 3$, $i \neq j$), give the root vector spanning each root space, and confirm the count $n(n-1) = 6$. Exhibit the root space decomposition $\mathfrak{sl}_3 = \mathfrak{h} \oplus \bigoplus_{i\neq j} \mathbb{C}E_{ij}$ and check that $\dim \mathfrak{h} = 2$ so that the dimensions add to $8 = \dim \mathfrak{sl}_3$.

3. Using $\mathfrak{sl}_3$, exhibit two roots $\alpha, \beta \in \Phi$ whose sum $\alpha + \beta$ is *not* a root and for which $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = 0$, and contrast this with a chained pair whose sum *is* a root and whose bracket is nonzero. (Hint: compare $[E_{12}, E_{13}]$ with $[E_{12}, E_{23}]$; the indices chain in only one of the two.) Explain why the vanishing bracket does not contradict proposition 7.2.5.

4. Prove that $\mathfrak{sl}_n$ has exactly $n(n-1)$ roots and that the dimension of $\mathfrak{sl}_n$ decomposes as $(n-1) + n(n-1) = n^2 - 1$, matching the two summands $\mathfrak{h}$ and $\bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$ of the root space decomposition. Which relation among the ε_i accounts for $\dim \mathfrak{h} = n - 1$ rather than n?

5. Prove directly, without invoking theorem 7.2.4, that for any subalgebra $\mathfrak{h}$ of any Lie algebra $\mathfrak{g}$ the zero-eigenspace $\mathfrak{g}_0 = \{X \mid [H, X] = 0 \ \forall H \in \mathfrak{h}\}$ equals the centralizer $C_\mathfrak{g}(\mathfrak{h})$, and that $\mathfrak{g}_0$ is a subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$ whenever $\mathfrak{h}$ is abelian. Where in theorem 7.2.4 is the sharper

equality $\mathfrak{g}_0 = \mathfrak{h}$ used, and what property of $\mathfrak{h}$ makes it hold?

6. Show that each root space $\mathfrak{g}_\alpha$ is invariant under $\mathrm{ad}H$ for every $H \in \mathfrak{h}$, and deduce that the decomposition $\mathfrak{g} = \bigoplus_\alpha \mathfrak{g}_\alpha$ is a decomposition of $\mathfrak{g}$ as an $\mathfrak{h}$-module (under the adjoint action). Which $\mathfrak{h}$-module structure does each $\mathfrak{g}_\alpha$ carry?

7.3 Properties of Roots and the Killing Form on the Cartan Subalgebra

The root space decomposition of theorem 7.2.4 split a semisimple $\mathfrak{g}$ into the Cartan subalgebra $\mathfrak{h}$ and the root spaces $\mathfrak{g}_\alpha$ indexed by the roots $\Phi \subseteq \mathfrak{h}^*$. That decomposition is a statement about how $\mathrm{ad}\mathfrak{h}$ acts. This section brings in the second structure carried by a semisimple algebra, the Killing form κ of definition 6.1.3, and shows that κ interacts with the decomposition in the tightest possible way: it pairs each root space $\mathfrak{g}_\alpha$ with the single opposite space $\mathfrak{g}_{-\alpha}$ and with nothing else. From this one fact everything the combinatorics of root systems will need follows. The Killing form restricts to a nondegenerate form on $\mathfrak{h}$, which transports to an inner product on the space spanned by the roots, and in that inner product the roots turn out to live in a genuine Euclidean space.

The payoff is a change of arena. The roots begin life as linear functionals on $\mathfrak{h}$, elements of the abstract dual $\mathfrak{h}^*$ with no metric attached. By the end of this section they sit inside a real vector space equipped with a positive-definite inner product, where angles and lengths make sense, and where the reflections and integrality conditions of the next two sections become Euclidean geometry. The mechanism that accomplishes this transport is the nondegeneracy of κ on $\mathfrak{h}$, and the mechanism behind that nondegeneracy is the orthogonality of root spaces for non-opposite roots.

Throughout, $\mathfrak{g}$ is a semisimple Lie algebra over $\mathbb{C}$, $\mathfrak{h}$ is a fixed Cartan subalgebra as produced in section 7.1, and $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$ is the root space decomposition, with $\mathfrak{g}_0 = \mathfrak{h}$ the centralizer of $\mathfrak{h}$. The Killing form $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y)$ is nondegenerate on all of $\mathfrak{g}$ by theorem 6.3.1, since $\mathfrak{g}$ is semisimple. The question is how that global nondegeneracy distributes across the pieces of the decomposition.

The starting observation is a trace computation. Fix roots α, β (or 0), take $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_\beta$, and ask what the operator $\mathrm{ad}X\,\mathrm{ad}Y$ does to the decomposition. By the grading $[\mathfrak{g}_\gamma, \mathfrak{g}_\delta] \subseteq \mathfrak{g}_{\gamma+\delta}$ of proposition 7.2.5, the operator $\mathrm{ad}Y$ sends $\mathfrak{g}_\gamma$ into $\mathfrak{g}_{\gamma+\beta}$ and then $\mathrm{ad}X$ sends that into $\mathfrak{g}_{\gamma+\beta+\alpha}$. The composite $\mathrm{ad}X\,\mathrm{ad}Y$ therefore shifts the grading by $\alpha + \beta$. An operator that moves every graded piece off itself has no diagonal entries, hence trace zero. The only way it can fix a graded piece, and so contribute to the trace, is if $\alpha + \beta = 0$.

Proposition 7.3.1: Killing-Orthogonality of Root Spaces

Let $\mathfrak{g}$ be semisimple over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$ and root space decomposition $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$. Write $\mathfrak{g}_0 = \mathfrak{h}$. Then:

1. For $\alpha, \beta \in \Phi \cup \{0\}$ with $\alpha + \beta \neq 0$, the spaces $\mathfrak{g}_\alpha$ and $\mathfrak{g}_\beta$ are κ-orthogonal: $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$.
2. For each $\alpha \in \Phi$, the Killing form restricts to a nondegenerate pairing $\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}$. In particular $\dim \mathfrak{g}_\alpha = \dim \mathfrak{g}_{-\alpha}$, and $-\alpha \in \Phi$ whenever $\alpha \in \Phi$.
3. The restriction $\kappa|_\mathfrak{h}$ of the Killing form to $\mathfrak{h}$ is nondegenerate.

***Proof*:**

1. *Orthogonality for non-opposite indices.* Let $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_\beta$ with $\alpha + \beta \neq 0$. For any $\gamma \in \Phi \cup \{0\}$, the grading of proposition 7.2.5 gives

$$(\mathrm{ad}X\,\mathrm{ad}Y)(\mathfrak{g}_\gamma) = [X,[Y,\mathfrak{g}_\gamma]] \subseteq [X,\mathfrak{g}_{\gamma+\beta}] \subseteq \mathfrak{g}_{\gamma+\alpha+\beta}$$

   where a summand $\mathfrak{g}_\delta$ is understood to be 0 when $\delta \notin \Phi \cup \{0\}$. Thus $\mathrm{ad}X\,\mathrm{ad}Y$ maps each summand $\mathfrak{g}_\gamma$ of the decomposition into the different summand $\mathfrak{g}_{\gamma+\alpha+\beta}$, since $\gamma+\alpha+\beta \neq \gamma$ because $\alpha+\beta \neq 0$. Choosing a basis of $\mathfrak{g}$ adapted to the direct sum decomposition, the matrix of $\mathrm{ad}X\,\mathrm{ad}Y$ has all its nonzero entries off the block diagonal; every diagonal entry is 0. Hence $\kappa(X,Y) = \mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y) = 0$. As X, Y were arbitrary in their spaces, $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$.

2. *Nondegenerate pairing of opposite spaces.* Fix $\alpha \in \Phi$. By part (1), $\mathfrak{g}_\alpha$ is κ-orthogonal to every summand $\mathfrak{g}_\beta$ with $\beta \neq -\alpha$, in particular to $\mathfrak{h} = \mathfrak{g}_0$ and to every $\mathfrak{g}_\beta$ with $\beta \in \Phi \setminus \{-\alpha\}$. Suppose $X \in \mathfrak{g}_\alpha$ were κ-orthogonal to all of $\mathfrak{g}_{-\alpha}$ as well. Then X would be κ-orthogonal to $\mathfrak{g}_{-\alpha}$ and to every other summand, hence to their direct sum, which is all of $\mathfrak{g}$. By nondegeneracy of κ on $\mathfrak{g}$ (theorem 6.3.1), $X = 0$. Therefore the pairing

$$\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}, \qquad (X,Y) \mapsto \kappa(X,Y)$$

   has trivial left radical; by symmetry of κ its right radical is trivial too, so the pairing is nondegenerate. A nondegenerate pairing between two finite-dimensional spaces forces them to have equal dimension, so $\dim \mathfrak{g}_\alpha = \dim \mathfrak{g}_{-\alpha}$. Since $\alpha \in \Phi$ means $\mathfrak{g}_\alpha \neq 0$, this common dimension is positive, so $\mathfrak{g}_{-\alpha} \neq 0$ and $-\alpha \in \Phi$.

3. *Nondegeneracy on $\mathfrak{h}$.* Apply part (1) with $\beta = 0$: $\mathfrak{h} = \mathfrak{g}_0$ is κ-orthogonal to every $\mathfrak{g}_\alpha$ with $\alpha \neq 0$, that is, to every root space. Suppose $H \in \mathfrak{h}$ is κ-orthogonal to all of $\mathfrak{h}$. Then H is orthogonal to $\mathfrak{h}$ and to each $\mathfrak{g}_\alpha$, hence to the whole of $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_\alpha \mathfrak{g}_\alpha$. Nondegeneracy of κ on $\mathfrak{g}$ again forces $H = 0$. Thus the radical of $\kappa|_\mathfrak{h}$ is zero and $\kappa|_\mathfrak{h}$ is nondegenerate, completing the proof.

The proposition is the structural bridge of the chapter, and its three parts play distinct roles. Part (1) is the mechanism: the grading forces $\mathrm{ad}X\,\mathrm{ad}Y$ to be trace-free unless the indices cancel, so the Killing form "sees" only opposite pairs. Part (2) turns this into the pairing that will, in the next section, let each root be matched with an element of $\mathfrak{g}_{-\alpha}$ to build an $\mathfrak{sl}_2$-triple; the equality $-\alpha \in \Phi$ is the first symmetry of the root set, recovered here from the symmetry of κ. Part (3) is what makes the whole theory metric: because κ is nondegenerate on $\mathfrak{h}$, the Cartan subalgebra and its dual are identified, and every linear functional on $\mathfrak{h}$, in particular every root, corresponds to a concrete vector $t_\alpha \in \mathfrak{h}$.

The identification in part (3) deserves to be named, since it is the object through which all subsequent geometry is expressed.

Definition 7.3.2: The Element t_α and the Form on $\mathfrak{h}^*$

Let $\mathfrak{g}$ be semisimple over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, and let $\kappa|_{\mathfrak{h}}$ be the nondegenerate form of proposition 7.3.1(3). For each $\alpha \in \mathfrak{h}^*$, the *$\kappa$-dual* of α is the unique element $t_\alpha \in \mathfrak{h}$ such that

$$\kappa(t_\alpha, H) = \alpha(H) \qquad \text{for all } H \in \mathfrak{h} \tag{7.1}$$

The assignment $\alpha \mapsto t_\alpha$ is the inverse of the isomorphism $\mathfrak{h} \to \mathfrak{h}^*$, $H \mapsto \kappa(H, -)$. The Killing form is transported to $\mathfrak{h}^*$ by declaring

$$(\alpha, \beta) := \kappa(t_\alpha, t_\beta) = \alpha(t_\beta) = \beta(t_\alpha) \tag{7.2}$$

a symmetric bilinear form on $\mathfrak{h}^*$.

Nondegeneracy of $\kappa|_{\mathfrak{h}}$ is exactly what guarantees the map $H \mapsto \kappa(H, -)$ from $\mathfrak{h}$ to $\mathfrak{h}^*$ is a bijection, so that t_α exists and is unique for every functional α; without part (3) of the preceding proposition the definition would be empty. The two equalities $\kappa(t_\alpha, t_\beta) = \alpha(t_\beta) = \beta(t_\alpha)$ in (7.2) follow by applying (7.1) once with $H = t_\beta$ and once with the roles reversed, using symmetry of κ; they show the transported form $(\cdot, \cdot)$ is well-defined and symmetric. The form $(\cdot, \cdot)$ is the object that will make $\mathfrak{h}^*$ a Euclidean space once restricted to the real span of the roots, and the coroots of the next section are built from the t_α.

The construction is best absorbed by running it on the standard example. For $\mathfrak{sl}_n$ the Cartan subalgebra is the diagonal, the roots are the differences $\varepsilon_i - \varepsilon_j$, and both t_α and the form $(\cdot, \cdot)$ can be written down in closed form.

Example 7.3.3: The Elements t_α and the Form for $\mathfrak{sl}_n$

Let $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$, with Cartan subalgebra $\mathfrak{h}$ the trace-zero diagonal matrices $H = \operatorname{diag}(a_1, \dots, a_n)$, $\sum_i a_i = 0$. The coordinate functionals $\varepsilon_i(H) = a_i$ satisfy $\varepsilon_1 + \cdots + \varepsilon_n = 0$ on $\mathfrak{h}$, and the roots are the $\varepsilon_i - \varepsilon_j$ for $i \neq j$, with root space $\mathfrak{g}_{\varepsilon_i - \varepsilon_j} = \mathbb{C}E_{ij}$ spanned by the elementary matrix. The Killing form on $\mathfrak{sl}_n$ is $\kappa(X, Y) = 2n \operatorname{tr}(XY)$, so on diagonal matrices $H = \operatorname{diag}(a_i)$ and $H' = \operatorname{diag}(b_i)$,

$$\kappa(H, H') = 2n \sum_{i=1}^{n} a_i b_i$$

The task is to find $t_{\varepsilon_i - \varepsilon_j} \in \mathfrak{h}$ representing $\alpha = \varepsilon_i - \varepsilon_j$. Working first inside all of $\mathfrak{gl}_n$ before projecting, the functional ε_i on the diagonal is $H \mapsto a_i$, and κ-duality against $2n \sum_k a_k b_k$ suggests the diagonal matrix $\frac{1}{2n} E_{ii}$. Projecting onto the trace-zero part, the element of $\mathfrak{h}$ representing $\varepsilon_i - \varepsilon_j$ is

$$t_{\varepsilon_i - \varepsilon_j} = \frac{1}{2n}\big(E_{ii} - E_{jj}\big)$$

Indeed for $H = \operatorname{diag}(a_1, \dots, a_n) \in \mathfrak{h}$,

$$\kappa(t_{\varepsilon_i - \varepsilon_j}, H) = 2n \cdot \frac{1}{2n} \operatorname{tr}\big((E_{ii} - E_{jj})H\big) = a_i - a_j = (\varepsilon_i - \varepsilon_j)(H)$$

so (7.1) holds. The transported form is then computed from (7.2). With $\alpha = \varepsilon_i - \varepsilon_j$ and $\beta = \varepsilon_k - \varepsilon_l$,

$$(\alpha, \beta) = \kappa(t_\alpha, t_\beta) = 2n \cdot \frac{1}{4n^2} \operatorname{tr}\big((E_{ii} - E_{jj})(E_{kk} - E_{ll})\big) = \frac{1}{2n}\big(\delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl}\big)$$

For a root paired with itself this gives $(\alpha,\alpha)=\frac{1}{2n}(1-0-0+1)=\frac{1}{n}$, a positive number, and for two roots sharing exactly one index with opposite signs (say $\varepsilon_i-\varepsilon_j$ and $\varepsilon_j-\varepsilon_k$ with i,j,k distinct) it gives $(\alpha,\beta)=\frac{1}{2n}(0-0-1+0)=-\frac{1}{2n}$. Every value is a real rational multiple of $\frac{1}{n}$, and $(\alpha,\alpha)=\frac{1}{n}>0$ for each root, the first sign of the positivity established in general below.

The example exhibits in coordinates what the general theory now proves in the abstract: the values (α,β) are real, and (α,α) is a positive real number for each root. The abstract argument does not have the luxury of the explicit basis E_{ij}; it must extract reality and positivity from the $\mathfrak{sl}_2$-structure attached to each root, which is the subject of section 7.4. The remaining results of this section assemble the facts about Φ and $(\cdot,\cdot)$ that are provable now, before that $\mathfrak{sl}_2$-machinery is built, and reduce the outstanding positivity claim to a single computation carried out there.

Before that, one structural fact about Φ is needed: the roots span $\mathfrak{h}^*$. If they did not, some direction in $\mathfrak{h}$ would be invisible to every root, and that direction would have to be central, which a semisimple algebra forbids. This is the second place where semisimplicity, rather than mere nondegeneracy of κ, is doing work.

Proposition 7.3.4: The Roots Span and the Form is Euclidean

Let $\mathfrak{g}$ be semisimple over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root set Φ, and the form $(\cdot,\cdot)$ on $\mathfrak{h}^*$ of definition 7.3.2. Then:

1. Φ spans $\mathfrak{h}^*$ over $\mathbb{C}$. Equivalently, no nonzero $H\in\mathfrak{h}$ satisfies $\alpha(H)=0$ for all $\alpha\in\Phi$.
2. $\alpha\in\Phi$ implies $-\alpha\in\Phi$, and the t_α for $\alpha\in\Phi$ span $\mathfrak{h}$.
3. Let $E=\operatorname{span}_{\mathbb{R}}\Phi\subseteq\mathfrak{h}^*$ be the real span of the roots. The form $(\cdot,\cdot)$ takes real values on E and is positive-definite there, so $(E,(\cdot,\cdot))$ is a Euclidean space of dimension equal to $\dim_{\mathbb{C}}\mathfrak{h}$.

***Proof*:**

1. *The roots span $\mathfrak{h}^*$.* Suppose not. Then the roots lie in a proper subspace of $\mathfrak{h}^*$, so there is a nonzero $H\in\mathfrak{h}$ annihilated by every root: $\alpha(H)=0$ for all $\alpha\in\Phi$. For each root space $\mathfrak{g}_\alpha$ and each $X\in\mathfrak{g}_\alpha$, the bracket is $[H,X]=\alpha(H)X=0$, so $\operatorname{ad}H$ vanishes on every root space. It also vanishes on $\mathfrak{h}=\mathfrak{g}_0$, because $\mathfrak{h}$ is abelian by proposition 7.1.7, so $[H,\mathfrak{h}]=0$. Since $\mathfrak{g}=\mathfrak{h}\oplus\bigoplus_\alpha\mathfrak{g}_\alpha$ is the whole algebra, $\operatorname{ad}H=0$, that is, $H\in\mathfrak{z}(\mathfrak{g})$. But a semisimple algebra has zero center by corollary 6.3.5 (its center is an abelian ideal, hence contained in $\operatorname{rad}\mathfrak{g}=0$), so $H=0$, contradicting $H\neq0$. Therefore Φ spans $\mathfrak{h}^*$.
2. *The t_α span $\mathfrak{h}$.* That $\alpha\in\Phi\Rightarrow-\alpha\in\Phi$ was shown in proposition 7.3.1(2). The map $\mathfrak{h}^*\to\mathfrak{h}$, $\alpha\mapsto t_\alpha$, is a linear isomorphism (it is the inverse of the isomorphism $H\mapsto\kappa(H,-)$ of definition 7.3.2), so it carries a spanning set of $\mathfrak{h}^*$ to a spanning set of $\mathfrak{h}$. By part (1) the roots span $\mathfrak{h}^*$, so the t_α for $\alpha\in\Phi$ span $\mathfrak{h}$.
3. *Reality and positive-definiteness on E.* The core is the formula
$$\kappa(H,H')=\sum_{\alpha\in\Phi}\alpha(H)\,\alpha(H')\qquad\text{for all } H,H'\in\mathfrak{h} \tag{7.3}$$
To prove it, evaluate the trace $\kappa(H,H')=\operatorname{tr}(\operatorname{ad}H\operatorname{ad}H')$ in a basis adapted to the root

space decomposition. On $\mathfrak{h} = \mathfrak{g}_0$ both $\mathrm{ad}H$ and $\mathrm{ad}H'$ act as 0 (as $\mathfrak{h}$ is abelian), so that block contributes nothing. On each root space $\mathfrak{g}_\alpha$, the operator $\mathrm{ad}H$ acts as the scalar $\alpha(H)$ and $\mathrm{ad}H'$ as $\alpha(H')$ (by definition of $\mathfrak{g}_\alpha$), so $\mathrm{ad}H\,\mathrm{ad}H'$ acts as the scalar $\alpha(H)\alpha(H')$, contributing $(\dim \mathfrak{g}_\alpha)\,\alpha(H)\alpha(H')$ to the trace. Summing over all summands gives

$$\kappa(H, H') = \sum_{\alpha \in \Phi} (\dim \mathfrak{g}_\alpha)\, \alpha(H)\, \alpha(H') \tag{7.4}$$

Each root space is one-dimensional, $\dim \mathfrak{g}_\alpha = 1$, by the $\mathfrak{sl}_2$-string argument of section 7.4, so the multiplicities drop out and (7.4) reduces to (7.3). The weaker form (7.4) already suffices for the arguments below, since each $\dim \mathfrak{g}_\alpha$ is a fixed positive integer, but the clean formula (7.3) is the one that reappears in the Weyl integration formula of Part III.

Take $H = t_\gamma$ and $H' = t_\delta$ for roots γ, δ. Since $\alpha(t_\gamma) = (\alpha, \gamma)$ by (7.2), (7.3) becomes

$$(\gamma, \delta) = \kappa(t_\gamma, t_\delta) = \sum_{\alpha \in \Phi} (\alpha, \gamma)(\alpha, \delta) \tag{7.5}$$

Specializing $\delta = \gamma$,

$$(\gamma, \gamma) = \sum_{\alpha \in \Phi} (\alpha, \gamma)^2 \tag{7.6}$$

Two facts drive the reality of the (α, γ) and the positivity of (γ, γ). The first is the nonvanishing $(\gamma, \gamma) \neq 0$ for every root γ, proved here directly, with no appeal to any later result. Choose $x \in \mathfrak{g}_\gamma$ and $y \in \mathfrak{g}_{-\gamma}$ with $\kappa(x, y) = 1$, possible because κ pairs $\mathfrak{g}_\gamma$ with $\mathfrak{g}_{-\gamma}$ nondegenerately (proposition 7.3.1). Then $[x, y] \in [\mathfrak{g}_\gamma, \mathfrak{g}_{-\gamma}] \subseteq \mathfrak{h}$ by proposition 7.2.5, and for every $H \in \mathfrak{h}$ the invariance of κ gives $\kappa([x, y], H) = \kappa(x, [y, H]) = \gamma(H)\kappa(x, y) = \kappa(t_\gamma, H)$, so $[x, y] = t_\gamma$ by nondegeneracy of $\kappa|_{\mathfrak{h}}$. Suppose $(\gamma, \gamma) = \gamma(t_\gamma) = 0$; then $[t_\gamma, x] = \gamma(t_\gamma)x = 0$ and $[t_\gamma, y] = -\gamma(t_\gamma)y = 0$, so $\mathfrak{s} = \mathrm{span}\{x, y, t_\gamma\}$ is a subalgebra whose derived algebra $[\mathfrak{s}, \mathfrak{s}] = \mathbb{C}t_\gamma$ is central in it, hence $\mathfrak{s}$ is solvable. Its image $\mathrm{ad}\mathfrak{s} \subseteq \mathfrak{gl}(\mathfrak{g})$ is then solvable, so by corollary 5.4.5 the derived algebra $\mathrm{ad}(\mathbb{C}t_\gamma)$ consists of nilpotent operators; in particular $\mathrm{ad}t_\gamma$ is nilpotent on $\mathfrak{g}$. But $t_\gamma \in \mathfrak{h}$ is semisimple, so $\mathrm{ad}t_\gamma$ is also diagonalizable, and a diagonalizable nilpotent operator is 0; thus $t_\gamma \in \mathfrak{z}(\mathfrak{g}) = 0$ and $\gamma = \kappa(t_\gamma, -) = 0$, a contradiction. Hence $(\gamma, \gamma) \neq 0$. The second fact is integrality: for all $\alpha, \gamma \in \Phi$ the ratio

$$m_{\alpha\gamma} := \frac{2(\alpha, \gamma)}{(\gamma, \gamma)} \in \mathbb{Z}$$

is an integer, established as theorem 7.4.6 in the next section. Two consequences flow from these, in order.

The value (γ, γ) is a positive rational number. Dividing (7.6) through by $(\gamma, \gamma)^2$ and substituting $(\alpha, \gamma) = \frac{1}{2} m_{\alpha\gamma}(\gamma, \gamma)$ gives

$$\frac{1}{(\gamma, \gamma)} = \sum_{\alpha \in \Phi} \left(\frac{(\alpha, \gamma)}{(\gamma, \gamma)} \right)^2 = \frac{1}{4} \sum_{\alpha \in \Phi} m_{\alpha\gamma}^2 \tag{7.7}$$

The right side of (7.7) is a sum of squares of integers, positive because the γ-term contributes $\frac{1}{4} m_{\gamma\gamma}^2 = \frac{1}{4} \cdot 4 = 1$. Therefore $1/(\gamma, \gamma)$ is a positive rational, so (γ, γ) is a positive rational; in particular $(\gamma, \gamma) > 0$ for every root γ.

The values (α, γ) are all real. With (γ, γ) now known to be a (positive, real) rational, the identity $(\alpha, \gamma) = \frac{1}{2} m_{\alpha\gamma}(\gamma, \gamma)$ exhibits each (α, γ) as a rational multiple of a rational, hence

a rational number, in particular real. By bilinearity $(\lambda, \mu) = \sum_{\gamma,\delta} c_\gamma d_\delta (\gamma, \delta)$ is real for $\lambda = \sum_\gamma c_\gamma \gamma$ and $\mu = \sum_\delta d_\delta \delta$ with real coefficients, so $(\cdot, \cdot)$ is real-valued on $E = \operatorname{span}_{\mathbb{R}} \Phi$. Positive-definiteness now follows. For $\lambda \in E$, write $\lambda = \sum_\gamma c_\gamma \gamma$ with $c_\gamma \in \mathbb{R}$; then $(\lambda, \lambda) = \sum_{\alpha \in \Phi} (\alpha, \lambda)^2$ by the same computation that gave (7.6) (apply (7.3) to $H = H' = t_\lambda$, where $t_\lambda = \sum_\gamma c_\gamma t_\gamma$), and each $(\alpha, \lambda) = \sum_\gamma c_\gamma (\alpha, \gamma)$ is real. Hence $(\lambda, \lambda) = \sum_\alpha (\alpha, \lambda)^2 \geq 0$, with equality iff $(\alpha, \lambda) = 0$ for every $\alpha \in \Phi$. But $(\alpha, \lambda) = \lambda(t_\alpha)$ vanishing for all α means λ annihilates the span of the t_α, which is all of $\mathfrak{h}$ by part (2), so $\lambda = 0$. Thus $(\cdot, \cdot)$ is positive-definite on E. Finally, $\dim_{\mathbb{R}} E = \dim_{\mathbb{C}} \mathfrak{h}^* = \dim_{\mathbb{C}} \mathfrak{h}$: the roots span $\mathfrak{h}^*$ over $\mathbb{C}$ by part (1), and the real span E has the same dimension over $\mathbb{R}$ because a $\mathbb{C}$-basis of $\mathfrak{h}^*$ chosen from Φ is $\mathbb{R}$-linearly independent once all pairwise forms (α, β) are real, so E is a real form of $\mathfrak{h}^*$. Therefore $(E, (\cdot, \cdot))$ is a Euclidean space of dimension $\dim_{\mathbb{C}} \mathfrak{h}$, completing the proof.

The proposition completes the change of arena promised at the start. The roots, which entered as elements of the abstract dual $\mathfrak{h}^*$, now sit in the Euclidean space $E = \operatorname{span}_{\mathbb{R}} \Phi$ carrying the positive-definite inner product $(\cdot, \cdot)$. In this space the length $|\alpha| = \sqrt{(\alpha, \alpha)}$ of a root and the angle between two roots are defined, and the entire combinatorial theory that follows is the study of the finite configuration $\Phi \subseteq E$ under these metric relations. The dimension of E is the rank of $\mathfrak{g}$, the common dimension of all Cartan subalgebras (theorem 13.2.7).

The dependencies in the proof that point forward deserve to be stated plainly. The positivity of (γ, γ) and the reality of the values (α, γ) rest on three inputs from the next section: the nonvanishing $(\gamma, \gamma) \neq 0$, the integrality $2(\alpha, \gamma)/(\gamma, \gamma) \in \mathbb{Z}$ of the Cartan integers, and the one-dimensionality $\dim \mathfrak{g}_\alpha = 1$. These are exactly the outputs of the $\mathfrak{sl}_2$-machinery of section 7.4: to each root α one attaches a copy of $\mathfrak{sl}_2$ acting on $\mathfrak{g}$, and the integrality of the weights of a finite-dimensional $\mathfrak{sl}_2$-module delivers the integers, the one-dimensionality, and the nonvanishing. The nondegenerate pairing $\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}$ of proposition 7.3.1(2) is precisely what makes that attachment possible, by giving a nonzero $f_\alpha \in \mathfrak{g}_{-\alpha}$ pairing nontrivially with a chosen $e_\alpha \in \mathfrak{g}_\alpha$. The order of exposition places the Euclidean structure here, where the Killing form makes it natural, and defers the two integrality inputs to the next section, where the $\mathfrak{sl}_2$ representation theory that proves them is developed in full.

The formula (7.3), $\kappa(H, H') = \sum_{\alpha \in \Phi} \alpha(H)\alpha(H')$, is worth isolating as the concrete description of the Killing form on $\mathfrak{h}$. It says $\kappa|_{\mathfrak{h}}$ is the sum over roots of the rank-one forms $\alpha \otimes \alpha$, and since $\Phi = -\Phi$ each root and its negative contribute the same term, so the sum is twice the sum over positive roots. This is the identity the reader is asked to verify for $\mathfrak{sl}_n$ below, and it is the computational heart of the Weyl integration formula in Part III, where the same expression reappears as the Jacobian of the exponential map on a maximal torus.

Exercise 7.3.1

1. For $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$ with diagonal Cartan subalgebra $\mathfrak{h}$ and roots $\alpha_{ij} = \varepsilon_i - \varepsilon_j$ $(i \neq j)$, compute $t_{\alpha_{12}}$ explicitly as a trace-zero diagonal matrix, verify $\kappa(t_{\alpha_{12}}, H) = \alpha_{12}(H)$ for $H = \operatorname{diag}(a_1, a_2, a_3) \in \mathfrak{h}$, and then compute the three values $(\alpha_{12}, \alpha_{ij})$ as α_{ij} ranges over the positive roots $\alpha_{12}, \alpha_{13}, \alpha_{23}$. Confirm $(\alpha_{12}, \alpha_{12}) > 0$ and read off the angle between α_{12} and α_{23}.

2. Prove directly from proposition 7.3.1 that for any root α and any nonzero $X \in \mathfrak{g}_\alpha$ there exists $Y \in \mathfrak{g}_{-\alpha}$ with $\kappa(X, Y) \neq 0$. Then show $[X, Y] = \kappa(X, Y)\, t_\alpha$, so that the bracket of opposite root vectors lands in $\mathbb{C} t_\alpha \subseteq \mathfrak{h}$. (Hint: $[X, Y] \in \mathfrak{g}_0 = \mathfrak{h}$ by proposition 7.2.5, and for $H \in \mathfrak{h}$ compute $\kappa([X, Y], H)$ using invariance of the Killing form.)

3. Using (7.6), show that $(\alpha,\alpha)\neq 0$ for every root α, granting the reality of the values (β,α). Explain why $(\alpha,\alpha)=0$ would force t_α to be annihilated by every root, and why that is impossible in a semisimple algebra.

4. Verify (7.3) directly for $\mathfrak{sl}_n$: show that for diagonal $H=\operatorname{diag}(a_1,\dots,a_n)$ and $H'=\operatorname{diag}(b_1,\dots,b_n)$ in $\mathfrak{h}$,
$$\sum_{i\neq j}(a_i-a_j)(b_i-b_j)=2n\sum_{i=1}^{n}a_ib_i$$
using $\sum_i a_i=\sum_i b_i=0$, and hence that $\kappa|_{\mathfrak{h}}$ is the sum $\sum_{\alpha\in\Phi}\alpha\otimes\alpha=2\sum_{\alpha\in\Phi^+}\alpha\otimes\alpha$ over the positive roots $\Phi^+=\{\varepsilon_i-\varepsilon_j : i<j\}$, matching $\kappa(H,H')=2n\sum_i a_ib_i$.

5. Suppose, contrary to proposition 7.3.4(1), that the roots of some semisimple $\mathfrak{g}$ failed to span $\mathfrak{h}^*$. Derive a contradiction by exhibiting a nonzero central element, and identify precisely which hypothesis on $\mathfrak{g}$ (semisimplicity, versus mere nondegeneracy of κ) is used. Then explain why an abelian Lie algebra, whose "Cartan subalgebra" is the whole algebra and whose root set is empty, is not a counterexample.

7.4 $\mathfrak{sl}_2$-Triples and the Cartan Integers

The root space decomposition of theorem 7.2.4 exhibited a semisimple Lie algebra as $\mathfrak{g}=\mathfrak{h}\oplus\bigoplus_{\alpha\in\Phi}\mathfrak{g}_\alpha$, and section 7.3 equipped the real span of the roots with a positive-definite inner product $(\cdot,\cdot)$. So far the roots are geometric data: a finite set of vectors in a Euclidean space. The present section extracts the arithmetic hidden inside that geometry. To each root α is attached a copy of $\mathfrak{sl}_2$ sitting inside $\mathfrak{g}$, and the whole algebra becomes a module over each of these copies. Because $\mathfrak{sl}_2$ has an unforgivingly rigid representation theory, in which the operator h acts with integer eigenvalues arranged in symmetric unbroken strings, that rigidity descends onto the roots: the numbers $2(\beta,\alpha)/(\alpha,\alpha)$ are forced to be integers, and the roots line up in unbroken strings. These integers, the *Cartan integers*, are the entire combinatorial content of a semisimple Lie algebra; Chapter 8 will show that they determine the algebra up to isomorphism.

The mechanism is the same one that governs the representation theory of angular momentum in quantum mechanics: raising and lowering operators, a highest-weight vector annihilated by the raising operator, and a ladder of weights descending by twos. The novelty here is that the ladder is not applied to an external representation but to $\mathfrak{g}$ acting on itself, so the eigenvalue constraints become constraints on the roots. This section builds the $\mathfrak{sl}_2$-triple, states precisely the weight-string facts it needs, and turns the ladder argument into the integrality and string theorems.

Fix a root $\alpha\in\Phi$. By proposition 7.3.1 the Killing form κ of definition 6.1.3 pairs $\mathfrak{g}_\alpha$ nondegenerately with $\mathfrak{g}_{-\alpha}$, and by proposition 7.3.4 the negative $-\alpha$ is again a root. This is exactly what is needed to produce, from a single choice of vector in $\mathfrak{g}_\alpha$, a matching vector in $\mathfrak{g}_{-\alpha}$ and a distinguished element of $\mathfrak{h}$ closing them into a bracket triple. The element of $\mathfrak{h}$ is the one associated to α through the Killing form, denoted t_α in definition 7.3.2 and characterized by $\kappa(t_\alpha,H)=\alpha(H)$ for all $H\in\mathfrak{h}$.

Theorem 7.4.1: The $\mathfrak{sl}_2$-Triple of a Root

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$ and root $\alpha \in \Phi$. Fix a nonzero $e_\alpha \in \mathfrak{g}_\alpha$. Then there is an $f_\alpha \in \mathfrak{g}_{-\alpha}$, normalized by $\kappa(e_\alpha, f_\alpha) = 2/(\alpha,\alpha)$, such that, setting $h_\alpha = [e_\alpha, f_\alpha] \in \mathfrak{h}$, the three elements satisfy

$$[h_\alpha, e_\alpha] = 2e_\alpha, \qquad [h_\alpha, f_\alpha] = -2f_\alpha, \qquad [e_\alpha, f_\alpha] = h_\alpha$$

Consequently the span $\mathfrak{s}_\alpha = \mathbb{C}e_\alpha \oplus \mathbb{C}h_\alpha \oplus \mathbb{C}f_\alpha$ is a three-dimensional subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}_2(\mathbb{C})$, with $e_\alpha, h_\alpha, f_\alpha$ playing the roles of the standard basis e, h, f. The element h_α depends only on α, not on the choice of e_α, and equals

$$h_\alpha = \frac{2t_\alpha}{(\alpha,\alpha)}$$

where $t_\alpha \in \mathfrak{h}$ is the element of definition 7.3.2. Moreover $\alpha(h_\alpha) = 2$.

Before proving this, it is worth isolating the one computation that makes the constants work out to 2. For any $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_{-\alpha}$, the bracket $[X, Y]$ lies in $\mathfrak{g}_0 = \mathfrak{h}$ by proposition 7.2.5, and it is not an arbitrary element of $\mathfrak{h}$: it points in the direction of t_α. The following lemma records this, and the whole construction rests on it.

Lemma 7.4.2: Brackets of Opposite Root Vectors

For $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_{-\alpha}$, one has

$$[X, Y] = \kappa(X, Y)\, t_\alpha$$

where $t_\alpha \in \mathfrak{h}$ is characterized by $\kappa(t_\alpha, H) = \alpha(H)$ for all $H \in \mathfrak{h}$.

***Proof*:**
By proposition 7.2.5 the bracket $[X, Y]$ lies in $\mathfrak{g}_{\alpha+(-\alpha)} = \mathfrak{g}_0 = \mathfrak{h}$. Since $\kappa|_{\mathfrak{h}}$ is nondegenerate by proposition 7.3.1, an element of $\mathfrak{h}$ is determined by its κ-pairings against all $H \in \mathfrak{h}$, so it suffices to show $\kappa([X, Y], H) = \kappa(X, Y)\,\kappa(t_\alpha, H) = \kappa(X, Y)\,\alpha(H)$ for every $H \in \mathfrak{h}$. Invariance of the Killing form (definition 6.1.1, established for κ in proposition 6.1.5) gives

$$\kappa([X, Y], H) = \kappa(X, [Y, H]) = -\kappa(X, [H, Y])$$

Now $Y \in \mathfrak{g}_{-\alpha}$, so $[H, Y] = -\alpha(H)\, Y$ by the definition of the root space $\mathfrak{g}_{-\alpha}$ (definition 7.2.1). Substituting,

$$\kappa([X, Y], H) = -\kappa\big(X, -\alpha(H)\, Y\big) = \alpha(H)\,\kappa(X, Y)$$

which is $\kappa(X, Y)\,\kappa(t_\alpha, H)$ as required. Nondegeneracy of $\kappa|_{\mathfrak{h}}$ then forces $[X, Y] = \kappa(X, Y)\, t_\alpha$, completing the proof.

The lemma reduces the whole triple to a single scalar, $\kappa(X, Y)$, and to the number $(\alpha,\alpha) = \kappa(t_\alpha, t_\alpha)$. By proposition 7.3.4 the number (α,α) is nonzero, so $2/(\alpha,\alpha)$ is a well-defined scalar, and this nonvanishing is exactly the hypothesis that lets one normalize the triple.

Proof Proof of theorem 7.4.1:
Step 1: producing f_α and computing h_α. Fix $0 \neq e_\alpha \in \mathfrak{g}_\alpha$. Because κ pairs $\mathfrak{g}_\alpha$ nondegenerately with $\mathfrak{g}_{-\alpha}$ (proposition 7.3.1), there is some $Y \in \mathfrak{g}_{-\alpha}$ with $\kappa(e_\alpha, Y) \neq 0$. Rescaling Y, choose $f_\alpha \in \mathfrak{g}_{-\alpha}$ with

$$\kappa(e_\alpha, f_\alpha) = \frac{2}{(\alpha, \alpha)}$$

which is possible since $(\alpha, \alpha) \neq 0$ by proposition 7.3.4. Set $h_\alpha = [e_\alpha, f_\alpha]$. By lemma 7.4.2,

$$h_\alpha = [e_\alpha, f_\alpha] = \kappa(e_\alpha, f_\alpha)\, t_\alpha = \frac{2}{(\alpha, \alpha)}\, t_\alpha$$

which is the claimed formula. In particular $h_\alpha \in \mathfrak{h}$ and, being a fixed positive multiple of t_α, it depends only on α and not on e_α: any other choice e'_α forces the matching f'_α through the same normalization, and the product $[e'_\alpha, f'_\alpha] = \kappa(e'_\alpha, f'_\alpha)\, t_\alpha = \frac{2}{(\alpha,\alpha)} t_\alpha$ again.

Step 2: the bracket relations. Compute $\alpha(h_\alpha)$ first, since the other two relations follow from it. Using $h_\alpha = \frac{2}{(\alpha,\alpha)} t_\alpha$ and $\alpha(H) = \kappa(t_\alpha, H)$,

$$\alpha(h_\alpha) = \frac{2}{(\alpha, \alpha)}\, \alpha(t_\alpha) = \frac{2}{(\alpha, \alpha)}\, \kappa(t_\alpha, t_\alpha) = \frac{2\,(\alpha, \alpha)}{(\alpha, \alpha)} = 2$$

Now $e_\alpha \in \mathfrak{g}_\alpha$ means $[H, e_\alpha] = \alpha(H)\, e_\alpha$ for every $H \in \mathfrak{h}$, so taking $H = h_\alpha$,

$$[h_\alpha, e_\alpha] = \alpha(h_\alpha)\, e_\alpha = 2e_\alpha$$

Likewise $f_\alpha \in \mathfrak{g}_{-\alpha}$ gives $[H, f_\alpha] = -\alpha(H)\, f_\alpha$, so

$$[h_\alpha, f_\alpha] = -\alpha(h_\alpha)\, f_\alpha = -2f_\alpha$$

Together with $[e_\alpha, f_\alpha] = h_\alpha$ from Step 1, all three relations hold.

Step 3: isomorphism with $\mathfrak{sl}_2$ and uniqueness. The three relations $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$ are precisely the structure equations of $\mathfrak{sl}_2(\mathbb{C})$ in its standard basis (example 6.3.2). The elements $e_\alpha, h_\alpha, f_\alpha$ are linearly independent: $e_\alpha \in \mathfrak{g}_\alpha$, $f_\alpha \in \mathfrak{g}_{-\alpha}$, and $h_\alpha \in \mathfrak{h}$ lie in three distinct summands of the root space decomposition (theorem 7.2.4), and $h_\alpha \neq 0$ because $\alpha(h_\alpha) = 2 \neq 0$. Hence the linear map sending $e \mapsto e_\alpha$, $h \mapsto h_\alpha$, $f \mapsto f_\alpha$ is a bijection of three-dimensional spaces carrying the structure constants of $\mathfrak{sl}_2$ to those of $\mathfrak{s}_\alpha = \mathbb{C}e_\alpha \oplus \mathbb{C}h_\alpha \oplus \mathbb{C}f_\alpha$, so it is a Lie algebra isomorphism $\mathfrak{s}_\alpha \cong \mathfrak{sl}_2(\mathbb{C})$, and $\mathfrak{s}_\alpha$ is in particular closed under the bracket, hence a subalgebra. The vector f_α is fixed by its defining normalization $\kappa(e_\alpha, f_\alpha) = 2/(\alpha, \alpha)$ together with the requirement $f_\alpha \in \mathfrak{g}_{-\alpha}$: once theorem 7.4.6 establishes $\dim \mathfrak{g}_{-\alpha} = 1$, this single linear condition determines f_α outright, so the triple $(e_\alpha, h_\alpha, f_\alpha)$ is uniquely determined by the choice of e_α. This completes the proof.

The subalgebra $\mathfrak{s}_\alpha$ is the single most important tool in the structure theory. Every constraint on the roots will be read off by regarding a suitable subspace of $\mathfrak{g}$ as a representation of $\mathfrak{s}_\alpha \cong \mathfrak{sl}_2$ and quoting the representation theory of $\mathfrak{sl}_2$. The distinguished element h_α deserves its own name, because it is what pairs with roots to produce integers.

Definition 7.4.3: Coroot

For a root $\alpha \in \Phi$, the *coroot* of α is the element

$$\alpha^\vee = h_\alpha = \frac{2t_\alpha}{(\alpha,\alpha)} \in \mathfrak{h}$$

of theorem 7.4.1. For any $\beta \in \mathfrak{h}^*$ one writes

$$\langle \beta, \alpha^\vee \rangle = \beta(h_\alpha) = \frac{2(\beta,\alpha)}{(\alpha,\alpha)}$$

for the value of β on the coroot; the equality of the two right-hand expressions holds because $\beta(t_\alpha) = \kappa(t_\beta, t_\alpha) = (\beta,\alpha)$, and multiplying by $2/(\alpha,\alpha)$ gives $\beta(h_\alpha) = \langle \beta, \alpha^\vee \rangle$. The quantities $\langle \beta, \alpha^\vee \rangle$ for $\alpha, \beta \in \Phi$ are the *Cartan integers*.

The pairing $\langle \beta, \alpha^\vee \rangle$ is linear in β but emphatically not symmetric in α and β: it carries a factor (α,α) in the denominator, so swapping the two roots rescales it by $(\beta,\beta)/(\alpha,\alpha)$. In particular $\langle \alpha, \alpha^\vee \rangle = 2(\alpha,\alpha)/(\alpha,\alpha) = 2$ always, recovering $\alpha(h_\alpha) = 2$. The name "integer" is at this point a promise, not a fact; theorem 7.4.6 discharges it. The running example shows the coroot.

Example 7.4.4: The $\mathfrak{sl}_2$-Triple for a Root of $\mathfrak{sl}_n$

Let $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$ with Cartan subalgebra $\mathfrak{h}$ the trace-zero diagonal matrices, and roots $\alpha_{ij} = \varepsilon_i - \varepsilon_j$ for $i \neq j$, where ε_k reads off the k-th diagonal entry (exercise 7.2.1). The root space $\mathfrak{g}_{\alpha_{ij}}$ is the line $\mathbb{C}E_{ij}$ spanned by the elementary matrix E_{ij}, since for a diagonal $H = \operatorname{diag}(a_1, \ldots, a_n)$ one has $[H, E_{ij}] = (a_i - a_j)E_{ij} = \alpha_{ij}(H)\, E_{ij}$. Take $e = E_{ij}$ and $f = E_{ji}$. Their bracket is

$$[E_{ij}, E_{ji}] = E_{ii} - E_{jj} =: H_{ij}$$

and direct computation gives $[H_{ij}, E_{ij}] = (1 - (-1))E_{ij} = 2E_{ij}$ and $[H_{ij}, E_{ji}] = -2E_{ji}$, exactly the triple relations of theorem 7.4.1. Thus $\alpha_{ij}^\vee = H_{ij} = E_{ii} - E_{jj}$, the coroot, is the diagonal matrix with $+1$ in slot i, -1 in slot j, and zeros elsewhere. The Cartan integer of another root α_{kl} against this coroot is

$$\langle \alpha_{kl}, \alpha_{ij}^\vee \rangle = \alpha_{kl}(H_{ij}) = (\varepsilon_k - \varepsilon_l)(H_{ij}) = (\delta_{ki} - \delta_{kj}) - (\delta_{li} - \delta_{lj})$$

an integer by inspection, taking a value in $\{0, \pm 1, \pm 2\}$: it is 2 when $(k,l) = (i,j)$, equals -2 when $(k,l) = (j,i)$, and is ± 1 when the index sets $\{k,l\}$ and $\{i,j\}$ share exactly one element in a compatible position, and 0 when the two roots are "disjoint" or "parallel" in the remaining cases. The subalgebra $\mathfrak{s}_{\alpha_{ij}} = \mathbb{C}E_{ij} \oplus \mathbb{C}H_{ij} \oplus \mathbb{C}E_{ji}$ is the copy of $\mathfrak{sl}_2$ acting on the i-th and j-th coordinate directions, fixing the others.

The $\mathfrak{sl}_n$ computation exhibits the integrality that the general theorem forces, and it does so without any representation theory: the Cartan integers are visibly integers because H_{ij} has integer diagonal entries and the roots evaluate to differences of those entries. The content of theorem 7.4.6 is that this integrality holds for every semisimple $\mathfrak{g}$, not only for the matrix realization. To prove it, the representation theory of $\mathfrak{sl}_2$ must be brought in.

What is needed from $\mathfrak{sl}_2$ is a precise statement about how the element h acts on any finite-dimensional module. The full classification of the irreducible representations of the algebraic group SL_2 is proved

in [10, Classification of Irreducible Representations]; the argument is short enough to give here in the form used, because the weight-string facts are the input to everything that follows.

Proposition 7.4.5: Weights of a Finite-Dimensional $\mathfrak{sl}_2$-Module

Let V be a nonzero finite-dimensional module over $\mathfrak{sl}_2(\mathbb{C})$ with standard basis e, h, f (relations $[h,e]=2e$, $[h,f]=-2f$, $[e,f]=h$). Then:

1. h acts diagonalizably on V with integer eigenvalues. The eigenvalues of h are called the *weights* of V, and the eigenspace $V_\mu = \{v \in V \mid hv = \mu v\}$ is the *weight space* of weight μ.
2. If $v \in V_\mu$ then $ev \in V_{\mu+2}$ and $fv \in V_{\mu-2}$: the operator e raises weight by 2 and f lowers it by 2.
3. If V is irreducible of dimension $m+1$, then the weights are exactly

$$m,\ m-2,\ m-4,\ \ldots,\ -m+2,\ -m$$

each occurring in a one-dimensional weight space; m is the *highest weight*. In general the multiset of weights of any V is symmetric about 0 and, for each weight μ, the operators e and f move v up and down the string without gaps.

***Proof*:**

Step 1: raising and lowering. For $v \in V_\mu$, the module relations give $h(ev) = e(hv) + [h,e]v = e(\mu v) + 2ev = (\mu+2)(ev)$, so $ev \in V_{\mu+2}$; symmetrically $h(fv) = (\mu-2)(fv)$, so $fv \in V_{\mu-2}$. This is part (2), and it holds for any V.

Step 2: integrality and diagonalizability of h. Since $\mathfrak{g} = \mathfrak{sl}_2$ is semisimple, the module V is a direct sum of irreducible submodules by Weyl's theorem (theorem 6.4.8), so it suffices to establish parts (1) and (3) for irreducible V; the general weight multiset is then the union of the irreducible strings, symmetric about 0 because each string is. Assume V is irreducible. The operator h has an eigenvalue $\lambda \in \mathbb{C}$ (the field is algebraically closed) with a nonzero eigenvector. Applying e repeatedly produces vectors in $V_{\lambda+2}, V_{\lambda+4}, \ldots$ by Step 1; these lie in distinct eigenspaces of h, so they are linearly independent, and finite-dimensionality forces $e^N v = 0$ for some N. Let $v_0 \neq 0$ be the last nonzero vector in this chain, a *highest-weight vector*: $hv_0 = \lambda_0 v_0$ for some $\lambda_0 \in \mathbb{C}$ and $ev_0 = 0$.

Step 3: the descending ladder. Define $v_k = f^k v_0$ for $k \geq 0$, with $v_{-1} = 0$. By Step 1, $hv_k = (\lambda_0 - 2k)v_k$. The key relation is computed by induction: for $k \geq 0$,

$$ev_k = k(\lambda_0 - k + 1)\, v_{k-1}$$

Indeed $ev_0 = 0$ handles $k = 0$, and assuming the formula for k,

$$ev_{k+1} = efv_k = fev_k + [e,f]v_k = f\big(k(\lambda_0 - k + 1)v_{k-1}\big) + hv_k$$

which equals $k(\lambda_0 - k + 1)v_k + (\lambda_0 - 2k)v_k = (k+1)(\lambda_0 - k)\, v_k$, the formula for $k+1$. The nonzero v_k are linearly independent (distinct h-eigenvalues), so finite-dimensionality forces a smallest m with $v_{m+1} = 0$ while $v_m \neq 0$. Applying e to $v_{m+1} = 0$ and using the displayed relation at $k = m+1$ gives $0 = ev_{m+1} = (m+1)(\lambda_0 - m)\, v_m$, and since $v_m \neq 0$ and $m+1 \neq 0$,

this forces $\lambda_0 = m$. Thus the highest weight λ_0 is the nonnegative integer m, and $v_0, v_1, \dots, v_m$ have h-eigenvalues $m, m-2, \dots, -m$, all integers.

Step 4: irreducibility fixes the module. The span $W = \langle v_0, \dots, v_m \rangle$ is stable under h (each v_k is an eigenvector), under f (as $fv_k = v_{k+1}$, with $v_{m+1} = 0$), and under e (by the displayed relation), so W is a nonzero submodule of the irreducible V, hence $W = V$. The weight spaces $V_{m-2k} = \mathbb{C}v_k$ are one-dimensional for $0 \le k \le m$, giving $\dim V = m + 1$ and the weight list $m, m-2, \dots, -m$ of part (3). In particular h is diagonalizable with integer eigenvalues, symmetric about 0, establishing part (1) and the symmetry claim, and the ladder $v_0, \dots, v_m$ visits every weight in the string without gaps, as required.

Three features of proposition 7.4.5 are the ones that transfer to roots. First, integrality: h has integer eigenvalues, never fractional ones. Second, symmetry: the weights are invariant under $\mu \mapsto -\mu$, which is the reflection about the origin that will become the reflection s_α in section 7.5. Third, the string is unbroken: if μ and $\mu + 2k$ are both weights, so is every $\mu + 2j$ between them, because the ladder operators connect them. Each of these becomes a statement about roots once the right subspace of $\mathfrak{g}$ is recognized as an $\mathfrak{s}_\alpha$-module.

The strategy is now transparent. Fix roots α, β with $\beta \neq \pm\alpha$. Collect the root spaces along the line $\beta + \mathbb{Z}\alpha$ into a single subspace and observe that it is stable under $\mathfrak{s}_\alpha = \mathfrak{sl}_2$, hence a module. The operator $h_\alpha = \alpha^\vee$ acts on $\mathfrak{g}_{\beta+k\alpha}$ by the scalar $(\beta + k\alpha)(h_\alpha) = \langle \beta, \alpha^\vee \rangle + 2k$, so the weights on this module are the numbers $\langle \beta, \alpha^\vee \rangle + 2k$. Everything about the roots $\beta + k\alpha$ is then dictated by proposition 7.4.5.

Theorem 7.4.6: Integrality of the Cartan Integers and the Root String

Let $\mathfrak{g}$ be semisimple over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$ and roots Φ. For $\alpha, \beta \in \Phi$:

1. The Cartan integer $\langle \beta, \alpha^\vee \rangle = \dfrac{2(\beta, \alpha)}{(\alpha, \alpha)}$ is an integer.
2. The reflected functional $\beta - \langle \beta, \alpha^\vee \rangle \alpha$ is again a root.
3. The *α-string through β* is an unbroken string: there are integers $p, q \ge 0$ such that $\beta + k\alpha \in \Phi$ for exactly the integers $-p \le k \le q$ (with β itself, at $k = 0$, always in the string), no gaps occur, and
$$p - q = \langle \beta, \alpha^\vee \rangle$$
4. Each root space is one-dimensional: $\dim \mathfrak{g}_\alpha = 1$ for every $\alpha \in \Phi$.
5. The only scalar multiples of α that are roots are $\pm\alpha$: if $c\alpha \in \Phi$ for $c \in \mathbb{C}$, then $c = \pm 1$.

The motivation for bundling five conclusions into one theorem is that they share a single proof: all follow from analyzing two $\mathfrak{s}_\alpha$-modules. Parts (1) through (3) come from the module strung along $\beta + \mathbb{Z}\alpha$; parts (4) and (5), which concern the line $\mathbb{Z}\alpha$ itself, come from the module $\bigoplus_k \mathfrak{g}_{k\alpha} \oplus \mathfrak{h}$ and a dimension count. Separating them into distinct theorems would obscure that they are one representation-theoretic fact seen from two angles.

***Proof*:**
Throughout, $\mathfrak{s}_\alpha = \mathbb{C}e_\alpha \oplus \mathbb{C}h_\alpha \oplus \mathbb{C}f_\alpha \cong \mathfrak{sl}_2$ is the triple of theorem 7.4.1, and $h_\alpha = \alpha^\vee$ acts on any $\mathfrak{g}_\gamma$ by the scalar $\gamma(h_\alpha) = \langle \gamma, \alpha^\vee \rangle$, since for $X \in \mathfrak{g}_\gamma$ one has $h_\alpha X = [h_\alpha, X] = \gamma(h_\alpha)X$.

Step 1: the β-string module. Assume first $\beta \neq \pm\alpha$ (the case $\beta = \pm\alpha$ is folded into Step 4). Set

$$M = \bigoplus_{k \in \mathbb{Z}} \mathfrak{g}_{\beta + k\alpha}$$

a finite direct sum, since only finitely many $\beta + k\alpha$ are roots. This M is an $\mathfrak{s}_\alpha$-module: by proposition 7.2.5, $[\mathfrak{g}_\alpha, \mathfrak{g}_{\beta+k\alpha}] \subseteq \mathfrak{g}_{\beta+(k+1)\alpha}$ and $[\mathfrak{g}_{-\alpha}, \mathfrak{g}_{\beta+k\alpha}] \subseteq \mathfrak{g}_{\beta+(k-1)\alpha}$, so e_α, f_α map M into M, and h_α preserves each summand. On the summand $\mathfrak{g}_{\beta+k\alpha}$, the element h_α acts by the scalar

$$(\beta + k\alpha)(h_\alpha) = \beta(h_\alpha) + k\,\alpha(h_\alpha) = \langle \beta, \alpha^\vee \rangle + 2k$$

using $\alpha(h_\alpha) = 2$ from theorem 7.4.1. Thus the weights of h_α on M are among the numbers $\langle \beta, \alpha^\vee \rangle + 2k$ for those k with $\mathfrak{g}_{\beta+k\alpha} \neq 0$.

Step 2: integrality. By proposition 7.4.5(1) every weight of h_α on the module M is an integer. Since β itself is a root, $k = 0$ contributes the weight $\langle \beta, \alpha^\vee \rangle$, which is therefore an integer. This is part (1) for $\beta \neq \pm\alpha$; for $\beta = \pm\alpha$ one has $\langle \pm\alpha, \alpha^\vee \rangle = \pm 2 \in \mathbb{Z}$ directly. Part (1) holds.

Step 3: the string is unbroken, and $p - q = \langle \beta, \alpha^\vee \rangle$. All weights of M have the form $\langle \beta, \alpha^\vee \rangle + 2k$, so they are congruent mod 2: they lie in a single $\mathfrak{sl}_2$-weight string in the sense that consecutive realized weights differ by 2. By proposition 7.4.5(3) the weight multiset of M is symmetric about 0 and, more to the point, the realized weights of an $\mathfrak{sl}_2$-module form no-gap strings: if μ and μ' are weights of M with $\mu < \mu'$ and $\mu \equiv \mu' \pmod 2$, then every $\mu + 2, \mu + 4, \ldots, \mu'$ is also a weight. Suppose $\beta + k\alpha$ and $\beta + k'\alpha$ are roots with $k < k'$; their weights $\langle \beta, \alpha^\vee \rangle + 2k$ and $\langle \beta, \alpha^\vee \rangle + 2k'$ are then both realized, so every intermediate weight $\langle \beta, \alpha^\vee \rangle + 2j$ for $k \leq j \leq k'$ is realized, meaning $\mathfrak{g}_{\beta+j\alpha} \neq 0$, so $\beta + j\alpha \in \Phi$. Hence the set of k with $\beta + k\alpha \in \Phi$ is an interval $\{-p, -p+1, \ldots, q\}$ of integers, containing 0; this is the unbroken α-string through β.

Now use symmetry. The highest weight of M is realized at $k = q$, giving weight $\langle \beta, \alpha^\vee \rangle + 2q$; the lowest at $k = -p$, giving weight $\langle \beta, \alpha^\vee \rangle - 2p$. The module M is irreducible, a fact established at the end of Step 4 once each summand $\mathfrak{g}_{\beta+k\alpha}$ is known to be one-dimensional; granting it, the top and bottom weights of the single string of M are negatives of each other by proposition 7.4.5(3), so the top weight equals minus the bottom weight:

$$\langle \beta, \alpha^\vee \rangle + 2q = -\big(\langle \beta, \alpha^\vee \rangle - 2p\big)$$

Solving, $2\langle \beta, \alpha^\vee \rangle = 2p - 2q$, that is $\langle \beta, \alpha^\vee \rangle = p - q$. This is part (3), and it also yields part (2): the reflection $\beta - \langle \beta, \alpha^\vee \rangle \alpha = \beta - (p-q)\alpha = \beta + (q-p)\alpha$, and since $-p \leq q - p \leq q$ (both $p, q \geq 0$), the index $q - p$ lies in the string interval, so $\beta + (q-p)\alpha \in \Phi$. Thus $s_\alpha(\beta) := \beta - \langle \beta, \alpha^\vee \rangle \alpha$ is a root, which is part (2). Reflecting $k \mapsto q - p - k$ across the string sends $\beta + k\alpha$ to $\beta + (q-p-k)\alpha$, exhibiting the string's symmetry.

Step 4: $\dim \mathfrak{g}_\alpha = 1$ and only $\pm\alpha$ are roots. The remaining two claims concern the line $\mathbb{C}\alpha$, and are extracted from a second module. Form

$$N = \mathfrak{h} \oplus \bigoplus_{c \in \mathbb{C},\ c\alpha \in \Phi} \mathfrak{g}_{c\alpha}$$

the sum of $\mathfrak{h}$ with all root spaces for roots proportional to α. This N is an $\mathfrak{s}_\alpha$-module: by proposition 7.2.5, $[\mathfrak{g}_{\pm\alpha}, \mathfrak{g}_{c\alpha}] \subseteq \mathfrak{g}_{(c\pm1)\alpha}$ (which is contained in N if $(c \pm 1)\alpha$ is a root and is zero

otherwise), while $[\mathfrak{g}_{\pm\alpha}, \mathfrak{h}] \subseteq \mathfrak{g}_{\pm\alpha} \subseteq N$ and $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathfrak{h} \subseteq N$ by lemma 7.4.2; so $e_\alpha, f_\alpha, h_\alpha$ all carry N into N. The element h_α acts on the summand $\mathfrak{g}_{c\alpha}$ by the scalar $(c\alpha)(h_\alpha) = 2c$, and on $\mathfrak{h}$ by 0, since $\mathfrak{h}$ is abelian by proposition 7.1.7 and $[H, H'] = 0$ for $H, H' \in \mathfrak{h}$. Thus the weight of h_α on $\mathfrak{g}_{c\alpha}$ is $2c$, an even number in the loose sense that it is twice c; in particular all weights coming from root spaces are of the form $2c$, and $\mathfrak{h}$ contributes weight 0 only.

Three submodules of N can be written down at once. First, the triple $\mathfrak{s}_\alpha = \mathbb{C}e_\alpha \oplus \mathbb{C}h_\alpha \oplus \mathbb{C}f_\alpha$ is an irreducible 3-dimensional submodule with weights $2, 0, -2$. Second, the kernel $\ker\alpha = \{H \in \mathfrak{h} \mid \alpha(H) = 0\}$, a codimension-1 subspace of $\mathfrak{h}$ (as $\alpha \neq 0$), is a submodule on which $\mathfrak{s}_\alpha$ acts trivially: for $H \in \ker\alpha$, the brackets $[e_\alpha, H] = -\alpha(H)e_\alpha = 0$, $[f_\alpha, H] = \alpha(H)f_\alpha = 0$, and $[h_\alpha, H] = 0$ all vanish, so $\mathbb{C}H$ is a trivial weight-0 submodule. Third, $\mathfrak{h} = \ker\alpha \oplus \mathbb{C}h_\alpha$ (the sum is direct because $\alpha(h_\alpha) = 2 \neq 0$), so $\ker\alpha$ and $\mathfrak{s}_\alpha$ together contain all of $\mathfrak{h}$, and $\ker\alpha \oplus \mathfrak{s}_\alpha$ is a submodule of N. By Weyl's theorem (theorem 6.4.8) N is completely reducible, so

$$N = \ker\alpha \oplus \mathfrak{s}_\alpha \oplus M'$$

as $\mathfrak{s}_\alpha$-modules, for some complementary submodule M'.

The key observation is that M' has no weight-0 vector. The weight-0 space of N is exactly $\mathfrak{h}$: the summand $\mathfrak{g}_{c\alpha}$ carries weight $2c$, which is 0 only if $c = 0$, but $0 \cdot \alpha = 0$ is not a root, so no root space sits in weight 0, leaving only $\mathfrak{h}$. Since $\mathfrak{h} = \ker\alpha \oplus \mathbb{C}h_\alpha \subseteq \ker\alpha \oplus \mathfrak{s}_\alpha$, the entire weight-0 space of N already lies in the first two summands, so M' contains no nonzero weight-0 vector. By proposition 7.4.5(3), a nonzero irreducible $\mathfrak{s}_\alpha$-module of highest weight m has a weight-0 vector precisely when m is even; hence every irreducible summand of M' has *odd* highest weight, and its weights $m, m-2, \ldots, -m$ are all odd integers. Consequently every weight of M' is an odd integer, and no even weight of N occurs in M': all even-weight vectors of N lie in $\ker\alpha \oplus \mathfrak{s}_\alpha$, whose even weights are exactly $\{-2, 0, 2\}$ (weight ± 2 on $\mathbb{C}e_\alpha, \mathbb{C}f_\alpha$ and weight 0 on $\ker\alpha \oplus \mathbb{C}h_\alpha$).

This determines the integer-scalar multiples. If $c \in \mathbb{Z}$ and $c\alpha \in \Phi$, then $\mathfrak{g}_{c\alpha}$ carries the even weight $2c$, which must lie in $\{-2, 0, 2\}$; as $c \neq 0$ (since $0 \notin \Phi$), this forces $c = \pm 1$, so among integer multiples only $\pm\alpha$ are roots and $2\alpha \notin \Phi$. Moreover the weight-2 space of N is even, hence lies in $\ker\alpha \oplus \mathfrak{s}_\alpha$, where it is exactly $\mathbb{C}e_\alpha$; but the weight-2 space of N contains $\mathfrak{g}_\alpha$, so $\mathfrak{g}_\alpha = \mathbb{C}e_\alpha$ is one-dimensional. Symmetrically $\dim\mathfrak{g}_{-\alpha} = 1$.

To rule out non-integer multiples, let $\beta_0 = c_0\alpha$ be a root proportional to α of shortest length (among the finitely many roots proportional to α, which exist as α itself is one). Running the entire argument of this step with β_0 in place of α, the integer-multiple conclusion gives that the only integer multiples of β_0 that are roots are $\pm\beta_0$. Every root proportional to α is proportional to β_0, say $r\beta_0$ for a real r; the Cartan integer $\langle r\beta_0, \beta_0^\vee\rangle = 2r$ is an integer by part (1), so $r \in \frac{1}{2}\mathbb{Z}$, and minimality of β_0 forbids $|r| < 1$, forcing r to be an integer of absolute value ≥ 1. By the integer-multiple conclusion for β_0, the only such roots are $\pm\beta_0$. Hence the roots proportional to α are exactly $\pm\beta_0$, and since α is one of them, $\alpha = \pm\beta_0$, so the roots proportional to α are precisely $\pm\alpha$. This is part (5) (both occur by proposition 7.3.4). As α ranged over an arbitrary root, $\dim\mathfrak{g}_\alpha = 1$ for every $\alpha \in \Phi$, which is part (4).

Finally, part (4) closes the small gap left in Step 3. Each realized weight space of the β-string module $M = \bigoplus_k \mathfrak{g}_{\beta+k\alpha}$ is now known to be one-dimensional (the summand $\mathfrak{g}_{\beta+k\alpha}$ has dimension 1 when $\beta + k\alpha$ is a root), and the weights $\langle\beta, \alpha^\vee\rangle + 2k$ are distinct across k, so M has all weight multiplicities equal to 1. A finite-dimensional $\mathfrak{sl}_2$-module whose realized weights form a single unbroken string each of multiplicity 1 is irreducible (a proper submodule would omit a weight or halve a multiplicity), so M is irreducible; its top and bottom weights are then negatives of each other by proposition 7.4.5(3), validating the symmetry step $\langle\beta, \alpha^\vee\rangle + 2q = -(\langle\beta, \alpha^\vee\rangle - 2p)$ used

to derive $p - q = \langle \beta, \alpha^\vee \rangle$. This completes the proof.

The theorem is the arithmetic heart of the subject. The two most consequential statements are integrality, part (1), and the one-dimensionality of root spaces, part (4). Integrality says the roots, as vectors in the Euclidean space $\operatorname{span}_{\mathbb{R}} \Phi$, satisfy the constraint that every pair α, β has $2(\beta, \alpha)/(\alpha, \alpha) \in \mathbb{Z}$; this is the single axiom that turns a finite spanning set of vectors into a *root system*, the object chapter 8 classifies. One-dimensionality of $\mathfrak{g}_\alpha$ says the root space decomposition is as tight as possible: apart from the Cartan subalgebra, $\mathfrak{g}$ is a direct sum of lines, one per root, so $\dim \mathfrak{g} = \dim \mathfrak{h} + |\Phi|$, and the algebra is reconstructed from the finite combinatorial datum $(\mathfrak{h}, \Phi)$ together with the structure constants of the brackets $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta]$.

The map $\beta \mapsto \beta - \langle \beta, \alpha^\vee \rangle \alpha$ of part (2) is the reflection s_α in the hyperplane orthogonal to α: it fixes $\ker \alpha$ and sends $\alpha \mapsto -\alpha$. Part (2) says this reflection permutes Φ, which is the seed of the Weyl group developed in section 7.5. The unbroken-string statement, part (3), is the computational tool that makes root systems finite and classifiable: knowing $p - q = \langle \beta, \alpha^\vee \rangle$ together with the bound on string lengths pins down which vectors can be roots at all.

The $\mathfrak{sl}_n$ realization exhibits every conclusion of the theorem concretely.

Example 7.4.7: Root Strings in $\mathfrak{sl}_n$

Continue with $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$, roots $\alpha_{ij} = \varepsilon_i - \varepsilon_j$, and coroots $\alpha_{ij}^\vee = E_{ii} - E_{jj}$ from example 7.4.4. Every root space $\mathfrak{g}_{\alpha_{ij}} = \mathbb{C} E_{ij}$ is one-dimensional, confirming theorem 7.4.6(4) by inspection, and $\dim \mathfrak{g} = (n^2 - 1) = (n - 1) + n(n - 1) = \dim \mathfrak{h} + |\Phi|$ since there are $n(n - 1)$ ordered pairs $i \neq j$. The multiples of a root α_{ij} in Φ are only $\pm \alpha_{ij} = \alpha_{ij}, \alpha_{ji}$, matching part (5): no $2\alpha_{ij} = 2\varepsilon_i - 2\varepsilon_j$ is a root, as roots have entry pattern $(+1, -1)$ not $(+2, -2)$.

The Cartan integers $\langle \alpha_{kl}, \alpha_{ij}^\vee \rangle = (\delta_{ki} - \delta_{kj}) - (\delta_{li} - \delta_{lj})$ from example 7.4.4 lie in $\{0, \pm 1, \pm 2\}$. Consider the $\alpha_{12} = \varepsilon_1 - \varepsilon_2$ string through $\beta = \alpha_{23} = \varepsilon_2 - \varepsilon_3$ (taking $n \geq 3$). The Cartan integer is

$$\langle \alpha_{23}, \alpha_{12}^\vee \rangle = \alpha_{23}(E_{11} - E_{22}) = (\varepsilon_2 - \varepsilon_3)(E_{11} - E_{22}) = (0 - 1) - (0 - 0) = -1$$

The candidates $\beta + k\alpha_{12}$ are $\alpha_{23} + k(\varepsilon_1 - \varepsilon_2) = \varepsilon_2 - \varepsilon_3 + k\varepsilon_1 - k\varepsilon_2$. For $k = 0$ this is the root $\varepsilon_2 - \varepsilon_3 = \alpha_{23}$; for $k = 1$ it is $\varepsilon_1 - \varepsilon_3 = \alpha_{13}$, again a root; for $k = -1$ it is ε_1-free with a $2\varepsilon_2$ pattern that is not of the form $\varepsilon_a - \varepsilon_b$, hence not a root. So the string is $\{\alpha_{23}, \alpha_{13}\}$, that is $k \in \{0, 1\}$, giving $p = 0$ (no room below) and $q = 1$ (one step up), and indeed $p - q = 0 - 1 = -1 = \langle \alpha_{23}, \alpha_{12}^\vee \rangle$, as theorem 7.4.6(3) predicts. The reflection $s_{\alpha_{12}}(\alpha_{23}) = \alpha_{23} - (-1)\alpha_{12} = \alpha_{23} + \alpha_{12} = \alpha_{13}$ is the other end of the string, a root as part (2) requires.

The example shows the string theorem functioning as a bookkeeping device: to find the α-string through β one computes a single Cartan integer, which fixes $p - q$, and then locates the endpoints. For $\mathfrak{sl}_n$ all strings have length 1 or 2, so the Cartan integers never exceed 2 in absolute value; the appearance of the value ± 3 is reserved for the exceptional algebra G_2, whose long strings are the source of its distinctive geometry. This bound on Cartan integers, that $\langle \beta, \alpha^\vee \rangle \langle \alpha, \beta^\vee \rangle \in \{0, 1, 2, 3\}$ for $\beta \neq \pm \alpha$, is the arithmetic that chapter 8 uses to enumerate all root systems; the exercises establish it.

Exercise 7.4.1

1. For $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$, the root system is A_2 with roots $\Phi = \{\pm \alpha_{12}, \pm \alpha_{23}, \pm \alpha_{13}\}$ where $\alpha_{13} = \alpha_{12} + \alpha_{23}$. Compute the full table of Cartan integers $\langle \beta, \alpha^\vee \rangle$ for all ordered pairs (β, α) of the two simple roots α_{12}, α_{23} and their sum α_{13}, using $\alpha_{ij}^\vee = E_{ii} - E_{jj}$. Verify that each is an

integer and identify the Cartan matrix $\left(\langle \alpha_i, \alpha_j^\vee \rangle\right)$ of the two simple roots.

2. Prove that for any two roots $\alpha, \beta \in \Phi$ with $\beta \neq \pm\alpha$, the product of Cartan integers satisfies

$$\langle \beta, \alpha^\vee \rangle \langle \alpha, \beta^\vee \rangle \in \{0, 1, 2, 3\}$$

Deduce that the two integers $\langle \beta, \alpha^\vee \rangle$ and $\langle \alpha, \beta^\vee \rangle$ have the same sign and that at least one of them has absolute value 1 whenever the product is nonzero.

3. The algebra $\mathfrak{so}_5(\mathbb{C})$ has rank 2 and a root system of type B_2 with four positive roots: two "long" roots and two "short" roots, with $(\text{long}, \text{long}) = 2(\text{short}, \text{short})$. Take a long root α and a short root β that is not orthogonal to it. Using the relation $\langle \beta, \alpha^\vee \rangle \langle \alpha, \beta^\vee \rangle = 2$ from the previous exercise, determine both Cartan integers $\langle \beta, \alpha^\vee \rangle$ and $\langle \alpha, \beta^\vee \rangle$, and describe the α-string through β and the β-string through α (their values of p, q).

4. The coroot construction $\gamma \mapsto \gamma^\vee = 2t_\gamma/(\gamma, \gamma)$ of definition 7.4.3 is defined for every nonzero $\gamma \in \operatorname{span}_{\mathbb{R}} \Phi$, not only for roots. Prove directly from the definition that it satisfies the scaling law

$$(c\gamma)^\vee = c^{-1}\gamma^\vee$$

for every nonzero scalar c, using linearity of $\gamma \mapsto t_\gamma$. Deduce the formal identity $\langle \beta, (2\alpha)^\vee \rangle = \frac{1}{2}\langle \beta, \alpha^\vee \rangle$, and explain why the fact that 2α is never a root (theorem 7.4.6(5)) does not obstruct this purely linear-algebraic computation.

5. Using theorem 7.4.6(3) and the bound of exercise exercise 7.4.1 (2), prove that every root string has length at most 4: if $\beta + k\alpha \in \Phi$ for $-p \leq k \leq q$ then $p + q \leq 3$. Give the type-A_2 and type-G_2 configurations realizing the extremes.

7.5 The Root System of $(\mathfrak{g}, \mathfrak{h})$ and the Weyl Group

The previous sections extracted, from a semisimple Lie algebra $\mathfrak{g}$ and a fixed Cartan subalgebra $\mathfrak{h}$, an increasingly rigid packet of data: the finite set Φ of roots inside the real span $E = \operatorname{span}_{\mathbb{R}} \Phi$ of proposition 7.3.4, a positive-definite inner product $(\cdot, \cdot)$ on E coming from the Killing form, and the integrality of the Cartan integers $\langle \beta, \alpha^\vee \rangle$ of theorem 7.4.6. Each of these facts was proved by feeding $\mathfrak{g}$ through the $\mathfrak{sl}_2$-triples of section 7.4. This section assembles them into a single object. To each root α the Cartan integers attach an orthogonal reflection s_α of E, and the group W generated by these reflections is the Weyl group of the pair $(\mathfrak{g}, \mathfrak{h})$.

The point of the assembly is a change of category. Once Φ is recognized as what will be called an *abstract root system* and W as its reflection group, the Lie algebra has been compressed to a finite piece of Euclidean combinatorial data. Every question about $\mathfrak{g}$ that the structure theory can answer is now a question about the configuration of vectors Φ in E and the finite group W permuting them. This is the datum that chapter 8 will study in the abstract, stripped of its Lie-algebra origin, and classify completely by Dynkin diagram. It is also, as Part III will show, the same combinatorial object that emerges from a compact group through its maximal torus.

The reflection attached to a root is forced on us by the integrality result of the previous section: theorem 7.4.6 already asserted that $\beta - \langle \beta, \alpha^\vee \rangle \alpha$ is again a root, and this expression is precisely the value of the reflection s_α at β.

Definition 7.5.1: Root Reflection

Let $\alpha \in \Phi$. The *reflection* associated to α is the linear map $s_\alpha \colon E \to E$ defined by

$$s_\alpha(v) = v - \langle v, \alpha^\vee \rangle \alpha = v - \frac{2(v,\alpha)}{(\alpha,\alpha)}\,\alpha$$

where $(\cdot,\cdot)$ is the inner product of definition 7.3.2 on E and $\langle v, \alpha^\vee \rangle = 2(v,\alpha)/(\alpha,\alpha)$ is its extension of the Cartan pairing to all of E. It fixes the hyperplane $\alpha^\perp = \{v \in E \mid (v,\alpha) = 0\}$ pointwise and sends α to $-\alpha$, so it is the orthogonal reflection of $(E,(\cdot,\cdot))$ in the hyperplane $\alpha^\perp$.

The formula in definition 7.5.1 is not an abstract choice of reflection. It is the unique orthogonal reflection whose fixed hyperplane is $\alpha^\perp$ and which negates α, and one checks the two defining properties directly. For $v = \alpha$ the coefficient $\langle \alpha, \alpha^\vee \rangle = 2(\alpha,\alpha)/(\alpha,\alpha) = 2$ gives $s_\alpha(\alpha) = \alpha - 2\alpha = -\alpha$. For $v \in \alpha^\perp$ the pairing $(v,\alpha) = 0$ makes the correction term vanish, so $s_\alpha(v) = v$. Since $E = \mathbb{R}\alpha \oplus \alpha^\perp$ orthogonally, these two behaviours determine s_α completely, and they exhibit it as the reflection through $\alpha^\perp$. In particular s_α is orthogonal, $s_\alpha^2 = \mathrm{id}$, and $\det s_\alpha = -1$, the last because s_α is the identity on the codimension-one subspace $\alpha^\perp$ and multiplication by -1 on the line $\mathbb{R}\alpha$.

Example 7.5.2: The Reflections of $\mathfrak{sl}_3$

Take $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$ with $\mathfrak{h}$ the diagonal trace-zero matrices, so that $\mathfrak{h}^*$ carries the functionals $\varepsilon_1, \varepsilon_2, \varepsilon_3$ reading off the diagonal entries, subject to $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$. The roots are $\Phi = \{\varepsilon_i - \varepsilon_j \mid i \neq j\}$, six in all, and the inner product induced by the Killing form is proportional to the standard one in which $\varepsilon_1, \varepsilon_2, \varepsilon_3$ are the orthogonal projections of an orthonormal basis of $\mathbb{R}^3$ onto the plane $\sum x_i = 0$. Write $\alpha = \varepsilon_1 - \varepsilon_2$. Every root has the same length, so $(\alpha,\alpha) = (\beta,\beta)$ for all roots, and the coroot pairing simplifies to $\langle \beta, \alpha^\vee \rangle = 2(\beta,\alpha)/(\alpha,\alpha)$. Computing on the root $\beta = \varepsilon_2 - \varepsilon_3$ gives $(\beta,\alpha) = -\tfrac{1}{2}(\alpha,\alpha)$, so $\langle \beta, \alpha^\vee \rangle = -1$ and

$$s_\alpha(\varepsilon_2 - \varepsilon_3) = (\varepsilon_2 - \varepsilon_3) + (\varepsilon_1 - \varepsilon_2) = \varepsilon_1 - \varepsilon_3$$

which is again a root. The reflection $s_{\varepsilon_1 - \varepsilon_2}$ is visibly the transposition swapping the indices 1 and 2: it fixes ε_3 and exchanges $\varepsilon_1 \leftrightarrow \varepsilon_2$, acting on Φ exactly as the permutation $(1\,2)$ acts on the labels. This is the first sign of the identification $W = S_3$ made precise below.

The example shows the reflections permuting the six roots of $\mathfrak{sl}_3$, and it shows each s_α realized as a transposition of coordinate labels. Taking all reflections together and closing under composition produces the group whose study occupies the rest of Part II and reappears in Part III.

Definition 7.5.3: The Weyl Group

The *Weyl group* of the pair $(\mathfrak{g}, \mathfrak{h})$ is the subgroup

$$W = W(\mathfrak{g}, \mathfrak{h}) = \langle s_\alpha \big| \alpha \in \Phi \rangle$$

of the orthogonal group $\mathrm{O}(E)$ of $(E,(\cdot,\cdot))$ generated by the reflections s_α of definition 7.5.1, one for each root $\alpha \in \Phi$.

That W is a subgroup of $\mathrm{O}(E)$ is immediate: each generator s_α is orthogonal, so every product of generators is orthogonal. What is not yet clear is that W is finite, or that it does anything intelligible

to Φ; both follow once the roots are recognized as an abstract root system, which is the next result. The Weyl group is a genuine invariant of $\mathfrak{g}$, independent of the choice of Cartan subalgebra $\mathfrak{h}$. Any two Cartan subalgebras are conjugate under an inner automorphism ψ of $\mathfrak{g}$ (theorem 13.2.7). If $\psi(\mathfrak{h}) = \mathfrak{h}'$, then ψ carries each root space $\mathfrak{g}_\alpha$ onto the root space of $\alpha' = \alpha \circ (\psi|_{\mathfrak{h}})^{-1}$, and because ψ preserves the Killing form (proposition 6.1.5), $\alpha \mapsto \alpha'$ extends to an isometry of the real spans of the two root sets. Conjugation by this isometry carries each s_α to $s_{\alpha'}$, so $W(\mathfrak{g}, \mathfrak{h}) \cong W(\mathfrak{g}, \mathfrak{h}')$. The present construction fixes one $\mathfrak{h}$ and works with the resulting W, and nothing below uses the independence of $\mathfrak{h}$.

The central result of the section collects the four properties of Φ proved across section 7.3 and section 7.4 into the exact list of axioms that the abstract theory of chapter 8 takes as its starting point. Nothing new is proved here: each axiom has already been established, and the content of the result is the recognition that these four facts are precisely what one needs to reconstruct the combinatorics of $\mathfrak{g}$ without ever mentioning $\mathfrak{g}$ again.

Theorem 7.5.4: The Roots Form a Root System

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$, let $\mathfrak{h}$ be a Cartan subalgebra, let $\Phi \subseteq \mathfrak{h}^*$ be the set of roots, and let $E = \operatorname{span}_{\mathbb{R}} \Phi$ with the inner product $(\cdot, \cdot)$ of definition 7.3.2. Then the pair (E, Φ) satisfies the axioms of an abstract root system:

1. *(R1)* Φ is finite, spans E, and $0 \notin \Phi$.
2. *(R2)* For $\alpha \in \Phi$, the only scalar multiples of α lying in Φ are α and $-\alpha$.
3. *(R3)* For every $\alpha \in \Phi$ the reflection s_α maps Φ to itself: $s_\alpha(\Phi) = \Phi$.
4. *(R4)* For all $\alpha, \beta \in \Phi$ the Cartan integer $\langle \beta, \alpha^\vee \rangle = 2(\beta, \alpha)/(\alpha, \alpha)$ is an integer.

***Proof*:**
Each axiom restates a result already proved. The four parts are verified in turn, matching the numbering of the statement.

1. *(R1) Finiteness, spanning, and* $0 \notin \Phi$. The set Φ is finite because the root space decomposition
$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$
of theorem 7.2.4 is a decomposition of the finite-dimensional space $\mathfrak{g}$ into nonzero summands indexed by Φ, so Φ is finite. It spans $E = \operatorname{span}_{\mathbb{R}} \Phi$ by the very definition of E, and the spanning statement of proposition 7.3.4 guarantees that this real span has dimension $\dim_{\mathbb{C}} \mathfrak{h}$, so E is not collapsed. Finally $0 \notin \Phi$ by definition 7.2.2, which defines Φ as the set of *nonzero* $\alpha \in \mathfrak{h}^*$ with $\mathfrak{g}_\alpha \neq 0$.
2. *(R2) Only* $\pm\alpha$ *are proportional roots.* This is one of the conclusions of theorem 7.4.6: for $\alpha \in \Phi$ the only multiples of α that lie in Φ are $\pm\alpha$. The argument there ran the $\mathfrak{sl}_2$-string through α and used $\dim \mathfrak{g}_\alpha = 1$ together with the fact that 2α cannot be a root, since $e_\alpha \in \mathfrak{g}_\alpha$ acting on the $\mathfrak{sl}_2$-triple would otherwise produce a weight that the module structure forbids.

3. *(R3) s_α preserves Φ.* Let $\alpha, \beta \in \Phi$. By theorem 7.4.6 the vector
$$s_\alpha(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \alpha$$
is again a root. Thus $s_\alpha(\beta) \in \Phi$ for every $\beta \in \Phi$, so $s_\alpha(\Phi) \subseteq \Phi$. Applying the same with the same α to the roots $s_\alpha(\beta)$, and using $s_\alpha^2 = \mathrm{id}$, gives $\beta = s_\alpha(s_\alpha(\beta)) \in s_\alpha(\Phi)$, so $\Phi \subseteq s_\alpha(\Phi)$. Hence $s_\alpha(\Phi) = \Phi$.

4. *(R4) Integrality of the Cartan integers.* This is the opening assertion of theorem 7.4.6: $\langle \beta, \alpha^\vee \rangle = 2(\beta, \alpha)/(\alpha, \alpha) \in \mathbb{Z}$ for all $\alpha, \beta \in \Phi$. It was obtained by identifying $\langle \beta, \alpha^\vee \rangle$ with the weight of β under the $\mathfrak{sl}_2$-triple $\{e_\alpha, h_\alpha, f_\alpha\}$ acting on $\bigoplus_k \mathfrak{g}_{\beta + k\alpha}$, and weights of finite-dimensional $\mathfrak{sl}_2$-modules are integers.

All four axioms hold, so (E, Φ) is a root system, completing the proof.

The four axioms are the entire input to chapter 8. That chapter takes a Euclidean space E and a finite set Φ satisfying (R1) through (R4) as its definition of an abstract root system, forgets that Φ came from a Lie algebra, and proves that up to isomorphism there are only the four infinite families A_n, B_n, C_n, D_n and the five exceptional systems E_6, E_7, E_8, F_4, G_2. The passage from $\mathfrak{g}$ to (E, Φ) is therefore the bridge across which the classification of semisimple Lie algebras is transported: a simple Lie algebra yields an irreducible root system, the classification pins the root system down by its Dynkin diagram, and a reconstruction theorem recovers $\mathfrak{g}$ from the diagram. The word *root system* is used here for the concrete pair (E, Φ); its abstract counterpart, and the proof that the concrete and abstract notions coincide, belong to the next chapter.

The reflection group W inherits its finiteness from the finiteness of Φ, and its action on E is completely determined by how it permutes the roots. This is what makes W a finite group rather than an unwieldy subgroup of $\mathrm{O}(E)$, and it is the reason W can be computed by combinatorics alone.

Proposition 7.5.5: The Weyl Group Permutes the Roots and Is Finite

Let $W = W(\mathfrak{g}, \mathfrak{h})$ be the Weyl group of definition 7.5.3. Then:

1. Every $w \in W$ maps Φ bijectively to Φ, so the action of W on E restricts to an action on the finite set Φ.
2. Every $w \in W$ preserves the inner product $(\cdot, \cdot)$ on E, that is, $W \subseteq \mathrm{O}(E)$.
3. The restriction homomorphism $W \to \mathrm{Sym}(\Phi)$ to the symmetric group on Φ is injective. Consequently W is finite, of order dividing $|\Phi|!$.

***Proof*:**
The three statements are proved in order.

1. *W permutes Φ.* By axiom (R3) of theorem 7.5.4, each generator s_α satisfies $s_\alpha(\Phi) = \Phi$, so each generator restricts to a bijection of the set Φ. The elements of W are finite products of generators and their inverses (each s_α is its own inverse), and a product of bijections of Φ is a bijection of Φ. Hence every $w \in W$ satisfies $w(\Phi) = \Phi$, and the linear action of W on E restricts to a permutation action on Φ.

2. *W preserves the inner product.* Each generator s_α is an orthogonal transformation of $(E, (\cdot,\cdot))$ by definition 7.5.1: it is the orthogonal reflection in the hyperplane $\alpha^\perp$, so $(s_\alpha u, s_\alpha v) = (u, v)$ for all $u, v \in E$. A composite of orthogonal maps is orthogonal, so every $w \in W$ preserves $(\cdot,\cdot)$, giving $W \subseteq \mathrm{O}(E)$.

3. *Injectivity into* $\mathrm{Sym}(\Phi)$ *and finiteness.* By part (1) there is a group homomorphism

$$\rho\colon W \to \mathrm{Sym}(\Phi), \qquad w \mapsto w|_\Phi$$

sending each w to the permutation it induces on the finite set Φ. Suppose $\rho(w) = \mathrm{id}$, so w fixes every root. By (R1) of theorem 7.5.4 the roots span E, and a linear map that fixes a spanning set is the identity on E; hence $w = \mathrm{id}$ in $\mathrm{O}(E)$. Therefore ρ is injective, exhibiting W as a subgroup of the finite group $\mathrm{Sym}(\Phi) \cong S_{|\Phi|}$. In particular W is finite, and its order divides $|\Phi|!$, completing the proof.

The injectivity in part (3) is the structural fact that tames the Weyl group. A priori a subgroup of $\mathrm{O}(E)$ generated by reflections could be infinite, as the reflection groups generated by two lines at an irrational angle already show. It is the requirement that the reflecting vectors form a root system, closed under its own reflections and spanning E, that pins W down to a finite group: the group cannot move any root outside the finite set Φ, and an orthogonal map that fixes Φ pointwise fixes all of E. The Weyl group is thus a finite group acting faithfully on a finite configuration of vectors, and this is exactly the kind of object that admits a complete combinatorial description.

With this proposition the reduction announced at the start of Part II is complete. A semisimple Lie algebra $\mathfrak{g}$, an infinite-dimensional object of continuous symmetries, has been compressed to the finite combinatorial datum (E, Φ, W): a finite set of vectors in a Euclidean space, together with the finite reflection group they generate. Everything the structure theory can say about $\mathfrak{g}$ is encoded here. chapter 8 studies (E, Φ) in the abstract and classifies it by Dynkin diagram, thereby classifying the simple Lie algebras; and when Part III returns to compact groups, the Weyl group $N(T)/T$ built there from a maximal torus T will be revealed as the same finite group W, so that the algebraic combinatorics of this chapter and the topological combinatorics of a compact group coincide.

Example 7.5.6: The A_2 Root System and $W = S_3$ for $\mathfrak{sl}_3$

The full combinatorial datum of $\mathfrak{sl}_3(\mathbb{C})$ can now be drawn. From example 7.5.2, E is the two-dimensional plane $\{(x_1, x_2, x_3) \mid \sum x_i = 0\}$ in $\mathbb{R}^3$, and $\Phi = \{\varepsilon_i - \varepsilon_j \mid i \neq j\}$ consists of six vectors. All six have the same length, and consecutive roots around the hexagon are $60°$ apart: for instance $\alpha = \varepsilon_1 - \varepsilon_2$ and $\gamma = \varepsilon_2 - \varepsilon_3$ satisfy $(\alpha,\gamma)/\sqrt{(\alpha,\alpha)(\gamma,\gamma)} = -\frac{1}{2}$, giving an angle of $120°$, while α and $\alpha + \gamma = \varepsilon_1 - \varepsilon_3$ make $60°$. The six roots are therefore the six vertices of a regular hexagon, spaced $60°$ apart. This is the root system called A_2 in chapter 8.

The Weyl group is computed from its permutation action. Each reflection $s_{\varepsilon_i - \varepsilon_j}$ acts on the labels $\{1, 2, 3\}$ as the transposition $(i\,j)$, as example 7.5.2 showed for $(1\,2)$. The group generated by the three transpositions $(1\,2), (1\,3), (2\,3)$ is the full symmetric group S_3, so

$$W(\mathfrak{sl}_3, \mathfrak{h}) = \langle s_{\varepsilon_1-\varepsilon_2}, s_{\varepsilon_1-\varepsilon_3}, s_{\varepsilon_2-\varepsilon_3} \rangle \cong S_3$$

of order 6. Geometrically, $W = S_3 \cong D_3$ is a subgroup of the full symmetry group D_6 (of order 12) of the regular hexagon of roots, of index 2. The axis fixed by s_α is the line $\alpha^\perp$, and since α is itself a vertex of the hexagon while $s_\alpha(\alpha) = -\alpha$, this axis is perpendicular to the vertex-to-vertex diagonal through α; for a regular hexagon that perpendicular runs through the midpoints of the pair of opposite edges, containing no vertex. The three reflections s_α are therefore the reflections

across the three edge-midpoint axes, and the products of pairs of reflections are the three rotations by $0°, 120°, 240°$. The remaining symmetries of the hexagon, the rotations by $60°$ and $300°$ and the point reflection $-\mathrm{id}$, do not lie in W: no element of the Weyl group of A_2 sends every root to its negative. The identification of the algebraic Weyl group S_3 with this index 2 reflection subgroup of the hexagon's symmetries is the template for the general correspondence between a root system and its Weyl group.

The example is the entire structure theory of $\mathfrak{sl}_3$ rendered as a picture: six vectors at the vertices of a hexagon, permuted by the symmetric group on three letters. Reading off the Dynkin diagram of A_2, which chapter 8 will do, recovers $\mathfrak{sl}_3$ up to isomorphism from this hexagon. The same reduction applied to $\mathfrak{sl}_2$ collapses to a single pair of opposite roots $\pm\alpha$ on a line, with $W = \{1, s_\alpha\}$ of order 2; applied to $\mathfrak{so}_5 \cong \mathfrak{sp}_4$ it produces the system $B_2 = C_2$, eight roots of two lengths whose Weyl group is dihedral of order 8. The exercises carry out these computations.

Exercise 7.5.1

1. Compute the Weyl group of $\mathfrak{sl}_2(\mathbb{C})$. Identify Φ, the space E, and show directly from definition 7.5.1 that $W = \{1, s_\alpha\}$ is cyclic of order 2. Verify that s_α negates the unique positive root and has determinant -1 as a map of the one-dimensional space E.

2. Let $\alpha \in \Phi$ be any root of a semisimple $\mathfrak{g}$. Prove directly from the formula in definition 7.5.1 that $s_\alpha^2 = \mathrm{id}$, so s_α has order 2 in W, and that $\det s_\alpha = -1$ as an element of $\mathrm{O}(E)$. Conclude that the sign homomorphism $\det\colon W \to \{\pm 1\}$ is surjective, so W always has a subgroup of index 2.

3. For $\mathfrak{sl}_3(\mathbb{C})$, verify in full that the three reflections $s_{\varepsilon_i - \varepsilon_j}$ act on the index set $\{1, 2, 3\}$ as the three transpositions, and hence that $W(\mathfrak{sl}_3, \mathfrak{h}) \cong S_3$. Then check that W acts transitively on the six roots of A_2, and identify the stabilizer of a single root.

4. The root system of $\mathfrak{so}_5(\mathbb{C})$ (equivalently $\mathfrak{sp}_4(\mathbb{C})$) is B_2: taking an orthonormal basis e_1, e_2 of $E = \mathbb{R}^2$, the roots are the four short roots $\pm e_1, \pm e_2$ and the four long roots $\pm e_1 \pm e_2$, eight in total. Using definition 7.5.1, compute the reflection s_{e_1} and the reflection $s_{e_1 - e_2}$ on these eight vectors, and show that the Weyl group of B_2 is the dihedral group of order 8 (the symmetry group of the square with vertices $\pm e_1 \pm e_2$).

5. Prove that for any nonzero semisimple $\mathfrak{g}$ the only vector of E fixed by every element of the Weyl group W is 0; that is, the space of W-invariants $E^W = \{v \in E \mid wv = v \text{ for all } w \in W\}$ is $\{0\}$. Deduce that no root is fixed by all of W.

8

Root Systems and the Classification

Chapter 7 attached to a semisimple Lie algebra a root system: a finite set of vectors in a Euclidean space, closed under the reflections it generates, with integral Cartan pairings. This chapter forgets the algebra altogether and studies the root system as a combinatorial object in its own right, then classifies it completely. The reward is one of the summits of the subject. The connected Dynkin diagrams, and with them the simple Lie algebras over $\mathbb{C}$, form an explicit finite list: four infinite families A_n, B_n, C_n, D_n and exactly five exceptions E_6, E_7, E_8, F_4, G_2. A classification of this rigidity, forced by nothing more than a handful of angle and length constraints among a few vectors, is what gives Lie theory its peculiar finiteness.

The argument proceeds in stages. A root system admits a base, a set of simple roots in terms of which every root is a combination with integer coefficients all of one sign; the Weyl group permutes the bases simply transitively, so the base is unique up to symmetry. The pairings among simple roots assemble into a Cartan matrix, drawn as a Dynkin diagram, and the diagram determines the root system. Classifying the admissible diagrams is then a finite combinatorial problem, of bounding angles, excluding cycles, and constraining branch nodes, which the chapter carries out, citing [10, Classification of Irreducible Root Systems] for the identical case-checks where the algebraic and combinatorial routes coincide. Serre's theorem completes the correspondence by reconstructing, from each Cartan matrix, a semisimple Lie algebra presented by explicit generators and relations, unique up to isomorphism; a semisimple Lie algebra is thus determined completely by its diagram. With the classification of the semisimple Lie algebras over $\mathbb{C}$ in hand, Part III returns from algebras to groups, now compact, to convert this structure theory into representation theory.

8.1 Abstract Root Systems: Axioms, Rank, and Duality

Chapter 7 ended by extracting from a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h}$ a finite configuration of vectors in a Euclidean space: the set Φ of roots inside $E = \operatorname{span}_{\mathbb{R}} \Phi$, carrying an inner product from the Killing form and closed under the reflections s_α. Theorem 7.5.4 distilled the properties of this configuration into four axioms. This chapter takes those axioms as a definition, severs the tie to $\mathfrak{g}$, and studies the resulting object on its own terms. The gain is that every question the structure theory can ask about a semisimple Lie algebra becomes a question about a finite set of vectors and the finite group of reflections it generates, and this Euclidean combinatorial question admits a complete answer by Dynkin diagram.

This first section fixes the language. The definition of an abstract root system is stated, together with its rank and its dual; the arithmetic that constrains two roots (the angle between them can only be $90°$, $120°$, $135°$, or $150°$, and their lengths only three ratios) is derived; the four rank-two systems $A_1 \times A_1$, A_2, B_2, G_2 are drawn explicitly; and the notion of irreducibility that mirrors the decomposition of a semisimple Lie algebra into simple ideals is introduced. Everything here is elementary linear algebra over $\mathbb{R}$, but it is the elementary linear algebra on which the entire classification turns.

The axioms are exactly the four facts proved for $(\mathfrak{g}, \mathfrak{h})$ in theorem 7.5.4, now read as a self-standing definition. A root system lives in a finite-dimensional real inner product space; the inner product is what lets one speak of reflections, angles, and lengths, and it is the reason the whole theory is Euclidean rather than linear.

Definition 8.1.1: Abstract Root System

Let E be a finite-dimensional real inner product space with inner product $(\cdot,\cdot)$. For a nonzero vector $\alpha \in E$ write $\alpha^\vee$ for the linear functional

$$\langle v, \alpha^\vee \rangle = \frac{2(v,\alpha)}{(\alpha,\alpha)}$$

and let $s_\alpha \colon E \to E$ be the reflection

$$s_\alpha(v) = v - \langle v, \alpha^\vee \rangle \alpha$$

fixing the hyperplane $\alpha^\perp = \{v \in E \mid (v,\alpha) = 0\}$ pointwise and sending α to $-\alpha$. A subset $\Phi \subseteq E$ is a *root system* if it satisfies:

(R1) Φ is finite, spans E, and $0 \notin \Phi$.

(R2) For each $\alpha \in \Phi$, the only scalar multiples of α in Φ are α and $-\alpha$; that is, $\mathbb{R}\alpha \cap \Phi = \{\alpha, -\alpha\}$.

(R3) For each $\alpha \in \Phi$, the reflection s_α maps Φ to itself: $s_\alpha(\Phi) = \Phi$.

(R4) For all $\alpha, \beta \in \Phi$, the number $\langle \beta, \alpha^\vee \rangle = 2(\beta,\alpha)/(\alpha,\alpha)$ is an integer.

The elements of Φ are called *roots*, and the integers $\langle \beta, \alpha^\vee \rangle$ are the *Cartan integers*.

Definition 8.1.2: Rank of a Root System

The *rank* of a root system $\Phi \subseteq E$ is $\operatorname{rank}\Phi = \dim_{\mathbb{R}} E$. Since (R1) requires Φ to span E, the rank equals the dimension of the real span of the roots.

These are precisely the four axioms verified in theorem 7.5.4 for the roots of a semisimple Lie algebra: (R1) is the finiteness of the root space decomposition together with the spanning of E; (R2) is the statement that $\pm\alpha$ are the only proportional roots; (R3) is that each reflection s_α permutes the roots; and (R4) is the integrality of the Cartan integers, which came from the weights of $\mathfrak{sl}_2$-modules. Every root system that arises in Lie theory arises this way. The point of the abstract definition is that one can now reason about Φ using only these four properties, and it will turn out that the four properties are so restrictive that Φ is determined, up to isomorphism, by a small labelled graph.

Axiom (R4) is the crux, and it is worth pausing on why it is an integrality statement at all. The quantity $\langle \beta, \alpha^\vee \rangle$ measures the projection of β onto the α-direction, scaled so that α itself has value 2; geometrically it counts, doubled, how many times α fits into that projection. That this count is always an integer is a rigidity condition of enormous force: it is what forbids two roots from meeting at, say, 50°, and it is the reason there are only finitely many root systems in each rank. The Weyl group $W = \langle s_\alpha \,|\, \alpha \in \Phi \rangle$, generated by the reflections, is by (R3) a group of permutations of the finite set Φ; since a linear map fixing the spanning set Φ is the identity, W embeds in the symmetric group on Φ and is therefore finite, exactly as proposition 7.5.5 showed for the Lie-algebra case.

Example 8.1.3: The Root System A_1

The smallest root system takes $E = \mathbb{R}$ with its standard inner product and $\Phi = \{\alpha, -\alpha\}$ for a single nonzero α, say $\alpha = 1 \in \mathbb{R}$. The axioms are immediate: Φ is finite, spans the line E, and omits 0, giving (R1); the only real multiples of α in Φ are $\pm\alpha$, giving (R2); the reflection $s_\alpha(v) = v - 2v = -v$ is multiplication by -1, which swaps α and $-\alpha$ and so preserves Φ, giving (R3); and $\langle \alpha, \alpha^\vee \rangle = 2$, $\langle -\alpha, \alpha^\vee \rangle = -2$ are integers, giving (R4). This root system has rank 1 and Weyl group $W = \{1, s_\alpha\} \cong \mathbb{Z}/2$. It is the root system of $\mathfrak{sl}_2(\mathbb{C})$, and it is the unique root system of rank 1: any rank-1 system spans a line, and (R2) forces it to be a single pair $\pm\alpha$, which after rescaling is this Φ. The name A_1 will be justified in section 8.4.

The reflection formula in definition 8.1.1 singles out for each root α a distinguished vector: the one obtained by inverting α's length squared. Collecting these produces a second root system, dual to the first, and the duality it sets up is one of the organizing symmetries of the whole theory.

Definition 8.1.4: Coroots and the Dual Root System

Let $\Phi \subseteq E$ be a root system. For each root $\alpha \in \Phi$ the *coroot* is the vector

$$\alpha^\vee = \frac{2\alpha}{(\alpha, \alpha)} \in E$$

so that $\langle v, \alpha^\vee \rangle = (v, \alpha^\vee)$ agrees with the functional of definition 8.1.1 under the identification of E with its dual by the inner product. The *dual root system* (or *inverse root system*) is

$$\Phi^\vee = \{\alpha^\vee \mid \alpha \in \Phi\}$$

the set of all coroots, regarded as a subset of the same space E.

The coroot inverts the root's length while keeping its direction: a long root has a short coroot and vice versa, since $\alpha^\vee$ has squared length $(\alpha^\vee, \alpha^\vee) = 4/(\alpha, \alpha)$. The pairing $\langle \beta, \alpha^\vee \rangle$ that appears in every axiom is the natural evaluation of the coroot $\alpha^\vee$ against the root β, which is why the same symbol serves both as a functional and, here, as a vector. The following proposition records that $\Phi^\vee$ is itself a root system and that dualizing twice returns the original; the involutivity is what makes duality a genuine symmetry rather than a one-way construction.

Proposition 8.1.5: The Dual Is a Root System

Let $\Phi \subseteq E$ be a root system. Then $\Phi^\vee$ is a root system in E of the same rank, and taking coroots twice recovers the original roots: $(\alpha^\vee)^\vee = \alpha$ for every $\alpha \in \Phi$. Moreover Φ and $\Phi^\vee$ have the same Weyl group, and $\langle \beta^\vee, (\alpha^\vee)^\vee \rangle = \langle \alpha, \beta^\vee \rangle$, so the Cartan integers of $\Phi^\vee$ are the transposes of those of Φ.

***Proof*:**

Step 1: Double dual. For a nonzero α, the coroot has squared length $(\alpha^\vee, \alpha^\vee) = (2\alpha/(\alpha,\alpha), 2\alpha/(\alpha,\alpha)) = 4(\alpha,\alpha)/(\alpha,\alpha)^2 = 4/(\alpha,\alpha)$. Hence

$$(\alpha^\vee)^\vee = \frac{2\alpha^\vee}{(\alpha^\vee, \alpha^\vee)} = \frac{2 \cdot 2\alpha/(\alpha,\alpha)}{4/(\alpha,\alpha)} = \alpha$$

so dualizing is an involution on vectors. In particular the map $\alpha \mapsto \alpha^\vee$ is a bijection $\Phi \to \Phi^\vee$ with inverse $\alpha^\vee \mapsto \alpha$.

Step 2: The reflection in $\alpha^\vee$ equals the reflection in α. Reflection in the coroot $\alpha^\vee$ is

$$s_{\alpha^\vee}(v) = v - \frac{2(v, \alpha^\vee)}{(\alpha^\vee, \alpha^\vee)} \alpha^\vee$$

Substituting $\alpha^\vee = 2\alpha/(\alpha,\alpha)$, so that $(v, \alpha^\vee) = 2(v,\alpha)/(\alpha,\alpha)$ and $(\alpha^\vee, \alpha^\vee) = 4/(\alpha,\alpha)$, the scalar coefficient becomes

$$\frac{2(v, \alpha^\vee)}{(\alpha^\vee, \alpha^\vee)} = \frac{2 \cdot 2(v,\alpha)/(\alpha,\alpha)}{4/(\alpha,\alpha)} = \frac{4(v,\alpha)/(\alpha,\alpha)}{4/(\alpha,\alpha)} = (v, \alpha)$$

and multiplying this scalar by $\alpha^\vee = 2\alpha/(\alpha,\alpha)$ gives

$$s_{\alpha^\vee}(v) = v - (v,\alpha) \cdot \frac{2\alpha}{(\alpha,\alpha)} = v - \frac{2(v,\alpha)}{(\alpha,\alpha)} \alpha = s_\alpha(v)$$

so $s_{\alpha^\vee} = s_\alpha$ as linear maps on E. Consequently the group generated by the reflections in the coroots is identical to the group generated by the reflections in the roots: $W(\Phi^\vee) = W(\Phi)$.

Step 3: $\Phi^\vee$ satisfies the axioms. (R1): $\Phi^\vee$ is the bijective image of the finite set Φ, hence finite; each $\alpha^\vee$ is a nonzero multiple of α, so $\Phi^\vee$ spans the same space E as Φ and omits 0. Thus $\operatorname{rank} \Phi^\vee = \dim E = \operatorname{rank} \Phi$. (R2): the multiples of $\alpha^\vee$ in $\Phi^\vee$ correspond under the bijection to multiples of α in Φ, and $(\lambda\alpha)^\vee = \lambda^{-1}\alpha^\vee$ for $\lambda \neq 0$, so $\mathbb{R}\alpha^\vee \cap \Phi^\vee = \{\pm\alpha^\vee\}$ follows from $\mathbb{R}\alpha \cap \Phi = \{\pm\alpha\}$. (R3): by Step 2, $s_{\alpha^\vee} = s_\alpha$ preserves Φ, and since s_α commutes with the length-inverting map $\beta \mapsto \beta^\vee$ (a reflection is orthogonal, so $s_\alpha(\beta)^\vee = s_\alpha(\beta^\vee)$), it also preserves $\Phi^\vee$. Explicitly $s_{\alpha^\vee}(\beta^\vee) = s_\alpha(\beta)^\vee \in \Phi^\vee$. (R4): the coroot of $\alpha^\vee$ inside $\Phi^\vee$ is $(\alpha^\vee)^\vee = \alpha$ by Step 1, so the Cartan integer of $\Phi^\vee$ pairing $\beta^\vee$ against $\alpha^\vee$ is

$$\langle \beta^\vee, (\alpha^\vee)^\vee \rangle = \frac{2(\beta^\vee, \alpha^\vee)}{(\alpha^\vee, \alpha^\vee)}$$

Direct computation gives $(\beta^\vee, \alpha^\vee) = (2\beta/(\beta,\beta), 2\alpha/(\alpha,\alpha)) = 4(\alpha,\beta)/[(\alpha,\alpha)(\beta,\beta)]$ and $(\alpha^\vee,\alpha^\vee) = 4/(\alpha,\alpha)$, hence

$$\langle \beta^\vee, (\alpha^\vee)^\vee \rangle = \frac{2 \cdot 4(\alpha,\beta)/[(\alpha,\alpha)(\beta,\beta)]}{4/(\alpha,\alpha)} = \frac{2(\alpha,\beta)}{(\beta,\beta)} = \langle \alpha, \beta^\vee \rangle$$

which is an integer by (R4) for Φ. So $\Phi^\vee$ satisfies all four axioms, and its Cartan integers are the transposes of those of Φ, completing the proof.

The interchange $\langle \beta^\vee, (\alpha^\vee)^\vee \rangle = \langle \alpha, \beta^\vee \rangle$ says that dualizing transposes the Cartan integers, which is why B_n and C_n are dual to each other in the Lie-algebra classification while A_n, D_n, and the exceptional types are self-dual. That the Weyl group is unchanged under duality is what allows one to compute W from whichever of Φ, $\Phi^\vee$ is more convenient; the two systems have the same reflections, the same chambers, and the same symmetry group, differing only in the lengths assigned to the roots. Duality will reappear whenever the theory distinguishes a root from its coroot, most visibly in the passage between weights and coweights in the representation theory of Part III.

The heart of the section is the arithmetic constraint that (R4) places on a pair of roots. Two roots cannot sit at an arbitrary angle: the product of their two Cartan integers is a small nonnegative integer, and this single fact pins the angle to one of four values and the ratio of squared lengths to one of three. The argument is short but the entire classification depends on it, so it is given in full.

Proposition 8.1.6: Elementary Properties of Roots

Let $\Phi \subseteq E$ be a root system and let $\alpha, \beta \in \Phi$ with $\beta \neq \pm\alpha$. Write θ for the angle between α and β, so that $(\alpha,\beta) = \sqrt{(\alpha,\alpha)(\beta,\beta)}\cos\theta$. Then:

1. The product of Cartan integers satisfies
$$\langle \alpha, \beta^\vee \rangle \langle \beta, \alpha^\vee \rangle = 4\cos^2\theta \in \{0, 1, 2, 3\}$$

2. Consequently, if α and β are not orthogonal and $(\beta,\beta) \geq (\alpha,\alpha)$, the possible values are exactly those in the following table, where α is the shorter root:

$\langle \alpha, \beta^\vee \rangle$	$\langle \beta, \alpha^\vee \rangle$	θ	$(\beta,\beta)/(\alpha,\alpha)$
1	1	60°	1
-1	-1	120°	1
1	2	45°	2
-1	-2	135°	2
1	3	30°	3
-1	-3	150°	3
0	0	90°	undetermined

3. *(Obtuse roots sum, acute roots differ.)* If $(\alpha,\beta) < 0$ then $\alpha + \beta \in \Phi$; if $(\alpha,\beta) > 0$ then $\alpha - \beta \in \Phi$.

***Proof*:**

Step 1: The product is $4\cos^2\theta$ *and lies in* $\{0,1,2,3\}$. Unwinding the definitions,

$$\langle\alpha,\beta^\vee\rangle\langle\beta,\alpha^\vee\rangle = \frac{2(\alpha,\beta)}{(\beta,\beta)}\cdot\frac{2(\beta,\alpha)}{(\alpha,\alpha)} = \frac{4(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} = 4\cos^2\theta$$

using $(\alpha,\beta)^2 = (\alpha,\alpha)(\beta,\beta)\cos^2\theta$. Since $0 \le \cos^2\theta \le 1$, the product lies in the real interval $[0,4]$. Both factors are integers by (R4), so the product is a nonnegative integer. The extreme value 4 occurs only when $\cos^2\theta = 1$, that is when α and β are parallel; but $\beta \neq \pm\alpha$ excludes this by (R2), so $\cos^2\theta < 1$ and the product is strictly less than 4. Hence it lies in $\{0,1,2,3\}$.

Step 2: Reading off the table. If the product is 0 then $\cos\theta = 0$, giving $\theta = 90°$ and $\langle\alpha,\beta^\vee\rangle = \langle\beta,\alpha^\vee\rangle = 0$; the length ratio is unconstrained. Now suppose the product is nonzero. The two integers $\langle\alpha,\beta^\vee\rangle$ and $\langle\beta,\alpha^\vee\rangle$ have the same sign (their product is positive) and their quotient is

$$\frac{\langle\beta,\alpha^\vee\rangle}{\langle\alpha,\beta^\vee\rangle} = \frac{2(\beta,\alpha)/(\alpha,\alpha)}{2(\alpha,\beta)/(\beta,\beta)} = \frac{(\beta,\beta)}{(\alpha,\alpha)}$$

so the ratio of squared lengths equals the ratio of the two Cartan integers. Ordering so that α is the shorter root, $|\langle\beta,\alpha^\vee\rangle| \ge |\langle\alpha,\beta^\vee\rangle|$, one factor has absolute value 1. Writing $\{|\langle\alpha,\beta^\vee\rangle|, |\langle\beta,\alpha^\vee\rangle|\} = \{1,m\}$ with $m \in \{1,2,3\}$ exhausts the three nonzero products $1,2,3$, and $4\cos^2\theta = m$ gives $\cos\theta = \pm\frac{1}{2}\sqrt{m}$. The plus sign forces $\theta \in \{60°, 45°, 30°\}$ (both Cartan integers positive), the minus sign $\theta \in \{120°, 135°, 150°\}$ (both negative); the length ratio is $(\beta,\beta)/(\alpha,\alpha) = m$ in each case. This is the table.

Step 3: Obtuse roots sum to a root. Suppose $(\alpha,\beta) < 0$. By Step 1 the integer $\langle\alpha,\beta^\vee\rangle\langle\beta,\alpha^\vee\rangle$ lies in $\{1,2,3\}$ and both factors are negative, so at least one of them equals -1; say $\langle\beta,\alpha^\vee\rangle = -1$ (the other case is symmetric). Then by (R3) the reflection s_α carries β to a root:

$$s_\alpha(\beta) = \beta - \langle\beta,\alpha^\vee\rangle\alpha = \beta - (-1)\alpha = \alpha + \beta \in \Phi$$

If instead $\langle\alpha,\beta^\vee\rangle = -1$, then $s_\beta(\alpha) = \alpha + \beta \in \Phi$ by the same computation. Either way $\alpha + \beta$ is a root. For the acute case $(\alpha,\beta) > 0$ with $\alpha \neq \beta$, apply the obtuse case to the pair $\alpha, -\beta$: here $(\alpha,-\beta) < 0$ and $-\beta \in \Phi$ by (R2), so $\alpha + (-\beta) = \alpha - \beta \in \Phi$, completing the proof.

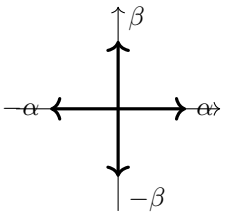
Proposition 8.1.6 is the single computation from which the finiteness of the classification descends. Its first part converts the integrality axiom into a bound on angles; its third part is the local move that lets one navigate around a root system, adding and subtracting roots to reach every root from a chosen few. The table is worth memorizing: two distinct non-proportional roots meet at exactly one of 30°, 45°, 60°, 90°, 120°, 135°, or 150°, and whenever they are not orthogonal their squared lengths stand in the ratio 1, 2, or 3. Three lengths, four obtuse angles: that is all the freedom a root system has locally, and the classification is the global bookkeeping of these local constraints. The obtuse-sum lemma in part (3) will be used at once to build the rank-two systems and, in section 8.2, to show that every root is a nonnegative or nonpositive integer combination of a chosen base.

With the angle table in hand, the rank-two root systems can be enumerated by drawing. In rank 2 every root system lives in the plane, and the possible angles between roots are so constrained that only four configurations survive. These four are worth seeing in full, because they are the smallest nonabelian cases and every higher-rank phenomenon already occurs among them.

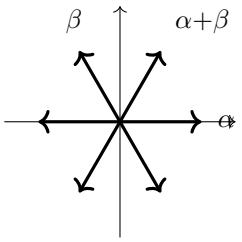
Example 8.1.7: The Rank-Two Root Systems

There are exactly four root systems of rank 2, up to isomorphism. Each is drawn below with all its roots; the shorter roots are drawn to a shorter length where the two lengths differ.

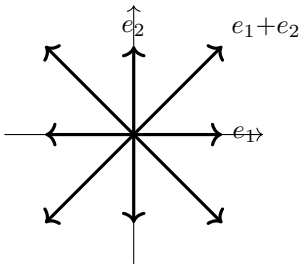
(1) $A_1 \times A_1$ (four roots, orthogonal). Take two orthogonal roots α, β of equal length and $\Phi = \{\pm\alpha, \pm\beta\}$. Since $(\alpha, \beta) = 0$, part (3) of proposition 8.1.6 produces no new roots, and the four vectors already satisfy the axioms. This is the orthogonal union of two copies of A_1, the root system of $\mathfrak{sl}_2 \times \mathfrak{sl}_2$. Its Weyl group is $\mathbb{Z}/2 \times \mathbb{Z}/2$, of order 4.



(2) A_2 (six roots, 60° apart). Take α, β of equal length with $\langle \alpha, \beta^\vee \rangle = \langle \beta, \alpha^\vee \rangle = -1$, so the angle is 120°. Since $(\alpha, \beta) < 0$, the sum $\alpha+\beta$ is a root by proposition 8.1.6(3), and $\Phi = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta)\}$ has six elements at the vertices of a regular hexagon, consecutive roots 60° apart. This is the root system of $\mathfrak{sl}_3$ (compare example 7.5.6); its Weyl group is S_3, of order 6.

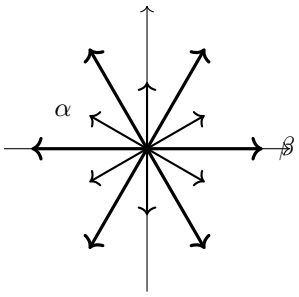


(3) $B_2 = C_2$ (eight roots, two lengths). Take a short root α and a long root β with $\langle \alpha, \beta^\vee \rangle = -1$, $\langle \beta, \alpha^\vee \rangle = -2$, so the angle is 135° and $(\beta, \beta) = 2(\alpha, \alpha)$. Closing under proposition 8.1.6(3) and the reflections gives four short roots $\pm\alpha$, $\pm(\alpha+\beta)$ and four long roots $\pm\beta$, $\pm(2\alpha+\beta)$: eight roots in all, the four short ones along the axes and the four long ones along the diagonals (taking $\alpha = e_1$, $\beta = -e_1 + e_2$ with e_1, e_2 orthonormal, the roots are $\pm e_1, \pm e_2$ and $\pm e_1 \pm e_2$). This is the root system of $\mathfrak{so}_5 \cong \mathfrak{sp}_4$; its Weyl group is dihedral of order 8. The dual $B_2^\vee$ is C_2, so the two coincide as unlabelled systems, which is the coincidence $B_2 = C_2$ named in proposition 8.4.5.



(4) G_2 (twelve roots, two lengths). Take a short root α and a long root β with $\langle \alpha, \beta^\vee \rangle = -1$, $\langle \beta, \alpha^\vee \rangle = -3$, so the angle is 150° and $(\beta, \beta) = 3(\alpha, \alpha)$. Closing up yields six short roots and six

long roots, twelve in all, arranged as the vertices of a regular hexagon (the long roots) with a second, smaller hexagon rotated by 30° (the short roots), so that consecutive roots are 30° apart. This is the root system of the exceptional Lie algebra $\mathfrak{g}_2$; its Weyl group is dihedral of order 12.



The exercises verify that these four exhaust the rank-2 possibilities: an angle table with only $90°, 120°, 135°, 150°$ available for the angle between two roots that are neither equal nor opposite leaves no room for a fifth configuration.

The four pictures are the entire theory in miniature. $A_1 \times A_1$ shows a reducible system, splitting into two orthogonal lines; A_2 is the simplest irreducible system with all roots of one length; and B_2 and G_2 exhibit the two-length phenomenon, with the ratio of squared lengths 2 and 3 respectively, the only ratios allowed by proposition 8.1.6. Read off the number of roots, one sees $4, 6, 8, 12$, and read off the Weyl-group orders, one sees $4, 6, 8, 12$; the equality of these two counts in rank 2 is a coincidence special to the plane, not a general law. What is general is that the whole configuration is recovered from two roots and their Cartan integers, which foreshadows the Cartan matrix of section 8.3.

The first of the four rank-two systems is visibly assembled from two smaller pieces sitting orthogonally, whereas the other three cannot be so split. This distinction, between a root system that decomposes and one that does not, is the combinatorial counterpart of the decomposition of a semisimple Lie algebra into simple ideals, and it is the last notion this section needs.

Definition 8.1.8: Irreducible Root System

A root system $\Phi \subseteq E$ is *reducible* if it can be written as a disjoint union $\Phi = \Phi_1 \sqcup \Phi_2$ of two nonempty subsets with $\Phi_1 \perp \Phi_2$, that is $(\alpha, \beta) = 0$ for all $\alpha \in \Phi_1$, $\beta \in \Phi_2$. It is *irreducible* if it is nonempty and not reducible, that is, it cannot be partitioned into two nonempty mutually orthogonal pieces.

Irreducibility is the indecomposability of the configuration under orthogonal splitting. If $\Phi = \Phi_1 \sqcup \Phi_2$ with the two parts orthogonal, then each Φ_i spans a subspace E_i, the two subspaces are orthogonal and together span E, and each Φ_i is a root system in its own E_i: the reflections s_α for $\alpha \in \Phi_1$ act trivially on E_2 and permute Φ_1, so the axioms hold on each piece separately. Iterating the split until no piece decomposes further expresses any root system as an orthogonal union of irreducible ones, and this decomposition is unique.

Proposition 8.1.9: Decomposition into Irreducibles

Every root system $\Phi \subseteq E$ is the orthogonal disjoint union

$$\Phi = \Phi_1 \sqcup \cdots \sqcup \Phi_r$$

of irreducible root systems $\Phi_i \subseteq E_i$, where $E = E_1 \perp \cdots \perp E_r$ is the corresponding orthogonal decomposition of E. The decomposition is unique up to reordering the summands.

***Proof*:**

Step 1: Existence. Define a relation on Φ by declaring $\alpha \sim \beta$ if there is a chain $\alpha = \gamma_0, \gamma_1, \ldots, \gamma_k = \beta$ of roots with $(\gamma_{i-1}, \gamma_i) \neq 0$ for each i. This is an equivalence relation (reflexive by the empty chain and $(\alpha, \alpha) \neq 0$, symmetric by reversing a chain, transitive by concatenation). Let $\Phi_1, \ldots, \Phi_r$ be the equivalence classes and $E_i = \operatorname{span}_{\mathbb{R}} \Phi_i$. If $\alpha \in \Phi_i$ and $\beta \in \Phi_j$ with $i \neq j$, then $(\alpha, \beta) = 0$, for otherwise $\alpha \sim \beta$ would put them in the same class. Hence $E_i \perp E_j$ for $i \neq j$, and since Φ spans E the subspaces E_i together span E; being mutually orthogonal they give $E = E_1 \perp \cdots \perp E_r$.

Step 2: Each class is a root system. Fix i and check the axioms for $\Phi_i \subseteq E_i$. (R1) is inherited: Φ_i is finite, spans E_i by definition, and omits 0. (R2) is inherited since it is a condition on each root separately. For (R3), let $\alpha, \beta \in \Phi_i$; then $s_\alpha(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \alpha$ is a root in Φ (axiom R3 for Φ), and it lies in Φ_i: if $(\alpha, \beta) = 0$ then $s_\alpha(\beta) = \beta \in \Phi_i$, while if $(\alpha, \beta) \neq 0$ then, using that s_α is orthogonal and $s_\alpha(\alpha) = -\alpha$,

$$(\alpha, s_\alpha(\beta)) = (s_\alpha(\alpha), s_\alpha(s_\alpha(\beta))) = (-\alpha, \beta) = -(\alpha, \beta) \neq 0$$

so $\alpha \sim s_\alpha(\beta)$ and hence $s_\alpha(\beta) \in \Phi_i$. Thus $s_\alpha(\Phi_i) = \Phi_i$. (R4) is a condition on pairs and is inherited. So each Φ_i is a root system, and it is irreducible: any orthogonal splitting of Φ_i would break a linking chain, contradicting that all of Φ_i is a single $\sim$-class.

Step 3: Uniqueness. Suppose $\Phi = \Psi_1 \sqcup \cdots \sqcup \Psi_s$ is another orthogonal decomposition into irreducibles. Roots in the same $\sim$-class must lie in the same Ψ_j: if $\alpha \sim \beta$ via a chain with nonzero consecutive inner products, then consecutive chain elements cannot lie in different orthogonal pieces, so the whole chain, and in particular β, lies in the piece containing α. Hence each Φ_i is contained in some Ψ_j. Conversely each Ψ_j, being irreducible, cannot be split, so it is a union of $\sim$-classes that must be a single class; thus each Ψ_j equals some Φ_i. The two decompositions therefore coincide up to reordering, completing the proof.

The decomposition into irreducibles is the root-system mirror of theorem 6.3.3, which wrote every semisimple Lie algebra as an orthogonal direct sum of simple ideals. Under the passage from $\mathfrak{g}$ to its root system, a direct sum $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$ of simple ideals sends the roots of each $\mathfrak{g}_i$ into mutually orthogonal subspaces, so the root system of $\mathfrak{g}$ is the orthogonal union of the root systems of the $\mathfrak{g}_i$, and a simple ideal yields an irreducible root system. The classification of semisimple Lie algebras therefore reduces to the classification of simple ones, which reduces to the classification of irreducible root systems, which section 8.4 carries out. This is why the abstract theory concentrates on irreducible systems: they are the atoms, and every root system is an orthogonal molecule of them.

Exercise 8.1.1

1. Prove that the only root systems of rank 2, up to isomorphism, are $A_1 \times A_1$, A_2, $B_2 = C_2$, and G_2. Argue as follows: pick two roots α, β that are neither equal nor opposite and span E, with α no longer than β; use proposition 8.1.6 to list the possible angles and length ratios; and for each case close up under reflection and proposition 8.1.6(3) to recover the full root system, checking that the four cases give exactly $4, 6, 8, 12$ roots.

2. For each of the rank-two systems in example 8.1.7, describe the dual root system $\Phi^\vee$ explicitly and identify its type. Verify $(\alpha^\vee)^\vee = \alpha$ on a long and a short root of B_2, and confirm that A_2 and G_2 are self-dual while $B_2^\vee = C_2$.

3. Using proposition 8.1.5, prove directly that the Weyl group of a root system Φ equals the Weyl group of its dual $\Phi^\vee$, by showing $s_{\alpha^\vee} = s_\alpha$ for every $\alpha \in \Phi$. Explain why this makes the assignment $\Phi \mapsto W(\Phi)$ insensitive to which of the two lengths the roots are assigned.

4. Show that no root system contains two roots α, β with $\beta \neq \pm\alpha$ meeting at an angle of $50°$, or indeed at any angle other than $30°, 45°, 60°, 90°, 120°, 135°, 150°$. Identify precisely which axiom of definition 8.1.1 rules out $50°$.

5. In the root system $A_2 = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta)\}$ of example 8.1.7, compute all Cartan integers $\langle \gamma, \delta^\vee \rangle$ for γ, δ ranging over the six roots, and verify axiom (R3) by checking that each reflection s_γ sends the set of six roots to itself.

6. Let $\Phi = \Phi_1 \sqcup \Phi_2$ be an orthogonal disjoint union of two nonempty root systems. Show Φ is a root system with Weyl group $W(\Phi) = W(\Phi_1) \times W(\Phi_2)$ and rank $\operatorname{rank} \Phi_1 + \operatorname{rank} \Phi_2$, and that Φ is reducible. Deduce that a root system is irreducible if and only if its Weyl group cannot be written as such a product coming from an orthogonal splitting of the roots.

8.2 Bases, Positive Roots, and Weyl Chambers

The axioms of definition 8.1.1 present a root system Φ as a symmetric configuration: every root α comes paired with $-\alpha$, and the whole set is stable under all of its own reflections. That symmetry is exactly what one wants for a clean set of axioms, but it obstructs any attempt to write down the system economically. A root system in rank n can have dozens of roots, yet it lives in an n-dimensional space, so most of those roots must be linear combinations of a small distinguished set. This section produces such a set: a *base*, or system of *simple roots*, a vector-space basis $\Delta = \{\alpha_1, \ldots, \alpha_n\}$ of E in terms of which every root expands with *integer coefficients all of the same sign*. The sign dichotomy splits Φ into positive and negative halves, and the base itself carries the entire combinatorial content of Φ: the finitely many Cartan integers $\langle \alpha_i, \alpha_j^\vee \rangle$ among the simple roots will turn out to determine the full system.

The construction is geometric. Choosing a hyperplane in E that misses every root splits Φ into the roots on its positive side and those on its negative side; the simple roots are the extremal, indecomposable positive roots. The hyperplanes to avoid are exactly the root hyperplanes $\alpha^\perp$, and the complement $E \setminus \bigcup_\alpha \alpha^\perp$ falls into finitely many open cones, the *Weyl chambers*. Bases correspond to chambers, and the Weyl group permutes the chambers in the sharpest possible way: simply transitively. This last fact organizes the whole theory. It shows that any two bases are related by a unique symmetry of Φ, that W is generated by the reflections in the simple roots alone, and that every root is a W-image of a simple one.

The definition of a base packages the two properties a distinguished spanning set must have: it is a

genuine vector-space basis, and it resolves every root into a same-sign integer combination.

Definition 8.2.1: Base of a Root System

Let Φ be a root system in the Euclidean space E. A *base* (or *simple system*) for Φ is a subset $\Delta = \{\alpha_1, \dots, \alpha_\ell\} \subseteq \Phi$ such that:

1. Δ is a basis of the vector space E;
2. every root $\beta \in \Phi$ can be written $\beta = \sum_{i=1}^{\ell} c_i \alpha_i$ with integer coefficients c_i that are either all ≥ 0 or all ≤ 0.

The elements of Δ are the *simple roots*, and $\ell = \dim E$ is the rank of Φ (definition 8.1.2).

Two requires sit inside the definition and pull in opposite directions. A basis of E is as small as a spanning set can be, forcing $|\Delta| = \dim E$; while the same-sign requirement forces Δ to be extremal enough that no root falls between the positive and negative cones it cuts out. It is not obvious that any set can satisfy both at once, and much of the section is devoted to producing one. Before that, the sign dichotomy the definition guarantees deserves a name of its own, since it is the organizing device of every subsequent computation.

Definition 8.2.2: Positive and Negative Roots, Height

Fix a base $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ of Φ. A root $\beta = \sum_i c_i \alpha_i$ is *positive* (relative to Δ) if all $c_i \geq 0$, and *negative* if all $c_i \leq 0$. Write Φ^+ for the set of positive roots and Φ^- for the set of negative roots, so that

$$\Phi = \Phi^+ \sqcup \Phi^-, \qquad \Phi^- = -\Phi^+$$

For a positive root $\beta = \sum_i c_i \alpha_i$, the *height* of β relative to Δ is the nonnegative integer

$$\operatorname{ht}(\beta) = \sum_{i=1}^{\ell} c_i$$

The simple roots are precisely the positive roots of height 1.

The decomposition $\Phi = \Phi^+ \sqcup \Phi^-$ is exhaustive and disjoint: every root is one or the other by the same-sign clause of definition 8.2.1, and no root is both, since a nonzero vector cannot have all coefficients simultaneously ≥ 0 and ≤ 0 unless it is zero, which is excluded because $0 \notin \Phi$. The equality $\Phi^- = -\Phi^+$ holds because negating a root flips the sign of every coefficient. Height stratifies the positive roots into finitely many levels; the simple roots occupy level one, and an induction on height is the standard engine for proving statements about all positive roots, since a positive root of height > 1 can always be lowered by subtracting a simple root (a fact isolated in lemma 8.2.5 below). The height also measures how far a positive root sits from the simple roots that generate it.

Example 8.2.3: The Base and Positive Roots of A_2

Take the root system A_2 of rank 2, realized as the six vectors $\varepsilon_i - \varepsilon_j$ $(1 \leq i \neq j \leq 3)$ in the plane $E = \{(x_1, x_2, x_3) \mid \sum x_i = 0\}$, with the inner product induced from $\mathbb{R}^3$ (example 8.1.7). Set

$$\alpha_1 = \varepsilon_1 - \varepsilon_2, \qquad \alpha_2 = \varepsilon_2 - \varepsilon_3$$

These two vectors are linearly independent and span the plane E, so they form a basis. The

remaining four roots expand as

$$\varepsilon_1 - \varepsilon_3 = \alpha_1 + \alpha_2, \qquad \varepsilon_2 - \varepsilon_1 = -\alpha_1, \qquad \varepsilon_3 - \varepsilon_2 = -\alpha_2, \qquad \varepsilon_3 - \varepsilon_1 = -\alpha_1 - \alpha_2$$

Every root is thus an integer combination of α_1, α_2 with coefficients all of one sign, so $\Delta = \{\alpha_1, \alpha_2\}$ is a base. The positive roots are

$$\Phi^+ = \{\alpha_1,\ \alpha_2,\ \alpha_1 + \alpha_2\}$$

three in number, with heights $1, 1, 2$. The simple roots α_1, α_2 are the two of height 1; the single root $\alpha_1 + \alpha_2$ of height 2 is the highest root of A_2. The negative roots $\Phi^- = -\Phi^+$ are the other three vertices of the hexagon, and $\Phi = \Phi^+ \sqcup \Phi^-$ recovers all six. Choosing instead $\alpha_1' = \varepsilon_1 - \varepsilon_2$ and $\alpha_2' = \varepsilon_3 - \varepsilon_1$ gives a different base, with a different positive system; the two bases are related by an element of the Weyl group, as the theory below will show.

The example exhibits the mechanism that the general construction abstracts. The choice of α_1, α_2 amounts to slicing the hexagon by a line through its center so that Φ^+ lies on one side; the two simple roots are the two positive roots nearest that line, the ones that cannot be split as a sum of two other positive roots. Making this precise for an arbitrary root system requires choosing the slicing hyperplane, which is the role of a regular element.

Existence of a base is the first substantial theorem. Its proof is a template worth stating in advance: pick a vector γ off all the root hyperplanes, declare a root positive when it has positive inner product with γ, and take the simple roots to be the positive roots that cannot be decomposed further. The work is in verifying that this recipe produces a genuine basis of E, and the key input is that distinct simple roots always meet at a non-acute angle.

Theorem 8.2.4: Existence of a Base

Every root system Φ in a Euclidean space E admits a base. Concretely, choose a vector $\gamma \in E$ with $(\gamma, \alpha) \neq 0$ for all $\alpha \in \Phi$ (a *regular* element; such γ exist because the finitely many hyperplanes $\alpha^\perp$ do not cover E). Set

$$\Phi^+(\gamma) = \{\alpha \in \Phi \mid (\gamma, \alpha) > 0\}$$

and let $\Delta(\gamma)$ be the set of *indecomposable* elements of $\Phi^+(\gamma)$, those $\alpha \in \Phi^+(\gamma)$ that cannot be written $\alpha = \beta_1 + \beta_2$ with $\beta_1, \beta_2 \in \Phi^+(\gamma)$. Then $\Delta(\gamma)$ is a base of Φ, and its positive system relative to $\Delta(\gamma)$ is exactly $\Phi^+(\gamma)$.

***Proof*:**
Existence of a regular γ comes first, then the base is built from it in four steps. Throughout, write $P = \Phi^+(\gamma)$ and $\Delta = \Delta(\gamma)$.

Step 0: A regular element exists. Each root α determines the hyperplane $\alpha^\perp = \{v \in E \mid (v, \alpha) = 0\}$, a proper subspace of E. A finite union of proper subspaces of a vector space over an infinite field cannot equal the whole space, so $E \setminus \bigcup_{\alpha \in \Phi} \alpha^\perp \neq \varnothing$; any γ in this complement satisfies $(\gamma, \alpha) \neq 0$ for all α. Since $\Phi = -\Phi$, replacing α by $-\alpha$ flips the sign of (γ, α), so $\Phi = P \sqcup (-P)$ with $-P = \{\alpha \mid (\gamma, \alpha) < 0\}$. This already splits the roots into a positive and a negative half; it remains to extract a basis.

Step 1: Every element of P is a nonnegative integer sum of elements of Δ. Suppose not, and

among the counterexamples choose $\alpha \in P$ with (γ, α) minimal; this is possible because $(\gamma, \cdot)$ takes finitely many values on the finite set P. Then $\alpha \notin \Delta$ (a simple root is trivially a sum of itself), so α is decomposable: $\alpha = \beta_1 + \beta_2$ with $\beta_1, \beta_2 \in P$. Pairing with γ,

$$(\gamma, \alpha) = (\gamma, \beta_1) + (\gamma, \beta_2)$$

and both summands on the right are strictly positive, so each is strictly less than (γ, α). By minimality neither β_1 nor β_2 is a counterexample, so each is a nonnegative integer sum of elements of Δ; adding these two expressions writes α as one as well, contradicting the choice of α. Hence no counterexample exists, and every root in P is a $\mathbb{Z}_{\geq 0}$-combination of Δ. Applying this to $-P = -\Phi^+(\gamma)$ shows every root in $-P$ is a $\mathbb{Z}_{\leq 0}$-combination, so Δ satisfies clause (2) of definition 8.2.1 and spans E (it spans Φ, which spans E by axiom (R1)).

Step 2: Distinct simple roots meet at a non-acute angle. The claim is that

$$(\alpha, \beta) \leq 0 \qquad \text{for distinct } \alpha, \beta \in \Delta$$

Suppose instead $(\alpha, \beta) > 0$ for some pair $\alpha \neq \beta$ in Δ. By the two-roots lemma of proposition 8.1.6, when two roots make a strictly acute angle their difference is again a root: $\alpha - \beta \in \Phi$. That root lies in P or in $-P$. If $\alpha - \beta \in P$, then $\alpha = \beta + (\alpha - \beta)$ exhibits α as a sum of two elements of P, contradicting the indecomposability of α; if $\alpha - \beta \in -P$, then $\beta - \alpha \in P$ and $\beta = \alpha + (\beta - \alpha)$ contradicts the indecomposability of β. Either way a contradiction, so $(\alpha, \beta) \leq 0$.

Step 3: Δ is linearly independent. Suppose $\sum_{\alpha \in \Delta} r_\alpha \alpha = 0$ with real coefficients r_α. Separate the indices by sign of the coefficient: let $S = \{\alpha \mid r_\alpha > 0\}$ and $T = \{\alpha \mid r_\alpha < 0\}$, and set

$$\varepsilon = \sum_{\alpha \in S} r_\alpha \alpha = \sum_{\beta \in T} (-r_\beta)\beta$$

the two sides being equal because the whole combination vanishes and the zero-coefficient terms drop out. Then

$$(\varepsilon, \varepsilon) = \Big(\sum_{\alpha \in S} r_\alpha \alpha,\ \sum_{\beta \in T} (-r_\beta)\beta \Big) = \sum_{\alpha \in S} \sum_{\beta \in T} r_\alpha (-r_\beta)\,(\alpha, \beta)$$

Every coefficient $r_\alpha(-r_\beta)$ is a product of two positive numbers, hence positive, while every $(\alpha, \beta) \leq 0$ by Step 2 (the indices S and T are disjoint, so $\alpha \neq \beta$ throughout). Thus $(\varepsilon, \varepsilon) \leq 0$, and since $(\cdot, \cdot)$ is positive-definite this forces $\varepsilon = 0$. But then $0 = (\gamma, \varepsilon) = \sum_{\alpha \in S} r_\alpha (\gamma, \alpha)$, a sum of strictly positive terms $r_\alpha(\gamma, \alpha) > 0$ unless S is empty. Hence $S = \varnothing$, and symmetrically $T = \varnothing$, so all $r_\alpha = 0$. Therefore Δ is linearly independent.

Step 4: Conclusion. By Steps 1 and 3, Δ is a linearly independent set that spans E, hence a basis of E, and every root is a same-sign integer combination of Δ. Thus Δ is a base in the sense of definition 8.2.1, and the positive roots relative to Δ are precisely those with nonnegative coefficients, which Step 1 identified with $P = \Phi^+(\gamma)$. This completes the proof.

The proof does more than assert existence: it constructs every base of Φ from a regular vector and characterizes the simple roots as the indecomposable positive roots. The linear-independence argument in Step 3 is the crux, and it runs entirely on the inequality $(\alpha_i, \alpha_j) \leq 0$ of Step 2 together with positive-definiteness. This non-acute-angle property of simple roots is not incidental; it is what makes the Cartan matrix of the next section have nonpositive off-diagonal entries, and it is the single

geometric fact from which the classification of Dynkin diagrams is ultimately squeezed. The theorem also shows there are at most as many bases as regular directions, and the count of bases is settled once the Weyl group is brought in.

One consequence of the height stratification is recorded now, since the simple-transitivity proof will lean on it. It says a positive root of height greater than one always has positive inner product with some simple root, so it can be lowered.

Lemma 8.2.5: Lowering by a Simple Root

Let Δ be a base with positive system Φ^+. If $\beta \in \Phi^+$ is not simple, then $(\beta, \alpha) > 0$ for at least one simple root $\alpha \in \Delta$, and for any such α the vector $\beta - \alpha$ is again a positive root, of height $\mathrm{ht}(\beta) - 1$.

***Proof*:**
Write $\beta = \sum_i c_i \alpha_i$ with all $c_i \geq 0$ and $\mathrm{ht}(\beta) = \sum_i c_i \geq 2$. If $(\beta, \alpha_i) \leq 0$ for every simple root α_i, then

$$(\beta, \beta) = \sum_i c_i (\beta, \alpha_i) \leq 0$$

which contradicts positive-definiteness since $\beta \neq 0$. So $(\beta, \alpha) > 0$ for some $\alpha = \alpha_j \in \Delta$. Because β is not proportional to α (it has height ≥ 2, so at least two distinct simple roots appear or one appears with coefficient ≥ 2, and in the latter case $\beta = c_j \alpha_j$ with $c_j \geq 2$ is impossible, as no such multiple of α_j lies in Φ by axiom (R2)), the two-roots lemma of proposition 8.1.6 applies: a strictly acute angle between the distinct roots β and α gives $\beta - \alpha \in \Phi$. Its expansion is $\beta - \alpha = (c_j - 1)\alpha_j + \sum_{i \neq j} c_i \alpha_i$. Some coefficient other than the j-th remains positive (else $\beta = c_j \alpha_j$ is a multiple of α_j, excluded), so this root is not negative; being a root that is not negative it is positive, and its height is $\mathrm{ht}(\beta) - 1$. This is what was claimed, completing the proof.

With a base in hand, the geometry organizing the choice comes into focus. The regular vectors are exactly those lying off all the root hyperplanes, and the connected pieces of that complement are the Weyl chambers. Two regular vectors in the same chamber sit on the same side of every hyperplane, so they produce the same positive system and hence the same base; and this is the first half of a bijection between chambers and bases.

Definition 8.2.6: Weyl Chamber

Let Φ be a root system in E. The *regular* vectors are those in

$$E_{\mathrm{reg}} = E \setminus \bigcup_{\alpha \in \Phi} \alpha^{\perp}$$

The *Weyl chambers* of Φ are the connected components of E_{reg}. Each chamber is an open convex cone. The chamber containing a regular vector γ is denoted $\mathfrak{C}(\gamma)$, and the base $\Delta(\gamma)$ produced from γ by theorem 8.2.4 is called the base associated to that chamber.

Each chamber is convex because it is an intersection of open half-spaces: fixing one regular γ, the sign of $(\cdot, \alpha)$ is constant on the component of γ for each root α, so the chamber is

$\{v \mid \operatorname{sgn}(v,\alpha) = \operatorname{sgn}(\gamma,\alpha) \text{ for all } \alpha \in \Phi\}$, a finite intersection of open half-spaces, and it is a cone because these conditions are invariant under positive scaling. The following proposition upgrades the observation that a chamber determines a base into a bijection.

Proposition 8.2.7: Chambers and Bases Correspond

The assignment $\mathfrak{C} \mapsto \Delta(\gamma)$, for any $\gamma \in \mathfrak{C}$, is a well-defined bijection from the set of Weyl chambers of Φ to the set of bases of Φ. Under it, the chamber corresponding to a base Δ is the open cone

$$\mathfrak{C}(\Delta) = \{v \in E \mid (v,\alpha) > 0 \text{ for all } \alpha \in \Delta\}$$

the *fundamental (dominant) chamber* of Δ.

***Proof*:**

Well-defined. If γ, γ' lie in the same chamber, then, as noted above, $\operatorname{sgn}(\gamma,\alpha) = \operatorname{sgn}(\gamma',\alpha)$ for every root α, so $\Phi^+(\gamma) = \Phi^+(\gamma')$ and hence $\Delta(\gamma) = \Delta(\gamma')$: the base depends only on the chamber.

Surjective. Every base arises from a regular vector. Given a base $\Delta = \{\alpha_1, \ldots, \alpha_\ell\}$, its positive system Φ^+ consists of the same-sign nonnegative roots; choose the vector $\gamma_\Delta \in E$ determined by $(\gamma_\Delta, \alpha_i) = 1$ for all i (possible since Δ is a basis of E). For any positive root $\beta = \sum_i c_i \alpha_i$ with all $c_i \geq 0$ and not all zero, $(\gamma_\Delta, \beta) = \sum_i c_i > 0$; for a negative root the value is < 0. So γ_Δ is regular, $\Phi^+(\gamma_\Delta) = \Phi^+$, and the indecomposable positive roots are exactly the simple roots α_i. A simple root cannot be $\beta_1 + \beta_2$ with both summands positive, since heights would add to give $\mathrm{ht} \geq 2$; conversely a non-simple positive root β is decomposable, since by lemma 8.2.5 some simple α has $\beta - \alpha \in \Phi^+$, whence $\beta = (\beta - \alpha) + \alpha$ splits β into two positive roots. Thus $\Delta(\gamma_\Delta) = \Delta$, and the chamber of γ_Δ maps to Δ.

Injective. If two chambers give the same base Δ, then both have the same positive system Φ^+ (the positive system relative to Δ), so every γ in either chamber satisfies $(\gamma,\alpha) > 0$ exactly for $\alpha \in \Phi^+$; the two chambers then satisfy identical sign conditions on every root, so they are the same component of E_{reg}.

The fundamental chamber. It remains to identify the chamber of Δ as $\mathfrak{C}(\Delta)$. A regular vector v has $\Phi^+(v) = \Phi^+$ if and only if $(v,\beta) > 0$ for all $\beta \in \Phi^+$. Since every positive root is a nonnegative combination of the simple roots, $(v,\alpha) > 0$ for all $\alpha \in \Delta$ forces $(v,\beta) > 0$ for all $\beta \in \Phi^+$; conversely $\Delta \subseteq \Phi^+$ gives the reverse implication. Hence the chamber is exactly $\{v \mid (v,\alpha) > 0 \text{ for all } \alpha \in \Delta\} = \mathfrak{C}(\Delta)$, as we sought to show.

The bijection turns questions about bases into questions about chambers, where geometry is available. Counting bases is now counting chambers, and understanding how bases transform under symmetries is understanding how W shuffles the chambers. The dominant chamber $\mathfrak{C}(\Delta)$ is the region where all simple roots are strictly positive; its closure is the fundamental domain for the W-action, and the reader will meet it again as the home of dominant weights when the representation theory is developed in Part III. The next theorem is the structural heart of the section.

The Weyl group acts on E by orthogonal transformations preserving Φ (theorem 7.5.4), hence it permutes the root hyperplanes, hence it permutes the chambers, hence (via the bijection just proved) it permutes the bases. The theorem states that this permutation action is as tight as an action can be: simply transitive. Every base is carried to every other by exactly one element of W. Three

consequences fall out of a single argument: transitivity gives that W moves any base to any other; the generation statement gives that the simple reflections alone already generate W; and simple transitivity (the uniqueness of the moving element) is what makes the whole picture rigid.

Theorem 8.2.8: W Acts Simply Transitively on Bases

Let Φ be a root system with Weyl group W, and fix a base Δ with positive system Φ^+. Write $s_i = s_{\alpha_i}$ for the *simple reflections*, the reflections in the simple roots $\alpha_i \in \Delta$. Then:

1. Each simple reflection s_i permutes the positive roots other than α_i: $s_i(\Phi^+ \setminus \{\alpha_i\}) = \Phi^+ \setminus \{\alpha_i\}$, while $s_i(\alpha_i) = -\alpha_i$.
2. W acts transitively on the set of bases of Φ (equivalently, on the set of Weyl chambers), and this action is simply transitive: the only $w \in W$ with $w(\Delta) = \Delta$ is $w = 1$.
3. W is generated by the simple reflections $s_1, \dots, s_\ell$.
4. Every root is W-conjugate to a simple root: for each $\beta \in \Phi$ there is $w \in W$ and $\alpha_i \in \Delta$ with $\beta = w(\alpha_i)$.

***Proof*:**

The four statements are proved in an order in which each feeds the next: (1) the key permutation lemma, then transitivity and generation together, then simple transitivity, then conjugacy.

Part (1): s_i permutes $\Phi^+ \setminus \{\alpha_i\}$. Let $\beta \in \Phi^+$ with $\beta \neq \alpha_i$, and write $\beta = \sum_j c_j \alpha_j$ with all $c_j \geq 0$. Since β is not a multiple of α_i (it is a positive root different from α_i, and $2\alpha_i \notin \Phi$), some coefficient c_k with $k \neq i$ is strictly positive. Now

$$s_i(\beta) = \beta - \langle \beta, \alpha_i^\vee \rangle \alpha_i = \sum_{j \neq i} c_j \alpha_j + \big(c_i - \langle \beta, \alpha_i^\vee \rangle\big)\alpha_i$$

The reflection changes only the coefficient of α_i; the coefficient $c_k > 0$ of α_k is untouched. A root all of whose coefficients relative to Δ have one sign, and one of whose coefficients (c_k) is strictly positive, must be a positive root. Hence $s_i(\beta) \in \Phi^+$. It is not α_i, because $s_i(\beta) = \alpha_i$ would give $\beta = s_i(\alpha_i) = -\alpha_i \notin \Phi^+$. Thus s_i maps $\Phi^+ \setminus \{\alpha_i\}$ into itself; being an involution, it maps this finite set bijectively onto itself. That $s_i(\alpha_i) = -\alpha_i$ is immediate from $\langle \alpha_i, \alpha_i^\vee \rangle = 2$.

A useful reformulation: let ρ be any vector with $(\rho, \alpha) > 0$ for all $\alpha \in \Phi^+$ (for instance the half-sum of the positive roots, or the γ_Δ of proposition 8.2.7). Applying s_i and using part (1), s_i permutes $\Phi^+ \setminus \{\alpha_i\}$ and sends α_i to $-\alpha_i$, so among the positive roots exactly one, namely α_i, is made negative by s_i. This single-root-flipping property is used repeatedly below.

Part (2), transitivity. Work with the subgroup $W' = \langle s_1, \dots, s_\ell \rangle$ generated by the simple reflections of the fixed base Δ, and let Δ' be any base, with associated regular vector γ' (its $\gamma_{\Delta'}$ from proposition 8.2.7). It suffices to find $w \in W'$ carrying γ' into the dominant chamber $\mathfrak{C}(\Delta)$ of Δ: then $w\gamma'$ and γ_Δ lie in the same chamber, so $w(\Delta') = \Delta$, and w^{-1} carries Δ to Δ'. Since w already lies in W', this proves that W' alone already acts transitively on the bases, the fact part (3) will upgrade to $W = W'$. Among all $u \in W'$ choose one maximizing the inner product $(u\gamma', \gamma_\Delta)$, where γ_Δ is the regular vector of Δ with $(\gamma_\Delta, \alpha_i) = 1$ for all simple α_i; the choice exists because the orbit $W'\gamma'$ is finite. The claim is $(u\gamma', \alpha_i) \geq 0$ for every simple root $\alpha_i \in \Delta$. If instead $(u\gamma', \alpha_i) < 0$ for some i, apply the simple reflection $s_i \in W'$; using that s_i is

orthogonal and self-inverse,

$$(s_i u\gamma', \gamma_\Delta) = (u\gamma', s_i\gamma_\Delta) = (u\gamma', \gamma_\Delta) - \langle \gamma_\Delta, \alpha_i^\vee\rangle (u\gamma', \alpha_i)$$

Here $\langle \gamma_\Delta, \alpha_i^\vee\rangle = 2(\gamma_\Delta, \alpha_i)/(\alpha_i, \alpha_i) > 0$ while $(u\gamma', \alpha_i) < 0$ by assumption, so the subtracted term is negative; thus $(s_i u\gamma', \gamma_\Delta) > (u\gamma', \gamma_\Delta)$, and $s_i u \in W'$ contradicts the maximality of u. Hence $(u\gamma', \alpha_i) \geq 0$ for all simple $\alpha_i \in \Delta$; as $u\gamma'$ is regular the inequalities are strict, so $u\gamma' \in \mathfrak{C}(\Delta)$. Taking $w = u$ completes the reduction, and $w(\Delta') = \Delta$ by proposition 8.2.7. Transitivity holds, and the transporting element lies in W'.

Part (3): generation. It remains to see $W' = W$, that is, every reflection s_β for $\beta \in \Phi$ lies in W'. By part (4), proved below independently of this point, $\beta = w(\alpha_i)$ for some $w \in W'$ and simple α_i. For any $w \in \mathrm{O}(E)$ and root α, orthogonality gives $s_{w\alpha} = w s_\alpha w^{-1}$: both sides fix $(w\alpha)^\perp = w(\alpha^\perp)$ pointwise and negate $w\alpha$, so they agree. Hence $s_\beta = s_{w\alpha_i} = w s_i w^{-1} \in W'$. Since W is generated by all the s_β (definition 7.5.3) and each lies in W', we get $W = W'$. Thus W is generated by the simple reflections. (Part (4)'s proof uses only transitivity by W' from part (2), so there is no circularity.)

Part (2), simple transitivity. Define the *length* of $w \in W$ as the least number $\mathrm{len}(w) = k$ of simple reflections in an expression $w = s_{i_1} \cdots s_{i_k}$ (finite by part (3), which is now available). Set alongside it the count

$$n(w) = |\Phi^+ \cap w^{-1}(\Phi^-)|$$

of positive roots that w sends to negative roots. The argument rests on the identity $n(w) = \mathrm{len}(w)$, which counts inversions by peeling reflections off a reduced expression. Fix a reduced $w = s_{i_1} \cdots s_{i_k}$ and set

$$\beta_j = (s_{i_{j+1}} \cdots s_{i_k})(\alpha_{i_j}) \qquad (1 \leq j \leq k)$$

with $\beta_k = \alpha_{i_k}$. By part (1), each simple reflection s_i flips the single positive root α_i and permutes $\Phi^+ \setminus \{\alpha_i\}$; peeling the reflections off one at a time shows that the β_j are k distinct positive roots, that $w(\beta_j) \in \Phi^-$ for each j, and that these are exactly the positive roots sent negative by w. (Distinctness uses that a repeated $\beta_j = \beta_{j'}$ would let one shorten the expression, contradicting reducedness.) Hence $n(w) = k = \mathrm{len}(w)$.

Now suppose $w(\Delta) = \Delta$. A base determines its positive system, so $w(\Phi^+) = \Phi^+$; that is, w sends no positive root to a negative one, and $n(w) = 0$. By the identity just proved, $\mathrm{len}(w) = 0$, and an element of length 0 is the empty product $w = 1$. Thus the only element fixing Δ is the identity: the transitive W-action on bases is simply transitive.

Part (4): every root is conjugate to a simple root. It suffices to show every root is simple for some base, since transitivity then moves that base, and the root inside it, onto Δ. Fix $\beta \in \Phi$ and build a regular vector v for which β becomes an indecomposable positive root, hence simple by theorem 8.2.4. For each root $\alpha \notin \mathbb{R}\beta$ its orthogonal projection $\pi(\alpha)$ onto the hyperplane $\beta^\perp$ is nonzero; choose $v_0 \in \beta^\perp$ with $(v_0, \pi(\alpha)) \neq 0$ for all such α, which is possible because the finitely many conditions $(v_0, \pi(\alpha)) = 0$ each cut out a proper subspace of $\beta^\perp$. Since $(v_0, \alpha) = (v_0, \pi(\alpha))$ for $v_0 \in \beta^\perp$, this makes $(v_0, \alpha) \neq 0$ for every $\alpha \notin \mathbb{R}\beta$. Now set $v = v_0 + \varepsilon\beta$. Then $(v, \beta) = \varepsilon(\beta, \beta)$, which is positive and tends to 0 as $\varepsilon \to 0^+$, while for $\alpha \notin \mathbb{R}\beta$, $(v, \alpha) = (v_0, \alpha) + \varepsilon(\beta, \alpha)$ tends to the nonzero limit (v_0, α). Fix $\varepsilon > 0$ small enough that v is regular and $0 < (v, \beta) < |(v, \alpha)|$ for every root $\alpha \notin \mathbb{R}\beta$.

With this v, $\beta \in \Phi^+(v)$ and β is indecomposable there: if $\beta = \beta_1 + \beta_2$ with $\beta_1, \beta_2 \in \Phi^+(v)$, then neither β_j is proportional to β (else the other summand is $0 \notin \Phi$), so $(v, \beta_j) > (v, \beta)$ for each j by the choice of v; but then $(v, \beta_1) + (v, \beta_2) > (v, \beta)$ contradicts $(v, \beta) = (v, \beta_1) + (v, \beta_2)$. Hence

β is a simple root of the base $\Delta(v)$. By transitivity (part (2)) there is $w \in W$ with $w(\Delta(v)) = \Delta$, and then $w(\beta)$ is a simple root of Δ; equivalently $\beta = w^{-1}(\alpha_i)$ for some simple α_i. This proves every root is W-conjugate to a simple root, completing the proof.

The theorem is the organizing principle of the entire subject. Its transitivity clause says the choice of base, which looked arbitrary, is a choice without consequences: any theorem proved for one base holds for all, since W carries one to another. The generation clause reduces the Weyl group, defined as generated by all $|\Phi|$ reflections, to the $\ell = \operatorname{rank}\Phi$ simple reflections; for A_2 this cuts three generators down to two, and in general it presents W as a Coxeter group on the simple reflections, the presentation that the Dynkin diagram will encode. The simple-transitivity clause makes the length function $\operatorname{len}(w) = n(w)$ well-behaved and gives W a unique longest element, met in the exercises. And the conjugacy clause says Φ has, up to the W-action, only ℓ roots, the simple ones, so all root-theoretic data is concentrated in the finite Cartan matrix among the simple roots.

That concentration is the payoff of the section. A root system in rank ℓ may contain many roots ($\ell^2 + \ell$ in the case of A_ℓ, and more for the other types), but the theorem shows that every one of them is a W-image of one of the ℓ simple roots, and that the positive roots are same-sign integer combinations of the simple roots governed entirely by the pairings $\langle \alpha_i, \alpha_j^\vee \rangle$. The full symmetric configuration Φ, together with its reflection group W, is thus reconstructible from the finite matrix $A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle$ of Cartan integers among the simple roots. That matrix, and the diagram that records it, are the subject of section 8.3; the classification of section 8.4 is then a classification of these matrices, and through them of the root systems and the semisimple Lie algebras that produced them.

Exercise 8.2.1

1. For $\mathfrak{sl}_n(\mathbb{C})$, whose root system is A_{n-1} with roots $\varepsilon_i - \varepsilon_j$ ($i \neq j$) in the hyperplane $\sum x_k = 0$ of $\mathbb{R}^n$, verify that
$$\Delta = \{\alpha_1, \dots, \alpha_{n-1}\}, \qquad \alpha_i = \varepsilon_i - \varepsilon_{i+1}$$
is a base. Show that the positive roots are exactly the $\varepsilon_i - \varepsilon_j$ with $i < j$, express $\varepsilon_i - \varepsilon_j$ ($i < j$) in the base, and compute its height. Deduce that $|\Phi^+| = \binom{n}{2}$.

2. Explain why the Weyl group W acts simply transitively on the set of Weyl chambers, and conclude that the number of Weyl chambers equals $|W|$. For A_{n-1}, where $W = S_n$ (example 7.5.6 for $n = 3$), deduce that there are $n!$ chambers, and interpret the chamber of a regular $\gamma = (\gamma_1, \dots, \gamma_n)$ (with distinct coordinates) as a total ordering of the coordinates.

3. A *longest element* of W is an element w_0 of maximal length $\operatorname{len}(w_0) = |\Phi^+|$. Using the identity $\operatorname{len}(w) = n(w) = |\Phi^+ \cap w^{-1}(\Phi^-)|$ from theorem 8.2.8, prove that a w_0 with $w_0(\Phi^+) = \Phi^-$ exists and is the unique element of maximal length, and that w_0 sends the base Δ to $-\Delta$. Explain why $w_0^2 = 1$.

4. Count the positive roots of A_2 and of B_2 directly from a base. For A_2 use the base of example 8.2.3; for B_2, realized with roots $\pm e_1, \pm e_2, \pm e_1 \pm e_2$ in $\mathbb{R}^2$, take $\Delta = \{\alpha_1, \alpha_2\}$ with $\alpha_1 = e_1 - e_2$ (long) and $\alpha_2 = e_2$ (short), and list all positive roots with their heights and their coefficients in the base.

5. Show that an irreducible root system (definition 8.1.8) has, relative to a base, a unique positive root of maximal height, the *highest root* $\tilde{\alpha}$, and that $(\tilde{\alpha}, \alpha) \geq 0$ for every simple root α. For A_2 and for B_2 (with the bases of the previous exercises), identify $\tilde{\alpha}$ and its coefficients in the base.

6. Prove that if α, β are distinct simple roots of a base Δ, then $(\alpha, \beta) \leq 0$, and deduce that $\langle \alpha, \beta^\vee \rangle \leq 0$. Give an example (from A_2, B_2, or G_2) where the value $\langle \alpha, \beta^\vee \rangle$ is 0, one where it is -1, and one where it is -2 or -3, and relate the value to the angle between the two simple roots.

8.3 Cartan Matrices and Dynkin Diagrams

The previous section reduced a root system to a much smaller packet of data: a base $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$, a set of $\ell = \dim E$ simple roots in terms of which every root is a $\mathbb{Z}$-combination with coefficients all of one sign. The Weyl group permutes the bases simply transitively (theorem 8.2.8), so a root system determines its base up to the symmetry W, and conversely the base determines the whole of Φ: every root is W-conjugate to a simple one, and W is generated by the simple reflections. What remains is to record the geometry *among* the simple roots, their lengths and mutual angles, in a form compact enough to be listed and classified. That record is the Cartan matrix, and its pictorial shorthand is the Dynkin diagram.

The passage from the root system to the Cartan matrix throws away everything about Φ except the pairings $\langle \alpha_i, \alpha_j^\vee \rangle$ of simple roots. The point of this section is that nothing is lost: the $\ell \times \ell$ integer matrix of these pairings determines the entire root system up to isomorphism, and hence determines the semisimple Lie algebra up to isomorphism. The infinite object Φ has been compressed to a small integer matrix, and by the end of the section the classification of simple Lie algebras will have been reduced to the classification of a certain finite family of decorated graphs.

The pairings that enter the matrix are exactly the Cartan integers, whose constraints proposition 8.1.6 has already pinned down: for $\beta \neq \pm\alpha$ the product $\langle \alpha, \beta^\vee \rangle \langle \beta, \alpha^\vee \rangle$ lies in $\{0, 1, 2, 3\}$. Applied to two simple roots, this severely limits the off-diagonal entries, and it is this limitation that makes the classification finite.

Definition 8.3.1: Cartan Matrix

Let Φ be a root system in E with base $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$. The *Cartan matrix* of Φ (relative to Δ) is the $\ell \times \ell$ integer matrix $A = (A_{ij})$ with entries the Cartan integers of the simple roots,

$$A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle = \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)}$$

for $1 \leq i, j \leq \ell$.

The convention on the indices matters and is fixed here once and for all: the (i, j) entry pairs α_i against the *coroot* of α_j, so the column index j is the root whose length appears in the denominator. The Cartan matrix is not symmetric in general precisely because (α_i, α_i) and (α_j, α_j) may differ, and the asymmetry records which of two adjacent simple roots is the longer. The entries obey a short list of constraints, all already available from the root-system axioms, which together make the matrix a highly constrained combinatorial object.

Proposition 8.3.2: Properties of the Cartan Matrix

Let $A = (A_{ij})$ be the Cartan matrix of a root system Φ with base $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$. Then:

1. $A_{ii} = 2$ for all i.
2. For $i \neq j$, $A_{ij} \in \{0, -1, -2, -3\}$.
3. $A_{ij} = 0$ if and only if $A_{ji} = 0$.
4. For $i \neq j$, the product $A_{ij}A_{ji} \in \{0, 1, 2, 3\}$, and $A_{ij}A_{ji} \leq 3$ with equality only when the two roots span a system of type G_2.
5. A is independent of the choice of base up to a simultaneous permutation of its rows and columns.

***Proof*:**
The five statements are proved in order.

1. *Diagonal entries.* By definition $A_{ii} = \langle \alpha_i, \alpha_i^\vee \rangle = 2(\alpha_i, \alpha_i)/(\alpha_i, \alpha_i) = 2$.
2. *Off-diagonal sign and size.* Fix $i \neq j$. Distinct simple roots satisfy $(\alpha_i, \alpha_j) \leq 0$ by the defining property of a base recorded in definition 8.2.1: two distinct simple roots are never at an acute angle. Hence $A_{ij} = 2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j) \leq 0$, and $A_{ij} \in \mathbb{Z}$ by axiom (R4) of definition 8.1.1. Since α_i and α_j are distinct simple roots they are linearly independent, so in particular $\alpha_i \neq \pm\alpha_j$, and proposition 8.1.6 applies: the possible values of A_{ij} for a pair of roots that are neither equal nor opposite are $0, \pm 1, \pm 2, \pm 3$. Combined with $A_{ij} \leq 0$ this gives $A_{ij} \in \{0, -1, -2, -3\}$.
3. *Vanishing is symmetric.* Both A_{ij} and A_{ji} are nonzero scalar multiples of the same number (α_i, α_j), namely $A_{ij} = 2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j)$ and $A_{ji} = 2(\alpha_j, \alpha_i)/(\alpha_i, \alpha_i)$, with $(\alpha_i, \alpha_j) = (\alpha_j, \alpha_i)$ by symmetry of the inner product. Therefore $A_{ij} = 0$ if and only if $(\alpha_i, \alpha_j) = 0$ if and only if $A_{ji} = 0$.
4. *The product of off-diagonal entries.* By proposition 8.1.6, for $\alpha_i \neq \pm\alpha_j$ the product of Cartan integers satisfies
$$A_{ij}A_{ji} = \langle \alpha_i, \alpha_j^\vee \rangle \langle \alpha_j, \alpha_i^\vee \rangle = 4\cos^2\theta_{ij} \in \{0, 1, 2, 3\}$$
where θ_{ij} is the angle between α_i and α_j; the value 4 is excluded because $4\cos^2\theta = 4$ forces $\cos^2\theta = 1$, hence $\alpha_i = \pm\alpha_j$. The extreme value $A_{ij}A_{ji} = 3$ arises precisely when $\{\alpha_i, \alpha_j\}$ generates a rank-two system with a 150° angle, which is the system G_2 of example 8.1.7.
5. *Independence of the base.* If Δ' is a second base, then by theorem 8.2.8 there is a (unique) $w \in W$ with $w(\Delta) = \Delta'$. Since w is an orthogonal transformation preserving Φ, it preserves all inner products, so $\langle w\alpha_i, (w\alpha_j)^\vee \rangle = \langle \alpha_i, \alpha_j^\vee \rangle$. Writing the simple roots of Δ' as $\alpha'_k = w\alpha_{\sigma(k)}$ for a permutation σ, the Cartan matrix relative to Δ' is $A'_{ij} = A_{\sigma(i)\sigma(j)}$, a simultaneous permutation of the rows and columns of A, completing the proof.

The Cartan matrix is thus an integer matrix with 2 on the diagonal, nonpositive integers off the

diagonal bounded below by -3, a symmetric pattern of zeros, and the crucial constraint that the product of a symmetric pair of off-diagonal entries never exceeds 3. It is the last constraint, inherited directly from the angle bound $4\cos^2\theta \leq 3$ of proposition 8.1.6, that forbids the wild configurations one might otherwise imagine and forces the eventual finite list. The independence in part (5) says the Cartan matrix is an invariant of the root system itself, not an artifact of the pair (root system, chosen base): reordering the simple roots permutes rows and columns together but changes nothing essential.

Before drawing anything, the smallest cases fix the pattern. In rank 2 the Cartan matrix is a 2×2 block, and the three irreducible rank-two systems A_2, B_2, G_2 are distinguished exactly by the value of the off-diagonal product $A_{12}A_{21} \in \{1, 2, 3\}$.

Example 8.3.3: The Rank-Two Cartan Matrices

The four rank-two root systems drawn in example 8.1.7 have the following Cartan matrices, computed directly from their roots.

For $A_1 \times A_1$ the two simple roots are orthogonal, $\alpha_1 \perp \alpha_2$, so both off-diagonal entries vanish:

$$A(A_1 \times A_1) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

For A_2 (the roots of $\mathfrak{sl}_3$) the two simple roots have equal length and lie at $120°$, so $(\alpha_1, \alpha_2) = -\frac{1}{2}(\alpha_1, \alpha_1)$ and both Cartan integers equal -1:

$$A(A_2) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

For B_2 take $\alpha_1 = e_1 - e_2$ (long, $(\alpha_1, \alpha_1) = 2$) and $\alpha_2 = e_2$ (short, $(\alpha_2, \alpha_2) = 1$), at $135°$ with $(\alpha_1, \alpha_2) = -1$. Then $A_{12} = 2(\alpha_1, \alpha_2)/(\alpha_2, \alpha_2) = -2$ and $A_{21} = 2(\alpha_2, \alpha_1)/(\alpha_1, \alpha_1) = -1$, so

$$A(B_2) = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$$

The asymmetry $A_{12} = -2 \neq -1 = A_{21}$ records that α_1 is the longer of the two roots.

For G_2 take α_1 long and α_2 short at $150°$ with length ratio $(\alpha_1, \alpha_1)/(\alpha_2, \alpha_2) = 3$. Then $A_{12} = -3$ and $A_{21} = -1$, giving

$$A(G_2) = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

In every case the off-diagonal product $A_{12}A_{21}$ is $0, 1, 2, 3$ respectively, matching the angles $90°, 120°, 135°, 150°$ through $A_{12}A_{21} = 4\cos^2\theta$.

The four matrices are the four possibilities for a rank-two block, and they are the atoms out of which every larger Cartan matrix is assembled: any two adjacent simple roots in a higher-rank system contribute one of these blocks. What distinguishes them is a single integer, the off-diagonal product $0, 1, 2, 3$, together with, in the unequal-length cases, the direction of the asymmetry. This is exactly the data a picture can carry, and it motivates the following pictorial encoding.

The matrix as written is efficient but opaque: reading off which system it describes means squinting at signs. The Dynkin diagram repackages the same information graphically, with one node per simple

root and a bond between two nodes whose thickness is the off-diagonal product and whose arrow, when the roots differ in length, points from the long root to the short.

Definition 8.3.4: Dynkin Diagram

Let Φ be a root system with base $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ and Cartan matrix A. The *Dynkin diagram* of Φ is the labelled graph with one node for each simple root α_i, in which nodes i and j (for $i \neq j$) are joined by

$$A_{ij}A_{ji} \in \{0, 1, 2, 3\}$$

bonds: no bond when $A_{ij} = 0$, and a single, double, or triple bond when $A_{ij}A_{ji}$ equals 1, 2, or 3. When the bond is double or triple, the two simple roots have different lengths, and the bond carries an arrow pointing from the node of the longer root toward the node of the shorter root.

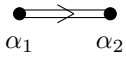
The diagram carries exactly the information of the Cartan matrix and no more. The number of bonds records the unordered product $A_{ij}A_{ji}$, hence the angle between the two simple roots; the arrow on a multiple bond records which of the two entries A_{ij}, A_{ji} is the more negative, hence which root is longer. From a diagram the Cartan matrix is reconstructed as follows: put 2 on the diagonal; for a single bond between i and j set $A_{ij} = A_{ji} = -1$; for a double bond with the arrow pointing toward j (so α_j is short) set the larger-magnitude entry $A_{ij} = -2$ and $A_{ji} = -1$; for a triple bond with arrow toward j set $A_{ij} = -3$, $A_{ji} = -1$. A single bond carries no arrow because the two roots then have equal length. The three basic irreducible rank-two diagrams display the three nontrivial cases.

Example 8.3.5: The Diagrams of A_2, B_2, and G_2

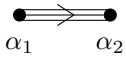
From the Cartan matrices of example 8.3.3, the rank-two diagrams are as follows. The system A_2 has off-diagonal product 1, hence a single unarrowed bond between two equal-length nodes:



The system B_2 has off-diagonal product 2, hence a double bond, with an arrow from the long root α_1 to the short root α_2:



The system G_2 has off-diagonal product 3, hence a triple bond, again with the arrow from the long root α_1 to the short root α_2:



The orthogonal system $A_1 \times A_1$ has no bond at all: two disjoint nodes, which is the pictorial statement that its two simple roots are perpendicular and its root system splits as a product.

The diagrams make visible what the matrices state numerically. A glance distinguishes the three irreducible rank-two systems, and the arrow's direction tells the two unequal-length cases apart from each other. The next result is the justification for the whole apparatus: this graph, a handful of nodes and bonds, loses nothing. It determines the root system completely.

The mechanism is reconstruction. The base and the Cartan integers together prescribe how each simple reflection acts on E, and closing the simple reflections into the Weyl group and sweeping

Δ around by W recovers all of Φ. An isomorphism of Cartan matrices gives compatible bases and identical simple-reflection actions, hence an isometry carrying one root system onto the other.

Theorem 8.3.6: The Cartan Matrix Determines the Root System

Let $\Phi \subseteq E$ and $\Phi' \subseteq E'$ be root systems with bases $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ and $\Delta' = \{\alpha'_1, \dots, \alpha'_\ell\}$. Suppose their Cartan matrices agree,

$$\langle \alpha_i, \alpha_j^\vee \rangle = \langle \alpha'_i, (\alpha'_j)^\vee \rangle \qquad \text{for all } i, j$$

Then the linear isomorphism $\varphi\colon E \to E'$ determined by $\varphi(\alpha_i) = \alpha'_i$ carries Φ onto Φ' and satisfies $\langle \varphi(\beta), \varphi(\gamma)^\vee \rangle = \langle \beta, \gamma^\vee \rangle$ for all $\beta, \gamma \in \Phi$. In particular a root system is determined up to isomorphism by its Cartan matrix, equivalently by its Dynkin diagram.

***Proof*:**
Since Δ is a basis of E and Δ' a basis of E', the assignment $\alpha_i \mapsto \alpha'_i$ extends uniquely to a linear isomorphism $\varphi\colon E \to E'$. The argument runs in three stages: transport the simple reflections, transport the Weyl group, and then sweep the base around to reach every root.

Step 1: The simple reflections are intertwined by φ. For each simple root α_i and any $v \in E$, the reflection acts by $s_{\alpha_i}(v) = v - \langle v, \alpha_i^\vee \rangle \alpha_i$. Evaluate on a simple root α_j and apply φ:

$$\varphi\big(s_{\alpha_i}(\alpha_j)\big) = \varphi(\alpha_j) - \langle \alpha_j, \alpha_i^\vee \rangle \varphi(\alpha_i) = \alpha'_j - \langle \alpha'_j, (\alpha'_i)^\vee \rangle \alpha'_i = s_{\alpha'_i}(\alpha'_j)$$

where the middle equality uses the hypothesis $\langle \alpha_j, \alpha_i^\vee \rangle = \langle \alpha'_j, (\alpha'_i)^\vee \rangle$. Both $\varphi \circ s_{\alpha_i}$ and $s_{\alpha'_i} \circ \varphi$ are linear maps $E \to E'$ that agree on the basis Δ, so they agree everywhere:

$$\varphi \circ s_{\alpha_i} = s_{\alpha'_i} \circ \varphi \qquad \text{for each } i$$

Step 2: The Weyl groups correspond. Conjugation by φ carries each generator s_{α_i} of $W = W(\Phi)$ to the generator $s_{\alpha'_i}$ of $W' = W(\Phi')$, since Step 1 gives $\varphi\, s_{\alpha_i}\, \varphi^{-1} = s_{\alpha'_i}$. By theorem 8.2.8 the simple reflections $\{s_{\alpha_i}\}$ generate W and $\{s_{\alpha'_i}\}$ generate W'. Therefore $w \mapsto \varphi w \varphi^{-1}$ is a group isomorphism $W \to W'$, and for every $w \in W$,

$$\varphi \circ w = (\varphi w \varphi^{-1}) \circ \varphi$$

with $\varphi w \varphi^{-1} \in W'$.

Step 3: Every root is reached, and pairings are preserved. By theorem 8.2.8 every root of Φ is W-conjugate to a simple root: given $\beta \in \Phi$ there are $w \in W$ and an index i with $\beta = w(\alpha_i)$. Then

$$\varphi(\beta) = \varphi(w(\alpha_i)) = (\varphi w \varphi^{-1})(\varphi(\alpha_i)) = (\varphi w \varphi^{-1})(\alpha'_i)$$

which is the image of the simple root $\alpha'_i \in \Phi'$ under the Weyl-group element $\varphi w \varphi^{-1} \in W'$, hence lies in Φ' because W' permutes Φ'. Thus $\varphi(\Phi) \subseteq \Phi'$. The same argument run with φ^{-1}, whose defining data $\alpha'_i \mapsto \alpha_i$ satisfies the transpose hypothesis, gives $\varphi^{-1}(\Phi') \subseteq \Phi$, so $\varphi(\Phi) = \Phi'$.

It remains to check that φ preserves the Cartan pairing on all of Φ, not just on Δ. The pairing $\langle x, y^\vee \rangle$ is *not* invariant under two independent Weyl elements applied to its two slots, so the reduction to simple roots cannot be done slot by slot. Instead read the pairing off the reflection,

extending Step 1 from simple to arbitrary roots. For $\beta \in \Phi$ the reflection s_β lies in W, and φ conjugates it into W': writing $\beta = w(\alpha_i)$ gives $s_\beta = w s_{\alpha_i} w^{-1}$, so by Step 2

$$\varphi\, s_\beta\, \varphi^{-1} = (\varphi w \varphi^{-1}) s_{\alpha_i'} (\varphi w \varphi^{-1})^{-1} = s_{\varphi(\beta)}$$

the last equality because a conjugate $u s_{\alpha_i'} u^{-1}$ of a reflection by $u \in W'$ is the reflection $s_{u(\alpha_i')}$ in the reflected root, and here $(\varphi w \varphi^{-1})(\alpha_i') = \varphi(w\alpha_i) = \varphi(\beta)$ by the displayed identity of the previous paragraph. Thus $\varphi\, s_\beta = s_{\varphi(\beta)}\, \varphi$ for *every* root β, not only the simple ones. Now compare the two reflection formulas: for any $\gamma \in \Phi$,

$$\varphi\big(s_\beta(\gamma)\big) = \varphi(\gamma) - \langle \gamma, \beta^\vee \rangle \varphi(\beta), \qquad s_{\varphi(\beta)}\big(\varphi(\gamma)\big) = \varphi(\gamma) - \langle \varphi(\gamma), \varphi(\beta)^\vee \rangle \varphi(\beta)$$

the first by applying φ to $s_\beta(\gamma) = \gamma - \langle \gamma, \beta^\vee \rangle \beta$, the second by the definition of $s_{\varphi(\beta)}$. The two left-hand sides agree by the intertwining just proved, and $\varphi(\beta) \neq 0$, so the coefficients of $\varphi(\beta)$ match:

$$\langle \varphi(\gamma), \varphi(\beta)^\vee \rangle = \langle \gamma, \beta^\vee \rangle$$

for all $\beta, \gamma \in \Phi$. Hence φ preserves the Cartan pairing throughout Φ, and φ is an isomorphism of root systems, as we sought to show.

The theorem is the linchpin of the classification. It says the map from root systems to Dynkin diagrams is injective on isomorphism classes: two root systems with the same diagram are the same. Combined with the fact, proved next section, that the map is also surjective onto a specific finite list of diagrams, this turns the classification of root systems into the purely graph-theoretic problem of listing the possible diagrams. The proof is worth pausing over, because it exhibits the exact sense in which the finite datum Δ plus the Cartan integers is a complete blueprint: the simple reflections are prescribed by the integers, the Weyl group is generated by the reflections, and the whole root system is the Weyl orbit of the base. Nothing about Φ escapes this reconstruction.

Two diagrams that are isomorphic as labelled graphs, arrows and multiplicities respected, yield equal Cartan matrices after reindexing, so the theorem is usually applied in its diagrammatic form: root systems are classified by their Dynkin diagrams up to graph isomorphism. The one remaining structural question is how the connectivity of the diagram reflects the structure of the root system, and the answer is exactly what one would hope.

Example 8.3.7: A_2 Reconstructed from Its Diagram

The reconstruction in theorem 8.3.6 can be watched explicitly for A_2. Its diagram is two nodes joined by a single bond, giving the Cartan matrix $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ of example 8.3.3. Take abstract simple roots α_1, α_2 in $E = \mathbb{R}^2$ with these Cartan integers. The simple reflections act by

$$s_1(\alpha_1) = -\alpha_1, \quad s_1(\alpha_2) = \alpha_2 - A_{21}\alpha_1 = \alpha_2 + \alpha_1$$

and symmetrically $s_2(\alpha_2) = -\alpha_2$, $s_2(\alpha_1) = \alpha_1 + \alpha_2$. Sweeping the base $\{\alpha_1, \alpha_2\}$ by the group $\langle s_1, s_2 \rangle$ generates

$$\pm\alpha_1, \quad \pm\alpha_2, \quad \pm(\alpha_1 + \alpha_2)$$

six vectors in all, on which the orbit closes. These are exactly the six roots of the hexagonal A_2 system, recovering $\Phi(\mathfrak{sl}_3)$ from the two-node diagram alone, with no reference to the Lie algebra.

The reconstruction recovers the roots of $\mathfrak{sl}_3$ from the barest possible data, two nodes and a bond, and it does so by the same Weyl-orbit sweep that the general theorem uses. The final result of the

section reads off the last piece of structural information carried by the diagram, its connectivity, and ties it back to the algebra.

A root system is called irreducible when it cannot be split into two mutually orthogonal nonempty pieces (definition 8.1.8); every root system is an orthogonal disjoint union of irreducible ones. On the diagram side, this splitting is visible as a splitting of the graph into connected components, because two simple roots are joined by a bond exactly when they are not orthogonal.

Proposition 8.3.8: Irreducible Root Systems Have Connected Diagrams

Let Φ be a root system with base Δ and Dynkin diagram D. Then Φ is irreducible if and only if D is connected. More generally, the decomposition of Φ into irreducible components corresponds bijectively to the decomposition of D into connected components: if $D = D_1 \sqcup \cdots \sqcup D_r$ are the connected components, then $\Phi = \Phi_1 \sqcup \cdots \sqcup \Phi_r$ is the decomposition into irreducibles, where Φ_k is the root system spanned by the simple roots in D_k.

***Proof*:**
It suffices to prove the component statement; the irreducibility equivalence is the case $r = 1$. Partition the base according to the connected components of the diagram: write $\Delta = \Delta_1 \sqcup \cdots \sqcup \Delta_r$, where Δ_k collects the simple roots whose nodes lie in the component D_k. Let $E_k = \operatorname{span}_{\mathbb{R}} \Delta_k$.

Step 1: Distinct components are orthogonal. If $\alpha \in \Delta_k$ and $\beta \in \Delta_m$ with $k \neq m$, then the nodes of α and β lie in different connected components of D, so there is no bond between them, which by definition 8.3.4 means $A_{\alpha\beta}A_{\beta\alpha} = 0$, hence $(\alpha, \beta) = 0$. Therefore $E_k \perp E_m$ for $k \neq m$, and $E = E_1 \perp \cdots \perp E_r$ is an orthogonal direct sum.

Step 2: Each E_k is W-stable and contains a full subsystem. For a simple root $\alpha \in \Delta_k$ and any $\beta \in \Delta_m$, the reflection $s_\alpha(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \alpha$ equals β when $m \neq k$ (the pairing vanishes by Step 1) and lies in E_k when $m = k$. Hence each s_α with $\alpha \in \Delta_k$ fixes E_m pointwise for $m \neq k$ and preserves E_k. Every root is W-conjugate to a simple root (theorem 8.2.8), and W is generated by the simple reflections; since every generator s_α preserves E_k, the subspace E_k is W-stable, so the orbit $W \cdot \alpha$ of any simple root $\alpha \in \Delta_k$ is contained in E_k. Thus every root lies in exactly one E_k, and setting $\Phi_k = \Phi \cap E_k$ gives $\Phi = \Phi_1 \sqcup \cdots \sqcup \Phi_r$ with the Φ_k pairwise orthogonal.

Step 3: Each Φ_k is irreducible. Each Φ_k is a root system in E_k with base Δ_k and Dynkin diagram D_k, which is connected by construction. If Φ_k split as an orthogonal union of two nonempty pieces, its base Δ_k would split accordingly into two mutually orthogonal nonempty subsets, forcing D_k to be disconnected, a contradiction. Hence each Φ_k is irreducible, and $\Phi = \bigsqcup_k \Phi_k$ is its decomposition into irreducible components, completing the proof.

The proposition closes the translation between root systems and diagrams. An irreducible root system is a connected diagram, and a general root system is a disjoint union of connected diagrams, matching its decomposition into irreducible pieces. Passing to the Lie algebra makes this the statement the classification is after. By theorem 6.3.3 a semisimple Lie algebra over $\mathbb{C}$ is an orthogonal direct sum of simple ideals, and a simple ideal contributes an irreducible root system, hence a single connected component of the diagram. Therefore each simple Lie algebra over $\mathbb{C}$ determines, and is determined by, a connected Dynkin diagram, and a semisimple Lie algebra corresponds to the disjoint union of the diagrams of its simple summands. That this correspondence is a bijection, every connected admissible diagram arising from a simple Lie algebra, is completed by the classification and realization of the next two sections.

This is the decisive reduction. Classifying the simple Lie algebras over $\mathbb{C}$ has become the problem of listing the connected Dynkin diagrams, a finite combinatorial question about a family of decorated graphs with 2's on the diagonal and off-diagonal products bounded by 3. That listing is the business of the next section, which shows the connected diagrams to be exactly the four infinite families A_n, B_n, C_n, D_n and the five exceptions E_6, E_7, E_8, F_4, G_2.

Exercise 8.3.1

1. Write down the Cartan matrices of A_2, B_2, and G_2 directly from their defining roots (take A_2 with two simple roots of equal length at $120°$; B_2 with a long root $e_1 - e_2$ and a short root e_2; G_2 with a long and a short root at $150°$ and length-squared ratio 3). For each, verify $A_{12}A_{21} = 4\cos^2\theta$ and identify the direction of the arrow in the Dynkin diagram.

2. Let Φ be the root system A_{n-1} of $\mathfrak{sl}_n$, with simple roots $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ for $1 \le i \le n-1$ (all of equal length, and $(\varepsilon_i, \varepsilon_j) = \delta_{ij}$ in $\mathbb{R}^n$). Show that the Cartan matrix is the $(n-1)\times(n-1)$ tridiagonal matrix with 2 on the diagonal and -1 on the two adjacent off-diagonals, and all other entries 0. Draw the resulting Dynkin diagram (a single chain of $n-1$ nodes).

3. Prove that the determinant of the Cartan matrix of any root system is a positive integer. (Suggestion: the Cartan matrix A satisfies $A = SD$ where S is the Gram matrix $S_{ij} = (\alpha_i, \alpha_j)$ and D is the diagonal matrix with entries $2/(\alpha_j, \alpha_j)$; use that the inner product is positive-definite on E.)

4. Let $\Phi^\vee$ be the dual root system of definition 8.1.4, with coroots $\alpha^\vee$ as its roots and the base $\{\alpha_1^\vee, \dots, \alpha_\ell^\vee\}$. Show that the Cartan matrix of $\Phi^\vee$ is the transpose of that of Φ, and conclude that the Dynkin diagram of $\Phi^\vee$ is obtained from that of Φ by reversing every arrow (so B_n and C_n are dual, while the simply-laced types and G_2, F_4 are self-dual as diagrams).

5. Suppose a root system Φ has a Dynkin diagram consisting of two disjoint single nodes (no bond). Identify Φ, its Cartan matrix, and the associated semisimple Lie algebra, and explain via proposition 8.3.8 why this Lie algebra is not simple.

6. A rank-three root system has Cartan matrix

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -1 & 2 \end{pmatrix}$$

   Determine its Dynkin diagram, including all bond multiplicities and arrow directions, and identify the type of the root system. Which nodes correspond to long roots and which to short?

8.4 The Classification Theorem

Everything in the chapter so far has been a systematic compression. A semisimple Lie algebra was reduced in chapter 7 to a root system (E, Φ), the root system was reduced by a choice of base to its Cartan matrix, and the Cartan matrix was drawn as a Dynkin diagram, a labelled graph on ℓ nodes. By theorem 8.3.6 this graph loses no information: it determines the root system, hence the Lie algebra, up to isomorphism. The question of which semisimple Lie algebras exist has therefore

become the question of which Dynkin diagrams occur, a question about finite labelled graphs with no Lie algebra visible anywhere in it.

This section answers that question completely. The connected admissible diagrams, those that can arise from an irreducible root system, form a short and rigid list: four infinite families and five exceptions, and nothing else. The proof is a sequence of elementary Euclidean inequalities, each ruling out a shape a diagram cannot have, until only these nine remain. The single input driving every step is the constraint $4(u_i, u_j)^2 \in \{0, 1, 2, 3\}$ on the angles between simple roots, itself a restatement of the Cartan-integer inequality $\langle \alpha, \beta^\vee \rangle \langle \beta, \alpha^\vee \rangle \in \{0, 1, 2, 3\}$ of proposition 8.1.6. That one arithmetic fact, propagated through the geometry of unit vectors in Euclidean space, is what forces the finiteness of the classification and with it the entire structure of the subject.

The argument mirrors the purely combinatorial case-checks carried out in *Algebraic Groups and Representations* [10, Classification of Irreducible Root Systems] for the identical problem on the algebraic-groups side. The two treatments arrive at the same list of diagrams because the datum being classified, the Cartan matrix of an irreducible root system, is the same object regardless of whether it was extracted from a Lie algebra by the root space decomposition or from a reductive group by its maximal torus. The structure of the argument is given here in full; the finite list of individually verifiable case-exclusions at the very end is cross-checked against that reference.

The classification is cleanest when phrased not for Dynkin diagrams directly but for the geometric configuration underneath them: a set of unit vectors whose pairwise inner products are constrained exactly as the simple roots are. Passing to unit vectors normalizes away the two root lengths, so that all the combinatorial content sits in the angles.

Definition 8.4.1: Admissible Set of Vectors

Let E be a Euclidean space with inner product $(\cdot, \cdot)$. A finite set $U = \{u_1, \dots, u_\ell\}$ of *unit* vectors in E is *admissible* if

1. the vectors $u_1, \dots, u_\ell$ are linearly independent, and
2. for every pair $i \neq j$ the inner product satisfies $(u_i, u_j) \leq 0$ and

$$4\,(u_i, u_j)^2 \in \{0, 1, 2, 3\}$$

The *diagram* of U is the graph on the nodes $u_1, \dots, u_\ell$ in which u_i and u_j are joined by $4(u_i, u_j)^2$ bonds.

The passage from a base $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ of an irreducible root system to an admissible set is a normalization. Setting $u_i = \alpha_i / |\alpha_i|$ produces unit vectors that are linearly independent, because a base is (theorem 8.2.4); and for $i \neq j$ the number

$$4\,(u_i, u_j)^2 = \frac{4\,(\alpha_i, \alpha_j)^2}{(\alpha_i, \alpha_i)(\alpha_j, \alpha_j)} = \langle \alpha_i, \alpha_j^\vee \rangle \langle \alpha_j, \alpha_i^\vee \rangle$$

is exactly the product of Cartan integers that proposition 8.1.6 showed lies in $\{0, 1, 2, 3\}$ for $\alpha_i \neq \pm \alpha_j$; and the inner product $(\alpha_i, \alpha_j) \leq 0$ for distinct simple roots by the defining property of a base (definition 8.2.1). The number of bonds $4(u_i, u_j)^2$ is precisely the bond count of the Dynkin diagram of definition 8.3.4, so the diagram of the admissible set U is the Dynkin diagram of Δ with its arrows erased. The arrow, which records which of the two roots is longer, is recoverable afterwards and plays no role in the classification of the underlying graph.

The reason for stripping to unit vectors is that admissibility is then a purely metric condition on a configuration in Euclidean space, and metric conditions are what the inequalities below exploit. The whole classification is the statement that very few connected graphs can be realized by a linearly independent set of unit vectors meeting at these prescribed angles.

Example 8.4.2: The Rank-Two Admissible Sets

Two unit vectors u_1, u_2 with $(u_1, u_2) \leq 0$ and $4(u_1, u_2)^2 \in \{0, 1, 2, 3\}$ realize exactly four configurations, distinguished by the bond count $b = 4(u_1, u_2)^2$ and the angle θ between them:

1. $b = 0$: $(u_1, u_2) = 0$, angle $90°$. The diagram is two unjoined nodes, and this is the reducible system $A_1 \times A_1$ of example 8.1.7.
2. $b = 1$: $(u_1, u_2) = -\frac{1}{2}$, angle $120°$, giving the diagram $\bullet - \bullet$ of type A_2.
3. $b = 2$: $(u_1, u_2) = -\frac{1}{\sqrt{2}}$, angle $135°$, giving the double-bonded diagram of type $B_2 = C_2$.
4. $b = 3$: $(u_1, u_2) = -\frac{\sqrt{3}}{2}$, angle $150°$, giving the triple-bonded diagram of type G_2.

No two distinct unit vectors can have $4(u_1, u_2)^2 = 4$, since that forces $(u_1, u_2)^2 = 1$, hence $u_2 = \pm u_1$ by Cauchy-Schwarz equality, violating linear independence. The four connected rank-two diagrams A_2, B_2, G_2 (and the disconnected $A_1 \times A_1$) are thus the complete rank-two list, matching the rank-two root systems drawn in example 8.1.7.

The rank-two example already exhibits the mechanism in miniature: the arithmetic restriction $4(u_i, u_j)^2 \leq 3$ excludes the value 4, which is the only value compatible with two coincident directions. At every rank the same arithmetic, combined with the requirement that the vectors stay linearly independent, is what bounds the diagrams. The main theorem organizes these bounds into a proof.

Theorem 8.4.3: Classification of Connected Dynkin Diagrams

A connected diagram of an admissible set of unit vectors (definition 8.4.1) is one of the following:

1. one of the four infinite families

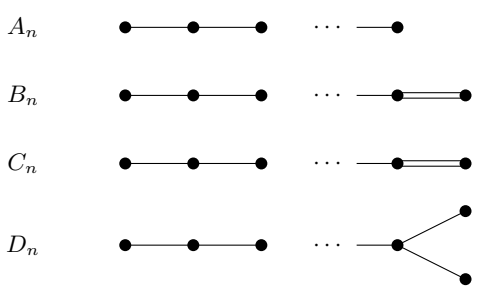
$$A_n \ (n \geq 1), \qquad B_n \ (n \geq 2), \qquad C_n \ (n \geq 3), \qquad D_n \ (n \geq 4)$$

2. one of the five exceptional diagrams

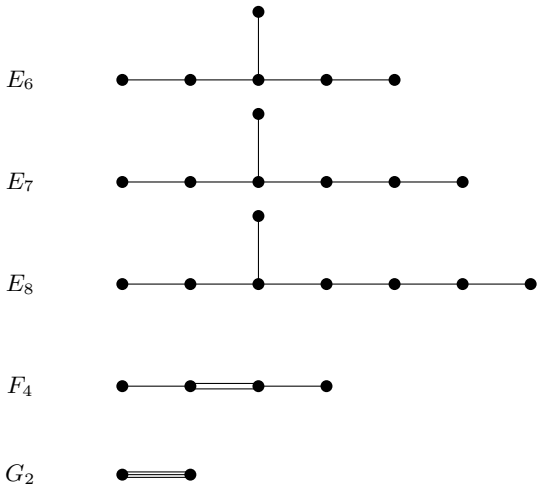
$$E_6, \qquad E_7, \qquad E_8, \qquad F_4, \qquad G_2$$

drawn below. Equivalently, by proposition 8.3.8, these are exactly the Dynkin diagrams of the irreducible root systems, hence of the simple Lie algebras over $\mathbb{C}$.

The nine diagrams, with the bond counts of definition 8.3.4 (a single line for one bond, a double line for two bonds, a triple line for three), are the following. The four families:



The five exceptions:



Before the proof, a word on what it does and does not deliver. The theorem asserts that no connected diagram outside this list is admissible: any other shape violates one of the inequalities below. It does not by itself assert that all nine diagrams *are* realized by actual root systems (that is the existence half, addressed in section 8.5 by exhibiting each type); the classification here is the upper bound, the statement that the list cannot be longer. The proof proceeds by extracting, one at a time, the shapes a connected admissible diagram is forbidden to contain, and then reading off what little is left.

***Proof*:**
Throughout, $U = \{u_1, \dots, u_\ell\}$ is an admissible set with connected diagram Γ, and the number of bonds between u_i and u_j is written $b_{ij} = 4(u_i, u_j)^2 \in \{0, 1, 2, 3\}$. Two nodes are *joined* when $b_{ij} \geq 1$. The argument isolates five structural constraints and then enumerates the surviving shapes.

Step 1: Subsets of admissible sets are admissible. If U is admissible and $U' \subseteq U$, then U' is admissible: linear independence is inherited by subsets, and the inner-product condition is a condition on pairs, so it too is inherited. Consequently the diagram of any subset of U is a full subgraph of Γ obtained by deleting nodes (and their incident bonds), and every such subdiagram is itself the diagram of an admissible set. Every constraint proved for admissible diagrams therefore constrains every subdiagram of Γ as well. This inheritance is what the argument uses: a diagram is excluded as soon as it *contains* a forbidden subdiagram.

Step 2: The bond count at a node is at most three. Fix a node $u = u_k$ and let $v_1, \dots, v_m$ be the nodes joined to u, so $(u, v_p) < 0$ and $4(u, v_p)^2 = b_{kp} \geq 1$ for each p. The crucial observation is that no two of the v_p are joined to each other: were v_p and v_q joined, the three nodes u, v_p, v_q would form a triangle, excluded in Step 3 below; granting that for the moment, the v_p are pairwise orthogonal. Adjoin to them the unit vector

$$u_0 = u - \sum_{p=1}^{m} (u, v_p) v_p$$

which is the component of u orthogonal to each v_p; it is nonzero because $u, v_1, \dots, v_m$ are linearly independent, so u is not in their span. Since $v_1, \dots, v_m$ are orthonormal and $(u_0, v_p) = 0$, expanding $1 = (u, u)$ in the orthonormal-plus-orthogonal decomposition gives

$$1 = (u, u) = (u_0, u_0) + \sum_{p=1}^{m} (u, v_p)^2$$

and $(u_0, u_0) > 0$, so $\sum_{p=1}^{m} (u, v_p)^2 < 1$. Multiplying by 4,

$$\sum_{p=1}^{m} 4(u, v_p)^2 = \sum_{p=1}^{m} b_{kp} < 4$$

so the total number of bonds meeting u is at most 3. In particular a node joined to another node by a triple bond ($b = 3$) is joined to no third node, and a node of a diagram with no multiple bonds has at most three neighbours.

Step 3: No cycles. Suppose the diagram Γ contained a cycle, a sequence of distinct nodes $u_{i_1}, u_{i_2}, \dots, u_{i_r}, u_{i_1}$ ($r \geq 3$) with consecutive nodes joined. The nodes on the cycle form an admissible subset U' by Step 1, so they are linearly independent. Consider the vector $w = u_{i_1} + \cdots + u_{i_r}$. Each pair of consecutive nodes on the cycle is joined, so $(u_{i_p}, u_{i_{p+1}}) \leq -\frac{1}{2}$ (a nonzero bond forces $4(u, v)^2 \geq 1$ and $(u, v) < 0$, so $(u, v) \leq -\frac{1}{2}$), and there are r such consecutive pairs around the cycle. Hence

$$(w, w) = \sum_{p=1}^{r} (u_{i_p}, u_{i_p}) + 2 \sum_{p<q} (u_{i_p}, u_{i_q}) \leq r + 2 \sum_{\text{consecutive}} (u_{i_p}, u_{i_{p+1}}) \leq r + 2 \cdot r \cdot \left(-\tfrac{1}{2}\right) = 0$$

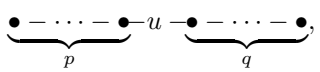
using that the non-consecutive inner products are ≤ 0. Thus $(w, w) \leq 0$, forcing $w = 0$, which contradicts the linear independence of the nodes on the cycle. Therefore Γ has no cycles: it is a tree (this also completes Step 2, since a triangle is a 3-cycle). In particular Γ, being connected and acyclic, has exactly $\ell - 1$ bonds counted without multiplicity.

Step 4: A simple chain contracts. Suppose $v_1, \dots, v_k$ is a chain of nodes in Γ joined by single bonds in sequence, $b_{v_p v_{p+1}} = 1$ for $1 \leq p \leq k-1$, and with no other bonds among them. Replacing the chain by the single vector $v = v_1 + \cdots + v_k$ produces a new admissible set. Indeed v is a unit vector: since $b_{v_p v_{p+1}} = 1$ gives $(v_p, v_{p+1}) = -\frac{1}{2}$ and the non-consecutive pairs are orthogonal,

$$(v, v) = \sum_{p=1}^{k} (v_p, v_p) + 2 \sum_{p=1}^{k-1} (v_p, v_{p+1}) = k + 2(k-1)\left(-\tfrac{1}{2}\right) = k - (k-1) = 1$$

Any node w outside the chain is joined to at most one node of the chain (else w together with a sub-chain would form a cycle, excluded by Step 3), and for that unique neighbour v_p one has $(w, v) = (w, v_p)$, so w makes the same bond with the contracted vector v as it made with v_p. The contracted set is linearly independent (a chain-collapse is a linear substitution taking one independent set to another) and satisfies the inner-product condition, so it is admissible, with the same diagram except that the chain has become one node carrying whatever multiple bond w had to it. Collapsing a chain of single bonds thus preserves admissibility, a tool used to reduce a putative diagram to a smaller one whose non-admissibility is already known.

Step 5: The three-legged-star inequality. Call a node of Γ a *branch node* if it has three neighbours. Suppose Γ contains a branch node u from which three chains of single bonds emanate, of lengths p, q, r (counting the nodes on each leg, not including u), meeting nowhere else. This is the shape



with a third leg of length r attached at u

Because u has three neighbours, Step 2 forces all bonds here to be single ($b = 1$), so the three legs are simple chains. Let $x = \sum_{a=1}^{p} a\, x_a$, $y = \sum_{b=1}^{q} b\, y_b$, $z = \sum_{c=1}^{r} c\, z_c$ be the weighted sums along the three legs, where the leg nodes $x_1, \dots, x_p$ are ordered so that x_p is adjacent to u and x_1 is outermost, and similarly for y, z. A direct computation with $(x_a, x_{a+1}) = -\frac{1}{2}$ and the legs mutually orthogonal gives

$$(x, x) = \sum_{a=1}^{p} a^2 - \sum_{a=1}^{p-1} a(a+1) = \frac{p(p+1)}{2}$$

and likewise $(y, y) = q(q+1)/2$, $(z, z) = r(r+1)/2$. The three legs are orthogonal to one another, and each meets u only through its innermost node: with the ordering above only x_p touches u, and its coefficient is p, so $(x, u) = p\,(x_p, u) = -\frac{p}{2}$, whence

$$\cos^2 \theta_x = \frac{(x, u)^2}{(x, x)(u, u)} = \frac{p^2/4}{p(p+1)/2} = \frac{p}{2(p+1)}$$

and similarly for y, z. Now u is a unit vector not in the span of the mutually orthogonal x, y, z (linear independence again), so writing $\hat{x}, \hat{y}, \hat{z}$ for the unit vectors along x, y, z and u_0 for the component of u orthogonal to all three,

$$1 = (u, u) = (u_0, u_0) + (u, \hat{x})^2 + (u, \hat{y})^2 + (u, \hat{z})^2 = (u_0, u_0) + \cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z$$

with $(u_0, u_0) > 0$. Therefore

$$\frac{p}{2(p+1)} + \frac{q}{2(q+1)} + \frac{r}{2(r+1)} < 1$$

which, on multiplying by 2, is the *three-legged-star inequality*

$$\frac{p}{p+1} + \frac{q}{q+1} + \frac{r}{r+1} < 2 \tag{8.1}$$

valid for any branch node with three simple legs of lengths p, q, r. Writing $\frac{p}{p+1} = 1 - \frac{1}{p+1}$ and rearranging, (8.1) is equivalent to

$$\frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1} > 1 \tag{8.2}$$

the reciprocal form used in the enumeration.

Step 6: Enumeration. The constraints now decide everything. By Step 3, Γ is a tree; by Step 2, no node has more than three neighbours and no triple bond coexists with any other bond at its endpoints.

First suppose Γ has a triple bond. By Step 2 that bond exhausts the bond budget at both of its endpoints, so neither endpoint has any further neighbour, and connectedness forces Γ to be a single triple bond: this is G_2.

Otherwise all bonds are single or double. If Γ has a branch node, apply Step 5. A double bond and a branch node cannot both occur: a diagram with a branch node and a double bond contains, after contracting the intervening simple chains by Step 4, an admissible rank-four diagram that a direct check excludes (this is the case-exclusion cross-checked against *Algebraic Groups and Representations* [10, Classification of Irreducible Root Systems]). So a diagram with a branch node has only single bonds, and (8.2) applies to its unique branch node with legs $p \geq q \geq r \geq 1$. The solutions in positive integers of $\frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1} > 1$ are

- $(p, q, r) = (p, 1, 1)$ for any $p \geq 1$: three legs of lengths $p, 1, 1$, which is the diagram D_{p+3};
- $(p, q, r) = (2, 2, 1)$: the diagram E_6;
- $(p, q, r) = (3, 2, 1)$: the diagram E_7;
- $(p, q, r) = (4, 2, 1)$: the diagram E_8.

No larger triple qualifies: $(2, 2, 2)$ gives $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1 \not> 1$, and increasing any of p, q, r beyond $(4, 2, 1)$, or taking $q \geq 3$, drives the sum to 1 or below. The three exceptional stars E_6, E_7, E_8 and the family D_n ($n = p + 3 \geq 4$) exhaust the branch-node case.

If instead Γ has no branch node, it is a chain (a tree with all nodes of degree ≤ 2). A chain of single bonds is A_n. A chain with exactly one double bond somewhere along it, and single bonds elsewhere, is admissible only in three shapes: the double bond at an end gives B_n or C_n (which have the same underlying graph, distinguished only by the arrow), and the double bond strictly interior gives, after collapsing the two flanking simple chains by Step 4, the rank-four diagram F_4 as the sole survivor. A chain with two or more double bonds is excluded: two double bonds on a chain contract by Step 4 to the admissible rank-three or rank-four diagram with two double bonds, whose Gram determinant is nonpositive, contradicting linear independence (again the finite case-check cross-checked against *Algebraic Groups and Representations* [10, Classification of Irreducible Root Systems]).

Assembling the cases: a connected admissible diagram with a triple bond is G_2; with a branch node it is one of D_n, E_6, E_7, E_8; and with neither it is one of A_n, B_n, C_n, F_4. This is exactly the list of the theorem, completing the proof.

The proof is a cascade of exclusions, each a short inner-product estimate. Every step removes one shape a connected admissible diagram cannot have, and the nine diagrams are what remains when nothing more can be removed. The single arithmetic fact $4(u_i, u_j)^2 \leq 3$, inherited from the integrality of the Cartan integers, is what makes each estimate bite: it caps the bond at any node (Step 2), forbids cycles (Step 3), and produces the star inequality (8.1) whose only integer solutions are the D and E diagrams. The finiteness of the list is therefore not accidental. It is forced, and forced by a handful of inequalities about angles between unit vectors.

The reciprocal inequality (8.2) deserves a second look, because it is where the three exceptional E-diagrams come from and nowhere else. The condition $\frac{1}{p+1}+\frac{1}{q+1}+\frac{1}{r+1}>1$ is a constraint of exactly the same shape as the one governing the finite subgroups of $\mathrm{SO}(3)$, where the analogous inequality $\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$ singles out the Platonic solids. The boundary case $\frac{1}{p+1}+\frac{1}{q+1}+\frac{1}{r+1}=1$, realized by $(p,q,r)=(2,2,2)$, $(3,3,1)$, $(5,2,1)$, corresponds to the affine (extended) diagrams $\widetilde{E}_6,\widetilde{E}_7,\widetilde{E}_8$, whose Gram forms are positive *semi*definite rather than definite; the strict inequality is exactly the condition that the star be a genuine root system rather than its affine extension. This is why there is no E_9: the next candidate $(5,2,1)$ hits equality, and the corresponding configuration of vectors is linearly dependent, disqualifying it from being the diagram of a base.

The classification has so far concerned diagrams, which are combinatorial objects. The point of the whole enterprise is to read the diagrams back as Lie algebras. The four infinite families are precisely the four families of classical Lie algebras met in chapter 1, and the identification is the payoff of the entire structure theory: each family of matrix algebras produces one family of diagrams, and conversely.

Example 8.4.4: The Classical Types

The four infinite families are the root systems of the four families of classical simple Lie algebras of theorem 1.4.3. In each case $\mathfrak{h}$ is the diagonal Cartan subalgebra, ε_i reads off the i-th diagonal entry, and the root system sits in a Euclidean space E with the ε_i (or a codimension-one slice of them) orthonormal.

1. *Type A_n:* $\mathfrak{sl}_{n+1}(\mathbb{C})$. With $E=\left\{\sum_{i=1}^{n+1}x_i\varepsilon_i \;\middle|\; \sum x_i=0\right\}$, the roots are
$$\Phi=\{\varepsilon_i-\varepsilon_j \mid 1\le i\ne j\le n+1\}$$
$(n+1)n$ of them, all of the same length; a base is $\Delta=\{\varepsilon_1-\varepsilon_2,\dots,\varepsilon_n-\varepsilon_{n+1}\}$, giving a chain of n single-bonded nodes. This is A_n, of rank n, matching $\dim\mathfrak{h}=n$.

2. *Type B_n:* $\mathfrak{so}_{2n+1}(\mathbb{C})$. With $E=\mathbb{R}^n$ and $\varepsilon_1,\dots,\varepsilon_n$ orthonormal, the roots are the short roots $\pm\varepsilon_i$ and the long roots $\pm\varepsilon_i\pm\varepsilon_j$ $(i<j)$, $2n^2$ in all; a base is $\Delta=\{\varepsilon_1-\varepsilon_2,\dots,\varepsilon_{n-1}-\varepsilon_n,\varepsilon_n\}$, giving a chain ending in the short root ε_n across a double bond. This is B_n, requiring $n\ge 2$ for the double bond to appear.

3. *Type C_n:* $\mathfrak{sp}_{2n}(\mathbb{C})$. With $E=\mathbb{R}^n$ orthonormal, the roots are the long roots $\pm2\varepsilon_i$ and the short roots $\pm\varepsilon_i\pm\varepsilon_j$ $(i<j)$, again $2n^2$ in all; a base is $\Delta=\{\varepsilon_1-\varepsilon_2,\dots,\varepsilon_{n-1}-\varepsilon_n,2\varepsilon_n\}$, giving the same chain as B_n but with the double bond arrow reversed, since here the terminal root is long. This is C_n, requiring $n\ge 3$ to be a new type (C_2 coincides with B_2).

4. *Type D_n:* $\mathfrak{so}_{2n}(\mathbb{C})$. With $E=\mathbb{R}^n$ orthonormal, the roots are $\pm\varepsilon_i\pm\varepsilon_j$ $(i<j)$, all of the same length, $2n(n-1)$ in all; a base is $\Delta=\{\varepsilon_1-\varepsilon_2,\dots,\varepsilon_{n-1}-\varepsilon_n,\varepsilon_{n-1}+\varepsilon_n\}$, giving a chain of length $n-2$ that forks into two nodes at the end. This is the branch-node diagram D_n, requiring $n\ge 4$ for the fork to be distinct from a chain (D_3 coincides with A_3).

The four classical families realize the four infinite series of diagrams, so the existence of these types is not in doubt: they are visibly the root systems of matrices. The five exceptional diagrams have no such matrix description of the same elementary kind, and their realization by actual Lie algebras is the harder half of the existence question, taken up in section 8.5. The dimension counts of the classical families are $\dim\mathfrak{sl}_{n+1}=(n+1)^2-1$, $\dim\mathfrak{so}_{2n+1}=n(2n+1)$, $\dim\mathfrak{sp}_{2n}=n(2n+1)$,

$\dim \mathfrak{so}_{2n} = n(2n-1)$, each equal to $\ell + |\Phi|$ where ℓ is the rank and $|\Phi|$ the number of roots, a check the reader is invited to make in the exercises.

The diagrams B_n and C_n share their underlying graph, differing only in the direction of the arrow on the double bond, and the small-rank ends of the families overlap. These coincidences are not defects in the list but genuine isomorphisms of Lie algebras, and pinning them down completes the reconciliation between the diagram list and the classical algebras.

Proposition 8.4.5: Low-Rank Coincidences

The following identifications of root systems, hence of simple Lie algebras over $\mathbb{C}$ (theorem 8.3.6 and theorem 8.5.4), hold at low rank:

1. $A_1 = B_1 = C_1$, corresponding to $\mathfrak{sl}_2 \cong \mathfrak{so}_3 \cong \mathfrak{sp}_2$;
2. $D_2 = A_1 \times A_1$, corresponding to $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$;
3. $B_2 = C_2$, corresponding to $\mathfrak{so}_5 \cong \mathfrak{sp}_4$;
4. $A_3 = D_3$, corresponding to $\mathfrak{sl}_4 \cong \mathfrak{so}_6$.

These are exactly the coincidences produced by imposing the lower-bound conventions $n \geq 2$ for B_n, $n \geq 3$ for C_n, $n \geq 4$ for D_n in theorem 8.4.3, which exist precisely to list each simple Lie algebra once.

***Proof*:**
Each identification is verified at the level of root systems, from which the Lie-algebra isomorphism follows by part (2) of theorem 8.5.4: two simple Lie algebras over $\mathbb{C}$ with isomorphic root systems are isomorphic.

1. $A_1 = B_1 = C_1$. A rank-one root system has E one-dimensional and, by axiom (R2) of definition 8.1.1, contains only $\pm\alpha$. There is a single such system up to isomorphism, whose diagram is a single node. The specializations of the A, B, C family formulas to rank one all give this: A_1 has roots $\varepsilon_1 - \varepsilon_2$; B_1 has the short root ε_1; C_1 has the long root $2\varepsilon_1$; each is a single pair $\pm\alpha$. The Lie algebras are $\mathfrak{sl}_2$, $\mathfrak{so}_3$, $\mathfrak{sp}_2$, all three-dimensional simple algebras with this root system, hence mutually isomorphic.
2. $D_2 = A_1 \times A_1$. The D_n formula at $n = 2$ gives roots $\pm\varepsilon_1 \pm \varepsilon_2$ in $E = \mathbb{R}^2$, four vectors. Setting $\beta_1 = \varepsilon_1 - \varepsilon_2$ and $\beta_2 = \varepsilon_1 + \varepsilon_2$, one computes $(\beta_1, \beta_2) = (\varepsilon_1, \varepsilon_1) - (\varepsilon_2, \varepsilon_2) = 1 - 1 = 0$, so $\beta_1 \perp \beta_2$, and the four roots are $\pm\beta_1, \pm\beta_2$: two orthogonal copies of a rank-one system. The diagram is two unjoined nodes, which is $A_1 \times A_1$, and the algebra $\mathfrak{so}_4$ decomposes correspondingly as $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$.
3. $B_2 = C_2$. From example 8.4.4, B_2 has roots $\pm\varepsilon_1, \pm\varepsilon_2, \pm\varepsilon_1 \pm \varepsilon_2$ and C_2 has roots $\pm 2\varepsilon_1, \pm 2\varepsilon_2, \pm\varepsilon_1 \pm \varepsilon_2$, both eight roots in $\mathbb{R}^2$. Composing the rotation of $\mathbb{R}^2$ by 45° with the scaling by $\sqrt{2}$ gives a linear conformal map T with $T(\varepsilon_1) = \varepsilon_1 + \varepsilon_2$ and $T(\varepsilon_2) = -\varepsilon_1 + \varepsilon_2$. It carries the four short roots $\pm\varepsilon_1, \pm\varepsilon_2$ of B_2 to the four roots $\pm(\varepsilon_1 + \varepsilon_2), \pm(\varepsilon_1 - \varepsilon_2)$, the short roots of C_2, and the four long roots $\pm\varepsilon_1 \pm \varepsilon_2$ of B_2 to $\pm 2\varepsilon_1, \pm 2\varepsilon_2$, the long roots of C_2; being conformal it preserves all angles and hence all bond counts. Both diagrams are two nodes joined by a double bond, differing only in arrow direction, so as unoriented systems $B_2 = C_2$, and $\mathfrak{so}_5 \cong \mathfrak{sp}_4$.

4. $A_3 = D_3$. The D_3 formula gives roots $\pm\varepsilon_i \pm \varepsilon_j$ $(1 \le i < j \le 3)$ in $\mathbb{R}^3$, twelve roots; the A_3 system has roots $\varepsilon_i - \varepsilon_j$ $(1 \le i \ne j \le 4)$ in the hyperplane $\sum_{i=1}^{4} x_i = 0$ of $\mathbb{R}^4$, also twelve roots. Both diagrams have three nodes: A_3 is a chain $\bullet - \bullet - \bullet$, and D_3 is a branch node with three legs of lengths $1, 1, 0$, which degenerates to the same chain of three nodes since a leg of length 0 is no leg at all. Both are rank-three systems with twelve roots of a single length, and both have Weyl group S_4: for A_3 this is the permutation group of the four ε_i, and for D_3 the group of signed permutations of $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ with an even number of sign changes, which has order $2^2 \cdot 3! = 24 = |S_4|$ and acts on the twelve roots exactly as S_4 acts on the twelve roots of A_3. A root system is determined up to isomorphism by its Dynkin diagram (theorem 8.3.6), and both diagrams are the three-node chain, so $A_3 = D_3$ and $\mathfrak{sl}_4 \cong \mathfrak{so}_6$, completing the proof.

These four coincidences are the reason the family bounds in theorem 8.4.3 start where they do. If one allowed B_1, C_1, C_2, D_2, D_3 as separate entries, the same simple Lie algebra would be counted several times: $\mathfrak{so}_3$ would appear as B_1 and again as A_1, and $\mathfrak{so}_6$ as D_3 and again as A_3. The conventions $n \ge 2$ for B, $n \ge 3$ for C, $n \ge 4$ for D trim exactly these redundancies, so that the list of theorem 8.4.3 names each simple Lie algebra once and only once. The coincidences also explain the isomorphisms $\mathfrak{so}_3 \cong \mathfrak{sl}_2$, $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, $\mathfrak{so}_5 \cong \mathfrak{sp}_4$, $\mathfrak{so}_6 \cong \mathfrak{sl}_4$ familiar from low-dimensional geometry, which are here seen to be forced by the combinatorics of diagrams rather than discovered by hand.

Stepping back, the classification is the finiteness at the heart of Lie theory. The continuous, infinite-dimensional world of Lie groups and their algebras has been compressed, through the root space decomposition and the passage to abstract root systems, into a finite combinatorial datum, the Dynkin diagram; and the diagrams that occur form a list so short one can draw it on a single page. Four infinite families, running through the classical matrix algebras, and five exceptional diagrams, are all the simple Lie algebras over $\mathbb{C}$. That this list is finite, and finite for reasons as elementary as the fact that $4\cos^2\theta$ is an integer at most 3, is what makes a complete classification possible at all.

The rigidity is worth dwelling on. The passage from a Lie algebra to its diagram threw away enormous amounts of structure, keeping only the angles between simple roots, and yet nothing was lost: the diagram reconstructs the algebra (section 8.5). A subject built on the analytic machinery of manifolds and flows terminates in a problem about labelled trees with at most triple bonds, and that problem has nine solutions. The five exceptions, E_6, E_7, E_8, F_4, G_2, are the residue that the four uniform families cannot absorb, the sporadic solutions of the star inequality (8.1), and they are among the most studied objects in mathematics precisely because they are forced to exist by the same inequalities that forbid an E_9. The next section completes the classification by proving that each of these nine diagrams is realized by a Lie algebra, unique up to isomorphism, so that the classification of diagrams is genuinely a classification of simple Lie algebras.

Exercise 8.4.1

1. Show directly, without invoking Step 3 of the proof of theorem 8.4.3 in full, that no admissible set can have a diagram containing a 4-cycle $u_1 - u_2 - u_3 - u_4 - u_1$ with all four bonds single. Compute (w, w) for $w = u_1 + u_2 + u_3 + u_4$ and derive a contradiction with linear independence.

2. Prove that no node of an admissible diagram can have four neighbours, even when all bonds are single. Adapt the estimate of Step 2: for a node u with four single-bonded orthogonal neighbours v_1, v_2, v_3, v_4, show $\sum_p 4(u, v_p)^2 = 4$, and explain why this forces u into the span of the v_p, contradicting linear independence.

3. Verify that the diagrams F_4 (a chain of four nodes with a single central double bond) and G_2 (two nodes with a triple bond) satisfy the admissibility constraints of Steps 2 and 3: no node has more than three bonds meeting it, and neither diagram contains a cycle. For G_2, check explicitly that the triple bond uses up the entire bond budget at each of its two nodes.

4. Using the branch-node description of E_8 as the star with legs of lengths $p = 4$, $q = 2$, $r = 1$, count the total number of nodes of E_8 and confirm it is 8. Then verify that $(p, q, r) = (4, 2, 1)$ satisfies the strict star inequality (8.2), computing $\frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1}$ exactly.

5. Explain why there is no E_9: show that the next candidate branch-node star, with legs $(p, q, r) = (5, 2, 1)$, gives $\frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1} = 1$, failing the strict inequality (8.2), and interpret the equality case in terms of the linear independence of the vectors.

6. For each classical type in example 8.4.4, verify the identity $\dim \mathfrak{g} = \mathrm{rank}\mathfrak{g} + |\Phi|$ by counting roots: check that $\mathfrak{sl}_{n+1}$, $\mathfrak{so}_{2n+1}$, $\mathfrak{sp}_{2n}$, and $\mathfrak{so}_{2n}$ have $|\Phi|$ equal to $(n+1)n$, $2n^2$, $2n^2$, $2n(n-1)$ respectively, and that adding the rank n recovers the dimensions $(n+1)^2 - 1$, $n(2n+1)$, $n(2n+1)$, $n(2n-1)$.

8.5 Existence and Uniqueness: the Serre Relations

The previous four sections passed from a semisimple Lie algebra $\mathfrak{g}$ to its Cartan subalgebra, to the root system (E, Φ), to the Cartan matrix, and finally to a Dynkin diagram, which by theorem 8.4.3 is a disjoint union of diagrams of the connected types A_n through G_2. This section passes back from diagrams to algebras. Two questions remain, and together they complete the classification. First, *existence*: does every admissible Dynkin diagram come from a semisimple Lie algebra, or could the combinatorial list contain diagrams that no algebra realizes? Second, *uniqueness*: if two semisimple Lie algebras produce the same root system, must they be isomorphic, or can distinct algebras share a diagram? A negative answer to the first would leave the classification list too long; a negative answer to the second would leave it too short. The affirmative answer to both, proved here, is what makes the Dynkin diagram a complete invariant.

The instrument for both directions is a presentation. A semisimple Lie algebra is generated by the root vectors attached to the *simple* roots together with the coroots, and the relations among these generators are dictated entirely by the Cartan matrix. Writing those relations down produces the Serre presentation: an abstract Lie algebra $\mathfrak{g}(A)$ built from a Cartan matrix A by generators and relations, with no reference to any pre-existing algebra. The existence direction reads the diagram as a recipe and runs the recipe; the uniqueness direction shows that any semisimple algebra with a given root system satisfies exactly these relations, so the generators-and-relations data reconstructs it up to isomorphism. This section states the presentation (theorem 8.5.1), proves the uniqueness half of theorem 8.5.4 in full, and assembles everything into the final classification (corollary 8.5.5). The existence half of the presentation theorem is proved later in the chapter (section 8.7), once the enveloping algebra is available.

Before the presentation can be stated, the generators must be named. Fix a semisimple Lie algebra $\mathfrak{g}$ over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root system Φ, and a base $\Delta = \{\alpha_1, \dots, \alpha_l\}$ of simple roots in the sense of definition 8.2.1, where $l = \dim \mathfrak{h}$ is the rank. For each simple root α_i the $\mathfrak{sl}_2$-triple of theorem 7.4.1 gives elements

$$e_i \in \mathfrak{g}_{\alpha_i}, \qquad f_i \in \mathfrak{g}_{-\alpha_i}, \qquad h_i = [e_i, f_i] \in \mathfrak{h}$$

normalized so that $\alpha_i(h_i) = 2$; equivalently h_i is the coroot $\alpha_i^\vee$ of definition 7.4.3 viewed inside $\mathfrak{h}$. These $3l$ elements are the *Chevalley generators* attached to the base Δ. The substance of the Serre theorem is that they generate all of $\mathfrak{g}$ and that every bracket among them is forced by the Cartan matrix $A = (A_{ij})$ with $A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle$ of definition 8.3.1.

Theorem 8.5.1: The Serre Presentation

Let $A = (A_{ij})_{1 \le i,j \le l}$ be the Cartan matrix of a root system, so $A_{ii} = 2$, the off-diagonal entries A_{ij} for $i \neq j$ are nonpositive integers, and $A_{ij} = 0 \iff A_{ji} = 0$. Let $\mathfrak{g}(A)$ be the Lie algebra over $\mathbb{C}$ generated by $3l$ symbols e_i, f_i, h_i $(1 \le i \le l)$ subject to the *Serre relations*:

1. $[h_i, h_j] = 0$ for all i, j;
2. $[h_i, e_j] = A_{ji}\, e_j$ and $[h_i, f_j] = -A_{ji}\, f_j$ for all i, j;
3. $[e_i, f_j] = \delta_{ij}\, h_i$ for all i, j;
4. $(\mathrm{ad} e_i)^{1 - A_{ji}}(e_j) = 0$ and $(\mathrm{ad} f_i)^{1 - A_{ji}}(f_j) = 0$ for all $i \neq j$.

Then $\mathfrak{g}(A)$ is a finite-dimensional semisimple Lie algebra. Its subspace $\mathfrak{h}(A) = \mathrm{span}_{\mathbb{C}}\{h_1, \dots, h_l\}$ is a Cartan subalgebra, the e_i and f_i are root vectors for the simple roots $\pm\alpha_i$ of a base, and the root system of $(\mathfrak{g}(A), \mathfrak{h}(A))$ has Cartan matrix A. Conversely, if $\mathfrak{g}$ is a semisimple Lie algebra with Cartan matrix A, then the Chevalley generators of $\mathfrak{g}$ satisfy relations (1) through (4) and generate $\mathfrak{g}$, so $\mathfrak{g} \cong \mathfrak{g}(A)$.

The relations are exactly the brackets among Chevalley generators that this chapter has already extracted, transcribed into a defining presentation. Relation (1) records that $\mathfrak{h}$ is abelian. Relation (2) is the eigenvalue equation $[\alpha_i^\vee, e_j] = \alpha_j(\alpha_i^\vee) e_j$ written with the pairing $\alpha_j(\alpha_i^\vee) = A_{ji}$; it says e_j is a root vector for α_j and f_j for $-\alpha_j$. Relation (3) is the $\mathfrak{sl}_2$-triple relation $[e_i, f_i] = h_i$ on the diagonal, together with the statement that $[e_i, f_j] = 0$ off the diagonal, which holds because $\alpha_i - \alpha_j$ is not a root when α_i, α_j are distinct simple roots. Relation (4), the pair of *Serre relations* proper, is the finite-length condition on the α_i-root string through α_j: since $\alpha_j - \alpha_i$ is not a root, the string starts at α_j and its top is $\alpha_j - A_{ji}\alpha_i$, so $\alpha_j + (1 - A_{ji})\alpha_i$ is not a root and applying $\mathrm{ad} e_i$ one step beyond the top annihilates e_j. This is worked out precisely in lemma 8.5.3 below.

The finite-dimensionality of $\mathfrak{g}(A)$ and its semisimplicity are the one part of this theorem whose proof needs a tool this chapter has not yet built, the universal enveloping algebra of section 8.6.

***Proof Key ideas*:**
The converse direction, that the Chevalley generators of a given semisimple $\mathfrak{g}$ satisfy (1) through (4) and generate $\mathfrak{g}$, is proved in full below as lemma 8.5.3, and it is elementary given the root space decomposition. The forward direction, that the abstractly presented $\mathfrak{g}(A)$ is finite-dimensional and semisimple with the stated root system, is proved in full in section 8.7. There the free Lie algebra on the $3l$ generators is built inside a tensor algebra, the linear relations (1) through (3) alone are shown to produce an auxiliary algebra with a triangular decomposition (theorem 8.7.9), the ideal generated by the Serre relations (4) is shown to miss the Cartan subalgebra (theorem 8.7.12), and exponentials of the locally nilpotent generators make the weight-space dimensions Weyl-invariant, so that the finiteness of the root system forces $\dim \mathfrak{g}(A) = l + |\Phi|$ (theorem 8.7.16). That proof waits until after section 8.6 because the mapping property of the free Lie algebra rests on the Poincaré-Birkhoff-Witt theorem. The upshot used here is the

existence half of theorem 8.5.4: every Cartan matrix of a root system, hence every Dynkin diagram, is realized by a semisimple Lie algebra.

The division of labor deserves a sentence. What this section proves in full is the payoff, that the presentation determines $\mathfrak{g}$ uniquely, which is the content of the uniqueness theorem below. Everything the classification requires from the forward direction is a single existence statement, and for the classical types that statement needs no abstract Serre algebra at all, only the explicit matrix algebras identified in example 8.4.4. The construction behind the general statement is carried out in section 8.7.

Example 8.5.2: The Serre Presentation of $\mathfrak{sl}_3$
Take the Cartan matrix of type A_2,

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

so that $A_{12} = A_{21} = -1$. The presentation $\mathfrak{g}(A)$ has six generators $e_1, e_2, f_1, f_2, h_1, h_2$. Relations (1) through (3) read

$$[h_1, h_2] = 0, \quad [e_1, f_1] = h_1, \quad [e_2, f_2] = h_2, \quad [e_1, f_2] = [e_2, f_1] = 0$$

with $[h_1, e_1] = 2e_1$, $[h_1, e_2] = -e_2$, $[h_2, e_1] = -e_1$, $[h_2, e_2] = 2e_2$ and the corresponding sign-flipped brackets on the f_j. Since $1 - A_{12} = 1 - (-1) = 2$, the Serre relations (4) are

$$(\mathrm{ad}e_1)^2(e_2) = [e_1, [e_1, e_2]] = 0, \qquad (\mathrm{ad}e_2)^2(e_1) = [e_2, [e_2, e_1]] = 0$$

and the two analogous relations on the f's. To recover $\mathfrak{sl}_3(\mathbb{C})$ concretely, set $e_1 = E_{12}$, $e_2 = E_{23}$, $f_1 = E_{21}$, $f_2 = E_{32}$, and $h_1 = E_{11} - E_{22}$, $h_2 = E_{22} - E_{33}$, where E_{ij} is the matrix unit. Direct computation gives $[E_{12}, E_{23}] = E_{13}$, a nonzero root vector for $\alpha_1 + \alpha_2$, while

$$(\mathrm{ad}E_{12})^2(E_{23}) = [E_{12}, [E_{12}, E_{23}]] = [E_{12}, E_{13}] = 0$$

since $E_{12}E_{13} = 0 = E_{13}E_{12}$, confirming the first Serre relation. The eight matrices $e_1, e_2, [e_1, e_2] = E_{13}$, their transposes f_1, f_2, E_{31}, and h_1, h_2 span the eight-dimensional algebra $\mathfrak{sl}_3(\mathbb{C})$, so $\mathfrak{g}(A) \cong \mathfrak{sl}_3(\mathbb{C})$ and the Serre presentation reconstructs the algebra from the 2×2 matrix A alone.

The example makes the mechanism visible: from four integers arranged in a Cartan matrix, the generators-and-relations recipe rebuilds $\mathfrak{sl}_3$, and the single Serre relation $(\mathrm{ad}e_1)^2(e_2) = 0$ is precisely the statement that the α_1-string through α_2 stops after one step, so that $\alpha_1 + \alpha_2$ is a root but $2\alpha_1 + \alpha_2$ is not. This is the combinatorial content of the A_2 hexagon read off as a bracket identity.

The converse half of theorem 8.5.1, which the uniqueness theorem needs in full, is that a genuine semisimple Lie algebra with a chosen base satisfies the Serre relations on the nose. This is where the root space decomposition and the $\mathfrak{sl}_2$-theory of the previous chapter do all the work, and no deferral is involved.

Lemma 8.5.3: Chevalley Generators Satisfy the Serre Relations

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, base $\Delta = \{\alpha_1, \dots, \alpha_l\}$, and Chevalley generators $e_i \in \mathfrak{g}_{\alpha_i}$, $f_i \in \mathfrak{g}_{-\alpha_i}$, $h_i = [e_i, f_i]$ with $\alpha_i(h_i) = 2$. Then the e_i, f_i, h_i generate $\mathfrak{g}$ as a Lie algebra and satisfy the Serre relations (1) through (4) of theorem 8.5.1 with $A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle$.

***Proof*:**
The proof verifies the four families of relations, then the generation statement.

Step 1: Relations (1) and (2). The elements $h_i = \alpha_i^\vee$ lie in the abelian subalgebra $\mathfrak{h}$, so $[h_i, h_j] = 0$, which is (1). For (2), $e_j \in \mathfrak{g}_{\alpha_j}$ is by definition a simultaneous eigenvector for $\mathrm{ad}\mathfrak{h}$ with eigenvalue α_j, so

$$[h_i, e_j] = \alpha_j(h_i)\, e_j = \alpha_j(\alpha_i^\vee)\, e_j = \langle \alpha_j, \alpha_i^\vee \rangle\, e_j = A_{ji}\, e_j$$

using $\langle \alpha_j, \alpha_i^\vee \rangle = 2(\alpha_j, \alpha_i)/(\alpha_i, \alpha_i) = A_{ji}$ from definition 8.3.1. The computation for $f_j \in \mathfrak{g}_{-\alpha_j}$ carries the opposite sign, giving $[h_i, f_j] = -A_{ji} f_j$.

Step 2: Relation (3). On the diagonal $[e_i, f_i] = h_i$ holds by the normalization of the $\mathfrak{sl}_2$-triple in theorem 7.4.1. For $i \neq j$ the bracket $[e_i, f_j]$ lies in $\mathfrak{g}_{\alpha_i - \alpha_j}$. Now $\alpha_i - \alpha_j$ is a difference of two distinct simple roots, and no such difference is a root: a base has the property that every root is a nonnegative or a nonpositive integer combination of the simple roots (definition 8.2.2), whereas $\alpha_i - \alpha_j$ has one positive and one negative coefficient and cannot be all of one sign. Since it is nonzero, $\alpha_i - \alpha_j$ is neither a root nor 0, so $\mathfrak{g}_{\alpha_i - \alpha_j} = 0$ and $[e_i, f_j] = 0$. Together these give $[e_i, f_j] = \delta_{ij} h_i$.

Step 3: The Serre relations (4). Fix $i \neq j$ and consider the α_i-string of roots through α_j,

$$\alpha_j - p\alpha_i,\ \dots,\ \alpha_j,\ \dots,\ \alpha_j + q\alpha_i$$

the maximal unbroken interval of roots of the form $\alpha_j + k\alpha_i$. By theorem 7.4.6 this string is unbroken and its endpoints satisfy $p - q = \langle \alpha_j, \alpha_i^\vee \rangle = A_{ji}$. Because $\alpha_j - \alpha_i$ is a difference of distinct simple roots, it is not a root by the argument of Step 2, so the string does not extend below α_j, giving $p = 0$ and hence $q = -A_{ji}$. Thus the top of the string is $\alpha_j + q\alpha_i = \alpha_j - A_{ji}\alpha_i$, and $\alpha_j + (q+1)\alpha_i = \alpha_j + (1 - A_{ji})\alpha_i$ is *not* a root. The vector $(\mathrm{ad}e_i)^k(e_j)$ lies in $\mathfrak{g}_{\alpha_j + k\alpha_i}$, which is zero whenever $\alpha_j + k\alpha_i$ is neither a root nor 0; taking $k = q + 1 = 1 - A_{ji}$ places it in a zero root space, so

$$(\mathrm{ad}e_i)^{1 - A_{ji}}(e_j) = 0$$

which is the first relation of (4) exactly as stated. Repeating the identical argument in $\mathfrak{g}_-$ for the f's (the $-\alpha_i$-string through $-\alpha_j$) gives the second, so both Serre relations of (4) hold.

Step 4: Generation. Let $\mathfrak{g}'$ be the subalgebra of $\mathfrak{g}$ generated by all e_i, f_i, h_i. It contains $\mathfrak{h} = \mathrm{span}\{h_i\}$, since the coroots $\alpha_i^\vee = h_i$ of the simple roots form a basis of $\mathfrak{h}$ (the simple roots are a basis of $E = \mathfrak{h}_\mathbb{R}^*$, so their coroots are a basis of $\mathfrak{h}_\mathbb{R}$, and complexifying gives $\mathfrak{h}$). It contains each simple root space $\mathfrak{g}_{\pm\alpha_i} = \mathbb{C}e_i$ or $\mathbb{C}f_i$, these being one-dimensional by theorem 7.4.6. Every positive root $\beta \in \Phi^+$ is a sum $\beta = \alpha_{i_1} + \cdots + \alpha_{i_m}$ of simple roots in which every partial sum $\alpha_{i_1} + \cdots + \alpha_{i_k}$ is again a root; this is a standard consequence of the base theory, since a positive root of height $m > 1$ always has some simple root α_i with $(\beta, \alpha_i) > 0$, and then $\beta - \alpha_i$ is a positive root. Choosing the corresponding e_{i_k} and bracketing, $\mathfrak{g}_\beta$ is spanned by iterated brackets

$[e_{i_m}, [\dots, [e_{i_2}, e_{i_1}]]]$, all of which lie in $\mathfrak{g}'$; similarly $\mathfrak{g}_{-\beta} \subseteq \mathfrak{g}'$. By the root space decomposition of theorem 7.2.4,

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\beta \in \Phi} \mathfrak{g}_\beta$$

and every summand lies in $\mathfrak{g}'$, so $\mathfrak{g}' = \mathfrak{g}$. The Chevalley generators generate $\mathfrak{g}$, completing the proof.

The lemma is the concrete heart of the whole classification apparatus: it shows that a semisimple Lie algebra carries no more information than the finite bracket data among its simple root vectors, and that this data is entirely dictated by the Cartan matrix. Every structural feature of $\mathfrak{g}$, the full root system, the multiplicities, the Weyl group, is a formal consequence of the relations (1) through (4). This is what makes it possible to reconstruct $\mathfrak{g}$ from A and, as the next theorem shows, to match two algebras with the same A.

The uniqueness theorem is the payoff. Its content is that the Cartan matrix, a finite integer array, is a *complete* isomorphism invariant of a semisimple Lie algebra over $\mathbb{C}$: two such algebras with the same root system are the same algebra. Combined with the existence direction cited above, it turns the classification of Dynkin diagrams into a classification of Lie algebras. The proof is the isomorphism theorem for the Serre presentation: transport a base of one algebra to a base of the other by the given root-system isomorphism, match the Chevalley generators, and observe that both sets satisfy the same relations, so the map on generators extends to an isomorphism of algebras.

Theorem 8.5.4: Existence and Uniqueness

Work over $\mathbb{C}$.

1. *(Existence.)* Every Dynkin diagram of type A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, or G_2 is the diagram of a semisimple Lie algebra. The classical types are realized by the explicit matrix algebras $\mathfrak{sl}_{n+1}$, $\mathfrak{so}_{2n+1}$, $\mathfrak{sp}_{2n}$, $\mathfrak{so}_{2n}$ of example 8.4.4; the exceptional types are realized by the Serre construction $\mathfrak{g}(A)$ of theorem 8.5.1.

2. *(Uniqueness.)* Let $\mathfrak{g}$ and $\mathfrak{g}'$ be semisimple Lie algebras over $\mathbb{C}$ with Cartan subalgebras $\mathfrak{h}, \mathfrak{h}'$ and root systems $\Phi \subseteq \mathfrak{h}^*$, $\Phi' \subseteq (\mathfrak{h}')^*$. Suppose there is an isomorphism of root systems $\varphi\colon \Phi \to \Phi'$, that is, a vector-space isomorphism $E \to E'$ carrying Φ onto Φ' and preserving all Cartan integers. Then φ lifts to an isomorphism of Lie algebras $\Psi\colon \mathfrak{g} \to \mathfrak{g}'$ with $\Psi(\mathfrak{h}) = \mathfrak{h}'$ inducing φ on roots. In particular a semisimple Lie algebra over $\mathbb{C}$ is determined up to isomorphism by its root system, equivalently by its Cartan matrix or Dynkin diagram.

***Proof*:**
The existence claim for the classical types is the identification of example 8.4.4: the matrix algebras $\mathfrak{sl}_{n+1}, \mathfrak{so}_{2n+1}, \mathfrak{sp}_{2n}, \mathfrak{so}_{2n}$ are semisimple by theorem 1.4.3 and have root systems of types A_n, B_n, C_n, D_n respectively, as recorded in example 8.4.4. The five exceptional diagrams are realized by the Serre algebra $\mathfrak{g}(A)$ of theorem 8.5.1, whose finite-dimensionality and semisimplicity are proved in theorem 8.7.16. This establishes part (1). The remainder of the proof is the uniqueness statement, the isomorphism theorem, which is given in full.

Step 1: Choose compatible bases. A root-system isomorphism carries positive systems to positive systems and bases to bases, because it preserves the Cartan integers and hence the reflections

s_α and the notion of a regular hyperplane. Fix a base $\Delta = \{\alpha_1, \dots, \alpha_l\}$ of Φ (theorem 8.2.4 guarantees one exists). Then $\Delta' = \{\alpha'_1, \dots, \alpha'_l\}$ with $\alpha'_i = \varphi(\alpha_i)$ is a base of Φ', and by hypothesis the Cartan integers agree:

$$A'_{ij} = \langle \alpha'_i, (\alpha'_j)^\vee \rangle = \langle \varphi(\alpha_i), \varphi(\alpha_j)^\vee \rangle = \langle \alpha_i, \alpha_j^\vee \rangle = A_{ij}$$

so $\mathfrak{g}$ and $\mathfrak{g}'$ have the *same* Cartan matrix $A = (A_{ij})$.

Step 2: Match the Chevalley generators. By theorem 7.4.1 choose Chevalley generators $e_i \in \mathfrak{g}_{\alpha_i}$, $f_i \in \mathfrak{g}_{-\alpha_i}$, $h_i = [e_i, f_i] = \alpha_i^\vee$ for $\mathfrak{g}$ relative to Δ. The coroots $h_i = \alpha_i^\vee$ form a basis of $\mathfrak{h}$, and since φ carries $\alpha_i^\vee$ to $(\alpha'_i)^\vee$ (a root-system isomorphism commutes with the coroot construction, as $\alpha^\vee$ is built from the inner-product data φ preserves), the assignment $h_i \mapsto h'_i$ where $h'_i = (\alpha'_i)^\vee$ is a linear isomorphism $\mathfrak{h} \to \mathfrak{h}'$.

Step 3: Fix the root-vector scaling. Each root space $\mathfrak{g}_{\alpha_i}$ is one-dimensional (theorem 7.4.6), so a Chevalley generator e'_i for $\mathfrak{g}'$ is determined only up to a nonzero scalar, and the matching must fix that freedom. Choose any $e'_i \in \mathfrak{g}_{\alpha'_i}$ nonzero; there is then a unique $f'_i \in \mathfrak{g}_{-\alpha'_i}$ with $[e'_i, f'_i] = h'_i$, because the normalization $\alpha'_i(h'_i) = 2$ of the $\mathfrak{sl}_2$-triple pins the bracket and the pairing of the one-dimensional spaces $\mathfrak{g}_{\alpha'_i}$ and $\mathfrak{g}_{-\alpha'_i}$ is nondegenerate (theorem 7.4.1). With this matched choice the assignment

$$e_i \mapsto e'_i, \qquad f_i \mapsto f'_i, \qquad h_i \mapsto h'_i$$

respects the diagonal relation, since $[e_i, f_i] = h_i$ maps to $[e'_i, f'_i] = h'_i$. No further scaling is needed: a rescaling $e'_i \mapsto \lambda_i e'_i$ is absorbed by the compensating $f'_i \mapsto \lambda_i^{-1} f'_i$, which preserves $[e'_i, f'_i] = h'_i$, so a consistent matching of triples always exists.

Step 4: The Serre relations are preserved. By lemma 8.5.3 applied to $\mathfrak{g}$, the generators e_i, f_i, h_i satisfy the Serre relations (1) through (4) for the Cartan matrix A. By the same lemma applied to $\mathfrak{g}'$, the generators e'_i, f'_i, h'_i satisfy the identical relations, since $\mathfrak{g}'$ has the *same* Cartan matrix A by Step 1. Every relation is a Lie-polynomial identity in the generators whose coefficients depend only on A: relation (2) reads $[h_i, e_j] = A_{ji} e_j$ on both sides with the same integers A_{ji}, relation (4) reads $(\mathrm{ad} e_i)^{1 - A_{ji}}(e_j) = 0$ with the same exponents, and (1), (3) are A-independent. Therefore the assignment on generators sends each defining relation of $\mathfrak{g}$ to a valid relation of $\mathfrak{g}'$.

Step 5: Extend to an isomorphism. Let $\mathfrak{g}(A)$ be the presented algebra of theorem 8.5.1, generated by symbols $\tilde{e}_i, \tilde{f}_i, \tilde{h}_i$ modulo the Serre relations for A. Both $\mathfrak{g}$ and $\mathfrak{g}'$ receive surjections from $\mathfrak{g}(A)$: by lemma 8.5.3 the assignment $\tilde{e}_i \mapsto e_i$, $\tilde{f}_i \mapsto f_i$, $\tilde{h}_i \mapsto h_i$ respects all defining relations and lands in a generating set, so it extends to a surjective homomorphism $\pi \colon \mathfrak{g}(A) \twoheadrightarrow \mathfrak{g}$ (proposition 8.7.5), and likewise $\pi' \colon \mathfrak{g}(A) \twoheadrightarrow \mathfrak{g}'$ using $\tilde{e}_i \mapsto e'_i$, $\tilde{f}_i \mapsto f'_i$, $\tilde{h}_i \mapsto h'_i$. By the forward direction of theorem 8.5.1 (proved in theorem 8.7.16), $\mathfrak{g}(A)$ is finite-dimensional and semisimple with root system of type A, of the same dimension as $\mathfrak{g}$ and $\mathfrak{g}'$; a surjection of vector spaces of equal finite dimension is an isomorphism, so both π and π' are isomorphisms. Then

$$\Psi = \pi' \circ \pi^{-1} \colon \mathfrak{g} \xrightarrow{\sim} \mathfrak{g}'$$

is an isomorphism of Lie algebras carrying $e_i \mapsto e'_i$, $f_i \mapsto f'_i$, $h_i \mapsto h'_i$. It sends $\mathfrak{h} = \mathrm{span}\{h_i\}$ onto $\mathfrak{h}' = \mathrm{span}\{h'_i\}$, hence carries the root space $\mathfrak{g}_{\alpha_i} = \mathbb{C} e_i$ onto $\mathfrak{g}'_{\alpha'_i} = \mathbb{C} e'_i$ and induces φ on the simple roots, and therefore on all of Φ since the simple roots generate. Thus Ψ lifts φ, establishing that $\mathfrak{g} \cong \mathfrak{g}'$ compatibly with the root-system isomorphism, as we sought to show.

The isomorphism theorem is the mechanism that converts the combinatorial classification into a

classification of algebras. Its structure is worth naming: the Serre-presented algebra $\mathfrak{g}(A)$ sits above both $\mathfrak{g}$ and $\mathfrak{g}'$ as a universal object, each of them a quotient, and the finite-dimensionality proved in theorem 8.7.16 forces both quotients to be isomorphisms. Uniqueness is then automatic. The only Lie-algebraic input is lemma 8.5.3, the verification that a semisimple algebra satisfies the Serre relations, and everything else is the formal behavior of generators and relations. This is the reason the Cartan matrix deserves to be called a complete invariant: not only can it be extracted from $\mathfrak{g}$, it can be run backward to recover $\mathfrak{g}$ uniquely.

One consequence deserves flagging before the classification is assembled. The isomorphism Ψ constructed above depends on the chosen bases and generators, and different choices give different isomorphisms differing by an automorphism of $\mathfrak{g}$. When $\mathfrak{g} = \mathfrak{g}'$ and φ is a symmetry of the Dynkin diagram, Ψ becomes an automorphism of $\mathfrak{g}$ realizing that symmetry: the diagram symmetries of A_n ($n \geq 2$), D_n, and E_6 are the source of the *outer* automorphisms of the corresponding simple algebras. The exercises develop this.

Everything assembled across Part II now collapses into a single statement. The structure theory began by isolating semisimplicity as the condition of vanishing radical (definition 5.5.5); it was then characterized by the nondegeneracy of the Killing form and the decomposition into simple ideals (theorem 6.3.3); the previous chapter attached to each simple ideal a root system (theorem 7.5.4); and this chapter classified the root systems by Dynkin diagram and showed the diagram recovers the algebra. The result is a complete and finite answer to a question about infinite-dimensional families of continuous symmetries.

Corollary 8.5.5: The Cartan-Killing Classification

The simple Lie algebras over $\mathbb{C}$ are classified, up to isomorphism, by the connected Dynkin diagrams: to each of

$$A_n\,(n \geq 1), \quad B_n\,(n \geq 2), \quad C_n\,(n \geq 3), \quad D_n\,(n \geq 4), \quad E_6,\ E_7,\ E_8,\ F_4,\ G_2$$

corresponds exactly one simple Lie algebra, and these are pairwise nonisomorphic and exhaust the simple Lie algebras over $\mathbb{C}$. A general semisimple Lie algebra over $\mathbb{C}$ is a direct sum of simple ideals and is classified, up to isomorphism, by the multiset of connected Dynkin diagrams of its summands, equivalently by its (not necessarily connected) Dynkin diagram.

***Proof*:**

By theorem 6.3.3 a semisimple Lie algebra $\mathfrak{g}$ is the direct sum $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$ of its simple ideals, and this decomposition is unique. The root system of $\mathfrak{g}$ is the orthogonal disjoint union of the root systems of the $\mathfrak{g}_k$, and by proposition 8.3.8 the simple summands correspond to the irreducible components of Φ, equivalently to the connected components of the Dynkin diagram. Thus $\mathfrak{g}$ is determined by the multiset of connected diagrams. For each connected diagram, theorem 8.4.3 lists the possibilities as exactly A_n through G_2; existence of a simple algebra with each such diagram is part (1) of theorem 8.5.4, and uniqueness is part (2).

It remains to separate the classes: isomorphic semisimple algebras have the same Dynkin diagram. Let $\Psi_0 \colon \mathfrak{g} \to \mathfrak{g}'$ be an isomorphism, and let $\mathfrak{h} \subseteq \mathfrak{g}$ and $\mathfrak{h}' \subseteq \mathfrak{g}'$ be the maximal toral subalgebras defining the root systems Φ and Φ'. Since $\operatorname{ad}(\Psi_0 X) = \Psi_0 \circ \operatorname{ad}X \circ \Psi_0^{-1}$, the operator $\operatorname{ad}(\Psi_0 X)$ is diagonalizable exactly when $\operatorname{ad}X$ is, so $\Psi_0(\mathfrak{h})$ is a maximal toral subalgebra of $\mathfrak{g}'$. By theorem 13.2.7 some $\varphi \in \operatorname{Int}(\mathfrak{g}')$ has $\varphi\Psi_0(\mathfrak{h}) = \mathfrak{h}'$.[a] The isomorphism $\Psi = \varphi\Psi_0$ satisfies

$[\Psi H, \Psi X] = \alpha(H)\,\Psi X$ for $H \in \mathfrak{h}$ and $X \in \mathfrak{g}_\alpha$, so it carries $\mathfrak{g}_\alpha$ onto $\mathfrak{g}'_{\alpha'}$ with $\alpha' = \alpha \circ (\Psi|_{\mathfrak{h}})^{-1}$, and $\alpha \mapsto \alpha'$ is a bijection $\Phi \to \Phi'$ given by a linear isomorphism of the duals. The trace computation of proposition 6.1.5(1) applies verbatim to Ψ, so $\kappa'(\Psi X, \Psi Y) = \kappa(X, Y)$. For $H \in \mathfrak{h}$ this gives $\kappa'(\Psi t_\alpha, \Psi H) = \kappa(t_\alpha, H) = \alpha(H) = \alpha'(\Psi H)$, so Ψ carries t_α to $t'_{\alpha'}$ (definition 7.3.2), and $(\alpha', \alpha') = \kappa'(t'_{\alpha'}, t'_{\alpha'}) = \kappa(t_\alpha, t_\alpha) = (\alpha, \alpha)$. Hence it carries the coroot $h_\alpha = 2t_\alpha/(\alpha, \alpha)$ to $h'_{\alpha'}$ (definition 7.4.3), and $\beta'(h'_{\alpha'}) = \beta(\Psi^{-1} h'_{\alpha'}) = \beta(h_\alpha)$ for all roots α, β. So $\alpha \mapsto \alpha'$ preserves Cartan integers and is an isomorphism of root systems. A linear isomorphism carrying Φ onto Φ' carries a base onto a base, because a base is defined by the expansion of the roots in it (definition 8.2.1), and the Cartan matrix does not depend on the base up to reordering (proposition 8.3.2(5)). Thus $\mathfrak{g}$ and $\mathfrak{g}'$ have the same Dynkin diagram. The diagrams on the list are pairwise distinct, so the algebras attached to them are pairwise nonisomorphic, and distinct multisets of connected diagrams likewise give nonisomorphic semisimple algebras. This exhausts and separates the isomorphism classes, completing the proof.

[a] This citation points forward to Part IV without circularity, because the proof of theorem 13.2.7 rests on the compact real form and the maximal torus theorem, and neither depends on the present corollary.

8.5.6 Remark: Automorphisms of the Group versus the Algebra Let G be a simply connected Lie group with Lie algebra $\mathfrak{g}$. A continuous automorphism of G is smooth (corollary 3.2.5), so it has a differential in $\mathrm{Aut}(\mathfrak{g})$. Conversely, theorem 3.3.7 integrates ψ and ψ^{-1} to mutually inverse automorphisms of G. So continuous automorphisms of G correspond bijectively to automorphisms of $\mathfrak{g}$, and for a complex simple $\mathfrak{g}$, exercise 8.5.1 (4) identifies the complex-linear ones modulo inner automorphisms with the symmetries of the Dynkin diagram. Forgetting the topology changes the answer. Every automorphism σ of the field $\mathbb{C}$ acts on $\mathrm{SL}_n(\mathbb{C})$ entry by entry. A continuous σ fixes $\mathbb{Q}$, hence $\mathbb{R}$, so it is the identity or complex conjugation. Conjugation gives a continuous automorphism of $\mathrm{SL}_n(\mathbb{C})$ that is not complex-linear on $\mathfrak{sl}_n(\mathbb{C})$. Every other σ, and such σ exist by the axiom of choice, gives an automorphism of the abstract group $\mathrm{SL}_n(\mathbb{C})$ that Lie theory cannot see. The classification of all automorphisms of the abstract group, and why field automorphisms are the only addition, is in the remark following [10, Isomorphism Theorem]; see also Borel and Tits [1].

This corollary is the summit of Part II. A semisimple Lie algebra over $\mathbb{C}$, an object that a priori could be any of an unclassifiable-looking continuum of bracket structures, turns out to be a finite direct sum drawn from a list of nine items, four infinite families and five exceptions, each labeled by a small diagram. The passage from "radical zero" to this finite picture is the arc the whole of Part II has traced: the negative, structureless definition of semisimplicity as vanishing radical has been converted, one theorem at a time, into a positive combinatorial classification in which every semisimple algebra is a transparent assembly of known pieces. The Killing form made semisimplicity detectable; the root space decomposition made it combinatorial; the root system axioms made it Euclidean; and the Dynkin diagram made it finite.

What Part II has *not* yet done is say what these algebras do. The classification catalogs the algebras but is silent on their representations, and it is representation theory that the structure theory was built to support. Part III takes up this task on the group side. There the same combinatorial data, now arising from a compact Lie group through its maximal torus rather than from a Lie algebra through its Cartan subalgebra, will organize the representation theory completely: the roots return as the weights of the adjoint representation, the Weyl group returns as $N(T)/T$, and the classification of this chapter becomes the classification of the compact groups whose representations the Peter-Weyl

theorem and the Weyl character formula compute. The finite picture built here is the coordinate system in which all of that computation takes place.

Exercise 8.5.1

1. Write out the complete list of Serre relations for the Cartan matrix of type A_2, and verify that the matrices $e_1 = E_{12}$, $e_2 = E_{23}$, $f_1 = E_{21}$, $f_2 = E_{32}$, $h_1 = E_{11} - E_{22}$, $h_2 = E_{22} - E_{33}$ in $\mathfrak{sl}_3(\mathbb{C})$ satisfy every one of them. In particular verify the two Serre relations $(\mathrm{ad}e_1)^2(e_2) = 0$ and $(\mathrm{ad}e_2)^2(e_1) = 0$ by direct matrix computation, and show that the eight matrices generated span $\mathfrak{sl}_3(\mathbb{C})$.

2. For $\mathfrak{sl}_n(\mathbb{C})$ with the standard Chevalley generators $e_i = E_{i,i+1}$, $f_i = E_{i+1,i}$, $h_i = E_{ii} - E_{i+1,i+1}$ $(1 \le i \le n-1)$, verify the relation $[e_i, f_j] = \delta_{ij} h_i$ directly from the matrix-unit multiplication rule $E_{ab}E_{cd} = \delta_{bc}E_{ad}$. Explain, in terms of the roots $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$, why $[e_i, f_j] = 0$ for $i \neq j$, by identifying the root space $\mathfrak{g}_{\alpha_i - \alpha_j}$ in which the bracket lies and explaining why it is zero.

3. Explain why the Serre presentation of a rank-l semisimple Lie algebra has only finitely many defining relations, counting them as a function of l. Contrast this with the fact that $\mathfrak{g}$ itself may have large dimension (for instance $\dim E_8 = 248$), and comment on what this says about the efficiency of the presentation.

4. A symmetry of a Dynkin diagram is a permutation of its nodes preserving the bonds. Determine the group of diagram symmetries for each of A_n $(n \ge 2)$, D_n $(n \ge 5)$, D_4, and E_6. Using theorem 8.5.4, explain how each such symmetry yields an automorphism of the corresponding simple Lie algebra, and why these are the sources of the outer automorphisms. (You may take as given that the inner automorphisms act trivially on the Dynkin diagram.)

5. The Cartan matrices of B_2 and C_2 are transposes of one another,
$$A(B_2) = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}, \qquad A(C_2) = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$$
Using theorem 8.5.1, write the Serre relations for each and exhibit an explicit isomorphism $\mathfrak{g}(A(B_2)) \cong \mathfrak{g}(A(C_2))$ by matching generators. Interpret the result in light of the low-rank coincidence $B_2 \cong C_2$.

6. Using only lemma 8.5.3 and the finite-dimensionality of $\mathfrak{g}(A)$ (theorem 8.7.16), give a one-paragraph argument that two semisimple Lie algebras over $\mathbb{C}$ with equal Cartan matrices are isomorphic, without reproducing the full proof of theorem 8.5.4. Identify precisely where the finite-dimensionality input is used.

8.6 The Universal Enveloping Algebra and the Poincaré-Birkhoff-Witt Theorem

The classification is a statement about brackets, but the representation theory it was built to support is a statement about operators, and operators multiply. A representation $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ (definition 6.4.1) lands in a space that carries more structure than a bracket: $\mathfrak{gl}(V) = \mathrm{End}(V)$ is an associative algebra, and the arguments of Part III lean on that extra multiplication constantly. The theorem of the highest weight will build an irreducible representation outward from a single

vector by applying products $\varphi(f_{\beta_1})\cdots\varphi(f_{\beta_k})$ of lowering operators, and a product of k operators is the image of no element of $\mathfrak{g}$ at all. Such expressions have a natural home, and it is not $\mathrm{End}(V)$ for any particular V: it is a single associative algebra manufactured from $\mathfrak{g}$ itself, through which every representation factors at once.

This section builds that algebra, the universal enveloping algebra $U(\mathfrak{g})$, and answers the two questions any such construction raises. The first is what maps out of it: the universal property below says that representations of $\mathfrak{g}$ are the same thing as modules over $U(\mathfrak{g})$, so the highest-weight modules of Part III become modules over a ring. The second is how large it is. Imposing the relation $XY - YX = [X, Y]$ could conceivably force further collapse, and the Poincaré-Birkhoff-Witt theorem says it does not: for any ordered basis of $\mathfrak{g}$ the ordered monomials form a basis of $U(\mathfrak{g})$, so the enveloping algebra is exactly as large as the commutation relations permit and $\mathfrak{g}$ sits inside it unharmed. The proof of that theorem is the substance of the section. Its payoff is threefold: the triangular decomposition of $U(\mathfrak{g})$ that drives the highest-weight theory of Part III, the free-algebra technique underlying the existence half of the Serre presentation (theorem 8.5.1), proved in the next section, and the algebraic tools Ado's theorem (theorem 3.4.3) uses, promised from the structure theory of this part.

The raw material is the tensor algebra of the underlying vector space, developed in the algebra prerequisite. To force the commutation relation, quotient by the smallest two-sided ideal containing its failures.

Definition 8.6.1: The Universal Enveloping Algebra

Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$. Let $\mathcal{T}(\mathfrak{g}) = \bigoplus_{k\geq 0} \mathcal{T}^k(\mathfrak{g})$ be the tensor algebra of the underlying vector space of $\mathfrak{g}$ ([2, Tensor Algebra, $\mathcal{T}(M)$ is an R-algebra]), and let $J(\mathfrak{g}) \subseteq \mathcal{T}(\mathfrak{g})$ be the two-sided ideal ([2, Ideals, Generating Ideals]) generated by the elements

$$X \otimes Y - Y \otimes X - [X, Y] \qquad X, Y \in \mathfrak{g}$$

The *universal enveloping algebra* of $\mathfrak{g}$ is the quotient algebra ([2, Quotient Ring])

$$U(\mathfrak{g}) := \mathcal{T}(\mathfrak{g}) \,/\, J(\mathfrak{g})$$

and the *canonical map* $\iota\colon \mathfrak{g} \to U(\mathfrak{g})$ is the composite of the inclusion $\mathfrak{g} = \mathcal{T}^1(\mathfrak{g}) \subseteq \mathcal{T}(\mathfrak{g})$ with the quotient map. By construction ι is linear and satisfies

$$\iota(X)\,\iota(Y) - \iota(Y)\,\iota(X) = \iota([X, Y]) \qquad X, Y \in \mathfrak{g} \tag{8.3}$$

The generators of $J(\mathfrak{g})$ are exactly the discrepancies between the two multiplications in play. In $\mathcal{T}(\mathfrak{g})$ the product of X and Y is the free concatenation $X \otimes Y$, which remembers nothing about the bracket. Setting $X \otimes Y - Y \otimes X - [X, Y]$ to zero declares that from now on the commutator of X and Y *is* their bracket, and quotienting by the ideal these elements generate imposes that declaration. The universal property below is the precise sense in which nothing further is imposed. Two warnings are in order. First, $U(\mathfrak{g})$ is not commutative unless $\mathfrak{g}$ is abelian: the relation prescribes the commutator, and prescribes it to be nonzero whenever the bracket is. Second, nothing in the construction makes it obvious that the quotient is nonzero, let alone that ι is injective. The ideal $J(\mathfrak{g})$ mixes degrees, since its generators have components in degrees two and one, so it could conceivably reach down and swallow $\mathcal{T}^1(\mathfrak{g}) = \mathfrak{g}$, or everything. Ruling that out is precisely the content of the Poincaré-Birkhoff-Witt theorem, and until it is proved the elements $\iota(X)$ must be handled without assuming they are distinct or nonzero.

What the construction does deliver immediately is the mapping property that gives the algebra its name: $U(\mathfrak{g})$ envelops every associative algebra picture of $\mathfrak{g}$.

Theorem 8.6.2: The Universal Property of the Enveloping Algebra

Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$, let A be a unital associative $\mathbb{C}$-algebra ([2, R-algebra]), and let $\varphi\colon \mathfrak{g} \to A$ be a linear map satisfying

$$\varphi([X,Y]) = \varphi(X)\varphi(Y) - \varphi(Y)\varphi(X) \qquad X, Y \in \mathfrak{g}$$

Then there is a unique homomorphism of unital $\mathbb{C}$-algebras $\Phi\colon U(\mathfrak{g}) \to A$ with $\Phi \circ \iota = \varphi$.

***Proof*:**

Existence. By the universal property of the tensor algebra ([2, $\mathcal{T}(M)$ is an R-algebra]), the linear map $\varphi\colon \mathfrak{g} \to A$ extends uniquely to a homomorphism of unital algebras $\widetilde{\Phi}\colon \mathcal{T}(\mathfrak{g}) \to A$ with $\widetilde{\Phi}|_{\mathfrak{g}} = \varphi$. On a generator of $J(\mathfrak{g})$,

$$\widetilde{\Phi}\big(X \otimes Y - Y \otimes X - [X,Y]\big) = \varphi(X)\varphi(Y) - \varphi(Y)\varphi(X) - \varphi([X,Y]) = 0$$

by the hypothesis on φ. An algebra homomorphism vanishing on the generators of a two-sided ideal vanishes on the whole ideal: a general element of $J(\mathfrak{g})$ is a finite sum of products agb with g a generator and $a, b \in \mathcal{T}(\mathfrak{g})$, and $\widetilde{\Phi}(agb) = \widetilde{\Phi}(a)\,\widetilde{\Phi}(g)\,\widetilde{\Phi}(b) = 0$. Hence $J(\mathfrak{g}) \subseteq \ker\widetilde{\Phi}$, and $\widetilde{\Phi}$ descends to a well-defined algebra homomorphism Φ on the quotient $U(\mathfrak{g}) = \mathcal{T}(\mathfrak{g})/J(\mathfrak{g})$ ([2, The first isomorphism theorem for Rings]), with $\Phi \circ \iota = \widetilde{\Phi}|_{\mathfrak{g}} = \varphi$.

Uniqueness. The algebra $\mathcal{T}(\mathfrak{g})$ is generated as a unital algebra by $\mathfrak{g} = \mathcal{T}^1(\mathfrak{g})$, and the quotient map is surjective, so $U(\mathfrak{g})$ is generated as a unital algebra by $\iota(\mathfrak{g})$. Two unital algebra homomorphisms out of $U(\mathfrak{g})$ agreeing on $\iota(\mathfrak{g})$ agree on all products of such elements and on 1, hence everywhere, completing the proof.

One consequence is worth extracting before anything else, because it settles the smallest of the worries raised above. The zero map $\mathfrak{g} \to \mathbb{C}$ satisfies the hypothesis of the theorem, both sides being zero, so there is a unital algebra homomorphism $U(\mathfrak{g}) \to \mathbb{C}$. It sends 1 to 1, so $1 \neq 0$ in $U(\mathfrak{g})$ and the enveloping algebra is not the zero ring. Whether ι is injective remains open. The second consequence is the one the section opened with, and it is the reason to care about $U(\mathfrak{g})$ at all: taking $A = \mathrm{End}(V)$ turns the hypothesis on φ into the definition of a representation.

Corollary 8.6.3: Representations Are Enveloping-Algebra Modules

Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$ and V a $\mathbb{C}$-vector space. The assignments $\Phi \mapsto \Phi \circ \iota$ and $\varphi \mapsto \Phi$ of theorem 8.6.2 are mutually inverse bijections between the unital $\mathbb{C}$-algebra homomorphisms $\Phi\colon U(\mathfrak{g}) \to \mathrm{End}(V)$ and the representations $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ of definition 6.4.1. Under this correspondence a subspace $W \subseteq V$ is a subrepresentation for φ if and only if it is invariant under $\Phi(U(\mathfrak{g}))$.

***Proof*:**
As vector spaces $\mathfrak{gl}(V)$ and $\mathrm{End}(V)$ coincide, and the commutator bracket of $\mathfrak{gl}(V)$ is computed by the associative multiplication of $\mathrm{End}(V)$. A linear map $\varphi\colon \mathfrak{g} \to \mathrm{End}(V)$ therefore satisfies the hypothesis of theorem 8.6.2 exactly when it is a Lie algebra homomorphism into $\mathfrak{gl}(V)$, that is, a representation. The theorem extends each such φ to a unique unital algebra homomorphism Φ with $\Phi \circ \iota = \varphi$, and conversely, for any unital algebra homomorphism Φ, the composite $\Phi \circ \iota$ is linear and carries the bracket to the commutator, because Φ is multiplicative and (8.3) holds in $U(\mathfrak{g})$. The two assignments are inverse by the uniqueness clause of the theorem. For the last claim: if W is invariant under $\Phi(U(\mathfrak{g}))$ it is in particular invariant under $\Phi(\iota(X)) = \varphi(X)$ for every X. Conversely, if $\varphi(X)W \subseteq W$ for every X, then W is invariant under every product of the operators $\varphi(X)$ and under scalars, and these span $\Phi(U(\mathfrak{g}))$ because $\iota(\mathfrak{g})$ and 1 generate $U(\mathfrak{g})$, completing the proof.

The corollary is the linearization promised in the opening paragraph: the bracket, a nonassociative operation, has been traded for module theory over an associative ring, with no loss of information about representations. The construction is also natural in $\mathfrak{g}$, in the strong sense that it is a functor.

Proposition 8.6.4: Functoriality of the Enveloping Algebra

Let $f\colon \mathfrak{g} \to \mathfrak{h}$ be a homomorphism of Lie algebras over $\mathbb{C}$. There is a unique homomorphism of unital $\mathbb{C}$-algebras $U(f)\colon U(\mathfrak{g}) \to U(\mathfrak{h})$ with $U(f) \circ \iota_{\mathfrak{g}} = \iota_{\mathfrak{h}} \circ f$. Moreover $U(\mathrm{id}_{\mathfrak{g}}) = \mathrm{id}_{U(\mathfrak{g})}$, and $U(g \circ f) = U(g) \circ U(f)$ for a further homomorphism $g\colon \mathfrak{h} \to \mathfrak{k}$.

***Proof*:**
The composite $\varphi := \iota_{\mathfrak{h}} \circ f\colon \mathfrak{g} \to U(\mathfrak{h})$ is linear, and for $X, Y \in \mathfrak{g}$

$$\varphi([X,Y]) = \iota_{\mathfrak{h}}\big(f([X,Y])\big) = \iota_{\mathfrak{h}}\big([f(X), f(Y)]\big) = \varphi(X)\varphi(Y) - \varphi(Y)\varphi(X)$$

using first that f preserves brackets (definition 1.4.5) and then the defining relation (8.3) in $U(\mathfrak{h})$. By theorem 8.6.2 there is a unique unital algebra homomorphism $U(f)$ with $U(f) \circ \iota_{\mathfrak{g}} = \varphi$, which is the first claim. The identity map $\mathrm{id}_{U(\mathfrak{g})}$ satisfies the property characterizing $U(\mathrm{id}_{\mathfrak{g}})$, and $U(g) \circ U(f)$ is a unital algebra homomorphism with $U(g) \circ U(f) \circ \iota_{\mathfrak{g}} = U(g) \circ \iota_{\mathfrak{h}} \circ f = \iota_{\mathfrak{k}} \circ g \circ f$, so it satisfies the property characterizing $U(g \circ f)$. Uniqueness in theorem 8.6.2 forces both equalities, completing the proof.

Two structural features of $U(\mathfrak{g})$ can be established directly from the construction, and both are needed for the Poincaré-Birkhoff-Witt theorem. The first records how much of the grading of the tensor algebra remains after the quotient. The ideal $J(\mathfrak{g})$ is not graded, because its generators mix degrees two and one, so $U(\mathfrak{g})$ inherits no grading. It inherits the next best thing, a filtration: the images of the tensors of degree at most k form an increasing exhaustion, and the failure of multiplication to respect degree strictly is confined to lower-order terms.

Proposition 8.6.5: The Filtration by Degree

Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$. For $k \geq 0$ let $U_k \subseteq U(\mathfrak{g})$ be the span of all products $\iota(Y_1)\cdots\iota(Y_j)$ with $j \leq k$ and $Y_1, \ldots, Y_j \in \mathfrak{g}$, the empty product being 1. Then:

1. $U_0 = \mathbb{C}1 \subseteq U_1 \subseteq U_2 \subseteq \cdots$, and $\bigcup_{k\geq 0} U_k = U(\mathfrak{g})$
2. $U_j U_k \subseteq U_{j+k}$ for all $j, k \geq 0$
3. $uv - vu \in U_{j+k-1}$ for all $u \in U_j$ and $v \in U_k$ with $j + k \geq 1$.

***Proof*:**
(1) The containments are immediate from the definition, since a product of at most j factors is a product of at most k factors for $j \leq k$. For the exhaustion: $\mathcal{T}(\mathfrak{g})$ is spanned by the simple tensors $Y_1 \otimes \cdots \otimes Y_j$ over all $j \geq 0$, the quotient map is surjective, and the image of such a tensor is the product $\iota(Y_1)\cdots\iota(Y_j) \in U_j$. Finally $U_0 = \mathbb{C}1$ by definition.

(2) A product of at most j factors times a product of at most k factors is a product of at most $j + k$ factors, and multiplication is bilinear.

(3) First take $j = 1$, so $u = \iota(X)$ with $X \in \mathfrak{g}$. For $k = 0$, v is a scalar and the commutator vanishes, landing in U_0. For $v = \iota(Y_1)\cdots\iota(Y_k)$ a product of k factors, expanding the commutator one factor at a time gives the Leibniz identity

$$\iota(X)\, v - v\, \iota(X) = \sum_{a=1}^{k} \iota(Y_1)\cdots\iota(Y_{a-1}) \big(\iota(X)\iota(Y_a) - \iota(Y_a)\iota(X)\big)\, \iota(Y_{a+1})\cdots\iota(Y_k)$$

which is verified by writing each summand as a difference of two words that move the factor $\iota(X)$ one step to the right, so that the sum telescopes. By (8.3) the middle factor of the a-th summand is $\iota([X, Y_a]) \in U_1$, so each summand is a product of k factors from $\iota(\mathfrak{g})$ and lies in U_k. By linearity the case $j = 1$ follows. For general j, write u as a linear combination of products $\iota(X_1)\cdots\iota(X_i)$ with $i \leq j$ and induct on i: with $u = \iota(X_1)\, u'$ and $u' = \iota(X_2)\cdots\iota(X_i)$,

$$uv - vu = \iota(X_1)\,(u'v - vu') + \big(\iota(X_1)\, v - v\, \iota(X_1)\big)\, u'$$

and the first term lies in $U_1 U_{i-1+k-1} \subseteq U_{i+k-1}$ by the inductive hypothesis and (2), while the second lies in $U_k U_{i-1} \subseteq U_{i+k-1}$ by the case $j = 1$. This completes the proof.

Part (3) is the precise sense in which $U(\mathfrak{g})$ is "commutative up to lower order": commutators do not vanish, but they drop one step down the filtration. That single fact powers the easy half of the Poincaré-Birkhoff-Witt theorem, because it lets any product be reordered at the cost of shorter terms. The second structural feature is the degenerate case in which no reordering cost arises at all.

Proposition 8.6.6: The Enveloping Algebra of an Abelian Lie Algebra

Let $\mathfrak{g}$ be an abelian Lie algebra over $\mathbb{C}$, so $[X, Y] = 0$ for all $X, Y \in \mathfrak{g}$, of dimension n with basis $x_1, \ldots, x_n$. Then $U(\mathfrak{g})$ is the symmetric algebra of $\mathfrak{g}$, and the assignment $\iota(x_i) \mapsto z_i$ extends to an isomorphism of $\mathbb{C}$-algebras

$$U(\mathfrak{g}) \cong \mathbb{C}[z_1, \ldots, z_n]$$

with the polynomial algebra in n variables.

***Proof*:**
When every bracket vanishes, the generators of $J(\mathfrak{g})$ reduce to the commutation tensors $X \otimes Y - Y \otimes X$, so $J(\mathfrak{g})$ is exactly the ideal defining the symmetric algebra and $U(\mathfrak{g}) = \mathcal{T}(\mathfrak{g})/J(\mathfrak{g}) = \mathcal{S}(\mathfrak{g})$ ([2, Symmetric Algebra]). For a vector space with basis $x_1, \ldots, x_n$ the symmetric algebra is the polynomial algebra on that basis, by the isomorphism carrying x_i to z_i ([2, The Symmetric Algebra of a Free Module]), completing the proof.

In the abelian case, then, the monomials $z_1^{a_1} \cdots z_n^{a_n}$ form a basis of $U(\mathfrak{g})$: every element is a polynomial in the images of the basis vectors, written with the factors in a fixed order, and no two such expressions coincide. The Poincaré-Birkhoff-Witt theorem is the statement that this picture survives for every $\mathfrak{g}$, with one correction required by noncommutativity. Since distinct basis vectors no longer commute, a chosen order of the factors must be fixed once and for all, and only the monomials respecting that order are counted.

Fix an ordered basis $x_1, \ldots, x_n$ of $\mathfrak{g}$. For a weakly increasing multi-index $I = (i_1 \le i_2 \le \cdots \le i_m)$ with entries in $\{1, \ldots, n\}$, of length $\ell(I) = m \ge 0$, the *ordered monomial* attached to I is the product

$$x_I := \iota(x_{i_1})\, \iota(x_{i_2}) \cdots \iota(x_{i_m}) \in U(\mathfrak{g})$$

with $x_\emptyset = 1$. The theorem is that these are a basis.

Theorem 8.6.7: The Poincaré-Birkhoff-Witt Theorem

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{C}$ with ordered basis $x_1, \ldots, x_n$. The ordered monomials x_I, as I runs over all weakly increasing multi-indices of all lengths $m \ge 0$, form a basis of the vector space $U(\mathfrak{g})$. Moreover, for each $k \ge 0$ the ordered monomials with $\ell(I) \le k$ form a basis of the filtration piece U_k of proposition 8.6.5.

The two halves of the claim, spanning and linear independence, are of entirely different depths. Spanning says that for every word in the basis vectors the reordering permitted by (8.3) terminates in ordered form, and it is an induction of the kind proposition 8.6.5(3) already rehearsed. Independence says that no linear combination of ordered monomials falls into the ideal $J(\mathfrak{g})$, and there is no way to verify that by manipulating the relations themselves: what is needed is a faithful witness, a concrete module in which the ordered monomials visibly act independently. The obvious candidate is the polynomial space of proposition 8.6.6, with x_I acting on the constant 1 by multiplication by $z_I := z_{i_1} \cdots z_{i_m}$. The delicate point is that $\mathfrak{g}$ is not abelian, so the action of x_i on an arbitrary monomial cannot simply be multiplication by z_i: correction terms of lower degree are forced, and the content of the following lemma is that the corrections can be chosen consistently, with the

Jacobi identity entering at exactly one point as the condition that makes consistency possible. This verification is the technical heart of the theorem, and it is carried out in full.

Lemma 8.6.8: The Polynomial Model of the Ordered Monomials

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{C}$ with ordered basis $x_1, \dots, x_n$, and let $S = \mathbb{C}[z_1, \dots, z_n]$. For a weakly increasing multi-index $I = (i_1 \le \cdots \le i_m)$ write $z_I = z_{i_1} \cdots z_{i_m}$, with $z_\emptyset = 1$, so that the z_I form a basis of S. Let $S_m \subseteq S$ be the span of the z_I with $\ell(I) \le m$, and write $i \le I$ to mean that I is empty or $i \le i_1$. Then there is a representation of $\mathfrak{g}$ on S, written $(X, p) \mapsto X \cdot p$, such that

1. $x_i \cdot z_I = z_i z_I$ whenever $i \le I$
2. $x_i \cdot z_I - z_i z_I \in S_m$ whenever $\ell(I) = m$.

***Proof*:**
A bilinear map $\mathfrak{g} \times S \to S$ is determined by the values $x_i \cdot z_I$ on pairs of basis elements, so the task is to define these values and to verify the module axiom

$$X \cdot (Y \cdot p) - Y \cdot (X \cdot p) = [X, Y] \cdot p \tag{8.4}$$

of definition 6.4.1, together with (1) and (2). Both sides of (8.4) are linear in X, in Y, and in p separately, so it suffices to verify it for $X = x_i$, $Y = x_j$, and $p = z_J$ a basis monomial. Conversely, once (8.4) is known for basis vectors and for all monomials in some S_m, linearity extends it to arbitrary $X, Y \in \mathfrak{g}$ and $p \in S_m$, and this extended form is the one the induction below uses. Throughout, an expression $[x_i, x_j] \cdot p$ means the bilinear extension applied to the bracket, expanded in the basis.

Construction. The values $x_i \cdot z_I$ are defined by induction on $m = \ell(I)$. For $m = 0$ set $x_i \cdot 1 := z_i$, as (1) requires. Let $m \ge 1$, suppose the values are defined for all monomials of length less than m and satisfy (1) and (2) there, and let $\ell(I) = m$. Write $I = (j, J)$ where $j = i_1$ is the smallest entry and $J = (i_2 \le \cdots \le i_m)$, so $j \le J$ and $\ell(J) = m - 1$. There are two cases.

- If $i \le j$, set $x_i \cdot z_I := z_i z_I$.
- If $i > j$, then by (2) at length $m - 1$ the element $w := x_i \cdot z_J - z_i z_J$ lies in S_{m-1}, and every term on the right of

$$x_i \cdot z_I := z_i z_I + x_j \cdot w + [x_i, x_j] \cdot z_J \tag{8.5}$$

  is already defined, since w and z_J both lie in S_{m-1}, where the action of every basis vector exists by the inductive hypothesis.

The recursion (8.5) is not arbitrary: it is forced by the axiom being aimed at. If (8.4) is to hold, then since $j \le J$ gives $x_j \cdot z_J = z_j z_J = z_I$ by (1), the value $x_i \cdot z_I$ must equal

$$x_i \cdot (x_j \cdot z_J) = x_j \cdot (x_i \cdot z_J) + [x_i, x_j] \cdot z_J = x_j \cdot (z_i z_J) + x_j \cdot w + [x_i, x_j] \cdot z_J$$

and the first term ought to be $z_j z_i z_J = z_i z_I$, which is what (8.5) declares. The work of the proof is to confirm that this forced definition is consistent, meaning that (8.4) really does hold in all configurations and not only in the one used to make the definition.

Property (1) holds at every length, by the first case of the construction: $i \leq I$ means exactly that I is empty or $i \leq j$.

Property (2) holds at every length, by induction: in the first case the difference is zero, and in the second it is $x_j \cdot w + [x_i, x_j] \cdot z_J$, which lies in S_m: applying (2) at lengths below m to each monomial of $w \in S_{m-1}$ and of $z_J \in S_{m-1}$, the action of a basis vector raises degree by at most one.

The module axiom. Denote by (C_m) the statement that (8.4) holds for all basis vectors x_i, x_j and all $p \in S_{m-1}$. Note that (C_m) is well posed: for $p \in S_{m-1}$, property (2) puts $x_j \cdot p$ in S_m, where the action of basis vectors is defined once the construction has reached length m. Since every polynomial lies in some S_{m-1}, the axiom in full is the conjunction of the (C_m), and (C_m) is proved by induction on m, with the convention $S_{-1} = 0$. The case (C_0) is vacuous. Fix $m \geq 1$, assume (C_{m-1}), and by linearity take $p = z_J$. If $\ell(J) < m - 1$ the claim is (C_{m-1}), so take $\ell(J) = m - 1$. Both sides of (8.4) change sign when i and j are exchanged, and both vanish when $i = j$, so it may be assumed that $j < i$. Two cases remain, according to whether j reaches the front of J.

Case 1: $j \leq J$. Then $x_j \cdot z_J = z_j z_J = z_{(j,J)}$ by (1), so the left side of the axiom is $x_i \cdot z_{(j,J)} - x_j \cdot (x_i \cdot z_J)$. The monomial $z_{(j,J)}$ has smallest entry $j < i$, so its value under x_i is given by the recursion (8.5), with the same $w = x_i \cdot z_J - z_i z_J$:

$$x_i \cdot z_{(j,J)} = z_i z_{(j,J)} + x_j \cdot w + [x_i, x_j] \cdot z_J$$

For the other term, $x_i \cdot z_J = z_i z_J + w$, and $z_i z_J$ is the basis monomial z_K for K the weakly increasing rearrangement of (i, J). Every entry of K is an entry of J or equals i, and $j \leq J$ with $j < i$, so $j \leq K$ and (1) applies at length m:

$$x_j \cdot (x_i \cdot z_J) = x_j \cdot z_K + x_j \cdot w = z_j z_i z_J + x_j \cdot w$$

Subtracting, and noting $z_i z_{(j,J)} = z_i z_j z_J = z_j z_i z_J$ in the commutative ring S, the terms $z_j z_i z_J$ and $x_j \cdot w$ cancel and the difference is exactly $[x_i, x_j] \cdot z_J$, which is (C_m) in this case.

Case 2: j does not reach the front of J. Write $J = (k, K)$ with k the smallest entry and $\ell(K) = m - 2$, so the failure of $j \leq J$ means $k < j < i$ and $k \leq K$. The strategy is to push everything down to z_K, where (C_{m-1}) applies, using Case 1 at length m for any pair whose smaller index is k. First, by (1) and (C_{m-1}),

$$x_j \cdot z_J = x_j \cdot (x_k \cdot z_K) = x_k \cdot (x_j \cdot z_K) + [x_j, x_k] \cdot z_K$$

and applying x_i gives

$$x_i \cdot (x_j \cdot z_J) = x_i \cdot \big(x_k \cdot (x_j \cdot z_K)\big) + x_i \cdot \big([x_j, x_k] \cdot z_K\big) \tag{8.6}$$

The first summand is handled by commuting x_i past x_k. Write $x_j \cdot z_K = z_j z_K + w'$ with $w' \in S_{m-2}$, by (2). The monomial $z_j z_K$ is $z_{K'}$ for K' the weakly increasing rearrangement of (j, K), of length $m - 1$, and $k \leq K'$ with $k < i$, so the pair (x_i, x_k) acting on $z_{K'}$ falls under Case 1 at length m, while the same pair acting on $w' \in S_{m-2}$ falls under (C_{m-1}). Adding the two instances,

$$x_i \cdot \big(x_k \cdot (x_j \cdot z_K)\big) = x_k \cdot \big(x_i \cdot (x_j \cdot z_K)\big) + [x_i, x_k] \cdot (x_j \cdot z_K)$$

and expanding the inner $x_i \cdot (x_j \cdot z_K)$ by (C_{m-1}) turns this into

$$x_i \cdot \big(x_k \cdot (x_j \cdot z_K)\big) = x_k \cdot \big(x_j \cdot (x_i \cdot z_K)\big) + x_k \cdot \big([x_i, x_j] \cdot z_K\big) + [x_i, x_k] \cdot (x_j \cdot z_K)$$

Substituting into (8.6),

$$x_i \cdot (x_j \cdot z_J) = x_k \cdot \big(x_j \cdot (x_i \cdot z_K)\big) + x_k \cdot \big([x_i, x_j] \cdot z_K\big) + [x_i, x_k] \cdot (x_j \cdot z_K) + x_i \cdot \big([x_j, x_k] \cdot z_K\big)$$

This identity was derived using only $k < i$, $k < j$, and $k \leq K$, so it holds verbatim with i and j exchanged. Subtracting the exchanged identity from the displayed one, the leading terms combine by (C_{m-1}), applied inside $x_k \cdot (\,\cdot\,)$ to the difference $x_j \cdot (x_i \cdot z_K) - x_i \cdot (x_j \cdot z_K) = -[x_i, x_j] \cdot z_K$, and the second terms are opposite, so the first two pairs contribute $x_k \cdot ([x_i, x_j] \cdot z_K)$ in total. The cross terms are handled by the extended form of (C_{m-1}), with one Lie slot occupied by a bracket:

$$[x_i, x_k] \cdot (x_j \cdot z_K) - x_j \cdot \big([x_i, x_k] \cdot z_K\big) = \big[[x_i, x_k], x_j\big] \cdot z_K$$

$$x_i \cdot \big([x_j, x_k] \cdot z_K\big) - [x_j, x_k] \cdot (x_i \cdot z_K) = \big[x_i, [x_j, x_k]\big] \cdot z_K$$

Assembling, the left side of (C_m) is

$$x_i \cdot (x_j \cdot z_J) - x_j \cdot (x_i \cdot z_J) = x_k \cdot \big([x_i, x_j] \cdot z_K\big) + \Big(\big[[x_i, x_k], x_j\big] + \big[x_i, [x_j, x_k]\big]\Big) \cdot z_K$$

while the right side, expanded by (1) and the extended (C_{m-1}) for the pair $([x_i, x_j], x_k)$, is

$$[x_i, x_j] \cdot z_J = [x_i, x_j] \cdot (x_k \cdot z_K) = x_k \cdot \big([x_i, x_j] \cdot z_K\big) + \big[[x_i, x_j], x_k\big] \cdot z_K$$

The difference of the two sides is therefore

$$\Big(\big[[x_i, x_k], x_j\big] + \big[x_i, [x_j, x_k]\big] - \big[[x_i, x_j], x_k\big]\Big) \cdot z_K = \Big(\big[x_j, [x_k, x_i]\big] + \big[x_i, [x_j, x_k]\big] + \big[x_k, [x_i, x_j]\big]\Big) \cdot z_K$$

after rewriting the first and third brackets by antisymmetry, and the inner sum is zero: it is precisely the Jacobi identity of definition 1.4.5 for the triple (x_i, x_j, x_k). This proves (C_m), and with it the module axiom (8.4) on all of S. The bilinear map is therefore a representation of $\mathfrak{g}$ on S satisfying (1) and (2), completing the proof.

The Jacobi identity appears exactly once, and not as a formality: it is the precise closing condition of the diamond of reorderings. Reducing $x_i x_j x_k$ to ordered form can begin with either of two swaps, the two routes produce correction terms assembled from iterated brackets, and the routes agree exactly because the bracket combination above vanishes. In an algebra whose bracket failed the Jacobi identity the two routes would genuinely disagree, the model of lemma 8.6.8 would not exist, and the ordered monomials would satisfy unplanned relations. With the model in hand, the theorem follows.

***Proof Proof of theorem 8.6.7*:**
Step 1: the ordered monomials with $\ell(I) \leq k$ span U_k. By proposition 8.6.5(1) and bilinearity of multiplication, U_k is spanned by the words $\iota(x_{j_1}) \cdots \iota(x_{j_l})$ with $l \leq k$ and arbitrary, not necessarily ordered, indices $j_1, \ldots, j_l$: expanding each factor of a product $\iota(Y_1) \cdots \iota(Y_j)$ in the basis produces such words. We argue by strong induction on l that every such word is a linear combination of ordered monomials of length at most l, the case $l = 0$ being the monomial 1. Fix l and induct on the number of *inversions* of the word, the pairs $a < b$ with $j_a > j_b$. If there are none, the word is an ordered monomial. Otherwise some adjacent pair satisfies $j_a > j_{a+1}$, and (8.3) rewrites the word as

$$\iota(x_{j_1}) \cdots \iota(x_{j_{a+1}})\, \iota(x_{j_a}) \cdots \iota(x_{j_l}) + \iota(x_{j_1}) \cdots \iota([x_{j_a}, x_{j_{a+1}}]) \cdots \iota(x_{j_l})$$

The first word has the same length and exactly one inversion fewer, since swapping an adjacent pair changes the relative order of that pair alone. The second term, after expanding the bracket in the basis, is a linear combination of words of length $l-1$, which the induction on length handles. Both inductions terminate, so U_k is spanned as claimed, and letting k run, the ordered monomials span $U(\mathfrak{g}) = \bigcup_k U_k$.

Step 2: the ordered monomials are linearly independent. Let $S = \mathbb{C}[z_1, \dots, z_n]$ carry the representation of lemma 8.6.8, and let $\Phi\colon U(\mathfrak{g}) \to \mathrm{End}(S)$ be the algebra homomorphism attached to it by corollary 8.6.3, so that $\Phi(\iota(X))\,p = X \cdot p$. We claim that

$$\Phi(x_I)(1) = z_I$$

for every weakly increasing I, by induction on $\ell(I)$. For $I = \emptyset$ both sides are 1. For $I = (i_1, I')$ with $I' = (i_2 \le \cdots \le i_m)$, multiplicativity of Φ and the inductive hypothesis give $\Phi(x_I)(1) = x_{i_1} \cdot \big(\Phi(x_{I'})(1)\big) = x_{i_1} \cdot z_{I'}$, and since $i_1 \le I'$, property (1) of the lemma evaluates this as $z_{i_1} z_{I'} = z_I$. Now suppose $\sum_I c_I x_I = 0$ is a finite linear relation among ordered monomials. Applying Φ and evaluating at 1 yields $\sum_I c_I z_I = 0$ in S. Distinct weakly increasing multi-indices give distinct commutative monomials z_I, and the monomials are linearly independent in the polynomial algebra, so every $c_I = 0$.

Steps 1 and 2 together make the ordered monomials with $\ell(I) \le k$ a basis of U_k for each k, and every finite set of ordered monomials lies in some U_k, so the full family is a basis of $U(\mathfrak{g})$, completing the proof.

Nothing in the section so far has used any property of $\mathbb{C}$ beyond its being a field: the construction of $U(\mathfrak{g})$, the universal property, and both halves of the Poincaré-Birkhoff-Witt proof are valid verbatim over any field. The generality matters, because Ado's theorem (theorem 3.4.3) concerns real Lie algebras. The proof in section 8.9 runs over $\mathbb{C}$ and descends to $\mathbb{R}$ by complexification, and exercise 8.9.1 (5) reruns it over an arbitrary field of characteristic zero, leaning on exactly this field-independence. A concrete computation shows the theorem in action.

Example 8.6.9: Ordered Monomials in $U(\mathfrak{sl}_2(\mathbb{C}))$

Let $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ with the ordered basis (f, h, e), chosen with the lowering operator first to match the conventions of Part III, and brackets $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$ (example 6.4.2). Writing elements of $U(\mathfrak{g})$ without ι, the ordered monomials are $f^a h^b e^c$ with $a, b, c \ge 0$, and theorem 8.6.7 says these form a basis. The filtration piece U_2, for instance, has as basis the ten monomials: the constant 1, then f, h, e, then $f^2, fh, fe, h^2, he, e^2$, matching the count $\binom{3+2}{2} = 10$. Any product reduces to the basis by repeated use of (8.3). For instance $ef = fe + [e, f] = fe + h$ is the reduction of the single out-of-order word of length two in e and f, and iterating,

$$ef^2 = (ef)f = (fe + h)f = f(ef) + hf = f(fe + h) + fh + [h, f] = f^2 e + 2fh - 2f$$

an identity in $U(\mathfrak{g})$ whose image under any representation φ is the operator identity $\varphi(e)\varphi(f)^2 = \varphi(f)^2\varphi(e) + 2\varphi(f)\varphi(h) - 2\varphi(f)$, the $k = 2$ case of the commutation formula for $\varphi(e)$ against powers of the lowering operator that the $\mathfrak{sl}_2$ theory of section 7.4 uses. Computations that previously had to be performed inside each representation separately are now single identities in one algebra, proved once.

The corollaries of the theorem are what the rest of the book uses. The first discharges the warning attached to definition 8.6.1.

Corollary 8.6.10: The Lie Algebra Embeds in Its Enveloping Algebra

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{C}$. The canonical map $\iota\colon \mathfrak{g} \to U(\mathfrak{g})$ is injective, and $U_1 = \mathbb{C}1 \oplus \iota(\mathfrak{g})$.

***Proof*:**
The ordered monomials of length at most one are $1, \iota(x_1), \dots, \iota(x_n)$, and by theorem 8.6.7 they form a basis of U_1, giving the direct sum decomposition. If $X = \sum_i c_i x_i$ satisfies $\iota(X) = \sum_i c_i \iota(x_i) = 0$, the independence forces every $c_i = 0$, so $X = 0$ and ι is injective, completing the proof.

From here on, $\mathfrak{g}$ is identified with its image $\iota(\mathfrak{g}) \subseteq U(\mathfrak{g})$ and the map ι is suppressed, so that (8.3) reads $XY - YX = [X, Y]$ for $X, Y \in \mathfrak{g}$. The identification is compatible with subalgebras in the strongest sense.

Corollary 8.6.11: Enveloping Algebras of Subalgebras

Let $\mathfrak{h} \subseteq \mathfrak{g}$ be a Lie subalgebra of a finite-dimensional Lie algebra over $\mathbb{C}$. The algebra homomorphism $U(\mathfrak{h}) \to U(\mathfrak{g})$ induced by the inclusion (proposition 8.6.4) is injective. Concretely, if an ordered basis of $\mathfrak{h}$ appears as a block of consecutive entries in an ordered basis $x_1, \dots, x_n$ of $\mathfrak{g}$ (an initial segment, a middle block, or a final segment: nothing below uses more than the relative order within the block), the map carries the Poincaré-Birkhoff-Witt basis of $U(\mathfrak{h})$ to the ordered monomials of $U(\mathfrak{g})$ with entries in that block, and $U(\mathfrak{h})$ is thereby identified with their span inside $U(\mathfrak{g})$.

***Proof*:**
The induced map sends $\iota_{\mathfrak{h}}(X)$ to $\iota_{\mathfrak{g}}(X)$ for $X \in \mathfrak{h}$, and is multiplicative, so it sends the ordered monomial of $U(\mathfrak{h})$ attached to a weakly increasing multi-index I with entries in $\{1, \dots, m\}$ to the ordered monomial of $U(\mathfrak{g})$ attached to the same I, which is legitimate because the extension preserves the order of the first m basis vectors. By theorem 8.6.7 applied to $\mathfrak{h}$, the source monomials are a basis of $U(\mathfrak{h})$, and by the same theorem applied to $\mathfrak{g}$ their images are distinct members of a basis of $U(\mathfrak{g})$, hence linearly independent. A linear map carrying a basis injectively into a linearly independent family is injective, completing the proof.

The final corollary is the form in which the highest-weight theory of Part III uses everything above, and it deserves its own statement. A choice of Cartan subalgebra and positive system splits a semisimple algebra into three subalgebras, and the enveloping algebra factors accordingly.

Corollary 8.6.12: The Triangular Decomposition of the Enveloping Algebra

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root system Φ, and positive system Φ^+ attached to a base (definition 8.2.1, definition 8.2.2). Set

$$\mathfrak{n}^+ = \bigoplus_{\alpha\in\Phi^+} \mathfrak{g}_\alpha, \qquad \mathfrak{n}^- = \bigoplus_{\alpha\in\Phi^+} \mathfrak{g}_{-\alpha}$$

Then:

1. $\mathfrak{n}^+$ and $\mathfrak{n}^-$ are Lie subalgebras of $\mathfrak{g}$, and as vector spaces
$$\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+$$
the *triangular decomposition* of $\mathfrak{g}$, and
2. the multiplication map
$$U(\mathfrak{n}^-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}^+) \longrightarrow U(\mathfrak{g}), \qquad u \otimes v \otimes w \longmapsto uvw$$
is an isomorphism of vector spaces. In particular $U(\mathfrak{g}) = U(\mathfrak{n}^-)\,U(\mathfrak{h})\,U(\mathfrak{n}^+)$.

***Proof*:**
(1) The root space decomposition (theorem 7.2.4) gives $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$, and $\Phi = \Phi^+ \sqcup \Phi^-$ with $\Phi^- = -\Phi^+$ (definition 8.2.2), so grouping the root spaces by sign yields the direct sum in (1). For the subalgebra claim, let $\alpha, \beta \in \Phi^+$. By proposition 7.2.5, $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$. The expansion of $\alpha + \beta$ in the base has all coefficients nonnegative and not all zero, being the sum of two such expansions, so $\alpha + \beta \neq 0$. If $\alpha + \beta$ is a root it is a positive root, and otherwise $\mathfrak{g}_{\alpha+\beta} = 0$. Either way the bracket lands in $\mathfrak{n}^+$, so $\mathfrak{n}^+$ is a subalgebra, and the argument for $\mathfrak{n}^-$ is identical with signs reversed.

(2) Choose a basis $y_1, \dots, y_p$ of $\mathfrak{n}^-$, a basis $h_1, \dots, h_l$ of $\mathfrak{h}$, and a basis $x_1, \dots, x_q$ of $\mathfrak{n}^+$, and order the concatenation

$$y_1 < \cdots < y_p < h_1 < \cdots < h_l < x_1 < \cdots < x_q$$

as an ordered basis of $\mathfrak{g}$, using (1). A Poincaré-Birkhoff-Witt monomial of $U(\mathfrak{g})$ for this order is precisely a product $M^- M^0 M^+$, where M^- is an ordered monomial in the y's, M^0 in the h's, and M^+ in the x's, because weakly increasing multi-indices list all y-entries first, then all h-entries, then all x-entries. By corollary 8.6.11 the factors M^-, M^0, M^+ run through the Poincaré-Birkhoff-Witt bases of $U(\mathfrak{n}^-)$, $U(\mathfrak{h})$, $U(\mathfrak{n}^+)$ inside $U(\mathfrak{g})$. The elementary tensors $M^- \otimes M^0 \otimes M^+$ form a basis of the tensor product ([2, Tensor Products of Free Modules], applied twice through the associativity of the tensor product), and the multiplication map, which is linear, carries this basis bijectively onto the Poincaré-Birkhoff-Witt basis of $U(\mathfrak{g})$ (theorem 8.6.7). A linear map carrying a basis bijectively onto a basis is an isomorphism, completing the proof.

The decomposition in (2) is an isomorphism of vector spaces and nothing more: the three factors do not commute with one another, and the content of the statement is that every element of $U(\mathfrak{g})$ is uniquely a sum of products with all lowering factors on the left, all Cartan factors in the middle, and all raising factors on the right. That normal form is the exact shape of the highest-weight

argument of Part III. Applied to a vector annihilated by $\mathfrak{n}^+$ and scaled by $\mathfrak{h}$, the factors M^0M^+ act by scalars and the submodule the vector generates is swept out by the lowering monomials M^- alone, which is the step the proof of theorem 12.1.3 takes on the complexification, with $\mathfrak{t}_{\mathbb{C}}$ in the role of $\mathfrak{h}$. The exercises carry this out. The same circle of ideas discharges obligations recorded earlier in the book. The existence half of the Serre presentation (theorem 8.5.1) analyzes the algebra on generators e_i, f_i, h_i through the free Lie algebra inside a tensor algebra, an analysis that begins with precisely the tools of this section and occupies section 8.7 next, and Ado's theorem (theorem 3.4.3) embeds an arbitrary Lie algebra into matrices by locating a finite-dimensional quotient of a $U(\mathfrak{g})$-module, so both stand downstream of the Poincaré-Birkhoff-Witt theorem. With the enveloping algebra in place, the last debt of the classification comes due, and the next section pays it.

Exercise 8.6.1

1. In $U(\mathfrak{sl}_2(\mathbb{C}))$ with the ordered basis (f, h, e) of example 8.6.9, reduce eh, e^2f, and hef to linear combinations of ordered monomials $f^a h^b e^c$.

2. Let $\Omega = ef + fe + \frac{1}{2}h^2 \in U(\mathfrak{sl}_2(\mathbb{C}))$. Write Ω in the ordered basis of example 8.6.9, prove that Ω is central in $U(\mathfrak{sl}_2(\mathbb{C}))$, and deduce that under every representation φ, with extension Φ as in corollary 8.6.3, the operator $\Phi(\Omega)$ commutes with the whole action, the mechanism of the Casimir element of definition 6.4.4. Compute the scalar by which Ω acts on the standard representation $\mathbb{C}^2$.

3. Let $\mathfrak{g}$ be n-dimensional. Using theorem 8.6.7, show that $\dim U_k = \binom{n+k}{k}$, and that the images of $u \in U_j$ and $v \in U_k$ in the quotients U_j/U_{j-1} and U_k/U_{k-1} multiply commutatively into U_{j+k}/U_{j+k-1}. Conclude that the associated graded algebra $\operatorname{gr} U(\mathfrak{g}) = \bigoplus_{k\geq 0} U_k/U_{k-1}$ (with $U_{-1} = 0$) is isomorphic to the polynomial algebra $\mathbb{C}[z_1, \dots, z_n]$, so that $U(\mathfrak{g})$ deforms the polynomial algebra without changing its size.

4. Show that there is a unique linear map $\tau\colon U(\mathfrak{g}) \to U(\mathfrak{g})$ with $\tau(1) = 1$, $\tau(X) = -X$ for $X \in \mathfrak{g}$, and $\tau(uv) = \tau(v)\tau(u)$ for all u, v, and that $\tau^2 = \mathrm{id}$. Compute τ on an ordered monomial. (Consider the opposite algebra $U(\mathfrak{g})^{\mathrm{op}}$, with the same underlying vector space and multiplication $u * v := vu$, and apply theorem 8.6.2.)

5. Let $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+$ be as in corollary 8.6.12, let V be a representation of $\mathfrak{g}$, and let $v \in V$ satisfy $X \cdot v = 0$ for all $X \in \mathfrak{n}^+$ and $H \cdot v = \lambda(H)\, v$ for all $H \in \mathfrak{h}$, for some $\lambda \in \mathfrak{h}^*$. Show that $U(\mathfrak{g})\, v = U(\mathfrak{n}^-)\, v$, so that the submodule generated by v is the span of the vectors obtained from v by lowering operators alone. Explain where this is used in the proof of theorem 12.1.3.

8.7 The Serre Construction: Existence

One debt remains from the classification. Section 8.5 proved the uniqueness half of theorem 8.5.4 in full, but the forward half of theorem 8.5.1, that the algebra $\mathfrak{g}(A)$ presented by a Cartan matrix A is finite-dimensional and semisimple with root system matching A, was only stated there, its proof postponed to the present section because the argument runs through the enveloping algebra. That tool is now in place, and this section settles the debt: it constructs $\mathfrak{g}(A)$ from the ground up and proves every claim the classification makes about it.

The construction proceeds in three stages, followed by a finiteness argument. The first stage manufactures the roomiest possible algebra on the $3l$ generators, the free Lie algebra, in which no

identity holds beyond those forced by bilinearity, antisymmetry, and the Jacobi identity. The second stage imposes only the *linear* relations (1) through (3) of theorem 8.5.1, producing an auxiliary algebra $\tilde{\mathfrak{g}}(A)$, and studies it through an explicit representation on a tensor algebra. The representation shows that the auxiliary algebra is no smaller than intended: the h_i span an l-dimensional abelian subalgebra, and $\tilde{\mathfrak{g}}(A)$ splits into a triangular sum exactly as a semisimple algebra does. The third stage imposes the Serre relations (4) and proves the decisive structural fact: the ideal they generate splits into its weight spaces and misses the Cartan subalgebra entirely, so the quotient $\mathfrak{g}(A)$ keeps the triangular shape. Finiteness then comes from symmetry. The operators $\mathrm{ad}e_i$ and $\mathrm{ad}f_i$ become locally nilpotent in $\mathfrak{g}(A)$, their exponentials realize the simple reflections as automorphisms, the dimensions of the weight spaces become invariants of the Weyl group acting on the root lattice, and the finiteness of the root system forces the finiteness of the algebra.

Throughout, fix an abstract root system Φ (definition 8.1.1) in the Euclidean space E, with base $\Delta = \{\alpha_1, \dots, \alpha_l\}$ (definition 8.2.1) and Cartan matrix $A = (A_{ij})$, where $A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle$ (definition 8.3.1). Write

$$Q = \mathbb{Z}\alpha_1 \oplus \cdots \oplus \mathbb{Z}\alpha_l \subseteq E$$

for the *root lattice* of integer combinations of the simple roots, which contains Φ by definition 8.2.1, and extend the height of definition 8.2.2 to all of Q by $\mathrm{ht}(\sum_i c_i\alpha_i) = \sum_i c_i$. Call an element of Q *sign-coherent* if its coefficients are all ≥ 0 or all ≤ 0, so that every root is sign-coherent. For $\lambda \in Q$ the pairings $\langle \lambda, \alpha_i^\vee \rangle$ are integers, being integer combinations of the Cartan integers $\langle \alpha_j, \alpha_i^\vee \rangle = A_{ji}$. Three small facts about generation in an arbitrary Lie algebra are used constantly below, and they are collected once.

Lemma 8.7.1: Generation by Iterated Brackets

Let $\mathfrak{g}$ be a Lie algebra over a field and $S \subseteq \mathfrak{g}$ a subset. Call an *iterated bracket* of S any element of S or any bracket $[u, v]$ of two iterated brackets of S.

1. The Lie subalgebra of $\mathfrak{g}$ generated by S is the linear span of the iterated brackets of S.
2. The ideal of $\mathfrak{g}$ generated by S is the linear span of the elements $[x_1, [x_2, [\cdots [x_m, s] \cdots]]]$ with $m \geq 0$, $x_1, \dots, x_m \in \mathfrak{g}$, and $s \in S$.
3. If S generates $\mathfrak{g}$ as a Lie algebra and $M \subseteq \mathfrak{g}$ is a subspace with $[s, M] \subseteq M$ for every $s \in S$, then M is an ideal of $\mathfrak{g}$.

***Proof*:**

(1) The span of the iterated brackets contains S, and it is closed under the bracket: the bracket of two spanning elements is again an iterated bracket by definition, and bilinearity extends this to the span. So the span is a subalgebra containing S. Conversely any subalgebra containing S contains every iterated bracket, by induction on the number of bracketings, hence contains the span. The span is therefore the smallest such subalgebra.

(2) Write M_0 for the span of the displayed elements. Bracketing on the left with any $y \in \mathfrak{g}$ carries a spanning element to another spanning element, so $[\mathfrak{g}, M_0] \subseteq M_0$ and M_0 is an ideal (definition 5.1.1), and it contains S as the case $m = 0$. Conversely an ideal containing S contains every displayed element, by induction on m, so M_0 is the smallest one.

(3) Set $N = \{x \in \mathfrak{g} \mid [x, M] \subseteq M\}$, a linear subspace containing S by hypothesis. It is a subalgebra: for $x, y \in N$ and $v \in M$ the Jacobi identity gives $[[x, y], v] = [x, [y, v]] - [y, [x, v]] \in M$.

By (1) applied to the generating set S, the subalgebra N contains all of $\mathfrak{g}$, so $[\mathfrak{g}, M] \subseteq M$ and M is an ideal, completing the proof.

The first stage of the construction is the algebra with no relations at all. The raw material is the same tensor algebra that built the enveloping algebra in section 8.6, used now not as a quotient but as an ambient container: inside any associative algebra the commutator is a Lie bracket, and the free Lie algebra is carved out of the tensor algebra as the smallest Lie subalgebra containing the generators.

Definition 8.7.2: The Free Lie Algebra

Let X be a finite set, let V_X be the $\mathbb{C}$-vector space with basis X, and let $\mathcal{T}(V_X)$ be its tensor algebra ([2, Tensor Algebra, $\mathcal{T}(M)$ is an R-algebra]). The commutator $[a, b] = ab - ba$ makes the associative algebra $\mathcal{T}(V_X)$ a Lie algebra: bilinearity and $[a, a] = 0$ are immediate, and the Jacobi identity expands into twelve associative products cancelling in pairs, exactly as for the matrix algebras of definition 1.4.5. The *free Lie algebra* on X is the Lie subalgebra

$$\mathfrak{L}(X) \subseteq \big(\mathcal{T}(V_X), [\cdot, \cdot]\big)$$

generated by the subset $X \subseteq V_X = \mathcal{T}^1(V_X)$.

By lemma 8.7.1(1), $\mathfrak{L}(X)$ is the span of the iterated commutators of the elements of X. The name is earned by a mapping property: a map of the generators into a Lie algebra extends to a homomorphism, uniquely, whenever the target can receive one at all. The proof is the first place the work of section 8.6 pays off in this section: for a target sitting inside an associative algebra the tensor algebra's own universal property does everything, and an arbitrary finite-dimensional target is placed inside an associative algebra by the Poincaré-Birkhoff-Witt embedding into its enveloping algebra.

Theorem 8.7.3: The Universal Property of the Free Lie Algebra

Let X be a finite set and let $\mathfrak{h}$ be a Lie algebra over $\mathbb{C}$ satisfying either of the following:

1. $\mathfrak{h}$ is a Lie subalgebra of an associative unital $\mathbb{C}$-algebra B equipped with the commutator bracket, or
2. $\mathfrak{h}$ is finite-dimensional.

Then every map of sets $f\colon X \to \mathfrak{h}$ extends to a unique homomorphism of Lie algebras $F\colon \mathfrak{L}(X) \to \mathfrak{h}$.

***Proof*:**
Uniqueness, common to both cases: two homomorphisms agreeing on X agree on every iterated bracket of X by induction on the number of bracketings, and these span $\mathfrak{L}(X)$ by lemma 8.7.1(1).

Case (1). Extend f linearly to $V_X \to B$ and apply the universal property of the tensor algebra ([2, $\mathcal{T}(M)$ is an R-algebra]) to obtain a homomorphism of unital algebras $\Phi\colon \mathcal{T}(V_X) \to B$ with $\Phi|_{V_X} = f$. An algebra homomorphism preserves commutators, $\Phi(ab-ba) = \Phi(a)\Phi(b) - \Phi(b)\Phi(a)$, so Φ is a homomorphism for the commutator Lie brackets on both sides. By induction on bracketings, Φ carries an iterated commutator of elements of X to the same iterated commutator of their images $f(x) \in \mathfrak{h}$, which lies in $\mathfrak{h}$ because $\mathfrak{h}$ is a Lie subalgebra of B. Hence the restriction

$F := \Phi|_{\mathfrak{L}(X)}$ lands in $\mathfrak{h}$ and is the required Lie algebra homomorphism.

Case (2). By corollary 8.6.10 the canonical map $\iota\colon \mathfrak{h} \to U(\mathfrak{h})$ is injective, and by (8.3) its image $\iota(\mathfrak{h})$ is a Lie subalgebra of $U(\mathfrak{h})$ under the commutator, with ι an isomorphism onto it. Case (1) applied to $\iota \circ f$ and $B = U(\mathfrak{h})$ produces a Lie homomorphism $F_1\colon \mathfrak{L}(X) \to \iota(\mathfrak{h})$, and $F := \iota^{-1} \circ F_1$ is the required extension, completing the proof.

With the free algebra in hand, "generators and relations" acquires its precise meaning: a presented algebra is the quotient of the free Lie algebra by the ideal generated by the relation elements. The second stage imposes the linear relations only.

Definition 8.7.4: The Auxiliary Algebra and the Serre Algebra

Let A be the Cartan matrix of Φ relative to Δ, and let $\mathfrak{L} = \mathfrak{L}(X)$ be the free Lie algebra on the $3l$ symbols $X = \{\tilde{e}_i, \tilde{f}_i, \tilde{h}_i \mid 1 \le i \le l\}$.

1. The *auxiliary algebra* $\tilde{\mathfrak{g}}(A)$ is the quotient of $\mathfrak{L}$ by the ideal I generated by the elements of the linear relations (1) through (3) of theorem 8.5.1,

$$[\tilde{h}_i, \tilde{h}_j], \qquad [\tilde{h}_i, \tilde{e}_j] - A_{ji}\,\tilde{e}_j, \qquad [\tilde{h}_i, \tilde{f}_j] + A_{ji}\,\tilde{f}_j, \qquad [\tilde{e}_i, \tilde{f}_j] - \delta_{ij}\,\tilde{h}_i$$

   for $1 \le i, j \le l$. Write $e_i, f_i, h_i \in \tilde{\mathfrak{g}}(A)$ for the images of the generators, $\tilde{\mathfrak{h}} = \operatorname{span}_{\mathbb{C}}\{h_1, \dots, h_l\}$, and $\tilde{\mathfrak{n}}^+$, $\tilde{\mathfrak{n}}^-$ for the Lie subalgebras of $\tilde{\mathfrak{g}}(A)$ generated by $e_1, \dots, e_l$ and by $f_1, \dots, f_l$ respectively.

2. The *Serre elements* of $\tilde{\mathfrak{g}}(A)$ are

$$\theta^+_{ij} = (\mathrm{ad} e_i)^{1-A_{ji}}(e_j), \qquad \theta^-_{ij} = (\mathrm{ad} f_i)^{1-A_{ji}}(f_j) \qquad (i \neq j)$$

   and the *Serre algebra* $\mathfrak{g}(A)$ is the quotient of $\tilde{\mathfrak{g}}(A)$ by the ideal K generated by all the Serre elements. The images of e_i, f_i, h_i in $\mathfrak{g}(A)$ are written with the same letters.

The definition packages the presentation of theorem 8.5.1 in two stages, linear relations first and the Serre relations (4) second. The next proposition records that the two-stage quotient is the presented algebra of that theorem and spells out what a homomorphism out of it amounts to. This mapping property is also what permits the surjections $\pi\colon \mathfrak{g}(A) \twoheadrightarrow \mathfrak{g}$ in Step 5 of the proof of theorem 8.5.4.

Proposition 8.7.5: Mapping Out of the Presented Algebras

Let $\mathfrak{h}$ be a Lie algebra over $\mathbb{C}$ as in theorem 8.7.3, and let $e'_i, f'_i, h'_i \in \mathfrak{h}$ $(1 \le i \le l)$ be given.

1. If these elements satisfy relations (1) through (3) of theorem 8.5.1, there is a unique homomorphism of Lie algebras $\tilde{\mathfrak{g}}(A) \to \mathfrak{h}$ with $e_i \mapsto e'_i$, $f_i \mapsto f'_i$, $h_i \mapsto h'_i$.

2. If they satisfy relations (1) through (4), the homomorphism of (1) kills K and induces a unique homomorphism $\mathfrak{g}(A) \to \mathfrak{h}$ with the same effect on the generators.

Moreover $\mathfrak{g}(A)$ is canonically isomorphic to the quotient of $\mathfrak{L}$ by the ideal generated by the elements of relations (1) through (4) all at once, so it is the Lie algebra presented by the generators e_i, f_i, h_i and the Serre relations, exactly as in theorem 8.5.1.

***Proof*:**
(1) By theorem 8.7.3 the assignment $\tilde{e}_i \mapsto e'_i$, $\tilde{f}_i \mapsto f'_i$, $\tilde{h}_i \mapsto h'_i$ extends to a homomorphism $F\colon \mathfrak{L} \to \mathfrak{h}$. By hypothesis F sends each generator of the ideal I to zero. By lemma 8.7.1(2) a general element of I is a combination of elements $[x_1, [\cdots [x_m, r]\cdots]]$ with r a generator, and F of such an element is $[F(x_1), [\cdots [F(x_m), F(r)]\cdots]] = 0$. So $I \subseteq \ker F$ and F descends to the quotient $\tilde{\mathfrak{g}}(A) = \mathfrak{L}/I$ (definition 5.1.1), with the stated effect on generators. Uniqueness holds because the generators generate.

(2) A homomorphism commutes with ad-powers, so the map of (1) sends θ^+_{ij} to $(\mathrm{ad}e'_i)^{1-A_{ji}}(e'_j) = 0$ and likewise θ^-_{ij} to zero, by relation (4) in $\mathfrak{h}$. It therefore kills the generators of K, hence K itself by lemma 8.7.1(2) again, and descends to $\mathfrak{g}(A)$.

For the final claim let $J \subseteq \mathfrak{L}$ be the ideal generated by the elements of relations (1) through (4), where the relation-(4) elements of $\mathfrak{L}$ are $(\mathrm{ad}\tilde{e}_i)^{1-A_{ji}}(\tilde{e}_j)$ and $(\mathrm{ad}\tilde{f}_i)^{1-A_{ji}}(\tilde{f}_j)$, whose images in $\tilde{\mathfrak{g}}(A)$ are the Serre elements. The composite $\mathfrak{L} \to \tilde{\mathfrak{g}}(A) \to \mathfrak{g}(A)$ kills every generator of J, hence J, and induces $\varphi\colon \mathfrak{L}/J \to \mathfrak{g}(A)$. In the other direction $\mathfrak{L} \to \mathfrak{L}/J$ kills the generators of I, so it induces a map $\tilde{\mathfrak{g}}(A) \to \mathfrak{L}/J$, which kills the Serre elements (their preimages lie in J) and induces $\psi\colon \mathfrak{g}(A) \to \mathfrak{L}/J$. Both composites $\varphi \circ \psi$ and $\psi \circ \varphi$ fix the images of the generators, which generate, so both are the identity, completing the proof.

Nothing so far shows the relations have not collapsed everything: for all that has been proved, $\tilde{\mathfrak{g}}(A)$ could be zero. Before building the representation that rules this out, it is worth recording the bookkeeping that organizes every argument to come: the algebra is graded by the root lattice, because every defining relation is homogeneous.

Proposition 8.7.6: The Root-Lattice Grading

Assign to the generators the degrees $\deg \tilde{e}_i = \alpha_i$, $\deg \tilde{f}_i = -\alpha_i$, and $\deg \tilde{h}_i = 0$ in the lattice Q.

1. There are direct sum decompositions

$$\mathfrak{L} = \bigoplus_{\lambda \in Q} \mathfrak{L}_\lambda, \qquad \tilde{\mathfrak{g}}(A) = \bigoplus_{\lambda \in Q} \tilde{\mathfrak{g}}(A)_\lambda, \qquad \mathfrak{g}(A) = \bigoplus_{\lambda \in Q} \mathfrak{g}(A)_\lambda$$

   in which the piece of degree λ is spanned by the iterated brackets of generators whose degrees sum to λ, the bracket adds degrees, and both quotient maps preserve degrees. The ideals I and K are graded: each is the direct sum of its intersections with the homogeneous pieces.

2. In $\tilde{\mathfrak{g}}(A)$, and likewise in $\mathfrak{g}(A)$, each h_i acts on the piece of degree λ by the scalar $\langle \lambda, \alpha_i^\vee \rangle$: that is, $[h_i, x] = \langle \lambda, \alpha_i^\vee \rangle\, x$ for x of degree λ.

***Proof*:**
(1) The tensor algebra $\mathcal{T}(V_X)$ has basis the monomials $x_{j_1} \otimes \cdots \otimes x_{j_m}$ in the generators ([2, Tensor Products of Free Modules], applied in each tensor degree), and assigning such a monomial the sum of the degrees of its letters makes $\mathcal{T}(V_X) = \bigoplus_{\lambda \in Q} \mathcal{T}(V_X)_\lambda$ a decomposition in which multiplication, being concatenation on monomials, adds degrees. Commutators of homogeneous elements are then homogeneous. Every iterated bracket of generators is homogeneous, and $\mathfrak{L}$ is

the span of these (lemma 8.7.1(1)). A subspace spanned by homogeneous elements is the direct sum of its homogeneous parts, since grouping any spanning combination by degree exhibits each degree component as a subcombination of spanning elements. This gives the decomposition of $\mathfrak{L}$ with $\mathfrak{L}_\lambda = \mathfrak{L} \cap \mathcal{T}(V_X)_\lambda$, and $[\mathfrak{L}_\lambda, \mathfrak{L}_\mu] \subseteq \mathfrak{L}_{\lambda+\mu}$.

Each generator of I is homogeneous: the elements $[\tilde{h}_i, \tilde{h}_j]$ and $[\tilde{e}_i, \tilde{f}_i] - \tilde{h}_i$ have degree 0, the elements $[\tilde{h}_i, \tilde{e}_j] - A_{ji}\tilde{e}_j$ and $[\tilde{h}_i, \tilde{f}_j] + A_{ji}\tilde{f}_j$ have degrees $\pm\alpha_j$, and $[\tilde{e}_i, \tilde{f}_j]$ for $i \neq j$ has degree $\alpha_i - \alpha_j$. By lemma 8.7.1(2), I is spanned by elements $[x_1, [\cdots [x_m, r] \cdots]]$. Expanding each x_k into homogeneous components and using multilinearity writes every such element as a sum of homogeneous ones lying in I, so I is spanned by homogeneous elements and is graded. The quotient then inherits the grading: the images of the $\mathfrak{L}_\lambda$ span $\tilde{\mathfrak{g}}(A)$, and if a sum $\sum_\lambda v_\lambda$ of homogeneous elements lies in I, then each $v_\lambda \in I \cap \mathfrak{L}_\lambda$ by gradedness of I, so the degree components of classes are well defined and the sum of the images is direct. The Serre elements are homogeneous of degrees $\pm\big((1 - A_{ji})\alpha_i + \alpha_j\big)$, so the identical argument grades K and $\mathfrak{g}(A)$, and both quotient maps preserve degrees by construction.

(2) The claim holds for the generators by relations (1) through (3): $[h_i, e_j] = A_{ji}e_j = \langle \alpha_j, \alpha_i^\vee \rangle e_j$, $[h_i, f_j] = -A_{ji}f_j = \langle -\alpha_j, \alpha_i^\vee \rangle f_j$, and $[h_i, h_j] = 0$. If $[h_i, u] = \langle \lambda, \alpha_i^\vee \rangle u$ and $[h_i, v] = \langle \mu, \alpha_i^\vee \rangle v$, the Jacobi identity gives

$$[h_i, [u, v]] = [[h_i, u], v] + [u, [h_i, v]] = \langle \lambda + \mu, \alpha_i^\vee \rangle \, [u, v]$$

so the claim passes to iterated brackets by induction, and these span each homogeneous piece. The same computation applies verbatim in $\mathfrak{g}(A)$, completing the proof.

One piece of linear algebra converts the grading into a genuine eigenspace decomposition: distinct lattice elements must give distinct functionals, and that is the nonsingularity of the Cartan matrix.

Lemma 8.7.7: The Cartan Matrix Is Nonsingular

Let Φ be a root system in E with base $\Delta = \{\alpha_1, \ldots, \alpha_l\}$. The linear map

$$E \longrightarrow \mathbb{R}^l, \qquad v \longmapsto \big(\langle v, \alpha_1^\vee \rangle, \ldots, \langle v, \alpha_l^\vee \rangle\big)$$

is an isomorphism. In particular the Cartan matrix A, whose j-th row $(A_{j1}, \ldots, A_{jl})$ is the image of α_j, is invertible, and distinct elements of Q have distinct images.

***Proof*:**
The space E has dimension l, since Δ is a basis (definition 8.2.1). If v lies in the kernel then $\langle v, \alpha_i^\vee \rangle = 2(v, \alpha_i)/(\alpha_i, \alpha_i) = 0$ for every i, so v is orthogonal to each α_i, hence to their span E, and $v = 0$ because the inner product is positive definite. An injective linear map between spaces of the same finite dimension is an isomorphism. The rows of A are the images of the basis Δ under an isomorphism, hence linearly independent, so A is invertible, completing the proof.

The representation that shows $\tilde{\mathfrak{g}}(A)$ is large enough acts on the tensor algebra of an l-dimensional space. The lowering generators act by left multiplication, the Cartan generators act diagonally, and the raising generators are defined by the recursion that relation (3) forces.

Proposition 8.7.8: The Tensor-Algebra Representation

Let V be an l-dimensional $\mathbb{C}$-vector space with basis $v_1, \dots, v_l$ and let $\mathcal{T} = \mathcal{T}(V)$ be its tensor algebra, with basis the monomials $v_{j_1} \otimes \cdots \otimes v_{j_t}$ for $t \geq 0$, the empty monomial being 1. There is a representation $\rho\colon \tilde{\mathfrak{g}}(A) \to \mathfrak{gl}(\mathcal{T})$ (definition 6.4.1) whose values on the generators are determined by: for every monomial w,

$$\rho(f_i)(w) = v_i \otimes w$$

$$\rho(h_i)(1) = 0, \qquad \rho(h_i)(v_{j_1} \otimes \cdots \otimes v_{j_t}) = -\big(A_{j_1 i} + \cdots + A_{j_t i}\big)\, v_{j_1} \otimes \cdots \otimes v_{j_t}$$

$$\rho(e_i)(1) = 0, \qquad \rho(e_i)(v_j \otimes w) = v_j \otimes \rho(e_i)(w) + \delta_{ij}\, \rho(h_i)(w)$$

Consequently $h_1, \dots, h_l$ are linearly independent in $\tilde{\mathfrak{g}}(A)$, so $\dim \tilde{\mathfrak{h}} = l$, and every e_i, f_i, h_i is nonzero.

***Proof*:**

Step 1: the endomorphisms. Define $\hat{f}_i, \hat{h}_i \in \mathrm{End}(\mathcal{T})$ on the monomial basis by the first two displays and extend linearly. Define $\hat{e}_i$ by induction on tensor degree, as zero in degree 0 and by the third display in degree $t \geq 1$, where the splitting of a monomial of positive degree as $v_j \otimes w$ is unique. For bookkeeping assign to the monomial $w = v_{j_1} \otimes \cdots \otimes v_{j_t}$ the weight $\mathrm{wt}(w) = -(\alpha_{j_1} + \cdots + \alpha_{j_t}) \in Q$. Then $\hat{h}_i$ acts on w by the scalar $\langle \mathrm{wt}(w), \alpha_i^\vee \rangle$, since $\langle \alpha_{j_s}, \alpha_i^\vee \rangle = A_{j_s i}$.

Step 2: the linear relations hold in $\mathrm{End}(\mathcal{T})$. Both $\hat{h}_i$ and $\hat{h}_j$ are diagonal on the monomial basis, so $[\hat{h}_i, \hat{h}_j] = 0$. For the pair $(\hat{h}_i, \hat{f}_j)$, on a monomial w of weight $\mathrm{wt}(w)$ the monomial $v_j \otimes w$ has weight $\mathrm{wt}(w) - \alpha_j$, so

$$[\hat{h}_i, \hat{f}_j](w) = \langle \mathrm{wt}(w) - \alpha_j, \alpha_i^\vee \rangle\, v_j \otimes w - \langle \mathrm{wt}(w), \alpha_i^\vee \rangle\, v_j \otimes w = -A_{ji}\, \hat{f}_j(w)$$

For the pair $(\hat{e}_i, \hat{f}_j)$, the defining recursion gives, on every monomial w,

$$[\hat{e}_i, \hat{f}_j](w) = \hat{e}_i(v_j \otimes w) - v_j \otimes \hat{e}_i(w) = \delta_{ij}\, \hat{h}_i(w)$$

For the pair $(\hat{h}_i, \hat{e}_j)$, first claim that $\hat{e}_j$ raises weight by α_j: on a monomial w of degree t, the value $\hat{e}_j(w)$ lies in the $\hat{h}_i$-eigenspace of eigenvalue $\langle \mathrm{wt}(w) + \alpha_j, \alpha_i^\vee \rangle$ for every i. This holds vacuously in degree 0, where $\hat{e}_j(1) = 0$. For $w = v_k \otimes u$ with u of degree $t-1$, the term $v_k \otimes \hat{e}_j(u)$ has weight $\mathrm{wt}(u) + \alpha_j - \alpha_k = \mathrm{wt}(w) + \alpha_j$ by induction, and the term $\delta_{jk}\, \hat{h}_j(u)$, present only when $k = j$, is a multiple of u, of weight $\mathrm{wt}(u) = \mathrm{wt}(w) + \alpha_j$ as well. Granting the claim,

$$[\hat{h}_i, \hat{e}_j](w) = \big(\langle \mathrm{wt}(w) + \alpha_j, \alpha_i^\vee \rangle - \langle \mathrm{wt}(w), \alpha_i^\vee \rangle\big)\, \hat{e}_j(w) = A_{ji}\, \hat{e}_j(w)$$

So the endomorphisms $\hat{e}_i, \hat{f}_i, \hat{h}_i$ satisfy relations (1) through (3) in the Lie algebra $\mathfrak{gl}(\mathcal{T})$.

Step 3: the representation. The Lie algebra $\mathfrak{gl}(\mathcal{T}) = \mathrm{End}(\mathcal{T})$ with the commutator bracket is a Lie subalgebra of the associative algebra $\mathrm{End}(\mathcal{T})$, so proposition 8.7.5(1) applies to the elements $\hat{e}_i, \hat{f}_i, \hat{h}_i$ and produces a homomorphism $\rho\colon \tilde{\mathfrak{g}}(A) \to \mathfrak{gl}(\mathcal{T})$ with $\rho(e_i) = \hat{e}_i$, $\rho(f_i) = \hat{f}_i$, $\rho(h_i) = \hat{h}_i$, which is a representation in the sense of definition 6.4.1.

Step 4: consequences. Suppose $\sum_i c_i h_i = 0$ in $\tilde{\mathfrak{g}}(A)$. Applying ρ and evaluating on the degree-one monomial v_j gives $-\sum_i c_i A_{ji} = 0$ for every j, that is, $Ac = 0$ for the coefficient vector c, and

lemma 8.7.7 forces $c = 0$. So the h_i are linearly independent and $\dim \tilde{\mathfrak{h}} = l$, and in particular $h_i \neq 0$. Each $f_i \neq 0$ because $\rho(f_i)(1) = v_i \neq 0$, and each $e_i \neq 0$ because $[e_i, f_i] = h_i \neq 0$, completing the proof.

With the h_i now known to be a basis of $\tilde{\mathfrak{h}}$, each $\lambda \in Q$ defines a linear functional $\lambda(\cdot)$ on $\tilde{\mathfrak{h}}$ by $\lambda(h_i) = \langle \lambda, \alpha_i^\vee \rangle$, and proposition 8.7.6(2) now reads: $[h, x] = \lambda(h)\, x$ for all $h \in \tilde{\mathfrak{h}}$ and all x of degree λ. The same notation serves in $\mathfrak{g}(A)$. The auxiliary algebra now splits along the grading into a negative part, a Cartan part, and a positive part, exactly the triangular shape of corollary 8.6.12, though here nothing is known yet to be finite-dimensional.

Theorem 8.7.9: The Triangular Decomposition of the Auxiliary Algebra

With the notation of definition 8.7.4:

1. $\tilde{\mathfrak{h}}$ is an l-dimensional abelian subalgebra of $\tilde{\mathfrak{g}}(A)$, and as vector spaces
$$\tilde{\mathfrak{g}}(A) = \tilde{\mathfrak{n}}^- \oplus \tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+$$
2. In terms of the grading of proposition 8.7.6: $\tilde{\mathfrak{g}}(A)_0 = \tilde{\mathfrak{h}}$, the subalgebra $\tilde{\mathfrak{n}}^+$ is the sum of the pieces $\tilde{\mathfrak{g}}(A)_\lambda$ over the nonzero λ with all coefficients ≥ 0, and $\tilde{\mathfrak{n}}^-$ is the corresponding sum over λ with all coefficients ≤ 0. In particular $\tilde{\mathfrak{g}}(A)_\lambda = 0$ whenever λ is not sign-coherent.
3. $\tilde{\mathfrak{g}}(A)_{\alpha_i} = \mathbb{C}e_i$ and $\tilde{\mathfrak{g}}(A)_{-\alpha_i} = \mathbb{C}f_i$ are one-dimensional, and $\tilde{\mathfrak{g}}(A)_{k\alpha_i} = 0$ for every integer k with $|k| \geq 2$.

***Proof*:**

Step 1: the three subspaces and their degrees. The span $\tilde{\mathfrak{h}}$ is abelian by relation (1) and l-dimensional by proposition 8.7.8, and it lies in $\tilde{\mathfrak{g}}(A)_0$. By lemma 8.7.1(1) the subalgebra $\tilde{\mathfrak{n}}^+$ is the span of the iterated brackets of $e_1, \ldots, e_l$. Such a bracket with t letters is homogeneous of degree a sum of t simple roots, which is nonzero with all coefficients ≥ 0. Hence $\tilde{\mathfrak{n}}^+$ lies in the sum of the pieces of nonzero degree with nonnegative coefficients, and symmetrically for $\tilde{\mathfrak{n}}^-$. The three supports are disjoint subsets of Q, so by the directness of the grading the sum $\tilde{\mathfrak{n}}^- + \tilde{\mathfrak{h}} + \tilde{\mathfrak{n}}^+$ is direct.

Step 2: the sum is everything. Write $\mathfrak{m} = \tilde{\mathfrak{n}}^- \oplus \tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+$. The bracket of each generator carries $\mathfrak{m}$ into $\mathfrak{m}$. Indeed $[e_k, \tilde{\mathfrak{n}}^+] \subseteq \tilde{\mathfrak{n}}^+$ and $[f_k, \tilde{\mathfrak{n}}^-] \subseteq \tilde{\mathfrak{n}}^-$, since these are subalgebras containing the relevant generator. Each iterated e-bracket m is homogeneous, so $[h_k, m]$ is a scalar multiple of m by proposition 8.7.6(2), giving $[h_k, \tilde{\mathfrak{n}}^\pm] \subseteq \tilde{\mathfrak{n}}^\pm$ and $[h_k, \tilde{\mathfrak{h}}] = 0$. Relations (2) give $[e_k, \tilde{\mathfrak{h}}] \subseteq \mathbb{C}e_k$ and $[f_k, \tilde{\mathfrak{h}}] \subseteq \mathbb{C}f_k$. Finally $[e_k, \tilde{\mathfrak{n}}^-] \subseteq \tilde{\mathfrak{h}} + \tilde{\mathfrak{n}}^-$, by induction on iterated f-brackets: the base case is $[e_k, f_j] = \delta_{kj} h_k \in \tilde{\mathfrak{h}}$, and for $m = [u, v]$ with u, v shorter iterated f-brackets,
$$[e_k, [u, v]] = [[e_k, u], v] + [u, [e_k, v]] \in [\tilde{\mathfrak{h}} + \tilde{\mathfrak{n}}^-,\, \tilde{\mathfrak{n}}^-] + [\tilde{\mathfrak{n}}^-,\, \tilde{\mathfrak{h}} + \tilde{\mathfrak{n}}^-] \subseteq \tilde{\mathfrak{n}}^-$$
using the containments already established. Symmetrically $[f_k, \tilde{\mathfrak{n}}^+] \subseteq \tilde{\mathfrak{h}} + \tilde{\mathfrak{n}}^+$. Since the generators generate $\tilde{\mathfrak{g}}(A)$, lemma 8.7.1(3) makes $\mathfrak{m}$ an ideal, and then every iterated bracket of generators lies in $\mathfrak{m}$ by induction on bracketings, the base case being the generators themselves and the inductive step being $[\tilde{\mathfrak{g}}(A), \mathfrak{m}] \subseteq \mathfrak{m}$. Hence $\mathfrak{m} = \tilde{\mathfrak{g}}(A)$.

Step 3: reading off the pieces. Fix $\lambda \in Q$ and take the degree-λ component of the decomposition

$\tilde{\mathfrak{g}}(A) = \tilde{\mathfrak{n}}^- \oplus \tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+$. Every element of $\tilde{\mathfrak{n}}^+$ is a combination of homogeneous iterated e-brackets, so its degree components lie in $\tilde{\mathfrak{n}}^+$ again, and likewise for the other two summands. Therefore $\tilde{\mathfrak{g}}(A)_\lambda$ equals the span of the degree-λ iterated e-brackets when $\lambda \neq 0$ has all coefficients ≥ 0, the span of the degree-λ iterated f-brackets when $\lambda \neq 0$ has all coefficients ≤ 0, $\tilde{\mathfrak{h}}$ when $\lambda = 0$, and 0 when λ is not sign-coherent. This proves (2), and (1) follows since the three supports exhaust the grading.

Step 4: the multiples of a simple root. An iterated e-bracket of degree α_i has exactly one letter, so $\tilde{\mathfrak{g}}(A)_{\alpha_i} = \mathbb{C}e_i$, which is one-dimensional since $e_i \neq 0$ (proposition 8.7.8). For $k \geq 2$, an iterated e-bracket of degree $k\alpha_i$ has all letters equal to e_i, because the simple roots are linearly independent. Moreover every iterated bracket of $t \geq 2$ letters all equal to a single element x vanishes, by induction: for $t = 2$ it is $[x, x] = 0$, and for $t \geq 3$ it is $[u, v]$ with u, v shorter brackets in x alone, one of which has at least two letters and so vanishes by induction. Hence $\tilde{\mathfrak{g}}(A)_{k\alpha_i} = 0$. The statements for $-\alpha_i$ and $-k\alpha_i$ are symmetric, completing the proof.

The third stage imposes the Serre relations. The danger is that the ideal K generated by the Serre elements might reach into $\tilde{\mathfrak{h}}$ and collapse the Cartan subalgebra that the tensor-algebra representation just secured. The safeguard is a symmetry and a computation. The symmetry swaps the two halves of the presentation.

Proposition 8.7.10: The Chevalley Involution

There is a unique automorphism ω of $\tilde{\mathfrak{g}}(A)$ with

$$\omega(e_i) = f_i, \qquad \omega(f_i) = e_i, \qquad \omega(h_i) = -h_i$$

It satisfies $\omega^2 = \mathrm{id}$, carries $\tilde{\mathfrak{g}}(A)_\lambda$ onto $\tilde{\mathfrak{g}}(A)_{-\lambda}$, and exchanges θ^+_{ij} with θ^-_{ij}. It preserves K and so descends to an automorphism of $\mathfrak{g}(A)$ with the same properties.

***Proof*:**
The free Lie algebra $\mathfrak{L}$ is a Lie subalgebra of the associative algebra $\mathcal{T}(V_X)$, so theorem 8.7.3 applies with target $\mathfrak{L}$ itself: the assignment $\tilde{e}_i \mapsto \tilde{f}_i$, $\tilde{f}_i \mapsto \tilde{e}_i$, $\tilde{h}_i \mapsto -\tilde{h}_i$ extends to a unique homomorphism $\tilde{\omega}\colon \mathfrak{L} \to \mathfrak{L}$. The composite $\tilde{\omega}^2$ fixes every generator, so $\tilde{\omega}^2 = \mathrm{id}$ by the uniqueness clause, and $\tilde{\omega}$ is an automorphism. It sends each generator of the ideal I to ± 1 times a generator of I:

$$[\tilde{h}_i, \tilde{h}_j] \mapsto [\tilde{h}_i, \tilde{h}_j], \qquad [\tilde{h}_i, \tilde{e}_j] - A_{ji}\tilde{e}_j \mapsto -\big([\tilde{h}_i, \tilde{f}_j] + A_{ji}\tilde{f}_j\big)$$

$$[\tilde{h}_i, \tilde{f}_j] + A_{ji}\tilde{f}_j \mapsto -\big([\tilde{h}_i, \tilde{e}_j] - A_{ji}\tilde{e}_j\big), \qquad [\tilde{e}_i, \tilde{f}_j] - \delta_{ij}\tilde{h}_i \mapsto -\big([\tilde{e}_j, \tilde{f}_i] - \delta_{ji}\tilde{h}_j\big)$$

the last using $\delta_{ij}\tilde{h}_i = \delta_{ji}\tilde{h}_j$. By lemma 8.7.1(2) every element of I is a combination of bracket words on these generators, and $\tilde{\omega}$ of such a word is the corresponding word on the images, which lies in I. So $\tilde{\omega}(I) \subseteq I$, with equality since $\tilde{\omega}^2 = \mathrm{id}$, and $\tilde{\omega}$ descends to an automorphism ω of $\tilde{\mathfrak{g}}(A)$ with $\omega^2 = \mathrm{id}$ and the stated effect on generators. Uniqueness again holds because the generators generate. Since $\tilde{\omega}$ negates the degrees of the generators, it carries degree-λ iterated brackets to degree-$(-\lambda)$ ones, so $\omega(\tilde{\mathfrak{g}}(A)_\lambda) = \tilde{\mathfrak{g}}(A)_{-\lambda}$. Because ω is a homomorphism it commutes with ad-powers, so

$$\omega(\theta^+_{ij}) = (\mathrm{ad} f_i)^{1-A_{ji}}(f_j) = \theta^-_{ij}, \qquad \omega(\theta^-_{ij}) = \theta^+_{ij}$$

Hence ω permutes the generating set of K, so $\omega(K) = K$ by lemma 8.7.1(2) once more, and ω descends to $\mathfrak{g}(A) = \tilde{\mathfrak{g}}(A)/K$ with the same properties, completing the proof.

The computation is the heart of the third stage: each Serre element is a highest-weight vector for the $\mathfrak{sl}_2$-relations of its own index, killed not only by the raising operator of that index but by *all* the raising generators at once. This is what will keep the ideal K inside the two nilpotent halves.

Lemma 8.7.11: The Serre Elements Are Killed by the Opposite Generators

In $\tilde{\mathfrak{g}}(A)$, for all $i \neq j$ and every k,

$$[e_k, \theta_{ij}^-] = 0, \qquad [f_k, \theta_{ij}^+] = 0$$

***Proof*:**
Throughout put $m = 1 - A_{ji}$, a positive integer since $A_{ji} \leq 0$ for $i \neq j$ (proposition 8.3.2). The Jacobi identity of definition 1.4.5, rewritten, says that ad is a homomorphism: $\mathrm{ad}([x,y]) = \mathrm{ad}x \circ \mathrm{ad}y - \mathrm{ad}y \circ \mathrm{ad}x$ for all x, y. The first identity is proved in two cases, according to whether k equals i.

Case $k \neq i$. Relation (3) gives $[e_k, f_i] = 0$, so the operators $\mathrm{ad}e_k$ and $\mathrm{ad}f_i$ commute, and

$$[e_k, \theta_{ij}^-] = \mathrm{ad}e_k\,(\mathrm{ad}f_i)^m(f_j) = (\mathrm{ad}f_i)^m\big([e_k, f_j]\big)$$

If $k \neq j$ then $[e_k, f_j] = 0$ and the bracket vanishes. If $k = j$ then $[e_j, f_j] = h_j$, and

$$(\mathrm{ad}f_i)(h_j) = -[h_j, f_i] = A_{ij}\, f_i, \qquad (\mathrm{ad}f_i)^2(h_j) = A_{ij}\,[f_i, f_i] = 0$$

If $m \geq 2$ the power $(\mathrm{ad}f_i)^m(h_j)$ therefore vanishes. If $m = 1$ then $A_{ji} = 0$, hence $A_{ij} = 0$ by proposition 8.3.2(3), and already $(\mathrm{ad}f_i)(h_j) = 0$.

Case $k = i$. Write $E = \mathrm{ad}e_i$, $F = \mathrm{ad}f_i$, $H = \mathrm{ad}h_i$, operators on $\tilde{\mathfrak{g}}(A)$. Since ad is a homomorphism, relations (2) and (3) give $[E, F] = \mathrm{ad}([e_i, f_i]) = H$ and $[H, F] = \mathrm{ad}([h_i, f_i]) = -2F$, using $A_{ii} = 2$. Two operator identities follow by induction on $m \geq 1$. First $[H, F^m] = -2mF^m$: the case $m = 1$ is given, and

$$[H, F^{m+1}] = [H, F]F^m + F[H, F^m] = -2F^{m+1} - 2mF^{m+1}$$

Second, the commutation rule

$$EF^m = F^mE + mF^{m-1}\big(H - (m-1)\big)$$

whose case $m = 1$ is $EF = FE + H$. Granting it for m,

$$EF^{m+1} = (FE + H)F^m = F\big(F^mE + mF^{m-1}(H - (m-1))\big) + F^mH - 2mF^m$$

which collects to $F^{m+1}E + (m+1)F^mH - m(m+1)F^m$, the identity for $m+1$. Now apply the commutation rule to f_j. Relation (3) gives $E(f_j) = [e_i, f_j] = 0$ since $i \neq j$, and relation (2) gives $H(f_j) = [h_i, f_j] = -A_{ji}f_j = (m-1)f_j$. Hence

$$[e_i, \theta_{ij}^-] = EF^m(f_j) = F^mE(f_j) + mF^{m-1}\big(H - (m-1)\big)(f_j) = 0 + 0 = 0$$

This proves $[e_k, \theta_{ij}^-] = 0$ for every k. Applying the Chevalley involution, $[f_k, \theta_{ij}^+] = \omega\big([e_k, \theta_{ij}^-]\big) = \omega(0) = 0$ by proposition 8.7.10, completing the proof.

The lemma in hand, the ideal generated by the Serre elements can be pinned down exactly: bracketing a θ^- with raising generators gives zero, with Cartan generators gives scalar multiples, and with lowering generators stays inside the lowering half. So the ideal is no bigger than the span of the lowering words, and in particular it has no component in degree 0.

Theorem 8.7.12: The Serre Ideal Misses the Cartan Subalgebra

Let K^- (resp. K^+) be the ideal of $\tilde{\mathfrak{g}}(A)$ generated by the elements θ^-_{ij} (resp. θ^+_{ij}), so that $K = K^+ + K^-$: the sum of the two ideals is an ideal containing every Serre element and contained in K, and K is the smallest such ideal, so the two agree. Then:

1. K^- is the span of the elements $(\mathrm{ad} f_{k_1}) \cdots (\mathrm{ad} f_{k_r})\, \theta^-_{ij}$ with $r \geq 0$, and $K^- \subseteq \tilde{\mathfrak{n}}^-$. Symmetrically $K^+ = \omega(K^-) \subseteq \tilde{\mathfrak{n}}^+$.
2. K is graded, and its homogeneous pieces vanish outside the degrees λ with $\mathrm{ht}(\lambda) \geq 2$ or $\mathrm{ht}(\lambda) \leq -2$. In particular $K \cap \tilde{\mathfrak{h}} = 0$ and $K \cap \mathbb{C}e_i = K \cap \mathbb{C}f_i = 0$.
3. Consequently, writing $\mathfrak{h}$, $\mathfrak{n}^\pm$ for the images of $\tilde{\mathfrak{h}}$, $\tilde{\mathfrak{n}}^\pm$ in $\mathfrak{g}(A) = \tilde{\mathfrak{g}}(A)/K$: the space $\mathfrak{h}$ is abelian of dimension l with basis $h_1, \ldots, h_l$, there is a triangular decomposition

$$\mathfrak{g}(A) = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+$$

with $\mathfrak{g}(A)_0 = \mathfrak{h}$, the pieces $\mathfrak{g}(A)_{\alpha_i} = \mathbb{C}e_i$ and $\mathfrak{g}(A)_{-\alpha_i} = \mathbb{C}f_i$ are one-dimensional, $\mathfrak{g}(A)_{k\alpha_i} = 0$ for $|k| \geq 2$, and $\mathfrak{g}(A)_\lambda = 0$ for every λ that is not sign-coherent.

***Proof*:**

(1) Let M be the span of the displayed lowering words. Then M contains every θ^-_{ij} (the case $r = 0$) and is contained in the ideal generated by them, its spanning elements being bracket words on the generators (lemma 8.7.1(2)). It remains to show M is an ideal, for then minimality gives $M = K^-$. By construction $[f_k, M] \subseteq M$. Each spanning word is homogeneous, since θ^-_{ij} is homogeneous and bracketing with f_k shifts degree, so $[h_k, M] \subseteq M$ by proposition 8.7.6(2). For $[e_k, M] \subseteq M$, induct on the length r of the word: the case $r = 0$ is lemma 8.7.11, and for a word $[f_{k_1}, u]$ with u of length $r - 1$,

$$[e_k, [f_{k_1}, u]] = [[e_k, f_{k_1}], u] + [f_{k_1}, [e_k, u]] = \delta_{kk_1}[h_k, u] + [f_{k_1}, [e_k, u]] \in M$$

using $[h_k, u] \in M$ and, inductively, $[e_k, u] \in M$. The generators of $\tilde{\mathfrak{g}}(A)$ all preserve M, so M is an ideal by lemma 8.7.1(3). Each spanning word lies in $\tilde{\mathfrak{n}}^-$, since θ^-_{ij} is an iterated f-bracket and $\tilde{\mathfrak{n}}^-$ is a subalgebra containing the f_k. Hence $K^- = M \subseteq \tilde{\mathfrak{n}}^-$. Applying the involution, $\omega(K^-)$ is an ideal (the image of an ideal under an automorphism) generated by the elements $\omega(\theta^-_{ij}) = \theta^+_{ij}$, so $K^+ = \omega(K^-)$, and it lies in $\omega(\tilde{\mathfrak{n}}^-) = \tilde{\mathfrak{n}}^+$, the image of the subalgebra generated by the f_k being the subalgebra generated by the $\omega(f_k) = e_k$.

(2) The word $(\mathrm{ad} f_{k_1}) \cdots (\mathrm{ad} f_{k_r})\, \theta^-_{ij}$ is homogeneous of degree $-\big((1-A_{ji})\alpha_i + \alpha_j\big) - (\alpha_{k_1} + \cdots + \alpha_{k_r})$, of height at most -2 because already $\mathrm{ht}\big((1 - A_{ji})\alpha_i + \alpha_j\big) = 2 - A_{ji} \geq 2$. So K^- is spanned by homogeneous elements of height ≤ -2, and $K^+ = \omega(K^-)$ by homogeneous elements of height ≥ 2. Their sum K is therefore graded with support in $|\,\mathrm{ht}\,| \geq 2$, and it meets neither $\tilde{\mathfrak{g}}(A)_0 \supseteq \tilde{\mathfrak{h}}$ nor the height-± 1 pieces $\mathbb{C}e_i$, $\mathbb{C}f_i$.

(3) Since K is graded, the quotient map preserves the grading and $\mathfrak{g}(A)_\lambda = \tilde{\mathfrak{g}}(A)_\lambda / (K \cap \tilde{\mathfrak{g}}(A)_\lambda)$

(proposition 8.7.6). At $\lambda = 0$ and $\lambda = \pm\alpha_i$ the intersection with K is zero by (2), so $\mathfrak{g}(A)_0$ is the isomorphic image of $\tilde{\mathfrak{g}}(A)_0 = \tilde{\mathfrak{h}}$ (theorem 8.7.9), abelian with basis $h_1, \dots, h_l$, and $\mathfrak{g}(A)_{\pm\alpha_i}$ are the one-dimensional images of $\mathbb{C}e_i$, $\mathbb{C}f_i$. The pieces at $k\alpha_i$ with $|k| \geq 2$ and at non-sign-coherent λ vanish already in $\tilde{\mathfrak{g}}(A)$ (theorem 8.7.9). Finally the images $\mathfrak{n}^{\pm}$ are the sums of the surviving pieces of nonzero sign-coherent degree, positive for $\mathfrak{n}^+$ and negative for $\mathfrak{n}^-$, so the decomposition $\tilde{\mathfrak{g}}(A) = \tilde{\mathfrak{n}}^- \oplus \tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+$ descends to the stated triangular decomposition of $\mathfrak{g}(A)$, direct by the grading, completing the proof.

Everything so far holds for the Cartan matrix of an arbitrary root system, and indeed the analysis to this point is the finite-type beginning of the theory of Kac-Moody algebras. What remains, and what uses finiteness of Φ, is to show that the surviving weights are exactly the roots. The route is symmetry: the Weyl group must act. Its generators will be exponentials, and exponentials need nilpotency.

Lemma 8.7.13: Local Nilpotency of the Generators

In $\mathfrak{g}(A)$ the operators $\mathrm{ad}e_i$ and $\mathrm{ad}f_i$ are *locally nilpotent*: for every $x \in \mathfrak{g}(A)$ there is an N with $(\mathrm{ad}e_i)^N x = 0$, and likewise for $\mathrm{ad}f_i$.

***Proof*:**
Fix i and write $D = \mathrm{ad}e_i$. The Jacobi identity says D is a derivation, $D[x,y] = [Dx, y] + [x, Dy]$, and induction with Pascal's rule gives the Leibniz binomial identity

$$D^n[x,y] = \sum_{k=0}^{n} \binom{n}{k} \left[D^k x,\ D^{n-k} y\right]$$

Let $N_i = \{x \in \mathfrak{g}(A) \mid D^n x = 0 \text{ for some } n\}$. This is a subspace: if $D^r x = 0$ and $D^s y = 0$ then $D^{\max(r,s)}$ kills any combination. It is a subalgebra: with r, s as above, every term of the Leibniz identity for $n = r + s$ has $k \geq r$ or $n - k \geq s$, so $D^{r+s}[x,y] = 0$. The generators all lie in N_i:

$$D(e_i) = 0, \qquad D^{1-A_{ji}}(e_j) = 0 \ (j \neq i), \qquad D^2(h_j) = -A_{ij}\,[e_i, e_i] = 0, \qquad D^3(f_j) = 0$$

The second is exactly the Serre relation (4), which holds in $\mathfrak{g}(A)$ because $\theta_{ij}^{+} \in K$. The third is $D(h_j) = -[h_j, e_i] = -A_{ij}e_i$ followed by $D(e_i) = 0$. The fourth is $D(f_j) = \delta_{ij}h_i$, then $D^2(f_j) = \delta_{ij}[e_i, h_i] = -2\delta_{ij}e_i$, then $D^3(f_j) = 0$. A subalgebra containing the generators is all of $\mathfrak{g}(A)$, so $N_i = \mathfrak{g}(A)$ and $\mathrm{ad}e_i$ is locally nilpotent. For $\mathrm{ad}f_i$, apply the involution of proposition 8.7.10: $(\mathrm{ad}f_i)^n x = \omega\big((\mathrm{ad}e_i)^n\,\omega(x)\big)$, so $(\mathrm{ad}f_i)^n x$ vanishes as soon as $(\mathrm{ad}e_i)^n$ kills $\omega(x)$, completing the proof.

Proposition 8.7.14: Exponentials and the Reflection Symmetry of the Weights

1. If $z \in \mathfrak{g}(A)$ is such that $\mathrm{ad}z$ is locally nilpotent, then $\exp(\mathrm{ad}z) = \sum_{n\geq 0} (\mathrm{ad}z)^n/n!$, a finite sum on every vector, is an automorphism of $\mathfrak{g}(A)$ with inverse $\exp(-\mathrm{ad}z)$.
2. For each i the automorphism
$$\tau_i = \exp(\mathrm{ad}e_i)\,\exp\big(\mathrm{ad}(-f_i)\big)\,\exp(\mathrm{ad}e_i)$$
satisfies $\tau_i(h) = h - \alpha_i(h)\,h_i$ for all $h \in \mathfrak{h}$, and
$$\tau_i\big(\mathfrak{g}(A)_\lambda\big) = \mathfrak{g}(A)_{s_i\lambda} \qquad (\lambda \in Q)$$
where $s_i = s_{\alpha_i}$ is the simple reflection of definition 8.1.1. In particular $\dim \mathfrak{g}(A)_\lambda = \dim \mathfrak{g}(A)_{w\lambda}$ for every w in the group generated by the simple reflections.

***Proof*:**
(1) On each x the sum is finite by local nilpotency, so $\exp(\mathrm{ad}z)$ is a well-defined linear map. Writing $D = \mathrm{ad}z$ and using the Leibniz binomial identity from the proof of lemma 8.7.13, all sums below being finite,
$$\exp(D)[x,y] = \sum_n \frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k}[D^k x, D^{n-k}y] = \sum_{k,j}\Big[\frac{D^k x}{k!},\frac{D^j y}{j!}\Big] = [\exp(D)x, \exp(D)y]$$
so $\exp(D)$ is a homomorphism. For invertibility, fix x and choose N with $D^N x = 0$. Then
$$\exp(D)\exp(-D)\,x = \sum_{n\geq 0}\frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k}D^k(-D)^{n-k}x = \sum_{n\geq 0}\frac{1}{n!}(D-D)^n x = x$$
the middle equality being the binomial theorem for the commuting operators D and $-D$, every term with $n \geq 1$ vanishing. The same holds with D and $-D$ exchanged, so $\exp(-D)$ is a two-sided inverse and $\exp(D)$ is an automorphism.

(2) All three factors of τ_i are automorphisms by (1) and lemma 8.7.13. First the effect on $\mathfrak{h}$. For $h \in \mathfrak{h}$ the grading (proposition 8.7.6(2)) gives $[h,e_i] = \alpha_i(h)e_i$ and $[h,f_i] = -\alpha_i(h)f_i$, so
$$(\mathrm{ad}e_i)(h) = -\alpha_i(h)\,e_i, \qquad \big(\mathrm{ad}(-f_i)\big)(h) = -\alpha_i(h)\,f_i$$
and the squares vanish, giving $\exp(\mathrm{ad}e_i)(h) = h - \alpha_i(h)e_i$ and $\exp(\mathrm{ad}(-f_i))(h) = h - \alpha_i(h)f_i$. On the elements e_i, h_i, f_i themselves,
$$\big(\mathrm{ad}(-f_i)\big)(e_i) = [e_i,f_i] = h_i, \qquad \big(\mathrm{ad}(-f_i)\big)(h_i) = [h_i,f_i] = -2f_i$$
so $\exp(\mathrm{ad}(-f_i))(e_i) = e_i + h_i - f_i$, the cube of $\mathrm{ad}(-f_i)$ killing e_i. Assembling, and using $\alpha_i(h_i) = \langle \alpha_i, \alpha_i^\vee\rangle = 2$,
$$\exp\big(\mathrm{ad}(-f_i)\big)\exp(\mathrm{ad}e_i)(h) = h - \alpha_i(h)f_i - \alpha_i(h)(e_i + h_i - f_i) = h - \alpha_i(h)(e_i + h_i)$$
and applying $\exp(\mathrm{ad}e_i)$, which fixes e_i and sends $h_i \mapsto h_i - 2e_i$,
$$\tau_i(h) = h - \alpha_i(h)e_i - \alpha_i(h)\big(e_i + h_i - 2e_i\big) = h - \alpha_i(h)\,h_i$$

as claimed. This formula is an involution of $\mathfrak{h}$: applying it twice returns h, by the same computation with $\alpha_i(h_i) = 2$. In particular $\tau_i(\mathfrak{h}) = \mathfrak{h}$ and τ_i^{-1} agrees with τ_i on $\mathfrak{h}$.

Now let $x \in \mathfrak{g}(A)_\lambda$. For every $h \in \mathfrak{h}$,

$$[h, \tau_i x] = \tau_i\big([\tau_i^{-1}h, x]\big) = \lambda\big(h - \alpha_i(h)h_i\big)\,\tau_i x = \big(\lambda(h) - \langle \lambda, \alpha_i^\vee\rangle\,\alpha_i(h)\big)\,\tau_i x$$

using $\lambda(h_i) = \langle \lambda, \alpha_i^\vee\rangle$, and the last scalar is $(s_i\lambda)(h)$ by the reflection formula $s_i\lambda = \lambda - \langle \lambda, \alpha_i^\vee\rangle\alpha_i$. Writing $\tau_i x = \sum_\mu y_\mu$ in the grading and comparing components, $(s_i\lambda)(h)\,y_\mu = \mu(h)\,y_\mu$ for all h, so $y_\mu = 0$ unless the functionals $\mu(\cdot)$ and $(s_i\lambda)(\cdot)$ coincide, which by lemma 8.7.7 forces $\mu = s_i\lambda$. Hence $\tau_i(\mathfrak{g}(A)_\lambda) \subseteq \mathfrak{g}(A)_{s_i\lambda}$. The inverse $\tau_i^{-1} = \exp(\mathrm{ad}(-e_i))\exp(\mathrm{ad}\,f_i)\exp(\mathrm{ad}(-e_i))$ has the same shape with e_i, f_i replaced by $-e_i, -f_i$, and these satisfy the identical bracket relations with h_i, so the same argument gives $\tau_i^{-1}(\mathfrak{g}(A)_{s_i\lambda}) \subseteq \mathfrak{g}(A)_\lambda$. The two restrictions compose to the identity in both orders, so τ_i maps $\mathfrak{g}(A)_\lambda$ isomorphically onto $\mathfrak{g}(A)_{s_i\lambda}$. For a product $w = s_{i_1}\cdots s_{i_r}$ the composite $\tau_{i_1}\circ\cdots\circ\tau_{i_r}$ carries $\mathfrak{g}(A)_\lambda$ isomorphically onto $\mathfrak{g}(A)_{w\lambda}$, giving the equality of dimensions and completing the proof.

The symmetry statement converts the finiteness question into combinatorics of the root lattice: which weights can a sign-coherent orbit contain? The answer is a lemma about root systems with no Lie algebra in it.

Lemma 8.7.15: One-Sign Orbits in the Root Lattice

Let Φ be a root system with base Δ, and let W' be the group generated by the simple reflections $s_1, \ldots, s_l$. Let $\lambda \in Q$ be nonzero, and suppose every element of the orbit $W'\lambda$ is sign-coherent. Then $w(\lambda) = c\,\alpha_i$ for some $w \in W'$, some index i, and some nonzero integer c. In particular every root is carried to a simple root by an element of W'.

***Proof*:**

Negating a sign-coherent element gives a sign-coherent element, so every element of $W'\lambda \cup (-W'\lambda)$ is sign-coherent, and none is zero since $\lambda \neq 0$. Taking $w = 1$ shows λ itself is sign-coherent, so one of $\pm\lambda$ has all coefficients ≥ 0, and the set P of elements of $W'\lambda \cup (-W'\lambda)$ with all coefficients ≥ 0 is nonempty. Heights of elements of P are positive integers, so there is a $\mu \in P$ of minimal height. Write $\mu = \sum_j c_j\alpha_j$ with all $c_j \geq 0$.

Since the inner product on E is positive definite and $\mu \neq 0$,

$$0 < (\mu, \mu) = \sum_j c_j\,(\mu, \alpha_j)$$

so some index j has $c_j > 0$ and $(\mu, \alpha_j) > 0$. Then $k = \langle \mu, \alpha_j^\vee\rangle$ is a positive integer, and $s_j\mu = \mu - k\alpha_j$ differs from μ only in the j-th coefficient, with $\mathrm{ht}(s_j\mu) = \mathrm{ht}(\mu) - k < \mathrm{ht}(\mu)$. The element $s_j\mu$ again lies in $W'\lambda \cup (-W'\lambda)$, so it is sign-coherent and nonzero. If all its coefficients were ≥ 0 it would lie in P with smaller height, contradicting minimality. So it has a strictly negative coefficient, which can only be the j-th one, and sign-coherence forces all its coefficients ≤ 0. But its coefficients away from j are the $c_m \geq 0$, so $c_m = 0$ for every $m \neq j$ and $\mu = c_j\alpha_j$ with $c_j \geq 1$. Unwinding, some $w \in W'$ has $w(\lambda) = \pm\mu = \pm c_j\alpha_j$, which is the claim with $c = \pm c_j$.

For the final statement let $\beta \in \Phi$. Every element of $W'\beta$ is a root, by axiom (R3) of definition 8.1.1 applied to each simple reflection in turn, and every root is sign-coherent (definition 8.2.1), so

the hypothesis holds and some $w \in W'$ has $w(\beta) = c\,\alpha_i$. Being a root, $c\,\alpha_i$ has $c = \pm 1$ by axiom (R2). If $c = -1$, compose with s_i, which sends $-\alpha_i$ to α_i. Either way some element of W' carries β to the simple root α_i, completing the proof.

The subgroup W' is in fact the whole Weyl group of Φ (theorem 8.2.8), though the proof above needs only the simple reflections. All the pieces are now on the table, and the existence theorem assembles them.

Theorem 8.7.16: Existence: the Serre Algebra of a Cartan Matrix

Let A be the Cartan matrix of a root system Φ with base $\Delta = \{\alpha_1, \dots, \alpha_l\}$, and let $\mathfrak{g}(A)$ be the Serre algebra of definition 8.7.4, the Lie algebra presented by the generators e_i, f_i, h_i and relations (1) through (4) of theorem 8.5.1. Then:

1. $\mathfrak{g}(A)_\lambda \neq 0$ exactly for $\lambda \in \Phi \cup \{0\}$, with $\dim \mathfrak{g}(A)_\lambda = 1$ for $\lambda \in \Phi$ and $\mathfrak{g}(A)_0 = \mathfrak{h}$. Consequently
$$\dim \mathfrak{g}(A) = l + |\Phi| < \infty$$
2. $\mathfrak{g}(A)$ is semisimple, $\mathfrak{h}$ is a Cartan subalgebra, the roots of the pair $(\mathfrak{g}(A), \mathfrak{h})$ are the functionals $\lambda(\cdot)$ for $\lambda \in \Phi$, the functionals $\alpha_1(\cdot), \dots, \alpha_l(\cdot)$ form a base of the resulting root system, and the Cartan matrix relative to that base is A.

In particular the forward half of theorem 8.5.1 holds.

***Proof*:**
Step 1: no weights survive outside $\Phi \cup \{0\}$. Let $\lambda \in Q$, $\lambda \notin \Phi \cup \{0\}$. If some $w \in W'$ makes $w(\lambda)$ non-sign-coherent, then $\mathfrak{g}(A)_{w\lambda} = 0$ by theorem 8.7.12(3), and $\dim \mathfrak{g}(A)_\lambda = \dim \mathfrak{g}(A)_{w\lambda} = 0$ by proposition 8.7.14(2). Otherwise the whole orbit $W'\lambda$ is sign-coherent, and lemma 8.7.15 produces $w \in W'$ with $w(\lambda) = c\,\alpha_i$, c a nonzero integer. If $c = \pm 1$ then $\lambda = w^{-1}(\pm\alpha_i)$ would be a root, since the simple reflections preserve Φ (axiom (R3)). So $|c| \geq 2$, and theorem 8.7.12(3) gives $\mathfrak{g}(A)_{c\alpha_i} = 0$, whence $\dim \mathfrak{g}(A)_\lambda = 0$ again by proposition 8.7.14(2).

Step 2: the roots survive, each with a line. For $\beta \in \Phi$, lemma 8.7.15 gives $w \in W'$ with $w(\beta) = \alpha_i$, so by proposition 8.7.14(2) and theorem 8.7.12(3),
$$\dim \mathfrak{g}(A)_\beta = \dim \mathfrak{g}(A)_{\alpha_i} = 1$$
Together with $\mathfrak{g}(A)_0 = \mathfrak{h}$ of dimension l, the grading now reads
$$\mathfrak{g}(A) = \mathfrak{h} \oplus \bigoplus_{\beta \in \Phi} \mathfrak{g}(A)_\beta, \qquad \dim \mathfrak{g}(A) = l + |\Phi|$$
which is part (1).

Step 3: weight spaces are eigenspaces, and ideals are graded. Two bookkeeping facts. First, distinct $\mu, \nu \in Q$ give distinct functionals on $\mathfrak{h}$: some $\langle \mu - \nu, \alpha_i^\vee \rangle \neq 0$ by lemma 8.7.7, that is, $\mu(h_i) \neq \nu(h_i)$. Second, every ideal J of $\mathfrak{g}(A)$ is graded. Let $x = \sum_{\mu \in S} x_\mu \in J$ with S finite and each x_μ a nonzero component. For each pair $\mu \neq \nu$ in S the locus of $h \in \mathfrak{h}$ with $\mu(h) = \nu(h)$ is a proper subspace, and finitely many proper subspaces cannot exhaust $\mathfrak{h}$: the product of the nonzero linear forms $\mu - \nu$ is a nonzero polynomial function on $\mathfrak{h}$, which does not vanish identically. Choose h with the values $\mu(h)$, $\mu \in S$, pairwise distinct. The iterates

$(\mathrm{ad}h)^k x = \sum_\mu \mu(h)^k x_\mu$ lie in J for all $k \geq 0$, and the Vandermonde matrix on the distinct nodes $\mu(h)$ is invertible, so each x_μ is a combination of the iterates and lies in J.

Step 4: no nonzero abelian ideal. Let J be an abelian ideal. By Step 3, $J = (J \cap \mathfrak{h}) \oplus \bigoplus_{\beta\in\Phi}(J \cap \mathfrak{g}(A)_\beta)$. Suppose $J \cap \mathfrak{g}(A)_\beta \neq 0$ for some $\beta \in \Phi$, so $\mathfrak{g}(A)_\beta \subseteq J$ by one-dimensionality. Choose $w \in W'$ with $w(\beta) = \alpha_j$ (lemma 8.7.15) and let T be the corresponding composite of the automorphisms τ_i, so that $T(\mathfrak{g}(A)_\beta) = \mathfrak{g}(A)_{\alpha_j} = \mathbb{C}e_j$ (proposition 8.7.14(2)). The image $T(J)$ is again an abelian ideal, and it contains e_j, hence also $h_j = [e_j, f_j] \in [T(J), \mathfrak{g}(A)] \subseteq T(J)$. But then $e_j, h_j \in T(J)$ with $[h_j, e_j] = 2e_j \neq 0$, contradicting abelianness. So $J \subseteq \mathfrak{h}$, and for $x \in J$ each bracket $[x, e_i] = \alpha_i(x)\, e_i$ lies in $J \cap \mathfrak{g}(A)_{\alpha_i} = 0$, forcing $\alpha_i(x) = 0$ for all i. The matrix of the functionals $\alpha_1(\cdot), \ldots, \alpha_l(\cdot)$ in the basis $h_1, \ldots, h_l$ is A itself, invertible by lemma 8.7.7, so these functionals span $\mathfrak{h}^*$ and $x = 0$. Hence $J = 0$.

Step 5: semisimplicity. Suppose $\operatorname{rad}\mathfrak{g}(A) \neq 0$ (definition 5.5.2), a nonzero solvable ideal. Its derived series (definition 5.1.3) descends to zero (definition 5.1.5), and each term is an ideal of $\mathfrak{g}(A)$: for an ideal $\mathfrak{a}$, the Jacobi identity gives $[\mathfrak{g}(A), [\mathfrak{a}, \mathfrak{a}]] \subseteq [[\mathfrak{g}(A), \mathfrak{a}], \mathfrak{a}] + [\mathfrak{a}, [\mathfrak{g}(A), \mathfrak{a}]] \subseteq [\mathfrak{a}, \mathfrak{a}]$, and induction extends this down the series. The last nonzero term is then a nonzero abelian ideal, contradicting Step 4. So $\operatorname{rad}\mathfrak{g}(A) = 0$ and $\mathfrak{g}(A)$ is semisimple (definition 5.5.5).

Step 6: $\mathfrak{h}$ is a Cartan subalgebra. It is abelian, hence nilpotent, its lower central series stopping at $[\mathfrak{h}, \mathfrak{h}] = 0$ (definition 5.2.3). For self-normalization (definition 7.1.4) let $x = \sum_\mu x_\mu$ satisfy $[x, \mathfrak{h}] \subseteq \mathfrak{h}$. For $\mu \neq 0$ in the support, choose $h \in \mathfrak{h}$ with $\mu(h) \neq 0$ (Step 3, comparing μ with $\nu = 0$). The degree-μ component of $[h, x]$ is $\mu(h)\, x_\mu$, while $[h, x] \in \mathfrak{h} = \mathfrak{g}(A)_0$ has zero component there, so $x_\mu = 0$. Hence $x \in \mathfrak{g}(A)_0 = \mathfrak{h}$. The same argument applied to an x commuting with $\mathfrak{h}$ gives $C_{\mathfrak{g}(A)}(\mathfrak{h}) = \mathfrak{h}$. Moreover $\mathfrak{h}$ is maximal toral. Each $h \in \mathfrak{h}$ acts on every $\mathfrak{g}(A)_\mu$ by a scalar (proposition 8.7.6(2)), so $\mathrm{ad}h$ is diagonalizable and $\mathfrak{h}$ is toral. A toral subalgebra containing $\mathfrak{h}$ is abelian (lemma 7.1.5, $\mathfrak{g}(A)$ being semisimple by Step 5), so it lies in $C_{\mathfrak{g}(A)}(\mathfrak{h}) = \mathfrak{h}$.

Step 7: the root system and its Cartan matrix. By Step 3 the simultaneous eigenspace of $\mathrm{ad}\mathfrak{h}$ for a functional $\varphi \in \mathfrak{h}^*$ (definition 7.2.1) is the sum of the pieces $\mathfrak{g}(A)_\lambda$ with $\lambda(\cdot) = \varphi$, and distinct λ give distinct functionals, so each eigenspace is a single piece. The roots of $(\mathfrak{g}(A), \mathfrak{h})$ in the sense of definition 7.2.2 are therefore exactly the functionals $\beta(\cdot)$ with $\beta \in \Phi$, with root spaces $\mathfrak{g}(A)_\beta$, and $\beta \mapsto \beta(\cdot)$ is injective and additive. Write $\bar\beta = \beta(\cdot)$. Every $\bar\beta$ is an integer combination of $\bar\alpha_1, \ldots, \bar\alpha_l$ with coefficients all of one sign, the expansion transported from definition 8.2.1 for Φ, and the $\bar\alpha_i$ are linearly independent (their matrix in the basis $h_1, \ldots, h_l$ is A) and hence a basis of the real span of the roots of the pair. The roots of the semisimple pair form an abstract root system (theorem 7.5.4), and $\{\bar\alpha_1, \ldots, \bar\alpha_l\}$ is a base for it in the sense of definition 8.2.1. Its Cartan matrix is computed from the coroots of definition 7.4.3. For the root $\bar\alpha_j$, the bracket space $[\mathfrak{g}(A)_{\alpha_j}, \mathfrak{g}(A)_{-\alpha_j}]$ is the line $\mathbb{C}\, t_{\bar\alpha_j}$ by lemma 7.4.2, and the coroot $h_{\bar\alpha_j} = \bar\alpha_j^\vee$ of theorem 7.4.1 is the unique element of that line on which $\bar\alpha_j$ takes the value 2. The element $h_j = [e_j, f_j]$ lies on the line, is nonzero, and has $\bar\alpha_j(h_j) = \langle \alpha_j, \alpha_j^\vee \rangle = 2$, so $h_{\bar\alpha_j} = h_j$. The Cartan matrix entry of definition 8.3.1 is therefore

$$\langle \bar\alpha_i, \bar\alpha_j^\vee \rangle = \bar\alpha_i(h_j) = \langle \alpha_i, \alpha_j^\vee \rangle = A_{ij}$$

This proves part (2). The forward half of theorem 8.5.1 follows: $\mathfrak{g}(A)$ is finite-dimensional and semisimple, $\mathfrak{h}(A) = \mathfrak{h}$ is a Cartan subalgebra, the e_i and f_i span the root spaces of the simple roots $\pm\bar\alpha_i$ of a base, and the root system of the pair has Cartan matrix A, completing the proof.

The existence half of theorem 8.5.4 now falls out. Each connected Dynkin diagram in the classification list is the diagram of an irreducible root system (theorem 8.4.3), that root system has a Cartan

matrix, and the theorem turns the matrix into a semisimple Lie algebra realizing the diagram. This is exactly the input the proof of theorem 8.5.4 required for the exceptional types, and the last outstanding promise of section 8.5 is discharged.

The construction rewards a look back. The linear relations alone produced an algebra with a perfectly good triangular decomposition but with nothing yet to bound its two wings. The Serre relations are exactly the finiteness conditions, and the mechanism by which they act is the one this chapter has used since the root strings first appeared: an $\mathfrak{sl}_2$-weight computation (lemma 8.7.11) showed each Serre element generates an ideal confined to one wing, and the Weyl symmetry then folded the surviving weights onto the finite root system. For integer matrices satisfying the same diagonal and sign conditions but coming from no root system, the same three-stage construction, run with an enlarged Cartan subalgebra where the matrix is singular, produces the Kac-Moody algebras, no longer finite-dimensional, a theory beyond the scope of this book. With existence proved, the classification of corollary 8.5.5 is proved in full. Its separation of the isomorphism classes rests on the conjugacy of maximal toral subalgebras (theorem 13.2.7), and the structure theory of this part is complete as an instrument. One refinement of that theory remains before the part closes. The isomorphism theorem (theorem 8.5.4) recovers $\mathfrak{g}$ from its Cartan matrix but fixes the root vectors only up to scaling, and the structure constants that record the brackets move with that scaling. The next section pins the scaling to an integral normalization, the Chevalley basis, whose real structure constants and opposite-sign pairing are what the compact real form of theorem 13.2.3 runs on. The oldest promise in the book, Ado's theorem, stated in Part I and deferred to this part, is discharged in the section that follows it.

Exercise 8.7.1

1. Let $l = 1$, so $\Phi = \{\alpha, -\alpha\}$ is the root system A_1 and $A = (2)$. Relation (4) is vacuous, so $K = 0$ and $\mathfrak{g}(A) = \tilde{\mathfrak{g}}(A)$. Using theorem 8.7.9, show that $\tilde{\mathfrak{n}}^+ = \mathbb{C}e$ and $\tilde{\mathfrak{n}}^- = \mathbb{C}f$, deduce $\dim \mathfrak{g}(A) = 3$, and exhibit an isomorphism $\mathfrak{g}(A) \cong \mathfrak{sl}_2(\mathbb{C})$. Check the count against theorem 8.7.16.

2. Still with $l = 1$, identify the tensor algebra of the one-dimensional space $V = \mathbb{C}v$ with the polynomial space on one variable by $v^{\otimes t} \leftrightarrow z^t$, and write out the representation of proposition 8.7.8: show that $\rho(f)(z^t) = z^{t+1}$, $\rho(h)(z^t) = -2t\, z^t$, and $\rho(e)(z^t) = -t(t-1)\, z^{t-1}$, and verify the relations $[\rho(h), \rho(e)] = 2\rho(e)$, $[\rho(h), \rho(f)] = -2\rho(f)$, and $[\rho(e), \rho(f)] = \rho(h)$ by direct computation on the monomials.

3. For the Cartan matrix of type A_2 (example 8.5.2), list the four Serre elements of definition 8.7.4, compute the degree in Q of each, and check that every degree has height of absolute value at least 2, as theorem 8.7.12(2) requires. Then use theorem 8.7.16 to compute $\dim \mathfrak{g}(A)$ and compare with the identification $\mathfrak{g}(A) \cong \mathfrak{sl}_3(\mathbb{C})$ of example 8.5.2.

4. Let E, F, H be linear operators on a vector space with $[E, F] = H$ and $[H, F] = -2F$. Prove the identities $[H, F^m] = -2m\, F^m$ and

$$EF^m = F^m E + mF^{m-1}\big(H - (m-1)\big) \qquad (m \geq 1)$$

   by induction, and deduce that $EF^m v = 0$ for any vector v with $Ev = 0$ and $Hv = (m-1)\, v$. Point to the exact place in the proof of lemma 8.7.11 where this configuration occurs.

5. In type A_2, let $\lambda = \alpha_1 + 2\alpha_2$. Show that $\lambda \notin \Phi \cup \{0\}$, compute $s_2(\lambda)$, and conclude from proposition 8.7.14 and theorem 8.7.12(3) that $\mathfrak{g}(A)_\lambda = 0$. Then verify the vanishing directly:

show that $\tilde{\mathfrak{g}}(A)_\lambda$ is spanned by the single Serre element $\theta_{21}^+ = (\mathrm{ad}e_2)^2(e_1)$, so that the quotient by K kills the whole piece.

8.8 The Chevalley Basis

The structure constants $N_{\alpha,\beta}$, defined by $[X_\alpha, X_\beta] = N_{\alpha,\beta}X_{\alpha+\beta}$ once a root vector X_α is chosen in each one-dimensional root space, carry the whole bracket of $\mathfrak{g}$ beyond the Cartan subalgebra. They are not canonical: rescaling a single X_α by a nonzero constant rescales every constant in which it appears, and a careless choice leaves the $N_{\alpha,\beta}$ arbitrary complex numbers. Chevalley's theorem is that the scalings can be fixed, once and for all, so that every $N_{\alpha,\beta}$ becomes an integer and opposite roots carry opposite constants. The resulting basis is defined over $\mathbb{Z}$, its brackets free of denominators, which is what lets a complex semisimple Lie algebra be reduced modulo a prime and is the starting point for the Chevalley groups over finite fields.

The application in Part IV asks for less than the full strength. What the compact real form needs is the reality of the constants and the sign relation $N_{-\alpha,-\beta} = -N_{\alpha,\beta}$, the two facts that force the generators rotated by i to close under the bracket. Both rest on a single symmetry of $\mathfrak{g}$, the involution that exchanges its raising and lowering halves and acts by -1 on the Cartan subalgebra.

Lemma 8.8.1: The Chevalley Involution of a Semisimple Lie Algebra

Let $\mathfrak{g}$ be semisimple over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$ and root system Φ. There is an automorphism θ of $\mathfrak{g}$ with $\theta^2 = \mathrm{id}$, with $\theta|_{\mathfrak{h}} = -\mathrm{id}$, and with $\theta(\mathfrak{g}_\alpha) = \mathfrak{g}_{-\alpha}$ for every $\alpha \in \Phi$.

***Proof*:**
Fix a base $\Delta = \{\alpha_1, \dots, \alpha_l\}$ of Φ (theorem 8.2.4) and Chevalley generators $e_i \in \mathfrak{g}_{\alpha_i}$, $f_i \in \mathfrak{g}_{-\alpha_i}$, $h_i = [e_i, f_i] = \alpha_i^\vee$ (theorem 7.4.1). The isomorphism theorem (theorem 8.5.4) identifies $\mathfrak{g}$ with the Serre algebra $\mathfrak{g}(A)$ of its Cartan matrix A through an isomorphism $\pi\colon \mathfrak{g}(A) \to \mathfrak{g}$ carrying the presented generators $\tilde{e}_i, \tilde{f}_i, \tilde{h}_i$ to e_i, f_i, h_i. The Chevalley involution ω of $\mathfrak{g}(A)$ (proposition 8.7.10) has $\omega(\tilde{e}_i) = \tilde{f}_i$, $\omega(\tilde{f}_i) = \tilde{e}_i$, and $\omega(\tilde{h}_i) = -\tilde{h}_i$, and it squares to the identity. Transporting it, $\theta = \pi \circ \omega \circ \pi^{-1}$ is an automorphism of $\mathfrak{g}$ with $\theta^2 = \mathrm{id}$, $\theta(e_i) = f_i$, $\theta(f_i) = e_i$, and $\theta(h_i) = -h_i$. The coroots $h_i = \alpha_i^\vee$ form a basis of $\mathfrak{h}$, so $\theta|_{\mathfrak{h}} = -\mathrm{id}$. Finally, for $x \in \mathfrak{g}_\alpha$ and $H \in \mathfrak{h}$, applying θ to $[H, x] = \alpha(H)\,x$ gives $[-H, \theta x] = \alpha(H)\,\theta x$, that is $[H, \theta x] = -\alpha(H)\,\theta x$, so $\theta x \in \mathfrak{g}_{-\alpha}$. Thus $\theta(\mathfrak{g}_\alpha) \subseteq \mathfrak{g}_{-\alpha}$, with equality because $\theta^2 = \mathrm{id}$, as we sought to show.

The involution collapses the choice of a root vector and its opposite into a single choice, since $X_{-\alpha}$ may be taken to be $-\theta(X_\alpha)$. One square root then spends the last scalar of freedom in each pair to arrange the coroot normalization, and the sign relation drops out of applying θ to a bracket. This is Chevalley's normalization. The reality and integrality of the constants follow from it together with the $\mathfrak{sl}_2$-strings of theorem 7.4.6.

Theorem 8.8.2: The Chevalley Basis

Let $\mathfrak{g}$ be semisimple over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root system Φ, and root space decomposition $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$ (theorem 7.2.4). Write $H_\alpha = \alpha^\vee$ for the coroot (definition 7.4.3) and $(\cdot, \cdot)$ for the inner product on the real span of Φ that the Killing form κ induces (proposition 7.3.4). Then root vectors $X_\alpha \in \mathfrak{g}_\alpha$, one for each $\alpha \in \Phi$, can be chosen so that:

1. $[X_\alpha, X_{-\alpha}] = H_\alpha$ for every α;
2. $\theta(X_\alpha) = -X_{-\alpha}$, where θ is the Chevalley involution of lemma 8.8.1;
3. the structure constants $N_{\alpha,\beta}$, defined by $[X_\alpha, X_\beta] = N_{\alpha,\beta} X_{\alpha+\beta}$ when $\alpha + \beta \in \Phi$ and $N_{\alpha,\beta} = 0$ otherwise, satisfy the sign relation $N_{-\alpha,-\beta} = -N_{\alpha,\beta}$ and the skew relation $N_{-\alpha,\beta} = -N_{\alpha,-\beta}$;
4. for $\alpha, \beta, \alpha + \beta \in \Phi$ with $\beta \neq \pm\alpha$, the constant $N_{\alpha,\beta}$ is a real integer, $N_{\alpha,\beta} = \pm(p+1)$, where $\beta - p\alpha, \ldots, \beta + q\alpha$ is the α-string through β (theorem 7.4.6).

The coroots $H_{\alpha_1}, \ldots, H_{\alpha_l}$ of a base, together with the X_α, form a *Chevalley basis* of $\mathfrak{g}$, a basis on which every structure constant is an integer: the $N_{\alpha,\beta}$ by (4), the brackets $[H_{\alpha_i}, X_\alpha] = \langle \alpha, \alpha_i^\vee \rangle X_\alpha$ by the Cartan integers, and $[X_\alpha, X_{-\alpha}] = H_\alpha$ because H_α lies in the integral span of the H_{α_i} (definition 7.4.3).

***Proof*:**

Write θ for the Chevalley involution (lemma 8.8.1), and recall that each root space is one-dimensional (theorem 7.4.6).

Step 1: the normalization (1) and (2). Fix a positive system Φ^+. For $\alpha \in \Phi^+$ choose any nonzero $Y_\alpha \in \mathfrak{g}_\alpha$. Then $\theta(Y_\alpha)$ is a nonzero element of $\mathfrak{g}_{-\alpha}$, and $[Y_\alpha, \theta Y_\alpha] \in [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}H_\alpha$ (theorem 7.4.1), say $[Y_\alpha, \theta Y_\alpha] = c_\alpha H_\alpha$. The scalar c_α is nonzero, because Y_α and θY_α are nonzero vectors of the one-dimensional spaces $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ whose bracket spans $\mathbb{C}H_\alpha$ (theorem 7.4.1). Choose $u_\alpha \in \mathbb{C}$ with $u_\alpha^2 = -c_\alpha$ and set

$$X_\alpha = u_\alpha^{-1} Y_\alpha, \qquad X_{-\alpha} = -\theta(X_\alpha)$$

Then $\theta(X_\alpha) = -X_{-\alpha}$ and $\theta(X_{-\alpha}) = -\theta^2(X_\alpha) = -X_\alpha$, so (2) holds for α and $-\alpha$, and

$$[X_\alpha, X_{-\alpha}] = -u_\alpha^{-2}[Y_\alpha, \theta Y_\alpha] = -u_\alpha^{-2} c_\alpha H_\alpha = -\frac{c_\alpha}{-c_\alpha} H_\alpha = H_\alpha$$

which is (1). Carrying this out for every $\alpha \in \Phi^+$, and taking $X_{-\alpha} = -\theta(X_\alpha)$ for the negative roots, produces one $X_\gamma \in \mathfrak{g}_\gamma$ for each $\gamma \in \Phi$ satisfying (1) and (2). From (1) and $H_\alpha = 2t_\alpha/(\alpha,\alpha)$, where $t_\alpha \in \mathfrak{h}$ represents α under κ, the pairing takes the value $\kappa(X_\alpha, X_{-\alpha}) = 2/(\alpha,\alpha) > 0$, used in Step 3.

Step 2: the sign and skew relations (3). Since θ is an automorphism with $\theta(X_\gamma) = -X_{-\gamma}$ for every γ (Step 1), applying it to $[X_\alpha, X_\beta] = N_{\alpha,\beta} X_{\alpha+\beta}$ gives

$$N_{-\alpha,-\beta} X_{-\alpha-\beta} = [X_{-\alpha}, X_{-\beta}] = [\theta X_\alpha, \theta X_\beta] = \theta[X_\alpha, X_\beta] = -N_{\alpha,\beta} X_{-\alpha-\beta}$$

so $N_{-\alpha,-\beta} = -N_{\alpha,\beta}$. Replacing α by $-\alpha$ turns this into $N_{\alpha,-\beta} = -N_{-\alpha,\beta}$, the skew relation. When $\alpha + \beta \notin \Phi$ every constant here vanishes and the relations are trivial.

Step 3: reality and the integer value (4). Fix $\alpha, \beta, \alpha+\beta \in \Phi$ with $\beta \neq \pm\alpha$, and let $\beta - p\alpha, \dots, \beta + q\alpha$ be the α-string through β, so $q \geq 1$ and $\langle \beta, \alpha^\vee \rangle = p - q$ (theorem 7.4.6). The subspace $M = \bigoplus_k \mathfrak{g}_{\beta+k\alpha}$ is the irreducible module over the $\mathfrak{sl}_2$-triple $\langle X_\alpha, X_{-\alpha}, H_\alpha \rangle$, with one-dimensional weight spaces and highest weight $m = p + q$ (theorem 7.4.6). The vector X_β sits q steps below the top of the ladder, so proposition 7.4.5 gives the operator $\mathrm{ad}X_{-\alpha}\,\mathrm{ad}X_\alpha$ the eigenvalue $q(m - q + 1) = q(p + 1)$ on the line $\mathfrak{g}_\beta$. Reading the same operator through the structure constants, $\mathrm{ad}X_{-\alpha}\,\mathrm{ad}X_\alpha(X_\beta) = N_{\alpha,\beta}N_{-\alpha,\alpha+\beta}X_\beta$, so

$$N_{\alpha,\beta}\, N_{-\alpha,\alpha+\beta} = q(p+1)$$

The Killing form is invariant, $\kappa([U,V],W) = \kappa([W,U],V)$, and by Step 1 satisfies $\kappa(X_\gamma, X_{-\gamma}) = 2/(\gamma,\gamma)$. Applying it with $U = X_{-\alpha}$, $V = X_{\alpha+\beta}$, $W = X_{-\beta}$,

$$N_{-\alpha,\alpha+\beta}\,\frac{2}{(\beta,\beta)} = \kappa\big([X_{-\alpha}, X_{\alpha+\beta}], X_{-\beta}\big) = \kappa\big([X_{-\beta}, X_{-\alpha}], X_{\alpha+\beta}\big) = N_{-\beta,-\alpha}\,\frac{2}{(\alpha+\beta,\alpha+\beta)}$$

By antisymmetry and the sign relation of Step 2, $N_{-\beta,-\alpha} = -N_{-\alpha,-\beta} = N_{\alpha,\beta}$, so $N_{-\alpha,\alpha+\beta} = N_{\alpha,\beta}\,(\beta,\beta)/(\alpha+\beta,\alpha+\beta)$. Substituting into the eigenvalue relation,

$$N_{\alpha,\beta}^2 = q(p+1)\,\frac{(\alpha+\beta,\alpha+\beta)}{(\beta,\beta)}$$

The right side is a positive rational number, since $q \geq 1$ and $(\cdot,\cdot)$ is positive-definite on the real span of Φ (proposition 7.3.4). Hence $N_{\alpha,\beta}$ is real, which is all that the compact real form of Part IV will require.

It remains to identify the right side with $(p+1)^2$, that is, to prove the root-string length identity $q\,(\alpha+\beta,\alpha+\beta) = (p+1)(\beta,\beta)$. Write $\ell(k) = (\beta+k\alpha, \beta+k\alpha)$. From $\langle \beta, \alpha^\vee \rangle = 2(\alpha,\beta)/(\alpha,\alpha) = p-q$ (theorem 7.4.6 and definition 7.4.3) one has $\ell(k) = (\beta,\beta) + (\alpha,\alpha)\,k(k+p-q)$, so $\ell(0) = (\beta,\beta)$ and $\ell(1) = (\alpha+\beta,\alpha+\beta) = (\beta,\beta) + (\alpha,\alpha)(p-q+1)$. Expanding,

$$q\,\ell(1) - (p+1)\,\ell(0) = (p-q+1)\big[q(\alpha,\alpha) - (\beta,\beta)\big]$$

so the identity is equivalent to $(p-q+1)\big[q(\alpha,\alpha) - (\beta,\beta)\big] = 0$. If $p = q-1$ the first factor vanishes. Otherwise $\ell(0) \neq \ell(1)$, so β and $\alpha+\beta$ have different lengths, and it is enough to show $(\beta,\beta) = q(\alpha,\alpha)$. The bottom root $\gamma = \beta - p\alpha$ has $\langle \gamma, \alpha^\vee \rangle = -(p+q)$, of absolute value at most 3 (proposition 8.1.6), so $p+q \leq 3$, and the pairs with $q \geq 1$ and $p \neq q-1$ are $(1,1)$, $(2,1)$, $(0,2)$, and $(0,3)$. For $(0,2)$ and $(0,3)$ the integer $\langle \beta, \alpha^\vee \rangle = p - q$ equals -2 and -3, and since the product of the two Cartan integers is at most 3 (proposition 8.1.6) the partner $\langle \alpha, \beta^\vee \rangle$ is -1, so the length ratio $(\beta,\beta)/(\alpha,\alpha) = \langle \beta, \alpha^\vee \rangle / \langle \alpha, \beta^\vee \rangle$ equals 2 and 3, in each case q. For $(2,1)$ the same product bound applied to γ, where $\langle \gamma, \alpha^\vee \rangle = -3$, makes α a short root of a system of length ratio 3, so β is no shorter than α and $(\beta,\beta)/(\alpha,\alpha) = 1/\langle \alpha, \beta^\vee \rangle \geq 1$ forces $\langle \alpha, \beta^\vee \rangle = 1$ and ratio $q = 1$. For the orthogonal pair $(1,1)$ the roots α and $\alpha+\beta$ are not orthogonal, with $\langle \alpha+\beta, \alpha^\vee \rangle = 2$, so $\langle \alpha, (\alpha+\beta)^\vee \rangle = 1$ and $(\alpha+\beta,\alpha+\beta) = 2(\alpha,\alpha)$, whence $(\beta,\beta) = (\alpha+\beta,\alpha+\beta) - (\alpha,\alpha) = (\alpha,\alpha) = q(\alpha,\alpha)$. In every case $q(\alpha,\alpha) = (\beta,\beta)$, so the length identity holds and $N_{\alpha,\beta}^2 = (p+1)^2$. Therefore $N_{\alpha,\beta} = \pm(p+1)$ is an integer, completing the proof.

The theorem separates what Part IV uses from what it does not. Reality of the constants and the sign relation of part (3) are all the compact real form of theorem 13.2.3 needs: rotating the coroots and

root vectors by i yields a real subalgebra precisely because the rotated brackets are real combinations of the rotated generators, and the sign relation pairs the two contributions of $\pm\alpha$ without leaving a stray factor of i. The integrality (4) is the sharper statement, the one that puts the entire algebra over $\mathbb{Z}$. It plays no further role in this book, but it is what the name Chevalley basis records, and it is the starting point for the theory of Chevalley groups. With the structure theory refined to its integral form, the oldest debt of Part I can be paid, and the next section constructs the faithful representation that Ado's theorem promises.

8.9 Ado's Theorem

Part I closed its central argument on credit. The existence half of Lie's third theorem (theorem 3.4.4) needed exactly one input beyond the integration theorems: Ado's theorem (theorem 3.4.3), the assertion that every finite-dimensional real Lie algebra admits a faithful finite-dimensional representation. The convergence proof for the Baker-Campbell-Hausdorff series (proposition 3.5.2) drew on the same assertion to reduce to matrices. Both times the theorem was stated and used as a black box, its proof promised to the structure theory of this part. That structure theory is now complete, and this closing section pays the debt. The tools are the ones assembled across Part II: the theorems of Engel and Lie (theorem 5.3.4, corollary 5.4.5), the radical and the nilradical (section 5.5), the Levi decomposition (theorem 6.5.4), and the enveloping algebra with its Poincaré-Birkhoff-Witt basis (section 8.6). The argument is purely algebraic. It draws on nothing from the group-theoretic development of Part I beyond the bare definitions, so the results that used the theorem there sit strictly downstream of this section and no circularity arises.

One representation comes for free, and seeing why it does not suffice locates the entire difficulty. The adjoint representation ad of definition 2.5.3 represents $\mathfrak{g}$ on itself, and its kernel is the center $\mathfrak{z}(\mathfrak{g})$ of proposition 5.2.4(3): $\operatorname{ad}X = 0$ says exactly that $[X, Y] = 0$ for every Y. For a semisimple algebra the center is a solvable ideal, hence zero, and ad is already faithful, settling the theorem in that case with no further work. The difficulty concentrates in the central and near-central directions, the ones every bracket kills. The Heisenberg algebra $\mathfrak{n}_3$ of example 5.2.5 shows the shape of the solution: ad kills its central element E_{13}, yet the defining representation on $\mathbb{C}^3$ is faithful, and there the central element acts by a nonzero *nilpotent* matrix. The general proof manufactures representations of exactly this kind. It proves more than faithfulness by design, keeping the action of the nilradical nilpotent at every stage, because that stronger property is what allows a representation to be pushed up an extension.

The route has four stages. First, for a nilpotent algebra, a faithful representation is produced inside the enveloping algebra itself, by truncating $U(\mathfrak{g})$ along the powers of the ideal generated by $\mathfrak{g}$. The Poincaré-Birkhoff-Witt theorem measures the truncation and certifies both its finite dimension and its faithfulness. Second, derivations of an algebra are extended to derivations of its enveloping algebra. Third comes the technical heart: a finite-dimensional representation of a solvable ideal $\mathfrak{a} \trianglelefteq \mathfrak{g}$ is extended to $\mathfrak{g}$ whenever $\mathfrak{g}$ splits as $\mathfrak{a}$ plus a complementary subalgebra, by replacing the representation with the left action of $U(\mathfrak{a})$ on a carefully stabilized finite-dimensional quotient and letting the complement act through derivations. Fourth, the stages are chained: from the nilradical to the radical one dimension at a time, and across the Levi decomposition in a single final step, after which a direct sum with ad restores faithfulness on the center.

Throughout the section the ground field is $\mathbb{C}$, in keeping with the convention of Part II, and every argument is valid verbatim over any algebraically closed field of characteristic zero. Algebraic closure

is invoked at exactly one point, flagged where it occurs, through the corollaries of Lie's theorem. The real case, the one Part I draws on, is recovered at the end by complexification, and the descent to an arbitrary field of characteristic zero is carried out in exercise 8.9.1 (5). As in section 8.6, a Lie algebra is identified with its image in its enveloping algebra (corollary 8.6.10) and the canonical map ι is suppressed.

The property the induction carries deserves a name before the work starts.

Definition 8.9.1: Nilrepresentation

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra with nilradical $\operatorname{nil}\mathfrak{g}$ (definition 5.5.3). A finite-dimensional representation φ of $\mathfrak{g}$ (definition 6.4.1) is a *nilrepresentation* if $\varphi(z)$ is a nilpotent operator for every $z \in \operatorname{nil}\mathfrak{g}$.

The refinement is not decoration. The extension step below hinges on a finiteness argument that is available only when the nilradical already acts nilpotently, so a plain faithful representation could not be pushed up an extension and the induction would not close. The first lemma records the assembly move used at the very end, and it is also the precise answer to the worry about the center raised above: the adjoint representation handles everything except the center, and a nilrepresentation faithful on the center handles the rest.

Lemma 8.9.2: Faithfulness from the Center

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and φ a finite-dimensional representation of $\mathfrak{g}$ on V with $\ker\varphi \cap \mathfrak{z}(\mathfrak{g}) = 0$. Define $\theta\colon \mathfrak{g} \to \mathfrak{gl}(V \oplus \mathfrak{g})$ by

$$\theta(X)(v, Y) = \big(\varphi(X)v,\ [X, Y]\big)$$

Then θ is a faithful finite-dimensional representation of $\mathfrak{g}$. Moreover $\mathfrak{z}(\mathfrak{g}) \subseteq \operatorname{nil}\mathfrak{g}$, and if φ is a nilrepresentation then so is θ.

***Proof*:**
The map θ acts componentwise as φ on V and as ad (definition 2.5.3) on $\mathfrak{g}$, and each component is a representation, so $\theta([X, X'])$ and $[\theta(X), \theta(X')]$ agree in each component and θ is a representation. Its kernel is $\ker\varphi \cap \ker\operatorname{ad}$, and $\ker\operatorname{ad} = \mathfrak{z}(\mathfrak{g})$ because $\operatorname{ad}X = 0$ means $[X, Y] = 0$ for all Y. Hence $\ker\theta = \ker\varphi \cap \mathfrak{z}(\mathfrak{g}) = 0$ and θ is faithful.

For the containment: the center is an ideal, since $[\mathfrak{g}, \mathfrak{z}(\mathfrak{g})] = 0 \subseteq \mathfrak{z}(\mathfrak{g})$, and as an abelian algebra it is nilpotent (definition 5.2.3, its lower central series stopping at the second term). A nilpotent ideal lies in the nilradical by the maximality in definition 5.5.3, so $\mathfrak{z}(\mathfrak{g}) \subseteq \operatorname{nil}\mathfrak{g}$.

Finally let φ be a nilrepresentation, write $\mathfrak{n} = \operatorname{nil}\mathfrak{g}$, and take $z \in \mathfrak{n}$. The operator $\varphi(z)$ is nilpotent by hypothesis, and $\operatorname{ad}z$ is nilpotent as well: $(\operatorname{ad}z)(\mathfrak{g}) = [z, \mathfrak{g}] \subseteq \mathfrak{n} = \mathfrak{n}^1$ because $\mathfrak{n}$ is an ideal, and if $(\operatorname{ad}z)^k(\mathfrak{g}) \subseteq \mathfrak{n}^k$ then $(\operatorname{ad}z)^{k+1}(\mathfrak{g}) = [z, (\operatorname{ad}z)^k(\mathfrak{g})] \subseteq [\mathfrak{n}, \mathfrak{n}^k] = \mathfrak{n}^{k+1}$ by the recursion of definition 5.2.1, so $(\operatorname{ad}z)^{c+1} = 0$ once $\mathfrak{n}^{c+1} = 0$. A block-diagonal operator with nilpotent blocks is nilpotent, so $\theta(z)$ is nilpotent and θ is a nilrepresentation, completing the proof.

The lemma is where the standard trap of the subject is disarmed, and it is worth pausing on. The adjoint representation kills the center, so the representation φ it is summed with must stay faithful there, and the constructions below are arranged so that, at every step of the climb, the representation

carried up from the solvable part stays injective on the nilradical, which contains the center by the lemma. Every faithfulness claim in the section reduces to this one point.

The first stage is the nilpotent case, and the construction lives entirely inside the enveloping algebra. The idea is the one already visible in proposition 8.6.6: for an abelian algebra, $U(\mathfrak{g})$ is a polynomial ring, and truncating at a fixed degree leaves a finite-dimensional space on which each variable acts as a nonzero nilpotent shift. The ideal implementing the truncation is generated by the algebra itself.

Definition 8.9.3: The Augmentation Ideal

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{C}$. The *augmentation ideal* $U_+ \subseteq U(\mathfrak{g})$ is the two-sided ideal generated by $\mathfrak{g}$ ([2, Ideals, Generating Ideals]).

The name records that U_+ is the kernel of the unital algebra homomorphism $U(\mathfrak{g}) \to \mathbb{C}$ attached to the zero representation, the homomorphism constructed after theorem 8.6.2. That description is not used below, but the spanning description is, constantly. Call a product $Y_1 Y_2 \cdots Y_l$ of elements of $\mathfrak{g}$ inside $U(\mathfrak{g})$ a *word* of length l, the empty word being 1.

Proposition 8.9.4: Words and the Powers of the Augmentation Ideal

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{C}$ with augmentation ideal U_+.

1. $U(\mathfrak{g})$ is spanned by the words of all lengths $l \geq 0$.
2. For every $m \geq 1$, the power $(U_+)^m$ is the span of the words of length at least m.
3. For $X \in \mathfrak{g}$ and $m \geq 0$, left multiplication by X carries $(U_+)^m$ into $(U_+)^{m+1}$, where $(U_+)^0 = U(\mathfrak{g})$.

***Proof*:**
(1) The tensor algebra $\mathcal{T}(\mathfrak{g})$ is spanned by the simple tensors $Y_1 \otimes \cdots \otimes Y_l$, the quotient map onto $U(\mathfrak{g})$ is surjective, and the image of such a tensor is the word $Y_1 \cdots Y_l$, exactly as in the proof of proposition 8.6.5(1).

(2) First take $m = 1$. A word of length $l \geq 1$ has the form uXv with $X \in \mathfrak{g}$ and u, v words, so it lies in the ideal generated by $\mathfrak{g}$. Conversely U_+ is spanned by the products uXv with $u, v \in U(\mathfrak{g})$ and $X \in \mathfrak{g}$, and expanding u and v into words by (1) writes each such product as a combination of words of length at least 1. For general m, the power $(U_+)^m$ is spanned by products $u_1 \cdots u_m$ with each $u_i \in U_+$, hence a combination of words of length at least 1 by the case $m = 1$, and multiplying out expresses $u_1 \cdots u_m$ as a combination of words of length at least m. Conversely a word of length $l \geq m$ factors as a product of m words of positive length, its first $m - 1$ letters taken singly and the remaining $l - m + 1$ letters lumped into the last factor, so it lies in $(U_+)^m$.

(3) Multiplying a word of length at least m on the left by X produces a word of length at least $m + 1$, and both sides of the claim are spans of such words by (1) and (2), completing the proof.

Truncating by $(U_+)^m$ kills every sufficiently long word, and by (3) each element of $\mathfrak{g}$ then acts nilpotently on the quotient, pushing vectors one step down a finite filtration. Finite-dimensionality of the quotient also comes cheaply, from the Poincaré-Birkhoff-Witt basis. The one delicate claim is faithfulness: it must be shown that no nonzero element of $\mathfrak{g}$ falls into $(U_+)^m$, and this is where

nilpotency of $\mathfrak{g}$ enters, through a bookkeeping device adapted to the lower central series. The point is that reordering a word can shorten it, a length-two word xy shedding the commutator $[x, y]$ of length one, so the length of a word is not a usable invariant on its own. What reordering does preserve is a weighted length, in which each letter counts with the depth at which it sits in the lower central series: shortening a word by a bracket merges the depths of the two letters it removes, leaving the total no smaller.

Lemma 8.9.5: Ordered Monomials by Depth

Let $\mathfrak{g}$ be a finite-dimensional nilpotent Lie algebra over $\mathbb{C}$ of nilpotency class c, so $\mathfrak{g}^{c+1} = 0$ (definition 5.2.3). Choose a basis $x_1, \dots, x_n$ *adapted to the lower central series*: a basis of $\mathfrak{g}^c$, extended to a basis of $\mathfrak{g}^{c-1}$, and so on up to $\mathfrak{g}^1 = \mathfrak{g}$. Define the *depth* w_j of x_j as the largest k with $x_j \in \mathfrak{g}^k$, and the depth of a word $x_{j_1} \cdots x_{j_l}$ in basis letters as $w_{j_1} + \cdots + w_{j_l}$. Then $1 \le w_j \le c$ for every j, the basis vectors of depth at least k form a basis of $\mathfrak{g}^k$, and:

1. every word in basis letters of depth at least m is a linear combination of ordered monomials of depth at least m
2. $\mathfrak{g} \cap (U_+)^{c+1} = 0$.

***Proof*:**

A vector adjoined while extending a basis of $\mathfrak{g}^{k+1}$ to one of $\mathfrak{g}^k$ lies in $\mathfrak{g}^k$ and not in $\mathfrak{g}^{k+1}$, since it is independent of a basis of $\mathfrak{g}^{k+1}$, so its depth is exactly k and the vectors of depth at least k are precisely those adjoined at stages $c, c-1, \dots, k$, which by construction form a basis of $\mathfrak{g}^k$. Depths are at most c because $\mathfrak{g}^{c+1} = 0$ contains no basis vector.

(1) Argue by strong induction on the length l, and for fixed length by induction on the number of inversions, exactly as in Step 1 of the proof of theorem 8.6.7. A word with no inversion is an ordered monomial of its own depth. Otherwise some adjacent pair is out of order, and (8.3) rewrites the word as the swapped word plus a word in which the two letters $x_{j_a}, x_{j_{a+1}}$ are replaced by the single letter $[x_{j_a}, x_{j_{a+1}}]$. The swapped word has the same length, the same depth, and one inversion fewer. For the bracket word, $[x_{j_a}, x_{j_{a+1}}] \in [\mathfrak{g}^{w_{j_a}}, \mathfrak{g}^{w_{j_{a+1}}}] \subseteq \mathfrak{g}^{w_{j_a} + w_{j_{a+1}}}$ by proposition 5.2.2(1), and expanding it in the adapted basis, whose vectors spanning $\mathfrak{g}^{w_{j_a} + w_{j_{a+1}}}$ all have depth at least $w_{j_a} + w_{j_{a+1}}$, produces words of length $l - 1$ and depth at least that of the original word. Both inductions terminate, proving (1).

(2) By proposition 8.9.4(2) and bilinearity of multiplication, $(U_+)^{c+1}$ is spanned by words in basis letters of length at least $c + 1$. Every letter has depth at least 1, so such words have depth at least $c + 1$, and by (1) the power $(U_+)^{c+1}$ lies in the span of the ordered monomials of depth at least $c + 1$. An ordered monomial of length 1 has depth at most c and the empty monomial has depth 0, so every ordered monomial in that span has length at least 2. Now let $X \in \mathfrak{g}$ satisfy $X \in (U_+)^{c+1}$. Expanding X in the basis writes it as a combination of ordered monomials of length 1, while the previous sentence writes it as a combination of ordered monomials of length at least 2. By the linear independence of the full family of ordered monomials (theorem 8.6.7), both combinations are zero, so $X = 0$ by corollary 8.6.10, completing the proof.

The nilpotent case of Ado's theorem now assembles itself.

Theorem 8.9.6: Ado's Theorem for Nilpotent Algebras

Let $\mathfrak{g}$ be a finite-dimensional nilpotent Lie algebra over $\mathbb{C}$ of nilpotency class c and dimension n, and set

$$V = U(\mathfrak{g}) \,/\, (U_+)^{c+1}$$

Then V is finite-dimensional, of dimension at most $\binom{n+c}{c}$, left multiplication descends to a representation φ of $\mathfrak{g}$ on V, the representation φ is faithful, and $\varphi(X)^{c+1} = 0$ for every $X \in \mathfrak{g}$. In particular every finite-dimensional nilpotent Lie algebra admits a faithful finite-dimensional representation by nilpotent operators, hence by strictly upper-triangular matrices in a suitable basis.

***Proof*:**

The representation. Left multiplication by $X \in \mathfrak{g}$ preserves the two-sided ideal $(U_+)^{c+1}$, so it descends to an operator $\varphi(X)$ on V, linearly in X. For $X, Y \in \mathfrak{g}$ and $u \in U(\mathfrak{g})$, the defining relation (8.3) gives $XYu - YXu = [X,Y]u$, so $\varphi([X,Y]) = \varphi(X)\varphi(Y) - \varphi(Y)\varphi(X)$ and φ is a representation in the sense of definition 6.4.1.

Finite dimension. By theorem 8.6.7 the ordered monomials span $U(\mathfrak{g})$, and every ordered monomial of length at least $c+1$ is a word of length at least $c+1$, hence lies in $(U_+)^{c+1}$ by proposition 8.9.4(2). So V is spanned by the images of the ordered monomials of length at most c, a basis of the filtration piece U_c, of which there are $\binom{n+c}{c}$, the count of exercise 8.6.1 (3). Hence $\dim V \leq \binom{n+c}{c}$.

Nilpotency. Let $V_m \subseteq V$ be the image of $(U_+)^m$, so that $V = V_0 \supseteq V_1 \supseteq \cdots \supseteq V_{c+1} = 0$. By proposition 8.9.4(3), $\varphi(X)V_m \subseteq V_{m+1}$ for every X and m, so $\varphi(X)^{c+1}V \subseteq V_{c+1} = 0$.

Faithfulness. If $\varphi(X) = 0$, then applying $\varphi(X)$ to the image of 1 shows $X \in (U_+)^{c+1}$, and lemma 8.9.5(2) forces $X = 0$.

Triangular form. The space V is nonzero: by lemma 8.9.5(1), the ideal $(U_+)^{c+1}$ lies in the span of the ordered monomials of positive depth, all of positive length, and 1 is linearly independent of these by theorem 8.6.7, so the image of 1 in V is nonzero. The image $\varphi(\mathfrak{g}) \subseteq \mathfrak{gl}(V)$ is therefore a Lie algebra of nilpotent operators on a nonzero finite-dimensional space, and the operator form of Engel's theorem (theorem 5.3.4) produces a basis in which every $\varphi(X)$ is strictly upper-triangular, completing the proof.

For an abelian algebra the theorem reproduces the classical picture. There $c = 1$, the quotient $V = U(\mathfrak{g})/(U_+)^2$ has the basis $1, x_1, \ldots, x_n$, and each X acts by sending 1 to X and every x_j to zero, a square-zero block matrix. Exercises 8.9.1 (1) and 8.9.1 (2) work this and the Heisenberg case out in coordinates. Two features of the construction deserve emphasis, because the rest of the proof leans on both. The representation space is a quotient of $U(\mathfrak{g})$ by a two-sided ideal, with $\mathfrak{g}$ acting by left multiplication, and the ideal is defined canonically from $\mathfrak{g}$, with no auxiliary choices. The second stage explains why canonicity matters: ideals of this kind are preserved by every derivation, which is exactly what an extension needs.

Lemma 8.9.7: Derivations Extend to the Enveloping Algebra

Let $\mathfrak{h}$ be a finite-dimensional Lie algebra over $\mathbb{C}$ and D a derivation of $\mathfrak{h}$, a linear map with $D[X,Y] = [DX,Y] + [X,DY]$ as in lemma 5.1.4. Then there is a unique derivation $\widetilde{D}$ of the associative algebra $U(\mathfrak{h})$, a linear map with $\widetilde{D}(uv) = \widetilde{D}(u)\,v + u\,\widetilde{D}(v)$, whose restriction to $\mathfrak{h}$ is D. Moreover:

1. $\widetilde{D}(1) = 0$, and on a word, $\widetilde{D}(Y_1 \cdots Y_l) = \sum_{a=1}^{l} Y_1 \cdots (DY_a) \cdots Y_l$
2. the assignment $D \mapsto \widetilde{D}$ is linear, and $[\widetilde{D}_1, \widetilde{D}_2] = \widetilde{[D_1, D_2]}$ for derivations D_1, D_2 of $\mathfrak{h}$.

***Proof*:**

Existence. Define a linear endomorphism $D_{\mathcal{T}}$ of the tensor algebra $\mathcal{T}(\mathfrak{h})$ ([2, Tensor Algebra]) by $D_{\mathcal{T}} = 0$ on $\mathcal{T}^0$ and

$$D_{\mathcal{T}}(Y_1 \otimes \cdots \otimes Y_l) = \sum_{a=1}^{l} Y_1 \otimes \cdots \otimes DY_a \otimes \cdots \otimes Y_l$$

on $\mathcal{T}^l$, which is well defined because the right side is multilinear in $Y_1, \ldots, Y_l$. On a product of two simple tensors the defining sum splits into the terms differentiating a factor of the first block and the terms differentiating a factor of the second, which is the Leibniz rule, and bilinearity extends it to all of $\mathcal{T}(\mathfrak{h})$, so $D_{\mathcal{T}}$ is a derivation of $\mathcal{T}(\mathfrak{h})$. It preserves the ideal $J(\mathfrak{h})$ of definition 8.6.1: on a generator,

$$D_{\mathcal{T}}\big(X\otimes Y - Y\otimes X - [X,Y]\big) = \big(DX\otimes Y - Y\otimes DX - [DX,Y]\big) + \big(X\otimes DY - DY\otimes X - [X,DY]\big)$$

using $D[X,Y] = [DX,Y] + [X,DY]$, a sum of two generators, and on a general element $a\,g\,b$ of $J(\mathfrak{h})$ with g a generator, the Leibniz rule gives $D_{\mathcal{T}}(a\,g\,b) = D_{\mathcal{T}}(a)\,g\,b + a\,D_{\mathcal{T}}(g)\,b + a\,g\,D_{\mathcal{T}}(b)$, each term in the two-sided ideal $J(\mathfrak{h})$. Hence $D_{\mathcal{T}}$ descends to a derivation $\widetilde{D}$ of $U(\mathfrak{h}) = \mathcal{T}(\mathfrak{h})/J(\mathfrak{h})$ restricting to D on $\mathfrak{h}$.

Uniqueness and (1). Any derivation δ of $U(\mathfrak{h})$ satisfies $\delta(1) = \delta(1\cdot 1) = 2\delta(1)$, so $\delta(1) = 0$, and the Leibniz rule determines δ on every word from its values on $\mathfrak{h}$, by induction on length. Words span $U(\mathfrak{h})$ by proposition 8.9.4(1), so a derivation is determined by its restriction to $\mathfrak{h}$, giving uniqueness, and the displayed formula in (1) follows by the same induction.

(2). For scalars α, β, the map $\alpha\widetilde{D}_1 + \beta\widetilde{D}_2$ is a derivation of $U(\mathfrak{h})$ restricting to $\alpha D_1 + \beta D_2$, so it equals the extension of $\alpha D_1 + \beta D_2$ by uniqueness. Similarly $[\widetilde{D}_1, \widetilde{D}_2]$ is a derivation of $U(\mathfrak{h})$, by the standard computation

$$[\widetilde{D}_1, \widetilde{D}_2](uv) = [\widetilde{D}_1, \widetilde{D}_2](u)\,v + u\,[\widetilde{D}_1, \widetilde{D}_2](v)$$

obtained by expanding both compositions with the Leibniz rule and cancelling the mixed terms, and its restriction to $\mathfrak{h}$ is $D_1D_2 - D_2D_1 = [D_1, D_2]$, itself a derivation of $\mathfrak{h}$ by the same cancellation. Uniqueness forces $[\widetilde{D}_1, \widetilde{D}_2] = \widetilde{[D_1, D_2]}$, completing the proof.

With derivations available upstairs, a representation of an ideal can be extended across a complementary subalgebra, provided the ideal of $U(\mathfrak{a})$ being acted on is stable under the relevant derivations.

The frame is the internal semidirect product of definition 6.5.1, and the statement isolates exactly what the two applications below need.

Proposition 8.9.8: Extending a Representation across a Semidirect Product

Let $\mathfrak{g} = \mathfrak{a} \rtimes \mathfrak{h}$ be an internal semidirect product of finite-dimensional Lie algebras over $\mathbb{C}$: an ideal $\mathfrak{a} \trianglelefteq \mathfrak{g}$ and a subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ with $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{h}$ as vector spaces (definition 6.5.1). For $Z \in \mathfrak{h}$ write $D_Z = \mathrm{ad}Z|_{\mathfrak{a}}$, a derivation of $\mathfrak{a}$. Let $J \subseteq U(\mathfrak{a})$ be a two-sided ideal with $\widetilde{D_Z}(J) \subseteq J$ for every $Z \in \mathfrak{h}$, and set $V = U(\mathfrak{a})/J$. Then

$$\rho(Y + Z)\,\big(u + J\big) = Yu + \widetilde{D_Z}(u) + J \qquad (Y \in \mathfrak{a},\ Z \in \mathfrak{h},\ u \in U(\mathfrak{a}))$$

is a well-defined representation of $\mathfrak{g}$ on V. For $Y \in \mathfrak{a}$, the operator $\rho(Y)$ is induced by left multiplication, and $\rho(Y) = 0$ exactly when $Y \in J$.

***Proof*:**

Well-definedness and linearity. For $Z \in \mathfrak{h}$ the map D_Z is a derivation of $\mathfrak{a}$: it preserves $\mathfrak{a}$ because $\mathfrak{a}$ is an ideal, and the Leibniz rule is the Jacobi identity read as in the proof of lemma 5.1.4. Left multiplication preserves the two-sided ideal J, and $\widetilde{D_Z}(J) \subseteq J$ by hypothesis, so both operators descend to V. The decomposition $X = Y + Z$ is linear in X, left multiplication is linear in Y, and $Z \mapsto \widetilde{D_Z}$ is linear by lemma 8.9.7(2), so ρ is linear.

The bracket, in three cases. Write λ_Y for left multiplication by $Y \in \mathfrak{a}$ on $U(\mathfrak{a})$. Every identity checked below among the λ's and $\widetilde{D}$'s descends to V. For $Y, Y' \in \mathfrak{a}$,

$$[\lambda_Y, \lambda_{Y'}]\,u = YY'u - Y'Yu = [Y, Y']\,u = \lambda_{[Y,Y']}\,u$$

by (8.3). For $Z \in \mathfrak{h}$ and $Y \in \mathfrak{a}$, the Leibniz rule gives

$$[\widetilde{D_Z}, \lambda_Y]\,u = \widetilde{D_Z}(Yu) - Y\,\widetilde{D_Z}(u) = \widetilde{D_Z}(Y)\,u = [Z, Y]\,u = \lambda_{[Z,Y]}\,u$$

and $[Z, Y] \in \mathfrak{a}$ because $\mathfrak{a}$ is an ideal, so the right side is $\rho([Z, Y])$. For $Z, Z' \in \mathfrak{h}$, lemma 8.9.7(2) gives $[\widetilde{D_Z}, \widetilde{D_{Z'}}] = [\widetilde{D_Z, D_{Z'}}]$, and $[D_Z, D_{Z'}] = D_{[Z,Z']}$ as derivations of $\mathfrak{a}$ by the Jacobi identity,

$$[D_Z, D_{Z'}](Y) = [Z, [Z', Y]] - [Z', [Z, Y]] = [[Z, Z'], Y]$$

with $[Z, Z'] \in \mathfrak{h}$ because $\mathfrak{h}$ is a subalgebra, so $[\rho(Z), \rho(Z')] = \rho([Z, Z'])$.

Assembly. For $X = Y + Z$ and $X' = Y' + Z'$, the bracket decomposes as

$$[X, X'] = \big([Y, Y'] + [Z, Y'] - [Z', Y]\big) + [Z, Z']$$

with the first group in $\mathfrak{a}$ and the last term in $\mathfrak{h}$, and expanding $[\rho(X), \rho(X')]$ bilinearly into the four commutators just computed matches $\rho([X, X'])$ term by term. Hence ρ is a representation.

The last claim. $\rho(Y)$ is induced by λ_Y by construction. If $Y \in J$ then $Yu \in J$ for every u, so $\rho(Y) = 0$. Conversely if $\rho(Y) = 0$, evaluating at the image of 1 gives $Y \in J$, completing the proof.

The proposition explains retroactively why the complement must be a subalgebra. Two derivation

operators bracket to the extension of an inner derivation when $[Z, Z']$ falls into $\mathfrak{a}$, and the extension of an inner derivation is $u \mapsto Yu - uY$, not left multiplication (exercise 8.9.1 (3)), so the case analysis above would break. The proposition also makes plain what remains to be arranged: a finite-dimensional quotient of $U(\mathfrak{a})$ whose defining ideal is stable under every derivation in sight. Kernels of representations have no reason to be stable, and the next two results repair this. The first is the structural input, and it is the one place in the entire section where algebraic closure of the field is used, through the corollaries of Lie's theorem.

Lemma 8.9.9: Derivations of a Solvable Algebra Map into the Nilradical

Let $\mathfrak{a}$ be a finite-dimensional solvable Lie algebra over $\mathbb{C}$, or over any algebraically closed field of characteristic zero, and let D be a derivation of $\mathfrak{a}$. Then $D(\mathfrak{a}) \subseteq \operatorname{nil} \mathfrak{a}$.

***Proof*:**

Form the semidirect product $\widetilde{\mathfrak{a}} = \mathfrak{a} \rtimes_\theta \mathbb{C}$ of definition 6.5.1, where $\mathbb{C}$ is the one-dimensional abelian Lie algebra and $\theta(t) = tD$, a homomorphism into the derivations of $\mathfrak{a}$ because the source is abelian and $[tD, sD] = 0$. Inside $\widetilde{\mathfrak{a}}$ the subspace $\mathfrak{a}$ is an ideal with one-dimensional abelian quotient, so $\widetilde{\mathfrak{a}}$ is an extension of a solvable algebra by a solvable ideal, hence solvable by proposition 5.1.6(2). Writing $e = (0, 1)$, the semidirect bracket gives $[e, (Y, 0)] = (DY, 0)$, so $D(\mathfrak{a}) \subseteq [\widetilde{\mathfrak{a}}, \widetilde{\mathfrak{a}}]$, and every bracket $[(Y, t), (Y', t')]$ has vanishing second component, so $\mathfrak{m} := [\widetilde{\mathfrak{a}}, \widetilde{\mathfrak{a}}] \subseteq \mathfrak{a}$.

By corollary 5.4.5(2), which is where algebraic closure and characteristic zero enter, the derived algebra $\mathfrak{m}$ of the solvable algebra $\widetilde{\mathfrak{a}}$ is nilpotent. It is an ideal of $\widetilde{\mathfrak{a}}$ by lemma 5.1.4, hence an ideal of $\mathfrak{a}$: $[\mathfrak{a}, \mathfrak{m}] \subseteq [\widetilde{\mathfrak{a}}, \mathfrak{m}] \subseteq \mathfrak{m}$. A nilpotent ideal of $\mathfrak{a}$ lies in the nilradical (definition 5.5.3), so $D(\mathfrak{a}) \subseteq \mathfrak{m} \subseteq \operatorname{nil} \mathfrak{a}$, completing the proof.

Specializing D to inner derivations, and feeding the lemma back through the radical, produces every structural fact the induction will need. The corollary is stated once and used three times below.

Corollary 8.9.10: The Nilradical along the Radical

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{C}$ with radical $\mathfrak{r} = \operatorname{rad}\mathfrak{g}$ and nilradical $\mathfrak{n} = \operatorname{nil} \mathfrak{g}$.

1. $[\mathfrak{g}, \mathfrak{r}] \subseteq \mathfrak{n}$, and $\operatorname{nil} \mathfrak{r} = \mathfrak{n}$.
2. Every subspace $\mathfrak{b}$ with $\mathfrak{n} \subseteq \mathfrak{b} \subseteq \mathfrak{r}$ is an ideal of $\mathfrak{g}$.
3. Every such $\mathfrak{b}$ is solvable with $\operatorname{nil} \mathfrak{b} = \mathfrak{n}$.

***Proof*:**

(1) For $X \in \mathfrak{g}$, the operator $\operatorname{ad}X$ preserves the ideal $\mathfrak{r}$ and restricts to a derivation of it, so lemma 8.9.9 applied to the solvable algebra $\mathfrak{r}$ gives $[X, \mathfrak{r}] \subseteq \operatorname{nil} \mathfrak{r}$. Hence $[\mathfrak{g}, \mathfrak{r}] \subseteq \operatorname{nil} \mathfrak{r}$, and it remains to identify $\operatorname{nil} \mathfrak{r}$ with $\mathfrak{n}$. The containment just proved shows $[\mathfrak{g}, \operatorname{nil} \mathfrak{r}] \subseteq \operatorname{nil} \mathfrak{r}$, so $\operatorname{nil} \mathfrak{r}$ is a nilpotent ideal of $\mathfrak{g}$ and $\operatorname{nil} \mathfrak{r} \subseteq \mathfrak{n}$ by definition 5.5.3. Conversely $\mathfrak{n} \subseteq \mathfrak{r}$ by (5.8), and $\mathfrak{n}$ is then a nilpotent ideal of $\mathfrak{r}$, so $\mathfrak{n} \subseteq \operatorname{nil} \mathfrak{r}$.

(2) $[\mathfrak{g}, \mathfrak{b}] \subseteq [\mathfrak{g}, \mathfrak{r}] \subseteq \mathfrak{n} \subseteq \mathfrak{b}$ by (1). In particular $\mathfrak{b}$ is a subalgebra.

(3) $\mathfrak{b}$ is a subalgebra of the solvable algebra $\mathfrak{r}$, hence solvable by proposition 5.1.6(1). The ideal $\mathfrak{n}$ of $\mathfrak{g}$ is a nilpotent ideal of $\mathfrak{b}$, so $\mathfrak{n} \subseteq \operatorname{nil} \mathfrak{b}$. Conversely, for $X \in \mathfrak{r}$ the operator $\operatorname{ad}X$ preserves $\mathfrak{b}$ by

(2) and restricts to a derivation of it, so lemma 8.9.9 applied to $\mathfrak{b}$ gives $[\mathfrak{r}, \mathfrak{b}] \subseteq \operatorname{nil} \mathfrak{b}$. In particular $[\mathfrak{r}, \operatorname{nil} \mathfrak{b}] \subseteq \operatorname{nil} \mathfrak{b}$, so $\operatorname{nil} \mathfrak{b}$ is a nilpotent ideal of $\mathfrak{r}$ and $\operatorname{nil} \mathfrak{b} \subseteq \operatorname{nil} \mathfrak{r} = \mathfrak{n}$ by (1), completing the proof.

Part (1) sharpens the picture of a general Lie algebra given by theorem 6.5.4: the semisimple factor carries the radical into the nilradical, so its induced action on $\mathfrak{r}/\mathfrak{n}$ is zero. Part (3) is the fact that keeps the meaning of "nilrepresentation" stable along the induction, since every intermediate algebra between $\mathfrak{n}$ and $\mathfrak{r}$ has the same nilradical. The last preparatory result is the technical heart of the whole proof. It takes an arbitrary finite-dimensional representation of a solvable algebra and replaces its kernel ideal by a smaller ideal of $U(\mathfrak{a})$ that every derivation preserves, though it enlarges the representation space by a controlled amount.

Lemma 8.9.11: The Stabilized Ideal

Let $\mathfrak{a}$ be a finite-dimensional solvable Lie algebra over $\mathbb{C}$, with nilradical $\mathfrak{n} = \operatorname{nil} \mathfrak{a}$, and let $I \subseteq U(\mathfrak{a})$ be a two-sided ideal such that

1. $U(\mathfrak{a})/I$ is finite-dimensional, and
2. the image in $U(\mathfrak{a})/I$ of every $z \in \mathfrak{n}$ is nilpotent.

Then there is a two-sided ideal $J \subseteq I$ satisfying (1) and (2) and additionally

3. $\widetilde{D}(J) \subseteq J$ for every derivation D of $\mathfrak{a}$.

***Proof*:**
If $I = U(\mathfrak{a})$, take $J = I$: all three conditions hold trivially. Assume I proper, write $A = U(\mathfrak{a})/I$ and $\pi\colon U(\mathfrak{a}) \to A$ for the quotient map, and set $N = \dim A \geq 1$. Let $Q \subseteq A$ be the two-sided ideal generated by the subspace $\pi(\mathfrak{n})$, let $K = \pi^{-1}(Q)$, a two-sided ideal of $U(\mathfrak{a})$ containing I, and define

$$J = K^N$$

The proof verifies, in order: $Q^N = 0$, hence $J \subseteq I$, then conditions (3), (1), and (2).

Step 1: any product of N elements of $\pi(\mathfrak{n})$ vanishes. Let $L\colon A \to \mathfrak{gl}(A)$ send a to left multiplication by a, a homomorphism of unital algebras. For $z, z' \in \mathfrak{n}$, the relation $zz' - z'z = [z, z']$ of (8.3) maps under π to $\pi(z)\pi(z') - \pi(z')\pi(z) = \pi([z, z'])$ with $[z, z'] \in \mathfrak{n}$, so $L(\pi(\mathfrak{n}))$ is a Lie subalgebra of $\mathfrak{gl}(A)$. Each of its elements is a nilpotent operator: by (2) there is a p with $\pi(z)^p = 0$, and $L_{\pi(z)}^p = L_{\pi(z)^p} = 0$. The operator form of Engel's theorem (theorem 5.3.4) yields a full flag $0 = A_0 \subsetneq A_1 \subsetneq \cdots \subsetneq A_N = A$ with $L(\pi(\mathfrak{n}))\, A_i \subseteq A_{i-1}$, so any composition of N operators from $L(\pi(\mathfrak{n}))$ is zero. Evaluating such a composition at $1 \in A$ shows $\pi(z_1)\pi(z_2)\cdots\pi(z_N) = 0$ for all $z_1, \ldots, z_N \in \mathfrak{n}$, and by linearity the same holds for any N elements of the subspace $\pi(\mathfrak{n})$.

Step 2: $Q^N = 0$, hence $J \subseteq I$. The power Q^N is spanned by the products $c_0 s_1 c_1 s_2 \cdots s_N c_N$ with each $s_i \in \pi(\mathfrak{n})$ and each $c_i \in A$. Since $U(\mathfrak{a})$ is spanned by words (proposition 8.9.4(1)) and π is a surjective algebra homomorphism, each c_i is a combination of products of elements of $\pi(\mathfrak{a})$, so Q^N is spanned by finite products whose factors come from $\pi(\mathfrak{a})$ and from $\pi(\mathfrak{n})$, at least N factors being of the second kind. Call the second-kind factors marked. The claim is that every such product is a combination of products beginning with N marked factors, which vanish by Step 1. Argue by strong induction on the number of factors, and for a fixed number of factors by induction on the number of pairs consisting of an unmarked factor standing anywhere to the left of a marked one. If there is no such pair, all marked factors stand at the front and the

product is zero by Step 1. Otherwise, take the leftmost marked factor preceded somewhere by an unmarked one. Its immediate predecessor must itself be unmarked, since a marked predecessor would stand further left with the same unmarked factor before it. So some unmarked $\pi(Y)$, $Y \in \mathfrak{a}$, stands immediately to the left of a marked $\pi(z)$, $z \in \mathfrak{n}$, and

$$\pi(Y)\pi(z) = \pi(z)\pi(Y) + \pi([Y,z])$$

by (8.3), with $[Y,z] \in \mathfrak{n}$ because the nilradical is an ideal. The first term has the same factors with one offending pair fewer, and the second has one factor fewer with the two factors merged into a single marked one, so both inductions close. Hence $Q^N = 0$, and applying π to $J = K^N$ gives $\pi(J) = Q^N = 0$, that is $J \subseteq \ker \pi = I$.

Step 3: condition (3). Let D be any derivation of $\mathfrak{a}$. By lemma 8.9.9, $D(\mathfrak{a}) \subseteq \mathfrak{n}$. On a word, lemma 8.9.7(1) gives $\widetilde{D}(Y_1 \cdots Y_l) = \sum_a Y_1 \cdots (DY_a) \cdots Y_l$, and each summand contains the factor $DY_a \in \mathfrak{n} \subseteq K$, so each summand lies in the two-sided ideal K. Together with $\widetilde{D}(1) = 0$ and proposition 8.9.4(1), this shows $\widetilde{D}(U(\mathfrak{a})) \subseteq K$. Now $J = K^N$ is spanned by products $u_1 \cdots u_N$ with every $u_i \in K$, and the Leibniz rule gives

$$\widetilde{D}(u_1 \cdots u_N) = \sum_{i=1}^{N} u_1 \cdots \widetilde{D}(u_i) \cdots u_N$$

in which every summand is again a product of N elements of K, since $\widetilde{D}(u_i) \in K$. Hence $\widetilde{D}(J) \subseteq J$.

Step 4: condition (1). First, $1 \notin K$: otherwise $1 = \pi(1) \in Q$, so $Q = A$ and $A = Q^N = 0$ by Step 2, contradicting the properness of I. Fix a basis $X_1, \ldots, X_n$ of $\mathfrak{a}$. For each i the powers $1, X_i, X_i^2, \ldots$ have linearly dependent images in the finite-dimensional space $U(\mathfrak{a})/K$, a quotient of A since $K \supseteq I$, so some nonzero polynomial p has $p(X_i) \in K$. Its top nonzero coefficient cannot sit in degree zero, since $1 \in K$ is excluded, so after scaling there is a monic p_i of degree $e_i \geq 1$ with $p_i(X_i) \in K$. Then $p_i(X_i)^N \in K^N = J$, and since powers of the single element X_i commute, $p_i(X_i)^N$ is a monic polynomial in X_i of degree $f_i = Ne_i$. Write $p_i(X_i)^N = X_i^{f_i} - r_i(X_i)$ with $\deg r_i < f_i$. Order the basis and consider the ordered monomials $X_1^{a_1} \cdots X_n^{a_n}$, which span $U(\mathfrak{a})$ by theorem 8.6.7. If $a_i \geq f_i$ for some i, then

$$X_1^{a_1} \cdots X_n^{a_n} = u \left(p_i(X_i)^N\right) v + u\, r_i(X_i)\, v$$

where $u = X_1^{a_1} \cdots X_i^{a_i - f_i}$ and $v = X_{i+1}^{a_{i+1}} \cdots X_n^{a_n}$. The first term lies in the two-sided ideal J, and the second is a combination of ordered monomials of strictly smaller total degree. Strong induction on the total degree shows every ordered monomial is, modulo J, a combination of the ordered monomials with $a_i < f_i$ for all i, of which there are at most $f_1 \cdots f_n$. Hence $\dim U(\mathfrak{a})/J \leq f_1 \cdots f_n$ is finite.

Step 5: condition (2). For $z \in \mathfrak{n}$, condition (2) for I gives $z^p \in I \subseteq K$ for some p, so $z^{pN} = (z^p)^N \in K^N = J$, and the image of z in $U(\mathfrak{a})/J$ is nilpotent, completing the proof.

Nothing in the lemma remembers the representation the ideal I came from beyond the two hypotheses, and that forgetfulness is its virtue: the stabilized ideal J is preserved by all derivations at once, including derivations arriving from an ambient algebra that has not yet entered the argument. The pieces now assemble into the solvable case.

Theorem 8.9.12: Ado's Theorem for Solvable Algebras

Every finite-dimensional solvable Lie algebra $\mathfrak{g}$ over $\mathbb{C}$ admits a faithful finite-dimensional nilrepresentation.

***Proof*:**
Write $\mathfrak{n} = \operatorname{nil}\mathfrak{g}$ and $m = \dim\mathfrak{g} - \dim\mathfrak{n}$. Since $\mathfrak{g}$ is a solvable ideal of itself, $\operatorname{rad}\mathfrak{g} = \mathfrak{g}$, so corollary 8.9.10 applies with $\mathfrak{r} = \mathfrak{g}$: choosing subspaces $\mathfrak{n} = \mathfrak{g}_0 \subset \mathfrak{g}_1 \subset \cdots \subset \mathfrak{g}_m = \mathfrak{g}$ with $\dim\mathfrak{g}_i = \dim\mathfrak{n} + i$, each $\mathfrak{g}_i$ is a solvable ideal of $\mathfrak{g}$ with $\operatorname{nil}\mathfrak{g}_i = \mathfrak{n}$, by parts (2) and (3). In particular $\mathfrak{g}_i \trianglelefteq \mathfrak{g}_{i+1}$, and picking $z_i \in \mathfrak{g}_{i+1} \setminus \mathfrak{g}_i$ exhibits each step as an internal semidirect product

$$\mathfrak{g}_{i+1} = \mathfrak{g}_i \rtimes \mathbb{C}z_i$$

with the line $\mathbb{C}z_i$ a subalgebra.

We construct, by induction on i, finite-dimensional representations σ_i of $\mathfrak{g}_i$ such that $\sigma_i(z)$ is nilpotent for every $z \in \mathfrak{n}$ and $\ker\sigma_i \cap \mathfrak{n} = 0$.

Base. Take σ_0 to be the representation of theorem 8.9.6 for the nilpotent algebra $\mathfrak{n}$: it is faithful and every element of $\mathfrak{n}$ acts nilpotently.

Step. Suppose σ_i is given, acting on V_i. Let $\Phi_i\colon U(\mathfrak{g}_i) \to \operatorname{End}(V_i)$ be the algebra homomorphism attached to σ_i by corollary 8.6.3, and set $I_i = \ker\Phi_i$. Then $U(\mathfrak{g}_i)/I_i$ embeds into $\operatorname{End}(V_i)$ as an algebra by the first isomorphism theorem ([2, The first isomorphism theorem for Rings]), so it is finite-dimensional, which is hypothesis (1) of lemma 8.9.11. For $z \in \mathfrak{n} = \operatorname{nil}\mathfrak{g}_i$, the embedding carries the image of z to $\sigma_i(z)$, which is nilpotent by the inductive hypothesis, and an element with nilpotent image under an injective algebra homomorphism is itself nilpotent, which is hypothesis (2). Since $\mathfrak{g}_i$ is solvable, lemma 8.9.11 produces a two-sided ideal $J_i \subseteq I_i$ satisfying (1), (2), and (3). Apply proposition 8.9.8 to the semidirect product $\mathfrak{g}_{i+1} = \mathfrak{g}_i \rtimes \mathbb{C}z_i$ and the ideal J_i, whose stability under $\widetilde{D}$ for $D = \operatorname{ad}z_i|_{\mathfrak{g}_i}$ is an instance of (3): the result is a representation σ_{i+1} of $\mathfrak{g}_{i+1}$ on $V_{i+1} = U(\mathfrak{g}_i)/J_i$, finite-dimensional by (1).

The two inductive properties persist. For $z \in \mathfrak{n} \subseteq \mathfrak{g}_i$, the operator $\sigma_{i+1}(z)$ is induced by left multiplication by z, and by (2) some power z^p lies in J_i, so $\sigma_{i+1}(z)^p$ is induced by left multiplication by z^p and vanishes. And if $z \in \mathfrak{n}$ has $\sigma_{i+1}(z) = 0$, then $z \in J_i \subseteq I_i$ by the last clause of proposition 8.9.8, so $\sigma_i(z) = \Phi_i(z) = 0$, and the inductive hypothesis forces $z = 0$.

Assembly. The representation σ_m of $\mathfrak{g}_m = \mathfrak{g}$ is finite-dimensional, every $z \in \mathfrak{n}$ acts nilpotently, and $\ker\sigma_m \cap \mathfrak{n} = 0$. Since $\mathfrak{z}(\mathfrak{g}) \subseteq \mathfrak{n}$ by lemma 8.9.2, the kernel of σ_m meets the center trivially, so lemma 8.9.2 upgrades $\theta = \sigma_m \oplus \operatorname{ad}$ to a faithful finite-dimensional nilrepresentation of $\mathfrak{g}$, completing the proof.

The general case is now a single further extension. The Levi decomposition replaces the one-dimensional steps of the climb by one semidirect product whose complement is the entire semisimple factor, and the stabilized ideal absorbs the new derivations without any change to the argument.

Theorem 8.9.13: Ado's Theorem over the Complex Numbers

Every finite-dimensional Lie algebra $\mathfrak{g}$ over $\mathbb{C}$, or over any algebraically closed field of characteristic zero, admits a faithful finite-dimensional nilrepresentation. In particular $\mathfrak{g}$ is isomorphic to a Lie subalgebra of $\mathfrak{gl}_N(K)$ for some N, where K is the ground field.

***Proof*:**
Write $\mathfrak{r} = \operatorname{rad}\mathfrak{g}$ and $\mathfrak{n} = \operatorname{nil}\mathfrak{g}$. The Levi decomposition (theorem 6.5.4) provides a semisimple subalgebra $\mathfrak{s}$ with $\mathfrak{g} = \mathfrak{r} \rtimes \mathfrak{s}$ as an internal semidirect product. By theorem 8.9.12 the solvable algebra $\mathfrak{r}$ has a faithful finite-dimensional nilrepresentation σ on a space V, and by corollary 8.9.10(1) its nilradical is $\operatorname{nil}\mathfrak{r} = \mathfrak{n}$, so $\sigma(z)$ is nilpotent for every $z \in \mathfrak{n}$.

Let $\Phi\colon U(\mathfrak{r}) \to \operatorname{End}(V)$ be the algebra homomorphism of corollary 8.6.3 and $I = \ker\Phi$. As in the proof of theorem 8.9.12, the quotient $U(\mathfrak{r})/I$ embeds into $\operatorname{End}(V)$, so it is finite-dimensional and the image of every $z \in \mathfrak{n}$ in it is nilpotent. Lemma 8.9.11 applied to the solvable algebra $\mathfrak{r}$ produces $J \subseteq I$ with properties (1) to (3), and proposition 8.9.8 applied to $\mathfrak{g} = \mathfrak{r} \rtimes \mathfrak{s}$, with the stability of J under $\widetilde{D}$ for every $D = \operatorname{ad}Z|_{\mathfrak{r}}$, $Z \in \mathfrak{s}$, again an instance of (3), yields a finite-dimensional representation ρ of $\mathfrak{g}$ on $U(\mathfrak{r})/J$.

The representation ρ is a nilrepresentation: every $z \in \mathfrak{n}$ lies in $\mathfrak{r}$ and acts by left multiplication, some power z^p lies in J by property (2), and $\rho(z)^p = 0$ follows. Moreover $\ker\rho \cap \mathfrak{r} = 0$: if $Y \in \mathfrak{r}$ has $\rho(Y) = 0$, the last clause of proposition 8.9.8 puts $Y \in J \subseteq I$, so $\sigma(Y) = \Phi(Y) = 0$, and faithfulness of σ gives $Y = 0$. Since $\mathfrak{z}(\mathfrak{g}) \subseteq \mathfrak{n} \subseteq \mathfrak{r}$ by lemma 8.9.2 and (5.8), the kernel of ρ meets the center trivially, and lemma 8.9.2 makes $\theta = \rho \oplus \operatorname{ad}$ a faithful finite-dimensional nilrepresentation of $\mathfrak{g}$. Choosing a basis of the representation space identifies $\mathfrak{g}$ with a Lie subalgebra of $\mathfrak{gl}_N(\mathbb{C})$, completing the proof.

It remains to descend to the real numbers, which is the case Part I uses. The descent is by complexification, the construction Part IV will develop systematically. Here only its first step is needed, and it is self-contained.

Corollary 8.9.14: Ado's Theorem over the Real Numbers

Every finite-dimensional real Lie algebra $\mathfrak{g}$ admits a faithful finite-dimensional representation: there is an injective Lie-algebra homomorphism $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(\mathbb{R})$ for some n. Together with theorem 8.9.13 for the complex clause, this proves theorem 3.4.3 in full.

***Proof*:**
Let $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$ be the complexification: the complex vector space $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$, with the bracket extended complex-bilinearly,

$$[X + iY,\ X' + iY'] = \big([X,X'] - [Y,Y']\big) + i\big([X,Y'] + [Y,X']\big)$$

Bilinearity, antisymmetry, and the Jacobi identity hold because both sides of each axiom are multilinear expressions agreeing on the real points, which span $\mathfrak{g}_{\mathbb{C}}$ over $\mathbb{C}$. The map $X \mapsto X + i0$ is an injective homomorphism of real Lie algebras $\mathfrak{g} \to \mathfrak{g}_{\mathbb{C}}$.

By theorem 8.9.13 there is a faithful representation of the complex Lie algebra $\mathfrak{g}_{\mathbb{C}}$ on some $\mathbb{C}^N$, and restricting it to $\mathfrak{g}$ gives an injective homomorphism of real Lie algebras $\mathfrak{g} \to \mathfrak{gl}_N(\mathbb{C})$. Finally,

regard $\mathbb{C}^N$ as $\mathbb{R}^{2N}$. Every complex-linear operator is real-linear, composition and addition are unchanged, and a nonzero complex-linear operator remains nonzero, so $\mathfrak{gl}_N(\mathbb{C})$ is a real Lie subalgebra of $\mathfrak{gl}_{2N}(\mathbb{R})$ and the composite $\mathfrak{g} \hookrightarrow \mathfrak{gl}_{2N}(\mathbb{R})$ is an injective homomorphism of real Lie algebras. Taking $n = 2N$ completes the proof.

With the corollary, the debt to Part I is retired. Lie's third theorem (theorem 3.4.4) and the convergence of the Baker-Campbell-Hausdorff series (proposition 3.5.2) now rest entirely on proofs carried out inside this book, and the equivalence between simply connected Lie groups and finite-dimensional real Lie algebras stands on its own ground. It is worth recording exactly how much of the field was used. Every construction of this section, the truncated enveloping algebra, the derivation extension, the stabilized ideal, and the two climbs, is valid over an arbitrary field of characteristic zero. Algebraic closure entered once, in lemma 8.9.9 through corollary 5.4.5(2), and exercise 8.9.1 (5) removes it by a scalar-extension argument, so that Ado's theorem holds over every field k of characteristic zero, with $\mathfrak{gl}_n(k)$ as the target. Characteristic zero, by contrast, is woven into this particular proof, entering through Lie's theorem and the Levi decomposition. The conclusion itself persists in positive characteristic, where an argument of Iwasawa running through restricted enveloping algebras produces the faithful representation, a route beyond the scope of this book.

The proof also repays a second look at where its weight sits. The enveloping algebra carried the entire construction: the nilpotent case is a truncation of $U(\mathfrak{g})$, the extension step is an action on a quotient of $U(\mathfrak{a})$, and every operator built along the way is a left multiplication or an induced derivation. This is the concrete meaning of the remark after corollary 8.6.12 that Ado's theorem locates a finite-dimensional quotient of a $U(\mathfrak{g})$-module. The structural theorems played the complementary role: Engel's theorem turned pointwise nilpotency into the vanishing of long products, Lie's corollaries confined derivations of a solvable algebra to its nilradical, and the Levi decomposition reduced the general case to the solvable one. With the debt paid, Part II closes, and Part III turns its instrument on the compact groups.

Exercise 8.9.1

1. Let $\mathfrak{g}$ be the abelian Lie algebra $\mathbb{C}^n$, of nilpotency class $c = 1$. Show that the space $V = U(\mathfrak{g})/(U_+)^2$ of theorem 8.9.6 has basis the images of $1, x_1, \dots, x_n$, compute the matrix of $\varphi(X)$ for $X = \sum_j c_j x_j$ in this basis, and verify directly that φ is faithful with $\varphi(X)^2 = 0$ for every X. Compare with the dimension bound $\binom{n+1}{1} = n + 1$.

2. Let $\mathfrak{g} = \mathfrak{n}_3$ be the Heisenberg algebra of example 5.2.5, with basis $P = E_{12}$, $Q = E_{23}$, $C = E_{13}$, with $[P, Q] = C$ and every other basis bracket zero, of class $c = 2$. Determine $\dim V$ for the representation of theorem 8.9.6, and show that $\varphi(C) \neq 0$ while $\mathrm{ad}C = 0$. Conclude that φ separates exactly the directions the adjoint representation loses.

3. Let $\mathfrak{h}$ be a finite-dimensional Lie algebra and $X \in \mathfrak{h}$, so that $D = \mathrm{ad}X$ is a derivation of $\mathfrak{h}$. Show that its extension of lemma 8.9.7 is $\widetilde{D}(u) = Xu - uX$, the commutator with X inside $U(\mathfrak{h})$. Explain, using this formula, why the case analysis in the proof of proposition 8.9.8 requires the complement $\mathfrak{h}$ to be a subalgebra.

4. Show that for any finite-dimensional Lie algebra $\mathfrak{g}$ the kernel of the adjoint representation is exactly $\mathfrak{z}(\mathfrak{g})$, and exhibit an algebra for which ad is not faithful. Then show that in the direct sum $\theta = \varphi \oplus \mathrm{ad}$ of lemma 8.9.2 neither summand can be dropped in general: find a $\mathfrak{g}$ and a nilrepresentation φ with $\ker\varphi \cap \mathfrak{z}(\mathfrak{g}) = 0$ for which φ alone is not faithful.

5. Let k be a field of characteristic zero and fix an algebraic closure $\overline{k}$ of k ([2, Algebraic Closure]). For a finite-dimensional Lie algebra $\mathfrak{g}$ over k, let $\mathfrak{g}_{\overline{k}} = \mathfrak{g} \otimes_k \overline{k}$ with the bracket extended $\overline{k}$-bilinearly.

    1. Show that for subspaces $\mathfrak{a}, \mathfrak{b} \subseteq \mathfrak{g}$, one has $[\mathfrak{a} \otimes_k \overline{k}, \mathfrak{b} \otimes_k \overline{k}] = [\mathfrak{a}, \mathfrak{b}] \otimes_k \overline{k}$ inside $\mathfrak{g}_{\overline{k}}$.
    2. Deduce that the derived series and the lower central series of $\mathfrak{g}_{\overline{k}}$ are the scalar extensions of those of $\mathfrak{g}$, so that $\mathfrak{g}$ is solvable, or nilpotent, exactly when $\mathfrak{g}_{\overline{k}}$ is.
    3. Conclude that lemma 8.9.9 holds over k: for $\mathfrak{a}$ solvable over k and D a derivation, $D(\mathfrak{a}) \subseteq \operatorname{nil} \mathfrak{a}$.
    4. Explain why every other step of this section is valid over k unchanged, and conclude that every finite-dimensional Lie algebra over k embeds into $\mathfrak{gl}_N(k)$ for some N.

Part III

Compact Lie Groups and Their Representations

The theme of this part is integration over a compact group, and what averaging gives. Chapter 9 constructs the invariant integral and draws the first consequence, complete reducibility; chapter 10 decomposes $L^2(G)$ into matrix coefficients; chapter 11 conjugates every element into a maximal torus and reduces integration over the group to integration over the torus; and chapter 12 closes the part with the character and dimension formulas, recovering by analysis what Part II found by algebra. The running example graduates with the theory: the irreducible representations of SU(2), their characters, and their tensor products are computed in full as each tool appears.

9

Compact Lie Groups and Invariant Integration

Part I built the dictionary between a Lie group and its Lie algebra, and Part II classified the semisimple Lie algebras by their root systems. Part III returns to groups, now compact, and cashes that structure theory into representation theory. Compactness is the single hypothesis that makes the analysis work: a compact group carries an invariant integral, and averaging over it forces every representation to be unitary and to split into irreducibles. Everything in this part, from the decomposition of $L^2(G)$ in the next chapter to the explicit character formula at the end, rests on the ability to integrate over the group, and this chapter assembles that foundation.

The chapter opens with the structural features that set compact groups apart: finitely many connected components, an exponential map that is onto the identity component, and a bi-invariant Riemannian metric. It then constructs the invariant integral, specializing the general existence of Haar measure cited to [15, Existence of Haar Measure] to the smooth compact case, where the measure is a bi-invariant volume form of total mass one. The integral powers the *unitarian trick*: averaging an arbitrary inner product produces an invariant one, so every finite-dimensional representation of a compact group is unitary and therefore a direct sum of irreducibles, the analytic counterpart of the algebraic complete-reducibility theorem (theorem 6.4.8) of chapter 6. A closer look at the compact semisimple groups, where the Killing form is negative-definite, yields Weyl's theorem that their fundamental groups are finite. The chapter closes with the two examples that anchor everything to follow, the circle and SU(2), whose irreducible representations are written down explicitly and will reappear at every later stage.

9.1 Compact Groups: First Facts and the Exponential Map

This section returns from the Lie algebra to the group, now assuming it is compact, and records the two facts that make the compact case so much better behaved than the general one. First, compactness is a strong finiteness constraint: it forces the group to have finitely many components and pins down the connected abelian ones completely. Second, and more striking, on a compact connected group the exponential map $\exp\colon \mathfrak{g} \to G$ is *surjective*, so every element is a genuine exponential $\exp(X)$ and not merely a product of exponentials. This last fact fails for the noncompact groups of Part I, and the reason it holds is geometric; the proof borrows one theorem from Riemannian geometry, Hopf-Rinow, and cites it to the companion volume rather than re-deriving it.

Definition 9.1.1: Compact Lie Group and Identity Component

A *compact Lie group* is a Lie group G whose underlying manifold is compact (equivalently, a Lie group that is compact as a topological space; Lie groups are Hausdorff by convention). The *identity component* $G^0 \subseteq G$ is the connected component of G containing the identity e.

The identity component is the piece of the group visible from the identity, and it is a group in its own right. Because multiplication and inversion are continuous, $G^0 \cdot G^0 \subseteq G^0$ and $(G^0)^{-1} = G^0$, so G^0 is a subgroup; it is open because a manifold is locally connected, so components are open, and closed because its complement is a union of the other (open) components. An open subgroup is automatically closed, and its cosets are the other components, all homeomorphic to G^0 by translation. Conjugation gHg^{-1} by any g is a homeomorphism fixing e, so it maps the component of e to itself: G^0 is a *normal* subgroup. The quotient G/G^0 is the discrete group of components, and when G is compact this quotient is a compact discrete group, hence finite. Thus G^0 is an open, closed, normal subgroup of finite index, and G is built from the connected group G^0 and the finite group G/G^0 of its components.

Example 9.1.2: The Orthogonal Group $\mathrm{O}(n)$

The orthogonal group $\mathrm{O}(n)$ is compact: it is a closed subset of the unit sphere of $M_n(\mathbb{R})$ in the operator norm, since $\|A\| = 1$ for orthogonal A, and it is closed as the solution set of $A^{\mathsf{T}}A = I$. It has exactly two components, distinguished by $\det = \pm 1$. The identity component is $\mathrm{O}(n)^0 = \mathrm{SO}(n)$, the rotations, and the other component is the coset of any reflection. Here $\mathrm{O}(n)/\mathrm{SO}(n) \cong \mathbb{Z}/2$, a finite group with two elements, and $\mathrm{O}(n) = \mathrm{SO}(n) \sqcup r\,\mathrm{SO}(n)$ for any reflection r. This is the smallest nontrivial instance of the component structure: a connected identity component $\mathrm{SO}(n)$ and a finite component group $\mathbb{Z}/2$.

Compactness alone already forces the finiteness of the component group and, for connected abelian groups, pins the group down completely. The next proposition collects the three basic structural facts that the rest of the chapter uses without further comment: how many components a compact group has, what the connected abelian ones are, and that passing to a closed subgroup stays inside the compact-Lie world.

Proposition 9.1.3: First Structural Facts on Compact Groups

Let G be a compact Lie group.

1. G has finitely many connected components; the identity component G^0 is an open, closed, normal subgroup of finite index, and G/G^0 is a finite group.
2. If G is connected and abelian, then G is a *torus*: there is an isomorphism of Lie groups $G \cong \mathbb{T}^n = (\mathbb{R}/\mathbb{Z})^n$ for some $n = \dim G$, and $\exp\colon \mathfrak{g} \to G$ is a surjective homomorphism whose kernel is a full lattice $\Lambda \cong \mathbb{Z}^n$ in $\mathfrak{g} \cong \mathbb{R}^n$.
3. Every closed subgroup $H \subseteq G$ is itself a compact Lie group, a Lie subgroup of G.

***Proof*:**

1. *Finitely many components.* The components of a manifold are open (a manifold is locally connected), so the connected components of G form an open cover of G by disjoint sets. Compactness forces this cover to be finite, so G has finitely many components. That G^0 is open, closed, and normal was shown in the discussion after definition 9.1.1: it is a subgroup because multiplication and inversion preserve the component of e, open because components of a manifold are open, closed because its complement is the union of the remaining open components, and normal because conjugation is a homeomorphism fixing e. The cosets of G^0 are the components, so their number, the index $[G : G^0]$, is finite, and G/G^0 is a finite group.
2. *Connected abelian groups are tori.* Assume G is connected and abelian. For an abelian group the Baker-Campbell-Hausdorff formula (theorem 3.5.1) collapses to $\exp(X)\exp(Y) = \exp(X + Y)$, since all brackets vanish; equivalently, differentiating $\exp(sX)\exp(tY) = \exp(sX + tY)$ recovers additivity. Hence $\exp\colon (\mathfrak{g}, +) \to G$ is a homomorphism of groups. Its image is an open subgroup, because exp is a local diffeomorphism near 0 (theorem 2.3.4), so the image contains an open neighborhood of e; an open subgroup of a connected group is the whole group, so exp is surjective. The kernel $\Lambda = \ker(\exp)$ is a subgroup of the vector group $(\mathfrak{g}, +) \cong \mathbb{R}^n$; because exp is a local diffeomorphism at 0, the kernel is discrete near 0, hence discrete. A discrete subgroup of $\mathbb{R}^n$ is a lattice $\Lambda \cong \mathbb{Z}^k$ spanning a k-dimensional subspace, and $G \cong \mathfrak{g}/\Lambda$ is compact if and only if the lattice is full ($k = n$); since G is compact, Λ is a full lattice and $G \cong \mathbb{R}^n/\mathbb{Z}^n = \mathbb{T}^n$. This is the closed-subgroup route of corollary 3.4.5 specialized to the abelian case: a connected abelian Lie group is $\mathbb{R}^n/\Lambda$, and compactness selects the full-lattice quotient.
3. *Closed subgroups are compact Lie subgroups.* Let $H \subseteq G$ be a closed subgroup. By the closed-subgroup theorem (theorem 3.2.1), H is an embedded Lie subgroup of G. A closed subset of a compact space is compact, so H is compact, and thus a compact Lie group, completing the proof.

Part (2) is the reason tori are the model abelian objects of the whole theory: a compact connected abelian Lie group is nothing more than a quotient of a vector space by a full lattice, and the exponential map is the quotient projection. The circle $S^1 = \mathbb{R}/\mathbb{Z}$ is the one-dimensional case, and the higher tori $\mathbb{T}^n$ are its products; chapter 11 will show that a maximal such torus sits inside every compact connected group and controls its representation theory. Part (3) keeps every construction

inside the category: stabilizers, centralizers, and kernels are all closed subgroups, so each is again a compact Lie group. The classical groups give the standard stock of examples.

Example 9.1.4: The Compact Classical Groups

The connected classical groups $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and $\mathrm{Sp}(n)$ of proposition 1.2.10 are all compact, as is the torus $\mathbb{T}^n$. Each is a closed and bounded subset of a matrix space, hence compact by Heine-Borel, and each is closed as the solution set of a system of polynomial equations in the matrix entries:

- $\mathrm{SO}(n) = \{A \in M_n(\mathbb{R}) \mid A^{\mathsf{T}}A = I,\ \det A = 1\}$, bounded because $\|A\| = 1$;
- $\mathrm{U}(n) = \{A \in M_n(\mathbb{C}) \mid A^*A = I\}$ and $\mathrm{SU}(n) = \{A \in \mathrm{U}(n) \mid \det A = 1\}$, bounded for the same reason;
- $\mathrm{Sp}(n) = \mathrm{U}(2n) \cap \mathrm{Sp}(2n, \mathbb{C})$, the compact symplectic group, bounded as a subset of $\mathrm{U}(2n)$.

By contrast $\mathrm{GL}(n, \mathbb{R})$, $\mathrm{SL}(n, \mathbb{R})$, and $\mathrm{Sp}(2n, \mathbb{R})$ are *not* compact: $\mathrm{SL}(n, \mathbb{R})$ contains the unbounded family $\mathrm{diag}(t, t^{-1}, 1, \ldots, 1)$ for $t > 0$, so it is not bounded. The running example $\mathrm{SU}(2) \cong S^3$ (proposition 1.5.1) is compact and connected, the prototype the rest of the chapter returns to.

The surjectivity of the exponential map on a compact connected group is the main result of the section, and its proof is Riemannian: a compact group carries a metric for which the one-parameter subgroups are geodesics, and on a complete manifold geodesics reach every point. Building that metric is the first step, and it rests on an averaging construction. Any inner product on $\mathfrak{g}$ can be made $\mathrm{Ad}(G)$-invariant by averaging it over G against the invariant integral, and an $\mathrm{Ad}(G)$-invariant inner product left-translates to a bi-invariant Riemannian metric. This uses the invariant integral constructed in the next section; the order is not circular, because the construction of Haar measure in section 9.2 makes no use of anything proved here.

Proposition 9.1.5: Bi-invariant Metrics and Geodesics

Let G be a compact Lie group.

1. There exists an $\mathrm{Ad}(G)$-invariant inner product $\langle -, - \rangle$ on $\mathfrak{g}$, obtained by averaging an arbitrary inner product over G against the invariant integral of section 9.2.
2. Left-translating $\langle -, - \rangle$ over G yields a Riemannian metric on G that is both left- and right-invariant (a *bi-invariant* metric).
3. For this metric, the geodesics through e are exactly the one-parameter subgroups $t \mapsto \exp(tX)$, $X \in \mathfrak{g}$; equivalently, the Riemannian exponential Exp_e coincides with the Lie-theoretic exponential exp.

***Proof*:**

1. *The invariant inner product.* Fix any inner product $(-, -)$ on the finite-dimensional real vector space $\mathfrak{g}$. The adjoint representation $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ of definition 2.5.1 acts on $\mathfrak{g}$;

define

$$\langle X, Y\rangle = \int_G \big(\mathrm{Ad}(g)X,\ \mathrm{Ad}(g)Y\big)\, dg$$

the average of the pullbacks $(\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y)$ over G against the normalized invariant integral of section 9.2. This is bilinear and symmetric because the integrand is, and positive-definite because $\langle X, X\rangle = \int_G (\mathrm{Ad}(g)X, \mathrm{Ad}(g)X)\, dg$ integrates a continuous nonnegative function that is strictly positive at $g = e$ when $X \neq 0$, so the integral is strictly positive. Invariance under $\mathrm{Ad}(h)$ follows from left-invariance of the integral: for any $h \in G$,

$$\langle \mathrm{Ad}(h)X, \mathrm{Ad}(h)Y\rangle = \int_G \big(\mathrm{Ad}(gh)X, \mathrm{Ad}(gh)Y\big)\, dg = \int_G \big(\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y\big)\, dg = \langle X, Y\rangle$$

using $\mathrm{Ad}(g)\,\mathrm{Ad}(h) = \mathrm{Ad}(gh)$ and the substitution $g \mapsto gh$ under the invariant integral. Thus $\langle -, -\rangle$ is $\mathrm{Ad}(G)$-invariant. This is the averaging construction of proposition 9.2.4 applied to the space of inner products; it is the same device that produces invariant Hermitian forms on representations in section 9.3.

2. *The bi-invariant metric.* Transport $\langle -, -\rangle$ around G by left translation: for $g \in G$ and tangent vectors $u, v \in T_gG$, set $\langle u, v\rangle_g = \langle (dL_{g^{-1}})_g u, (dL_{g^{-1}})_g v\rangle$, where $L_{g^{-1}}$ is left translation by g^{-1}. This is a Riemannian metric, left-invariant by construction. It is also right-invariant: right translation R_h acts on $\mathfrak{g} = T_eG$ (after correcting by left translation to the identity) through $\mathrm{Ad}(h^{-1})$, and $\langle -, -\rangle$ was built to be $\mathrm{Ad}(G)$-invariant; since $\mathrm{Ad}(h^{-1})$ preserves $\langle -, -\rangle$, the transported metric it induces on each T_gG is unchanged by R_h, so the metric is right-invariant. A left- and right-invariant metric is bi-invariant, and inversion $g \mapsto g^{-1}$ is then an isometry as well, since its differential at e is $-\mathrm{id}$, an isometry of $\langle -, -\rangle$.

3. *Geodesics are one-parameter subgroups.* For a bi-invariant metric on a Lie group, the Levi-Civita connection satisfies $\nabla_X Y = \frac{1}{2}[X, Y]$ on left-invariant vector fields, so a left-invariant field X has $\nabla_X X = \frac{1}{2}[X, X] = 0$: its integral curves are geodesics. The integral curve of the left-invariant field X through e is the one-parameter subgroup $t \mapsto \exp(tX)$ by the very definition of exp in theorem 2.3.4. Conversely every geodesic through e arises this way, since geodesics through a point are determined by their initial velocity $X \in T_eG = \mathfrak{g}$, and $t \mapsto \exp(tX)$ already provides one for each X. Hence the Riemannian exponential $\mathrm{Exp}_e(X)$, which follows the geodesic with initial velocity X for unit time, equals $\exp(X)$. The two computations that a bi-invariant metric produces the connection $\nabla_X Y = \frac{1}{2}[X, Y]$ and that its geodesics are one-parameter subgroups are standard Riemannian geometry; see [7, The connection of a bi-invariant metric, Geodesics and curvature of a bi-invariant metric], completing the proof.

The proposition converts a group-theoretic question, whether every element is an exponential, into a geometric one about geodesics, and the geometry is favorable because the metric is bi-invariant. The identification $\mathrm{Exp}_e = \exp$ is the crucial bridge: the Lie exponential, defined by flowing along one-parameter subgroups, and the Riemannian exponential, defined by following geodesics, are literally the same map on a compact group with its bi-invariant metric. This is what lets a theorem about geodesic reachability be read as a theorem about exponentials.

Theorem 9.1.6: Surjectivity of the Exponential Map

Let G be a compact *connected* Lie group with Lie algebra $\mathfrak{g}$. Then the exponential map $\exp\colon \mathfrak{g} \to G$ is surjective: every $g \in G$ can be written $g = \exp(X)$ for some $X \in \mathfrak{g}$.

***Proof*:**
Equip G with the bi-invariant Riemannian metric of proposition 9.1.5. Because G is compact, it is a compact Riemannian manifold, hence complete as a metric space, so it is geodesically complete: every geodesic extends to all of $\mathbb{R}$. By the Hopf-Rinow theorem ([7, Hopf-Rinow]), a connected, complete Riemannian manifold has the property that any two points are joined by a length-minimizing geodesic. Applying this to the pair (e, g), there is a geodesic from e to g; writing $X \in T_eG = \mathfrak{g}$ for its initial velocity (rescaled so that it reaches g at time 1), one has $g = \mathrm{Exp}_e(X)$.

By proposition 9.1.5(3) the geodesics through e are exactly the one-parameter subgroups $t \mapsto \exp(tX)$, so the Riemannian exponential coincides with the Lie exponential: $\mathrm{Exp}_e(X) = \exp(X)$. Hence $g = \exp(X)$, and since $g \in G$ was arbitrary, exp is surjective, completing the proof.

The theorem is the first genuine payoff of compactness at the level of the group rather than the algebra. On a general Lie group exp need not be onto (it misses part of $\mathrm{SL}(2,\mathbb{R})$, as the closing discussion records), and one must work with products $\exp(X_1)\cdots\exp(X_k)$ to reach every element of the identity component. Compactness removes this obstruction entirely: a single exponential suffices. Two concrete instances show the abstract statement, and both are computations one can carry out by hand.

Example 9.1.7: exp onto SU(2) and onto U(n)
Two compact groups exhibit the surjectivity concretely.

First, $\mathrm{SU}(2)$. Its Lie algebra $\mathfrak{su}(2)$ consists of the traceless skew-Hermitian 2×2 matrices. Every $X \in \mathfrak{su}(2)$ has the form $X = \left(\begin{smallmatrix} ia & b+ic \\ -b+ic & -ia \end{smallmatrix}\right)$ with $a, b, c \in \mathbb{R}$, and setting $r = \sqrt{a^2+b^2+c^2}$ one computes $X^2 = -r^2 I$, so the exponential series sums in closed form to

$$\exp(X) = \cos r \cdot I + \frac{\sin r}{r} X$$

(with the convention $\frac{\sin r}{r} = 1$ at $r = 0$). This is exactly the concrete surjection of proposition 1.5.3: as X ranges over $\mathfrak{su}(2)$ the right-hand side ranges over all of $\mathrm{SU}(2) \cong S^3$, so $\exp\colon \mathfrak{su}(2) \to \mathrm{SU}(2)$ is onto.

Second, $\mathrm{U}(n)$. The spectral theorem for normal operators (see [14, Normal Compact, Then Diagonalizable, Spectrum of a Self-Adjoint Operator]) diagonalizes any unitary $U \in \mathrm{U}(n)$: there is an orthonormal basis in which $U = V\,\mathrm{diag}(e^{i\theta_1},\dots,e^{i\theta_n})\,V^{-1}$ with $V \in \mathrm{U}(n)$ and $\theta_j \in \mathbb{R}$. Set $A = V\,\mathrm{diag}(i\theta_1,\dots,i\theta_n)\,V^{-1}$. Then $A^* = -A$, so $A \in \mathfrak{u}(n)$ is skew-Hermitian, and $\exp(A) = V\,\mathrm{diag}(e^{i\theta_1},\dots,e^{i\theta_n})\,V^{-1} = U$ because exp commutes with conjugation and acts on a diagonal matrix entrywise. Hence every unitary is the exponential of a skew-Hermitian matrix: $\exp\colon \mathfrak{u}(n) \to \mathrm{U}(n)$ is surjective. Restricting to determinant 1 (equivalently $\mathrm{tr}A = 0$) gives the same statement for $\mathrm{SU}(n)$.

The contrast with the noncompact case locates precisely where compactness is used. On $\mathrm{SL}(2,\mathbb{R})$ the exponential map is *not* surjective. The matrix $\left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$ lies in $\mathrm{SL}(2,\mathbb{R})$ but is not $\exp(X)$ for any

$X \in \mathfrak{sl}(2,\mathbb{R})$: a traceless X has characteristic polynomial $t^2 + \det X$, so its eigenvalues $\pm\lambda$ are real ($\det X < 0$), purely imaginary ($\det X > 0$), or zero ($\det X = 0$), and these three cases are exhaustive. Correspondingly $\exp(X)$ has eigenvalues $e^{\pm\lambda}$, a reciprocal pair of positive reals when λ is real or a conjugate pair $e^{\pm i\mu}$ on the unit circle when $\lambda = i\mu$ is imaginary, and neither type can produce the repeated eigenvalue -1 with a nontrivial Jordan block. The obstruction is exactly the failure of the previous argument: $\mathrm{SL}(2,\mathbb{R})$ is noncompact, so it carries no bi-invariant metric, its one-parameter subgroups are not the geodesics of any complete invariant metric, and Hopf-Rinow has nothing to act on. Surjectivity of exp is thus a diagnostic for compactness of the connected part, not a formal feature of Lie groups.

Exercise 9.1.1

1. Let G be a connected compact abelian Lie group of dimension n. Prove directly, without quoting proposition 9.1.3(2), that $\exp\colon \mathfrak{g} \to G$ is a surjective group homomorphism, that its kernel Λ is a discrete subgroup of $\mathfrak{g} \cong \mathbb{R}^n$, and that Λ is a full lattice (rank n). Conclude $G \cong \mathbb{T}^n$. Where is compactness used, and where is commutativity used?

2. Show that $\exp\colon \mathfrak{gl}(n,\mathbb{C}) \to \mathrm{GL}(n,\mathbb{C})$ is surjective, so every invertible complex matrix is an exponential. (Hint: reduce to Jordan blocks and exponentiate a logarithm of each block; a unipotent block $I + N$ with N nilpotent has logarithm given by the terminating series $\log(I+N) = N - \frac{1}{2}N^2 + \cdots$.) Explain why this does *not* make $\mathrm{GL}(n,\mathbb{C})$ compact, and reconcile this with theorem 9.1.6.

3. Verify carefully that the averaged form $\langle X, Y\rangle = \int_G (\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y)\, dg$ of proposition 9.1.5(1) is positive-definite, spelling out why the integral of a continuous nonnegative function that is strictly positive at one point is strictly positive. Then show that a compact group's Killing form (definition 6.1.3) is one source of an $\mathrm{Ad}(G)$-invariant *symmetric bilinear* form, and explain why it need not be an inner product (positive-definite) in general.

4. Give an explicit pair $X \neq X' \in \mathfrak{su}(2)$ with $\exp(X) = \exp(X')$, so that exp is surjective but not injective on the compact group $\mathrm{SU}(2)$. Deduce from surjectivity-without-injectivity that exp cannot be a diffeomorphism $\mathfrak{g} \to G$ for any compact connected G of positive dimension. (Hint: use the closed form of example 9.1.7 for two diagonal matrices of different "lengths" r.)

5. Prove that the matrix $A = \left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$ is not in the image of $\exp\colon \mathfrak{sl}(2,\mathbb{R}) \to \mathrm{SL}(2,\mathbb{R})$. (Hint: classify the eigenvalues of $\exp(X)$ for $\mathrm{tr}X = 0$ into the cases real reciprocal pair, unit-circle conjugate pair, and repeated eigenvalue $+1$, and check that A, with its repeated eigenvalue -1 and nondiagonalizable Jordan form, matches none.)

6. Let G be a compact Lie group and $H \subseteq G$ a closed subgroup. Using proposition 9.1.3(3), show that H is a compact Lie group, and that the centralizer $Z_G(S) = \{g \in G \mid gs = sg \text{ for all } s \in S\}$ of any subset $S \subseteq G$ is a compact Lie subgroup. Give the analogous statement for the kernel of a Lie group homomorphism $\varphi\colon G \to K$.

9.2 Haar Measure on Compact Groups

The previous section put a bi-invariant Riemannian metric on a compact group by averaging an inner product over G, and that averaging quietly used the one analytic tool this chapter really needs: a way to integrate a function over the group that does not care where the group is centered. This

section builds that tool. A compact Lie group carries a canonical measure, unique once its total mass is fixed at 1, that is invariant under left and right translation and under inversion, and integration against it turns every construction that a finite group performs by averaging over its elements into one a continuous group performs by integrating over itself. The measure is the normalized Haar measure, and the invariant integral it defines is used throughout the rest of Part III.

Two facts organize the section. First, the measure exists and is unique: a compact group is small enough that a left-invariant volume form can be built by hand and normalized, and an averaging argument shows no two invariant probability measures can differ. Second, the invariant integral has an averaging operator hiding inside it. Given any continuous linear action of G on a finite-dimensional space, integrating the orbit of a vector produces a G-invariant vector, and this single construction, applied to inner products, is what the unitarian trick of section 9.3 and the compact-real-form theory of section 9.4 will run over and over.

On a compact *Lie* group the measure is concrete enough to construct directly from a volume form, and the compactness makes invariance from the right and under inversion automatic once left-invariance is fixed. The theorem records all three invariances at once and fixes the normalization that makes the measure canonical.

Theorem 9.2.1: Haar Measure on a Compact Lie Group

Let G be a compact Lie group. There is a unique Borel probability measure μ on G that is left-invariant: $\mu(hE) = \mu(E)$ for every Borel set $E \subseteq G$ and every $h \in G$. This measure, the *normalized Haar measure*, is automatically right-invariant, $\mu(Eh) = \mu(E)$, and inversion-invariant, $\mu(E^{-1}) = \mu(E)$.

***Proof*:**
The existence and uniqueness of a left-invariant Radon probability measure on any compact (indeed any locally compact) topological group is proved in full in [15, Existence of Haar Measure, Uniqueness of Haar Measure]; the argument below specializes that result to the smooth compact case, where the measure comes from a volume form, and then derives the two further invariances, which hold for compact groups but not for locally compact groups in general. Throughout, a left-invariant Borel probability measure is used interchangeably with the positive normalized left-invariant linear functional $f \mapsto \int_G f\, d\mu$ on $C(G)$, the two being equivalent for a compact metric space by the Riesz representation theorem ([15, Riesz-Markov Representation]).

Step 1: existence via a left-invariant volume form. Let $n = \dim G$ and fix a nonzero alternating n-form $\omega_e \in \bigwedge^n \mathfrak{g}^*$ on the Lie algebra $\mathfrak{g} = T_eG$; the space $\bigwedge^n \mathfrak{g}^*$ is one-dimensional, so ω_e is unique up to a scalar. Translate it over the group by left multiplication: for $g \in G$, let $L_g \colon G \to G$ be $x \mapsto gx$, and define an n-form ω on G by

$$\omega_g = (L_{g^{-1}})^*\, \omega_e$$

so that ω is left-invariant, $(L_h)^*\omega = \omega$ for all $h \in G$, by construction. Because ω_g is nonzero at every point, ω is a volume form, and it defines a positive Borel measure by integration of continuous functions against the associated density $|\omega|$,

$$\nu(f) = \int_G f\, |\omega|, \qquad f \in C(G)$$

(the density $|\omega|$ is used so that the measure does not depend on an orientation). Left-invariance of ω gives left-invariance of ν: for $h \in G$ the change of variables $x \mapsto hx$ has $(L_h)^*\omega = \omega$, so

$\nu(f \circ L_h) = \nu(f)$. Since G is compact, $\nu(1) = \int_G |\omega|$ is finite and positive, and rescaling ω_e arranges $\nu(1) = 1$. The resulting μ is a left-invariant Borel probability measure.

Step 2: uniqueness by the Fubini averaging trick. Let μ and ν be two left-invariant Borel probability measures on G; it suffices to show $\int_G f\,d\mu = \int_G f\,d\nu$ for every $f \in C(G)$. Fix $f \in C(G)$ and integrate the continuous function $(g,h) \mapsto f(g^{-1}h)$ over $G \times G$ against the product measure $\mu \times \nu$. Fubini's theorem lets the two iterated integrals be equated:

$$\int_G \Big(\int_G f(g^{-1}h)\,d\nu(h) \Big) d\mu(g) = \int_G \Big(\int_G f(g^{-1}h)\,d\mu(g) \Big) d\nu(h)$$

Evaluate each side by a single left-translation substitution. On the left, fix g and substitute $h \mapsto gh$; since $h \mapsto gh$ is a left translation, left-invariance of ν gives $\int_G f(g^{-1}(gh))\,d\nu(h) = \int_G f(h)\,d\nu(h) = \int_G f\,d\nu$, a constant in g, and integrating it against μ (total mass 1) returns $\int_G f\,d\nu$. On the right, fix h and substitute $g \mapsto hg$; left-invariance of μ gives $\int_G f((hg)^{-1}h)\,d\mu(g) = \int_G f(g^{-1}h^{-1}h)\,d\mu(g) = \int_G f(g^{-1})\,d\mu(g)$, again constant in h, and integrating against ν returns $\int_G f(g^{-1})\,d\mu(g)$. Equating the two evaluations,

$$\int_G f\,d\nu = \int_G f(g^{-1})\,d\mu(g) \tag{9.1}$$

for every $f \in C(G)$. Applying (9.1) with $\nu = \mu$ gives $\int_G f\,d\mu = \int_G f(g^{-1})\,d\mu(g)$: any left-invariant probability measure is inversion-invariant. Substituting this back into (9.1) yields $\int_G f\,d\nu = \int_G f(g^{-1})\,d\mu(g) = \int_G f\,d\mu$, so $\mu = \nu$ and the left-invariant probability measure is unique.

Step 3: right-invariance from uniqueness. Inversion-invariance, $\mu(E^{-1}) = \mu(E)$, is already recorded in Step 2: applied to the normalized Haar measure μ, the identity $\int_G f\,d\mu = \int_G f(g^{-1})\,d\mu(g)$ is exactly inversion-invariance of μ at the level of integrals, hence of the measure. For right-invariance, fix $h \in G$ and define the right-translated measure $\mu_h(E) = \mu(Eh)$. Right translation commutes with left translation, so μ_h is again left-invariant, $\mu_h(gE) = \mu(gEh) = \mu(Eh) = \mu_h(E)$, and $\mu_h(G) = \mu(Gh) = \mu(G) = 1$, so μ_h is a left-invariant probability measure. By the uniqueness of Step 2, $\mu_h = \mu$, which is right-invariance $\mu(Eh) = \mu(E)$, completing the proof.

The construction says two different things depending on which step one reads. Step 1 is a smooth-manifold fact: a left-invariant top-degree form always exists on a Lie group, because a single alternating form on the tangent space at the identity spreads to the whole group by translation, and on a compact group its total integral is finite and can be normalized to 1. Steps 2 and 3 are where compactness is used. On a general locally compact group the left-invariant Haar measure need not be right-invariant; the failure is recorded by the modular function, and groups on which the two coincide are called unimodular. A compact group is always unimodular, and the reason is exactly Step 3: right translation carries a left-invariant probability measure to another left-invariant probability measure, and uniqueness forces the two to agree. The same mechanism makes the measure inversion-invariant. So the single normalized Haar measure on a compact group is invariant under everything one could want, and the choice of normalization $\mu(G) = 1$ makes it canonical rather than merely unique up to scale.

The uniqueness argument of Step 2 sets the pattern the later averaging arguments follow. Two invariant measures are compared by feeding both into a single object, the double integral of $(g,h) \mapsto f(g^{-1}h)$, and Fubini collapses that object to $\int f\,d\nu$ read one way and to $\int f(g^{-1})\,d\mu(g)$ read the other;

inversion-invariance then identifies the latter with $\int f\,d\mu$. Nothing about the measures is used except invariance and total mass 1. For a finite group the same statement reads: averaging does not depend on how the averaging is organized. It is the reason the invariant integral defined next is well-defined without any choices.

With existence and uniqueness in hand, the measure is repackaged as a linear functional on continuous functions, the invariant integral, whose defining feature is a battery of invariances inherited directly from theorem 9.2.1. This is the object the rest of the book uses; one rarely returns to the underlying set function.

Definition 9.2.2: The Invariant Integral

Let G be a compact Lie group with normalized Haar measure μ of theorem 9.2.1. The *invariant integral* is the linear functional

$$f \longmapsto \int_G f(g)\,dg := \int_G f\,d\mu, \qquad f \in C(G)$$

on the space $C(G)$ of continuous complex-valued functions on G. It is linear, positive (if $f \geq 0$ then $\int_G f(g)\,dg \geq 0$, with equality forcing $f \equiv 0$ for continuous nonnegative f), and normalized,

$$\int_G 1\,dg = 1$$

and it is invariant under left and right translation and under inversion: for every $h \in G$,

$$\int_G f(hg)\,dg = \int_G f(gh)\,dg = \int_G f(g^{-1})\,dg = \int_G f(g)\,dg$$

The three invariances are the translation- and inversion-invariance of the underlying measure transcribed into the language of functions: $\int_G f(hg)\,dg = \int_G f\,dg$ is left-invariance, $\int_G f(gh)\,dg = \int_G f\,dg$ is right-invariance, and $\int_G f(g^{-1})\,dg = \int_G f\,dg$ is inversion-invariance. The notation dg suppresses the measure entirely, and this is deliberate: because μ is canonical there is only one integral to write, and the invariances mean one may substitute $g \mapsto hg$, $g \mapsto gh$, or $g \mapsto g^{-1}$ inside the integral without changing its value, exactly as one reindexes a finite average $\frac{1}{|G|}\sum_g$ over a finite group. Every manipulation in the sections that follow is one of these three substitutions.

The measure becomes concrete on the running examples, where the volume form of the existence proof is written down explicitly.

Example 9.2.3: Haar Measure on S^1, on $\mathbb{T}^n$, and on $\mathrm{SU}(2)$

1. *The circle* $S^1 = \mathrm{U}(1)$. Coordinatize S^1 by the angle $\theta \in [0, 2\pi)$, so $g = e^{i\theta}$. Left translation by $e^{i\alpha}$ is $\theta \mapsto \theta + \alpha \pmod{2\pi}$, a rigid rotation of the coordinate, so the left-invariant density is a constant multiple of $d\theta$. Normalizing to total mass 1 forces

$$\int_{S^1} f(g)\,dg = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta$$

since $\frac{1}{2\pi}\int_0^{2\pi} d\theta = 1$. This is the averaging that underlies Fourier series: the Fourier coefficient $\widehat{f}(n) = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})e^{-in\theta}\,d\theta$ is exactly $\int_{S^1} f(g)\overline{\chi_n(g)}\,dg$ for the character $\chi_n(e^{i\theta}) = e^{in\theta}$.

2. *The torus* $\mathbb{T}^n = (S^1)^n$. The Haar measure of a product of compact groups is the product of the Haar measures, because the product measure is left-invariant with total mass 1, so uniqueness (theorem 9.2.1) identifies it as the Haar measure. Hence on $\mathbb{T}^n$,
$$\int_{\mathbb{T}^n} f(g)\,dg = \frac{1}{(2\pi)^n}\int_0^{2\pi}\cdots\int_0^{2\pi} f(e^{i\theta_1},\dots,e^{i\theta_n})\,d\theta_1\cdots d\theta_n$$

3. *The group* $\mathrm{SU}(2) = S^3$. By proposition 1.5.1, $\mathrm{SU}(2)$ is diffeomorphic to the unit sphere $S^3 \subseteq \mathbb{C}^2 = \mathbb{R}^4$ via $\left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a}\end{smallmatrix}\right) \mapsto (a,b)$ with $|a|^2+|b|^2=1$. Left and right multiplication by a fixed element of $\mathrm{SU}(2)$ act on $\mathbb{C}^2=\mathbb{R}^4$ as orthogonal transformations, since they preserve the Hermitian form and hence the underlying real inner product, so they preserve the round surface measure of S^3. The normalized round measure, the standard surface measure divided by the total area $2\pi^2$ of S^3, is therefore left- and right-invariant with total mass 1, and by uniqueness it is the Haar measure of $\mathrm{SU}(2)$. In Euler-angle coordinates $g = \begin{pmatrix} e^{i(\psi+\varphi)/2}\cos\frac{\vartheta}{2} & ie^{i(\psi-\varphi)/2}\sin\frac{\vartheta}{2} \\ ie^{-i(\psi-\varphi)/2}\sin\frac{\vartheta}{2} & e^{-i(\psi+\varphi)/2}\cos\frac{\vartheta}{2}\end{pmatrix}$ with $\vartheta\in[0,\pi]$, $\varphi\in[0,2\pi)$, $\psi\in[0,4\pi)$, the invariant integral is
$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{16\pi^2}\int_0^{\pi}\int_0^{2\pi}\int_0^{4\pi} f(g)\,\sin\vartheta\,d\psi\,d\varphi\,d\vartheta$$
the factor $\frac{1}{16\pi^2}$ normalizing the total mass to 1.

The three examples show the same measure from three angles. On S^1 it is the uniform average that Fourier analysis runs on; on $\mathbb{T}^n$ it is the product, which is why the analysis of a torus factors coordinate by coordinate; and on $\mathrm{SU}(2)$ it is the round measure on a sphere, the first case where the group is nonabelian and the Weyl integration formula of chapter 11 will later reduce this four-dimensional integral to a one-dimensional integral over a maximal torus. In every case the defining property is that translating the integrand leaves the integral unchanged.

The invariant integral manufactures invariant objects on require. Feed it the orbit of a vector under a representation and it returns a fixed vector; feed it an inner product and it returns an invariant inner product. The construction is a single averaging operator, worth stating once cleanly, because every later use is an instance of it.

Proposition 9.2.4: The Averaging Operator

Let G be a compact Lie group acting continuously and linearly on a finite-dimensional complex vector space V, written $(g, v) \mapsto g \cdot v$. Define the *averaging operator* $A\colon V \to V$ by

$$A(v) = \int_G g \cdot v \, dg$$

the integral taken coordinatewise in any basis. Then:

1. $A(v)$ is G-invariant: $h \cdot A(v) = A(v)$ for all $h \in G$ and all $v \in V$.
2. A is G-equivariant and is a projection of V onto the subspace $V^G = \{v \in V \mid g \cdot v = v \text{ for all } g \in G\}$ of invariants; that is, A is linear, $A|_{V^G} = \mathrm{id}$, and $A(V) = V^G$, so $A^2 = A$.
3. If h_0 is any Hermitian inner product on V, then

$$h(u, v) = \int_G h_0(g \cdot u,\, g \cdot v) \, dg$$

is a G-invariant Hermitian inner product on V: $h(g \cdot u,\, g \cdot v) = h(u, v)$ for all $g \in G$.

***Proof*:**
The integrals converge because G is compact and the integrands are continuous V-valued (respectively scalar-valued) functions of g: a continuous function on a compact space is bounded, so each coordinate integral is a finite integral against the finite measure μ.

1. Fix $h \in G$. The action is linear, so $h \cdot (-)$ is a fixed linear map on V and passes through the integral coordinatewise:

$$h \cdot A(v) = h \cdot \int_G g \cdot v \, dg = \int_G h \cdot (g \cdot v) \, dg = \int_G (hg) \cdot v \, dg$$

The substitution $g \mapsto hg$ is left translation, and by the left-invariance of definition 9.2.2 the last integral equals $\int_G g \cdot v \, dg = A(v)$. Hence $h \cdot A(v) = A(v)$, so $A(v) \in V^G$.

2. Linearity of A is linearity of the integral. If $v \in V^G$ then $g \cdot v = v$ for all g, so $A(v) = \int_G v \, dg = v \int_G 1 \, dg = v$ by normalization, giving $A|_{V^G} = \mathrm{id}$. By part (1) the image $A(V)$ lies in V^G, and $A|_{V^G} = \mathrm{id}$ gives $A(V) \supseteq A(V^G) = V^G$, so $A(V) = V^G$. Then $A^2(v) = A(A(v)) = A(v)$ because $A(v) \in V^G$ is fixed by A, so A is a projection onto V^G. For equivariance, fix $g_0 \in G$ and expand

$$A(g_0 \cdot v) = \int_G g \cdot (g_0 \cdot v) \, dg = \int_G (gg_0) \cdot v \, dg$$

The substitution $g \mapsto gg_0^{-1}$ is right translation, and by the right-invariance of definition 9.2.2 the last integral equals $\int_G g \cdot v \, dg = A(v)$. Since $A(v) \in V^G$ is fixed by g_0, this gives $A(g_0 \cdot v) = A(v) = g_0 \cdot A(v)$; thus A commutes with the action.

3. Sesquilinearity of h is inherited from h_0 under the linear integral. For positivity, $h(v, v) = \int_G h_0(g \cdot v, g \cdot v) \, dg$ integrates a nonnegative continuous function, so $h(v, v) \geq 0$, and it

vanishes only if $h_0(g\cdot v, g\cdot v)=0$ for all g; taking $g=e$ gives $h_0(v,v)=0$, so $v=0$. Thus h is a Hermitian inner product. For invariance, fix $h_1\in G$:

$$h(h_1\cdot u,\, h_1\cdot v)=\int_G h_0\big(g\cdot(h_1\cdot u),\, g\cdot(h_1\cdot v)\big)\,dg=\int_G h_0\big((gh_1)\cdot u,\,(gh_1)\cdot v\big)\,dg$$

The substitution $g\mapsto gh_1$ is right translation, and by the right-invariance of definition 9.2.2 the last integral equals $\int_G h_0(g\cdot u,\, g\cdot v)\,dg=h(u,v)$, as we sought to show.

Part (1) is averaging in its rawest form: the average of an orbit is a fixed point, because translating the average only reshuffles the orbit being averaged, and the integral cannot tell the difference. Part (2) upgrades this to the observation that A is a projection onto the invariants, so the fixed subspace V^G is always a direct summand split off by an equivariant map. Part (3) is the version that does the real work downstream: it turns any inner product into a G-invariant one, its construction drawing on the right-invariance of the integral, while part (1) rests on left-invariance, which together is why the proposition needed both directions of invariance. This is the single construction reused in proposition 9.1.5 to produce the $\mathrm{Ad}(G)$-invariant inner product on $\mathfrak{g}$ that gave the bi-invariant metric, in section 9.3 to make every representation unitary, and in section 9.4 to control the Killing form of a compact group. It is proved once here and cited each of those times.

The connection to finite groups is exact. Over a finite group G, the representation theory of [3] runs on the averaging operator $\frac{1}{|G|}\sum_{g\in G} g\cdot v$, which projects onto invariants, and on the averaged inner product $\frac{1}{|G|}\sum_{g\in G} h_0(g\cdot u, g\cdot v)$, which is G-invariant. The invariant integral $\int_G(-)\,dg$ is the same operator with the normalized sum $\frac{1}{|G|}\sum_g$ replaced by integration against normalized Haar measure. Every theorem that finite-group representation theory proves by averaging over the group, Maschke's theorem, the existence of invariant inner products, Schur orthogonality, has a compact-group analogue proved by the same argument with the integral in place of the sum. The one new ingredient compactness gives is the integral itself: the finiteness of G that made the sum available is replaced by the compactness of G that makes the Haar integral available, and nothing else in the arguments changes. That is why compact groups behave, representation-theoretically, like finite groups.

Exercise 9.2.1

1. Using the round-measure normalization of example 9.2.3(3), compute

$$\int_{\mathrm{SU}(2)} |a|^2\,dg$$

where $g=\left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a}\end{smallmatrix}\right)\in \mathrm{SU}(2)$, that is, the average over $\mathrm{SU}(2)$ of the squared modulus of the upper-left matrix coefficient. (Hint: $|a|^2+|b|^2=1$ on S^3, and by symmetry $\int|a|^2\,dg=\int|b|^2\,dg$.)

2. The *modular function* of a Lie group G is the homomorphism $\Delta\colon G\to(0,\infty)$ measuring the failure of a left Haar measure to be right-invariant, $\mu(Eh)=\Delta(h)^{-1}\mu(E)$. Prove that a compact group is *unimodular*, that is, $\Delta\equiv 1$, using that Δ is a continuous homomorphism and G is compact.

3. Let $H\subsetneq G$ be a proper closed subgroup of a connected compact Lie group G. Prove that the Haar measure of H inside G is zero, $\mu(H)=0$. (Consider the dimension of H as a submanifold; if $\dim H=\dim G$, use connectedness.)

4. A *class function* on G is a continuous function f constant on conjugacy classes, $f(hgh^{-1}) = f(g)$ for all g, h. Show that the averaging operator applied over the conjugation action, $(\mathcal{A}f)(g) = \int_G f(xgx^{-1})\,dx$, produces a class function for arbitrary $f \in C(G)$, and that $\mathcal{A}f = f$ when f is already a class function.

5. Prove that the Haar measure of a product $G_1 \times G_2$ of compact Lie groups is the product $\mu_1 \times \mu_2$ of their Haar measures, as asserted for $\mathbb{T}^n$ in example 9.2.3(2). Where is uniqueness (theorem 9.2.1) used?

6. Deduce the inversion-invariance $\int_G f(g^{-1})\,dg = \int_G f(g)\,dg$ from left- and right-invariance, by showing that $f \mapsto \int_G f(g^{-1})\,dg$ is a left-invariant probability integral and applying uniqueness (theorem 9.2.1) once.

9.3 The Unitarian Trick and Complete Reducibility

The invariant integral of section 9.2 gives a way to replace any object attached to a representation by a G-invariant one, by averaging it over the group. Applied to an inner product, this single device forces the entire representation theory of a compact group into the same rigid shape as the representation theory of a finite group: every representation is unitary, every subrepresentation is complemented, and the irreducibles are the indecomposable atoms out of which all representations are built. This section carries out that argument. It is the analytic counterpart of the algebraic complete reducibility of theorem 6.4.8, with the invariant integral in place of the Casimir element.

Throughout, G is a compact Lie group and $\varphi\colon G \to \mathrm{GL}(V)$ a finite-dimensional representation over $\mathbb{C}$, meaning a continuous (hence smooth) homomorphism into the group of invertible linear operators on a finite-dimensional complex vector space V. A *subrepresentation* is a subspace $W \subseteq V$ with $\varphi(g)W \subseteq W$ for all $g \in G$, and V is *irreducible* if $V \neq 0$ and its only subrepresentations are 0 and V. A linear map $T\colon V \to V'$ between representations is *G-equivariant* (an *intertwiner*) if $T\varphi(g) = \varphi'(g)T$ for all g. This is the same vocabulary as for Lie-algebra modules (definition 6.4.1), now for the group.

The first result is the unitarian trick itself. A Hermitian inner product on V is G-invariant when $\langle \varphi(g)v, \varphi(g)w\rangle = \langle v, w\rangle$ for all $g \in G$ and all $v, w \in V$, which says exactly that every operator $\varphi(g)$ is unitary for that inner product. There is no reason an arbitrary inner product should be invariant, but averaging one over G repairs this.

Theorem 9.3.1: The Unitarian Trick

Let G be a compact Lie group and $\varphi\colon G \to \mathrm{GL}(V)$ a finite-dimensional complex representation. Then V admits a G-invariant Hermitian inner product; for this inner product $\varphi(g) \in \mathrm{U}(V)$ for every $g \in G$, so φ is a *unitary* representation.

***Proof*:**
Start from any Hermitian inner product $\langle -, -\rangle_0$ on V; one exists because $V \cong \mathbb{C}^n$ carries the standard inner product, which transports to V by a choice of basis. This $\langle -, -\rangle_0$ need not be

G-invariant, so average it over G against normalized Haar measure (theorem 9.2.1). Define

$$\langle v, w\rangle = \int_G \langle \varphi(g)v, \varphi(g)w\rangle_0 \, dg$$

The integrand $g \mapsto \langle \varphi(g)v, \varphi(g)w\rangle_0$ is continuous in g, being a composition of the smooth map φ with the continuous inner product, so the integral is defined and finite over the compact group G.

This $\langle -, -\rangle$ is a Hermitian inner product. Linearity in the first slot and conjugate-linearity in the second are inherited from $\langle -, -\rangle_0$ under the integral, and conjugate symmetry $\langle w, v\rangle = \overline{\langle v, w\rangle}$ likewise passes through. For positive-definiteness, if $v \neq 0$ then $\varphi(g)v \neq 0$ for every g (each $\varphi(g)$ is invertible), so $\langle \varphi(g)v, \varphi(g)v\rangle_0 > 0$ for all g; the integrand is a strictly positive continuous function on G, hence $\langle v, v\rangle > 0$. This is the general averaging construction of proposition 9.2.4 applied to the inner product.

Invariance is where the group law and the invariance of the integral meet. Fix $h \in G$. Then

$$\langle \varphi(h)v, \varphi(h)w\rangle = \int_G \langle \varphi(g)\varphi(h)v, \varphi(g)\varphi(h)w\rangle_0 \, dg = \int_G \langle \varphi(gh)v, \varphi(gh)w\rangle_0 \, dg$$

using that φ is a homomorphism, so $\varphi(g)\varphi(h) = \varphi(gh)$. The integrand is now the function $g \mapsto \langle \varphi(gh)v, \varphi(gh)w\rangle_0$, which is the right translate by h of the original integrand $g \mapsto \langle \varphi(g)v, \varphi(g)w\rangle_0$. By the right-invariance of the Haar integral (definition 9.2.2), $\int_G f(gh)\, dg = \int_G f(g)\, dg$, so the last integral equals $\int_G \langle \varphi(g)v, \varphi(g)w\rangle_0 \, dg = \langle v, w\rangle$. Thus $\langle \varphi(h)v, \varphi(h)w\rangle = \langle v, w\rangle$ for every h, which is exactly the statement that each $\varphi(h)$ preserves $\langle -, -\rangle$, so $\varphi(h) \in \mathrm{U}(V)$, completing the proof.

The mechanism is worth isolating, because the same three lines will return in every averaging argument of this chapter. An arbitrary structure on V (here an inner product) is transported by each $\varphi(g)$ and the transports are averaged; the result is invariant because right-translating the averaging variable by h just permutes the group and leaves the integral fixed. Positive-definiteness survives because the measure is positive with full support, so the integral of a nonnegative continuous function vanishes only when the function itself vanishes; the averaged form is thus not just bilinear and invariant but positive-definite, an invariant inner product. Compactness enters twice and essentially: it is what makes the total mass finite (so the average exists) and what lets the mass be normalized to a probability measure (so averaging is idempotent and returns an invariant form unchanged). For a noncompact group there is no finite invariant measure to average against, and the trick fails; this is the first sign that noncompact representation theory is a different subject.

The name "unitarian trick" is due to Weyl, who used it to transfer results between a compact group and its complexification. Once φ is realized inside $\mathrm{U}(V)$, the geometry of the unitary group becomes available, and the immediate payoff is that orthogonal complements are respected by the action.

Example 9.3.2: An Invariant Inner Product for a Rotation Action

Take $G = \mathrm{SO}(2)$, the circle of rotations, acting on $V = \mathbb{C}^2$ by the defining representation $\varphi(R_\theta) = R_\theta$, where $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$. Here the standard Hermitian inner product on $\mathbb{C}^2$ is already invariant, since each R_θ is a real orthogonal, hence unitary, matrix. The averaging

construction reproduces it: for the standard $\langle -,-\rangle_0$,

$$\langle v,w\rangle = \frac{1}{2\pi}\int_0^{2\pi}\langle R_\theta v, R_\theta w\rangle_0\, d\theta = \frac{1}{2\pi}\int_0^{2\pi}\langle v,w\rangle_0\, d\theta = \langle v,w\rangle_0$$

because $\langle R_\theta v, R_\theta w\rangle_0 = \langle v,w\rangle_0$ is constant in θ. Averaging a non-invariant product is more instructive: give $\mathbb{C}^2$ instead the skewed inner product $\langle v,w\rangle_0 = v_1\overline{w_1}+2v_2\overline{w_2}$, for which $R_{\pi/2}$ is not unitary (it sends the short axis to the long one). Averaging over θ produces $\langle v,w\rangle = \frac{3}{2}(v_1\overline{w_1}+v_2\overline{w_2})$, a scalar multiple of the standard product, which is invariant. The average has forgotten the skew and returned the round inner product the circle preserves.

With every representation made unitary, complete reducibility is now a short step: taking orthogonal complements decomposes any representation into irreducible pieces. This is the analytic form of Weyl's theorem, and it is the reason the representation theory of compact groups is tractable at all.

Theorem 9.3.3: Complete Reducibility for Compact Groups

Let G be a compact Lie group. Every finite-dimensional complex representation V of G is a direct sum of irreducible subrepresentations: there are irreducible subrepresentations $W_1,\dots,W_r \subseteq V$ with $V = W_1\oplus\cdots\oplus W_r$. Equivalently, every subrepresentation of V has a G-invariant complement.

***Proof*:**
Fix a G-invariant Hermitian inner product $\langle -,-\rangle$ on V, which exists by theorem 9.3.1. The key observation is that for such an inner product, the orthogonal complement of a subrepresentation is again a subrepresentation.

Let $W \subseteq V$ be a subrepresentation and let $W^\perp = \{v \in V \mid \langle v,w\rangle = 0 \text{ for all } w \in W\}$ be its orthogonal complement. For $v \in W^\perp$, $g \in G$, and $w \in W$,

$$\langle \varphi(g)v, w\rangle = \langle \varphi(g^{-1})\varphi(g)v, \varphi(g^{-1})w\rangle = \langle v, \varphi(g^{-1})w\rangle = 0$$

The first equality is the invariance of $\langle -,-\rangle$ under $\varphi(g^{-1})$; the last holds because $\varphi(g^{-1})w \in W$, as W is a subrepresentation, and $v \perp W$. Hence $\varphi(g)v \in W^\perp$ for every g, so $W^\perp$ is a subrepresentation. Since $V = W\oplus W^\perp$ as inner-product spaces and both summands are G-invariant, this is a decomposition $V = W\oplus W^\perp$ of representations.

The full decomposition follows by induction on $\dim V$. If $V = 0$ the empty direct sum works. If $V \neq 0$, choose a nonzero subrepresentation of least dimension; it is irreducible, since a proper nonzero subrepresentation would have strictly smaller dimension. Call it W_1 and split $V = W_1\oplus W_1^\perp$ as above. The complement $W_1^\perp$ is a representation of strictly smaller dimension, so by the inductive hypothesis it is a direct sum $W_2\oplus\cdots\oplus W_r$ of irreducible subrepresentations. Then $V = W_1\oplus W_2\oplus\cdots\oplus W_r$ is the required decomposition, completing the proof.

The proof is the Weyl-reducibility argument of theorem 6.4.8 run analytically. There the semisimple Lie algebra had no group to average over, and the Casimir element gave the invariant projection that split off a summand; here compactness gives the invariant inner product directly, and orthogonal projection is the splitting. The two proofs establish the same conclusion, that every finite-dimensional representation decomposes into irreducibles, by two routes that later converge: a compact semisimple group G and its complexified Lie algebra $\mathfrak{g}_{\mathbb{C}}$ have the same finite-dimensional representations (a

correspondence made precise in the theory of maximal tori and highest weights that follows), so complete reducibility for one transfers to the other. Weyl's original proof of the algebraic theorem went through exactly this transfer, deducing the algebraic statement from the compact one by the unitarian trick, before the purely algebraic Casimir proof was found.

Complete reducibility says the irreducibles are the atoms of the theory: to understand all finite-dimensional representations of a compact group it suffices to understand the irreducible ones and how they assemble. What remains is to control the assembly, and the tool for that is Schur's lemma, which measures equivariant maps between irreducibles. The reader has met its finite-group form already ([3]); the compact case is identical, because the only inputs are irreducibility and the algebraic closedness of $\mathbb{C}$.

Corollary 9.3.4: Schur's Lemma for Compact Groups

Let G be a compact Lie group and let V, V' be finite-dimensional complex representations of G.

1. If V and V' are irreducible, then any G-equivariant map $T\colon V \to V'$ is either zero or an isomorphism.
2. If V is irreducible, then any G-equivariant endomorphism $T\colon V \to V$ is a scalar: $T = \lambda \operatorname{id}_V$ for some $\lambda \in \mathbb{C}$.

***Proof*:**

Part (1). Let $T\colon V \to V'$ be G-equivariant, so $T\varphi(g) = \varphi'(g)T$ for all g. The kernel $\ker T \subseteq V$ is a subrepresentation: if $Tv = 0$ then $T(\varphi(g)v) = \varphi'(g)(Tv) = 0$, so $\varphi(g)v \in \ker T$. The image $T(V) \subseteq V'$ is a subrepresentation too, because $\varphi'(g)(Tv) = T(\varphi(g)v) \in T(V)$. Since V is irreducible, $\ker T$ is 0 or V; since V' is irreducible, $T(V)$ is 0 or V'. If $T \neq 0$ then $\ker T \neq V$ forces $\ker T = 0$ and $T(V) \neq 0$ forces $T(V) = V'$, so T is injective and surjective, hence an isomorphism.

Part (2). Let $T\colon V \to V$ be a G-equivariant endomorphism of the irreducible representation V. Because V is a nonzero finite-dimensional vector space over the algebraically closed field $\mathbb{C}$, the operator T has an eigenvalue $\lambda \in \mathbb{C}$. The operator $T - \lambda \operatorname{id}_V$ is again G-equivariant (a scalar operator commutes with everything, and T commutes with the action by hypothesis), and it has nonzero kernel, the λ-eigenspace. By part (1), applied with $V' = V$, an equivariant endomorphism of an irreducible representation is zero or an isomorphism; since $T - \lambda \operatorname{id}_V$ is not an isomorphism (its kernel is nonzero), it must be zero. Hence $T = \lambda \operatorname{id}_V$, completing the proof.

Part (2) is where the field being $\mathbb{C}$ is used: the eigenvalue λ must exist inside the field for the argument to close, and over $\mathbb{R}$ it can fail. The rotation representation of example 9.3.2, viewed as a real two-dimensional irreducible representation of $\mathrm{SO}(2)$, has the rotation $R_{\pi/2}$ itself as an equivariant endomorphism (rotations commute), and $R_{\pi/2}$ is not a real scalar; complexifying restores the scalars $\pm i$ and splits the representation into two complex lines. This is why the theory is cleanest over $\mathbb{C}$, and why finite-dimensional complex representations are the default object of study.

Schur's lemma turns complete reducibility into a bookkeeping statement about multiplicities. If V_λ is an irreducible representation, the space of equivariant maps $\operatorname{Hom}_G(V_\lambda, V)$ measures how many copies of V_λ occur inside V: by part (1) each nonzero such map lands isomorphically onto a copy of V_λ, and Schur's lemma makes the count exact. Writing $V = \bigoplus_i W_i$ as in theorem 9.3.3 and grouping

the summands isomorphic to a fixed irreducible V_λ, one obtains the *isotypic component*

$$V[\lambda] \cong V_\lambda \otimes_{\mathbb{C}} \operatorname{Hom}_G(V_\lambda, V)$$

in which $\operatorname{Hom}_G(V_\lambda, V)$ is the *multiplicity space*: its dimension is the number of copies of V_λ in the decomposition, and Schur's lemma guarantees this dimension is well defined, independent of the chosen decomposition. The direct sum of the isotypic components over the distinct irreducibles is all of V. This isotypic decomposition is the finite-group picture reproduced verbatim ([3]), and its development for the infinite-dimensional regular representation on $L^2(G)$ is the content of the Peter-Weyl theorem of the next chapter, where the multiplicity of each irreducible turns out to be its own dimension.

For a compact group, then, every finite-dimensional representation is unitary, decomposes into irreducibles, and its irreducible constituents occur with well-defined multiplicities counted by equivariant maps. This is the platform on which Peter-Weyl and the character theory of the following chapters are built: having reduced all representations to irreducibles, the remaining work is to list the irreducibles and to compute with their characters, and both become possible because complete reducibility has already been secured by integration.

Exercise 9.3.1

1. A vector $v \in V$ is *G-invariant* if $\varphi(g)v = v$ for all $g \in G$. Show that the set V^G of G-invariant vectors is a subrepresentation on which G acts trivially, and prove that if $V^G \neq 0$ then V contains a trivial one-dimensional subrepresentation that splits off as a direct summand. Deduce that a representation with no nonzero invariant vectors has no trivial summand.

2. With V any finite-dimensional representation of the compact group G, define $P\colon V \to V$ by $Pv = \int_G \varphi(g)v\,dg$. Show that P is a G-equivariant projection onto the invariant subspace V^G, that is, $P^2 = P$, $P(V) = V^G$, and $P\varphi(g) = \varphi(g)P = P$ for all g. This is the averaging operator of proposition 9.2.4 in its projection form.

3. Let $\varphi\colon G \to \mathrm{U}(V)$ and $\psi\colon G \to \mathrm{U}(V')$ be unitary representations of a compact group G, with invariant inner products $\langle -,-\rangle_V$ and $\langle -,-\rangle_{V'}$. Show that the tensor product representation $\varphi\otimes\psi$ on $V\otimes V'$, defined by $(\varphi\otimes\psi)(g) = \varphi(g)\otimes\psi(g)$, is unitary for the inner product determined by $\langle v\otimes v', w\otimes w'\rangle = \langle v,w\rangle_V \langle v',w'\rangle_{V'}$. Conclude that a tensor product of finite-dimensional representations of a compact group is completely reducible.

4. Let G be a finite group, regarded as a compact Lie group of dimension zero, and let $V = \mathbb{C}[G]$ be the vector space of complex-valued functions on G with G acting by $(\varphi(g)f)(x) = f(g^{-1}x)$ (the left regular representation). Write the invariant integral $\int_G f\,dg$ explicitly as a normalized sum, exhibit a G-invariant Hermitian inner product on V by averaging, and verify directly that the constant functions form a trivial subrepresentation with a G-invariant complement.

5. Consider the noncompact group $G = \mathbb{R}$ under addition, with the representation $\varphi\colon \mathbb{R} \to \mathrm{GL}_2(\mathbb{C})$ given by $\varphi(t) = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$. Verify that φ is a homomorphism. Show that $W = \operatorname{span}\big((1,0)^{\mathsf{T}}\big)$ is a subrepresentation, and prove that it has *no* $\varphi(\mathbb{R})$-invariant complement in $\mathbb{C}^2$. Explain why this does not contradict theorem 9.3.3, and identify precisely which step of the unitarian trick fails for $G = \mathbb{R}$.

6. Give a finite-dimensional *real* representation of a compact group that is irreducible over $\mathbb{R}$ but whose equivariant endomorphism algebra is larger than $\mathbb{R}$, and explain how this is consistent with corollary 9.3.4, which asserts scalar intertwiners only over $\mathbb{C}$.

9.4 Compact Semisimple Groups: the Killing Form and the Fundamental Group

The three preceding sections attached to a compact group two invariant objects: the invariant integral of section 9.2 and, through it, the $\mathrm{Ad}(G)$-invariant inner product on $\mathfrak{g}$ of proposition 9.1.5. This section turns those objects onto the Killing form κ of definition 6.1.3, the intrinsic bilinear form that measured semisimplicity in Part II. Compactness forces a sign: the averaged inner product makes each $\mathrm{ad}X$ skew, and a skew operator squares to something with nonpositive trace. The Killing form of a compact group is therefore negative semidefinite, and its degeneracy is exactly the center of $\mathfrak{g}$. This sign controls the global topology of the group. A compact group whose Killing form is negative-definite carries a bi-invariant metric of positive Ricci curvature, and Bonnet-Myers then forces its universal cover to be compact, so its fundamental group is finite. The algebraic condition of semisimplicity, read through a metric, becomes a finiteness statement about a topological invariant.

The negative sign also isolates a distinguished real form. A real Lie algebra with negative-definite Killing form is the Lie algebra of a compact group, and every complex semisimple Lie algebra arises by complexifying exactly one such real algebra up to isomorphism, its compact real form.

The sign computation rests on a linear-algebra fact about the averaged inner product. If an operator preserves an inner product infinitesimally, it is skew-symmetric with respect to that inner product, hence diagonalizable over $\mathbb{C}$ with purely imaginary eigenvalues. Applying this to $\mathrm{ad}X$ turns the trace $\kappa(X,X) = \mathrm{tr}\big((\mathrm{ad}X)^2\big)$ into a sum of squares of imaginary numbers, which is ≤ 0. The equality case is where the center enters.

Proposition 9.4.1: Negative-Definiteness of the Killing Form

Let G be a compact Lie group with Lie algebra $\mathfrak{g}$, and let κ be the Killing form of definition 6.1.3. Then κ is negative semidefinite: $\kappa(X,X) \leq 0$ for every $X \in \mathfrak{g}$, with $\kappa(X,X) = 0$ if and only if X lies in the center $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X,Y] = 0 \text{ for all } Y\}$. Consequently κ is negative-definite if and only if $\mathfrak{z}(\mathfrak{g}) = 0$, that is, if and only if $\mathfrak{g}$ is semisimple.

***Proof*:**

By proposition 9.1.5 the averaging construction of proposition 9.2.4 produces an $\mathrm{Ad}(G)$-invariant inner product $\langle -,- \rangle$ on $\mathfrak{g}$: for every $g \in G$ the operator $\mathrm{Ad}(g)$ is orthogonal, $\langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle = \langle X, Y \rangle$. Fix $X \in \mathfrak{g}$ and differentiate this identity along the one-parameter subgroup $g = \exp(tX)$. Since $\mathrm{Ad}(\exp(tX)) = \exp(t\,\mathrm{ad}X)$ by the naturality of exp (definition 2.5.1), the curve $t \mapsto \exp(t\,\mathrm{ad}X)$ consists of orthogonal operators, and differentiating $\langle \exp(t\,\mathrm{ad}X)Y, \exp(t\,\mathrm{ad}X)Z \rangle = \langle Y, Z \rangle$ at $t = 0$ gives

$$\langle (\mathrm{ad}X)Y, Z \rangle + \langle Y, (\mathrm{ad}X)Z \rangle = 0$$

for all $Y, Z \in \mathfrak{g}$. Thus $\mathrm{ad}X$ is skew-symmetric with respect to $\langle -,- \rangle$: its adjoint is $-\mathrm{ad}X$.

A skew-symmetric operator on a real inner-product space is normal, so it is diagonalizable over $\mathbb{C}$ and its eigenvalues are purely imaginary. Indeed, if $A^* = -A$ and $Av = \mu v$ for a nonzero v in the complexification, then $\mu\langle v, v \rangle = \langle Av, v \rangle = \langle v, A^*v \rangle = -\langle v, Av \rangle = -\overline{\mu}\langle v, v \rangle$, forcing $\mu = -\overline{\mu}$, so $\mu \in i\mathbb{R}$. Write the eigenvalues of $\mathrm{ad}X$ as $ic_1, \ldots, ic_m$ with $c_j \in \mathbb{R}$ (repeated with multiplicity, $m = \dim \mathfrak{g}$). Because $\mathrm{ad}X$ is diagonalizable, so is $(\mathrm{ad}X)^2$, with eigenvalues $(ic_j)^2 = -c_j^2$. The

Killing form is the trace of this operator:

$$\kappa(X,X) = \operatorname{tr}\big((\mathrm{ad}X)^2\big) = \sum_{j=1}^{m}(ic_j)^2 = -\sum_{j=1}^{m} c_j^2 \le 0$$

which establishes the semidefinite bound.

For the equality case, $\kappa(X,X) = -\sum_j c_j^2 = 0$ holds if and only if every $c_j = 0$, that is if and only if all eigenvalues of $\mathrm{ad}X$ vanish. Since $\mathrm{ad}X$ is diagonalizable, vanishing eigenvalues force $\mathrm{ad}X = 0$, which says $[X,Y] = 0$ for all Y, that is $X \in \mathfrak{z}(\mathfrak{g})$; conversely $X \in \mathfrak{z}(\mathfrak{g})$ gives $\mathrm{ad}X = 0$ and $\kappa(X,X) = 0$. Hence the null vectors of the quadratic form $X \mapsto \kappa(X,X)$ are exactly the center.

Finally, κ is negative-definite precisely when $X \mapsto \kappa(X,X)$ has no nonzero null vector, which by the equality case means $\mathfrak{z}(\mathfrak{g}) = 0$. It remains to link trivial center to semisimplicity for the Lie algebra of a compact group, and we do so as a chain of biconditionals. If $\mathfrak{z}(\mathfrak{g}) = 0$ then κ is negative-definite, hence nondegenerate, so Cartan's criterion for semisimplicity (theorem 6.3.1) makes $\mathfrak{g}$ semisimple; conversely a semisimple algebra has $\mathfrak{z}(\mathfrak{g}) = 0$, since a nonzero center would be a nonzero abelian, hence solvable, ideal. Thus κ negative-definite $\iff \mathfrak{z}(\mathfrak{g}) = 0 \iff \mathfrak{g}$ semisimple, completing the proof.

The proposition inverts the role the Killing form played in Part II. There κ was a computational device whose nondegeneracy detected semisimplicity over an arbitrary base; here, for a compact group, its *sign* is fixed once and for all, and definiteness rather than mere nondegeneracy is the semisimplicity criterion. The mechanism is the averaged inner product: it is the extra structure a general semisimple algebra lacks and a compact group gives, and it converts the abstract statement "$\mathrm{ad}X$ is a derivation" into the metric statement "$\mathrm{ad}X$ is skew," from which the sign is automatic. The center is the kernel because $\mathrm{ad}X = 0$ is exactly the condition that X commute with everything.

The negative sign, not the nondegeneracy alone, is what the geometry of the next theorem uses: a bi-invariant metric built from $-\kappa$ has curvature of a definite sign, and that sign is positive precisely because κ is negative. The abelian case marks the boundary of the statement.

Example 9.4.2: The Killing Form of a Torus and of $\mathfrak{su}(2)$

For the circle $S^1 = \mathrm{U}(1)$, or any torus $\mathbb{T}^n$, the Lie algebra $\mathfrak{g} = \mathbb{R}^n$ is abelian, so $[X,Y] = 0$ for all X, Y and $\mathrm{ad}X = 0$ identically. Hence $\kappa(X,Y) = \operatorname{tr}(\mathrm{ad}X\,\mathrm{ad}Y) = 0$ for every pair: the Killing form vanishes. Every X is a null vector because every X is central, and κ is negative semidefinite but as far from negative-definite as possible. A torus is compact but not semisimple, and the vanishing Killing form records exactly that.

At the opposite extreme sits $\mathfrak{su}(2)$, the Lie algebra of the compact group $\mathrm{SU}(2)$ (proposition 1.5.1). In the half-Pauli basis E_1, E_2, E_3 with $[E_1,E_2] = E_3$, $[E_2,E_3] = E_1$, $[E_3,E_1] = E_2$, the operator $\mathrm{ad}E_1$ sends $E_1 \mapsto 0$, $E_2 \mapsto E_3$, $E_3 \mapsto -E_2$, so in the basis (E_1,E_2,E_3)

$$\mathrm{ad}E_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad (\mathrm{ad}E_1)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

whence $\kappa(E_1,E_1) = \operatorname{tr}\big((\mathrm{ad}E_1)^2\big) = -2$. By the symmetry of the bracket relations the same holds for E_2 and E_3, and the mixed products $(\mathrm{ad}E_i)(\mathrm{ad}E_j)$ for $i \neq j$ are traceless because each $\mathrm{ad}E_k$ carries every basis vector to a multiple of a *different* basis vector, so the composite has zero diagonal. Hence $\kappa = -2\,\mathrm{Id}$ in this basis: the Killing form of $\mathfrak{su}(2)$ is $\kappa(X,Y) = -2\langle X,Y\rangle_{\text{Euclid}}$, negative-definite. The eigenvalues of $\mathrm{ad}E_1$ are $0, i, -i$, matching the proof: $\kappa(E_1,E_1) = i^2 + (-i)^2 = -2$.

The center is 0, and $\mathfrak{su}(2)$ is semisimple, indeed simple.

The sign that proposition 9.4.1 extracts is not only a fact about individual compact groups; run in reverse, it isolates a distinguished real form inside each complex semisimple Lie algebra. A complex semisimple Lie algebra $\mathfrak{g}_{\mathbb{C}}$, classified by its root system in chapter 8, is an abstract combinatorial object with no group attached a priori. The proposition says that whenever $\mathfrak{g}_{\mathbb{C}}$ is the complexification of the Lie algebra of a compact group, that real algebra has negative-definite Killing form. This picks out one real form as the "compact" one.

Definition 9.4.3: Compact Real Form

Let $\mathfrak{g}_{\mathbb{C}}$ be a complex semisimple Lie algebra. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g}_0 \subseteq \mathfrak{g}_{\mathbb{C}}$ such that $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus i\,\mathfrak{g}_0$ as real vector spaces, so that $\mathfrak{g}_{\mathbb{C}} \cong \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$ is its complexification. A real form $\mathfrak{g}_0$ is a *compact real form* of $\mathfrak{g}_{\mathbb{C}}$ if the Killing form of $\mathfrak{g}_0$ is negative-definite. The Killing form of $\mathfrak{g}_0$ is the restriction to $\mathfrak{g}_0$ of the Killing form κ of $\mathfrak{g}_{\mathbb{C}}$, because an $\mathbb{R}$-basis of $\mathfrak{g}_0$ is a $\mathbb{C}$-basis of $\mathfrak{g}_{\mathbb{C}}$, and for $X, Y \in \mathfrak{g}_0$ the matrix of $\mathrm{ad}X\,\mathrm{ad}Y$ in it is the same on $\mathfrak{g}_0$ as on $\mathfrak{g}_{\mathbb{C}}$. So the condition says that $\kappa|_{\mathfrak{g}_0}$ is negative-definite. Equivalently (by proposition 9.4.1 and lemma 9.4.4), $\mathfrak{g}_0$ is the Lie algebra of a compact semisimple Lie group.

The equivalence in the definition runs both ways. If $\mathfrak{g}_0 = \mathrm{Lie}(G)$ for a compact semisimple group G, then proposition 9.4.1 makes its Killing form negative-definite. Conversely, a real Lie algebra with negative-definite Killing form is semisimple and is the Lie algebra of a compact group, its adjoint group (lemma 9.4.4). Every complex semisimple Lie algebra has a compact real form, and it is unique up to conjugacy. The existence and uniqueness are proved in full in Part IV, where the compact real form is constructed from a Chevalley basis of the root space decomposition (theorem 7.2.4) by a sign adjustment on the root vectors. The general real-forms theory of Part IV begins by measuring how far an arbitrary real form sits from this one. For $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}_2(\mathbb{C})$ the compact real form is $\mathfrak{su}(2)$ of example 9.4.2, whose complexification $\mathfrak{su}(2) \oplus i\,\mathfrak{su}(2) = \mathfrak{sl}_2(\mathbb{C})$ recovers the whole complex algebra. The noncompact real form $\mathfrak{sl}_2(\mathbb{R})$ of the same $\mathfrak{sl}_2(\mathbb{C})$ appears in Part IV as its noncompact sibling.

Lemma 9.4.4: The Adjoint Group of a Compact Lie Algebra

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{R}$ or $\mathbb{C}$. Each operator $\exp(\mathrm{ad}X)$ with $X \in \mathfrak{g}$ is an automorphism of $\mathfrak{g}$, and the subgroup of $\mathrm{Aut}(\mathfrak{g})$ these operators generate is the *adjoint group* $\mathrm{Int}(\mathfrak{g})$, whose elements are the *inner automorphisms* of $\mathfrak{g}$. If $\mathfrak{g}_0$ is a real Lie algebra with negative-definite Killing form, then:

1. $\mathrm{Aut}(\mathfrak{g}_0)$ is a compact subgroup of $\mathrm{GL}(\mathfrak{g}_0)$, and its identity component is $\mathrm{Int}(\mathfrak{g}_0)$.
2. $\mathrm{Int}(\mathfrak{g}_0)$ is a compact connected Lie group with Lie algebra $\mathrm{ad}(\mathfrak{g}_0) \subseteq \mathfrak{gl}(\mathfrak{g}_0)$, and $\mathrm{ad}\colon \mathfrak{g}_0 \to \mathrm{ad}(\mathfrak{g}_0)$ is an isomorphism of Lie algebras.

***Proof*:**
Step 1: exponentials of derivations are automorphisms. Let D be a derivation of $\mathfrak{g}$ and $X, Y \in \mathfrak{g}$. Induction on n, applying the derivation rule once at each step and Pascal's rule to the coefficients,

gives the Leibniz formula

$$D^n[X,Y] = \sum_{k=0}^{n} \binom{n}{k} [D^k X, D^{n-k} Y]$$

Fix a norm on $\mathfrak{g}$. The bracket is bilinear on a finite-dimensional space, so $\|[A,B]\| \le C\|A\|\|B\|$ for a constant C, and the exponential series of D converges absolutely (proposition 1.3.3). Hence the double series $\sum_{k,l\ge 0}[D^k X/k!,\, D^l Y/l!]$ converges absolutely and may be summed in any order. Grouped by $n = k + l$, the Leibniz formula turns it into $\sum_n D^n[X,Y]/n! = \exp(D)[X,Y]$, and summed over k and l separately it is $[\exp(D)X, \exp(D)Y]$. As $\exp(D)$ is invertible with inverse $\exp(-D)$ (proposition 1.3.4(2)), it is an automorphism. The Jacobi identity says that $\mathrm{ad}X$ is a derivation, so each $\exp(\mathrm{ad}X)$ is an automorphism.

Step 2: $\mathrm{Aut}(\mathfrak{g}_0)$ *is compact.* The Killing form κ of $\mathfrak{g}_0$ is definite, hence nondegenerate, so $\mathfrak{g}_0$ is semisimple (theorem 6.3.1). The group $\mathrm{Aut}(\mathfrak{g}_0)$ of invertible A with $A[X,Y] = [AX, AY]$ for all X, Y is closed in $\mathrm{GL}(\mathfrak{g}_0)$, each condition being the vanishing of a continuous function of A. Every automorphism preserves κ (proposition 6.1.5(1)), so $\mathrm{Aut}(\mathfrak{g}_0)$ lies in the group of linear maps preserving the inner product $-\kappa$, which in a basis orthonormal for $-\kappa$ is $\mathrm{O}(n)$, $n = \dim \mathfrak{g}_0$, a closed and bounded subset of $M_n(\mathbb{R})$. Being closed in the compact group $\mathrm{O}(n)$, $\mathrm{Aut}(\mathfrak{g}_0)$ is compact, and it is a Lie group by theorem 3.2.1.

Step 3: its Lie algebra is $\mathrm{ad}(\mathfrak{g}_0)$. By theorem 3.2.1 the Lie algebra of $\mathrm{Aut}(\mathfrak{g}_0)$ consists of the $D \in \mathfrak{gl}(\mathfrak{g}_0)$ with $\exp(tD) \in \mathrm{Aut}(\mathfrak{g}_0)$ for all $t \in \mathbb{R}$. For such D, differentiating $\exp(tD)[X,Y] = [\exp(tD)X, \exp(tD)Y]$ at $t = 0$ (proposition 1.3.4(1)) gives $D[X,Y] = [DX, Y] + [X, DY]$, so D is a derivation. Conversely, if D is a derivation then so is each tD, and Step 1 puts $\exp(tD)$ in $\mathrm{Aut}(\mathfrak{g}_0)$. The Lie algebra is therefore $\mathrm{Der}(\mathfrak{g}_0)$, which equals $\mathrm{ad}(\mathfrak{g}_0)$ by corollary 6.3.5(3), and the same corollary makes $\mathrm{ad}\colon \mathfrak{g}_0 \to \mathrm{ad}(\mathfrak{g}_0)$ an isomorphism.

Step 4: the identity component is $\mathrm{Int}(\mathfrak{g}_0)$. The identity component $\mathrm{Aut}(\mathfrak{g}_0)^0$ is open and closed in $\mathrm{Aut}(\mathfrak{g}_0)$ (proposition 9.1.3(1)), so it is a compact connected Lie group with the same Lie algebra $\mathrm{ad}(\mathfrak{g}_0)$. By corollary 2.3.5 it is generated by the exponentials $\exp(\mathrm{ad}X)$ with $X \in \mathfrak{g}_0$, and these generate $\mathrm{Int}(\mathfrak{g}_0)$ by definition. Hence $\mathrm{Aut}(\mathfrak{g}_0)^0 = \mathrm{Int}(\mathfrak{g}_0)$, which proves (1) and, with Step 3, (2), completing the proof.

The definiteness of the Killing form now feeds directly into topology. A bi-invariant metric on a compact semisimple group has curvature controlled by κ, and Weyl's theorem reads that curvature bound through the Bonnet-Myers theorem to conclude that the group is, up to a finite cover, simply connected. The precise input is a curvature formula for bi-invariant metrics, which belongs to the companion volume on Riemannian geometry and is cited rather than reproved here.

The formula is the following. For the bi-invariant metric of proposition 9.1.5, whose geodesics through e are one-parameter subgroups, the sectional and Ricci curvatures are computed from the bracket alone. The Ricci tensor of a bi-invariant metric is $\mathrm{Ric}(X,X) = -\frac{1}{4}\kappa(X,X)$ ([7, Ricci curvature of a bi-invariant metric]). When κ is negative-definite this makes Ric positive-definite, and a positive lower bound on Ricci curvature is exactly the hypothesis of Bonnet-Myers.

Theorem 9.4.5: Weyl's Finiteness Theorem

Let G be a compact semisimple Lie group. Then the fundamental group $\pi_1(G)$ is finite. Equivalently, the universal cover $\widetilde{G}$ of proposition 3.4.1 is compact.

***Proof*:**
Since $\mathfrak{g}$ is semisimple its center is trivial, so by proposition 9.4.1 the Killing form κ is negative-definite. The universal cover $\widetilde{G}$ is a Lie group with the same Lie algebra $\mathfrak{g}$ (proposition 3.4.1), connected and simply connected. The strategy is to prove $\widetilde{G}$ compact directly, using a bi-invariant metric of positive Ricci curvature, and then to read off finiteness of $\pi_1(G)$ from the covering.

Step 1: the bi-invariant metric and its curvature. Equip $\mathfrak{g}$ with the positive-definite inner product $\langle X, Y\rangle = -\kappa(X, Y)$, which is Ad-invariant because κ is invariant under Ad (the Killing form satisfies $\kappa(\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y) = \kappa(X, Y)$ from proposition 6.1.5). By proposition 9.1.5 this inner product extends to a bi-invariant Riemannian metric on $\widetilde{G}$ (the construction of that proposition uses only an Ad-invariant inner product on $\mathfrak{g}$, which $\widetilde{G}$ and G share). For any bi-invariant metric the Ricci curvature is $\mathrm{Ric}(X, X) = -\frac{1}{4}\kappa(X, X)$ on the Lie algebra ([7, Ricci curvature of a bi-invariant metric], which also records the normalization $\mathrm{Ric} = \frac{1}{4}\, g$ that this step derives). Because $\langle X, X\rangle = -\kappa(X, X)$ by construction,

$$\mathrm{Ric}(X, X) = -\tfrac{1}{4}\kappa(X, X) = \tfrac{1}{4}\langle X, X\rangle$$

so the Ricci curvature is bounded below by the positive multiple $\frac{1}{4}$ of the metric: $\mathrm{Ric} \geq \frac{1}{4}g$ as quadratic forms.

Step 2: compactness of $\widetilde{G}$ via Bonnet-Myers. A bi-invariant metric on a Lie group is geodesically complete: its geodesics through e are the one-parameter subgroups $t \mapsto \exp(tX)$, defined for all $t \in \mathbb{R}$ (proposition 9.1.5), and left translation carries these to geodesics through every point, so every geodesic extends to all of $\mathbb{R}$. Thus $\widetilde{G}$ is a complete, simply connected Riemannian manifold with $\mathrm{Ric} \geq \frac{1}{4}g > 0$. The Bonnet-Myers theorem ([7, Bonnet-Myers]) states that a complete Riemannian manifold of dimension n whose Ricci curvature satisfies $\mathrm{Ric} \geq (n-1)k\, g$ for a constant $k > 0$ has diameter at most $\pi/\sqrt{k}$ and is therefore compact. Take $n = \dim \widetilde{G}$ and $k = 1/(4(n-1)) > 0$, so that $(n-1)k\, g = \frac{1}{4}g$ and the bound of Step 1 is exactly $\mathrm{Ric} \geq (n-1)k\, g$; here $n \geq 3$ because a compact semisimple group has no summand of dimension less than 3, so $n - 1 > 0$ and k is a genuine positive constant. Applying Bonnet-Myers to $\widetilde{G}$ with this k, $\widetilde{G}$ has finite diameter, and a complete manifold of finite diameter is compact. Hence $\widetilde{G}$ is compact.

Step 3: finiteness of $\pi_1(G)$. The universal covering homomorphism $p\colon \widetilde{G} \to G$ (proposition 3.4.1) is a surjective Lie group homomorphism whose kernel $\ker p$ is a discrete normal subgroup, and $\pi_1(G) \cong \ker p$ (the deck group of a covering group is the kernel of the covering homomorphism, identified with π_1 of the base). Since $\widetilde{G}$ is compact by Step 2 and $\ker p$ is a discrete subgroup of a compact group, $\ker p$ is finite: it is a closed discrete subset of a compact space, hence compact and discrete, hence finite. Therefore $\pi_1(G) \cong \ker p$ is finite, as we sought to show.

Weyl's theorem is a statement about a topological invariant proved entirely by analysis and geometry. Nothing in the conclusion mentions curvature or metrics, only the fundamental group; yet the argument routes through a Riemannian metric, computes its curvature from the Lie bracket, and invokes a comparison theorem about geodesics. Negative-definiteness of the Killing form is what makes the Ricci curvature positive, and positive Ricci curvature is what Bonnet-Myers converts into a finite diameter. A compact semisimple group is thus never far from simply connected, its fundamental group a finite obstruction that a finite cover removes.

The theorem also explains why the classification of compact semisimple groups is a refinement, not a coarsening, of the classification of complex semisimple Lie algebras. Each complex semisimple Lie algebra $\mathfrak{g}_{\mathbb{C}}$ has a simply connected compact group $\widetilde{G}$ realizing its compact real form, and every

compact group with that Lie algebra is a quotient $\widetilde{G}/\Gamma$ by a finite central subgroup Γ; the finitely many such Γ index the finitely many compact groups sharing the algebra. Because π_1 is finite, the tower of these quotients is finite, and the center $Z(\widetilde{G})$, which contains every such Γ, is itself finite. The running example makes the tower visible.

Example 9.4.6: The Fundamental Groups of SU(2), SO(3), and SU(n)

The group SU(2) is diffeomorphic to the 3-sphere S^3 (proposition 1.5.1), which is simply connected, so $\pi_1(\mathrm{SU}(2)) = 0$. Its Lie algebra $\mathfrak{su}(2)$ is semisimple (example 9.4.2), and SU(2) is therefore its own universal cover: it is the simply connected compact group $\widetilde{G}$ for the root system A_1. Weyl's theorem applies vacuously, $\pi_1(\mathrm{SU}(2)) = 0$ being finite.

The double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5 has kernel $\{\pm I\}$, a central subgroup of order 2. Hence $\mathrm{SO}(3) \cong \mathrm{SU}(2)/\{\pm I\}$ is a quotient of the simply connected SU(2) by a finite central subgroup, and its fundamental group is

$$\pi_1(\mathrm{SO}(3)) \cong \{\pm I\} \cong \mathbb{Z}/2\mathbb{Z}$$

as computed in corollary 1.5.6. This is finite, in accordance with Weyl's theorem, and it exhibits the full tower for A_1: the simply connected SU(2) at the top and its unique nontrivial quotient SO(3) below, indexed by the two subgroups $\{I\}$ and $\{\pm I\}$ of the order-2 center $Z(\mathrm{SU}(2)) = \{\pm I\}$.

The special unitary groups SU(n) are all simply connected: $\pi_1(\mathrm{SU}(n)) = 0$ for every $n \geq 2$. This follows from the fibration $\mathrm{SU}(n-1) \hookrightarrow \mathrm{SU}(n) \to S^{2n-1}$ (theorem 4.1.3) and induction from the base case $\mathrm{SU}(2) = S^3$, the odd spheres S^{2n-1} being simply connected for $n \geq 2$. Since $\mathfrak{su}(n)$ is semisimple, each SU(n) is its own universal cover, the simply connected compact group for the root system A_{n-1}. Its center is $Z(\mathrm{SU}(n)) = \{\zeta I \mid \zeta^n = 1\} \cong \mathbb{Z}/n\mathbb{Z}$, finite as Weyl's theorem requires, and the quotients $\mathrm{SU}(n)/\Gamma$ by the subgroups $\Gamma \subseteq \mathbb{Z}/n\mathbb{Z}$ enumerate the compact connected groups with Lie algebra $\mathfrak{su}(n)$, the largest quotient $\mathrm{SU}(n)/Z(\mathrm{SU}(n)) = \mathrm{PSU}(n)$ being the adjoint group.

The examples display the two ends of the phenomenon. For a simply connected group like SU(n) the fundamental group is trivial and Weyl's theorem says nothing new, but the theorem still governs the finite center, whose subgroups index the other compact groups with the same algebra. For a non-simply-connected group like SO(3) the fundamental group is a nonzero finite group, and it is exactly the kernel of the cover from the simply connected model, a finite piece of central data. The compactness of the universal cover is the fact that makes the fundamental group finite in both cases, converting the analytic conclusion of Bonnet-Myers into an arithmetic statement about a group of deck transformations.

Exercise 9.4.1

1. Verify directly that the Killing form of $\mathfrak{su}(2)$ is negative-definite by computing $\kappa(X, X)$ for a general $X = aE_1 + bE_2 + cE_3$ in the half-Pauli basis, using the bracket relations $[E_1, E_2] = E_3$, $[E_2, E_3] = E_1$, $[E_3, E_1] = E_2$. Express the answer as a negative multiple of $a^2 + b^2 + c^2$, and identify the eigenvalues of $\mathrm{ad}X$ as predicted by proposition 9.4.1.

2. Let $\mathbb{T}^n$ be an n-torus with Lie algebra $\mathbb{R}^n$. Show that its Killing form is identically zero, and explain why this is consistent with proposition 9.4.1 (negative semidefinite, not negative-definite). What does this say about whether a compact abelian group can be semisimple?

3. Prove that the center $Z(G)$ of a compact *semisimple* connected Lie group G is finite. Deduce

it from Weyl's finiteness theorem applied to $\widetilde{G}$, using that $Z(\widetilde{G})$ is a discrete subgroup of the compact $\widetilde{G}$ and that $Z(G)$ is a quotient of $Z(\widetilde{G})$ by the covering kernel.

4. Show that $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}_n(\mathbb{C})$. Verify that $\mathfrak{su}(n) \oplus i\,\mathfrak{su}(n) = \mathfrak{sl}_n(\mathbb{C})$ as real vector spaces by writing an arbitrary traceless complex matrix Z as $Z = X + iY$ with $X, Y \in \mathfrak{su}(n)$, and confirm that $\mathfrak{su}(n)$ has negative-definite Killing form (you may cite that $\mathfrak{su}(n)$ is the Lie algebra of the compact semisimple group $\mathrm{SU}(n)$).

5. The proof of theorem 9.4.5 uses the bi-invariant-metric Ricci formula $\mathrm{Ric}(X,X) = -\frac{1}{4}\kappa(X,X)$. Explain in one paragraph why the negative sign in κ (proposition 9.4.1) is exactly what makes the Ricci curvature positive, and why a compact group that is *not* semisimple, such as a torus, fails to have positive Ricci curvature for this metric.

6. Using the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, exhibit explicitly the finite fundamental group $\pi_1(\mathrm{SO}(3))$ as a subgroup of $\mathrm{SU}(2)$, and explain how the compactness of $\mathrm{SU}(2) = S^3$ forces this subgroup to be finite in accordance with theorem 9.4.5.

9.5 Representations of Compact Groups: S^1 and SU(2)

Complete reducibility (theorem 9.3.3) reduces the representation theory of a compact group to the description of its irreducible representations: once every finite-dimensional representation is a direct sum of irreducibles, the entire finite-dimensional theory is known as soon as the irreducibles are listed and their combination rules understood. This section carries out that listing for the two groups on which everything downstream is modelled, the circle $S^1 = \mathrm{U}(1)$ and the running example $\mathrm{SU}(2)$. Each is small enough that every irreducible, every weight, and every character can be written down explicitly, and each already exhibits the three features that the rest of Part III raises to an arbitrary compact group: a labelling of irreducibles by weights, an orthogonality of characters, and a character formula expressed as a ratio.

The circle is the abelian model. Its irreducibles are one-dimensional, indexed by a single integer, and the decomposition of a representation into them is the finite-dimensional case of the Fourier expansion of a function on the circle.

Example 9.5.1: The Irreducible Representations of S^1

Regard $S^1 = \mathrm{U}(1) = \{z \in \mathbb{C} \mid |z| = 1\}$ as a compact connected abelian Lie group under multiplication. For each $n \in \mathbb{Z}$ the map

$$\chi_n \colon S^1 \to \mathbb{C}^\times, \qquad \chi_n(z) = z^n$$

is a continuous homomorphism, hence a one-dimensional complex representation on $\mathbb{C}$ (acting by $z \cdot v = z^n v$). These are pairwise inequivalent (distinct n give distinct traces z^n), and they are the *only* irreducible representations of S^1: every irreducible complex representation of S^1 is isomorphic to some χ_n. By theorem 9.3.3, every finite-dimensional representation of S^1 is a direct sum of the χ_n.

Three points require verification: that these are representations, that they exhaust the irreducibles, and that they are one-dimensional. The first is immediate, since $\chi_n(zw) = (zw)^n = z^n w^n = \chi_n(z)\chi_n(w)$. For the third, an irreducible complex representation of an abelian group is one-dimensional: if V is

irreducible then, because the group is abelian, every operator $\rho(g)$ commutes with the whole action and so, by Schur's lemma for compact groups (corollary 9.3.4), acts as a scalar; every subspace of V is then invariant, so irreducibility forces $\dim V = 1$. A one-dimensional representation of S^1 is a continuous homomorphism $S^1 \to \mathbb{C}^\times$; since S^1 is compact, its image is a compact subgroup of $\mathbb{C}^\times$, hence lies in $\mathrm{U}(1) = S^1$. It remains to see that a continuous homomorphism $\chi\colon S^1 \to S^1$ is $z \mapsto z^n$ for a unique $n \in \mathbb{Z}$. Composing with the covering $t \mapsto e^{it}$ gives a continuous homomorphism $\mathbb{R} \to S^1$, which by the classification of one-parameter subgroups (theorem 2.3.1) has the form $t \mapsto e^{iat}$ for some $a \in \mathbb{R}$; requiring it to descend through the kernel $2\pi\mathbb{Z}$ forces $a = n \in \mathbb{Z}$, so $\chi(e^{it}) = e^{int}$, that is $\chi = \chi_n$. The irreducible representations of S^1 are therefore exactly the $\{\chi_n \mid n \in \mathbb{Z}\}$.

The characters χ_n are the pure oscillations $z^n = e^{in\theta}$ on $S^1 = \{e^{i\theta} \mid \theta \in \mathbb{R}/2\pi\mathbb{Z}\}$, and they are orthonormal for the normalized invariant integral of definition 9.2.2: writing Haar measure as $dz = \frac{1}{2\pi}\, d\theta$,

$$\int_{S^1} \chi_m(z)\,\overline{\chi_n(z)}\, dz = \frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)\theta}\, d\theta = \delta_{mn}$$

The full force of this observation belongs to the next chapter: the family $\{z^n \mid n \in \mathbb{Z}\}$ is not only orthonormal but *complete* in $L^2(S^1)$, so that every square-integrable function on the circle expands in a convergent series $\sum_n c_n z^n$. This is the classical theory of Fourier series, and it is the Peter-Weyl theorem for the group S^1: the irreducible characters form an orthonormal basis of L^2 of the group. The finite-dimensional statement recorded here, that every finite-dimensional representation is a finite sum of the χ_n, is its trigonometric-polynomial case, and the passage from finite sums to L^2-convergent series is the analytic content chapter 10 gives for a general compact group.

The circle settles the abelian case. The first nonabelian example, and the one that governs the whole of highest-weight theory, is SU(2). Its irreducibles are built by hand from the standard two-dimensional representation, and the $\mathfrak{sl}_2$-theory of chapter 7 identifies them completely. The construction realizes each irreducible as a space of homogeneous polynomials on which the group acts by substituting variables.

Proposition 9.5.2: The Irreducible Representations of SU(2)

For each integer $n \geq 0$ let V_n be the space of homogeneous polynomials of degree n in two variables x, y over $\mathbb{C}$, with basis the monomials $x^n, x^{n-1}y, \ldots, y^n$, so $\dim V_n = n+1$. The group SU(2) acts on V_n by linear substitution of variables: for $g = \left(\begin{smallmatrix} a & b \\ -\bar{b} & \bar{a} \end{smallmatrix}\right) \in \mathrm{SU}(2)$ and $p \in V_n$,

$$(g \cdot p)(x, y) = p\big(ax - \bar{b}\,y,\ bx + \bar{a}\,y\big)$$

that is, g acts on the variables by the substitution $(x, y) \mapsto (x, y)g$ (equivalently $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto g^T \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$), extended to polynomials. Each V_n is irreducible, and $V_n \cong V_m$ as SU(2)-representations only if $n = m$. Every finite-dimensional irreducible complex representation of SU(2) is isomorphic to exactly one V_n; there is one irreducible of each dimension $n+1 = 1, 2, 3, \ldots$, and $V_n = \mathrm{Sym}^n(\mathbb{C}^2)$ is the n-th symmetric power of the standard representation $V_1 = \mathbb{C}^2$.

Symmetric powers are the candidates for a structural reason. The standard representation $V_1 = \mathbb{C}^2$ is SU(2) acting on column vectors by matrix multiplication (proposition 1.5.1 identifies SU(2) with the unit quaternions acting on $\mathbb{C}^2$). Its symmetric powers $\mathrm{Sym}^n(\mathbb{C}^2)$ are the degree-n homogeneous polynomials in the two coordinate functions x, y of $\mathbb{C}^2$, and the induced action is substitution of variables. That these symmetric powers are already irreducible, and already all the irreducibles, is the content of the proposition. The proof runs through the complexified Lie algebra, where the

raising and lowering operators of the $\mathfrak{sl}_2$-theory make the weight structure visible.

***Proof*:**
Differentiate the action to obtain the action of the Lie algebra $\mathfrak{su}(2)$, and complexify to $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}_2(\mathbb{C})$, on which the standard basis e, h, f of theorem 7.4.6 operates by the raising, weight, and lowering operators computed below.

Step 1: the maximal torus and the weights. Let $T = \left\{ t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta}) \;\middle|\; \theta \in \mathbb{R} \right\} \subseteq \mathrm{SU}(2)$ be the diagonal maximal torus. The torus element t_θ has $a = e^{i\theta}$ and $b = 0$ in the action formula of the proposition, so under the substitution action it sends $x \mapsto e^{i\theta}x$ and $y \mapsto e^{-i\theta}y$; hence on the monomial $x^{n-k}y^k$,

$$t_\theta \cdot (x^{n-k}y^k) = (e^{i\theta}x)^{n-k}(e^{-i\theta}y)^k = e^{i(n-2k)\theta}\, x^{n-k}y^k$$

Adopting the sign convention that the *weight* of a vector on which t_θ acts by $e^{im\theta}$ is the integer m, the monomial $x^{n-k}y^k$ has weight $n-2k$. As k runs from 0 to n, the weights are

$$n,\ n-2,\ n-4,\ \ldots,\ -n$$

each occurring exactly once, an unbroken string symmetric about 0 with steps of 2. Differentiating, the element $h = \operatorname{diag}(1,-1)$ of $\mathfrak{sl}_2(\mathbb{C})$ (the generator of the complexified torus) acts on $x^{n-k}y^k$ by the scalar $n-2k$: the weight is the h-eigenvalue, and each weight space is one-dimensional, spanned by a single monomial.

Step 2: the raising and lowering operators. The remaining basis elements $e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ of $\mathfrak{sl}_2(\mathbb{C})$ act as first-order differential operators. Differentiating the substitution action gives, on V_n,

$$e = x\,\frac{\partial}{\partial y}, \qquad f = y\,\frac{\partial}{\partial x}$$

so that $e\cdot(x^{n-k}y^k) = k\,x^{n-k+1}y^{k-1}$ raises the weight by 2 and $f\cdot(x^{n-k}y^k) = (n-k)\,x^{n-k-1}y^{k+1}$ lowers it by 2, matching the bracket relations $[h,e] = 2e$, $[h,f] = -2f$ of theorem 7.4.6. The highest-weight monomial x^n is annihilated by e, and repeated application of f walks down the entire monomial ladder $x^n \to x^{n-1}y \to \cdots \to y^n$, since $f\cdot(x^{n-k}y^k) = (n-k)\,x^{n-k-1}y^{k+1}$ has nonzero coefficient for each $k < n$.

Step 3: irreducibility. Let $0 \neq W \subseteq V_n$ be a subrepresentation. Because W is stable under the torus T, it is a sum of weight spaces: fix $\theta = \pi/(n+1)$, for which the $n+1$ eigenvalues $e^{i(n-2k)\theta}$, $k = 0, \ldots, n$, are pairwise distinct (the exponents $n-2k$ are distinct integers of absolute value at most n, so the corresponding angles are distinct modulo 2π). An operator with distinct eigenvalues preserves a subspace only if that subspace is spanned by eigenvectors, and since each weight space of V_n is one-dimensional and spanned by a monomial, W is spanned by a subset of the monomials $x^{n-k}y^k$. In particular W contains at least one monomial, say $x^{n-k_0}y^{k_0}$. Applying e repeatedly raises it to a nonzero multiple of the top monomial x^n, which therefore lies in W; applying f repeatedly to x^n produces a nonzero multiple of every monomial $x^{n-k}y^k$ in turn, by Step 2. Hence W contains all $n+1$ monomials, so $W = V_n$, and V_n is irreducible.

Step 4: completeness. Let V be any finite-dimensional irreducible complex representation of SU(2). Restricting to T and using that T is abelian, V decomposes into T-weight spaces, so the complexified generator h acts diagonalizably with integer eigenvalues (the weights). Let m be the largest weight occurring, and let $v \neq 0$ be a vector of weight m. Then $e\cdot v$ has weight $m+2$; since m is the largest weight, $e\cdot v = 0$, so v is a highest-weight vector. The relations $[h,f] = -2f$,

$[e, f] = h$ of theorem 7.4.6 give, exactly as in the $\mathfrak{sl}_2$-representation theory there, that the vectors $v, f \cdot v, f^2 \cdot v, \ldots$ have weights $m, m-2, m-4, \ldots$ and span a nonzero subrepresentation of V; by irreducibility they span all of V. The ladder terminates: some $f^{r+1} \cdot v = 0$ while $f^r \cdot v \neq 0$, and the Cartan-integer computation of theorem 7.4.6, evaluating $ef^{r+1} \cdot v = 0$ against the string, forces $m = r \geq 0$ and gives $\dim V = r + 1 = m + 1$. The resulting module, with weights $m, m-2, \ldots, -m$ each once, is isomorphic to V_m by matching highest-weight vectors and comparing the action of e, h, f on the two ladders. Thus $V \cong V_m$ with $m \geq 0$, and the correspondence $n \mapsto V_n$ is a bijection from $\mathbb{Z}_{\geq 0}$ onto the isomorphism classes of irreducibles, one in each dimension $n + 1$, completing the proof.

The proposition is the analytic realization of the abstract highest-weight classification. Chapter 7 attached to $\mathfrak{sl}_2(\mathbb{C})$ a list of irreducible modules indexed by a nonnegative integer, the highest weight, built by the raising and lowering operators of theorem 7.4.6; here those same modules appear concretely as spaces of homogeneous polynomials, and the group SU(2), not only its Lie algebra, acts on them. That the passage from $\mathfrak{sl}_2(\mathbb{C})$-modules back to SU(2)-representations loses nothing, that every algebra module integrates to a group representation, is guaranteed by SU(2) being simply connected (proposition 3.4.1): a representation of the Lie algebra of a simply-connected group integrates to a representation of the group. The weights $n, n-2, \ldots, -n$ of V_n record the module, and the pattern they exhibit, an unbroken string symmetric about the origin, is the pattern that maximal-tori and root-system theory generalizes to every compact group in the chapters ahead.

With the irreducibles in hand, their characters are a short computation. The character of a representation is the function $g \mapsto \operatorname{tr} \rho(g)$; for a compact group it is a class function, constant on conjugacy classes, and every element of SU(2) is conjugate to a diagonal one, so a character is determined by its values on the maximal torus T. The character of V_n on T is a sum of the weight exponentials, and summing the geometric series produces the ratio that names the general theory.

Proposition 9.5.3: The Character of V_n

Let $\chi_n = \operatorname{tr}(\cdot \,|V_n)$ be the character of the irreducible representation V_n of proposition 9.5.2. On the maximal torus element $t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta})$,

$$\chi_n(t_\theta) = \sum_{k=0}^{n} e^{i(n-2k)\theta} = e^{in\theta} + e^{i(n-2)\theta} + \cdots + e^{-in\theta}$$

and for $\theta \notin \pi\mathbb{Z}$ this sum has the closed form

$$\chi_n(t_\theta) = \frac{\sin\big((n+1)\theta\big)}{\sin\theta}$$

At $\theta \in \pi\mathbb{Z}$ the value is the limit $\chi_n(t_0) = n + 1 = \dim V_n$ and $\chi_n(t_\pi) = (-1)^n(n+1)$.

***Proof*:**

By Step 1 of the proof of proposition 9.5.2, the monomial $x^{n-k}y^k$ is a T-eigenvector with eigenvalue $e^{i(n-2k)\theta}$ under t_θ. The character is the trace of t_θ on V_n, the sum of these eigenvalues over the basis of monomials:

$$\chi_n(t_\theta) = \sum_{k=0}^{n} e^{i(n-2k)\theta}$$

This is a finite geometric progression with first term $e^{in\theta}$, ratio $e^{-2i\theta}$, and $n+1$ terms. Factor out the symmetric middle to expose the ratio: the sum equals

$$\chi_n(t_\theta) = \frac{e^{i(n+1)\theta} - e^{-i(n+1)\theta}}{e^{i\theta} - e^{-i\theta}}$$

valid for $e^{i\theta} \neq e^{-i\theta}$, that is $\theta \notin \pi\mathbb{Z}$. To confirm the closed form, multiply the claimed right-hand side by the denominator: $\big(e^{i\theta} - e^{-i\theta}\big)\sum_{k=0}^{n} e^{i(n-2k)\theta}$ telescopes, since

$$\big(e^{i\theta} - e^{-i\theta}\big)e^{i(n-2k)\theta} = e^{i(n-2k+1)\theta} - e^{i(n-2k-1)\theta}$$

and summing over k from 0 to n cancels all interior terms, leaving $e^{i(n+1)\theta} - e^{-i(n+1)\theta}$; dividing back gives the displayed ratio. By Euler's formula $e^{i\varphi} - e^{-i\varphi} = 2i\sin\varphi$, both numerator and denominator carry a factor $2i$, which cancels:

$$\chi_n(t_\theta) = \frac{2i\sin\big((n+1)\theta\big)}{2i\sin\theta} = \frac{\sin\big((n+1)\theta\big)}{\sin\theta}$$

At $\theta \in \pi\mathbb{Z}$ the denominator vanishes; the sum form is continuous, giving $\chi_n(t_0) = \sum_{k=0}^{n} 1 = n+1$ and $\chi_n(t_\pi) = \sum_{k=0}^{n} e^{i(n-2k)\pi} = \sum_{k=0}^{n}(-1)^n = (-1)^n(n+1)$, which are also the limits of $\sin((n+1)\theta)/\sin\theta$. This gives the two forms of the character, completing the proof.

The shape of this formula, an alternating exponential sum in the numerator divided by a fixed factor in the denominator, is the template for the whole subject. The numerator $e^{i(n+1)\theta} - e^{-i(n+1)\theta}$ is an antisymmetric combination of exponentials attached to the highest weight n shifted by 1; the denominator $e^{i\theta} - e^{-i\theta}$ is the same antisymmetric combination for the smallest possible shift, and is independent of n. This denominator is the *Weyl denominator* of SU(2), the antisymmetrization of a single exponential over the two-element Weyl group $\{\pm 1\}$ (definition 7.5.3) acting by $\theta \mapsto -\theta$ on the torus. The Weyl character formula of chapter 12 is precisely the statement that for *every* compact connected group the character of the irreducible with highest weight λ is a ratio of this form: an alternating sum over the Weyl group in the numerator, built from λ shifted by a fixed vector, divided by the same alternating sum built from the shift alone. The SU(2) computation is the base case from which that general formula is read off, and the general proof, for all its machinery, produces nothing more than the pattern already present here.

These characters also make the orthogonality of the previous discussion concrete. The functions χ_n are orthonormal for the invariant integral on SU(2) once that integral is pushed to the torus, and the pushforward carries the factor $\sin^2\theta$ that reappears as the Weyl integration formula in Chapter 11. Taken together, S^1 and SU(2) already display the entire pattern of the compact theory: irreducibles indexed by weights, an orthonormal family of characters, and a character written as a ratio whose denominator depends only on the group. Peter-Weyl (chapter 10) upgrades the orthonormal family to an orthonormal basis of $L^2(G)$, and the Weyl character formula (chapter 12) upgrades the ratio to every compact connected group; both take the two computations of this section as their template. The final exercise verifies the orthogonality directly and records the Clebsch-Gordan rule $V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1}$, the multiplication table of the representation ring of SU(2), which the addition of weights makes a one-line computation.

Exercise 9.5.1

1. Show that the character of the standard two-dimensional representation $V_1 = \mathbb{C}^2$ of SU(2)

is $\chi_1(t_\theta) = e^{i\theta} + e^{-i\theta} = 2\cos\theta$, and confirm that this equals both $\operatorname{tr}\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ and the value $\sin(2\theta)/\sin\theta$ from proposition 9.5.3.

2. Using that the character of a tensor product is the product of characters, and that the character of a direct sum is the sum, prove the Clebsch-Gordan decomposition

$$V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1} \qquad (n \geq 1)$$

by checking the identity $\chi_1(t_\theta)\,\chi_n(t_\theta) = \chi_{n+1}(t_\theta) + \chi_{n-1}(t_\theta)$ of characters and invoking that a finite-dimensional representation of SU(2) is determined up to isomorphism by its character (a consequence of complete reducibility, theorem 9.3.3).

3. Assume the invariant integral of a class function f on SU(2) is given in terms of the torus angle by

$$\int_{\mathrm{SU}(2)} f\,dg = \frac{2}{\pi}\int_0^{\pi} f(t_\theta)\,\sin^2\theta\,d\theta$$

(the Weyl integration formula for SU(2), proved in chapter 11). Verify the character orthogonality relation

$$\int_{\mathrm{SU}(2)} \chi_m(g)\,\overline{\chi_n(g)}\,dg = \delta_{mn}$$

for the characters χ_m, χ_n of proposition 9.5.3, and explain in one sentence why this is the SU(2) instance of the Schur orthogonality that Peter-Weyl generalizes.

4. Deduce from proposition 9.5.2 that SU(2) has exactly one irreducible representation of each dimension $1, 2, 3, \ldots$, and contrast this with a finite abelian group, whose irreducibles are all one-dimensional. What feature of SU(2), as opposed to S^1, is responsible for the higher-dimensional irreducibles?

5. For the circle S^1, show that the tensor product of the representations χ_m and χ_n of example 9.5.1 is χ_{m+n}, so that the representation ring of S^1 is the group ring of $\mathbb{Z}$. Contrast the addition-of-weights rule here with the Clebsch-Gordan rule of exercise 9.5.1 (2) for SU(2).

6. Identify the adjoint representation of SU(2) on its Lie algebra $\mathfrak{su}(2) \cong \mathbb{R}^3$, complexified to $\mathbb{C}^3$, with one of the V_n. Compute its character on t_θ from the fact that $\mathrm{Ad}(t_\theta)$ acts on $\mathfrak{su}(2)_{\mathbb{C}}$ with eigenvalues $e^{2i\theta}, 1, e^{-2i\theta}$, and match the result to proposition 9.5.3.

10

The Peter-Weyl Theorem

Chapter 9 made every finite-dimensional representation of a compact group unitary and completely reducible, so that the irreducible representations are the indivisible atoms of the theory. The Peter-Weyl theorem is the statement that these atoms account for all of the group's harmonic analysis at once. The matrix coefficients of the irreducible representations, the functions $g \mapsto \langle \varphi(g)v, w \rangle$, are dense in the continuous functions on G and form an orthogonal basis of $L^2(G)$; the Hilbert space $L^2(G)$ decomposes as the completed direct sum, over the irreducible representations, of the blocks they span. It is the exact generalization of the Fourier series on the circle, where the irreducibles are the characters $z \mapsto z^n$ and the theorem is the completeness of the trigonometric system, and it is the analytic result on which all of the character theory to come depends.

The chapter proceeds from the matrix coefficients and the Schur orthogonality relations, which fix how those coefficients pair under the invariant integral, to the theorem itself. The proof applies the spectral theorem for compact self-adjoint operators, cited to [14], to a convolution operator on $L^2(G)$: its eigenspaces are finite-dimensional and translation-stable, and assembling them recovers the decomposition of $L^2(G)$ into isotypic blocks. The characters, the traces of the irreducibles, then form an orthonormal basis of the class functions, so that the multiplicity of an irreducible in any representation is computed by an inner product. Two structural consequences close the chapter: every compact Lie group admits a faithful finite-dimensional representation, hence is a matrix group, and the group can be reconstructed from its category of representations, the Tannaka-Krein duality that the maximal-torus and character theory of the next chapters make fully computable.

10.1 Matrix Coefficients and Representative Functions

The complete reducibility of section 9.3 reduces the representation theory of a compact group to its irreducibles, but it says nothing about how the representations are related to functions on the group itself. The bridge is built by reading off the matrix entries of a representation as the group

element varies: each choice of two vectors in the representation space produces a scalar function on G, and the collection of all such functions, over all finite-dimensional representations, is an algebra of functions rich enough to remember the whole group. This section constructs that algebra, the representative functions $R(G)$, and establishes its closure properties. The Peter-Weyl theorem of section 10.3 will show that $R(G)$ is dense in $C(G)$, so that these functions capture every continuous function on G; the present task is to build $R(G)$ and to see that it is closed under the operations, products, conjugation, and translation, that make the density theorem possible.

Throughout, G is a compact Lie group and every representation (φ, V) is finite-dimensional and, by theorem 9.3.1, equipped with a G-invariant Hermitian inner product $\langle -, - \rangle$, so that each $\varphi(g)$ is unitary. The inner product is conjugate-linear in its second slot.

Definition 10.1.1: Matrix Coefficient

Let (φ, V) be a finite-dimensional unitary representation of the compact Lie group G, and let $v, w \in V$. The associated *matrix coefficient* is the function

$$\varphi_{v,w} \colon G \to \mathbb{C}, \qquad \varphi_{v,w}(g) = \langle \varphi(g)v, w \rangle$$

Fixing an orthonormal basis $e_1, \dots, e_d$ of V (with $d = \dim V$), the *entry functions* of φ are the matrix coefficients

$$\varphi_{ij} \colon G \to \mathbb{C}, \qquad \varphi_{ij}(g) = \langle \varphi(g)e_j, e_i \rangle$$

which are the (i, j) entries of the unitary matrix representing $\varphi(g)$ in that basis.

Each matrix coefficient is a continuous function on G: the map $g \mapsto \varphi(g)$ is continuous into $\mathrm{GL}(V)$ (indeed smooth, since φ is a Lie group homomorphism), and the inner product is continuous, so their composition $g \mapsto \langle \varphi(g)v, w \rangle$ is continuous. Thus every $\varphi_{v,w}$ lies in $C(G)$, the algebra of continuous complex-valued functions on G. The name is literal: once a basis is chosen, $\varphi(g)$ is a $d \times d$ matrix $(\varphi_{ij}(g))_{i,j}$, and the entry functions φ_{ij} record how each matrix entry varies as g ranges over the group. A general matrix coefficient $\varphi_{v,w}$ is a linear combination of these entries: writing $v = \sum_j v_j e_j$ and $w = \sum_i w_i e_i$, linearity of $\varphi(g)$ and sesquilinearity of the inner product give

$$\varphi_{v,w}(g) = \Big\langle \varphi(g) \sum_j v_j e_j, \sum_i w_i e_i \Big\rangle = \sum_{i,j} v_j \overline{w_i}\, \varphi_{ij}(g)$$

so the entry functions φ_{ij} span the same space of functions as the matrix coefficients $\varphi_{v,w}$. A single representation of dimension d thus contributes at most d^2 linearly independent functions to the picture.

The prototype is the circle, where the matrix coefficients are exactly the characters that Fourier analysis already knows.

Example 10.1.2: Matrix Coefficients of the Circle

Let $G = S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$, the circle group. Its irreducible representations were listed in example 9.5.1: for each $n \in \mathbb{Z}$ the one-dimensional representation $\chi_n \colon S^1 \to \mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$ sends $z \mapsto z^n$, and these are all the irreducibles. Because each χ_n acts on the one-dimensional space $V = \mathbb{C}$, a matrix coefficient is determined by two scalars $v, w \in \mathbb{C}$, and with the standard inner product $\langle a, b \rangle = a\bar{b}$ on $\mathbb{C}$,

$$(\chi_n)_{v,w}(z) = \langle \chi_n(z)v, w \rangle = \langle z^n v, w \rangle = v\overline{w}\, z^n$$

The nonzero matrix coefficients of χ_n are therefore the scalar multiples of the single entry function $z \mapsto z^n$, the pure exponential. Ranging over all $n \in \mathbb{Z}$, the matrix coefficients span the trigonometric polynomials $\{\sum_n c_n z^n \mid c_n \in \mathbb{C},\ \text{finitely many nonzero}\}$. Writing $z = e^{i\theta}$, these are the finite Fourier series $\sum_n c_n e^{in\theta}$. The Peter-Weyl assertion that these functions are dense in $C(S^1)$ is the classical statement that trigonometric polynomials approximate continuous periodic functions uniformly, so Peter-Weyl for S^1 is Fourier analysis.

For a compact group the matrix coefficients of every irreducible must be assembled, not just those of a single one, and the totality of what they span is the object of interest.

Definition 10.1.3: Representative Functions

The space $R(G)$ of *representative functions* on the compact Lie group G is the linear span, inside $C(G)$, of all matrix coefficients of all finite-dimensional representations of G:

$$R(G) = \operatorname{span}_{\mathbb{C}} \{\varphi_{v,w} \mid (\varphi, V) \text{ a finite-dimensional representation},\ v, w \in V\}$$

An element of $R(G)$ is thus a finite sum of matrix coefficients, possibly drawn from different representations.

By the remark following definition 10.1.1, one may equally describe $R(G)$ as the span of all entry functions φ_{ij} of all finite-dimensional representations, since the matrix coefficients of a fixed representation and its entry functions span the same subspace of $C(G)$. A representative function is one that comes from a matrix representation: a function on G that can be written down as an entry, or a combination of entries, of some finite-dimensional matrix realization of the group. The circle example shows that for S^1 these are the trigonometric polynomials; the general theory will show that for any compact group $R(G)$ is just as large relative to $C(G)$.

The reason $R(G)$ deserves a name of its own, rather than being a shapeless collection of functions, is that it is closed under all the natural operations one can perform on functions on a group. It is an algebra, it is stable under complex conjugation, and it is invariant under translating the argument on either side. Each of these closure properties comes from a corresponding operation on representations, and tracking that correspondence is the content of the next result.

10.1.4 Convention: Left and Right Translation For $h \in G$ and $f \in C(G)$, the *left translate* $L_h f$ and *right translate* $R_h f$ are the functions $(L_h f)(g) = f(h^{-1}g)$ and $(R_h f)(g) = f(gh)$. The inverse in L_h makes $h \mapsto L_h$ a left action of G on $C(G)$, and $h \mapsto R_h$ a right action. Both send $C(G)$ to $C(G)$, since precomposition with the continuous map $g \mapsto h^{-1}g$ (or $g \mapsto gh$) preserves continuity.

The closure of $R(G)$ under products and conjugation rests on two constructions from the representation theory of the previous chapters: the tensor product of two representations and the dual of a representation. The tensor product $V \otimes W$ carries the representation $(\varphi \otimes \psi)(g) = \varphi(g) \otimes \psi(g)$ (unitary by exercise 9.3.1 (3)), and by theorem 9.3.3 it decomposes into irreducibles. The dual V^* carries the contragredient representation, defined below inside the proof. With these in hand, the closure properties fall out by direct computation.

Proposition 10.1.5: The Algebra of Representative Functions

Let G be a compact Lie group. The space $R(G)$ of representative functions is a unital subalgebra of $C(G)$ with the following properties.

1. *Constants and characters.* $R(G)$ contains the constant functions and the character $\chi_V = \operatorname{tr}\varphi$ of every finite-dimensional representation (φ, V).
2. *Products.* A product of matrix coefficients is a matrix coefficient of the tensor product: $\varphi_{v,w} \cdot \psi_{v',w'} = (\varphi \otimes \psi)_{v\otimes v',\, w\otimes w'}$. Hence $R(G)$ is closed under multiplication.
3. *Conjugation.* The complex conjugate of a matrix coefficient of V is a matrix coefficient of the dual representation V^*. Hence $R(G)$ is closed under complex conjugation.
4. *Translation.* $R(G)$ is invariant under left and right translation: for $h \in G$,
$$L_h\varphi_{v,w} = \varphi_{v,\,\varphi(h)w}, \qquad R_h\varphi_{v,w} = \varphi_{\varphi(h)v,\,w}$$
so each translate of a matrix coefficient of V is again a matrix coefficient of V.

***Proof*:**
The four parts are established in turn, each converting an operation on functions into an operation on representations.

Part (1): Constants and characters. The trivial representation is the one-dimensional representation $\mathbf{1}\colon G \to \mathrm{GL}_1(\mathbb{C})$ with $\mathbf{1}(g) = 1$ for all g. Its matrix coefficients are $\mathbf{1}_{v,w}(g) = \langle 1\cdot v, w\rangle = v\overline{w}$, the constant function with value $v\overline{w}$; letting v, w range over $\mathbb{C}$ produces every constant. So $R(G)$ contains the constants, and in particular the constant 1, which is the unit of $C(G)$; this makes $R(G)$ unital once multiplicative closure is shown in part (2). For characters, fix an orthonormal basis $e_1, \dots, e_d$ of V and use that the trace of the matrix $(\varphi_{ij}(g))$ is the sum of its diagonal entries:
$$\chi_V(g) = \operatorname{tr}\varphi(g) = \sum_{i=1}^{d} \langle \varphi(g)e_i, e_i\rangle = \sum_{i=1}^{d} \varphi_{ii}(g)$$
Each φ_{ii} is a matrix coefficient, so their finite sum χ_V lies in $R(G)$.

Part (2): Products. Let (φ, V) and (ψ, W) be finite-dimensional unitary representations, and give $V \otimes W$ the inner product determined on decomposable tensors by $\langle v\otimes v', w\otimes w'\rangle = \langle v, w\rangle_V \langle v', w'\rangle_W$, for which $\varphi\otimes\psi$ is unitary (exercise 9.3.1 (3)). For $v, w \in V$ and $v', w' \in W$,
$$(\varphi\otimes\psi)_{v\otimes v',\, w\otimes w'}(g) = \langle (\varphi\otimes\psi)(g)(v\otimes v'),\, w\otimes w'\rangle = \langle \varphi(g)v \otimes \psi(g)v',\, w\otimes w'\rangle$$
and by the definition of the tensor inner product this factors as
$$\langle \varphi(g)v, w\rangle_V \, \langle \psi(g)v', w'\rangle_W = \varphi_{v,w}(g)\, \psi_{v',w'}(g)$$
Thus $\varphi_{v,w}\cdot\psi_{v',w'} = (\varphi\otimes\psi)_{v\otimes v',\, w\otimes w'}$ is a matrix coefficient of the representation $V\otimes W$, hence lies in $R(G)$. A product of two arbitrary elements of $R(G)$ is a finite sum of such products, obtained by distributing the multiplication over the two defining sums, so $R(G)$ is closed under multiplication. Together with part (1) this makes $R(G)$ a unital subalgebra of $C(G)$.

Part (3): Conjugation. The dual space $V^* = \operatorname{Hom}_{\mathbb{C}}(V, \mathbb{C})$ carries the *contragredient* representation φ^*, defined by $(\varphi^*(g)\xi)(u) = \xi(\varphi(g^{-1})u)$ for $\xi \in V^*$ and $u \in V$; this is a representation

because $\varphi^*(gh)\xi = \xi \circ \varphi(h^{-1}g^{-1}) = \xi \circ \varphi(h^{-1}) \circ \varphi(g^{-1}) = \varphi^*(g)\varphi^*(h)\xi$. The invariant inner product on V gives a conjugate-linear isomorphism $V \to V^*$ sending $u \mapsto u^\flat = \langle -, u\rangle$; unitarity of φ makes this isomorphism G-equivariant, since for $u \in V$ and $g \in G$,

$$\varphi^*(g)(u^\flat) = \langle \varphi(g^{-1})(-), u\rangle = \langle -, \varphi(g)u\rangle = (\varphi(g)u)^\flat$$

using $\langle \varphi(g^{-1})x, u\rangle = \langle x, \varphi(g)u\rangle$, the invariance of the inner product. Transporting the inner product of V to V^* through this isomorphism makes φ^* unitary, and its matrix coefficients compute as follows. For $v, w \in V$,

$$\overline{\varphi_{v,w}(g)} = \overline{\langle \varphi(g)v, w\rangle} = \langle w, \varphi(g)v\rangle = \langle \varphi(g^{-1})w, v\rangle = (\varphi^*)_{v^\flat,\, w^\flat}(g)$$

The second equality is conjugate symmetry of the inner product, the third is invariance, and the last identifies the result as the matrix coefficient of the contragredient at the vectors $v^\flat, w^\flat \in V^*$: unwinding the definition, $(\varphi^*)_{v^\flat,w^\flat}(g) = \langle \varphi^*(g)v^\flat, w^\flat\rangle_{V^*} = \langle (\varphi(g)v)^\flat, w^\flat\rangle_{V^*} = \langle w, \varphi(g)v\rangle$, using the G-equivariance $\varphi^*(g)v^\flat = (\varphi(g)v)^\flat$ established above and the transport identity $\langle u^\flat, u'^\flat\rangle_{V^*} = \langle u', u\rangle_V$. Hence $\overline{\varphi_{v,w}}$ is a matrix coefficient of V^*, so $R(G)$ is closed under complex conjugation.

Part (4): Translation. For $h, g \in G$ and $v, w \in V$, the right translate is

$$(R_h\varphi_{v,w})(g) = \varphi_{v,w}(gh) = \langle \varphi(gh)v, w\rangle = \langle \varphi(g)\varphi(h)v, w\rangle = \varphi_{\varphi(h)v,\, w}(g)$$

using that φ is a homomorphism, so $\varphi(gh) = \varphi(g)\varphi(h)$. The left translate is

$$(L_h\varphi_{v,w})(g) = \varphi_{v,w}(h^{-1}g) = \langle \varphi(h^{-1}g)v, w\rangle = \langle \varphi(g)v, \varphi(h)w\rangle = \varphi_{v,\, \varphi(h)w}(g)$$

where the third equality moves $\varphi(h^{-1}) = \varphi(h)^{-1}$ across the inner product using unitarity, $\langle \varphi(h)^{-1}\varphi(g)v, w\rangle = \langle \varphi(g)v, \varphi(h)w\rangle$. In both cases the translate is a matrix coefficient of the same representation V at translated vectors, so $R(G)$ is invariant under both left and right translation, completing the proof.

On a compact Hausdorff space such as G, a subalgebra of the continuous complex functions that contains the constants, is closed under complex conjugation and separates points is dense in the uniform norm, by the complex Stone-Weierstrass theorem,[1] and proposition 10.1.5 gives three of these four hypotheses outright: $R(G)$ is a subalgebra (parts 1 and 2), it is conjugation-closed (part 3), and it contains the constants (part 1). The fourth hypothesis, that $R(G)$ separates points, says that the finite-dimensional representations distinguish every pair of group elements, and proposition 10.1.5 does not give it. Theorem 10.3.4(1) gives the density of $R(G)$ in $C(G)$, and its proof uses convolution operators directly. Point-separation follows from that density. When G has a faithful finite-dimensional unitary representation, its matrix coefficients separate points at once, and Stone-Weierstrass gives the density without the convolution operators. The translation invariance of part (4) plays a different role: it makes $R(G)$ a representation of $G \times G$ under left and right translation, and this bi-translation structure is what organizes the Peter-Weyl decomposition of $L^2(G)$ into blocks indexed by the irreducibles.

The translation invariance also carries a finiteness worth isolating, since it is the defining feature of representative functions in the abstract theory: the span of all left and right translates of a single matrix coefficient is finite-dimensional. Part (4) shows $L_h\varphi_{v,w} = \varphi_{v,\varphi(h)w}$ and $R_h\varphi_{v,w} = \varphi_{\varphi(h)v,w}$,

[1] [9, Complex Stone-Weierstrass Theorem].

and as h ranges over G the vectors $\varphi(h)v$ and $\varphi(h)w$ range within the finite-dimensional space V; so every translate of $\varphi_{v,w}$ lies in the finite-dimensional space $\{\varphi_{x,y} \mid x, y \in V\}$ of matrix coefficients of V. A function on G whose two-sided translates span a finite-dimensional space is exactly what one means abstractly by a representative function, and this recovers $R(G)$ without reference to a chosen representation. The circle shows this: the translates of $z \mapsto z^n$ are $z \mapsto (h^{-1}z)^n = h^{-n}z^n$, all scalar multiples of z^n, spanning the one-dimensional space $\mathbb{C}z^n$.

For a nonabelian example, the entry functions of SU(2) exhibit the matrix coefficients as polynomial functions of the matrix entries.

Example 10.1.6: Matrix Coefficients of SU(2)

Write an element of SU(2) as $g = \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix}$ with $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$. The irreducible representations V_n of SU(2), for $n \geq 0$, act on the space of homogeneous polynomials of degree n in two variables x, y, with g acting by substituting the entries of g; this is the symmetric-power representation of proposition 9.5.2, of dimension $n+1$. Take the monomial basis $x^n, x^{n-1}y, \ldots, y^n$ of V_n. The action of g on a basis monomial $x^{n-k}y^k$ is obtained by the linear substitution $x \mapsto ax - \overline{b}y$, $y \mapsto bx + \overline{a}y$ (from the entries of g, following proposition 9.5.2) followed by expansion, so

$$\varphi(g)\big(x^{n-k}y^k\big) = (ax - \overline{b}y)^{n-k}(bx + \overline{a}y)^k$$

Expanding the two binomials and collecting the coefficient of each monomial $x^{n-i}y^i$ expresses the entry function $\varphi_{ij}(g)$ as a polynomial in $a, b, \overline{a}, \overline{b}$, homogeneous of total degree n. For $n = 1$, with V_1 the defining representation on $\mathbb{C}^2 = \operatorname{span}(x, y)$, the entry functions are $a, b, -\overline{b}, \overline{a}$ themselves, the four entries of g. For $n = 2$, with basis x^2, xy, y^2, one entry function is the coefficient of x^2 in $\varphi(g)x^2 = (ax - \overline{b}y)^2 = a^2x^2 - 2a\overline{b}\,xy + \overline{b}^2y^2$, namely $\varphi_{11}(g) = a^2$, and the middle diagonal entry is the coefficient of xy in $\varphi(g)(xy) = (ax - \overline{b}y)(bx + \overline{a}y) = ab\,x^2 + (|a|^2 - |b|^2)xy - \overline{a}\,\overline{b}\,y^2$, namely $|a|^2 - |b|^2$; the last diagonal entry is the coefficient of y^2 in $\varphi(g)y^2 = (bx + \overline{a}y)^2 = b^2x^2 + 2\overline{a}b\,xy + \overline{a}^2y^2$, namely $\overline{a}^2$. Summing the three diagonal entries recovers the character $\chi_{V_2}(g) = a^2 + (|a|^2 - |b|^2) + \overline{a}^2$, in agreement with the character formula of proposition 9.5.3. Every entry function of every V_n is a polynomial in $a, b, \overline{a}, \overline{b}$, so $R(\mathrm{SU}(2))$ consists of the polynomial functions in these four coordinates, restricted to the sphere $|a|^2 + |b|^2 = 1$.

The SU(2) example shows point-separation directly: the coordinate functions a and b are themselves entry functions of the defining representation V_1, and two distinct elements of SU(2) differ in at least one of a, b, so already the matrix coefficients of V_1 separate the points of SU(2). This is the concrete mechanism the Peter-Weyl theorem promotes to arbitrary compact groups: finite-dimensional representations whose entries, taken together, form coordinates fine enough to distinguish group elements, and consequently a $R(G)$ large enough to recover all of $C(G)$ and, through it, the group G itself.

Exercise 10.1.1

1. Let (φ, V) and (φ', V') be isomorphic finite-dimensional unitary representations, and let $T\colon V \to V'$ be a G-equivariant unitary isomorphism. Show that for all $v, w \in V$, $\varphi_{v,w} = \varphi'_{Tv,Tw}$. Deduce that the set of matrix coefficients of a representation, and hence its contribution to $R(G)$, depends only on the isomorphism class of the representation, not on any chosen basis or realization.

2. Let $\chi\colon G \to \mathbb{C}^\times$ be a one-dimensional representation of G (a continuous homomorphism into

the nonzero complex numbers, necessarily landing in the unit circle since G is compact). Determine all matrix coefficients of χ, and show that the entry functions of the tensor powers $\chi^{\otimes n}$ are the powers $g \mapsto \chi(g)^n$. Conclude that the matrix coefficients of a one-dimensional representation and all its tensor powers form the subalgebra of $C(G)$ generated by χ.

3. Let $f = \varphi_{v,w}$ be a matrix coefficient of a d-dimensional representation (φ, V). Prove that the span of all left and right translates $\{L_{h_1} R_{h_2} f \mid h_1, h_2 \in G\}$ is a finite-dimensional subspace of $C(G)$, of dimension at most d^2. Identify this span in terms of the matrix coefficients of V.

4. Suppose the finite-dimensional representations of a compact group G separate points, meaning that for any two distinct $g_1, g_2 \in G$ there is a representation (φ, V) with $\varphi(g_1) \neq \varphi(g_2)$. Show that then the representative functions $R(G)$ separate the points of G: for distinct g_1, g_2 there is $f \in R(G)$ with $f(g_1) \neq f(g_2)$. Explain why the converse also holds, so that "enough representations" and "$R(G)$ separates points" are equivalent conditions.

5. Using proposition 10.1.5, express the product of the two entry functions φ_{11} and φ_{22} of a two-dimensional representation (φ, V) as a single matrix coefficient of $V \otimes V$, naming the two vectors of $V \otimes V$ involved. Then, for the circle representations χ_m, χ_n of example 9.5.1, verify that the product of their matrix coefficients is a matrix coefficient of χ_{m+n}, consistently with $\chi_m \otimes \chi_n = \chi_{m+n}$.

6. Show that a real-valued function $f \in R(G)$ satisfies $f = \overline{f}$, and use proposition 10.1.5(3) to conclude that f is a matrix coefficient of a representation of the form $V \oplus V^*$. Deduce that $R(G)$ is the complexification of the real algebra $R(G) \cap C(G, \mathbb{R})$ of real-valued representative functions.

10.2 Schur Orthogonality for Compact Groups

The matrix coefficients of section 10.1 are functions on G, and G carries a Haar integral. The question of this section is what the integral does to them, and the answer is as rigid as it could be: the matrix coefficients of the irreducible representations are mutually orthogonal in $L^2(G)$, and the ones coming from a single irreducible have a common length determined by nothing but its dimension. This is the compact-group form of Schur orthogonality, the relation that in the finite-group case ([3]) makes the character table an orthonormal object. The proof is the averaging argument of section 9.3 applied once more, now to a linear map between two irreducibles: averaging a map over G forces it to intertwine, and Schur's lemma (corollary 9.3.4) then pins the averaged map to zero or a scalar. Reading off matrix entries turns that dichotomy into the orthogonality relations.

Throughout, V_λ and V_μ are finite-dimensional irreducible unitary representations of the compact Lie group G, with actions $\varphi^\lambda \colon G \to \mathrm{U}(V_\lambda)$ and $\varphi^\mu \colon G \to \mathrm{U}(V_\mu)$ (every finite-dimensional representation is unitary by theorem 9.3.1). Fix orthonormal bases, so that $\varphi^\lambda(g) = (\varphi^\lambda_{ij}(g))$ is a unitary matrix for each g, with matrix coefficients $\varphi^\lambda_{ij}(g) = \langle \varphi^\lambda(g) e_j, e_i \rangle$ (definition 10.1.1), and write $d_\lambda = \dim V_\lambda$. Unitarity is used in one specific form: $\varphi^\lambda(g)^{-1} = \varphi^\lambda(g)^*$, which in entries reads $\varphi^\lambda_{ij}(g^{-1}) = \overline{\varphi^\lambda_{ji}(g)}$. The $L^2(G)$ inner product is $\langle f_1, f_2 \rangle = \int_G f_1(g)\overline{f_2(g)}\, dg$ against normalized Haar measure (theorem 9.2.1).

The engine is a single averaging construction. Given any linear map between the two representation spaces, averaging its conjugates by the group actions produces a map that commutes with both actions, and Schur's lemma classifies such maps completely. Making this precise is the whole proof.

Theorem 10.2.1: Schur Orthogonality Relations

Let V_λ and V_μ be irreducible unitary representations of a compact Lie group G, with matrix coefficients φ^λ_{ij} and φ^μ_{kl} in fixed orthonormal bases, and set $d_\lambda = \dim V_\lambda$.

1. If V_λ and V_μ are inequivalent, then every matrix coefficient of V_λ is orthogonal in $L^2(G)$ to every matrix coefficient of V_μ:
$$\int_G \varphi^\lambda_{ij}(g)\,\overline{\varphi^\mu_{kl}(g)}\,dg = 0$$
for all i, j and all k, l.
2. Within a single irreducible V_λ,
$$\int_G \varphi^\lambda_{ij}(g)\,\overline{\varphi^\lambda_{kl}(g)}\,dg = \frac{1}{d_\lambda}\,\delta_{ik}\,\delta_{jl}$$

***Proof*:**
Let $A\colon V_\mu \to V_\lambda$ be any linear map. Average its conjugates by the two group actions: define
$$B = \int_G \varphi^\lambda(g)\,A\,\varphi^\mu(g)^{-1}\,dg$$
the integral of the operator-valued continuous function $g \mapsto \varphi^\lambda(g)\,A\,\varphi^\mu(g)^{-1}$, taken entrywise against Haar measure. The argument has two steps: first show B intertwines, then read the orthogonality relations off a well-chosen A.

Step 1: B is an intertwiner. Fix $h \in G$. Using that φ^λ and φ^μ are homomorphisms,
$$\varphi^\lambda(h)\,B = \int_G \varphi^\lambda(h)\varphi^\lambda(g)\,A\,\varphi^\mu(g)^{-1}\,dg = \int_G \varphi^\lambda(hg)\,A\,\varphi^\mu(hg)^{-1}\varphi^\mu(h)\,dg$$
where the second equality inserts $\varphi^\mu(g)^{-1} = \varphi^\mu(hg)^{-1}\varphi^\mu(h)$ and combines $\varphi^\lambda(h)\varphi^\lambda(g) = \varphi^\lambda(hg)$. Pull the constant operator $\varphi^\mu(h)$ out to the right of the integral and substitute $g' = hg$; by the left-invariance of the Haar integral (definition 9.2.2), $\int_G F(hg)\,dg = \int_G F(g')\,dg'$, so
$$\varphi^\lambda(h)\,B = \left(\int_G \varphi^\lambda(g')\,A\,\varphi^\mu(g')^{-1}\,dg'\right)\varphi^\mu(h) = B\,\varphi^\mu(h)$$
Thus $\varphi^\lambda(h)\,B = B\,\varphi^\mu(h)$ for every h, so $B\colon V_\mu \to V_\lambda$ is G-equivariant.

Step 2: apply Schur's lemma and read off entries. By corollary 9.3.4, the intertwiner B is constrained by whether the two irreducibles are equivalent.

If V_λ and V_μ are inequivalent, then part (1) of corollary 9.3.4 forces $B = 0$ for every choice of A. Take $A = E_{jl}$, the elementary map sending the basis vector $e_l \in V_\mu$ to $e_j \in V_\lambda$ and all other basis vectors to 0, so that A has matrix entries $(E_{jl})_{pq} = \delta_{pj}\delta_{ql}$. The (i,k) entry of the integrand is
$$\left(\varphi^\lambda(g)\,E_{jl}\,\varphi^\mu(g)^{-1}\right)_{ik} = \sum_{p,q} \varphi^\lambda_{ip}(g)\,(E_{jl})_{pq}\,\varphi^\mu_{qk}(g^{-1}) = \varphi^\lambda_{ij}(g)\,\varphi^\mu_{lk}(g^{-1})$$

Unitarity of φ^μ gives $\varphi^\mu_{lk}(g^{-1}) = \overline{\varphi^\mu_{kl}(g)}$, so the (i,k) entry of B is

$$B_{ik} = \int_G \varphi^\lambda_{ij}(g)\,\overline{\varphi^\mu_{kl}(g)}\,dg$$

and $B = 0$ means every such integral vanishes. This is part (1).

If instead $V_\mu = V_\lambda$ (the same irreducible), then part (2) of corollary 9.3.4 forces $B = c\,\mathrm{id}$ for a scalar c depending on A. Taking the trace of the defining integral and using that the trace is linear and conjugation-invariant, $\mathrm{tr}B = \int_G \mathrm{tr}\big(\varphi^\lambda(g)\,A\,\varphi^\lambda(g)^{-1}\big)\,dg = \int_G \mathrm{tr}(A)\,dg = \mathrm{tr}(A)$, since Haar measure is normalized. On the other hand $\mathrm{tr}B = \mathrm{tr}(c\,\mathrm{id}) = c\,d_\lambda$, so $c = \mathrm{tr}(A)/d_\lambda$ and

$$\int_G \varphi^\lambda(g)\,A\,\varphi^\lambda(g)^{-1}\,dg = \frac{\mathrm{tr}(A)}{d_\lambda}\,\mathrm{id}$$

Now take $A = E_{jl}$ again, for which $\mathrm{tr}(E_{jl}) = \delta_{jl}$. Exactly the entry computation above (with $\mu = \lambda$) gives $B_{ik} = \int_G \varphi^\lambda_{ij}(g)\,\overline{\varphi^\lambda_{kl}(g)}\,dg$, while the scalar form gives $B_{ik} = \frac{\delta_{jl}}{d_\lambda}\,\delta_{ik}$. Equating the two,

$$\int_G \varphi^\lambda_{ij}(g)\,\overline{\varphi^\lambda_{kl}(g)}\,dg = \frac{1}{d_\lambda}\,\delta_{ik}\,\delta_{jl}$$

which is part (2), completing the proof.

The single averaging construction $A \mapsto \int_G \varphi^\lambda(g)\,A\,\varphi^\mu(g)^{-1}\,dg$ carries both statements. It manufactures an intertwiner out of any linear map, and Schur's lemma then leaves only two possibilities: for inequivalent irreducibles the intertwiner must vanish, which is orthogonality across distinct irreducibles; for a single irreducible it must be a scalar, whose value the trace computes, which is the normalized orthogonality inside one irreducible. The pattern of the compact theory recurs here in miniature: an arbitrary object is averaged into an invariant one, and invariance is so restrictive that it determines the object up to a constant. The constant $1/d_\lambda$ is the one new piece of information, and it reappears in Peter-Weyl as the multiplicity with which each irreducible occurs in $L^2(G)$.

The relations say precisely that the functions φ^λ_{ij}, as λ ranges over the inequivalent irreducibles and (i,j) over basis-index pairs, form an *orthogonal* family in $L^2(G)$, each of squared length $1/d_\lambda$. Rescaling repairs the length: the functions $\sqrt{d_\lambda}\,\varphi^\lambda_{ij}$ are an orthonormal family. Whether that family is a basis, that is, whether it is also complete, is the content of the Peter-Weyl theorem in section 10.3; Schur orthogonality gives the orthogonality for free, and the analytic work that remains is completeness.

Summing the diagonal relations over the basis collapses the matrix coefficients to characters, and orthogonality descends to them. Recall from section 10.1 that the character of V_λ is the sum of its diagonal matrix coefficients, $\chi_\lambda = \sum_i \varphi^\lambda_{ii}$; it is the trace $\chi_\lambda(g) = \mathrm{tr}\varphi^\lambda(g)$, a class function independent of the chosen basis.

Corollary 10.2.2: Orthonormality of Irreducible Characters

Let $\{V_\lambda\}$ be a family of pairwise inequivalent irreducible unitary representations of the compact Lie group G, with characters $\chi_\lambda = \mathrm{tr}\varphi^\lambda$. Then the characters are orthonormal in $L^2(G)$:

$$\langle \chi_\lambda, \chi_\mu \rangle = \int_G \chi_\lambda(g)\,\overline{\chi_\mu(g)}\,dg = \delta_{\lambda\mu}$$

***Proof*:**

Expand each character as the sum of its diagonal matrix coefficients, $\chi_\lambda = \sum_i \varphi^\lambda_{ii}$ and $\chi_\mu = \sum_k \varphi^\mu_{kk}$, and use linearity of the integral:

$$\langle \chi_\lambda, \chi_\mu \rangle = \sum_i \sum_k \int_G \varphi^\lambda_{ii}(g)\,\overline{\varphi^\mu_{kk}(g)}\,dg$$

If V_λ and V_μ are inequivalent, every term vanishes by part (1) of theorem 10.2.1, so $\langle \chi_\lambda, \chi_\mu \rangle = 0$. If $\mu = \lambda$, part (2) gives $\int_G \varphi^\lambda_{ii}(g)\,\overline{\varphi^\lambda_{kk}(g)}\,dg = \frac{1}{d_\lambda}\,\delta_{ik}\delta_{ik} = \frac{1}{d_\lambda}\,\delta_{ik}$, so only the $i = k$ terms survive and

$$\langle \chi_\lambda, \chi_\lambda \rangle = \sum_i \frac{1}{d_\lambda} = \frac{d_\lambda}{d_\lambda} = 1$$

Both cases together read $\langle \chi_\lambda, \chi_\mu \rangle = \delta_{\lambda\mu}$, as we sought to show.

This is the compact-group analogue of the first orthogonality relation of finite-group character theory ([3]), with the normalized sum $\frac{1}{|G|}\sum_g$ over a finite group replaced by the Haar integral $\int_G dg$. The diagonal length $1/d_\lambda$ of the individual matrix coefficients is exactly compensated by the d_λ diagonal terms in the character, so the character comes out with unit length regardless of dimension. Characters are the basis-independent part of the matrix coefficients: they forget the basis and remember only the isomorphism class of the representation, and the orthonormality just proved says that inequivalent irreducibles leave distinct, mutually perpendicular traces on $L^2(G)$. The full strength of this observation, that the irreducible characters form an orthonormal *basis* of the space of class functions, waits on Peter-Weyl and is developed in section 10.4; orthonormality is the half available now.

Two worked cases ground the relations. The circle recovers the elementary Fourier orthogonality, and SU(2) recovers a weighted trigonometric identity that the character formula of section 10.1 already put in place.

Example 10.2.3: Fourier Orthogonality on the Circle

Let $G = S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$, the circle group, whose irreducible unitary representations are the one-dimensional characters $\chi_n(z) = z^n$ for $n \in \mathbb{Z}$ (example 9.5.1). Each is one-dimensional, so $d_n = 1$, and the single matrix coefficient of χ_n is χ_n itself. Normalized Haar measure on S^1 is $dz = \frac{1}{2\pi}\,d\theta$ under $z = e^{i\theta}$. The Schur relations (theorem 10.2.1) here read

$$\int_{S^1} z^m\,\overline{z^n}\,dz = \frac{1}{2\pi}\int_0^{2\pi} e^{im\theta}e^{-in\theta}\,d\theta = \frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)\theta}\,d\theta = \delta_{mn}$$

since the integral of $e^{ik\theta}$ over a full period is 2π when $k = 0$ and 0 otherwise. This is the orthogonality of the Fourier characters, and the factor $1/d_n = 1$ matches the fact that the

exponentials $e^{in\theta}$ are already orthonormal without rescaling. Because $\chi_n = \chi_n$ as both character and matrix coefficient, the character orthonormality of corollary 10.2.2 is the same computation. Completeness of this family, that every L^2 function on the circle is its Fourier series, is Peter-Weyl for S^1.

For a nonabelian example the running group SU(2) shows the dimension factor at work and turns the abstract orthogonality into a concrete trigonometric identity.

Example 10.2.4: Character Orthogonality for SU(2)

The irreducible representations of SU(2) are the spaces V_n of homogeneous polynomials of degree n in two variables, with $\dim V_n = n+1$ (proposition 9.5.2). A class function on SU(2) depends only on the conjugacy class, which is determined by an eigenvalue angle $\theta \in [0, \pi]$: every element is conjugate to $\mathrm{diag}(e^{i\theta}, e^{-i\theta})$. The character of V_n evaluated on this class is

$$\chi_n(\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}$$

the formula established in proposition 9.5.3. The Weyl integration formula for SU(2), proved in the next chapter, expresses the Haar integral of a class function f as

$$\int_{\mathrm{SU}(2)} f \, dg = \frac{2}{\pi} \int_0^{\pi} f(\theta) \, \sin^2\theta \, d\theta$$

the weight $\sin^2\theta$ being the density of the conjugacy classes. Against this weight the orthonormality of corollary 10.2.2 becomes a trigonometric identity. For $\langle \chi_m, \chi_n \rangle$ (all characters are real here, so the conjugate is inert),

$$\langle \chi_m, \chi_n \rangle = \frac{2}{\pi} \int_0^{\pi} \frac{\sin((m+1)\theta)}{\sin\theta} \, \frac{\sin((n+1)\theta)}{\sin\theta} \, \sin^2\theta \, d\theta = \frac{2}{\pi} \int_0^{\pi} \sin((m+1)\theta) \sin((n+1)\theta) \, d\theta$$

The $\sin^2\theta$ from the measure cancels the two $\sin\theta$ denominators, leaving a plain product of sines. The product-to-sum identity $\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b))$ turns the integrand into $\frac{1}{2}\big(\cos((m-n)\theta) - \cos((m+n+2)\theta)\big)$, and integrating $\cos(k\theta)$ over $[0, \pi]$ gives π when $k = 0$ and 0 for any nonzero integer k. Since $m + n + 2 \geq 2 > 0$, the second cosine always integrates to 0, and the first contributes π exactly when $m = n$. Hence

$$\langle \chi_m, \chi_n \rangle = \frac{2}{\pi} \cdot \frac{\pi}{2} \, \delta_{mn} = \delta_{mn}$$

confirming corollary 10.2.2 directly. The $\sin^2\theta$ weight is precisely what is needed to convert the ratios of sines into orthogonal trigonometric functions; this interplay between the character formula and the integration weight is the mechanism the Weyl character formula generalizes to every compact group.

Exercise 10.2.1

1. Deduce from theorem 10.2.1 that the rescaled matrix coefficients $\sqrt{d_\lambda}\, \varphi^\lambda_{ij}$, as λ ranges over the inequivalent irreducibles and (i, j) over pairs of basis indices, form an orthonormal family in $L^2(G)$. Explain why this is an orthonormal *system* but not obviously an orthonormal *basis*,

and name the theorem whose task is to close that gap.

2. Let V be a finite-dimensional unitary representation of a compact group G with character $\chi = \chi_V$. Using complete reducibility (theorem 9.3.3), write $V \cong \bigoplus_\lambda V_\lambda^{\oplus m_\lambda}$ with multiplicities $m_\lambda \geq 0$, and show that $\langle \chi, \chi \rangle = \sum_\lambda m_\lambda^2$. Conclude that V is irreducible if and only if $\langle \chi, \chi \rangle = 1$.

3. Show that the multiplicity of the trivial representation in a finite-dimensional representation V equals $\int_G \chi_V(g)\, dg$, and interpret this integral as the dimension of the space V^G of G-invariant vectors. Verify the two agree using the averaging projection $P = \int_G \varphi(g)\, dg$ onto V^G.

4. For an irreducible V_λ of dimension d_λ and a representation (φ, V), define the operator $P_\lambda = d_\lambda \int_G \overline{\chi_\lambda(g)}\, \varphi(g)\, dg$ on V. Show that P_λ is G-equivariant and that on an irreducible summand $V_\mu \subseteq V$ it acts as the identity when $\mu = \lambda$ and as 0 when $\mu \neq \lambda$. Conclude that P_λ is the projection onto the λ-isotypic component of V.

5. Using the character formula $\chi_n(\theta) = \sin((n+1)\theta)/\sin\theta$ for SU(2) (proposition 9.5.3) and the orthogonality of corollary 10.2.2, compute $\langle \chi_1 \chi_1, \chi_n \rangle$ for each n, and thereby decompose the tensor product $V_1 \otimes V_1$ into irreducibles. Confirm the answer is the Clebsch-Gordan decomposition $V_1 \otimes V_1 \cong V_2 \oplus V_0$.

6. Explain where the proof of theorem 10.2.1 uses that the representations are over $\mathbb{C}$ rather than $\mathbb{R}$, by pointing to the step in corollary 9.3.4 that requires an eigenvalue in the field. Give the two-dimensional rotation representation of SO(2) over $\mathbb{R}$ as an example whose real matrix coefficients do not satisfy the scalar-intertwiner conclusion, and note what complexifying repairs.

10.3 The Peter–Weyl Theorem

Schur orthogonality (theorem 10.2.1) showed that the normalized matrix coefficients $\sqrt{d_\lambda}\, \varphi_{ij}^\lambda$ of the irreducible representations form an orthonormal *system* in $L^2(G)$. Orthonormality is half of a basis; what remains is completeness, the statement that these functions span a dense subspace, so that every L^2 function is approximated arbitrarily well by finite combinations of matrix coefficients. This is the Peter–Weyl theorem, and its proof is where the analysis of the chapter concentrates. Schur orthogonality was linear algebra averaged over G, but completeness requires an infinite-dimensional input, the spectral theorem for compact self-adjoint operators, applied to a convolution operator built from the group. The output is a complete accounting of $L^2(G)$ as a representation of G: it is the Hilbert-space direct sum of one copy of $\mathrm{End}(V_\lambda)$ for each irreducible λ, and the group acts on the λ-block by left and right translation on the two tensor factors of $V_\lambda \otimes V_\lambda^*$.

Two facts are cited rather than proved, both from the Hilbert-space theory of a prerequisite. The first is that $L^2(G)$, formed against the normalized Haar measure of theorem 9.2.1, is a Hilbert space in which $C(G)$ is a dense subspace ([14, L^p is a Banach Space, Hilbert Spaces], and the density is measure theory: [15, $C_c(X)$ Dense in L^p], with $C_c = C$ on the compact G). The second is the spectral theorem: a compact self-adjoint operator on a Hilbert space has an orthonormal eigenbasis, its eigenvalues are real and accumulate only at 0, and each nonzero eigenvalue has a finite-dimensional eigenspace ([14, Normal Compact, Then Diagonalizable, Spectrum of a Self-Adjoint Operator]). The group-theoretic content, that the eigenspaces are stable under translation and therefore consist of matrix coefficients, and that varying the operator sweeps out all of $L^2(G)$, is proved here in full.

The device that couples the group to the operator theory is convolution. For $f, k \in C(G)$ the *convolution* $f * k$ is the function

$$(f * k)(x) = \int_G f(xg^{-1})\, k(g)\, dg = \int_G f(g)\, k(g^{-1}x)\, dg \tag{10.1}$$

the second form obtained by the substitution $g \mapsto x^{-1}g$ and the invariance of Haar measure (definition 9.2.2). Convolution is an operation $C(G) \times C(G) \to C(G)$: the integrand is continuous in g for each fixed x, and continuity in x follows because f is uniformly continuous on the compact group G, so $x \mapsto f(xg^{-1})$ varies uniformly with x. Fixing k and letting f vary defines the *convolution operator*

$$R_k \colon C(G) \to C(G), \qquad R_k f = f * k \tag{10.2}$$

whose spectral theory will produce the matrix coefficients. The one algebraic property that makes it useful is that it commutes with left translation.

Proposition 10.3.1: The Convolution Operator

Let G be a compact Lie group and $k \in C(G)$. The convolution operator $R_k f = f * k$ of (10.2) has the following properties.

1. R_k commutes with left translation: writing $(L_h f)(x) = f(h^{-1}x)$, one has $R_k(L_h f) = L_h(R_k f)$ for every $h \in G$.
2. R_k is bounded on $L^2(G)$, with $\|R_k f\|_2 \le \|k\|_1 \, \|f\|_2$, and extends to a bounded operator on the Hilbert space $L^2(G)$.
3. If k satisfies $k(g) = \overline{k(g^{-1})}$, then R_k is self-adjoint on $L^2(G)$.
4. R_k is a compact operator on $L^2(G)$.

***Proof*:**

1. *Commuting with left translation.* Using the first form of (10.1),

$$\begin{aligned} \big(R_k(L_h f)\big)(x) &= \int_G (L_h f)(xg^{-1})\, k(g)\, dg \\ &= \int_G f(h^{-1}xg^{-1})\, k(g)\, dg \\ &= (R_k f)(h^{-1}x) \\ &= \big(L_h(R_k f)\big)(x) \end{aligned}$$

   The convolution variable g is untouched by the left translation, which acts only on the argument x; this is the point of convolving on the right. Right translation ($x \mapsto xh$) does not pass through, because it reaches into the kernel argument.

2. *Boundedness.* For $f \in C(G)$, the second form of (10.1) and the Cauchy-Schwarz inequality give, for each x,

$$|(f * k)(x)|^2 = \left| \int_G f(g)\, k(g^{-1}x)\, dg \right|^2 \le \Big(\int_G |k(g^{-1}x)|\, dg \Big) \Big(\int_G |f(g)|^2 |k(g^{-1}x)|\, dg \Big)$$

The first factor equals $\|k\|_1$ by invariance of Haar measure. Integrating over x and interchanging the order of integration, using $\int_G |k(g^{-1}x)|\,dx = \|k\|_1$ for each fixed g, gives $\|f*k\|_2^2 \le \|k\|_1^2\,\|f\|_2^2$. Since $C(G)$ is dense in $L^2(G)$, R_k extends by continuity to a bounded operator on $L^2(G)$.

3. *Self-adjointness.* For $f_1, f_2 \in C(G)$, unwind the inner product and substitute $y = xg^{-1}$ to move the kernel across:

$$\langle R_k f_1, f_2\rangle = \int_G \int_G f_1(xg^{-1})\,k(g)\,\overline{f_2(x)}\,dg\,dx = \int_G \int_G f_1(y)\,k(g)\,\overline{f_2(yg)}\,dg\,dy$$

Replacing g by g^{-1} and using $k(g^{-1}) = \overline{k(g)}$ turns this into $\int_G \int_G f_1(y)\,\overline{k(g)\,f_2(yg^{-1})}\,dg\,dy = \langle f_1, R_k f_2\rangle$, since $(R_k f_2)(y) = \int_G f_2(yg^{-1})\,k(g)\,dg$. Hence $R_k = R_k^*$.

4. *Compactness.* The operator R_k is an integral operator with continuous kernel: writing $K(x,g) = k(g^{-1}x)$, one has $(R_k f)(x) = \int_G f(g)\,K(x,g)\,dg$, and K is continuous on the compact space $G \times G$, hence bounded and square-integrable there. An integral operator on the L^2 space of a finite measure space whose kernel lies in $L^2(G \times G)$ is Hilbert-Schmidt, and a Hilbert-Schmidt operator is compact ([14, Hilbert-Schmidt Operators As Integral Operators, Hilbert-Schmidt Operators Are Compact]). Therefore R_k is compact, completing the proof.

The four properties divide the labor cleanly. Commutation with left translation (part 1) is the group theory: it forces each eigenspace of R_k to be a left-translation-invariant subspace, and a finite-dimensional such subspace, as the next lemma shows, is spanned by matrix coefficients. Self-adjointness and compactness (parts 3 and 4) are the analysis: they permit the spectral theorem, which gives an orthonormal eigenbasis with finite-dimensional nonzero eigenspaces. Boundedness (part 2) is the technical glue letting the operator, defined on continuous functions, act on the Hilbert space $L^2(G)$. A single well chosen kernel k produces one finite bundle of matrix coefficients; the theorem comes from letting k range over an approximate identity, so that these bundles exhaust everything.

The bridge from analysis back to representation theory is the observation that a finite-dimensional space of continuous functions stable under translation is made of matrix coefficients. This converts the abstract eigenspaces of R_k into the concrete functions φ_{ij}^{λ} of definition 10.1.1.

Lemma 10.3.2: Translation-Invariant Function Spaces

Let $W \subseteq C(G)$ be a finite-dimensional subspace stable under left translation, that is, $L_h f \in W$ for every $f \in W$ and $h \in G$. Then every $f \in W$ is a representative function: $W \subseteq R(G)$.

***Proof*:**
Left translation makes W a finite-dimensional representation of G. Define $\rho\colon G \to \mathrm{GL}(W)$ by $\rho(h) = L_h$, so $(\rho(h)f)(x) = f(h^{-1}x)$; this is a homomorphism because $L_{h_1}L_{h_2} = L_{h_1h_2}$, and it is continuous because $(h,f) \mapsto L_h f$ is jointly continuous by the uniform continuity of functions in W on the compact group G. Thus (ρ, W) is a finite-dimensional continuous representation of G.

Fix a basis $f_1, \dots, f_m$ of W and write $\rho(h)f_j = \sum_i \rho_{ij}(h) f_i$, so the matrix entries $\rho_{ij}\colon G \to \mathbb{C}$ are continuous. Evaluation at the identity is a linear functional $\varepsilon\colon W \to \mathbb{C}$, $\varepsilon(f) = f(e)$. For

$f \in W$,

$$f(x) = (L_{x^{-1}} f)(e) = \varepsilon\big(\rho(x^{-1})f\big)$$

because $(L_{x^{-1}} f)(e) = f(x)$. Expanding $f = \sum_j c_j f_j$ and using $\rho(x^{-1})f_j = \sum_i \rho_{ij}(x^{-1})f_i$,

$$f(x) = \sum_{i,j} c_j\, \rho_{ij}(x^{-1})\, \varepsilon(f_i) = \sum_{i,j} c_j\, f_i(e)\, \rho_{ij}(x^{-1})$$

This exhibits f as a finite linear combination of the functions $x \mapsto \rho_{ij}(x^{-1})$, which are the matrix coefficients of the contragredient representation $h \mapsto \rho(h^{-1})^{\mathsf{T}}$ in the sense of definition 10.1.1. Hence $f \in R(G)$, and since $f \in W$ was arbitrary, $W \subseteq R(G)$, completing the proof.

Any device that manufactures finite-dimensional translation-invariant subspaces of $C(G)$ automatically manufactures matrix coefficients, and the convolution operator R_k is such a device: its nonzero eigenspaces are finite-dimensional (compactness) and left-translation-invariant (commutation), so each is a space of representative functions. The remaining question is whether *enough* matrix coefficients arise this way to be dense, and the answer comes from choosing the kernels k well.

The kernels are chosen to form an approximate identity: a family concentrating its mass near the identity element, so that convolving with k perturbs a function by an arbitrarily small amount. Such a family exists on any compact group.

Lemma 10.3.3: Approximate Identities on a Compact Group

Let G be a compact Lie group. For every neighborhood U of the identity e there is a continuous function $k\colon G \to \mathbb{R}$ with $k \geq 0$, $k(g) = k(g^{-1})$, $\int_G k(g)\, dg = 1$, and k vanishing outside U. For such a k and any $f \in C(G)$,

$$\|f * k - f\|_\infty \leq \sup_{g \in U} \sup_{x \in G} |f(xg) - f(x)| \tag{10.3}$$

and the right side tends to 0 as U shrinks to $\{e\}$, so $f * k \to f$ uniformly.

***Proof*:**

Existence. Choose a symmetric neighborhood $V = V^{-1}$ of e with $V \subseteq U$, arranged by replacing V with $V \cap V^{-1}$. Pick a continuous bump function $b\colon G \to [0, \infty)$ with $b(e) > 0$ and $b = 0$ outside V (a coordinate bump on a chart around e, extended by zero, using that G is a manifold). Symmetrize by $b_0(g) = b(g) + b(g^{-1})$, which is continuous, nonnegative, symmetric, positive at e, and supported in $V = V^{-1} \subseteq U$. Its integral $c = \int_G b_0\, dg$ is positive because b_0 is a nonnegative continuous function positive near e and Haar measure has full support. Then $k = b_0/c$ has all four listed properties.

The estimate. Using $\int_G k = 1$ and the first form of (10.1),

$$(f * k)(x) - f(x) = \int_G \big(f(xg^{-1}) - f(x)\big)\, k(g)\, dg$$

The integrand vanishes unless $g \in U$, where k is supported. There, writing $y = xg^{-1}$ gives $f(xg^{-1}) - f(x) = f(y) - f(yg)$, a difference of f against its right translate by g, so

$$|f(xg^{-1}) - f(x)| = |f(y) - f(yg)| \leq \sup_{z \in G} |f(z) - f(zg)|$$

uniformly in x (with $z = y$ ranging over G). Since $k \geq 0$ has total mass 1, pulling this bound out of the integral yields (10.3). Finally f is uniformly continuous on the compact group, so $\sup_{x \in G} |f(xg) - f(x)| \to 0$ as $g \to e$; hence the right side of (10.3) tends to 0 as $U \to \{e\}$, and $f * k \to f$ uniformly, completing the proof.

An approximate identity acts like a delta function that convolution can integrate against: convolving with k replaces f by an average of its translates over the small neighborhood U, and as U shrinks this average returns f. The uniform estimate (10.3) is what upgrades the L^2 density proved next into density in the uniform norm on $C(G)$, the stronger of the two conclusions of the theorem.

Everything is now assembled: convolution operators are compact and self-adjoint, their eigenspaces are matrix coefficients, and an approximate identity guarantees these operators together see every function. The Peter–Weyl theorem records the consequences.

Theorem 10.3.4: The Peter–Weyl Theorem

Let G be a compact Lie group with normalized Haar measure, and let $\widehat{G}$ index the isomorphism classes of irreducible unitary representations V_λ, of dimensions d_λ, with matrix coefficients φ_{ij}^{λ} in orthonormal bases. Then:

1. The algebra $R(G)$ of representative functions is dense in $C(G)$ for the uniform norm, and dense in $L^2(G)$.
2. The normalized matrix coefficients $\left\{ \sqrt{d_\lambda}\, \varphi_{ij}^{\lambda} : \lambda \in \widehat{G},\ 1 \leq i, j \leq d_\lambda \right\}$ form an orthonormal basis of $L^2(G)$.
3. As a unitary representation of $G \times G$ under left and right translation, $L^2(G)$ decomposes as the Hilbert-space direct sum

$$L^2(G) = \widehat{\bigoplus_{\lambda \in \widehat{G}}} \operatorname{End}(V_\lambda) \ \cong \ \widehat{\bigoplus_{\lambda \in \widehat{G}}} V_\lambda \otimes V_\lambda^*$$

the λ-summand being the span of $\{\varphi_{ij}^{\lambda}\}$, on which $G \times G$ acts by right translation on the first tensor factor and left translation on the second.

***Proof*:**

The three statements are proved in the order (2), then (1), then (3), with the completeness of the matrix coefficients as the central step.

Step 1: the matrix coefficients span a dense subspace of $L^2(G)$. Let $\mathcal{M} \subseteq L^2(G)$ be the closed linear span of all matrix coefficients, that is, the L^2-closure of $R(G)$. The goal is $\mathcal{M} = L^2(G)$; suppose instead $\mathcal{M} \neq L^2(G)$, so its orthogonal complement $\mathcal{M}^{\perp}$ contains a nonzero function f_0. The strategy is to produce from f_0 a nonzero matrix coefficient lying inside $\mathcal{M}^{\perp}$, which contradicts $\mathcal{M}^{\perp} \perp \mathcal{M}$.

By lemma 10.3.3 choose an approximate identity k, real and symmetric so that $k = \overline{k(\cdot^{-1})}$ and the hypothesis of proposition 10.3.1(3) holds, with U small enough that $\|R_k f_0 - f_0\|_2 < \|f_0\|_2$. For continuous f_0 this follows from (10.3); for a general $f_0 \in L^2$ first approximate it in L^2 by a continuous function. Then $R_k f_0 \neq 0$, since $R_k f_0 = 0$ would force $\|f_0\|_2 = \|R_k f_0 - f_0\|_2 < \|f_0\|_2$.

The operator R_k is compact and self-adjoint on $L^2(G)$ (proposition 10.3.1), so by the spectral theorem ([14, Normal Compact, Then Diagonalizable]) it has an orthonormal eigenbasis, real eigenvalues accumulating only at 0, and a finite-dimensional eigenspace for each nonzero eigenvalue. It preserves $\mathcal{M}^\perp$: for $f \in \mathcal{M}^\perp$ and a matrix coefficient $\psi = \varphi^\lambda_{ij}$, self-adjointness gives $\langle R_k f, \psi\rangle = \langle f, R_k\psi\rangle$, and $R_k\psi = \psi * k \in R(G) \subseteq \mathcal{M}$ because $(\psi * k)(x) = \int_G \varphi^\lambda_{ij}(xg^{-1})\,k(g)\,dg = \sum_m \varphi^\lambda_{im}(x) \int_G \varphi^\lambda_{mj}(g^{-1})\,k(g)\,dg$ is a finite combination of the matrix coefficients φ^λ_{im} of the same representation, so $\langle f, R_k\psi\rangle = 0$; hence $R_k f \in \mathcal{M}^\perp$. The restriction $R_k|_{\mathcal{M}^\perp}$ is again compact and self-adjoint, and it is nonzero since $R_k f_0 \neq 0$. Therefore it has a nonzero eigenvalue μ with a finite-dimensional eigenspace $E_\mu \subseteq \mathcal{M}^\perp$.

This eigenspace is left-translation-invariant: if $R_k f = \mu f$ then $R_k(L_h f) = L_h(R_k f) = \mu\, L_h f$ by proposition 10.3.1(1), so $L_h f \in E_\mu$. Every element of E_μ is continuous, being $\mu^{-1}R_k f = \mu^{-1}(f * k)$, a convolution. By lemma 10.3.2, the finite-dimensional left-translation-invariant space E_μ of continuous functions satisfies $E_\mu \subseteq R(G)$: it consists of matrix coefficients. But $E_\mu \subseteq \mathcal{M}^\perp$ is orthogonal to all matrix coefficients, forcing $E_\mu = 0$, contradicting $\mu \neq 0$. Hence $\mathcal{M}^\perp = 0$ and $\mathcal{M} = L^2(G)$: the matrix coefficients are dense in $L^2(G)$.

Step 2: statement (2), the orthonormal basis. By Schur orthogonality (theorem 10.2.1) the functions $\sqrt{d_\lambda}\,\varphi^\lambda_{ij}$ are orthonormal in $L^2(G)$: coefficients of inequivalent irreducibles are orthogonal, and within V_λ one has $\int_G \varphi^\lambda_{ij}\,\overline{\varphi^\lambda_{kl}}\,dg = \frac{1}{d_\lambda}\delta_{ik}\delta_{jl}$. Their span is exactly $R(G)$: every representative function is a finite combination of matrix coefficients, and complete reducibility (theorem 9.3.3) writes any matrix coefficient of any finite-dimensional representation in terms of those of the irreducibles. By Step 1 this span is dense in $L^2(G)$. An orthonormal system with dense span is an orthonormal basis ([14, Orthonormal Basis Condition]), which is statement (2).

Step 3: statement (1), density in $C(G)$. Density of $R(G)$ in $L^2(G)$ is Step 1. For the uniform norm, let $f \in C(G)$ and $\varepsilon > 0$. Choose an approximate identity k with $\|f * k - f\|_\infty < \varepsilon/2$ by lemma 10.3.3, and approximate f in L^2 by a representative function $p \in R(G)$. Then

$$\|R_k f - R_k p\|_\infty = \|(f-p) * k\|_\infty \leq \|f - p\|_2\,\|k\|_2$$

by Cauchy-Schwarz applied to $\big((f-p)*k\big)(x) = \int_G (f-p)(xg^{-1})k(g)\,dg$. Choosing $\|f-p\|_2 < \varepsilon/(2\|k\|_2)$ makes this at most $\varepsilon/2$. Since $R_k p = p * k \in R(G)$ ($R(G)$ is closed under R_k, being left-translation-invariant), the representative function $R_k p$ satisfies

$$\|f - R_k p\|_\infty \leq \|f - R_k f\|_\infty + \|R_k f - R_k p\|_\infty < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon$$

so $R(G)$ is dense in $C(G)$ for the uniform norm, which is statement (1).

Step 4: statement (3), the $G \times G$ decomposition. Group the orthonormal basis of Step 2 by λ. For fixed λ, the d_λ^2 functions $\{\varphi^\lambda_{ij}\}$ span a subspace $M_\lambda \subseteq L^2(G)$ of dimension d_λ^2. The group $G \times G$ acts on $L^2(G)$ by $\big((a,b)\cdot f\big)(x) = f(a^{-1}xb)$, combining left and right translation. On $\varphi^\lambda_{ij}(x) = \langle \rho_\lambda(x)e_j, e_i\rangle$,

$$\big((a,b)\cdot\varphi^\lambda_{ij}\big)(x) = \langle \rho_\lambda(a^{-1}xb)e_j, e_i\rangle = \sum_{p,q} \overline{\rho_\lambda(a)_{pi}}\;\rho_\lambda(b)_{qj}\;\varphi^\lambda_{pq}(x)$$

using $\rho_\lambda(a^{-1}) = \rho_\lambda(a)^*$ for the unitary ρ_λ. Thus M_λ is stable under $G\times G$, the action on the row index i is the contragredient representation ρ_λ^*, and the action on the column index j is ρ_λ; the identification $\varphi^\lambda_{ij} \leftrightarrow e_j \otimes e_i^*$ is an isomorphism of $G\times G$-representations $M_\lambda \cong V_\lambda \otimes V_\lambda^* \cong \mathrm{End}(V_\lambda)$. The M_λ are mutually orthogonal by Schur orthogonality and their union is a basis by Step 2, so $L^2(G)$ is the Hilbert-space direct sum of the M_λ, which is statement (3), completing the proof.

The single function space $L^2(G)$ contains a complete record of the representation theory of G. Reading the decomposition through left translation, each irreducible appears inside the left-regular representation with multiplicity exactly its dimension. On the λ-block, left translation acts through the second tensor factor V_λ^*, so $\mathrm{End}(V_\lambda) \cong (V_\lambda^*)^{\oplus d_\lambda}$ as a left G-module (the d_λ columns of a matrix each carry a copy of V_λ^*). As λ ranges over $\widehat{G}$ so does λ^*, and $\dim V_\lambda^* = d_\lambda$, so the multiplicities are unchanged by the dualization. So the regular representation of a compact group sees every irreducible, and sees each one as many times as its dimension. This is the infinite-dimensional completion of the isotypic picture from theorem 9.3.3: there a finite-dimensional representation broke into isotypic blocks with finite multiplicities; here the regular representation breaks into all isotypic blocks at once, with the multiplicity of V_λ equal to $\dim V_\lambda$.

The name Peter–Weyl attaches to the 1927 paper of Fritz Peter and Hermann Weyl that first proved density of representative functions for a general compact group; the convolution-operator proof given here, isolating the compact self-adjoint operator and reading off its eigenspaces, is the modern streamlining. The role of compactness is total: it gives the finite Haar measure that makes $L^2(G)$ and the averaging available, it makes the integral kernel continuous on a compact product and hence Hilbert-Schmidt, and it lets the approximate identity be a genuine element of $C(G)$. On a noncompact group none of this survives intact, the irreducibles are typically infinite-dimensional, and the discrete index $\widehat{G}$ becomes a continuous parameter, the beginning of a different theory.

The most concrete instance is the circle, where Peter–Weyl is exactly the completeness of Fourier series.

Example 10.3.5: Peter–Weyl for the Circle is Fourier Series

Take $G = S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$, written $z = e^{i\theta}$, with normalized Haar measure $\frac{1}{2\pi}\,d\theta$. Its irreducible representations are all one-dimensional, indexed by $n \in \mathbb{Z}$, with $\rho_n(e^{i\theta}) = e^{in\theta}$ on $\mathbb{C}$ (example 9.5.1); here $d_n = 1$ for every n, and the single matrix coefficient of ρ_n is the character $\chi_n(\theta) = e^{in\theta}$. Peter–Weyl asserts that $\{\sqrt{d_n}\, e^{in\theta}\}_{n\in\mathbb{Z}} = \{e^{in\theta}\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $L^2(S^1)$, which is the classical theorem that the exponentials $e^{in\theta}$ form a complete orthonormal system, so every $f \in L^2(S^1)$ has a convergent Fourier expansion $f = \sum_{n\in\mathbb{Z}} \hat{f}(n)\, e^{in\theta}$ with $\hat{f}(n) = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta$. The decomposition of statement (3) reads $L^2(S^1) = \widehat{\bigoplus}_{n\in\mathbb{Z}}\mathbb{C}$, one line per frequency, and the $S^1 \times S^1$ action on the n-th line is $(a,b)\cdot e^{in\theta} = a^{-n}b^n e^{in\theta}$, the outer tensor $\rho_n^* \otimes \rho_n$ on the one-dimensional $\mathrm{End}(\mathbb{C})$. Every statement of the theorem is a fact about Fourier series that the compact abelian case makes visible; the general theorem is what remains of Fourier analysis when the group is nonabelian and the frequencies $e^{in\theta}$ are replaced by matrix coefficients φ_{ij}^{λ}.

For SU(2) the same theorem is the completeness of the matrix coefficients of the irreducible representations V_m of proposition 9.5.2. There $\widehat{G} = \{V_m : m \geq 0\}$ with $d_m = \dim V_m = m+1$, so $L^2(\mathrm{SU}(2)) = \widehat{\bigoplus}_{m\geq 0}\mathrm{End}(V_m)$, and the normalized coefficients $\sqrt{m+1}\,\varphi_{ij}^{V_m}$ form an orthonormal basis. The multiplicity of V_m in the regular representation is $m+1$, its own dimension, and this decomposition is the decomposition underlying harmonic analysis on the 3-sphere $\mathrm{SU}(2) \cong S^3$.

The decomposition of the regular representation gives, for any representation, a canonical way to isolate the part built from a single irreducible. The λ-isotypic component collects all the copies of V_λ, and there is an explicit projection onto it computed by averaging against the character χ_λ.

Definition 10.3.6: Isotypic Component and Its Projection

Let (φ, V) be a finite-dimensional representation of a compact Lie group G, and let $\lambda \in \widehat{G}$ be an irreducible with character χ_λ and dimension d_λ. The *λ-isotypic component* $V[\lambda] \subseteq V$ is the sum of all subrepresentations of V isomorphic to V_λ; equivalently, the image of the natural map $V_\lambda \otimes \mathrm{Hom}_G(V_\lambda, V) \to V$. The *isotypic projection* $P_\lambda \colon V \to V$ is

$$P_\lambda = d_\lambda \int_G \overline{\chi_\lambda(g)}\, \varphi(g)\, dg \tag{10.4}$$

the operator averaging φ against $d_\lambda \overline{\chi_\lambda}$.

The projection formula (10.4) is the averaging device of proposition 9.2.4 tuned to a single irreducible. Averaging $\varphi(g)$ against the constant function 1 projects onto the invariant vectors, the trivial-isotypic component; weighting instead by $d_\lambda \overline{\chi_\lambda}$ projects onto the V_λ-isotypic component. That P_λ is the claimed projection is a short Schur-orthogonality computation, carried out in the exercises.

Example 10.3.7: The Isotypic Projection on a Sum of Irreducibles

Let $V = V_\lambda \oplus V_\mu$ be a direct sum of two inequivalent irreducibles of a compact group G, with $\lambda \neq \mu$. Compute P_λ from (10.4). On the summand V_λ, where the representation is ρ_λ, the matrix entries of P_λ are

$$d_\lambda \int_G \overline{\chi_\lambda(g)}\, \varphi^\lambda_{ij}(g)\, dg = d_\lambda \sum_k \int_G \overline{\varphi^\lambda_{kk}(g)}\, \varphi^\lambda_{ij}(g)\, dg = d_\lambda \sum_k \tfrac{1}{d_\lambda} \delta_{ik}\delta_{jk} = \delta_{ij}$$

by theorem 10.2.1, since $\chi_\lambda = \sum_k \varphi^\lambda_{kk}$; hence $P_\lambda|_{V_\lambda} = \mathrm{id}_{V_\lambda}$. On the summand V_μ, every entry is $d_\lambda \int_G \overline{\chi_\lambda}\, \varphi^\mu_{ij}\, dg = 0$ because coefficients of inequivalent irreducibles are orthogonal (theorem 10.2.1), so $P_\lambda|_{V_\mu} = 0$. Therefore P_λ is the projection onto $V_\lambda = V[\lambda]$ along V_μ, as the definition claims.

The isotypic components refine the decomposition into irreducibles of theorem 9.3.3 into a canonical one: the individual irreducible summands W_i in $V = W_1 \oplus \cdots \oplus W_r$ are not unique, since an isotypic block of multiplicity greater than one can be split into lines in many ways, but the isotypic component $V[\lambda]$ is intrinsic, being the sum of *all* copies of V_λ, and $V = \bigoplus_\lambda V[\lambda]$ is the canonical decomposition. The projections P_λ pick these blocks out, and computing them requires nothing but the character χ_λ. This is the mechanism the next section develops: characters detect everything about a representation, and Peter–Weyl is what guarantees they detect enough.

Exercise 10.3.1

1. Verify in detail that Peter–Weyl for $G = S^1$ is the completeness of the Fourier system. Using example 9.5.1 that the irreducibles of S^1 are the characters $\chi_n(e^{i\theta}) = e^{in\theta}$ for $n \in \mathbb{Z}$, each one-dimensional, write out what statements (1), (2), and (3) of theorem 10.3.4 say and identify each with a classical statement about Fourier series on the circle.

2. Show that the left-regular representation of a compact Lie group G on $L^2(G)$ contains each irreducible with multiplicity exactly equal to its dimension. Use that $\mathrm{End}(V_\lambda) \cong V_\lambda \otimes V_\lambda^*$ carries the left-translation action through the second tensor factor only, so as a left G-module it is $(V_\lambda^*)^{\oplus d_\lambda}$; then note that as λ runs over $\widehat{G}$ so does λ^*, with $\dim V_\lambda^* = d_\lambda$.

3. Argue that the "delta function" δ_e at the identity has the Peter–Weyl expansion $\delta_e = \sum_{\lambda \in \widehat{G}} d_\lambda \, \chi_\lambda$, in the sense that convolution against the partial sums recovers every f. Set $e_\lambda = d_\lambda \chi_\lambda$; show that $R_{e_\lambda} : f \mapsto f * e_\lambda$ acts as the orthogonal projection onto the λ-block M_λ of $L^2(G)$, and conclude that $\sum_\lambda e_\lambda$ acts as the identity, which is the meaning of $\delta_e = \sum_\lambda d_\lambda \chi_\lambda$.

4. Prove that the operator $P_\lambda = d_\lambda \int_G \overline{\chi_\lambda(g)}\, \varphi(g)\, dg$ of definition 10.3.6 is the projection onto the λ-isotypic component of a finite-dimensional representation (φ, V): show that P_λ is G-equivariant, that it acts as the identity on any subrepresentation isomorphic to V_λ and kills every subrepresentation isomorphic to an inequivalent V_μ, and deduce $P_\lambda^2 = P_\lambda$.

5. Suppose G admits a faithful finite-dimensional representation (φ, V), so that $\varphi \colon G \to \mathrm{U}(V)$ is injective. Show that the matrix coefficients of φ and their complex conjugates generate a subalgebra of $R(G)$ that is dense in $C(G)$ by the complex Stone-Weierstrass theorem.[2] Identify its three hypotheses on the subalgebra (contains the constants, closed under conjugation, separates points) with properties of matrix coefficients proved in proposition 10.1.5, and explain which one uses faithfulness.

6. Explain, in one paragraph, precisely which step of the proof of theorem 10.3.4 fails for the noncompact group $G = (\mathbb{R}, +)$, and relate this to the fact that the irreducible unitary representations of $\mathbb{R}$ are the one-dimensional characters $x \mapsto e^{i\xi x}$ indexed by the *continuous* parameter $\xi \in \mathbb{R}$, so that "$L^2(\mathbb{R}) = \bigoplus_\xi \mathbb{C}$" becomes the Fourier *transform* (an integral) rather than a Fourier *series* (a sum).

10.4 Characters and the Ring of Representations

The Peter-Weyl theorem (theorem 10.3.4) placed the matrix coefficients of the irreducibles as an orthonormal basis of $L^2(G)$. This section extracts from each representation a single function, its character, and shows that the characters carry all the information about a representation that matters up to isomorphism. The character is the trace of the representing operators; because trace is conjugation-invariant, it is a class function, and the orthogonality of matrix coefficients collapses to an orthogonality of characters. The payoff is that decomposing a representation into irreducibles becomes a computation of inner products, and the collection of all representations, added and multiplied, becomes a commutative ring in which the irreducible characters are a basis. For the circle and SU(2) this ring is a polynomial ring, and its multiplication table is the Clebsch-Gordan rule.

The character discards everything about a representation except the eigenvalue data of each group element. That this suffices to recover the representation is the content of the section, and it is the analogue of the finite-group character theory the reader has already met ([3]), now over the compact group's L^2.

[2] [9, Complex Stone-Weierstrass Theorem].

Definition 10.4.1: Character of a Representation

Let G be a compact Lie group and (φ, V) a finite-dimensional complex representation. Its *character* is the function

$$\chi_V : G \to \mathbb{C}, \qquad \chi_V(g) = \operatorname{tr}\big(\varphi(g)\big)$$

the trace of the operator $\varphi(g)$ on V. A *class function* is a function $f : G \to \mathbb{C}$ constant on conjugacy classes, that is $f(hgh^{-1}) = f(g)$ for all $g, h \in G$.

The character is a class function, and its behaviour under the standard operations on representations is dictated by the corresponding operations on traces. These are the properties that make the character a usable invariant.

Proposition 10.4.2: Elementary Properties of Characters

Let G be a compact Lie group and let V, W be finite-dimensional complex representations of G. Then:

1. χ_V is a class function, and $\chi_V(e) = \dim V$;
2. $\chi_{V\oplus W} = \chi_V + \chi_W$ and $\chi_{V\otimes W} = \chi_V\,\chi_W$;
3. $\chi_{V^*} = \overline{\chi_V}$, where V^* is the dual (contragredient) representation;
4. two representations with the same character are isomorphic.

***Proof*:**

Part (1). For $g, h \in G$ the trace is conjugation-invariant, so $\chi_V(hgh^{-1}) = \operatorname{tr}\big(\varphi(h)\varphi(g)\varphi(h)^{-1}\big) = \operatorname{tr}\big(\varphi(g)\big) = \chi_V(g)$, using $\operatorname{tr}(ABA^{-1}) = \operatorname{tr}(B)$; thus χ_V is a class function. At the identity $\varphi(e) = \mathrm{id}_V$, so $\chi_V(e) = \operatorname{tr}(\mathrm{id}_V) = \dim V$.

Part (2). On the direct sum $V\oplus W$ the operator $(\varphi\oplus\psi)(g)$ is block diagonal with blocks $\varphi(g), \psi(g)$, so its trace is the sum of the block traces, $\chi_{V\oplus W}(g) = \operatorname{tr}\varphi(g) + \operatorname{tr}\psi(g) = \chi_V(g) + \chi_W(g)$. On the tensor product the trace of a tensor of operators is the product of the traces, $\operatorname{tr}\big(\varphi(g)\otimes\psi(g)\big) = \operatorname{tr}\varphi(g)\operatorname{tr}\psi(g)$, so $\chi_{V\otimes W}(g) = \chi_V(g)\,\chi_W(g)$.

Part (3). Make V unitary by the unitarian trick (theorem 9.3.1), so that each $\varphi(g)$ is a unitary operator, with eigenvalues on the unit circle. The dual representation acts on V^* by $\varphi^*(g) = \big(\varphi(g)^{-1}\big)^{\mathsf{T}}$, whose eigenvalues are the inverses of those of $\varphi(g)$. If the eigenvalues of $\varphi(g)$ are $\lambda_1, \dots, \lambda_d$ with $|\lambda_i| = 1$, then those of $\varphi^*(g)$ are $\lambda_i^{-1} = \overline{\lambda_i}$, so

$$\chi_{V^*}(g) = \sum_i \overline{\lambda_i} = \overline{\sum_i \lambda_i} = \overline{\chi_V(g)}$$

giving $\chi_{V^*} = \overline{\chi_V}$.

Part (4). By complete reducibility (theorem 9.3.3) write $V \cong \bigoplus_\lambda V_\lambda^{\oplus m_\lambda}$, the sum over the distinct irreducibles V_λ with multiplicities $m_\lambda \geq 0$. By part (2) the character is $\chi_V = \sum_\lambda m_\lambda \chi_\lambda$. The irreducible characters χ_λ are orthonormal in $L^2(G)$ (corollary 10.2.2), hence linearly independent, so the coefficients m_λ are recovered from χ_V as $m_\lambda = \langle \chi_V, \chi_\lambda \rangle$. Two representations with equal

characters therefore have equal isotypic multiplicities m_λ for every λ, and so are isomorphic, completing the proof.

The four properties turn the assignment $V \mapsto \chi_V$ into a map from representations to class functions that converts $\oplus$ into $+$ and $\otimes$ into $\times$. Part (4) is the reason the character is worth studying: a representation, an object with a great deal of internal linear-algebraic structure, is pinned down up to isomorphism by a single function on the group. The mechanism behind part (4) is the orthonormality of corollary 10.2.2; the character determines the multiplicities because the irreducible characters are an orthonormal, hence independent, family, and the multiplicities are their coordinates. The dual formula in part (3) records that for a unitary representation the character of the contragredient is the complex conjugate, which is why the L^2 inner product $\langle \chi_V, \chi_W \rangle = \int_G \chi_V \, \overline{\chi_W} \, dg$ is the natural pairing on characters, and why a self-dual representation has a real character.

Example 10.4.3: Characters of the Circle and of SU(2)

For the circle S^1 the irreducible $\chi_n(z) = z^n$ of example 9.5.1 is one-dimensional, so its character equals the representation itself, the function $z \mapsto z^n$. The tensor product rule of proposition 10.4.2 reads $\chi_m \chi_n = \chi_{m+n}$, the addition of weights, and duality reads $\chi_n^* = \overline{\chi_n} = \chi_{-n}$, since $\overline{z^n} = z^{-n}$ on $|z| = 1$.

For SU(2) the irreducible V_n has character, on the torus element $t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta})$,

$$\chi_n(t_\theta) = e^{in\theta} + e^{i(n-2)\theta} + \cdots + e^{-in\theta} = \frac{\sin\big((n+1)\theta\big)}{\sin\theta}$$

by proposition 9.5.3. Here $\chi_n(e) = \chi_n(t_0) = n + 1 = \dim V_n$, confirming part (1); the character is real, so each V_n is self-dual, $V_n^* \cong V_n$; and the tensor rule gives $\chi_1 \chi_n = \chi_{n+1} + \chi_{n-1}$, the Clebsch-Gordan identity of exercise 9.5.1 (2). Every element of SU(2) is conjugate to some t_θ, so a class function on SU(2) is determined by its values on the torus, and the displayed formula determines χ_n completely.

With the character established as a class function determining its representation, the next result explains why class functions are the right ambient space: the irreducible characters are not only an orthonormal family in it, they are an orthonormal *basis*. Completeness is the harmonic-analytic content, and it is Peter-Weyl read on the invariants of the conjugation action.

Theorem 10.4.4: Irreducible Characters as a Basis of Class Functions

Let G be a compact Lie group and let $L^2(G)^G \subseteq L^2(G)$ be the Hilbert space of square-integrable class functions, the functions invariant under conjugation $f(hgh^{-1}) = f(g)$. Then the irreducible characters $\{\chi_\lambda\}$, indexed by the isomorphism classes of irreducible representations, form an orthonormal basis of $L^2(G)^G$. Consequently, for a finite-dimensional representation V:

1. the multiplicity of V_λ in V is $\langle \chi_V, \chi_\lambda \rangle = \int_G \chi_V \, \overline{\chi_\lambda} \, dg$;
2. $\langle \chi_V, \chi_V \rangle = \sum_\lambda m_\lambda^2$ is the sum of the squared multiplicities;
3. V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.

***Proof*:**

Orthonormality. The χ_λ are class functions by proposition 10.4.2(1), so they lie in $L^2(G)^G$, and they are orthonormal there by corollary 10.2.2: $\langle \chi_\lambda, \chi_\mu \rangle = \delta_{\lambda\mu}$.

Completeness. It suffices to show that a class function $f \in L^2(G)^G$ orthogonal to every χ_λ is zero, since an orthonormal family whose orthogonal complement is zero is a basis. Peter-Weyl (theorem 10.3.4) gives the orthogonal decomposition of the full space

$$L^2(G) = \widehat{\bigoplus_\lambda} \operatorname{End}(V_\lambda)$$

into the isotypic blocks spanned by the matrix coefficients $\{\varphi^\lambda_{ij}\}$ of the irreducibles, with $\{\sqrt{d_\lambda}\,\varphi^\lambda_{ij}\}$ an orthonormal basis, where $d_\lambda = \dim V_\lambda$. Expand f in this basis: writing $f = \sum_\lambda \sum_{i,j} c^\lambda_{ij} \varphi^\lambda_{ij}$ in L^2, the component of f in the block of λ is $\sum_{i,j} c^\lambda_{ij} \varphi^\lambda_{ij}$. The claim is that on the block of λ, invariance under conjugation forces this component to be a multiple of $\chi_\lambda = \sum_i \varphi^\lambda_{ii}$.

To see this, average over conjugation. For $f \in L^2(G)^G$ and any $h \in G$ the function $g \mapsto f(hgh^{-1})$ equals f, so f is fixed by the averaging operator (proposition 9.2.4)

$$(Pf)(g) = \int_G f(hgh^{-1})\, dh = f(g)$$

which projects $L^2(G)$ onto $L^2(G)^G$. Applying P to a single matrix coefficient φ^λ_{ij} and using $\varphi^\lambda(hgh^{-1}) = \varphi^\lambda(h)\varphi^\lambda(g)\varphi^\lambda(h)^{-1}$,

$$\big(P\varphi^\lambda_{ij}\big)(g) = \int_G \sum_{k,l} \varphi^\lambda(h)_{ik}\, \varphi^\lambda_{kl}(g)\, \varphi^\lambda(h^{-1})_{lj}\, dh = \sum_{k,l} \Big(\int_G \varphi^\lambda(h)_{ik}\, \varphi^\lambda(h^{-1})_{lj}\, dh \Big) \varphi^\lambda_{kl}(g)$$

The averaged coefficient in parentheses is computed by Schur orthogonality (theorem 10.2.1): it equals $\frac{1}{d_\lambda}\delta_{ij}\delta_{kl}$. Hence $P\varphi^\lambda_{ij} = \frac{1}{d_\lambda}\delta_{ij} \sum_k \varphi^\lambda_{kk} = \frac{1}{d_\lambda}\delta_{ij}\chi_\lambda$. Every matrix coefficient projects onto a multiple of χ_λ under P, so the subspace of *class* functions inside the block of λ is exactly the line $\mathbb{C}\,\chi_\lambda$. Since f is a class function, $f = Pf$, and applying P termwise to the expansion of f collapses each block to its χ_λ component:

$$f = Pf = \sum_\lambda \Big(\tfrac{1}{d_\lambda} \sum_i c^\lambda_{ii} \Big) \chi_\lambda = \sum_\lambda \langle f, \chi_\lambda \rangle\, \chi_\lambda$$

the last equality because $\{\chi_\lambda\}$ is orthonormal. If f is orthogonal to every χ_λ then every coefficient $\langle f, \chi_\lambda \rangle$ vanishes, so $f = 0$. The orthogonal complement of $\{\chi_\lambda\}$ in $L^2(G)^G$ is zero, and the χ_λ are an orthonormal basis.

Consequences. Write $V \cong \bigoplus_\lambda V_\lambda^{\oplus m_\lambda}$, so $\chi_V = \sum_\lambda m_\lambda \chi_\lambda$ by proposition 10.4.2(2). Part (1) is the orthonormal expansion of a vector: $\langle \chi_V, \chi_\lambda \rangle = \sum_\mu m_\mu \langle \chi_\mu, \chi_\lambda \rangle = m_\lambda$. Part (2) is the Parseval identity for the same expansion: $\langle \chi_V, \chi_V \rangle = \sum_{\lambda,\mu} m_\lambda m_\mu \langle \chi_\lambda, \chi_\mu \rangle = \sum_\lambda m_\lambda^2$. For part (3), the sum $\sum_\lambda m_\lambda^2$ of nonnegative integers equals 1 if and only if exactly one m_λ is 1 and the rest are 0, that is $V \cong V_\lambda$ is irreducible, completing the proof.

The theorem is the harmonic analysis of class functions. Peter-Weyl decomposed all of $L^2(G)$ into matrix-coefficient blocks; restricting to functions that survive conjugation-averaging, exactly one

dimension survives in each block, the trace direction χ_λ, and these trace directions assemble into a complete orthonormal system on the space of class functions. For the circle this recovers the completeness of $\{z^n\}$ in $L^2(S^1)$, the classical Fourier basis, now seen as the character basis of the class functions of an abelian group where every function is a class function. For SU(2), where the class functions are the even functions of the torus angle θ, the theorem asserts that the characters $\sin((n+1)\theta)/\sin\theta$ are an orthonormal basis of those, a statement equivalent to the completeness of $\{\sin(k\theta)\}$ on $(0,\pi)$ after clearing the Weyl factor.

The three numerical consequences are the working tools of the subject. Part (1) turns the decomposition of a representation into an integral: to find how many copies of V_λ sit inside V, integrate χ_V against $\overline{\chi_\lambda}$. Part (3) is the sharpest test for irreducibility available, a single number, the value $\langle \chi_V, \chi_V \rangle$, which equals 1 precisely when V is irreducible. These are the compact-group forms of the finite-group tools of [3], with the normalized integral in place of the averaged sum over the group.

Example 10.4.5: Irreducibility Test for $V_1 \otimes V_1$

The tensor square $V_1 \otimes V_1$ of the standard representation of SU(2) has character χ_1^2, where $\chi_1(t_\theta) = 2\cos\theta$ by exercise 9.5.1 (1). Its self-inner-product is computed against the Weyl integration formula for SU(2) (exercise 9.5.1 (3)),

$$\langle \chi_1^2, \chi_1^2 \rangle = \frac{2}{\pi} \int_0^\pi (2\cos\theta)^4 \, \sin^2\theta \, d\theta = 2$$

so by theorem 10.4.4(3) the representation $V_1 \otimes V_1$ is *not* irreducible: the value $2 = 1^2 + 1^2$ signals two distinct irreducible constituents, each of multiplicity one. Indeed $\chi_1^2 = (2\cos\theta)^2 = e^{2i\theta} + 2 + e^{-2i\theta} = \chi_2 + \chi_0$, so $V_1 \otimes V_1 \cong V_2 \oplus V_0$, the Clebsch-Gordan rule with $n = 1$. The inner products $\langle \chi_1^2, \chi_2 \rangle = 1$ and $\langle \chi_1^2, \chi_0 \rangle = 1$ read off the two multiplicities directly.

The character has now become a coordinate system: a representation is a vector of nonnegative integers, its multiplicities, and the operations $\oplus$ and $\otimes$ act on these vectors additively and multiplicatively. Packaging all representations of G into a single algebraic object, the representation ring, makes the characters genuine ring elements and the Clebsch-Gordan rules the multiplication table.

Definition 10.4.6: The Representation Ring

Let G be a compact Lie group. The *representation ring* $\mathcal{R}(G)$ is the free abelian group on the isomorphism classes $[V_\lambda]$ of irreducible finite-dimensional complex representations, with addition induced by direct sum, $[V] + [W] = [V \oplus W]$, and multiplication induced by tensor product, $[V] \cdot [W] = [V \otimes W]$, extended bilinearly. Its elements are the *virtual characters*, the formal differences $[V] - [W]$; the irreducible classes $\{[V_\lambda]\}$ are a $\mathbb{Z}$-basis. The symbol $\mathcal{R}(G)$ denotes this representation ring, not to be confused with the algebra of representative functions $R(G)$ of definition 10.1.3.

The character map embeds this ring into the class functions and identifies its multiplication with pointwise multiplication of characters.

Proposition 10.4.7: The Character Map is a Ring Embedding

The assignment $[V] \mapsto \chi_V$ extends to an injective ring homomorphism

$$\chi\colon \mathcal{R}(G) \hookrightarrow L^2(G)^G, \qquad \sum_\lambda n_\lambda [V_\lambda] \mapsto \sum_\lambda n_\lambda \chi_\lambda$$

from the representation ring to the class functions, carrying $+$ to $+$ and the ring product to pointwise multiplication of functions. Its image is the $\mathbb{Z}$-span of the irreducible characters, a subring of $L^2(G)^G$.

***Proof*:**
Additivity $\chi_{V\oplus W} = \chi_V + \chi_W$ and multiplicativity $\chi_{V\otimes W} = \chi_V \chi_W$ are proposition 10.4.2(2), so the map on generators extends to a ring homomorphism from $\mathcal{R}(G)$ to the ring of class functions under pointwise operations. It is injective because the irreducible characters χ_λ are linearly independent over $\mathbb{Z}$ (indeed over $\mathbb{C}$), being orthonormal by corollary 10.2.2: a virtual character $\sum_\lambda n_\lambda [V_\lambda]$ in the kernel has $\sum_\lambda n_\lambda \chi_\lambda = 0$, forcing every $n_\lambda = 0$. The image is by construction the $\mathbb{Z}$-linear span of the χ_λ, closed under multiplication because a product $\chi_\lambda \chi_\mu = \chi_{V_\lambda \otimes V_\mu}$ is again a nonnegative integer combination of irreducible characters (the isotypic decomposition of $V_\lambda \otimes V_\mu$), so it is a subring, as we sought to show.

The representation ring makes the bookkeeping of representations into algebra: the whole finite-dimensional representation theory of G is the study of the commutative ring $\mathcal{R}(G)$, whose additive structure records how representations decompose and whose multiplicative structure records the tensor product. The embedding into class functions is its concrete face, realizing each virtual representation as an actual function on the group; the ring product, abstractly the tensor product, becomes ordinary multiplication of functions, so questions about tensor decompositions become questions about factoring products of characters. For the two running examples the ring is a polynomial ring in one variable.

Example 10.4.8: The Representation Ring of SU(2)

For the circle, the classes $[\chi_n]$, $n \in \mathbb{Z}$, multiply by $[\chi_m][\chi_n] = [\chi_{m+n}]$ (exercise 9.5.1 (5)), so $\mathcal{R}(S^1) = \mathbb{Z}[z, z^{-1}]$ is the ring of Laurent polynomials, with $[\chi_n] \mapsto z^n$; this is the group ring $\mathbb{Z}[\mathbb{Z}]$ of the character lattice of S^1.

For SU(2), set $\chi_1 = z + z^{-1}$, writing $z = e^{i\theta}$ for the torus coordinate, so that $\chi_1(t_\theta) = 2\cos\theta$. The Clebsch-Gordan rule of proposition 9.5.2,

$$\chi_1 \chi_n = \chi_{n+1} + \chi_{n-1}$$

is a recursion that expresses each χ_n as a polynomial in χ_1: rearranged as $\chi_{n+1} = \chi_1\chi_n - \chi_{n-1}$ with $\chi_0 = 1$ and $\chi_1 = \chi_1$, it gives $\chi_2 = \chi_1^2 - 1$, $\chi_3 = \chi_1^3 - 2\chi_1$, and in general $\chi_n = U_n(\chi_1/2)$, the Chebyshev polynomial of the second kind evaluated at $\cos\theta$, since $\chi_n(t_\theta) = \sin((n+1)\theta)/\sin\theta$ is by definition $U_n(\cos\theta)$. Because each χ_n is a degree-n polynomial in χ_1 with leading coefficient 1, the characters $\{\chi_0, \chi_1, \chi_2, \dots\}$ and the powers $\{1, \chi_1, \chi_1^2, \dots\}$ span the same $\mathbb{Z}$-module, and no polynomial relation holds among them (the χ_n are $\mathbb{Z}$-independent by proposition 10.4.7). Hence

$$\mathcal{R}\big(\mathrm{SU}(2)\big) = \mathbb{Z}[\chi_1]$$

is a polynomial ring in the single variable χ_1, the character of the standard representation. The

irreducible characters are the images of the Chebyshev basis; the ring is free on χ_1, and its arithmetic is the arithmetic of these polynomials.

The polynomial description of $\mathcal{R}(\mathrm{SU}(2))$ closes the loop opened in chapter 9. The Clebsch-Gordan rule was recorded there as the multiplication table of the irreducibles; here it is a recursion generating a polynomial ring, with the irreducible characters realized as the Chebyshev polynomials in $\chi_1 = z+z^{-1}$. The pattern generalizes: for a compact connected group the representation ring is a polynomial ring on the fundamental representations, a statement that becomes transparent once the maximal torus reduces characters to symmetric Laurent polynomials in the torus coordinates, developed in the chapters ahead through the theory of maximal tori and the Weyl character formula.

The character has turned representation theory into the linear algebra of an inner-product space. Decomposing a representation is projecting its character onto the orthonormal basis $\{\chi_\lambda\}$ of class functions; the multiplicity of each irreducible is a coordinate, an inner product computed by an integral. This is the same picture as for finite groups, with the sum over the group replaced by the invariant integral and the finite list of irreducibles replaced by the countable family Peter-Weyl produced, now spanning the L^2 class functions rather than a finite-dimensional space. The computational payoff is the reduction of every representation-theoretic question to an integral against characters. What makes that integral tractable is that a character is determined by its values on a maximal torus, where it becomes a symmetric Laurent polynomial in the torus coordinates, a reduction the following chapters make explicit through the maximal torus and the Weyl integration formula.

Exercise 10.4.1

1. Let $V = V_1$ be the standard two-dimensional representation of $\mathrm{SU}(2)$, with character $\chi_1(t_\theta) = e^{i\theta}+e^{-i\theta}$. Compute the character of $V\otimes V$ directly as χ_1^2, expand it as a sum of the exponentials $e^{ik\theta}$, and read off the decomposition $V\otimes V \cong V_2 \oplus V_0$ into irreducibles by matching to the characters χ_n of proposition 9.5.3. Confirm the answer against example 10.4.5.

2. Let V be a finite-dimensional representation of a compact group G, decomposed as $V \cong \bigoplus_\lambda V_\lambda^{\oplus m_\lambda}$. Show that the number of irreducible constituents of V counted with multiplicity, namely $\sum_\lambda m_\lambda$, equals $\sum_\lambda \langle \chi_V, \chi_\lambda\rangle$, and that the number of *distinct* irreducible constituents is the number of λ with $\langle \chi_V, \chi_\lambda\rangle \neq 0$. State carefully why $\sum_\lambda m_\lambda$ is not in general equal to $\langle \chi_V, \chi_V\rangle$, and identify the representations for which the two coincide.

3. Let $T = T^n = (S^1)^n$ be an n-dimensional torus, with character lattice $X^*(T) = \mathrm{Hom}(T, S^1) \cong \mathbb{Z}^n$. Show that the irreducible representations of T are the characters $\chi_\mu(t) = t^\mu$ for $\mu \in X^*(T)$, that their tensor product rule is $\chi_\mu\chi_\nu = \chi_{\mu+\nu}$, and deduce that the representation ring is the group ring of the character lattice, $R(T) = \mathbb{Z}[X^*(T)] \cong \mathbb{Z}[z_1^{\pm1}, \dots, z_n^{\pm1}]$, the ring of Laurent polynomials in n variables.

4. Call a representation V of a compact group G *real* if $V = W \otimes_{\mathbb{R}} \mathbb{C}$ for a real representation W (a representation on a real vector space W, complexified). Show that a real representation has a real-valued character. Show conversely, using proposition 10.4.2(3), that a representation with real character is self-dual, $V \cong V^*$, and explain in one sentence why the converse to the first statement nonetheless fails, that a real character need not force a real structure.

5. The adjoint representation of $\mathrm{SU}(2)$ on its complexified Lie algebra $\mathfrak{su}(2)_{\mathbb{C}}$ has character $\chi_{\mathrm{Ad}}(t_\theta) = 1+2\cos 2\theta$ (exercise 9.5.1 (6)). Compute $\langle \chi_{\mathrm{Ad}}, \chi_{\mathrm{Ad}}\rangle$ using the $\mathrm{SU}(2)$ integration formula of exercise 9.5.1 (3), and conclude from theorem 10.4.4(3) that the adjoint representation

is irreducible. Identify which V_n it is.

6. In the representation ring $\mathcal{R}(\mathrm{SU}(2)) = \mathbb{Z}[\chi_1]$, the virtual character $\chi_2 - \chi_0 = \chi_1^2 - 2$ is a difference of actual characters. Explain why no finite-dimensional representation of $\mathrm{SU}(2)$ has $\chi_2 - \chi_0$ as its character, and characterize precisely which elements of $\mathcal{R}(\mathrm{SU}(2)) = \mathbb{Z}[\chi_1]$ are characters of actual representations rather than virtual ones.

10.5 Consequences: Linearity of Compact Groups and Tannaka-Krein

The Peter-Weyl theorem (theorem 10.3.4) says that the matrix coefficients of the irreducible representations exhaust the harmonic analysis of a compact group. Two structural statements follow, and they pull in opposite directions. The first is a rigidity: a compact Lie group, defined intrinsically as a manifold with a group law, is always a group of matrices, so the abstract category of compact Lie groups is no larger than the concrete category of closed subgroups of unitary groups. The second is a reconstruction: not only is the group linear, it is nothing but its representations, in that the group can be recovered from the tensor category $\mathrm{Rep}(G)$, or equivalently from the algebra $R(G)$ of representative functions. This section proves the first in full and states the second with its construction, deferring the categorical machinery to prose.

Both rest on the same output of Peter-Weyl: the representative functions separate the points of G. If $g \neq h$, some matrix coefficient $\varphi_{v,w}$ of some finite-dimensional representation takes different values at g and h, because the coefficients are dense in $C(G)$ and a continuous function separating g from h exists. Point separation by matrix coefficients is exactly the statement that the finite-dimensional representations, taken together, do not collapse any pair of group elements, and that is the seed of both the faithful representation and the reconstruction.

The first consequence realizes a compact Lie group as a matrix group. A single irreducible representation need not be faithful (the trivial representation is the extreme case), but Peter-Weyl guarantees enough representations to distinguish every pair of elements, and a compactness argument collapses these into one faithful representation of finite dimension.

Theorem 10.5.1: Compact Lie Groups Are Linear

Let G be a compact Lie group. Then G admits a faithful finite-dimensional complex representation. Consequently G is isomorphic, as a Lie group, to a closed subgroup of the unitary group $\mathrm{U}(n)$ for some n; a compact Lie group is a matrix group.

***Proof*:**
Step 1: the finite-dimensional representations separate points. For a finite-dimensional representation $\varphi\colon G \to \mathrm{GL}(V)$ let $\ker\varphi$ be its kernel. The claim is that the intersection

$$N = \bigcap_{\varphi} \ker\varphi$$

over all finite-dimensional representations φ is trivial. Suppose $g \in N$, so $\varphi(g) = \mathrm{id}$ for every finite-dimensional φ. Then every matrix coefficient $\varphi_{v,w}(g) = \langle \varphi(g)v, w\rangle = \langle v, w\rangle$ equals its value at the identity, so $\varphi_{v,w}(g) = \varphi_{v,w}(e)$ for every representative function. By theorem 10.3.4

the representative functions are dense in $C(G)$, so they separate points; a continuous function on G taking equal values at g and e for *every* representative function, together with density, forces $g = e$. Indeed if $g \neq e$, choose by Urysohn's lemma a continuous f with $f(g) \neq f(e)$, approximate it uniformly by a representative function ψ to within $\frac{1}{3}|f(g) - f(e)|$, and then $\psi(g) \neq \psi(e)$, contradicting $g \in N$. Hence $N = \{e\}$.

Step 2: finitely many representations already separate points. Each kernel $\ker\varphi$ is a closed normal subgroup of G: it is the preimage of $\{\mathrm{id}\}$ under the continuous homomorphism φ, hence closed, and normal as the kernel of a homomorphism. By theorem 3.2.1 each $\ker\varphi$ is therefore an embedded Lie subgroup. Consider the family of finite intersections $\ker\varphi_1 \cap \cdots \cap \ker\varphi_r$, which are again closed embedded Lie subgroups; order them by inclusion. A finite intersection is itself the kernel of the direct sum $\varphi_1 \oplus \cdots \oplus \varphi_r$, so the family is exactly the kernels of the finite-dimensional representations, closed under passing to a common refinement.

Choose a finite intersection $K = \ker\varphi_1 \cap \cdots \cap \ker\varphi_r$ of least dimension, and among those of that dimension one with the fewest connected components. If $K \neq \{e\}$, then since $N = \bigcap_\varphi \ker\varphi = \{e\}$ by Step 1, some representation ψ has $K \not\subseteq \ker\psi$, so $K \cap \ker\psi$ is a proper closed subgroup of K. A proper closed subgroup has either strictly smaller dimension, or the same dimension and strictly fewer connected components (its identity component is an open subgroup of K° of the same dimension, hence equal to K°, so a proper containment can only drop components). Either way $K \cap \ker\psi = \ker\varphi_1 \cap \cdots \cap \ker\varphi_r \cap \ker\psi$ is a finite intersection strictly below K in the (dimension, component count) ordering, contradicting minimality. Hence $K = \{e\}$, that is, $\varphi_1, \ldots, \varphi_r$ already have trivial common kernel.

Step 3: assemble a faithful representation. Set $\Phi = \varphi_1 \oplus \cdots \oplus \varphi_r$ on $V = V_1 \oplus \cdots \oplus V_r$. Its kernel is $\ker\varphi_1 \cap \cdots \cap \ker\varphi_r = \{e\}$, so Φ is a faithful finite-dimensional representation of G. This proves the first assertion.

Step 4: realize G inside $\mathrm{U}(n)$. By the unitarian trick (theorem 9.3.1) the space V carries a G-invariant Hermitian inner product, for which every $\Phi(g)$ is unitary; fixing an orthonormal basis of $V \cong \mathbb{C}^n$ ($n = \dim V$) gives a continuous injective homomorphism $\Phi\colon G \to \mathrm{U}(n)$. The image $\Phi(G)$ is a subgroup of $\mathrm{U}(n)$, and it is compact, being the continuous image of the compact space G; a compact subset of the Hausdorff group $\mathrm{U}(n)$ is closed, so $\Phi(G)$ is a closed subgroup. By theorem 3.2.1 it is an embedded Lie subgroup of $\mathrm{U}(n)$. Finally $\Phi\colon G \to \Phi(G)$ is a continuous bijective homomorphism from a compact group to a Hausdorff group, hence a homeomorphism, and a bijective Lie group homomorphism whose inverse is continuous is a Lie group isomorphism. Thus $G \cong \Phi(G)$ is a closed subgroup of $\mathrm{U}(n)$, completing the proof.

The theorem removes any daylight between the abstract and the concrete definition of a compact Lie group. Chapter 1 began with matrix groups, the closed subgroups of $\mathrm{GL}(n, \mathbb{C})$, and chapter 2 replaced them with intrinsic Lie groups, manifolds carrying a compatible group law; the worry was that the intrinsic definition might be strictly more general. For compact groups it is not. Every compact Lie group is, up to isomorphism, a closed subgroup of some $\mathrm{U}(n)$, so the two starting points of the book coincide in the compact case, and the running example $\mathrm{SU}(2)$ was a matrix group from the outset for a structural reason, not by accident of presentation. The mechanism is worth isolating: linearity comes entirely from point separation by matrix coefficients (Step 1), which is Peter-Weyl, together with the noetherian-style descent on closed subgroups (Step 2) that is special to Lie groups, where dimension and component count give a well-founded ordering.

Compactness is essential twice, in the same two places it was essential for the unitarian trick. It gives the invariant integral behind Peter-Weyl, without which the matrix coefficients need not separate

points, and it makes the image $\Phi(G)$ automatically closed, without which the embedding could fail to be a Lie subgroup. A noncompact Lie group can fail to be linear: the universal cover $\widetilde{\mathrm{SL}(2,\mathbb{R})}$ has no faithful finite-dimensional representation at all, because any such representation factors through a finite cover, so the compact hypothesis cannot be dropped.

Example 10.5.2: The Standard Embedding of SU(2)

For $G = \mathrm{SU}(2)$ the faithful representation of theorem 10.5.1 can be taken to be the single standard representation $V_1 = \mathbb{C}^2$ of proposition 9.5.2, on which $g = \left(\begin{smallmatrix} a & b \\ -\bar{b} & \bar{a} \end{smallmatrix}\right)$ acts by matrix multiplication. This representation is faithful: $\Phi(g) = \mathrm{id}$ on $\mathbb{C}^2$ means $g = I$ as a matrix. It is already unitary, since $\mathrm{SU}(2) \subseteq \mathrm{U}(2)$ by definition, so no averaging is needed, and the embedding $\Phi\colon \mathrm{SU}(2) \hookrightarrow \mathrm{U}(2)$ is the inclusion. Here $n = 2$ and the theorem's construction returns the tautological realization, matching the fact that $\mathrm{SU}(2)$ entered the book as a matrix group. Among the higher irreducibles V_n the faithful ones are exactly those with n odd, since the central element $-I$ acts on $V_n = \mathrm{Sym}^n(\mathbb{C}^2)$ by $(-1)^n$; the even ones $V_2, V_4, \dots$ have kernel the center $\{\pm I\}$, and V_1 is the economical faithful choice. The trivial V_0 has kernel all of $\mathrm{SU}(2)$.

For a finite group G, regarded as a compact Lie group of dimension zero, the same argument specializes to a classical fact. Point separation is immediate because $C(G)$ is all functions on the finite set G, and the descent of Step 2 terminates because G is finite; the faithful representation can be taken to be the regular representation on $\mathbb{C}[G]$, which is faithful since $\varphi(g)f = f$ for the indicator functions forces $g = e$. This recovers Cayley's theorem in its linear form, that every finite group embeds in some $\mathrm{GL}(n,\mathbb{C})$, from the same principle. The finite-group version needs no analysis, but Peter-Weyl is the statement that makes the argument go through for a positive-dimensional group, where "all functions" must be replaced by "continuous functions" and density of the representative functions replaces their exhaustiveness.

The second consequence sharpens linearity into an equivalence. The group is not just *a* matrix group; it is determined, as a group, by the totality of its representations. This is the content of Tannaka-Krein duality, which recovers a compact group from its representation category, and it is the precise sense in which, for a compact group, "the group" and "its representation theory" are two names for one object. The reconstruction is built from the representative functions, which by proposition 10.1.5 form an algebra, and in fact a Hopf algebra, whose structure maps encode the group law.

To state the duality, package the representations into two equivalent pieces of data. On the categorical side is the forgetful functor $\omega\colon \mathrm{Rep}(G) \to \mathrm{Vect}_{\mathbb{C}}$ sending a finite-dimensional representation (φ, V) to its underlying vector space V and an intertwiner to itself; a group element $g \in G$ acts on this functor by the family $\{\varphi(g)\colon V \to V\}$, one operator for each object, compatible with all intertwiners and with tensor products. On the algebraic side is the Hopf algebra $R(G)$ of representative functions (definition 10.1.3), whose comultiplication $\Delta\colon R(G) \to R(G) \otimes R(G)$ encodes the group law by the formula

$$(\Delta f)(g,h) = f(gh) \tag{10.5}$$

Its counit is evaluation at e, and its antipode is $(Sf)(g) = f(g^{-1})$. The two packagings are dual: a point of G is at once a tensor-compatible family of operators on ω and an algebra homomorphism $R(G) \to \mathbb{C}$ given by evaluation.

Theorem 10.5.3: Tannaka-Krein Duality

Let G be a compact Lie group.

1. *(Categorical form.)* The map sending $g \in G$ to the family of operators $\big(\varphi(g)\big)_{(\varphi,V)\in \mathrm{Rep}(G)}$ is an isomorphism of G onto the group of natural transformations η of the forgetful functor $\omega\colon \mathrm{Rep}(G) \to \mathrm{Vect}_{\mathbb{C}}$ that are invertible and *tensor-preserving*, that is, natural families $\eta_V\colon V \to V$ satisfying $\eta_{V\otimes W} = \eta_V \otimes \eta_W$ and $\eta_{\mathbf{1}} = \mathrm{id}$ on the trivial representation.
2. *(Hopf-algebraic form.)* The map sending $g \in G$ to evaluation $\mathrm{ev}_g\colon R(G) \to \mathbb{C}$, $f \mapsto f(g)$, is a homeomorphism of G onto the group of $\mathbb{C}$-algebra homomorphisms $R(G) \to \mathbb{C}$ that respect the complex conjugation, with group law induced by the comultiplication (10.5).

The two forms are the categorical and the algebraic face of one statement, and the passage between them is Peter-Weyl. The full categorical proof of part (1) belongs to the general Tannakian formalism, developed for finite and compact groups in [3]; the argument below establishes the reconstruction map and shows where Peter-Weyl makes it bijective, deferring the naturality bookkeeping of the tensor-preservation condition to that reference.

***Proof Key ideas*:**
The reconstruction map itself is unambiguous: send $g \in G$ to ev_g on $R(G)$ (part 2), equivalently to the family $(\varphi(g))_{(\varphi,V)}$ acting on the forgetful functor (part 1). That each ev_g is an algebra homomorphism respecting conjugation is immediate, and (10.5) shows that the group law on evaluations matches multiplication in G: $\mathrm{ev}_g \star \mathrm{ev}_h = \mathrm{ev}_{gh}$, where $\star$ is the convolution product dual to Δ. Peter-Weyl gives the two nontrivial properties.

Injectivity. The map is injective because $R(G)$ separates points (theorem 10.3.4): if $g \neq h$ then some $f \in R(G)$ has $f(g) \neq f(h)$, so $\mathrm{ev}_g \neq \mathrm{ev}_h$. This is the same point separation that gave the faithful representation in theorem 10.5.1.

Surjectivity. Every conjugation-respecting algebra homomorphism $\chi\colon R(G) \to \mathbb{C}$ is ev_g for some $g \in G$. Peter-Weyl makes $R(G)$ dense in $C(G)$, and χ is bounded for the uniform norm (an algebra homomorphism to $\mathbb{C}$ preserving conjugation is a $*$-homomorphism, hence norm one), so χ extends to a character of the commutative C^*-algebra $C(G)$; by Gelfand-Naimark the characters of $C(G)$ are exactly the evaluations at points of the compact space G, so $\chi = \mathrm{ev}_g$. That evaluations are all of the spectrum uses only that G is compact Hausdorff and that $R(G)$ is a dense conjugation-closed unital subalgebra, which is precisely the analytic content of Peter-Weyl.

For part (1), a tensor-preserving invertible natural transformation η of ω acts on each irreducible V_λ by an operator η_λ, and naturality across all intertwiners together with the tensor condition forces the family (η_λ) to define an algebra homomorphism $R(G) \to \mathbb{C}$ of the type classified in part (2); the point g it corresponds to satisfies $\eta_V = \varphi(g)$ on every V. The verification that these two descriptions match, and that tensor-preservation corresponds to multiplicativity for Δ, is the Tannakian bookkeeping deferred to [3]. This gives the reconstruction in both forms, as claimed.

The duality is the converse direction to theorem 10.5.1. Linearity says the group injects into its representations; reconstruction says nothing is lost, that the representations remember the group exactly. The characters carry the class-function part of this memory: by the character basis theorem (theorem 10.4.4) a representation is determined up to isomorphism by its character (definition 10.4.1),

so the whole tensor category is already encoded in the character table together with the tensor product rule $\chi_{V\otimes W} = \chi_V \chi_W$. Tannaka-Krein promotes this from "the representations are classified by characters" to "the group is classified by its representations," recovering not just the isomorphism classes but the group itself.

The analogy with commutative duality theory is exact and worth naming. For an abelian compact group, the irreducible representations are one-dimensional, the characters form a group $\widehat{G}$, the Pontryagin dual, and reconstruction is the Pontryagin duality $G \cong \widehat{\widehat{G}}$ of [16, Pontryagin Duality Theorem]. For a nonabelian compact group the irreducibles are higher-dimensional and no longer form a group, so the dual object is not another group but the whole representation category, or the Hopf algebra $R(G)$; Tannaka-Krein is the nonabelian Pontryagin duality, with the loss of the group structure on the dual compensated by the categorical structure on $\mathrm{Rep}(G)$.

Example 10.5.4: Reconstructing the Circle from Its Characters

For $G = S^1$ the representative functions are the finite Laurent polynomials $\sum_n c_n z^n$, spanned by the characters $\chi_n \colon z \mapsto z^n$ of example 9.5.1, and $R(S^1) = \mathbb{C}[z, z^{-1}]$ as an algebra. The Hopf structure is $\Delta(z) = z \otimes z$ (since $(zw)^n = z^n w^n$ encodes $\chi_n(gh) = \chi_n(g)\chi_n(h)$), counit $z \mapsto 1$, and antipode $z \mapsto z^{-1}$. A conjugation-respecting algebra homomorphism $\chi \colon \mathbb{C}[z, z^{-1}] \to \mathbb{C}$ is determined by the single value $w = \chi(z) \in \mathbb{C}^\times$, and the conjugation condition $\chi(\bar{z}) = \overline{\chi(z)}$ with $\bar{z} = z^{-1}$ forces $\bar{w} = w^{-1}$, that is, $|w| = 1$. Thus the algebra homomorphisms are exactly the evaluations ev_w at points $w \in S^1$, and the group they form under convolution, with $w_1 w_2$ the product, is S^1 itself. The circle is reconstructed from the ring $\mathbb{C}[z, z^{-1}]$ of its representative functions as the unitary points of its spectrum, which is the S^1 instance of theorem 10.5.3, and the same computation is the statement that $\widehat{\widehat{S^1}} = S^1$: the dual of $\mathbb{Z} = \widehat{S^1}$ is again S^1.

Peter-Weyl thereby closes the loop the previous chapter opened. Chapter 9 forced every representation of a compact group to decompose into irreducibles, reducing the study of all representations to the study of the atoms; Peter-Weyl showed the atoms fill $L^2(G)$ and separate the points of G; and the present section shows that the atoms determine G completely, as a matrix group and then as the spectrum of its own representative functions. A compact group is no more than its representations, since it embeds in a unitary group built from them, and no less, since it is recovered from them. What remains is to make the atoms explicit. The maximal torus of the next chapters reduces every irreducible to its restriction to a torus, where it becomes a sum of characters indexed by weights, and the Weyl character formula then writes down each irreducible character in closed form; the reconstruction of theorem 10.5.3 thereby becomes completely computable for a connected compact group, since its input, the full list of characters, is computed explicitly.

Exercise 10.5.1

1. Let G be a compact Lie group with the property that its only finite-dimensional complex representation is a direct sum of trivial representations (equivalently, G has no nontrivial irreducible representation). Prove that G is the trivial group. Where does the argument use theorem 10.3.4?

2. Let G be a finite group. Show directly that the left regular representation on $\mathbb{C}[G]$, with $(\varphi(g)f)(x) = f(g^{-1}x)$, is faithful and finite-dimensional, and explain how this is the specialization of theorem 10.5.1 to a group of dimension zero. Which step of the theorem's proof becomes trivial for a finite group?

3. Verify that the standard representation $V_1 = \mathbb{C}^2$ realizes $\mathrm{SU}(2)$ as a closed subgroup of $\mathrm{U}(2)$ in the sense of theorem 10.5.1: that the action is faithful, unitary, and has closed image. Then show that V_1 is the irreducible representation of smallest dimension that is faithful, and that $V_2 = \mathrm{Sym}^2(\mathbb{C}^2)$ is *not* faithful, identifying its kernel.

4. Using proposition 10.1.5, verify that the representative functions $R(G)$ form a Hopf algebra: define the comultiplication by $(\Delta f)(g, h) = f(gh)$ and check that Δf lies in $R(G) \otimes R(G)$ when $f = \varphi_{v,w}$ is a matrix coefficient, exhibit the counit and the antipode, and interpret each of the three structure maps as encoding a piece of the group structure (multiplication, the identity, inversion).

5. Let G be a compact *abelian* Lie group. Explain why every irreducible complex representation of G is one-dimensional, why the characters then form a group $\widehat{G}$ under pointwise multiplication, and how theorem 10.5.3 specializes to the Pontryagin duality statement $G \cong \widehat{\widehat{G}}$ in this case.

6. The universal cover $\widetilde{\mathrm{SL}(2,\mathbb{R})}$ of $\mathrm{SL}(2,\mathbb{R})$ is a connected Lie group with infinite center, and every finite-dimensional representation of it factors through a finite quotient of the center, hence is not faithful. Explain, referring to the proof of theorem 10.5.1, precisely which step fails for this group, and why compactness cannot be dropped from the hypothesis.

11

Maximal Tori and the Weyl Group

The Peter-Weyl theorem reduced the harmonic analysis of a compact group to its irreducible characters, but computing a character means understanding the conjugacy classes on which it is constant. The maximal torus is the instrument for that. In a compact connected group every element lies in a maximal torus, and all maximal tori are conjugate, so a class function is determined by its restriction to a single maximal torus T, a restriction invariant under the finite Weyl group $W = N(T)/T$. The whole representation theory of a compact group is thereby transported to the torus, where it becomes the combinatorics of characters of T acted on by W, and the abstract root systems of Part II reappear, now attached to a group rather than to a complex Lie algebra.

This chapter proves the maximal-torus conjugacy theorem by a degree argument: the map $G/T \times T \to G$ sending (gT, t) to gtg^{-1} has nonzero degree, which forces it to be onto and identifies the Weyl group as the order of its generic fibre. Complexifying the adjoint action of T splits $\mathfrak{g}_{\mathbb{C}}$ into root spaces, reproducing the root space decomposition of chapter 7 for the compact group, and $N(T)/T$ is finite and realizes the reflection group of that root system. The Jacobian of the conjugation map turns out to be the squared Weyl denominator, and integrating it yields the Weyl integration formula, which rewrites an integral over G as an integral over T against that density, the computational engine behind the character formula of chapter 12. The chapter ends by reading the fundamental group and the center of G off the lattices of weights and coroots, completing the passage from the group to its lattice data.

11.1 Tori and the Maximal Torus Theorem

For a compact group the analogue of a Cartan subalgebra is a maximal torus, a largest connected abelian closed subgroup. This section proves the maximal torus theorem: any two maximal tori are conjugate and their union is everything. The theorem reduces the analysis of G to that of T: because every element is conjugate into a fixed torus T, a class function on G is determined by its values on

T, and the abelian harmonic analysis of T replaces the noncommutative analysis of G.

The theorem is proved by a degree count. The conjugation map $q\colon G/T \times T \to G$, $(gT, t) \mapsto gtg^{-1}$, sends the product of a flag manifold and a torus to the group. Its image is the union of all conjugates of T, and its degree is the number of ways a generic element is hit, the order of the Weyl group. A map of nonzero degree between compact manifolds of equal dimension is surjective, and surjectivity of q is exactly the statement that every element is conjugate into T. The topological input, that a smooth map of nonzero degree is onto, is cited to [13, Basic Properties of the Degree]; the differential computation that pins the degree down is carried out here.

Definition 11.1.1: Torus, Maximal Torus, and Rank

Let G be a compact Lie group. A *torus* in G is a connected abelian closed subgroup $S \subseteq G$. By proposition 9.1.3, such an S is a compact connected abelian Lie group, hence isomorphic to the standard torus $\mathbb{T}^k = (\mathbb{R}/\mathbb{Z})^k$ with $k = \dim S$. A *maximal torus* of G is a torus T contained in no strictly larger torus. The *rank* of G is the dimension of a maximal torus,

$$\operatorname{rank} G = \dim T$$

which the maximal torus theorem (theorem 11.1.3) shows is independent of the choice of T.

Maximal tori exist for an elementary reason. The Lie algebra of any torus in G is an abelian subalgebra of $\mathfrak{g}$, and conversely every abelian subalgebra of $\mathfrak{g}$ integrates to a connected abelian subgroup whose closure is again a torus. Choosing an abelian subalgebra of largest dimension and closing up the subgroup it generates produces a maximal torus; since $\mathfrak{g}$ is finite-dimensional, a subalgebra of maximal dimension exists, so G has at least one maximal torus. The content of the theorem is that they are all conjugate.

Every $X \in \mathfrak{g}$ lies in some maximal abelian subalgebra, and $\exp(\mathbb{R}X)$ has closure a torus, so every one-parameter subgroup sits inside a maximal torus. At the level of algebras a maximal abelian subalgebra containing a given element always exists, and the theorem promotes this to the group and adds conjugacy.

Example 11.1.2: The Diagonal Torus in SU(2)

In SU(2) the diagonal matrices

$$T = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \;\middle|\; \theta \in \mathbb{R} \right\}$$

form a maximal torus. They are closed (the diagonal and unitarity conditions are closed), connected (parametrized by θ), and abelian (diagonal matrices commute), so T is a torus, isomorphic to $\mathbb{T}^1 = \mathrm{U}(1)$ by $\theta \mapsto e^{i\theta}$. It is maximal because the centralizer of a nonscalar element $\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ with $e^{i\theta} \neq e^{-i\theta}$ is exactly the diagonal subgroup T; nothing outside T commutes with all of T, so no larger abelian subgroup exists. Hence $\operatorname{rank}\mathrm{SU}(2) = 1$.

That every element of SU(2) is conjugate into T is the spectral theorem in disguise. A unitary matrix is diagonalizable by a unitary change of basis, and for $A \in \mathrm{SU}(2)$ with eigenvalues $e^{\pm i\theta}$ (they multiply to $\det A = 1$, so they are inverse to each other on the unit circle) there is $g \in \mathrm{SU}(2)$ with $gAg^{-1} = \operatorname{diag}(e^{i\theta}, e^{-i\theta}) \in T$. The maximal torus theorem is the coordinate-free form of this diagonalization.

The reduction of SU(2) to its diagonal torus is the model for the general theorem. The obstacle in general is that "diagonalize" has no a priori meaning; the substitute is a topological argument forcing the conjugation map to be onto. One paragraph fixes the standing hypothesis and the two facts the proof uses.

Throughout, G is compact and *connected*, and T is a fixed maximal torus with Lie algebra $\mathfrak{t}$. The proof needs two inputs. First, a torus has a *topological generator*: an element $t_0 \in T$ whose integer powers $\{t_0^n \mid n \in \mathbb{Z}\}$ are dense in T. This is the Kronecker density theorem, that a rotation of $\mathbb{T}^k$ by an angle vector with rationally independent coordinates generates a dense subgroup, and every torus $\mathbb{T}^k$ has such an element. Second, the differential of conjugation is the adjoint representation: for $t \in G$ the derivative at the identity of $x \mapsto txt^{-1}$ is $\mathrm{Ad}(t)$ on $\mathfrak{g}$.

Theorem 11.1.3: The Maximal Torus Theorem

Let G be a compact connected Lie group and $T \subseteq G$ a maximal torus. Then:

1. Every element of G is conjugate to an element of T; that is,
$$G = \bigcup_{g \in G} g\,T\,g^{-1}$$
2. Any two maximal tori of G are conjugate. In particular every maximal torus has dimension $\dim T$, so the rank of definition 11.1.1 is well defined.
3. The exponential map $\exp\colon \mathfrak{g} \to G$ is surjective, reproving theorem 9.1.6.

***Proof*:**
Consider the smooth map

$$q\colon G/T \times T \longrightarrow G, \qquad q(gT, t) = g\,t\,g^{-1}$$

It is well defined: if $g' = gh$ with $h \in T$, then $g'tg'^{-1} = ghth^{-1}g^{-1} = gtg^{-1}$ because T is abelian, so the value depends only on the coset gT. The quotient G/T is a compact manifold by theorem 4.1.3, so $G/T \times T$ is a compact manifold of dimension

$$\dim(G/T) + \dim T = (\dim G - \dim T) + \dim T = \dim G$$

equal to the dimension of the target. Both domain and target are compact and orientable, so the mapping degree $\deg(q)$ is defined ([13, The Degree Is Well Defined and Homotopy Invariant]). The strategy is to show $\deg(q) \neq 0$ by exhibiting a regular value with a nonzero, computable number of preimages, then to invoke that a smooth map of nonzero degree between closed oriented manifolds of equal dimension is surjective.

Step 1: the differential of q at a generating point. Fix a topological generator $t_0 \in T$, so $\overline{\{t_0^n \mid n \in \mathbb{Z}\}} = T$. Compute dq at (eT, t_0). Identify $T_{eT}(G/T) \cong \mathfrak{g}/\mathfrak{t}$ and, via left translation by t_0, identify $T_{t_0}T \cong \mathfrak{t}$ and $T_{t_0}G \cong \mathfrak{g}$, so

$$T_{(eT,t_0)}(G/T \times T) \cong \mathfrak{g}/\mathfrak{t} \oplus \mathfrak{t}$$

Two families of curves realize the two summands. Moving in the T-direction, $t = t_0 \exp(sH)$ with $H \in \mathfrak{t}$ gives $q(eT, t) = t_0 \exp(sH)$, whose velocity translated back by t_0^{-1} is H; the differential

is the identity on the $\mathfrak{t}$ summand. Moving in the G/T-direction, $g = \exp(sX)$ with $X \in \mathfrak{g}$ representing $X + \mathfrak{t}$ gives

$$q(\exp(sX)\,T, t_0) = \exp(sX)\,t_0\,\exp(-sX)$$

Differentiating at $s = 0$ and translating back by t_0^{-1} on the left, the velocity is $\mathrm{Ad}(t_0^{-1})X - X = (\mathrm{Ad}(t_0^{-1}) - 1)X$. Assembling the two directions, in the identification $T_{t_0}G \cong \mathfrak{g}$ the differential is

$$dq_{(eT,t_0)}(X + \mathfrak{t},\, H) = (\mathrm{Ad}(t_0^{-1}) - 1)X + H \tag{11.1}$$

acting as $\mathrm{Ad}(t_0^{-1}) - 1$ on $\mathfrak{g}/\mathfrak{t}$ and as the identity on $\mathfrak{t}$.

Step 2: t_0 is a regular value. The operator $\mathrm{Ad}(t_0^{-1})$ fixes $\mathfrak{t}$ pointwise: since $t_0 \in T$ and T is abelian, conjugation by t_0 fixes T, so its derivative fixes $\mathfrak{t}$. Hence $\mathrm{Ad}(t_0^{-1}) - 1$ vanishes on $\mathfrak{t}$ and descends to an operator A on $\mathfrak{g}/\mathfrak{t}$; the claim is that A is invertible. Complexify. The commuting family $\{\mathrm{Ad}(t) \big| t \in T\}$ decomposes $\mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}$ into simultaneous eigenspaces, on each of which T acts by a character (a *root* when nontrivial, taken up in section 11.3); the claim is that no trivial character occurs on the quotient, equivalently that the T-fixed subspace $\mathfrak{g}_0 = \{X \in \mathfrak{g} \mid \mathrm{Ad}(t)X = X \text{ for all } t \in T\}$ is exactly $\mathfrak{t}$. That $\mathfrak{t} \subseteq \mathfrak{g}_0$ is clear, since $\mathrm{Ad}(t)$ fixes $\mathfrak{t}$ pointwise; the reverse inclusion is where the maximality of T enters. Let $X \in \mathfrak{g}_0$. Then $t\exp(sX)t^{-1} = \exp(s\ \mathrm{Ad}(t)X) = \exp(sX)$ for every $t \in T$ and $s \in \mathbb{R}$, so the one-parameter subgroup $\exp(\mathbb{R}X)$ commutes with T; the product $\exp(\mathbb{R}X) \cdot T$ is therefore a connected abelian subgroup, and its closure is a torus containing T. By maximality of T this closure equals T, so $\exp(\mathbb{R}X) \subseteq T$ and $X \in \mathfrak{t}$. Hence $\mathfrak{g}_0 = \mathfrak{t}$, and the fixed subspace of T on $\mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}$ is zero, so every character occurring there is nontrivial. On the eigenspace with character α, the operator $\mathrm{Ad}(t_0^{-1})$ acts by $\alpha(t_0)^{-1}$, a value on the unit circle, so A acts by $\alpha(t_0)^{-1} - 1$. This scalar is nonzero unless $\alpha(t_0) = 1$; but if $\alpha(t_0) = 1$ then $\alpha \equiv 1$ on the dense set $\{t_0^n\}$, hence on all of T, contradicting that α is nontrivial. Therefore A has no zero eigenvalue and is invertible. By (11.1) the full differential $dq_{(eT,t_0)}$ is invertible, being the identity on $\mathfrak{t}$ and the invertible A on $\mathfrak{g}/\mathfrak{t}$. Thus q is a local diffeomorphism at (eT, t_0), and t_0 is a regular value.

Step 3: counting the preimages of t_0. A point $(gT, t) \in q^{-1}(t_0)$ satisfies $gtg^{-1} = t_0$, so $t = g^{-1}t_0g \in T$. Since $\overline{\{t_0^n\}} = T$ and conjugation is a continuous automorphism,

$$g^{-1}Tg = g^{-1}\,\overline{\{t_0^n\}}\,g = \overline{\{(g^{-1}t_0g)^n\}} = \overline{\{t^n\}} \subseteq T$$

so $g^{-1}Tg \subseteq T$, and equality holds by dimension. Thus $g \in N(T)$, the normalizer of T. Two normalizer elements give the same coset gT precisely when they differ by an element of T, so the admissible cosets are indexed by $N(T)/T$, and each coset determines the torus element $t = g^{-1}t_0g$. Hence

$$|q^{-1}(t_0)| = |N(T)/T|$$

This is finite: $N(T)/T$ is a compact Lie group whose identity component is trivial. Its Lie algebra is $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{t})/\mathfrak{t}$, where $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{t}) = \{X \in \mathfrak{g} \mid [X, \mathfrak{t}] \subseteq \mathfrak{t}\}$ is the normalizer of $\mathfrak{t}$; here the normalizer reduces to the centralizer, since an X with a nonzero root component along some α would satisfy $[\mathfrak{t}, X] \not\subseteq \mathfrak{t}$, so $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{g}_0 = \mathfrak{t}$ by Step 2. Thus its Lie algebra is zero, and a compact group with trivial identity component is finite. Write $|W| := |N(T)/T|$; it is the order of the Weyl group of section 11.2.

Step 4: nonzero degree and surjectivity. At each preimage q is a local diffeomorphism, by the Step 2 computation applied at (gT, t), where $t = g^{-1}t_0g$ is again a topological generator of T (a conjugate of a generator generates the conjugate torus). The degree is the signed count of

preimages of a regular value, each counted with the sign of the determinant of dq in oriented bases; no cancellation occurs because these local signs all agree. Indeed, by (11.1) the differential at a preimage (gT, t) is the identity on $\mathfrak{t}$ and $\mathrm{Ad}(t^{-1}) - 1$ on $\mathfrak{g}/\mathfrak{t}$, so the local sign is the sign of $\det\big(\mathrm{Ad}(t^{-1}) - 1\big|_{\mathfrak{g}/\mathfrak{t}}\big)$. Complexifying and decomposing $\mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}$ into the root eigenspaces of T (Step 2), this determinant is

$$\det\big(\mathrm{Ad}(t^{-1}) - 1\big|_{\mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}}\big) = \prod_{\alpha} \big(\alpha(t)^{-1} - 1\big)$$

the product over all roots α. The roots occur in pairs $\pm\alpha$, and each root value $\alpha(t)$ lies on the unit circle, so $\alpha(t)^{-1} = \overline{\alpha(t)}$; pairing α with $-\alpha$ gives

$$\big(\alpha(t)^{-1} - 1\big)\big(\alpha(t) - 1\big) = \big(\overline{\alpha(t)} - 1\big)\big(\alpha(t) - 1\big) = |\alpha(t) - 1|^2$$

Hence the determinant is $\prod_{\alpha>0}|\alpha(t) - 1|^2$, a positive real number since each $\alpha(t) \neq 1$ (Step 2: α is nontrivial and t generates T). The real determinant on $\mathfrak{g}/\mathfrak{t}$ equals this positive complex value, so $\det dq > 0$ at every preimage, and the local signs are therefore all equal, with no cancellation. Therefore

$$\deg(q) = \pm\,|N(T)/T| = \pm\,|W| \neq 0$$

A smooth map of nonzero degree between closed oriented manifolds of equal dimension is surjective ([13, Basic Properties of the Degree]): a point outside the image would lie in the open complement of the compact image, hence be a regular value with empty preimage, forcing $\deg(q) = 0$. Since $\deg(q) \neq 0$, the map q is surjective, and its image is $\bigcup_g gTg^{-1}$. This proves part (1).

Step 5: conjugacy of maximal tori. Let S be any maximal torus and $s_0 \in S$ a topological generator. By part (1) there is $g \in G$ with $gs_0g^{-1} \in T$. Then

$$gSg^{-1} = g\,\overline{\{s_0^n\}}\,g^{-1} = \overline{\{(gs_0g^{-1})^n\}} \subseteq T$$

so the maximal torus gSg^{-1} is contained in the torus T; maximality forces $gSg^{-1} = T$, whence $S = g^{-1}Tg$. Both tori then have the same dimension, so $\mathrm{rank}G = \dim T$ is well defined, proving part (2).

Step 6: surjectivity of exp. Given $g \in G$, part (1) gives $h \in G$ and $t \in T$ with $g = hth^{-1}$. Since T is a connected abelian compact group, its own exponential $\exp\colon \mathfrak{t} \to T$ is surjective (proposition 9.1.3), so $t = \exp(H)$ for some $H \in \mathfrak{t} \subseteq \mathfrak{g}$. By naturality of exp under the conjugation automorphism,

$$g = h\exp(H)h^{-1} = \exp(\mathrm{Ad}(h)H)$$

so g is in the image of $\exp\colon \mathfrak{g} \to G$. Hence exp is surjective, reproving theorem 9.1.6 and completing the proof.

The theorem is the analytic counterpart of the algebraic fact that all Cartan subalgebras of a complex semisimple Lie algebra are conjugate under its inner automorphisms, and theorem 13.2.7 deduces that fact from it through the compact real form. Here the conjugating elements lie in the group G, and the argument is topological, a dimension count and a degree. The map q has image the union of conjugates of T, so its surjectivity, extracted from the nonvanishing of its degree, is the conjugacy statement. The order $|W| = |N(T)/T|$, which surfaced as the number of preimages of a generic point, is the same Weyl group that acts on the root system, and section 11.2 takes it up in its own right.

Surjectivity of exp falls out as a corollary, but the theorem gives more than theorem 9.1.6 did. The Hopf-Rinow proof produced *some* X with $g = \exp X$; the present one locates X inside a conjugate of the fixed abelian $\mathfrak{t}$, so the exponential is onto through the maximal torus. Every element sits on a one-parameter subgroup lying in a maximal torus, which is what makes the torus the right coordinate system for the group.

The consequential corollary is the reduction of class functions to the torus. A class function is constant on conjugacy classes, and part (1) says every class meets T, so a class function is pinned down by its restriction to T.

Corollary 11.1.4: Every Conjugacy Class Meets T

Let G be a compact connected Lie group and T a maximal torus. Then every conjugacy class of G intersects T. Consequently a continuous class function $f\colon G \to \mathbb{C}$, one satisfying $f(gxg^{-1}) = f(x)$ for all g, x, is determined by its restriction $f|_T$: if two continuous class functions agree on T, they agree on G.

***Proof*:**
Let $x \in G$ with conjugacy class $C = \{gxg^{-1} \mid g \in G\}$. By part (1) of theorem 11.1.3 there is $g \in G$ with $gxg^{-1} \in T$, so $C \cap T \neq \emptyset$. Now let f be a continuous class function and $x \in G$ arbitrary. Choosing g with $t := gxg^{-1} \in T$, the class-function property gives

$$f(x) = f(gxg^{-1}) = f(t) = (f|_T)(t)$$

so $f(x)$ is computed from $f|_T$ at any torus point conjugate to x. If $f_1|_T = f_2|_T$ for two class functions, then $f_1(x) = f_1(t) = f_2(t) = f_2(x)$ for every x, so $f_1 = f_2$, as we sought to show.

This is the reduction that powers the character theory to come. The characters of definition 10.4.1 are class functions, an orthonormal basis of the class functions by theorem 10.4.4, and the corollary recovers each from its restriction to T. On T a character is a finite combination of the characters of the abelian group T, a Laurent polynomial in the torus coordinates, so computing a character on all of G reduces to computing a Laurent polynomial on T. The restriction is not injective on functions, because distinct points of T can be conjugate in G; the residual identification is the action of the Weyl group, so class functions correspond not to all functions on T but to the W-invariant ones. That refinement is proposition 11.2.4, once the Weyl group is in hand in section 11.2, and the Weyl character formula of chapter 12 then computes these W-invariant functions explicitly.

The mechanism specializes to the two families that anchor the chapter, the unitary and special unitary groups, where the maximal torus theorem is literally the spectral theorem.

Example 11.1.5: The Diagonal Torus in $\mathrm{U}(n)$ and $\mathrm{SU}(n)$
In $\mathrm{U}(n)$ the diagonal unitary matrices

$$T = \{\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) \mid \theta_1, \dots, \theta_n \in \mathbb{R}\}$$

form a maximal torus, isomorphic to $\mathbb{T}^n = (\mathrm{U}(1))^n$. It is closed, connected, and abelian, so a torus; it is maximal because a matrix commuting with every diagonal matrix having distinct entries is itself diagonal, so the centralizer of a generic element of T is T, leaving no room for a larger torus. Hence $\operatorname{rank} \mathrm{U}(n) = n$. The maximal torus theorem here is the spectral theorem for unitary matrices: every $A \in \mathrm{U}(n)$ is unitarily diagonalizable, so there is $g \in \mathrm{U}(n)$ with

$gAg^{-1} = \operatorname{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) \in T$. Conjugacy of maximal tori is the statement that any two orthonormal eigenbases differ by a unitary change of basis.

For $\mathrm{SU}(n)$ the maximal torus is the diagonal subgroup cut by the determinant-one condition,

$$T = \left\{ \operatorname{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) \;\middle|\; \textstyle\sum_j \theta_j \in 2\pi\mathbb{Z} \right\}$$

an $(n-1)$-dimensional torus, so $\operatorname{rank}\mathrm{SU}(n) = n - 1$: the single constraint $\prod_j e^{i\theta_j} = 1$ drops the dimension by one. A special unitary matrix is again unitarily diagonalizable, its eigenvalues multiplying to $\det = 1$, so it is conjugate in $\mathrm{SU}(n)$ to a diagonal matrix in T. For $n = 2$ this recovers example 11.1.2: the constraint $\theta_1 + \theta_2 \in 2\pi\mathbb{Z}$ makes $\theta_2 = -\theta_1$ on the identity component, giving $\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ and $\operatorname{rank}\mathrm{SU}(2) = 1$.

The two examples display the theorem as a coordinate-free spectral theorem and fix the rank computations $\operatorname{rank}\mathrm{U}(n) = n$ and $\operatorname{rank}\mathrm{SU}(n) = n - 1$ that the exercises extend to the orthogonal groups. The diagonal torus is the maximal torus because it is its own centralizer, the group-level analogue of a Cartan subalgebra being self-centralizing (proposition 7.1.7). The self-centralizing property returns in section 11.2 as the identity $C_G(T) = T$, from which the finiteness of the Weyl group follows.

Exercise 11.1.1

1. Let G be a compact Lie group and $X \in \mathfrak{g}$. Show that the closure $\overline{\exp(\mathbb{R}X)}$ of the one-parameter subgroup is a torus in G. Deduce that $\exp X$ lies in some maximal torus, and combine this with theorem 11.1.3(3) to give a second proof that every element of a compact connected G lies in a maximal torus.

2. Compute the rank of each of $\mathrm{SU}(n)$, $\mathrm{SO}(2n)$, and $\mathrm{SO}(2n+1)$ by exhibiting a maximal torus explicitly. For the orthogonal groups use block-diagonal matrices built from 2×2 rotation blocks $R(\theta) = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, and confirm that $\operatorname{rank}\mathrm{SO}(2n) = \operatorname{rank}\mathrm{SO}(2n+1) = n$.

3. Prove that the centralizer $C_G(T) = \{g \in G \mid gt = tg \text{ for all } t \in T\}$ of a maximal torus T in a compact connected group G equals T itself. (Hint: take $g \in C_G(T)^0$, place it in a maximal torus S of the compact connected group $C_G(T)^0$, and show $S \cdot T$ is a torus containing T, so $S \subseteq T$ by maximality.)

4. Suppose G is a compact connected Lie group of rank 1, so $\dim T = 1$. Using that the complexified adjoint action decomposes $\mathfrak{g}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_\alpha \mathfrak{g}_\alpha$ and that a rank-one root system has a single positive root with one-dimensional root spaces (theorem 7.4.6), argue that $\dim G \in \{1, 3\}$, and conclude that G is one of $\mathrm{U}(1)$, $\mathrm{SU}(2)$, or $\mathrm{SO}(3)$.

5. In the proof of theorem 11.1.3 the number of preimages of a topological generator t_0 under q was $|N(T)/T|$. Explain why a non-generating regular value $t_1 \in T$, one whose powers are not dense, can have more preimages, or fail to be a regular value at all, and why neither possibility affects the degree computation. (Hint: the degree is a single integer computed at *any* one regular value; the generator is chosen to make the count clean.)

11.2 The Weyl Group $N(T)/T$

The conjugacy theorem of section 11.1 reduced a compact connected group G to a single maximal torus T, but not to a coordinate system: the conjugation carrying an element into T is far from unique. An element already in T can be moved to a different element of T by conjugating with something in G that preserves T as a set, and these residual symmetries are the ambiguity left over after the reduction. This section isolates them as a finite group, the Weyl group, and shows that it is the whole ambiguity: two elements of T are conjugate in G precisely when the Weyl group relates them. Its finiteness is what makes T a genuine coordinate torus rather than an unwieldy slice.

The symmetries preserving T form the normalizer of T in G. The torus itself sits inside the normalizer and acts trivially by conjugation, since T is abelian, so the group acting effectively on T is the quotient of the normalizer by T.

Definition 11.2.1: The Weyl Group of a Compact Group

Let G be a compact connected Lie group and $T \leq G$ a maximal torus (definition 11.1.1). The *normalizer* of T in G is the closed subgroup

$$N(T) = \{g \in G \mid gTg^{-1} = T\}$$

of all elements whose conjugation preserves T. The *Weyl group* of G (with respect to T) is the quotient group

$$W = W(G,T) = N(T)/T$$

Each coset $wT \in W$ acts on T by conjugation, $t \mapsto wtw^{-1}$, well-defined because T acts trivially on itself by conjugation, and this action is by automorphisms of the Lie group T. Differentiating at the identity gives a linear action of W on the Lie algebra $\mathfrak{t} = \mathrm{Lie}(T)$, and dualizing gives an action on the character lattice $X^*(T) = \mathrm{Hom}(T, \mathrm{U}(1))$ and on any other W-invariant structure attached to T.

The definition packages three things at once. First, W is a group, being the quotient of $N(T)$ by the normal subgroup T; normality is immediate, since $N(T)$ is precisely the stabilizer of T under conjugation and T is normal in its own stabilizer. Second, W *acts*, and an element of $N(T)$ acting trivially on T is precisely one centralizing T, so the action of W on T is faithful exactly when every element centralizing T already lies in T, that is when the centralizer $Z_G(T)$ equals T. This holds: an element centralizing all of T has an adjoint action fixing $\mathfrak{t}$ pointwise, so its infinitesimal directions lie in the joint zero-eigenspace of $\mathrm{ad}(\mathfrak{t})$, which maximality forces to be $\mathfrak{t}$ itself (the eigenspace computation of theorem 11.2.3); the connected centralizer $Z_G(T)^0$ therefore has Lie algebra $\mathfrak{t}$ and equals T, and in a compact connected group the centralizer of a torus is connected, so $Z_G(T) = T$. Hence no coset of T other than the identity acts trivially and W embeds in the automorphisms of T. Third, the action propagates by functoriality to $\mathfrak{t}$, to $X^*(T)$, and to everything built from these, so that a single finite group governs the symmetries of the torus, its Lie algebra, and its character theory at once. This is the same group $W = \langle s_\alpha \rangle$ that chapter 7 attached to the abstract root system of $\mathfrak{g}_{\mathbb{C}}$ (definition 7.5.3); the identification of the two is completed in section 11.3, once the roots of (G,T) are in hand.

Before proving anything about W, the running example shows it where every object is a 2×2 matrix and the group is as small as it can be.

Example 11.2.2: The Weyl Group of SU(2)

For $G = \mathrm{SU}(2)$ the maximal torus is the diagonal circle

$$T = \left\{ d(\theta) = \mathrm{diag}(e^{i\theta}, e^{-i\theta}) \;\middle|\; \theta \in \mathbb{R} \right\}$$

of section 11.1. To compute $N(T)$, ask which $g \in \mathrm{SU}(2)$ conjugates T into itself. Conjugation by g preserves T if and only if it permutes the two diagonal entries $e^{\pm i\theta}$, since these are the eigenvalues of $d(\theta)$ and g acts on eigenspaces. The identity coset leaves the entries fixed; the only other possibility swaps them, realized by

$$w = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in \mathrm{SU}(2)$$

which satisfies

$$w\, d(\theta)\, w^{-1} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \left(\begin{smallmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{smallmatrix}\right) \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{smallmatrix}\right) = d(-\theta)$$

so w normalizes T and acts on it by the reflection $\theta \mapsto -\theta$. The normalizer is $N(T) = T \cup wT$, and the Weyl group is

$$W(\mathrm{SU}(2), T) = N(T)/T = \{1, \bar{w}\} \cong \mathbb{Z}/2\mathbb{Z}$$

generated by the class $\bar{w}$ of w. On the Lie algebra $\mathfrak{t} \cong \mathbb{R}$, coordinatized by θ, the nontrivial element acts as $\theta \mapsto -\theta$, the reflection through the origin; on the character lattice $X^*(T) \cong \mathbb{Z}$, where the character $d(\theta) \mapsto e^{in\theta}$ is the integer n, it acts as $n \mapsto -n$. This is the reflection s_α (definition 7.5.1) in the single positive root α of A_1, recovering the abstract Weyl group of $\mathfrak{sl}_2(\mathbb{C})$ from chapter 7.

The example already exhibits the two facts the general theory must establish: the normalizer $N(T)$ is a finite union of cosets of T (here two), and the Weyl group is finite (here of order two). Both follow from a single structural statement, that T is the entire identity component of its own normalizer. The reason is infinitesimal. An element of the Lie algebra of $N(T)$ is a direction in G along which T is preserved; the only such directions are $\mathfrak{t}$ itself, because any component of such a direction lying outside $\mathfrak{t}$ sits in a root space, where T acts by a nonzero rotation, and the condition of preserving $\mathfrak{t}$ forces that component to vanish. The maximality of T is what leaves no room for a direction fixed by T beyond $\mathfrak{t}$, so nothing outside $\mathfrak{t}$ survives.

The thread is that maximality of T has an infinitesimal consequence, and the infinitesimal consequence controls the whole normalizer. Chapter 7 recorded that maximality of a Cartan subalgebra forces it to be self-centralizing. The compact-group statement is the analogous fact for the normalizer, phrased through the adjoint action.

Theorem 11.2.3: Finiteness of the Weyl Group

Let G be a compact connected Lie group and $T \leq G$ a maximal torus. Then the identity component of the normalizer is T itself,

$$N(T)^0 = T$$

Consequently the Weyl group $W = N(T)/T$ is the group of connected components of the compact group $N(T)$, and it is finite.

***Proof*:**
The proof has two movements: first that $N(T)$ is a compact Lie group, so that W is its component group, and then that this component group is finite because $N(T)^0 = T$.

Step 1: $N(T)$ is a compact Lie group. The normalizer $N(T) = \{g \mid gTg^{-1} = T\}$ is closed in G: if $g_k \to g$ with each $g_k T g_k^{-1} = T$, then for every $t \in T$ the limit $gtg^{-1} = \lim_k g_k t g_k^{-1}$ lies in the closed set T, and the same applied to g^{-1} gives $gTg^{-1} = T$, so $g \in N(T)$. A closed subgroup of the compact group G is compact, and by the closed-subgroup theorem (theorem 3.2.1) it is an embedded Lie subgroup. Its identity component $N(T)^0$ is a compact connected Lie group containing the connected subgroup T, since T normalizes itself and is connected.

Step 2: the Lie algebra of $N(T)$ is $\mathfrak{t}$. Write $\mathfrak{n} = \mathrm{Lie}(N(T))$. An element $X \in \mathfrak{g}$ lies in $\mathfrak{n}$ if and only if the one-parameter subgroup $\exp(sX)$ normalizes T for all s, that is $\exp(sX)\, T\, \exp(-sX) = T$. Differentiating the containment $\exp(sX)\, T\, \exp(-sX) \subseteq T$ shows $\mathrm{Ad}(\exp(sX))$ preserves $\mathfrak{t}$ for all s, and differentiating once more at $s = 0$ gives $[\,X, \mathfrak{t}\,] \subseteq \mathfrak{t}$; conversely $[\,X, \mathfrak{t}\,] \subseteq \mathfrak{t}$ integrates back to the group statement because $\mathfrak{t}$ is the Lie algebra of the closed connected subgroup T. Thus

$$\mathfrak{n} = \mathfrak{n}_{\mathfrak{g}}(\mathfrak{t}) = \{X \in \mathfrak{g} \mid [\,X, H\,] \in \mathfrak{t} \text{ for all } H \in \mathfrak{t}\}$$

is the normalizer of $\mathfrak{t}$ in $\mathfrak{g}$. It remains to show $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{t}$.

Decompose $\mathfrak{g}$ into eigenspaces for the action of T. Fix an $\mathrm{Ad}(G)$-invariant inner product on $\mathfrak{g}$, obtained by averaging any inner product over G against the invariant integral (proposition 9.2.4); such a product exists for every compact G, whether or not $\mathfrak{g}$ is semisimple. For each $H \in \mathfrak{t}$ the operator $\mathrm{ad}(H)$ is skew-symmetric with respect to this inner product, because $\mathrm{Ad}(\exp(sH))$ is orthogonal and $\mathrm{ad}(H)$ is its infinitesimal generator. A commuting family of skew-symmetric operators is simultaneously diagonalizable over $\mathbb{C}$ with purely imaginary eigenvalues, so $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ splits as a direct sum of joint eigenspaces

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha} \mathfrak{g}_{\alpha}$$

where $\mathfrak{t}_{\mathbb{C}}$ is the joint zero-eigenspace, the vectors killed by every $\mathrm{ad}(H)$, and each $\mathfrak{g}_{\alpha}$ is a joint eigenspace on which $\mathrm{ad}(H)$ acts by the scalar $i\,\alpha(H)$ for a nonzero real linear functional α on $\mathfrak{t}$. That the zero-eigenspace is exactly $\mathfrak{t}_{\mathbb{C}}$, no larger, is the maximality of T: an element of $\mathfrak{g}$ commuting with all of $\mathfrak{t}$ would generate, together with $\mathfrak{t}$, an abelian subalgebra, hence a torus containing T, which by maximality forces the element into $\mathfrak{t}$.

Now suppose $X \in \mathfrak{n}_{\mathfrak{g}}(\mathfrak{t})$, and write its complexification as $X = X_0 + \sum_{\alpha} X_{\alpha}$ with $X_0 \in \mathfrak{t}_{\mathbb{C}}$ and $X_{\alpha} \in \mathfrak{g}_{\alpha}$. For $H \in \mathfrak{t}$,

$$[\,X, H\,] = -[\,H, X\,] = -\sum_{\alpha} i\,\alpha(H)\, X_{\alpha}$$

since $\mathrm{ad}(H)$ kills X_0 and scales each X_{α} by $i\,\alpha(H)$. The normalizer condition $[\,X, H\,] \in \mathfrak{t}_{\mathbb{C}}$ requires this sum to lie in $\mathfrak{t}_{\mathbb{C}}$; but the terms $i\,\alpha(H)\, X_{\alpha}$ lie in the spaces $\mathfrak{g}_{\alpha}$, which meet $\mathfrak{t}_{\mathbb{C}}$ only in 0. Hence $\alpha(H)\, X_{\alpha} = 0$ for every α and every $H \in \mathfrak{t}$. Choosing H with $\alpha(H) \neq 0$, possible because $\alpha \neq 0$, forces $X_{\alpha} = 0$ for every α, so $X = X_0 \in \mathfrak{t}_{\mathbb{C}}$. Intersecting with the real form gives $X \in \mathfrak{t}$. Therefore $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{t}$, and $\mathrm{Lie}(N(T)) = \mathfrak{t}$.

Step 3: finiteness. The identity component $N(T)^0$ is the connected Lie subgroup of G with Lie algebra $\mathrm{Lie}(N(T)) = \mathfrak{t}$, which is the connected subgroup T itself. Thus $N(T)^0 = T$. The Weyl group is therefore the component group

$$W = N(T)/T = N(T)/N(T)^0 = \pi_0(N(T))$$

The components of $N(T)$ are the cosets of the open subgroup $N(T)^0 = T$, and there are finitely many because $N(T)$ is compact while each coset is open, so a compact space is covered by finitely many of these disjoint open cosets. Hence W is finite, completing the proof.

The argument shows that the only obstruction to $N(T)$ being T is discrete, a finite set of components, and that the discreteness is forced by maximality of T through the absence of a zero eigenvalue off $\mathfrak{t}$. The eigenspace decomposition appearing in the middle of the proof is not incidental: it is the root space decomposition of $\mathfrak{g}_{\mathbb{C}}$ that section 11.3 develops in full, and the functionals α are the roots of (G, T). The proof of finiteness has thus already met the roots, using only that they are nonzero on $\mathfrak{t}$, which is all maximality provides. That they form a root system, and that W acts on them as the reflection group, is the business of the next section.

Finiteness of W turns the residual conjugation ambiguity of the maximal torus theorem into a finite group, and the next result identifies that ambiguity precisely. The maximal torus theorem said every conjugacy class meets T; the question left open is how many times, and along which internal symmetries of T the intersection points are related. The answer is that they are related exactly by W.

The point is that conjugacy in G, restricted to T, collapses to a finite equivalence relation. Two elements of T that are G-conjugate need not be equal, but the discrepancy is governed by the finite group W, so classifying conjugacy classes of G becomes the finite combinatorial problem of classifying W-orbits on T.

Proposition 11.2.4: Conjugacy in T is Weyl Conjugacy

Let G be a compact connected Lie group and $T \leq G$ a maximal torus. Two elements $t_1, t_2 \in T$ are conjugate in G if and only if they are conjugate under the Weyl group: there exists $w \in W = N(T)/T$ with $t_2 = w t_1 w^{-1}$. Consequently the conjugacy classes of G are in bijection with the orbits of W on T, and restriction $f \mapsto f|_T$ is a bijection

$$\{\text{class functions on } G\} \xrightarrow{\sim} \{W\text{-invariant functions on } T\}$$

***Proof*:**

One direction is immediate. If $w \in N(T)$ represents a Weyl group element and $t_2 = w t_1 w^{-1}$, then t_1 and t_2 are conjugate in G by $w \in G$, and the value $w t_1 w^{-1}$ depends only on the coset wT because T is abelian: replacing w by $w\tau$ with $\tau \in T$ gives $w\tau t_1 \tau^{-1} w^{-1} = w t_1 w^{-1}$. So W-conjugacy implies G-conjugacy.

The forward direction. Suppose $g t_1 g^{-1} = t_2$ for some $g \in G$; the task is to replace g by an element of $N(T)$. Set $t = t_2$ and consider the centralizer

$$Z_G(t) = \{x \in G \mid xtx^{-1} = t\}$$

a closed, hence compact, subgroup of G, and let $Z = Z_G(t)^0$ be its identity component, a compact connected Lie group. Both t_1 and t_2 lie in T, and T is abelian, so T centralizes $t = t_2$; thus $T \subseteq Z_G(t)$, and since T is connected, $T \subseteq Z$. On the other hand $g t_1 g^{-1} = t_2 = t$ shows g conjugates t_1 to t, and t_1 is centralized by T, so gTg^{-1} centralizes $g t_1 g^{-1} = t$; that is, $gTg^{-1} \subseteq Z_G(t)$, and again by connectedness $gTg^{-1} \subseteq Z$.

The two subgroups T and gTg^{-1} are both maximal tori of the compact connected group Z. Each is a torus in Z, and each is maximal already in the larger group G, hence a fortiori maximal in

Z: a torus of Z strictly containing T would be a torus of G strictly containing T, contradicting maximality in G. Applying the maximal torus theorem (theorem 11.1.3) *inside* the compact connected group Z, the two maximal tori T and gTg^{-1} of Z are conjugate in Z: there is $z \in Z$ with

$$z\,(gTg^{-1})\,z^{-1} = T$$

Then the element $w := zg$ satisfies $wTw^{-1} = zgTg^{-1}z^{-1} = T$, so $w \in N(T)$. Moreover, since $z \in Z = Z_G(t)^0 \subseteq Z_G(t)$ centralizes $t = t_2$,

$$wt_1w^{-1} = z\,(gt_1g^{-1})\,z^{-1} = z\,t_2\,z^{-1} = t_2$$

Thus the class $\bar{w} \in W$ satisfies $\bar{w}\cdot t_1 = t_2$, and t_1, t_2 are W-conjugate. This proves the equivalence.

The corollary on class functions. A class function on G is a function constant on G-conjugacy classes. By the maximal torus theorem every conjugacy class meets T (corollary 11.1.4), so a class function is determined by its restriction to T, giving injectivity of $f \mapsto f|_T$. The restriction is W-invariant: for $t \in T$ and $w \in W$, the elements t and wtw^{-1} are G-conjugate, so $f(t) = f(wtw^{-1})$, meaning $f|_T$ is constant on W-orbits. Conversely, given a W-invariant function h on T, define $\tilde{h}$ on G by $\tilde{h}(g) = h(t)$ where t is any element of T conjugate to g; this is well-defined precisely because two elements of T conjugate in G are W-conjugate by the equivalence just proved, and h is constant on W-orbits, so the choice of t does not matter. The function $\tilde{h}$ is a class function restricting to h, giving surjectivity. Hence restriction is a bijection between class functions on G and W-invariant functions on T, and the conjugacy classes of G correspond to the W-orbits on T, as we sought to show.

The proposition converts a question about conjugacy in the full group G, a manifold of possibly large dimension, into a question about the action of a *finite* group W on the torus T. Everything invariant under conjugation, characters foremost among them, is thereby reduced to something W-invariant on T: the characters χ_λ of definition 10.4.1, being class functions, are exactly the W-invariant functions on T they restrict to, and the Weyl character formula of chapter 12 will write each of them as an explicit W-symmetric expression in the coordinates of T. The bijection also anticipates the Weyl integration formula of section 11.4: an integral of a class function over G can be pushed down to an integral over T, and the finite-to-one nature of the conjugation map, with its $|W|$-element generic fibre, is the source of the factor $1/|W|$ appearing there.

The forward direction turns on one move. The obstruction to promoting an arbitrary conjugating element g to one normalizing T was resolved by passing to the centralizer of the target and applying the maximal torus theorem there, one dimension down. The same "conjugate inside the centralizer" move computes the Weyl group of $\mathrm{U}(n)$ below (example 11.2.5), where the centralizer of a diagonal element with distinct entries is again the diagonal torus, and it recurs in the reduction to the centralizer of a torus in section 11.3.

The abstract statement is best absorbed through the family $\mathrm{U}(n)$ and $\mathrm{SU}(n)$, where the Weyl group is the symmetric group and its action on T is literal permutation of coordinates.

Example 11.2.5: The Weyl Group of $\mathrm{U}(n)$ and $\mathrm{SU}(n)$

For $G = \mathrm{U}(n)$ the maximal torus is the diagonal subgroup

$$T = \{\mathrm{diag}(z_1, \ldots, z_n) \mid z_j \in \mathrm{U}(1)\}$$

of section 11.1. An element $g \in \mathrm{U}(n)$ normalizes T if and only if conjugation by g permutes the diagonal entries, that is, if and only if g carries each coordinate axis $\mathbb{C}e_j$ to some coordinate axis

$\mathbb{C}e_{\sigma(j)}$; such a g is a permutation matrix times a diagonal unitary. Every permutation $\sigma \in S_n$ is realized by the permutation matrix P_σ with $P_\sigma e_j = e_{\sigma(j)}$, which is unitary and satisfies

$$P_\sigma \operatorname{diag}(z_1, \dots, z_n)\, P_\sigma^{-1} = \operatorname{diag}(z_{\sigma^{-1}(1)}, \dots, z_{\sigma^{-1}(n)})$$

permuting the diagonal entries by σ. The normalizer is $N(T) = \{P_\sigma D \mid \sigma \in S_n,\ D \in T\}$, the monomial unitary matrices, and modding by the diagonal T leaves exactly the permutations:

$$W(\mathrm{U}(n), T) = N(T)/T \cong S_n$$

acting on $T \cong (\mathrm{U}(1))^n$ by permuting the n circle factors, and on $\mathfrak{t} \cong \mathbb{R}^n$ and $X^*(T) \cong \mathbb{Z}^n$ by permuting coordinates.

For $G = \mathrm{SU}(n)$ the maximal torus is the codimension-one subtorus $z_1 \cdots z_n = 1$, and the same permutation matrices work, except that P_σ must be scaled to have determinant 1. Since $\det(P_\sigma) = \operatorname{sgn}(\sigma) = \pm 1$, it suffices to rescale a single nonzero entry of P_σ from 1 to $\operatorname{sgn}(\sigma)$; this multiplies the determinant by $\operatorname{sgn}(\sigma)$, landing the matrix in $\mathrm{SU}(n)$ while it still induces the same permutation of the diagonal. The class in $N(T)/T$ is unchanged by this adjustment, so

$$W(\mathrm{SU}(n), T) \cong S_n$$

as well, of order $n!$, acting by permuting the coordinates $\varepsilon_1, \dots, \varepsilon_n$ of $\mathfrak{t}$ subject to $\varepsilon_1 + \cdots + \varepsilon_n = 0$. This is the Weyl group of the root system A_{n-1}, whose roots $\varepsilon_i - \varepsilon_j$ and reflections $s_{\varepsilon_i - \varepsilon_j}$ (the transpositions $(i\,j)$) are exactly those found for $\mathfrak{sl}_n(\mathbb{C})$ in chapter 7. The case $n = 2$ recovers example 11.2.2: $S_2 = \mathbb{Z}/2\mathbb{Z}$ swapping $z_1 \leftrightarrow z_2$, that is $\theta \mapsto -\theta$ on the circle $z_1 = e^{i\theta} = z_2^{-1}$.

The two examples display the same principle at two sizes: the Weyl group is the group of those permutations of the "eigenvalue slots" of the torus that are realized by conjugation by an element of G, and for the unitary groups every permutation is realized, so $W = S_n$. That $|W(\mathrm{SU}(n))| = n!$ and that W acts by permuting coordinates are the facts a computation in chapter 12 leans on when it symmetrizes over the Weyl group. The identification of W with the abstract reflection group of the root system, glimpsed here through $s_{\varepsilon_i - \varepsilon_j} = (i\,j)$, is proved in general in section 11.3.

Exercise 11.2.1

1. Show that the Weyl group of $\mathrm{SO}(3)$ is $\mathbb{Z}/2\mathbb{Z}$. Take the maximal torus T to be the group of rotations about a fixed axis (the z-axis), $T = \{R_z(\theta) \mid \theta \in \mathbb{R}\} \cong \mathrm{U}(1)$, and exhibit an element of $\mathrm{SO}(3)$ normalizing T and acting on it by $\theta \mapsto -\theta$. Explain why no element of $\mathrm{SO}(3)$ can act on T in any other nontrivial way, and relate the result to $W(\mathrm{SU}(2)) = \mathbb{Z}/2\mathbb{Z}$ via the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

2. Prove that $|W(\mathrm{SU}(n), T)| = n!$, where T is the diagonal maximal torus. Identify W with S_n acting by permuting the diagonal entries, and check that the permutation matrices (adjusted to determinant 1) exhaust $N(T)/T$ by arguing that any element normalizing T must permute the coordinate axes $\mathbb{C}e_1, \dots, \mathbb{C}e_n$.

3. A *Weyl chamber* is a connected component of the complement in $\mathfrak{t}$ of the union of the reflection hyperplanes $\ker \alpha$ of the roots. Using the abstract fact that W acts simply transitively on the Weyl chambers (theorem 8.2.8), show that the number of Weyl chambers equals $|W|$. For $\mathrm{SU}(3)$, whose root system is A_2 with $|W| = 6$, confirm that the six chambers are the six sectors cut out of the plane $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$ by the three lines $\varepsilon_i = \varepsilon_j$.

4. Call $t \in T$ *regular* if it lies on no reflection hyperplane, equivalently if $\log t \in \mathfrak{t}$ satisfies $\alpha(\log t) \neq 0$ for every root α. Show that a regular element has trivial stabilizer in W: if $w \in W$ fixes a regular t, then $w = 1$. Conclude that the W-orbit of a regular element has exactly $|W|$ elements, so by proposition 11.2.4 the conjugacy class of a regular t meets T in exactly $|W|$ points.

5. The character lattice of the diagonal torus $T \leq \mathrm{U}(n)$ is $X^*(T) \cong \mathbb{Z}^n$, with $(m_1, \ldots, m_n)$ corresponding to $\mathrm{diag}(z_1, \ldots, z_n) \mapsto z_1^{m_1} \cdots z_n^{m_n}$. Write out the action of $W = S_n$ on $X^*(T)$ explicitly, and verify that the character $\det = (1, 1, \ldots, 1)$ is W-invariant while $(1, 0, \ldots, 0)$ is not. Interpret the W-invariance of det as a statement about class functions using proposition 11.2.4.

6. Show by the $\mathrm{SU}(2)$ example that the normalizer $N(T)$ need not be the direct product $T \times W$. For $w = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, compute w^2 and determine the order of w in $\mathrm{SU}(2)$; deduce that the extension $1 \to T \to N(T) \to W \to 1$ does not split through this lift. Explain why $N(T)$ is nonetheless a compact Lie group with exactly two components.

11.3 Roots of a Compact Group and $\mathfrak{g}_{\mathbb{C}}$

The maximal torus T of definition 11.1.1 captured G up to conjugacy: every element lies in a conjugate of T (theorem 11.1.3) and every class function is determined by its restriction to T (proposition 11.2.4). What the torus does not yet reveal is the *transverse* structure of G, the directions in which G fails to be abelian. Those directions are visible on the Lie algebra, where T acts by the adjoint representation. This section decomposes $\mathfrak{g}$ under that action. The obstruction is that a rotation of a real plane has no real eigenvectors, so the eigenspaces live only after complexifying; the payoff is that the complexified Lie algebra $\mathfrak{g}_{\mathbb{C}}$ splits into weight spaces indexed by a finite set of linear functionals on $\mathfrak{t}$, the roots, and this decomposition is *exactly* the root space decomposition of $\mathfrak{g}_{\mathbb{C}}$ from Part II, applied to its semisimple part.

The identification has a structural consequence worth stating at the outset. The Weyl group $W = N(T)/T$ of definition 11.2.1, defined group-theoretically as a normalizer quotient, will turn out to equal the abstract Weyl group $\langle s_\alpha \rangle$ of definition 7.5.3, defined combinatorially as the reflection group of the root system. One group has an analytic origin, the other a purely combinatorial one; they coincide because both act as the symmetries of Φ, and every root reflection is realized by an actual element of $N(T)$ through the copy of $\mathrm{SU}(2)$ that the root carries.

Fix a compact connected Lie group G with maximal torus T, and write $\mathfrak{g} = \mathrm{Lie}(G)$, $\mathfrak{t} = \mathrm{Lie}(T)$. The adjoint action of T on $\mathfrak{g}$ is by orthogonal transformations for the invariant inner product of proposition 9.1.5, so each $\mathrm{Ad}(t)$ is a rotation; a nontrivial rotation of a real vector space has no eigenvectors, only invariant 2-planes. To diagonalize the whole family $\{\mathrm{Ad}(t)\}_{t \in T}$ simultaneously the eigenvalues must be allowed to be complex, and for that the arena must be the complexification.

Definition 11.3.1: Roots of a Compact Group

Let $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ be the complexified Lie algebra, with $\mathfrak{t}_{\mathbb{C}} = \mathfrak{t} \otimes_{\mathbb{R}} \mathbb{C}$ its Cartan subalgebra. The torus T acts on $\mathfrak{g}_{\mathbb{C}}$ by the *complexified adjoint representation* $\mathrm{Ad}_{\mathbb{C}} \colon T \to \mathrm{GL}(\mathfrak{g}_{\mathbb{C}})$, the $\mathbb{C}$-linear extension of Ad. Because T is compact the representation is completely reducible, and because T is abelian every irreducible summand is one-dimensional, a character of T (the character theory of theorem 10.4.4); so $\mathfrak{g}_{\mathbb{C}}$ is a direct sum of characters: there is a finite set of characters $\alpha \in X^*(T) = \mathrm{Hom}(T, \mathrm{U}(1))$ and a decomposition

$$\mathfrak{g}_{\mathbb{C}} = \bigoplus_{\alpha} \mathfrak{g}_{\alpha}, \qquad \mathfrak{g}_{\alpha} = \{X \in \mathfrak{g}_{\mathbb{C}} \mid \mathrm{Ad}_{\mathbb{C}}(t)X = \alpha(t)\, X \text{ for all } t \in T\}$$

into simultaneous eigenspaces, the *weight spaces*. Each character α is determined by its differential $d\alpha \colon \mathfrak{t} \to i\mathbb{R}$, which extends $\mathbb{C}$-linearly to a functional in $\mathfrak{t}_{\mathbb{C}}^*$; identify α with this functional, so that $\mathrm{ad}(H)X = \alpha(H)\, X$ for $X \in \mathfrak{g}_{\alpha}$ and $H \in \mathfrak{t}_{\mathbb{C}}$. The symbol α thus carries two linked normalizations: on the group it is the U(1)-valued character $t \mapsto \alpha(t)$, and on the algebra it is the functional $H \mapsto \alpha(H) = d\alpha(H) \in i\mathbb{R}$ for $H \in \mathfrak{t}$; the two agree via $\alpha(\exp H) = e^{\alpha(H)}$. A third, purely real normalization, the angle $-i\, d\alpha(H) \in \mathbb{R}$, appears in proposition 11.3.4 and is flagged there. The *roots* of (G, T) are the *nonzero* characters α occurring in $\mathfrak{g}_{\mathbb{C}}$, that is, those with $\alpha \neq 0$ and $\mathfrak{g}_{\alpha} \neq 0$. Write $\Phi \subseteq X^*(T)$ for the set of roots.

The definition is the compact-group reading of a familiar mechanism. On the Lie-algebra side of Part II a Cartan subalgebra $\mathfrak{h}$ acted on $\mathfrak{g}$ by simultaneously diagonalizable operators because $\mathfrak{h}$ was toral; here $\mathfrak{t}$ plays that role, and simultaneous diagonalizability comes instead from the compactness of the group T, through the character theory of theorem 10.4.4. The two inputs, an abstract Cartan subalgebra and a compact torus, produce the same output: a weight decomposition. The nonzero weights are the roots, and the zero weight space is the part of $\mathfrak{g}_{\mathbb{C}}$ on which T acts trivially, which is the centralizer of $\mathfrak{t}$; because T is a *maximal* torus this centralizer is exactly $\mathfrak{t}_{\mathbb{C}}$, so $\mathfrak{g}_0 = \mathfrak{t}_{\mathbb{C}}$. The next result records this and identifies the outcome with the Part II decomposition.

Two features of the compact setting distinguish the roots from bare weights. First, the roots come in pairs: complex conjugation on $\mathfrak{g}_{\mathbb{C}}$ (fixing $\mathfrak{g}$) sends $\mathfrak{g}_{\alpha}$ to $\mathfrak{g}_{-\alpha}$, since $\mathrm{Ad}(t)$ is real, so $\alpha \in \Phi$ forces $-\alpha \in \Phi$. Second, each root space is one-dimensional, a fact inherited from the Part II theory once the identification is made. Both are visible already in the running example.

Example 11.3.2: The Root of SU(2)

Let $G = \mathrm{SU}(2)$, with maximal torus $T = \left\{ t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta}) \;\middle|\; \theta \in \mathbb{R} \right\}$ of rank 1. The Lie algebra $\mathfrak{su}(2)$ consists of the traceless skew-Hermitian 2×2 matrices, and its complexification is $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}_2(\mathbb{C})$, all traceless complex matrices (proposition 1.5.1 identifies the group; the complexification is the statement that $\mathfrak{su}(2)$ is a compact real form). Take the standard basis

$$H = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right), \qquad E = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \qquad F = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$$

so $\mathfrak{t}_{\mathbb{C}} = \mathbb{C}H$. Conjugating by t_θ gives $\mathrm{Ad}(t_\theta)E = t_\theta E t_\theta^{-1} = e^{2i\theta}E$ and $\mathrm{Ad}(t_\theta)F = e^{-2i\theta}F$, while $\mathrm{Ad}(t_\theta)H = H$. Define the character $\alpha \colon T \to \mathrm{U}(1)$ by $\alpha(t_\theta) = e^{2i\theta}$; then

$$\mathfrak{sl}_2(\mathbb{C}) = \underbrace{\mathbb{C}H}_{\mathfrak{t}_{\mathbb{C}}} \oplus \underbrace{\mathbb{C}E}_{\mathfrak{g}_{\alpha}} \oplus \underbrace{\mathbb{C}F}_{\mathfrak{g}_{-\alpha}}$$

is the weight decomposition. The roots are $\Phi = \{\alpha, -\alpha\}$, a single positive root α and its negative, each root space a line, and conjugation carries $\mathfrak{g}_{\alpha}$ to $\mathfrak{g}_{-\alpha}$. As a functional on $\mathfrak{t}$, α reads

$d\alpha(s\,iH) = 2is$ for $s \in \mathbb{R}$, so $\alpha(H) = 2$: this is the single root of type A_1, matching $\varepsilon_1 - \varepsilon_2$ under the diagonal identification of example 8.4.4.

The example exhibits everything the general theorem asserts in the smallest case: the zero weight space is the complexified torus, the nonzero weights come in a conjugate pair, and each root space is a line. The theorem now says that this pattern is the root space decomposition of Part II verbatim, so that Φ is an abstract root system and its combinatorial Weyl group is the analytic one.

Theorem 11.3.3: The Root Space Decomposition of $\mathfrak{g}_{\mathbb{C}}$

Let G be a compact connected Lie group with maximal torus T and roots Φ. Then $\mathfrak{g}_0 = \mathfrak{t}_{\mathbb{C}}$, and

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

is the simultaneous eigenspace decomposition of $\mathrm{ad}(\mathfrak{t}_{\mathbb{C}})$. Writing $\mathfrak{g}_{\mathbb{C}} = \mathfrak{z}_{\mathbb{C}} \oplus \mathfrak{g}_{\mathbb{C}}^{\mathrm{ss}}$ for the splitting into its center and its semisimple part, the pair $(\mathfrak{g}_{\mathbb{C}}^{\mathrm{ss}}, \mathfrak{t}_{\mathbb{C}} \cap \mathfrak{g}_{\mathbb{C}}^{\mathrm{ss}})$ satisfies the root space decomposition theorem 7.2.4: the display *is* that decomposition, Φ is the root system of theorem 7.5.4, and each $\dim_{\mathbb{C}} \mathfrak{g}_\alpha = 1$. Consequently the Weyl group $W = N(T)/T$ of definition 11.2.1 coincides with the abstract Weyl group of definition 7.5.3, the group generated by the reflections s_α of definition 7.5.1: the natural map

$$N(T)/T \longrightarrow \mathrm{Aut}(\Phi), \qquad nT \longmapsto \left(\alpha \mapsto \alpha \circ \mathrm{Ad}(n)^{-1}\right)$$

is an injective homomorphism whose image is exactly $\langle s_\alpha : \alpha \in \Phi \rangle$.

***Proof*:**

Step 1: the zero weight space is $\mathfrak{t}_{\mathbb{C}}$. By definition 11.3.1 the zero weight space is $\mathfrak{g}_0 = \{X \in \mathfrak{g}_{\mathbb{C}} \mid \mathrm{Ad}(t)X = X \text{ for all } t \in T\}$, the fixed space of $\mathrm{Ad}(T)$. Differentiating the fixing condition along one-parameter subgroups of T shows $\mathfrak{g}_0 = \{X \in \mathfrak{g}_{\mathbb{C}} \mid [H, X] = 0 \text{ for all } H \in \mathfrak{t}\}$, the complexified centralizer $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{t})_{\mathbb{C}}$: a vector fixed by the connected group T is annihilated by its Lie algebra $\mathfrak{t}$, and conversely $\exp(\mathfrak{t})$ generates T, so annihilation by $\mathfrak{t}$ forces fixing by T. It therefore suffices to show $\mathfrak{z}_{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{t}$. The inclusion $\mathfrak{t} \subseteq \mathfrak{z}_{\mathfrak{g}}(\mathfrak{t})$ is clear since $\mathfrak{t}$ is abelian. For the reverse, let $Z = \mathfrak{z}_{\mathfrak{g}}(\mathfrak{t})$ and let $Z^0 \subseteq G$ be the connected subgroup with Lie algebra Z. Then T is a central torus in the compact connected group Z^0, so a maximal torus T' of Z^0 containing T is abelian and centralizes T; but a connected abelian subgroup of G containing the maximal torus T must equal T, so $T' = T$. Thus Z^0 has T as a maximal torus contained in its center, forcing Z^0 abelian with $\mathrm{Lie}(Z^0) = \mathfrak{t}$; hence $Z = \mathfrak{t}$, giving $\mathfrak{g}_0 = \mathfrak{t}_{\mathbb{C}}$ and the displayed decomposition.

Step 2: identification with the Part II decomposition. The subalgebra $\mathfrak{t}_{\mathbb{C}} \subseteq \mathfrak{g}_{\mathbb{C}}$ is abelian, and by Step 1 it is self-centralizing: $\mathfrak{z}_{\mathfrak{g}_{\mathbb{C}}}(\mathfrak{t}_{\mathbb{C}}) = \mathfrak{g}_0 = \mathfrak{t}_{\mathbb{C}}$. Moreover $\mathrm{ad}(H)$ is diagonalizable for each $H \in \mathfrak{t}_{\mathbb{C}}$, being diagonalized by the weight decomposition. A self-centralizing subalgebra of ad-diagonalizable elements is a Cartan subalgebra (definition 7.1.4), so $\mathfrak{t}_{\mathbb{C}}$ is a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$. Now $\mathfrak{g}_{\mathbb{C}}$ need not be semisimple, since G may have a central torus, but its center $\mathfrak{z}_{\mathbb{C}} = \mathfrak{z}(\mathfrak{g})_{\mathbb{C}} \subseteq \mathfrak{t}_{\mathbb{C}}$ splits off: $\mathfrak{g}_{\mathbb{C}} = \mathfrak{z}_{\mathbb{C}} \oplus \mathfrak{g}_{\mathbb{C}}^{\mathrm{ss}}$ with $\mathfrak{g}_{\mathbb{C}}^{\mathrm{ss}}$ semisimple, because the Killing form of $\mathfrak{g}$ is negative-semidefinite with kernel exactly $\mathfrak{z}$ (proposition 9.4.1), so it is nondegenerate on a complement of the center. The central summand contributes only to the zero weight space, so the nonzero weight spaces $\mathfrak{g}_\alpha$ and their labels α are unchanged by discarding $\mathfrak{z}_{\mathbb{C}}$. Applied to $(\mathfrak{g}_{\mathbb{C}}^{\mathrm{ss}}, \mathfrak{t}_{\mathbb{C}} \cap \mathfrak{g}_{\mathbb{C}}^{\mathrm{ss}})$, the theorem theorem 7.2.4 gives precisely the decomposition of the display,

with $\dim_{\mathbb{C}} \mathfrak{g}_\alpha = 1$ for each α and Φ satisfying the root-system axioms of theorem 7.5.4. Thus the analytically defined Φ is the algebraically defined root system, and the coroots $\alpha^\vee$ of definition 7.4.3 and the reflections s_α of definition 7.5.1 are available.

Step 3: the map $N(T)/T \to \mathrm{Aut}(\Phi)$ is a well-defined injection. If $n \in N(T)$ then $\mathrm{Ad}(n)$ preserves $\mathfrak{t}$, hence $\mathfrak{t}_{\mathbb{C}}$, and permutes the weight spaces: for $X \in \mathfrak{g}_\alpha$ and $t \in T$,

$$\mathrm{Ad}(t)\big(\mathrm{Ad}(n)X\big) = \mathrm{Ad}(n)\ \mathrm{Ad}(n^{-1}tn)X = \mathrm{Ad}(n)\big(\alpha(n^{-1}tn)X\big) = \big(\alpha \circ \mathrm{Ad}(n)^{-1}\big)(t) \cdot \mathrm{Ad}(n)X$$

so $\mathrm{Ad}(n)$ carries $\mathfrak{g}_\alpha$ to $\mathfrak{g}_{w(\alpha)}$ where $w(\alpha) = \alpha \circ \mathrm{Ad}(n)^{-1}$. Since $\mathrm{Ad}(n)$ is an automorphism of $\mathfrak{g}_{\mathbb{C}}$ fixing $\mathfrak{t}_{\mathbb{C}}$, it permutes Φ and preserves the inner product on $\mathfrak{t}^*$ (that inner product is $\mathrm{Ad}(G)$-invariant, proposition 9.1.5), so $w \in \mathrm{Aut}(\Phi)$. The action depends only on the coset nT: if $n \in T$ then $\mathrm{Ad}(n)$ acts trivially on $\mathfrak{t}_{\mathbb{C}}$, so $w = \mathrm{id}$. For injectivity, suppose $nT \mapsto \mathrm{id}$, so $\mathrm{Ad}(n)$ fixes every $\alpha \in \Phi$. The roots span $\mathfrak{t}^*$ modulo the annihilator of the semisimple part, and $\mathrm{Ad}(n)$ already fixes the central directions (for connected G the group $\mathrm{Ad}(G)$ acts trivially on $\mathfrak{z}(\mathfrak{g})$, since $\mathrm{ad}X$ annihilates the center for every X), so $\mathrm{Ad}(n)$ fixes $\mathfrak{t}^*$ pointwise, hence fixes $\mathfrak{t}$ pointwise. Then n centralizes T, so $n \in C_G(T)$, and injectivity reduces to the identity $C_G(T) = T$, that a maximal torus is self-centralizing. An element acting trivially on $\mathfrak{t}$ need not lie in T a priori, so n must be located inside T directly; the maximal-torus theorem does this in two steps. First, n lies in *some* maximal torus: because n centralizes T, the left translation L_n on G/T commutes with the T-action, and L_n is homotopic to the identity (G being connected), so its Lefschetz number equals $\chi(G/T) = |W| > 0$ (the fixed-point count underlying theorem 11.1.3); hence the fixed set $(G/T)^n$ is nonempty, and a fixed coset gT gives $g^{-1}ng \in T$, so n lies in the maximal torus $S = gTg^{-1}$. Second, S can be conjugated onto T by an element centralizing n: the centralizer $C_G(n)$ is a compact group containing both the connected torus T (which centralizes n, since $n \in C_G(T)$) and the maximal torus $S \ni n$, so T and S are both maximal tori of the identity component $C_G(n)^0$; by conjugacy of maximal tori there (theorem 11.1.3) there is $h \in C_G(n)^0$ with $S = hTh^{-1}$. Then $h^{-1}nh \in h^{-1}Sh = T$, and $h^{-1}nh = n$ because h centralizes n, so $n \in T$ and nT is trivial.

Step 4: every reflection s_α is realized in $N(T)$. Fix $\alpha \in \Phi$. By Step 2 the root α carries the $\mathfrak{sl}_2$-triple $(e_\alpha, h_\alpha, f_\alpha)$ of theorem 7.4.6 inside $\mathfrak{g}_{\mathbb{C}}$, spanning a subalgebra $\mathfrak{s}_\alpha \cong \mathfrak{sl}_2(\mathbb{C})$. This triple can be chosen compatibly with the compact form: take $e_\alpha \in \mathfrak{g}_\alpha$ and $f_\alpha = -\sigma(e_\alpha) \in \mathfrak{g}_{-\alpha}$, where σ is the conjugation fixing $\mathfrak{g}$; then $X_\alpha := e_\alpha - f_\alpha$ and $Y_\alpha := i(e_\alpha + f_\alpha)$ together with ih_α lie in $\mathfrak{g}$ and span a subalgebra isomorphic to $\mathfrak{su}(2)$, the compact real form of $\mathfrak{s}_\alpha$. Exponentiating gives a homomorphism $\varphi_\alpha\colon \mathrm{SU}(2) \to G$ whose image G_α is a compact connected subgroup with this $\mathfrak{su}(2)$ as Lie algebra. The circle $\theta \mapsto \varphi_\alpha\big(\mathrm{diag}(e^{i\theta}, e^{-i\theta})\big)$ lies in T, since its Lie algebra $\mathbb{R}\, ih_\alpha \subseteq \mathfrak{t}$ ($h_\alpha \in \mathfrak{t}_{\mathbb{C}}$), and the nontrivial normalizer element of $\mathrm{SU}(2)$, the class of $w_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, maps to $n_\alpha = \varphi_\alpha(w_0) \in G$. Conjugation by w_0 inverts the diagonal torus of $\mathrm{SU}(2)$, so $\mathrm{Ad}(n_\alpha)$ acts by -1 on the line $\mathbb{R}\, ih_\alpha$ and fixes the orthogonal complement $\ker \alpha \subseteq \mathfrak{t}$ pointwise (that complement centralizes $\mathfrak{s}_\alpha$, so G_α acts trivially on it). Hence $n_\alpha \in N(T)$, and its action on $\mathfrak{t}^*$ is the reflection fixing $\ker \alpha$ and negating α, which is s_α of definition 7.5.1.

Step 5: the image is exactly $\langle s_\alpha \rangle$. Step 4 gives $\langle s_\alpha : \alpha \in \Phi \rangle \subseteq \mathrm{im}\big(N(T)/T \to \mathrm{Aut}(\Phi)\big)$. For the reverse inclusion, the abstract Weyl group $\langle s_\alpha \rangle$ acts simply transitively on the Weyl chambers of Φ (theorem 8.2.8). Given w in the image, compose it with a suitable element of $\langle s_\alpha \rangle$ so that the composite fixes a chosen chamber C; then the composite fixes the associated set of positive roots, hence the base of simple roots, hence every root (each root is an integer combination of simple roots), and by the injectivity of Step 3 the composite is the identity coset. Therefore $w \in \langle s_\alpha \rangle$, and the image is exactly the abstract Weyl group, as we sought to show.

The combinatorial data extracted algebraically in Part II, the root system and its Weyl group, is recovered analytically from the group G and its torus T with no new choices. Every statement proved for the abstract root system now applies to (G,T): the roots form a root system of one of the classified types, the number of them is $\dim G - \operatorname{rank} G$ (the roots account for $\mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}$, of dimension $\dim G - \dim T$), and the Weyl group is finite and generated by reflections. The two definitions of W, one a normalizer quotient and one a reflection group, are not merely abstractly isomorphic; the isomorphism is the concrete map that reads a normalizer element off its action on the roots, and its inverse builds a normalizer element from a reflection using the copy of $\mathrm{SU}(2)$ each root carries. That copy of $\mathrm{SU}(2)$, the subgroup G_α of the proof, is the geometric content of a root: it is where the noncommutativity of G in the α-direction is concentrated.

The identification also fixes the arithmetic. The coroots $\alpha^\vee$ of definition 7.4.3 live in $\mathfrak{t}$ (indeed $ih_\alpha \in \mathfrak{t}$ in the proof), the Cartan integers $\langle \beta, \alpha^\vee \rangle$ of theorem 7.4.6 are the same integers as before, and the pairing between $X^*(T)$ and $X_*(T)$ that the next section develops is the pairing between weights and coweights already present in Part II. The compact group adds a lattice refinement to the picture, the subject of section 11.5, but the underlying root system is unchanged.

The complex decomposition is the efficient bookkeeping, but the group G is real, and for integration on G (the subject of section 11.4) the real form of the decomposition is what enters. Pairing each root space with its conjugate produces real invariant 2-planes on which T acts by rotations, and the angle of the rotation is the root read as a real functional.

Proposition 11.3.4: The Real Root Space Decomposition

Let G be a compact connected Lie group with maximal torus T, and fix a set $\Phi^+ \subseteq \Phi$ of positive roots (a choice of definition 8.2.1). For each $\alpha \in \Phi^+$ set

$$\mathfrak{m}_\alpha = (\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}) \cap \mathfrak{g}$$

the real points of the sum of the two conjugate root spaces. Then each $\mathfrak{m}_\alpha$ is a 2-dimensional real subspace of $\mathfrak{g}$, and

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{m}_\alpha$$

is an orthogonal direct sum for the invariant inner product. On $\mathfrak{m}_\alpha$ the torus acts by a rotation of angle $\alpha(t)$, where here α is read as the *real* angle: the character of definition 11.3.1 is $t \mapsto e^{i\alpha(t)}$, so $\alpha(t) \in \mathbb{R}/2\pi\mathbb{Z}$, and correspondingly $\alpha(H) = -i\, d\alpha(H) \in \mathbb{R}$ for $H \in \mathfrak{t}$ (matching $\alpha(H) = 2$ in example 11.3.2 once H is scaled into $\mathfrak{t}$). Choosing the basis $U_\alpha = e_\alpha - f_\alpha$, $V_\alpha = i(e_\alpha + f_\alpha)$ of theorem 11.3.3,

$$\operatorname{Ad}(t)\begin{pmatrix} U_\alpha \\ V_\alpha \end{pmatrix} = \begin{pmatrix} \cos\alpha(t) & \sin\alpha(t) \\ -\sin\alpha(t) & \cos\alpha(t) \end{pmatrix}\begin{pmatrix} U_\alpha \\ V_\alpha \end{pmatrix}$$

so for $H \in \mathfrak{t}$ the operator $\operatorname{ad}(H)$ restricted to $\mathfrak{m}_\alpha$ has eigenvalues $\pm i\,\alpha(H)$.

***Proof*:**
Complex conjugation σ on $\mathfrak{g}_{\mathbb{C}}$ has fixed set $\mathfrak{g}$ and satisfies $\sigma(\mathfrak{g}_\alpha) = \mathfrak{g}_{-\alpha}$, because $\operatorname{Ad}(t)$ is real: if $\operatorname{Ad}(t)X = \alpha(t)X$ then $\operatorname{Ad}(t)\sigma(X) = \sigma(\operatorname{Ad}(t)X) = \overline{\alpha(t)}\,\sigma(X) = \alpha(t)^{-1}\sigma(X)$, and $\alpha(t)^{-1} = (-\alpha)(t)$ since $\alpha(t) \in \mathrm{U}(1)$. Thus $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$ is σ-stable, a complex 2-space with a conjugation, so its real points form a real 2-space $\mathfrak{m}_\alpha$ whose complexification is $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$.

Explicitly, with $f_\alpha = -\sigma(e_\alpha)$ as in theorem 11.3.3, the vectors $U_\alpha = e_\alpha - f_\alpha$ and $V_\alpha = i(e_\alpha + f_\alpha)$ are σ-fixed, hence lie in $\mathfrak{g}$, and span $\mathfrak{m}_\alpha$ over $\mathbb{R}$.

The root spaces for distinct $\pm\alpha$ pairs are orthogonal: for $\alpha \neq \pm\beta$ the spaces $\mathfrak{g}_{\pm\alpha}$ and $\mathfrak{g}_{\pm\beta}$ are eigenspaces of the orthogonal action $\mathrm{Ad}(T)$ for distinct characters, and eigenspaces of an orthogonal action for distinct characters are orthogonal; likewise each $\mathfrak{m}_\alpha$ is orthogonal to the zero-weight part $\mathfrak{t}$. Since the $\mathfrak{m}_\alpha$ ($\alpha \in \Phi^+$) together with $\mathfrak{t}$ account for all of $\mathfrak{g}_{\mathbb{C}}$ after complexifying, and they are real subspaces of $\mathfrak{g}$, they exhaust $\mathfrak{g}$: this is the displayed orthogonal decomposition, of real dimension $\dim\mathfrak{t} + 2\,|\Phi^+| = \mathrm{rank}G + |\Phi| = \dim G$.

For the action, fix $t \in T$ and write $c = \alpha(t) \in \mathbb{R}/2\pi\mathbb{Z}$, so $\mathrm{Ad}(t)e_\alpha = e^{ic}e_\alpha$ and $\mathrm{Ad}(t)f_\alpha = e^{-ic}f_\alpha$. Then

$$\mathrm{Ad}(t)U_\alpha = e^{ic}e_\alpha - e^{-ic}f_\alpha = \cos c\,(e_\alpha - f_\alpha) + i\sin c\,(e_\alpha + f_\alpha) = \cos c\,U_\alpha + \sin c\,V_\alpha$$

and similarly $\mathrm{Ad}(t)V_\alpha = -\sin c\,U_\alpha + \cos c\,V_\alpha$, the claimed rotation. Differentiating at $t = \exp(sH)$, where $c = s\,\alpha(H) + o(s)$, gives $\mathrm{ad}(H)U_\alpha = \alpha(H)V_\alpha$ and $\mathrm{ad}(H)V_\alpha = -\alpha(H)U_\alpha$, so $\mathrm{ad}(H)|_{\mathfrak{m}_\alpha}$ is the infinitesimal rotation with eigenvalues $\pm i\,\alpha(H)$, completing the proof.

The proposition is the geometry the integration formula uses. A compact group is, infinitesimally, its torus together with a family of rotating 2-planes, one for each positive root, and T turns each plane at its own angular speed α. This is the precise sense in which the roots measure the noncommutativity of G: were G abelian, $\mathfrak{g}$ would be all torus, with no rotating planes and no roots. The eigenvalue $\mathrm{ad}(H)|_{\mathfrak{m}_\alpha} = \pm i\,\alpha(H)$ is exactly what section 11.4 needs to compute the Jacobian of the conjugation map $q\colon G/T \times T \to G$, whose Jacobian is the determinant of $\mathrm{Ad}(t^{-1}) - 1$ on $\mathfrak{g}/\mathfrak{t} = \bigoplus_\alpha \mathfrak{m}_\alpha$; on each 2-plane $\mathrm{Ad}(t^{-1}) - 1$ is a rotation by $-\alpha(t)$ minus the identity, and the density $|\delta(t)|^2$ of the Weyl integration formula is assembled plane by plane from these rotations.

Example 11.3.5: The Real Decomposition for SU(2) and the Roots of U(n)

1. For $G = \mathrm{SU}(2)$, example 11.3.2 gives the single positive root α with $\alpha(t_\theta) = e^{2i\theta}$, so there is one 2-plane $\mathfrak{m}_\alpha = (\mathbb{C}E \oplus \mathbb{C}F) \cap \mathfrak{su}(2)$, spanned by $U_\alpha = E - F = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $V_\alpha = i(E+F) = \left(\begin{smallmatrix} 0 & i \\ i & 0 \end{smallmatrix}\right)$, the off-diagonal directions. The decomposition $\mathfrak{su}(2) = \mathfrak{t} \oplus \mathfrak{m}_\alpha = \mathbb{R}\,iH \oplus \mathfrak{m}_\alpha$ splits the algebra into the diagonal circle and its rotating complement. The torus t_θ turns $\mathfrak{m}_\alpha$ by the angle $\alpha(t_\theta) = 2\theta$: as θ runs once around T the plane rotates twice, and $\mathrm{ad}(iH)$ has eigenvalues $\pm i\,\alpha(iH) = \pm 2i$ on $\mathfrak{m}_\alpha$, matching the weights ± 2 of example 11.3.2.

2. For $G = \mathrm{U}(n)$ the maximal torus is the diagonal $T = \left\{\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) \;\middle|\; \theta_j \in \mathbb{R}\right\}$, of rank n, with $\mathfrak{t} = \{\mathrm{diag}(i\theta_1, \dots, i\theta_n) \mid \theta_j \in \mathbb{R}\}$. Complexifying, $\mathfrak{u}(n)_{\mathbb{C}} = \mathfrak{gl}_n(\mathbb{C})$, and conjugating the matrix unit E_{jk} ($j \neq k$) by $t = \mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})$ gives $\mathrm{Ad}(t)E_{jk} = e^{i(\theta_j - \theta_k)}E_{jk}$. Hence the characters are $\varepsilon_j - \varepsilon_k$ with $\varepsilon_j(t) = e^{i\theta_j}$, and the roots are

$$\Phi = \{\varepsilon_j - \varepsilon_k \mid 1 \leq j \neq k \leq n\}, \qquad \mathfrak{g}_{\varepsilon_j - \varepsilon_k} = \mathbb{C}E_{jk}$$

   the type A_{n-1} system of example 8.4.4 (for $\mathrm{SU}(n)$ one restricts to $\sum\theta_j = 0$, removing the central direction; the roots are unchanged). Each positive root $\varepsilon_j - \varepsilon_k$ ($j < k$) contributes the real 2-plane spanned by $E_{jk} - E_{kj}$ and $i(E_{jk} + E_{kj})$, and there are $\binom{n}{2}$ of them, so $\dim \mathrm{U}(n) = n + 2\binom{n}{2} = n^2$, as it must. The rotation group $\mathrm{SO}(3)$, the image of $\mathrm{SU}(2)$ under the double cover of corollary 1.5.6, shares the root system of $\mathrm{SU}(2)$: it has rank 1 and a single positive root, its adjoint action being the same 3-dimensional rotation, so $\mathrm{SU}(2)$ and

SO(3) have the same $\Phi = \{\pm\alpha\}$ of type A_1 and differ only in the lattice data of the next section.

The examples close the loop with Part II. The roots of $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ are the $\varepsilon_j - \varepsilon_k$ that the Lie-algebra computation of example 8.4.4 produced from $\mathfrak{sl}_n(\mathbb{C})$, reached now by conjugating diagonal matrices in the compact group rather than by bracketing in the complex algebra; the two routes agree because theorem 11.3.3 identified the decompositions. That $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ share a root system while differing as groups is the first sign that Φ does not see the group on the nose, only its Lie algebra: the finer invariant separating $\mathrm{SU}(2)$ from $\mathrm{SO}(3)$ is the character lattice $X^*(T) \subseteq \mathfrak{t}^*$, larger for $\mathrm{SU}(2)$ (which is simply connected, so its character lattice is the full weight lattice P) than for the adjoint group $\mathrm{SO}(3)$ (whose character lattice is the root lattice Q), and section 11.5 makes that lattice the classifying datum for compact groups with a fixed root system.

Exercise 11.3.1

1. Let $G = \mathrm{SU}(3)$, with maximal torus the diagonal $T = \{\mathrm{diag}(e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3}) \mid \theta_1 + \theta_2 + \theta_3 = 0\}$. Compute the roots of (G, T) by conjugating the matrix units E_{jk}, and verify that $\Phi = \{\varepsilon_j - \varepsilon_k \mid 1 \le j \ne k \le 3\}$ is the root system of type A_2: exhibit the six roots as the vertices of a regular hexagon in the plane $\{\sum x_j\varepsilon_j \mid \sum x_j = 0\}$, and identify a set of positive roots and a base.

2. Using proposition 11.3.4, show that for $H \in \mathfrak{t}$ the operator $\mathrm{ad}(H)$ on $\mathfrak{g}$ is skew-symmetric for the invariant inner product, and that its eigenvalues on $\mathfrak{g}_{\mathbb{C}}$ are 0 (with multiplicity $\mathrm{rank}G$) together with the numbers $\pm i\,\alpha(H)$ for $\alpha \in \Phi^+$. Deduce that $\mathrm{ad}(H)$ is invertible on $\mathfrak{g}/\mathfrak{t}$ precisely when $\alpha(H) \ne 0$ for every root, and interpret this as H being a regular element of $\mathfrak{t}$ (one lying on no root hyperplane). Contrast this with the group-level condition on $t = \exp(H)$: the operator $\mathrm{Ad}(t) - 1$ is invertible on $\mathfrak{g}/\mathfrak{t}$ precisely when $e^{i\alpha(H)} \ne 1$, that is $\alpha(H) \notin 2\pi\mathbb{Z}$, for every root.

3. Prove that the number of roots of a compact connected group G equals $\dim G - \mathrm{rank}G$. Check the formula on $\mathrm{SU}(2)$ (dimension 3, rank 1, so 2 roots), on $\mathrm{U}(n)$ and $\mathrm{SU}(n)$, and on $\mathrm{SO}(5)$ (type B_2).

4. For each root α let $\alpha^\vee$ be the coroot of definition 7.4.3, so that $ih_\alpha \in \mathfrak{t}$. Show that the vectors ih_α span $\mathfrak{t} \cap \mathfrak{g}^{\mathrm{ss}}$ over $\mathbb{R}$, and that for a semisimple G their $\mathbb{R}$-span is all of $\mathfrak{t}$. Verify this for $\mathrm{SU}(2)$, where the single coroot gives iH spanning the rank-1 torus.

5. Explain why $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ have the same root system even though they are not isomorphic as groups. Which datum of the pair (G, T), unchanged in passing to Φ, distinguishes them? (Answer at the level of example 11.3.5; the full classifying invariant is developed in section 11.5.)

6. Let σ be complex conjugation on $\mathfrak{g}_{\mathbb{C}}$ fixing $\mathfrak{g}$. Prove that $\sigma(\mathfrak{g}_\alpha) = \mathfrak{g}_{-\alpha}$ for every root α, and conclude that $\Phi = -\Phi$. Explain where compactness of G is used (specifically that $\mathrm{Ad}(t)$ is orthogonal, so its eigenvalues lie on the unit circle).

11.4 The Weyl Integration Formula

The three preceding sections built the machinery that transports the harmonic analysis of G to its maximal torus: every conjugacy class meets T (corollary 11.1.4), two elements of T are conjugate

in G exactly when they are W-conjugate (proposition 11.2.4), and the adjoint action of T on $\mathfrak{g}_{\mathbb{C}}$ splits into the root spaces of theorem 11.3.3. What remains is to move an *integral* of a class function from G to T, not just its values. A class function is constant on each fibre of the conjugation map $q\colon G/T \times T \to G$, $(gT, t) \mapsto gtg^{-1}$, whose nonzero degree carried the maximal-torus theorem (theorem 11.1.3); pushing the Haar measure of G forward along q turns $\int_G f\,dg$ into an integral over T, weighted by the Jacobian of q. That Jacobian is the content of this section. It is the determinant of $\mathrm{Ad}(t^{-1}) - 1$ on $\mathfrak{g}/\mathfrak{t}$, and the root space decomposition evaluates it explicitly as a product over the roots, the squared modulus of a single function on T called the Weyl denominator.

The formula that results reduces every integral of a class function to a one-dimensional integral over the torus against this explicit density. Through it the character theory of chapter 12 becomes computable: orthogonality of characters, dimensions, and multiplicities all become integrals over T that the Weyl denominator makes elementary.

The density is assembled from the roots. Fix a choice of positive roots Φ^+ inside the root system Φ of (G, T) (definition 11.3.1); each root $\alpha \in X^*(T)$ is a character of T, written multiplicatively as $t \mapsto e^{\alpha}(t)$, a homomorphism $T \to \mathrm{U}(1)$. Half-integral characters $e^{\alpha/2}$ need not be defined on T itself, but the antisymmetric *differences* $e^{\alpha/2} - e^{-\alpha/2}$ combine into a function whose square is defined on T, and it is the square that carries the analytic content.

Definition 11.4.1: The Weyl Denominator

Let G be a compact connected Lie group with maximal torus T, roots Φ, and a chosen set of positive roots Φ^+ (definition 11.3.1). The *Weyl denominator* is the function on T

$$\delta(t) = \prod_{\alpha \in \Phi^+} \left(e^{\alpha/2}(t) - e^{-\alpha/2}(t)\right)$$

where $e^{\pm\alpha/2}(t)$ denote the values of the half-root characters (defined up to a common sign whose square is unambiguous). Its squared modulus is

$$|\delta(t)|^2 = \prod_{\alpha \in \Phi^+} |e^{\alpha/2}(t) - e^{-\alpha/2}(t)|^2 = \prod_{\alpha \in \Phi} \left(1 - e^{-\alpha}(t)\right) = \det\left(\mathrm{Ad}(t^{-1}) - 1 \,\middle|\, \mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}\right)$$

a well-defined nonnegative function on T independent of the choice of Φ^+ and of the signs of the half-roots.

The three expressions for $|\delta|^2$ are the same quantity read in three registers, and the identity among them is where the root space decomposition enters. On the root space $\mathfrak{g}_\alpha$ the torus element t acts by $\mathrm{Ad}(t) = e^{\alpha}(t)$ (theorem 11.3.3), so $\mathrm{Ad}(t^{-1}) - 1$ acts by $e^{-\alpha}(t) - 1$; taking the determinant over $\mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}} = \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$ multiplies these eigenvalues, giving $\prod_{\alpha\in\Phi}(e^{-\alpha}(t) - 1) = \prod_{\alpha\in\Phi}(1 - e^{-\alpha}(t))$, the sign cancelling because $|\Phi|$ is even (roots come in pairs $\pm\alpha$). Pairing each positive root α with its negative and factoring $e^{\alpha/2}$ from the first factor and $e^{-\alpha/2}$ from the second turns the pair into

$$(1 - e^{-\alpha})(1 - e^{\alpha}) = \left(e^{\alpha/2} - e^{-\alpha/2}\right)\left(e^{-\alpha/2} - e^{\alpha/2}\right) = |e^{\alpha/2} - e^{-\alpha/2}|^2$$

using that $e^{-\alpha/2} = \overline{e^{\alpha/2}}$ on T (values on $\mathrm{U}(1)$), so $|e^{\alpha/2} - e^{-\alpha/2}|^2 = (e^{\alpha/2} - e^{-\alpha/2})\overline{(e^{\alpha/2} - e^{-\alpha/2})}$. The middle product over Φ^+ is exactly $|\delta|^2$, and the whole quantity is real and nonnegative because each paired factor is a squared modulus. The determinant expression is the one the change of variables will produce; the product over positive roots is the one that is computable.

The density has a transparent meaning in torus coordinates. Writing $t = \exp(\theta)$ for $\theta \in \mathfrak{t}$ and evaluating a root as the real linear functional $\alpha(\theta)$ (so $e^{\pm\alpha/2}(t) = e^{\pm i\alpha(\theta)/2}$), each positive-root factor becomes

$$|e^{i\alpha(\theta)/2} - e^{-i\alpha(\theta)/2}|^2 = \big(2\sin\tfrac{\alpha(\theta)}{2}\big)^2 = 4\sin^2\tfrac{\alpha(\theta)}{2}$$

so that $|\delta(t)|^2 = \prod_{\alpha\in\Phi^+} 4\sin^2\big(\alpha(\theta)/2\big)$. The Weyl denominator is thus a product of $\sin^2$ terms, one per positive root, each vanishing exactly where its root does. This vanishing locus is the set of *singular* elements, those lying on more than one maximal torus; on the complementary regular set $|\delta|^2 > 0$, and it is on the regular set that q is a genuine covering of the prescribed order.

Example 11.4.2: The Weyl Denominator of SU(2)

For $G = \mathrm{SU}(2)$ the maximal torus is $T = \big\{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta}) \;\big|\; \theta \in \mathbb{R}\big\}$, and the complexified adjoint action of T on $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}_2(\mathbb{C})$ has the two roots $\pm\alpha$, where $\alpha(t_\theta) = e^{2i\theta}$ acts on the root vector $E = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ by $\mathrm{Ad}(t_\theta)E = e^{2i\theta}E$ (definition 11.3.1). There is a single positive root α, so the Weyl denominator is the one-factor product

$$\delta(t_\theta) = e^{\alpha/2}(t_\theta) - e^{-\alpha/2}(t_\theta) = e^{i\theta} - e^{-i\theta} = 2i\sin\theta$$

and its squared modulus is

$$|\delta(t_\theta)|^2 = |e^{i\theta} - e^{-i\theta}|^2 = 4\sin^2\theta$$

Read as a determinant, $\mathfrak{sl}_2(\mathbb{C})/\mathfrak{t}_{\mathbb{C}} = \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$ is two-dimensional, $\mathrm{Ad}(t_\theta^{-1}) - 1$ acts by $e^{-2i\theta} - 1$ on $\mathfrak{g}_\alpha$ and $e^{2i\theta} - 1$ on $\mathfrak{g}_{-\alpha}$, and the product $(e^{-2i\theta} - 1)(e^{2i\theta} - 1) = 2 - 2\cos 2\theta = 4\sin^2\theta$ agrees. The half-root $e^{\alpha/2}(t_\theta) = e^{i\theta}$ happens to be a genuine character of T here, so δ itself, not only $|\delta|^2$, is defined; this is the exponential-sum denominator of the SU(2) character formula $\chi_n(t_\theta) = \sin((n+1)\theta)/\sin\theta$ of proposition 9.5.3, whose denominator $\sin\theta$ is $\delta(t_\theta)/2i$. The density $4\sin^2\theta$ vanishes exactly at $\theta \in \pi\mathbb{Z}$, the two singular elements $\pm I$, which lie on every maximal torus.

The example already exhibits the pattern: the character denominator and the integration density are the same object, so integrating a product of characters against $|\delta|^2$ collapses to integrating exponential sums against each other, which the orthogonality of exponentials makes trivial. Before cashing this in, the Jacobian identity has to be established for a general G.

The route from G to T is the conjugation map. The maximal-torus theorem realized $q\colon G/T \times T \to G$, $(gT, t) \mapsto gtg^{-1}$, as a surjection of nonzero degree (theorem 11.1.3); over the regular set it is a covering whose fibre has order $|W|$ (proposition 11.2.4), because the maximal tori through a regular element are permuted simply transitively by W. The change-of-variables formula converts the Haar integral on G into an integral over $G/T \times T$ against the absolute Jacobian determinant of q, and for a class function the G/T factor integrates out. Computing that Jacobian is the one analytic step, and the root space decomposition does it in closed form.

Theorem 11.4.3: The Weyl Integration Formula

Let G be a compact connected Lie group with maximal torus T and Weyl group $W = N(T)/T$, and let dg and dt be the normalized Haar measures on G and T (total mass 1). For every continuous class function f on G,

$$\int_G f(g)\,dg = \frac{1}{|W|}\int_T f(t)\,|\delta(t)|^2\,dt$$

where $|\delta(t)|^2 = \det\big(\mathrm{Ad}(t^{-1}) - 1 \,\big|\, \mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}\big)$ is the squared Weyl denominator of definition 11.4.1.

***Proof*:**

The proof computes the Jacobian of q, applies the change-of-variables formula with its generic fibre count, and normalizes the constant by testing against $f = 1$.

Step 1: the differential of q. Let $q\colon G/T \times T \to G$ be $q(gT, t) = gtg^{-1}$, well-defined because $t \in T$ is fixed by conjugation by T (theorem 11.1.3). Fix a regular $t \in T$ and a point (gT, t); by left-invariance of the target Haar measure it suffices to compute the differential at $g = e$, trivializing both source and target by translation. The tangent space of G/T at the base point is $\mathfrak{g}/\mathfrak{t}$, the tangent space of T at t is $\mathfrak{t}$, and the tangent space of G at t is identified with $\mathfrak{g}$ by right translation $R_{t^{-1}}$. Decompose a tangent vector to $G/T \times T$ as $(X + \mathfrak{t}, H)$ with $X \in \mathfrak{g}$ (modulo $\mathfrak{t}$) and $H \in \mathfrak{t}$. The curve $s \mapsto \exp(sX)\,t\,\exp(-sX)$ moving in the G/T direction has derivative at $s = 0$, after right-translating by t^{-1},

$$\frac{d}{ds}\Big|_0 \big(\exp(sX)\,t\,\exp(-sX)\,t^{-1}\big) = X - \mathrm{Ad}(t)X = \big(1 - \mathrm{Ad}(t)\big)X$$

while the curve $s \mapsto t\exp(sH)$ moving in the T direction contributes $H \in \mathfrak{t}$. Hence in the splitting $\mathfrak{g} = \mathfrak{t} \oplus (\mathfrak{g}/\mathfrak{t})$ the differential of q (after right translation by t^{-1}) is

$$dq_{(gT,t)}(X + \mathfrak{t}, H) = H + \big(1 - \mathrm{Ad}(t)\big)X$$

block-triangular, with the identity on the $\mathfrak{t}$ block and $1 - \mathrm{Ad}(t)$ on the $\mathfrak{g}/\mathfrak{t}$ block. The operator $1 - \mathrm{Ad}(t)$ preserves $\mathfrak{t}$ (acting as 0 there, since $\mathrm{Ad}(t)$ fixes $\mathfrak{t}$ pointwise) and descends to $\mathfrak{g}/\mathfrak{t}$; on that quotient it is invertible precisely because t is regular, so no root satisfies $e^{\alpha}(t) = 1$.

Step 2: the Jacobian is $|\delta(t)|^2$. The absolute Jacobian determinant of q at (gT, t) is therefore

$$|\det dq_{(gT,t)}| = |\det\big(1 - \mathrm{Ad}(t) \,\big|\, \mathfrak{g}/\mathfrak{t}\big)|$$

Complexifying to $\mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}} = \bigoplus_{\alpha\in\Phi} \mathfrak{g}_{\alpha}$ diagonalizes $\mathrm{Ad}(t)$: it acts by $e^{\alpha}(t)$ on $\mathfrak{g}_{\alpha}$ (theorem 11.3.3), so

$$\det\big(1 - \mathrm{Ad}(t) \,\big|\, \mathfrak{g}_{\mathbb{C}}/\mathfrak{t}_{\mathbb{C}}\big) = \prod_{\alpha\in\Phi} \big(1 - e^{\alpha}(t)\big)$$

This is real and equals its own conjugate, because $\Phi = -\Phi$ pairs each factor $1 - e^{\alpha}(t)$ with $1 - e^{-\alpha}(t) = \overline{1 - e^{\alpha}(t)}$ (values on $\mathrm{U}(1)$). By the computation following definition 11.4.1 the product equals $|\delta(t)|^2 = \prod_{\alpha\in\Phi^+} 4\sin^2(\alpha(\theta)/2) \geq 0$. The determinant on the real space $\mathfrak{g}/\mathfrak{t}$ is the same nonnegative number, so $|\det dq_{(gT,t)}| = |\delta(t)|^2$, independent of g.

Step 3: change of variables with the generic fibre. The map q is smooth and surjective (theorem 11.1.3); its critical set is the preimage of the singular elements, a set of measure zero

by Sard's theorem ([13, Sard's Theorem]), on whose complement q restricts to a covering. By proposition 11.2.4 a regular $g_0 = gtg^{-1}$ determines t up to the action of W, so the fibre $q^{-1}(g_0)$ over a regular value has exactly $|W|$ points, one for each of the $|W|$ Weyl translates of t. The change-of-variables formula for a smooth map between oriented manifolds of the same dimension ([13]) then gives, for any continuous φ on G,

$$\int_{G/T\times T} \varphi\big(q(gT,t)\big)\,|\det dq_{(gT,t)}|\,d\mu = |W|\int_G \varphi(g)\,dg$$

where $d\mu$ is the product of the G-invariant probability measure on G/T and the Haar measure dt on T, and the factor $|W|$ is the constant fibre count. Substituting the Jacobian of Step 2,

$$\int_{G/T}\int_T \varphi(gtg^{-1})\,|\delta(t)|^2\,dt\,d(gT) = |W|\int_G \varphi(g)\,dg$$

Step 4: reduction for a class function. Take $\varphi = f$ a class function. Then $f(gtg^{-1}) = f(t)$ is independent of gT, so the inner integrand does not depend on the G/T variable, and integrating out the G/T factor against the probability measure on G/T leaves its total mass 1:

$$\int_{G/T}\int_T f(t)\,|\delta(t)|^2\,dt\,d(gT) = \Big(\int_{G/T} d(gT)\Big)\int_T f(t)\,|\delta(t)|^2\,dt = \int_T f(t)\,|\delta(t)|^2\,dt$$

Combining with Step 3, $|W|\int_G f\,dg = \int_T f(t)\,|\delta(t)|^2\,dt$, that is

$$\int_G f(g)\,dg = \frac{1}{|W|}\int_T f(t)\,|\delta(t)|^2\,dt$$

Step 5: the normalization is automatic. It remains to check that the constant is exactly $1/|W|$ and not some other multiple of the two normalized Haar measures. Testing the derived identity on the class function $f \equiv 1$, the left side is $\int_G 1\,dg = 1$ because dg has total mass 1, so the identity forces $\frac{1}{|W|}\int_T|\delta(t)|^2\,dt = 1$, that is $\int_T|\delta(t)|^2\,dt = |W|$. This is not an extra assumption but a consequence: the change-of-variables constant was pinned to the fibre count $|W|$ in Step 3, and the probability measure on G/T used there is the one induced by the normalized Haar measures via the fibration $T \to G \to G/T$, so no free scalar remains. The identity $\int_T|\delta|^2\,dt = |W|$ is thus both a corollary of the formula and its consistency check, completing the proof.

The formula completes the reduction to the torus. Every integral of a class function over the whole group, a quantity that in principle requires integrating over a manifold of the group's full dimension, collapses to an integral over the torus T (of dimension $\operatorname{rank} G$) weighted by the explicit density $|\delta|^2$. The finite factor $1/|W|$ corrects for overcounting: the conjugation map wraps $|W|$ copies of the regular part of T onto the regular part of G, and the density is the Jacobian recording how much the wrapping stretches. The Weyl denominator makes such integrals tractable because it is built from the same exponential sums as the characters themselves.

The formula also relocates the character inner product. For characters χ_λ, χ_μ the Schur orthogonality $\int_G \chi_\lambda\overline{\chi_\mu}\,dg = \delta_{\lambda\mu}$ becomes, on the torus,

$$\int_G \chi_\lambda(g)\,\overline{\chi_\mu(g)}\,dg = \frac{1}{|W|}\int_T \chi_\lambda(t)\,\overline{\chi_\mu(t)}\,|\delta(t)|^2\,dt$$

so orthogonality of the group characters is orthogonality of their torus restrictions against the weight $|\delta|^2$. When each $\chi_\lambda(t)\,\delta(t)$ is an antisymmetric exponential sum, as the Weyl character formula (chapter 12) will show, the weighted orthogonality on T reduces to orthogonality of the exponential characters of the torus, which the harmonic analysis of $\mathrm{U}(1)^r$ makes elementary. The Weyl integration formula is the bridge that carries the abstract orthogonality of Peter-Weyl (theorem 10.4.4) onto the torus, where it can be verified by hand.

Example 11.4.4: The Weyl Integration Formula for SU(2) and Character Orthonormality

For $G = \mathrm{SU}(2)$ the maximal torus is $T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta}) \mid \theta \in [0, 2\pi)\}$, the Weyl group is $W = \mathbb{Z}/2$ acting by $t_\theta \mapsto t_{-\theta}$ (definition 11.2.1), and the squared Weyl denominator is $|\delta(t_\theta)|^2 = 4\sin^2\theta$ (example 11.4.2). The normalized Haar measure on the circle T is $dt = d\theta/2\pi$, so the Weyl integration formula reads

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{|W|}\int_T f(t)\,|\delta(t)|^2\,dt = \frac{1}{2}\cdot\frac{1}{2\pi}\int_0^{2\pi} f(t_\theta)\cdot 4\sin^2\theta\,d\theta$$

for a class function f. Since the integrand is even in θ (both $f(t_\theta) = f(t_{-\theta})$ by W-invariance and $\sin^2\theta$ are even), the integral over $[0, 2\pi)$ is twice the integral over $[0, \pi]$, and the constants collapse:

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{2}{\pi}\int_0^{\pi} f(t_\theta)\,\sin^2\theta\,d\theta$$

recovering the density used in proposition 9.5.3 and in the chapter 9 exercises. This makes the irreducible characters $\chi_n(t_\theta) = \sin((n+1)\theta)/\sin\theta$ orthonormal. Each χ_n is real, so $\overline{\chi_n} = \chi_n$, and

$$\int_{\mathrm{SU}(2)} \chi_m\,\overline{\chi_n}\,dg = \frac{2}{\pi}\int_0^{\pi} \frac{\sin((m+1)\theta)}{\sin\theta}\cdot\frac{\sin((n+1)\theta)}{\sin\theta}\cdot\sin^2\theta\,d\theta = \frac{2}{\pi}\int_0^{\pi} \sin\big((m+1)\theta\big)\sin\big((n+1)\theta\big)\,d\theta$$

the two $\sin\theta$ denominators cancelling the density $\sin^2\theta$ exactly. The product-to-sum identity $2\sin a\sin b = \cos(a-b) - \cos(a+b)$ gives

$$\frac{2}{\pi}\int_0^{\pi} \sin\big((m+1)\theta\big)\sin\big((n+1)\theta\big)\,d\theta = \frac{1}{\pi}\int_0^{\pi}\Big(\cos\big((m-n)\theta\big) - \cos\big((m+n+2)\theta\big)\Big)\,d\theta$$

The integral $\int_0^{\pi}\cos(k\theta)\,d\theta$ equals π for $k = 0$ and 0 for any nonzero integer k. Here $m+n+2 \geq 2 > 0$ always, so the second term vanishes, while the first contributes π exactly when $m = n$ and 0 otherwise. Hence

$$\int_{\mathrm{SU}(2)} \chi_m\,\overline{\chi_n}\,dg = \delta_{mn}$$

the characters of SU(2) are orthonormal, the SU(2) instance of Schur orthogonality obtained by pushing the integral to the torus and cancelling the Weyl density against the two character denominators. The Weyl denominator is precisely what makes this cancellation, and hence the whole computation, elementary.

The mechanism the SU(2) computation displays is general. The density $|\delta|^2$ is the modulus-squared of an antisymmetric sum, and each irreducible character on T, by the Weyl character formula (chapter 12), is such a sum divided by δ; multiplying two characters by the density $|\delta|^2 = \delta\overline{\delta}$ clears

both denominators and leaves a product of numerator exponential sums, whose torus integral is computed by the orthogonality of the characters of T. This is why the character formula and the integration formula belong to the same circle of ideas: the denominator of the one is the density of the other, and their interplay reduces every question about compact-group characters to the linear algebra of the weight lattice, developed in the next section.

Exercise 11.4.1

1. Carry out the SU(2) orthonormality computation of example 11.4.4 in full detail for the pair $m = 1$, $n = 3$: write out $\chi_1(t_\theta) = \sin(2\theta)/\sin\theta$ and $\chi_3(t_\theta) = \sin(4\theta)/\sin\theta$, form the integrand of $\int_{\mathrm{SU}(2)} \chi_1\overline{\chi_3}\,dg$ against the density $\frac{2}{\pi}\sin^2\theta$, and evaluate the resulting $\frac{1}{\pi}\int_0^\pi(\cos 2\theta - \cos 6\theta)\,d\theta$ to confirm the value $0 = \delta_{13}$.

2. Show that the singular set of G (the elements lying on more than one maximal torus) has Haar measure zero, so the regular set has full measure. Argue that on T the singular locus is the zero set $\{t \in T \mid |\delta(t)|^2 = 0\}$, a finite union of proper subtori (one $\ker\alpha$ per root), and that its image under the conjugation map q is a set of measure zero in G by Sard's theorem ([13, Sard's Theorem]). Conclude that dropping the singular set changes no integral in theorem 11.4.3.

3. Prove that $|\delta(t)|^2$ vanishes at $t \in T$ if and only if t is singular, that is, if and only if $e^\alpha(t) = 1$ for some root α. Use the product formula $|\delta(t)|^2 = \prod_{\alpha\in\Phi}(1 - e^{-\alpha}(t))$ of definition 11.4.1 and the fact that a product of complex numbers is zero iff one factor is. Interpret the condition $e^\alpha(t) = 1$ as t lying in the kernel of the root character $\alpha\colon T \to \mathrm{U}(1)$.

4. Verify the normalization identity $\int_T |\delta(t)|^2\,dt = |W|$ established in Step 5 of the proof of theorem 11.4.3 directly for SU(2): compute $\int_T 4\sin^2\theta\,\frac{d\theta}{2\pi}$ over $\theta \in [0, 2\pi)$ and check it equals $|W| = 2$. Explain in one sentence why this equality is the special case $f \equiv 1$ of the Weyl integration formula and why it fixes the constant $1/|W|$.

5. Write down the Weyl integration formula for $\mathrm{U}(n)$. The maximal torus is $T = \{\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) \mid \theta_j \in \mathbb{R}\}$, the Weyl group is $W = S_n$ (so $|W| = n!$), and the roots are $\alpha_{jk}(t) = e^{i(\theta_j - \theta_k)}$ for $j \neq k$. Show that the squared Weyl denominator is the squared Vandermonde

$$|\delta(t)|^2 = \prod_{j<k} |e^{i\theta_j} - e^{i\theta_k}|^2$$

and conclude that for a class function f,

$$\int_{\mathrm{U}(n)} f\,dg = \frac{1}{n!}\int_{[0,2\pi)^n} f(\theta)\prod_{j<k}|e^{i\theta_j} - e^{i\theta_k}|^2\,\frac{d\theta_1\cdots d\theta_n}{(2\pi)^n}$$

6. Explain, using corollary 11.1.4 and proposition 11.2.4, why the Weyl integration formula requires f to be a class function and cannot hold for arbitrary continuous f. Identify precisely which step of the proof of theorem 11.4.3 uses the class-function hypothesis, and describe what the formula would compute for a general f.

11.5 Weights, Coroots, and the Fundamental Group

The root system Φ reconstructs the Lie algebra $\mathfrak{g}_{\mathbb{C}}$ but not the group: the same $\mathfrak{g}_{\mathbb{C}}$ underlies $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$, which differ only in their global topology. This section records the one further datum that pins the group down. It is a lattice, the character lattice $X^*(T)$, sitting between two lattices that the root system alone determines, the root lattice Q and the weight lattice P. Its position between them reads off the center and the fundamental group of G, and the finitely many admissible positions enumerate the finitely many compact connected groups sharing the Lie algebra $\mathfrak{g}$, refining the finiteness of theorem 9.4.5 into an exact count.

The torus T is where these lattices live, so the section begins with the two lattices canonically attached to any torus: the characters and the cocharacters, dual to one another under a perfect integer pairing.

Definition 11.5.1: Character, Cocharacter, Weight, and Root Lattices

Let T be a compact torus of rank r, with Lie algebra $\mathfrak{t}$ and exponential $\exp\colon \mathfrak{t} \to T$.

1. The *character lattice* is $X^*(T) = \mathrm{Hom}(T, \mathrm{U}(1))$, the group of continuous homomorphisms $\chi\colon T \to \mathrm{U}(1)$ under pointwise multiplication.
2. The *cocharacter lattice* is $X_*(T) = \mathrm{Hom}(\mathrm{U}(1), T)$, the group of continuous homomorphisms $\lambda\colon \mathrm{U}(1) \to T$.
3. The *pairing* $\langle -, -\rangle\colon X^*(T) \times X_*(T) \to \mathbb{Z}$ sends (χ, λ) to the integer n for which $\chi(\lambda(z)) = z^n$ for all $z \in \mathrm{U}(1)$.

Now let G be a compact connected Lie group with maximal torus T and root system $\Phi \subseteq X^*(T)$ (definition 11.3.1), and let $\alpha^\vee$ be the coroot of α (definition 7.4.3). The *root lattice* is the subgroup $Q = \mathbb{Z}\Phi \subseteq X^*(T)$ generated by the roots, and the *weight lattice* is

$$P = \{\lambda \in \mathfrak{t}^* \mid \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \text{ for every } \alpha \in \Phi\}$$

the forms taking integer values on every coroot. When $\mathfrak{g}$ is semisimple, so that the roots span $\mathfrak{t}^*$, the constraints cut P down to a genuine full-rank lattice; when they do not span (for instance a torus, where $\Phi = \varnothing$ and $P = \mathfrak{t}^*$; see exercise 11.5.1 (2)), the word "lattice" for P is a convenient abuse and the finiteness of P/Q below is asserted only in the semisimple case.

Each of $X^*(T)$ and $X_*(T)$ is a free abelian group of rank r, and the pairing identifies each as the $\mathbb{Z}$-dual of the other. The mechanism is the exponential. The kernel $\ker(\exp\colon \mathfrak{t} \to T)$ is a full-rank lattice $\Lambda \subseteq \mathfrak{t}$, since $T = \mathfrak{t}/\Lambda \cong (\mathbb{R}/\mathbb{Z})^r$; a cocharacter $\lambda\colon \mathrm{U}(1) \to T$ is the same as a lift $z = e^{i\theta} \mapsto \exp(\theta X)$ for a vector $X \in \mathfrak{t}$ with $\exp(2\pi X) = e$ (well-definedness under $\theta \mapsto \theta + 2\pi$, which fixes z), so $X_*(T)$ is exactly the lattice $\frac{1}{2\pi}\Lambda$ scaled to unit period. Dually a character $\chi\colon T \to \mathrm{U}(1)$ is $\exp(X) \mapsto e^{2\pi i \mu(X)}$ for a form $\mu \in \mathfrak{t}^*$ that is integer-valued on $\frac{1}{2\pi}\Lambda$, so $X^*(T)$ is the dual lattice. Under these identifications the pairing $\langle \chi, \lambda \rangle$ is the value $\mu(X) \in \mathbb{Z}$, and it is perfect: it exhibits $X^*(T) = \mathrm{Hom}(X_*(T), \mathbb{Z})$. The differentials $d\chi_e = 2\pi i\, \mu$ let $X^*(T)$ be read either as homomorphisms to the circle or, once $2\pi i$ is stripped, as the lattice of forms $\mu \in \mathfrak{t}^*$ integral on $\frac{1}{2\pi}\Lambda$; the two viewpoints are used interchangeably below, and a root $\alpha \in \Phi$ is a character of T in the first and a form in $\mathfrak{t}^*$ in the second.

The root and weight lattices are the two extremes cut out by the root system. The root lattice Q is the smallest lattice containing the roots; the weight lattice P is the largest lattice on which the coroots are integer-valued, so $\langle P, \alpha^\vee \rangle \subseteq \mathbb{Z}$ for every α. That the roots themselves lie in P is the integrality of the Cartan integers $\langle \alpha, \beta^\vee \rangle \in \mathbb{Z}$ (theorem 7.4.6), giving the inclusion $Q \subseteq P$. Both are intrinsic to Φ, hence to the Lie algebra alone; the character lattice $X^*(T)$ of the actual group is squeezed between them.

Example 11.5.2: The Lattices of SU(2) and SO(3)

The maximal torus of SU(2) is $T = \left\{ t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta}) \;\middle|\; \theta \in \mathbb{R} \right\}$, a circle of rank 1. Write $\varepsilon \colon T \to \mathrm{U}(1)$ for the character $t_\theta \mapsto e^{i\theta}$; then every character is a power $\varepsilon^n \colon t_\theta \mapsto e^{in\theta}$, so $X^*(T) = \mathbb{Z}\varepsilon \cong \mathbb{Z}$. Identify $\mathfrak{t}^* \cong \mathbb{R}$ by $\varepsilon \leftrightarrow 1$. The single positive root is $\alpha = 2\varepsilon$, because the adjoint action of t_θ on the raising operator of $\mathfrak{sl}_2(\mathbb{C})$ multiplies it by $e^{2i\theta}$ (theorem 11.3.3); its coroot satisfies $\langle \alpha, \alpha^\vee \rangle = 2$. Thus $Q = \mathbb{Z}\alpha = 2\mathbb{Z}$ in the coordinate where $X^*(T) = \mathbb{Z}$, while the weight lattice is $P = \{ m \in \mathbb{Z} \mid \langle m, \alpha^\vee \rangle \in \mathbb{Z} \} = \mathbb{Z}$, since $\langle \varepsilon, \alpha^\vee \rangle = 1$. The chain is

$$Q = 2\mathbb{Z} \subseteq X^*(T) = \mathbb{Z} = P$$

so SU(2) realizes the top lattice $X^*(T) = P$, with index $[P : Q] = 2$.

The rotation group $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ has the same Lie algebra and the same root $\alpha = 2\varepsilon$, but a different torus: its maximal torus $T' = T/\{\pm I\}$ is the circle of rotations, and a character of T' is a character of T trivial on $-I = t_\pi$. Since $\varepsilon^n(t_\pi) = e^{in\pi} = (-1)^n$, only the even powers descend, so $X^*(T') = \mathbb{Z}\alpha = 2\mathbb{Z} = Q$. Now the chain reads

$$Q = 2\mathbb{Z} = X^*(T') \subseteq P = \mathbb{Z}$$

so SO(3) realizes the bottom lattice $X^*(T') = Q$. The two groups with algebra $\mathfrak{su}(2)$ are the two lattices between Q and P, and $[P : Q] = 2$ counts them exactly.

The example shows the position of $X^*(T)$ carrying real information: SU(2) sits at P, SO(3) at Q, and no other lattice lies strictly between. The next result establishes the pattern in general, that $X^*(T)$ always lands between the two extremes and that the extremes are realized by the simply connected and adjoint groups.

Proposition 11.5.3: The Lattice Chain and the Isogeny Type

Let G be a compact connected Lie group with maximal torus T, root lattice Q, and weight lattice P of definition 11.5.1. Then

$$Q \subseteq X^*(T) \subseteq P$$

and the quotient P/Q is finite (theorem 9.4.5). When $\mathfrak{g}$ is semisimple, the two extremes are realized:

1. if $G = \widetilde{G}$ is simply connected, then $X^*(T) = P$: every integral weight is a character of T;
2. if G is adjoint (that is, $G = \mathrm{Int}(\mathfrak{g}) = G/Z(G)$ acting by Ad), then $X^*(T) = Q$: the characters of T are exactly the root lattice.

A general compact connected G with semisimple Lie algebra $\mathfrak{g}$ therefore corresponds to a lattice $X^*(T)$ with $Q \subseteq X^*(T) \subseteq P$, and the finite group P/Q parametrizes the possibilities.

***Proof*:**

Step 1: the inclusions. Every root α is a character of T by definition 11.3.1, so $\Phi \subseteq X^*(T)$ and hence $Q = \mathbb{Z}\Phi \subseteq X^*(T)$. For the upper inclusion, let $\chi \in X^*(T)$ be any character, with differential $\mu = d\chi_e/(2\pi i) \in \mathfrak{t}^*$. The coroot $\alpha^\vee$ is the cocharacter $\alpha^\vee\colon \mathrm{U}(1) \to T$ landing in the maximal torus of the SU(2) subgroup generated by the root spaces $\mathfrak{g}_{\pm\alpha}$ (theorem 11.3.3), so $\chi \circ \alpha^\vee\colon \mathrm{U}(1) \to \mathrm{U}(1)$ is a character of the circle, namely $z \mapsto z^{\langle \chi, \alpha^\vee \rangle}$ with $\langle \chi, \alpha^\vee \rangle \in \mathbb{Z}$. Thus μ pairs integrally with every coroot, so $\mu \in P$, giving $X^*(T) \subseteq P$. Finiteness of P/Q is the finiteness of theorem 9.4.5 read on the lattice side, and is proved again below when P/Q is identified with the central and fundamental-group data.

Step 2: the simply connected extreme. Let $G = \widetilde{G}$ be simply connected with semisimple $\mathfrak{g}$. The claim is $X^*(T) = P$, that every $\mu \in P$ integrates to a character of T; the cocharacter equality $X_*(T) = Q^\vee$ will follow by duality. Step 1 gives $X^*(T) \subseteq P$, so only the reverse inclusion $P \subseteq X^*(T)$ is at issue, and it is where simple connectivity enters. Fix $\mu \in P$. The theory of the finite-dimensional representations of the semisimple $\mathfrak{g}$ produces a representation of $\mathfrak{g}$ on which μ occurs as a weight of $\mathfrak{t}$ (concretely the irreducible module of extreme weight μ; this is the highest-weight construction taken up in chapter 12, used here only to exhibit a single representation realizing μ). Because $\widetilde{G}$ is simply connected, every representation of its Lie algebra $\mathfrak{g}$ integrates to a representation of the group (corollary 3.4.5); the resulting representation of $\widetilde{G}$ restricts to T with μ among the characters of T appearing in it, so $\mu \in X^*(T)$. Since $\mu \in P$ was arbitrary, $P \subseteq X^*(T)$, and with Step 1 this gives $X^*(T) = P$. It is exactly simple connectivity that removes the constraint "μ trivial on some Γ" that a nontrivial covering would impose; without it the argument would only realize those μ trivial on the deck group. Now dualize under the perfect pairing $X^*(T) = \mathrm{Hom}(X_*(T), \mathbb{Z})$ (equivalently $X_*(T) = \mathrm{Hom}(X^*(T), \mathbb{Z})$): the annihilator description of P is

$$P = \{\mu \in \mathfrak{t}^* \mid \langle \mu, \alpha^\vee \rangle \in \mathbb{Z}\ \forall \alpha\} = \mathrm{Hom}(Q^\vee, \mathbb{Z})$$

so the lattice dual to $X^*(T) = P$ is $X_*(T) = \mathrm{Hom}(P, \mathbb{Z}) = Q^\vee$, the coroot lattice. Thus the simply connected group has cocharacter lattice exactly $Q^\vee$, and every integral weight is a character of its maximal torus.

Step 3: the adjoint extreme. Let G be adjoint, acting faithfully by Ad on $\mathfrak{g}_{\mathbb{C}}$. Its representations are exactly the representations of $\mathfrak{g}$ on which the center acts trivially. A character of the adjoint torus T_{ad} is a weight of such a representation; the adjoint representation itself has weights $\Phi \cup \{0\}$, and its tensor powers carry weights in $\mathbb{Z}\Phi = Q$, so every character lies in Q. Conversely each root is a character of T_{ad}. Hence $X^*(T_{\mathrm{ad}}) = Q$. Dually the cocharacter lattice of the adjoint group is the coweight lattice $P^\vee = \{H \in \mathfrak{t} \mid \langle \alpha, H\rangle \in \mathbb{Z}\ \forall \alpha\}$, the largest lattice on which the roots are integer-valued, and $X^*(T_{\mathrm{ad}}) = \mathrm{Hom}(P^\vee, \mathbb{Z}) = Q$.

Step 4: the intermediate case. An arbitrary compact connected G with semisimple $\mathfrak{g}$ is a quotient $\widetilde{G}/\Gamma$ by a finite central subgroup $\Gamma \subseteq Z(\widetilde{G})$ (theorem 9.4.5). Its maximal torus is $\widetilde{T}/\Gamma$, and a character of it is a character of $\widetilde{T}$ trivial on Γ; so $X^*(T)$ is the subgroup of $P = X^*(\widetilde{T})$ of weights killing Γ. As Γ ranges over the finitely many subgroups of $Z(\widetilde{G})$, the lattice $X^*(T)$ ranges over lattices between Q (when $\Gamma = Z(\widetilde{G})$, the adjoint case) and P (when $\Gamma = \{e\}$, the simply connected case), and P/Q parametrizes them, completing the proof.

The proposition turns the informal statement that "the group is more than its Lie algebra" into a finite invariant. Everything the root system gives, the two lattices $Q \subseteq P$ and their finite quotient, is intrinsic to $\mathfrak{g}$; the extra choice that specifies the group is a single lattice $X^*(T)$ wedged between them. The two extremes have names, the simply connected group at P and the adjoint group at Q, and every other compact group with the same algebra sits at an intermediate lattice, one for each subgroup of P/Q. The next theorem makes the correspondence exact by identifying the two quotients $X^*(T)/Q$ and $P/X^*(T)$, from below and from above, with the center and the fundamental group.

The mechanism is again the exponential map and the universal cover. The kernel of exp on $\mathfrak{t}$ is the cocharacter lattice up to the factor 2π; the fundamental group of G measures how far that kernel exceeds the minimal coroot lattice, and the center measures how far the character lattice exceeds the root lattice. These are the two finite quotients dual to the placement of $X^*(T)$, and they are exchanged when $Q^\vee$ and Q swap roles.

Theorem 11.5.4: The Fundamental Group and the Center

Let G be a compact connected semisimple Lie group with maximal torus T, root lattice Q, weight lattice P, and coroot lattice $Q^\vee = \mathbb{Z}\{\alpha^\vee \mid \alpha \in \Phi\}$. Then there are canonical isomorphisms

$$\pi_1(G) \cong X_*(T)/Q^\vee, \qquad Z(G) \cong \mathrm{Hom}\big(X^*(T)/Q,\ \mathrm{U}(1)\big)$$

both finite. The fundamental group and the center are the two finite quotients dual to the position of $X^*(T)$ between Q and P, and the compact connected groups with Lie algebra $\mathfrak{g}$ are in bijection with the subgroups of P/Q (corollary 3.4.5, refining theorem 9.4.5).

***Proof*:**

Step 1: the fundamental group as a cocharacter quotient. Let $p\colon \widetilde{G} \to G$ be the universal cover (proposition 3.4.1), a homomorphism of compact groups (theorem 9.4.5) with $\ker p \cong \pi_1(G)$ central and finite. Restricting to maximal tori, $\widetilde{T} = p^{-1}(T)^0$ covers T with the same kernel $\ker p$, and both tori have Lie algebra $\mathfrak{t}$. On $\mathfrak{t}$ the two exponential maps satisfy $\exp_T = p \circ \exp_{\widetilde{T}}$, and p

collapses the extra kernel, so

$$\ker(\exp_{\widetilde{T}}) \subseteq \ker(\exp_T) \subseteq \mathfrak{t}$$

Identifying $\ker(\exp_{\widetilde{T}}) = 2\pi\, Q^\vee$ (the simply connected cocharacter lattice, Step 2 of proposition 11.5.3) and $\ker(\exp_T) = 2\pi\, X_*(T)$, the deck group is the quotient of kernel lattices,

$$\pi_1(G) \cong \ker p \cong \ker(\exp_T)/\ker(\exp_{\widetilde{T}}) \cong X_*(T)/Q^\vee$$

finite because $Q^\vee$ has finite index in the coweight lattice $P^\vee \supseteq X_*(T)$, with $[P^\vee : Q^\vee] = |P/Q|$ by duality of the pairing.

Step 2: the center as a character quotient. A central element commutes with the connected G, hence with T, and by maximality lies in T, so $Z(G) \subseteq T$. An element $t \in T$ is central if and only if $\mathrm{Ad}(t)$ fixes every root vector, that is $\alpha(t) = 1$ for every $\alpha \in \Phi$, since $\mathrm{Ad}(t)$ acts on $\mathfrak{g}_\alpha$ by $\alpha(t)$ (theorem 11.3.3). Thus

$$Z(G) = \{t \in T \mid \alpha(t) = 1 \text{ for all } \alpha \in \Phi\} = \bigcap_{\alpha\in\Phi} \ker \alpha$$

the simultaneous kernel of the roots. By Pontryagin duality for the torus, $T = \mathrm{Hom}(X^*(T), \mathrm{U}(1))$, and a point t is a homomorphism $X^*(T) \to \mathrm{U}(1)$; the condition $\alpha(t) = 1$ for all roots says this homomorphism is trivial on $Q = \mathbb{Z}\Phi$, hence factors through $X^*(T)/Q$. Therefore

$$Z(G) = \mathrm{Hom}\big(X^*(T)/Q,\ \mathrm{U}(1)\big)$$

Since $X^*(T)/Q$ is a finite abelian group (it lies inside P/Q), its Pontryagin dual is finite abelian of the same order, so $Z(G)$ is finite with $|Z(G)| = [X^*(T) : Q]$.

Step 3: duality and the count. The two quotients sit on opposite sides of $X^*(T)$, and the perfect pairing $\langle -, - \rangle \colon X^*(T) \times X_*(T) \to \mathbb{Z}$ matches each quotient with a quotient of the *dual* lattice, not with the other quotient of G. Passing to $\mathbb{Q}/\mathbb{Z} \hookrightarrow \mathrm{U}(1)$, the pairing identifies $X_*(T)/Q^\vee$ with the Pontryagin dual of $P/X^*(T)$, since the annihilator of $Q^\vee$ is P and the annihilator of $X^*(T)$ is $X_*(T)$; symmetrically it identifies $X^*(T)/Q$ with the Pontryagin dual of $P^\vee/X_*(T)$. Thus $\pi_1(G) = X_*(T)/Q^\vee$ is dual to $P/X^*(T)$ (not to $X^*(T)/Q$), while $Z(G)$ is dual to $X^*(T)/Q$; the two are distinct in general, as $\mathrm{SO}(3)$ already shows with $\pi_1 = \mathbb{Z}/2$ but $Z = 0$. The pairing does force the orders: $|Z(G)| = [X^*(T) : Q]$ and $|\pi_1(G)| = [X_*(T) : Q^\vee] = [P : X^*(T)]$, so their product is $[P : Q]$, and $Z(G)$ and $\pi_1(G)$ trade off as $X^*(T)$ slides between Q and P. Finally each compact connected G with algebra $\mathfrak{g}$ is $\widetilde{G}/\Gamma$ for a unique $\Gamma \subseteq Z(\widetilde{G}) = P/Q$ (Step 4 of proposition 11.5.3), and the assignment $G \mapsto X^*(T)/Q$ is a bijection between such groups and the subgroups of P/Q, refining corollary 3.4.5 and theorem 9.4.5, completing the proof.

The theorem completes the dictionary from a compact group to its combinatorics. Where the root system determines the Lie algebra, and the Lie algebra determines the group up to the finite central data of theorem 9.4.5, that finite data is now named exactly: the fundamental group is $X_*(T)/Q^\vee$, the excess of the actual cocharacter lattice over the minimal coroot lattice, and the center is $\mathrm{Hom}(X^*(T)/Q, \mathrm{U}(1))$, dual to the excess of the character lattice over the root lattice. Within a single group G the two are not Pontryagin-dual to each other, only order-constrained: $|Z(G)| \cdot |\pi_1(G)| = |P/Q|$ is fixed, so a group near the top of the chain (with $X^*(T)$ close to P) has small fundamental group and large center, and a group near the bottom has large fundamental group and small center. The simply connected group has $\pi_1 = 0$ and the full center $Z(\widetilde{G}) = P/Q$; the adjoint group has trivial center and the full fundamental group $\pi_1 = P/Q$; each intermediate

quotient trades one for the other. The genuine Pontryagin exchange of Z with π_1 is not within G but across the dual group, taken up next.

This exchange is the compact-group case of Langlands duality. Passing from the root system Φ to its dual system $\Phi^\vee = \{\alpha^\vee\}$ swaps roots for coroots, hence Q for $Q^\vee$ and P for $P^\vee$, and therefore swaps the two quotients $X^*(T)/Q$ and $X_*(T)/Q^\vee$: the center of G becomes the fundamental group of the group $G^\vee$ built from the dual root system, and vice versa. The simply connected form of G corresponds to the adjoint form of $G^\vee$. The full development belongs to the Langlands dual group and the representation theory of chapter 12, but the lattice bookkeeping here already exhibits the symmetry.

Example 11.5.5: The Isogeny Classes of Type A_1 and Type A_{n-1}

Type A_1: $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$. With $P = \mathbb{Z}$, $Q = 2\mathbb{Z}$ from example 11.5.2, the quotient is $P/Q = \mathbb{Z}/2$. The simply connected group $\mathrm{SU}(2)$ has $X^*(T) = P$, so

$$Z(\mathrm{SU}(2)) = \mathrm{Hom}(P/Q, \mathrm{U}(1)) = \mathrm{Hom}(\mathbb{Z}/2, \mathrm{U}(1)) \cong \mathbb{Z}/2, \qquad |\pi_1(\mathrm{SU}(2))| = [P : X^*(T)] = 1$$

matching $Z(\mathrm{SU}(2)) = \{\pm I\}$ and $\mathrm{SU}(2) = S^3$ simply connected. The adjoint group $\mathrm{SO}(3)$ has $X^*(T') = Q$, so

$$Z(\mathrm{SO}(3)) = \mathrm{Hom}(Q/Q, \mathrm{U}(1)) = 0, \qquad \pi_1(\mathrm{SO}(3)) \cong P/Q \cong \mathbb{Z}/2, \quad |\pi_1(\mathrm{SO}(3))| = [P : Q] = 2$$

recovering the known $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2$. The two subgroups of $P/Q = \mathbb{Z}/2$ index the two groups, exactly as theorem 11.5.4 predicts.

Type A_{n-1}: $\mathrm{SU}(n)$ and $\mathrm{PU}(n)$. The root system of $\mathfrak{su}(n)$ is A_{n-1}, with $P/Q \cong \mathbb{Z}/n$. The simply connected group is $\mathrm{SU}(n)$: it is its own universal cover, so $\pi_1(\mathrm{SU}(n)) = 0$ and its center is the full

$$Z(\mathrm{SU}(n)) = \{\zeta I \mid \zeta^n = 1\} \cong \mu_n \cong \mathbb{Z}/n$$

the group of n-th roots of unity, dual to $P/Q = \mathbb{Z}/n$. The adjoint group is the projective unitary group $\mathrm{PU}(n) = \mathrm{SU}(n)/\mu_n$: its center is trivial and $\pi_1(\mathrm{PU}(n)) = P/Q = \mathbb{Z}/n$. The full family is $\mathrm{SU}(n)/\mu_d$ for each divisor $d \mid n$, indexed by the subgroups $\mu_d \subseteq \mu_n$, that is by the subgroups of $\mathbb{Z}/n$, with center $\mu_{n/d}$ and fundamental group $\mathbb{Z}/d$; the extremes $d = 1$ and $d = n$ recover $\mathrm{SU}(n)$ and $\mathrm{PU}(n)$, and the proper divisors sit strictly between. For n prime only the two extremes exist; for $n = 4$ the divisor $d = 2$ gives an intermediate group $\mathrm{SU}(4)/\{\pm I\}$ with center μ_2 and $\pi_1 = \mathbb{Z}/2$.

The isogeny class of a compact semisimple group is thus read off a single lattice. Its finitely many forms are the lattices between Q and P; the center of each is the dual of $X^*(T)/Q$ and the fundamental group is $P/X^*(T)$; and the two are exchanged by the dual root system. Nothing about the group's global topology remains hidden once its character lattice is located in the chain. The dictionary from the group to its combinatorial data, begun with the maximal torus in section 11.1 and continued through the Weyl group and the root system, is now complete for the count established here: within a fixed semisimple $\mathfrak{g}$ the isogeny class is exactly the position of $X^*(T)$ in the chain. That the full quadruple $(X^*(T), \Phi, X_*(T), \Phi^\vee)$, the root datum, determines the compact connected group up to isomorphism and that every root datum arises, the classification-by-root-data theorem, is the wider statement this section prepares but does not prove in full.

Exercise 11.5.1

1. Using theorem 11.5.4, show that the center of $\mathrm{SU}(n)$ is $\mu_n = \{\zeta I \mid \zeta^n = 1\} \cong \mathbb{Z}/n$. Take the simply connected type-A_{n-1} facts $X^*(T) = P$ and $P/Q \cong \mathbb{Z}/n$, and identify the resulting characters with the scalar matrices ζI directly.

2. Let $T = \mathbb{T}^n = (\mathbb{R}/\mathbb{Z})^n$ be an n-torus, which has no roots. Show that $X^*(T) \cong \mathbb{Z}^n$ and $X_*(T) \cong \mathbb{Z}^n$, that the pairing is the standard dot product, and that (since $\Phi = \varnothing$) the root lattice is $Q = 0$ and the weight lattice is all of $\mathfrak{t}^* \cong \mathbb{R}^n$. Conclude that a torus imposes no constraint of the form $Q \subseteq X^*(T) \subseteq P$ and that $\pi_1(\mathbb{T}^n) = \mathbb{Z}^n$ is infinite, consistent with $\mathbb{T}^n$ not being semisimple.

3. For the unitary group $\mathrm{U}(n)$, whose maximal torus is the diagonal $T = \left\{\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) \;\middle|\; \theta_j \in \mathbb{R}\right\}$, show that $X^*(T) = \mathbb{Z}^n$ with the coordinate characters $\varepsilon_j \colon \mathrm{diag}(e^{i\theta_k}) \mapsto e^{i\theta_j}$ as a basis, and that the roots are $\varepsilon_j - \varepsilon_k$ for $j \neq k$. Compute the root lattice $Q = \{\sum a_j \varepsilon_j \in \mathbb{Z}^n \mid \sum a_j = 0\}$ and explain why $\mathrm{U}(n)$ is not semisimple (its center is the full circle of scalars, positive-dimensional), so theorem 11.5.4 does not apply to it directly but does apply to its semisimple part $\mathrm{SU}(n)$.

4. The dual root system $\Phi^\vee = \{\alpha^\vee\}$ swaps $Q \leftrightarrow Q^\vee$ and $P \leftrightarrow P^\vee$. Using theorem 11.5.4, explain in a short paragraph why passing to the Langlands dual group $G^\vee$ (built from $\Phi^\vee$) exchanges the center and the fundamental group: $Z(G) \cong \pi_1(G^\vee)$ and $\pi_1(G) \cong Z(G^\vee)$. Illustrate with type A_1, where $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ are dual to each other.

5. Let $G = \widetilde{G}/\Gamma$ for a subgroup $\Gamma \subseteq Z(\widetilde{G}) = P/Q$, so that $X^*(T)$ is the lattice of weights killing Γ. Prove that $|Z(G)| \cdot |\pi_1(G)| = |P/Q|$, and deduce that a group is simply connected if and only if it has the largest possible center among the forms of its Lie algebra, and adjoint if and only if it has trivial center.

6. The group $\mathrm{SO}(2n)$ has root system D_n, with $P/Q \cong \mathbb{Z}/4$ when n is odd and $P/Q \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ when n is even. Its simply connected cover is the spin group $\mathrm{Spin}(2n)$. Explain, using theorem 11.5.4, why $\pi_1(\mathrm{SO}(2n)) = \mathbb{Z}/2$ (so $\mathrm{SO}(2n)$ is neither simply connected nor adjoint), and identify $\mathrm{Spin}(2n)$ and the adjoint group $\mathrm{PSO}(2n)$ as the two extremes of the chain, with $\mathrm{SO}(2n)$ strictly between them.

12

The Weyl Character Formula

The maximal-torus and Weyl-integration machinery of chapter 11 reduced every character of a compact connected group to a Weyl-invariant function on a maximal torus. This chapter identifies those characters exactly, completing the program begun with Peter-Weyl. The theorem of the highest weight classifies the irreducible representations by their dominant weights, one for each lattice point of the dominant Weyl chamber, and the Weyl character formula writes each irreducible character as a ratio of alternating sums over the Weyl group: an antisymmetrized exponential in the numerator, the Weyl denominator beneath it. It is the exact generalization of the $\sin((n+1)\theta)/\sin\theta$ formula for $\mathrm{SU}(2)$, and it is the analytic summit of the book, the point at which the root-system combinatorics of Part II and the integration theory of Part III combine to compute every irreducible character in closed form.

The highest-weight theorem is proved by joining the algebraic highest-weight theory of the semisimple Lie algebra to the compact-group integration: every dominant integral weight is the highest weight of a unique irreducible, and distinct dominant weights give inequivalent irreducibles. The character formula then follows from the Weyl integration formula and the orthonormality of characters, once the alternating sum $A_{\lambda+\rho} = \sum_{w \in W} \operatorname{sgn}(w)\, e^{w(\lambda+\rho)}$ is seen to be divisible by the Weyl denominator A_ρ, with quotient the character χ_λ. Evaluating the formula in the limit at the identity yields the Weyl dimension formula, a product over the positive roots; Kostant's formula expands the weight multiplicities through the partition function; the Borel-Weil theorem realizes each irreducible geometrically as the holomorphic sections of a line bundle on the flag manifold, with the Bott refinement cited to [10]; and the chapter, with Part III, closes on the explicit representation theory of $\mathrm{SU}(n)$ and $\mathrm{SO}(n)$, where the characters become Schur polynomials.

12.1 Dominant Weights and the Theorem of the Highest Weight

Peter-Weyl (theorem 10.3.4) decomposed $L^2(G)$ into the matrix coefficients of the irreducible representations, so that classifying the irreducibles is the remaining content of the harmonic analysis of a compact connected group. Each irreducible is determined by its character (theorem 10.4.4), and the character is a class function, hence a W-invariant function on the maximal torus T (proposition 11.2.4), assembled from the weights by which T acts. The task of this section is to convert that classification into a discrete labelling: to attach to each irreducible a single weight of T, the largest weight it contains, and to show that this label runs exactly once over the lattice points of a cone.

The weights themselves live in the character lattice of chapter 11, and the root system of $\mathfrak{g}_{\mathbb{C}}$ singles out among them a convex cone of *dominant* weights. That cone is the index set for the irreducibles.

Definition 12.1.1: Weights, Dominant Weights, and the Dominance Order

Let G be a compact connected Lie group with maximal torus T, character lattice $X^*(T) = \mathrm{Hom}(T, \mathrm{U}(1))$ (definition 11.5.1), root system $\Phi \subseteq X^*(T)$ (theorem 11.3.3), and a base $\Delta \subseteq \Phi$ with positive system Φ^+ (definition 8.2.1).

1. A *weight* of a finite-dimensional representation $\rho\colon G \to \mathrm{GL}(V)$ is a character $\mu \in X^*(T)$ occurring in the restriction $\rho|_T$: writing $V = \bigoplus_\mu V[\mu]$ for the decomposition of V into the joint eigenspaces

$$V[\mu] = \{v \in V \mid \rho(t)v = \mu(t)\, v \text{ for all } t \in T\}$$

   the weights of V are those μ with $V[\mu] \neq 0$, and $\dim V[\mu]$ is the *multiplicity* of μ.

2. An *integral weight* is any element of $X^*(T)$, regarded via its differential as a form $\mu \in \mathfrak{t}^*$ integral on the coroots, $\langle \mu, \alpha^\vee \rangle \in \mathbb{Z}$ for every $\alpha \in \Phi$ (definition 7.4.3). A weight is *dominant* if it lies in the closed dominant Weyl chamber,

$$\langle \lambda, \alpha^\vee \rangle \geq 0 \qquad \text{for every } \alpha \in \Phi^+$$

   and the set of dominant integral weights is denoted $P^+ \subseteq X^*(T)$.

3. The *dominance order* on $\mathfrak{t}^*$ is the partial order

$$\mu \leq \lambda \quad \Longleftrightarrow \quad \lambda - \mu \in \textstyle\sum_{\alpha \in \Phi^+} \mathbb{Z}_{\geq 0}\, \alpha$$

   that is, $\lambda - \mu$ is a nonnegative integer combination of positive roots.

The three notions play distinct roles. The weights of V record how T acts, decomposing V into lines and planes on which the torus rotates at definite speeds; the multiplicities $\dim V[\mu]$ are the raw combinatorial data the character packages, since $\chi_V|_T = \sum_\mu \dim V[\mu]\, e^\mu$. Dominance selects one representative from each W-orbit of weights: the closed chamber is a fundamental domain for W acting on $\mathfrak{t}^*$ (theorem 8.2.8), so every weight is W-conjugate to exactly one dominant one, and the highest weight of an irreducible will turn out to be the dominant member of its top W-orbit. The dominance order, finally, makes precise what "largest" means: it is a genuine partial order, not a total one, but it is total enough on the weights of a single irreducible to have a unique maximum.

These three are the entire vocabulary of the classification.

The running example makes each notion a computation with integers.

Example 12.1.2: Weights and Dominant Weights for SU(2) and SU(n)

For $G = \mathrm{SU}(2)$ the torus is the diagonal circle $T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta}) \mid \theta \in \mathbb{R}\}$, and $X^*(T) = \mathbb{Z}\varepsilon \cong \mathbb{Z}$ with $\varepsilon\colon t_\theta \mapsto e^{i\theta}$ (example 11.5.2). The single positive root is $\alpha = 2\varepsilon$, with coroot satisfying $\langle \varepsilon, \alpha^\vee \rangle = 1$. Identify a weight $m\varepsilon$ with the integer m; then $\langle m\varepsilon, \alpha^\vee \rangle = m$, so the dominant weights are

$$P^+ = \{m \in \mathbb{Z} \mid m \geq 0\} = \mathbb{Z}_{\geq 0}$$

the nonnegative integers. The irreducible $V_n = \mathrm{Sym}^n(\mathbb{C}^2)$ of proposition 9.5.2 has weights $n, n-2, \ldots, -n$ (the exponents by which t_θ scales the monomials $x^{n-k}y^k$), each of multiplicity one; its largest weight in the dominance order is $n = n\varepsilon$, which is dominant, and the assignment $n \mapsto V_n$ already realizes the bijection $P^+ \to \{\text{irreducibles}\}$ that the general theorem asserts. The dominance order here is $m \leq n$ iff $n - m \in \{0, 2, 4, \ldots\}$, so on the weights $\{n, n-2, \ldots, -n\}$ of V_n it is a chain with top n.

For $G = \mathrm{SU}(n)$ the torus is the diagonal $\{\mathrm{diag}(z_1, \ldots, z_n) \mid \prod z_j = 1\}$, with $X^*(T) = \mathbb{Z}^n/\mathbb{Z}(1, \ldots, 1)$ coordinatized by $\varepsilon_1, \ldots, \varepsilon_n$ (the ε_j reading off the j-th diagonal entry) subject to $\varepsilon_1 + \cdots + \varepsilon_n = 0$, positive roots $\varepsilon_i - \varepsilon_j$ for $i < j$, and coroots pairing so that $\langle \mu, (\varepsilon_i - \varepsilon_j)^\vee \rangle = \mu_i - \mu_j$. A weight $\lambda = \sum_j \lambda_j \varepsilon_j$ is dominant precisely when $\lambda_i - \lambda_j \geq 0$ for all $i < j$, that is,

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$$

so the dominant weights of SU(n) are the weakly decreasing integer sequences modulo the common shift $(1, \ldots, 1)$, normalized (say) to $\lambda_n = 0$: the partitions $\lambda_1 \geq \cdots \geq \lambda_{n-1} \geq 0$, drawn as Young diagrams. The case $n = 2$ recovers $P^+ = \mathbb{Z}_{\geq 0}$, the single part λ_1.

The example exhibits the shape of the answer twice over: for SU(2) the dominant weights are the nonnegative integers, one irreducible apiece, and for SU(n) they are the Young diagrams. The general theorem below promotes this from a computation to a structural fact for every compact connected G, and its proof is where the algebraic structure theory of Part II is finally spent on the representation theory of Part III.

The classification rests on the interaction of two structures on a representation: the action of the whole torus T, which splits V into weight spaces, and the action of the individual root $\mathfrak{sl}_2$-triples, which move a vector between weight spaces λ and $\lambda \pm \alpha$. A vector of maximal weight is one that every raising operator kills, and the sole surviving analysis is that of a single $\mathfrak{sl}_2$, already carried out in theorem 7.4.6. That is the entire engine of the proof.

Theorem 12.1.3: The Theorem of the Highest Weight

Let G be a compact connected Lie group with maximal torus T, positive roots Φ^+, and dominant integral weights P^+.

1. Every finite-dimensional irreducible representation V of G has a unique *highest weight*: a weight λ of V maximal in the dominance order. This λ is dominant integral, its weight space $V[\lambda]$ is one-dimensional, and V is generated by $V[\lambda]$ under the action of G.
2. Two irreducibles with the same highest weight are isomorphic, and distinct dominant weights label inequivalent irreducibles.
3. Every $\lambda \in P^+$ is the highest weight of an irreducible V_λ.

Consequently $\lambda \mapsto V_\lambda$ is a bijection from P^+ onto the set of isomorphism classes of irreducible representations of G.

***Proof*:**
Complexify: a representation of the compact group G differentiates to a representation of $\mathfrak{g}_{\mathbb{C}}$, and the analysis of chapter 7 applies. For each root α fix an $\mathfrak{sl}_2$-triple $(e_\alpha, h_\alpha, f_\alpha) \subseteq \mathfrak{g}_{\mathbb{C}}$ with $h_\alpha = \alpha^\vee$, so that e_α maps $V[\mu]$ into $V[\mu+\alpha]$ (raising) and f_α into $V[\mu-\alpha]$ (lowering), because $[h, e_\alpha] = \alpha(h)\, e_\alpha$ for $h \in \mathfrak{t}_{\mathbb{C}}$. The proof runs in five movements.

Step 1: existence of a highest weight. The weights of V are finite in number, since V is finite-dimensional and $V = \bigoplus_\mu V[\mu]$. Choose a linear functional $\xi\colon \mathfrak{t}^* \to \mathbb{R}$ that is positive on every positive root, $\xi(\alpha) > 0$ for all $\alpha \in \Phi^+$ (a regular direction; such ξ exists because Φ^+ lies in an open half-space, definition 8.2.1). Among the finitely many weights of V, pick one, λ, on which ξ is maximal. This λ is maximal in the dominance order too: if $\mu > \lambda$ were a weight then $\mu - \lambda$ would be a nonzero nonnegative combination of positive roots, forcing $\xi(\mu) > \xi(\lambda)$, contradicting maximality of $\xi(\lambda)$. So a dominance-maximal weight exists.

Step 2: the highest weight is dominant, and annihilated by every raising operator. Let $0 \neq v \in V[\lambda]$ with λ chosen as in Step 1. For each positive root α, the vector $e_\alpha v$ lies in $V[\lambda+\alpha]$; but $\lambda+\alpha > \lambda$ in the dominance order and λ is maximal, so $\lambda + \alpha$ is not a weight, giving $V[\lambda+\alpha] = 0$ and hence $e_\alpha v = 0$. Thus every raising operator annihilates v. Now restrict to the $\mathfrak{sl}_2$-triple of a fixed $\alpha \in \Phi^+$. Since $e_\alpha v = 0$, the vector v is a highest-weight vector for this $\mathfrak{sl}_2$, and $h_\alpha v = \langle \lambda, \alpha^\vee \rangle\, v$. The representation theory of $\mathfrak{sl}_2$, which underlies the root-string analysis of theorem 7.4.6, forces the eigenvalue of h_α on a highest-weight vector of a finite-dimensional module to be a nonnegative integer: applying f_α repeatedly to v produces vectors of h_α-eigenvalue $\langle \lambda, \alpha^\vee \rangle, \langle \lambda, \alpha^\vee \rangle - 2, \ldots$, and finite-dimensionality plus the trace identity $\operatorname{tr}(h_\alpha) = 0$ on the cyclic $\mathfrak{sl}_2$-submodule pin the top eigenvalue to $\langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}_{\geq 0}$. As α ranges over Φ^+ this says $\lambda \in P^+$.

Step 3: $\dim V[\lambda] = 1$ and V is generated by $V[\lambda]$. Let V' be the $\mathfrak{g}_{\mathbb{C}}$-submodule generated by a single highest-weight vector $v \in V[\lambda]$. Because $\mathfrak{g}_{\mathbb{C}} = \mathfrak{n}^- \oplus \mathfrak{t}_{\mathbb{C}} \oplus \mathfrak{n}^+$ with $\mathfrak{n}^\pm$ the sums of the positive and negative root spaces, and $\mathfrak{n}^+ v = 0$ (Step 2) while $\mathfrak{t}_{\mathbb{C}} v = \mathbb{C}v$, the Poincaré-Birkhoff-Witt theorem (theorem 8.6.7; the triangular form of corollary 8.6.12 applies to the semisimple part of $\mathfrak{g}_{\mathbb{C}}$, with the central summand absorbed into the ordered basis of $\mathfrak{t}_{\mathbb{C}}$) writes V' as the span of the vectors $f_{\beta_1} \cdots f_{\beta_k} v$ obtained by lowering. Every such vector has weight $\lambda - (\beta_1 + \cdots + \beta_k)$ with each $\beta_i \in \Phi^+$, so it lies below λ; in particular the only contribution to weight exactly λ is v itself, giving $V'[\lambda] = \mathbb{C}v$. Now V is irreducible and $V' \neq 0$ is a nonzero submodule, so $V' = V$,

whence $V[\lambda] = V'[\lambda] = \mathbb{C}v$ is one-dimensional and V is generated by $V[\lambda]$. Uniqueness of the highest weight follows: any two dominance-maximal weights λ, λ' each carry a highest-weight vector generating all of V, so all weights of V lie both $\leq \lambda$ and $\leq \lambda'$; applying this with the two roles exchanged gives $\lambda \leq \lambda'$ and $\lambda' \leq \lambda$, hence $\lambda = \lambda'$. This proves part (1).

Step 4: same highest weight forces isomorphism. Let V, W be irreducibles with the same highest weight λ, and let $v \in V[\lambda]$, $w \in W[\lambda]$ be highest-weight vectors. In the direct sum $V \oplus W$ the vector $u = (v, w)$ has weight λ and is annihilated by every e_α, so it is a highest-weight vector; let $U \subseteq V \oplus W$ be the submodule it generates. The two projections $p_V : U \to V$ and $p_W : U \to W$ are G-maps sending u to the generators v and w; since V and W are generated by v and w (Step 3), both projections are surjective, hence, V and W being irreducible, isomorphisms onto their images V and W. Neither can be the zero map (each hits a generator), and by Schur's lemma an irreducible submodule of $V \oplus W$ projecting nontrivially to both factors makes $V \cong U \cong W$. Concretely, p_V is an isomorphism $U \xrightarrow{\sim} V$ and p_W an isomorphism $U \xrightarrow{\sim} W$, so $p_W \circ p_V^{-1} : V \xrightarrow{\sim} W$ is the desired G-isomorphism.

Step 5: distinct dominant weights give inequivalent irreducibles. If $V_\lambda \cong V_{\lambda'}$ then they have the same set of weights, hence the same dominance-maximal weight, so $\lambda = \lambda'$. Equivalently, and independently of the generation argument, the characters χ_λ are orthonormal (theorem 10.4.4), so $V_\lambda \cong V_{\lambda'}$ forces $\chi_\lambda = \chi_{\lambda'}$; reading off the top weight of each character recovers $\lambda = \lambda'$. Either way $\lambda \mapsto V_\lambda$ is injective on P^+. This proves part (2).

Existence. It remains to realize every dominant $\lambda \in P^+$ as a highest weight (part (3)). This is given by the Borel-Weil theorem of section 12.4: for λ dominant the space of holomorphic sections $H^0(G/T, L_\lambda)$ of the associated line bundle is an irreducible representation of G with highest weight λ (theorem 12.4.3), so V_λ exists. An equivalent purely algebraic construction, realizing the irreducible of highest weight λ as the unique irreducible submodule of the induced module $H^0(\lambda)$, is carried out in [10, Classification of Irreducible Representations]. With existence in hand, $\lambda \mapsto V_\lambda$ is a bijection from P^+ onto the isomorphism classes of irreducibles, completing the proof.

The theorem is the compact-group form of the highest-weight classification, and it is the exact point where the two halves of the book meet. Part II reduced a semisimple Lie algebra to the combinatorial data of its root system; Part III reduced a compact group to its maximal torus and characters. The highest-weight theorem fuses them: the irreducibles of G, which Peter-Weyl produced as an abstract but uncounted family, are now enumerated by the lattice points of a single convex cone, the dominant chamber, one irreducible for each. Representation theory has become the combinatorics of P^+. A dominant weight is a finite piece of integer data, a point of P^+ or a Young diagram; every question about the irreducible it labels, its dimension, its character, its tensor decompositions, is thereby a question about that data, to be answered by the explicit formulas of the sections that follow.

The cone P^+ also fixes the arithmetic role of the lattice chain $Q \subseteq X^*(T) \subseteq P$ of proposition 11.5.3. The dominant weights of G are the dominant elements of its own character lattice $X^*(T)$, not of the full weight lattice P: the simply connected group has the most irreducibles (all of P^+), the adjoint group the fewest (the dominant elements of Q), and a general G an intermediate set, so the same Lie algebra supports different representation theories according to the global topology recorded by $\pi_1(G)$ (theorem 11.5.4).

Exercise 12.1.1

1. List the irreducible representations of SU(2) by highest weight, and for each give its dimension

and its full set of weights. Confirm that $\lambda \mapsto V_\lambda$ is a bijection from $P^+ = \mathbb{Z}_{\geq 0}$ onto the irreducibles by matching with the classification $V_n = \mathrm{Sym}^n(\mathbb{C}^2)$ of proposition 9.5.2.

2. The *fundamental weights* $\omega_1, \ldots, \omega_{n-1}$ of $\mathrm{SU}(n)$ are the dominant integral weights dual to the simple coroots, $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$. For $\mathrm{SU}(3)$, with simple roots $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$, write ω_1 and ω_2 in the ε-coordinates (normalized to $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$), verify that each is dominant, and identify V_{ω_1} and V_{ω_2} among the standard representation $\mathbb{C}^3$, its dual, and $\Lambda^2\mathbb{C}^3$.

3. Let V and W be irreducibles of G with highest weights λ and μ. Show that $\lambda + \mu$ is a weight of $V \otimes W$, that it is the unique dominance-maximal weight of $V \otimes W$, and that its weight space is one-dimensional. Deduce that the irreducible $V_{\lambda+\mu}$ occurs in $V \otimes W$ with multiplicity exactly one (the "Cartan component").

4. The *contragredient* (dual) representation V_λ^* acts on the dual space by $(\rho^*(g)\varphi)(v) = \varphi(\rho(g^{-1})v)$. Show that the weights of V_λ^* are the negatives of the weights of V_λ, and conclude that V_λ^* is irreducible with highest weight $-w_0(\lambda)$, where w_0 is the longest element of W (exercise 8.2.1 (3)). Verify for $\mathrm{SU}(2)$ that $V_n^* \cong V_n$, and explain in terms of w_0 why every $\mathrm{SU}(2)$ representation is self-dual.

5. Prove that the dominance order $\leq$ is a genuine partial order on $X^*(T)$ (reflexive, antisymmetric, transitive), and that when restricted to the weights of a single irreducible V_λ it has λ as unique maximum. Show by the example of $\mathrm{SU}(3)$ and the adjoint representation $V_{\alpha_1+\alpha_2}$ (the eight-dimensional representation on $\mathfrak{su}(3)_\mathbb{C}$) that $\leq$ is *not* a total order: exhibit two weights of $V_{\alpha_1+\alpha_2}$ that are incomparable.

6. Step 1 of theorem 12.1.3 chose a regular direction ξ with $\xi(\alpha) > 0$ for all $\alpha \in \Phi^+$. Show that the resulting maximal weight λ is independent of the choice of such ξ, so long as ξ is regular for the fixed positive system Φ^+. What changes if ξ is chosen positive on a *different* positive system? Relate the answer to the fact that the highest weights for different choices of Φ^+ form a single W-orbit.

12.2 The Weyl Character Formula

The theorem of the highest weight (section 12.1) names the irreducibles V_λ but does not compute them. This section writes their characters χ_λ in closed form, the datum that by theorem 10.4.4 determines V_λ up to isomorphism and by proposition 11.2.4 is a W-invariant function on the torus T. The answer is a ratio of two antisymmetrized exponential sums, one built from λ and one universal, and the tool that produces it is the Weyl integration formula (theorem 11.4.3), which turns the orthonormality of characters into an elementary computation on the torus.

The strategy is to guess the numerator and denominator separately. The denominator is a fixed function on T, independent of λ, whose square is the density $|\delta|^2$ of the integration formula; the numerator is the antisymmetrization of $e^{\lambda+\rho}$, shifted by a universal weight ρ that repairs the failure of λ to be regular. Both are W-anti-invariant, so their ratio is W-invariant, hence a candidate class function; the integration formula then forces the candidate to be exactly χ_λ.

Two pieces of universal data enter. The first is the weight built from the positive roots.

Definition 12.2.1: The Weyl Vector

Let G be a compact connected Lie group with maximal torus T, root system $\Phi \subset X^*(T)$, and a chosen set of positive roots Φ^+ (definition 11.3.1). The *Weyl vector* is the half-sum of the positive roots,

$$\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$$

a vector in the real vector space $X^*(T) \otimes \mathbb{R}$ (an integral weight in $X^*(T)$ when G is simply connected, and always a weight of a covering group). It is characterized among all vectors by the property that $\langle \rho, \alpha_i^\vee \rangle = 1$ for every simple root α_i, where $\alpha_i^\vee$ is the coroot of definition 7.4.3.

The Weyl vector is the correction term that makes the highest-weight combinatorics symmetric. A dominant weight λ may lie on a wall of the dominant chamber, where $\langle \lambda, \alpha^\vee \rangle = 0$ for some positive root α and the stabilizer of λ in W is nontrivial; adding ρ pushes λ strictly into the interior, since $\langle \lambda + \rho, \alpha^\vee \rangle = \langle \lambda, \alpha^\vee \rangle + 1 \geq 1 > 0$ for every simple coroot. Thus $\lambda + \rho$ is always *regular* (strictly dominant), with trivial W-stabilizer, even when λ is not. The characterization $\langle \rho, \alpha_i^\vee \rangle = 1$ is the reason ρ appears throughout the theory: it is the unique weight taking the value 1 on every simple coroot, the vector that a reflection s_{α_i} moves by exactly $-\alpha_i$.

The characterizing property is worth checking on the running example before it is used.

Example 12.2.2: The Weyl Vector of SU(2) and SU(3)

For $G = \mathrm{SU}(2)$ there is a single positive root α (the root of A_1, with $\alpha(t_\theta) = e^{2i\theta}$ on the diagonal torus, as in example 11.4.2), so the Weyl vector is

$$\rho = \frac{1}{2}\alpha$$

Its value on the single coroot is $\langle \rho, \alpha^\vee \rangle = \frac{1}{2}\langle \alpha, \alpha^\vee \rangle = \frac{1}{2} \cdot 2 = 1$, confirming the characterization. In the integer coordinate $n \leftrightarrow (t_\theta \mapsto e^{in\theta})$ on $X^*(T) \cong \mathbb{Z}$, the root α is the integer 2 and $\rho = \frac{1}{2}\alpha$ is the value 1; the shift $\lambda \mapsto \lambda + \rho$ sends the highest weight n of the irreducible V_n (proposition 9.5.2) to $n+1$, the origin of the shift in the character formula $\chi_n = \sin((n+1)\theta)/\sin\theta$.

For $G = \mathrm{SU}(3)$, whose root system is A_2 with positive roots $\alpha_1, \alpha_2, \alpha_1 + \alpha_2$ (the simple roots and their sum), the Weyl vector is

$$\rho = \frac{1}{2}\big(\alpha_1 + \alpha_2 + (\alpha_1 + \alpha_2)\big) = \alpha_1 + \alpha_2$$

One checks $\langle \rho, \alpha_1^\vee \rangle = \langle \alpha_1 + \alpha_2, \alpha_1^\vee \rangle = 2 + (-1) = 1$, using the A_2 Cartan integers $\langle \alpha_1, \alpha_1^\vee \rangle = 2$ and $\langle \alpha_2, \alpha_1^\vee \rangle = -1$ (definition 8.3.1), and symmetrically $\langle \rho, \alpha_2^\vee \rangle = 1$. In the standard coordinates $\rho = \varepsilon_1 - \varepsilon_3$ up to the trace-zero convention, and ρ equals the sum $\omega_1 + \omega_2$ of the two fundamental weights, since a fundamental weight ω_i is defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$ and ρ takes the value 1 on every simple coroot.

The second piece of data antisymmetrizes an exponential over the Weyl group. Recall that the group ring $\mathbb{Z}[X^*(T)]$ has $\mathbb{Z}$-basis the formal exponentials e^μ for $\mu \in X^*(T)$, with multiplication $e^\mu e^\nu = e^{\mu+\nu}$, and that W acts on it by $w \cdot e^\mu = e^{w\mu}$; the character χ_λ restricted to T is such an element, the sum of e^μ over the weights μ of V_λ counted with multiplicity.

Definition 12.2.3: The Alternating Sum

Let $\mu \in X^*(T)$ be a weight and let $\operatorname{sgn}(w) = \det(w \,|\, \mathfrak{t})$ be the sign character of W, equal to $(-1)^{\ell(w)}$ where $\ell(w)$ is the length of w as a word in the simple reflections. The *alternating sum* attached to μ is

$$A_\mu = \sum_{w \in W} \operatorname{sgn}(w)\, e^{w\mu}$$

an element of $\mathbb{Z}[X^*(T)]$ (or of the character ring of a covering torus when μ is only half-integral). It is W*-anti-invariant*: for every $v \in W$,

$$v \cdot A_\mu = \sum_{w \in W} \operatorname{sgn}(w)\, e^{vw\mu} = \operatorname{sgn}(v) \sum_{w \in W} \operatorname{sgn}(vw)\, e^{vw\mu} = \operatorname{sgn}(v)\, A_\mu$$

the reindexing $w \mapsto vw$ permuting W and sgn being a homomorphism.

Anti-invariance is the defining feature the whole section exploits. An anti-invariant element changes sign under an odd Weyl element and is fixed by an even one, exactly as a determinant behaves under row permutations, and this forces a strong divisibility: any anti-invariant element vanishes on every reflection hyperplane and is therefore divisible by the product of the corresponding linear factors. Two anti-invariant elements with the same such factors and the same size therefore agree up to a scalar. The alternating sum A_μ is the canonical anti-invariant built from μ, and when μ is regular its $|W|$ terms $e^{w\mu}$ are distinct, so A_μ has $|W|$ nonzero terms with unit coefficients ± 1; when μ lies on a wall, a reflection fixing μ pairs each term with its negative and $A_\mu = 0$.

The first payoff identifies the alternating sum of ρ itself with the Weyl denominator, closing the gap between the two guesses.

Theorem 12.2.4: The Weyl Denominator Formula

Let G be a compact connected Lie group with maximal torus T and positive roots Φ^+. The alternating sum of the Weyl vector factors as a product over the positive roots,

$$A_\rho = \sum_{w \in W} \operatorname{sgn}(w)\, e^{w\rho} = \prod_{\alpha \in \Phi^+} \left(e^{\alpha/2} - e^{-\alpha/2}\right)$$

This product is the Weyl denominator δ of definition 11.4.1, so $A_\rho = \delta$; in particular $|A_\rho|^2 = |\delta|^2$ is the density of the Weyl integration formula.

***Proof*:**

Write $\Delta = \prod_{\alpha \in \Phi^+} (e^{\alpha/2} - e^{-\alpha/2})$ for the product and $A_\rho = \sum_w \operatorname{sgn}(w) e^{w\rho}$ for the alternating sum. The proof shows both are W-anti-invariant, that Δ divides A_ρ, and that a comparison of the single top term forces $A_\rho = \Delta$.

Step 1: both sides are anti-invariant with top term e^ρ. Anti-invariance of A_ρ is definition 12.2.3. For Δ, it suffices to check the generating reflections. A simple reflection $s_i = s_{\alpha_i}$ permutes the positive roots other than α_i among themselves and sends α_i to $-\alpha_i$ (theorem 8.2.8); the factor $e^{\alpha_i/2} - e^{-\alpha_i/2}$ is sent to $e^{-\alpha_i/2} - e^{\alpha_i/2} = -(e^{\alpha_i/2} - e^{-\alpha_i/2})$, while the remaining factors are permuted, so $s_i \cdot \Delta = -\Delta = \operatorname{sgn}(s_i)\Delta$. Since the simple reflections generate W and sgn

is a homomorphism, Δ is anti-invariant. Expanding the product, the term picking $e^{\alpha/2}$ from every factor gives $e^{\frac{1}{2}\sum_{\alpha>0}\alpha} = e^{\rho}$, and every other choice replaces some $e^{\alpha/2}$ by $e^{-\alpha/2}$, lowering the exponent; thus the highest term of Δ in the dominance order is e^{ρ} with coefficient 1. The highest term of A_ρ is likewise e^{ρ}: the terms $e^{w\rho}$ satisfy $w\rho \preceq \rho$ with equality only for $w = 1$ (since ρ is strictly dominant), and $\operatorname{sgn}(1) = 1$.

Step 2: Δ divides A_ρ in $\mathbb{Z}[X^*(T)]$ up to the half-lattice. Both Δ and A_ρ are anti-invariant. A general principle governs such elements: if f is W-anti-invariant, then $s_\alpha \cdot f = -f$ for every reflection, so f vanishes when e^{α} acts as the identity, and in the polynomial ring $\mathbb{Z}[X^*(T)]$ (a Laurent polynomial ring, hence a unique factorization domain after passing to the half-lattice) each factor $e^{\alpha/2} - e^{-\alpha/2}$ divides f; the factors for distinct positive roots are coprime, so their product Δ divides f. Applying this to $f = A_\rho$ gives $A_\rho = \Delta \cdot h$ for some $h \in \mathbb{Z}[X^*(T)]$.

Step 3: the quotient is a constant, equal to 1. The quotient $h = A_\rho/\Delta$ is W-invariant, being the ratio of two anti-invariant elements. Compare degrees: by Step 1 both A_ρ and Δ have highest term e^{ρ}, so their quotient h has highest term $e^0 = 1$ with coefficient 1, and h has no term of positive height. A W-invariant element whose only top term is the constant 1, with all other exponents strictly lower, must be the constant 1: any nonzero exponent ν appearing in h would have its whole W-orbit appear (by invariance), and the dominant representative of that orbit would exceed the constant term in height, contradicting that 1 is the top term. Hence $h = 1$ and $A_\rho = \Delta$. Comparing with definition 11.4.1, the product $\Delta = \prod_{\alpha>0}(e^{\alpha/2} - e^{-\alpha/2})$ is exactly the Weyl denominator δ, so $A_\rho = \delta$ and $|A_\rho|^2 = |\delta|^2$, as we sought to show.

The identity $A_\rho = \delta$ says the fixed denominator of the character formula, which was defined analytically as a Jacobian in definition 11.4.1, is the same object as the algebraically defined alternating sum A_ρ, so that the whole formula lives inside the single ring $\mathbb{Z}[X^*(T)]$ of exponential sums on the torus. Read one way, it is a purely combinatorial factorization: antisymmetrizing e^{ρ} over W produces the product $\prod_{\alpha>0}(e^{\alpha/2} - e^{-\alpha/2})$, a Weyl-group analog of the Vandermonde factorization $\det(x_i^{j-1}) = \prod_{i<j}(x_j - x_i)$. Read the other way, it certifies that the density $|\delta|^2$ of the integration formula is $|A_\rho|^2$, so that integrating an exponential sum against the Weyl density is integrating it against $A_\rho\overline{A_\rho}$, which the orthogonality of torus characters makes elementary. Both readings are used in the proof of the character formula.

Everything is now in place. The candidate for χ_λ is the ratio of the alternating sum of $\lambda + \rho$ over the alternating sum of ρ, and the argument that this candidate is correct rests on two facts the integration formula gives: the candidates are orthonormal as λ ranges over dominant weights, and the true character χ_λ, multiplied by A_ρ, is an anti-invariant integral combination of the same alternating sums with leading term $A_{\lambda+\rho}$. Matching the two under the constraint $\|\chi_\lambda\| = 1$ pins the equality.

Theorem 12.2.5: The Weyl Character Formula

Let G be a compact connected Lie group with maximal torus T, Weyl group W, and Weyl vector ρ. For each dominant integral weight $\lambda \in P^+$, the character of the irreducible representation V_λ of highest weight λ (theorem 12.1.3) is, on the regular set of T,

$$\chi_\lambda = \frac{A_{\lambda+\rho}}{A_\rho} = \frac{\sum_{w\in W} \operatorname{sgn}(w)\, e^{w(\lambda+\rho)}}{\sum_{w\in W} \operatorname{sgn}(w)\, e^{w\rho}} = \frac{\sum_{w\in W} \operatorname{sgn}(w)\, e^{w(\lambda+\rho)}}{\prod_{\alpha\in\Phi^+}\left(e^{\alpha/2} - e^{-\alpha/2}\right)}$$

The three expressions agree by the Weyl denominator formula (theorem 12.2.4).

***Proof*:**
The proof has three movements: the ratios $A_{\mu+\rho}/A_\rho$ for dominant μ are orthonormal for the character inner product; the product $A_\rho\chi_\lambda$ is an anti-invariant integral combination of the $A_{\mu+\rho}$ with leading term $A_{\lambda+\rho}$; and the norm condition $\|\chi_\lambda\| = 1$ forces $A_\rho\chi_\lambda = A_{\lambda+\rho}$.

Step 1: orthonormality of the $A_{\mu+\rho}/A_\rho$. For a weight ν, write $\langle e^\nu, e^{\nu'}\rangle_T = \int_T e^\nu \overline{e^{\nu'}}\, dt = \delta_{\nu\nu'}$, the orthonormality of the distinct characters of the torus T under its normalized Haar measure. For two dominant weights μ, μ', the elements $\mu+\rho$ and $\mu'+\rho$ are strictly dominant, so each has trivial W-stabilizer and the $|W|$ exponentials $e^{w(\mu+\rho)}$ are distinct. Using $A_\rho = \delta$ from theorem 12.2.4, the ratio $A_{\mu+\rho}/A_\rho$ is a class function on G (both numerator and denominator are anti-invariant, so the ratio is W-invariant), and the Weyl integration formula (theorem 11.4.3) computes its inner product with $A_{\mu'+\rho}/A_\rho$ as

$$\left\langle \frac{A_{\mu+\rho}}{A_\rho}, \frac{A_{\mu'+\rho}}{A_\rho} \right\rangle_G = \frac{1}{|W|}\int_T \frac{A_{\mu+\rho}}{A_\rho}\overline{\left(\frac{A_{\mu'+\rho}}{A_\rho}\right)}\,|\delta|^2\,dt = \frac{1}{|W|}\int_T A_{\mu+\rho}\,\overline{A_{\mu'+\rho}}\,dt$$

the density $|\delta|^2 = A_\rho\overline{A_\rho}$ cancelling the two denominators A_ρ and $\overline{A_\rho}$ exactly. Expanding the two alternating sums,

$$\begin{aligned}\int_T A_{\mu+\rho}\,\overline{A_{\mu'+\rho}}\,dt &= \sum_{w,w'\in W}\operatorname{sgn}(w)\operatorname{sgn}(w')\int_T e^{w(\mu+\rho)}\,\overline{e^{w'(\mu'+\rho)}}\,dt \\ &= \sum_{w,w'\in W}\operatorname{sgn}(w)\operatorname{sgn}(w')\,\delta_{w(\mu+\rho),\,w'(\mu'+\rho)}\end{aligned}$$

A term survives only when $w(\mu+\rho) = w'(\mu'+\rho)$, that is when $\mu+\rho$ and $\mu'+\rho$ lie in the same W-orbit; since both are strictly dominant, and each W-orbit meets the strictly dominant chamber exactly once, this forces $\mu = \mu'$ and then $w = w'$ (trivial stabilizer). When $\mu = \mu'$ the surviving terms are the $|W|$ diagonal pairs $w = w'$, each contributing $\operatorname{sgn}(w)^2 = 1$, for a total of $|W|$; when $\mu \neq \mu'$ no term survives and the total is 0. Dividing by $|W|$,

$$\left\langle \frac{A_{\mu+\rho}}{A_\rho}, \frac{A_{\mu'+\rho}}{A_\rho} \right\rangle_G = \delta_{\mu\mu'}$$

so the family $\{A_{\mu+\rho}/A_\rho\}_{\mu\in P^+}$ is orthonormal.

Step 2: $A_\rho\chi_\lambda$ is an integral combination of the $A_{\mu+\rho}$ with leading term $A_{\lambda+\rho}$. The character χ_λ, restricted to T, is the integral exponential sum $\sum_\mu m_\lambda(\mu)\,e^\mu$, where $m_\lambda(\mu) \in \mathbb{Z}_{\geq 0}$ is the multiplicity of the weight μ in V_λ; it is W-invariant because χ_λ is a class function (proposition 11.2.4) and the weight multiplicities are W-invariant. The product $A_\rho\chi_\lambda$ is therefore W-anti-invariant, being the product of the anti-invariant A_ρ and the invariant χ_λ, and it is an integral combination of exponentials e^ν. Now group the exponentials of any anti-invariant integral element by W-orbit. On a wall orbit (some s_α fixing ν) the anti-invariance forces the coefficient to vanish, since s_α must both fix the exponent and negate its coefficient; on a regular orbit the coefficients are determined by the one at the strictly dominant representative ν_0, and the orbit contributes exactly that coefficient times A_{ν_0}. Hence every anti-invariant integral element is a $\mathbb{Z}$-combination $\sum_\mu c_\mu A_{\mu+\rho}$ of the alternating sums $A_{\mu+\rho}$ for dominant μ (writing the strictly dominant representative as $\mu+\rho$ with μ dominant). In particular

$$A_\rho\chi_\lambda = \sum_{\mu\in P^+} c_\mu\,A_{\mu+\rho}, \qquad c_\mu \in \mathbb{Z}$$

The leading term identifies c_λ. The highest weight of χ_λ is λ with multiplicity $m_\lambda(\lambda) = 1$ (theorem 12.1.3), and the highest exponent of A_ρ is ρ (theorem 12.2.4, Step 1), so the highest exponent of the product $A_\rho\chi_\lambda$ is $\lambda+\rho$, with coefficient $1\cdot 1 = 1$. On the right side the exponent $\lambda+\rho$ occurs only in the term $A_{\lambda+\rho}$ (as its leading term, coefficient 1), all other $A_{\mu+\rho}$ having strictly lower leading exponents $\mu+\rho \prec \lambda+\rho$. Matching coefficients gives $c_\lambda = 1$, so

$$A_\rho\chi_\lambda = A_{\lambda+\rho} + \sum_{\mu\prec\lambda} c_\mu\, A_{\mu+\rho}$$

Step 3: the norm condition removes the lower terms. Divide by A_ρ: since $A_\rho = \delta$ is nonzero on the regular set, Step 2 gives

$$\chi_\lambda = \frac{A_{\lambda+\rho}}{A_\rho} + \sum_{\mu\prec\lambda} c_\mu \frac{A_{\mu+\rho}}{A_\rho}$$

an expansion of χ_λ in the orthonormal family of Step 1. Take squared norms. The character of an irreducible is a unit vector, $\|\chi_\lambda\|_G^2 = \int_G |\chi_\lambda|^2\, dg = 1$ by Schur orthogonality (theorem 10.4.4), and by orthonormality of the $A_{\mu+\rho}/A_\rho$ the squared norm of the right side is the sum of squared integer coefficients,

$$1 = \|\chi_\lambda\|_G^2 = 1 + \sum_{\mu\prec\lambda} c_\mu^2$$

Every $c_\mu \in \mathbb{Z}$, so $\sum_{\mu\prec\lambda} c_\mu^2 = 0$ forces $c_\mu = 0$ for all $\mu \prec \lambda$. Therefore

$$\chi_\lambda = \frac{A_{\lambda+\rho}}{A_\rho}$$

on the regular set, and the three displayed forms agree by theorem 12.2.4, completing the proof.

The formula writes every irreducible character of a compact connected group as a single antisymmetrized exponential sum, $A_{\lambda+\rho} = \sum_w \operatorname{sgn}(w) e^{w(\lambda+\rho)}$, divided by the universal Weyl denominator $A_\rho = \prod_{\alpha>0}(e^{\alpha/2} - e^{-\alpha/2})$, a ratio of $|W|$-term sums that depends on the representation only through the shift $\lambda \mapsto \lambda+\rho$ in the numerator. Everything the earlier chapters assembled is cashed here at once: the root-system combinatorics of Part II gives the positive roots, the Weyl group, and the sign character; the highest-weight theorem gives the leading term $A_{\lambda+\rho}$ and its multiplicity one; and the integration formula of section 11.4 gives the orthonormality that removes every lower term. The proof is a linear-algebra argument on the weight lattice, carried out entirely inside the ring of exponential sums on the torus, made elementary by the single identity $|\delta|^2 = A_\rho \overline{A_\rho}$.

What makes the formula usable is that neither the numerator nor the denominator refers to the representation as a space of vectors. Both are finite sums over the finite group W, computed from the weight λ and the root data alone, so the entire character, and with it the dimension and every weight multiplicity, is recovered without ever constructing V_λ. The next section evaluates the ratio at the identity to extract $\dim V_\lambda$ (theorem 12.3.1) and expands the reciprocal $1/A_\rho$ to read off the multiplicities $m_\lambda(\mu)$ (theorem 12.3.6); both are consequences of the closed form obtained here.

The smallest case is the one the formula was built to generalize.

Example 12.2.6: The Weyl Character Formula for SU(2)

For $G = \mathrm{SU}(2)$ the maximal torus is $T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta}) \mid \theta \in \mathbb{R}\}$, the Weyl group is $W = \{1, s_\alpha\} \cong \mathbb{Z}/2$ acting by $\theta \mapsto -\theta$ (example 11.2.2), and the single positive root α gives $\rho = \frac{1}{2}\alpha$ (example 12.2.2). The irreducible V_n has highest weight $\lambda = n$ in the integer coordinate on $X^*(T)$ (proposition 9.5.2), so $\lambda + \rho = n + 1$ in units where $e^\lambda(t_\theta) = e^{in\theta}$ and ρ contributes the half-shift making $e^{\lambda+\rho}(t_\theta) = e^{i(n+1)\theta}$. The alternating sum in the numerator has two terms, indexed by 1 and s_α with signs $+1$ and -1:

$$A_{\lambda+\rho}(t_\theta) = e^{(n+1)i\theta} - e^{-(n+1)i\theta}$$

the second term being $\mathrm{sgn}(s_\alpha)\, e^{s_\alpha(\lambda+\rho)} = -e^{-i(n+1)\theta}$ since s_α negates the weight. The denominator is the same sum with $n = 0$,

$$A_\rho(t_\theta) = e^{i\theta} - e^{-i\theta}$$

which is $\delta(t_\theta) = 2i\sin\theta$ from example 11.4.2. The character formula therefore gives

$$\chi_n(t_\theta) = \frac{A_{\lambda+\rho}}{A_\rho} = \frac{e^{(n+1)i\theta} - e^{-(n+1)i\theta}}{e^{i\theta} - e^{-i\theta}} = \frac{2i\sin\big((n+1)\theta\big)}{2i\sin\theta} = \frac{\sin\big((n+1)\theta\big)}{\sin\theta}$$

recovering exactly the SU(2) character formula of proposition 9.5.3, obtained there by hand from the explicit weight string $-n, -n+2, \ldots, n-2, n$ of V_n. The general formula reproduces that string automatically: the geometric-series identity $\frac{\sin((n+1)\theta)}{\sin\theta} = e^{in\theta} + e^{i(n-2)\theta} + \cdots + e^{-in\theta}$ is the expansion of the ratio, exhibiting the $n+1$ weights of V_n each with multiplicity one. The whole content of the Weyl character formula, for the running example, is this cancellation of the two-term Vandermonde $e^{i\theta} - e^{-i\theta}$ against the two-term numerator.

The SU(2) computation is the entire theory in miniature: an antisymmetrized numerator over a two-element Weyl group, divided by the same sum at $\lambda = 0$, collapsing to the ratio of sines. The passage to a general G replaces the two-term sums by $|W|$-term sums and the single positive root by the full set Φ^+, but the mechanism, cancel the denominator A_ρ against the density and match leading terms, is unchanged. The next section pushes the same ratio to its two numerical consequences, and Part IV will carry the compact-group representation theory established here into the noncompact setting, where characters become distributions and the clean closed form of this section acquires a boundary of singular directions.

Exercise 12.2.1

1. Recover the SU(2) character formula from the general one in full detail. Take $W = \{1, s_\alpha\}$ acting on $X^*(T) \cong \mathbb{Z}$ by $n \mapsto -n$, $\rho = 1$ (in the coordinate where the root α is 2), and the highest weight $\lambda = n$. Write out $A_{\lambda+\rho}(t_\theta)$ and $A_\rho(t_\theta)$ as two-term sums, take the ratio, and simplify to $\sin((n+1)\theta)/\sin\theta$. Then expand this ratio as a finite geometric-type sum to read off the $n+1$ weights of V_n with their multiplicities, confirming agreement with proposition 9.5.2.

2. Verify the Weyl denominator formula $A_\rho = \prod_{\alpha>0}(e^{\alpha/2} - e^{-\alpha/2})$ directly for the root system A_2 (the case $G = \mathrm{SU}(3)$). The positive roots are $\alpha_1, \alpha_2, \alpha_1 + \alpha_2$, the Weyl group is $W = S_3$ (order 6), and $\rho = \alpha_1 + \alpha_2$ (example 12.2.2). Introduce coordinates $x = e^{\varepsilon_1}, y = e^{\varepsilon_2}, z = e^{\varepsilon_3}$ with $xyz = 1$ (so that $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$), expand both the six-term alternating sum $\sum_{w\in S_3} \mathrm{sgn}(w) e^{w\rho}$ and the three-factor product, and check that both equal the Vandermonde $\prod_{i<j}(x_i - x_j)$ in $x_1 = x, x_2 = y, x_3 = z$ exactly (the constraint $xyz = 1$ makes the two agree

on the nose, with no separating monomial).

3. Compute the character of the fundamental representation $V_{\omega_1} = \mathbb{C}^3$ of SU(3) (highest weight $\omega_1 = \varepsilon_1$, the standard representation) from the Weyl character formula, and confirm that on the diagonal torus $\mathrm{diag}(x, y, z)$ with $xyz = 1$ it equals $x + y + z$, the sum of the three eigenvalues. Identify the three weights $\varepsilon_1, \varepsilon_2, \varepsilon_3$ of $\mathbb{C}^3$ in the numerator expansion.

4. Compute the character of the adjoint representation of SU(3) (highest weight the highest root $\theta = \alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$) from the Weyl character formula, and confirm that on $\mathrm{diag}(x, y, z)$ it equals $\sum_{i \neq j} x_i x_j^{-1} + 2$, the character of the eight-dimensional adjoint representation, matching $\dim \mathfrak{su}(3)_{\mathbb{C}} = 8$: the six off-diagonal weights $\varepsilon_i - \varepsilon_j$ $(i \neq j)$ each with multiplicity one, and the zero weight with multiplicity two (the rank).

5. Show that the numerator $A_{\lambda+\rho}$ of the Weyl character formula is nonzero for every dominant λ, and explain why the formula would fail if one replaced $\lambda + \rho$ by λ alone (i.e. dropped the ρ-shift). Concretely, prove that $A_\mu = 0$ whenever μ lies on a wall of the dominant chamber (some $\langle \mu, \alpha^\vee \rangle = 0$), by exhibiting the reflection s_α that fixes μ and pairs each term $\mathrm{sgn}(w)\, e^{w\mu}$ with the term indexed by ws_α, namely $\mathrm{sgn}(ws_\alpha)\, e^{ws_\alpha \mu} = -\mathrm{sgn}(w)\, e^{w\mu}$. Deduce that the ρ-shift, which makes $\lambda + \rho$ strictly dominant, is exactly what keeps the numerator from collapsing.

6. Prove that under the action of $w \in W$ the numerator transforms by $w \cdot A_{\lambda+\rho} = \mathrm{sgn}(w)\, A_{\lambda+\rho}$ and the denominator by $w \cdot A_\rho = \mathrm{sgn}(w)\, A_\rho$, and conclude that the ratio $\chi_\lambda = A_{\lambda+\rho}/A_\rho$ is W-invariant, hence a genuine class function via proposition 11.2.4. Explain in one sentence why this W-invariance is a necessary consistency check on the formula, given that χ_λ is the restriction to T of a class function on G.

12.3 The Weyl Dimension Formula and Kostant Multiplicities

The character formula of section 12.2 is a single expression carrying all the numerical data of an irreducible representation. Its value and its Taylor coefficients along the torus encode two invariants the reader will want in closed form: the dimension of V_λ, read off by evaluating the character at the identity, and the multiplicity of each weight μ in V_λ, read off as the coefficient of e^μ in the character's expansion. This section extracts both. The dimension emerges from a limit that resolves the $0/0$ one meets on naive substitution; the multiplicities emerge from expanding the reciprocal of the Weyl denominator as a partition generating function.

The dimension of a representation is the trace of the identity operator, so $\dim V_\lambda = \chi_\lambda(e)$. Substituting the identity $t = e$ directly into the Weyl character formula $\chi_\lambda = A_{\lambda+\rho}/A_\rho$ (theorem 12.2.5) gives $0/0$: every formal exponential satisfies $e^\mu(e) = 1$, so the numerator becomes $\sum_{w \in W} \mathrm{sgn}(w) = 0$ and the denominator likewise $\sum_{w \in W} \mathrm{sgn}(w) = 0$, since W has equally many elements of each sign as soon as it contains a reflection. The ratio is genuinely $0/0$ at the identity, not undefined for a superficial reason. The resolution is to approach the identity along a one-parameter subgroup $\exp(sx)$ for a *regular* $x \in \mathfrak{t}$, where the denominator vanishes to a definite order and one can cancel the common zero. This is the same L'Hôpital phenomenon that turns $\sin((n+1)\theta)/\sin\theta$ into $n + 1$ as $\theta \to 0$ for SU(2), and the general computation is its exact higher-rank analogue.

Theorem 12.3.1: The Weyl Dimension Formula

Let G be a compact connected Lie group with maximal torus T, positive roots Φ^+, and Weyl vector $\rho = \frac{1}{2}\sum_{\alpha>0}\alpha$ (definition 12.2.1). For a dominant integral weight $\lambda \in P^+$ (definition 12.1.1), the irreducible representation V_λ of highest weight λ (theorem 12.1.3) has dimension

$$\dim V_\lambda = \prod_{\alpha>0} \frac{\langle \lambda+\rho, \alpha^\vee\rangle}{\langle \rho, \alpha^\vee\rangle}$$

the product running over the positive roots $\alpha \in \Phi^+$, with $\alpha^\vee$ the coroot of α (definition 7.4.3) and $\langle -,-\rangle$ the pairing of weights with coroots. Equivalently, in terms of a W-invariant inner product $(-,-)$ on $\mathfrak{t}^*$,

$$\dim V_\lambda = \prod_{\alpha>0} \frac{(\lambda+\rho, \alpha)}{(\rho, \alpha)}$$

***Proof*:**

Fix a regular element $x \in \mathfrak{t}$, meaning $\alpha(x) \neq 0$ for every root α; such x exist because the finitely many hyperplanes $\ker\alpha$ do not exhaust $\mathfrak{t}$. Evaluate the character along the curve $t_s = \exp(sx)$, $s \in \mathbb{R}$, which approaches the identity as $s \to 0$. On this curve a formal exponential e^μ becomes the exponential of a real variable,

$$e^\mu(\exp(sx)) = e^{s\,\mu(x)}$$

where $\mu(x) = \langle \mu, x\rangle$ is the value of the linear functional μ on x. Continuity of χ_λ gives $\dim V_\lambda = \chi_\lambda(e) = \lim_{s\to 0}\chi_\lambda(t_s)$, so it suffices to compute the limit of $A_{\lambda+\rho}(t_s)/A_\rho(t_s)$ as $s \to 0$.

Step 1: the alternating sums as antisymmetrized exponentials. Writing $\nu = \lambda + \rho$, the numerator evaluated on t_s is

$$A_\nu(t_s) = \sum_{w\in W} \operatorname{sgn}(w)\, e^{w\nu}(\exp(sx)) = \sum_{w\in W} \operatorname{sgn}(w)\, e^{s\,(w\nu)(x)}$$

and using W-invariance of the pairing, $(w\nu)(x) = \langle w\nu, x\rangle = \langle \nu, w^{-1}x\rangle$; relabelling the sum by $w \mapsto w^{-1}$ (which preserves sgn) gives the symmetric form

$$A_\nu(t_s) = \sum_{w\in W} \operatorname{sgn}(w)\, e^{s\,\langle \nu, wx\rangle}$$

The same holds for the denominator with ν replaced by ρ.

Step 2: order of vanishing at $s = 0$. Expand each exponential as a power series in s. Then

$$A_\nu(t_s) = \sum_{k\geq 0} \frac{s^k}{k!} \underbrace{\sum_{w\in W} \operatorname{sgn}(w)\, \langle \nu, wx\rangle^k}_{c_k(\nu)}$$

The coefficient $c_k(\nu) = \sum_w \operatorname{sgn}(w)\,\langle \nu, wx\rangle^k$ is, as a function of the vector x, a polynomial of degree k that is *alternating* under W: replacing x by $w_0 x$ multiplies the sum by $\operatorname{sgn}(w_0)$, since relabelling $w \mapsto ww_0^{-1}$ restores the sum up to that sign. An alternating polynomial vanishes on

every reflection hyperplane $\ker\alpha$, hence is divisible by the linear form α for each positive root α; the product $\prod_{\alpha>0}\alpha$ of these $N := |\Phi^+|$ distinct forms therefore divides c_k. Consequently $c_k(\nu) = 0$ for every $k < N$, and $A_\nu(t_s)$ vanishes to order at least N at $s = 0$. The leading coefficient is $c_N(\nu)/N!$, and since c_N is an alternating polynomial of degree exactly N it is a constant multiple of the minimal alternating polynomial $\prod_{\alpha>0}\alpha$,

$$c_N(\nu) = \kappa \cdot \prod_{\alpha>0}\langle \nu, \alpha^\vee\rangle$$

for a constant κ independent of ν; here the product is written in terms of the coroots so that κ depends only on x and the normalization, not on which weight ν is fed in. The identical statement holds for the denominator with $\nu = \rho$.

Step 3: the limit. Both $A_{\lambda+\rho}(t_s)$ and $A_\rho(t_s)$ vanish to order exactly N at $s = 0$, with leading terms $\frac{s^N}{N!}\,c_N(\lambda+\rho)$ and $\frac{s^N}{N!}\,c_N(\rho)$; the order is exactly N because $c_N(\lambda+\rho)$ and $c_N(\rho)$ are both nonzero, as $\lambda+\rho$ is strictly dominant (a dominant λ shifted by the strictly dominant ρ). The higher-order terms are killed in the ratio as $s \to 0$, and the powers of s and the factors $N!$ cancel, leaving

$$\dim V_\lambda = \lim_{s\to 0}\frac{A_{\lambda+\rho}(t_s)}{A_\rho(t_s)} = \frac{c_N(\lambda+\rho)}{c_N(\rho)} = \frac{\kappa\prod_{\alpha>0}\langle\lambda+\rho,\alpha^\vee\rangle}{\kappa\prod_{\alpha>0}\langle\rho,\alpha^\vee\rangle} = \prod_{\alpha>0}\frac{\langle\lambda+\rho,\alpha^\vee\rangle}{\langle\rho,\alpha^\vee\rangle}$$

the constant κ and the choice of regular x dropping out entirely. That $c_N(\rho) \neq 0$ is exactly the statement that ρ is strictly dominant, so $\langle\rho,\alpha^\vee\rangle > 0$ for every positive root, which holds because $\langle\rho,\alpha_i^\vee\rangle = 1$ on each simple coroot while every positive coroot is a nonnegative integer combination of simple coroots (exercise 12.3.1 (4)). Rewriting the pairing $\langle\nu,\alpha^\vee\rangle = 2(\nu,\alpha)/(\alpha,\alpha)$ through the inner product, the common factors $2/(\alpha,\alpha)$ cancel between numerator and denominator, giving the equivalent form $\prod_{\alpha>0}(\lambda+\rho,\alpha)/(\rho,\alpha)$, completing the proof.

The formula is a ratio of two products over the positive roots, one built from $\lambda+\rho$ and one from ρ alone. Reading it as a polynomial in λ, the numerator $\prod_{\alpha>0}\langle\lambda+\rho,\alpha^\vee\rangle$ is a polynomial of degree $N = |\Phi^+|$ in the coordinates of λ, so dimensions grow polynomially in the highest weight, with degree equal to the number of positive roots. The denominator is the value of that polynomial at $\lambda = 0$, namely $\prod_{\alpha>0}\langle\rho,\alpha^\vee\rangle$; the numerator at $\lambda = 0$ is the same product, so $\dim V_0$, the ratio of the numerator at $\lambda = 0$ to this denominator, equals 1, the dimension of the trivial representation, and the formula is correctly normalized. The appearance of the ρ-shift $\lambda \mapsto \lambda+\rho$, the same shift that governs the character formula, recurs throughout Weyl's theory: dimensions and characters both become clean when weights are measured from ρ rather than from 0.

The smallest instance is the running example, where the product is a single factor and the formula collapses to the count already known from Peter-Weyl.

Example 12.3.2: Dimensions for SU(2)

For $G = \mathrm{SU}(2)$ the root system is A_1 with a single positive root α, and the maximal torus is the diagonal circle of section 12.2. The Weyl vector is $\rho = \frac{1}{2}\alpha$, and the dominant integral weights are $\lambda_n = \frac{n}{2}\alpha$ for $n = 0, 1, 2, \ldots$, indexed so that V_{λ_n} is the irreducible of highest weight n (the $(n+1)$-dimensional representation $\mathrm{Sym}^n(\mathbb{C}^2)$ of proposition 9.5.2). The single coroot is $\alpha^\vee$,

normalized by $\langle \alpha, \alpha^\vee \rangle = 2$, so $\langle \rho, \alpha^\vee \rangle = \frac{1}{2}\langle \alpha, \alpha^\vee \rangle = 1$ and

$$\langle \lambda_n + \rho, \alpha^\vee \rangle = \left\langle \tfrac{n+1}{2}\alpha, \alpha^\vee \right\rangle = \tfrac{n+1}{2} \cdot 2 = n+1$$

The Weyl dimension formula, a single factor here, gives

$$\dim V_{\lambda_n} = \frac{\langle \lambda_n + \rho, \alpha^\vee \rangle}{\langle \rho, \alpha^\vee \rangle} = \frac{n+1}{1} = n+1$$

matching $\dim \mathrm{Sym}^n(\mathbb{C}^2) = n+1$ exactly. The same count falls out of the character formula's $\theta \to 0$ limit: $\sin((n+1)\theta)/\sin\theta \to (n+1)$, the L'Hôpital computation that the general theorem generalizes.

One step up in rank, the three positive roots of A_2 turn the formula into a product of three factors, and it recovers the dimensions of the fundamental and adjoint representations of SU(3) that the reader has met as 3, $\bar{3}$, and 8.

Example 12.3.3: Fundamental and Adjoint Dimensions for SU(3)

Take $G = \mathrm{SU}(3)$, with root system A_2. In the standard coordinates the simple roots are $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$, the positive roots are α_1, α_2, and $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$, so $N = |\Phi^+| = 3$, and the Weyl vector is $\rho = \frac{1}{2}(\alpha_1 + \alpha_2 + (\alpha_1 + \alpha_2)) = \alpha_1 + \alpha_2$. The pairings on the simple coroots are $\langle \rho, \alpha_1^\vee \rangle = \langle \rho, \alpha_2^\vee \rangle = 1$ and, since α_1, α_2 each pair to 1 against $(\alpha_1 + \alpha_2)^\vee$, one finds $\langle \rho, (\alpha_1 + \alpha_2)^\vee \rangle = 2$. Write a dominant weight in the fundamental-weight basis as $\lambda = a\varpi_1 + b\varpi_2$ with $a, b \geq 0$, characterized by $\langle \lambda, \alpha_1^\vee \rangle = a$ and $\langle \lambda, \alpha_2^\vee \rangle = b$, whence $\langle \lambda, (\alpha_1 + \alpha_2)^\vee \rangle = a + b$. Then $\langle \lambda + \rho, \alpha_1^\vee \rangle = a + 1$, $\langle \lambda + \rho, \alpha_2^\vee \rangle = b + 1$, and $\langle \lambda + \rho, (\alpha_1 + \alpha_2)^\vee \rangle = a + b + 2$, so the dimension formula reads

$$\dim V_{a,b} = \frac{(a+1)}{1} \cdot \frac{(b+1)}{1} \cdot \frac{(a+b+2)}{2} = \frac{(a+1)(b+1)(a+b+2)}{2}$$

The three named cases fall out at once. The standard representation $\mathbb{C}^3 = V_{1,0}$ has $\dim V_{1,0} = \frac{2 \cdot 1 \cdot 3}{2} = 3$; its dual $\bar{3} = V_{0,1}$ has $\dim V_{0,1} = \frac{1 \cdot 2 \cdot 3}{2} = 3$; and the adjoint representation is $V_{1,1}$, the highest root being $\alpha_1 + \alpha_2$, with

$$\dim V_{1,1} = \frac{2 \cdot 2 \cdot 4}{2} = 8$$

matching $\dim \mathfrak{su}(3)_{\mathbb{C}} = \dim \mathfrak{sl}_3(\mathbb{C}) = 8$. The formula thus reproduces the entire low-dimensional representation theory of SU(3) from the single polynomial in (a, b).

The dimension is only the value of the character at one point. The full character carries more: expanded as a sum $\chi_\lambda = \sum_\mu m_\lambda(\mu)\, e^\mu$ over the weight lattice, its coefficient $m_\lambda(\mu)$ is the multiplicity of μ as a weight of V_λ, the dimension of the μ-weight space. The dimension formula sums these, $\dim V_\lambda = \sum_\mu m_\lambda(\mu)$; Kostant's formula computes each one separately. The mechanism is to invert the Weyl denominator: writing $1/A_\rho$ as a generating series in the positive roots turns the ratio $A_{\lambda+\rho}/A_\rho$ into an explicit alternating sum over the Weyl group of a counting function.

To state it, one first names the counting function. Every weight that appears in a representation is a highest weight minus a sum of positive roots, so the number of ways of writing a given weight as such a sum is the natural bookkeeping device.

Definition 12.3.4: The Kostant Partition Function

Let Φ^+ be the set of positive roots. The *Kostant partition function* $\mathcal{P}\colon X^*(T) \to \mathbb{Z}_{\geq 0}$ assigns to a weight η the number of ways of writing η as a sum of positive roots with nonnegative integer multiplicities,

$$\mathcal{P}(\eta) = \#\Big\{(k_\alpha)_{\alpha>0} \in \mathbb{Z}_{\geq 0}^{\Phi^+} \;\Big|\; \eta = \sum_{\alpha>0} k_\alpha\, \alpha\Big\}$$

with the conventions $\mathcal{P}(0) = 1$ (the empty sum) and $\mathcal{P}(\eta) = 0$ when η is not a nonnegative integer combination of positive roots. Equivalently, $\mathcal{P}$ is the generating function identity

$$\prod_{\alpha>0} \frac{1}{1 - e^{-\alpha}} = \sum_{\eta} \mathcal{P}(\eta)\, e^{-\eta}$$

in the completed group ring, the product of geometric series $(1 - e^{-\alpha})^{-1} = \sum_{k\geq 0} e^{-k\alpha}$.

The partition function counts, for each weight η, the lattice points in the polytope of nonnegative solutions to $\sum k_\alpha \alpha = \eta$; it is the root-system analogue of the classical partition function counting ways to write an integer as a sum of parts. The generating-function form is the one that feeds directly into the character formula: it is precisely the expansion of the reciprocal Weyl denominator, once A_ρ is written as a product.

Example 12.3.5: The Partition Function for SU(3)

For SU(3) the positive roots are α_1, α_2, and $\alpha_1 + \alpha_2$. A weight $\eta = p\,\alpha_1 + q\,\alpha_2$ with $p, q \geq 0$ is a sum of positive roots as $\eta = k_1\alpha_1 + k_2\alpha_2 + k_3(\alpha_1 + \alpha_2)$, so $k_1 + k_3 = p$ and $k_2 + k_3 = q$ with all $k_i \geq 0$. The free parameter is k_3, ranging over $0 \leq k_3 \leq \min(p, q)$, and each choice determines k_1, k_2; hence

$$\mathcal{P}(p\,\alpha_1 + q\,\alpha_2) = \min(p, q) + 1$$

for $p, q \geq 0$, and $\mathcal{P} = 0$ off the cone. For instance $\mathcal{P}(\alpha_1 + \alpha_2) = \min(1, 1) + 1 = 2$: the decompositions are $\alpha_1 + \alpha_2$ and $(\alpha_1 + \alpha_2)$ itself, the two ways of building the highest root.

With the partition function in hand, inverting the Weyl denominator produces the multiplicity formula. The full justification of the generating-series manipulation belongs to the algebraic theory of the universal enveloping algebra and Verma modules; the compact-group argument reproduces the identity at the level of formal characters, and the convergence and formal-identity bookkeeping is cited to the algebraic development.

Theorem 12.3.6: Kostant's Multiplicity Formula

Let V_λ be the irreducible representation of dominant highest weight λ. The multiplicity of a weight μ in V_λ, that is the dimension of its μ-weight space, is

$$m_\lambda(\mu) = \sum_{w \in W} \operatorname{sgn}(w)\, \mathcal{P}\big(w(\lambda + \rho) - (\mu + \rho)\big)$$

where $\mathcal{P}$ is the Kostant partition function (definition 12.3.4) and ρ the Weyl vector.

***Proof Key ideas*:**
Start from the Weyl character formula $\chi_\lambda = A_{\lambda+\rho}/A_\rho$ (theorem 12.2.5) and the Weyl denominator formula $A_\rho = \prod_{\alpha>0}(e^{\alpha/2} - e^{-\alpha/2})$ (theorem 12.2.4). Factor e^ρ out of the denominator, using $\rho = \frac{1}{2}\sum_{\alpha>0}\alpha$:

$$A_\rho = e^\rho \prod_{\alpha>0}\left(1 - e^{-\alpha}\right)$$

so that, inverting in the completed group ring,

$$\frac{1}{A_\rho} = e^{-\rho}\prod_{\alpha>0}\frac{1}{1-e^{-\alpha}} = e^{-\rho}\sum_{\eta}\mathcal{P}(\eta)\,e^{-\eta}$$

by the generating-function form of definition 12.3.4. Multiplying by the numerator $A_{\lambda+\rho} = \sum_{w\in W}\operatorname{sgn}(w)\,e^{w(\lambda+\rho)}$ gives

$$\chi_\lambda = \sum_{w\in W}\operatorname{sgn}(w)\sum_{\eta}\mathcal{P}(\eta)\,e^{w(\lambda+\rho)-\rho-\eta}$$

Reading off the coefficient of e^μ requires $w(\lambda+\rho)-\rho-\eta = \mu$, that is $\eta = w(\lambda+\rho)-(\mu+\rho)$, so

$$m_\lambda(\mu) = \sum_{w\in W}\operatorname{sgn}(w)\,\mathcal{P}\big(w(\lambda+\rho)-(\mu+\rho)\big)$$

as claimed. The steps use formal manipulations in the completed group ring $\mathbb{Z}[[X^*(T)]]$: the geometric-series inverses of $1-e^{-\alpha}$ converge in the topology where $e^{-\alpha}$ is small in the dominant direction, and the product over $\alpha>0$ multiplies out to the stated single series. The full verification that these formal series manipulations are legitimate, together with the identification of χ_λ with the formal character of the corresponding irreducible highest-weight module, is the content of the algebraic highest-weight and Verma-module theory, cited to [10, The Weyl Character Formula]; the identity of formal characters proved there specializes to the compact-group χ_λ by the highest-weight classification of theorem 12.1.3, completing the argument.

The formula reads the multiplicity as a signed count: the naive guess $\mathcal{P}(\lambda-\mu)$, counting the ways to descend from the highest weight λ to μ by subtracting positive roots, is the $w=1$ term, and the remaining Weyl-group terms are the inclusion-exclusion corrections that account for the overcounting. The correction terms are precisely what the ρ-shift and the alternating sign encode: without them the partition count overshoots, because not every descent path from λ to μ lands on an actual weight of the irreducible, and the anti-invariance of $A_{\lambda+\rho}$ prunes the excess. The dimension formula is the sum over μ of these multiplicities: adding $m_\lambda(\mu)$ over all μ recovers $\dim V_\lambda$, whose closed product form the limit argument extracted directly.

The two theorems of this section are two readings of one generating identity. The Weyl character formula $\chi_\lambda = A_{\lambda+\rho}/A_\rho$ is a single element of the group ring; its *value at the identity*, computed by the regular-direction limit, is the dimension $\dim V_\lambda$, and the *coefficient of* e^μ in its expansion, computed by inverting the denominator, is the multiplicity $m_\lambda(\mu)$. One formula, evaluated two ways, yields both the dimension and the weight multiplicities. This closes the computational program of Part III: the root-system combinatorics of Part II and the integration theory of chapter 11 have combined into explicit polynomials for dimensions and explicit signed sums for multiplicities. Part IV turns from these compact-group computations to the noncompact real forms, where the same root

data reorganizes into the Cartan and Iwasawa decompositions.

Exercise 12.3.1

1. Use the SU(3) dimension polynomial $\dim V_{a,b} = \frac{1}{2}(a+1)(b+1)(a+b+2)$ to compute the dimensions of the irreducibles with highest weights $(a,b) = (2,0)$, $(0,2)$, $(2,1)$, and $(3,0)$. Identify $V_{2,0}$ as $\mathrm{Sym}^2(\mathbb{C}^3)$ and check the dimension against $\binom{3+2-1}{2}$.

2. For SU(4) the positive roots are $\varepsilon_i - \varepsilon_j$ for $1 \le i < j \le 4$ (six of them). Write the highest weight of the standard representation $\mathbb{C}^4$ in the coordinates $\lambda = \varepsilon_1$ (modulo the trace relation) and use the Weyl dimension formula to verify $\dim V = 4$. Then verify $\dim \Lambda^2\mathbb{C}^4 = 6$ by the same formula, taking the highest weight to be $\varepsilon_1 + \varepsilon_2$, and confirm both against the hook-content count $\binom{4}{1}$ and $\binom{4}{2}$ for the one-box and two-box columns.

3. In the adjoint representation $V_{1,1}$ of SU(3) (dimension 8), the zero weight $\mu = 0$ is the weight of the Cartan subalgebra $\mathfrak{t}_{\mathbb{C}}$, which is two-dimensional. Compute $m_{V_{1,1}}(0)$ from Kostant's formula $m_\lambda(0) = \sum_{w \in W} \mathrm{sgn}(w)\, \mathcal{P}(w(\lambda+\rho) - \rho)$ using $\lambda + \rho = 2\alpha_1 + 2\alpha_2$ (from $\rho = \alpha_1 + \alpha_2$), the SU(3) partition count $\mathcal{P}(p\alpha_1 + q\alpha_2) = \min(p,q) + 1$ from example 12.3.5, and $|W| = 6$; confirm the answer is 2.

4. Show that $\langle \rho, \alpha_i^\vee \rangle = 1$ for every simple root α_i, and deduce that $\langle \rho, \alpha^\vee \rangle > 0$ for every positive root α. Explain why this is exactly what makes the denominator $\prod_{\alpha>0} \langle \rho, \alpha^\vee \rangle$ in the Weyl dimension formula nonzero.

5. Argue from the Weyl dimension formula that, for a fixed compact simple G with N positive roots, $\dim V_\lambda$ is a polynomial in the coordinates of λ of degree exactly N. What does this say about the growth rate of dimensions of irreducibles as the highest weight moves out along a ray in the dominant chamber?

6. For SU(2) with highest weight λ_n (the $(n+1)$-dimensional irreducible), the weights are $n, n-2, \dots, -n$, each with multiplicity 1. Verify this from Kostant's formula: with $W = \{1, s\}$, $\mathrm{sgn}(s) = -1$, and the single positive root α so that $\mathcal{P}(k\alpha) = 1$ for $k \ge 0$ and 0 otherwise, compute $m_{\lambda_n}(\mu)$ for a weight $\mu = m\,\frac{\alpha}{2}$ and confirm it is 1 exactly when $m \in \{n, n-2, \dots, -n\}$.

12.4 The Borel-Weil Theorem

The theorem of the highest weight (theorem 12.1.3) produced V_λ only abstractly, and the Weyl character formula (theorem 12.2.5) computed its character without displaying the vector space. The Borel-Weil theorem gives the missing model: it realizes V_λ as the holomorphic sections of a line bundle on a compact complex manifold attached to G, the flag manifold, so that the representation, its highest weight, and the G-action all become geometric. Existence of V_λ, taken on faith in section 12.1, is thereby recovered by construction: for λ dominant the section space is nonzero, and it is exactly V_λ.

The construction rests on two complex-geometric facts about the flag manifold that belong to the algebraic-groups side of the theory. Both are cited to [10, G/B is Projective, Basic Properties of $\mathcal{L}(\lambda)$], where the flag variety $G_{\mathbb{C}}/B$ is built and its equivariant line bundles are classified; this section takes them as given and does the representation-theoretic work, identifying the section space as a G-module.

The homogeneous space G/T carries more structure than the smooth-manifold quotient of chapter 4. Complexifying G to a complex reductive group $G_{\mathbb{C}}$ and choosing a Borel subgroup $B \leq G_{\mathbb{C}}$, the quotient G/T acquires a complex structure through the diffeomorphism $G/T \cong G_{\mathbb{C}}/B$, and $G_{\mathbb{C}}/B$ is a smooth projective complex variety. To each character of T one attaches a holomorphic line bundle over it, and the dominant characters are exactly those whose bundle has sections. The definition packages the space and its bundles.

Definition 12.4.1: The Flag Manifold and the Line Bundle L_λ

Let G be a compact connected Lie group with maximal torus T, complexification $G_{\mathbb{C}}$, and a Borel subgroup $B \leq G_{\mathbb{C}}$ containing the complexified torus $T_{\mathbb{C}}$, chosen so that the roots of B are the positive roots Φ^+. The *flag manifold* of G is the compact homogeneous space

$$G/T \;\cong\; G_{\mathbb{C}}/B$$

which is a projective complex manifold ([10, G/B is Projective]). A weight $\lambda \in X^*(T)$ extends to a holomorphic character $\lambda\colon B \to \mathbb{C}^\times$, trivial on the unipotent radical of B. The associated *holomorphic line bundle* is the quotient

$$L_\lambda \;=\; \big(G_{\mathbb{C}} \times \mathbb{C}\big)/\!\sim, \qquad (gb, z) \sim \big(g,\, \lambda(b)\, z\big) \quad (b \in B)$$

a holomorphic line bundle over $G_{\mathbb{C}}/B$ on which $G_{\mathbb{C}}$, and hence G, acts by left translation $h \cdot [g, z] = [hg, z]$, covering the action on the base. Its holomorphic sections form the complex vector space $H^0(G/T, L_\lambda)$, a representation of G.

The convention deserves a word, since sign choices proliferate here and only one makes the theorem read cleanly. A holomorphic section $s\colon G_{\mathbb{C}}/B \to L_\lambda$ is the same datum as a holomorphic function $f\colon G_{\mathbb{C}} \to \mathbb{C}$ that is B-equivariant of weight λ, meaning

$$f(gb) = \lambda(b)^{-1} f(g) \qquad \text{for all } g \in G_{\mathbb{C}},\ b \in B$$

the twisting by λ^{-1} being exactly what makes the pair $(g, f(g))$ well-defined in the quotient L_λ. The group G acts on such functions by $(h \cdot f)(g) = f(h^{-1}g)$, left translation, and this is the G-module structure on $H^0(G/T, L_\lambda)$. The whole content of Borel-Weil is that this space of twisted functions is finite-dimensional, and irreducible with highest weight λ, exactly when λ is dominant.

Before the general statement, the smallest case shows every object: $G = \mathrm{SU}(2)$, where the flag manifold is the Riemann sphere and the line bundles are the familiar $\mathcal{O}(n)$.

Example 12.4.2: The Flag Manifold of $\mathrm{SU}(2)$ is $\mathbb{C}\mathrm{P}^1$

For $G = \mathrm{SU}(2)$ the maximal torus is the diagonal circle T, and the flag manifold is

$$\mathrm{SU}(2)/T \;\cong\; \mathbb{C}\mathrm{P}^1$$

the complex projective line, identified in chapter 4 with the 2-sphere $S^2 \cong \mathrm{SU}(2)/\mathrm{U}(1)$. On the complex side $G_{\mathbb{C}} = \mathrm{SL}(2,\mathbb{C})$, the Borel B is the upper-triangular subgroup, and $\mathrm{SL}(2,\mathbb{C})/B \cong \mathbb{C}\mathrm{P}^1$ sends a coset to the line it stabilizes, matching $\mathrm{SU}(2)/T \cong \mathbb{C}\mathrm{P}^1$.

The characters of $T = \big\{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta}) \;\big|\; \theta \in \mathbb{R}\big\}$ are the integers: $t_\theta \mapsto e^{in\theta}$ is the weight $n \in X^*(T) \cong \mathbb{Z}$. Writing λ_n for this weight, the associated line bundle is the n-th power of the hyperplane bundle,

$$L_{\lambda_n} \;\cong\; \mathcal{O}(n)$$

the unique holomorphic line bundle on $\mathbb{C}\mathrm{P}^1$ of degree n. The positive root of A_1 is α with $\lambda_n = \frac{n}{2}\alpha$ under the standard identification, so the dominant weights are $n \geq 0$, and $\mathcal{O}(n)$ is exactly the bundle with holomorphic sections precisely when $n \geq 0$.

On $\mathbb{C}\mathrm{P}^1$ the degree n line bundle $\mathcal{O}(n)$ has global holomorphic sections only for $n \geq 0$, and for $n \geq 0$ the section space is $(n+1)$-dimensional, so it must be the irreducible V_n of the same dimension. The identification of the sections with homogeneous polynomials, carried out below, makes this precise and shows the highest weight is n. Everything the general theorem asserts is visible here, with the flag manifold a sphere and the sections polynomials.

With the flag manifold and its bundles in place, the motivating question is which line bundle produces which representation: the bundle attached to a dominant weight produces the irreducible of that highest weight, and no others produce anything. The representation-theoretic core of the proof is short. The section space is a G-module, its T-weights are visible from the equivariance condition, a highest-weight line is picked out by holomorphicity along the unipotent directions, and irreducibility follows from the theorem of the highest weight once that line is shown to generate. The complex-analytic inputs, finite-dimensionality of H^0 and the vanishing $H^0(G/T, L_\lambda) = 0$ for non-dominant λ, are cited.

Theorem 12.4.3: The Borel-Weil Theorem

Let G be a compact connected Lie group with maximal torus T and flag manifold $G/T \cong G_{\mathbb{C}}/B$, and let $\lambda \in X^*(T)$ be a weight. Then:

1. If λ is dominant, the space of holomorphic sections $H^0(G/T, L_\lambda)$ is, as a representation of G, isomorphic to the irreducible V_λ of highest weight λ:
$$H^0(G/T, L_\lambda) \;\cong\; V_\lambda$$
2. If λ is not dominant, then $H^0(G/T, L_\lambda) = 0$.

***Proof*:**
Realize sections as twisted functions, as in the discussion after definition 12.4.1: a holomorphic section of L_λ is a holomorphic function $f\colon G_{\mathbb{C}} \to \mathbb{C}$ satisfying $f(gb) = \lambda(b)^{-1} f(g)$ for $b \in B$, and G acts by $(h \cdot f)(g) = f(h^{-1}g)$. Call this space $\mathcal{H}_\lambda = H^0(G/T, L_\lambda)$.

Step 1: the section space is a G-representation, and it is finite-dimensional. That $\mathcal{H}_\lambda$ is a complex vector space closed under the G-action is immediate from the linearity of the equivariance condition and of left translation: if f_1, f_2 satisfy the condition so does any linear combination, and $(h \cdot f)(gb) = f(h^{-1}gb) = \lambda(b)^{-1} f(h^{-1}g) = \lambda(b)^{-1}(h \cdot f)(g)$, so $h \cdot f$ is again a section. Finite-dimensionality of $\mathcal{H}_\lambda = H^0$ of a holomorphic line bundle on the compact complex manifold G/T is a theorem of complex geometry (Cartan-Serre finiteness), cited to [10]. A finite-dimensional continuous representation of a compact group is completely reducible (theorem 9.3.3), so $\mathcal{H}_\lambda$ decomposes into irreducibles; the remaining steps identify which.

Step 2: the weights of $\mathcal{H}_\lambda$ and a highest-weight line. Restrict the G-action to the torus T and decompose $\mathcal{H}_\lambda$ into T-weight spaces. The Borel B has a Bruhat decomposition $G_{\mathbb{C}} = \bigsqcup_{w \in W} BwB$; its open dense cell is B^-B, where B^- is the opposite Borel, and a holomorphic section is determined by its restriction to this open cell, hence by its values on the unipotent radical N^- of B^-, since $g = n^- t b$ generically with $n^- \in N^-$, $t \in T_{\mathbb{C}}$, $b \in B$. The torus factor t is redundant,

being already carried by $B = T_{\mathbb{C}}N$; it is displayed only to make the weight bookkeeping below visible, and the equivariance condition guarantees the resulting formulas depend only on the product tb. The equivariance condition $f(gb) = \lambda(b)^{-1}f(g)$ fixes the dependence on the B-factor; what remains free is the restriction $f|_{N^-}$, a holomorphic function on the affine space N^-.

Among all sections there is a distinguished one, the constant along N^-: define f_λ on the open cell by $f_\lambda(n^-tb) = \lambda(tb)^{-1}$ using $\lambda(tb) = \lambda(t)\lambda(b)$, and extend holomorphically. This f_λ is a well-defined holomorphic section (its holomorphic extension across the lower-dimensional cells is where dominance of λ enters, cited to [10, Properties of the Induced Module]), and it is a T-weight vector: for $s \in T \subseteq T_{\mathbb{C}}$,

$$(s \cdot f_\lambda)(n^-tb) = f_\lambda(s^{-1}n^-tb) = f_\lambda\big((s^{-1}n^-s)\,s^{-1}t\,b\big) = \lambda(s^{-1}t\,b)^{-1} = \lambda(s)\,f_\lambda(n^-tb)$$

using that $s^{-1}n^-s \in N^-$ (the torus normalizes N^-) so it is absorbed into the N^--factor on which f_λ is constant. Thus f_λ has T-weight λ.

This weight λ is the highest weight occurring in $\mathcal{H}_\lambda$. Every T-weight μ of a section satisfies $\mu \leq \lambda$ in the dominance order: expanding a section along N^-, whose Lie algebra is $\bigoplus_{\alpha>0}\mathfrak{g}_{-\alpha}$, the monomials in the negative-root coordinates lower the weight from λ by sums of positive roots, so every weight is λ minus a nonnegative combination of positive roots, that is $\mu \leq \lambda$. Moreover the λ-weight space is one-dimensional, spanned by f_λ: a section of weight λ is constant along N^- (any nonconstant negative-root monomial strictly lowers the weight), hence a scalar multiple of f_λ.

Step 3: f_λ is a highest-weight vector and generates. The line $\mathbb{C}f_\lambda$ is annihilated by the raising operators: for a positive root α, the root vector $e_\alpha \in \mathfrak{g}_\alpha \subseteq \mathrm{Lie}(B)$ acts on f_λ by differentiating the right-invariant realization, and on the B-equivariant function of highest weight it acts by zero, because raising would produce a weight $\lambda + \alpha \not\leq \lambda$, impossible by Step 2. Thus f_λ is a highest-weight vector of weight λ (theorem 12.1.3). The cyclic G-submodule it generates is therefore isomorphic to V_λ: a highest-weight vector of dominant weight λ generates a copy of the irreducible V_λ, by the theorem of the highest weight.

It remains to see this copy is everything, that $\mathcal{H}_\lambda = V_\lambda$ with no further summands. Suppose $\mathcal{H}_\lambda \cong V_\lambda \oplus M$ with $M \neq 0$. Then M, being a nonzero representation of a compact group, contains an irreducible, hence a highest-weight vector of some dominant weight μ; that vector is a section of T-weight μ. By Step 2 every weight of $\mathcal{H}_\lambda$ is $\leq \lambda$, so $\mu \leq \lambda$, and μ is itself a highest weight, hence dominant. But the λ-weight space of $\mathcal{H}_\lambda$ is one-dimensional (Step 2), already exhausted by $f_\lambda \in V_\lambda$, so M has no vector of weight λ, forcing $\mu < \lambda$ strictly. Ruling out such an M with strictly smaller highest weight is exactly where the weight combinatorics stop short. Reading a section off its restriction to the big cell realizes it as an arbitrary holomorphic function on the affine space N^-, and the weights of *all* such functions run through λ minus every nonnegative combination of positive roots without bound. The N^--expansion by itself therefore does not truncate the multiplicities to those of χ_λ. The extra input needed is holomorphic extension across the lower-dimensional cells, which keeps only those weight vectors whose expansion extends to a global holomorphic section; this forces the character of $\mathcal{H}_\lambda$ to equal χ_λ, so that $\dim \mathcal{H}_\lambda = \dim V_\lambda$ and hence $M = 0$. This extension-and-vanishing statement is the complex-geometric input cited to [10, The Borel-Weil Theorem], which states the isomorphism under its own bundle convention as $H^0 \cong L(\lambda)^\vee$, equivalently $H^0(-w_0\lambda) \cong L(\lambda)$, a relabeling by $\lambda \mapsto -w_0\lambda$ that leaves the dimension count used here unchanged; equivalently it is the $w = 1$ case of Borel-Weil-Bott (box 12.4.5), itself only stated and cited here, which gives $H^0(G/T, L_\lambda) \cong V_\lambda$ outright. Hence $\mathcal{H}_\lambda \cong V_\lambda$, proving part (1).

Step 4: vanishing for non-dominant λ. If λ is not dominant, then λ is not the highest weight of any irreducible, and the candidate section f_λ fails to extend holomorphically across the boundary of the open cell: the would-be weight-λ function has a pole along a lower cell precisely because $\langle \lambda, \alpha^\vee \rangle < 0$ for some simple α. That $H^0(G/T, L_\lambda) = 0$ for non-dominant λ is the vanishing statement cited to [10, The Borel-Weil Theorem]; in representation-theoretic terms, a nonzero section would carry a dominant highest weight $\leq \lambda$, impossible when λ itself is not dominant and dominates every weight present. This gives part (2), completing the proof.

The theorem settles what section 12.1 left open. There the existence of V_λ was an assertion, proved by an algebraic construction or deferred; here it is a geometric object, the space of holomorphic sections of an explicit line bundle on an explicit manifold. The dominant weights are singled out not by a chamber condition imposed from outside but by the intrinsic geometry: their bundles are the ones with sections, and non-dominant bundles have none. The highest-weight line, abstract in theorem 12.1.3, becomes the concrete section f_λ constant along the big cell, a B-eigenvector picked out by holomorphicity. The theorem lives most naturally on the complex algebraic side, which is why the complex-analytic inputs are cited to [10] rather than reproved: the flag variety, its projectivity, and the section-vanishing dichotomy are algebraic-geometric statements about $G_{\mathbb{C}}/B$.

The running example is the identification promised above, worked in full.

Example 12.4.4: Sections of $\mathcal{O}(n)$ on $\mathbb{C}\mathrm{P}^1$ Recover V_n

Take $G = \mathrm{SU}(2)$, so $G/T \cong \mathbb{C}\mathrm{P}^1$ and $L_{\lambda_n} \cong \mathcal{O}(n)$ (example 12.4.2). In homogeneous coordinates $[x : y]$ on $\mathbb{C}\mathrm{P}^1$, a holomorphic section of $\mathcal{O}(n)$ is a holomorphic function of the two coordinates $(x, y) \in \mathbb{C}^2 \setminus \{0\}$ that is homogeneous of degree n,

$$s(cx, cy) = c^n \, s(x, y) \qquad \text{for all } c \in \mathbb{C}^\times$$

The holomorphic such functions are exactly the homogeneous polynomials of degree n in x and y, with basis

$$x^n, \; x^{n-1}y, \; \ldots, \; xy^{n-1}, \; y^n$$

an $(n+1)$-dimensional space, and there are none for $n < 0$ since a negative-degree homogeneous holomorphic function on $\mathbb{C}^2 \setminus \{0\}$ cannot extend across the origin. Thus for $n \geq 0$,

$$H^0\big(\mathbb{C}\mathrm{P}^1, \mathcal{O}(n)\big) \;=\; \{\text{homogeneous degree-}n\text{ polynomials in } x, y\} \;=\; \mathrm{Sym}^n(\mathbb{C}^2)$$

which is exactly the irreducible $V_n = \mathrm{Sym}^n(\mathbb{C}^2)$ of proposition 9.5.2. The SU(2)-action by substitution of variables agrees with the left-translation action on sections, and the highest-weight vector is x^n, of weight n under $t_\theta \colon x \mapsto e^{i\theta}x$. The general Borel-Weil theorem thereby recovers, in its smallest case, the entire list of irreducibles of SU(2) and their highest weights, with the flag manifold a sphere and V_λ a space of polynomials.

The SU(2) case is the model to carry: sections of a positive line bundle on the sphere are the polynomials, and the polynomials are the irreducible. The general flag manifold replaces $\mathbb{C}\mathrm{P}^1$ by $G_{\mathbb{C}}/B$ and $\mathcal{O}(n)$ by L_λ, but the mechanism is unchanged, a highest-weight section constant along the big cell, generating the irreducible. That the sphere case is literally polynomials is why SU(2) is the right first encounter with the theorem.

Borel-Weil computes the space of holomorphic sections, the zeroth cohomology $H^0(G/T, L_\lambda)$, and finds it nonzero exactly for dominant λ. It says nothing about higher cohomology $H^i(G/T, L_\lambda)$ for

$i > 0$, nor about what a non-dominant λ does beyond killing H^0. Bott's refinement answers both at once, and the answer is governed by the same ρ-shifted Weyl action that appeared in the character formula (definition 12.2.1). The statement is recorded here and cited to [10, The Borel–Weil–Bott Theorem]; its proof, which is cohomological, belongs to the algebraic-groups treatment.

12.4.5 The Borel-Weil-Bott Theorem Let $\lambda \in X^*(T)$ be a weight and ρ the Weyl vector (definition 12.2.1). Two cases occur, according to the position of $\lambda + \rho$:

- If $\lambda + \rho$ lies on a wall, that is $\langle \lambda + \rho, \alpha^\vee \rangle = 0$ for some root α, then *all* cohomology vanishes: $H^i(G/T, L_\lambda) = 0$ for every $i \geq 0$.
- If $\lambda + \rho$ is regular, there is a unique $w \in W$ making $w(\lambda + \rho)$ dominant; writing $\mu = w(\lambda + \rho) - \rho$ (a dominant weight) and $\ell(w)$ for the length of w, the cohomology is concentrated in the single degree $\ell(w)$:

$$H^i(G/T, L_\lambda) \cong \begin{cases} V_\mu & i = \ell(w) \\ 0 & i \neq \ell(w) \end{cases}$$

When λ is dominant, $\lambda + \rho$ is already dominant, $w = 1$, $\ell(w) = 0$, and $\mu = \lambda$, recovering Borel-Weil (theorem 12.4.3): the cohomology sits in degree 0 and equals V_λ. The proof is cohomological (Serre duality along the $\mathbb{CP}^1$-fibrations of $G_{\mathbb{C}}/B$ plus induction on the length of w through the SL(2) associated to each simple root); see [10, The Borel–Weil–Bott Theorem].

Bott's picture reorganizes the whole family of line bundles into three regimes. A dominant λ gives an irreducible in degree 0, the Borel-Weil case. A weight λ with $\lambda + \rho$ regular but not dominant still produces an irreducible, but shifted into a higher cohomological degree equal to the number of walls the Weyl element w crosses to make $\lambda + \rho$ dominant, and the representation is V_μ for the ρ-shifted dominant weight μ. A weight with $\lambda + \rho$ singular, lying on a wall, contributes nothing in any degree. The ρ-shift is the same actor as in the denominator of the Weyl character formula: dominance of $\lambda + \rho$, not of λ, is the true dividing line, and ρ measures the distance from the wall. The same ρ-shift thus governs both the character formula and the cohomology of the line bundles.

For SU(2) the three regimes are visible on $\mathbb{CP}^1$. Here $\rho = \frac{1}{2}\alpha$ corresponds to the integer 1, so $\lambda_n + \rho$ corresponds to $n + 1$. For $n \geq 0$ this is regular and dominant, $w = 1$, and $H^0(\mathbb{CP}^1, \mathcal{O}(n)) = V_n$ in degree 0. For $n = -1$ the shift gives $n + 1 = 0$, singular, and indeed $\mathcal{O}(-1)$ has no cohomology in any degree: $H^0(\mathbb{CP}^1, \mathcal{O}(-1)) = H^1(\mathbb{CP}^1, \mathcal{O}(-1)) = 0$. For $n \leq -2$ the shift $n + 1 \leq -1$ is regular but negative, the reflection s_α makes it positive with $\ell(s_\alpha) = 1$, and the cohomology reappears in degree 1: $H^1(\mathbb{CP}^1, \mathcal{O}(n)) \cong V_{-n-2}$, matching Serre duality $H^1(\mathbb{CP}^1, \mathcal{O}(n)) \cong H^0(\mathbb{CP}^1, \mathcal{O}(-n-2))^*$ on the sphere.

This completes the analytic construction of the irreducibles. Part IV leaves compactness behind: complexifying the Lie algebra and choosing a Cartan involution splits a noncompact real group into its maximal compact part and a Euclidean complement, and the flag manifold reappears there as the geometry underlying the principal series and the boundary of a symmetric space.

Exercise 12.4.1

1. On $\mathbb{CP}^1$ with homogeneous coordinates $[x : y]$, write out an explicit basis for $H^0(\mathbb{CP}^1, \mathcal{O}(3))$ as homogeneous polynomials, and confirm its dimension against the Weyl dimension formula for

V_3. Identify the highest-weight vector and its weight under $t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$, and compute the T-weight of each basis element.

2. Explain, using the homogeneous-function description of sections of $\mathcal{O}(n)$ on $\mathbb{C}\mathrm{P}^1$, why $H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}(n)) = 0$ for every $n < 0$. Relate this to the general Borel-Weil statement (theorem 12.4.3) that $H^0(G/T, L_\lambda) = 0$ for non-dominant λ, and identify which non-dominant weights of $\mathrm{SU}(2)$ the negative integers n correspond to.

3. Using the Borel-Weil-Bott picture (box 12.4.5), determine all cohomology $H^i(\mathbb{C}\mathrm{P}^1, \mathcal{O}(-1))$ of the tautological bundle. Explain in terms of the ρ-shift why this bundle is the unique one on $\mathbb{C}\mathrm{P}^1$ with vanishing cohomology in every degree, and describe the general wall condition it instantiates.

4. For $\mathrm{SU}(2)$, compute $H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}(-4))$ using Bott's theorem: find the Weyl element w making $\lambda + \rho$ dominant, its length $\ell(w)$, the shifted weight $\mu = w(\lambda + \rho) - \rho$, and hence the representation appearing in degree $\ell(w)$. Check the answer against Serre duality $H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}(n)) \cong H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}(-n-2))^*$.

5. Show that the highest-weight line in $H^0(G/T, L_\lambda)$ is a B-eigenvector, that is, a common eigenvector for the whole Borel B acting on sections. Argue that its weight is λ and that it is annihilated by every positive root vector e_α, and explain why this pins down the line up to scalar (use that the λ-weight space is one-dimensional).

6. Suppose, hypothetically, that $H^0(G/T, L_\lambda)$ were *reducible* for some dominant λ, say $H^0(G/T, L_\lambda) \cong V_\lambda \oplus M$ with $M \neq 0$. Using only the theorem of the highest weight (theorem 12.1.3) and the one-dimensionality of the λ-weight space, and without invoking the Weyl character formula, show that any such M must have a highest weight strictly below λ, so a second copy of V_λ is ruled out. Explain why these two inputs alone do not eliminate M entirely, and identify what the proof of theorem 12.4.3 gives to close the gap. (This isolates the representation-theoretic half of the irreducibility argument.)

12.5 Worked Representations of $\mathrm{SU}(n)$ and $\mathrm{SO}(n)$

The highest-weight theorem (theorem 12.1.3) and the Weyl character formula (theorem 12.2.5) are stated for an abstract compact connected group; this section runs them on the two classical families the book has carried since chapter 1, and in doing so closes Part III. For $\mathrm{SU}(n)$ the dominant weights are partitions, the fundamental representations are the exterior powers of the standard $\mathbb{C}^n$, and the Weyl character formula turns out to be a formula the reader may already know under another name: Jacobi's bialternant, whose ratio is the Schur polynomial. For $\mathrm{SO}(n)$ the tensor representations are again built from exterior powers, but the full picture requires the double cover $\mathrm{Spin}(n)$ and the spin representations that live only there. The abstract summit of the chapter becomes, in these examples, a piece of classical combinatorics.

Everything below reuses the concrete Weyl-group and root data already computed for these groups: $W(\mathrm{SU}(n)) \cong S_n$ acting by permuting coordinates (example 11.2.5), the root system A_{n-1} with roots $\varepsilon_i - \varepsilon_j$, and the diagonal maximal torus of chapter 11. No new structure theory is needed, only the bookkeeping of applying the general formulas.

Example 12.5.1: Representations of SU(n)

Let $G = \mathrm{SU}(n)$ with its diagonal maximal torus $T = \{\mathrm{diag}(z_1, \dots, z_n) \mid z_j \in \mathrm{U}(1),\ z_1 \cdots z_n = 1\}$. Its character lattice is $X^*(T) = \mathbb{Z}^n/\mathbb{Z}(1, \dots, 1)$, and a weight is written $\lambda = (\lambda_1, \dots, \lambda_n)$ with the convention that adding a constant $(c, \dots, c)$ to all entries gives the same character of T. The Weyl group is S_n (example 11.2.5), permuting the entries λ_i; the positive roots are $\varepsilon_i - \varepsilon_j$ for $i < j$, so a weight is dominant precisely when its entries weakly decrease. Normalizing the last entry to 0, the dominant integral weights P^+ (definition 12.1.1) are exactly the partitions

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{n-1} \geq \lambda_n = 0$$

with $\lambda_i \in \mathbb{Z}$, that is the Young diagrams with at most $n-1$ rows. Each such partition λ names one irreducible V_λ, and theorem 12.1.3 makes $\lambda \mapsto V_\lambda$ a bijection onto the irreducibles.

The standard representation and its exterior powers. The standard representation is $\mathbb{C}^n$ itself, on which $\mathrm{SU}(n)$ acts by matrix multiplication. The diagonal torus acts on the basis vector e_i by $\mathrm{diag}(z_1, \dots, z_n) \cdot e_i = z_i e_i$, so the weights of $\mathbb{C}^n$ are $\varepsilon_1, \dots, \varepsilon_n$, and the highest is $\varepsilon_1 = (1, 0, \dots, 0)$: the standard representation is $V_{(1,0,\dots,0)}$, the partition of a single box. Its dual $(\mathbb{C}^n)^*$ has weights $-\varepsilon_i$, with highest weight $-\varepsilon_n = (1, 1, \dots, 1, 0)$ after normalization, the column of $n-1$ boxes; for $\mathrm{SU}(2)$ the standard representation is self-dual, but for $n \geq 3$ the standard and its dual are inequivalent.

The exterior power $\Lambda^k(\mathbb{C}^n)$ has as a weight basis the wedges $e_{i_1} \wedge \cdots \wedge e_{i_k}$ for $i_1 < \cdots < i_k$, on which the torus acts by $z_{i_1} \cdots z_{i_k}$; the weights are the sums $\varepsilon_{i_1} + \cdots + \varepsilon_{i_k}$, the highest being $\varepsilon_1 + \cdots + \varepsilon_k = (1, \dots, 1, 0, \dots, 0)$ with k ones. This is the dominant weight ω_k of the single column of k boxes, and $\Lambda^k(\mathbb{C}^n)$ is the irreducible V_{ω_k}. These $n-1$ representations $\Lambda^1(\mathbb{C}^n), \dots, \Lambda^{n-1}(\mathbb{C}^n)$ are the *fundamental representations* of $\mathrm{SU}(n)$, one for each node of the A_{n-1} Dynkin diagram; every dominant weight is a nonnegative combination $\lambda = \sum_k a_k \omega_k$ of the fundamental weights, and V_λ appears inside the corresponding tensor product $\bigotimes_k (\Lambda^k \mathbb{C}^n)^{\otimes a_k}$ as the highest-weight constituent. The top power $\Lambda^n(\mathbb{C}^n)$ is the trivial representation, since $\mathrm{SU}(n)$ acts on it by $\det = 1$.

The symmetric power $\mathrm{Sym}^m(\mathbb{C}^n)$ has weight basis the monomials $e_{i_1} \cdots e_{i_m}$ with $i_1 \leq \cdots \leq i_m$, highest weight $m\varepsilon_1 = (m, 0, \dots, 0)$, the single row of m boxes; it is the irreducible $V_{(m,0,\dots,0)}$. For $\mathrm{SU}(2)$ these are the complete list: $\mathrm{Sym}^m(\mathbb{C}^2)$ is the spin-$m/2$ irreducible of proposition 9.5.2, of dimension $m+1$.

The dimension. The Weyl dimension formula (theorem 12.3.1) gives $\dim V_\lambda$ as a product over the positive roots $\varepsilon_i - \varepsilon_j$, $i < j$. Writing $\ell_i = \lambda_i + (n-i)$ for the strictly decreasing sequence $\lambda + \rho$ shifted, so that $\langle \lambda + \rho, (\varepsilon_i - \varepsilon_j)^\vee \rangle = \ell_i - \ell_j$ and $\langle \rho, (\varepsilon_i - \varepsilon_j)^\vee \rangle = j - i$, the formula reads

$$\dim V_\lambda = \prod_{1 \leq i < j \leq n} \frac{\ell_i - \ell_j}{j - i}$$

the ratio of two Vandermonde products in the shifted parts. This is the hook-content count of the Young diagram λ in disguise: rewriting the product as $\prod_{(i,j) \in \lambda} \frac{n+j-i}{h(i,j)}$, where (i, j) runs over the boxes of λ, $j - i$ is the content of a box, and $h(i, j)$ is its hook length, expresses the same number combinatorially.

The character. The Weyl character formula computes χ_λ on the torus, evaluated at $\mathrm{diag}(z_1, \dots, z_n)$, as the ratio of alternating sums $A_{\lambda+\rho}/A_\rho$. Written in the z_i this is the symmetric function

$$\chi_\lambda(z_1, \dots, z_n) = s_\lambda(z_1, \dots, z_n)$$

the *Schur polynomial* of the partition λ, a symmetric polynomial of degree $|\lambda| = \sum_i \lambda_i$ in $z_1, \dots, z_n$. That χ_λ is exactly s_λ, with the Weyl character formula reproducing the classical bialternant definition of s_λ, is the content of proposition 12.5.3.

The example collapses the abstract classification onto a picture the reader can draw: a dominant weight for $\mathrm{SU}(n)$ is a Young diagram, the fundamental representations are the exterior powers, and the character is a Schur polynomial. The two most useful special cases are the ones just isolated, the column $\Lambda^k(\mathbb{C}^n) = V_{\omega_k}$ and the row $\mathrm{Sym}^m(\mathbb{C}^n) = V_{(m,0,\dots,0)}$, because every irreducible sits inside tensor products of these and the decomposition of such tensor products is governed by the same Schur-polynomial multiplication. The bijection with partitions is the reason the representation theory of $\mathrm{SU}(n)$ and the combinatorics of Young diagrams are one subject.

The orthogonal groups run the same machine, but with a wrinkle that has no analogue for $\mathrm{SU}(n)$: the integer-weight representations do not exhaust the irreducibles. Half of them, the spin representations, are visible only after passing to the double cover.

Example 12.5.2: Tensor and Spin Representations of $\mathrm{SO}(n)$

Let $G = \mathrm{SO}(n)$ acting on $\mathbb{R}^n$, or on its complexification $\mathbb{C}^n$. The standard representation $\mathbb{C}^n$ is irreducible for $n \geq 3$, and its exterior powers $\Lambda^k(\mathbb{C}^n)$ furnish a family of representations exactly as for the unitary groups. Here the invariant symmetric form on $\mathbb{C}^n$ makes $\Lambda^k(\mathbb{C}^n)$ isomorphic to $\Lambda^{n-k}(\mathbb{C}^n)$ by Hodge duality, so the distinct exterior powers run only up to $k = \lfloor n/2 \rfloor$; and in the middle degree, when $n = 2m$ is even, $\Lambda^m(\mathbb{C}^n)$ splits into the self-dual and anti-self-dual halves $\Lambda^m_+ \oplus \Lambda^m_-$. These exterior powers, and the irreducible constituents cut out of their tensor products by removing traces with respect to the invariant form, are the *tensor representations* of $\mathrm{SO}(n)$: exactly the irreducibles whose highest weight is an *integer* weight, a genuine character of the diagonal maximal torus of $\mathrm{SO}(n)$.

The tensor representations do not exhaust the dominant weights. The weight lattice of $\mathrm{SO}(n)$ contains, beyond the integer weights, the *half-integer* weights $\frac{1}{2}(\pm\varepsilon_1 \pm \cdots \pm \varepsilon_m)$ (for $\mathrm{SO}(2m)$ or $\mathrm{SO}(2m+1)$), which are dominant integral weights of the root system yet are not characters of the torus of $\mathrm{SO}(n)$ itself: the element $\frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_m)$ is a well-defined weight of the Lie algebra but exponentiates only to a character of a torus one cover up. The irreducibles carrying these highest weights are the *spin representations*, and they are representations not of $\mathrm{SO}(n)$ but of its simply connected double cover $\mathrm{Spin}(n)$, the group whose existence is forced by $\pi_1(\mathrm{SO}(n)) = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 3$.

The smallest case is the running example. For $n = 3$ the double cover is $\mathrm{Spin}(3) = \mathrm{SU}(2)$, and the covering map is the two-to-one homomorphism $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5. The integer-weight irreducibles of $\mathrm{SO}(3)$ are the odd-dimensional $\mathrm{Sym}^{2j}(\mathbb{C}^2)$, of dimensions $1, 3, 5, \dots$, which descend to genuine representations of $\mathrm{SO}(3)$; the half-integer-weight irreducibles are the even-dimensional $\mathrm{Sym}^{2j+1}(\mathbb{C}^2)$, of dimensions $2, 4, 6, \dots$, which are representations of $\mathrm{SU}(2) = \mathrm{Spin}(3)$ that do not descend to $\mathrm{SO}(3)$, because the central element $-I \in \mathrm{SU}(2)$, which maps to the identity of $\mathrm{SO}(3)$, acts by -1. The two-dimensional standard representation $\mathbb{C}^2$ of $\mathrm{SU}(2)$ is the basic spinor, the spin-$\frac{1}{2}$ representation: the smallest spin representation of $\mathrm{Spin}(3)$, corresponding to the highest weight $\frac{1}{2}\varepsilon_1$. Thus the physicist's distinction between integer-spin and half-integer-spin representations is exactly the split between representations of $\mathrm{SO}(3)$ and the strictly larger stock of representations of its double cover.

For general n the spin representations are built from the Clifford algebra of the quadratic form on $\mathbb{R}^n$: $\mathrm{Spin}(n)$ embeds in the units of the Clifford algebra, and the spinor module of that

algebra restricts to the spin representation (one irreducible spin representation for n odd, two half-spin representations S^+, S^- for n even). On the algebraic side the spin representations appear among the fundamental representations of Spin(n), labelled by their highest weights ([10, Fundamental Representations]). The Clifford-algebra construction itself lies beyond both books, and the SU(2) $\to$ SO(3) picture here is the first instance of the general theory.

The two examples exhibit the same abstract theory realized two ways. For SU(n) every dominant weight is a character of the torus, so every irreducible is a tensor representation and the classification by Young diagrams is complete; the subtlety of SO(n) is that its fundamental group is nontrivial, so its weight lattice is strictly larger than its character lattice, and the missing irreducibles are recovered only on the simply connected cover. This is the representation-theoretic face of the π_1 and covering-group analysis of chapter 4: which weights are attainable on a given group in an isogeny class, as opposed to on its universal cover, is exactly the lattice question $X^*(T) \subseteq P$ that theorem 11.5.4 answered.

It remains to justify the identification of the SU(n) character with the Schur polynomial, promised in example 12.5.1. The Weyl character formula, specialized to A_{n-1}, is a determinant of monomials over a Vandermonde determinant, and this ratio is precisely the classical bialternant that defines the Schur polynomial.

Proposition 12.5.3: The Weyl Character Formula as Jacobi's Bialternant

Let $\lambda = (\lambda_1 \geq \cdots \geq \lambda_n \geq 0)$ be a partition, regarded as a dominant weight of U(n) (or of SU(n) modulo $(1, \dots, 1)$). The character of the irreducible V_λ, evaluated on the diagonal torus at $\mathrm{diag}(z_1, \dots, z_n)$, is the ratio of alternating (skew-symmetric) polynomials

$$\chi_\lambda(z_1, \dots, z_n) = \frac{\det\left(z_i^{\lambda_j + n - j}\right)_{1 \leq i,j \leq n}}{\det\left(z_i^{n-j}\right)_{1 \leq i,j \leq n}} = s_\lambda(z_1, \dots, z_n)$$

Jacobi's bialternant formula for the Schur polynomial s_λ. The denominator $\det(z_i^{n-j}) = \prod_{i<j}(z_i - z_j)$ is the Vandermonde determinant, and it equals the Weyl denominator of theorem 12.2.4 written multiplicatively in the z_i.

***Proof*:**

The Weyl character formula (theorem 12.2.5) gives $\chi_\lambda = A_{\lambda+\rho}/A_\rho$, where $A_\mu = \sum_{w \in W} \mathrm{sgn}(w)\, e^{w\mu}$ and $W = S_n$ (example 11.2.5). Coordinatize the torus by $z_i = e^{\varepsilon_i}$, so that the formal exponential $e^{\sum_i m_i \varepsilon_i}$ becomes the monomial $z_1^{m_1} \cdots z_n^{m_n}$. For a weight $\mu = (\mu_1, \dots, \mu_n)$ and $w = \sigma \in S_n$ acting by permuting coordinates, $e^{\sigma\mu} = \prod_i z_i^{\mu_{\sigma^{-1}(i)}}$, so the alternating sum is the fully expanded determinant

$$A_\mu = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_i z_i^{\mu_{\sigma^{-1}(i)}} = \det\left(z_i^{\mu_j}\right)_{1 \leq i,j \leq n}$$

by the Leibniz expansion of the determinant. Here $\rho = \frac{1}{2}\sum_{i<j}(\varepsilon_i - \varepsilon_j) = (n-1, n-2, \dots, 1, 0)$ in these coordinates, half the sum of the positive roots $\varepsilon_i - \varepsilon_j$ (definition 12.2.1); so the shifted weight has entries $(\lambda + \rho)_j = \lambda_j + (n - j)$. Substituting $\mu = \lambda + \rho$ into the determinant identity gives the numerator $A_{\lambda+\rho} = \det(z_i^{\lambda_j + n - j})$, and $\mu = \rho$ gives the denominator $A_\rho = \det(z_i^{n-j})$.

The denominator is the Vandermonde determinant. The matrix (z_i^{n-j}) has (i,j) entry z_i^{n-j}, so

its columns run through the powers $n-1, n-2, \dots, 0$ of each z_i; this is the transpose of the standard Vandermonde matrix, whence

$$A_\rho = \det\big(z_i^{\,n-j}\big) = \prod_{1 \le i < j \le n} (z_i - z_j)$$

The right-hand product is $\prod_{\alpha>0}(z^{\alpha/2} - z^{-\alpha/2})$ up to the overall monomial $\prod_i z_i^{-(n-1)/2}$ that shifts ρ to be symmetric about 0: with $z^{\varepsilon_i - \varepsilon_j} = z_i/z_j$ one has $z_i - z_j = (z_i z_j)^{1/2}(z^{(\varepsilon_i-\varepsilon_j)/2} - z^{-(\varepsilon_i-\varepsilon_j)/2})$, and collecting the prefactors over all $i<j$ produces exactly $\prod_i z_i^{(n-1)/2}$. Hence A_ρ agrees with the Weyl denominator $\prod_{\alpha>0}(e^{\alpha/2} - e^{-\alpha/2})$ of theorem 12.2.4 up to this monomial, which cancels between numerator and denominator of χ_λ.

The quotient $\det(z_i^{\lambda_j+n-j})/\det(z_i^{n-j})$ is, by definition, the Schur polynomial $s_\lambda(z_1, \dots, z_n)$: it is a symmetric polynomial, because the two determinants are each alternating and their ratio is therefore invariant under permuting the z_i, and it has degree $|\lambda|$. This is Jacobi's bialternant, so $\chi_\lambda = s_\lambda$, as we sought to show.

The proposition is the punchline of the chapter for the classical groups. The Weyl character formula, an abstract ratio of anti-invariant lattice sums, becomes for $\mathrm{SU}(n)$ the nineteenth-century formula that Jacobi and Cauchy wrote for the Schur functions, and the Weyl denominator becomes the Vandermonde. The bridge runs both ways: the representation theory gives a conceptual proof that the bialternant ratio is a polynomial (it is a character, hence a genuine class function), while the symmetric-function theory gives the combinatorial engine, the Pieri and Littlewood-Richardson rules for multiplying Schur functions, that computes tensor-product decompositions of $\mathrm{SU}(n)$-representations. Of the full symmetric-function development, the algebraic companion proves the Littlewood-Richardson rule ([10, Littlewood–Richardson Rule]). The Jacobi-Trudi identities, the Robinson-Schensted correspondence, and Schur-Weyl duality between $\mathrm{SU}(n)$ and the symmetric group S_m on $(\mathbb{C}^n)^{\otimes m}$ are classical symmetric-function theory beyond both books.

With this, the Weyl character formula has been cashed into the classical representation theory of the two families that motivated the subject, and Part III is complete: from invariant integration and Peter-Weyl through maximal tori and the Weyl integration formula to the character and dimension formulas, the representation theory of a compact connected group is now computable in closed form. Part IV removes compactness. The same Cartan-subalgebra and root-system data that organized the compact groups will reappear, complexified and then re-realified through a Cartan involution, to control the noncompact real forms; the running thread $\mathrm{SU}(2)$ acquires its noncompact sibling $\mathrm{SL}(2,\mathbb{R})$, and the character theory of this section is the compact prototype against which the harder noncompact representation theory is measured. The bridge to symmetric spaces in chapter 15 and to [10] for the algebraic route stands ready.

Exercise 12.5.1

1. Show that the character of the fundamental representation $\Lambda^k(\mathbb{C}^n)$ of $\mathrm{SU}(n)$, evaluated at $\mathrm{diag}(z_1, \dots, z_n)$, is the elementary symmetric polynomial

$$\chi_{\Lambda^k}(z_1, \dots, z_n) = e_k(z_1, \dots, z_n) = \sum_{i_1 < \cdots < i_k} z_{i_1} \cdots z_{i_k}$$

by computing the trace of $\mathrm{diag}(z_1, \dots, z_n)$ acting on the weight basis $e_{i_1} \wedge \cdots \wedge e_{i_k}$. Identify this with the Schur polynomial s_{ω_k} for the single-column partition $\omega_k = (1, \dots, 1, 0, \dots, 0)$

with k ones.

2. Use the Weyl dimension formula for A_2 to compute $\dim V_\lambda$ for the $\mathrm{SU}(3)$ representation with Young diagram $\lambda = (2,1,0)$. The positive roots are $\varepsilon_1 - \varepsilon_2$, $\varepsilon_2 - \varepsilon_3$, $\varepsilon_1 - \varepsilon_3$, and $\rho = (2,1,0)$; form the shifted parts $\ell_i = \lambda_i + (3-i)$ and evaluate $\dim V_\lambda = \prod_{i<j} \frac{\ell_i - \ell_j}{j-i}$. Which classical representation of $\mathrm{SU}(3)$ is this, and what is its dimension?

3. Explain why the two-dimensional standard representation $\mathbb{C}^2$ of $\mathrm{SU}(2)$ is *not* a representation of $\mathrm{SO}(3)$, whereas the three-dimensional $\mathrm{Sym}^2(\mathbb{C}^2)$ is. Use the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5: determine how the central element $-I \in \mathrm{SU}(2)$ acts on $\mathrm{Sym}^m(\mathbb{C}^2)$, and deduce that $\mathrm{Sym}^m(\mathbb{C}^2)$ descends to $\mathrm{SO}(3)$ exactly when m is even. Which of these are tensor representations and which are spin representations?

4. Compute the Schur polynomial $s_\lambda(z_1, z_2, z_3)$ for the two-box partition $\lambda = (2,0,0)$ (a single row of two boxes) via Jacobi's bialternant (proposition 12.5.3), and check that it equals the character of $\mathrm{Sym}^2(\mathbb{C}^3)$, the complete homogeneous symmetric polynomial $h_2(z_1,z_2,z_3) = \sum_{i \le j} z_i z_j$. Repeat for $\lambda = (1,1,0)$ (a single column of two boxes) and confirm you recover $e_2(z_1,z_2,z_3)$, the character of $\Lambda^2(\mathbb{C}^3)$.

5. Show that the standard representation $\mathbb{C}^n$ of $\mathrm{SU}(n)$ is self-dual if and only if $n = 2$. Compute the highest weight of the dual $(\mathbb{C}^n)^*$ in the normalization $\lambda_n = 0$, and compare it to the highest weight ε_1 of $\mathbb{C}^n$; conclude that the two are equal only for $n = 2$, so that for $n \ge 3$ the standard and its dual are inequivalent fundamental representations.

6. Verify by direct expansion that $\det(z_i^{n-j})_{1 \le i,j \le n} = \prod_{i<j}(z_i - z_j)$ for $n = 3$, and identify this determinant with the value of the Weyl denominator A_ρ of $\mathrm{SU}(3)$. Explain, using proposition 11.2.4, why the vanishing of this Vandermonde on the locus $z_i = z_j$ corresponds to the non-regular (Weyl-fixed) elements of the maximal torus.

Part IV

Real Forms and Symmetric Spaces

This closing part studies the real semisimple group as it stands, compact or not. Its three chapters share one engine, the interplay between a real form and its complexification. Chapter 13 studies how a real Lie algebra sits inside its complexification, constructs the compact real form, and proves the Cartan decomposition, splitting the group into a maximal compact subgroup and a Euclidean factor; chapter 14 refines the splitting to the Iwasawa decomposition $G = KAN$ through the restricted roots; and chapter 15 reads the same decompositions geometrically, where they become the theory of Riemannian symmetric spaces and the bridge to [7].

13

Real Forms and the Cartan Decomposition

Every structure theorem of Part II and Part III was purchased with an assumption: the Lie algebra was complex in Part II, the group compact in Part III. Compactness made the Killing form negative-definite (proposition 9.4.1), representations unitary, and the exponential surjective. Part IV removes that assumption. A real semisimple Lie group need not be compact, its Killing form is indefinite, and its representation theory becomes the deep and unfinished subject of harmonic analysis on noncompact groups. The organizing discovery is that a noncompact group is not shapeless: it is a compact group times a Euclidean space, and the two factors are visible already on the Lie algebra. That is the Cartan decomposition, the subject of this chapter and the structural foundation of everything that follows.

The mechanism is complexification. A real Lie algebra $\mathfrak{g}_0$ and its compact sibling both complexify to the same complex Lie algebra $\mathfrak{g}_{\mathbb{C}}$, so the classification of chapter 8 already contains every real group; what a real form records is *how* $\mathfrak{g}_0$ sits inside $\mathfrak{g}_{\mathbb{C}}$, encoded by an antilinear involution. Among all real forms of a complex semisimple $\mathfrak{g}_{\mathbb{C}}$ exactly one is compact, and this chapter first constructs it, from the root space decomposition (theorem 7.2.4) by a sign adjustment on the root vectors, and proves it unique up to conjugacy, discharging the promise made in chapter 9. Aligning an arbitrary real form with the compact one produces the *Cartan involution* θ, whose eigenspace decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ splits the algebra into a maximal compact part $\mathfrak{k}$, on which the Killing form is negative-definite, and a complementary Euclidean part $\mathfrak{p}$, on which it is positive-definite.

The decomposition integrates to the group. For a connected semisimple real group G the map $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, is a diffeomorphism, so $G = K \exp(\mathfrak{p})$ is a maximal compact subgroup times a contractible Euclidean factor; K is unique up to conjugacy, and G deformation-retracts onto it, so the topology of a noncompact semisimple group is entirely carried by its maximal compact. For matrix groups this is the polar decomposition, an orthogonal or unitary matrix times a positive-

definite symmetric or Hermitian one, and the running example shows it: $\mathrm{SL}(2,\mathbb{R}) = \mathrm{SO}(2)\exp(\mathfrak{p})$, the noncompact sibling of the $\mathrm{SU}(2)$ that has threaded the book since chapter 1, with the hyperbolic plane $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ appearing as the first noncompact symmetric space, taken up in full in chapter 15. The chapter closes on $\mathrm{SL}(n,\mathbb{R})$, $\mathrm{SU}(p,q)$, and $\mathrm{SL}(2,\mathbb{R})$ worked end to end, the Cartan decomposition made explicit in coordinates; chapter 14 then refines it into the finer $G = KAN$ of Iwasawa, and chapter 15 reads the whole picture as the geometry of a symmetric space.

13.1 Complexification and Real Forms

A real semisimple Lie algebra $\mathfrak{g}_0$ is recovered from its complexification $\mathfrak{g}_{\mathbb{C}}$ by remembering one extra piece of data: the antilinear map that fixes $\mathfrak{g}_0$ pointwise. This section makes that idea precise. It defines the complexification of a real Lie algebra, defines a real form of a complex Lie algebra together with its associated conjugation, and proves that the two notions are interchangeable, real forms of $\mathfrak{g}_{\mathbb{C}}$ being the same thing as conjugations of $\mathfrak{g}_{\mathbb{C}}$. The running example is the pair $\mathfrak{sl}_n(\mathbb{R})$ and $\mathfrak{su}(n)$, two real forms of the single complex algebra $\mathfrak{sl}_n(\mathbb{C})$, one noncompact and one compact.

The first construction goes from real to complex. Tensoring a real Lie algebra with $\mathbb{C}$ enlarges the scalars to the complex numbers while keeping the bracket, and the result is a complex Lie algebra of the same dimension.

Definition 13.1.1: Complexification of a Real Lie Algebra

Let $\mathfrak{g}_0$ be a real Lie algebra. Its *complexification* is the complex vector space

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$$

with the $\mathbb{C}$-bilinear extension of the bracket of $\mathfrak{g}_0$, defined on the summands by

$$[X \otimes z,\, Y \otimes w] = [X,Y] \otimes zw \qquad (X,Y \in \mathfrak{g}_0,\ z,w \in \mathbb{C})$$

and extended $\mathbb{C}$-bilinearly. Writing X for $X \otimes 1$ and iY for $Y \otimes i$, every element of $\mathfrak{g}_{\mathbb{C}}$ is uniquely $X + iY$ with $X, Y \in \mathfrak{g}_0$, so $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus i\,\mathfrak{g}_0$ as real vector spaces and $\dim_{\mathbb{C}} \mathfrak{g}_{\mathbb{C}} = \dim_{\mathbb{R}} \mathfrak{g}_0$. The map

$$\sigma_0 \colon \mathfrak{g}_{\mathbb{C}} \to \mathfrak{g}_{\mathbb{C}}, \qquad \sigma_0(X + iY) = X - iY$$

is the *natural conjugation* of $\mathfrak{g}_{\mathbb{C}}$ across $\mathfrak{g}_0$; it is antilinear, satisfies $\sigma_0^2 = \mathrm{id}$, and its fixed set is exactly $\mathfrak{g}_0$.

The bracket formula is forced: once the bracket is $\mathbb{R}$-bilinear on $\mathfrak{g}_0$, the only $\mathbb{C}$-bilinear extension to $\mathfrak{g}_{\mathbb{C}}$ is the one given, and the Jacobi identity for $\mathfrak{g}_{\mathbb{C}}$ follows from that of $\mathfrak{g}_0$ by $\mathbb{C}$-multilinearity, so $\mathfrak{g}_{\mathbb{C}}$ is a complex Lie algebra. The natural conjugation records how $\mathfrak{g}_0$ sits inside $\mathfrak{g}_{\mathbb{C}}$: it is complex conjugation on the second tensor factor, $\sigma_0(X \otimes z) = X \otimes \bar{z}$, and its fixed set $\{X + iY \,|\, \sigma_0(X + iY) = X + iY\}$ is the set of elements with $Y = 0$, namely $\mathfrak{g}_0$. This conjugation, an antilinear symmetry of $\mathfrak{g}_{\mathbb{C}}$, carries exactly the information of the real form and can be reconstructed from it.

The complexification of the running example is the algebra whose structure theory Parts II and III computed.

Example 13.1.2: The Complexification of $\mathfrak{su}(n)$

Let $\mathfrak{g}_0 = \mathfrak{su}(n) = \{X \in \mathfrak{gl}_n(\mathbb{C}) \mid X^* = -X,\ \mathrm{tr}X = 0\}$, the traceless skew-Hermitian matrices, regarded as a real Lie algebra (it is a real, not complex, subalgebra of $\mathfrak{gl}_n(\mathbb{C})$, since multiplying a skew-Hermitian matrix by i makes it Hermitian). Its complexification is $\mathfrak{sl}_n(\mathbb{C})$. Indeed, given a traceless complex matrix Z, split it into skew-Hermitian and Hermitian parts,

$$Z = \tfrac{1}{2}(Z - Z^*) + \tfrac{1}{2}(Z + Z^*) = X + iY, \qquad X = \tfrac{1}{2}(Z - Z^*), \quad Y = -\tfrac{i}{2}(Z + Z^*)$$

where X is skew-Hermitian and $Y = -\frac{i}{2}(Z + Z^*)$ is skew-Hermitian as well (the Hermitian matrix $Z + Z^*$ times $-i$ is skew-Hermitian), both traceless because $\mathrm{tr}Z = 0$ and $\mathrm{tr}(Z^*) = \overline{\mathrm{tr}Z} = 0$. Thus $Z = X + iY$ with $X, Y \in \mathfrak{su}(n)$, and the splitting is unique since a matrix that is both skew-Hermitian and Hermitian is zero. Hence $\mathfrak{sl}_n(\mathbb{C}) = \mathfrak{su}(n) \oplus i\,\mathfrak{su}(n)$, so $\mathfrak{sl}_n(\mathbb{C})$ is the complexification of $\mathfrak{su}(n)$, and the natural conjugation is $\sigma_0(Z) = X - iY = \frac{1}{2}(Z - Z^*) - \frac{1}{2}(Z + Z^*) = -Z^*$, the negative conjugate-transpose. This is the analytic origin of the compact real form $\mathfrak{su}(n)$ of definition 9.4.3; the dimension count $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1 = \dim_{\mathbb{C}} \mathfrak{sl}_n(\mathbb{C})$ confirms the equality of dimensions.

The reverse construction goes from complex to real. Rather than start with a real algebra and enlarge, one starts with a complex algebra $\mathfrak{g}_{\mathbb{C}}$ and asks which real subalgebras have it as their complexification. Such a subalgebra is a *real form*, and the point of the definition is that it comes packaged with an antilinear symmetry that pins it down.

Definition 13.1.3: Real Form and Conjugation

Let $\mathfrak{g}_{\mathbb{C}}$ be a complex Lie algebra. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g}_0 \subseteq \mathfrak{g}_{\mathbb{C}}$ such that

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus i\,\mathfrak{g}_0$$

as real vector spaces, so that $\mathfrak{g}_{\mathbb{C}} \cong \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$ is the complexification of $\mathfrak{g}_0$ (definition 13.1.1). A *conjugation* of $\mathfrak{g}_{\mathbb{C}}$ is a map $\sigma\colon \mathfrak{g}_{\mathbb{C}} \to \mathfrak{g}_{\mathbb{C}}$ that is

1. *antilinear:* $\sigma(cX) = \bar{c}\,\sigma(X)$ and $\sigma(X + Y) = \sigma(X) + \sigma(Y)$ for $c \in \mathbb{C}$, $X, Y \in \mathfrak{g}_{\mathbb{C}}$;
2. *involutive:* $\sigma^2 = \mathrm{id}$;
3. a *Lie algebra homomorphism:* $\sigma([X, Y]) = [\sigma(X), \sigma(Y)]$.

The *conjugation associated to the real form* $\mathfrak{g}_0$ is the map $\sigma(X + iY) = X - iY$ $(X, Y \in \mathfrak{g}_0)$, whose fixed set $\mathfrak{g}_{\mathbb{C}}^{\sigma} = \{Z \in \mathfrak{g}_{\mathbb{C}} \big| \sigma(Z) = Z\}$ is $\mathfrak{g}_0$.

A conjugation is the algebraic abstraction of complex conjugation: antilinear and involutive like $z \mapsto \bar{z}$ on $\mathbb{C}$, and bracket-preserving so that it is an automorphism of the underlying real Lie algebra (it is not $\mathbb{C}$-linear, so not an automorphism of $\mathfrak{g}_{\mathbb{C}}$ as a complex algebra). The definition packages two objects, a real form $\mathfrak{g}_0$ and its conjugation σ, and asserts nothing yet about how tightly they are linked; the theorem below shows they determine each other. A real form of a complex *semisimple* $\mathfrak{g}_{\mathbb{C}}$ recovers exactly the notion used in definition 9.4.3, where the real form was required additionally to have negative-definite Killing form to be *compact*; here the algebra $\mathfrak{g}_{\mathbb{C}}$ need not be semisimple and the real form carries no positivity condition.

The equivalence of the two data is what lets the classification of real forms proceed by classifying

conjugations.

Theorem 13.1.4: Real Forms and Conjugations

Let $\mathfrak{g}_{\mathbb{C}}$ be a complex Lie algebra. The assignments

$$\mathfrak{g}_0 \longmapsto \sigma \quad \text{(the conjugation across } \mathfrak{g}_0\text{)}, \qquad \sigma \longmapsto \mathfrak{g}_{\mathbb{C}}^{\sigma} \quad \text{(the fixed set)}$$

are mutually inverse bijections between the real forms of $\mathfrak{g}_{\mathbb{C}}$ and the conjugations of $\mathfrak{g}_{\mathbb{C}}$.

***Proof*:**
The proof runs in two directions and closes the loop by checking the two composites are identities.

Step 1: a real form gives a conjugation. Let $\mathfrak{g}_0$ be a real form, so $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus i\,\mathfrak{g}_0$ as real vector spaces, and define $\sigma(X + iY) = X - iY$ for $X, Y \in \mathfrak{g}_0$. This is well defined because the decomposition $Z = X + iY$ is unique: $\mathfrak{g}_0 \cap i\,\mathfrak{g}_0 = 0$ forces the real and imaginary parts of Z to be determined. The map is $\mathbb{R}$-linear and satisfies $\sigma(iZ) = -i\,\sigma(Z)$: writing $Z = X + iY$, one has $iZ = -Y + iX$, so $\sigma(iZ) = -Y - iX = -i(X - iY) = -i\,\sigma(Z)$, and combined with $\mathbb{R}$-linearity this gives antilinearity $\sigma(cZ) = \bar{c}\,\sigma(Z)$ for all $c \in \mathbb{C}$. It is involutive since $\sigma^2(X + iY) = \sigma(X - iY) = X + iY$. It preserves the bracket: for $X_1 + iY_1$ and $X_2 + iY_2$ with all $X_j, Y_j \in \mathfrak{g}_0$, expanding by $\mathbb{C}$-bilinearity of the bracket of $\mathfrak{g}_{\mathbb{C}}$ and using that $[\mathfrak{g}_0, \mathfrak{g}_0] \subseteq \mathfrak{g}_0$ (so $\mathfrak{g}_0$ is a real subalgebra) gives

$$[X_1 + iY_1,\, X_2 + iY_2] = \big([X_1, X_2] - [Y_1, Y_2]\big) + i\big([X_1, Y_2] + [Y_1, X_2]\big)$$

whose real part lies in $\mathfrak{g}_0$ and imaginary part in $\mathfrak{g}_0$; applying σ negates the imaginary part, and the identical expansion of $[\sigma(X_1 + iY_1), \sigma(X_2 + iY_2)] = [X_1 - iY_1, X_2 - iY_2]$ produces the same real part and the negated imaginary part, so $\sigma[Z_1, Z_2] = [\sigma Z_1, \sigma Z_2]$. Thus σ is a conjugation, and by construction its fixed set is $\{X + iY \,\big|\, Y = 0\} = \mathfrak{g}_0$.

Step 2: a conjugation gives a real form. Let σ be a conjugation and set $\mathfrak{g}_0 = \mathfrak{g}_{\mathbb{C}}^{\sigma}$, the fixed set. It is a real subspace, being the kernel of the $\mathbb{R}$-linear map $\sigma - \mathrm{id}$. It is closed under the bracket: if $\sigma(X) = X$ and $\sigma(Y) = Y$ then $\sigma[X, Y] = [\sigma X, \sigma Y] = [X, Y]$, so $[X, Y] \in \mathfrak{g}_0$, making $\mathfrak{g}_0$ a real Lie subalgebra. To see $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus i\,\mathfrak{g}_0$, use the two projections

$$P_+(Z) = \tfrac{1}{2}\big(Z + \sigma(Z)\big), \qquad P_-(Z) = \tfrac{1}{2i}\big(Z - \sigma(Z)\big)$$

Antilinearity gives $\sigma(P_+ Z) = \frac{1}{2}(\sigma Z + Z) = P_+ Z$, so $P_+(Z) \in \mathfrak{g}_0$; and $\sigma(P_- Z) = \sigma\big(\frac{1}{2i}(Z - \sigma Z)\big) = \overline{\big(\frac{1}{2i}\big)}(\sigma Z - Z) = -\frac{1}{2i}(\sigma Z - Z) = P_-(Z)$, so $P_-(Z) \in \mathfrak{g}_0$ too. Then

$$Z = P_+(Z) + i\,P_-(Z)$$

exhibits Z as an element of $\mathfrak{g}_0 + i\,\mathfrak{g}_0$, so the sum is all of $\mathfrak{g}_{\mathbb{C}}$. The sum is direct: if $W \in \mathfrak{g}_0 \cap i\,\mathfrak{g}_0$, write $W = iU$ with $U \in \mathfrak{g}_0$; then $\sigma(W) = W$ (since $W \in \mathfrak{g}_0$) and $\sigma(W) = \sigma(iU) = -i\,\sigma(U) = -iU = -W$ (since $U \in \mathfrak{g}_0$), forcing $W = -W$, so $W = 0$. Hence $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus i\,\mathfrak{g}_0$ and $\mathfrak{g}_0 = \mathfrak{g}_{\mathbb{C}}^{\sigma}$ is a real form.

Step 3: the composites are the identity. Starting from a real form $\mathfrak{g}_0$, Step 1 produces the conjugation σ across $\mathfrak{g}_0$, and its fixed set is $\mathfrak{g}_0$ again, so $\sigma \mapsto \mathfrak{g}_{\mathbb{C}}^{\sigma}$ undoes $\mathfrak{g}_0 \mapsto \sigma$. Starting from a conjugation σ, Step 2 produces the real form $\mathfrak{g}_0 = \mathfrak{g}_{\mathbb{C}}^{\sigma}$, and the conjugation across it sends $P_+(Z) + i\,P_-(Z) \mapsto P_+(Z) - i\,P_-(Z)$, which equals $\sigma(Z)$ because $\sigma(Z) = P_+(Z) - i\,P_-(Z)$ by the very definition of the projections; so $\mathfrak{g}_0 \mapsto \sigma$ undoes $\sigma \mapsto \mathfrak{g}_{\mathbb{C}}^{\sigma}$. The two assignments are therefore mutually inverse bijections, as we sought to show.

The theorem trades a subspace for a symmetry. A real form is a choice of real subalgebra spanning $\mathfrak{g}_{\mathbb{C}}$ over $\mathbb{C}$, which looks like extra structure imposed on $\mathfrak{g}_{\mathbb{C}}$; the theorem says it is nothing more than an antilinear involutive automorphism of $\mathfrak{g}_{\mathbb{C}}$, a symmetry that $\mathfrak{g}_{\mathbb{C}}$ carries within itself. The dictionary is exact: the real form is the fixed set of the symmetry, and the symmetry is reflection across the real form. This is the reformulation that governs the rest of the chapter. Classifying the real forms of a fixed $\mathfrak{g}_{\mathbb{C}}$ becomes classifying its conjugations, and since two conjugations conjugate by an automorphism of $\mathfrak{g}_{\mathbb{C}}$ produce isomorphic real forms, the classification is really of conjugations up to conjugacy in $\mathrm{Aut}(\mathfrak{g}_{\mathbb{C}})$. The distinguished conjugation the chapter singles out is the one whose fixed set is the compact real form; measuring an arbitrary conjugation against it is what produces the Cartan decomposition in section 13.3.

The construction is concrete for the classical case, where the two basic real forms of $\mathfrak{sl}_n(\mathbb{C})$ arise from two visibly different conjugations.

Example 13.1.5: The Two Real Forms $\mathfrak{sl}_n(\mathbb{R})$ and $\mathfrak{su}(n)$ of $\mathfrak{sl}_n(\mathbb{C})$

Take $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}_n(\mathbb{C})$, the traceless complex $n \times n$ matrices.

The *split* conjugation is entrywise complex conjugation,

$$\sigma(X) = \bar{X}, \qquad (\bar{X})_{jk} = \overline{X_{jk}}$$

It is antilinear because $\overline{cX} = \bar{c}\,\bar{X}$, involutive because $\overline{\bar{X}} = X$, and a bracket homomorphism because $\overline{XY} = \bar{X}\,\bar{Y}$ gives $\overline{[X,Y]} = [\bar{X},\bar{Y}]$; it preserves tracelessness since $\mathrm{tr}\bar{X} = \overline{\mathrm{tr}X} = 0$. Its fixed set is the traceless *real* matrices,

$$\mathfrak{sl}_n(\mathbb{C})^{\sigma} = \left\{X \in \mathfrak{sl}_n(\mathbb{C}) \;\middle|\; \bar{X} = X\right\} = \mathfrak{sl}_n(\mathbb{R})$$

so by theorem 13.1.4 the real form attached to entrywise conjugation is $\mathfrak{sl}_n(\mathbb{R})$. Explicitly $\mathfrak{sl}_n(\mathbb{C}) = \mathfrak{sl}_n(\mathbb{R}) \oplus i\,\mathfrak{sl}_n(\mathbb{R})$, splitting a traceless complex matrix into its real and imaginary parts.

The *compact* conjugation is the negative conjugate-transpose,

$$\tau(X) = -X^* = -\bar{X}^{\mathsf{T}}$$

It is antilinear because $(cX)^* = \bar{c}\,X^*$ so $\tau(cX) = \bar{c}\,\tau(X)$; involutive because $\tau^2(X) = -(-X^*)^* = X$; and a bracket homomorphism because $(XY)^* = Y^*X^*$ gives $[X,Y]^* = [Y^*,X^*] = -[X^*,Y^*]$, hence $\tau[X,Y] = -[X,Y]^* = [X^*,Y^*] = [-X^*,-Y^*] = [\tau X,\tau Y]$; and it preserves tracelessness. Its fixed set is

$$\mathfrak{sl}_n(\mathbb{C})^{\tau} = \{X \in \mathfrak{sl}_n(\mathbb{C}) \mid -X^* = X\} = \{X \mid X^* = -X,\ \mathrm{tr}X = 0\} = \mathfrak{su}(n)$$

the traceless skew-Hermitian matrices, so the real form attached to negative conjugate-transpose is $\mathfrak{su}(n)$, the compact real form of definition 9.4.3 (the compact groups $\mathrm{SU}(n)$ of chapter 9). The decomposition $\mathfrak{sl}_n(\mathbb{C}) = \mathfrak{su}(n) \oplus i\,\mathfrak{su}(n)$ is that of example 13.1.2.

The two conjugations σ and τ are different maps, and they fix different subalgebras. For $n = 2$ they specialize to the running dichotomy: $\mathfrak{sl}_2(\mathbb{R})$, the real traceless 2×2 matrices, is the noncompact real form of $\mathfrak{sl}_2(\mathbb{C})$, while $\mathfrak{su}(2)$, the traceless skew-Hermitian matrices, is the compact real form, the Lie algebra of the running example $\mathrm{SU}(2)$. Both complexify to the same $\mathfrak{sl}_2(\mathbb{C})$, but $\mathfrak{sl}_2(\mathbb{R})$ has an indefinite Killing form (a Cartan decomposition with a nontrivial $\mathfrak{p}$, taken up in section 13.3) whereas $\mathfrak{su}(2)$ has negative-definite Killing form. The difference between the two conjugations is precisely the difference between the noncompact group $\mathrm{SL}(2,\mathbb{R})$ and the compact group $\mathrm{SU}(2)$.

A real form is thus not extra data adjoined to $\mathfrak{g}_{\mathbb{C}}$ from outside, but a symmetry that $\mathfrak{g}_{\mathbb{C}}$ already possesses. The complex algebra $\mathfrak{sl}_n(\mathbb{C})$ knows about both $\mathfrak{sl}_n(\mathbb{R})$ and $\mathfrak{su}(n)$ because it carries both conjugations σ and τ; choosing a real form is choosing which conjugation to remember. The program of the chapter is to organize these choices. Among all conjugations of a complex semisimple $\mathfrak{g}_{\mathbb{C}}$, one class fixes a compact real form, and the existence and uniqueness of that class is the content of section 13.2; comparing a given real form's conjugation with the compact one, so that the two commute, is what section 13.3 does to split a noncompact algebra into its maximal compact part and a Euclidean complement. The single antilinear involution introduced here is the seed of every decomposition in Part IV.

Exercise 13.1.1

1. Verify in full that entrywise complex conjugation $\sigma(X) = \bar{X}$ is a conjugation of $\mathfrak{sl}_n(\mathbb{C})$ with fixed set $\mathfrak{sl}_n(\mathbb{R})$. Check antilinearity, $\sigma^2 = \mathrm{id}$, the bracket identity $\sigma[X,Y] = [\sigma X, \sigma Y]$, and preservation of tracelessness, then identify $\mathfrak{sl}_n(\mathbb{C})^{\sigma}$ and exhibit the direct-sum decomposition $\mathfrak{sl}_n(\mathbb{C}) = \mathfrak{sl}_n(\mathbb{R}) \oplus i\,\mathfrak{sl}_n(\mathbb{R})$ explicitly.

2. Verify in full that $\tau(X) = -X^*$ is a conjugation of $\mathfrak{sl}_n(\mathbb{C})$ with fixed set $\mathfrak{su}(n)$. In particular confirm the bracket identity $\tau[X,Y] = [\tau X, \tau Y]$ using $(XY)^* = Y^*X^*$, and confirm that τ preserves tracelessness. Conclude that $\mathfrak{su}(n)$ is a real form of $\mathfrak{sl}_n(\mathbb{C})$, and compute $\dim_{\mathbb{R}} \mathfrak{su}(n)$ to check it against $\dim_{\mathbb{C}} \mathfrak{sl}_n(\mathbb{C})$.

3. Exhibit a real form of $\mathfrak{sl}_2(\mathbb{C})$ different from both $\mathfrak{sl}_2(\mathbb{R})$ and $\mathfrak{su}(2)$ by writing down its conjugation, or explain why every real form of $\mathfrak{sl}_2(\mathbb{C})$ is isomorphic to one of these two. (Hint: a real form of $\mathfrak{sl}_2(\mathbb{C})$ is a three-dimensional real Lie algebra whose complexification is $\mathfrak{sl}_2(\mathbb{C})$; over $\mathbb{R}$ the three-dimensional simple Lie algebras are $\mathfrak{sl}_2(\mathbb{R})$ and $\mathfrak{su}(2)$, with $\mathfrak{sl}_2(\mathbb{R}) \cong \mathfrak{so}(2,1)$ and $\mathfrak{su}(2) \cong \mathfrak{so}(3)$.)

4. Let σ and σ' be two conjugations of a complex Lie algebra $\mathfrak{g}_{\mathbb{C}}$. Show that $\varphi = \sigma \circ \sigma'$ is a $\mathbb{C}$-linear automorphism of $\mathfrak{g}_{\mathbb{C}}$ (in particular $\varphi \in \mathrm{Aut}(\mathfrak{g}_{\mathbb{C}})$), so that any two conjugations differ by an automorphism, $\sigma = \varphi \circ \sigma'$ with $\varphi \in \mathrm{Aut}(\mathfrak{g}_{\mathbb{C}})$; hence the conjugations lie inside the coset $\mathrm{Aut}(\mathfrak{g}_{\mathbb{C}}) \cdot \sigma'$ once one conjugation σ' exists. Show conversely that for $\varphi \in \mathrm{Aut}(\mathfrak{g}_{\mathbb{C}})$ the composite $\varphi \circ \sigma'$ is antilinear and bracket-preserving, and is a conjugation if and only if $\sigma' \varphi \sigma' = \varphi^{-1}$ (equivalently φ commutes with σ'-conjugation in the way that makes $(\varphi\sigma')^2 = \mathrm{id}$), so the conjugations are exactly the involutive members of the coset, a proper subset in general.

5. Confirm directly that the complexification of $\mathfrak{su}(2)$ is $\mathfrak{sl}_2(\mathbb{C})$. Using the basis $H = \left(\begin{smallmatrix} i & 0 \\ 0 & -i \end{smallmatrix}\right)$, $E_1 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $E_2 = \left(\begin{smallmatrix} 0 & i \\ i & 0 \end{smallmatrix}\right)$ of $\mathfrak{su}(2)$, show that H, E_1, E_2 are $\mathbb{C}$-linearly independent in $\mathfrak{gl}_2(\mathbb{C})$ and $\mathbb{C}$-span $\mathfrak{sl}_2(\mathbb{C})$, so that $\mathfrak{su}(2) \otimes_{\mathbb{R}} \mathbb{C} = \mathfrak{sl}_2(\mathbb{C})$, and write the natural conjugation σ_0 of $\mathfrak{sl}_2(\mathbb{C})$ across $\mathfrak{su}(2)$ in matrix terms.

6. Prove, purely from the axioms of a conjugation, that the fixed set $\mathfrak{g}_{\mathbb{C}}^{\sigma}$ of any conjugation σ is a real Lie subalgebra of $\mathfrak{g}_{\mathbb{C}}$, and that $i\,\mathfrak{g}_{\mathbb{C}}^{\sigma}$ is the -1-eigenspace of σ (the set of Z with $\sigma(Z) = -Z$). Conclude that σ is determined by its $+1$-eigenspace alone.

13.2 Compact Real Forms: Existence and Uniqueness

Among the real forms of a complex semisimple Lie algebra assembled in section 13.1, one stands apart. The classification of theorem 13.1.4 presents the real forms of $\mathfrak{g}_{\mathbb{C}}$ as an unstructured list,

indexed by the conjugations of $\mathfrak{g}_{\mathbb{C}}$; nothing yet distinguishes $\mathfrak{su}(n)$ from $\mathfrak{sl}_n(\mathbb{R})$ inside the family of real forms of $\mathfrak{sl}_n(\mathbb{C})$. The distinguishing datum is the sign of the Killing form. A real form on which κ is negative-definite is the Lie algebra of a compact group (proposition 9.4.1), and definition 9.4.3 named such a form the *compact real form*. That definition, made in chapter 9, came with a promise: every complex semisimple $\mathfrak{g}_{\mathbb{C}}$ carries exactly one compact real form up to conjugacy. This section discharges it.

Existence is a construction. The root space decomposition of $\mathfrak{g}_{\mathbb{C}}$ (theorem 7.2.4) exhibits $\mathfrak{g}_{\mathbb{C}}$ as a Cartan subalgebra with root spaces attached, and the $\mathfrak{sl}_2$-triples of theorem 7.4.6 let one normalize the root vectors so that all structure constants become real. Rotating those real generators by i in the correct pattern produces a real subalgebra on which κ is negative-definite: the compact real form. Uniqueness is a conjugacy argument: any two compact real forms give two conjugations of $\mathfrak{g}_{\mathbb{C}}$ whose composition is a positive automorphism, and its real powers conjugate one form onto the other. Together they make "the compact real form" a well-defined object attached to $\mathfrak{g}_{\mathbb{C}}$, the fixed reference against which every other real form is measured in the sections that follow.

The construction is cleanest to see on $\mathfrak{sl}_2(\mathbb{C})$ before it is stated in general, so the smallest case comes first.

Example 13.2.1: The Compact Real Form of $\mathfrak{sl}_2(\mathbb{C})$

The complex Lie algebra $\mathfrak{sl}_2(\mathbb{C})$ has the standard basis

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

with brackets $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$ (the $\mathfrak{sl}_2$-triple of theorem 7.4.6). There is one positive root α, with $E \in \mathfrak{g}_\alpha$, $F \in \mathfrak{g}_{-\alpha}$, and $H = H_\alpha$ the coroot. The real span $\mathbb{R}H \oplus \mathbb{R}E \oplus \mathbb{R}F$ is the split real form $\mathfrak{sl}_2(\mathbb{R})$, on which κ is indefinite. Rotating instead by i in the pattern the general construction will prescribe, set

$$u_1 = iH, \qquad u_2 = E - F, \qquad u_3 = i(E + F)$$

Explicitly these are

$$u_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad u_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad u_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

each of which is skew-Hermitian and traceless, so $\mathfrak{u} = \mathbb{R}u_1 \oplus \mathbb{R}u_2 \oplus \mathbb{R}u_3 = \mathfrak{su}(2)$. A direct computation of the brackets gives

$$[u_1, u_2] = 2u_3, \qquad [u_2, u_3] = 2u_1, \qquad [u_3, u_1] = 2u_2$$

so $\mathfrak{u}$ closes under the bracket; it is the Lie algebra of SU(2), whose Killing form $\kappa(X, Y) = 4\mathrm{tr}(XY)$ is negative-definite (proposition 9.4.1, since SU(2) is compact semisimple). Meanwhile $\mathfrak{g}_{\mathbb{C}} = \mathfrak{u} \oplus i\mathfrak{u}$: the six matrices $u_1, u_2, u_3, iu_1, iu_2, iu_3$ span $\mathfrak{sl}_2(\mathbb{C})$ over $\mathbb{R}$. Thus $\mathfrak{su}(2)$ is a compact real form of $\mathfrak{sl}_2(\mathbb{C})$, the compact sibling of the split form $\mathfrak{sl}_2(\mathbb{R})$.

The rotation $H \mapsto iH$, $E - F$, $i(E + F)$ is the pattern of the general construction: the coroot directions H_α are multiplied by i, and each pair $(X_\alpha, X_{-\alpha})$ of opposite root vectors contributes the real combination $X_\alpha - X_{-\alpha}$ and the imaginary combination $i(X_\alpha + X_{-\alpha})$. Everything hinges on choosing the root vectors so that the resulting real span closes under the bracket, and that is

a statement about the structure constants $N_{\alpha,\beta}$ defined by $[X_\alpha, X_\beta] = N_{\alpha,\beta} X_{\alpha+\beta}$. The following normalization, due to Chevalley, makes them real and skew in a sense that forces closure.

Lemma 13.2.2: Chevalley Normalization of the Root Vectors

Let $\mathfrak{g}_{\mathbb{C}}$ be a complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$, root system Φ, and root space decomposition $\mathfrak{g}_{\mathbb{C}} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$ (theorem 7.2.4). For each root α let $H_\alpha \in \mathfrak{h}$ be the coroot (definition 7.4.3). Then one can choose root vectors $X_\alpha \in \mathfrak{g}_\alpha$, one for each $\alpha \in \Phi$, such that

1. $[X_\alpha, X_{-\alpha}] = H_\alpha$ and $\kappa(X_\alpha, X_{-\alpha}) = \dfrac{\kappa(H_\alpha, H_\alpha)}{2}$ for every α;
2. $[H, X_\alpha] = \alpha(H)\, X_\alpha$ for every $H \in \mathfrak{h}$;
3. the structure constants $N_{\alpha,\beta}$ defined by $[X_\alpha, X_\beta] = N_{\alpha,\beta}\, X_{\alpha+\beta}$ (when $\alpha+\beta \in \Phi$, and $N_{\alpha,\beta} = 0$ otherwise) are all *real*, and satisfy the sign relation

$$N_{-\alpha,-\beta} = -N_{\alpha,\beta}$$

***Proof*:**

Property (2) is the definition of a root vector (theorem 7.2.4), and $\dim \mathfrak{g}_\alpha = 1$ (theorem 7.4.6) means X_α is determined up to a nonzero complex scalar. The proof normalizes these scalars in two stages: first to arrange (1), then to arrange the reality and skew relation in (3).

Step 1: normalizing so that $[X_\alpha, X_{-\alpha}] = H_\alpha$. Fix any nonzero $Y_\alpha \in \mathfrak{g}_\alpha$ and $Y_{-\alpha} \in \mathfrak{g}_{-\alpha}$. Since κ is nondegenerate and pairs $\mathfrak{g}_\alpha$ with $\mathfrak{g}_{-\alpha}$ (theorem 7.2.4), $\kappa(Y_\alpha, Y_{-\alpha}) \neq 0$, and the invariance of κ gives $[Y_\alpha, Y_{-\alpha}] = \kappa(Y_\alpha, Y_{-\alpha})\, t_\alpha$, where $t_\alpha \in \mathfrak{h}$ is the element representing α under κ. The coroot is $H_\alpha = 2t_\alpha/\kappa(t_\alpha, t_\alpha)$ (definition 7.4.3). Rescaling Y_α by a complex constant to X_α and setting $X_{-\alpha}$ a matching multiple of $Y_{-\alpha}$, the single scalar of freedom (one rescaling, with $X_{-\alpha}$ then pinned by the pairing) can be spent to arrange $[X_\alpha, X_{-\alpha}] = H_\alpha$; this in turn determines $\kappa(X_\alpha, X_{-\alpha}) = \kappa(H_\alpha, H_\alpha)/2$, since $[X_\alpha, X_{-\alpha}] = \kappa(X_\alpha, X_{-\alpha})\, t_\alpha$ and $H_\alpha = 2t_\alpha/\kappa(t_\alpha, t_\alpha)$ force $\kappa(X_\alpha, X_{-\alpha}) = 2/\kappa(t_\alpha, t_\alpha) = \kappa(H_\alpha, H_\alpha)/2$. This is exactly the $\mathfrak{sl}_2$-triple normalization of theorem 7.4.6, in which $\{X_\alpha, X_{-\alpha}, H_\alpha\}$ spans a copy of $\mathfrak{sl}_2(\mathbb{C})$. This fixes each pair $X_\alpha, X_{-\alpha}$ up to a common sign ambiguity that Step 2 exploits.

Step 2: reality and the skew relation. It remains to see that the X_α can be chosen, compatibly with the normalization (1), so that every $N_{\alpha,\beta}$ is real. This is the Chevalley basis theorem. By theorem 8.8.2 applied to $\mathfrak{g}_{\mathbb{C}}$, there is a system of root vectors satisfying (1) whose structure constants are all real, indeed integers $N_{\alpha,\beta} = \pm(p+1)$ for the α-string $\beta - p\alpha, \ldots, \beta + q\alpha$ through β (theorem 7.4.6), and which satisfy $N_{-\alpha,-\beta} = -N_{\alpha,\beta}$. Taking the X_α to be such a Chevalley basis realizes (1) and (3) together, the skew relation $N_{-\alpha,\beta} = -N_{\alpha,-\beta}$ being that sign relation with $-\alpha$ in place of α. Property (2) restates that each X_α lies in $\mathfrak{g}_\alpha$, as we sought to show.

The reality of the $N_{\alpha,\beta}$ is the whole point. Once the structure constants are real, the real vector space spanned by the rotated generators is closed under the bracket, because the brackets of those generators are real combinations of generators of the same shape. The sign relation $N_{-\alpha,-\beta} = -N_{\alpha,\beta}$ is what pairs the two contributions of $\pm\alpha$ correctly, so that a bracket of two elements of the candidate real form lands back inside it rather than picking up an unwanted factor of i. The next theorem assembles the rotated generators and verifies both the closure and the sign of the Killing form.

Theorem 13.2.3: Existence of a Compact Real Form

Every complex semisimple Lie algebra $\mathfrak{g}_{\mathbb{C}}$ has a compact real form (definition 9.4.3): a real form $\mathfrak{u}$ with $\kappa|_{\mathfrak{u}}$ negative-definite. Explicitly, with root vectors X_α normalized as in lemma 13.2.2,

$$\mathfrak{u} = \Big(\sum_{\alpha\in\Phi} \mathbb{R}\, iH_\alpha\Big) \oplus \sum_{\alpha>0} \Big(\mathbb{R}\,(X_\alpha - X_{-\alpha}) \oplus \mathbb{R}\, i(X_\alpha + X_{-\alpha})\Big)$$

is a real subalgebra of $\mathfrak{g}_{\mathbb{C}}$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{u} \oplus i\mathfrak{u}$ and $\kappa|_{\mathfrak{u}}$ negative-definite.

***Proof*:**
Write $\mathfrak{t}_0 = \sum_\alpha \mathbb{R}\, iH_\alpha$ for the first summand and, for each positive root α, set

$$A_\alpha = X_\alpha - X_{-\alpha}, \qquad B_\alpha = i(X_\alpha + X_{-\alpha})$$

so that $\mathfrak{u} = \mathfrak{t}_0 \oplus \bigoplus_{\alpha>0}(\mathbb{R}A_\alpha \oplus \mathbb{R}B_\alpha)$. The proof checks in turn that $\mathfrak{u}$ is a real subalgebra, that $\mathfrak{g}_{\mathbb{C}} = \mathfrak{u} \oplus i\mathfrak{u}$, and that $\kappa|_{\mathfrak{u}}$ is negative-definite.

Step 1: $\mathfrak{u}$ is a real form as a vector space. The elements iH_α (spanning $\mathfrak{t}_0$), together with the A_α and B_α for $\alpha > 0$, are $\mathbb{R}$-linearly independent: the iH_α lie in $i\mathfrak{h}$, while A_α, B_α have components only in $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$, and the root space decomposition is a direct sum. Counting real dimensions, $\dim_{\mathbb{R}} \mathfrak{t}_0 = \dim_{\mathbb{C}} \mathfrak{h} = \ell$ (the rank, since the H_α span $\mathfrak{h}$), and each positive root contributes 2, for a total of $\ell + 2\cdot|\Phi^+| = \ell + |\Phi| = \dim_{\mathbb{C}} \mathfrak{g}_{\mathbb{C}}$. To see $\mathfrak{g}_{\mathbb{C}} = \mathfrak{u} \oplus i\mathfrak{u}$, note $iA_\alpha = i(X_\alpha - X_{-\alpha})$ and $iB_\alpha = -(X_\alpha + X_{-\alpha})$, so $\mathbb{R}$-combinations of $A_\alpha, B_\alpha, iA_\alpha, iB_\alpha$ recover X_α and $X_{-\alpha}$, while $H_\alpha = -i(iH_\alpha)$ lies in $i\mathfrak{t}_0$; hence $\mathfrak{u} + i\mathfrak{u}$ contains a full $\mathbb{C}$-basis and the real dimension count $2\dim_{\mathbb{C}} \mathfrak{g}_{\mathbb{C}}$ forces the sum to be direct.

Step 2: $\mathfrak{u}$ is closed under the bracket. There are three types of bracket to check, using the reality of the $N_{\alpha,\beta}$ from lemma 13.2.2. Throughout, $\alpha(H_\beta) \in \mathbb{Z}$ is a Cartan integer (theorem 7.4.6), hence real.

For the toral part, $\mathfrak{t}_0$ is abelian since $\mathfrak{h}$ is, so $[\mathfrak{t}_0, \mathfrak{t}_0] = 0$. For a toral generator against a root pair, use $[H_\beta, X_{\pm\alpha}] = \pm\alpha(H_\beta)X_{\pm\alpha}$:

$$[iH_\beta, A_\alpha] = i\,\alpha(H_\beta)(X_\alpha + X_{-\alpha}) = \alpha(H_\beta)\, B_\alpha \in \mathfrak{u}$$

and similarly $[iH_\beta, B_\alpha] = i\cdot i\alpha(H_\beta)(X_\alpha - X_{-\alpha}) = -\alpha(H_\beta)\, A_\alpha \in \mathfrak{u}$, both with real coefficients.

For two root pairs $\alpha, \beta > 0$ with $\alpha \neq \beta$, expand

$$[A_\alpha, A_\beta] = [X_\alpha - X_{-\alpha},\ X_\beta - X_{-\beta}] = N_{\alpha,\beta}X_{\alpha+\beta} - N_{\alpha,-\beta}X_{\alpha-\beta} - N_{-\alpha,\beta}X_{-\alpha+\beta} + N_{-\alpha,-\beta}X_{-\alpha-\beta}$$

Using $N_{-\alpha,-\beta} = -N_{\alpha,\beta}$ and $N_{-\alpha,\beta} = -N_{\alpha,-\beta}$ (the skew relation, lemma 13.2.2), the four terms group into

$$[A_\alpha, A_\beta] = N_{\alpha,\beta}\,(X_{\alpha+\beta} - X_{-(\alpha+\beta)}) - N_{\alpha,-\beta}\,(X_{\alpha-\beta} - X_{-(\alpha-\beta)})$$

which is $N_{\alpha,\beta}A_{\alpha+\beta} - N_{\alpha,-\beta}A_{\alpha-\beta}$ under the standing convention $A_\gamma = X_\gamma - X_{-\gamma}$ for all γ (so $A_{-\gamma} = -A_\gamma$, and $B_\gamma = i(X_\gamma + X_{-\gamma})$ with $B_{-\gamma} = B_\gamma$), a real combination of the generators, with the roots that fail to lie in Φ contributing zero. The mixed and B-brackets are identical in form:

$$[A_\alpha, B_\beta] = N_{\alpha,\beta}\, B_{\alpha+\beta} + N_{\alpha,-\beta}\, B_{\alpha-\beta}, \qquad [B_\alpha, B_\beta] = -N_{\alpha,\beta}\, A_{\alpha+\beta} - N_{\alpha,-\beta}\, A_{\alpha-\beta}$$

again real combinations of generators of $\mathfrak{u}$, the extra $i^2 = -1$ in $[B_\alpha, B_\beta]$ producing the sign that keeps the result inside $\mathfrak{u}$.

Finally, for $\alpha = \beta$ the diagonal bracket lands in the torus,

$$[A_\alpha, B_\alpha] = [X_\alpha - X_{-\alpha},\ i(X_\alpha + X_{-\alpha})] = i\big([X_\alpha, X_{-\alpha}] - [X_{-\alpha}, X_\alpha]\big) = 2i\,[X_\alpha, X_{-\alpha}] = 2\,iH_\alpha \in \mathfrak{t}_0$$

the cross terms $[X_\alpha, X_\alpha] = [X_{-\alpha}, X_{-\alpha}] = 0$ vanishing. Every bracket of generators is a real combination of generators, so $\mathfrak{u}$ is a real subalgebra.

Step 3: $\kappa|_\mathfrak{u}$ is negative-definite. By lemma 13.2.2(1), $\kappa(X_\alpha, X_{-\alpha}) = \kappa(H_\alpha, H_\alpha)/2 > 0$ is real and positive, since κ is positive on the real span of the coroots (proposition 7.3.4); write $c_\alpha = \kappa(X_\alpha, X_{-\alpha}) > 0$. The Killing form is ad-invariant and vanishes between root spaces $\mathfrak{g}_\alpha, \mathfrak{g}_\beta$ unless $\beta = -\alpha$, and between $\mathfrak{h}$ and any root space. Compute on the generators. On the torus, $\kappa(iH_\alpha, iH_\beta) = -\kappa(H_\alpha, H_\beta)$, and κ is positive-definite on the real span of the H_α (proposition 7.3.4), so κ is *negative*-definite on $\mathfrak{t}_0 = i \cdot$ (that real span). On a root pair,

$$\kappa(A_\alpha, A_\alpha) = \kappa(X_\alpha - X_{-\alpha},\ X_\alpha - X_{-\alpha}) = -2\kappa(X_\alpha, X_{-\alpha}) = -2c_\alpha < 0$$

using $\kappa(X_\alpha, X_\alpha) = \kappa(X_{-\alpha}, X_{-\alpha}) = 0$ (root spaces are κ-isotropic), and likewise

$$\kappa(B_\alpha, B_\alpha) = i^2\,\kappa(X_\alpha + X_{-\alpha},\ X_\alpha + X_{-\alpha}) = -\,2\kappa(X_\alpha, X_{-\alpha}) = -2c_\alpha < 0$$

Cross terms vanish: $\kappa(iH_\alpha, A_\beta) = 0$ and $\kappa(iH_\alpha, B_\beta) = 0$ because $\mathfrak{h} \perp \mathfrak{g}_{\pm\beta}$, while $\kappa(A_\alpha, B_\alpha) = i\,\kappa(X_\alpha - X_{-\alpha}, X_\alpha + X_{-\alpha}) = i(\kappa(X_\alpha, X_{-\alpha}) - \kappa(X_{-\alpha}, X_\alpha)) = 0$ by symmetry of κ, and $\kappa(A_\alpha, A_\beta) = \kappa(A_\alpha, B_\beta) = 0$ for $\alpha \neq \beta$ since the underlying roots do not pair. The summands $\mathfrak{t}_0$ and the planes $\mathbb{R}A_\alpha \oplus \mathbb{R}B_\alpha$ (for $\alpha > 0$) are therefore mutually κ-orthogonal, and κ is negative-definite on each, so $\kappa|_\mathfrak{u}$ is negative-definite. By definition 9.4.3, $\mathfrak{u}$ is a compact real form of $\mathfrak{g}_\mathbb{C}$, as we sought to show.

The theorem confirms in general what example 13.2.1 exhibited for $\mathfrak{sl}_2(\mathbb{C})$: rotating the coroots by i and antisymmetrizing each root pair produces a real form on which κ is negative-definite. The mechanism is the Weyl unitarian trick read backwards. In chapter 9 the trick averaged an inner product over a compact group to force complete reducibility. Here no group with Lie algebra $\mathfrak{u}$ is available yet, and the construction manufactures $\mathfrak{u}$ directly from the root data, the negative-definiteness of κ being what lemma 9.4.4 converts into the compactness of a group. The subalgebra $\mathfrak{u}$ integrates to the compact connected group $\mathrm{Int}(\mathfrak{u})$ (lemma 9.4.4), so $\mathfrak{u}$ deserves its name. Its universal cover, the simply connected integration of corollary 3.4.5, is compact as well, because $\pi_1(\mathrm{Int}(\mathfrak{u}))$ is finite (theorem 9.4.5).

One structural point in the construction is worth isolating, because it recurs throughout Part IV. The real span of the Chevalley basis $\{H_\alpha, X_\alpha\}$ is also a real form, the *split* real form, on which κ is indefinite. The two forms differ only by where the factors of i are inserted, yet one is compact and one is maximally noncompact. The compact form is singled out purely by the sign of κ, and the next result shows that sign pins it down completely, up to the natural symmetry of $\mathfrak{g}_\mathbb{C}$.

13.2.4 Warning: the naive real span need not be a compact form It is tempting to build a real form by taking the $\mathbb{R}$-span of a basis of $\mathfrak{g}_{\mathbb{C}}$ and hoping the Killing form comes out negative-definite. It generally does not. The $\mathbb{R}$-span of the Chevalley basis $\{H_\alpha\} \cup \{X_\alpha : \alpha \in \Phi\}$ is the *split* real form (the $\mathfrak{sl}_n(\mathbb{R})$-type algebra for $\mathfrak{sl}_n(\mathbb{C})$), and κ there is *indefinite*: it is positive on the coroots H_α and on the symmetric combinations $X_\alpha + X_{-\alpha}$, negative on the antisymmetric $X_\alpha - X_{-\alpha}$. Only the specific rotation of theorem 13.2.3, with i inserted on the coroots and on the symmetric root combinations, flips every sign the same way and yields negative-definiteness. The insertion of i turns real eigenvalues of ad into imaginary ones, which is what compactness requires.

Uniqueness is the second half of the well-definedness. A complex semisimple $\mathfrak{g}_{\mathbb{C}}$ may have many compact real forms as literal subalgebras, one for each Cartan subalgebra and each choice of Chevalley signs, but they are all carried onto one another by inner automorphisms of $\mathfrak{g}_{\mathbb{C}}$. The proof is a clean piece of linear algebra: two compact forms give two conjugations, their composite is a positive-definite self-adjoint automorphism for the inner product of one form, and its real powers furnish a path of inner automorphisms conjugating one conjugation to the other.

Theorem 13.2.5: Uniqueness of the Compact Real Form

Let $\mathfrak{g}_{\mathbb{C}}$ be a complex semisimple Lie algebra. Any two compact real forms of $\mathfrak{g}_{\mathbb{C}}$ are conjugate by an element of $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$, the group of inner automorphisms (lemma 9.4.4). Equivalently, if τ, τ' are the conjugations (definition 13.1.3) of two compact real forms $\mathfrak{u} = \mathfrak{g}_{\mathbb{C}}^{\tau}$ and $\mathfrak{u}' = \mathfrak{g}_{\mathbb{C}}^{\tau'}$, there is $\varphi \in \mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$ with $\varphi(\mathfrak{u}) = \mathfrak{u}'$.

***Proof*:**

Let τ, τ' be the conjugations of $\mathfrak{g}_{\mathbb{C}}$ fixing $\mathfrak{u}, \mathfrak{u}'$ respectively; each is an antilinear involutive automorphism (theorem 13.1.4). The idea is to build from $\mathfrak{u}$ a positive-definite Hermitian inner product on $\mathfrak{g}_{\mathbb{C}}$, express the composite $\tau\tau'$ as a positive operator for it, and slide $\tau\tau'$ to the identity along its real powers.

Step 1: a Hermitian inner product from the compact form. Define

$$\langle X, Y \rangle = -\kappa(X, \tau Y)$$

for $X, Y \in \mathfrak{g}_{\mathbb{C}}$. This is a positive-definite Hermitian form. It is sesquilinear because κ is $\mathbb{C}$-bilinear and τ is antilinear. It is Hermitian symmetric because κ is symmetric and τ-invariant ($\kappa(\tau X, \tau Y) = \overline{\kappa(X, Y)}$, as τ is an automorphism intertwining complex conjugation). It is positive-definite because for $X \in \mathfrak{u}$ one has $\tau X = X$, so $\langle X, X \rangle = -\kappa(X, X) > 0$ by definition 9.4.3 (negative-definiteness of κ on $\mathfrak{u}$), and $\mathfrak{g}_{\mathbb{C}} = \mathfrak{u} \oplus i\mathfrak{u}$ with $\langle iX, iX \rangle = -\kappa(iX, \tau(iX)) = -\kappa(iX, -iX) = -\kappa(X, X) > 0$ likewise.

Step 2: $\rho = \tau\tau'$ is positive-definite self-adjoint. Set $\rho = \tau\tau'$, a complex-linear automorphism of $\mathfrak{g}_{\mathbb{C}}$, being the composite of the two antilinear maps τ, τ'. Two facts about ρ drive the argument. First, the intertwining relation

$$\rho\,\tau = \tau\,\rho^{-1}$$

which holds because $\rho\tau = \tau\tau'\tau$ and, using $\tau^2 = (\tau')^2 = \mathrm{id}$, $\tau'\tau = (\tau\tau')^{-1} = \rho^{-1}$, so $\rho\tau = \tau(\tau'\tau) = \tau\rho^{-1}$. Second, ρ is self-adjoint and positive-definite for $\langle -, - \rangle$. Both use the κ-invariance of a conjugation, $\kappa(\tau X, \tau Y) = \overline{\kappa(X, Y)}$ (and the same for τ'), valid because a conjugation is a

κ-preserving automorphism intertwining complex conjugation. For self-adjointness, compute

$$\langle \rho X, Y\rangle = -\kappa(\tau\tau' X, \tau Y) = -\overline{\kappa(\tau' X, Y)} = -\kappa(X, \tau' Y) = \langle X, \rho Y\rangle$$

the third equality from $\kappa(\tau' X, Y) = \kappa(Y, \tau' X) = \overline{\kappa(\tau' Y, X)} = \overline{\kappa(X, \tau' Y)}$ (symmetry of κ and τ'-invariance), and the last from $\tau\rho Y = \tau\tau\tau' Y = \tau' Y$. For positivity, let $\langle X, Y\rangle' = -\kappa(X, \tau' Y)$ be the analogous positive-definite form of the compact form $\mathfrak{u}'$. Then for $X \neq 0$,

$$\langle \rho X, X\rangle = -\kappa(\tau\tau' X, \tau X) = -\overline{\kappa(\tau' X, X)} = -\kappa(X, \tau' X) = \langle X, X\rangle' > 0$$

using τ-invariance in the second equality, the reality of $\kappa(\tau' X, X) = \kappa(X, \tau' X)$ (both are $-\langle X, X\rangle' \in \mathbb{R}$) in the third, and the positive-definiteness of $\langle -, -\rangle'$ in the last. Thus ρ is a positive-definite self-adjoint operator, with real positive eigenvalues.

Step 3: the real powers of ρ conjugate τ' to τ. Being positive-definite self-adjoint, ρ has a well-defined real one-parameter family of powers ρ^s ($s \in \mathbb{R}$), obtained by raising each positive eigenvalue to the power s on its eigenspace. Write $\mathfrak{g}^\lambda$ for the λ-eigenspace of ρ, so that $\mathfrak{g}_{\mathbb{C}} = \bigoplus_\lambda \mathfrak{g}^\lambda$ with every $\lambda > 0$. For $X \in \mathfrak{g}^\lambda$ and $Y \in \mathfrak{g}^\mu$, the automorphism property gives $\rho[X, Y] = [\rho X, \rho Y] = \lambda\mu\,[X, Y]$, so $[X, Y] \in \mathfrak{g}^{\lambda\mu}$ (which is 0 when $\lambda\mu$ is not an eigenvalue). The operator $D = \log\rho$, acting by $\log\lambda$ on $\mathfrak{g}^\lambda$, therefore satisfies $D[X, Y] = (\log\lambda + \log\mu)[X, Y] = [DX, Y] + [X, DY]$ on such pairs, and by bilinearity D is a derivation of $\mathfrak{g}_{\mathbb{C}}$, with $\rho^s = \exp(sD)$. Since $\mathfrak{g}_{\mathbb{C}}$ is semisimple, every derivation is inner (corollary 6.3.5), so $D = \mathrm{ad}Z$ for some $Z \in \mathfrak{g}_{\mathbb{C}}$, and $\rho^s = \exp(\mathrm{ad}(sZ))$ is an automorphism lying in $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$ (lemma 9.4.4).

The intertwining relation of Step 2 extends to all real powers: from $\rho\tau = \tau\rho^{-1}$ and the functional calculus (both sides act by the same real function of ρ after conjugating by the antilinear τ), $\rho^s\,\tau = \tau\,\rho^{-s}$ for every $s \in \mathbb{R}$. Set $\varphi = \rho^{-1/2} \in \mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$. Then

$$\varphi\,\tau\,\varphi^{-1} = \rho^{-1/2}\,\tau\,\rho^{1/2} = \tau\,\rho^{1/2}\,\rho^{1/2} = \tau\,\rho = \tau\,\tau\tau' = \tau'$$

using $\rho^{-1/2}\tau = \tau\rho^{1/2}$ in the second step and $\tau^2 = \mathrm{id}$ in the last. An automorphism intertwining τ and τ' carries the fixed set $\mathfrak{u} = \mathfrak{g}_{\mathbb{C}}^\tau$ onto $\mathfrak{u}' = \mathfrak{g}_{\mathbb{C}}^{\tau'}$, so $\varphi(\mathfrak{u}) = \mathfrak{u}'$ with $\varphi \in \mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$, completing the proof.

The two theorems together justify the definite article: "*the* compact real form" of a complex semisimple $\mathfrak{g}_{\mathbb{C}}$ is well-defined, a single $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$-conjugacy class of subalgebras. The assignment $\mathfrak{g}_{\mathbb{C}} \mapsto \mathfrak{u}$ is functorial in the sense that an isomorphism $\mathfrak{g}_{\mathbb{C}} \cong \mathfrak{g}'_{\mathbb{C}}$ carries the compact form of one to a compact form of the other, and uniqueness makes this canonical up to inner automorphism. This is why the compact real form is the natural base point for the study of real forms: every real form of $\mathfrak{g}_{\mathbb{C}}$ can be positioned relative to the fixed $\mathfrak{u}$, and the measurement of that position is exactly the Cartan involution constructed in section 13.3. The alignment of an arbitrary conjugation σ with the conjugation τ of the compact form (theorem 13.2.5 provides the freedom to make $\sigma\tau = \tau\sigma$) is the construction section 13.3 rests on.

The concrete payoff is the identification of the compact real form of each classical complex Lie algebra, which the uniqueness theorem certifies is the compact form.

Example 13.2.6: $\mathfrak{su}(n)$ as the Compact Real Form of $\mathfrak{sl}_n(\mathbb{C})$

The complex Lie algebra $\mathfrak{sl}_n(\mathbb{C})$ of traceless $n \times n$ complex matrices has two familiar real forms among its subalgebras (chapter 9):

- $\mathfrak{sl}_n(\mathbb{R})$, the traceless *real* matrices, with conjugation $\sigma(X) = \overline{X}$ (entrywise complex conjugation);
- $\mathfrak{su}(n)$, the traceless skew-Hermitian matrices $X = -X^*$, with conjugation $\tau(X) = -X^*$.

Both satisfy $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \oplus i\mathfrak{g}_0$, so both are real forms of $\mathfrak{sl}_n(\mathbb{C})$. They are told apart by the Killing form $\kappa(X, Y) = 2n\,\mathrm{tr}(XY)$. On $\mathfrak{su}(n)$, a skew-Hermitian $X = -X^*$ has $\mathrm{tr}(X^2) = -\mathrm{tr}(XX^*) = -\|X\|^2 \leq 0$, with equality only at $X = 0$, so $\kappa(X, X) = 2n\,\mathrm{tr}(X^2) < 0$: the Killing form is negative-definite, and $\mathfrak{su}(n)$ is the compact real form (proposition 9.4.1, $\mathrm{SU}(n)$ being compact). On $\mathfrak{sl}_n(\mathbb{R})$, taking $X = \mathrm{diag}(1, -1, 0, \ldots, 0)$ gives $\kappa(X, X) = 2n \cdot 2 > 0$ while a skew-symmetric Y gives $\kappa(Y, Y) < 0$, so κ is *indefinite*: $\mathfrak{sl}_n(\mathbb{R})$ is a noncompact real form. By theorem 13.2.5, every compact real form of $\mathfrak{sl}_n(\mathbb{C})$ is Int-conjugate to $\mathfrak{su}(n)$, so $\mathfrak{su}(n)$ is *the* compact real form. For $n = 2$ this recovers example 13.2.1, with $\mathfrak{su}(2)$ compact and $\mathfrak{sl}_2(\mathbb{R})$ noncompact, the running contrast of Part IV.

The same reasoning identifies the compact real forms of the other classical series. The complex orthogonal algebra $\mathfrak{so}(n, \mathbb{C})$ has compact real form $\mathfrak{so}(n) = \mathfrak{so}(n, \mathbb{R})$, the real skew-symmetric matrices, on which κ is negative-definite because $\mathrm{SO}(n)$ is compact; the noncompact real forms are the indefinite orthogonal algebras $\mathfrak{so}(p, q)$ with $p + q = n$, $p, q > 0$. The complex symplectic algebra $\mathfrak{sp}(2n, \mathbb{C})$ has compact real form $\mathfrak{sp}(n) = \mathfrak{sp}(2n, \mathbb{C}) \cap \mathfrak{u}(2n)$, the Lie algebra of the compact symplectic group. In every case the compact form is the intersection of the complex algebra with the skew-Hermitian matrices, the "$-X^*$" involution being the conjugation τ of the compact real form. This uniformity, that the compact form is cut out by $X = -X^*$ across the classical types, is the concrete face of the abstract existence and uniqueness proved above, and it is the fixed frame the Cartan decomposition will rotate against in section 13.3.

In Part II the rank, the Weyl group and the root system of a complex semisimple Lie algebra were computed from one fixed maximal toral subalgebra $\mathfrak{h}$, and their independence of $\mathfrak{h}$ was deferred to the conjugacy of Cartan subalgebras. The compact real form proves that conjugacy. The construction of theorem 13.2.3 accepts an arbitrary $\mathfrak{h}$, and the real span of the rotated coroots iH_α is a maximal abelian subalgebra of the resulting $\mathfrak{u}$ whose complexification is $\mathfrak{h}$. Theorem 13.2.5 moves one compact form onto another, and inside a single compact form the maximal torus theorem moves one maximal abelian subalgebra onto another.

Theorem 13.2.7: Conjugacy of Cartan Subalgebras

Let $\mathfrak{g}_{\mathbb{C}}$ be a complex semisimple Lie algebra. For any two maximal toral subalgebras $\mathfrak{h}_1, \mathfrak{h}_2$ of $\mathfrak{g}_{\mathbb{C}}$ there is $\psi \in \mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$ with $\psi(\mathfrak{h}_1) = \mathfrak{h}_2$. Consequently any two Cartan subalgebras of $\mathfrak{g}_{\mathbb{C}}$ are conjugate under $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$, and all of them have the same dimension, the rank of $\mathfrak{g}_{\mathbb{C}}$.

***Proof*:**
The assertions about Cartan subalgebras follow from the first one, because the Cartan subalgebras of $\mathfrak{g}_{\mathbb{C}}$ are exactly its maximal toral subalgebras (proposition 7.1.7) and ψ is a linear isomorphism. Let $\mathfrak{h}_1, \mathfrak{h}_2$ be maximal toral. By proposition 7.1.7 each is abelian, self-centralizing and a Cartan subalgebra, so each has a root space decomposition (theorem 7.2.4).

Step 1: each $\mathfrak{h}_i$ is the complexification of a maximal abelian subalgebra of a compact real form. Fix i, choose root vectors X_α for $(\mathfrak{g}_{\mathbb{C}}, \mathfrak{h}_i)$ as in lemma 13.2.2, and let $\mathfrak{u}_i$ be the compact real form of theorem 13.2.3 built from them, with first summand $\mathfrak{t}_i = \sum_\alpha \mathbb{R}\, iH_\alpha$. The t_α span $\mathfrak{h}_i$

(proposition 7.3.4(2)) and each H_α is a nonzero multiple of t_α (definition 7.4.3), so the H_α span $\mathfrak{h}_i$ over $\mathbb{C}$ and $\mathfrak{h}_i = \mathfrak{t}_i + i\mathfrak{t}_i$. The subspace $\mathfrak{t}_i$ is an abelian subalgebra, as $\mathfrak{h}_i$ is abelian. To see that it is maximal abelian in $\mathfrak{u}_i$, let $X \in \mathfrak{u}_i$ commute with $\mathfrak{t}_i$. The bracket is $\mathbb{C}$-bilinear, so X commutes with $\mathfrak{t}_i + i\mathfrak{t}_i = \mathfrak{h}_i$, and $X \in C_{\mathfrak{g}_\mathbb{C}}(\mathfrak{h}_i) = \mathfrak{h}_i$ (theorem 7.2.4). In the notation $A_\alpha = X_\alpha - X_{-\alpha}$ and $B_\alpha = i(X_\alpha + X_{-\alpha})$ of the proof of theorem 13.2.3, write $X = T + \sum_{\alpha>0}(a_\alpha A_\alpha + b_\alpha B_\alpha)$ with $T \in \mathfrak{t}_i$ and $a_\alpha, b_\alpha \in \mathbb{R}$. For $\alpha > 0$ the component of X in $\mathfrak{g}_\alpha$ is $(a_\alpha + ib_\alpha)X_\alpha$, and it vanishes because $X \in \mathfrak{h}_i$. Since a_α and b_α are real, $a_\alpha = b_\alpha = 0$, so $X = T \in \mathfrak{t}_i$. Thus $\mathfrak{t}_i$ is its own centralizer in $\mathfrak{u}_i$. An abelian subalgebra of $\mathfrak{u}_i$ containing $\mathfrak{t}_i$ lies in that centralizer, so it equals $\mathfrak{t}_i$, and $\mathfrak{t}_i$ is maximal abelian.

Step 2: matching the compact forms. By theorem 13.2.5 there is $\varphi \in \mathrm{Int}(\mathfrak{g}_\mathbb{C})$ with $\varphi(\mathfrak{u}_1) = \mathfrak{u}_2$. As a $\mathbb{C}$-linear automorphism, φ restricts to an isomorphism of real Lie algebras $\mathfrak{u}_1 \to \mathfrak{u}_2$, so $\mathfrak{t}_1' = \varphi(\mathfrak{t}_1)$ is a maximal abelian subalgebra of $\mathfrak{u}_2$, and $\varphi(\mathfrak{h}_1) = \mathfrak{t}_1' + i\mathfrak{t}_1'$ by Step 1. It remains to carry $\mathfrak{t}_1'$ onto $\mathfrak{t}_2$ inside $\mathfrak{u}_2$.

Step 3: a compact group with Lie algebra $\mathfrak{u}_2$. The Killing form of $\mathfrak{u}_2$ is negative-definite (definition 9.4.3), so by lemma 9.4.4 the adjoint group $U = \mathrm{Int}(\mathfrak{u}_2) \subseteq \mathrm{GL}(\mathfrak{u}_2)$ is a compact connected Lie group with Lie algebra $\mathrm{ad}(\mathfrak{u}_2)$, and $\mathrm{ad}\colon \mathfrak{u}_2 \to \mathrm{ad}(\mathfrak{u}_2)$ is an isomorphism. Hence $\mathfrak{a}_1 = \mathrm{ad}(\mathfrak{t}_1')$ and $\mathfrak{a}_2 = \mathrm{ad}(\mathfrak{t}_2)$ are maximal abelian subalgebras of the Lie algebra of U.

Step 4: a maximal abelian subalgebra is the Lie algebra of a maximal torus. Let $\mathfrak{a}$ be a maximal abelian subalgebra of $\mathrm{ad}(\mathfrak{u}_2)$ and S the closure in U of $\exp(\mathfrak{a})$. The bracket of $\mathfrak{gl}(\mathfrak{u}_2)$ is the commutator, so elements of $\mathfrak{a}$ commute as operators. Hence $\exp(D)\exp(E) = \exp(D+E) = \exp(E)\exp(D)$ for $D, E \in \mathfrak{a}$ (proposition 1.3.5), and $\exp(\mathfrak{a})$ is an abelian subgroup of U, connected as the continuous image of the vector space $\mathfrak{a}$. Its closure S is a closed connected subgroup, and it is abelian because the commutator map $(x, y) \mapsto xyx^{-1}y^{-1}$ is continuous and equals the identity on the dense subset $\exp(\mathfrak{a}) \times \exp(\mathfrak{a})$ of $S \times S$. So S is a torus (definition 11.1.1). It is compact, hence closed in $\mathrm{GL}(\mathfrak{u}_2)$, and its Lie algebra $\mathfrak{s}$, the set of D with $\exp(tD) \in S$ for all t (theorem 3.2.1), contains $\mathfrak{a}$. It is abelian: for $D, E \in \mathfrak{s}$ the elements $\exp(sD)$ and $\exp(tE)$ of S commute, so $\exp(sD)\exp(tE)\exp(-sD) = \exp(tE)$, and differentiating in t and then in s at 0 (proposition 1.3.4(1)) gives $DE - ED = 0$. Since $S \subseteq U$, every $D \in \mathfrak{s}$ has $\exp(tD) \in U$ for all t, so $\mathfrak{s} \subseteq \mathrm{ad}(\mathfrak{u}_2)$, the Lie algebra of U (theorem 3.2.1). Maximality of $\mathfrak{a}$ among abelian subalgebras of $\mathrm{ad}(\mathfrak{u}_2)$ then forces $\mathfrak{s} = \mathfrak{a}$. Finally S is a maximal torus. If $S' \subseteq U$ is a torus containing S, its Lie algebra lies in $\mathrm{ad}(\mathfrak{u}_2)$ for the same reason, is abelian by the same argument, and contains $\mathfrak{a}$, so it equals $\mathfrak{a}$, and S', being connected, is generated by $\exp(\mathfrak{a}) \subseteq S$ (corollary 2.3.5), so $S' = S$.

Step 5: conjugating $\mathfrak{t}_1'$ onto $\mathfrak{t}_2$. Step 4 gives maximal tori S_1, S_2 of U with Lie algebras $\mathfrak{a}_1, \mathfrak{a}_2$. By theorem 11.1.3(2) there is $u \in U$ with $S_2 = uS_1u^{-1}$. For $D \in \mathfrak{a}_1$ and $t \in \mathbb{R}$, $\exp(t\,uDu^{-1}) = u\exp(tD)u^{-1} \in S_2$ (proposition 1.3.4(3)), so $uDu^{-1} \in \mathfrak{a}_2$ by theorem 3.2.1. Thus $u\mathfrak{a}_1u^{-1} \subseteq \mathfrak{a}_2$, and the same argument with u^{-1} gives the reverse inclusion. Since u is an automorphism of $\mathfrak{u}_2$, $u[X, u^{-1}Y] = [uX, Y]$, that is $u \circ \mathrm{ad}X \circ u^{-1} = \mathrm{ad}(uX)$ for $X \in \mathfrak{u}_2$. Hence $\mathrm{ad}(u(\mathfrak{t}_1')) = u\mathfrak{a}_1u^{-1} = \mathfrak{a}_2 = \mathrm{ad}(\mathfrak{t}_2)$, and injectivity of ad gives $u(\mathfrak{t}_1') = \mathfrak{t}_2$.

Step 6: back to $\mathfrak{g}_\mathbb{C}$. By definition of the adjoint group, and since $\exp(\mathrm{ad}X)^{-1} = \exp(\mathrm{ad}(-X))$, there are $X_1, \dots, X_k \in \mathfrak{u}_2$ with $u = \exp(\mathrm{ad}X_1)\cdots\exp(\mathrm{ad}X_k)$, each $\mathrm{ad}X_j$ acting on $\mathfrak{u}_2$. Let $u_\mathbb{C}(X + iY) = uX + i\,uY$ for $X, Y \in \mathfrak{u}_2$ be the $\mathbb{C}$-linear extension of u to $\mathfrak{g}_\mathbb{C} = \mathfrak{u}_2 \oplus i\mathfrak{u}_2$. The $\mathbb{C}$-linear extension of $\mathrm{ad}X_j$ on $\mathfrak{u}_2$ is $\mathrm{ad}X_j$ on $\mathfrak{g}_\mathbb{C}$, because the bracket of $\mathfrak{g}_\mathbb{C}$ is $\mathbb{C}$-bilinear and restricts to that of $\mathfrak{u}_2$. Complex-linear extension respects sums, products and limits of operators, so $u_\mathbb{C} = \exp(\mathrm{ad}X_1)\cdots\exp(\mathrm{ad}X_k)$ with each $\mathrm{ad}X_j$ now acting on $\mathfrak{g}_\mathbb{C}$, and $u_\mathbb{C} \in \mathrm{Int}(\mathfrak{g}_\mathbb{C})$. Set $\psi = u_\mathbb{C}\varphi \in \mathrm{Int}(\mathfrak{g}_\mathbb{C})$. By Steps 2, 5 and 1,

$$\psi(\mathfrak{h}_1) = u_\mathbb{C}(\mathfrak{t}_1' + i\mathfrak{t}_1') = u(\mathfrak{t}_1') + i\,u(\mathfrak{t}_1') = \mathfrak{t}_2 + i\mathfrak{t}_2 = \mathfrak{h}_2$$

completing the proof.

The conjugacy turns the choices of Part II into invariants of $\mathfrak{g}_{\mathbb{C}}$. The rank is the common dimension of the Cartan subalgebras. An element of $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$ carrying $\mathfrak{h}_1$ to $\mathfrak{h}_2$ preserves the Killing form and carries root spaces to root spaces, so it induces an isomorphism of the two root systems, and with it an isomorphism of the Weyl groups and an equality of the Dynkin diagrams. The last of these separates the algebras of the Cartan-Killing classification (corollary 8.5.5).

Exercise 13.2.1

1. Run the Weyl construction of theorem 13.2.3 for $\mathfrak{sl}_2(\mathbb{C})$ from scratch. Starting from the standard triple H, E, F with $[H,E] = 2E$, $[H,F] = -2F$, $[E,F] = H$, form $u_1 = iH$, $u_2 = E - F$, $u_3 = i(E+F)$, and compute the three brackets $[u_1,u_2]$, $[u_2,u_3]$, $[u_3,u_1]$ directly. Verify that the resulting real span is closed under the bracket, that each u_j is skew-Hermitian and traceless, and hence that $\mathfrak{u} = \mathfrak{su}(2)$. Confirm $\kappa(u_j,u_j) < 0$ for each j using $\kappa(X,Y) = 4\mathrm{tr}(XY)$ on $\mathfrak{sl}_2(\mathbb{C})$.

2. Show that the naive real span of the Chevalley basis need not be a compact form, in two ways. First, exhibit for $\mathfrak{sl}_2(\mathbb{C})$ an element of the real span $\mathbb{R}H \oplus \mathbb{R}E \oplus \mathbb{R}F = \mathfrak{sl}_2(\mathbb{R})$ on which κ is positive and one on which κ is negative, proving $\kappa|_{\mathfrak{sl}_2(\mathbb{R})}$ is indefinite. Second, replace $u_3 = i(E+F)$ in exercise 13.2.1 (1) by the "wrong rotation" $v_3 = E + F$ (no i), and show that $\mathbb{R}(iH) \oplus \mathbb{R}(E-F) \oplus \mathbb{R}(E+F)$ is *not* closed under the bracket, so it is not even a subalgebra. Which structure-constant relation from lemma 13.2.2 fails to be exploited?

3. Identify the compact real form of $\mathfrak{so}(n,\mathbb{C})$ (the complex Lie algebra of skew-symmetric $n \times n$ complex matrices). Show that $\mathfrak{so}(n) = \mathfrak{so}(n,\mathbb{R})$, the real skew-symmetric matrices, is a real form of $\mathfrak{so}(n,\mathbb{C})$ with $\mathfrak{so}(n,\mathbb{C}) = \mathfrak{so}(n) \oplus i\,\mathfrak{so}(n)$, and that κ is negative-definite on it (you may cite that $\mathrm{SO}(n)$ is compact via proposition 9.4.1). Then exhibit a noncompact real form $\mathfrak{so}(p,q)$ with $p+q=n$, $p,q>0$, and explain in one sentence why it cannot be conjugate to $\mathfrak{so}(n)$.

4. Explain why $\mathfrak{sl}_2(\mathbb{C})$ has exactly the two real forms $\mathfrak{su}(2)$ and $\mathfrak{sl}_2(\mathbb{R})$ up to conjugacy. Use the following facts, which you may take as given: a real form of a complex semisimple algebra is determined by the sign of its Killing form together with the "defect" (the signature of κ on a maximal abelian subalgebra of the noncompact part), and for rank one there is a single such invariant. Argue that the compact form (negative-definite κ) and the split form (maximally indefinite κ) exhaust the possibilities when the rank is one, so that $\mathfrak{su}(2)$ and $\mathfrak{sl}_2(\mathbb{R})$ are the only real forms up to isomorphism.

5. In the uniqueness proof (theorem 13.2.5), the Hermitian form $\langle X,Y\rangle = -\kappa(X,\tau Y)$ was built from the compact form's conjugation τ. Verify in detail that it is positive-definite: check that $\langle X,X\rangle > 0$ for every nonzero $X = X_1 + iX_2$ with $X_1, X_2 \in \mathfrak{u}$, using $\tau(X_1+iX_2) = X_1 - iX_2$ and the negative-definiteness of $\kappa|_{\mathfrak{u}}$ from definition 9.4.3. Explain why the same construction with τ replaced by the conjugation of a *noncompact* real form fails to give a positive-definite form.

6. The uniqueness theorem asserts conjugacy by an *inner* automorphism, not merely by some automorphism of $\mathfrak{g}_{\mathbb{C}}$. Locate the step in the proof of theorem 13.2.5 where innerness is established, and explain the role of "all derivations of a semisimple Lie algebra are inner" (corollary 6.3.5) in guaranteeing that $\varphi = \rho^{-1/2}$ lies in $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$.

13.3 Cartan Involutions and the Cartan Decomposition

The compact real form $\mathfrak{u}$ of $\mathfrak{g}_{\mathbb{C}}$ from section 13.2 gives a datum the noncompact case lacks. A general real form $\mathfrak{g}_0$ of $\mathfrak{g}_{\mathbb{C}}$ is not compact, so its Killing form κ is indefinite and there is no invariant inner product on $\mathfrak{g}_0$ to work with. But $\mathfrak{g}_0$ and $\mathfrak{u}$ live in the same complexification, and if the two can be aligned inside $\mathfrak{g}_{\mathbb{C}}$ the compact form gives the missing positive-definite datum: comparing $\mathfrak{g}_0$ against $\mathfrak{u}$ produces an involution θ of $\mathfrak{g}_0$ whose ± 1-eigenspaces split $\mathfrak{g}_0$ into a part where κ is negative-definite and a part where it is positive-definite. This involution and this splitting are the subject of the section. They reduce every question about the noncompact group G to a compact piece K and a Euclidean complement.

The order of business is to define the splitting axiomatically, extract its structure, and then prove that it always exists. The axiom is positivity of a modified Killing form; the structure is a bracket-graded orthogonal decomposition; and the existence proof aligns $\mathfrak{g}_0$ with a compact form, using the uniqueness of theorem 13.2.5.

Definition 13.3.1: Cartan Involution

Let $\mathfrak{g}_0$ be a real semisimple Lie algebra with Killing form κ. A *Cartan involution* of $\mathfrak{g}_0$ is an involutive automorphism $\theta\colon \mathfrak{g}_0 \to \mathfrak{g}_0$, that is a Lie algebra automorphism with $\theta^2 = \mathrm{id}$, for which the symmetric bilinear form

$$B_\theta(X, Y) = -\kappa(X, \theta Y)$$

is positive-definite on $\mathfrak{g}_0$. Write $\mathfrak{k}$ for the $(+1)$-eigenspace of θ and $\mathfrak{p}$ for the (-1)-eigenspace, so that $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ as a vector space.

The two conditions pull in opposite directions. An automorphism θ of order two always splits $\mathfrak{g}_0$ into ± 1-eigenspaces, and the bilinear form B_θ is automatically symmetric, since κ is symmetric and θ is κ-orthogonal (an automorphism preserves κ, so $\kappa(X, \theta Y) = \kappa(\theta X, \theta^2 Y) = \kappa(\theta X, Y)$). What is not automatic is positivity: κ itself is indefinite on a noncompact $\mathfrak{g}_0$, and B_θ repairs the sign only if θ is chosen to flip the directions on which κ is positive. On the $(+1)$-eigenspace $\mathfrak{k}$ the form B_θ equals $-\kappa$, so positivity of B_θ forces κ negative-definite there; on the (-1)-eigenspace $\mathfrak{p}$ it equals $+\kappa$, forcing κ positive-definite there. The Cartan involution is thus the exact device that separates the compact directions of $\mathfrak{g}_0$ (where $\kappa < 0$) from the noncompact ones (where $\kappa > 0$).

The running example shows the abstraction before any theorem is proved.

Example 13.3.2: The Cartan Involution of $\mathfrak{sl}_n(\mathbb{R})$

Let $\mathfrak{g}_0 = \mathfrak{sl}_n(\mathbb{R})$, the traceless real $n \times n$ matrices, a real form of $\mathfrak{sl}_n(\mathbb{C})$ (section 13.1). Its Killing form is $\kappa(X, Y) = 2n \operatorname{tr}(XY)$, the restriction of the $\mathfrak{gl}_n$ form (example 6.1.8). Define

$$\theta(X) = -X^{\mathsf{T}}$$

This is $\mathbb{R}$-linear, preserves the trace-zero condition, and is an automorphism: $\theta[X, Y] = -[X, Y]^{\mathsf{T}} = -(Y^{\mathsf{T}}X^{\mathsf{T}} - X^{\mathsf{T}}Y^{\mathsf{T}}) = X^{\mathsf{T}}Y^{\mathsf{T}} - Y^{\mathsf{T}}X^{\mathsf{T}} = [-X^{\mathsf{T}}, -Y^{\mathsf{T}}] = [\theta X, \theta Y]$, using $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$. It is involutive, since $\theta^2(X) = -(-X^{\mathsf{T}})^{\mathsf{T}} = X$. The eigenspaces are

$$\mathfrak{k} = \{X \mid X^{\mathsf{T}} = -X\} = \mathfrak{so}(n), \qquad \mathfrak{p} = \{X \mid X^{\mathsf{T}} = X,\ \operatorname{tr} X = 0\} = \mathrm{Sym}_0(n)$$

the antisymmetric matrices and the traceless symmetric matrices. To check positivity, compute

$$B_\theta(X,Y) = -\kappa(X,\theta Y) = -2n\,\mathrm{tr}(X(-Y^{\mathsf{T}})) = 2n\,\mathrm{tr}(XY^{\mathsf{T}})$$

Now $\mathrm{tr}(XY^{\mathsf{T}}) = \sum_{i,j} X_{ij}Y_{ij}$ is the standard Euclidean inner product on the space of matrices, so $B_\theta(X,X) = 2n\sum_{i,j} X_{ij}^2 > 0$ for $X \neq 0$. Thus θ is a Cartan involution, $\mathfrak{k} = \mathfrak{so}(n)$ is the compact rotation subalgebra, and $\mathfrak{p}$ is the Euclidean complement of traceless symmetric matrices. For $n = 2$ this reads $\mathfrak{k} = \mathfrak{so}(2) = \mathbb{R}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\mathfrak{p}$ the two-dimensional space of traceless symmetric 2×2 matrices, the split relevant to the running example $\mathfrak{sl}_2(\mathbb{R})$.

The eigenspace decomposition of the example is forced by the definition, and its structure is uniform across all real semisimple algebras. The bracket behaves like a $\mathbb{Z}/2$-grading, and the Killing form respects the grading with the signs the definition requires.

Theorem 13.3.3: The Cartan Decomposition

Let $\mathfrak{g}_0$ be a real semisimple Lie algebra with a Cartan involution θ, and let $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ be the decomposition into the $(+1)$- and (-1)-eigenspaces of θ. Then:

1. The bracket satisfies $[\mathfrak{k},\mathfrak{k}] \subseteq \mathfrak{k}$, $[\mathfrak{k},\mathfrak{p}] \subseteq \mathfrak{p}$, and $[\mathfrak{p},\mathfrak{p}] \subseteq \mathfrak{k}$; in particular $\mathfrak{k}$ is a subalgebra.
2. The Killing form κ is negative-definite on $\mathfrak{k}$, positive-definite on $\mathfrak{p}$, and $\mathfrak{k} \perp \mathfrak{p}$ with respect to κ.
3. For $X \in \mathfrak{k}$ the operator $\mathrm{ad}X$ is skew-symmetric with respect to B_θ, and for $X \in \mathfrak{p}$ it is symmetric.
4. $\mathfrak{k}$ is a maximal compactly embedded subalgebra: it is maximal among the subalgebras of $\mathfrak{g}_0$ on which κ is negative-definite, in the sense that no subalgebra properly containing $\mathfrak{k}$ carries a negative-definite Killing form.

***Proof*:**
The argument runs through the four statements in turn; the first two are the substance and the last two are consequences.

1. *Bracket relations.* Because θ is an automorphism, $\theta[X,Y] = [\theta X, \theta Y]$ for all X, Y. On eigenvectors this reads off the grading. If $X, Y \in \mathfrak{k}$, then $\theta[X,Y] = [\theta X, \theta Y] = [X,Y]$, so $[X,Y]$ has eigenvalue $+1$ and lies in $\mathfrak{k}$. If $X \in \mathfrak{k}$ and $Y \in \mathfrak{p}$, then $\theta[X,Y] = [X,-Y] = -[X,Y]$, so $[X,Y] \in \mathfrak{p}$. If $X, Y \in \mathfrak{p}$, then $\theta[X,Y] = [-X,-Y] = [X,Y]$, so $[X,Y] \in \mathfrak{k}$. These are exactly the three relations, with eigenvalues multiplying: θ acts by $(-1)^a(-1)^b$ on $[\mathfrak{g}^a, \mathfrak{g}^b]$ where $\mathfrak{g}^+ = \mathfrak{k}$, $\mathfrak{g}^- = \mathfrak{p}$. In particular $[\mathfrak{k},\mathfrak{k}] \subseteq \mathfrak{k}$ makes $\mathfrak{k}$ a subalgebra.
2. *Signs and orthogonality of κ.* By hypothesis $B_\theta(X,Y) = -\kappa(X,\theta Y)$ is positive-definite. On $\mathfrak{k}$ the involution acts as the identity, so for $X \in \mathfrak{k}$,

$$B_\theta(X,X) = -\kappa(X,\theta X) = -\kappa(X,X)$$

and positivity of B_θ gives $\kappa(X,X) < 0$ for $X \neq 0$: the Killing form is negative-definite on $\mathfrak{k}$. On $\mathfrak{p}$ the involution acts as $-\mathrm{id}$, so for $X \in \mathfrak{p}$,

$$B_\theta(X,X) = -\kappa(X,-X) = \kappa(X,X)$$

and positivity gives $\kappa(X,X) > 0$ for $X \neq 0$: the Killing form is positive-definite on $\mathfrak{p}$. For orthogonality, take $X \in \mathfrak{k}$ and $Y \in \mathfrak{p}$. Then $\theta Y = -Y$, so

$$B_\theta(X,Y) = -\kappa(X,\theta Y) = \kappa(X,Y)$$

On the other hand B_θ is the positive-definite inner product for which θ is an isometry, and $\mathfrak{k}$ and $\mathfrak{p}$ are the eigenspaces of the self-adjoint involution θ for distinct eigenvalues $+1$ and -1; distinct eigenspaces of a B_θ-self-adjoint operator are B_θ-orthogonal, so $B_\theta(X,Y) = 0$. Combining, $\kappa(X,Y) = B_\theta(X,Y) = 0$, so $\mathfrak{k} \perp \mathfrak{p}$ under κ as well.

3. *Skew-symmetry and symmetry of* $\mathrm{ad}X$. The Killing form is invariant, $\kappa([Z,X],Y) + \kappa(X,[Z,Y]) = 0$ for all Z,X,Y (definition 6.1.3), which says $\mathrm{ad}Z$ is κ-skew for every Z. Transport this across $B_\theta(X,Y) = -\kappa(X,\theta Y)$. For $Z \in \mathfrak{g}_0$ and $X,Y \in \mathfrak{g}_0$,

$$B_\theta(\mathrm{ad}Z\,X,Y) = -\kappa([Z,X],\theta Y) = \kappa(X,[Z,\theta Y]) = \kappa(X,\theta[\theta Z,Y])$$

using κ-skewness of $\mathrm{ad}Z$ and then $[Z,\theta Y] = \theta[\theta Z,\theta^2 Y] = \theta[\theta Z,Y]$. The right side is $-B_\theta(X,[\theta Z,Y]) = -B_\theta(X,\mathrm{ad}(\theta Z)\,Y)$. Thus the B_θ-adjoint of $\mathrm{ad}Z$ is $-\mathrm{ad}(\theta Z)$. For $Z = X \in \mathfrak{k}$ we have $\theta X = X$, so the adjoint of $\mathrm{ad}X$ is $-\mathrm{ad}X$: skew-symmetric. For $Z = X \in \mathfrak{p}$ we have $\theta X = -X$, so the adjoint of $\mathrm{ad}X$ is $+\mathrm{ad}X$: symmetric.

4. *Maximality of* $\mathfrak{k}$. First, κ is negative-definite on $\mathfrak{k}$ by part (2), so $\mathfrak{k}$ is one subalgebra with the property in question. Suppose $\mathfrak{k}'$ is a subalgebra with $\mathfrak{k} \subseteq \mathfrak{k}'$ on which κ is negative-definite, and suppose for contradiction that $\mathfrak{k}' \neq \mathfrak{k}$. Pick $X \in \mathfrak{k}' \setminus \mathfrak{k}$ and decompose $X = X_\mathfrak{k} + X_\mathfrak{p}$ with $X_\mathfrak{k} \in \mathfrak{k}$ and $X_\mathfrak{p} \in \mathfrak{p}$; since $X \notin \mathfrak{k}$, the component $X_\mathfrak{p}$ is nonzero. Because $\mathfrak{k} \subseteq \mathfrak{k}'$, the difference $X_\mathfrak{p} = X - X_\mathfrak{k}$ lies in $\mathfrak{k}'$ as well. But $X_\mathfrak{p} \in \mathfrak{p}$, where part (2) gives $\kappa(X_\mathfrak{p},X_\mathfrak{p}) > 0$, contradicting the negative-definiteness of κ on $\mathfrak{k}'$. Hence no subalgebra properly containing $\mathfrak{k}$ carries a negative-definite Killing form, so $\mathfrak{k}$ is maximal among subalgebras on which κ is negative-definite, a maximal compactly embedded subalgebra, completing the proof.

The decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ is available in every noncompact real group. The subalgebra $\mathfrak{k}$ is compact in the strong sense of part (2): the restriction $\kappa|_\mathfrak{k}$ is negative-definite, and by part (3) each $\mathrm{ad}X$ ($X \in \mathfrak{k}$) is B_θ-skew, hence $\mathrm{ad}_\mathfrak{k}X$ is skew on the invariant inner product $B_\theta|_\mathfrak{k}$, which forces the intrinsic Killing form of $\mathfrak{k}$ to be negative-definite as well; so by the sign criterion of proposition 9.4.1 the subgroup it generates is compact (up to center), the maximal compact directions of G. The complement $\mathfrak{p}$ is not a subalgebra, $[\mathfrak{p},\mathfrak{p}] \subseteq \mathfrak{k}$ landing back in $\mathfrak{k}$ rather than in $\mathfrak{p}$, and κ is positive-definite there, so $\mathfrak{p}$ is a Euclidean complement whose exponential will turn out to be a closed cell rather than a subgroup. Part (3) feeds section 13.4: the operators $\mathrm{ad}X$ for $X \in \mathfrak{p}$ are symmetric with respect to the inner product B_θ, hence diagonalizable with real eigenvalues, which is what makes the global map $\exp\colon \mathfrak{p} \to G$ a diffeomorphism onto a closed submanifold. Part (4) says $\mathfrak{k}$ is not just a compact subalgebra but a maximal one, no larger subalgebra keeps the Killing form negative-definite; the maximal compact subgroup it generates turns out to be unique up to conjugacy, but that is the global statement proved in section 13.4.

The definition presumes a Cartan involution exists. For $\mathfrak{sl}_n(\mathbb{R})$ one was exhibited by hand, but the general existence is the substantive theorem, and it is where the compact real form of section 13.2 earns its place: the Cartan involution of $\mathfrak{g}_0$ is what the conjugation onto a compact form of $\mathfrak{g}_\mathbb{C}$ induces on $\mathfrak{g}_0$.

Theorem 13.3.4: Existence and Uniqueness of Cartan Involutions

Every real semisimple Lie algebra $\mathfrak{g}_0$ admits a Cartan involution. Any two Cartan involutions of $\mathfrak{g}_0$ are conjugate by an inner automorphism: if θ_1, θ_2 are Cartan involutions then $\theta_2 = \varphi\theta_1\varphi^{-1}$ for some $\varphi \in \mathrm{Int}(\mathfrak{g}_0)$.

The proof rests on aligning two conjugations of $\mathfrak{g}_{\mathbb{C}}$. Recall the setup. The complexification $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$ carries the conjugation σ of the real form $\mathfrak{g}_0$, the antilinear involutive automorphism fixing $\mathfrak{g}_0$ (definition 13.1.3). By theorem 13.2.3, $\mathfrak{g}_{\mathbb{C}}$ also has a compact real form $\mathfrak{u}$, with its own conjugation τ. If σ and τ happened to commute, then $\theta = \sigma\tau$ would be a linear involution of $\mathfrak{g}_{\mathbb{C}}$ preserving $\mathfrak{g}_0$, and its restriction to $\mathfrak{g}_0$ would be the Cartan involution. The content of the proof is that $\mathfrak{u}$ can always be moved by an inner automorphism so that this commuting holds. This alignment is isolated as a lemma.

Lemma 13.3.5: Alignment of the Compact Form

Let $\mathfrak{g}_0$ be a real form of $\mathfrak{g}_{\mathbb{C}}$ with conjugation σ, and let $\mathfrak{u}$ be a compact real form of $\mathfrak{g}_{\mathbb{C}}$ with conjugation τ. Then $\mathfrak{u}$ can be replaced by an $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$-conjugate whose conjugation τ' commutes with σ, that is $\sigma\tau' = \tau'\sigma$.

The proof is a spectral argument on a single operator, the square $\rho = \omega^2$ of the composite $\omega = \sigma\tau$. Its structural steps are given in full; the one delicate part, the functional-calculus estimate identifying the exact power that aligns the forms, is a computation with no conceptual content and is indicated as such (it is the eigenvalue bookkeeping for the self-adjoint positive operator ρ, using only the spectral theorem for ρ and the intertwining $\sigma\rho^t\sigma^{-1} = \rho^{-t}$).

***Proof Key ideas*:**
The composite $\omega = \sigma\tau$ is a complex-linear automorphism of $\mathfrak{g}_{\mathbb{C}}$, being the product of two antilinear maps. The compact form $\mathfrak{u}$ gives the Hermitian inner product $B_\tau(Z, W) = -\kappa(Z, \tau W)$, positive-definite on $\mathfrak{g}_{\mathbb{C}}$ (the same computation as in definition 13.3.1, now over $\mathbb{C}$: κ is negative-definite on $\mathfrak{u}$ since $\mathfrak{u}$ is compact, so B_τ is positive). The three steps are: ω is B_τ-self-adjoint, so its square $\rho = \omega^2$ is B_τ-positive-definite; the real powers ρ^t are inner automorphisms; and a specific power of ρ carries $\mathfrak{u}$ to a σ-stable compact form.

Step 1: ω is B_τ-self-adjoint, and $\rho = \omega^2$ is B_τ-positive-definite. A direct check on the definition of B_τ, using that both σ and τ preserve the Killing form and that $\tau^2 = \mathrm{id}$, shows $B_\tau(\omega Z, W) = B_\tau(Z, \omega W)$, so ω is B_τ-self-adjoint. Here ω itself is *not* positive: it is only self-adjoint, hence has real eigenvalues but of both signs in general (concretely, for $\mathfrak{g}_0 = \mathfrak{sl}_2(\mathbb{R}) \subseteq \mathfrak{sl}_2(\mathbb{C})$ against the compact form $\mathfrak{su}(2)$, one computes $B_\tau(\omega Z, Z) = -\kappa(Z, \sigma Z)$, positive on $X_\alpha - X_{-\alpha}$ but negative on iH). This is the point of departure from the compact-versus-compact uniqueness argument of theorem 13.2.5, where both conjugations are compact and the analogous composite is genuinely positive; here σ is a general conjugation and only ω's self-adjointness survives. The remedy is to pass to the square. Because ω is B_τ-self-adjoint, $\rho = \omega^2 = \omega^*\omega$ satisfies

$$B_\tau(\rho Z, Z) = B_\tau(\omega Z, \omega Z) > 0 \quad \text{for } Z \neq 0$$

so ρ is B_τ-self-adjoint and positive-definite. Hence ρ has a spectral decomposition $\mathfrak{g}_{\mathbb{C}} = \bigoplus_\mu \mathfrak{g}_{\mathbb{C}}^\mu$ into eigenspaces for positive real eigenvalues μ, and its real powers ρ^t are well defined by functional calculus.

Step 2: the real powers ρ^t are inner automorphisms. Since $\rho = \omega^2$ is the square of an automorphism it is an automorphism, and being positive it has a logarithm $D = \log \rho$, defined by functional calculus (acting as $\log \mu$ on the μ-eigenspace). This D is a derivation of $\mathfrak{g}_{\mathbb{C}}$: differentiating $\rho[X,Y] = [\rho X, \rho Y]$ across the spectral decomposition gives $D[X,Y] = [DX,Y] + [X,DY]$. Since $\mathfrak{g}_{\mathbb{C}}$ is semisimple every derivation is inner (corollary 6.3.5), so $D = \mathrm{ad}Z_0$ for some $Z_0 \in \mathfrak{g}_{\mathbb{C}}$, and $\rho^t = e^{tD} = e^{t\mathrm{ad}Z_0} = \mathrm{Ad}(\exp(tZ_0)) \in \mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$ for every real t. In particular $\rho^{1/4}$ is an inner automorphism.

Step 3: a power of ρ aligns the compact form. The two conjugations interact with ω through the exact operator identities $\sigma\omega\sigma^{-1} = \omega^{-1}$ and $\tau\omega\tau^{-1} = \omega^{-1}$, verified by $\sigma(\sigma\tau)\sigma = \tau\sigma = \omega^{-1}$ and $\tau(\sigma\tau)\tau = \tau\sigma = \omega^{-1}$. Squaring, $\sigma\rho\sigma^{-1} = (\sigma\omega\sigma^{-1})^2 = \omega^{-2} = \rho^{-1}$, and likewise $\tau\rho\tau^{-1} = \rho^{-1}$. Because σ and τ are antilinear and ρ has positive real spectrum, these identities pass to real powers, $\sigma\rho^t\sigma^{-1} = \rho^{-t}$ and likewise for τ. The functional-calculus computation, tracking how σ permutes the eigenspaces $\mathfrak{g}_{\mathbb{C}}^{\mu} \leftrightarrow \mathfrak{g}_{\mathbb{C}}^{1/\mu}$, then shows that the inner automorphism $\varphi = \rho^{1/4}$ conjugates τ to $\tau' = \varphi\tau\varphi^{-1}$ commuting with σ; equivalently it carries $\mathfrak{u}$ to a compact real form $\mathfrak{u}' = \varphi(\mathfrak{u})$ stable under σ, that is $\sigma(\mathfrak{u}') = \mathfrak{u}'$. This is the standard eigenvalue estimate for such positive operators and is the only computational step; its content is that the fourth root of the *square* ρ is the exact power balancing the $\mu \leftrightarrow 1/\mu$ symmetry that σ imposes on the spectrum. Since $\varphi \in \mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$, the form $\mathfrak{u}'$ is an inner conjugate of $\mathfrak{u}$ with $\sigma\tau' = \tau'\sigma$, completing the proof.

With the alignment in hand the main theorem follows.

***Proof of theorem 13.3.4*:**
Existence. Choose a compact real form $\mathfrak{u}$ of $\mathfrak{g}_{\mathbb{C}}$ (theorem 13.2.3) with conjugation τ, and by lemma 13.3.5 arrange $\sigma\tau = \tau\sigma$, where σ is the conjugation of $\mathfrak{g}_0$. Set $\theta = \sigma\tau = \tau\sigma$, a complex-linear automorphism of $\mathfrak{g}_{\mathbb{C}}$ with $\theta^2 = \sigma\tau\sigma\tau = \sigma^2\tau^2 = \mathrm{id}$ (using commutativity and $\sigma^2 = \tau^2 = \mathrm{id}$). Because θ commutes with σ, it preserves the fixed set $\mathfrak{g}_0 = \mathfrak{g}_{\mathbb{C}}^{\sigma}$: if $\sigma X = X$ then $\sigma(\theta X) = \theta(\sigma X) = \theta X$. Its restriction $\theta_0 = \theta|_{\mathfrak{g}_0}$ is therefore an involutive automorphism of $\mathfrak{g}_0$. It remains to check that $B_{\theta_0}(X,Y) = -\kappa(X,\theta_0 Y)$ is positive-definite on $\mathfrak{g}_0$. For $X \in \mathfrak{g}_0$ write $X = X_+ + X_-$ in the ± 1-eigenspaces of θ_0; these are $\mathfrak{k} = \mathfrak{g}_0 \cap \mathfrak{u}$ and $\mathfrak{p} = \mathfrak{g}_0 \cap i\mathfrak{u}$, because on $\mathfrak{u}$ the map τ is the identity so $\theta = \sigma$ acts as σ, and $\theta = +1$ on $\mathfrak{g}_0 \cap \mathfrak{u}$ while $\theta = -1$ on $\mathfrak{g}_0 \cap i\mathfrak{u}$. On $\mathfrak{k} \subseteq \mathfrak{u}$ the Killing form κ is negative-definite (as $\mathfrak{u}$ is compact, definition 9.4.3), so $B_{\theta_0} = -\kappa > 0$ there; on $\mathfrak{p} \subseteq i\mathfrak{u}$ the form κ is positive-definite (the Killing form of $\mathfrak{g}_{\mathbb{C}}$ satisfies $\kappa(iZ,iW) = -\kappa(Z,W)$, flipping the sign on $i\mathfrak{u}$ relative to $\mathfrak{u}$), so $B_{\theta_0} = +\kappa > 0$ there. Since $\mathfrak{k}$ and $\mathfrak{p}$ are κ-orthogonal, B_{θ_0} is positive-definite on all of $\mathfrak{g}_0$, and θ_0 is a Cartan involution.

Uniqueness up to conjugacy. Let θ_1, θ_2 be two Cartan involutions of $\mathfrak{g}_0$, with decompositions $\mathfrak{g}_0 = \mathfrak{k}_i \oplus \mathfrak{p}_i$. Complexifying, each θ_i extends to a complex-linear involution of $\mathfrak{g}_{\mathbb{C}}$, and since it preserves $\mathfrak{g}_0 = \mathfrak{g}_{\mathbb{C}}^{\sigma}$ it commutes with σ. The composite $\tau_i = \sigma\theta_i = \theta_i\sigma$ is then an antilinear involution whose fixed algebra is a compact real form $\mathfrak{u}_i = \mathfrak{k}_i \oplus i\mathfrak{p}_i$ of $\mathfrak{g}_{\mathbb{C}}$ (its Killing form is negative-definite: $\kappa < 0$ on $\mathfrak{k}_i$ by theorem 13.3.3(2), and $\kappa < 0$ on $i\mathfrak{p}_i$ since $\kappa > 0$ on $\mathfrak{p}_i$ and $\kappa(iZ,iZ) = -\kappa(Z,Z)$). Here τ_i is the conjugation of $\mathfrak{u}_i$, and it commutes with σ because θ_i does; equivalently, both $\mathfrak{u}_i$ are σ-stable.

The point of uniqueness is that the conjugator between $\mathfrak{u}_1$ and $\mathfrak{u}_2$ can be found inside $\mathrm{Int}(\mathfrak{g}_0)$, not merely $\mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$, and this is where σ-stability of both forms is spent. Both τ_1, τ_2 are compact conjugations, so the composite $\omega = \tau_2\tau_1$ is a complex-linear automorphism, self-adjoint and positive-definite for the Hermitian form B_{τ_1}: this is exactly the compact-versus-compact situation of theorem 13.2.5, in which the composite of two compact conjugations *is* genuinely positive (both τ_i give negative-definite κ), so no passage to a square is needed. The positive operator

$\rho = \omega$ therefore has real powers $\rho^t \in \mathrm{Int}(\mathfrak{g}_{\mathbb{C}})$ by the argument of Steps 2 and 3 of lemma 13.3.5, and $\varphi = \rho^{1/2}$ satisfies $\varphi\tau_1\varphi^{-1} = \tau_2$, so $\varphi(\mathfrak{u}_1) = \mathfrak{u}_2$. The decisive extra fact is that ρ commutes with σ: since τ_1, τ_2 each commute with σ, so does $\omega = \tau_2\tau_1$, hence so does every element of its functional calculus, in particular $\varphi = \rho^{1/2}$. An inner automorphism of $\mathfrak{g}_{\mathbb{C}}$ commuting with σ preserves $\mathfrak{g}_0 = \mathfrak{g}_{\mathbb{C}}^{\sigma}$ and restricts there to an inner automorphism of $\mathfrak{g}_0$; thus $\varphi|_{\mathfrak{g}_0} \in \mathrm{Int}(\mathfrak{g}_0)$. Finally $\varphi(\mathfrak{u}_1) = \mathfrak{u}_2$ gives $\varphi(\mathfrak{k}_1) = \varphi(\mathfrak{g}_0 \cap \mathfrak{u}_1) = \mathfrak{g}_0 \cap \mathfrak{u}_2 = \mathfrak{k}_2$ and likewise $\varphi(\mathfrak{p}_1) = \mathfrak{p}_2$; matching the ± 1-eigenspaces of θ_1 and θ_2 yields $\varphi\theta_1\varphi^{-1} = \theta_2$. Thus any two Cartan involutions are inner-conjugate, as we sought to show.

The theorem completes the reduction. A noncompact real form has no canonical inner product, but it has a Cartan involution, canonical up to inner conjugacy, and the involution gives both the compact subalgebra $\mathfrak{k}$ and the positive form B_θ that were missing. The construction is the precise sense in which the Cartan involution comes from the compact form: aligning $\mathfrak{g}_0$ with $\mathfrak{u}$ inside $\mathfrak{g}_{\mathbb{C}}$ and reading off the composite $\sigma\tau$ back on $\mathfrak{g}_0$. The eigenspaces are $\mathfrak{k} = \mathfrak{g}_0 \cap \mathfrak{u}$ and $\mathfrak{p} = \mathfrak{g}_0 \cap i\mathfrak{u}$, so $\mathfrak{k}$ is literally the compact form's intersection with $\mathfrak{g}_0$ and $\mathfrak{p}$ is the part of $\mathfrak{g}_0$ that lies along the imaginary directions of $\mathfrak{u}$.

One numerical reading of the decomposition is worth recording, because it measures noncompactness directly. The signature of κ on $\mathfrak{g}_0$, the number of positive eigenvalues minus the number of negative ones in an orthogonal basis, is

$$\mathrm{sig}(\kappa) = \dim \mathfrak{p} - \dim \mathfrak{k}$$

by theorem 13.3.3(2): κ is positive-definite on the $\dim \mathfrak{p}$-dimensional $\mathfrak{p}$ and negative-definite on the $\dim \mathfrak{k}$-dimensional $\mathfrak{k}$. When $\mathfrak{g}_0$ is compact, $\mathfrak{p} = 0$, $\theta = \mathrm{id}$, and the signature is $-\dim \mathfrak{g}_0$, maximally negative; the further $\mathfrak{g}_0$ is from compact, the larger $\dim \mathfrak{p}$ and the larger the signature. The Cartan decomposition thus reads off exactly how far a real form sits from its compact sibling: $\dim \mathfrak{p}$ counts the noncompact directions, and $\dim \mathfrak{k}$ the compact ones.

A second concrete instance, the indefinite unitary groups, shows the same mechanism with a Hermitian rather than symmetric transpose.

Example 13.3.6: The Cartan Involution of $\mathfrak{su}(p,q)$

Let $\mathfrak{su}(p,q)$ be the Lie algebra of the group $\mathrm{SU}(p,q)$ preserving a Hermitian form of signature (p,q) on $\mathbb{C}^{p+q}$; concretely, with $I_{p,q} = \mathrm{diag}(I_p, -I_q)$, it consists of the trace-zero complex matrices X with $X^* I_{p,q} + I_{p,q} X = 0$, where $X^* = \overline{X}^{\mathsf{T}}$ is the conjugate transpose. A Cartan involution is

$$\theta(X) = -X^*$$

the negative conjugate transpose. It is $\mathbb{R}$-linear, involutive ($\theta^2(X) = (X^*)^* = X$), and an automorphism by the same transpose identity as in example 13.3.2, now with the conjugate transpose. Its eigenspaces are

$$\mathfrak{k} = \{X \in \mathfrak{su}(p,q) \mid X^* = -X\}, \qquad \mathfrak{p} = \{X \in \mathfrak{su}(p,q) \mid X^* = X\}$$

Writing a matrix in (p,q)-block form, the condition $X^* I_{p,q} + I_{p,q} X = 0$ forces the diagonal blocks to be skew-Hermitian and the off-diagonal blocks to be related by conjugate transpose; the θ-fixed part $\mathfrak{k}$ is then the block-diagonal skew-Hermitian trace-zero matrices,

$$\mathfrak{k} = \mathfrak{s}\big(\mathfrak{u}(p) \oplus \mathfrak{u}(q)\big) = \{\mathrm{diag}(A, D) \mid A \in \mathfrak{u}(p),\ D \in \mathfrak{u}(q),\ \mathrm{tr}A + \mathrm{tr}D = 0\}$$

so $K = \mathrm{S}(\mathrm{U}(p) \times \mathrm{U}(q))$ at the group level, and $\mathfrak{p}$ is the space of off-diagonal Hermitian blocks. Positivity of $B_\theta(X,Y) = -\kappa(X, \theta Y)$ proportional to $\mathrm{tr}(XY^*)$ follows exactly as for $\mathfrak{sl}_n(\mathbb{R})$, since

$\operatorname{tr}(XX^*) = \sum_{i,j}|X_{ij}|^2 > 0$. The signature of κ on $\mathfrak{su}(p,q)$ is $\dim\mathfrak{p} - \dim\mathfrak{k} = 2pq - (p^2+q^2-1)$, the excess of the $2pq$ off-diagonal real directions over the compact block-diagonal ones.

The two examples $\mathfrak{sl}_n(\mathbb{R})$ and $\mathfrak{su}(p,q)$ exhibit the same pattern with the two natural conjugations of matrices: transpose gives the split form $\mathfrak{sl}_n(\mathbb{R})$ with $\mathfrak{k} = \mathfrak{so}(n)$, and conjugate transpose gives the Hermitian form $\mathfrak{su}(p,q)$ with $\mathfrak{k} = \mathfrak{s}(\mathfrak{u}(p)\oplus\mathfrak{u}(q))$. In each case θ is the negative of the relevant involution, $\mathfrak{k}$ is the compact subalgebra it fixes, and $\mathfrak{p}$ its Euclidean (-1)-eigenspace. Section 13.4 integrates θ to a group involution Θ with $K = G^\Theta$ and turns the vector decomposition $\mathfrak{g}_0 = \mathfrak{k}\oplus\mathfrak{p}$ into the global Cartan decomposition $G = K\exp(\mathfrak{p})$, and chapter 15 reads $\mathfrak{p}$ as the tangent space of the associated Riemannian symmetric space G/K.

Exercise 13.3.1

1. Verify in full that $\theta(X) = -X^{\mathsf{T}}$ is a Cartan involution of $\mathfrak{sl}_n(\mathbb{R})$. Show it is an involutive automorphism, identify its ±1-eigenspaces as $\mathfrak{so}(n)$ and the traceless symmetric matrices $\mathrm{Sym}_0(n)$, and confirm that $B_\theta(X,Y) = 2n\operatorname{tr}(XY^{\mathsf{T}})$ is positive-definite. Check directly that the bracket relations $[\mathfrak{k},\mathfrak{k}]\subseteq\mathfrak{k}$, $[\mathfrak{k},\mathfrak{p}]\subseteq\mathfrak{p}$, $[\mathfrak{p},\mathfrak{p}]\subseteq\mathfrak{k}$ hold by computing $[A,S]$ and $[S,S']$ for A antisymmetric and S,S' symmetric.

2. For $\mathfrak{g}_0 = \mathfrak{sl}_2(\mathbb{R})$, compute $\dim\mathfrak{k}$ and $\dim\mathfrak{p}$ explicitly by exhibiting bases of $\mathfrak{so}(2)$ and of the traceless symmetric 2×2 matrices, and deduce the signature $\operatorname{sig}(\kappa) = \dim\mathfrak{p} - \dim\mathfrak{k}$ of the Killing form of $\mathfrak{sl}_2(\mathbb{R})$. Confirm your answer against the Killing form $\kappa(X,Y) = 4\operatorname{tr}(XY)$ of $\mathfrak{sl}_2$ by diagonalizing it on the standard basis $\{H,E,F\}$.

3. Let $G = \mathrm{SL}(n,\mathbb{R})$ and define $\Theta\colon G\to G$ by $\Theta(g) = (g^{\mathsf{T}})^{-1}$. Show that Θ is a Lie group automorphism with $\Theta^2 = \mathrm{id}$, that its differential at the identity is the Cartan involution $\theta(X) = -X^{\mathsf{T}}$ of example 13.3.2, and that its fixed-point group $G^\Theta = \{g \mid (g^{\mathsf{T}})^{-1} = g\}$ is exactly $\mathrm{SO}(n)$. Conclude that $K = G^\Theta$ is the compact subgroup with Lie algebra $\mathfrak{k} = \mathfrak{so}(n)$.

4. Prove that for any Cartan involution θ of a real semisimple $\mathfrak{g}_0$, the B_θ-adjoint of $\mathrm{ad}Z$ is $-\mathrm{ad}(\theta Z)$ for every $Z\in\mathfrak{g}_0$. Deduce that $\mathrm{ad}X$ is B_θ-skew-symmetric for $X\in\mathfrak{k}$ and B_θ-symmetric for $X\in\mathfrak{p}$, and explain why this makes each $\mathrm{ad}X$ ($X\in\mathfrak{p}$) diagonalizable over $\mathbb{R}$.

5. Show that any two Cartan involutions θ_1,θ_2 of $\mathfrak{g}_0$ are conjugate by an inner automorphism. You may cite the uniqueness of the compact real form (theorem 13.2.5) and the alignment lemma (lemma 13.3.5); the task is to assemble them into the statement $\theta_2 = \varphi\theta_1\varphi^{-1}$ with $\varphi\in\mathrm{Int}(\mathfrak{g}_0)$, identifying which compact real forms θ_1,θ_2 produce.

6. Suppose $\mathfrak{g}_0$ is a compact real semisimple Lie algebra, so κ is negative-definite. Show that the identity map $\theta = \mathrm{id}$ is a Cartan involution, that $\mathfrak{k} = \mathfrak{g}_0$ and $\mathfrak{p} = 0$, and that the signature of κ is $-\dim\mathfrak{g}_0$. Interpret $\dim\mathfrak{p} = 0$ as the statement that a compact form has no noncompact directions.

13.4 The Global Cartan Decomposition and Maximal Compact Subgroups

The Cartan decomposition of section 13.3 splits the Lie algebra: $\mathfrak{g}_0 = \mathfrak{k}\oplus\mathfrak{p}$, with θ-eigenvalues $+1$ on $\mathfrak{k}$ and -1 on $\mathfrak{p}$. This section integrates that splitting to the group. The bracket relations $[\mathfrak{k},\mathfrak{k}]\subseteq\mathfrak{k}$,

$[\mathfrak{k},\mathfrak{p}] \subseteq \mathfrak{p}$, $[\mathfrak{p},\mathfrak{p}] \subseteq \mathfrak{k}$ (theorem 13.3.3) already show $\mathfrak{k}$ is a subalgebra but $\mathfrak{p}$ is not; the group-level statement must therefore be asymmetric, and it is: $K = \exp(\mathfrak{k})$ is a genuine subgroup while $\exp(\mathfrak{p})$ is only a submanifold, and the product map $K \times \mathfrak{p} \to G$ is a diffeomorphism. All of the topology of a noncompact semisimple group concentrates in the compact factor K, and the complementary directions $\mathfrak{p}$ form a contractible Euclidean cell, so $\pi_1(G) \cong \pi_1(K)$.

The mechanism is the polar decomposition of matrices, promoted from linear algebra to a statement about G by realizing G faithfully so that θ becomes inverse conjugate-transpose. Every element then factors as a rotation times a positive-definite stretch, and the stretch is the exponential of a symmetric matrix.

The proof of the global decomposition rests on the spectral theorem for positive-definite Hermitian matrices, which the argument uses twice: once to factor a matrix as unitary times positive-definite, and once to identify the positive-definite matrices with the exponentials of Hermitian ones. Both are standard linear algebra, recorded here as a lemma so the group-theoretic argument reads cleanly.

Lemma 13.4.1: Matrix Polar Decomposition

Let V be a finite-dimensional real or complex inner product space, and write $\mathrm{U}(V)$ for its orthogonal or unitary group, $\mathcal{H}(V)$ for the space of self-adjoint (symmetric or Hermitian) operators, and $\mathcal{P}(V) \subset \mathcal{H}(V)$ for the positive-definite self-adjoint operators.

1. The exponential map $\exp\colon \mathcal{H}(V) \to \mathcal{P}(V)$ is a diffeomorphism, with inverse the operator logarithm.
2. Every invertible operator g factors uniquely as $g = up$ with $u \in \mathrm{U}(V)$ and $p \in \mathcal{P}(V)$, given by $p = (g^*g)^{1/2}$ and $u = gp^{-1}$. This is the polar decomposition of [4, chapter 11.3], where existence and the uniqueness of both factors for invertible g are proved; the proof is repeated below in the notation used here.

***Proof*:**

Part 1. A self-adjoint operator X is diagonalizable in an orthonormal eigenbasis with real eigenvalues $\lambda_1, \dots, \lambda_n$ (the spectral theorem, [4, The Spectral Theorem for Self-Adjoint Operators over $\mathbb{R}$, The Spectral Theorem for Normal Operators]). In that basis $\exp X$ is diagonal with entries $e^{\lambda_i} > 0$, so $\exp X$ is positive-definite self-adjoint, and every element of $\mathcal{P}(V)$ arises this way: a positive-definite p has an orthonormal eigenbasis with eigenvalues $\mu_i > 0$, and $X = \log p$, diagonal with entries $\log \mu_i$ in the same basis, is self-adjoint with $\exp X = p$. The assignment $p \mapsto \log p$ is smooth, being given on each spectral projection by the smooth function log on $(0,\infty)$, and inverts exp; hence exp is a diffeomorphism $\mathcal{H}(V) \to \mathcal{P}(V)$.

Part 2. The operator g^*g is self-adjoint and positive-definite, since $\langle g^*g\, v, v\rangle = \langle gv, gv\rangle > 0$ for $v \neq 0$ (using that g is invertible), so its unique positive-definite square root $p = (g^*g)^{1/2}$ exists by Part 1 (take the square root eigenvalue by eigenvalue). Set $u = gp^{-1}$. Then

$$u^*u = p^{-1}g^*g\,p^{-1} = p^{-1}p^2p^{-1} = 1$$

so $u \in \mathrm{U}(V)$, and $g = up$ by construction. For uniqueness, if $g = u'p'$ with $u' \in \mathrm{U}(V)$ and $p' \in \mathcal{P}(V)$, then $g^*g = p'u'^*u'p' = p'^2$, so p' is a positive-definite square root of g^*g; the positive-definite square root is unique (the eigenvalue square root $\mu \mapsto \sqrt{\mu}$ is single-valued on $(0,\infty)$), forcing $p' = p$ and then $u' = gp'^{-1} = u$, completing the proof.

The lemma is the entire analytic input to the global decomposition. Part 1 says the positive-definite matrices are, as a manifold, just a copy of the Euclidean space $\mathcal{H}(V)$ under exp: no topology, contractible, a cell. Part 2 says every invertible matrix is a rotation times a stretch, uniquely, exactly as a nonzero complex number factors as a phase times a positive modulus. Realizing G inside $\mathrm{GL}(V)$ so that θ is $g \mapsto (g^*)^{-1}$ turns the two eigenspaces $\mathfrak{k}$ and $\mathfrak{p}$ into the skew-adjoint and self-adjoint matrices in $\mathfrak{g}_0$, and then K lands in $\mathrm{U}(V)$ and $\exp\mathfrak{p}$ in $\mathcal{P}(V)$; the group statement is the lemma restricted to G.

That realization is the one nontrivial reduction, and it is given by the Cartan involution itself.

Theorem 13.4.2: The Global Cartan Decomposition

Let G be a connected semisimple real Lie group with Lie algebra $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ the Cartan decomposition attached to a Cartan involution θ (theorem 13.3.3), and let K be the connected subgroup with Lie algebra $\mathfrak{k}$. Then the map

$$\Phi\colon K \times \mathfrak{p} \longrightarrow G, \qquad (k, X) \longmapsto k \exp X$$

is a diffeomorphism. In particular $G = K\exp(\mathfrak{p})$, every $g \in G$ factors uniquely as $g = k\exp X$ with $k \in K$ and $X \in \mathfrak{p}$, and K is closed in G and equals the fixed-point group G^{Θ} of the global Cartan involution Θ integrating θ.

***Proof*:**

The strategy is to realize G faithfully as a matrix group on which θ becomes inverse conjugate-transpose, so that lemma 13.4.1 applies, and then check that the polar factors of an element of G lie back in G.

Step 1: a θ-adapted faithful representation. Because G is semisimple, the Killing form κ is non-degenerate (theorem 6.3.1), and the adjoint representation $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g}_0)$ has discrete kernel $Z(G)$. Equip $\mathfrak{g}_0$ with the inner product $B_\theta(X, Y) = -\kappa(X, \theta Y)$, which is positive-definite by the definition of a Cartan involution (definition 13.3.1). For $X \in \mathfrak{g}_0$ the operator $\mathrm{ad}X$ has adjoint $-\mathrm{ad}(\theta X)$ with respect to B_θ: invariance of κ (definition 6.1.3) gives $\kappa([X, Y], Z) = -\kappa(Y, [X, Z])$, and unwinding the definition of B_θ turns this into $B_\theta(\mathrm{ad}X\, Y, Z) = -B_\theta(Y, \mathrm{ad}(\theta X)Z)$. Hence for $X \in \mathfrak{k}$ (where $\theta X = X$) the operator $\mathrm{ad}X$ is skew-adjoint, and for $X \in \mathfrak{p}$ (where $\theta X = -X$) it is self-adjoint, with respect to the fixed inner product B_θ. Passing to the group, $\mathrm{Ad}(\exp X) = \exp(\mathrm{ad}X)$ is orthogonal for $X \in \mathfrak{k}$ and positive-definite symmetric for $X \in \mathfrak{p}$. Thus in the orthonormal frame of $(\mathfrak{g}_0, B_\theta)$ the adjoint representation sends θ to $g \mapsto (g^{\mathsf{T}})^{-1}$: writing $V = (\mathfrak{g}_0, B_\theta)$, one has $\mathrm{Ad}(\Theta(g)) = (\mathrm{Ad}(g)^{\mathsf{T}})^{-1}$ on $\mathrm{Ad}(G)$, where Θ is the automorphism of G with differential θ. Concretely, in the θ-adapted realization $V = (\mathfrak{g}_0, B_\theta)$ just constructed, Θ is the restriction to G of the matrix map $g \mapsto (g^{\mathsf{T}})^{-1}$ (inverse conjugate-transpose in the orthonormal frame of V), an automorphism of $\mathrm{GL}(V)$ preserving G whose differential on $\mathfrak{g}_0$ is θ; on $\mathrm{Ad}(G)$ it acts by this transpose-inverse. Replacing G by $\mathrm{Ad}(G)$ loses only the discrete center, which the decomposition tolerates (Step 5); until then work with $\bar{G} = \mathrm{Ad}(G) \subseteq \mathrm{GL}(V)$, $\bar{K} = \mathrm{Ad}(K)$, and θ-adapted inner product on V.

Step 2: the polar factors of the group. The group $\bar{G} = \mathrm{Ad}(G)$ is closed in $\mathrm{GL}(V)$: it is the identity component of the automorphism group $\mathrm{Aut}(\mathfrak{g}_0)$, a real algebraic subgroup of $\mathrm{GL}(V)$ cut out by the polynomial bracket-preservation equations, hence itself closed (Step 3 records these defining equations). It is stable under $g \mapsto (g^{\mathsf{T}})^{-1}$, that is $\Theta(\bar{G}) = \bar{G}$. Its Lie algebra $\mathrm{ad}(\mathfrak{g}_0) \subseteq \mathfrak{gl}(V)$ decomposes as $\mathrm{ad}\mathfrak{k} \oplus \mathrm{ad}\mathfrak{p}$ into skew-symmetric and symmetric matrices, by Step 1. Write $\mathrm{O}(V)$

for the orthogonal group of V and $\mathcal{P}(V)$, $\mathcal{H}(V)$ as in lemma 13.4.1. The subgroup $\bar{G} \cap \mathrm{O}(V)$ is closed in $\bar{G}$; its Lie algebra is $\mathrm{ad}\mathfrak{g}_0 \cap (\text{skew-symmetric}) = \mathrm{ad}\mathfrak{k}$, so its identity component is the connected subgroup $\mathrm{Ad}(K) = \bar{K}$ with that Lie algebra. Likewise $\exp(\mathrm{ad}\mathfrak{p}) \subseteq \bar{G} \cap \mathcal{P}(V)$, since $\mathrm{ad}\mathfrak{p}$ is symmetric. Steps 3 and 4 show these inclusions are equalities: $\bar{G} \cap \mathcal{P}(V) = \exp(\mathrm{ad}\mathfrak{p})$ and $\bar{G} \cap \mathrm{O}(V) = \bar{K}$.

Step 3: the polar factorization on $\bar{G}$. Let $g \in \bar{G}$. By lemma 13.4.1 write $g = up$ with $u \in \mathrm{O}(V)$ and $p = (g^{\mathsf{T}}g)^{1/2} \in \mathcal{P}(V)$. Since $\bar{G}$ is Θ-stable, $\Theta(g) = (g^{\mathsf{T}})^{-1} \in \bar{G}$, hence $g^{\mathsf{T}} = \Theta(g)^{-1} \in \bar{G}$ and $g^{\mathsf{T}}g = \Theta(g)^{-1}g \in \bar{G}$, so $g^{\mathsf{T}}g \in \bar{G} \cap \mathcal{P}(V)$. Write $q = g^{\mathsf{T}}g$ and $q = \exp Y$ with $Y = \log q$ the symmetric logarithm; the claim is that the whole real curve $t \mapsto q^t = \exp(tY)$ lies in $\bar{G}$, whence $Y \in \mathrm{ad}\mathfrak{g}_0$. This is the one delicate analytic point, and integer powers alone do not deliver it: it must be forced from the *defining equations* of $\bar{G}$. The group $\bar{G} = \mathrm{Ad}(G)$ is the identity component of the automorphism group $\mathrm{Aut}(\mathfrak{g}_0) = \{a \in \mathrm{GL}(V) \mid a[\xi,\eta] = [a\xi, a\eta] \text{ for all } \xi,\eta\}$, which is a real algebraic subgroup of $\mathrm{GL}(V)$: it is cut out by the real polynomial equations $P_j(a) = 0$ in the entries of a expressing $a[e_r, e_s] = [ae_r, ae_s]$ on a basis $\{e_r\}$ of $\mathfrak{g}_0$. Since $\bar{G}$ is the identity component of $\mathrm{Aut}(\mathfrak{g}_0)$ it satisfies these same equations. Diagonalize $q = R\,\mathrm{diag}(\mu_i)R^{\mathsf{T}}$ ($R \in \mathrm{O}(V)$, $\mu_i > 0$); then each entry of $q^t = R\,\mathrm{diag}(\mu_i^t)R^{\mathsf{T}}$ is an exponential sum $\sum_i c_i\, e^{t\lambda_i}$ with $\lambda_i = \log \mu_i$ real, so each polynomial $P_j(q^t)$, being a polynomial in these entries, is again a real generalized Dirichlet polynomial $\sum_m d_m e^{t\rho_m}$ in t with finitely many real frequencies ρ_m. This function vanishes at every integer t, since $q^t = q^n \in \bar{G} \subseteq \mathrm{Aut}(\mathfrak{g}_0)$ there; but a generalized Dirichlet polynomial that is not identically zero has only finitely many real zeros, so $P_j(q^t) \equiv 0$ for all real t. Hence $q^t \in \mathrm{Aut}(\mathfrak{g}_0)$ for every $t \in \mathbb{R}$, and since $t \mapsto q^t$ is a continuous curve starting at $q^0 = I \in \bar{G}$ and staying in $\mathrm{Aut}(\mathfrak{g}_0)$, it lies in the identity component $\bar{G}$ throughout. Thus $t \mapsto q^t$ is a one-parameter subgroup of $\bar{G}$, and its generator $Y = \log q \in \mathrm{ad}\mathfrak{g}_0 = \mathrm{Lie}(\bar{G})$. Being symmetric, $Y \in \mathrm{ad}\mathfrak{g}_0 \cap (\text{symmetric}) = \mathrm{ad}\mathfrak{p}$ by Step 1. In particular $p = (g^{\mathsf{T}}g)^{1/2} = q^{1/2} \in \bar{G} \cap \mathcal{P}(V)$ and $\log p = \frac{1}{2}Y \in \mathrm{ad}\mathfrak{p}$, so $p = \exp(\mathrm{ad}X)$ for a unique $X \in \mathfrak{p}$; then $u = gp^{-1} \in \bar{G} \cap \mathrm{O}(V)$. This proves the reverse inclusions of Step 2 for the positive factors, and exhibits every $g \in \bar{G}$ as $g = u\exp(\mathrm{ad}X)$ with $u \in \bar{G} \cap \mathrm{O}(V)$ and $X \in \mathfrak{p}$. The factorization is unique: if $u\exp(\mathrm{ad}X) = u'\exp(\mathrm{ad}X')$ then, by the uniqueness in lemma 13.4.1, $u = u'$ and $\exp(\mathrm{ad}X) = \exp(\mathrm{ad}X')$, whence $\mathrm{ad}X = \mathrm{ad}X'$ (injectivity of exp on symmetric matrices, lemma 13.4.1(1)) and $X = X'$ (injectivity of ad on the semisimple $\mathfrak{g}_0$). Thus the polar map

$$(\bar{G} \cap \mathrm{O}(V)) \times \mathfrak{p} \longrightarrow \bar{G}, \qquad (u, X) \longmapsto u\exp(\mathrm{ad}X)$$

is a bijection; it is smooth with smooth inverse $g \mapsto \big(g\,(g^{\mathsf{T}}g)^{-1/2},\, X\big)$, where $X \in \mathfrak{p}$ is determined by $\mathrm{ad}X = \frac{1}{2}\log(g^{\mathsf{T}}g)$ (the maps $g \mapsto (g^{\mathsf{T}}g)^{\pm 1/2}$ and log are smooth by lemma 13.4.1, and ad is a linear isomorphism onto $\mathrm{ad}\mathfrak{p}$), hence a diffeomorphism onto $\bar{G}$.

Step 4: the orthogonal factor is connected, so $\bar{G} \cap \mathrm{O}(V) = \bar{K}$. This is the point at which the argument must not assume what it is proving. The equality $\bar{G} \cap \mathrm{O}(V) = \bar{K}$ says the orthogonal factor is *connected*, and connectedness is forced independently, without reference to any factorization of G itself. The group $\bar{G} = \mathrm{Ad}(G)$ is connected, being the continuous image of the connected G. By Step 3 it is diffeomorphic to $(\bar{G} \cap \mathrm{O}(V)) \times \mathfrak{p}$, a product whose second factor $\mathfrak{p} \cong \mathbb{R}^{\dim \mathfrak{p}}$ is connected; a product of two spaces is connected only if each factor is, so $\bar{G} \cap \mathrm{O}(V)$ is connected. A connected group equals its own identity component, which by Step 2 is $\bar{K}$. Hence $\bar{G} \cap \mathrm{O}(V) = \bar{K}$, and Step 3 becomes $\bar{G} = \bar{K}\exp(\mathrm{ad}\mathfrak{p})$ with the polar map $\bar{K} \times \mathfrak{p} \to \bar{G}$ a diffeomorphism. An element $u \in \bar{K}$ has trivial positive polar factor, so $\Theta(u) = (u^{\mathsf{T}})^{-1} = u$; conversely a Θ-fixed $g = u\exp(\mathrm{ad}X)$ in $\bar{G}$ has $u\exp(-\mathrm{ad}X) = \Theta(g) = g = u\exp(\mathrm{ad}X)$, forcing $X = 0$ by uniqueness, so $\bar{K}$ is exactly the Θ-fixed subgroup $\bar{G}^{\Theta}$ of $\bar{G}$.

Step 5: lifting to G. It remains to lift $\bar{G} = \bar{K}\exp(\mathrm{ad}\mathfrak{p})$ through the covering $\mathrm{Ad}\colon G \to \bar{G}$, whose kernel is the discrete center $Z(G)$. The map Φ is injective: if $k\exp X = k'\exp X'$ with $k, k' \in K$ and $X, X' \in \mathfrak{p}$, applying Ad and the uniqueness of the polar map on $\bar{G}$ (Step 3, with $\bar{K}$ in place of $\bar{G} \cap \mathrm{O}(V)$) gives $\mathrm{Ad}(k) = \mathrm{Ad}(k')$ and $\exp(\mathrm{ad}X) = \exp(\mathrm{ad}X')$, hence $\mathrm{ad}X = \mathrm{ad}X'$ and $X = X'$ as in Step 3, and then $k = k'$. The differential of Φ at $(e, 0)$ is the identity $\mathfrak{k} \oplus \mathfrak{p} \to \mathfrak{g}_0$, an isomorphism, and left translation by k together with the group structure make Φ a local diffeomorphism at every point.

For surjectivity, the one central fact needed is that Θ fixes $Z(G)$ pointwise, equivalently $Z(G) \subseteq G^{\Theta} = K$. This does not rely on any factorization of $g \in G$. A central z commutes with the connected G, hence with the compactly embedded K (whose Lie algebra $\mathfrak{k}$ is a maximal compactly embedded subalgebra, theorem 13.3.3); the connected abelian group generated by z and a maximal torus of K is again compactly embedded and contains that torus, so by maximality it lies in K, giving $z \in K$ and thus $Z(G) \subseteq K$. Since $\bar{K} = \bar{G}^{\Theta}$ (Step 4) and $\mathrm{Ad}(K) = \bar{K}$, the fixed subgroup G^{Θ} has identity component K and contains $Z(G)$, so $\Theta(z) = z$ for every $z \in Z(G)$. Now let $g \in G$. Its image $\mathrm{Ad}(g) = \bar{k}\exp(\mathrm{ad}X)$ with $\bar{k} = \mathrm{Ad}(k_0)$, $k_0 \in K$, $X \in \mathfrak{p}$; then $\mathrm{Ad}\big(g(k_0\exp X)^{-1}\big) = 1$, so $g(k_0\exp X)^{-1} = z \in Z(G)$. As $z \in K$ and z is central, $g = z\,k_0\exp X = (zk_0)\exp X \in K\exp(\mathfrak{p})$, proving surjectivity. A bijective local diffeomorphism is a diffeomorphism ([13, Properties of Local Diffeomorphisms]), so $\Phi\colon K \times \mathfrak{p} \to G$ is a diffeomorphism. Finally K is closed, being the Ad-preimage of the closed $\bar{K}$ (it equals G^{Θ}, closed as the fixed set of the continuous Θ), and $K = G^{\Theta}$, completing the proof.

The theorem is the group-level Cartan decomposition: G is diffeomorphic to $K \times \mathbb{R}^{\dim \mathfrak{p}}$, a compact factor times a Euclidean space. The compact factor K carries all the topology; the Euclidean factor $\exp(\mathfrak{p})$ is a contractible cell, diffeomorphic to $\mathfrak{p}$ under exp. This is the precise sense in which removing compactness changes nothing topologically. The noncompact directions are a featureless cell glued onto a compact group. The identity $K = G^{\Theta}$ records that K is intrinsic, not a choice: it is the fixed group of the global Cartan involution, so any two Cartan involutions, being conjugate (theorem 13.3.4), yield conjugate K.

The first corollary of the product being a diffeomorphism is that K is as large a compact subgroup as G admits, and that this pins down the topology of G completely.

Theorem 13.4.3: Maximal Compact Subgroups

Let G be a connected semisimple real Lie group with Cartan decomposition $G = K\exp(\mathfrak{p})$ as in theorem 13.4.2. Then:

1. K is a maximal compact subgroup of G.
2. Every compact subgroup of G is conjugate to a subgroup of K; in particular any two maximal compact subgroups are conjugate.
3. The map $(k, X, t) \mapsto k\exp((1-t)X)$ is a deformation retraction of G onto K, so G is homotopy equivalent to K; consequently $\pi_1(G) \cong \pi_1(K)$. When K is moreover semisimple this group is finite (theorem 9.4.5); in general K is compact but only reductive, with a central torus, and $\pi_1(G)$ can be infinite.

***Proof*:**
Part 3, first: the retraction. Define $r\colon G \times [0,1] \to G$ by $r(k\exp X, t) = k\exp((1-t)X)$, using the unique factorization of theorem 13.4.2; in the coordinates $G \cong K \times \mathfrak{p}$ this is $(k, X, t) \mapsto (k, (1-t)X)$, continuous. At $t = 0$ it is the identity, at $t = 1$ it lands in $K = K \times \{0\}$, and it fixes K pointwise throughout, so r is a deformation retraction of G onto K. A deformation retraction is a homotopy equivalence, so $\pi_1(G) \cong \pi_1(K)$. The compact connected group K is reductive; *when* it is semisimple, theorem 9.4.5 makes $\pi_1(K)$ finite, but K carries in general a central torus contributing a free abelian summand to $\pi_1(K)$, so $\pi_1(G)$ need not be finite. The example of $\mathrm{SL}(2,\mathbb{R})$ below, where $K = \mathrm{SO}(2)$ is a pure torus and $\pi_1(G) \cong \mathbb{Z}$, shows the finiteness fails.

Part 2: conjugation into K. The argument turns on a fixed-point property of the space $P = \exp(\mathfrak{p}) \cong \mathfrak{p}$, which is established here by an averaging construction internal to the matrix model, requiring no geometry beyond theorem 13.4.2. In that model P is the set $\bar{G} \cap \mathcal{P}(V)$ of positive-definite elements of $\bar{G}$, on which $\bar{G}$ acts by $g \cdot p = gp\,\Theta(g)^{-1} = gp\,g^{\mathsf{T}}$; this action is transitive with stabilizer of the base point $1 \in P$ equal to $\bar{K}$ (an element fixes 1 iff $gg^{\mathsf{T}} = 1$, that is $g \in \bar{G} \cap \mathrm{O}(V) = \bar{K}$), so $\bar{G}/\bar{K} \cong P$. Two elementary facts about P carry the argument, both consequences of Step 3 of theorem 13.4.2: first, P is closed under the midpoint operation $m(p,q) = p^{1/2}(p^{-1/2}qp^{-1/2})^{1/2}p^{1/2}$, since each of $p^{\pm 1/2}$ and $(p^{-1/2}qp^{-1/2})^{1/2}$ is a real power of an element of P and hence again in $\bar{G}$ (Step 3), while the product is positive-definite; second, the function

$$d(p,q)^2 = \operatorname{tr}\bigl(\log(p^{-1/2}qp^{-1/2})\bigr)^2$$

is, for each fixed q, a strictly convex function of $p \in P$ along these midpoint geodesics, its unique critical point being $p = q$ (the log of a positive matrix being computed eigenvalue by eigenvalue, this reduces to the strict convexity of $s \mapsto (\log s)^2$ composed with the geodesic, a one-variable check).

Let $H \subseteq G$ be a compact subgroup and $\bar{H} = \mathrm{Ad}(H) \subseteq \bar{G}$ its (compact) image. Averaging over the normalized Haar measure of H, the function $p \mapsto \int_H d(\bar{h}\cdot p,\, 1)^2\, dh$ is strictly convex on P and proper, so it has a unique minimizer $p_* \in P$; because the metric d is $\bar{G}$-invariant and Haar measure is invariant, replacing p by $\bar{h}_0 \cdot p$ permutes the integrand, so p_* is fixed by every $\bar{h}_0 \in \bar{H}$. The invariance $d(g\cdot p, g\cdot q) = d(p,q)$ holds because $d(p,q)^2$ depends only on the eigenvalues of $p^{-1}q$, and $(g\cdot p)^{-1}(g\cdot q) = (g^{\mathsf{T}})^{-1}(p^{-1}q)g^{\mathsf{T}}$ is similar to $p^{-1}q$, so those eigenvalues are unchanged. Writing $p_* = g\cdot 1$ with $g \in \bar{G}$, the stabilizer of p_* is $g\bar{K}g^{-1}$, so $\bar{H} \subseteq g\bar{K}g^{-1}$. Lifting through Ad (whose kernel $Z(G) \subseteq K$ lies in every conjugate of K) and choosing $g_0 \in G$ with $\mathrm{Ad}(g_0) = g$ gives $g_0^{-1}Hg_0 \subseteq K$. The averaging here is the same construction as in exercise 13.4.1 (5), which runs it explicitly for $H \subseteq \mathrm{GL}(n,\mathbb{R})$ and $K = \mathrm{O}(n)$.

Part 1 and the rest of Part 2. Suppose $H \supseteq K$ is a compact subgroup. By Part 2, $H \subseteq gKg^{-1}$ for some $g \in G$, so $K \subseteq H \subseteq gKg^{-1}$. Now K and gKg^{-1} are diffeomorphic compact connected groups, hence of equal dimension, and K is a closed subgroup of the connected gKg^{-1} of the same dimension; a closed subgroup of full dimension contains the identity component, and gKg^{-1} is connected, so $K = gKg^{-1}$ and therefore $H = K$. Thus K is maximal compact, proving Part 1. For the conjugacy of maximal compacts, let K' be any maximal compact subgroup; by Part 2 it conjugates into K, say $g^{-1}K'g \subseteq K$, and maximality of K' (hence of the compact $g^{-1}K'g$) forces $g^{-1}K'g = K$, so K' and K are conjugate, completing the proof.

The theorem completes the topological picture. Not only is G diffeomorphic to $K \times \mathbb{R}^{\dim \mathfrak{p}}$, but K is intrinsic up to conjugacy, and every compact subgroup sits inside a conjugate of it. The fundamental group, the homotopy type, indeed all the homotopy invariants of the noncompact G are those of its

maximal compact K, which is where Part III already computed them: $\pi_1(G)$ is not a new invariant of noncompact groups but $\pi_1(K)$ transported across the deformation retraction. Whether it is finite depends on K: for a compact *semisimple* K the finiteness of theorem 9.4.5 applies, but the maximal compact of a noncompact semisimple G is only compact reductive, and its central torus can make $\pi_1(K)$ infinite, as $K = \mathrm{SO}(2)$ inside $\mathrm{SL}(2,\mathbb{R})$ shows below. The convexity computed by hand in Part 2 is a consequence of the nonpositive curvature of $P = \exp(\mathfrak{p})$; chapter 15 recasts the same P as the symmetric space G/K, where this midpoint convexity becomes the Cartan-Hadamard property and the averaging fixed point becomes the Riemannian center of mass.

Specializing the global decomposition to the classical groups recovers the polar decomposition of matrices in its familiar form, now read as a structural fact about the group.

Corollary 13.4.4: The Classical Polar Decomposition

1. Every invertible real matrix factors uniquely as $g = kp$ with $k \in \mathrm{O}(n)$ orthogonal and p positive-definite symmetric; writing $p = \exp X$ with X symmetric gives $\mathrm{GL}(n,\mathbb{R}) = \mathrm{O}(n)\exp(\mathrm{Sym}(n))$.
2. $\mathrm{SL}(n,\mathbb{R}) = \mathrm{SO}(n)\exp(\mathrm{Sym}_0(n))$, where $\mathrm{Sym}_0(n)$ is the traceless symmetric matrices; every $g \in \mathrm{SL}(n,\mathbb{R})$ is $\mathrm{SO}(n)$ times the exponential of a traceless symmetric matrix, uniquely.
3. Every invertible complex matrix factors uniquely as $g = up$ with $u \in \mathrm{U}(n)$ unitary and p positive-definite Hermitian; and $\mathrm{SU}(p,q) = \mathrm{S}(\mathrm{U}(p)\times\mathrm{U}(q))\exp(\mathfrak{p})$, where $\mathrm{S}(\mathrm{U}(p)\times\mathrm{U}(q))$ is the maximal compact and $\mathfrak{p}$ the corresponding self-adjoint part.

***Proof*:**
Parts 1 and 3 are lemma 13.4.1(2) restated (real and complex cases), with the positive factor written as $\exp X$ by lemma 13.4.1(1). Part 2 is theorem 13.4.2 for $G = \mathrm{SL}(n,\mathbb{R})$ with the Cartan involution $\theta(X) = -X^{\mathsf{T}}$: then $\mathfrak{k} = \mathfrak{so}(n)$, so $K = \mathrm{SO}(n)$, and $\mathfrak{p} = \mathrm{Sym}_0(n)$; equivalently, given $g \in \mathrm{SL}(n,\mathbb{R})$, the polar factor $p = (g^{\mathsf{T}}g)^{1/2}$ has determinant $\det(g^{\mathsf{T}}g)^{1/2} = |\det g| = 1$ and is positive-definite symmetric, so $p = \exp X$ with $X = \log p$ symmetric and $\mathrm{tr}X = \log\det p = 0$, while $k = gp^{-1} \in \mathrm{SO}(n)$. The $\mathrm{SU}(p,q)$ factorization is theorem 13.4.2 for that group; its Cartan decomposition is computed in example 13.5.2. Uniqueness in each part is the uniqueness of the polar decomposition, completing the proof.

The corollary is the reason the abstract theorem feels familiar: it is the same polar decomposition one meets in linear algebra, $g = kp$ with k a rotation and p a positive stretch, promoted to the assertion that the rotation part sweeps out the maximal compact K and the stretch part sweeps out the Euclidean cell $\exp(\mathfrak{p})$. For $\mathrm{GL}(n,\mathbb{R})$, which is reductive rather than semisimple, the same statement holds with $K = \mathrm{O}(n)$ (the reductive case adds only a central $\mathbb{R}^\times$ of scalars, which contributes a further Euclidean log-direction); here $\mathrm{O}(n)$ is disconnected, its two components detecting $\det = \pm 1$, so it falls outside the literal hypothesis of theorem 13.4.3(1), which was stated for the connected K of a semisimple group, even though it remains maximal compact. The semisimple restriction $\mathrm{SL}(n,\mathbb{R})$ is the clean case where $K = \mathrm{SO}(n)$ is connected.

The running example is $\mathrm{SL}(2,\mathbb{R})$, where every ingredient is a 2×2 matrix and the decomposition can be watched directly.

Example 13.4.5: The Cartan Decomposition of $\mathrm{SL}(2,\mathbb{R})$

For $G = \mathrm{SL}(2,\mathbb{R})$ with $\theta(X) = -X^{\mathsf{T}}$, the eigenspaces are

$$\mathfrak{k} = \mathfrak{so}(2) = \left\{ \left(\begin{smallmatrix} 0 & -\vartheta \\ \vartheta & 0 \end{smallmatrix}\right) \;\middle|\; \vartheta \in \mathbb{R} \right\}, \qquad \mathfrak{p} = \left\{ \left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right) \;\middle|\; a, b \in \mathbb{R} \right\}$$

so $\mathfrak{p}$ is the traceless symmetric matrices, $\dim \mathfrak{p} = 2$. Exponentiating $\mathfrak{k}$ gives $K = \mathrm{SO}(2) = \left\{ k_\varphi = \left(\begin{smallmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{smallmatrix}\right) \;\middle|\; \varphi \in \mathbb{R} \right\}$, the circle. Theorem 13.4.2 then reads

$$\mathrm{SL}(2,\mathbb{R}) = \mathrm{SO}(2)\exp(\mathfrak{p}) \cong S^1 \times \mathbb{R}^2$$

Concretely, given $g \in \mathrm{SL}(2,\mathbb{R})$, form $p = (g^{\mathsf{T}} g)^{1/2}$, a positive-definite symmetric matrix of determinant 1, so $p = \exp\left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$ for a unique traceless symmetric exponent, and $k = gp^{-1} \in \mathrm{SO}(2)$. For instance $g = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 1/2 \end{smallmatrix}\right)$ is already positive-definite symmetric, so $k = I$ and $g = \exp\left(\begin{smallmatrix} \log 2 & 0 \\ 0 & -\log 2 \end{smallmatrix}\right)$; the shear $g = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has $g^{\mathsf{T}} g = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, whose positive square root is the symmetric part p and $k = gp^{-1}$ the rotation.

The retraction $(k_\varphi, X, t) \mapsto k_\varphi \exp((1-t)X)$ collapses the $\mathbb{R}^2$ of $\mathfrak{p}$ to the origin, retracting $\mathrm{SL}(2,\mathbb{R})$ onto the circle $\mathrm{SO}(2)$. Hence

$$\pi_1(\mathrm{SL}(2,\mathbb{R})) \cong \pi_1(\mathrm{SO}(2)) \cong \mathbb{Z}$$

so $\mathrm{SL}(2,\mathbb{R})$ has infinite cyclic fundamental group, and its universal cover $\widetilde{\mathrm{SL}(2,\mathbb{R})}$ is an infinite-sheeted cover (proposition 3.4.1) that is not a matrix group, the noncompact counterpart of the finite double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5.

The $\mathrm{SL}(2,\mathbb{R})$ computation exhibits the whole theorem in two dimensions: the compact $\mathrm{SO}(2)$ carries the single circle's worth of topology, the plane $\mathfrak{p}$ contributes nothing, and the fundamental group is that of the circle. The contrast with the compact sibling $\mathrm{SU}(2) \cong S^3$ is instructive. $\mathrm{SU}(2)$ is simply connected because its maximal torus is a circle but the group is a 3-sphere, whereas $\mathrm{SL}(2,\mathbb{R})$, sharing the complexified Lie algebra $\mathfrak{sl}_2(\mathbb{C})$, retracts onto its maximal compact $\mathrm{SO}(2)$ and inherits that circle's $\pi_1 = \mathbb{Z}$. The noncompact real form and the compact real form of the same complex algebra thus have opposite topological character, one contractible-times-a-sphere, the other contractible-times-a-circle, a difference invisible at the Lie-algebra level and made precise only by the global decomposition.

The infinite fundamental group of $\mathrm{SL}(2,\mathbb{R})$ produces the tower of nontrivial covers, among them the metaplectic double cover, whose genuine (non-matrix) representations include the discrete series of noncompact groups. The decomposition $G = K\exp(\mathfrak{p})$ is the first of the two structural factorizations that organize a noncompact semisimple group. Chapter 14 refines the noncompact directions $\exp(\mathfrak{p})$ by introducing a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$ and the restricted roots, producing the finer Iwasawa decomposition $G = KAN$, and chapter 15 reads the coset space $G/K \cong \exp(\mathfrak{p})$ as a Riemannian symmetric space with $\mathfrak{p}$ as its tangent model at the base point.

Exercise 13.4.1

1. Compute the polar decomposition $g = kp$ of the matrix $g = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$: form $g^{\mathsf{T}} g$, find its positive-definite square root p, and hence $k = gp^{-1}$. Check that $k \in \mathrm{SO}(2)$ and that $p = \exp X$ for a traceless symmetric X.
2. Prove directly that $\exp\colon \mathfrak{p} \to \exp(\mathfrak{p})$ is a diffeomorphism onto the positive-definite symmetric

matrices of $\mathrm{SL}(n,\mathbb{R})$ (i.e. lemma 13.4.1(1) restricted to traceless symmetric matrices), by exhibiting the inverse and checking smoothness of both maps via the spectral decomposition.

3. Show that the map $r\colon \mathrm{SL}(2,\mathbb{R})\times[0,1]\to\mathrm{SL}(2,\mathbb{R})$, $r(k\exp X,t)=k\exp((1-t)X)$, is a well-defined deformation retraction onto $\mathrm{SO}(2)$, using the uniqueness of the factorization from theorem 13.4.2. Deduce $\pi_1(\mathrm{SL}(2,\mathbb{R}))\cong\mathbb{Z}$.

4. Prove that $\mathrm{O}(n)$ is a maximal compact subgroup of $\mathrm{GL}(n,\mathbb{R})$: show that any subgroup of $\mathrm{GL}(n,\mathbb{R})$ strictly containing $\mathrm{O}(n)$ is noncompact, by exhibiting an unbounded one-parameter family inside it.

5. Let $H\subseteq\mathrm{GL}(n,\mathbb{R})$ be a compact subgroup. Prove that H is conjugate into $\mathrm{O}(n)$ by averaging: put $\langle u,v\rangle_H=\int_H\langle hu,hv\rangle\,dh$ for the standard inner product $\langle-,-\rangle$ and normalized Haar measure on H, show $\langle-,-\rangle_H$ is an H-invariant inner product, and conclude that in an orthonormal basis for $\langle-,-\rangle_H$ every element of H is orthogonal. Relate this to theorem 13.4.3(2).

6. Verify the inclusion $Z(G)\subseteq K$ of theorem 13.4.2 in the concrete cases $G=\mathrm{SL}(n,\mathbb{R})$ and $G=\mathrm{SL}(2,\mathbb{R})$: compute the center directly (the scalar matrices ζI with $\zeta^n=1$ and $\det=1$; for $n=2$ this is $\{\pm I\}$) and check by hand that each central element is orthogonal, hence lies in $K=\mathrm{SO}(n)$, so its $\exp(\mathfrak{p})$-factor is trivial. Confirm that this is consistent with $\Theta(z)=z$ for $\theta(X)=-X^{\mathsf T}$.

13.5 Examples: $\mathrm{SL}(n,\mathbb{R})$, $\mathrm{SU}(p,q)$, $\mathrm{SL}(2,\mathbb{R})$

The general theory now has a bill to pay. The Cartan involution of section 13.3 and the global decomposition $G=K\exp(\mathfrak{p})$ of section 13.4 were proved once, abstractly, for every connected semisimple real group. This section cashes them into three families where the involution is a visible operation on matrices, the eigenspaces $\mathfrak{k}$ and $\mathfrak{p}$ are read off by inspection, and the diffeomorphism $K\times\mathfrak{p}\to G$ is the classical polar decomposition. The three examples are chosen to exhibit the full range of the theory: $\mathrm{SL}(n,\mathbb{R})$, where $\mathfrak{p}$ is the traceless symmetric matrices; $\mathrm{SU}(p,q)$, where the compact part K is a product of two unitary groups and the quotient is a Hermitian symmetric space; and $\mathrm{SL}(2,\mathbb{R})$, the running noncompact sibling of $\mathrm{SU}(2)$, worked end to end down to its action on the upper half-plane. Each closes with a symmetric space G/K whose geometry chapter 15 takes up in full. The first example is the one every general statement of the chapter was written against.

Example 13.5.1: The Cartan Decomposition of $\mathrm{SL}(n,\mathbb{R})$

Let $G=\mathrm{SL}(n,\mathbb{R})$, with Lie algebra $\mathfrak{g}_0=\mathfrak{sl}(n,\mathbb{R})$, the real traceless $n\times n$ matrices. The map

$$\theta(X)=-X^{\mathsf T}$$

is an automorphism of $\mathfrak{g}_0$: it is linear, it preserves the trace-zero condition, and it respects the bracket, since $\theta[X,Y]=-[X,Y]^{\mathsf T}=-(Y^{\mathsf T}X^{\mathsf T}-X^{\mathsf T}Y^{\mathsf T})=[-X^{\mathsf T},-Y^{\mathsf T}]=[\theta X,\theta Y]$. It is involutive because $(X^{\mathsf T})^{\mathsf T}=X$. That θ is a Cartan involution (definition 13.3.1) is the computation $B_\theta(X,Y)=-\kappa(X,\theta Y)=2n\,\mathrm{tr}(XY^{\mathsf T})$, using the Killing form $\kappa(X,Y)=2n\,\mathrm{tr}(XY)$ of $\mathfrak{sl}(n,\mathbb{R})$ (example 6.1.8); on the diagonal $B_\theta(X,X)=2n\,\mathrm{tr}(XX^{\mathsf T})=2n\sum_{i,j}X_{ij}^2>0$ is the sum of the squares of the entries, positive-definite.

The ± 1-eigenspaces of θ split $\mathfrak{g}_0$ by the symmetry of a matrix. On the $+1$-eigenspace $-X^{\mathsf{T}} = X$, so $X^{\mathsf{T}} = -X$: these are the antisymmetric matrices, automatically traceless, and

$$\mathfrak{k} = \mathfrak{so}(n) = \left\{ X \in \mathfrak{gl}(n,\mathbb{R}) \;\middle|\; X^{\mathsf{T}} = -X \right\}$$

is the Lie algebra of the rotation group. On the -1-eigenspace $-X^{\mathsf{T}} = -X$, so $X^{\mathsf{T}} = X$: the symmetric matrices, cut down to trace zero by membership in $\mathfrak{sl}$, giving

$$\mathfrak{p} = \mathrm{Sym}_0(n) = \left\{ X \in \mathfrak{gl}(n,\mathbb{R}) \;\middle|\; X^{\mathsf{T}} = X,\ \mathrm{tr} X = 0 \right\}$$

the traceless symmetric matrices. The bracket relations of theorem 13.3.3 are visible: a bracket of two antisymmetric matrices is antisymmetric ($[\mathfrak{k},\mathfrak{k}] \subseteq \mathfrak{k}$), a bracket of an antisymmetric and a symmetric matrix is symmetric ($[\mathfrak{k},\mathfrak{p}] \subseteq \mathfrak{p}$), and a bracket of two symmetric matrices is antisymmetric ($[\mathfrak{p},\mathfrak{p}] \subseteq \mathfrak{k}$), each a one-line transpose check.

Dimensions confirm the split. An antisymmetric matrix is determined by its strictly upper triangle, so $\dim \mathfrak{k} = \binom{n}{2} = \frac{n(n-1)}{2}$; a traceless symmetric matrix is determined by its upper triangle including the diagonal, minus the one trace relation, so $\dim \mathfrak{p} = \binom{n+1}{2} - 1 = \frac{n(n+1)}{2} - 1$. Their sum is $\frac{n(n-1)}{2} + \frac{n(n+1)}{2} - 1 = n^2 - 1 = \dim \mathfrak{sl}(n,\mathbb{R})$, as it must be.

On the group, $K = \exp(\mathfrak{k})$ is the connected rotation group $\mathrm{SO}(n)$, and theorem 13.4.2 makes

$$\mathrm{SO}(n) \times \mathrm{Sym}_0(n) \to \mathrm{SL}(n,\mathbb{R}), \qquad (k, X) \mapsto k \exp X$$

a diffeomorphism, so $\mathrm{SL}(n,\mathbb{R}) = \mathrm{SO}(n)\exp(\mathrm{Sym}_0(n))$. Concretely this is the polar decomposition (corollary 13.4.4): every $g \in \mathrm{SL}(n,\mathbb{R})$ factors uniquely as $g = k\,p$ with $k \in \mathrm{SO}(n)$ orthogonal and $p = \exp X$ a positive-definite symmetric matrix of determinant one, obtained from g by $p = (g^{\mathsf{T}} g)^{1/2}$ and $k = gp^{-1}$. The maximal compact subgroup is $\mathrm{SO}(n)$, and $\mathrm{SL}(n,\mathbb{R})$ deformation-retracts onto it (theorem 13.4.3), so $\pi_1(\mathrm{SL}(n,\mathbb{R})) = \pi_1(\mathrm{SO}(n))$, which is $\mathbb{Z}/2$ for $n \geq 3$ and $\mathbb{Z}$ for $n = 2$.

The example is the chapter in miniature: an involution one can write down, two eigenspaces distinguished by a single sign, and a global factorization that is the matrix polar decomposition. The nonnegative quantity that drives everything is $\mathrm{tr}(XX^{\mathsf{T}})$, the squared Frobenius norm; its positive-definiteness is exactly the condition $B_\theta > 0$ that makes $-X^{\mathsf{T}}$ a Cartan involution, and its restriction to $\mathfrak{p}$ is the inner product that will make $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$ a Riemannian symmetric space. That quotient has dimension $\dim \mathfrak{p} = \frac{n(n+1)}{2} - 1$: the tangent space at the identity coset is $\mathfrak{p}$, and $K = \mathrm{SO}(n)$ collapses the compact directions. For $n = 2$ the count gives $\dim \mathfrak{p} = 2$, the hyperbolic plane; chapter 15 identifies the space in general as the space of positive-definite symmetric $n \times n$ matrices of determinant one, one of the noncompact symmetric spaces of type A.

The second example keeps a Hermitian form fixed rather than a real bilinear one, and the compact part K acquires internal structure.

Example 13.5.2: The Cartan Decomposition of $\mathrm{SU}(p,q)$

Fix the diagonal signature matrix

$$I_{p,q} = \begin{pmatrix} I_p & 0 \\ 0 & -I_q \end{pmatrix}$$

with p entries $+1$ and q entries -1, defining the Hermitian form $\langle v, w \rangle_{p,q} = w^* I_{p,q} v$ on $\mathbb{C}^{p+q}$ of

signature (p,q). The *special indefinite unitary group*

$$\mathrm{SU}(p,q) = \{g \in \mathrm{SL}(p+q,\mathbb{C}) \mid g^* I_{p,q}\, g = I_{p,q}\}$$

is the group of determinant-one complex linear maps preserving this form; for $q = 0$ it is the compact $\mathrm{SU}(p+q)$, and for $q \geq 1$ it is noncompact. Its Lie algebra is

$$\mathfrak{su}(p,q) = \{X \in \mathfrak{sl}(p+q,\mathbb{C}) \mid X^* I_{p,q} + I_{p,q} X = 0\}$$

the traceless matrices X with $X^* I_{p,q} = -I_{p,q} X$. The map

$$\theta(X) = -X^*$$

is a Cartan involution: it is an involutive automorphism of $\mathfrak{su}(p,q)$ (the defining condition $X^* I_{p,q} = -I_{p,q} X$ is preserved because $(-X^*)^* I_{p,q} = -X I_{p,q} = I_{p,q} X^* = -I_{p,q}(-X^*)$ after using the condition twice), and $B_\theta(X,Y) = -\kappa(X,\theta Y)$ is a positive multiple of $\mathrm{tr}(XX^*) = \sum_{i,j} |X_{ij}|^2$, positive-definite. Equivalently θ is conjugation by the signature matrix, $\theta(X) = I_{p,q}\, X\, I_{p,q}$, since $X^* = -I_{p,q} X I_{p,q}$ on $\mathfrak{su}(p,q)$; the two descriptions of θ agree on the algebra.

Writing a matrix in (p,q)-block form $X = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, the eigenspaces of $\theta = \mathrm{Ad}(I_{p,q})$ are the block-diagonal and block-off-diagonal parts. The $+1$-eigenspace is

$$\mathfrak{k} = \left\{ \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} \;\middle|\; A \in \mathfrak{u}(p),\ D \in \mathfrak{u}(q),\ \mathrm{tr}A + \mathrm{tr}D = 0 \right\} = \mathfrak{s}(\mathfrak{u}(p) \oplus \mathfrak{u}(q))$$

the block-diagonal skew-Hermitian matrices of total trace zero, and the -1-eigenspace is

$$\mathfrak{p} = \left\{ \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix} \;\middle|\; B \in \mathbb{C}^{p \times q} \right\}$$

the off-diagonal blocks, parametrized freely by the $p \times q$ complex matrix B (the condition $X \in \mathfrak{su}(p,q)$ forces the lower block to be B^*). Thus $\dim_{\mathbb{R}} \mathfrak{p} = 2pq$. The maximal compact subgroup is

$$K = \mathrm{S}\big(\mathrm{U}(p) \times \mathrm{U}(q)\big) = \left\{ \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix} \;\middle|\; k_1 \in \mathrm{U}(p),\ k_2 \in \mathrm{U}(q),\ \det k_1 \det k_2 = 1 \right\}$$

and theorem 13.4.2 gives $\mathrm{SU}(p,q) = K \exp(\mathfrak{p})$, the polar decomposition of a form-preserving matrix into a block-unitary part and the exponential of an off-diagonal Hermitian part.

The contrast with $\mathrm{SL}(n,\mathbb{R})$ is instructive. There the maximal compact was a single simple group $\mathrm{SO}(n)$; here it is $\mathrm{S}(\mathrm{U}(p) \times \mathrm{U}(q))$, a product of two unitary groups sharing a determinant constraint, and $\mathfrak{p}$ carries a complex structure, being the space of $p \times q$ complex matrices. That complex structure is not incidental. The center of K contains a circle acting on $\mathfrak{p} = \mathbb{C}^{p \times q}$ by scalar multiplication, and this circle action makes the symmetric space $\mathrm{SU}(p,q)/\mathrm{S}(\mathrm{U}(p) \times \mathrm{U}(q))$ a *Hermitian* symmetric space, one carrying an invariant complex structure compatible with its metric. For $p = q = 1$ it is the unit disk; in general it is realized as a bounded symmetric domain in $\mathbb{C}^{pq}$, the higher-rank analogue of the disk. Chapter 15 develops these bounded-domain models, of which the $\mathrm{SU}(p,q)$ family is the type-A prototype; the point to carry forward is that the Cartan decomposition already sees the complex structure, encoded in the circle in the center of K.

The running example deserves the fullest treatment. Everything computes by hand.

Example 13.5.3: The Cartan Decomposition of $\mathrm{SL}(2,\mathbb{R})$

Let $G = \mathrm{SL}(2,\mathbb{R})$, with Lie algebra $\mathfrak{g}_0 = \mathfrak{sl}(2,\mathbb{R})$ of traceless real 2×2 matrices. A basis is

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

with brackets $[H,E] = 2E$, $[H,F] = -2F$, $[E,F] = H$. The Cartan involution is $\theta(X) = -X^{\mathsf{T}}$, as for $\mathrm{SL}(n,\mathbb{R})$ (example 13.5.1). Its $+1$-eigenspace is the antisymmetric matrices,

$$\mathfrak{k} = \mathfrak{so}(2) = \mathbb{R}\,W, \qquad W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = E - F$$

one-dimensional, and its -1-eigenspace is the traceless symmetric matrices,

$$\mathfrak{p} = \mathrm{Sym}_0(2) = \mathbb{R}H \oplus \mathbb{R}(E+F) = \left\{ \begin{pmatrix} a & b \\ b & -a \end{pmatrix} \;\middle|\; a,b \in \mathbb{R} \right\}$$

two-dimensional, so $\dim\mathfrak{k} = 1$, $\dim\mathfrak{p} = 2$, summing to $3 = \dim\mathfrak{sl}(2,\mathbb{R})$. The compact factor is $K = \exp(\mathfrak{k}) = \mathrm{SO}(2)$, the rotations $k_\vartheta = \left(\begin{smallmatrix} \cos\vartheta & \sin\vartheta \\ -\sin\vartheta & \cos\vartheta \end{smallmatrix}\right) = \exp(\vartheta W)$, and theorem 13.4.2 gives $\mathrm{SL}(2,\mathbb{R}) = \mathrm{SO}(2)\exp(\mathfrak{p})$.

The three basis vectors organize the whole structure of $\mathrm{SL}(2,\mathbb{R})$. The abelian subspace $\mathbb{R}H \subseteq \mathfrak{p}$ exponentiates to the diagonal subgroup $A = \{\mathrm{diag}(e^t, e^{-t}) \mid t \in \mathbb{R}\}$, which chapter 14 will identify as the A-factor of the Iwasawa decomposition $G = KAN$; the compact $W = E - F$ generates $K = \mathrm{SO}(2)$; and the nilpotent E generates the upper-triangular unipotent subgroup N. The next proposition records the full structure and the action on the upper half-plane, with $\mathrm{SO}(2)$ appearing again as the stabilizer of a point.

Proposition 13.5.4: The Structure of $\mathrm{SL}(2,\mathbb{R})$

Let $G = \mathrm{SL}(2,\mathbb{R})$ act on the upper half-plane $\mathfrak{H} = \{z \in \mathbb{C} \mid \mathrm{Im}\, z > 0\}$ by Möbius transformations,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az+b}{cz+d}$$

Then:

1. the action is well-defined (it preserves $\mathfrak{H}$) and transitive;
2. the stabilizer of $i \in \mathfrak{H}$ is exactly $K = \mathrm{SO}(2)$, so evaluation at i induces a diffeomorphism $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2) \xrightarrow{\sim} \mathfrak{H}$;
3. the Cartan decomposition $\mathrm{SL}(2,\mathbb{R}) = \mathrm{SO}(2)\exp(\mathfrak{p})$ recovers this identification: $\exp(\mathfrak{p})$ moves i across all of $\mathfrak{H}$ while $\mathrm{SO}(2)$ fixes it.

***Proof*:**

The three parts are proved in turn.

Part (1): well-definedness and transitivity. For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$ and $z \in \mathbb{C}$ with $cz + d \neq 0$,

a direct computation of the imaginary part gives

$$\operatorname{Im}\left(\frac{az+b}{cz+d}\right) = \frac{\operatorname{Im}\big((az+b)\overline{(cz+d)}\big)}{|cz+d|^2} = \frac{(ad-bc)\operatorname{Im} z}{|cz+d|^2} = \frac{\operatorname{Im} z}{|cz+d|^2}$$

using $ad-bc = \det g = 1$ and the fact that a,b,c,d are real. Thus $\operatorname{Im} z > 0$ forces $\operatorname{Im}(g\cdot z) > 0$ (and $cz+d \neq 0$, since $cz+d=0$ with c,d real and $\operatorname{Im} z>0$ would need $c=d=0$, impossible in $\mathrm{SL}(2,\mathbb{R})$), so the action preserves $\mathfrak{H}$. It is a group action because Möbius composition matches matrix multiplication. For transitivity, given $z = x+iy \in \mathfrak{H}$ with $y>0$, the matrix

$$g_z = \begin{pmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R})$$

sends i to $g_z \cdot i = \sqrt{y}\, i \cdot \sqrt{y} + x = x+iy = z$, so the orbit of i is all of $\mathfrak{H}$.

Part (2): the stabilizer of i. Suppose $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ fixes i, so $\frac{ai+b}{ci+d} = i$, that is $ai+b = i(ci+d) = -c+di$. Matching real and imaginary parts gives $b=-c$ and $a=d$. Together with $\det g = ad-bc = a^2+c^2 = 1$, this says

$$g = \begin{pmatrix} a & -c \\ c & a \end{pmatrix}, \qquad a^2+c^2 = 1$$

which is exactly $\mathrm{SO}(2)$, with $a = \cos\vartheta$, $c = -\sin\vartheta$. Hence $\mathrm{Stab}_G(i) = \mathrm{SO}(2) = K$. By the orbit-stabilizer diffeomorphism (theorem 4.2.3) applied to the transitive action of part (1), evaluation $g \mapsto g\cdot i$ descends to a diffeomorphism $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2) \xrightarrow{\sim} \mathfrak{H}$.

Part (3): matching the Cartan decomposition. By part (2) the quotient map factors through the coset space G/K, and by theorem 13.4.2 every coset gK has a unique representative in $\exp(\mathfrak{p})$: applying the theorem to g^{-1} and inverting gives the opposite-order factorization $g = \exp(X)\,k'$ with $X \in \mathfrak{p}$ and $k' \in \mathrm{SO}(2)$, so that $gK = \exp(X)\,k'K = \exp(X)K$, the $\exp(\mathfrak{p})$-factor now sitting on the left where it represents the coset directly. Since $\mathrm{SO}(2)$ fixes i and evaluation $\exp(X) \mapsto \exp(X)\cdot i$ is a bijection $\exp(\mathfrak{p}) \to \mathfrak{H}$ (the composite of the diffeomorphisms $\exp(\mathfrak{p}) \cong G/K \cong \mathfrak{H}$), the Euclidean factor $\exp(\mathfrak{p})$ sweeps out $\mathfrak{H}$ from the basepoint i while K holds i fixed. Concretely, $\exp(tH) = \operatorname{diag}(e^t, e^{-t})$ sends i to $e^{2t}i$, moving i along the imaginary axis, and $\exp(s(E+F))$ moves it off the axis, the two together covering $\mathfrak{H}$, as we sought to show.

The proposition realizes the abstract quotient G/K of theorem 13.4.3 as a concrete geometric object: $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ *is* the upper half-plane, with the Cartan decomposition giving global coordinates in which $\mathrm{SO}(2)$ is the isotropy at i and $\exp(\mathfrak{p})$ the transverse Euclidean directions. This is the smallest nontrivial noncompact symmetric space, and it is the same $\mathfrak{H}$ that carries the hyperbolic metric of constant curvature -1; chapter 15 equips it with the invariant metric coming from the Killing form on $\mathfrak{p}$ and identifies the isometry group as $\mathrm{PSL}(2,\mathbb{R})$. The imaginary-axis motion $i \mapsto e^{2t}i$ is the one-parameter group A, so the Iwasawa refinement $G = KAN$ of chapter 14 reads, on $\mathfrak{H}$, as: rotate about i (K), dilate along the axis (A), translate horizontally (N).

The last thread ties $\mathrm{SL}(2,\mathbb{R})$ back to the compact running example. The two groups $\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SU}(2)$ share the complexification $\mathfrak{sl}(2,\mathbb{C})$, and they are its two real forms met so far: $\mathrm{SU}(2)$ compact, $\mathrm{SL}(2,\mathbb{R})$ split noncompact.

Example 13.5.5: $\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SU}(2)$ as Real Forms of $\mathfrak{sl}(2,\mathbb{C})$

The complexification of both $\mathfrak{sl}(2,\mathbb{R})$ and $\mathfrak{su}(2)$ is $\mathfrak{sl}(2,\mathbb{C})$ (definition 13.1.1), so each is a real form of $\mathfrak{sl}(2,\mathbb{C})$ in the sense of definition 13.1.3. They are inequivalent: the Killing form $\kappa(X,Y) = 4\,\mathrm{tr}(XY)$ restricts to $\mathfrak{su}(2)$ as a negative-definite form, marking it the compact real form (definition 9.4.3), whereas on $\mathfrak{sl}(2,\mathbb{R})$ it has signature $(2,1)$: positive-definite on $\mathfrak{p} = \mathrm{Sym}_0(2)$ (dimension 2) and negative-definite on $\mathfrak{k} = \mathfrak{so}(2)$ (dimension 1), by theorem 13.3.3. The two conjugations of $\mathfrak{sl}(2,\mathbb{C})$ fixing them, $\sigma_{\mathbb{R}}(X) = \bar{X}$ (fixing $\mathfrak{sl}(2,\mathbb{R})$) and $\tau(X) = -X^*$ (fixing $\mathfrak{su}(2)$), commute, and their composite $\sigma_{\mathbb{R}}\tau(X) = -\overline{X^*} = -X^{\mathsf{T}}$ is precisely the Cartan involution θ of $\mathfrak{sl}(2,\mathbb{R})$, exhibiting the alignment of theorem 13.3.4 in the smallest case.

At the group level the compact/noncompact duality shows up in the covering groups. $\mathrm{SU}(2)$ double covers $\mathrm{SO}(3)$ (theorem 1.5.5), realizing $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ as the compact quotient. Its noncompact analogue is

$$\mathrm{SO}(2,1)^0 \cong \mathrm{PSL}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})/\{\pm I\}$$

the identity component of the indefinite orthogonal group $\mathrm{SO}(2,1)$: $\mathrm{SL}(2,\mathbb{R})$ acts on the 3-dimensional space $\mathfrak{sl}(2,\mathbb{R})$ by the adjoint representation, preserving the Killing form of signature $(2,1)$, and the induced map $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SO}(2,1)^0$ is a double cover with kernel $\{\pm I\}$, the noncompact twin of $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

The pair $\mathrm{SU}(2)$, $\mathrm{SL}(2,\mathbb{R})$ gives two real forms of a single complex Lie algebra: one is a compact 3-sphere group, the other an unbounded group acting on the hyperbolic plane, and the Cartan involution is exactly the bookkeeping that converts one into the other. This is the running thread of the book closing on itself. The $\mathrm{SU}(2)$ that appeared in chapter 1 as the double cover of $\mathrm{SO}(3)$, carried through the maximal-torus and character theory of Part III, reappears here as the compact real form of $\mathfrak{sl}(2,\mathbb{C})$, with $\mathrm{SL}(2,\mathbb{R})$ its noncompact sibling and $\mathrm{SO}(2,1)^0 \cong \mathrm{PSL}(2,\mathbb{R})$ the exact analogue of the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

Three noncompact families have now been organized by a single mechanism. In each case a visible involution split the algebra into a compact part $\mathfrak{k}$ and a Euclidean part $\mathfrak{p}$, the group factored as a maximal compact times a contractible Euclidean piece, and the quotient G/K emerged as a symmetric space: $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$, the space of positive-definite unimodular symmetric matrices; $\mathrm{SU}(p,q)/\mathrm{S}(\mathrm{U}(p)\times\mathrm{U}(q))$, a bounded symmetric domain; and $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2) = \mathfrak{H}$, the hyperbolic plane. The Cartan decomposition has done its structural work, isolating the compact directions and exhibiting the rest as Euclidean. Two refinements remain. Chapter 14 subdivides the Euclidean factor itself, decomposing $\exp(\mathfrak{p})$ along restricted roots into the finer $G = KAN$ of Iwasawa, which for $\mathrm{SL}(2,\mathbb{R})$ is the rotation-dilation-translation factorization of the half-plane. Chapter 15 reads the quotient G/K geometrically, putting on it the invariant metric that the Killing form gives on $\mathfrak{p}$ and recognizing the compact/noncompact duality of $\mathrm{SU}(2)$ and $\mathrm{SL}(2,\mathbb{R})$ as the central symmetry of the theory of symmetric spaces.

Exercise 13.5.1

1. Polar-decompose the element $g = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$. That is, find the factorization $g = k\,p$ with $k \in \mathrm{SO}(2)$ and p positive-definite symmetric of determinant one, using $p = (g^{\mathsf{T}}g)^{1/2}$ and $k = gp^{-1}$. Verify that $p = \exp X$ for some $X \in \mathfrak{p} = \mathrm{Sym}_0(2)$, confirming $g \in \mathrm{SO}(2)\exp(\mathfrak{p})$.

2. Verify directly that $\mathrm{SO}(2)$ is the stabilizer of $i \in \mathfrak{H}$ under the Möbius action of $\mathrm{SL}(2,\mathbb{R})$: show that $\left(\begin{smallmatrix} \cos\vartheta & \sin\vartheta \\ -\sin\vartheta & \cos\vartheta \end{smallmatrix}\right)\cdot i = i$ for every ϑ, and conversely that a matrix fixing i must have the form

$\left(\begin{smallmatrix} a & -c \\ c & a \end{smallmatrix}\right)$ with $a^2 + c^2 = 1$.

3. Compute the dimension of the symmetric space $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$ for general n, and evaluate it for $n = 2, 3, 4$. Confirm that it equals $\dim \mathfrak{p} = \frac{n(n+1)}{2} - 1$, and check the identity $\dim \mathrm{SL}(n,\mathbb{R}) - \dim \mathrm{SO}(n) = \dim \mathfrak{p}$.

4. Show that $\mathrm{SU}(1,1)$ is isomorphic to $\mathrm{SL}(2,\mathbb{R})$. Explicitly, verify that the Cayley matrix $C = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$ conjugates one group to the other, $C\,\mathrm{SL}(2,\mathbb{R})\,C^{-1} = \mathrm{SU}(1,1)$, and explain how the corresponding Möbius transformation carries the upper half-plane model $\mathfrak{H}$ to the unit-disk model $\mathbb{D} = \{w \in \mathbb{C} \mid |w| < 1\}$, with $\mathrm{SU}(1,1)$ acting on $\mathbb{D}$.

5. Determine the Cartan decomposition of $\mathrm{SL}(2,\mathbb{C})$, viewed as a real Lie group of real dimension 6. Take the Cartan involution $\theta(X) = -X^{*}$; identify $\mathfrak{k}$, show that $K = \mathrm{SU}(2)$, and identify $\mathfrak{p}$. Conclude $\mathrm{SL}(2,\mathbb{C}) = \mathrm{SU}(2)\exp(\mathfrak{p})$ and compute $\dim_{\mathbb{R}} \mathfrak{p}$.

6. For $\mathrm{SU}(p,q)$, verify that $\dim_{\mathbb{R}} \mathfrak{p} = 2pq$ and that $\dim_{\mathbb{R}} \mathfrak{k} = p^2 + q^2 - 1$, and check that the two sum to $\dim_{\mathbb{R}} \mathfrak{su}(p,q) = (p+q)^2 - 1$.

14

The Iwasawa Decomposition

The Cartan decomposition of chapter 13 split a noncompact semisimple group as $G = K\exp(\mathfrak{p})$, a maximal compact times a Euclidean factor. That factorization is symmetric but coarse: the Euclidean part $\exp(\mathfrak{p})$ is not a subgroup, and the decomposition sees the noncompact directions only as an undifferentiated block. This chapter refines it. Diagonalizing the adjoint action of a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$ resolves $\mathfrak{p}$ into *restricted roots*, and reassembling the pieces produces the Iwasawa decomposition $G = KAN$: a compact group K, an abelian group $A = \exp\mathfrak{a}$, and a nilpotent group N, whose product map is a diffeomorphism. Unlike the Cartan factors, all three are subgroups, and the solvable part AN acts simply transitively on the symmetric space G/K, making KAN the coordinate system in which noncompact harmonic analysis is done.

The restricted roots are the real analogue of the roots of chapter 11, with one decisive sign change. There a maximal torus acted on $\mathfrak{g}_{\mathbb{C}}$ through skew operators, so its eigenvalues were imaginary and the arena was the complexification; here the maximal abelian $\mathfrak{a}$ lies in $\mathfrak{p}$, where ad is symmetric for the Cartan form, so its eigenvalues are *real* and the decomposition happens already over $\mathbb{R}$. The restricted roots form a root system Σ on $\mathfrak{a}^*$, possibly nonreduced, and a choice of positive restricted roots assembles the positive root spaces into the nilpotent $\mathfrak{n}$. Exponentiated, this gives the Lie-algebra Iwasawa decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}$, and integrating it yields $G = KAN$.

From the Iwasawa data one reads off the coarse subgroup structure of G. The centralizer $M = Z_K(\mathfrak{a})$ assembles with A and N into the minimal parabolic subgroup $P = MAN$, whose conjugates and overgroups are the parabolic subgroups, and the double cosets of P organize G into the Bruhat decomposition $G = \bigsqcup_w PwP$ indexed by the restricted Weyl group. For $\mathrm{SL}(n,\mathbb{R})$ every piece is elementary linear algebra: the Iwasawa decomposition is the Gram-Schmidt process, the parabolic subgroups are the block-triangular stabilizers of flags, and Bruhat is the LU decomposition with row swaps. The running example $\mathrm{SL}(2,\mathbb{R})$ is worked in full, its KAN coordinates realizing the upper half-plane $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ and its elements sorted into elliptic, hyperbolic, and parabolic by trace. The chapter closes on the bridge the whole apparatus was built for: a noncompact semisimple group has no nontrivial finite-dimensional unitary representation, so its unitary theory is infinite-

dimensional, and the minimal parabolic $P = MAN$ is exactly what the principal series is induced from. That representation theory belongs to the sequel volumes; chapter 15 first reads the Cartan and Iwasawa data geometrically, as a Riemannian symmetric space.

14.1 Restricted Roots and the Restricted Root Space Decomposition

The Cartan decomposition of chapter 13 split a real semisimple Lie algebra into $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ and lifted the splitting to the global product $G = K\exp(\mathfrak{p})$ (theorem 13.3.3, theorem 13.4.2). That decomposition is symmetric: $\mathfrak{k}$ carries the compact directions and $\mathfrak{p}$ the Euclidean complement, with no further structure imposed on $\mathfrak{p}$. This section refines the picture inside $\mathfrak{p}$. The operators $\mathrm{ad}H$ for $H \in \mathfrak{p}$ are symmetric for the inner product B_θ, hence simultaneously diagonalizable over $\mathbb{R}$ once the H commute; choosing a maximal commuting family inside $\mathfrak{p}$ diagonalizes ad and cuts $\mathfrak{g}_0$ into joint eigenspaces. The eigenvalues are the *restricted roots*, and the joint eigenspace decomposition is the analytic heart of the Iwasawa decomposition assembled in section 14.2.

The construction runs parallel to the root space decomposition of chapter 7, with one decisive change of sign. There a Cartan subalgebra $\mathfrak{h}$ of the complex algebra diagonalized ad with complex eigenvalues; here the diagonalizing subspace lies in $\mathfrak{p}$, where ad is symmetric rather than skew, so the eigenvalues are real. The maximal commuting subspace of $\mathfrak{p}$ plays the role of the Cartan subalgebra, and its dimension is a new invariant measuring how far G is from compact.

Definition 14.1.1: Maximal Abelian Subspace and Real Rank

Let $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ be the Cartan decomposition of a real semisimple Lie algebra with Cartan involution θ (theorem 13.3.3). A *maximal abelian subspace* $\mathfrak{a} \subseteq \mathfrak{p}$ is a subspace on which the bracket vanishes, $[\mathfrak{a}, \mathfrak{a}] = 0$, and which is not properly contained in any larger abelian subspace of $\mathfrak{p}$. The *real rank* of $\mathfrak{g}_0$ (or of G) is the dimension $\dim \mathfrak{a}$.

A maximal abelian subspace exists for the trivial reason that dimension is bounded: start from any line $\mathbb{R}H \subseteq \mathfrak{p}$ (abelian because one-dimensional) and enlarge it whenever a commuting vector lies outside; the process stops in at most $\dim \mathfrak{p}$ steps. Real rank is well defined only if the answer does not depend on which $\mathfrak{a}$ is chosen, and this is the content of the conjugacy statement below. Before it, a caution on the name. The subspace $\mathfrak{a}$ is abelian as a Lie algebra, but it is *not* a Cartan subalgebra of $\mathfrak{g}_0$: it sits inside $\mathfrak{p}$, which is not a subalgebra, and its complexification need not be a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$. It is the largest *split* torus one can find, the piece of a maximal torus that acts by real eigenvalues, and the real rank counts exactly those directions.

Proposition 14.1.2: Conjugacy of Maximal Abelian Subspaces

Any two maximal abelian subspaces of $\mathfrak{p}$ are conjugate under the adjoint action of K: if $\mathfrak{a}, \mathfrak{a}' \subseteq \mathfrak{p}$ are both maximal abelian, there is $k \in K$ with $\mathrm{Ad}(k)\mathfrak{a} = \mathfrak{a}'$. In particular $\dim \mathfrak{a} = \dim \mathfrak{a}'$, so the real rank is well defined.

***Proof*:**

The inner product B_θ restricts to a positive-definite form on $\mathfrak{p}$, and $K = G^\Theta$ acts on $\mathfrak{p}$ by Ad preserving this form (theorem 13.4.2), so $\mathrm{Ad}(k)|_\mathfrak{p}$ is orthogonal for each $k \in K$. Fix regular elements: an element $H \in \mathfrak{a}$ is *regular* in $\mathfrak{p}$ when its centralizer $\mathfrak{z}_\mathfrak{p}(H) = \{X \in \mathfrak{p} \mid [H, X] = 0\}$ is as small as possible, and for such an H one has $\mathfrak{z}_\mathfrak{p}(H) = \mathfrak{a}$, since $\mathfrak{a}$ is a maximal abelian subspace containing H and any $X \in \mathfrak{p}$ commuting with a regular H commutes with all of $\mathfrak{a}$. Choose regular elements $H \in \mathfrak{a}$ and $H' \in \mathfrak{a}'$.

Consider the smooth function $f\colon K \to \mathbb{R}$, $f(k) = B_\theta(\mathrm{Ad}(k)H, H')$. Since K is compact, f attains a maximum at some $k_0 \in K$; replacing H by $H_0 = \mathrm{Ad}(k_0)H \in \mathfrak{p}$, the function $k \mapsto B_\theta(\mathrm{Ad}(k)H_0, H')$ is maximized at the identity, so its derivative vanishes: for every $X \in \mathfrak{k}$,

$$0 = \frac{d}{dt}\Big|_0 B_\theta(\mathrm{Ad}(\exp tX)H_0, H') = B_\theta([X, H_0], H')$$

Now feed in the particular element $X = [H_0, H']$, which lies in $\mathfrak{k}$ because $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$ (theorem 13.3.3(1)). The operator $\mathrm{ad}(H_0)$ is B_θ-symmetric, since $H_0 \in \mathfrak{p}$ (theorem 13.3.3(3)), so with $Y = [H_0, H']$,

$$0 = B_\theta([X, H_0], H') = B_\theta([Y, H_0], H') = -B_\theta([H_0, Y], H') = -B_\theta(Y, [H_0, H']) = -B_\theta(Y, Y)$$

using $B_\theta([H_0, Y], H') = B_\theta(Y, [H_0, H'])$ from the symmetry of $\mathrm{ad}(H_0)$ and $Y = [H_0, H']$. Positive-definiteness of B_θ then forces $Y = [H_0, H'] = 0$: the maximizing H_0 commutes with H'.

The subspace $\mathrm{Ad}(k_0)\mathfrak{a}$ is again maximal abelian in $\mathfrak{p}$, being the $\mathrm{Ad}(K)$-image of the maximal abelian $\mathfrak{a}$. It contains $H_0 = \mathrm{Ad}(k_0)H$, whose regularity makes $\mathfrak{z}_\mathfrak{p}(H_0) = \mathrm{Ad}(k_0)\mathfrak{a}$; and $[H_0, H'] = 0$ places H' in $\mathfrak{z}_\mathfrak{p}(H_0) = \mathrm{Ad}(k_0)\mathfrak{a}$. Since H' is regular in $\mathfrak{p}$, every element of $\mathrm{Ad}(k_0)\mathfrak{a}$ commutes with H', so $\mathrm{Ad}(k_0)\mathfrak{a} \subseteq \mathfrak{z}_\mathfrak{p}(H') = \mathfrak{a}'$; and $\mathrm{Ad}(k_0)\mathfrak{a}$, being maximal abelian, cannot sit properly inside the abelian $\mathfrak{a}'$, so $\mathrm{Ad}(k_0)\mathfrak{a} = \mathfrak{a}'$. The two maximal abelian subspaces are therefore K-conjugate, completing the proof.

The conjugacy is the exact analogue, inside $\mathfrak{p}$, of the maximal-torus conjugacy theorem of chapter 11 (theorem 11.1.3): where a compact group has all its maximal tori conjugate, a noncompact group has all its maximal split abelian subspaces conjugate, and in both cases the dimension of the diagonalizing piece is an invariant. Real rank is a coarse measurement. It ranges from 0, when $\mathfrak{p}$ contains no nonzero abelian subspace, meaning $\mathfrak{p} = 0$ and G is compact, up to the full $\dim \mathfrak{p}$ for the maximally noncompact *split* real forms, where a maximal abelian subspace of $\mathfrak{p}$ is a full Cartan subalgebra.

With $\mathfrak{a}$ fixed, the family $\{\mathrm{ad}(H)\}_{H \in \mathfrak{a}}$ is a commuting family of B_θ-symmetric operators on $\mathfrak{g}_0$. The spectral theorem diagonalizes them simultaneously over $\mathbb{R}$, and the joint eigenspaces are indexed by the linear functionals recording the eigenvalues.

Definition 14.1.3: Restricted Roots and Restricted Root Spaces

Fix a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$. For a linear functional $\lambda \in \mathfrak{a}^*$, the *restricted root space* is

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g}_0 \mid [H, X] = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}$$

the joint eigenspace on which each $\mathrm{ad}(H)$ acts by the scalar $\lambda(H)$. A nonzero $\lambda \in \mathfrak{a}^*$ with $\mathfrak{g}_\lambda \neq 0$ is a *restricted root*, and the set of all restricted roots is denoted $\Sigma \subseteq \mathfrak{a}^*$.

The definition mirrors the root space decomposition of a complex semisimple algebra (theorem 7.2.4), reading $\mathfrak{a}$ in place of the Cartan subalgebra $\mathfrak{h}$ and Σ in place of the root system Φ. The difference is that $\mathfrak{a}^*$ is a real vector space and the eigenvalues $\lambda(H)$ are real numbers, not the imaginary values that a compact torus produces. The functional $\lambda = 0$ is excluded from Σ but its eigenspace $\mathfrak{g}_0^0 = \mathfrak{z}_{\mathfrak{g}_0}(\mathfrak{a})$, the centralizer of $\mathfrak{a}$, is where the constant part of the decomposition lives. That Σ is finite, that the $\mathfrak{g}_\lambda$ span $\mathfrak{g}_0$, and that Σ obeys root-system axioms are the content of the theorem below.

Theorem 14.1.4: The Restricted Root Space Decomposition

Let $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ be the Cartan decomposition of a real semisimple Lie algebra, $\mathfrak{a} \subseteq \mathfrak{p}$ a maximal abelian subspace, and $\Sigma \subseteq \mathfrak{a}^*$ the set of restricted roots. Then:

1. The set Σ is finite, and there is an orthogonal direct sum decomposition
$$\mathfrak{g}_0 = \mathfrak{g}_0^0 \oplus \bigoplus_{\lambda \in \Sigma} \mathfrak{g}_\lambda, \qquad \mathfrak{g}_0^0 = \mathfrak{m} \oplus \mathfrak{a}$$
where $\mathfrak{m} = \mathfrak{z}_{\mathfrak{k}}(\mathfrak{a})$ is the centralizer of $\mathfrak{a}$ in $\mathfrak{k}$, orthogonal for the inner product B_θ.
2. The bracket grades the decomposition: $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subseteq \mathfrak{g}_{\lambda+\mu}$ for all $\lambda, \mu \in \mathfrak{a}^*$ (with $\mathfrak{g}_\nu = 0$ when $\nu \notin \Sigma \cup \{0\}$).
3. The Cartan involution sends $\mathfrak{g}_\lambda$ to $\mathfrak{g}_{-\lambda}$; in particular $\lambda \in \Sigma$ if and only if $-\lambda \in \Sigma$, and $\dim \mathfrak{g}_\lambda = \dim \mathfrak{g}_{-\lambda}$.
4. The set Σ is a root system in the subspace of $\mathfrak{a}^*$ it spans, possibly non-reduced, with restricted Weyl group generated by the reflections s_λ.

***Proof*:**

Step 1: simultaneous diagonalization over $\mathbb{R}$. For $H \in \mathfrak{a} \subseteq \mathfrak{p}$, the operator $\mathrm{ad}(H)$ is symmetric with respect to the positive-definite inner product B_θ, by theorem 13.3.3(3). A symmetric operator on a finite-dimensional real inner product space is diagonalizable over $\mathbb{R}$ with real eigenvalues (the real spectral theorem). Moreover the family $\{\mathrm{ad}(H)\}_{H \in \mathfrak{a}}$ commutes: $[\mathrm{ad}(H), \mathrm{ad}(H')] = \mathrm{ad}([H, H']) = 0$ because $\mathfrak{a}$ is abelian. A commuting family of symmetric operators is simultaneously diagonalizable, so $\mathfrak{g}_0$ decomposes into joint real eigenspaces. Each joint eigenspace assigns to every $H \in \mathfrak{a}$ its eigenvalue, and this assignment $H \mapsto$ (eigenvalue) is linear in H, hence a functional $\lambda \in \mathfrak{a}^*$; the eigenspace is exactly $\mathfrak{g}_\lambda$. Thus
$$\mathfrak{g}_0 = \bigoplus_{\lambda} \mathfrak{g}_\lambda$$
summed over the finite set of λ with $\mathfrak{g}_\lambda \neq 0$, and Σ is this set with $\lambda = 0$ removed. Finiteness of Σ is immediate: a direct sum of nonzero subspaces of the finite-dimensional $\mathfrak{g}_0$ has finitely many summands.

Step 2: orthogonality and the zero eigenspace. Distinct joint eigenspaces of a commuting family of B_θ-symmetric operators are B_θ-orthogonal: if $\lambda \neq \mu$, pick H with $\lambda(H) \neq \mu(H)$, and for $X \in \mathfrak{g}_\lambda$, $Y \in \mathfrak{g}_\mu$,
$$\lambda(H) B_\theta(X, Y) = B_\theta(\mathrm{ad}(H)X, Y) = B_\theta(X, \mathrm{ad}(H)Y) = \mu(H) B_\theta(X, Y)$$

forces $B_\theta(X, Y) = 0$. Hence the decomposition of Step 1 is B_θ-orthogonal. It remains to identify $\mathfrak{g}_0^0$. By definition $\mathfrak{g}_0^0 = \mathfrak{z}_{\mathfrak{g}_0}(\mathfrak{a})$ is θ-stable: θ stabilizes $\mathfrak{a}$, acting as -1 there ($\theta H = -H$ for $H \in \mathfrak{a} \subseteq \mathfrak{p}$), and since θ is an automorphism $\theta[H, X] = [\theta H, \theta X] = [-H, \theta X]$, so $X \in \mathfrak{g}_0^0$ iff $[H, X] = 0$ iff $[-H, \theta X] = 0$ iff $\theta X \in \mathfrak{g}_0^0$. A θ-stable subspace splits into its ± 1-eigenparts, so $\mathfrak{g}_0^0 = (\mathfrak{g}_0^0 \cap \mathfrak{k}) \oplus (\mathfrak{g}_0^0 \cap \mathfrak{p})$. The first summand is $\mathfrak{z}_{\mathfrak{k}}(\mathfrak{a}) = \mathfrak{m}$. The second is $\mathfrak{z}_{\mathfrak{p}}(\mathfrak{a})$, which equals $\mathfrak{a}$: any $X \in \mathfrak{p}$ commuting with all of $\mathfrak{a}$ makes $\mathfrak{a} + \mathbb{R}X$ an abelian subspace of $\mathfrak{p}$ containing $\mathfrak{a}$, so by maximality $X \in \mathfrak{a}$. Therefore $\mathfrak{g}_0^0 = \mathfrak{m} \oplus \mathfrak{a}$, and combining with the orthogonal splitting over Σ gives the decomposition of part (1).

Step 3: the bracket grading. Let $X \in \mathfrak{g}_\lambda$ and $Y \in \mathfrak{g}_\mu$. For $H \in \mathfrak{a}$ the Jacobi identity gives

$$[H, [X, Y]] = [[H, X], Y] + [X, [H, Y]] = \lambda(H)[X, Y] + \mu(H)[X, Y] = (\lambda + \mu)(H)\,[X, Y]$$

so $[X, Y] \in \mathfrak{g}_{\lambda+\mu}$. This proves part (2). When $\lambda + \mu \notin \Sigma \cup \{0\}$ the target space is zero, so the bracket vanishes.

Step 4: the Cartan involution flips the sign. Let $X \in \mathfrak{g}_\lambda$. Since θ is an automorphism and $\theta H = -H$ for $H \in \mathfrak{a} \subseteq \mathfrak{p}$,

$$[H, \theta X] = -[\theta H, \theta X] = -\theta[H, X] = -\theta(\lambda(H)X) = -\lambda(H)\,\theta X$$

so $\theta X \in \mathfrak{g}_{-\lambda}$. Thus θ maps $\mathfrak{g}_\lambda$ into $\mathfrak{g}_{-\lambda}$, and applying θ again shows it is a bijection $\mathfrak{g}_\lambda \to \mathfrak{g}_{-\lambda}$; in particular $-\lambda \in \Sigma$ whenever $\lambda \in \Sigma$, and the two eigenspaces have equal dimension. This is part (3), and it contrasts with the compact torus of chapter 11, where the analogous involution was complex conjugation and the eigenvalues were purely imaginary; here the eigenvalues are real and θ reverses their sign.

Step 5: Σ is a root system. Equip $\mathfrak{a}$ with the inner product $\langle \cdot, \cdot \rangle = B_\theta|_{\mathfrak{a}}$, and transport it to $\mathfrak{a}^*$; let $V \subseteq \mathfrak{a}^*$ be the span of Σ. Three axioms must be checked (theorem 7.5.4 for the abstract notion). *Finiteness and spanning*: Σ is finite by Step 1, and it spans V by definition of V; no nonzero $H \in \mathfrak{a}$ can have $\lambda(H) = 0$ for all $\lambda \in \Sigma$ unless H is central, and semisimplicity of $\mathfrak{g}_0$ forbids a nonzero central element, so Σ spans $\mathfrak{a}^*$ and $V = \mathfrak{a}^*$. *Reflection integrality*: for $\lambda \in \Sigma$ pick a nonzero $E \in \mathfrak{g}_\lambda$; then E, $\theta E \in \mathfrak{g}_{-\lambda}$, and $H_\lambda = [E, \theta E]$ span an $\mathfrak{sl}_2$-triple with $H_\lambda \in \mathfrak{a}$, exactly as in the complex case (chapter 7). Representation theory of this $\mathfrak{sl}_2$ acting on $\mathfrak{g}_0$ forces $\mu(H_\lambda) \in \mathbb{Z}$ for every $\mu \in \Sigma$ and forces Σ to be stable under the reflection

$$s_\lambda(\mu) = \mu - \frac{2\langle \mu, \lambda \rangle}{\langle \lambda, \lambda \rangle}\,\lambda$$

the same argument that made Φ a root system, now applied to the real eigenvalues. These are the root-system axioms, so Σ is a root system with Weyl group generated by the s_λ, completing the proof of part (4).

The one point of departure from the complex theory is that Σ need not be *reduced*: an $\mathfrak{sl}_2$-triple can appear with 2λ also a root, because the multiplicity of the ± 2-weight need not vanish when $\mathfrak{g}_\lambda$ has dimension greater than one. So Σ is an abstract root system in $\mathfrak{a}^*$, possibly of the non-reduced type BC, completing the proof.

The decomposition is the tool that makes the noncompact structure computable. The centralizer $\mathfrak{g}_0^0 = \mathfrak{m} \oplus \mathfrak{a}$ is the analogue of the Cartan subalgebra: $\mathfrak{a}$ is the split part on which ad acts by real eigenvalues, and $\mathfrak{m} = \mathfrak{z}_{\mathfrak{k}}(\mathfrak{a})$ is a compact remainder on which $\mathfrak{a}$ acts trivially, the piece a compact torus would absorb but a split torus cannot see. Every restricted root λ comes paired with $-\lambda$ by

theorem 14.1.4(3), and choosing a system of positive roots Σ^+ singles out the nilpotent subalgebra $\mathfrak{n} = \bigoplus_{\lambda>0} \mathfrak{g}_\lambda$ that section 14.2 exponentiates to the N factor of $G = KAN$. The grading of part (2) is what makes $\mathfrak{n}$ nilpotent: iterated brackets push λ ever further positive until the root space is zero.

Two features distinguish restricted roots from the complex roots of chapter 11. First, the eigenvalues are real, forced by $\mathfrak{a} \subseteq \mathfrak{p}$ and the symmetry of $\mathrm{ad}(\mathfrak{p})$; on a compact maximal torus ad was skew and the eigenvalues imaginary, the contrast that separates a compact torus from a split one. Second, the root spaces $\mathfrak{g}_\lambda$ can have dimension greater than one, and 2λ can be a root, so Σ can be non-reduced. The restricted roots are the complex roots projected onto the real split subspace $\mathfrak{a}$: several complex roots collapse onto one restricted root, inflating its multiplicity.

The split real form is the case where no collapsing happens, and $\mathfrak{sl}_n(\mathbb{R})$ is the model.

Example 14.1.5: Restricted Roots of $\mathfrak{sl}_n(\mathbb{R})$

Let $\mathfrak{g}_0 = \mathfrak{sl}_n(\mathbb{R})$ with Cartan involution $\theta(X) = -X^{\mathsf{T}}$, so $\mathfrak{k} = \mathfrak{so}(n)$ and $\mathfrak{p} = \mathrm{Sym}_0(n)$ is the traceless symmetric matrices (example 13.5.1). A maximal abelian subspace of $\mathfrak{p}$ is

$$\mathfrak{a} = \{\mathrm{diag}(a_1, \ldots, a_n) \mid a_i \in \mathbb{R},\ \textstyle\sum_i a_i = 0\}$$

the traceless *real diagonal* matrices: these commute, they are symmetric, and no symmetric matrix outside the diagonal commutes with a diagonal matrix having distinct entries, so $\mathfrak{a}$ is maximal abelian. Thus the real rank of $\mathfrak{sl}_n(\mathbb{R})$ is $n-1$, the full dimension of $\mathfrak{p}$ along the diagonal, and $\mathfrak{sl}_n(\mathbb{R})$ is a split real form.

Let $\varepsilon_i \in \mathfrak{a}^*$ read off the i-th diagonal entry, $\varepsilon_i(\mathrm{diag}(a_1, \ldots, a_n)) = a_i$. For the matrix unit E_{ij} $(i \neq j)$ and $H = \mathrm{diag}(a_1, \ldots, a_n) \in \mathfrak{a}$,

$$[H, E_{ij}] = (a_i - a_j)E_{ij} = (\varepsilon_i - \varepsilon_j)(H)\, E_{ij}$$

so $E_{ij} \in \mathfrak{g}_{\varepsilon_i - \varepsilon_j}$, and the restricted roots are

$$\Sigma = \{\varepsilon_i - \varepsilon_j \mid i \neq j\}$$

the root system of type A_{n-1}, each root space $\mathfrak{g}_{\varepsilon_i - \varepsilon_j} = \mathbb{R}E_{ij}$ one-dimensional. The zero eigenspace is $\mathfrak{g}_0^0 = \mathfrak{a}$ itself: a diagonal matrix commutes with $\mathfrak{a}$, and $\mathfrak{m} = \mathfrak{z}_{\mathfrak{k}}(\mathfrak{a}) = 0$, because a skew-symmetric matrix commuting with a diagonal matrix of distinct entries is itself diagonal, hence zero. So for the split form $\mathfrak{m}$ vanishes, the restricted root system coincides with the ordinary root system, and the decomposition $\mathfrak{g}_0 = \mathfrak{a} \oplus \bigoplus_{i \neq j} \mathbb{R}E_{ij}$ is the familiar decomposition of $\mathfrak{sl}_n$ into diagonal, upper-triangular, and lower-triangular parts.

For $n = 2$ the picture collapses to a single positive root, the smallest instance and the running example of the chapter.

Example 14.1.6: Real Rank One: $\mathfrak{sl}_2(\mathbb{R})$

For $\mathfrak{sl}_2(\mathbb{R})$, the maximal abelian subspace is the one-dimensional

$$\mathfrak{a} = \mathbb{R}H, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

so the real rank is 1: $\mathfrak{sl}_2(\mathbb{R})$ is the smallest real-rank-one group. Writing $\alpha \in \mathfrak{a}^*$ for the functional $\alpha(tH) = 2t$, the standard nilpotent generators $E = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $F = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ satisfy $[H, E] = 2E$ and

$[H, F] = -2F$, so

$$\mathfrak{g}_\alpha = \mathbb{R}E, \qquad \mathfrak{g}_{-\alpha} = \mathbb{R}F, \qquad \Sigma = \{\alpha, -\alpha\}$$

a single positive restricted root α with its negative. Here $\mathfrak{m} = \mathfrak{z}_{\mathfrak{k}}(\mathfrak{a}) = 0$, since $\mathfrak{k} = \mathfrak{so}(2)$ is one-dimensional and its generator does not commute with H. The Cartan involution sends $E \mapsto \theta E = -E^{\mathsf{T}} = -F$, exhibiting $\theta\mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$ concretely. The whole decomposition is $\mathfrak{sl}_2(\mathbb{R}) = \mathbb{R}E \oplus \mathbb{R}H \oplus \mathbb{R}F$, the split form of type A_1.

The indefinite unitary groups show the non-reduced phenomenon, where a restricted root and its double both appear.

Example 14.1.7: A Non-Reduced System: $\mathfrak{su}(p, q)$

Let $\mathfrak{g}_0 = \mathfrak{su}(p, q)$ with $p \geq q$, whose Cartan decomposition has $\mathfrak{k} = \mathrm{s}(\mathfrak{u}(p) \oplus \mathfrak{u}(q))$ and $\mathfrak{p}$ the off-diagonal Hermitian blocks (example 13.5.2). A maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$ has dimension q, so the real rank is q rather than the full rank $p+q-1$ of the complex algebra: the split directions are only q, and the excess $p - q$ shows up inside $\mathfrak{m}$. The restricted root system is

$$\Sigma = \{\pm\varepsilon_i \pm \varepsilon_j \ (i < j),\ \pm\varepsilon_i,\ \pm 2\varepsilon_i \mid 1 \leq i, j \leq q\}$$

of type BC_q when $p > q$, the non-reduced system in which both a short root ε_i and its double $2\varepsilon_i$ occur. The presence of ε_i and $2\varepsilon_i$ together is exactly the non-reducedness that theorem 14.1.4(4) permits: several complex roots of $\mathfrak{su}(p, q)_{\mathbb{C}} = \mathfrak{sl}_{p+q}(\mathbb{C})$ project onto the same restricted root, and the multiplicities $\dim \mathfrak{g}_{\varepsilon_i}$ can exceed one. When $p = q$ the short roots $\pm\varepsilon_i$ drop out and the system reduces to type C_q; the smallest non-reduced case is $\mathfrak{su}(2, 1)$, with $q = 1$, whose restricted roots $\{\pm\varepsilon_1, \pm 2\varepsilon_1\}$ form the rank-one non-reduced system BC_1.

The three examples calibrate the invariant. The split form $\mathfrak{sl}_n(\mathbb{R})$ has real rank equal to the complex rank and $\mathfrak{m} = 0$; the compact form (real rank zero) sits at the opposite extreme with $\mathfrak{a} = 0$ and Σ empty; and $\mathfrak{su}(p, q)$ interpolates, with real rank $\min(p, q)$ measuring the number of split directions and a nonzero $\mathfrak{m}$ absorbing the compact remainder. Real rank is the precise measurement of noncompactness: it is zero exactly when G is compact and Σ is empty, and it grows to the complex rank as the real form approaches the maximally split one. The restricted root system Σ, together with the multiplicities $\dim \mathfrak{g}_\lambda$, is the combinatorial data that will carry the whole analysis of the noncompact group, exactly as the ordinary root system carried the compact theory.

Exercise 14.1.1

1. For $\mathfrak{sl}_n(\mathbb{R})$ with $\mathfrak{a}$ the traceless real diagonal matrices, verify in full that $\Sigma = \{\varepsilon_i - \varepsilon_j \mid i \neq j\}$ is the restricted root system of type A_{n-1}. Compute $[H, E_{ij}]$ for $H = \mathrm{diag}(a_1, \ldots, a_n) \in \mathfrak{a}$, identify each restricted root space, and confirm that $\dim \mathfrak{g}_{\varepsilon_i - \varepsilon_j} = 1$ and that $\mathfrak{g}_0^0 = \mathfrak{a}$.

2. For $\mathfrak{sl}_2(\mathbb{R})$ with $\mathfrak{a} = \mathbb{R}H$, $H = \mathrm{diag}(1, -1)$, compute the restricted roots explicitly. Show that $\Sigma = \{\alpha, -\alpha\}$ with $\alpha(tH) = 2t$, that $\mathfrak{g}_\alpha = \mathbb{R}E$ and $\mathfrak{g}_{-\alpha} = \mathbb{R}F$, and verify that $\theta E = -F$ so that $\theta\mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$. Conclude that $\mathfrak{sl}_2(\mathbb{R})$ has real rank one.

3. Prove that a real semisimple Lie algebra $\mathfrak{g}_0$ has real rank zero if and only if it is compact. (Use that real rank zero means $\mathfrak{p}$ contains no nonzero abelian subspace, and that $\mathfrak{p}$ is abelian as soon as it is nonzero along a single line.)

4. Let $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ be a Cartan decomposition with inner product B_θ. Prove directly that $\mathrm{ad}(H)$

is B_θ-symmetric for every $H \in \mathfrak{a} \subseteq \mathfrak{p}$, and explain why this forces the eigenvalues $\lambda(H)$ to be real. Contrast with the compact torus of chapter 11, where ad is B_θ-skew and the eigenvalues are imaginary.

5. Identify $\mathfrak{m} = \mathfrak{z}_{\mathfrak{k}}(\mathfrak{a})$ for $\mathfrak{sl}_n(\mathbb{R})$ and for $\mathfrak{su}(p,q)$. Show $\mathfrak{m} = 0$ for $\mathfrak{sl}_n(\mathbb{R})$ by proving that a skew-symmetric matrix commuting with a real diagonal matrix of distinct entries is zero, and describe why $\mathfrak{m} \neq 0$ for $\mathfrak{su}(p,q)$ when $p > q$.

6. Exhibit the non-reduced BC_1 restricted root system of $\mathfrak{su}(2,1)$. Show that the real rank is 1, that $\Sigma = \{\pm\varepsilon_1, \pm 2\varepsilon_1\}$, and identify which restricted root spaces have dimension greater than one, contrasting with the reduced A_1 system of $\mathfrak{sl}_2(\mathbb{R})$.

14.2 The Iwasawa Decomposition $G = KAN$

The restricted root space decomposition of section 14.1 sorts $\mathfrak{g}_0$ into the centralizer $\mathfrak{m} \oplus \mathfrak{a}$ and the restricted root spaces $\mathfrak{g}_\lambda$. A choice of positive restricted roots collapses this data into three subalgebras that fit together in a single product. The Cartan decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ of chapter 13 splits $\mathfrak{g}_0$ into a compact part and a Euclidean complement, but $\mathfrak{p}$ is not a subalgebra and $\exp(\mathfrak{p})$ is not a subgroup. The Iwasawa decomposition trades that symmetric splitting for one whose three factors are all subgroups: a compact K, an abelian $A = \exp \mathfrak{a}$, and a nilpotent N assembled from the positive restricted root spaces. The section builds the nilpotent piece, proves the algebra splits as $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}$, and integrates the splitting to the group identity $G = KAN$.

The device that selects a positive half of Σ is the same device that ordered an abstract root system: a base Δ (definition 8.2.1), or equivalently a choice of Weyl chamber in $\mathfrak{a}^*$. Since Σ is a root system in $\mathfrak{a}^*$ (theorem 14.1.4), everything from the abstract theory applies verbatim.

Definition 14.2.1: Iwasawa Data

Let $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ be a Cartan decomposition of a real semisimple Lie algebra, $\mathfrak{a} \subseteq \mathfrak{p}$ a maximal abelian subspace (definition 14.1.1), and $\Sigma \subseteq \mathfrak{a}^*$ the restricted roots (definition 14.1.3). Fix a choice of *positive restricted roots* $\Sigma^+ \subseteq \Sigma$: the roots taking positive values on a fixed regular $H_0 \in \mathfrak{a}$, so that $\Sigma = \Sigma^+ \sqcup (-\Sigma^+)$ (the analogue for Σ of the base Δ of definition 8.2.1). The associated *nilpotent subalgebra* is

$$\mathfrak{n} = \bigoplus_{\lambda \in \Sigma^+} \mathfrak{g}_\lambda$$

the sum of the positive restricted root spaces, and $\mathfrak{a} \oplus \mathfrak{n}$ is the associated *solvable subalgebra*.

Two facts make the definition consistent. First, $\mathfrak{n}$ is a subalgebra: the grading $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subseteq \mathfrak{g}_{\lambda+\mu}$ (theorem 14.1.4) sends a bracket of positive root spaces into $\mathfrak{g}_{\lambda+\mu}$, which is a summand of $\mathfrak{n}$ when $\lambda + \mu \in \Sigma^+$ and is zero when $\lambda + \mu \notin \Sigma$, so $[\mathfrak{n}, \mathfrak{n}] \subseteq \mathfrak{n}$. Second, $\mathfrak{n}$ is nilpotent. Order Σ^+ by height (the sum of coefficients in the base expansion); a bracket strictly raises height, so iterating the bracket eventually raises the height past that of the highest root and lands in zero. Concretely, if $\lambda_{\max}$ is a maximal positive root then the lower central series terminates once the accumulated height exceeds $\mathrm{ht}(\lambda_{\max})$, so $\mathfrak{n}$ is nilpotent. Adjoining $\mathfrak{a}$ keeps it solvable but not nilpotent: $[\mathfrak{a}, \mathfrak{g}_\lambda] = \mathfrak{g}_\lambda$ for $\lambda \neq 0$ (each $H \in \mathfrak{a}$ acts on $\mathfrak{g}_\lambda$ by the scalar $\lambda(H)$, nonzero for suitable H), so $\mathfrak{a}$ acts on $\mathfrak{n}$ semisimply with nonzero weights, and $[\mathfrak{a} \oplus \mathfrak{n}, \mathfrak{a} \oplus \mathfrak{n}] = \mathfrak{n}$, whose next derived term $[\mathfrak{n}, \mathfrak{n}]$ is a proper subspace; the derived series descends to zero, so $\mathfrak{a} \oplus \mathfrak{n}$ is solvable.

The running example puts the abstraction on the ground before any theorem.

Example 14.2.2: The Nilpotent Subalgebra of $\mathfrak{sl}_n(\mathbb{R})$

Take $\mathfrak{g}_0 = \mathfrak{sl}_n(\mathbb{R})$ with the Cartan involution $\theta(X) = -X^{\mathsf{T}}$ of example 13.5.1, so $\mathfrak{k} = \mathfrak{so}(n)$ and $\mathfrak{p}$ the traceless symmetric matrices. A maximal abelian $\mathfrak{a} \subseteq \mathfrak{p}$ is the space of traceless real diagonal matrices $H = \operatorname{diag}(h_1, \ldots, h_n)$, $\sum h_i = 0$. For the coordinate functional $\varepsilon_i(H) = h_i$, the matrix unit E_{ij} $(i \neq j)$ satisfies $[H, E_{ij}] = (h_i - h_j)E_{ij} = (\varepsilon_i - \varepsilon_j)(H)\, E_{ij}$, so $\mathfrak{g}_{\varepsilon_i - \varepsilon_j} = \mathbb{R}E_{ij}$ and $\Sigma = \{\varepsilon_i - \varepsilon_j \,|\, i \neq j\}$ is the type A_{n-1} system, with $\mathfrak{m} = Z_{\mathfrak{k}}(\mathfrak{a}) = 0$. Choosing $\Sigma^+ = \{\varepsilon_i - \varepsilon_j \,|\, i < j\}$ gives

$$\mathfrak{n} = \bigoplus_{i<j} \mathbb{R}E_{ij} = \{X \mid X \text{ strictly upper triangular}\}$$

the strictly upper triangular matrices, and $\mathfrak{a} \oplus \mathfrak{n}$ is the traceless upper triangular matrices. The grading $[E_{ij}, E_{jk}] = E_{ik}$ raises the index gap $j - i$, and any product of n strictly upper triangular matrices vanishes, exhibiting the nilpotency directly. This is the split $\mathfrak{sl}_n(\mathbb{R}) = \mathfrak{so}(n) \oplus \mathfrak{a} \oplus \mathfrak{n}$ into antisymmetric plus traceless diagonal plus strictly upper triangular, the infinitesimal form of the decomposition the section is after.

The splitting the example exhibits by inspection is a general theorem. It is a vector space direct sum, not a direct sum of ideals: $\mathfrak{a} \oplus \mathfrak{n}$ is a subalgebra and $\mathfrak{k}$ is a subalgebra, but neither is an ideal and they overlap nowhere except at the origin.

Theorem 14.2.3: The Iwasawa Decomposition of $\mathfrak{g}_0$

With the notation of definition 14.2.1, there is a vector space direct sum

$$\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n}$$

in which $\mathfrak{k}$ is a subalgebra, $\mathfrak{a} \oplus \mathfrak{n}$ is a subalgebra, and $\mathfrak{k} \cap (\mathfrak{a} \oplus \mathfrak{n}) = 0$. It is not a direct sum of ideals: none of the three summands is an ideal of $\mathfrak{g}_0$.

***Proof*:**

That $\mathfrak{k}$ is a subalgebra is theorem 13.3.3(1), and that $\mathfrak{a} \oplus \mathfrak{n}$ is a subalgebra was checked after definition 14.2.1. Two things remain: the intersection $\mathfrak{k} \cap (\mathfrak{a} \oplus \mathfrak{n})$ is zero, and the three subspaces span. Once both hold the sum is direct by dimension.

Step 1: the intersection is zero. The Cartan involution θ acts as $+1$ on $\mathfrak{k}$ and -1 on $\mathfrak{p}$. On $\mathfrak{a} \subseteq \mathfrak{p}$ it is -1, and on $\mathfrak{g}_\lambda$ it carries $\mathfrak{g}_\lambda$ isomorphically to $\mathfrak{g}_{-\lambda}$ (theorem 14.1.4). For $\lambda \in \Sigma^+$ the space $\mathfrak{g}_{-\lambda}$ lies in $\mathfrak{n}^- = \bigoplus_{\lambda>0} \mathfrak{g}_{-\lambda}$, and $\mathfrak{n} \cap \mathfrak{n}^- = 0$ because they involve disjoint restricted root spaces. Suppose $X \in \mathfrak{k} \cap (\mathfrak{a} \oplus \mathfrak{n})$. Then $\theta X = X$ since $X \in \mathfrak{k}$. Write $X = H + \sum_{\lambda>0} X_\lambda$ with $H \in \mathfrak{a}$ and $X_\lambda \in \mathfrak{g}_\lambda$. Applying θ,

$$X = \theta X = -H + \sum_{\lambda>0} \theta X_\lambda$$

with $-H \in \mathfrak{a}$ and $\theta X_\lambda \in \mathfrak{g}_{-\lambda}$. Comparing the two expressions for X across the direct sum $\mathfrak{a} \oplus \bigoplus_{\lambda \neq 0} \mathfrak{g}_\lambda$ of theorem 14.1.4 forces $H = -H$, so $H = 0$. For the root spaces, fix $\lambda > 0$ and read off the $\mathfrak{g}_\lambda$-component of both sides: on the left it is X_λ, while on the right $-H$ contributes nothing and each $\theta X_\mu \in \mathfrak{g}_{-\mu}$ has zero $\mathfrak{g}_\lambda$-component (as $-\mu < 0 < \lambda$), so the $\mathfrak{g}_\lambda$-component of θX vanishes. Hence $X_\lambda = 0$ for every λ, so $X = 0$ and the intersection is trivial.

Step 2: the dimension count. The restricted root space decomposition gives

$$\dim \mathfrak{g}_0 = \dim \mathfrak{m} + \dim \mathfrak{a} + \sum_{\lambda \in \Sigma} \dim \mathfrak{g}_\lambda = \dim \mathfrak{m} + \dim \mathfrak{a} + 2 \sum_{\lambda > 0} \dim \mathfrak{g}_\lambda$$

using $\theta \colon \mathfrak{g}_\lambda \xrightarrow{\sim} \mathfrak{g}_{-\lambda}$, so $\dim \mathfrak{g}_\lambda = \dim \mathfrak{g}_{-\lambda}$ and the roots pair off. By definition $\dim \mathfrak{n} = \sum_{\lambda>0} \dim \mathfrak{g}_\lambda$. The remaining term is $\dim \mathfrak{k}$. Compute it from θ: an element of $\mathfrak{k}$ is a θ-fixed vector, and applying the projection $\frac{1}{2}(1+\theta)$ to $\mathfrak{g}_0$ produces the fixed subspace. On $\mathfrak{m} \oplus \mathfrak{a} = \mathfrak{g}_0^0$ the involution θ fixes $\mathfrak{m}$ and negates $\mathfrak{a}$, so the fixed part of $\mathfrak{g}_0^0$ is exactly $\mathfrak{m}$. On each pair $\mathfrak{g}_\lambda \oplus \mathfrak{g}_{-\lambda}$ (for $\lambda > 0$) the map $(1+\theta)$ sends $X_\lambda \mapsto X_\lambda + \theta X_\lambda$, and the image

$$(1+\theta)\mathfrak{g}_\lambda = \{X + \theta X \mid X \in \mathfrak{g}_\lambda\}$$

is a subspace of $\mathfrak{k}$ isomorphic to $\mathfrak{g}_\lambda$ (the map $X \mapsto X + \theta X$ is injective on $\mathfrak{g}_\lambda$ since $\theta X \in \mathfrak{g}_{-\lambda}$ can cancel $X \in \mathfrak{g}_\lambda$ only if both vanish). These fixed subspaces are independent across distinct $\lambda > 0$, so

$$\mathfrak{k} = \mathfrak{m} \oplus \bigoplus_{\lambda>0} (1+\theta)\mathfrak{g}_\lambda, \qquad \dim \mathfrak{k} = \dim \mathfrak{m} + \sum_{\lambda>0} \dim \mathfrak{g}_\lambda$$

Adding the three dimensions,

$$\begin{aligned} \dim \mathfrak{k} + \dim \mathfrak{a} + \dim \mathfrak{n} &= \Big(\dim \mathfrak{m} + \sum_{\lambda>0} \dim \mathfrak{g}_\lambda\Big) + \dim \mathfrak{a} + \sum_{\lambda>0} \dim \mathfrak{g}_\lambda \\ &= \dim \mathfrak{m} + \dim \mathfrak{a} + 2 \sum_{\lambda>0} \dim \mathfrak{g}_\lambda \\ &= \dim \mathfrak{g}_0 \end{aligned}$$

The three subspaces have dimensions summing to $\dim \mathfrak{g}_0$ and, by Step 1, $\mathfrak{k}$ meets $\mathfrak{a} \oplus \mathfrak{n}$ only in 0. Since $\mathfrak{a} \cap \mathfrak{n} = 0$ as well (distinct restricted root spaces, one of them $\mathfrak{g}_0^0$), the sum $\mathfrak{k} + \mathfrak{a} + \mathfrak{n}$ is a direct sum of the correct total dimension, hence all of $\mathfrak{g}_0$.

Step 3: no summand is an ideal. Were $\mathfrak{k}$ an ideal, $[\mathfrak{p}, \mathfrak{k}] \subseteq \mathfrak{k}$, but theorem 13.3.3(1) gives $[\mathfrak{p}, \mathfrak{k}] \subseteq \mathfrak{p}$, so $\mathfrak{k}$ an ideal would force $[\mathfrak{p}, \mathfrak{k}] = 0$, impossible for a semisimple $\mathfrak{g}_0$ with nonzero $\mathfrak{p}$. Were $\mathfrak{a} \oplus \mathfrak{n}$ an ideal, it would be a solvable ideal, hence contained in the radical, which is zero in the semisimple $\mathfrak{g}_0$; but $\mathfrak{a} \oplus \mathfrak{n} \neq 0$. And $\mathfrak{n}$ is not an ideal since $[\mathfrak{k}, \mathfrak{n}]$ is not contained in $\mathfrak{n}$ (bracketing a positive root vector with $\mathfrak{k}$ produces $\mathfrak{m} \oplus \mathfrak{a}$ and negative root components, as Step 2's $(1+\theta)\mathfrak{g}_\lambda \subseteq \mathfrak{k}$ makes visible). Thus the decomposition is a vector space direct sum of subalgebras, not of ideals, completing the proof.

The theorem is the exact structural break from the Cartan decomposition. There the two summands $\mathfrak{k}$ and $\mathfrak{p}$ were θ-eigenspaces, orthogonal under B_θ and matched in a symmetric way; here the three summands $\mathfrak{k}$, $\mathfrak{a}$, $\mathfrak{n}$ are subalgebras but the decomposition is skew, no orthogonality and no θ-symmetry among them. What is gained is that all three are subalgebras, and hence will exponentiate to subgroups. The nilpotent $\mathfrak{n}$ is new: it is assembled from the positive restricted root spaces $\mathfrak{g}_\lambda$, each of which lies in neither $\mathfrak{k}$ nor $\mathfrak{p}$ (indeed θ carries $\mathfrak{g}_\lambda$ to $\mathfrak{g}_{-\lambda}$), and it had no counterpart in the compact theory, where the analogous ad-eigenvalues were imaginary and produced rotations rather than a nilpotent shear. The identity $\mathfrak{k} = \mathfrak{m} \oplus \bigoplus_{\lambda>0}(1+\theta)\mathfrak{g}_\lambda$ from Step 2 is the bookkeeping that makes the dimension count balance, and it records the sense in which $\mathfrak{k}$ is a diagonal copy of the root spaces: each positive root contributes one θ-symmetric combination $X + \theta X$ to $\mathfrak{k}$, one θ-antisymmetric combination to $\mathfrak{p}$, and its raw self X to $\mathfrak{n}$.

Integrating the algebra splitting to the group is the substance of the chapter. The three subalgebras integrate to closed subgroups K, $A = \exp \mathfrak{a}$, $N = \exp \mathfrak{n}$, and the claim is that multiplication assembles them into all of G with no redundancy.

The abelian and nilpotent factors are the tractable ones. Since $\mathfrak{a}$ is abelian and consists of $\mathbb{R}$-diagonalizable operators, $A = \exp \mathfrak{a}$ is a closed subgroup isomorphic to $(\mathbb{R}^r, +)$ with $r = \dim \mathfrak{a}$ the real rank, and $\exp\colon \mathfrak{a} \to A$ is a diffeomorphism. Since $\mathfrak{n}$ is nilpotent, $\exp\colon \mathfrak{n} \to N$ is a diffeomorphism onto a closed, simply connected nilpotent Lie group (the exponential of a nilpotent Lie algebra is a polynomial diffeomorphism, a standard fact recorded for matrix groups in chapter 1 and in general in [13]); its inverse is given by the terminating logarithm series. The delicate factor is the product.

Theorem 14.2.4: The Iwasawa Decomposition $G = KAN$

Let G be a connected semisimple Lie group with finite center, Cartan involution θ, and maximal compact subgroup K (theorem 13.4.3). With $A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$ as above, the multiplication map

$$K \times A \times N \longrightarrow G, \qquad (k, a, n) \longmapsto kan$$

is a diffeomorphism. Every $g \in G$ factors uniquely as $g = kan$ with $k \in K$, $a \in A$, $n \in N$, and the factors depend smoothly on g.

***Proof*:**

The strategy is to reduce the global statement to a linear-algebra fact by letting AN act on the symmetric space G/K. The map is smooth because multiplication in G is smooth; the work is that it is a bijection with smooth inverse. This is proved by showing the solvable group $S = AN$ acts simply transitively on G/K.

Step 1: $S = AN$ is a closed subgroup, and G/K is its target. By theorem 14.2.3 the subalgebra $\mathfrak{s} = \mathfrak{a} \oplus \mathfrak{n}$ is solvable, and $\mathfrak{n}$ is an ideal of $\mathfrak{s}$ with $\mathfrak{s}/\mathfrak{n} \cong \mathfrak{a}$ abelian; so $S = \exp \mathfrak{s} = AN$ is a connected solvable subgroup, and $\exp\colon \mathfrak{s} \to S$ is a diffeomorphism (a simply connected solvable group with the given ideal structure), giving S the manifold structure of $\mathfrak{a} \times \mathfrak{n}$. Since $\mathfrak{s} \cap \mathfrak{k} = 0$ (theorem 14.2.3), the orbit map $S \to G/K$, $s \mapsto sK$, is an immersion at the identity: its differential is the composite $\mathfrak{s} \hookrightarrow \mathfrak{g}_0 \twoheadrightarrow \mathfrak{g}_0/\mathfrak{k} = T_{eK}(G/K)$, injective because $\mathfrak{s} \cap \mathfrak{k} = 0$, and surjective because $\dim \mathfrak{s} = \dim \mathfrak{a} + \dim \mathfrak{n} = \dim \mathfrak{g}_0 - \dim \mathfrak{k} = \dim(G/K)$ by the dimension count of theorem 14.2.3. So the orbit map is a local diffeomorphism at eK, and by homogeneity (left translation by S) at every point of the orbit.

Step 2: S acts simply transitively on G/K. It suffices to prove the orbit map $\varphi\colon S \to G/K$ is a bijection; injectivity is simple transitivity and surjectivity is transitivity. The cleanest route realizes φ as Gram-Schmidt in a faithful representation, and the general case reduces to it. Embed $G \hookrightarrow \mathrm{GL}(V)$ with V a real vector space carrying a θ-adapted inner product, an inner product for which the Cartan involution is $g \mapsto (g^{\mathsf{T}})^{-1}$, so that $K = G \cap \mathrm{O}(V)$, A acts by positive diagonal operators in a suitable basis, and N acts by unitriangular operators (such a representation exists for any G as above, obtained by Gram-Schmidt against a K-invariant inner product on a faithful representation, see [13]; for the classical groups the defining representation already is one). Under this embedding G/K is realized as an orbit of positive-definite symmetric forms, and factoring $g = kan$ is precisely applying Gram-Schmidt to the columns of g: the orthonormalization produces the orthogonal frame, which is the factor k (equivalently, k^{-1} triangularizes g), the norms of the successive projections produce the positive diagonal a, and the

coefficients expressing the original vectors in the orthonormal frame produce the unitriangular n. Gram-Schmidt succeeds and is unique for every invertible g, so φ is a bijection. The $\mathrm{SL}(n,\mathbb{R})$ instance below carries out this orthonormalization explicitly and is the model for the general argument.

Step 3: the multiplication map is a diffeomorphism. By Step 2 the orbit map $\varphi\colon S \to G/K$ is a bijective local diffeomorphism (Step 1), hence a diffeomorphism. Consider $\mu\colon K \times S \to G$, $(k,s) \mapsto ks$. Surjectivity: given $g \in G$, the class $gK \in G/K$ equals $\varphi(s) = sK$ for a unique $s \in S$ by Step 2, so $s^{-1}g \in K$ and $g = s(s^{-1}g) \in SK$; thus $G = SK$, and inverting, $G = G^{-1} = (SK)^{-1} = K^{-1}S^{-1} = KS$, so every g has the form ks with $k \in K$, $s \in S$. Injectivity: if $k_1 s_1 = k_2 s_2$ then $k_2^{-1}k_1 = s_2 s_1^{-1} \in K \cap S$, and $K \cap S = \{e\}$ already from Step 2, since $s \in K \cap S$ gives $\varphi(s) = sK = eK = \varphi(e)$, so $s = e$ by the injectivity of φ; hence $k_1 = k_2$ and $s_1 = s_2$. So μ is a smooth bijection; its differential at (e,e) is $(\Xi, Z) \mapsto \Xi + Z$ on $\mathfrak{k} \times \mathfrak{s} \to \mathfrak{g}_0$, an isomorphism by theorem 14.2.3, so μ is a local diffeomorphism at (e,e) and, by homogeneity, everywhere; a bijective local diffeomorphism is a diffeomorphism ([13, Properties of Local Diffeomorphisms]). Finally $S = AN$ with $A \times N \to S$, $(a,n) \mapsto an$, a diffeomorphism (Step 1, exp on the semidirect product $\mathfrak{s} = \mathfrak{a} \ltimes \mathfrak{n}$), so

$$K \times A \times N \xrightarrow{\sim} K \times S \xrightarrow{\mu} G$$

is a composite of diffeomorphisms. Uniqueness of the three factors and their smooth dependence on g are exactly the statement that this composite is a diffeomorphism, completing the proof.

The three factors sit at three different registers. The compact K carries the rotational directions, the abelian A the radial or dilation directions, and the nilpotent N the shear directions. Unlike the Cartan factor $\exp(\mathfrak{p})$ of theorem 13.4.2, which is a submanifold but not a subgroup, all three Iwasawa factors are genuine subgroups. A general element of G is thereby written as a product of elements each lying in a well-understood subgroup, and the product coordinates are global (defined for every g, with no cells or lower-dimensional exceptions, in contrast to the Bruhat decomposition of section 14.3). The subgroup $S = AN$ acts simply transitively on the symmetric space G/K, so G/K is diffeomorphic to S, hence to the Euclidean space $\mathfrak{a} \times \mathfrak{n}$; the noncompact symmetric space is a solvable Lie group in disguise. This is the coordinate system on which noncompact harmonic analysis is built: the Iwasawa projection $g \mapsto a(g) \in A$ (the middle factor) is the source of the spherical functions and the Harish-Chandra c-function of the sequel theory, and the induced representations from the minimal parabolic $P = MAN$ (section 14.3) are constructed on functions transforming under exactly this A-factor.

The concrete anchor is the running example, where the abstract simply-transitive action is literally the Gram-Schmidt process.

Example 14.2.5: $\mathrm{SL}(n,\mathbb{R}) = \mathrm{SO}(n) \cdot A \cdot N$ as Gram-Schmidt

Let $G = \mathrm{SL}(n,\mathbb{R})$, $K = \mathrm{SO}(n)$, A the positive diagonal matrices of determinant one, and N the upper unitriangular matrices (ones on the diagonal, arbitrary entries above). These are the groups $\exp\mathfrak{k}$, $\exp\mathfrak{a}$, $\exp\mathfrak{n}$ of example 14.2.2: exp of a traceless diagonal matrix is a positive diagonal matrix of determinant one, and exp of a strictly upper triangular matrix is upper unitriangular (the series terminates). The factorization $g = kan$ of a matrix $g \in \mathrm{SL}(n,\mathbb{R})$ is the Gram-Schmidt orthonormalization of its columns.

Write $g = (v_1 | \cdots | v_n)$ by columns. Gram-Schmidt produces an orthonormal frame $q_1, \ldots, q_n$ and an upper triangular change of basis: setting $u_1 = v_1$ and $u_j = v_j - \sum_{i<j} \langle v_j, q_i \rangle\, q_i$ with

$q_j = u_j/\|u_j\|$, one has

$$v_j = \sum_{i \le j} r_{ij} q_i, \qquad r_{jj} = \|u_j\| > 0$$

so $g = QR$ with $Q = (q_1 \big| \cdots \big| q_n)$ orthogonal and $R = (r_{ij})$ upper triangular with positive diagonal. This is the QR decomposition. Split $R = DR'$ where $D = \mathrm{diag}(r_{11}, \ldots, r_{nn})$ is the positive diagonal and $R' = D^{-1}R$ is upper unitriangular, so $g = QDR'$ with $Q \in \mathrm{O}(n)$, D positive diagonal, $R' \in N$. Determinants pin down the two remaining conditions at once: $1 = \det g = \det Q \cdot \det D \cdot 1$ with $\det D = \prod_j r_{jj} > 0$, so $\det Q > 0$, forcing $\det Q = +1$ and hence $Q \in \mathrm{SO}(n)$; and then $\det D = 1/\det Q = 1$, so $D \in A$. Thus $g = Q\,D\,R'$ reads $g = kan$ with $k = Q \in \mathrm{SO}(n)$, $a = D \in A$, $n = R' \in N$, with no further normalization. Uniqueness of Gram-Schmidt (the orthonormal frame and the positive norms are determined by g) is uniqueness of the Iwasawa factorization, and its smoothness in g is the smoothness of the orthonormalization formulas. This is theorem 14.2.4 made completely explicit.

The two-dimensional case is small enough to write every factor by hand.

Example 14.2.6: The Iwasawa Factorization of $\mathrm{SL}(2,\mathbb{R})$

For $G = \mathrm{SL}(2,\mathbb{R})$ the three subgroups are

$$K = \mathrm{SO}(2) = \left\{ k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix} \right\}, \quad A = \left\{ a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} \right\}, \quad N = \left\{ n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \right\}$$

These are exp of $\mathfrak{so}(2) = \mathbb{R}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, of $\mathfrak{a} = \mathbb{R}\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, and of $\mathfrak{n} = \mathbb{R}\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, the rank-one split of $\mathfrak{sl}_2(\mathbb{R})$ (Σ^+ a single positive restricted root, $\mathfrak{m} = 0$). To factor a given $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $ad - bc = 1$, orthonormalize the first column $v_1 = \binom{a}{c}$: its norm is $\rho = \sqrt{a^2 + c^2}$, so $q_1 = \rho^{-1}\binom{a}{c}$ and the rotation carrying $\binom{1}{0}$ to q_1 is k_φ with $\cos\varphi = a/\rho$, $\sin\varphi = c/\rho$. Then

$$k_\varphi^{-1} g = \begin{pmatrix} \rho & * \\ 0 & \rho^{-1} \end{pmatrix}$$

is upper triangular with positive diagonal (the bottom-left entry is $-ac/\rho + ca/\rho = 0$ and the diagonal product is $\det g = 1$, forcing the lower-right to be ρ^{-1}). Setting $e^t = \rho$ and reading off the strictly upper entry gives $k_\varphi^{-1} g = a_t n_x$ with $x = (ab + cd)/\rho^2$, hence

$$g = k_\varphi\, a_t\, n_x, \qquad e^t = \sqrt{a^2 + c^2}, \quad \tan\varphi = c/a, \quad x = \frac{ab + cd}{a^2 + c^2}$$

the explicit KAN coordinates. It is the AN-part that acts simply transitively on $\mathfrak{H} = G/K$: apply $a_t n_x$ to the basepoint i of the upper half-plane (example 13.5.3), innermost factor first, so $n_x \cdot i = x + i$ and then $a_t \cdot (x + i) = e^{2t}(x + i) = e^{2t}x + i\,e^{2t}$. Thus $a_t n_x \cdot i = e^{2t}x + i\,e^{2t}$, and every $x' + iy' \in \mathfrak{H}$ is reached from i by a unique $a_t n_x$, with $y' = e^{2t}$ and $x' = e^{2t}x$: this is the simple transitivity of AN on $\mathfrak{H}$. The compact factor k_φ then acts on the point $a_t n_x \cdot i$, moving it off i in general (it is K that fixes i, not k_φ acting after $a_t n_x$), which is why the orbit map used is $a_t n_x \cdot i$ rather than $g \cdot i$. The full treatment on $\mathfrak{H}$ (geodesics as A-orbits, horocycles as N-orbits) is section 14.4.

Exercise 14.2.1

1. Carry out the Iwasawa factorization $g = kan$ of $g = \left(\begin{smallmatrix} 2 & 1 \\ 2 & 2 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$ (check $\det g = 2\cdot 2 - 1\cdot 2 = 2$; first rescale to $\mathrm{SL}(2,\mathbb{R})$ by dividing by $\sqrt{2}$, or work in $\mathrm{GL}(2,\mathbb{R})^+$ and note A becomes all positive diagonal). Using the formulas of example 14.2.6, compute e^t, φ, and x explicitly, and verify $k_\varphi a_t n_x = g$ by multiplying the three matrices.

2. Let $\mathfrak{n}$ be a nilpotent Lie algebra of nilpotency class c (so the $(c+1)$-fold iterated bracket vanishes). Show that the Baker-Campbell-Hausdorff series (theorem 3.5.1) is a finite polynomial on $\mathfrak{n}$, deduce that $\exp\colon \mathfrak{n} \to N$ is a polynomial map with polynomial inverse log, and conclude that N is a simply connected nilpotent Lie group diffeomorphic to $\mathfrak{n}$. Illustrate with $\mathfrak{n} = \{\text{strictly upper triangular } 3\times 3 \text{ matrices}\}$, the Heisenberg algebra.

3. Verify directly from definition 14.2.1 that $\mathfrak{a} \oplus \mathfrak{n}$ is a solvable subalgebra: show $[\mathfrak{a}\oplus\mathfrak{n}, \mathfrak{a}\oplus\mathfrak{n}] = \mathfrak{n}$ and that the derived series of $\mathfrak{n}$ terminates, so the derived series of $\mathfrak{a}\oplus\mathfrak{n}$ reaches 0. Why is $\mathfrak{a}\oplus\mathfrak{n}$ *not* nilpotent when $\Sigma \neq \varnothing$?

4. Show that $\exp\colon \mathfrak{n} \to N$ is a diffeomorphism onto a closed subgroup by the following steps: (i) exp is injective on $\mathfrak{n}$ (use that log is a polynomial inverse, from the previous question); (ii) exp is a local diffeomorphism (its differential at 0 is the identity); (iii) the image is closed (it is the set of unipotent elements of a fixed form, cut out by polynomial equations). Contrast with $\exp\colon \mathfrak{p} \to \exp(\mathfrak{p})$ of theorem 13.4.2, which is a diffeomorphism onto a submanifold that is *not* a subgroup.

5. Express $\dim N$ in terms of the positive restricted roots: show $\dim N = \sum_{\lambda\in\Sigma^+} \dim \mathfrak{g}_\lambda$ (the sum of the restricted root multiplicities). Compute it for $\mathfrak{sl}_n(\mathbb{R})$ (all multiplicities 1, $|\Sigma^+| = \binom{n}{2}$, so $\dim N = \binom{n}{2}$) and check $\dim K + \dim A + \dim N = \dim \mathrm{SL}(n,\mathbb{R}) = n^2 - 1$ using $\dim \mathrm{SO}(n) = \binom{n}{2}$ and $\dim A = n-1$.

6. Prove the crux of theorem 14.2.3 in the concrete case $\mathfrak{sl}_2(\mathbb{R})$: show by hand that $\mathfrak{so}(2)\cap(\mathfrak{a}\oplus\mathfrak{n}) = 0$, where $\mathfrak{a}\oplus\mathfrak{n}$ is the traceless upper triangular matrices, and confirm $\dim\mathfrak{so}(2)+\dim\mathfrak{a}+\dim\mathfrak{n} = 1+1+1 = 3 = \dim\mathfrak{sl}_2(\mathbb{R})$.

14.3 Parabolic Subgroups and the Bruhat Decomposition

The Iwasawa decomposition $G = KAN$ of section 14.2 names a distinguished solvable subgroup AN and leaves the centralizer $M = Z_K(\mathfrak{a})$ to one side. Assembling all three of M, A, and N into a single subgroup $P = MAN$ produces the minimal parabolic, the noncompact analogue of a Borel subgroup, and the coset space G/P is the real flag manifold. This section builds P and the larger standard parabolics that sit between it and G, records the restricted Weyl group $W = N_K(\mathfrak{a})/M$ acting on $\mathfrak{a}$, and organizes G into finitely many P-double-cosets indexed by W: the Bruhat decomposition.

Two threads run alongside the abstract statements. The restricted Weyl group repeats, for the restricted root system Σ, the chamber geometry that theorem 8.2.8 established for an ordinary root system, and the Bruhat cells for $\mathrm{SL}(n,\mathbb{R})$ are the strata of Gaussian elimination, the classes of matrices sharing a pattern of leading nonzero minors. The finest cell-closure combinatorics belong to the algebraic flag variety, developed over an algebraically closed field in the algebraic-groups companion, and are cited there rather than repeated here.

Definition 14.3.1: Minimal and Standard Parabolic Subgroups

Let G be a connected semisimple Lie group with Iwasawa decomposition $G = KAN$ (theorem 14.2.4) attached to a choice of positive restricted roots Σ^+, and let $M = Z_K(\mathfrak{a})$ be the centralizer of $\mathfrak{a}$ in K. The *minimal parabolic subgroup* is the product

$$P = MAN$$

Let $\Delta \subseteq \Sigma^+$ be the set of simple restricted roots (a base for Σ, in the sense of definition 8.2.1). For each subset $\Theta \subseteq \Delta$ let Σ_Θ be the set of restricted roots that are integer combinations of Θ, put $\mathfrak{a}_\Theta = \bigcap_{\lambda\in\Theta} \ker\lambda \subseteq \mathfrak{a}$, and set

$$\mathfrak{n}_\Theta = \bigoplus_{\lambda\in\Sigma^+\setminus\Sigma_\Theta} \mathfrak{g}_\lambda, \qquad \mathfrak{m}_\Theta = \mathfrak{m} \oplus \mathfrak{a}_\Theta^\perp \oplus \bigoplus_{\lambda\in\Sigma_\Theta} \mathfrak{g}_\lambda$$

where $\mathfrak{a}_\Theta^\perp$ is the orthocomplement of $\mathfrak{a}_\Theta$ in $\mathfrak{a}$. The *standard parabolic subgroup* P_Θ is the subgroup with Lie algebra $\mathfrak{m}_\Theta \oplus \mathfrak{a}_\Theta \oplus \mathfrak{n}_\Theta$; equivalently, P_Θ is generated by P together with the restricted root subgroups $\exp\mathfrak{g}_{-\lambda}$ for $\lambda \in \Theta$. The homogeneous space G/P is the *(real) flag manifold* of G.

The extremes fix the meaning of the interpolation. For $\Theta = \varnothing$ the set Σ_Θ is empty, $\mathfrak{a}_\Theta = \mathfrak{a}$, $\mathfrak{n}_\Theta = \mathfrak{n}$, and $\mathfrak{m}_\Theta = \mathfrak{m}$, so $P_\varnothing = MAN = P$: the minimal parabolic. For $\Theta = \Delta$ every positive restricted root lies in Σ_Θ, so $\mathfrak{n}_\Theta = 0$ and $\mathfrak{a}_\Theta = 0$, leaving $\mathfrak{m}_\Theta = \mathfrak{g}_0$, so $P_\Delta = G$. Each intermediate Θ reinstates the restricted root subgroups $\exp\mathfrak{g}_{-\lambda}$ for $\lambda \in \Theta$, enlarging P by adjoining the negative directions those simple roots span, and the assignment $\Theta \mapsto P_\Theta$ is an order-preserving bijection from subsets of Δ onto the subgroups of G containing P. The name *parabolic* is inherited from the algebraic theory, where the analogue of $P_\varnothing$ is a Borel subgroup and the analogous quotient G/P is the projective flag variety; the real quotient G/P is compact because $G = KP$ (from $G = KAN$ and $M \subseteq K$), so $G/P = K/M$ is a continuous image of the compact group K.

The running example makes P and the standard parabolics visible as block-triangular matrices.

Example 14.3.2: Parabolic Subgroups of $\mathrm{SL}(n,\mathbb{R})$

Take $G = \mathrm{SL}(n,\mathbb{R})$, with $K = \mathrm{SO}(n)$, A the positive diagonal matrices of determinant one, and N the upper unitriangular matrices, as in example 13.5.1 and the Iwasawa decomposition of section 14.2. Here $\mathfrak{a}$ is the traceless real diagonal matrices, and $M = Z_K(\mathfrak{a})$ is the group of diagonal matrices in $\mathrm{SO}(n)$, namely the diagonal sign matrices $\mathrm{diag}(\varepsilon_1,\dots,\varepsilon_n)$ with $\varepsilon_i = \pm 1$ and $\prod_i \varepsilon_i = 1$, a finite group $(\mathbb{Z}/2)^{n-1}$. The minimal parabolic is

$$P = MAN = \{g \in \mathrm{SL}(n,\mathbb{R}) \mid g \text{ is upper triangular}\}$$

the full upper-triangular subgroup: A gives the positive diagonal, N the unit upper-triangular part, and M the diagonal signs that let the diagonal entries be any nonzero reals of product one. The simple restricted roots are $\Delta = \{\varepsilon_i - \varepsilon_{i+1}\}_{i=1}^{n-1}$, and a subset $\Theta \subseteq \Delta$ corresponds to an ordered partition of n: omitting the simple root $\varepsilon_i - \varepsilon_{i+1}$ from Θ places a block boundary between rows i and $i+1$. The standard parabolic P_Θ is then the group of block upper-triangular matrices for that partition, the matrices preserving the flag $0 \subseteq V_1 \subseteq \cdots \subseteq V_r = \mathbb{R}^n$ whose steps have the block dimensions. For $n = 3$ the four proper cases are: $\Theta = \varnothing$, the upper-triangular P (flag $V_1 \subseteq V_2$, dimensions $1,1,1$); $\Theta = \{\varepsilon_1 - \varepsilon_2\}$, block sizes $2,1$ (a single plane V_2); $\Theta = \{\varepsilon_2 - \varepsilon_3\}$, block sizes

$1,2$ (a single line V_1); and $\Theta = \Delta$, all of $\mathrm{SL}(3,\mathbb{R})$. The flag manifold $G/P = \mathrm{SO}(3)/M$ is the manifold of complete flags in $\mathbb{R}^3$.

The finite group $W = N_K(\mathfrak{a})/M$ that indexes the double cosets is the exact restricted analogue of the Weyl group of a compact group (definition 11.2.1) and of the abstract Weyl group of a root system. Its action on $\mathfrak{a}$ is the reflection action of the restricted root system, and the proof reruns the chamber argument of theorem 8.2.8 with K-conjugation giving the reflections.

Proposition 14.3.3: The Restricted Weyl Group

Let G be a connected semisimple Lie group with maximal abelian $\mathfrak{a} \subseteq \mathfrak{p}$, restricted root system $\Sigma \subseteq \mathfrak{a}^*$, and $M = Z_K(\mathfrak{a})$, $M^* = N_K(\mathfrak{a})$ the centralizer and normalizer of $\mathfrak{a}$ in K. The quotient

$$W = N_K(\mathfrak{a})/Z_K(\mathfrak{a}) = M^*/M$$

is a finite group, the *restricted Weyl group*. Its natural action on $\mathfrak{a}$ (and dually on $\mathfrak{a}^*$) is faithful, preserves Σ, and coincides with the action of the Weyl group $W(\Sigma)$ of the restricted root system: W is generated by the reflections s_λ in the restricted roots $\lambda \in \Sigma$, and it permutes the Weyl chambers of $\mathfrak{a}$ simply transitively.

***Proof*:**
The argument has three parts: the action is well defined and faithful; each restricted-root reflection is realized by an element of M^*; and no element of M^* acts on $\mathfrak{a}$ outside $W(\Sigma)$. Finiteness then follows.

Step 1: a faithful action on $\mathfrak{a}$. An element $k \in M^* = N_K(\mathfrak{a})$ satisfies $\mathrm{Ad}(k)\mathfrak{a} = \mathfrak{a}$, so $\mathrm{Ad}(k)$ restricts to a linear automorphism of $\mathfrak{a}$; this gives a homomorphism $M^* \to \mathrm{GL}(\mathfrak{a})$. Its kernel is exactly the set of $k \in M^*$ acting trivially on $\mathfrak{a}$, that is $\{k \in N_K(\mathfrak{a}) \mid \mathrm{Ad}(k)H = H \ \forall H \in \mathfrak{a}\} = Z_K(\mathfrak{a}) = M$. Thus $W = M^*/M$ acts faithfully on $\mathfrak{a}$. Because $\mathrm{Ad}(k)$ is an automorphism of $\mathfrak{g}_0$ fixing $\mathfrak{a}$ setwise, it carries the restricted root space $\mathfrak{g}_\lambda$ to $\mathfrak{g}_{\lambda\circ\mathrm{Ad}(k)^{-1}}$, so it permutes Σ; hence W acts on $\mathfrak{a}^*$ preserving Σ, and this action lands in the finite group of permutations of the finite set Σ that extend to linear maps, so once Steps 2 and 3 identify the image with $W(\Sigma)$ the finiteness is automatic.

Step 2: every reflection s_λ *is realized in* M^*. Fix a restricted root $\lambda \in \Sigma$ and a nonzero $H_\lambda \in \mathfrak{a}$ with $\mu(H_\lambda) = \langle \mu, \lambda \rangle$ for all μ, the coroot direction. Choose $0 \neq X \in \mathfrak{g}_\lambda$. Because $\theta\mathfrak{g}_\lambda = \mathfrak{g}_{-\lambda}$ (theorem 14.1.4), the vector $\theta X \in \mathfrak{g}_{-\lambda}$ is nonzero, and the $\mathfrak{sl}_2$-triple built from X, θX, and their bracket spans a subalgebra $\mathfrak{s} \cong \mathfrak{sl}_2(\mathbb{R})$ meeting $\mathfrak{a}$ in $\mathbb{R}H_\lambda$. Since $\theta(X + \theta X) = \theta X + X = X + \theta X$, the element $Y = X + \theta X$ lies in the $(+1)$-eigenspace $\mathfrak{k}$. A direct computation in $\mathfrak{s} \cong \mathfrak{sl}_2(\mathbb{R})$, the rotation-by-$\pi$ computation for the compact $\mathrm{SU}(2)$ inside it, produces $k = \exp(\frac{\pi}{2}\widehat{Y}) \in \exp\mathfrak{k} \subseteq K$ for the suitably normalized $\widehat{Y} \in \mathfrak{k} \cap \mathfrak{s}$, with $\mathrm{Ad}(k)|_{\mathfrak{a}} = s_\lambda$, the reflection fixing $\ker\lambda$ and negating H_λ. Since $\mathrm{Ad}(k)$ preserves $\mathfrak{a}$, we have $k \in N_K(\mathfrak{a}) = M^*$, and its class in W acts as s_λ. Thus every restricted-root reflection lies in the image of $W \to \mathrm{GL}(\mathfrak{a})$, so $W \supseteq W(\Sigma)$.

Step 3: no element acts outside $W(\Sigma)$. Let $w \in W$ be represented by $k \in M^*$, and let C be a fixed (open) Weyl chamber of $\mathfrak{a}$ for Σ. Then $\mathrm{Ad}(k)C$ is again an open chamber, since $\mathrm{Ad}(k)$ preserves Σ and hence the chamber walls $\ker\lambda$. By theorem 8.2.8 applied to the restricted root system Σ (which is a genuine root system in $\mathfrak{a}^*$ by theorem 14.1.4), the Weyl group $W(\Sigma)$ acts simply transitively on chambers, so there is a unique $\sigma \in W(\Sigma)$ with $\sigma C = \mathrm{Ad}(k)C$. (When Σ is non-reduced, of type BC, theorem 8.2.8 still applies: passing from each pair $\lambda, 2\lambda$ to the shorter

root leaves the walls $\ker\lambda$ and the reflections unchanged, so the chamber geometry and Weyl group of BC_n coincide with those of the reduced B_n.) By Step 2 the reflection generators of $W(\Sigma)$ are realized in W, so σ is represented by some $k' \in M^*$; then $k'^{-1}k$ represents $\sigma^{-1}w \in W$ and fixes the chamber C setwise. An element of W fixing an open chamber fixes it pointwise on a neighborhood, hence acts trivially on $\mathfrak{a}$, and by faithfulness $\sigma^{-1}w = 1$, so $w = \sigma \in W(\Sigma)$. Combined with Step 2, $W = W(\Sigma)$: it is finite, generated by the reflections s_λ, and acts simply transitively on chambers, completing the proof.

The restricted Weyl group is the combinatorial control on the noncompact directions, exactly as the ordinary Weyl group controls a maximal torus. It is generated by reflections, its order is the number of restricted Weyl chambers, and the identification $W = W(\Sigma)$ means every fact proved for abstract root systems in section 14.1 transfers to it: a longest element, a length function, and a unique open chamber up to the W-action. For $\mathrm{SL}(n,\mathbb{R})$ the restricted root system is $\Sigma = \{\varepsilon_i - \varepsilon_j\}$ of type A_{n-1}, so $W = S_n$, the symmetric group realized inside $K = \mathrm{SO}(n)$ by signed permutation matrices of determinant one modulo the diagonal signs M. The single class of a transposition $(i\,i{+}1)$ is the reflection $s_{\varepsilon_i - \varepsilon_{i+1}}$, and $|W| = n!$ counts the chambers, one for each ordering of the diagonal entries.

With P and W in hand, the double cosets PwP organize the whole group. The counting is finite and indexed by W, and the disjointness is the substantive claim.

Theorem 14.3.4: The Bruhat Decomposition

Let G be a connected semisimple Lie group with minimal parabolic $P = MAN$ and restricted Weyl group $W = N_K(\mathfrak{a})/M$ (proposition 14.3.3). For $w \in W$ choose any representative $\dot{w} \in N_K(\mathfrak{a})$; the double coset $P\dot{w}P$ depends only on w, written PwP. Then G is the disjoint union of these double cosets:

$$G = \bigsqcup_{w \in W} PwP$$

Distinct $w \neq w'$ give disjoint cells, $PwP \cap Pw'P = \varnothing$. The cell Pw_0P for the longest element $w_0 \in W$ is open and dense in G, and its image in G/P is the *big cell*, an open dense subset diffeomorphic to N; its complement is the union of the lower-dimensional cells PwP/P for $w \neq w_0$.

The full statement carries a finer layer, the closure order among the cells, which is the combinatorial content of the Bruhat order. That closure combinatorics, a statement about the algebraic flag variety over an algebraically closed field, is deferred to the algebraic-groups companion [10, Closure Order of Bruhat Cells], and the argument below establishes the partition into cells, the indexing by W, and the openness of the big cell, deferring only the description of $\overline{PwP}$ in terms of the Bruhat order to that source.

***Proof Key ideas*:**
The proof runs through the geometry of the flag manifold G/P: the group N acts on G/P with finitely many orbits, indexed by W, and the double cosets PwP are the preimages of these orbits. The full closure combinatorics of the orbits is the part deferred to [10, Closure Order of Bruhat Cells].

Step 1: the double cosets are indexed by W. A class $w \in W = M^*/M$ has representatives $\dot{w}$ and $\dot{w}m$ for $m \in M$. Since $M \subseteq P$, the double cosets coincide: $P(\dot{w}m)P = P\dot{w}(mP) = P\dot{w}P$. Thus $P\dot{w}P$ depends only on the class $w = \dot{w}M$, giving a well-defined map $W \to \{P\text{-double cosets}\}$, $w \mapsto PwP$.

Step 2: surjectivity, from the N-orbits on G/P. Project $G \to G/P$. A P-double coset PgP is the preimage of the P-orbit of gP under the left P-action on G/P; since $P = MAN$ and MA is contained in the stabilizer structure of the base point, the orbits of P on G/P are refined by the orbits of its unipotent part N together with the finite Weyl data. The Iwasawa decomposition gives $G = KAN = KP$, so $G/P = K/M$ is compact and K acts transitively; the restricted-root grading of $\mathfrak{n}$ makes N act on $G/P = K/M$ with a cell for each chamber, that is one N-orbit for each $w \in W$. Concretely, since $N \subseteq P$ fixes the base point $o = eP$, its orbit $N \cdot o = \{o\}$ is the single-point $w = e$ cell, and the M^*-translates $\dot{w} \cdot o$ for $w \in W$ are representatives of the finitely many N-orbits; their P-saturations $P\dot{w}P$ exhaust G. Hence $G = \bigcup_{w \in W} PwP$.

Step 3: disjointness. This is the substantive claim; we argue it in full, deferring only the closure combinatorics to [10, Closure Order of Bruhat Cells]. The invariant that separates the cells is a fixed point on the flag manifold. Work on G/P with base point $o = eP$, and consider the action of $A = \exp \mathfrak{a}$. The A-fixed points of G/P are exactly the finitely many cosets $\dot{w} \cdot o$ for $w \in W$: a coset gP is A-fixed iff $A \subseteq gPg^{-1}$, iff $g^{-1}Ag \subseteq P$; since A is a maximal $\mathbb{R}$-split torus and any two such are P-conjugate (the conjugacy of maximal abelian subspaces of $\mathfrak{p}$, definition 14.1.1, passed to the group), $g^{-1}Ag$ and A are conjugate in P, so we may take $g \in N_G(A)$, and $N_G(A)$ meets each class of $N_K(\mathfrak{a})/M = W$ in a single P-coset, giving the A-fixed set $\{\dot{w} \cdot o \mid w \in W\}$, of size $|W|$ by Step 1.

Now the N-orbit $N \cdot (\dot{w} \cdot o)$ contains the single A-fixed point $\dot{w} \cdot o$ among these: if $n\dot{w} \cdot o = \dot{w'} \cdot o$ were a second one, then $\dot{w} \cdot o$ and $\dot{w'} \cdot o$ would lie in one N-orbit, but A normalizes N, so A would permute the finite A-fixed set inside that orbit while fixing $\dot{w} \cdot o$; the only N-translate of an A-fixed point that is again A-fixed is the point itself (an A-fixed point x has $n \cdot x$ A-fixed iff $ana^{-1} \cdot x = n \cdot x$ for all a, iff $n^{-1}ana^{-1}$ fixes x for all a, and letting $a \to \infty$ along the positive chamber contracts n to e, forcing $n \cdot x = x$). Hence each N-orbit on G/P contains exactly one of the points $\dot{w} \cdot o$, so the map $w \mapsto N \cdot (\dot{w} \cdot o)$ is injective. Since $P\dot{w}P/P$ is the P-saturation $MAN \cdot (\dot{w} \cdot o) = N \cdot (\dot{w} \cdot o)$ (because MA fixes each $\dot{w} \cdot o$: A by the above and $M = Z_K(\mathfrak{a})$ commuting with $\dot{w}$ up to M), distinct w give distinct orbits and hence distinct double cosets: $PwP \cap Pw'P = \varnothing$ for $w \neq w'$. Thus the union of Step 2 is disjoint.

Step 4: the big cell. The longest element $w_0 \in W$ is the unique element sending the positive chamber to its negative, hence $\mathrm{Ad}(\dot{w}_0)\Sigma^+ = \Sigma^-$ and $\mathrm{Ad}(\dot{w}_0)\mathfrak{n} = \bar{\mathfrak{n}} = \theta\mathfrak{n}$, the opposite nilpotent. The product map

$$\overline{N} \times M \times A \times N \longrightarrow G, \qquad (\bar{n}, m, a, n) \mapsto \bar{n}\, man$$

where $\overline{N} = \theta N = \exp \bar{\mathfrak{n}}$, is a diffeomorphism onto an open subset, because its differential at the identity is the direct-sum decomposition $\mathfrak{g}_0 = \bar{\mathfrak{n}} \oplus \mathfrak{m} \oplus \mathfrak{a} \oplus \mathfrak{n}$ (the restricted root space decomposition of theorem 14.1.4, regrouped), and $\overline{N} \cap P = \{e\}$. Its image $\overline{N}P$ is open, and its image in G/P is $\overline{N} \cdot o$, the open dense $\overline{N}$-orbit of the base point $o = eP$; this is the top cell of the *opposite* Bruhat decomposition $\bigsqcup_w \overline{P}wP$ (the $\overline{P}e\overline{P}$ cell for the opposite minimal parabolic $\overline{P} = MA\overline{N}$), the LDU locus. It only *contains* o (take $\bar{n} = e$), which is the $w = e$ cell $PeP/P = N \cdot o = \{o\}$ of the standard decomposition (Step 3); being open dense, $\overline{N} \cdot o$ is not that point, and it is not Pw_0P either. Translating by $\dot{w}_0$ carries $\overline{N}P$ to the w_0 cell: $Pw_0P = P\dot{w}_0P$ contains $\dot{w}_0\,\overline{N}P = (\dot{w}_0\overline{N}\dot{w}_0^{-1})\dot{w}_0P = N\dot{w}_0P$, so its image in G/P is $N \cdot (\dot{w}_0 \cdot o) = \dot{w}_0\overline{N} \cdot o$, swept out by N from the point $\dot{w}_0 \cdot o$ (which is $\neq o$). This image is open and diffeomorphic to $N \cong \mathbb{R}^{\dim \mathfrak{n}}$ and is the big cell. Its complement is a finite union of the strictly lower-dimensional cells PwP/P, $w \neq w_0$, each an N-orbit of dimension $\sum_{\lambda>0,\, w\lambda<0} \dim \mathfrak{g}_\lambda < \dim \mathfrak{n}$ (for split G, where all restricted-root multiplicities are 1, this dimension is $\ell(w) < \ell(w_0)$); each such $w \neq w_0$ leaves at least one positive-root direction unflipped, so the cell omits that direction and has strictly smaller dimension. A proper finite union of lower-dimensional submanifolds has empty

interior, so the big cell is dense. The description of how these smaller cells fit together in the closure $\overline{PwP}$ is the Bruhat-order combinatorics deferred to [10, Closure Order of Bruhat Cells], completing the proof of the decomposition into $|W|$ cells with the stated big cell.

The Bruhat decomposition cuts a group of positive dimension into finitely many pieces, one for each element of the finite reflection group W, with a single top-dimensional piece carrying almost all of the group. For the flag manifold G/P this is a cell decomposition into $|W|$ affine cells, the real analogue of the Schubert cell decomposition of a projective flag variety, and it is the starting point for computing the topology of G/P and for organizing the induced representations of section 14.5, which are built by inducing from P and analyzed cell by cell. The next example shows the decomposition: for $\mathrm{SL}(n,\mathbb{R})$ the Bruhat class is Gaussian elimination, the cell of a matrix recording which pivots its elimination requires.

Example 14.3.5: Bruhat Decomposition and the *LU* Decomposition for $\mathrm{SL}(n,\mathbb{R})$

For $G = \mathrm{SL}(n,\mathbb{R})$ the minimal parabolic P is the upper-triangular subgroup and $W = S_n$ (example 14.3.2). The Bruhat decomposition $G = \bigsqcup_{w\in S_n} PwP$ is the classification of matrices by the row permutation their Gaussian elimination needs. Every $g \in \mathrm{SL}(n,\mathbb{R})$ factors as a double-coset normal form

$$g = b_1\, \dot{w}\, b_2$$

with b_1, b_2 upper triangular and $\dot{w}$ a permutation matrix; the permutation $w \in S_n$ that brings the nonzero pivots into position is the Bruhat class. The identity element e lies in $w = e$: its cell is $PeP = P$, the full upper-triangular subgroup itself, whose image in G/P is the single base point o, the small end of the order, of codimension $\dim \mathfrak{n}$ in G. This should not be confused with the LDU locus of matrices $g = n_-\, d\, n_+$ (n_- lower unitriangular, d diagonal, n_+ upper unitriangular) admitting Gaussian elimination without row swaps, the condition that all leading principal minors be nonzero. That locus is instead the open dense set $\overline{N}P$, the top cell of the *opposite* decomposition $\bigsqcup_w \overline{P}wP$; it contains e (whence o) but, being codimension 0, is not the e-cell P. The big cell of the standard decomposition is Pw_0P for w_0 the reversal permutation, and its criterion is the reversed one: $g \in Pw_0P$ precisely when the lower-left (antidiagonal) minors are nonzero, that is when the $k \times k$ submatrix in the bottom-left corner (rows $n-k+1,\dots,n$, columns $1,\dots,k$) is invertible for every k. Each cell is cut out by the nonvanishing of finitely many minors, so Pw_0P is open, and its complement lies in the vanishing loci of those minors, so it is dense, as the theorem asserts. Counting the cells recovers $|W| = n!$, one for each pivot pattern.

The rank-one running example collapses the count to two cells, the smallest nontrivial instance, and matches the two pieces of a projective line.

Example 14.3.6: The Two Cells of $\mathrm{SL}(2,\mathbb{R})$

For $G = \mathrm{SL}(2,\mathbb{R})$ the restricted root system has rank one, $W = \{1, w\}$ with w the single nontrivial reflection, so $|W| = 2$ and the Bruhat decomposition has exactly two cells:

$$\mathrm{SL}(2,\mathbb{R}) = P \sqcup PwP$$

Here P is the upper-triangular subgroup and $w = w_0$ is represented by $\dot{w} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SO}(2)$. The flag manifold is $G/P = \mathrm{SO}(2)/M = \mathbb{R}\mathrm{P}^1$, the projective line, on which $\mathrm{SL}(2,\mathbb{R})$ acts by Möbius transformations of the real line $\mathbb{R} \cup \{\infty\}$. The two cells match the two pieces of $\mathbb{R}\mathrm{P}^1$: the identity coset $eP \in G/P$ is the point $[1:0] = \infty$, whose stabilizer is P (the upper-triangular matrices fix ∞), so P/P is the single point at infinity; and the big cell PwP/P is its complement, the affine line

$\mathbb{R} = \mathbb{RP}^1 \setminus \{\infty\}$, swept out by $N = \{\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)\}$ acting on the point $\dot{w}_0 \cdot \infty = 0$ by $x \mapsto \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) \cdot 0 = x$, so $N \cong \mathbb{R}$ parametrizes the affine line diffeomorphically. A matrix $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ lies in the big cell exactly when its lower-left entry c is nonzero, and in the small cell P when $c = 0$ (upper triangular): the two Bruhat cells are $\{c \neq 0\}$ (open dense) and $\{c = 0\} = P$ (a codimension-one locus), matching the affine line and the point at infinity. The rival condition $\{a \neq 0\}$ picks out instead the $\overline{N}P$ locus of matrices with nonzero leading minor a, the LDU-decomposable matrices; this is the top cell of the *opposite* decomposition (the $\overline{P}e\overline{P}$ cell for $\overline{P} = MA\overline{N}$), open dense of codimension 0, not the e-cell $PeP = P = \{c = 0\}$. It contains e (whence the base point ∞), but the two open dense sets differ: on G/P the $\overline{N}P$ locus is the orbit $\overline{N} \cdot \infty = \mathbb{RP}^1 \setminus \{0\}$, the complement of $0 = \dot{w}_0 \cdot \infty$, whereas the big cell $\{c \neq 0\}$ is $\mathbb{RP}^1 \setminus \{\infty\}$; they must not be confused.

The parabolic subgroups interpolate between the minimal P and all of G, one for each subset of the simple restricted roots, and the Bruhat decomposition partitions G into $|W|$ cells indexed by the restricted Weyl group. Together they are the finite combinatorial data underlying both the geometry of the flag manifold G/P, whose cells are the Bruhat cells, and the representation theory of section 14.5, whose principal series are induced from characters of P and decomposed along the same cellular structure.

Exercise 14.3.1

1. Identify all subgroups of $\mathrm{SL}(3,\mathbb{R})$ containing the minimal parabolic P (the upper-triangular subgroup), and match each to a subset Θ of the two simple restricted roots $\Delta = \{\varepsilon_1 - \varepsilon_2,\ \varepsilon_2 - \varepsilon_3\}$. For each, describe the block-triangular shape and the flag in $\mathbb{R}^3$ it stabilizes.

2. Verify the two-cell Bruhat decomposition of $\mathrm{SL}(2,\mathbb{R})$ directly on $\mathbb{RP}^1$. Show that the upper-triangular P is the stabilizer of $\infty = [1:0]$ under the Möbius action of $\mathrm{SL}(2,\mathbb{R})$ on $\mathbb{R} \cup \{\infty\}$, that the big cell is the affine line $\mathbb{R}$, and that $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ lies in the big cell iff $a \neq 0$.

3. Compute the restricted Weyl group of $\mathfrak{sl}_n(\mathbb{R})$, and exhibit the reflection $s_{\varepsilon_i - \varepsilon_j}$ as a signed permutation matrix in $\mathrm{SO}(n)$ modulo the diagonal-sign group M. Deduce $|W| = n!$ and identify the longest element w_0.

4. Show that a parabolic subgroup is self-normalizing: $N_G(P) = P$ for the minimal parabolic $P = MAN$ (and more generally $N_G(P_\Theta) = P_\Theta$). Deduce that distinct standard parabolics are never conjugate, and that the map $\Theta \mapsto P_\Theta$ is injective.

5. For a connected semisimple G whose restricted root system has type A_2 (for instance $\mathrm{SL}(3,\mathbb{R})$), count the Bruhat cells $|W|$ and list them by the length function on $W = S_3$. Which cell is the big cell, and what is its dimension as a subset of G/P?

14.4 The Structure of $\mathrm{SL}(2,\mathbb{R})$ in Full

The running example has carried a fragment of each general theorem: its Cartan involution and decomposition in chapter 13, its restricted root in section 14.1, its Iwasawa factorization in section 14.2, and its two Bruhat cells in section 14.3. This section assembles those fragments into a single picture. The group $\mathrm{SL}(2,\mathbb{R})$ has real rank one, so it is the smallest noncompact semisimple group and the place where the entire chapter comes down to explicit matrices, with every subgroup a concrete family of 2×2 matrices and every decomposition an explicit motion on the upper half-plane $\mathfrak{H} = \{z \in \mathbb{C} \mid \mathrm{Im} z > 0\}$.

The task here is to run the three decompositions end to end on $\mathfrak{H}$ and to read off, from the trace of an element, which one-parameter subgroup it belongs to.

Recall the action. The group $\mathrm{SL}(2,\mathbb{R})$ acts on $\mathfrak{H}$ by Möbius transformations,

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot z = \frac{az+b}{cz+d}$$

which preserves the upper half-plane because $\mathrm{Im}\left(\frac{az+b}{cz+d}\right) = \frac{\mathrm{Im}z}{|cz+d|^2}$ for $ad-bc=1$. The action is transitive, and the stabilizer of the base point i is the rotation group $\mathrm{SO}(2)$: a matrix fixes i iff $\frac{ai+b}{ci+d} = i$, that is $ai+b = -c+di$, forcing $a=d$ and $b=-c$, which together with $a^2+b^2=1$ is exactly $\mathrm{SO}(2)$. By orbit-stabilizer (theorem 4.2.3) this identifies $\mathfrak{H}$ with the symmetric space $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$, the model on which the whole chapter is read.

Example 14.4.1: The Iwasawa Decomposition of $\mathrm{SL}(2,\mathbb{R})$

For $G = \mathrm{SL}(2,\mathbb{R})$ the three Iwasawa factors are

$$K = \mathrm{SO}(2) = \left\{ k_\varphi = \left(\begin{smallmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{smallmatrix}\right) \;\middle|\; \varphi \in \mathbb{R} \right\}, \qquad A = \left\{ a_t = \left(\begin{smallmatrix} e^t & 0 \\ 0 & e^{-t} \end{smallmatrix}\right) \;\middle|\; t \in \mathbb{R} \right\}$$

$$N = \left\{ n_x = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) \mid x \in \mathbb{R} \right\}$$

Here K is the maximal compact subgroup of chapter 13, $A = \exp\mathfrak{a}$ for $\mathfrak{a} = \mathbb{R}\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ the maximal abelian subspace of $\mathfrak{p}$, and $N = \exp\mathfrak{n}$ for the single positive restricted root space $\mathfrak{n} = \mathbb{R}\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ found in section 14.1. The Iwasawa decomposition $G = KAN$ (theorem 14.2.4) writes each g uniquely as $g = k_\varphi a_t n_x$, which for 2×2 matrices is exactly the Gram-Schmidt process: k_φ orthonormalizes the columns, a_t records their lengths, and n_x records their shear.

Reading the factorization through the action on $\mathfrak{H}$ makes the three factors geometric. The factor A scales, and N translates:

$$a_t \cdot i = e^{2t} i, \qquad n_x \cdot i = i + x$$

so a_t moves i up the imaginary axis to height e^{2t}, and n_x slides it horizontally by x. Applying the two in the order that appears in $g = kan$ acting on the base point (with K fixing i),

$$k_\varphi a_t n_x \cdot i = k_\varphi a_t \cdot (i+x) = k_\varphi \cdot (e^{2t}x + e^{2t} i)$$

does not yet land at a general point, because k_φ then rotates about i. To reach an arbitrary $x+iy$ one instead reads the coordinates off AN: the subgroup AN (upper triangular, positive diagonal) sends i to

$$a_t n_x \cdot i = a_t \cdot (i+x) = e^{2t}x + e^{2t} i$$

Setting $y = e^{2t}$ and renaming the horizontal coordinate, the point of $\mathfrak{H}$ with coordinates $x+iy$ $(y>0)$ is $\left(\begin{smallmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{smallmatrix}\right) \cdot i$, a single element of AN. Thus AN acts *simply transitively* on $\mathfrak{H}$: every point is hit exactly once, and the pair $(x,y) = (x, e^{2t})$ is a global coordinate system, the Iwasawa or horocyclic coordinates on $\mathfrak{H}$.

The two one-parameter subgroups A and N are also flows with a geometric meaning that the symmetric-space picture of chapter 15 will generalize. The subgroup A acts on the imaginary axis by $a_t \cdot (iy) = e^{2t}(iy)$, which is the geodesic flow along the imaginary axis: the vertical line from 0 to ∞ is the unique geodesic of $\mathfrak{H}$ through i fixed by A, and a_t translates along it at speed 2 in the hyperbolic metric $\frac{dx^2+dy^2}{y^2}$ (the hyperbolic arc length of the path $t \mapsto a_t \cdot i = e^{2t} i$ from $t=0$ to t is

$\int_0^t \frac{d}{dt'} \log e^{2t'}\, dt' = 2t$). The subgroup N acts by $n_x \cdot z = z + x$, the horizontal translations, which preserve each horizontal line $\mathrm{Im} z = y$; these lines are the horocycles based at the boundary point ∞, the limits of hyperbolic circles as their centers run off to ∞. So the Iwasawa decomposition splits the geometry of $\mathfrak{H}$ into rotation about i (from K), geodesic flow along the vertical axis (from A), and horocyclic flow toward ∞ (from N).

With the factors fixed, the three decompositions of the chapter can be placed side by side. Each factors $\mathrm{SL}(2,\mathbb{R})$ using the same building blocks K, A, N, but in a different pattern, and each answers a different geometric question about $\mathfrak{H}$ and its boundary.

Proposition 14.4.2: The Three Decompositions of $\mathrm{SL}(2,\mathbb{R})$

Let $G = \mathrm{SL}(2,\mathbb{R})$, with $K = \mathrm{SO}(2)$, A, N as in example 14.4.1, $M = \{\pm I\} = Z_K(\mathfrak{a})$, $P = MAN$ the group of upper triangular matrices of positive-or-negative sign on the diagonal, and $w = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ a representative of the nontrivial element of the restricted Weyl group W. Then G admits three factorizations:

1. *Cartan (polar), $G = KAK$.* Every g is $g = k_1 a_s k_2$ with $k_1, k_2 \in K$ and $a_s \in A$, $s \geq 0$ (theorem 13.4.2); on $\mathfrak{H}$ this is the polar coordinate presentation of a point about the base point i, with $2s$ the hyperbolic distance $d(i, g \cdot i)$ (so s is half the distance, since $a_s \cdot i = e^{2s} i$) and k_1 the direction.

2. *Iwasawa, $G = KAN$.* Every g is uniquely $g = kan$ (theorem 14.2.4); on $\mathfrak{H}$ this is the horocyclic coordinate system, in which the point $g \cdot i = x + iy$ is recorded by the unique element of AN that carries i to it (AN acts simply transitively on $\mathfrak{H}$). This element is in general *not* the an-part of g: since k moves $an \cdot i$ off $g \cdot i$ unless k fixes $an \cdot i$, the horocyclic coordinate is read off AN acting on i to reach $g \cdot i$, separately from the residual rotation k.

3. *Bruhat, $G = P \sqcup PwP$.* As a disjoint union G is the minimal parabolic P together with the single other double coset PwP (theorem 14.3.4); on the boundary $\partial\mathfrak{H} = \mathbb{R}\mathrm{P}^1$, the coset P is the stabilizer of ∞ and the two cells are the point $\{\infty\}$ and its complement, the affine line $\mathbb{R} \subseteq \mathbb{R}\mathrm{P}^1$.

***Proof*:**
Each factorization is the specialization to $G = \mathrm{SL}(2,\mathbb{R})$ of a general theorem of the chapter, together with its reading on $\mathfrak{H}$; the content here is the identification of the factors, which the general proofs leave abstract.

1. The global Cartan decomposition $G = K \exp(\mathfrak{p})$ of theorem 13.4.2 combines with the polar form of corollary 13.4.4: every symmetric positive-definite element $\exp X$ ($X \in \mathfrak{p}$) is $k(\exp X')k^{-1}$ with $X' \in \mathfrak{a}$, since any $X \in \mathfrak{p}$ is K-conjugate into the maximal abelian $\mathfrak{a}$ (all maximal abelian subspaces of $\mathfrak{p}$ are K-conjugate, and $\mathfrak{a}$ is one-dimensional). Hence $g = k \exp X = k(k' a_s k'^{-1}) = (kk') a_s (k'^{-1})$ is of the form $k_1 a_s k_2$ with $s \geq 0$. On $\mathfrak{H}$, the orbit $A \cdot i = \left\{ e^{2s} i \;\middle|\; s \in \mathbb{R} \right\}$ is the imaginary axis, and K rotates it about i to sweep out all of $\mathfrak{H}$; the parameter s is half the hyperbolic distance from i because $d(i, e^{2s} i) = \int_1^{e^{2s}} \frac{dy}{y} = 2s$, matching the geodesic-flow reading of A.

2. Uniqueness of $g = kan$ is theorem 14.2.4, and the horocyclic reading is example 14.4.1: AN

acts simply transitively on $\mathfrak{H}$, so $g \cdot i$ determines an uniquely and k is whatever rotation remains.

3. By theorem 14.3.4 the double cosets are indexed by $W = \{1, w\}$, so $G = P \sqcup PwP$ with exactly two cells. The parabolic P consists of the upper triangular matrices in $\mathrm{SL}(2,\mathbb{R})$; a matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ fixes the boundary point $\infty \in \mathbb{R}\mathrm{P}^1$ under the Möbius action iff $c = 0$, that is iff it is upper triangular, so $P = \mathrm{Stab}_G(\infty)$. The identification $G/P = \mathbb{R}\mathrm{P}^1 = \partial\mathfrak{H}$ then sends the coset eP to ∞ and the big cell PwP, where $c \neq 0$, to the points $\frac{a}{c} \in \mathbb{R}$; the two Bruhat cells are the fixed point $\{\infty\}$ and the affine line $\mathbb{R} = \mathbb{R}\mathrm{P}^1 \setminus \{\infty\}$, completing the proof.

The three decompositions answer three different questions, and the contrast is the payoff of collecting them. The Cartan factorization KAK is the polar picture: it measures a point of $\mathfrak{H}$ by its distance s from the base point and a direction, so it is the natural coordinate for anything radially symmetric about i, and it is the decomposition underlying the Cartan (Harish-Chandra) polar integration formula that integrates a K-bi-invariant function over G against the Jacobian of the KAK map (see [16]). The Iwasawa factorization KAN is the horocyclic picture: it gives a genuine global coordinate chart (x, y) on $\mathfrak{H}$ in which AN is a group acting simply transitively, so it is the natural coordinate for anything adapted to the boundary point ∞, and it is what induced representations are built from in section 14.5. The Bruhat factorization is the boundary picture: it stratifies not $\mathfrak{H}$ but its boundary circle $\mathbb{R}\mathrm{P}^1$, into the fixed point of P and the open cell on which N acts by translation, and this stratification is what organizes the asymptotics of functions on $\mathfrak{H}$ as z approaches the boundary. One group, three factorizations, three geometries: interior polar, interior horocyclic, and boundary.

The same three subgroups K, A, N also classify the elements of $\mathrm{SL}(2,\mathbb{R})$ up to conjugacy, and here the classifying invariant is visible on the matrix itself. The characteristic polynomial of $g \in \mathrm{SL}(2,\mathbb{R})$ is $\lambda^2 - (\mathrm{tr}g)\lambda + 1$, whose roots are determined by the single number $\mathrm{tr}g$; the trichotomy of that number is the trichotomy of the group.

Example 14.4.3: Elliptic, Hyperbolic, and Parabolic Elements

A nonidentity element $g \in \mathrm{SL}(2,\mathbb{R})$ falls into one of three classes according to $|\mathrm{tr}g|$, matching the three one-parameter subgroups and their fixed points on $\mathfrak{H} \cup \partial\mathfrak{H}$:

1. *Elliptic,* $|\mathrm{tr}g| < 2$. The eigenvalues $\frac{\mathrm{tr}g \pm \sqrt{(\mathrm{tr}g)^2 - 4}}{2}$ are a complex conjugate pair $e^{\pm i\theta}$ on the unit circle, so g is conjugate to a rotation $k_\theta \in K = \mathrm{SO}(2)$. It has exactly one fixed point in $\mathfrak{H}$ and none on the boundary: g rotates $\mathfrak{H}$ about that interior point. The element k_θ itself fixes i.

2. *Hyperbolic,* $|\mathrm{tr}g| > 2$. The eigenvalues are real and distinct, $\pm e^{\pm t}$ with $t \neq 0$ (the sign is that of $\mathrm{tr}g$: for $\mathrm{tr}g > 2$ the eigenvalues are $e^{\pm t}$, for $\mathrm{tr}g < -2$ they are $-e^{\pm t}$), so g is conjugate to a diagonal $\pm a_t \in \pm A$. It has no fixed point in $\mathfrak{H}$ and two fixed points on $\partial\mathfrak{H} = \mathbb{R}\mathrm{P}^1$, the two eigendirections; g translates along the geodesic joining them. The element a_t fixes 0 and ∞ and translates along the imaginary axis.

3. *Parabolic,* $|\mathrm{tr}g| = 2$ *(and* $g \neq \pm I$*).* The eigenvalues coincide, both $+1$ or both -1, and g is not diagonalizable, so g is conjugate to $\pm n_x \in \pm N$ with $x \neq 0$. It has no fixed point in $\mathfrak{H}$ and exactly one on $\partial\mathfrak{H}$, the single eigendirection; g fixes that boundary point and slides the horocycles based there. The element n_x fixes ∞ and translates the horocycles $\mathrm{Im}z = y$.

The trichotomy is exactly the Jordan classification of a determinant-one 2×2 real matrix, read

off by the discriminant $(\mathrm{tr}g)^2 - 4$, and it lines up with the Iwasawa factors because K, A, N are the three model one-parameter subgroups. Every nonidentity conjugacy class of $\mathrm{SL}(2,\mathbb{R})$ meets one of the standard families: a rotation k_θ (elliptic), a signed diagonal $\pm a_t$ (hyperbolic), or a signed unipotent $\pm n_x$ (parabolic), with the boundary case $|\mathrm{tr}g| = 2$ splitting off the parabolic elements as the degenerate limit between the elliptic and hyperbolic regimes. The number of fixed points on $\mathfrak{H} \cup \partial\mathfrak{H}$ counts them as well: one interior fixed point (elliptic), two boundary fixed points (hyperbolic), one boundary fixed point (parabolic). This is the source of the classification of isometries of the hyperbolic plane, and it is the rank-one case of the general theory of semisimple, split, and unipotent elements that the structure theory of Part II attached to a general Lie algebra.

Every abstract object of the chapter has now appeared concretely in $\mathrm{SL}(2,\mathbb{R})$. The Cartan decomposition split it into $\mathrm{SO}(2)$ and a Euclidean complement; the restricted root system was the single root of type A_1, with $\mathfrak{n}$ its one positive root space; the Iwasawa decomposition was Gram-Schmidt and gave horocyclic coordinates on $\mathfrak{H}$; the minimal parabolic P was the upper triangular matrices, with the two-element restricted Weyl group and the two Bruhat cells on $\mathbb{RP}^1$. Because $\mathrm{SL}(2,\mathbb{R})$ has real rank one, none of these has any hidden higher-rank complexity: $\mathfrak{a}$ is a line, Σ has one positive root, W has two elements, and the flag manifold G/P is a circle. Larger groups replace each of these by its higher-dimensional analogue, but the geometry stays the same in outline, and the upper half-plane $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ is the prototype of the Riemannian symmetric space G/K that chapter 15 builds in general, with A the geodesic flow, N the horocyclic flow, and the Bruhat cells the boundary at infinity.

Exercise 14.4.1

1. Compute the Iwasawa factorization $g = k_\varphi a_t n_x$ of $g = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 3 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$ by Gram-Schmidt on its columns. Separately compute the point $g \cdot i = x' + iy'$ of $\mathfrak{H}$ directly from the Möbius action, confirm the height by $y' = |ci + d|^{-2}$, and explain why the A-parameter of g as a matrix differs from the A-parameter of the point $g \cdot i$ (that is, why $y' \neq e^{2t}$ here).

2. Classify each of the following elements of $\mathrm{SL}(2,\mathbb{R})$ as elliptic, hyperbolic, or parabolic by its trace, and find its fixed point(s) on $\mathfrak{H} \cup \partial\mathfrak{H}$:
$$A_1 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right), \qquad A_2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right), \qquad A_3 = \left(\begin{smallmatrix} 1 & 3 \\ 0 & 1 \end{smallmatrix}\right)$$

3. Show directly from the Möbius action that $K = \mathrm{SO}(2)$ is the full stabilizer of $i \in \mathfrak{H}$ in $\mathrm{SL}(2,\mathbb{R})$, and conclude via orbit-stabilizer (theorem 4.2.3) that $\mathfrak{H} \cong \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ as a homogeneous space.

4. Show that $A = \{a_t \mid t \in \mathbb{R}\}$ is the stabilizer, inside the identity component of the setwise stabilizer, of the geodesic from 0 to ∞ (the imaginary axis) in $\mathfrak{H}$: prove a_t maps the imaginary axis to itself and translates along it, and that any element of $\mathrm{SL}(2,\mathbb{R})$ fixing both boundary points 0 and ∞ is a diagonal matrix, hence lies in $\pm A$.

5. The Cayley transform $c(z) = \frac{z-i}{z+i}$ carries $\mathfrak{H}$ to the unit disk $\mathbb{D} = \{w \in \mathbb{C} \mid |w| < 1\}$. Show that conjugating the $\mathrm{SL}(2,\mathbb{R})$-action by c realizes the disk model of hyperbolic geometry, with symmetry group $\mathrm{SU}(1,1) = \left\{ \left(\begin{smallmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{smallmatrix}\right) \;\middle|\; |\alpha|^2 - |\beta|^2 = 1 \right\}$, and that $\mathrm{SU}(1,1)$ is conjugate to $\mathrm{SL}(2,\mathbb{R})$ inside $\mathrm{SL}(2,\mathbb{C})$. Which subgroup of $\mathrm{SU}(1,1)$ corresponds to $K = \mathrm{SO}(2)$, and what is the fixed point of that subgroup in $\mathbb{D}$?

6. Explain why $|\mathrm{tr}g|$ (rather than $\mathrm{tr}g$ itself) is the conjugacy invariant used in the trichotomy, by showing that conjugation preserves $\mathrm{tr}g$ but that the two classes $\mathrm{tr}g = c$ and $\mathrm{tr}g = -c$ are

exchanged by $g \mapsto -g$ (multiplication by $-I \in Z(\mathrm{SL}(2,\mathbb{R}))$), so the geometry on $\mathfrak{H}$, where $\pm I$ act identically, depends only on $|\mathrm{tr} g|$.

14.5 Toward the Representation Theory of Noncompact Groups

The parabolic $P = MAN$ built in section 14.3 was constructed as a piece of group structure, but its use is in representation theory. The compact groups of Part III were understood completely through their finite-dimensional representations: Peter-Weyl decomposed $L^2(G)$ into finite-dimensional blocks, and the Weyl character formula computed every irreducible character. None of that survives when G is noncompact and semisimple with no compact factor. The finite-dimensional unitary representations vanish (proposition 14.5.1 below), so the harmonic analysis of G must be carried by infinite-dimensional representations, and the first family of them, the principal series, is manufactured directly from the minimal parabolic $P = MAN$ of this chapter. This closing section states the obstruction precisely, then names the object that replaces the finite-dimensional theory, which is where the Iwasawa decomposition is used.

Only one result here is proved in full; the rest is a signpost. The principal series and the finer families it belongs to are the subject of a companion volume, and the harmonic analysis that decomposes $L^2(G)$ into them is deferred there. The point of the section is orientation, not development: to close Part IV by showing where its structural output feeds.

The compact case worked because a compact group has a finite-dimensional faithful unitary representation and enough of them to separate points. The first fact to record is that a semisimple group with no compact factor has essentially none: every finite-dimensional unitary representation is trivial. This is the exact sense in which the representation theory must go infinite-dimensional, and it falls straight out of the Cartan decomposition of chapter 13.

Proposition 14.5.1: No Finite-Dimensional Unitary Representations

Let G be a connected semisimple Lie group with finite center and no compact factor (equivalently, no nonzero compact simple ideal in $\mathfrak{g}_0$), and let $\pi\colon G \to \mathrm{U}(V)$ be a finite-dimensional unitary representation on a complex Hilbert space V. Then π is trivial: $\pi(g) = \mathrm{id}_V$ for all $g \in G$.

***Proof*:**

Let $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ be the Cartan decomposition of the Lie algebra of G (theorem 13.3.3), with $\mathfrak{k}$ the compact part and $\mathfrak{p}$ the Euclidean complement on which the Killing form is positive-definite.

Step 1: the image of $d\pi$ lands in a compact Lie algebra. Fix the G-invariant Hermitian inner product on V for which π is unitary, and write $d\pi\colon \mathfrak{g}_0 \to \mathfrak{gl}(V)$ for the differential. For each $X \in \mathfrak{g}_0$ the one-parameter group $\pi(\exp tX) = \exp(t\, d\pi(X))$ is unitary for all real t; differentiating $\pi(\exp tX)^* \pi(\exp tX) = \mathrm{id}_V$ at $t = 0$ gives $d\pi(X)^* + d\pi(X) = 0$, so $d\pi(X)$ is skew-Hermitian. Thus $d\pi(\mathfrak{g}_0) \subseteq \mathfrak{u}(V)$, the Lie algebra of skew-Hermitian operators, which is the Lie algebra of the *compact* unitary group $\mathrm{U}(V)$. The image $d\pi(\mathfrak{g}_0)$ is a Lie subalgebra of $\mathfrak{u}(V)$.

Step 2: the image is a compact Lie algebra. Set $\mathfrak{h} = d\pi(\mathfrak{g}_0) \subseteq \mathfrak{u}(V)$, the Lie algebra of the connected subgroup $H = \pi(G)^\circ \subseteq \mathrm{U}(V)$. The trace form $\langle A, B\rangle = \mathrm{Retr}(A^* B)$ is a positive-definite $\mathrm{Ad}(\mathrm{U}(V))$-invariant inner product on $\mathfrak{u}(V)$, hence $\mathrm{Ad}(H)$-invariant on $\mathfrak{h}$ since

$H \subseteq \mathrm{U}(V)$; so every $\mathrm{ad}Y$ ($Y \in \mathfrak{h}$) is skew for it and hence has purely imaginary eigenvalues, giving $\kappa_{\mathfrak{h}}(Y,Y) = \mathrm{tr}\big((\mathrm{ad}_{\mathfrak{h}}Y)^2\big) \le 0$ (proposition 9.4.1). Thus the Killing form of $\mathfrak{h}$ is negative-semidefinite, so $\mathfrak{h}$ is of compact type: none of its simple factors has indefinite Killing form.

Step 3: no noncompact factor survives, so $d\pi = 0$. As a quotient of the semisimple algebra $\mathfrak{g}_0$, the image $\mathfrak{h} = \mathfrak{g}_0/\ker d\pi$ is a sum of a subset of the simple ideals of $\mathfrak{g}_0$ (corollary 6.3.5). By hypothesis $\mathfrak{g}_0$ has no compact factor, so every simple ideal of $\mathfrak{g}_0$ has indefinite Killing form. But Step 2 forces every simple factor of $\mathfrak{h}$ to have negative-definite Killing form, that is to be compact. A simple algebra cannot be both, so no simple ideal of $\mathfrak{g}_0$ survives in $\mathfrak{h}$: the image $\mathfrak{h} = 0$, and $d\pi = 0$. Hence π is trivial on the identity component of G, which is G by connectedness, completing the proof.

For a compact G the finite-dimensional unitary representations are abundant, faithful, and computed exactly by the Weyl character formula; for a semisimple G with no compact factor they collapse to the trivial one, and any unitary representation that sees the noncompact directions of G at all must be infinite-dimensional. The mechanism is a sign clash. A finite-dimensional unitary representation lands inside the compact group $\mathrm{U}(V)$, whose Lie algebra $\mathfrak{u}(V)$ has negative-semidefinite Killing form, so the image of $\mathfrak{g}_0$ is of compact type; but $\mathfrak{g}_0$ is noncompact, its Killing form positive-definite on the whole of $\mathfrak{p}$, and a noncompact simple factor has no room inside a compact $\mathfrak{u}(V)$. The very positive-definiteness of κ on $\mathfrak{p}$ that made $\exp\colon \mathfrak{p} \to G$ a diffeomorphism onto a noncompact cell in chapter 13 is what obstructs a finite-dimensional unitary image. Finite-dimensional *non*-unitary representations do exist in abundance, the defining representation of $\mathrm{SL}(2,\mathbb{R})$ on $\mathbb{R}^2$ among them; the proposition rules out only unitarity, which is exactly what harmonic analysis requires.

The replacement is built by induction from the parabolic, and this is where the chapter's structure theory is spent. A character of $P = MAN$ that is trivial on the nilpotent N and unitary on A induces up to a unitary representation of G on a space of functions on G/P, and these are the principal series. The construction uses nothing but the Iwasawa and parabolic data assembled in section 14.2 and section 14.3.

Definition 14.5.2: The Spherical Principal Series

Let G be a connected semisimple Lie group with minimal parabolic $P = MAN$ (definition 14.3.1), and let $\mathfrak{a}$ be the Lie algebra of A. For a linear functional $\nu \in \mathfrak{a}^*$, let $\chi_\nu\colon P \to \mathbb{C}^\times$ be the character

$$\chi_\nu(man) = a^{\,i\nu+\rho} = e^{(i\nu+\rho)(\log a)}$$

trivial on M and N, where $\rho = \frac{1}{2}\sum_{\lambda\in\Sigma^+}(\dim \mathfrak{g}_\lambda)\,\lambda$ is the half-sum of the positive restricted roots weighted by multiplicity. The *spherical principal series* representation $\pi_\nu = \mathrm{Ind}_P^G \chi_\nu$ acts on the Hilbert space

$$\mathcal{H}_\nu = \big\{ f\colon G \to \mathbb{C} \;\big|\; f(gman) = \chi_\nu(man)^{-1} f(g),\ f|_K \in L^2(K) \big\}$$

by left translation, $(\pi_\nu(g)f)(x) = f(g^{-1}x)$. For real $\nu \in \mathfrak{a}^*$ the representation π_ν is unitary.

The definition packs three ingredients into one formula, and each is a product of this chapter. The parabolic $P = MAN$ is the group on which the inducing character lives; the shift by ρ is the half-sum of positive restricted roots from the Iwasawa datum of definition 14.2.1, present so that the induced action is unitary rather than bounded; and the underlying space G/P, which section 14.3 identified as the real flag manifold, is the geometric carrier of the functions. The functions transform on the

right by χ_ν and so descend to sections of a line bundle over G/P; the ρ-shift is precisely the twist that makes the natural G-invariant pairing on such sections a Hilbert space inner product, so that π_ν preserves it. The construction is the noncompact, infinite-dimensional analogue of the Borel-Weil realization of highest-weight modules in the compact theory, with the compact flag variety G/B replaced by the real flag manifold G/P.

Every symbol in the definition, M, A, N, the positive restricted roots, the half-sum ρ, the flag manifold G/P, was constructed in section 14.1, section 14.2, and section 14.3. The full harmonic analysis, when π_ν is irreducible, when two of them are equivalent, and how $L^2(G)$ decomposes into principal series together with the other tempered families, is the content of the Plancherel theorem for real reductive groups; that development is the province of the companion volume *Representation Theory of Real Reductive Groups* [6], with the general theory of induced representations and Haar-measure normalizations in *Harmonic Analysis* [16]. What matters here is only that the object exists and is assembled from Iwasawa data.

The running example shows the three ingredients and, with it, the two other families of representations that round out the picture for the smallest noncompact group.

Example 14.5.3: The Representations of $\mathrm{SL}(2,\mathbb{R})$

For $G = \mathrm{SL}(2,\mathbb{R})$ the Iwasawa decomposition is $G = KAN$ with $K = \mathrm{SO}(2)$, $A = \{\mathrm{diag}(e^t, e^{-t})\}$, and N the upper unitriangular matrices (example 14.4.1). The restricted root system has a single positive root λ with $\dim \mathfrak{g}_\lambda = 1$, so $\rho = \frac{1}{2}\lambda$ and $M = \{\pm I\}$. The minimal parabolic $P = MAN$ is the group of upper triangular matrices, and $G/P = \mathbb{R}\mathrm{P}^1$ is the projective line, the boundary circle of the upper half-plane $\mathfrak{H}$.

Three families of irreducible unitary representations organize the theory, all named here and developed in [6].

1. The *principal series* π_ν, $\nu \in \mathbb{R}$, realized on $L^2(\mathbb{R}\mathrm{P}^1)$ by the boundary action of G on functions on $G/P = \mathbb{R}\mathrm{P}^1$, twisted by χ_ν. These are the $\mathrm{SL}(2,\mathbb{R})$ case of definition 14.5.2, unitary and irreducible for every real ν. The lone reducibility in the unitary principal series belongs not to this spherical family but to the nonspherical (odd) series at $\nu = 0$, where the induced representation splits into the two limits of discrete series.

2. The *discrete series* $D_n^{\pm}$, $n \geq 2$, the representations that occur discretely in $L^2(G)$, realized on holomorphic (and antiholomorphic) functions on $\mathfrak{H}$ with an $\mathrm{SL}(2,\mathbb{R})$-invariant norm. They have no analogue for compact groups and exist precisely because $\mathfrak{H}$ is a noncompact complex domain.

3. The *complementary series* C_s, $0 < s < 1$, a one-parameter family carrying a nonstandard invariant inner product on the same function space as the principal series, unitary in a range where the principal-series pairing degenerates.

These are organized by their $K = \mathrm{SO}(2)$-types. Restricting a representation to the maximal compact $K = \mathrm{SO}(2)$ and decomposing into characters $e^{in\theta}$ of the circle sorts the vectors into Fourier modes indexed by $n \in \mathbb{Z}$. The spherical principal series π_ν of definition 14.5.2, with χ_ν trivial on $M = \{\pm I\}$, contains every even K-type once, $n \in 2\mathbb{Z}$; the parity is forced by M, not by ν, and the odd-K-type ladder belongs to the other (nonspherical) principal series, induced from the nontrivial character of M. The discrete series D_n^+ contains the K-types $n, n+2, n+4, \ldots$ bounded below, which is what makes it "discrete" and holomorphic; the complementary series matches the spherical principal series in K-types. The $\mathrm{SO}(2)$-decomposition plays the role here

that the weight decomposition of Part III plays for compact groups, and its K-type patterns distinguish the three families of the unitary dual of $\mathrm{SL}(2,\mathbb{R})$. The geometric reading of $\mathfrak{H} = G/K$ that underlies the discrete series is taken up in chapter 15.

The example closes the chapter and the Part. The three families of $\mathrm{SL}(2,\mathbb{R})$ representations, principal, discrete, complementary, are the model on which the general theory of real reductive groups is built, and every one of them is organized by the Iwasawa and Cartan data developed in Part IV: the principal series by the parabolic $P = MAN$, the discrete series by the symmetric space $\mathfrak{H} = G/K$ coming from the Cartan decomposition, the K-types by the maximal compact K. The structural results of Part IV, the Cartan decomposition of chapter 13, the Iwasawa decomposition and its parabolics here, are the input to noncompact harmonic analysis and, beyond it, the Langlands program. Two continuations follow directly. The immediate one is geometric: the Cartan datum $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ reads as the tangent data of a Riemannian symmetric space G/K, and that reading is the subject of chapter 15, the next and final chapter. The representation theory proper, the principal series and its companions, the Plancherel decomposition, and the automorphic forms that live on arithmetic quotients, belongs to the companion volumes *Representation Theory of Real Reductive Groups* [6] and *Harmonic Analysis* [16].

Exercise 14.5.1

1. Show that a torus $\mathbb{T}^n = (\mathbb{R}/\mathbb{Z})^n$, and in particular the circle $\mathrm{SO}(2)$, has an abundance of finite-dimensional unitary representations: exhibit for each $\mathbf{m} \in \mathbb{Z}^n$ the one-dimensional unitary character $\chi_{\mathbf{m}}(\theta_1,\dots,\theta_n) = e^{2\pi i(m_1\theta_1+\cdots+m_n\theta_n)}$, and explain why proposition 14.5.1 does not apply. Which hypothesis of the proposition fails for a torus, and why does compactness rescue the finite-dimensional theory?

2. Prove directly, without invoking proposition 14.5.1 in the abstract, that $\mathfrak{sl}_2(\mathbb{R})$ has no nontrivial finite-dimensional representation $d\pi\colon \mathfrak{sl}_2(\mathbb{R}) \to \mathfrak{gl}(V)$ that is unitary in the sense that $d\pi(X)$ is skew-Hermitian for a fixed Hermitian inner product on V and every $X \in \mathfrak{sl}_2(\mathbb{R})$. Use the standard basis H, E, F with $[H,E] = 2E$, $[H,F] = -2F$, $[E,F] = H$. Show that $d\pi(H)$ is diagonalizable with integer eigenvalues by the $\mathfrak{sl}_2$ weight theory, that skew-Hermiticity forces those eigenvalues to be purely imaginary, and conclude $d\pi(H) = 0$; then deduce $d\pi(E) = d\pi(F) = 0$ from the bracket relations.

3. Describe the $\mathrm{SL}(2,\mathbb{R})$ principal series π_ν concretely as functions on the boundary $\mathbb{R}\mathrm{P}^1$ of the upper half-plane. Identify $\mathbb{R}\mathrm{P}^1$ with G/P, describe the action of $g \in \mathrm{SL}(2,\mathbb{R})$ on a function $f\colon \mathbb{R}\mathrm{P}^1 \to \mathbb{C}$ (as a fractional linear transformation on the circle together with the χ_ν-twist), and explain why the twist by $\rho = \frac{1}{2}\lambda$ is needed for the $L^2(\mathbb{R}\mathrm{P}^1)$ inner product to be G-invariant.

4. Explain why the discrete series D_n^+ of $\mathrm{SL}(2,\mathbb{R})$ cannot be finite-dimensional. Using the description of D_n^+ as having $K = \mathrm{SO}(2)$-types $n, n+2, n+4, \dots$ unbounded above, argue that a representation with infinitely many distinct K-types must be infinite-dimensional, and connect this to proposition 14.5.1: a nontrivial unitary representation of $\mathrm{SL}(2,\mathbb{R})$ has no upper bound on its K-types.

5. Identify the $K = \mathrm{SO}(2)$-types of the $\mathrm{SL}(2,\mathbb{R})$ spherical principal series π_ν of definition 14.5.2. Restrict π_ν acting on $L^2(\mathbb{R}\mathrm{P}^1)$ to $K = \mathrm{SO}(2)$, note that $\mathrm{SO}(2)$ acts on $\mathbb{R}\mathrm{P}^1$ = boundary circle by rotation, decompose L^2 of the circle into its Fourier modes $e^{in\theta}$, and state which $n \in \mathbb{Z}$ occur, each with multiplicity one. Check that the central element $-I \in M$ acts trivially on

$\mathcal{H}_\nu$ (because χ_ν is trivial on M), for every ν, so that only even n survive; conclude that the parity is fixed by the M-character, not by ν, and note that the odd ladder is the nonspherical principal series induced from the other character of M.

6. Contrast the two halves of the book in one paragraph. For a compact group G, the finite-dimensional unitary irreducibles are the whole story (Peter-Weyl, the Weyl character formula); for a semisimple G with no compact factor they are trivial (proposition 14.5.1). State what replaces the finite-dimensional theory, name the parabolic $P = MAN$ as the object the replacement is built on, and explain in a sentence why the same maximal torus / maximal compact subgroup that indexed the compact theory now indexes the K-types of the infinite-dimensional representations.

15

Symmetric Spaces

The Cartan decomposition of chapter 13 was an algebraic statement, $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$, and the Iwasawa refinement of chapter 14 organized the group around it. This final chapter reads the same data geometrically. The Euclidean summand $\mathfrak{p}$ is the tangent space of a Riemannian manifold on which G acts by isometries, the quotient G/K by a maximal compact subgroup, and this manifold is a *symmetric space*: at every point there is an isometry reversing geodesics through it. The Cartan involution θ is the geodesic symmetry at the base point, the bracket $[[\,\cdot\,,\cdot\,],\cdot\,]$ on $\mathfrak{p}$ is the Riemann curvature, and the compact and noncompact real forms of chapter 13 become the compact and noncompact symmetric spaces, dual to one another with curvatures of opposite sign. The dictionary between a Lie group and its Lie algebra, which has run the length of the book, closes here on a third column: the geometry the algebra encodes.

The correspondence is exact. A simply connected Riemannian symmetric space is the same datum as an *orthogonal symmetric Lie algebra*, a Lie algebra with an involution whose fixed subalgebra is compactly embedded, with an invariant inner product on the complementary eigenspace; the space is recovered as G/K and its metric, geodesics, and curvature are read off the bracket. Passing an orthogonal symmetric Lie algebra to its dual, replacing $\mathfrak{m}$ by $i\mathfrak{m}$ inside the complexification, exchanges compact type with noncompact type and flips the sign of the curvature, so the round sphere and the hyperbolic space, the Grassmannian and its noncompact partner, come in dual pairs built from the same $\mathfrak{k}$. The classification of symmetric spaces is thereby reduced to the classification of real simple Lie algebras and their involutions from chapter 13, the theorem of Cartan that the geometry of chapter 13's real forms is the geometry of a short list of model spaces.

Two structural facts owed from the differential-geometry side are given here, where the Lie theory makes them cheap. That the isometry group of a Riemannian manifold is a Lie group is the Myers-Steenrod theorem, proved by realizing the isometry group as a closed subgroup through the closed-subgroup theorem of chapter 3; and the splitting of a general symmetric space into Euclidean, compact, and noncompact factors follows from the decomposition of its orthogonal symmetric Lie algebra. The running example is the upper half-plane $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ of

chapter 13 and chapter 14, now seen as the hyperbolic plane of curvature -1, dual to the round sphere $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$; it scales up to hyperbolic space, spheres, the Grassmannians, and the Siegel upper half-space that carries the moduli of abelian varieties. The chapter closes on the locally symmetric quotients $\Gamma\backslash G/K$, the arithmetic geometry on which automorphic forms live, and hands the reader to the sequel volumes and to the closing chapter's map of where the subject goes next.

15.1 Riemannian Symmetric Spaces and the Isometry Group

The Cartan decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$ of chapter 13 has so far been read as pure Lie-algebra structure: $\mathfrak{k}$ the compact directions, $\mathfrak{p}$ the Euclidean complement, glued back into the global product $G = K\exp(\mathfrak{p})$. The geometric reading, which this chapter develops, is that $\mathfrak{p}$ is a *tangent space* and G/K a curved manifold whose curvature is computed from the bracket on $\mathfrak{p}$. Before that reading can begin, the geometry itself must be pinned down: what kind of manifold G/K is, and what symmetry singles it out. This section gives the definition and its first structural consequence. A Riemannian symmetric space is a manifold carrying, at every point, a reflection that reverses geodesics through that point; the reflection at a base point will turn out to be exactly the involution θ of chapter 13, tying the geometry back to the Cartan decomposition.

The local task here is twofold. First, define the class of manifolds precisely, taking the Riemannian primitives (metric, geodesic, isometry) as given and cited to the differential-geometry layer. Second, prove that the isometry group of such a manifold is a Lie group. This second fact, the Myers-Steenrod theorem, is what lets a symmetric space be presented as a homogeneous space G/K with G a Lie group, so that the algebraic machinery of the previous chapters applies. Its proof is a showcase of the closed-subgroup theorem theorem 3.2.1, and it is the theorem that the Riemannian-geometry sequel of chapter 13 forward-cites to this book.

The notions of Riemannian metric, geodesic, and isometry are developed in the differential-geometry layer and are used here without re-derivation. A *Riemannian metric* on a smooth manifold M is a smoothly varying inner product g_p on each tangent space T_pM; an *isometry* is a diffeomorphism $\varphi\colon M \to M$ with $g_{\varphi(p)}(d\varphi_p X, d\varphi_p Y) = g_p(X,Y)$ for all tangent vectors; a *geodesic* is a curve of locally shortest length, equivalently a curve with zero acceleration for the Levi-Civita connection. These are the primitives of [7], cited and not rebuilt. The one construction this section adds on top of them is the geodesic symmetry.

Definition 15.1.1: Riemannian Symmetric Space

Let (M, g) be a connected Riemannian manifold. A *geodesic symmetry* at a point $p \in M$ is an isometry $s_p\colon M \to M$ that fixes p and reverses tangent directions there:

$$s_p(p) = p, \qquad (ds_p)_p = -\mathrm{id}_{T_pM}$$

If a geodesic symmetry exists at every point and is only required to be defined on a neighborhood of that point, M is *locally symmetric*; if each s_p extends to a (global) isometry of M, then M is a *(globally) Riemannian symmetric space*.

The name geodesic symmetry earns its meaning from the exponential map. A geodesic γ through p with $\gamma(0) = p$ and $\gamma'(0) = v$ is carried by any isometry φ fixing p to the geodesic through p with initial velocity $d\varphi_p v$, because isometries take geodesics to geodesics. For s_p, whose differential at p

is $-\mathrm{id}$, this means $s_p(\gamma(t)) = \gamma(-t)$: the symmetry reflects each geodesic through p in the point p, sending $\gamma(t)$ to $\gamma(-t)$ along the same curve traced backwards. A geodesic symmetry, when it exists, is therefore unique, since its effect on every geodesic through p is forced, and those geodesics fill a neighborhood of p. What is *not* automatic is that this local reflection extends to a global isometry, and it is exactly this global extension that the definition requires and that makes M symmetric.

The first example is the running $\mathfrak{H}$.

Example 15.1.2: The Upper Half-Plane $\mathfrak{H}$

Let $\mathfrak{H} = \{z \in \mathbb{C} \mid \mathrm{Im} z > 0\}$ be the upper half-plane with the hyperbolic metric of constant curvature -1, on which $\mathrm{SL}(2,\mathbb{R})$ acts by Möbius transformations with $\mathrm{SO}(2)$ fixing i (proposition 13.5.4). The geodesics of $\mathfrak{H}$ are the vertical half-lines and the semicircles meeting the real axis at right angles. At the base point i, the map

$$s_i(z) = -\frac{1}{z}$$

is a geodesic symmetry. It fixes i, since $-1/i = i$; it is an isometry of $\mathfrak{H}$, being the Möbius transformation attached to $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$, which preserves the hyperbolic metric; and its differential at i is $-\mathrm{id}$, because s_i carries the geodesic vertical axis through i to itself with reversed orientation. The vertical axis $\{iy \mid y > 0\}$ is fixed setwise and $s_i(iy) = -1/(iy) = i/y$, so s_i interchanges iy and i/y: it reflects the axis in i, exactly reversing the geodesic. Thus $\mathfrak{H}$ is a Riemannian symmetric space with $s_i \colon z \mapsto -1/z$ the symmetry at i, and the symmetry at any other point is obtained by conjugating s_i by an isometry carrying that point to i.

The round sphere and hyperbolic space are the two constant-curvature companions.

Example 15.1.3: Spheres and Hyperbolic Space

The round sphere $S^n \subseteq \mathbb{R}^{n+1}$ with its standard metric is symmetric: at a point p, the geodesic symmetry s_p is the reflection of $\mathbb{R}^{n+1}$ fixing the line $\mathbb{R}p$ and negating its orthogonal complement, restricted to S^n. This is an orthogonal map, hence an isometry of S^n; it fixes p; and on $T_pS^n = p^\perp$ it acts as $-\mathrm{id}$. Geometrically s_p rotates the sphere by a half-turn about the axis through p and $-p$, carrying each great circle through p to itself with reversed orientation.

Hyperbolic space H^n is symmetric by the same pattern in the hyperboloid model $\{x \in \mathbb{R}^{n,1} \mid \langle x, x\rangle = -1,\ x_0 > 0\}$: the geodesic symmetry at p is the Lorentz reflection fixing $\mathbb{R}p$ and negating $p^\perp$, restricted to H^n. For $n = 2$ this is the $\mathfrak{H}$ of example 15.1.2 in a different model, and s_p again reverses every geodesic through p.

These examples share a structural feature worth isolating before it is proved in general: in each, the symmetries s_p move points around freely enough that any point can be carried to any other. On S^n a half-turn about one axis followed by a half-turn about another rotates the sphere by an arbitrary amount as the axes vary; on $\mathfrak{H}$ the maps s_i and s_z compose to a Möbius transformation translating i to a new point. This is the transitivity that proposition 15.1.5 extracts. First, though, the isometry group must be shown to be a Lie group at all.

The passage from a Riemannian manifold to the algebra of the previous chapters runs through its isometry group. For a symmetric space to become a homogeneous space G/K in the sense of chapter 4, the group G acting on it must be a Lie group, and the natural candidate is the full isometry group of M. That this group, defined purely metrically, carries a compatible smooth structure is the Myers-Steenrod theorem. The proof rests on one input from the differential-geometry layer and one from this book. The input from [7, An isometry is determined by its 1-jet at one point] is

rigidity: an isometry of a connected Riemannian manifold is completely determined by its value and differential at a single point. The input from this book is the closed-subgroup theorem theorem 3.2.1, which upgrades a closed subgroup of a Lie group to an embedded Lie subgroup. Rigidity realizes the isometry group as such a closed subgroup, and the closed-subgroup theorem then gives the smooth structure.

Theorem 15.1.4: Myers-Steenrod: The Isometry Group is a Lie Group

Let (M, g) be a connected Riemannian manifold. The group $\mathrm{Isom}(M, g)$ of isometries of M, equipped with the compact-open topology, is a Lie group, and its action $\mathrm{Isom}(M, g) \times M \to M$ is smooth. For each $p \in M$ the isotropy subgroup $\mathrm{Isom}(M, g)_p$ of isometries fixing p is compact.

***Proof Key ideas*:**
The argument has two ingredients kept sharply separate: a rigidity statement cited from [7, An isometry is determined by its 1-jet at one point], and the Lie-group structure built here from the closed-subgroup theorem.

Step 1: rigidity (cited). The one differential-geometry input is that an isometry of a connected Riemannian manifold is determined by its 1-jet at any single point. Precisely: if $\varphi, \psi \colon M \to M$ are isometries with $\varphi(p) = \psi(p)$ and $d\varphi_p = d\psi_p$ for some $p \in M$, then $\varphi = \psi$ on all of M. This is a self-contained theorem of [7, An isometry is determined by its 1-jet at one point], proved there from the fact that isometries commute with the exponential map, so that $\varphi(\exp_p v) = \exp_{\varphi(p)}(d\varphi_p v)$; connectedness then propagates agreement from a neighborhood of p to all of M. The result is upstream in that book and does not depend on the present one, so citing it introduces no circularity, even though [7] in turn cites the present theorem for its development of symmetric spaces. Rigidity is used as a black box; it is not reproved here.

Step 2: pinning an isometry by one orthonormal frame. Fix $p \in M$ and let $\mathcal{F}(M)$ be the orthonormal frame bundle of M, whose points are the pairs (q, u) with $q \in M$ and $u \colon \mathbb{R}^n \xrightarrow{\sim} T_q M$ a linear isometry (with respect to g_q and the standard inner product on $\mathbb{R}^n$). An isometry φ acts on frames by $\varphi \cdot (q, u) = (\varphi(q), d\varphi_q \circ u)$, and this action preserves orthonormality because $d\varphi_q$ is a linear isometry $T_q M \to T_{\varphi(q)} M$. Fix one reference frame (p, u_0). By rigidity, the map

$$\Phi \colon \mathrm{Isom}(M, g) \to \mathcal{F}(M), \qquad \Phi(\varphi) = \varphi \cdot (p, u_0) = \big(\varphi(p), d\varphi_p \circ u_0\big)$$

is *injective*: two isometries agreeing at the frame (p, u_0) agree in value and differential at p, hence are equal. So an isometry is completely recorded by the single frame it sends (p, u_0) to, and $\mathrm{Isom}(M, g)$ is identified with its image $\Phi(\mathrm{Isom}(M, g)) \subseteq \mathcal{F}(M)$, a subset of the orthonormal frame bundle.

Step 3: the image is closed, and inherits a manifold structure. The orthonormal frame bundle $\mathcal{F}(M)$ is a smooth manifold, a principal $\mathrm{O}(n)$-bundle over M ([7], [13, The Orthonormal Frame Bundle]). The compact-open topology on $\mathrm{Isom}(M, g)$ makes Φ a homeomorphism onto its image, and that image is closed in $\mathcal{F}(M)$: a limit of frames $\varphi_k \cdot (p, u_0)$ arising from isometries is again the image of an isometry, because the metric-preserving equations passing to the limit still describe an isometry (an Ascoli-type equicontinuity argument, the isometries being 1-Lipschitz for the induced distance). The image is moreover a subgroup under the frame action. Thus $\mathrm{Isom}(M, g)$ is realized as a closed subgroup of the Lie group of frame transformations acting on $\mathcal{F}(M)$.

Step 4: applying the closed-subgroup theorem. A closed subgroup of a Lie group is an embedded

Lie subgroup, by the closed-subgroup theorem theorem 3.2.1. Applied to the closed subgroup $\mathrm{Isom}(M,g)$ of frame transformations, this endows it with a smooth manifold structure making it a Lie group, with the compact-open topology as its manifold topology. Smoothness of the action $\mathrm{Isom}(M,g)\times M\to M$ follows because the action on frames is smooth and M is recovered as the base of $\mathcal{F}(M)$; equivalently, the evaluation $\varphi\mapsto\varphi(q)$ is smooth for each q, being a composite of the smooth frame projection with Φ.

Step 5: compactness of the isotropy. An isometry fixing p is determined by its differential $d\varphi_p\in\mathrm{O}(T_pM)\cong\mathrm{O}(n)$, again by rigidity, so Φ restricts to an injection of the isotropy $\mathrm{Isom}(M,g)_p$ into the compact group $\mathrm{O}(n)$ (the frames at the single point p). Its image is closed, hence compact, and Φ is a homeomorphism onto it, so $\mathrm{Isom}(M,g)_p$ is compact. This is the fact that will make the stabilizer K compact in proposition 15.1.5, completing the proof.

The theorem is the bridge from metric geometry to Lie theory. An isometry group is defined by a condition, preservation of distances, with no smoothness built in; Myers-Steenrod says the condition secretly carries a smooth structure, and a rigid one, since a single frame pins an isometry down. The mechanism is that rigidity turns the infinite-dimensional-looking set of isometries into a subset of the finite-dimensional frame bundle, and the closed-subgroup theorem does the rest. Every appearance of a symmetric space as a quotient G/K in the rest of this chapter rests on this theorem, which is why [7] defers its proof here rather than duplicating it.

With the isometry group known to be a Lie group, the defining reflections of a symmetric space can be put to work. The geodesic symmetries do more than reflect: composing two of them produces a *transvection*, a map that slides the manifold along a geodesic, and enough transvections move any point to any other. This transitivity presents M as a homogeneous space, and the compactness of the isotropy, established above, makes the presentation one to which the coset-manifold theorem applies.

Proposition 15.1.5: A Symmetric Space is Homogeneous and Complete

Let M be a globally Riemannian symmetric space. Then:

1. M is geodesically complete;
2. the group $G=\mathrm{Isom}(M)^\circ$ (the identity component of the isometry group) acts transitively on M, so M is homogeneous;
3. fixing a base point $o\in M$ and setting $K=\mathrm{Stab}_G(o)$, the map $gK\mapsto g\cdot o$ is a diffeomorphism $M\cong G/K$, with G a Lie group and K compact.

***Proof*:**
Step 1: transvections translate along geodesics. For points $p,q\in M$ let s_p,s_q be the global geodesic symmetries, and call the composite $\tau=s_qs_p$ a *transvection*. Suppose p and q lie on a common geodesic γ, parametrized so that $\gamma(0)=p$ and $\gamma(a)=q$. The reflection s_p in p sends $\gamma(t)$ to $\gamma(-t)$, and the reflection s_q in q sends $\gamma(t)$ to $\gamma(2a-t)$, since it reverses γ about the point $\gamma(a)$. Composing, $\tau=s_qs_p$ sends $\gamma(t)\mapsto s_q(\gamma(-t))=\gamma(2a-(-t))=\gamma(t+2a)$: the transvection slides the geodesic forward by twice the separation a. The transvection τ is an isometry, being a composite of two isometries, and it translates γ by the fixed amount $2a$, moving $p=\gamma(0)$ to $\gamma(2a)$.

Step 2: completeness. A geodesic γ defined on an interval can always be extended: reflecting in

an endpoint $\gamma(b)$ by the global isometry $s_{\gamma(b)}$ carries the already-defined portion $\gamma([b-\varepsilon, b])$ to a portion $\gamma([b, b+\varepsilon])$ on the far side, prolonging the geodesic past b. Since a global geodesic symmetry exists at every point, every geodesic extends indefinitely in both directions, so M is geodesically complete.

Step 3: transitivity. Let $p, q \in M$. By completeness and the Hopf-Rinow theorem ([7, Hopf-Rinow]), any two points are joined by a minimizing geodesic γ with $\gamma(0) = p$ and $\gamma(1) = q$. The transvection along γ through the midpoint $m = \gamma(\frac{1}{2})$, namely $\tau = s_m s_p$, sends $p = \gamma(0)$ to $\gamma(1) = q$ by the translation computation of Step 1 (with separation $a = \frac{1}{2}$, translating by 1). Each transvection lies in the identity component $G = \mathrm{Isom}(M)^\circ$, because $t \mapsto s_{\gamma(t/2)} s_p$ is a continuous path of isometries from the identity ($t = 0$ gives $s_p s_p = \mathrm{id}$) to τ ($t = 1$). Hence G carries p to any q, and the action of G on M is transitive: M is homogeneous.

Step 4: the coset presentation. By theorem 15.1.4 the full isometry group is a Lie group; its identity component G is then a connected Lie group, and it acts transitively and smoothly on M by Step 3. Fix $o \in M$ and let $K = \mathrm{Stab}_G(o)$ be the isotropy at o. The orbit-stabilizer theorem theorem 4.2.3 identifies the orbit map with a bijection $G/K \to M$, $gK \mapsto g \cdot o$, and since $K = G \cap \mathrm{Isom}(M)_o$ is closed in G, the quotient-manifold theorem theorem 4.1.3 makes G/K a smooth manifold and the bijection a diffeomorphism $G/K \cong M$. The isotropy K is a closed subgroup of G contained in the compact isotropy $\mathrm{Isom}(M)_o$ of theorem 15.1.4, hence compact. Thus $M \cong G/K$ with G a Lie group and K compact, completing the proof.

A condition stated point by point, that each point admit a reflecting isometry, globalizes to a single group G acting transitively, so that the local data of the reflections is repackaged as the homogeneous space G/K studied since chapter 4. The compactness of K is not incidental: it is what will let an $\mathrm{Ad}(K)$-invariant inner product be averaged onto $\mathfrak{m} \cong T_oM$ in the next section, giving the invariant metric from purely algebraic data. The transvection $s_q s_p$, the product of two reflections, behaves like a translation exactly as the product of two reflections in Euclidean geometry is a translation. Symmetric spaces are, in this precise sense, the curved manifolds on which a reflection group acts the way the Euclidean reflection group acts on flat space. On $\mathfrak{H}$ the machinery is explicit. The symmetry $s_i \colon z \mapsto -1/z$ and the symmetry s_{2i} at $2i$ compose to a Möbius transformation carrying i up the imaginary axis, and running the base point along the axis sweeps out a one-parameter group of transvections, the dilations $z \mapsto e^t z$, which is exactly the A factor of the Iwasawa decomposition $G = KAN$ met in example 14.4.1. The presentation $\mathfrak{H} \cong \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ of proposition 13.5.4 is thus the instance of proposition 15.1.5 for the smallest noncompact symmetric space, with $G = \mathrm{PSL}(2,\mathbb{R})$ the identity component of the isometry group and $K = \mathrm{SO}(2)/\{\pm I\}$ the compact isotropy at i.

The reflection s_o at the base point is more than a device for producing transvections. It is an involutive isometry of M fixing o, so its differential at o is an involution of T_oM, and conjugation by s_o is an involutive automorphism of G. That automorphism, transported to the Lie algebra, will be the involution θ whose ± 1-eigenspaces recover the Cartan decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{m}$ of chapter 13. The geometry of the symmetric space and the algebra of the Cartan involution are, in this way, the same object seen from two sides, and making that identification precise is the work of section 15.2.

Exercise 15.1.1

1. Verify in full that $s_i \colon z \mapsto -1/z$ is the geodesic symmetry of $\mathfrak{H}$ at i. Show that s_i fixes i, that it is the isometry attached to $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$, that $s_i \circ s_i = \mathrm{id}$, and that s_i interchanges iy and i/y on the vertical geodesic axis, so that $(ds_i)_i = -\mathrm{id}$ on $T_i\mathfrak{H}$.

2. Let G be a connected Lie group with a bi-invariant Riemannian metric (an inner product on $\mathfrak{g}$

invariant under $\mathrm{Ad}(G)$, extended by left translation). Prove that G is a Riemannian symmetric space, with geodesic symmetry at the identity e given by the inversion $s_e(g) = g^{-1}$, and the symmetry at any other point obtained by conjugating s_e by a left translation. (Recall that for a bi-invariant metric the geodesics through e are the one-parameter subgroups $t \mapsto \exp(tX)$, and that s_e sends $\exp(tX)$ to $\exp(-tX)$.)

3. Prove that the transvections $s_q s_p$ generate a group acting transitively on a globally symmetric space M. Show that for any two points there is a transvection carrying one to the other, and that the transvections lie in the identity component $\mathrm{Isom}(M)^\circ$.

4. Show that S^n and H^n are Riemannian symmetric spaces by exhibiting the geodesic symmetry at a point explicitly. For $S^n \subseteq \mathbb{R}^{n+1}$, take the orthogonal reflection fixing $\mathbb{R}p$ and negating $p^\perp$; for H^n in the hyperboloid model, take the corresponding Lorentz reflection. In each case verify that the map is an isometry fixing p with differential $-\mathrm{id}$ on T_pM.

5. Explain why the isotropy subgroup $K = \mathrm{Stab}_G(o)$ of the isometry group at a base point o is compact. Trace the argument through theorem 15.1.4: an isometry fixing o is determined by its differential $d\varphi_o \in \mathrm{O}(T_oM)$, so K embeds in the compact group $\mathrm{O}(n)$ as a closed subgroup.

6. Show that a geodesic symmetry at p, if it exists, is unique. Use that its differential at p is forced to be $-\mathrm{id}$ and that an isometry is determined by its 1-jet at a point (theorem 15.1.4, Step 1). Conclude that on a symmetric space the assignment $p \mapsto s_p$ is well defined.

15.2 Orthogonal Symmetric Lie Algebras and the Correspondence

The geodesic symmetries of section 15.1 made a symmetric space M into a homogeneous space G/K and produced, at the base point o, an involution $\theta = \mathrm{Ad}(s_o)$ of the isometry algebra $\mathfrak{g}_0 = \mathrm{Lie}(G)$. That involution carries all the local data: its -1-eigenspace is the tangent space T_oM, its $+1$-eigenspace is the isotropy algebra $\mathfrak{k}$, and the compactness of K constrains how $\mathfrak{k}$ sits inside $\mathfrak{g}_0$. This section names the resulting algebraic object, the *orthogonal symmetric Lie algebra*, and proves that it is a complete substitute for the symmetric space: passing to the pair $(\mathfrak{g}_0, \theta)$ loses nothing, and every such pair comes from a symmetric space. The correspondence converts a problem in Riemannian geometry into one in the structure theory of chapter 13. It also makes curvature computable, since it turns the Riemann tensor into a bracket.

The pair $(\mathfrak{g}_0, \theta)$ is exactly the Cartan decomposition data of chapter 13 with one relaxation: $\mathfrak{g}_0$ need not be semisimple, and θ need not be a Cartan involution in the strict sense of definition 13.3.1. What is retained is that the $+1$-eigenspace is compactly embedded, so that its group is compact and can serve as an isotropy group.

Definition 15.2.1: Orthogonal Symmetric Lie Algebra

An *orthogonal symmetric Lie algebra* is a pair $(\mathfrak{g}_0, \theta)$ consisting of a real Lie algebra $\mathfrak{g}_0$ and an involutive automorphism $\theta\colon \mathfrak{g}_0 \to \mathfrak{g}_0$ (so $\theta^2 = \mathrm{id}$, $\theta \neq \mathrm{id}$), such that the $+1$-eigenspace $\mathfrak{k}$ is *compactly embedded*: there is an inner product on $\mathfrak{g}_0$ for which every $\mathrm{ad}(X)$ with $X \in \mathfrak{k}$ is skew-symmetric, equivalently the connected subgroup of $\mathrm{Aut}(\mathfrak{g}_0)$ generated by $\{\exp(\mathrm{ad}X) : X \in \mathfrak{k}\}$ has compact closure. Write

$$\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{m}$$

for the eigenspace decomposition, where $\mathfrak{k}$ is the $+1$-eigenspace and $\mathfrak{m}$ the -1-eigenspace of θ. The pair is *effective* if $\mathfrak{k}$ contains no nonzero ideal of $\mathfrak{g}_0$.

The bracket relations of a Cartan decomposition hold verbatim, forced by θ being an automorphism. If X, Y lie in the $+1$-eigenspace then $\theta[X,Y] = [\theta X, \theta Y] = [X,Y]$, so $[X,Y]$ lies in $\mathfrak{k}$; if $X \in \mathfrak{k}$ and $Y \in \mathfrak{m}$ then $\theta[X,Y] = [X,-Y] = -[X,Y]$, so $[X,Y] \in \mathfrak{m}$; and if $X, Y \in \mathfrak{m}$ then $\theta[X,Y] = [-X,-Y] = [X,Y]$, so $[X,Y] \in \mathfrak{k}$. In symbols,

$$[\mathfrak{k},\mathfrak{k}] \subseteq \mathfrak{k}, \qquad [\mathfrak{k},\mathfrak{m}] \subseteq \mathfrak{m}, \qquad [\mathfrak{m},\mathfrak{m}] \subseteq \mathfrak{k}$$

the same three inclusions proved for the Cartan decomposition in theorem 13.3.3. The first makes $\mathfrak{k}$ a subalgebra; the second says $\mathfrak{m}$ is a $\mathfrak{k}$-module under ad, the *isotropy representation*; the third is the one that will read as curvature. Effectiveness records a geometric normalization: an ideal of $\mathfrak{g}_0$ contained in $\mathfrak{k}$ would act trivially on $\mathfrak{m} \cong T_oM$ and so contribute nothing to the geometry, an isometry group larger than needed. Removing it forces G to act on M as faithfully as the tangent data allows.

The running example is the pair behind the hyperbolic plane, read off from the Cartan decomposition of $\mathrm{SL}(2,\mathbb{R})$ computed in chapter 13.

Example 15.2.2: The Orthogonal Symmetric Lie Algebra of $\mathfrak{H}$

Take $\mathfrak{g}_0 = \mathfrak{sl}(2,\mathbb{R})$ with the involution $\theta(X) = -X^{\mathsf{T}}$, the Cartan involution of example 13.5.3. Its $+1$-eigenspace is $\mathfrak{k} = \mathfrak{so}(2) = \mathbb{R}W$, $W = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and its -1-eigenspace is $\mathfrak{m} = \mathrm{Sym}_0(2)$, the traceless symmetric matrices, spanned by

$$X_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The subalgebra $\mathfrak{k}$ is compactly embedded because $\exp(\mathbb{R}\,\mathrm{ad}W)$ generates $\mathrm{SO}(2)$ acting on $\mathfrak{g}_0$, a compact group; concretely $\mathrm{ad}W$ is skew for the inner product $\langle X, Y\rangle = \mathrm{tr}(XY^{\mathsf{T}})$ on $\mathfrak{sl}(2,\mathbb{R})$. The pair is effective: the only ideals of the simple algebra $\mathfrak{sl}(2,\mathbb{R})$ are 0 and $\mathfrak{sl}(2,\mathbb{R})$, and neither lies in the proper subspace $\mathfrak{k}$. The three inclusions are visible on the basis, $[W, X_1] = -2X_2 \in \mathfrak{m}$, $[W, X_2] = 2X_1 \in \mathfrak{m}$, and $[X_1, X_2] = 2W \in \mathfrak{k}$. This last bracket, a symmetric-times-symmetric landing back in $\mathfrak{k}$, is the one that will produce the curvature -1 of the hyperbolic plane.

The example shows the abstraction: an orthogonal symmetric Lie algebra is precisely the tangent data of a symmetric space, packaged so that no geometry is mentioned. The theorem below shows that the packaging is lossless. It is a bijection between Riemannian manifolds with symmetries on one side and Lie algebras with involutions on the other.

Theorem 15.2.3: The Symmetric Space Correspondence

There is a bijection between

1. isomorphism classes of simply connected Riemannian globally symmetric spaces, and
2. isomorphism classes of effective orthogonal symmetric Lie algebras $(\mathfrak{g}_0, \theta)$ equipped with an $\mathrm{Ad}(K)$-invariant inner product on the -1-eigenspace $\mathfrak{m}$,

where K is the connected subgroup with Lie algebra $\mathfrak{k}$. Under the bijection, a symmetric space M with base point o gives $\mathfrak{g}_0 = \mathrm{Lie}(\mathrm{Isom}(M)^\circ)$, $\theta = \mathrm{Ad}(s_o)$, and $\mathfrak{m} \cong T_oM$ with the inner product g_o; conversely, a pair $(\mathfrak{g}_0, \theta)$ with inner product gives the symmetric space $M = G/K$ built from the simply connected G with Lie algebra $\mathfrak{g}_0$.

***Proof*:**
The two constructions are described and then shown to be mutually inverse.

Step 1: from a symmetric space to a pair. Let M be a simply connected Riemannian globally symmetric space with base point o and geodesic symmetry s_o. By proposition 15.1.5 the identity component $G = \mathrm{Isom}(M)^\circ$ acts transitively, $K = \mathrm{Stab}_G(o)$ is compact, and $M = G/K$ (theorem 4.2.3). Set $\mathfrak{g}_0 = \mathrm{Lie}(G)$. Conjugation by s_o is an automorphism of G fixing o, so it stabilizes K and differentiates to an automorphism $\theta = \mathrm{Ad}(s_o)$ of $\mathfrak{g}_0$. Since $s_o^2 = \mathrm{id}$, $\theta^2 = \mathrm{Ad}(s_o^2) = \mathrm{id}$: the involution property comes for free from the involutivity of the geodesic symmetry. The $+1$-eigenspace of θ is $\mathfrak{k} = \mathrm{Lie}(K)$: an element of $\mathfrak{g}_0$ is fixed by $\mathrm{Ad}(s_o)$ exactly when its one-parameter group commutes with s_o, and on the isotropy algebra s_o acts by id because $d(s_o)_o = -\mathrm{id}$ conjugates the isotropy representation trivially. The -1-eigenspace $\mathfrak{m}$ is identified with T_oM by the differential of the orbit map $g \mapsto g \cdot o$: this differential kills $\mathfrak{k}$ and restricts to an isomorphism $\mathfrak{m} \xrightarrow{\sim} T_oM$, because $d(s_o)_o = -\mathrm{id}$ means $\mathrm{Ad}(s_o)$ acts by -1 on the tangent directions, singling out $\mathfrak{m}$. Compact embedding of $\mathfrak{k}$ is immediate: K is compact, so $\mathrm{Ad}(K)|_{\mathfrak{g}_0}$ has compact closure and preserves any $\mathrm{Ad}(K)$-invariant inner product, for which $\mathrm{ad}(\mathfrak{k})$ is skew. Effectiveness holds because $G = \mathrm{Isom}(M)^\circ$ acts almost effectively: an ideal of $\mathfrak{g}_0$ inside $\mathfrak{k}$ would generate a normal subgroup of G fixing o and acting trivially on T_oM, hence trivially on M by corollary 3.4.5, contradicting that G is a group of isometries with trivial pointwise-fixing kernel. Finally the metric g_o on $T_oM \cong \mathfrak{m}$ is $\mathrm{Ad}(K)$-invariant, since K acts by isometries fixing o. This produces the pair together with its inner product.

Step 2: from a pair to a symmetric space. Let $(\mathfrak{g}_0, \theta)$ be an effective orthogonal symmetric Lie algebra with an $\mathrm{Ad}(K)$-invariant inner product $\langle -, - \rangle$ on $\mathfrak{m}$. Let G be the simply connected Lie group with Lie algebra $\mathfrak{g}_0$ (corollary 3.4.5) and $K = \exp(\mathfrak{k})$ the connected subgroup with Lie algebra $\mathfrak{k}$; compact embedding makes K have compact closure, and for the correspondence one takes K closed, so $M = G/K$ is a smooth manifold by the quotient-manifold theorem (theorem 4.1.3). The tangent space at $o = eK$ is $T_oM \cong \mathfrak{g}_0/\mathfrak{k} \cong \mathfrak{m}$, and transporting $\langle -, - \rangle$ around by G defines a G-invariant Riemannian metric: the invariance of the inner product under $\mathrm{Ad}(K)$ is exactly the compatibility needed for the metric to be well-defined on cosets, as $\mathfrak{m}$ is $\mathrm{Ad}(K)$-stable by the inclusion $[\mathfrak{k}, \mathfrak{m}] \subseteq \mathfrak{m}$. The involution θ integrates to an automorphism σ of G (using simple connectivity, corollary 3.4.5) with $\sigma(K) = K$, so it descends to a map $s_o \colon G/K \to G/K$ fixing o. Its differential at o is $\theta|_{\mathfrak{m}} = -\mathrm{id}$, so s_o is a geodesic symmetry at o. Translating by G produces a geodesic symmetry $s_p = g s_o g^{-1}$ at every point $p = g \cdot o$, and each is an isometry because G acts by isometries and s_o is an isometry, being induced by the

metric-preserving σ. Thus $M = G/K$ is globally symmetric, and it is simply connected because G is simply connected and K is connected. This produces the symmetric space.

Step 3: the constructions are inverse. Starting from M, Step 1 produces $(\mathfrak{g}_0, \theta)$ with $\mathfrak{g}_0 = \mathrm{Lie}(\mathrm{Isom}(M)^\circ)$; feeding this into Step 2 rebuilds G/K with $G = \mathrm{Isom}(M)^\circ$ and $K = \mathrm{Stab}_G(o)$, which is M by proposition 15.1.5, with the same metric because the inner product transported is the original g_o. Conversely, starting from an effective pair $(\mathfrak{g}_0, \theta)$, Step 2 builds $M = G/K$; applying Step 1 to M returns $\mathrm{Lie}(\mathrm{Isom}(M)^\circ)$, which equals $\mathfrak{g}_0$ precisely because effectiveness makes G act as the full identity component of the isometry group: no larger isometry algebra can appear (any extra isometry would enlarge $\mathfrak{k}$ by an ideal, forbidden by effectiveness), and none smaller (transvections already generate G). The involutions match because θ was integrated to σ and then differentiated back. Hence the two passages are mutually inverse, establishing the bijection, as we sought to show.

Everything geometric about a simply connected symmetric space, its dimension, its isometry group, its curvature, its rank, is encoded in the finite-dimensional linear-algebra datum $(\mathfrak{g}_0, \theta, \langle -, - \rangle)$, and every such datum is realized. The simple-connectivity and effectiveness conditions are the bookkeeping that makes the bijection exact rather than approximate: without simple connectivity one would have to track a discrete quotient, and without effectiveness one would over-count isometry algebras by ideals acting trivially. With both imposed, the classification of symmetric spaces becomes the classification of effective orthogonal symmetric Lie algebras, the task section 15.3 takes up.

The payoff to cash immediately is curvature. The correspondence would be a formal equivalence if it did not compute something, and what it computes is the Riemann tensor, which collapses to a double bracket on $\mathfrak{m}$. The differential-geometric notions of the Levi-Civita connection, geodesics, and the Riemann curvature tensor are developed in [7]; what is proved here is that on a symmetric space they are all given algebraically by the bracket.

Proposition 15.2.4: Curvature of a Symmetric Space

Let $M = G/K$ be the Riemannian symmetric space of an orthogonal symmetric Lie algebra $(\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{m}, \theta)$ with $\mathrm{Ad}(K)$-invariant inner product $\langle -, - \rangle$ on $\mathfrak{m} \cong T_oM$. Then:

1. the geodesics of M through o are the curves $t \mapsto \exp(tX) \cdot o$ for $X \in \mathfrak{m}$;
2. the Riemann curvature tensor at o is purely algebraic,
$$R(X, Y)Z = -[[X, Y], Z] \qquad (X, Y, Z \in \mathfrak{m})$$
3. the sectional curvature of the plane spanned by orthonormal $X, Y \in \mathfrak{m}$ is
$$K(X, Y) = -\langle [[X, Y], Y], X \rangle$$

***Proof*:**
The starting point is the invariant connection, computed in [7, The Levi-Civita connection of an invariant metric, Connection and geodesics of a naturally reductive metric]: the containment $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{k}$ makes G/K naturally reductive, and on the fields along orbits obtained by translating tangent vectors at o the covariant derivative is
$$\nabla_X Y = \tfrac{1}{2}[X, Y]_{\mathfrak{m}}$$

the $\mathfrak{m}$-component of the bracket, for X, Y tangent at o; the companion formula for the action fields carries the opposite sign, and in the symmetric case both vanish by the same containment, which is all Step 2 uses.

Step 1: geodesics. The map s_o is an isometry fixing o with $d(s_o)_o = -\mathrm{id}$. A geodesic γ with $\gamma(0) = o$ and $\dot\gamma(0) = X$ is carried by s_o to a geodesic with the same initial point and initial velocity $-X$, so $s_o(\gamma(t)) = \gamma(-t)$. Now let $\gamma(t) = \exp(tX) \cdot o$ with $X \in \mathfrak{m}$. The one-parameter group $g_t = \exp(tX)$ satisfies $\sigma(g_t) = \exp(t\,\theta X) = \exp(-tX) = g_t^{-1}$, since $\theta X = -X$ on $\mathfrak{m}$; hence the transvection $s_{\gamma(s/2)} s_o$ translates along γ, and γ is an orbit of the one-parameter group of transvections $\exp(tX)$. A curve that is an orbit of a one-parameter group of transvections is a geodesic, because at each of its points the geodesic symmetry reverses it, and a curve reversed by the symmetry at each point has vanishing geodesic curvature. This gives part (1): the geodesics through o are exactly the $\exp(tX) \cdot o$ for $X \in \mathfrak{m}$.

Step 2: the connection vanishes at o on $\mathfrak{m}$. For $X, Y \in \mathfrak{m}$, the bracket $[X, Y]$ lies in $\mathfrak{k}$ by $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{k}$, so its $\mathfrak{m}$-component is zero. Thus the invariant connection gives

$$\nabla_X Y = \tfrac{1}{2}[X, Y]_{\mathfrak{m}} = 0$$

at o, for all $X, Y \in \mathfrak{m}$: the Levi-Civita connection of invariant fields on $\mathfrak{m}$ vanishes at the base point. This is the algebraic expression of the symmetry $\nabla R = 0$ that characterizes a symmetric space.

Step 3: the curvature tensor. The Riemann tensor is $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$. Evaluate at o on invariant fields extending $X, Y, Z \in \mathfrak{m}$. By Step 2 the terms $\nabla_X \nabla_Y Z$ and $\nabla_Y \nabla_X Z$ vanish at o: the inner $\nabla_Y Z$ is $\mathfrak{k}$-valued near o, so differentiating and projecting leaves no $\mathfrak{m}$-contribution at o beyond the bracket with the $\mathfrak{k}$-part. Carrying this out, the only surviving term comes from $\nabla_{[X,Y]} Z$ with $[X, Y] \in \mathfrak{k}$: for $W \in \mathfrak{k}$ and $Z \in \mathfrak{m}$ the invariant connection acts by the isotropy representation, $\nabla_W Z = [W, Z]$. This is the linear isotropy ($\mathfrak{k}$-)action of $\mathfrak{k}$ on the tangent space $\mathfrak{m}$, proved in [7, The isotropy representation of a reductive homogeneous space, The Levi-Civita connection of an invariant metric], and it carries no factor of $\frac{1}{2}$, in contrast to the $\nabla_X Y = \frac{1}{2}[X, Y]_{\mathfrak{m}}$ formula above that governs the $\mathfrak{m} \times \mathfrak{m}$ case. Thus

$$R(X, Y)Z = -\nabla_{[X,Y]} Z = -[[X, Y], Z]$$

using that the second-derivative terms cancel by Step 2 and the antisymmetry of the bracket. This is part (2). The sign is fixed by the convention of [7, Convention: The curvature sign convention]; the double bracket makes R algebraic and $\nabla R = 0$ automatic, since R is built from the constant tensor $[[-, -], -]$.

Step 4: sectional curvature. For orthonormal $X, Y \in \mathfrak{m}$, the sectional curvature of their span is $K(X, Y) = \langle R(X, Y)Y, X \rangle$. Substituting part (2),

$$K(X, Y) = \langle R(X, Y)Y, X \rangle = \langle -[[X, Y], Y], X \rangle = -\langle [[X, Y], Y], X \rangle$$

which is part (3), completing the proof.

The formula is the whole content of the correspondence made computational. Sectional curvature, an analytic invariant defined through the second fundamental form of geodesic variations, is a single triple bracket paired against the inner product. Two consequences are immediate. First, $\nabla R = 0$: the curvature tensor is the constant tensor $-[[-, -], -]$, so it is parallel, which is the intrinsic characterization of a symmetric space, adopted as the definition of a locally symmetric

space in [7, Locally symmetric space]. Second, the sign of the curvature is governed by the sign of $\langle[[X,Y],Y],X\rangle$, which by the invariance of the Killing form is a sign of κ on $\mathfrak{m}$: this is the seed of the compact/noncompact trichotomy of section 15.3, where $\kappa|_{\mathfrak{m}} < 0$ gives $K \geq 0$ and $\kappa|_{\mathfrak{m}} > 0$ gives $K \leq 0$.

The place to watch the formula work is the running example, where the bracket is a 2×2 matrix computation and the answer is the constant curvature -1 of the hyperbolic plane.

Example 15.2.5: The Curvature of $\mathfrak{H}$ from the Bracket

Return to $(\mathfrak{sl}(2,\mathbb{R}),\theta)$ of example 15.2.2, with $\mathfrak{m} = \mathrm{Sym}_0(2)$ spanned by

$$X_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The inner product on $\mathfrak{m}$ is $\langle X,Y\rangle = \frac{1}{2}\mathrm{tr}(XY)$, chosen to normalize the metric; on the basis $\langle X_1,X_1\rangle = \frac{1}{2}\mathrm{tr}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = 1$, $\langle X_2,X_2\rangle = \frac{1}{2}\mathrm{tr}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = 1$, and $\langle X_1,X_2\rangle = \frac{1}{2}\mathrm{tr}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) = 0$, so X_1, X_2 are orthonormal. Compute the brackets. First

$$[X_1,X_2] = X_1X_2 - X_2X_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = 2W$$

which lies in $\mathfrak{k} = \mathbb{R}W$ as required. Then

$$[[X_1,X_2],X_2] = [2W,X_2] = 2(WX_2 - X_2W) = 2\left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\right) = 2 \cdot 2X_1 = 4X_1$$

Pairing against X_1,

$$K(X_1,X_2) = -\langle[[X_1,X_2],X_2],X_1\rangle = -\langle 4X_1,X_1\rangle = -4\langle X_1,X_1\rangle = -4$$

with the inner product $\langle X,Y\rangle = \frac{1}{2}\mathrm{tr}(XY)$. Rescaling the metric by the constant factor 4, that is taking $\langle X,Y\rangle = 2\mathrm{tr}(XY)$, multiplies sectional curvature by $\frac{1}{4}$ and gives $K(X_1,X_2) = -1$: the space $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ carries, in the invariant metric normalized this way, constant curvature -1. This identifies it as the hyperbolic plane H^2, the model of constant negative curvature. The geodesics $t \mapsto \exp(tX_1)\cdot i = e^{2t}i$ are the imaginary axis, matching the A-orbit description of example 14.4.1 and the Möbius picture of proposition 13.5.4: dilation along the axis is a geodesic through i.

The computation is the correspondence in miniature. A curvature, which classically requires the metric, the connection, and a variation of geodesics, is here three matrix brackets and one trace. Contrast the compact dual $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$. There $\mathfrak{m}$ is again two-dimensional, spanned by two of the rotation generators, but $\mathfrak{m}$ sits inside the *compact* algebra $\mathfrak{so}(3)$, where the Killing form is negative-definite; the same double-bracket computation with $[\mathfrak{m},\mathfrak{m}] \subseteq \mathfrak{k}$ produces $K = +1$ rather than -1, the constant positive curvature of the round sphere. The single sign difference between $\mathfrak{sl}(2,\mathbb{R})$ and $\mathfrak{so}(3)$, the sign of κ on $\mathfrak{m}$, distinguishes the hyperbolic plane from the sphere. That sign flip is the compact/noncompact duality, developed in full in section 15.3.

The correspondence has converted Riemannian geometry into Lie algebra. The metric is an $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{m}$; the geodesics are one-parameter subgroups; the curvature is a double bracket. Every question about a symmetric space is now a question about the pair $(\mathfrak{g}_0,\theta)$, and classifying symmetric spaces is classifying orthogonal symmetric Lie algebras, which reduces by

section 15.3 to the real forms and involutions of chapter 13.

Exercise 15.2.1

1. For $(\mathfrak{sl}(2,\mathbb{R}),\theta)$ with $\mathfrak{m} = \mathrm{Sym}_0(2)$ spanned by $X_1 = \mathrm{diag}(1,-1)$ and $X_2 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, compute $[X_1, X_2]$, then $[[X_1, X_2], X_2]$, and confirm from $K(X_1, X_2) = -\langle [[X_1, X_2], X_2], X_1 \rangle$ that the curvature is negative. State the metric normalization for which $K = -1$, identifying $\mathfrak{H}$ as the hyperbolic plane.

2. Let $\mathfrak{g}_0 = \mathfrak{so}(3)$ with $\theta = \mathrm{Ad}(\mathrm{diag}(1,1,-1))$, so $\mathfrak{k} = \mathfrak{so}(2)$ (rotations of the first two coordinates) and $\mathfrak{m}$ is spanned by the two generators Y_1, Y_2 mixing the third coordinate with the first two. Compute $[Y_1, Y_2] \in \mathfrak{k}$ and $[[Y_1, Y_2], Y_2]$, and verify that $K(Y_1, Y_2) = +1$ in the metric normalized to unit sphere radius, so $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ has constant curvature $+1$, the compact dual of $\mathfrak{H}$.

3. Let G be a compact connected Lie group with Lie algebra $\mathfrak{g}$ and a bi-invariant metric (proposition 9.1.5). Consider $\mathfrak{g}_0 = \mathfrak{g} \oplus \mathfrak{g}$ with the swap involution $\theta(X, Y) = (Y, X)$. Identify $\mathfrak{k}$ and $\mathfrak{m}$, check that $(\mathfrak{g}_0, \theta)$ is an orthogonal symmetric Lie algebra, and show that the associated symmetric space is G itself, realized as $(G \times G)/\Delta G$. Using the curvature formula, show its sectional curvature is ≥ 0.

4. Prove directly from θ being an involutive automorphism with ± 1-eigenspaces $\mathfrak{k}, \mathfrak{m}$ that $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}$, $[\mathfrak{k}, \mathfrak{m}] \subseteq \mathfrak{m}$, and $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{k}$. Explain why the third inclusion, and not $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{m}$, is what makes the curvature formula $R(X, Y)Z = -[[X, Y], Z]$ take values in $\mathfrak{m}$.

5. For $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$, explain the identification $\mathfrak{m} \cong T_o\mathfrak{H}$ at $o = i$ concretely: show that the orbit map $g \mapsto g \cdot i$ has differential at e killing $\mathfrak{k} = \mathfrak{so}(2)$ and restricting to an isomorphism $\mathfrak{m} \to T_i\mathfrak{H}$, and compute the images of X_1, X_2 as tangent vectors at i.

6. Show that an orthogonal symmetric Lie algebra $(\mathfrak{g}_0, \theta)$ fails to be effective exactly when $\mathfrak{k}$ contains a nonzero ideal $\mathfrak{i}$ of $\mathfrak{g}_0$, and that in that case the connected subgroup $\exp(\mathfrak{i})$ acts trivially on $\mathfrak{m} \cong T_oM$. Deduce that effectiveness is the condition that G act on $M = G/K$ with discrete pointwise-fixing kernel.

15.3 The Compact/Noncompact Duality and the Cartan Classification

The correspondence of section 15.2 translated a symmetric space $M = G/K$ into the algebraic pair $(\mathfrak{g}_0, \theta)$ with $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{m}$, and proposition 15.2.4 turned the Riemann curvature into the bracket $R(X, Y)Z = -[[X, Y], Z]$ on $\mathfrak{m}$. One invariant governs that formula: the sign of the Killing form on $\mathfrak{m}$. When κ is negative-definite on $\mathfrak{m}$ the sectional curvature is nonnegative; when positive-definite, nonpositive; when the bracket $[\mathfrak{m}, \mathfrak{m}]$ vanishes, the curvature is zero. This trichotomy sorts every symmetric space into a compact type, a noncompact type, and a Euclidean type, and it pairs the first two by an operation that lives entirely inside the complexification: replacing $\mathfrak{m}$ by $i\mathfrak{m}$. That operation is the compact/noncompact duality, and it flips the curvature sign, so a compact-type space of positive curvature is mirrored by a noncompact-type space of negative curvature built on the same $\mathfrak{k}$. With the duality in hand the classification halves: every noncompact irreducible symmetric space is the dual of a compact one, and the compact ones reduce to the Cartan-Killing classification

of chapter 13. This section states the duality with a full proof and the Cartan classification in full.

The sign trichotomy is visible already in the two anchors of section 15.2: $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ has $\kappa|_{\mathfrak{m}} > 0$ and constant curvature -1, while its compact partner $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ has $\kappa|_{\mathfrak{m}} < 0$ and constant curvature $+1$. The duality makes this partnership structural rather than coincidental. To state it, the dual pair must first be built inside the complexification.

Definition 15.3.1: The Dual of an Orthogonal Symmetric Lie Algebra

Let $(\mathfrak{g}_0, \theta)$ be an orthogonal symmetric Lie algebra (definition 15.2.1) with eigenspace decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{m}$, and let $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$ be its complexification. The *dual* of $(\mathfrak{g}_0, \theta)$ is the real subspace

$$\mathfrak{g}_0^* = \mathfrak{k} \oplus i\mathfrak{m} \subseteq \mathfrak{g}_{\mathbb{C}}$$

obtained by keeping $\mathfrak{k}$ and replacing $\mathfrak{m}$ by $i\mathfrak{m}$, equipped with the involution θ^* that is $+1$ on $\mathfrak{k}$ and -1 on $i\mathfrak{m}$.

The pair $(\mathfrak{g}_0, \theta)$ is of *compact type* if the Killing form κ is negative-definite on $\mathfrak{m}$, of *noncompact type* if κ is positive-definite on $\mathfrak{m}$, and of *Euclidean type* if $[\mathfrak{m}, \mathfrak{m}] = 0$. From the curvature formula of proposition 15.2.4 and κ-invariance, the sectional curvature of the associated symmetric space is respectively $K \geq 0$, $K \leq 0$, and $K \equiv 0$.

The three types are exhaustive on the irreducible pieces: an irreducible orthogonal symmetric Lie algebra is of exactly one of the three types, a fact recorded in proposition 15.3.5 below. The naming matches the geometry. A compact-type space is compact (its transvection group is a compact semisimple group), a noncompact-type space is noncompact and diffeomorphic to a Euclidean space, and a Euclidean-type space is a flat torus or a Euclidean space itself. The link between the sign of $\kappa|_{\mathfrak{m}}$ and the curvature follows from proposition 15.2.4 together with the invariance of the Killing form: for $X, Y \in \mathfrak{m}$ with $\langle X, Y\rangle = 0$ and $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{k}$, the sectional curvature $K(X, Y)$ is a positive multiple of $-\langle [[X, Y], Y], X\rangle = \langle [X, Y], [X, Y]\rangle_{\mathfrak{k}}$, and the sign of that bracket length is exactly the sign of $\kappa|_{\mathfrak{m}}$ transported through the identity $\langle [X, Y], [X, Y]\rangle_{\mathfrak{k}} = -\kappa([X, Y], [X, Y])$ on the compact directions.

Before the duality theorem, the dual must be checked to be a real Lie algebra, and its type must be seen to flip. Both follow from a single sign computation inside $\mathfrak{g}_{\mathbb{C}}$.

Theorem 15.3.2: The Compact/Noncompact Duality

Let $(\mathfrak{g}_0, \theta)$ be an orthogonal symmetric Lie algebra with dual $(\mathfrak{g}_0^*, \theta^*)$ as in definition 15.3.1. Then:

1. $\mathfrak{g}_0^* = \mathfrak{k} \oplus i\mathfrak{m}$ is a real Lie subalgebra of $\mathfrak{g}_{\mathbb{C}}$, and θ^* is an involutive automorphism of it with eigenspace decomposition $\mathfrak{g}_0^* = \mathfrak{k} \oplus i\mathfrak{m}$.
2. The Killing form satisfies $\kappa^*|_{i\mathfrak{m}} = -\kappa|_{\mathfrak{m}}$ under the identification $X \mapsto iX$; hence $(\mathfrak{g}_0, \theta)$ is of compact type if and only if $(\mathfrak{g}_0^*, \theta^*)$ is of noncompact type, and conversely.
3. Duality is involutive, $(\mathfrak{g}_0^*)^* = \mathfrak{g}_0$, and it negates the curvature: the algebraic curvature tensor of $\mathfrak{g}_0^*$ on $i\mathfrak{m}$ is the negative of that of $\mathfrak{g}_0$ on $\mathfrak{m}$ under $X \mapsto iX$. It therefore restricts to an involutive bijection between compact-type and noncompact-type irreducible orthogonal symmetric Lie algebras, exchanging $K \geq 0$ with $K \leq 0$.

***Proof*:**
Step 1: the dual is a real subalgebra. The brackets on $\mathfrak{g}_0$ respect the θ-grading: since θ is an automorphism with $+1$-eigenspace $\mathfrak{k}$ and -1-eigenspace $\mathfrak{m}$,

$$[\mathfrak{k},\mathfrak{k}] \subseteq \mathfrak{k}, \qquad [\mathfrak{k},\mathfrak{m}] \subseteq \mathfrak{m}, \qquad [\mathfrak{m},\mathfrak{m}] \subseteq \mathfrak{k}$$

the three inclusions being the statements that a bracket of two $+1$-eigenvectors is a $+1$-eigenvector, a bracket of a $+1$ with a -1 is a -1, and a bracket of two -1-eigenvectors is a $+1$. Now compute the three brackets on $\mathfrak{g}_0^* = \mathfrak{k} \oplus i\mathfrak{m}$ inside $\mathfrak{g}_{\mathbb{C}}$, where the bracket is $\mathbb{C}$-bilinear:

$$[\mathfrak{k},\mathfrak{k}] \subseteq \mathfrak{k}, \qquad [\mathfrak{k}, i\mathfrak{m}] = i[\mathfrak{k},\mathfrak{m}] \subseteq i\mathfrak{m}, \qquad [i\mathfrak{m}, i\mathfrak{m}] = i^2[\mathfrak{m},\mathfrak{m}] = -[\mathfrak{m},\mathfrak{m}] \subseteq \mathfrak{k}$$

The first bracket is unchanged. The mixed bracket pulls the single factor of i through the $\mathbb{C}$-bilinear bracket and lands back in $i\mathfrak{m}$. The third bracket is the decisive one: two factors of i produce $i^2 = -1$, so $[i\mathfrak{m}, i\mathfrak{m}] = -[\mathfrak{m},\mathfrak{m}]$, which still lies in $\mathfrak{k}$ because $\mathfrak{k}$ is an $\mathbb{R}$-subspace closed under negation. All three brackets land inside $\mathfrak{k} \oplus i\mathfrak{m}$, so $\mathfrak{g}_0^*$ is closed under the bracket, and it is a real subspace of $\mathfrak{g}_{\mathbb{C}}$; hence it is a real Lie subalgebra. The map θ^*, being $+1$ on $\mathfrak{k}$ and -1 on $i\mathfrak{m}$, squares to the identity and respects each of the three brackets above by the same sign bookkeeping that made θ an automorphism, so θ^* is an involutive automorphism with the stated eigenspaces. This proves part (1).

Step 2: the Killing form flips sign on $\mathfrak{m}$. The Killing form of $\mathfrak{g}_{\mathbb{C}}$ is $\mathbb{C}$-bilinear and restricts on the real form $\mathfrak{g}_0$ to κ, and on the real form $\mathfrak{g}_0^*$ to κ^* (the Killing form of a real form is the restriction of the complex Killing form, because ad and the trace are computed identically over the two fields on a shared basis). For $X, Y \in \mathfrak{m}$, the vectors $iX, iY \in i\mathfrak{m}$ satisfy

$$\kappa^*(iX, iY) = \kappa_{\mathbb{C}}(iX, iY) = i^2\, \kappa_{\mathbb{C}}(X, Y) = -\kappa(X, Y)$$

using $\mathbb{C}$-bilinearity of $\kappa_{\mathbb{C}}$ to extract $i^2 = -1$ and the fact that $\kappa_{\mathbb{C}}(X, Y) = \kappa(X, Y)$ for $X, Y \in \mathfrak{g}_0$. So κ^* on $i\mathfrak{m}$ is the negative of κ on $\mathfrak{m}$ under $X \mapsto iX$. If $\kappa|_{\mathfrak{m}} < 0$ (compact type), then $\kappa^*|_{i\mathfrak{m}} > 0$ (noncompact type), and conversely; this is part (2). The same identity restricted to $\mathfrak{k}$ leaves κ unchanged, since $\kappa^*(X, Y) = \kappa_{\mathbb{C}}(X, Y) = \kappa(X, Y)$ for $X, Y \in \mathfrak{k}$ with no factor of i: the compact directions $\mathfrak{k}$ are shared by the pair, and only the transverse directions $\mathfrak{m}$ change sign.

Step 3: involutivity and the curvature flip. Applying the construction twice sends $\mathfrak{m} \mapsto i\mathfrak{m} \mapsto i(i\mathfrak{m}) = i^2\mathfrak{m} = -\mathfrak{m} = \mathfrak{m}$ as an $\mathbb{R}$-subspace, so $(\mathfrak{g}_0^*)^* = \mathfrak{k} \oplus \mathfrak{m} = \mathfrak{g}_0$ and duality is involutive. For the curvature, proposition 15.2.4 gives the algebraic curvature tensor on the tangent space $\mathfrak{m}$ as $R(X, Y)Z = -[[X, Y], Z]$. On $\mathfrak{g}_0^*$ the corresponding tensor on $i\mathfrak{m}$ is $R^*(iX, iY)(iZ) = -[[iX, iY], iZ]$, and the bracket count gives three factors of i inside a single iterated bracket that begins with $[i\mathfrak{m}, i\mathfrak{m}] = -[\mathfrak{m},\mathfrak{m}]$:

$$[[iX, iY], iZ] = [\,-[X, Y]\,,\, iZ\,] = -\,i\,[[X, Y], Z] = -i\,\big(-R(X, Y)Z\big) = i\,R(X, Y)Z$$

where the inner bracket $[iX, iY] = -[X, Y] \in \mathfrak{k}$ carries the i^2, and the outer bracket $[\mathfrak{k}, i\mathfrak{m}]$ carries one further i. Hence

$$R^*(iX, iY)(iZ) = -[[iX, iY], iZ] = -\,i\,R(X, Y)Z = i\,\big(-R(X, Y)Z\big)$$

so under the identification $W \mapsto iW$ of $\mathfrak{m}$ with $i\mathfrak{m}$, the curvature R^* is exactly $-R$. The sectional curvatures therefore satisfy $K^* = -K$: nonnegative curvature becomes nonpositive and back. Combined with part (2), which exchanges the compact and noncompact types, duality is an involutive bijection between compact-type and noncompact-type pairs, and it exchanges the curvature signs, completing the proof.

The mechanism is a single change of sign. Passing from $\mathfrak{m}$ to $i\mathfrak{m}$ inserts one factor of i per $\mathfrak{m}$-vector, and every structural quantity counts those factors: the bracket $[\mathfrak{m},\mathfrak{m}]$ carries two and flips sign, so a compact-type algebra whose $[\mathfrak{m},\mathfrak{m}]$ pointed one way in $\mathfrak{k}$ acquires the opposite bracket, negating the curvature; the Killing form $\kappa|_{\mathfrak{m}}$ is a quadratic form and carries two, flipping the definiteness. This is the same sign that produced the Cartan involution of chapter 13: there, complexifying a compact real form and twisting by θ manufactured a noncompact one, and here that twist is read geometrically as passing to the dual symmetric space. The two real forms $\mathfrak{g}_0$ and $\mathfrak{g}_0^*$ sit inside the same $\mathfrak{g}_{\mathbb{C}}$ and share the same maximal compact $\mathfrak{k}$; what distinguishes them is the sign the bracket carries on the complementary directions.

The dual construction is transparent on the running example, where it converts the negatively curved half-plane into the positively curved sphere.

Example 15.3.3: The Half-Plane Dual to the Sphere

Take $\mathfrak{g}_0 = \mathfrak{sl}_2(\mathbb{R})$ with the Cartan involution $\theta(X) = -X^{\mathsf{T}}$, so $\mathfrak{k} = \mathfrak{so}(2) = \mathbb{R}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\mathfrak{m} = \mathrm{Sym}_0(2)$ is the traceless symmetric matrices (example 13.5.3). The associated symmetric space is $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2) = H^2$, of constant curvature -1, and $\kappa|_{\mathfrak{m}} > 0$: the pair is of noncompact type. A basis of $\mathfrak{m}$ is

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

whose bracket $[A,B] = AB - BA = -2\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ lands in $\mathfrak{k} = \mathfrak{so}(2)$, as $[\mathfrak{m},\mathfrak{m}] \subseteq \mathfrak{k}$ requires.

The dual replaces $\mathfrak{m}$ by $i\mathfrak{m} = \{iA, iB\}$ inside $\mathfrak{sl}_2(\mathbb{C})$. The new bracket $[iA, iB] = i^2[A,B] = -[A,B] = +2\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has flipped sign, and the real subalgebra $\mathfrak{g}_0^* = \mathfrak{so}(2) \oplus i\mathrm{Sym}_0(2)$ is a compact real form of $\mathfrak{sl}_2(\mathbb{C})$. Concretely $\mathfrak{g}_0^* \cong \mathfrak{su}(2) \cong \mathfrak{so}(3)$: the skew generator of $\mathfrak{so}(2)$ together with $iA = \left(\begin{smallmatrix} i & 0 \\ 0 & -i \end{smallmatrix}\right)$ and $iB = \left(\begin{smallmatrix} 0 & i \\ i & 0 \end{smallmatrix}\right)$ spans a copy of $\mathfrak{su}(2)$ (these are i times Hermitian, hence skew-Hermitian and traceless). The dual symmetric space is $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$, of constant curvature $+1$. The Killing form has flipped from positive to negative on the tangent directions, the curvature from -1 to $+1$, and the shared $\mathfrak{k} = \mathfrak{so}(2)$ is the common maximal compact of both $\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SU}(2)$. This is the smallest instance of the duality: the hyperbolic plane and the round sphere are dual symmetric spaces, mirror images across the sign of the bracket.

The half-plane and the sphere are the rank-one, dimension-two model, and the duality scales to all constant-curvature spaces: $H^n = \mathrm{SO}(n,1)^\circ/\mathrm{SO}(n)$ is dual to $S^n = \mathrm{SO}(n+1)/\mathrm{SO}(n)$, both on $\mathfrak{k} = \mathfrak{so}(n)$. The next example records the family of algebras where the split real form and its compact dual are paired.

Example 15.3.4: The Split and Compact Duals on $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$

Take $\mathfrak{g}_0 = \mathfrak{sl}_n(\mathbb{R})$ with $\theta(X) = -X^{\mathsf{T}}$, so $\mathfrak{k} = \mathfrak{so}(n)$ and $\mathfrak{m} = \mathrm{Sym}_0(n)$ (example 13.5.1). Here $\kappa|_{\mathfrak{m}} > 0$, so the symmetric space $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$ is of noncompact type, of nonpositive curvature; its rank is $n-1$, the dimension of a maximal abelian subspace of $\mathfrak{m}$ (definition 14.1.1). The dual replaces $\mathfrak{m} = \mathrm{Sym}_0(n)$ by $i\mathrm{Sym}_0(n)$ inside $\mathfrak{sl}_n(\mathbb{C})$, and

$$\mathfrak{g}_0^* = \mathfrak{so}(n) \oplus i\mathrm{Sym}_0(n) = \mathfrak{su}(n)$$

the compact real form of $\mathfrak{sl}_n(\mathbb{C})$: a matrix in $\mathfrak{so}(n)$ is real skew, a matrix in $i\mathrm{Sym}_0(n)$ is i times real symmetric hence imaginary Hermitian, and together they span the traceless skew-Hermitian matrices $\mathfrak{su}(n)$. The dual symmetric space is $\mathrm{SU}(n)/\mathrm{SO}(n)$, of compact type and nonnegative

curvature, again of rank $n-1$. The two spaces $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$ and $\mathrm{SU}(n)/\mathrm{SO}(n)$ share the complexification $\mathfrak{sl}_n(\mathbb{C})$ and the maximal compact $\mathfrak{so}(n)$; the sign of the bracket on $\mathfrak{m}$ is all that separates the noncompact space of nonpositive curvature from its compact positively curved mirror.

The duality reduces the classification to its compact-type half plus a splitting into irreducible factors. The splitting is the algebraic form of the de Rham decomposition, and it isolates the three types cleanly.

Proposition 15.3.5: The Trichotomy of Simply Connected Symmetric Spaces

Every simply connected Riemannian symmetric space M is a Riemannian product

$$M = M_0 \times M_+ \times M_-$$

of a Euclidean space $M_0 = \mathbb{R}^d$, a compact-type symmetric space M_+, and a noncompact-type symmetric space M_-, each factor uniquely determined up to order and each itself a product of irreducible symmetric spaces. Correspondingly the orthogonal symmetric Lie algebra $(\mathfrak{g}_0, \theta)$ decomposes as an orthogonal direct sum

$$\mathfrak{g}_0 = \mathfrak{g}_0^{\mathrm{eucl}} \oplus \mathfrak{g}_0^{+} \oplus \mathfrak{g}_0^{-}$$

of a Euclidean ideal (with $[\mathfrak{m},\mathfrak{m}] = 0$ on its $\mathfrak{m}$-part), a compact-type semisimple ideal, and a noncompact-type semisimple ideal, each θ-stable, and each of the last two a sum of irreducible orthogonal symmetric Lie algebras.

***Proof Key ideas*:**
The Killing form κ of the semisimple part of $\mathfrak{g}_0$ is nondegenerate, and θ is an isometry for $B_\theta = -\kappa(\,\cdot\,, \theta\,\cdot\,)$; the $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{m}$ makes $\mathfrak{m}$ an orthogonal $\mathrm{Ad}(K)$-module. First split off the Euclidean part: let $\mathfrak{m}_0 = \{X \in \mathfrak{m} \mid [X,\mathfrak{m}] = 0\}$ be the flat directions, on which the curvature $R(X,Y)Z = -[[X,Y],Z]$ vanishes; the ideal $\mathfrak{g}_0^{\mathrm{eucl}} = [\mathfrak{m}_0, \mathfrak{k}] \oplus \mathfrak{m}_0$ (adjusted to be a θ-stable ideal) carries $[\mathfrak{m}_0, \mathfrak{m}_0] = 0$ and is the Euclidean factor. On the complementary semisimple ideal, κ is nondegenerate on $\mathfrak{m}$, and it splits orthogonally into the subspace where κ is negative-definite and the subspace where it is positive-definite; these are $\mathrm{Ad}(K)$-invariant because $\mathrm{Ad}(K)$ preserves κ (the operators $\mathrm{ad}(\mathfrak{k})$ are κ-skew), so they are the $\mathfrak{m}$-parts of θ-stable ideals $\mathfrak{g}_0^{+}$ (compact type) and $\mathfrak{g}_0^{-}$ (noncompact type). A semisimple Lie algebra is a direct sum of simple ideals (theorem 13.3.3 and the structure theory of chapter 13), and each simple θ-stable summand gives an irreducible orthogonal symmetric Lie algebra, so $\mathfrak{g}_0^{\pm}$ are each sums of irreducibles.

On the geometric side, an orthogonal direct sum decomposition of $(\mathfrak{g}_0, \theta)$ into θ-stable ideals corresponds under theorem 15.2.3 to a Riemannian product of the associated simply connected symmetric spaces, since the invariant metric on $\mathfrak{m} = \mathfrak{m}^{\mathrm{eucl}} \oplus \mathfrak{m}^{+} \oplus \mathfrak{m}^{-}$ is the orthogonal sum of the invariant metrics on the factors and the curvature respects the splitting. Uniqueness up to order is proved on the algebra side: the flat directions and the sign of κ on $\mathfrak{m}$ cut out the three blocks canonically, the splitting of the semisimple part into simple θ-stable ideals is unique by theorem 6.3.3, and the correspondence transports the unique algebraic decomposition to the factors of M. The manifold-level counterpart, for an arbitrary presentation of a complete manifold as a product of a flat factor and holonomy-irreducible factors, is [7, Uniqueness of the

de Rham decomposition]; that the irreducible symmetric factors here are holonomy-irreducible is proved in neither book, which is why this proof runs on the algebra, completing the proof.

A general symmetric space is a product, so nothing is lost by studying the three pure types separately, and the Euclidean type is elementary: it is a flat space $\mathbb{R}^d$ or a torus, with $\mathfrak{m}$ abelian and the group G a semidirect product of $\mathbb{R}^d$ with K. The interesting geometry lives in the compact and noncompact types, and the duality of theorem 15.3.2 pairs those two, so the classification need only enumerate the irreducible compact-type pairs and then read off the noncompact ones as their duals. That enumeration is the Cartan classification.

The compact-type irreducibles come in two kinds, according to whether the transvection group is simple or a product of two copies of a simple group. Cartan's theorem names both and asserts that the list of classical and exceptional simple Lie algebras of chapter 13 controls everything.

Theorem 15.3.6: The Cartan Classification of Symmetric Spaces

Every irreducible orthogonal symmetric Lie algebra of compact type is of exactly one of the following two kinds, and every irreducible one of noncompact type is the dual (definition 15.3.1) of one of these.

1. *Type I.* $(\mathfrak{g}_0, \theta)$ with $\mathfrak{g}_0 = \mathfrak{u}$ a compact simple Lie algebra and θ an involutive automorphism of $\mathfrak{u}$; then $\mathfrak{k}$ is the fixed-point subalgebra $\mathfrak{u}^\theta$ and $M = U/K$ with U the compact simple group. The classical type I compact symmetric spaces are the families

$$\mathrm{SU}(n)/\mathrm{SO}(n), \quad \mathrm{SU}(2n)/\mathrm{Sp}(n), \quad \mathrm{SU}(p+q)/\mathrm{S}(\mathrm{U}(p)\times\mathrm{U}(q)),$$

$$\mathrm{SO}(p+q)/(\mathrm{SO}(p)\times\mathrm{SO}(q)), \quad \mathrm{Sp}(n)/\mathrm{U}(n), \quad \mathrm{Sp}(p+q)/(\mathrm{Sp}(p)\times\mathrm{Sp}(q))$$

   together with the exceptional spaces attached to the involutions of $\mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8, \mathfrak{f}_4, \mathfrak{g}_2$, in Cartan's labels EI through EIX, FI, FII, and G.

2. *Type II.* $(\mathfrak{g}\oplus\mathfrak{g}, \sigma)$ with $\mathfrak{g}$ a compact simple Lie algebra and $\sigma(X,Y) = (Y,X)$ the swap; then $\mathfrak{k} = \{(X,X)\} \cong \mathfrak{g}$ is the diagonal and $\mathfrak{m} = \{(X,-X)\}$, and the symmetric space is the compact simple group G itself, $M = (G\times G)/\Delta G \cong G$, with its bi-invariant metric.

The noncompact irreducible symmetric spaces are exactly the duals: for type II the dual is $G_{\mathbb{C}}/G$, the complexification modulo its compact form, and for type I the dual is the noncompact symmetric space G^*/K attached to the noncompact real form determined by θ.

***Proof Key ideas*:**
The full case-by-case enumeration is combinatorial and is deferred to [10]; the structural reduction, which is the content worth exhibiting, runs as follows.

Reduction to real simple Lie algebras and involutions. By proposition 15.3.5 an irreducible compact-type pair $(\mathfrak{g}_0, \theta)$ has $\mathfrak{g}_0$ semisimple with $\kappa|_{\mathfrak{m}} < 0$, so $\mathfrak{g}_0$ is a compact semisimple Lie algebra, and irreducibility forces $\mathfrak{g}_0$ to be either simple or a sum of two isomorphic simple ideals swapped by θ. If $\mathfrak{g}_0 = \mathfrak{u}$ is simple, θ is an involutive automorphism of $\mathfrak{u}$ and the pair is type I; if $\mathfrak{g}_0 = \mathfrak{g}\oplus\mathfrak{g}$ with θ swapping the factors, the pair is type II. The type II case collapses immediately: $\mathfrak{k} = \Delta\mathfrak{g}$ is the diagonal and $M = (G\times G)/\Delta G \cong G$ as computed in the statement, so type II is exactly the compact simple groups viewed as symmetric spaces. Type I therefore reduces to the classification of involutive automorphisms of a compact simple Lie algebra $\mathfrak{u}$ up to conjugacy.

Involutions match noncompact real forms. An involution θ of a compact simple $\mathfrak{u}$ is the same datum as a Cartan involution of a noncompact real form of $\mathfrak{u}_{\mathbb{C}}$: given θ, the algebra $\mathfrak{u}^{\theta} \oplus i\mathfrak{u}^{-\theta}$ (fixed part plus i times anti-fixed part) is a noncompact real form of $\mathfrak{u}_{\mathbb{C}}$ with Cartan decomposition $\mathfrak{k} \oplus \mathfrak{p}$, exactly the dual of theorem 15.3.2; conversely each noncompact real form furnishes such a θ. This is the correspondence proved in chapter 13 between real forms, Cartan involutions, and the compact form. So the type I compact symmetric spaces are in bijection with the conjugacy classes of involutions of $\mathfrak{u}$, equivalently with the noncompact real forms of $\mathfrak{u}_{\mathbb{C}}$, equivalently with the Vogan/Satake diagrams that decorate the Dynkin diagram of $\mathfrak{u}_{\mathbb{C}}$ with the involution data.

The finite list. Running through the simple Dynkin diagrams $A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, G_2$ of chapter 13 and their involutions produces the finite list. For A_n ($\mathfrak{u} = \mathfrak{su}(n+1)$) the involutions give $\mathrm{SU}(n+1)/\mathrm{SO}(n+1)$, $\mathrm{SU}(2m)/\mathrm{Sp}(m)$, and $\mathrm{SU}(p+q)/\mathrm{S}(\mathrm{U}(p) \times \mathrm{U}(q))$; for B_n, D_n ($\mathfrak{so}$) the real Grassmannians $\mathrm{SO}(p+q)/(\mathrm{SO}(p) \times \mathrm{SO}(q))$; for C_n ($\mathfrak{sp}$) the spaces $\mathrm{Sp}(n)/\mathrm{U}(n)$ and $\mathrm{Sp}(p+q)/(\mathrm{Sp}(p) \times \mathrm{Sp}(q))$; and the exceptional diagrams yield the finitely many exceptional symmetric spaces EI through G. The verification that this list is complete and irredundant is deferred to [10]. Applying the duality of theorem 15.3.2 to every entry produces the noncompact irreducibles, completing the classification.

The classification completes the structural program of the book. The Cartan-Killing classification of chapter 13 enumerated the simple Lie algebras by Dynkin diagram; the real forms were then classified by the involutions decorating those diagrams; and now the symmetric spaces are classified by the same data read once more, this time as the tangent geometry of a homogeneous space. A symmetric space is determined by a compact simple Lie algebra and an involution, or equivalently by a real form of a complex simple Lie algebra, so the entire geometry, the curvature, the rank, the duality, is encoded in a decorated Dynkin diagram. Type II identifies a compact Lie group with a symmetric space, so the compact-group theory of chapter 11 and chapter 12 is subsumed as a special case, and type I plus the duality accounts for everything else. The geometry of a symmetric space is thus entirely determined by a real form and an involution, closing the arc from a Lie group, through its Lie algebra, to its geometry.

The two kinds are visible on the anchors already computed. The half-plane and the sphere illustrate type I: $S^2 = \mathrm{SU}(2)/\mathrm{SO}(2)$ arises from the involution $\theta(X) = \bar{X}$ of $\mathfrak{su}(2)$ (fixed subalgebra $\mathfrak{so}(2)$), and $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ is its noncompact dual, the two paired by example 15.3.3. The compact group $\mathrm{SU}(2)$ itself, viewed as $(\mathrm{SU}(2) \times \mathrm{SU}(2))/\Delta\mathrm{SU}(2) \cong S^3$, illustrates type II: its dual is $\mathrm{SL}(2,\mathbb{C})/\mathrm{SU}(2) = H^3$, hyperbolic 3-space. The classification says every irreducible symmetric space is one of these two shapes.

Exercise 15.3.1

1. Let $(\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{m}, \theta)$ be an orthogonal symmetric Lie algebra. Verify in full that $\mathfrak{g}_0^* = \mathfrak{k} \oplus i\mathfrak{m} \subseteq \mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra by checking the three brackets $[\mathfrak{k},\mathfrak{k}]$, $[\mathfrak{k}, i\mathfrak{m}]$, $[i\mathfrak{m}, i\mathfrak{m}]$, and identify precisely where the sign flip $[i\mathfrak{m}, i\mathfrak{m}] = -[\mathfrak{m},\mathfrak{m}]$ occurs. Explain why the flip forces the third bracket to remain inside $\mathfrak{k}$.

2. Prove that the Killing forms satisfy $\kappa^*(iX, iY) = -\kappa(X, Y)$ for $X, Y \in \mathfrak{m}$, using $\mathbb{C}$-bilinearity of the Killing form of $\mathfrak{g}_{\mathbb{C}}$. Deduce that compact type and noncompact type are exchanged by duality, and that κ is unchanged on the shared $\mathfrak{k}$.

3. Show that $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ and $S^2 = \mathrm{SU}(2)/\mathrm{SO}(2)$ are dual symmetric spaces. Starting from $\mathfrak{k} = \mathfrak{so}(2)$ and $\mathfrak{m} = \mathrm{Sym}_0(2)$, verify that $\mathfrak{so}(2) \oplus i\mathrm{Sym}_0(2) \cong \mathfrak{su}(2)$, that the bracket on $\mathfrak{m}$ flips sign, and that the curvature passes from -1 to $+1$.

4. Classify the rank-one symmetric spaces. Show that the constant-curvature spaces S^n (compact, $K = +1$), $\mathbb{R}^n$ (flat), and H^n (noncompact, $K = -1$) are rank one, and that the projective spaces $\mathbb{C}\mathrm{P}^n$, $\mathbb{H}\mathbb{P}^n$, and the Cayley plane are the remaining compact rank-one type I spaces. State the duality partner of each compact one.

5. Identify the type (I or II, compact or noncompact) of each of the following: $\mathrm{SU}(n)/\mathrm{SO}(n)$; a compact simple Lie group G regarded as a symmetric space; $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$; the complexification $\mathrm{SL}(n,\mathbb{C})/\mathrm{SU}(n)$.

6. Confirm that the curvature sign matches the type in each case of the previous question. For $\mathrm{SU}(n)/\mathrm{SO}(n)$ show $\kappa|_{\mathfrak{m}} < 0$ and $K \geq 0$; for $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$ show $\kappa|_{\mathfrak{m}} > 0$ and $K \leq 0$; and explain why type II compact groups have $K \geq 0$ while their duals $G_{\mathbb{C}}/G$ have $K \leq 0$.

15.4 Hyperbolic Space, Spheres, Grassmannians, and the Siegel Upper Half-Space

This section runs the classical families through the machine of section 15.2 and section 15.3. Each example is a real form from chapter 13 together with an involution, presented as a homogeneous space G/K with its curvature and its dual read off the bracket. The running thread $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ reappears twice over, first as the hyperbolic plane H^2 and finally as the genus-one Siegel space, tying the taxonomy back to a single object the reader already knows in full.

The first family is the constant-curvature pair: hyperbolic space and its dual sphere, the rank-one spaces where a single number, the curvature, determines the geometry.

Example 15.4.1: Hyperbolic Space and the Round Sphere
Fix $n \geq 2$. The *sphere* and *hyperbolic space* form a dual pair of rank-one symmetric spaces with the same isotropy group $\mathrm{SO}(n)$.

The sphere. The round sphere is the compact-type space

$$S^n = \mathrm{SO}(n+1)/\mathrm{SO}(n)$$

with $G = \mathrm{SO}(n+1)$ acting by rotations and $K = \mathrm{SO}(n)$ the stabilizer of the base point $o = (0,\dots,0,1)$. The Cartan decomposition of $\mathfrak{so}(n+1)$ for the involution $\theta(X) = JXJ$ with $J = \mathrm{diag}(I_n, -1)$ is

$$\mathfrak{so}(n+1) = \mathfrak{so}(n) \oplus \mathfrak{m}, \qquad \mathfrak{m} = \left\{ \begin{pmatrix} 0 & v \\ -v^{\mathsf{T}} & 0 \end{pmatrix} \;\middle|\; v \in \mathbb{R}^n \right\}$$

where $\mathfrak{so}(n)$ sits in the upper-left block. The subspace $\mathfrak{m} \cong \mathbb{R}^n$ is the tangent space T_oS^n, and the bracket $[\mathfrak{m},\mathfrak{m}] \subseteq \mathfrak{so}(n)$ is nonzero, so by proposition 15.2.4 the sectional curvature is positive; normalizing the invariant metric makes it constantly $+1$.

Hyperbolic space. The noncompact-type dual replaces $\mathfrak{m}$ by $i\mathfrak{m}$ inside the complexification, which at the level of real forms exchanges $\mathfrak{so}(n+1)$ for the indefinite orthogonal algebra $\mathfrak{so}(n,1)$. Concretely,

$$H^n = \mathrm{SO}(n,1)^\circ/\mathrm{SO}(n)$$

is the identity component of the group of isometries of the Lorentzian form $x_1^2 + \cdots + x_n^2 - x_{n+1}^2$ acting on the upper sheet of the hyperboloid, with $K = \mathrm{SO}(n)$ again the isotropy. Its Cartan

decomposition is

$$\mathfrak{so}(n,1) = \mathfrak{so}(n) \oplus i\mathfrak{m}, \qquad i\mathfrak{m} = \left\{ \begin{pmatrix} 0 & v \\ v^{\mathsf{T}} & 0 \end{pmatrix} \;\middle|\; v \in \mathbb{R}^n \right\}$$

differing from the sphere only in the sign of the transposed block, which is precisely the sign flip $\mathfrak{m} \mapsto i\mathfrak{m}$ of definition 15.3.1. The bracket $[\,i\mathfrak{m}, i\mathfrak{m}\,]$ carries the opposite sign, so the curvature is constantly -1 by theorem 15.3.2.

Rank and the running example. Both spaces have real rank one: a maximal abelian subspace of $\mathfrak{m}$ (or of $i\mathfrak{m}$) is a single line, since any two of the displayed matrices with linearly independent v fail to commute unless one is a multiple of the other. The $n = 2$ case is the hyperbolic plane

$$H^2 = \mathrm{SO}(2,1)^\circ/\mathrm{SO}(2) = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2) = \mathfrak{H}$$

reconciling the two descriptions of the hyperbolic plane: the hyperboloid model $\mathrm{SO}(2,1)^\circ/\mathrm{SO}(2)$ and the upper-half-plane model $\mathfrak{H}$ of example 13.5.3. The identification is the exceptional isomorphism $\mathrm{SO}(2,1)^\circ \cong \mathrm{PSL}(2,\mathbb{R})$, under which the Lorentz action on the hyperboloid becomes the Möbius action on $\mathfrak{H}$, and the constant curvature -1 matches the value computed directly for $\mathfrak{H}$ in section 15.2.

The pair S^n and H^n is the model of duality at its simplest: same K, same rank, curvatures $+1$ and -1, differing only in the sign carried by the $\mathfrak{m}$-bracket. A subspace of $\mathfrak{m}$ that is one-dimensional cannot record more than an overall scale, which is why rank one forces constant curvature. Everything the reader has learned about $\mathfrak{H}$ transfers: geodesics are the orbits of the abelian factor A from the Iwasawa decomposition $\mathrm{SL}(2,\mathbb{R}) = KAN$ (example 14.4.1), and the identity component of the isometry group is $\mathrm{SO}(n,1)^\circ$ for H^n, in agreement with the Myers-Steenrod description of section 15.1. The moment the tangent space acquires a second independent flat direction, curvature stops being a single number, and the geometry becomes higher-rank. The Grassmannians are the first family where this happens.

Example 15.4.2: Real, Complex, and Quaternionic Grassmannians

Fix $0 < k < n$. The *real Grassmannian* $\mathrm{Gr}_k(\mathbb{R}^n)$ of k-dimensional subspaces of $\mathbb{R}^n$ is the compact-type symmetric space

$$\mathrm{Gr}_k(\mathbb{R}^n) = \mathrm{SO}(n)/\big(\mathrm{SO}(k) \times \mathrm{SO}(n-k)\big)$$

with $\mathrm{SO}(n)$ acting transitively on k-planes and the isotropy $\mathrm{SO}(k) \times \mathrm{SO}(n-k)$ fixing the splitting $\mathbb{R}^n = \mathbb{R}^k \oplus \mathbb{R}^{n-k}$ at the base point. The involution is $\theta(X) = JXJ$ with $J = \mathrm{diag}(I_k, -I_{n-k})$, whose fixed algebra is $\mathfrak{k} = \mathfrak{so}(k) \oplus \mathfrak{so}(n-k)$ and whose -1-eigenspace is

$$\mathfrak{m} = \left\{ \begin{pmatrix} 0 & B \\ -B^{\mathsf{T}} & 0 \end{pmatrix} \;\middle|\; B \in \mathbb{R}^{k \times (n-k)} \right\}$$

the $k \times (n-k)$ block of off-diagonal rotations, so $\dim \mathrm{Gr}_k(\mathbb{R}^n) = k(n-k)$.

The noncompact dual. Replacing $\mathfrak{m}$ by $i\mathfrak{m}$ passes to the indefinite form and gives

$$\mathrm{SO}(k, n-k)^\circ/\big(\mathrm{SO}(k) \times \mathrm{SO}(n-k)\big)$$

the noncompact-type dual, whose points are the positive-definite k-planes for the form of signature $(k, n-k)$; its curvature is nonpositive where the Grassmannian's is nonnegative.

Rank. A maximal abelian subspace of $\mathfrak{m}$ is obtained by taking B to run over the diagonal $k\times(n-k)$ matrices, and the largest such diagonal has $\min(k,n-k)$ independent entries. Hence

$$\operatorname{rank}\operatorname{Gr}_k(\mathbb{R}^n)=\min(k,n-k)$$

The rank counts the independent principal angles between two k-planes: two subspaces in general position meet at $\min(k,n-k)$ distinct angles, and these angles are exactly the coordinates on a maximal flat. When $k=1$ the rank is one and $\operatorname{Gr}_1(\mathbb{R}^n)=\mathbb{R}\mathrm{P}^{n-1}$ recovers a constant-curvature space; for $2\le k\le n-2$ the rank exceeds one and the Grassmannian is a higher-rank symmetric space, the first such family in the classification.

Complex and quaternionic analogues. Replacing the real ground field throughout gives the two companion families. The complex Grassmannian is

$$\operatorname{Gr}_k(\mathbb{C}^n)=\mathrm{SU}(n)/\mathrm{S}\big(\mathrm{U}(k)\times\mathrm{U}(n-k)\big)$$

with noncompact dual $\mathrm{SU}(k,n-k)/\mathrm{S}(\mathrm{U}(k)\times\mathrm{U}(n-k))$, and the quaternionic Grassmannian is $\operatorname{Gr}_k(\mathbb{H}^n)=\mathrm{Sp}(n)/\big(\mathrm{Sp}(k)\times\mathrm{Sp}(n-k)\big)$ with dual $\mathrm{Sp}(k,n-k)/\big(\mathrm{Sp}(k)\times\mathrm{Sp}(n-k)\big)$. All three have rank $\min(k,n-k)$; the complex family is Hermitian symmetric, a point taken up in proposition 15.4.4.

The Grassmannians exhibit rank as a geometric invariant. Where the sphere and hyperbolic space could pack their entire geometry into one curvature number, a Grassmannian of rank r carries an r-dimensional flat totally geodesic torus (compact type) or Euclidean space (noncompact type) through each point, and the r principal angles between subspaces are the coordinates on it. The real rank of chapter 14 and the rank of the symmetric space are the same integer: $\min(k,n-k)$ is exactly the dimension of a maximal abelian subspace of $\mathfrak{m}$, which is a maximal abelian subspace of the $\mathfrak{p}$ of section 14.1 for the noncompact dual. Grassmannians are thus the family that shows higher rank before the general theory requires it.

The last family carries an extra structure the previous ones lack: an invariant complex structure. A symmetric space whose tangent space is a complex vector space, compatibly with the metric and the curvature, is called Hermitian symmetric, and the Siegel upper half-space is the archetype.

Example 15.4.3: The Siegel Upper Half-Space

Fix $g\ge 1$. The *Siegel upper half-space* of genus g is

$$\mathfrak{H}_g=\big\{Z=X+iY\in\operatorname{Sym}_g(\mathbb{C})\;\big|\;Y=\operatorname{Im}Z\text{ positive definite}\big\}$$

the symmetric complex $g\times g$ matrices with positive-definite imaginary part. The real symplectic group $\mathrm{Sp}(2g,\mathbb{R})$ acts on $\mathfrak{H}_g$ by the generalized Möbius transformations

$$\begin{pmatrix}A&B\\C&D\end{pmatrix}\cdot Z=(AZ+B)(CZ+D)^{-1}$$

and the action is transitive with the stabilizer of the base point $Z=iI_g$ equal to the unitary group $\mathrm{U}(g)$, embedded in $\mathrm{Sp}(2g,\mathbb{R})$ as $\big\{\left(\begin{smallmatrix}A&-B\\B&A\end{smallmatrix}\right)\;\big|\;A+iB\in\mathrm{U}(g)\big\}$. Hence

$$\mathfrak{H}_g=\mathrm{Sp}(2g,\mathbb{R})/\mathrm{U}(g)$$

a noncompact-type symmetric space of dimension $g(g+1)$ over $\mathbb{R}$, complex dimension $g(g+1)/2$. Its compact dual is $\mathrm{Sp}(g)/\mathrm{U}(g)$, the space of compatible complex structures on a symplectic vector space, obtained by the substitution $\mathfrak{m}\mapsto i\mathfrak{m}$ as before.

The genus-one case. For $g = 1$ a symmetric 1×1 complex matrix is a single complex number, and positive-definite imaginary part means positive imaginary part, so

$$\mathfrak{H}_1 = \{z \in \mathbb{C} \mid \mathrm{Im} z > 0\} = \mathfrak{H}$$

is the ordinary upper half-plane. The group is $\mathrm{Sp}(2, \mathbb{R}) = \mathrm{SL}(2, \mathbb{R})$, since a 2×2 matrix is symplectic exactly when it has determinant one, and $\mathrm{U}(1) = \mathrm{SO}(2)$ under the embedding above; the coset space is $\mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2) = \mathfrak{H}$ of example 13.5.3, and the Möbius formula reduces to the classical $z \mapsto (az + b)/(cz + d)$. The running example is thus the $g = 1$ member of the Siegel family.

The Siegel space carries a feature invisible in the previous examples: because its base point stabilizer $\mathrm{U}(g)$ has a positive-dimensional center, the tangent space inherits a complex structure that the isotropy preserves. This is the Hermitian symmetric condition, and the center of K is exactly the diagnostic for it.

Proposition 15.4.4: Hermitian Symmetric Spaces and the Center of the Isotropy

Let $M = G/K$ be an irreducible symmetric space of compact or noncompact type with $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{m} \cong T_o M$. Then M is *Hermitian symmetric*, that is, it carries a G-invariant complex structure $J \colon \mathfrak{m} \to \mathfrak{m}$ with $J^2 = -\mathrm{id}$, orthogonal and parallel, if and only if the center of K is one-dimensional. In that case $J = \mathrm{ad}(Z)$ for a suitable central element Z of $\mathfrak{k}$, the generator of the central circle in K. The Siegel space $\mathfrak{H}_g = \mathrm{Sp}(2g, \mathbb{R})/\mathrm{U}(g)$ and the complex Grassmannian $\mathrm{Gr}_k(\mathbb{C}^n) = \mathrm{SU}(n)/\mathrm{S}(\mathrm{U}(k) \times \mathrm{U}(n-k))$ are Hermitian symmetric; the real Grassmannian $\mathrm{Gr}_k(\mathbb{R}^n)$ with $3 \le k \le n-3$ is not.

***Proof*:**
A G-invariant almost complex structure on $M = G/K$ is the same as an $\mathrm{Ad}(K)$-equivariant endomorphism $J \colon \mathfrak{m} \to \mathfrak{m}$ with $J^2 = -\mathrm{id}$: invariance under G reduces to invariance at the base point, where J must commute with the isotropy representation $\mathrm{Ad}|_K$ on $\mathfrak{m}$. On an irreducible symmetric space this isotropy representation is a real irreducible representation of K, so by Schur's lemma its commutant is one of the three real division algebras $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$. A J with $J^2 = -\mathrm{id}$ is a square root of $-\mathrm{id}$ in this commutant, and the three cases are different. When the commutant is $\mathbb{R}$ there is no such J at all, and M admits no invariant complex structure. When it is $\mathbb{H}$ there is a whole 2-sphere of square roots of $-\mathrm{id}$ (the imaginary quaternions of unit norm), so J exists but is far from unique; such an M is not Hermitian but *quaternion-Kähler*, and the quaternionic Grassmannian $\mathrm{Gr}_k(\mathbb{H}^n)$ of example 15.4.2 is the model case. The Hermitian case is the middle one: M is Hermitian precisely when the commutant is *exactly* $\mathbb{C}$, neither $\mathbb{R}$ nor $\mathbb{H}$, for then $\pm J$ are the only square roots of $-\mathrm{id}$ and the invariant complex structure is unique up to conjugation. That copy of $\mathbb{C}$ is given by a one-dimensional central subalgebra of $\mathfrak{k}$, because a central element $Z \in \mathfrak{z}(\mathfrak{k})$ acts on $\mathfrak{m}$ through $\mathrm{ad}(Z)$, which commutes with all of $\mathrm{Ad}(K)$ and can be normalized to satisfy $\mathrm{ad}(Z)^2 = -\mathrm{id}$. Conversely, a commutant strictly larger than $\mathbb{R}$ that is not the full $\mathbb{H}$ is exactly $\mathbb{C}$, and its skew part is one-dimensional and central; the $\mathbb{H}$ case has $\mathfrak{z}(\mathfrak{k})$ still zero-dimensional, since the quaternionic isotropy $\mathrm{Sp}(k) \times \mathrm{Sp}(n-k)$ is semisimple with finite center. Thus M is Hermitian if and only if $\mathfrak{z}(\mathfrak{k})$ is one-dimensional, equivalently the center of K is one-dimensional, and $J = \mathrm{ad}(Z)$ for the normalized generator Z; the criterion correctly excludes the quaternion-Kähler spaces, whose center is finite. Any J so obtained is orthogonal, being ad of an element that acts skew-symmetrically for the $\mathrm{Ad}(K)$-invariant inner product, and

it is parallel by [7, Invariant tensor fields on a symmetric space are parallel]: the G-invariant J is invariant under every transvection, since G contains them all (along geodesics through o they are the one-parameter groups $\exp(tX)$ of proposition 15.2.4, and their G-conjugates cover the rest). The remaining step, that a parallel almost complex structure is integrable, making J a genuine complex structure, is proved in neither this book nor [7] and is taken as given here.

For $\mathfrak{H}_g$ the isotropy is $K = \mathrm{U}(g)$, whose center is the circle of scalar unitaries $\{e^{i\varphi} I_g \mid \varphi \in \mathbb{R}\}$, one-dimensional, so $\mathfrak{H}_g$ is Hermitian; the induced J is the multiplication by i on the tangent space $\mathrm{Sym}_g(\mathbb{C})$ of symmetric complex matrices. For the complex Grassmannian the isotropy $\mathrm{S}(\mathrm{U}(k) \times \mathrm{U}(n-k))$ likewise has a one-dimensional center, the diagonal circle acting by opposite phases on the two blocks, so it too is Hermitian. For the real Grassmannian with $3 \le k \le n-3$ both factors $\mathrm{SO}(k)$ and $\mathrm{SO}(n-k)$ have $k, n-k \ge 3$, so neither is $\mathrm{SO}(2)$ and each has finite center; the center of $\mathrm{SO}(k) \times \mathrm{SO}(n-k)$ is then finite, of dimension zero, so no invariant complex structure exists, completing the proof. The excluded boundary $k = 2$ (or $n - k = 2$) is different: there an $\mathrm{SO}(2)$ factor contributes a one-dimensional center and the real Grassmannian $\mathrm{Gr}_2(\mathbb{R}^n)$ becomes the Hermitian complex quadric.

The Hermitian symmetric spaces of noncompact type are the *bounded symmetric domains*: each embeds as a bounded open subset of a complex vector space, holomorphically and equivariantly, generalizing the realization of $\mathfrak{H}$ as the unit disk by the Cayley transform. The Siegel space $\mathfrak{H}_g$ is the bounded symmetric domain of the symplectic family, and it is the base of an arithmetic geometry. Holomorphic functions on $\mathfrak{H}_g$ transforming suitably under an arithmetic subgroup of $\mathrm{Sp}(2g, \mathbb{R})$ are the *Siegel modular forms*, the genus-g generalization of the classical modular forms on $\mathfrak{H}$; and the quotient of $\mathfrak{H}_g$ by the integral symplectic group $\mathrm{Sp}(2g, \mathbb{Z})$ is the moduli space of principally polarized abelian varieties of dimension g, each point recording a complex torus $\mathbb{C}^g/(\mathbb{Z}^g + Z\mathbb{Z}^g)$ with its polarization. The genus-one instance $\mathfrak{H}_1 = \mathfrak{H}$ is the upper half-plane whose quotient parametrizes elliptic curves, the running example seen from the top of the tower.

These three families exhibit the entire taxonomy of section 15.3. The sphere and hyperbolic space are the rank-one constant-curvature spaces at the extremes of the trichotomy $M_0 \times M_+ \times M_-$ of proposition 15.3.5, one of positive and one of negative curvature. The Grassmannians are the first higher-rank family, with rank $\min(k, n-k)$ measuring the dimension of a maximal flat and the number of independent angles between subspaces. The Siegel space and the complex Grassmannians are the Hermitian symmetric domains, distinguished by a one-dimensional center of the isotropy K and carrying an invariant complex structure. Each is a real form of a complex semisimple Lie algebra together with an involution, in the sense of chapter 13, and each comes paired with a compact dual through the single algebraic operation $\mathfrak{m} \mapsto i\mathfrak{m}$. The Cartan classification of theorem 15.3.6 is nothing but the statement that these families, together with the exceptional ones, exhaust the irreducible possibilities.

Exercise 15.4.1

1. Write down the Cartan decomposition $\mathfrak{so}(n,1) = \mathfrak{so}(n) \oplus \mathfrak{m}$ for hyperbolic space $H^n = \mathrm{SO}(n,1)^\circ/\mathrm{SO}(n)$. Identify $\mathfrak{m}$ explicitly as a space of matrices, verify that $\mathfrak{m} \cong T_o H^n \cong \mathbb{R}^n$ as a vector space, and check that $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{so}(n)$. Contrast the sign of the transposed block with the compact dual $\mathfrak{so}(n+1) = \mathfrak{so}(n) \oplus \mathfrak{m}'$.

2. Compute the rank of $\mathrm{Gr}_2(\mathbb{R}^4) = \mathrm{SO}(4)/(\mathrm{SO}(2) \times \mathrm{SO}(2))$. Exhibit a maximal abelian subspace of $\mathfrak{m}$ explicitly using diagonal blocks, and confirm that rank $= \min(2,2) = 2$, so that $\mathrm{Gr}_2(\mathbb{R}^4)$ is a higher-rank symmetric space. What is its dimension?

3. Show that the genus-one Siegel upper half-space $\mathfrak{H}_1$ equals the ordinary upper half-plane $\mathfrak{H}$. Verify that a symmetric 1×1 complex matrix with positive-definite imaginary part is a complex number with positive imaginary part, and that the Möbius formula $Z \mapsto (AZ+B)(CZ+D)^{-1}$ reduces to $z \mapsto (az+b)/(cz+d)$.

4. Prove that $\mathrm{Sp}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})$. Show that a 2×2 real matrix M preserves the standard symplectic form $\omega(u,v) = u^{\mathsf{T}}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)v$ if and only if $\det M = 1$, and deduce that $\mathrm{U}(1) = \mathrm{SO}(2)$ under the embedding $A + iB \mapsto \left(\begin{smallmatrix} A & -B \\ B & A \end{smallmatrix}\right)$.

5. Using proposition 15.4.4, determine which of the following symmetric spaces are Hermitian by inspecting the center of the isotropy group K: (a) $S^n = \mathrm{SO}(n+1)/\mathrm{SO}(n)$; (b) $\mathrm{Gr}_k(\mathbb{C}^n) = \mathrm{SU}(n)/\mathrm{S}(\mathrm{U}(k)\times\mathrm{U}(n-k))$; (c) $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$ for $n \geq 3$; (d) $\mathfrak{H}_g = \mathrm{Sp}(2g,\mathbb{R})/\mathrm{U}(g)$. Justify each answer by identifying the center of K.

6. For the dual pair $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ and $H^2 = \mathrm{SO}(2,1)^\circ/\mathrm{SO}(2)$, verify directly from the curvature formula $R(X,Y)Z = -[[X,Y],Z]$ of proposition 15.2.4 that the sectional curvatures have opposite signs. Take a basis of $\mathfrak{m}$ and compute the single bracket $[[X,Y],Y]$ in each case, exhibiting the sign flip $\mathfrak{m} \mapsto i\mathfrak{m}$.

15.5 Locally Symmetric Spaces and Where the Structure Leads

The correspondence of section 15.2 and the duality of section 15.3 built the globally symmetric spaces: those on which each geodesic symmetry s_p extends to a global isometry, so that $M = G/K$ is a homogeneous space of its own isometry group. The last move of the chapter is to relax the global condition to a local one and pass to discrete quotients. The correspondence delivered the simply connected models G/K of each type; the compact examples with interesting topology, and the arithmetic examples that carry number theory, are quotients $\Gamma\backslash G/K$ of those models by a discrete group. This section records that passage, then closes the main text by reading the book back through the object it has arrived at.

This is a signpost, not a development: the geometry of the quotients and the harmonic analysis of functions on them belong to the sequel volumes named below, and the section states only what the structure built here becomes on the far side of that passage.

A globally symmetric space is symmetric at every point in the strongest sense. Weakening the requirement so that the geodesic symmetry s_p need only be an isometry of a neighborhood of p, not of all of M, produces a strictly larger class, the one that discrete quotients of symmetric spaces populate. The differential-geometric characterization is a single tensor equation, and it is what makes the class recognizable in practice.

Proposition 15.5.1: Locally Symmetric Spaces

Let M be a connected complete Riemannian manifold, with Levi-Civita connection ∇ and Riemann curvature tensor R ([7]). The following are equivalent.

1. The curvature is parallel: $\nabla R = 0$.
2. Each geodesic symmetry s_p (the map reversing geodesics through p, $\exp_p(v) \mapsto \exp_p(-v)$) is a local isometry at p.
3. The universal cover $\widetilde{M}$ is a globally symmetric space G/K, and $M = \Gamma\backslash G/K$ for a discrete group $\Gamma \subseteq \mathrm{Isom}(\widetilde{M})$ of isometries acting freely and properly discontinuously.

A manifold satisfying these conditions is a *locally symmetric space*.

***Proof Key ideas*:**
The equivalence of (1) and (2) is the local statement of the symmetric-space condition and is proved on the Riemannian side in [7, Geodesic symmetries and parallel curvature]. The geodesic symmetry s_p fixes p and has differential $-\mathrm{id}$ at p; by the naturality of the exponential map it is a local isometry exactly when it preserves the curvature tensor, and since $d(s_p)_p = -\mathrm{id}$ acts on the $(1,4)$-tensor ∇R by the sign $(-1)^5 = -1$ while an isometry must preserve it, the two forces balance if and only if ∇R vanishes at p. Running p over M gives $\nabla R = 0$ throughout, and conversely $\nabla R = 0$ makes each s_p curvature-preserving, hence a local isometry by [7, Cartan's local isometry theorem], applied to the locally symmetric M with both manifolds equal to M and $F = -\mathrm{id}$ on T_pM.

For (2) $\Rightarrow$ (3), completeness and $\nabla R = 0$ let the local symmetries be continued along paths: the universal cover $\widetilde{M}$ is complete, simply connected, and locally symmetric, and a simply connected complete locally symmetric space is globally symmetric, since the local isometries s_p extend to global ones by analytic continuation along the simply connected $\widetilde{M}$. Thus $\widetilde{M} = G/K$ is globally symmetric with $G = \mathrm{Isom}(\widetilde{M})^\circ$ (theorem 15.1.4, proposition 15.1.5). The deck group $\Gamma = \pi_1(M)$ acts on $\widetilde{M}$ by isometries, freely and properly discontinuously because it is a deck group, and $M = \Gamma\backslash\widetilde{M} = \Gamma\backslash G/K$. For (3) $\Rightarrow$ (1), the condition $\nabla R = 0$ is local and passes to and from a covering: G/K is globally symmetric, hence locally symmetric with $\nabla R = 0$ (section 15.2), and $\nabla R = 0$ is preserved by the local isometry $\widetilde{M} \to M$, completing the proof.

A globally symmetric space is thus the simply connected model, and every locally symmetric space is one of its discrete quotients. The tensor equation $\nabla R = 0$ is the working definition: it is checkable on any given manifold without reference to a group, and it says that the curvature looks the same from every point, transported parallelly. The passage to $\Gamma\backslash G/K$ is where the subject meets number theory. When G is the real points of a semisimple algebraic group defined over $\mathbb{Q}$ and Γ is an arithmetic subgroup, the quotient $\Gamma\backslash G/K$ is an *arithmetic manifold*, and its geometry, volume, and spectrum encode arithmetic invariants of Γ.

The smallest arithmetic example is the running example of the chapter, quotiented by its integer points.

Example 15.5.2: The Modular Curve $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$

Take $G/K = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2) = \mathfrak{H}$, the upper half-plane with its curvature -1 metric (example 13.5.3, example 15.4.1), and let $\Gamma = \mathrm{SL}(2,\mathbb{Z})$ act by fractional linear transformations $z \mapsto \dfrac{az+b}{cz+d}$. The action is properly discontinuous, so the quotient

$$Y(1) = \mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$$

is a complete locally symmetric space in the extended, orbifold sense, of constant curvature -1 away from its two special points. It is not quite a manifold: the elliptic points i (fixed by $z \mapsto -1/z$, of order 2) and $\rho = e^{2\pi i/3}$ (of order 3) have nontrivial finite stabilizers in Γ, so $Y(1)$ is a hyperbolic *orbifold* rather than a smooth surface, with cone singularities of angles π and $2\pi/3$. Away from these points Γ acts freely and $Y(1)$ is a genuine hyperbolic surface.

The quotient has *finite* hyperbolic area, computed from the standard fundamental domain

$$\mathcal{F} = \left\{ z \in \mathfrak{H} \;\middle|\; |z| \geq 1,\ |\mathrm{Re}\, z| \leq \tfrac{1}{2} \right\}$$

as $\mathrm{area}(\mathcal{F}) = \displaystyle\int_{\mathcal{F}} \frac{dx\,dy}{y^2} = \frac{\pi}{3}$, so $Y(1)$ is a finite-volume hyperbolic orbifold. It is noncompact: the region $\mathrm{Im}\, z \to \infty$ is a *cusp*, a single end of finite area but infinite diameter, the quotient of a horoball by the parabolic subgroup generated by $z \mapsto z+1$. Compactifying by adding one point at the cusp gives $X(1) = \overline{\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}}$, topologically a sphere, the coarse moduli space of elliptic curves over $\mathbb{C}$.

The modular curve is the point at which the four Parts of this book converge, and each of them appears in it. The symmetric space $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ is the geometry of chapter 13 and this chapter: a globally symmetric space of noncompact type, curvature -1, dual to the sphere S^2. The group $\mathrm{SL}(2,\mathbb{R})$ is the noncompact real form whose Cartan and Iwasawa decompositions organized Part IV; its maximal compact $K = \mathrm{SO}(2)$ is the running compact torus of Part III, and its Lie algebra $\mathfrak{sl}_2(\mathbb{R})$ is the smallest object of the structure theory of Part II. The discrete group $\mathrm{SL}(2,\mathbb{Z})$ is the arithmetic input, and functions on the quotient $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ are modular forms.

Beneath the four Parts run three subject-threads, and a symmetric space is their meeting point. The first is Lie theory: the group G and its algebra $\mathfrak{g}_0$, the Cartan decomposition $\mathfrak{g}_0 = \mathfrak{k} \oplus \mathfrak{p}$, and the involution θ, all of Part II and Part IV. The second is Riemannian geometry: the metric on G/K, its geodesics as orbits of one-parameter subgroups, and its curvature computed as a bracket in proposition 15.2.4, the geometry whose full development, the isometry rigidity, the comparison theorems, and the classification of spaces by curvature sign, is the subject of *Riemannian Geometry* [7]. The third is representation theory and harmonic analysis: functions on $\Gamma\backslash G/K$, decomposed under G, are automorphic forms, and this is the province of *Representation Theory of Real Reductive Groups* [6] and *Harmonic Analysis* [16]. The Cartan decomposition of chapter 13 has been read three ways in the course of the book: first as an algebraic splitting of $\mathfrak{g}_0$ into eigenspaces of θ, then as a group decomposition $G = K\exp(\mathfrak{p})$ refined by the Iwasawa product $G = KAN$ of chapter 14, and now as a geometry, the tangent data of the symmetric space G/K with curvature the bracket on $\mathfrak{m} \cong T_oM$ (chapter 13's $\mathfrak{p}$, in the notation of this chapter).

The modular curve is where this geometry carries its arithmetic. Functions on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ transforming suitably under Γ are *modular forms*, whose Fourier coefficients encode arithmetic data, the number of representations of an integer by a quadratic form, the divisor sums of Eisenstein series, the Hecke eigenvalues that match Galois representations. The Langlands program is, in one of its faces, the

assertion that the automorphic forms on arithmetic quotients $\Gamma\backslash G/K$ of symmetric spaces, for general G, correspond to representations of Galois groups, so that the harmonic analysis of $\Gamma\backslash G/K$ becomes a bridge between the analytic world of automorphic forms and the arithmetic world of number fields. The symmetric space G/K built in this chapter is the geometric stage on which that correspondence is set.

The representation theory of the symmetric spaces built here, the automorphic forms on their arithmetic quotients and the Plancherel decomposition of $L^2(\Gamma\backslash G)$, is where the sequel volumes begin. The Closing chapter, chapter 16, lays out the map of further reading, pairing each direction, the representation theory of noncompact real reductive groups, harmonic analysis on symmetric spaces, the Riemannian geometry proper, and the Langlands program, with the companion volume and the staple external text that carry it forward. The book has followed one dictionary, between a Lie group and its Lie algebra, from the matrix exponential of Part I to the geometry of G/K; the symmetric space is where that dictionary hands the reader to the next subject.

Exercise 15.5.1

1. Let G/K be a globally symmetric space and $\Gamma \subseteq G$ a discrete subgroup acting freely and properly discontinuously on G/K. Show that the quotient $\Gamma\backslash G/K$ is a locally symmetric space: it inherits a complete Riemannian metric with $\nabla R = 0$, and the covering map $G/K \to \Gamma\backslash G/K$ is a local isometry. Explain why the geodesic symmetries s_p descend to local, but in general not global, isometries of the quotient.

2. Exhibit the modular curve $Y(1) = \mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ as a locally symmetric space. Identify $\mathfrak{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ as its universal-cover globally symmetric space, describe the geometry of $Y(1)$ as a finite-area hyperbolic orbifold, and locate its two orbifold points and its single cusp. Compute the hyperbolic area of the standard fundamental domain $\mathcal{F}$ and confirm it is finite.

3. Explain why $\nabla R = 0$ is equivalent to the geodesic symmetries being local isometries. Show that at a point p, an isometry fixing p with differential $-\mathrm{id}$ must preserve the tensor ∇R, and that $-\mathrm{id}$ acts on ∇R (a tensor of odd total degree) by the scalar -1; conclude that $\nabla R = 0$ at p is forced, and conversely that $\nabla R = 0$ makes s_p curvature-preserving, hence a local isometry.

4. Give a compact locally symmetric space of noncompact type. Show that a closed hyperbolic surface $\Sigma_g = \Gamma\backslash\mathfrak{H}$ of genus $g \geq 2$, for Γ a torsion-free cocompact discrete subgroup of $\mathrm{SL}(2,\mathbb{R})$, is a compact locally symmetric space with $\nabla R = 0$ and constant curvature -1, whose universal cover is the globally symmetric space $\mathfrak{H}$. Explain why the genus must be at least 2 (relate it to the Gauss-Bonnet theorem and the sign of the curvature).

5. Let M be a complete locally symmetric space. Prove that its universal cover $\widetilde{M}$ is a globally symmetric space, and that $M = \Gamma\backslash\widetilde{M}$ for $\Gamma = \pi_1(M)$ a discrete group of isometries acting freely and properly discontinuously. Explain which of the three types of definition 15.3.1 the factors of $\widetilde{M}$ may have, and why a compact M with $\nabla R = 0$ and negative-definite curvature forces $\widetilde{M}$ to be of noncompact type.

6. In one paragraph, explain how the modular curve $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ draws together the four Parts of the book. Name where the symmetric space $\mathfrak{H}$, the group $\mathrm{SL}(2,\mathbb{R})$ and its Cartan/Iwasawa structure, the maximal compact $\mathrm{SO}(2)$, the Lie algebra $\mathfrak{sl}_2(\mathbb{R})$, and the arithmetic group $\mathrm{SL}(2,\mathbb{Z})$ each enter, and state what functions on the quotient are.

16

What Next?

The book has followed one dictionary through four parts. Part I built the correspondence between a Lie group and its Lie algebra, turning differential-geometric questions about the group into linear-algebraic questions about the bracket. Part II classified the semisimple Lie algebras by their root systems, reducing the structure theory to combinatorics. Part III cashed that combinatorics into the representation theory of compact groups, culminating in the Weyl character formula, which computes every irreducible character in closed form. Part IV removed compactness, organizing a noncompact real group by its Cartan and Iwasawa decompositions and reading the result geometrically as a symmetric space. What remains is to say where each of these threads is picked up.

The most direct continuation is the representation theory this book stopped short of. Chapter 14 constructed the principal series from the minimal parabolic but deferred the harmonic analysis: which representations are irreducible, which are equivalent, and how $L^2(G)$ decomposes into them. That is the theory of infinite-dimensional representations of real reductive groups, built on $(\mathfrak{g}, K)$-modules, the discrete series, and the Plancherel theorem, and it is the subject of [6]. The classic account is Knapp's *Representation Theory of Semisimple Groups*, and the Plancherel theorem for $\mathrm{SL}(2,\mathbb{R})$ worked in full is the concrete case to master first.

The harmonic analysis of the symmetric spaces of chapter 15 is a parallel road. Spherical functions, the Harish-Chandra transform, and the Plancherel theorem for G/K generalize the Fourier analysis of chapter 10 from the compact to the noncompact setting; this is developed in [16], with Helgason's *Groups and Geometric Analysis* the standard reference. The Riemannian geometry of symmetric spaces in its own right, geodesics, curvature, holonomy, and the fine classification, is carried further in [7] and in Helgason's *Differential Geometry, Lie Groups, and Symmetric Spaces*, the book from which much of chapter 15 descends.

The structure theory of Part II and Part III has an algebraic twin. Everything done here over $\mathbb{R}$ and $\mathbb{C}$ with analytic tools has a counterpart for algebraic groups over an arbitrary field, where the root datum replaces the root system and the Bruhat decomposition is a statement about a group scheme; that is the route of [10], which also develops the Borel-Weil-Bott theorem that chapter 12

only stated. The two developments meet again in the theory of p-adic groups, where the compact maximal is replaced by a maximal compact open subgroup and the Iwasawa decomposition becomes the statement $G = BK$ over a local field.

Beyond these lies the destination that motivates much of the modern subject. The locally symmetric spaces $\Gamma\backslash G/K$ at the end of chapter 15 are the arithmetic quotients on which automorphic forms live, and the representation theory of G is the local input to the Langlands program, the conjectural dictionary between automorphic representations and Galois representations. The first case, $G = \mathrm{GL}_2$, is the subject of [5], and the higher-dimensional arithmetic geometry of the Siegel upper half-space from chapter 15 opens into [8]. The Lie theory of this book is the ground floor of that structure: every automorphic representation is, at each archimedean place, a representation of a real reductive group, decomposed by exactly the Cartan and Iwasawa data assembled in Part IV.

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